

PROJECT MANUAL

for

LIDL 1517 – NORCROSS, GA
Intersection of Jimmy Carter Boulevard and Shelby Drive
Norcross, GA 30093

for



LIDL US, LLC
3500 South Clark Street
Arlington, Virginia 22202

ISSUED FOR PERMIT CYCLE 1

November 7, 2025

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SECTION 002113 – INSTRUCTIONS TO BIDDERS

1.1 INSTRUCTIONS TO BIDDERS

- A. AIA Document A701, "Instructions to Bidders," is hereby incorporated into the Procurement and Contracting Requirements by reference.

- 1. A copy of AIA Document A701, "Instructions to Bidders," is bound to owner's document, part of the bid package.

END OF DOCUMENT 002113

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SECTION 003100 - GENERAL CONDITIONS

The standard document A201–2017, “General Conditions Of The Contract For Construction” as published by the AIA (American Institute of Architects) in its entirety is hereby established as a part of owner’s document, incorporated in the bid package.

END OF DOCUMENT 003100

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SECTION 003110 - SUPPLEMENTARY CONDITIONS

ARTICLE 1 – GENERAL PROVISIONS

Refer to Owner's contract document.

ARTICLE 7 - CHANGES IN THE WORK

Refer to Owner's contract document.

ARTICLE 8 - TIME

Refer to owner's contract document.

ARTICLE 13 - MISCELLANEOUS PROVISIONS

Refer to owner's contract document.

END OF DOCUMENT 003110

SECTION 010200 – GENERAL SITEWORK REQUIREMENTS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SITEWORK LAYOUT**A. Monuments and Benchmarks**

1. Maintain all monuments, property corners, bench marks and other reference points.
2. If these are disturbed or destroyed during construction operations, have them replaced by a licensed surveyor within the project state. This replacement shall be at no additional expense to the Contract.

B. Laying out the Work.

1. Locate all existing bench marks and other reference points.
2. Protect these points throughout construction.
3. Layout work utilizing these reference points.

C. Record Drawings

1. Maintain a record of the horizontal and vertical locations of all underground utilities and piping, both newly installed and existing encountered.
2. Maintain a record of any variations of the work.
3. Record Drawings shall be certified by a registered Land Surveyor within the project state.
4. Maintain record drawings throughout construction, recording information as work is performed. Record Drawings shall be available for periodic inspection during the course of the project.
5. Submit final, complete record drawings at Substantial Completion.

1.3 EASEMENTS

- A. Verify the acquisition of all off-site easements and Rights-of-Way prior to the start of off-site construction. This may be done by contacting the Architect.
- B. Restore all off-site easements to the condition existing prior to the start of work.

1.4 MAINTENANCE OF TRAFFIC

- A. Maintain vehicular and pedestrian traffic across the frontage of this project. Comply with all applicable safety requirements.

1.5 CORRELATION OF CONSTRUCTION DOCUMENTS

- A. Review construction documents thoroughly prior to the start of construction.
- B. Report any conflict or discrepancy discovered in the Construction Documents to the Architect and/or Engineer prior to the start of construction.
- C. Report any conflict or discrepancy discovered between the Construction Documents and state and local governmental regulations to the Architect prior to the start of construction.

1.6 PROJECT CONDITIONS

- A. The conditions existing at the time of inspection for bidding purposes will be maintained by the Owner to the extent practical. However, minor variations may occur due to natural occurrences prior to the start of clearing work. The general contractor (GC) will visit the site during the bid so the minor discrepancies are covered in the bid.

- B. The location of existing underground utilities indicated is approximate only. Field locate all existing underground utilities in the area of work, regardless of whether or not they are indicated. Call "Miss Utility" prior to the start of demolition work for assistance in the location of existing underground utilities.
- C. Should charted, uncharted or incorrectly charted utilities be encountered during demolition, contact the Architect and/or Engineer immediately for instructions. Cooperate with Owner and utility companies to keep services and facilities in operation.

PART 2 - PRODUCTS

Not Applicable

PART 3 – EXECUTION

3.1 PROJECT CLEAN UP

- A. Clean site as construction progresses. Do not allow trash or other waste materials to accumulate. In addition to keeping the site clean, two full interior and exterior project clean ups will be completed prior to the handover to the Facilities Manager (FM) and the Opening. Project clean ups will be completed per item B below:
- B. Prior to requesting the punch-list inspection, clean the site to the following requirements:
 - 1. Power wash all walks and pavements.
 - 2. The remainder of the site shall be broom clean.
 - 3. Remove all trash and debris.

3.2 EXISTING FACILITIES

- A. Preserve existing signs, markers, guardrails, fences and other amenities in their original condition unless written permission is obtained for their removal and replacement.
- B. Replace damaged items at no additional cost to the Contract.

END OF SECTION 010200

SECTION 011000 – SUMMARY**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Work covered by the Contract Documents.
 - 2. Use of premises.
 - 3. Owner's occupancy requirements.
 - 4. Specification formats and conventions.
 - 5. Drawing conventions
 - 6. Precaution and Safety Requirements
 - 7. Miscellaneous Provisions; Mockups

1.3 PROJECT INFORMATION

- A. Project Identification: LIDL 1517 Norcross, GA
 - Store Location: Intersection of Jimmy Carter Boulevard and Shelby Drive,
Norcross, GA 30093
- B. Owner: LIDL US, LLC
 - 1. Owner's Representative: Tori Bauers
 - 2. Address: 3500 South Clark Street, Arlington, VA 22202
 - 3. Phone: 703.927.4890
 - 4. Email: Tori.Bauers@lidl.us
- C. Architect: Progressive Architecture Engineering & Planning I, LLC
 - 1. Architect's Representative: Alex Hengstebeck, Project Manager
 - 2. Address: 1811 4 Mile Road NE, Grand Rapids, MI 49525
 - 3. Phone: 614.893.1159
 - 4. Email: ahengstebeck@weareprogressive.com

1.4 WORK COVERED BY CONTRACT DOCUMENTS

- A. The Work of Project is defined by the Contract Documents and consists of the following:
 - 1. Work Description: Complete construction of a new one-story, stand-alone building for a fully complete grocery store. Primary building construction materials include cast-in-place concrete foundation and slab-on-grade, CMU walls (single wythe and cavity wall assemblies), structural steel system (framing members, joists and deck), single ply membrane roofing, polished concrete floors, interior partitions and finishes. Work includes all associated site, landscaping, mechanical, electrical, plumbing, fire protection and others as indicated on contract documents.

- B. Type of Contract: Project will be constructed under a single prime contract.
- C. The contractor shall pay legally required sales, consumer and use taxes.
- D. The Owner shall pay for the building permit(s).
- E. Except for permits and fees that are the responsibility of the Contractor under the Contract Documents, including those required under Section 3.7.1, the Owner shall secure and pay for necessary approvals, easements, assessments and charges required for construction, use or occupancy of permanent structures or for permanent changes in existing facilities.

1.5 WORK PERFORMED BY OWNER'S VENDORS

- A. Cooperate fully with Owner, so work may be carried out smoothly, without interfering with or delaying Work under this Contract or work by Owner's Vendors. Coordinate the Work of this Contract with work performed by Owner's Vendors.
- B. Concurrent Work: Owner's Vendors will perform certain construction operations at Project site. Those operations will be conducted simultaneously with Work under this Contract
- C. Subsequent Work: Owner's Vendors will perform additional work at site after Substantial Completion. Completion of that work will depend on successful completion of preparatory Work under this Contract.

1.6 USE OF PREMISES

- A. General: Contractor shall have full use of Project site for construction operations during construction period. Contractor's use of Project site is limited only by Owner's right to perform work or to retain other contractors on portions of Project.

1.7 OWNER'S OCCUPANCY REQUIREMENTS

- A. Owner Occupancy of Completed Areas of Construction: Owner reserves the right to occupy and to place and install equipment in completed areas of building, before Substantial Completion, provided such occupancy does not interfere with completion of the Work. Such placement of equipment and partial occupancy shall not constitute acceptance of the total Work.
 - 1. Architect will prepare a Certificate of Substantial Completion for each specific portion of the Work to be occupied before Owner occupancy.
 - 2. Obtain a Certificate of Occupancy from authorities having jurisdiction before Owner occupancy.
 - 3. Before partial Owner occupancy, mechanical and electrical systems shall be fully operational, and required tests and inspections shall be successfully completed. On occupancy, Owner will operate and maintain mechanical and electrical systems serving occupied portions of building.
 - 4. On occupancy, Owner will assume responsibility for maintenance and custodial service for occupied portions of building.

1.8 CONSTRUCTION TIME

- A. The Contractor, must provide schedule before the Contract is awarded, shall prepare and submit for the Owner's and Architect's information a Contractor's construction schedule for the Work. The schedule shall not exceed time limits current under the Contract Documents, shall be revised at appropriate intervals as required by the conditions of the Work and Project, shall be

related to the entire Project to the extent required by the Contract Documents, and shall provide for expeditious and practicable execution of the Work.

- B. The Contractor shall prepare a submittal schedule, promptly after being awarded the Contract and thereafter as necessary to maintain a current submittal schedule and shall submit the schedule(s) for the Architect's approval. The Architect's Owner's approval. The Owner's approval shall not unreasonably be delayed or withheld. The submittal schedule shall (1) be coordinated with the Contractor's construction schedule, Project Schedule, and (2) allow the Architect and Owner reasonable time to review submittals. If the Contractor fails to submit a submittal schedule, the Contractor shall not be entitled to any increase in Contract Sum or extension of Contract Time based on the time required for review of submittals.
- C. The Contractor shall perform the Work in general accordance with the most recent schedules submitted to the Owner and Architect.
- D. During the course of the Work, not less often than weekly, and at Owner's request, from time to time, the Contractor shall submit to the Architect and Owner a current, updated version of the Project Schedule. The Project Schedule shall include the Milestones set forth in the Bid Documents. Each updated Project Schedule shall provide the same detail as the initial Project Schedule, unless approved otherwise by the Owner. The Project Schedule shall conform to any changes in the Contract Time set out in Change Orders. The Project Schedule shall not be extended without the prior written approval of the Owner in each instance. The Project Schedule shall also: (1) provide a graphic representation of all activities and events that will occur during performance of the Work; (2) identify each phase of construction and occupancy; and (3) set forth critical path dates to ensure the timely and orderly Final Completion of the Work in accordance with the requirements of the Contract Documents.

1.9 WORK RESTRICTIONS

- A. Work Restrictions, General: Comply with restrictions on construction operations. The Contractor shall assume full responsibility for protection and safekeeping of all products under this Contract, which are stored on the site.
 - 1. Comply with limitations on use of public streets and other requirements of authorities having jurisdiction.
- B. On-Site Work Hours: Limit work within the entire project site to normal business working hours of 7:00 a.m. to 5:00 p.m., Monday through Friday, except as otherwise indicated.
 - 1. Weekend hours and early morning hours: Only with jurisdictional approval.
- C. Existing Utility Interruptions: Do not interrupt utilities serving any adjacent properties.
- D. Noise, Vibration, and Odors: Coordinate operations that may result in high levels of noise, vibration, odors, or other disruption to adjacent properties. The levels of objectionable noise, vibration, odors or other disruptions will be at the sole determination of the Owner and Project Manager.
- E. Nonsmoking Building: The use of tobacco products is not permitted within the building or within 25 feet (8 m) of entrances, operable windows, or outdoor air intakes.
- F. Controlled Substances: Use of controlled substances within the building or on the project site is not permitted. The possession, under the influence, and use of drugs and/or alcohol while on the project property is strictly forbidden by the Owner.

1.10 SPECIFICATION FORMATS AND CONVENTIONS

- A. Specification Format: The Specifications are organized into Divisions and Sections using the division format and CSI/CSC's "MasterFormat" numbering system.
 - 1. Division 01: Sections in Division 01 govern the execution of the Work of all Sections in the Specifications.
- B. Specification Content: The Specifications use certain conventions for the style of language and the intended meaning of certain terms, words, and phrases when used in particular situations. These conventions are as follows:
 - 1. Abbreviated Language: Language used in the Specifications and other Contract Documents is abbreviated. Words and meanings shall be interpreted as appropriate. Words implied, but not stated, shall be inferred as the sense requires. Singular words shall be interpreted as plural and plural words shall be interpreted as singular where applicable as the context of the Contract Documents indicates.
 - 2. Imperative mood and streamlined language are generally used in the Specifications. Requirements expressed in the imperative mood are to be performed by Contractor. Occasionally, the indicative or subjunctive mood may be used in the Section Text for clarity to describe responsibilities that must be fulfilled indirectly by Contractor or by others when so noted.
 - a. The words "shall," "shall be," or "shall comply with," depending on the context, are implied where a colon (:) is used within a sentence or phrase.
 - 3. Specification requirements are to be performed by Contractor unless specifically stated otherwise.

1.11 DRAWING CONVENTIONS

- A. Division 1 General Requirements: Requirements of Sections in Division 1 apply to the Work of all Sections in the Specifications.
- B. Drawing Coordination: Requirements for materials and products identified on the Drawings are described in detail in the Specifications. One or more of the following are used on the Drawings to identify materials and products:
 - 1. Terminology: Materials and products are identified by the typical generic terms used in the individual Specifications Sections.

1.12 PRECAUTION AND SAFETY REQUIREMENTS

- A. Fire Protection: Provide and maintain an adequate number of hand fire extinguishers at convenient and appropriate locations during construction. Avoid all accumulations of flammable debris by removing rubbish promptly. Take all other precautions necessary to prevent fire. Supervise closely the storage of paint materials and other combustible products.
- B. Accident Prevention and Safety: Comply with all applicable laws, ordinances, rules, regulations and orders of governing authorities having jurisdiction for the safety of persons and property to protect them from damage, injury or loss. Erect and maintain, as required for safety and protection, including fences, railings, barricades, lighting, posting of danger signs and other warnings against hazards. Where prevention of construction accidents is not regulated by code or ordinance, comply with AGC's "Manual of Accident Prevention in Construction." Contractor

shall be solely responsible for initiating, maintaining and supervising all safety precautions and programs in connection with the Project. All scaffolds shall be built in accordance with all requirements of local, state, and Federal laws and regulations.

1. Project safety is the responsibility of the Contractor.

1.13 MISCELLANEOUS PROVISIONS; MOCKUPS

- A. Exterior and Interior Mockups: Construct a mockup of the typical exterior / interior construction per each division, as directed by the Architect.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 011000

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SECTION 012110 – SITEWORK ALLOWANCES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section specifies administrative and procedural requirements regarding allowances and unit prices associated with sitework.
- B. This Section includes the following unit price rates:
1. Additional Excavation
 2. Additional Excavation in Trenches
 3. Additional Excavation in Footings.
 4. Mass Rock Removal
 5. Trench Rock Removal
 6. Subsurface Drain
 7. Unit cost for open graded aggregate, dense graded aggregate, and large open graded aggregate
 8. Woven Geotextile Fabric
 9. Hourly rate for skilled labor (electrician/plumber/etc)
 10. Hourly rate for foreman and unskilled labor
- C. Hourly rate for Foreman and unskilled labor This Section includes the following lump-sum allowances:
1. Firelane Striping and Signage
 2. Lump sum for classified site

PART 2 – PRODUCTS (Not Applicable)**PART 3 - EXECUTION****3.1 SCHEDULE OF UNIT PRICE ALLOWANCES**

- A. Additional Excavation
1. Provide in the Base Bid an allowance of XXXX c.y. for excavation of material, where authorized or directed, below or in addition to the levels required for the Work. Dispose of excavated material in an approved location on-site. Backfill with imported structural fill material compacted per specifications. Credit or additions to the Contract Price for actual quantities removed and replaced (based on volume of material cut) shall be made per the Unit Prices contained in the Bid Form. Include in the unit price the cost of quantity verification by a Surveyor Licensed in the project state.

B. Additional Excavation in Trenches

1. Provide in the Base Bid an allowance of XXXX c.y. for excavation of material in trenches and footings, where authorized or directed, below or in addition to the levels required for the Work. Dispose of excavated material in an approved location on-site. Backfill with stone compacted per specifications. Credit or additions to the Contract Price for actual quantities removed and replaced (based on volume of material cut) shall be made per the Unit Prices contained in the Bid Form. Include in the unit price the cost of quantity verification by a Surveyor Licensed in the project state

C. Additional Excavation in Footings

1. Provide in the Base Bid an allowance of XXXX c.y. for excavation of material in trenches and footings, where authorized or directed, below or in addition to the levels required for the Work. Dispose of excavated material in an approved location on-site. Backfill with stone compacted per specifications. Credit or additions to the Contract Price for actual quantities removed and replaced (based on volume of material cut) shall be made per the Unit Prices contained in the Bid Form. Include in the unit price the cost of quantity verification by a Surveyor Licensed in the project state. Any required lowering of footings will be done as a negotiated price.

D. Mass Rock Excavation

1. Provide in the Base Bid an allowance of XXXX c.y. for excavation of rock. Dispose of excavated rock material in an approved location on-site. Backfill with imported fill material compacted per specifications. Credit or additions to the Contract Price for actual quantities removed and replaced (per the pay limits established in the Specifications) shall be made per the Unit Prices contained in the Bid Form. Include in the unit price the cost of quantity verification by a Surveyor Licensed in the project state.

E. Rock Excavation in Trenches

1. Provide in the Base Bid an allowance of XXXX c.y. for excavation of rock in trenches. Dispose of excavated rock material in an approved location on-site. Backfill excavation with stone. Credit or additions to the Contract Price for actual quantities removed and replaced (per the pay limits established in the Specifications) shall be made per the Unit Prices contained in the Bid Form. Include in the unit price the cost of quantity verification by a Surveyor Licensed in the project state .

F. Rock excavation at Footings

1. Provide in the Base Bid an allowance of XXXX c.y. for excavation of rock in footings. Dispose of excavated rock material in an approved location on-site. Backfill excavation with stone. Credit or additions to the Contract Price for actual quantities removed and replaced (per the pay limits established in the Specifications) shall be made per the Unit Prices contained in the Bid Form. Include in the unit price the cost of quantity verification by a Surveyor Licensed in the project state.

G. Subsurface Drain

1. Provide in the Base Bid an allowance of XXXX l.f. of subsurface drain as detailed on Drawing C-X.

H. Aggregate

1. Provide in the base bid an allowance of XXXX c.y. of aggregate needed for subgrade preparation. Credit or additions to the Contract Price for actual quantities removed and replaced (per the pay limits established in the Specifications) shall be made per the Unit Prices contained in the Bid Form. Include in the unit price the cost of quantity verification by a Surveyor Licensed in the project state.

I. Woven Geotextile Fabric, in place (S.Y.)

3.2 SCHEDULE OF LUMP SUM ALLOWANCES

A. Firelane Striping and Signage

1. Provide in the Base Bid an allowance of \$ XXXX l.f. for firelane striping and signage as directed by the Architect and or Engineer.

B. Classified Site

1. Provide in the base bid an allowance of \$ XXXX for classified site condition as directed by the Owner

3.3 ADMINISTRATION OF SITEWORK UNIT PRICE ALLOWANCES

A. Unit Prices for each allowance shall be given on the Bid Form.

1. The Owner reserves the right to negotiate said Unit Prices prior to the award of Contract. The Unit Prices are subject to negotiation even after the award of the contract if it varies greatly from the RS Means Standards.

B. Allowances required by this Section shall be included in the Base Bid amount.

C. Allowances required by this Section shall be indicated on the Schedule of Values and shall be determined by multiplying the quantity indicated by the unit price given on the Bid Form.

D. Submit invoices or surveyor's certificate, as required, with pay requests that involve the Unit Price Allowances.

E. Credit unused amount of Unit Price Allowance (if any) to Owner by Change Order at Project Closeout.

3.4 ADMINISTRATION OF LUMP-SUM ALLOWANCES

A. Conform to the requirements of the General Conditions.

END OF SECTION 012110

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SECTION 012600 – CONTRACT MODIFICATION PROCEDURES

PART 1 - GENERAL

Refer to owner's contract document.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 012600

SECTION 012900 - PAYMENT PROCEDURES

PART 1 - GENERAL

Refer to Owner's contract document.

PART 3 - PRODUCTS (Not Used)

PART 2 - EXECUTION (Not Used)

END OF SECTION 012900

SECTION 013100 - PROJECT MANAGEMENT AND COORDINATION**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following:
 - 1. Coordination.
 - 2. Digital project management.
 - 3. Project meetings.
 - 4. Requests for Interpretation (RFIs).
- B. See Division 01 Section "Execution" for procedures for coordinating general installation and field-engineering services, including establishment of benchmarks and control points.
- C. See Division 01 Section 019113 "General Commissioning Requirements" for coordinating the Work with Owner's Commissioning Authority.

1.3 DEFINITIONS

- A. RFI: Request from Contractor seeking interpretation or clarification of the Contract Documents.

1.4 COORDINATION

- A. Coordination: Coordinate construction operations included in different Sections of the Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations included in different sections that depend on each other for proper installation, connection, and operation.
 - 1. Schedule construction operations in sequence required to obtain the best results where installation of one part of the Work depends on installation of other components, before or after its own installation.
 - 2. Coordinate installation of different components with other contractors to ensure maximum accessibility for required maintenance, service, and repair.
 - 3. Make adequate provisions to accommodate items scheduled for later installation.
 - 4. Where availability of space is limited, coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair of all components, including mechanical and electrical.
- B. Prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.
 - 1. Prepare similar memoranda for Owner and separate contractors if coordination of their Work is required.

- C. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities and activities of other contractors to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to, the following:

1. Preparation of Contractor's Construction Schedule.
2. Preparation of the Schedule of Values or milestone payment.
3. Installation and removal of temporary facilities and controls.
4. Delivery and processing of submittals.
5. Progress meetings.
6. Preinstallation conferences.
7. Project closeout activities.
8. Startup and adjustment of systems.
9. Project closeout activities.

1.5 DIGITAL PROJECT MANAGEMENT PROCEDURES

- A. Use of Architect's Digital Data Files: Digital data files of Architect's will be provided by Architect for Contractor's use during construction.

1. Digital data files may be used by Contractor in preparing coordination drawings, Shop Drawings, and Project record Drawings.
2. Architect makes no representations as to the accuracy or completeness of digital data files as they relate to Contract Drawings.
3. Contractor shall execute a data licensing agreement in the form of Agreement included in Project Manual.

- B. Web-Based Project Software: ProjectMates, Owner to provide, administer, and use web-based Project software site for purposes of hosting and managing Project communication and documentation until Final Completion.

1. Web-based Project software site includes the following features:
 - a. Compilation of Project data, including Contractor, subcontractors, Architect, architect's consultants, Owner, and other entities involved in Project. Include names of individuals and contact information.
 - b. Access control for each entity for each workflow process, to determine entity's digital rights to access, modify or delete documents.
 - c. Document workflow planning, allowing customization of workflow between project entities.
 - d. Creation, logging, tracking, and notification for Project communications required in other Specification Sections, including, but not limited to, RFIs, submittals, Minor Changes in the Work, Construction Change Directives, and Change Orders.
 - e. Track status of each Project communication in real time, and log time and date when responses are provided.
 - f. Processing and tracking of payment applications.
 - g. Creating and distributing meeting minutes.
 - h. Document management for Drawings, Specifications, and coordination drawings, including revision control.
2. At completion of Project, provide digital archive in format that is readable by common desktop software applications in format acceptable to Architect. Provide data in locked format to prevent further changes.

- C. PDF Document Preparation: Where PDFs are required to be submitted to Architect, prepare as follows:
1. Assemble complete submittal package into a single indexed file incorporating submittal requirements of a single Specification Section and transmittal form with links enabling navigation to each item.
 2. Name file with submittal number or other unique identifier, including revision identifier.
 3. Certifications: Where digitally submitted certificates and certifications are required, provide a digital signature with digital certificate on where indicated.

1.6 PROJECT MEETINGS

- A. General: Schedule and conduct meetings and conferences at Project site, unless otherwise indicated.
1. Attendees: Inform participants and others involved, and individuals whose presence is required, of date and time of each meeting. Notify Owner and Architect of scheduled meeting dates and times.
 2. Agenda: Prepare the meeting agenda. Distribute the agenda to all invited attendees.
 3. Minutes: Record significant discussions and agreements achieved. Distribute the meeting minutes to everyone concerned, including Owner and Architect, within three days of the meeting.
- B. Preconstruction Conference / Project Kick-Off: Schedule a preconstruction conference before starting construction, at a time convenient to Owner and Architect, but no later than 15 days after execution of the Agreement. Hold the conference at Project site or another convenient location. Conduct the meeting to review responsibilities and personnel assignments.
1. Attendees: Authorized representatives of Owner, Architect, and their consultants; Contractor and its superintendent; major subcontractors; suppliers; and other concerned parties shall attend the conference. All participants at the conference shall be familiar with Project and authorized to conclude matters relating to the Work.
 2. Agenda: Discuss items of significance that could affect progress, including the following:
 - a. Tentative construction schedule.
 - b. Phasing.
 - c. Critical work sequencing and long-lead items.
 - d. Designation of key personnel and their duties.
 - e. Procedures for processing field decisions and Change Orders.
 - f. Procedures for RFIs.
 - g. Procedures for testing and inspecting.
 - h. Procedures for processing Applications for Payment.
 - i. Distribution of the Contract Documents.
 - j. Submittal procedures.
 - k. Preparation of Record Documents.
 - l. Use of the premises and existing building.
 - m. Work restrictions.
 - n. Owner's occupancy requirements.
 - o. Responsibility for temporary facilities and controls.
 - p. Construction waste management and recycling.
 - q. Parking availability.
 - r. Office, work, and storage areas.
 - s. Equipment deliveries and priorities.
 - t. First aid.
 - u. Security.

- v. Progress cleaning.
 - w. Working hours.
- 3. Minutes: Record and distribute meeting minutes.
- C. Sustainable Design Requirements Coordination Conference: Owner will schedule and conduct a sustainable design coordination conference before starting construction, at a time convenient to Owner, Architect, and Contractor.
 - 1. Attendees: Authorized representatives of Owner, Owner's Commissioning Authority, Architect, and their consultants; Contractor and its superintendent and sustainable design coordinator; major subcontractors; suppliers; and other concerned parties shall attend the conference. Participants at the conference shall be familiar with Project and authorized to conclude matters relating to the Work.
 - 2. Agenda: Discuss items of significance that could affect meeting sustainable design requirements, including the following:
 - a. Sustainable design Project checklist.
 - b. General requirements for sustainable design-related procurement and documentation.
 - c. Project closeout requirements and sustainable design certification procedures.
 - d. Role of sustainable design coordinator.
 - e. Construction waste management.
 - f. Construction operations and sustainable design requirements and restrictions.
 - 3. Minutes: Entity responsible for conducting meeting will record and distribute meeting minutes.
- D. Preinstallation Conferences: Conduct a preinstallation conference at Project site before each construction activity that requires coordination with other construction.
 - 1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by the installation and its coordination or integration with other materials and installations that have preceded or will follow, shall attend the meeting. Advise Architect of scheduled meeting dates.
 - 2. Agenda: Review progress of other construction activities and preparations for the particular activity under consideration, including requirements for the following:
 - a. The Contract Documents.
 - b. Options.
 - c. Related RFIs.
 - d. Related Change Orders.
 - e. Purchases.
 - f. Deliveries.
 - g. Submittals.
 - h. Review of mockups.
 - i. Possible conflicts.
 - j. Compatibility problems.
 - k. Time schedules.
 - l. Weather limitations.
 - m. Manufacturer's written recommendations.
 - n. Warranty requirements.
 - o. Compatibility of materials.
 - p. Acceptability of substrates.
 - q. Temporary facilities and controls.

- r. Space and access limitations.
 - s. Regulations of authorities having jurisdiction.
 - t. Testing and inspecting requirements.
 - u. Installation procedures.
 - v. Coordination with other work.
 - w. Required performance results.
 - x. Protection of adjacent work.
 - y. Protection of construction and personnel.
 - 3. Record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.
 - 4. Reporting: Distribute minutes of the meeting to each party present and to parties who should have been present.
 - 5. Do not proceed with installation if the conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to performance of the Work and reconvene the conference at earliest feasible date.
- E. Progress Meetings: Conduct progress meetings at weekly intervals.
- 1. Attendees: In addition to representatives of Owner and Architect, each contractor, subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented at these meetings. All participants at the conference shall be familiar with Project and authorized to conclude matters relating to the Work.
 - 2. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Include topics for discussion as appropriate to status of Project.
 - a. Contractor's Construction Schedule: Review progress since the last meeting. Determine whether each activity is on time, ahead of schedule, or behind schedule, in relation to Contractor's Construction Schedule. Determine how construction behind schedule will be expedited; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities will be completed within the Contract Time.
 - 1) Review schedule for next period.
 - b. Review present and future needs of each entity present, including the following:
 - 1) Interface requirements.
 - 2) Sequence of operations.
 - 3) Status of submittals.
 - 4) Deliveries.
 - 5) Off-site fabrication.
 - 6) Access.
 - 7) Site utilization.
 - 8) Temporary facilities and controls.
 - 9) Work hours.
 - 10) Hazards and risks.
 - 11) Progress cleaning.
 - 12) Quality and work standards.
 - 13) Status of correction of deficient items.
 - 14) Field observations.
 - 15) RFIs.
 - 16) Status of proposal requests.

- 17) Pending changes.
 - 18) Status of Change Orders.
 - 19) Pending claims and disputes.
 - 20) Documentation of information for payment requests.
3. Minutes: Record the meeting minutes.
 4. Reporting: Distribute minutes of the meeting to each party present and to parties who should have been present.
 - a. Schedule Updating: Revise Contractor's Construction Schedule after each progress meeting where revisions to the schedule have been made or recognized. Issue revised schedule concurrently with the report of each meeting.

1.7 REQUESTS FOR INTERPRETATION (RFIs)

- A. Procedure: Immediately on discovery of the need for interpretation of the Contract Documents, and if not possible to request interpretation at Project meeting, prepare and submit an RFI in the form specified.
 1. RFIs shall originate with Contractor. RFIs submitted by entities other than Contractor will be returned with no response.
 2. Coordinate and submit RFIs in a prompt manner so as to avoid delays in Contractor's work or work of subcontractors.
- B. Content of the RFI: Include a detailed, legible description of item needing interpretation and the following:
 1. Project name.
 2. Date.
 3. Name of Contractor.
 4. Name of Architect.
 5. RFI number, numbered sequentially.
 6. Specification Section number and title and related paragraphs, as appropriate.
 7. Drawing number and detail references, as appropriate.
 8. Field dimensions and conditions, as appropriate.
 9. Contractor's suggested solution(s). If Contractor's solution(s) impact the Contract Time or the Contract Sum, Contractor shall state impact in the RFI.
 10. Contractor's signature.
 11. Attachments: Include drawings, descriptions, measurements, photos, Product Data, Shop Drawings, and other information necessary to fully describe items needing interpretation.
- C. Hard-Copy RFIs: Contractor's standard form.
 1. Identify each page of attachments with the RFI number and sequential page number.
- D. Architect's Action: Architect will review each RFI, determine action required, and return it. Allow seven working days for Architect's response for each RFI. RFIs received after 1:00 p.m. will be considered as received the following working day.
 1. The following RFIs will be returned without action:
 - a. Requests for approval of submittals.
 - b. Requests for approval of substitutions.

- c. Requests for coordination information already indicated in the Contract Documents.
 - d. Requests for adjustments in the Contract Time or the Contract Sum.
 - e. Requests for interpretation of Architect's actions on submittals.
 - f. Incomplete RFIs or RFIs with numerous errors.
- 2. Architect's action may include a request for additional information, in which case Architect's time for response will start again.
- 3. Architect's action on RFIs that may result in a change to the Contract Time or the Contract Sum may be eligible for Contractor to submit Change Proposal according to Division 01 Section "Contract Modification Procedures."
 - a. If Contractor believes the RFI response warrants change in the Contract Time or the Contract Sum, notify Architect in writing within 5 days of receipt of the RFI response.
- E. On receipt of Architect's action, update the RFI log and immediately distribute the RFI response to affected parties. Review response and notify Architect within five days if Contractor disagrees with response.
- F. RFI Log: Prepare, maintain, and submit a tabular log of RFIs organized by the RFI number. Submit log weekly. Include the following:
 - 1. Project name.
 - 2. Name and address of Contractor.
 - 3. Name and address of Architect.
 - 4. RFI number including RFIs that were dropped and not submitted.
 - 5. RFI description.
 - 6. Date the RFI was submitted.
 - 7. Date Architect's response was received.
 - 8. Identification of related Minor Change in the Work, Construction Change Directive, and Proposal Request, as appropriate.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 013100

SECTION 013200 - CONSTRUCTION PROGRESS DOCUMENTATION**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for documenting the progress of construction during performance of the Work, including the following:
 - 1. Contractor's Construction Schedule.
 - 2. Submittals Schedule.
 - 3. Construction reports.
 - 4. Field condition reports.
- B. See Division 01 Section "Payment Procedures" for submitting the Schedule of Values.

1.3 DEFINITIONS

- A. Activity: A discrete part of a project that can be identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources.
 - 1. Critical activities are activities on the critical path. They must start and finish on the planned early start and finish times.
 - 2. Predecessor Activity: An activity that precedes another activity in the network.
 - 3. Successor Activity: An activity that follows another activity in the network.
- B. CPM: Critical path method, which is a method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine the critical path of Project and when activities can be performed.
- C. Critical Path: The longest connected chain of interdependent activities through the network schedule that establishes the minimum overall Project duration and contains no float.
- D. Float: The measure of leeway in starting and completing an activity.
 - 1. Float time is not for the exclusive use or benefit of either Owner or Contractor, but is a jointly owned, expiring Project resource available to both parties as needed to meet schedule milestones and Contract completion date.
- E. Fragnet: A partial or fragmentary network that breaks down activities into smaller activities for greater detail.
- F. Major Area: A story of construction, a separate building, or a similar significant construction element.

1.4 SUBMITTALS

- A. Format for Submittals: Submit required submittals in the following format:

1. PDF electronic file.
 - B. Submittals Schedule: Arrange the following information in a tabular format:
 1. Scheduled date for first submittal.
 2. Specification Section number and title.
 3. Submittal category (action or informational).
 4. Name of subcontractor.
 5. Description of the Work covered.
 6. Scheduled date for Architect's final release.
 - C. Contractor's Construction Schedule: Submit initial schedule showing entire schedule for entire construction period.
 1. Submit an electronic copy of schedule, using software indicated and labeled to comply with requirements for submittals. Include type of schedule (Initial or Updated) and date on label.
 - D. CPM Reports: Concurrent with CPM schedule, submit each of the following reports. Format for each activity in reports shall contain activity number, activity description, cost and resource loading, original duration, remaining duration, early start date, early finish date, late start date, late finish date, and total float in calendar days.
 1. Activity Report: List of all activities sorted by activity number and then early start date, or actual start date if known.
 2. Logic Report: List of preceding and succeeding activities for all activities, sorted in ascending order by activity number and then early start date, or actual start date if known.
 3. Total Float Report: List of all activities sorted in ascending order of total float.
 - E. Construction Reports: Maintain copy of all reports at Contractor's field office for Architect's and Owner's review.
 - F. Field Condition Reports: Submit two copies at time of discovery of differing conditions.
- 1.5 COORDINATION
- A. Coordinate preparation and processing of schedules and reports with performance of construction activities and with scheduling and reporting of separate contractors.
 - B. Coordinate Contractor's Construction Schedule with the Schedule of Values, list of subcontracts, Submittals Schedule, progress reports, payment requests, and other required schedules and reports.
 1. Secure time commitments for performing critical elements of the Work from parties involved.
 2. Coordinate each construction activity in the network with other activities and schedule them in proper sequence.

PART 2 - PRODUCTS

2.1 CONTRACTOR'S CONSTRUCTION SCHEDULE, GENERAL

- A. Gantt-Chart Schedule: Submit a comprehensive, fully developed, horizontal, Gantt-chart-type, Contractor's Construction Schedule within 30 days of date established for the Notice to Proceed.
 - 1. Time Frame: Extend schedule from date established for the Notice to Proceed to date of Final Completion.
 - a. Contract completion date shall not be changed by submission of a schedule that shows an early completion date, unless specifically authorized by Change Order.
- B. Preparation: Indicate each significant construction activity separately. Identify first workday of each week with a continuous vertical line.
- C. Activities: Treat each story or separate area as a separate numbered activity for each principal element of the Work. Comply with the following:
 - 1. Activity Duration: Define activities so no activity is longer than 20 days, unless specifically allowed by Architect.
 - 2. Procurement Activities: Include procurement process activities for the following long lead items and major items, requiring a cycle of more than 60 days, as separate activities in schedule. Procurement cycle activities include, but are not limited to, submittals, approvals, purchasing, fabrication, and delivery.
 - 3. Startup and Testing Time: Include not less than 30 days for startup and testing.
 - 4. Substantial Completion: Indicate completion in advance of date established for Substantial Completion and allow time for Architect's administrative procedures necessary for certification of Substantial Completion.
 - 5. Final Completion: Indicate completion in advance date established for Final Completion and allow time for the General Contractor's procedures necessary for items included but not related to final billing, warranty, final release of liens, as-builts, etc.
- D. Constraints: Include constraints and work restrictions indicated in the Contract Documents and as follows in schedule and show how the sequence of the Work is affected.
 - 1. Work Restrictions: Show the effect of the following items on the schedule:
 - a. Coordination with existing construction.
 - b. Limitations of continued occupancies.
 - c. Uninterruptible services.
 - d. Partial occupancy before Substantial Completion.
 - e. Use of premises restrictions.
 - f. Provisions for future construction.
 - g. Seasonal variations.
 - h. Environmental control.
 - 2. Work Stages: Indicate important stages of construction for each major portion of the Work.
 - 3. Other Constraints: Owner's on-going on-site business operations.
- E. Milestones: Include milestones indicated in the Contract Documents in schedule, including, but not limited to, the Notice to Proceed, Substantial Completion, and Final Completion.
- F. Contract Modifications: For each proposed contract modification and concurrent with its submission, prepare a time-impact analysis using fragnets to demonstrate the effect of the proposed change on the overall project schedule.

2.2 REPORTS

- A. Construction Reports: Prepare a two-week look ahead and one week look back report recording the following information concerning events at Project site:
1. List of subcontractors at Project site.
 2. Equipment at Project site.
 3. Material deliveries.
 4. High and low temperatures and general weather conditions.
 5. Accidents.
 6. Stoppages, delays, shortages, and losses.
 7. Meter readings and similar recordings.
 8. Orders and requests of authorities having jurisdiction.
 9. Services connected and disconnected.
 10. Equipment or system tests and startups.
- B. Field Condition Reports: Immediately on discovery of a difference between field conditions and the Contract Documents, prepare and submit a detailed report. Submit with a request for interpretation. Include a detailed description of the differing conditions, together with recommendations for changing the Contract Documents.
- C. Weather Delays: Prepare weather delay report, should the project encounter weather delays which is more than regional normal. This should include historical record of the weather pattern and what was encountered during the life cycle of the project.

PART 3 - EXECUTION

3.1 CONTRACTOR'S CONSTRUCTION SCHEDULE

- A. Contractor's Construction Schedule Updating: At weekly intervals, update overall schedule to reflect actual construction progress and activities. Issue schedule one week before each regularly scheduled progress meeting.
1. Revise schedule immediately after each meeting or other activity where revisions have been recognized or made. Issue updated schedule concurrently with the report of each such meeting.
 2. Include a report with updated schedule that indicates every change, including, but not limited to, changes in logic, durations, actual starts and finishes, and activity durations.
 3. As the Work progresses, indicate Actual Completion percentage for each activity.
- B. Distribution: Distribute copies of approved schedule to Architect Owner, separate contractors, testing and inspecting agencies, and other parties identified by Contractor with a need-to-know schedule responsibility.
1. Post copies in Project meeting rooms and temporary field offices.
 2. When revisions are made, distribute updated schedules to the same parties and post in the same locations. Delete parties from distribution when they have completed their assigned portion of the Work and are no longer involved in performance of construction activities.

END OF SECTION 013200

SECTION 013300 - SUBMITTAL PROCEDURES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other submittals.
- B. Related Sections:
 - 1. See Division 01 Section "Construction Progress Documentation" for submitting schedules and reports, including Contractor's Construction Schedule.
 - 2. See Division 01 Section "Quality Requirements" for submitting test and inspection reports and for mockup requirements.
 - 3. See Division 01 Section "Closeout Procedures" for submitting warranties.
 - 4. See Division 01 Section "Project Record Documents" for submitting Record Drawings, Record Specifications, and Record Product Data.
 - 5. See Division 01 Section "Operation and Maintenance Data" for submitting operation and maintenance manuals.
 - 6. See Division 01 Section "Demonstration and Training" for submitting videotapes of demonstration of equipment and training of Owner's personnel.

1.3 DEFINITIONS

- A. Action Submittals: Written and graphic information that requires Architect's responsive action.
- B. Informational Submittals: Written information that does not require Architect's responsive action. Submittals may be rejected for not complying with requirements.
- C. Conformance Submittals: Written and graphic information is required. However, electronic submission via Owner's Project Management software platform will not be reviewed by the Architect, Engineer or Owner.

1.4 SUBMITTAL PROCEDURES

- A. Coordination: Coordinate preparation and processing of submittals with performance of construction activities.
 - 1. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
 - 2. Coordinate transmittal of different types of submittals for related parts of the Work so processing will not be delayed because of need to review submittals concurrently for coordination.
 - a. Architect reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.

- B. Submittals Schedule: Comply with requirements in Division 01 Section "Construction Progress Documentation" for list of submittals and time requirements for scheduled performance of related construction activities.
- C. Processing Time: Allow enough time for submittal review, including time for resubmittals, as follows. Time for review shall commence on Architect's receipt of submittal. No extension of the Contract Time will be authorized because of failure to transmit submittals enough in advance of the Work to permit processing, including resubmittals.
1. Initial Review: Allow 7 calendar days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required. Architect will advise Contractor when a submittal being processed must be delayed for coordination.
 2. Intermediate Review: If intermediate submittal is necessary, process it in same manner as initial submittal.
 3. Resubmittal Review: Allow 7 calendar days for review of each resubmittal.
- D. Identification: Place a permanent label or title block on each submittal for identification.
1. Indicate name of firm or entity that prepared each submittal on label or title block.
 2. Provide a space approximately 6 by 8 inches on label or beside title block to record Contractor's review and approval markings and action taken by Architect.
 3. Include the following information on label for processing and recording action taken:
 - a. Project name.
 - b. Date.
 - c. Name and address of Architect.
 - d. Name and address of Contractor.
 - e. Name and address of subcontractor.
 - f. Name and address of supplier.
 - g. Name of manufacturer.
 - h. Submittal number or other unique identifier, including revision identifier.
 - 1) Submittal number shall use Specification Section number followed by a decimal point and then a sequential number (e.g., 061000.01). Resubmittals shall include an alphabetic suffix after another decimal point (e.g., 061000.01.A).
 - i. Number and title of appropriate Specification Section.
 - j. Drawing number and detail references, as appropriate.
 - k. Location(s) where product is to be installed, as appropriate.
 - l. Other necessary identification.
- E. Deviations: Highlight, encircle, or otherwise specifically identify deviations from the Contract Documents on submittals.
- F. Additional Copies: Unless additional copies are required for final submittal, and unless Architect observes noncompliance with provisions in the Contract Documents, initial submittal may serve as final submittal.
1. Additional copies submitted for maintenance manuals will not be marked with action taken and will be returned.

- G. Transmittal: Package each submittal individually and appropriately for transmittal and handling. Transmit each submittal using a transmittal form. Architect will discard submittals received from sources other than Contractor.
 - 1. Transmittal Form: Use Contractor's standard form.
- H. Resubmittals: Make resubmittals in same form and number of copies as initial submittal.
 - 1. Note date and content of previous submittal.
 - 2. Note date and content of revision in label or title block and clearly indicate extent of revision.
 - 3. Resubmit submittals until they are marked "APPROVED", "NO EXCEPTIONS TAKEN", "APPROVED AS NOTED", or similar language.
- I. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, authorities having jurisdiction, and others as necessary for performance of construction activities. Show distribution on transmittal forms.
- J. Use for Construction: Use only final submittals with mark indicating "APPROVED", "NO EXCEPTIONS TAKEN", "APPROVED AS NOTED", "NOTE MARKINGS", or similar language as taken by Architect.

1.5 CONTRACTOR'S USE OF ARCHITECT'S CAD FILES

- A. General: At Contractor's written request, copies of Architect's Floor Plan in DWF or PDF file format will be provided to Contractor for Contractor's use in connection with Project, subject to the following conditions:
 - 1. Contractor must submit to Architect signed Electronic File Transfer Agreement prior to release of electronic files. Agreement form is provided by the Architect and is attached to the end of this section.

PART 2 - PRODUCTS

2.1 ACTION SUBMITTALS

- A. See "Submittal List" at the end of this section for products that qualify as Action Submittals.
- B. General: Prepare and submit Action Submittals required by individual Specification Sections.
 - 1. Submit electronic submittals via email as PDF electronic files.
 - a. Architect will return annotated file. Annotate and retain one copy of file as an electronic Project record document file.
- C. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.
 - 1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
 - 2. Mark each copy of each submittal to show which products and options are applicable.
 - 3. Include the following information, as applicable:
 - a. Manufacturer's written recommendations.

- b. Manufacturer's product specifications.
 - c. Manufacturer's installation instructions.
 - d. Manufacturer's catalog cuts.
 - e. Wiring diagrams showing factory-installed wiring.
 - f. Printed performance curves.
 - g. Operational range diagrams.
 - h. Compliance with specified referenced standards.
 - i. Testing by recognized testing agency.
 - 4. Submit PDF electronic files.
- D. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data.
 - 1. Preparation: Fully illustrate requirements in the Contract Documents. Include the following information, as applicable:
 - a. Dimensions.
 - b. Identification of products.
 - c. Fabrication and installation drawings.
 - d. Roughing-in and setting diagrams.
 - e. Wiring diagrams showing field-installed wiring, including power, signal, and control wiring.
 - f. Shopwork manufacturing instructions.
 - g. Templates and patterns.
 - h. Schedules.
 - i. Notation of coordination requirements.
 - j. Notation of dimensions established by field measurement.
 - k. Relationship to adjoining construction clearly indicated.
 - l. Seal and signature of professional engineer if specified.
 - m. Wiring Diagrams: Differentiate between manufacturer-installed and field-installed wiring.
 - 2. Submit PDF electronic files.
- E. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other elements and for a comparison of these characteristics between submittal and actual component as delivered and installed.
 - 1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.
 - 2. Identification: Attach label on unexposed side of Samples that includes the following:
 - a. Generic description of Sample.
 - b. Product name and name of manufacturer.
 - c. Sample source.
 - d. Number and title of appropriate Specification Section.
 - 3. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout the course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
 - 4. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available.

- a. Number of Samples: Submit one full set(s) of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Architect will return submittal with options selected.
- 5. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from same material to be used for the Work, cured and finished in manner specified, and physically identical with material or product proposed for use, and that show full range of color and texture variations expected. Samples include, but are not limited to, the following: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.
 - a. Number of Samples: Submit three sets of Samples. Architect will retain one Sample set; remainder will be returned. Mark up and retain one returned Sample set as a Project Record Sample.
- F. Submittals Schedule: Comply with requirements specified in Division 01 Section "Construction Progress Documentation."
- G. Application for Payment: Comply with requirements specified in Division 01 Section "Payment Procedures."
- H. Schedule of Values: Comply with requirements specified in Division 01 Section "Payment Procedures."
- I. Subcontract List: Prepare a written summary identifying individuals or firms proposed for each portion of the Work, including those who are to furnish products or equipment fabricated to a special design.
 - 1. Submit PDF electronic file of the subcontractor list, unless otherwise directed.

2.2 CONFORMANCE SUBMITTALS

- A. See "Submittal List" at the end of this section for products that qualify as Conformance Submittals.
- B. For record purposes, submit documentation, product data, shop drawings, etc. to the Architect or Engineer for products to be incorporated into the work as specified. Architect will stamp "Action Not Required" and return to Contractor. Submit PDF electronic files via Owner's Project Management software platform for Owner's record.
- C. At Project Closeout, provide a summary of all products installed as Conformance Products. Indicate the name of manufacturer, product and finish incorporated into the work.
- D. Sign and submit the "Product Conformance Summary" to the Architect. No further action is required.
- E. IMPORTANT: Non-complying products found on the "Product Conformance Summary" which were incorporated into the work without obtaining proper approval for the substitution must be replaced immediately with specified products at no cost to the Owner, no change to the construction schedule and no operational impacts to the Owner.

1. No Substitutions will be allowed for non-complying products – they must be replaced with specified products; i.e., no “after installation” approvals.
- F. If the Contractor proposes to use a product that is NOT as specified, a “Substitution Request” must be submitted for the Architect’s review and Owner’s approval prior to installation.
1. If the Owner approves the substitution, all submittal requirements noted in the applicable specification section must be met.
 2. Approving the “Substitution Request” does not alleviate the submittal requirements.
 3. Provide substitution submittals to the Architect for review and Owner for approval following procedures for “Action Submittals”.

2.3 INFORMATIONAL SUBMITTALS

- A. General: Prepare and submit Informational Submittals required by other Specification Sections.
1. Submit PDF electronic files of each submittal, unless otherwise indicated.
 2. Certificates and Certifications: Provide a notarized statement that includes signature of entity responsible for preparing certification. Certificates and certifications shall be signed by an officer or other individual authorized to sign documents on behalf of that entity.
 3. Test and Inspection Reports: Comply with requirements specified in Division 01 Section "Quality Requirements."
- B. Coordination Drawings: Comply with requirements specified in Division 01 Section "Project Management and Coordination."
- C. Contractor's Construction Schedule: Comply with requirements specified in Division 01 Section "Construction Progress Documentation."
- D. Qualification Data: Prepare written information that demonstrates capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.
- E. Welding Certificates: Prepare written certification that welding procedures and personnel comply with requirements in the Contract Documents. Submit record of Welding Procedure Specification (WPS) and Procedure Qualification Record (PQR) on AWS forms. Include names of firms and personnel certified.
- F. Installer Certificates: Prepare written statements on manufacturer's letterhead certifying that Installer complies with requirements in the Contract Documents and, where required, is authorized by manufacturer for this specific Project.
- G. Manufacturer Certificates: Prepare written statements on manufacturer's letterhead certifying that manufacturer complies with requirements in the Contract Documents. Include evidence of manufacturing experience where required.
- H. Product Certificates: Prepare written statements on manufacturer's letterhead certifying that product complies with requirements in the Contract Documents.
- I. Material Certificates: Prepare written statements on manufacturer's letterhead certifying that material complies with requirements in the Contract Documents.

- J. Material Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in the Contract Documents.
- K. Product Test Reports: Prepare written reports indicating current product produced by manufacturer complies with requirements in the Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.
- L. Research/Evaluation Reports: Prepare written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project.
- M. Preconstruction Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in the Contract Documents.
- N. Compatibility Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.
- O. Field Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in the Contract Documents.
- P. Maintenance Data: Prepare written and graphic instructions and procedures for operation and normal maintenance of products and equipment. Comply with requirements specified in Division 01 Section "Operation and Maintenance Data."
- Q. Design Data: Prepare written and graphic information, including, but not limited to, performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.
- R. Manufacturer's Instructions: Prepare written or published information that documents manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer.
- S. Manufacturer's Field Reports: Prepare written information documenting factory-authorized service representative's tests and inspections. Include the following, as applicable:
 - 1. Statement on condition of substrates and their acceptability for installation of product.
 - 2. Summary of installation procedures being followed, whether they comply with requirements and, if not, what corrective action was taken.
 - 3. Results of operational and other tests and a statement of whether observed performance complies with requirements.
- T. Insurance Certificates and Bonds: Prepare written information indicating current status of insurance or bonding coverage. Include name of entity covered by insurance or bond, limits of coverage, amounts of deductibles, if any, and term of the coverage.

- U. Material Safety Data Sheets (MSDSs): Submit information directly to Owner; do not submit to Architect.

- 1. Architect will not review submittals that include MSDSs and will return them for resubmittal.

2.4 DELEGATED DESIGN

- A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.

- 1. If criteria indicated are not sufficient to perform services or certification required, submit a written request for additional information to Architect.

- B. Delegated-Design Submittal: In addition to Shop Drawings, Product Data, and other required submittals, submit three copies of a statement, signed and sealed by the responsible design professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional.

- 1. Indicate that products and systems comply with performance and design criteria in the Contract Documents. Include list of codes, loads, and other factors used in performing these services.

PART 3 - EXECUTION

3.1 CONTRACTOR'S REVIEW

- A. Review each submittal and check for coordination with other Work of the Contract and for compliance with the Contract Documents. Note corrections and field dimensions. Mark with approval stamp before submitting to Architect.
- B. Approval Stamp: Stamp each submittal with a uniform, approval stamp. Include Project name and location, submittal number, Specification Section title and number, name of reviewer, date of Contractor's approval, and statement certifying that submittal has been reviewed, checked, and approved for compliance with the Contract Documents.

3.2 ARCHITECT'S ACTION

- A. The Architect's review is for general conformance with the design intent expressed in the Contract Documents. Approval of a particular element does not constitute approval of an assembly of which the element is a part.
- B. Markings or comments shall not be construed as relieving the Contractor from compliance with the Contract Documents nor permitting Contractor's deviation from Contract Documents.
- C. The Contractor remains solely responsible for submittal details, dimensions, accuracy, and quantities, installation and performance of equipment and systems designed by the contractor, for selecting fabrication processes, for techniques of assembly, for construction means, methods, sequence and procedures and for performing the Work in a safe manner
- D. General: Architect will not review submittals that do not bear Contractor's approval stamp and will return them without action.

1. Action Submittals: Architect will review each submittal, make marks to indicate corrections or modifications required, and return it.
- E. Action Submittals: Architect will review each submittal, make marks to indicate corrections or modifications required, and return it.
- F. Informational Submittals: Architect will review each submittal and will not return it, or will return it if it does not comply with requirements. Architect will forward each submittal to appropriate party.
- G. Conformance Submittals: Architect will not review conformance submittal. If/When submitted via Owner's Project Management software platform, conformance submittal will not be reviewed or returned. Contractor, fabricator, vendor, or installer is responsible for conformance with design drawings and specifications.
- H. Partial submittals are not acceptable, will be considered nonresponsive, and will be returned without review.
- I. Submittals not required by the Contract Documents may not be reviewed and may be discarded.
- J. Please note: The Architect and Engineer's shop drawing review responsibilities include only two (2) reviews for any shop drawing submittal including initial review and review of one (1) resubmittal. Any Architectural and/or Engineering shop drawing review time beyond the above listed two (2) reviews will be charged to the Owner at the Architect and/or Engineer's standard hourly rates for all hours associated with the processing and review of said shop drawing submission. The Owner may request a credit to the contract sum by the General Contractor in the amount of this charge for additional reviews.

END OF SECTION 013300

SEE SUBMITTAL LIST AT THE END OF THIS SECTION FOR CONFORMANCE AND ACTION SUBMITTAL ITEMS.

SEE FOLLOWING ATTACHMENT FOR ELECTRONIC FILE TRANSFER AGREEMENT FORM.

SUBMITTAL LIST		
SPECIFICATION SECTION	CONFORMANCE	ACTION
031500 – Concrete Accessories	X (except items noted as Action Submittal)	Shop Drawings
033000 – Cast-In-Place Concrete	X (except items noted as Action Submittal)	Mix Designs Rebar Shop Drawings
033010 – Cast-In-Place Interior Concrete Slab-On-Grade	X (except items noted as Action Submittal)	Mix Designs Rebar Shop Drawings
033543 – Polished Concrete Finish and Joint Filler	X	
042000 – Unit Masonry	X (except items noted as Action Submittal)	Mix Designs for Mortar & Grout Rebar Shop Drawings Certifications
042516 – Thin Brick Panel System	X	
042600 – Adhered Masonry Veneer	X	
047200 – Cast Stone Masonry	X	
051200 – Structural Steel Framing	X (except items noted as Action Submittal)	Shop Drawings
052100 – Steel Joist Framing	X (except items noted as Action Submittal)	Shop Drawings
053100 – Steel Decking	X (except items noted as Action Submittal)	Shop Drawings
054000 – Cold-Formed Metal Framing	X	
055000 – Metal Fabrications	X	
055119 – Metal Grating Stairs	X	
055213 – Pipe and Tube Railings	X	
055313 – Bar Gratings	X	
061000 – Rough Carpentry	X	
061600 – Sheathing	X	
064116 – Plastic-Laminate-Faced Architectural Cabinets	X	

SUBMITTAL LIST		
SPECIFICATION SECTION	CONFORMANCE	ACTION
071113 – Bituminous Dampproofing	X	
071900 – Water Repellents	X	
072100 – Thermal Insulation	X	
072726 – Fluid-Applied Membrane Air Barriers	X	
074213.23 – Metal Composite Material Wall Panels	X	
074243 – Composite Wall Panels	X	
074293 – Metal Soffit Systems	X	
075419 – Polyvinyl-Chloride (PVC) Roofing	X	
075423 – Thermoplastic Polyolefin (TPO) Roofing	X	
076200 – Sheet Metal Flashing and Trim	X	
077100 – Roof Specialties	X	
077200 – Roof Accessories	X	
078413 – Penetration Firestopping	X	
079200 – Joint Sealants	X	
081113 – Hollow Metal Doors and Frames	X	
083213 – Aluminum-Framed Sliding Glass Doors	X	
083483 – Traffic Doors	X	
083613 – Sectional Doors	X	
084113 – Aluminum-Framed Storefronts		X
087100 – Door Hardware		X
088000 – Glazing		X
089119 – Louvers	X	

SUBMITTAL LIST		
SPECIFICATION SECTION	CONFORMANCE	ACTION
092116 – Non-Structural Metal Framing	X	
092900 – Gypsum Board	X	
095113 – Acoustical Panel Ceilings	X	
096513 – Resilient Base and Accessories	X	
097720 – Fiberglass Reinforced Wall Panels	X	
099113 – Exterior Painting	X	
099123 – Interior Painting	X	
099600 – High-Performance Coatings	X	
102600 – Wall Surface Protection	X	
102600.01 – Wall and Door Protection	X	
102800 – Toilet, Bath and Laundry Accessories		X
104416 – Fire Extinguishers	X	
107313 – Metal Canopies	X	
111316 – Loading Dock Seals and Shelters	X	
111319 – Stationary Loading Dock Equipment	X	
122413 – Roller Window Shades		X
123661.16 – Solid Surfacing Countertops	X	
210517 – Sleeves and Sleeve Seals for Fire Suppression Piping	X	
210518 – Escutcheons for Fire Suppression Piping	X	

SUBMITTAL LIST		
SPECIFICATION SECTION	CONFORMANCE	ACTION
210523 – General Duty Valves for Water Based Fire Suppression Piping	X	
210548 – Vibration and Seismic Controls for Fire Suppression Piping and Equipment	X	
210553 – Identification for Fire Suppression Piping and Equipment	X	
211119 – Fire Department Connections	X	
211313 – Wet Pipe Sprinkler Systems		X
213113 – Electric-Drive, Centrifugal Fire Pumps	X	
213400 – Pressure Maintenance Pumps	X	
220517 – Sleeves and Sleeve Seals for Plumbing Piping	X	
220518 – Escutcheons for Plumbing Piping	X	
220523 – General-Duty Valves for Plumbing Piping	X	
220529 – Hangers and Supports for Plumbing Piping and Equipment	X	
220548 – Vibration and Seismic Controls for Plumbing Piping and Equipment	X	
220553 – Identification for Plumbing Piping and Equipment	X	
220719 – Plumbing Piping Insulation	X	
220800 – Commissioning of Plumbing	X	
221116 – Domestic Water Piping	X	
221119 – Domestic Water Piping Specialties	X	
221123 – Domestic Water Pumps	X	
221123.13 – Domestic Water Packaged Booster Pumps	X	

SUBMITTAL LIST		
SPECIFICATION SECTION	CONFORMANCE	ACTION
221316 – Sanitary Waste and Vent Piping	X	
221319 – Sanitary Waste Piping Specialties	X	
221319.13 – Sanitary Drains	X	
221323 – Grease Waste Interceptors	X	
223300 – Electric, Domestic Water Heaters	X	
224213.13 – Commercial Water Closets		X
224216.13 – Commercial Lavatories		X
224216.16 – Commercial Sinks		X
224716 – Drinking Fountains	X	
230503 – General Provisions for HVAC Work	X	
230505 – Basic HVAC Materials and Methods	X	
230513 – Common Motor Requirements for HVAC Equipment	X	
230517 – Sleeves and Sleeve Seals for HVAC Piping	X	
230518 – Escutcheons for HVAC Piping	X	
230529 – Hangers and Supports for HVAC Piping and Equipment	X	
230548 – Vibration Controls for HVAC	X	
230553 – Identification for HVAC Piping and Equipment	X	
230593 – Testing, Adjusting and Balancing for HVAC	X	
230700 – HVAC Insulation		X
230800 – Commissioning of HVAC		X
230920 – Building Automation System, General	X	

SUBMITTAL LIST		
SPECIFICATION SECTION	CONFORMANCE	ACTION
230921 – Building Automation System – DDC Interface and Communication Devices	X	
230922 – Building Automation System – DDC Materials	X	
230923 – Building Automation System, General Sequence	X	
231123 – Natural Gas Piping	X	
232300 – Refrigerant Piping	X	
233113 – Metal Ducts		X
233300 – Air Duct Accessories	X	
233423 – HVAC Power Ventilators	X	
233713 – Diffusers, Registers and Grilles		X
238119 – Self-Contained Air Conditioners		X
238126 – Split-System Air Conditioners		X
260503 – General Requirements for Electrical Systems	X	
260505 – Basic Electrical Materials and Methods	X	
260507 – Temporary Power and Communication Systems	X	
260511 – Conductors and Cables for Electrical Systems	X	
260519 – Low-Voltage Electrical Power Conductors and Cables	X	
260526 – Grounding and Bonding for Electrical Systems	X	
260529 – Hangers and Supports for Electrical Systems	X	

SUBMITTAL LIST		
SPECIFICATION SECTION	CONFORMANCE	ACTION
260533 – Raceways and Boxes for Electrical Systems	X	
260536 – Cable Trays for Electrical Systems	X	
260543 – Underground Ducts and Raceways for Electrical Systems	X	
260553 – Identification for Electrical Systems	X	
260573 – Overcurrent Protective Device Coordination Study	X	
260800 – Commissioning for Electrical Stand Alone Version	X	
260813 – Testing of Electrical Systems	X	
260923 – Lighting Control Devices	X	
262126 – Cold Storage Frost Heave Prevention Systems	X	
262200 – Low-Voltage Transformers		X
262413.70 – Integrated Power Center		X
262416 – Panelboards		X
262713 – Electricity Metering	X	
262716 – Indoor Service Poles	X	
262726 – Wiring Devices	X	
262813 – Fuses		X
262816 – Enclosed Switches and Circuit Breakers		X
262913 – Enclosed Controllers		X
262923 – Variable Frequency Motor Controllers		X

SUBMITTAL LIST		
SPECIFICATION SECTION	CONFORMANCE	ACTION
263353 – Static Uninterruptible Power Supply	X	
263600 – Transfer Switches	X	
264113 – Lightning Protection for Structures	X	
264300 – Surge Protection Devices	X	
265100 – Interior Lighting		X
265600 – Exterior Lighting		X
270500 – Common Work Results for Communications Systems	X	
270526 – Grounding and Bonding for Communications Systems	X	
270528 – Pathways for Communications	X	
271100 – Communications Equipment Room Fittings	X	
271500 – Communications Horizontal Cabling	X	
283100 – Fire Detection and Alarm		X
311000 – Site Clearing	X	
312000 – Earthwork		X
312500 – Erosion Control		X
313116 – Termite Control	X	
321216 – Asphalt Pavement		X
321313 – Site Concrete		X
321700 – Pavement Markings, Signs and Specialties	X	
323113 – Chain Link Fences and Gates (Interior)	X	

SUBMITTAL LIST		
SPECIFICATION SECTION	CONFORMANCE	ACTION
323113 – Chain Link Fences and Gates (Exterior)	X	
328000 – Irrigation System		X
329200 - Lawns and Grasses		X
329223 - Sod		X
329300 - Exterior Plants		X
331000 - Exterior Water System		X
333000 - Sanitary Sewerage		X
334100 - Storm Drainage		X
334613 - Foundation Drainage Systems		X

MG2

8444 Westpark Dr, Ste 120
McLean, VA 22102703 564 8484
MG2.com

ELECTRONIC FILE TRANSFER AGREEMENT

This Agreement is made in reference to the construction of **LIDL #### – XXXXX, City, State** (MG2 Project No. XX-XXXX-XXX), and for **LIDL US, LLC** hereafter referred to as Owner.

At your request, MG2 Corporation (MG2) will provide electronic files for your convenience and use in the preparation of shop drawings related to **LIDL #### – XXXXX, City, State** subject to the following terms and conditions:

Data contained on these electronic files is part of MG2's instruments of service and shall not be used by you or anyone else receiving this data through or from you for any purpose other than as a convenience in the preparation of shop drawings for the referenced project. Any use or reuse by you or by others will be at your sole risk and without liability or legal exposure to MG2.

These electronic files are not contract documents. Significant differences may exist between these electronic files and corresponding hard copy contract documents due to addenda, change orders or other revisions. Because of the possibility the information and data delivered in machine readable form may be altered, whether inadvertently or otherwise, MG2 reserves the right to retain hard copy originals of the electronic documentation delivered to the contractor, in machine readable form, which the original shall be referred to and shall govern in the event of any inconsistency between the two. You bear sole responsibility for determining if any conflict exists.

You agree to assume all risks associated therewith, and to the fullest extent permitted by law to hold harmless and indemnify the Owner, MG2 and its consultants from and against all claims, liabilities, losses, damages, and costs, including but not limited to attorney's fees, arising therefrom or in connection therewith.

By your use of these electronic files, you are not relieved of your duty to fully comply with the contract documents, including and without limitation, the need to check confirm and coordinate all dimensions and details, take field measurements, verify field conditions and coordinate your work with that of other contractors for the project.

Under no circumstances shall delivery of the electronic files for use by you be deemed a sale by MG2, and MG2 makes no warranties, either expressed or implied, of merchantability and fitness for any particular purpose. In no event shall MG2 be liable for any loss of profit or any consequential damages.

Please indicate your agreement to the terms and conditions outlined above with your signature below.

Name: _____ Date: _____

Company: _____



SECTION 014000 - QUALITY REQUIREMENTS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for quality assurance and quality control.
- B. Testing and inspecting services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with the Contract Document requirements.
 - 1. Specified tests, inspections, and related actions do not limit Contractor's other quality-assurance and quality-control procedures that facilitate compliance with the Contract Document requirements.
 - 2. Requirements for Contractor to provide quality-assurance and quality-control services required by Architect, Owner, or authorities having jurisdiction are not limited by provisions of this Section.
- C. See applicable Divisions 02 through 33 Sections for specific test and inspection requirements.

1.3 DEFINITIONS

- A. Quality-Assurance Services: Activities, actions, and procedures performed before and during execution of the Work to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.
- B. Quality-Control Services: Tests, inspections, procedures, and related actions during and after execution of the Work to evaluate that actual products incorporated into the Work and completed construction comply with requirements. Services do not include contract enforcement activities performed by Architect.
- C. Mockups: Full-size, physical assemblies that are constructed on-site. Mockups are used to verify selections made under sample submittals, to demonstrate aesthetic effects and, where indicated, qualities of materials and execution, and to review construction, coordination, testing, or operation; they are not Samples. Approved mockups establish the standard by which the Work will be judged.
- D. Preconstruction Testing: Tests and inspections that are performed specifically for the Project before products and materials are incorporated into the Work to verify performance or compliance with specified criteria.
- E. Product Testing: Tests and inspections that are performed by an NRTL, an NVLAP, or a testing agency qualified to conduct product testing and acceptable to authorities having jurisdiction, to establish product performance and compliance with industry standards.
- F. Source Quality-Control Testing: Tests and inspections that are performed at the source, i.e., plant, mill, factory, or shop.

- G. Field Quality-Control Testing: Tests and inspections that are performed on-site for installation of the Work and for completed Work.
- H. Testing Agency: An entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.
- I. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, Subcontractor, or Sub-subcontractor, to perform a particular construction operation, including installation, erection, application, and similar operations.
 - 1. Using a term such as "carpentry" does not imply that certain construction activities must be performed by accredited or unionized individuals of a corresponding generic name, such as "carpenter." It also does not imply that requirements specified apply exclusively to tradespeople of the corresponding generic name.
- J. Experienced: When used with an entity, "experienced" means having successfully completed a minimum of five previous projects similar in size and scope to this Project; being familiar with special requirements indicated; and having complied with requirements of authorities having jurisdiction.

1.4 CONFLICTING REQUIREMENTS

- A. General: If compliance with two or more standards is specified and the standards establish different or conflicting requirements for minimum quantities or quality levels, comply with the most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Architect for a decision before proceeding.
- B. Minimum Quantity or Quality Levels: The quantity or quality level shown or specified shall be the minimum provided or performed. The actual installation may comply exactly with the minimum quantity or quality specified, or it may exceed the minimum within reasonable limits. To comply with these requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Architect for a decision before proceeding.

1.5 SUBMITTALS

- A. Qualification Data: For testing agencies specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include proof of qualifications in the form of a recent report on the inspection of the testing agency by a recognized authority.
- B. Reports: Prepare and submit certified written reports that include the following:
 - 1. Date of issue.
 - 2. Project title and number.
 - 3. Name, address, and telephone number of testing agency.
 - 4. Dates and locations of samples and tests or inspections.
 - 5. Names of individuals making tests and inspections.
 - 6. Description of the Work and test and inspection method.
 - 7. Identification of product and Specification Section.
 - 8. Complete test or inspection data.
 - 9. Test and inspection results and an interpretation of test results.
 - 10. Record of temperature and weather conditions at time of sample taking and testing and inspecting.
 - 11. Comments or professional opinion on whether tested or inspected Work complies with the Contract Document requirements.

12. Name and signature of laboratory inspector.
13. Recommendations on retesting and re-inspecting.

- C. Permits, Licenses, and Certificates: For Owner's records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of the Work.

1.6 QUALITY ASSURANCE

- A. General: Qualifications paragraphs in this Article establish the minimum qualification levels required; individual Specification Sections specify additional requirements.
- B. Installer Qualifications: A firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent to that indicated for this Project, whose work has resulted in construction with a record of successful in-service performance.
- C. Manufacturer Qualifications: A firm experienced in manufacturing products or systems similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- D. Fabricator Qualifications: A firm experienced in producing products similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- E. Professional Engineer Qualifications: A professional engineer who is legally qualified to practice in jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of the system, assembly, or product that are similar to those indicated for this Project in material, design, and extent.
- F. Specialists: Certain sections of the Specifications require that specific construction activities shall be performed by entities who are recognized experts in those operations. Specialists shall satisfy qualification requirements indicated and shall be engaged for the activities indicated.
1. Requirement for specialists shall not supersede building codes and regulations governing the Work.
- G. Testing Agency Qualifications: An NRTL, an NVLAP, or an independent agency with the experience and capability to conduct testing and inspecting indicated, as documented according to ASTM E 548; and with additional qualifications specified in individual Sections; and where required by authorities having jurisdiction, that is acceptable to authorities.
1. NRTL: A nationally recognized testing laboratory according to 29 CFR 1910.7.
 2. NVLAP: A testing agency accredited according to NIST's National Voluntary Laboratory Accreditation Program.
- H. Factory-Authorized Service Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of manufacturer's products that are similar in material, design, and extent to those indicated for this Project.

- I. Mockups: Before installing portions of the Work requiring mockups, build mockups for each form of construction and finish required to comply with the following requirements, using materials indicated for the completed Work:
 1. Build mockups in location and of size indicated or, if not indicated, as directed by Architect and/or Owner.
 2. Notify Architect seven days in advance of dates and times when mockups will be constructed.
 3. Demonstrate the proposed range of aesthetic effects and workmanship.
 4. Obtain Architect's approval of mockups before ordering product(s), fabrication, or construction.
 5. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
 6. Demolish and remove mockups when directed, unless otherwise indicated.

1.7 QUALITY CONTROL

- A. Owner Responsibilities: Where quality-control services are indicated as Owner's responsibility, Owner will engage a qualified testing agency to perform these services.
 1. Owner will furnish Contractor with names, addresses, and telephone numbers of testing agencies engaged and a description of types of testing and inspecting they are engaged to perform.
 2. Costs for retesting and reinspecting construction that replaces or is necessitated by work that failed to comply with the Contract Documents will be charged to Contractor.
- B. Tests and inspections not explicitly assigned to Owner are Contractor's responsibility. Unless otherwise indicated, provide quality-control services specified and those required by authorities having jurisdiction. Perform quality-control services required of Contractor by authorities having jurisdiction, whether specified or not.
 1. Where services are indicated as Contractor's responsibility, engage a qualified testing agency to perform these quality-control services.
 - a. Contractor shall not employ same entity engaged by Owner, unless agreed to in writing by Owner.
 2. Notify testing agencies at least 24 hours in advance of time when Work that requires testing or inspecting will be performed.
 3. Where quality-control services are indicated as Contractor's responsibility, submit a certified written report, in duplicate, of each quality-control service.
 4. Testing and inspecting requested by Contractor and not required by the Contract Documents are Contractor's responsibility.
 5. Submit additional copies of each written report directly to authorities having jurisdiction, when they so direct.
- C. Manufacturer's Field Services: Where indicated, engage a factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing as specified in Division 01 Section "Submittal Procedures."
- D. Retesting/Reinspecting: Regardless of whether original tests or inspections were Contractor's responsibility, provide quality-control services, including retesting and reinspecting, for construction that replaced Work that failed to comply with the Contract Documents.

- E. Testing Agency Responsibilities: Cooperate with Architect and Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.
1. Notify Architect and Contractor promptly of irregularities or deficiencies observed in the Work during performance of its services.
 2. Determine the location from which test samples will be taken and in which in-situ tests are conducted.
 3. Conduct and interpret tests and inspections and state in each report whether tested and inspected work complies with or deviates from requirements.
 4. Submit a certified written report, in duplicate, of each test, inspection, and similar quality-control service through Contractor.
 5. Do not release, revoke, alter, or increase the Contract Document requirements or approve or accept any portion of the Work.
 6. Do not perform any duties of Contractor.
- F. Associated Services: Cooperate with agencies performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide the following:
1. Access to the Work.
 2. Incidental labor and facilities necessary to facilitate tests and inspections.
 3. Adequate quantities of representative samples of materials that require testing and inspecting. Assist agency in obtaining samples.
 4. Facilities for storage and field curing of test samples.
 5. Delivery of samples to testing agencies.
 6. Preliminary design mix proposed for use for material mixes that require control by testing agency.
 7. Security and protection for samples and for testing and inspecting equipment at Project site.
- G. Coordination: Coordinate sequence of activities to accommodate required quality-assurance and -control services with a minimum of delay and to avoid necessity of removing and replacing construction to accommodate testing and inspecting.
1. Schedule times for tests, inspections, obtaining samples, and similar activities.

1.8 SPECIAL TESTS AND INSPECTIONS

- A. Special Tests and Inspections: Owner will engage a qualified testing agency to conduct special tests and inspections required by authorities having jurisdiction as the responsibility of Owner, and as follows:
- B. Special Tests and Inspections: Conducted by a qualified testing agency as required by authorities having jurisdiction, as indicated in individual Specification Sections, and as follows:
1. Verifying that manufacturer maintains detailed fabrication and quality-control procedures and reviewing the completeness and adequacy of those procedures to perform the Work.
 2. Notifying Architect and Contractor promptly of irregularities and deficiencies observed in the Work during performance of its services.
 3. Submitting a certified written report of each test, inspection, and similar quality-control service to Architect with copy to Contractor and to authorities having jurisdiction.
 4. Submitting a final report of special tests and inspections at Substantial Completion which includes a list of unresolved deficiencies.
 5. Interpreting tests and inspections and stating in each report whether tested and inspected work complies with or deviates from the Contract Documents.

6. Retesting and reinspecting corrected work.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 REPAIR AND PROTECTION

- A. General: On completion of testing, inspecting, sample taking, and similar services, repair damaged construction and restore substrates and finishes.
 1. Comply with the Contract Document requirements for Division 01 Section "Cutting and Patching."
- B. Protect construction exposed by or for quality-control service activities.
- C. Repair and protection are Contractor's responsibility, regardless of the assignment of responsibility for quality-control services.

END OF SECTION 014000

SECTION 014200 - REFERENCES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 DEFINITIONS

- A. General: Basic Contract definitions are included in the Conditions of the Contract.
- B. "Approved": When used to convey Architect's action on Contractor's submittals, applications, and requests, "approved" is limited to Architect's duties and responsibilities as stated in the Conditions of the Contract.
- C. "Directed": A command or instruction by Architect. Other terms including "requested," "authorized," "selected," "required," and "permitted" have the same meaning as "directed."
- D. "Indicated": Requirements expressed by graphic representations or in written form on Drawings, in Specifications, and in other Contract Documents. Other terms including "shown," "noted," "scheduled," and "specified" have the same meaning as "indicated."
- E. "Regulations": Laws, ordinances, statutes, and lawful orders issued by authorities having jurisdiction, and rules, conventions, and agreements within the construction industry that control performance of the Work.
- F. "Furnish": Supply and deliver to Project site, ready for unloading, unpacking, assembly, installation, and similar operations.
- G. "Install": Unload, temporarily store, unpack, assemble, erect, place, anchor, apply, work to dimension, finish, cure, protect, clean, and similar operations at Project site.
- H. "Provide": Furnish and install, complete and ready for the intended use.
- I. "Project Site": Space available for performing construction activities. The extent of Project site is shown on Drawings and may or may not be identical with the description of the land on which Project is to be built.

1.3 INDUSTRY STANDARDS

- A. Applicability of Standards: Unless the Contract Documents include more stringent requirements, applicable construction industry standards have the same force and effect as if bound or copied directly into the Contract Documents to the extent referenced. Such standards are made a part of the Contract Documents by reference.
- B. Publication Dates: Comply with standards in effect as of date of the Contract Documents unless otherwise indicated.
- C. Copies of Standards: Each entity engaged in construction on Project should be familiar with industry standards applicable to its construction activity. Copies of applicable standards are not bound with the Contract Documents.

1. Where copies of standards are needed to perform a required construction activity, obtain copies directly from publication source.

1.4 ABBREVIATIONS AND ACRONYMS

- A. Industry Organizations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities indicated in Gale's "Encyclopedia of Associations: National Organizations of the U.S." or in Columbia Books' "National Trade & Professional Associations of the United States."

- B. Industry Organizations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list.

1. AABC - Associated Air Balance Council; www.aabc.com.
2. AAMA - American Architectural Manufacturers Association; www.aamanet.org.
3. AASHTO - American Association of State Highway and Transportation Officials; www.transportation.org.
4. ABMA - American Bearing Manufacturers Association; www.americanbearings.org.
5. ABMA - American Boiler Manufacturers Association; www.abma.com.
6. ACI - American Concrete Institute; (Formerly: ACI International); www.concrete.org.
7. ACPA - American Concrete Pipe Association; www.concrete-pipe.org.
8. AEIC - Association of Edison Illuminating Companies, Inc. (The); www.aeic.org.
9. AF&PA - American Forest & Paper Association; www.afandpa.org.
10. AGA - American Gas Association; www.aga.org.
11. AHAM - Association of Home Appliance Manufacturers; www.aham.org.
12. AHRI - Air-Conditioning, Heating, and Refrigeration Institute (The); www.ahrinet.org.
13. AI - Asphalt Institute; www.asphaltinstitute.org.
14. AIA - American Institute of Architects (The); www.aia.org.
15. AISC - American Institute of Steel Construction; www.aisc.org.
16. AISI - American Iron and Steel Institute; www.steel.org.
17. AMCA - Air Movement and Control Association International, Inc.; www.amca.org.
18. ANSI - American National Standards Institute; www.ansi.org.
19. APA - APA - The Engineered Wood Association; www.apawood.org.
20. APA - Architectural Precast Association; www.archprecast.org.
21. API - American Petroleum Institute; www.api.org.
22. ARI - Air-Conditioning & Refrigeration Institute; (See AHRI).
23. ARI - American Refrigeration Institute; (See AHRI).
24. ASCE - American Society of Civil Engineers; www.asce.org.
25. ASCE/SEI - American Society of Civil Engineers/Structural Engineering Institute; (See ASCE).
26. ASHRAE - American Society of Heating, Refrigerating and Air-Conditioning Engineers; www.ashrae.org.
27. ASME - ASME International; (American Society of Mechanical Engineers); www.asme.org.
28. ASSE - American Society of Safety Engineers (The); www.asse.org.
29. ASSE - American Society of Sanitary Engineering; www.asse-plumbing.org.
30. ASTM - ASTM International; www.astm.org.
31. ATIS - Alliance for Telecommunications Industry Solutions; www.atis.org.
32. AWI - Architectural Woodwork Institute; www.awinet.org.
33. AWMAC - Architectural Woodwork Manufacturers Association of Canada; www.awmac.com.
34. AWPA - American Wood Protection Association; www.awpa.com.
35. AWS - American Welding Society; www.aws.org.
36. AWWA - American Water Works Association; www.awwa.org.
37. BHMA - Builders Hardware Manufacturers Association; www.buildershardware.com.

38. BIA - Brick Industry Association (The); www.gobrick.com.
39. BICSI - BICSI, Inc.; www.bicsi.org.
40. BIFMA - BIFMA International; (Business and Institutional Furniture Manufacturer's Association); www.bifma.org.
41. BISSC - Baking Industry Sanitation Standards Committee; www.bissc.org.
42. CDA - Copper Development Association; www.copper.org.
43. CEA - Canadian Electricity Association; www.electricity.ca.
44. CEA - Consumer Electronics Association; www.ce.org.
45. CFSEI - Cold-Formed Steel Engineers Institute; www.cfsei.org.
46. CGA - Compressed Gas Association; www.cganet.com.
47. Cisca - Ceilings & Interior Systems Construction Association; www.cisca.org.
48. CISPI - Cast Iron Soil Pipe Institute; www.cispi.org.
49. CLFMI - Chain Link Fence Manufacturers Institute; www.chainlinkinfo.org.
50. CPA - Composite Panel Association; www.pbmdf.com.
51. CRI - Carpet and Rug Institute (The); www.carpet-rug.org.
52. CRRC - Cool Roof Rating Council; www.coolroofs.org.
53. CRSI - Concrete Reinforcing Steel Institute; www.crsi.org.
54. CSA - Canadian Standards Association; www.csa.ca.
55. CSA - CSA International; (Formerly: IAS - International Approval Services); www.csa-international.org.
56. CSI - Construction Specifications Institute (The); www.csinet.org.
57. CTI - Cooling Technology Institute; (Formerly: Cooling Tower Institute); www.cti.org.
58. CWC - Composite Wood Council; (See CPA).
59. DASMA - Door and Access Systems Manufacturers Association; www.dasma.com.
60. DHI - Door and Hardware Institute; www.dhi.org.
61. ECA - Electronic Components Association; (See ECIA).
62. ECAMA - Electronic Components Assemblies & Materials Association; (See ECIA).
63. ECIA - Electronic Components Industry Association; www.eciaonline.org.
64. EIA - Electronic Industries Alliance; (See TIA).
65. EIMA - EIFS Industry Members Association; www.eima.com.
66. EJMA - Expansion Joint Manufacturers Association, Inc.; www.ejma.org.
67. ESD - ESD Association; (Electrostatic Discharge Association); www.esda.org.
68. ESTA - Entertainment Services and Technology Association; (See PLASA).
69. EVO - Efficiency Valuation Organization; www.evo-world.org.
70. FCI - Fluid Controls Institute; www.fluidcontrolsintstitute.org.
71. FM Approvals - FM Approvals LLC; www.fmglobal.com.
72. FM Global - FM Global; (Formerly: FMG - FM Global); www.fmglobal.com.
73. FRSA - Florida Roofing, Sheet Metal & Air Conditioning Contractors Association, Inc.; www.floridarroof.com.
74. FSA - Fluid Sealing Association; www.fluidsealing.com.
75. FSC - Forest Stewardship Council U.S.; www.fscus.org.
76. GA - Gypsum Association; www.gypsum.org.
77. GANA - Glass Association of North America; www.glasswebsite.com.
78. GS - Green Seal; www.greenseal.org.
79. HI - Hydraulic Institute; www.pumps.org.
80. HII/GAMA - Hydronics Institute/Gas Appliance Manufacturers Association; (See AHRI).
81. HMMA - Hollow Metal Manufacturers Association; (See NAAMM).
82. HPVA - Hardwood Plywood & Veneer Association; www.hpva.org.
83. HPW - H. P. White Laboratory, Inc.; www.hpwhite.com.
84. IAPSC - International Association of Professional Security Consultants; www.iapsc.org.
85. IAS - International Accreditation Service; www.iasonline.org.
86. IAS - International Approval Services; (See CSA).
87. ICBO - International Conference of Building Officials; (See ICC).
88. ICC - International Code Council; www.iccsafe.org.
89. ICEA - Insulated Cable Engineers Association, Inc.; www.icea.net.

90. ICPA - International Cast Polymer Alliance; www.icpa-hq.org.
91. ICRI - International Concrete Repair Institute, Inc.; www.icri.org.
92. IEC - International Electrotechnical Commission; www.iec.ch.
93. IEEE - Institute of Electrical and Electronics Engineers, Inc. (The); www.ieee.org.
94. IES - Illuminating Engineering Society; (Formerly: Illuminating Engineering Society of North America); www.ies.org.
95. IESNA - Illuminating Engineering Society of North America; (See IES).
96. IEST - Institute of Environmental Sciences and Technology; www.iest.org.
97. IGMA - Insulating Glass Manufacturers Alliance; www.igmaonline.org.
98. IGSHPA - International Ground Source Heat Pump Association; www.igshpa.okstate.edu.
99. Intertek - Intertek Group; (Formerly: ETL SEMCO; Intertek Testing Service NA); www.intertek.com.
100. ISA - International Society of Automation (The); (Formerly: Instrumentation, Systems, and Automation Society); www.isa.org.
101. ISAS - Instrumentation, Systems, and Automation Society (The); (See ISA).
102. ISFA - International Surface Fabricators Association; (Formerly: International Solid Surface Fabricators Association); www.isfanow.org.
103. ISO - International Organization for Standardization; www.iso.org.
104. ISSFA - International Solid Surface Fabricators Association; (See ISFA).
105. ITU - International Telecommunication Union; www.itu.int/home.
106. KCMA - Kitchen Cabinet Manufacturers Association; www.kcma.org.
107. LMA - Laminating Materials Association; (See CPA).
108. LPI - Lightning Protection Institute; www.lightning.org.
109. MBMA - Metal Building Manufacturers Association; www.mbma.com.
110. MCA - Metal Construction Association; www.metalconstruction.org.
111. MFMA - Metal Framing Manufacturers Association, Inc.; www.metalframingmfg.org.
112. MHIA - Material Handling Industry of America; www.mhia.org.
113. MPPA - Moulding & Millwork Producers Association; www.wmmpa.com.
114. MPI - Master Painters Institute; www.paintinfo.com.
115. MSS - Manufacturers Standardization Society of The Valve and Fittings Industry Inc.; www.mss-hq.org.
116. NAAMM - National Association of Architectural Metal Manufacturers; www.naamm.org.
117. NACE - NACE International; (National Association of Corrosion Engineers International); www.nace.org.
118. NADCA - National Air Duct Cleaners Association; www.nadca.com.
119. NAIMA - North American Insulation Manufacturers Association; www.naima.org.
120. NBI - New Buildings Institute; www.newbuildings.org.
121. NCAA - National Collegiate Athletic Association (The); www.ncaa.org.
122. NCMA - National Concrete Masonry Association; www.ncma.org.
123. NEBB - National Environmental Balancing Bureau; www.nebb.org.
124. NECA - National Electrical Contractors Association; www.necanet.org.
125. NeLMA - Northeastern Lumber Manufacturers Association; www.nelma.org.
126. NEMA - National Electrical Manufacturers Association; www.nema.org.
127. NETA - InterNational Electrical Testing Association; www.netaworld.org.
128. NFHS - National Federation of State High School Associations; www.nfhs.org.
129. NFPA - National Fire Protection Association; www.nfpa.org.
130. NFPA - NFPA International; (See NFPA).
131. NFRC - National Fenestration Rating Council; www.nfrc.org.
132. NHLA - National Hardwood Lumber Association; www.nhla.com.
133. NLGA - National Lumber Grades Authority; www.nlga.org.
134. NOMMA - National Ornamental & Miscellaneous Metals Association; www.nomma.org.
135. NRCA - National Roofing Contractors Association; www.nrca.net.
136. NRMCA - National Ready Mixed Concrete Association; www.nrmca.org.
137. NSF - NSF International; www.nsf.org.
138. NSPE - National Society of Professional Engineers; www.nspe.org.

139. NSSGA - National Stone, Sand & Gravel Association; www.nssga.org.
140. NTMA - National Terrazzo & Mosaic Association, Inc. (The); www.ntma.com.
141. PCI - Precast/Prestressed Concrete Institute; www.pci.org.
142. PDI - Plumbing & Drainage Institute; www.pdionline.org.
143. PLASA - PLASA; (Formerly: ESTA - Entertainment Services and Technology Association); www.plasa.org.
144. RCSC - Research Council on Structural Connections; www.boltcouncil.org.
145. RFCI - Resilient Floor Covering Institute; www.rfci.com.
146. RIS - Redwood Inspection Service; www.redwoodinspection.com.
147. SAE - SAE International; www.sae.org.
148. SCTE - Society of Cable Telecommunications Engineers; www.scte.org.
149. SDI - Steel Deck Institute; www.sdi.org.
150. SDI - Steel Door Institute; www.steeldoor.org.
151. SEFA - Scientific Equipment and Furniture Association (The); www.sefalabs.com.
152. SEI/ASCE - Structural Engineering Institute/American Society of Civil Engineers; (See ASCE).
153. SIA - Security Industry Association; www.siaonline.org.
154. SJI - Steel Joist Institute; www.steeljoist.org.
155. SMA - Screen Manufacturers Association; www.smainfo.org.
156. SMACNA - Sheet Metal and Air Conditioning Contractors' National Association; www.smacna.org.
157. SMPTE - Society of Motion Picture and Television Engineers; www.smpte.org.
158. SPFA - Spray Polyurethane Foam Alliance; www.sprayfoam.org.
159. SPIB - Southern Pine Inspection Bureau; www.spib.org.
160. SPRI - Single Ply Roofing Industry; www.spri.org.
161. SRCC - Solar Rating & Certification Corporation; www.solar-rating.org.
162. SSINA - Specialty Steel Industry of North America; www.ssina.com.
163. SSPC - SSPC: The Society for Protective Coatings; www.sspc.org.
164. STI - Steel Tank Institute; www.steeltank.com.
165. SWI - Steel Window Institute; www.steelwindows.com.
166. SWPA - Submersible Wastewater Pump Association; www.swpa.org.
167. TCNA - Tile Council of North America, Inc.; www.tileusa.com.
168. TEMA - Tubular Exchanger Manufacturers Association, Inc.; www.tema.org.
169. TIA - Telecommunications Industry Association (The); (Formerly: TIA/EIA - Telecommunications Industry Association/Electronic Industries Alliance); www.tiaonline.org.
170. TIA/EIA - Telecommunications Industry Association/Electronic Industries Alliance; (See TIA).
171. TMS - The Masonry Society; www.masonrysociety.org.
172. UL - Underwriters Laboratories Inc.; www.ul.com.
173. UNI - Uni-Bell PVC Pipe Association; www.uni-bell.org.
174. USGBC - U.S. Green Building Council; www.usgbc.org.
175. USITT - United States Institute for Theatre Technology, Inc.; www.usitt.org.
176. WASTEC - Waste Equipment Technology Association; www.wastec.org.
177. WCLIB - West Coast Lumber Inspection Bureau; www.wclib.org.
178. WCMA - Window Covering Manufacturers Association; www.wcmanet.org.
179. WDMA - Window & Door Manufacturers Association; www.wdma.com.
180. WI - Woodwork Institute; www.wicnet.org.
181. WSRCA - Western States Roofing Contractors Association; www.wsrca.com.
182. WWPA - Western Wood Products Association; www.wwpa.org.

C. Code Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list.

1. IAPMO - International Association of Plumbing and Mechanical Officials; www.iapmo.org.

2. ICC - International Code Council; www.iccsafe.org.
 3. ICC-ES - ICC Evaluation Service, LLC; www.icc-es.org.
- D. Federal Government Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list.
1. COE - Army Corps of Engineers; www.usace.army.mil.
 2. CPSC - Consumer Product Safety Commission; www.cpsc.gov.
 3. DOC - Department of Commerce; National Institute of Standards and Technology; www.nist.gov.
 4. DOD - Department of Defense; www.quicksearch.dla.mil.
 5. DOE - Department of Energy; www.energy.gov.
 6. EPA - Environmental Protection Agency; www.epa.gov.
 7. FAA - Federal Aviation Administration; www.faa.gov.
 8. FG - Federal Government Publications; www.gpo.gov.
 9. GSA - General Services Administration; www.gsa.gov.
 10. HUD - Department of Housing and Urban Development; www.hud.gov.
 11. LBL - Lawrence Berkeley National Laboratory; Environmental Energy Technologies Division; www.eetd.lbl.gov.
 12. OSHA - Occupational Safety & Health Administration; www.osha.gov.
 13. SD - Department of State; www.state.gov.
 14. TRB - Transportation Research Board; National Cooperative Highway Research Program; The National Academies; www.trb.org.
 15. USDA - Department of Agriculture; Agriculture Research Service; U.S. Salinity Laboratory; www.ars.usda.gov.
 16. USDA - Department of Agriculture; Rural Utilities Service; www.usda.gov.
 17. USDJ - Department of Justice; Office of Justice Programs; National Institute of Justice; www.ojp.usdoj.gov.
 18. USP - U.S. Pharmacopeial Convention; www.usp.org.
 19. USPS - United States Postal Service; www.usps.com.
- E. Standards and Regulations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the standards and regulations in the following list.
1. CFR - Code of Federal Regulations; Available from Government Printing Office; www.gpo.gov/fdsys.
 2. DOD - Department of Defense; Military Specifications and Standards; Available from DLA Document Services; www.quicksearch.dla.mil.
 3. DSCC - Defense Supply Center Columbus; (See FS).
 4. FED-STD - Federal Standard; (See FS).
 5. FS - Federal Specification; Available from DLA Document Services; www.quicksearch.dla.mil.
 - a. Available from Defense Standardization Program; www.dsp.dla.mil.
 - b. Available from General Services Administration; www.gsa.gov.
 - c. Available from National Institute of Building Sciences/Whole Building Design Guide; www.wbdg.org/ccb.
 6. MILSPEC - Military Specification and Standards; (See DOD).
 7. USAB - United States Access Board; www.access-board.gov.
 8. USATBCB - U.S. Architectural & Transportation Barriers Compliance Board; (See USAB).

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 014200

SECTION 015000 - TEMPORARY FACILITIES AND CONTROLS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes requirements for temporary utilities, support facilities, and security and protection facilities.
- B. See Division 01 Section "Execution" for progress cleaning requirements.
- C. See applicable Divisions 02 through 33 Sections for temporary heat, ventilation, and humidity requirements for products in those Sections.

1.3 DEFINITIONS

- A. Permanent Enclosure: As determined by Architect, permanent or temporary roofing is complete, insulated, and weathertight; exterior walls are insulated and weathertight; and all openings are closed with permanent construction or substantial temporary closures.

1.4 USE CHARGES

- A. General: Cost or use charges for temporary facilities shall be included in the Contract Sum. Allow other entities to use temporary services and facilities without cost, including, but not limited to, Architect, testing agencies, and authorities having jurisdiction.

1.5 SUBMITTALS

- A. Site Plan: Show temporary facilities, utility hookups, staging areas, and parking areas for construction personnel.

1.6 QUALITY ASSURANCE

- A. Electric Service: Comply with NECA, NEMA, and UL standards and regulations for temporary electric service. Install service to comply with NFPA 70.
- B. Tests and Inspections: Arrange for authorities having jurisdiction to test and inspect each temporary utility before use. Obtain required certifications and permits.

1.7 PROJECT CONDITIONS

- A. Temporary Use of Permanent Facilities: Installer of each permanent service shall assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before Owner's acceptance, regardless of previously assigned responsibilities.

PART 2 - PRODUCTS**2.1 MATERIALS**

- A. Pavement: See Civil Drawings.
- B. Portable Chain-Link Fencing: Minimum 2-inch, 9-gage, galvanized steel, chain-link fabric fencing; minimum 6 feet high with galvanized steel pipe posts; minimum 2-3/8-inch- OD line posts and 2-7/8-inch- OD corner and pull posts, with 1-5/8-inch- OD top and bottom rails. Provide concrete bases for supporting posts.
- C. Lumber and Plywood: Comply with requirements in Division 06 Section "Rough Carpentry."
- D. Gypsum Board: Minimum 1/2 inch thick by 48 inches wide by maximum available lengths; regular-type panels with tapered edges. Comply with ASTM C 36/C 36M.
- E. Insulation: Unfaced mineral-fiber blanket, manufactured from glass, slag wool, or rock wool; with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively.

2.2 TEMPORARY FACILITIES

- A. Field Offices, General: Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.
- B. Storage and Fabrication Sheds: Provide sheds sized, furnished, and equipped to accommodate materials and equipment for construction operations.

2.3 EQUIPMENT

- A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.
- B. HVAC Equipment: Provide vented, self-contained, liquid-propane-gas or fuel-oil heaters with individual space thermostatic control.
 - 1. Use of gasoline-burning space heaters, open-flame heaters, or salamander-type heating units is prohibited.
 - 2. Heating Units: Listed and labeled for type of fuel being consumed, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
 - 3. Permanent HVAC System: If Owner authorizes use of permanent HVAC system for temporary use during construction, provide filter with MERV of 8 at each return air grille in system and remove at end of construction.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.
 - 1. Locate facilities to limit site disturbance as specified in Division 01 Section "Summary."
- B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.2 TEMPORARY UTILITY INSTALLATION

- A. General: Install temporary service or connect to existing service.
 - 1. Arrange with utility company, Owner, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services.
- B. Sewers and Drainage: Provide temporary utilities to remove effluent lawfully.
 - 1. Connect temporary sewers to municipal system as directed by authorities having jurisdiction.
- C. Water Service: Install water service and distribution piping in sizes and pressures adequate for construction.
- D. Sanitary Facilities: Provide temporary toilets, wash facilities, and drinking water for use of construction personnel. Comply with authorities having jurisdiction for type, number, location, operation, and maintenance of fixtures and facilities.
- E. Heating and Cooling: Provide temporary heating and cooling required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of low temperatures or high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed.
- F. Ventilation and Humidity Control: Provide temporary ventilation required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed. Coordinate ventilation requirements to produce ambient condition required and minimize energy consumption.
- G. Electric Power Service: Provide electric power service and distribution system of sufficient size, capacity, and power characteristics required for construction operations.
 - 1. Install electric power service overhead, unless otherwise indicated.
- H. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, inspections, and traffic conditions.
 - 1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.
- I. Telephone Service: Provide temporary telephone service in common-use facilities for use by all construction personnel.
 - 1. Provide additional telephone lines for the following:
 - a. Provide a dedicated telephone line for each facsimile machine and computer in each field office.
 - 2. At each telephone, post a list of important telephone numbers including police and fire departments, Contractor's home office, Architect's office, Owner's office and Principal subcontractors' field and home offices.
 - 3. Provide superintendent with cellular telephone or portable two-way radio for use when away from field office.

- J. Electronic Communication Service: Provide a desktop computer in the field office adequate for use by Architect and Owner to access project electronic documents and maintain electronic communications including electronic mail. Provide the following:
1. Printer: "All-in-one" unit equipped with printer server, combining color printing, photocopying, scanning, and faxing, or separate units for each of these three functions.
 2. Internet Service: Broadband modem, router and ISP, equipped with hardware firewall.
 3. Internet Security: Integrated software, providing software firewall, virus, spyware, phishing, and spam protection in a combined application.

3.3 SUPPORT FACILITIES INSTALLATION

- A. General: Comply with the following:
1. Provide incombustible construction for offices, shops, and sheds located within construction area or within 30 feet of building lines. Comply with NFPA 241.
 2. Maintain support facilities until near Substantial Completion. Remove before Substantial Completion. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to Owner.
- B. Temporary Roads and Paved Areas: Construct and maintain temporary roads and paved areas adequate for construction operations. Locate temporary roads and paved areas within construction limits indicated on Drawings.
1. Provide dust-control treatment that is nonpolluting and nontracking. Reapply treatment as required to minimize dust.
- C. Temporary Use of Permanent Roads and Paved Areas: Locate temporary roads and paved areas in same location as permanent roads and paved areas. Construct and maintain temporary roads and paved areas adequate for construction operations. Extend temporary roads and paved areas, within construction limits indicated, as necessary for construction operations.
1. Coordinate elevations of temporary roads and paved areas with permanent roads and paved areas.
 2. Prepare subgrade and install subbase and base for temporary roads and paved areas according to site work requirements.
 3. Recondition base after temporary use, including removing contaminated material, regrading, proofrolling, compacting, and testing.
 4. Delay installation of final course of permanent hot-mix asphalt pavement until immediately before Substantial Completion. Repair hot-mix asphalt base-course pavement before installation of final course according to Section 321216 "Asphalt Paving."
- D. Traffic Controls: Comply with requirements of authorities having jurisdiction.
1. Protect any existing site improvements to remain including curbs, pavement, and utilities.
 2. Maintain access for fire-fighting equipment and access to any fire hydrants.
- E. Parking: Provide temporary parking areas for construction personnel. Cost of any and all temporary parking for construction personnel is solely the responsibility of the general contractor.

- F. Dewatering Facilities and Drains: Comply with requirements of authorities having jurisdiction. Maintain Project site, excavations, and construction free of water.
 - 1. Dispose of rainwater in a lawful manner that will not result in flooding Project or adjoining properties nor endanger permanent Work or temporary facilities.
 - 2. Remove snow and ice as required to minimize accumulations.
- G. Project Identification and Temporary Signs: Provide Project identification and other signs. Install signs where indicated to inform public and individuals seeking entrance to Project. Unauthorized signs are not permitted.
 - 1. Provide temporary, directional signs for construction personnel and visitors.
 - 2. Maintain and touchup signs so they are legible at all times.
 - 3. Installation of "Now Hiring" and marketing signs. Signs to be provided by the Owner.
- H. Waste Disposal Facilities: Comply with requirements specified in Section 017419 "Construction Waste Management and Disposal" and authorities having jurisdiction. Provide waste-collection containers in sizes adequate to handle waste from construction operations. Comply with Division 01 Section "Execution" for progress cleaning requirements.
- I. Lifts and Hoists: Provide facilities necessary for hoisting materials and personnel.
 - 1. Truck cranes and similar devices used for hoisting materials are considered "tools and equipment" and not temporary facilities.
- J. Temporary Stairs: Until permanent stairs or ladders are available, provide temporary stairs where ladders are not adequate.

3.4 SECURITY AND PROTECTION FACILITIES INSTALLATION

- A. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and that minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.
- B. Temporary Erosion and Sedimentation Control: Provide measures to prevent soil erosion and discharge of soil-bearing water runoff and airborne dust to adjacent properties and walkways, according to requirements of authorities having jurisdiction.
- C. Stormwater Control: Comply with authorities having jurisdiction. Provide barriers in and around excavations and subgrade construction to prevent flooding by runoff of stormwater from heavy rains.
- D. Tree and Plant Protection: Install temporary fencing located as indicated or outside the drip line of trees to protect vegetation from damage from construction operations. Protect tree root systems from damage, flooding, and erosion.
- E. Pest Control: Engage pest-control service to recommend practices to minimize attraction and harboring of rodents, roaches, and other pests and to perform extermination and control procedures at regular intervals so Project will be free of pests and their residues at Substantial Completion. Obtain extended warranty for Owner. Perform control operations lawfully, using environmentally safe materials.

- F. Site Enclosure Fence: Before construction operations begin, furnish and install site enclosure fence in a manner that will prevent people and animals from easily entering site except by entrance gates.
 - 1. Extent of Fence: As required to enclose entire Project site or portion determined sufficient to accommodate construction operations.
 - 2. Maintain security by limiting number of keys and restricting distribution to authorized personnel. Provide Owner with one set of keys.
- G. Security Enclosure and Lockup: Install substantial temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security.
- H. Barricades, Warning Signs, and Lights: Comply with requirements of authorities having jurisdiction for erecting structurally adequate barricades, including warning signs and lighting.
- I. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.
 - 1. Where heating or cooling is needed and permanent enclosure is not complete, insulate temporary enclosures.
- J. Temporary Fire Protection: Install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241.
 - 1. Prohibit smoking in construction areas.
 - 2. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition according to requirements of authorities having jurisdiction.
 - 3. Develop and supervise an overall fire-prevention and fire-protection program for personnel at Project site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
 - 4. Provide temporary standpipes and hoses for fire protection. Hang hoses with a warning sign stating that hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.

3.5 MOISTURE AND MOLD CONTROL

- A. Contractor's Moisture Protection Plan: Avoid trapping water in finished work. Document visible signs of mold that may appear during construction.
- B. Exposed Construction Phase: Before installation of weather barriers, when materials are subject to wetting and exposure and to airborne mold spores, protect materials from water damage and keep porous and organic materials from coming into prolonged contact with concrete.
- C. Partially Enclosed Construction Phase: After installation of weather barriers but before full enclosure and conditioning of building, when installed materials are still subject to infiltration of moisture and ambient mold spores, protect as follows:
 - 1. Do not load or install drywall or other porous materials or components, or items with high organic content, into partially enclosed building.
 - 2. Keep interior spaces reasonably clean and protected from water damage.

3. Discard or replace water-damaged and wet material.
4. Discard, replace, or clean stored or installed material that begins to grow mold.
5. Perform work in a sequence that allows any wet materials adequate time to dry before enclosing the material in drywall or other interior finishes.

D. Controlled Construction Phase of Construction: After completing and sealing of the building enclosure but prior to the full operation of permanent HVAC systems, maintain as follows:

1. Control moisture and humidity inside building by maintaining effective dry-in conditions.
2. Remove materials that cannot be completely restored to their manufactured moisture level within 48 hours.

3.6 OPERATION, TERMINATION, AND REMOVAL

A. Supervision: Enforce strict discipline in use of temporary facilities. To minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.

B. Maintenance: Maintain facilities in good operating condition until removal.

1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on a 24-hour basis where required to achieve indicated results and to avoid possibility of damage.

C. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.

D. Termination and Removal: Remove each temporary facility (including contractor's field office) when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than three (3) weeks prior to opening. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.

1. Materials and facilities that constitute temporary facilities are property of Contractor. Owner reserves right to take possession of Project identification signs.
2. At Substantial Completion, clean and renovate permanent facilities used during construction period. Comply with final cleaning requirements specified in Division 01 Section "Closeout Procedures."

END OF SECTION 015000

SECTION 016000 - PRODUCT REQUIREMENTS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for selection of products for use in Project; product delivery, storage, and handling; manufacturers' standard warranties on products; special warranties; product substitutions; and comparable products.
- B. See Division 01 Section "Closeout Procedures" for submitting warranties for Contract closeout.
- C. See applicable Divisions 02 through 33 Sections for specific requirements for warranties on products and installations specified to be warranted.

1.3 DEFINITIONS

- A. Products: Items purchased for incorporating into the Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes the terms "material," "equipment," "system," and terms of similar intent.
 - 1. Named Products: Items identified by manufacturer's product name, including make or model number or other designation shown or listed in manufacturer's published product literature, that is current as of date of the Contract Documents.
 - 2. New Products: Items that have not previously been incorporated into another project or facility, except that products consisting of recycled-content materials are allowed, unless explicitly stated otherwise. Products salvaged or recycled from other projects are not considered new products.
 - 3. Comparable Product: Product that is demonstrated and approved through submittal process, or where indicated as a product substitution, to have the indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product.
- B. Substitutions: Changes in products, materials, equipment, and methods of construction from those required by the Contract Documents and proposed by Contractor.
- C. Basis-of-Design Product Specification: Where a specific manufacturer's product is named and accompanied by the words "basis of design," including make or model number or other designation, to establish the significant qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics for purposes of evaluating comparable products of other named manufacturers.

1.4 SUBMITTALS

- A. Substitution Requests: Submit three copies of each request for consideration. Identify product or fabrication or installation method to be replaced. Include Specification Section number and title and Drawing numbers and titles.

1. Substitution Request Form: Use CSI Form 13.1A or contractor's standard form containing the same requirements as the CSI Form.
 2. Documentation: Show compliance with requirements for substitutions and the following, as applicable:
 - a. Statement indicating why specified material or product cannot be provided.
 - b. Coordination information, including a list of changes or modifications needed to other parts of the Work and to construction performed by Owner and separate contractors, that will be necessary to accommodate proposed substitution.
 - c. Detailed comparison of significant qualities of proposed substitution with those of the Work specified. Significant qualities may include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
 - d. Product Data, including drawings and descriptions of products and fabrication and installation procedures.
 - e. Samples, where applicable or requested.
 - f. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners.
 - g. Material test reports from a qualified testing agency indicating and interpreting test results for compliance with requirements indicated.
 - h. Research/evaluation reports evidencing compliance with building code in effect for Project, from a model code organization acceptable to authorities having jurisdiction.
 - i. Detailed comparison of Contractor's Construction Schedule using proposed substitution with products specified for the Work, including effect on the overall Contract Time. If specified product or method of construction cannot be provided within the Contract Time, include letter from manufacturer, on manufacturer's letterhead, stating lack of availability or delays in delivery.
 - j. Cost information, including a proposal of change, if any, in the Contract Sum.
 - k. Contractor's certification that proposed substitution complies with requirements in the Contract Documents and is appropriate for applications indicated.
 - l. Contractor's waiver of rights to additional payment or time that may subsequently become necessary because of failure of proposed substitution to produce indicated results.
 3. Architect's Action: If necessary, Architect will request additional information or documentation for evaluation within 5 days of receipt of a request for substitution. Architect will notify Contractor of acceptance or rejection of proposed substitution within 7 days of receipt of request, or 5 days of receipt of additional information or documentation, whichever is later.
 - a. Form of Acceptance: Change Order.
 - b. Use product specified if Architect cannot make a decision on use of a proposed substitution within time allocated.
- B. Comparable Product Requests: Submit three copies of each request for consideration. Identify product or fabrication or installation method to be replaced. Include Specification Section number and title and Drawing numbers and titles.
1. Architect's Action: If necessary, Architect will request additional information or documentation for evaluation within one week of receipt of a comparable product request. Architect will notify Contractor of approval or rejection of proposed comparable product request within 7 days of receipt of request, or 5 days of receipt of additional information or documentation, whichever is later.

- a. Form of Approval: As specified in Division 01 Section "Submittal Procedures."
 - b. Use product specified if Architect cannot make a decision on use of a comparable product request within time allocated.
- C. Basis-of-Design Product Specification Submittal: Comply with requirements in Division 01 Section "Submittal Procedures." Show compliance with requirements.

1.5 QUALITY ASSURANCE

- A. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, product selected shall be compatible with products previously selected, even if previously selected products were also options.

1.6 PRODUCT DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle products using means and methods that will prevent damage, deterioration, and loss, including theft. Comply with manufacturer's written instructions.
- B. Delivery and Handling:
- 1. Schedule delivery to minimize long-term storage at Project site and to prevent overcrowding of construction spaces.
 - 2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration and temperature, theft, and other losses.
 - 3. Deliver products to Project site in an undamaged condition in manufacturer's original sealed container or other packaging system, complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
 - 4. Inspect products on delivery to ensure compliance with the Contract Documents and to ensure that products are undamaged and properly protected.
- C. Storage:
- 1. Store products to allow for inspection and measurement of quantity or counting of units.
 - 2. Store materials in a manner that will not endanger Project structure.
 - 3. Store products that are subject to damage by the elements, under cover in a weathertight enclosure above ground, with ventilation adequate to prevent condensation.
 - 4. Store cementitious products and materials on elevated platforms.
 - 5. Store foam plastic from exposure to sunlight, except to extent necessary for period of installation and concealment.
 - 6. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
 - 7. Protect stored products from damage and liquids from freezing. Provide additional wrapping in addition to any manufacturer's packaging to ensure protection of products from damage.
 - 8. Continue to protect products from damage after installation. Maintain protective wrapping where practical until final project inspection for substantial completion.
 - 9. Any products damaged during delivery or storage shall be replaced with new products. Patching and repair of damaged products is not acceptable.

1.7 PRODUCT WARRANTIES

- A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by the Contract Documents. Manufacturer's disclaimers and limitations on

product warranties do not relieve Contractor of obligations under requirements of the Contract Documents.

1. Manufacturer's Warranty: Preprinted written warranty published by individual manufacturer for a particular product and specifically endorsed by manufacturer to Owner.
 2. Special Warranty: Written warranty required by or incorporated into the Contract Documents, either to extend time limit provided by manufacturer's warranty or to provide more rights for Owner.
- B. Special Warranties: Prepare a written document that contains appropriate terms and identification, ready for execution. Submit a draft for approval before final execution.
1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.
 2. Specified Form: When specified forms are included with the Specifications, prepare a written document using appropriate form properly executed.
 3. Refer to Divisions 2 through 16 Sections for specific content requirements and particular requirements for submitting special warranties.
- C. Submittal Time: Comply with requirements in Division 01 Section "Closeout Procedures."
- D. Product Registration: General contractor shall register the equipment after it is commissioned and turned on.

PART 2 - PRODUCTS

2.1 PRODUCT SELECTION PROCEDURES

- A. General Product Requirements: Provide products that comply with the Contract Documents, that are undamaged and, unless otherwise indicated, that are new at time of installation.
1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
 2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types that have been produced and used successfully in similar situations on other projects.
 3. Owner reserves the right to limit selection to products with warranties not in conflict with requirements of the Contract Documents.
 4. Where products are accompanied by the term "as selected," Architect will make selection.
 5. Where products are accompanied by the term "match sample," sample to be matched is Architect's.
 6. Descriptive, performance, and reference standard requirements in the Specifications establish "salient characteristics" of products.
- B. Product Selection Procedures:
1. Product: Where Specifications name a single product and manufacturer, provide the named product that complies with requirements.
 2. Manufacturer/Source: Where Specifications name a single manufacturer or source, provide a product by the named manufacturer or source that complies with requirements.
 3. Products: Where Specifications include a list of names of both products and manufacturers, provide one of the products listed that complies with requirements.

4. Manufacturers: Where Specifications include a list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements.
5. Product Options: Where Specifications indicate that sizes, profiles, and dimensional requirements on Drawings are based on a specific product or system, provide the specified product or system. Comply with provisions in Part 2 "Product Substitutions" Article for consideration of an unnamed product or system.
6. Basis-of-Design Product: Where Specifications name a product and include a list of manufacturers, provide the specified product or a comparable product by one of the other named manufacturers. Drawings and Specifications indicate sizes, profiles, dimensions, and other characteristics that are based on the product named. Comply with provisions in Part 2 "Comparable Products" Article for consideration of an unnamed product by the other named manufacturers.
7. Visual Matching Specification: Where Specifications require matching an established Sample, select a product that complies with requirements and matches Architect's sample. Architect's decision will be final on whether a proposed product matches.
 - a. If no product available within specified category matches and complies with other specified requirements, comply with provisions in Part 2 "Product Substitutions" Article for proposal of product.
8. Visual Selection Specification: Where Specifications include the phrase "as selected from manufacturer's colors, patterns, textures" or a similar phrase, select a product that complies with other specified requirements.
 - a. Standard Range: Where Specifications include the phrase "standard range of colors, patterns, textures" or similar phrase, Architect will select color, pattern, density, or texture from manufacturer's product line that does not include premium items.
 - b. Full Range: Where Specifications include the phrase "full range of colors, patterns, textures" or similar phrase, Architect will select color, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

2.2 PRODUCT SUBSTITUTIONS

- A. Timing: Architect will consider requests for substitution if received within 60 days after the Notice of Award. Requests received after that time may be considered or rejected at discretion of Architect.
- B. Conditions: Architect will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, Architect will return requests without action, except to record noncompliance with these requirements:
 1. Requested substitution offers Owner a substantial advantage in cost, time, energy conservation, or other considerations, after deducting additional responsibilities Owner must assume. Owner's additional responsibilities may include compensation to Architect for redesign and evaluation services, increased cost of other construction by Owner, and similar considerations.
 2. Requested substitution does not require extensive revisions to the Contract Documents.
 3. Requested substitution is consistent with the Contract Documents and will produce indicated results.
 4. Substitution request is fully documented and properly submitted.
 5. Requested substitution will not adversely affect Contractor's Construction Schedule.
 6. Requested substitution has received necessary approvals of authorities having jurisdiction.

7. Requested substitution is compatible with other portions of the Work.
8. Requested substitution has been coordinated with other portions of the Work.
9. Requested substitution provides specified warranty.

2.3 COMPARABLE PRODUCTS

- A. Conditions: Architect will consider Contractor's request for comparable product when the following conditions are satisfied. If the following conditions are not satisfied, Architect will return requests without action, except to record noncompliance with these requirements:
1. Evidence that the proposed product does not require extensive revisions to the Contract Documents that it is consistent with the Contract Documents and will produce the indicated results, and that it is compatible with other portions of the Work.
 2. Detailed comparison of significant qualities of proposed product with those named in the Specifications. Significant qualities include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
 3. Evidence that proposed product provides specified warranty.
 4. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners, if requested.
 5. Samples, if requested.

PART 3 - EXECUTION (Not Used)

END OF SECTION 016000

SECTION 016400 - OWNER FURNISHED PRODUCTS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes:
 - 1. Owner's responsibilities.
 - 2. Contractor's responsibilities.
 - 3. Products furnished to the site and paid for by Owner.

1.3 OWNER'S RESPONSIBILITIES

- A. Arrange for and deliver Owner reviewed shop drawings, product data, and samples, to Contractor.
- B. Arrange and pay for product delivery to Contractor's warehouse facility or the project site as required by the Project schedule and available storage space. Coordinate delivery location with Contractor.
- C. Submit claims for transportation damage and replace damaged, defective, or deficient items reported by Contractor on receipt.
- D. Arrange for manufacturers' warranties, inspections and field service.

1.4 CONTRACTOR'S RESPONSIBILITIES

- A. Determine quantities required for Owner furnished Products.
- B. Review Owner provided shop drawings, product data, and samples.
- C. Receive and unload and store products at Project site; inspect for completeness and damage.
- D. Submit report of missing or damaged Products to Owner within two days of receipt of Products.
- E. If storage is done at Contractor's warehouse, the Contractor will be responsible for loading, transportation to construction area, unloading in preparation for installation. Receive, handle, store, install and finish products at Project site.
- F. Repair or replace items damaged after receipt.
- G. Deliver to Owner's dedicated warehouse the unused materials and remnants for Owner to store as maintenance materials. Obtain signed receipt from Owner for unused materials delivered to Owner.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Coordinate delivery schedule and delivery location with Owner to expedite Project.

- B. Receive, store and handle Owner furnished products under provisions of Section 01600.

PART 2 - PRODUCTS - Not Used

PART 3 - EXECUTION

3.1 OWNER FURNISHED PRODUCTS SCHEDULE

- A. Items furnished and installed by Owner (FO/IO):
 - 1. See Responsibility Matrix on drawings.
- B. Items furnished by Owner and installed by Contractor (FO/IO):
 - 1. See Responsibility Matrix on drawings.

END OF SECTION 016400

SECTION 017300 – EXECUTION**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes general procedural requirements governing execution of the Work including, but not limited to, the following:
 - 1. Construction layout.
 - 2. Field engineering and surveying.
 - 3. General installation of products.
 - 4. Progress cleaning.
 - 5. Starting and adjusting.
 - 6. Protection of installed construction.
 - 7. Correction of the Work.
- B. See Division 01 Section "Closeout Procedures" for submitting final property survey with Project Record Documents, recording of Owner-accepted deviations from indicated lines and levels, and final cleaning.

1.3 SUBMITTALS

- A. Layout Plan: Show new and existing perimeter, column grids, and other building location requirements.
- B. Certificates: Submit certificate signed by professional engineer certifying that location and elevation of improvements comply with requirements.
- C. Landfill Receipts: Submit copy of receipts issued by a landfill facility, licensed to accept hazardous materials, for hazardous waste disposal.
- D. Certified Surveys: Submit two copies signed by professional engineer.
- E. Final Property Survey: Submit 10 copies showing the Work performed and record survey data.

1.4 QUALITY ASSURANCE

- A. Land Surveyor Qualifications: A professional land surveyor who is legally qualified to practice in jurisdiction where Project is located and who is experienced in providing land-surveying services of the kind indicated.

PART 2 - PRODUCTS (Not Used)**PART 3 - EXECUTION****3.1 EXAMINATION**

- A. Existing Conditions: The existence and location of site improvements, utilities, and other construction indicated as existing are not guaranteed. Before beginning work, investigate and verify the existence and location of mechanical and electrical systems and other construction affecting the Work.
 - 1. Before construction, verify the location and points of connection of utility services.
- B. Existing Utilities: The existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Before beginning sitework, investigate and verify the existence and location of underground utilities and other construction affecting the Work.
 - 1. Before construction, verify the location and invert elevation at points of connection of sanitary sewer, storm sewer, and water-service piping; and underground electrical services.
 - 2. Furnish location data for work related to Project that must be performed by public utilities serving Project site.
- C. Acceptance of Conditions: Examine substrates, areas, and conditions, with Installer or Applicator present where indicated, for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.
 - 1. Verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.
 - 2. Examine roughing-in for mechanical and electrical systems to verify actual locations of connections before equipment and fixture installation.
 - 3. Examine walls, floors, and roofs for suitable conditions where products and systems are to be installed.
 - 4. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Existing Utility Information: Furnish information to local utility and Owner that is necessary to adjust, move, or relocate existing utility structures, utility poles, lines, services, or other utility appurtenances located in or affected by construction. Coordinate with authorities having jurisdiction.
- B. Field Measurements: Take field measurements as required to fit the Work properly. Recheck measurements before installing each product. Where portions of the Work are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying the Work.
- C. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.
- D. Review of Contract Documents and Field Conditions: Immediately on discovery of the need for clarification of the Contract Documents, submit a request for information to Architect. Include a detailed description of problem encountered, together with recommendations for changing the Contract Documents.

3.3 CONSTRUCTION LAYOUT

- A. Verification: Before proceeding to lay out the Work, verify layout information shown on Drawings, in relation to the property survey and existing benchmarks. If discrepancies are discovered, notify Architect promptly.
- B. General: Engage a professional engineer to lay out the Work using accepted surveying practices.
 - 1. Establish benchmarks and control points to set lines and levels at each story of construction and elsewhere as needed to locate each element of Project.
 - 2. Establish dimensions within tolerances indicated. Do not scale Drawings to obtain required dimensions.
 - 3. Inform installers of lines and levels to which they must comply.
 - 4. Check the location, level and plumb, of every major element as the Work progresses.
 - 5. Notify Architect when deviations from required lines and levels exceed allowable tolerances.
 - 6. Close site surveys with an error of closure equal to or less than the standard established by authorities having jurisdiction.
- C. Site Improvements: Locate and lay out site improvements, including pavements, grading, fill and topsoil placement, utility slopes, and invert elevations.
- D. Building Lines and Levels: Locate and lay out control lines and levels for structures, building foundations, column grids, and floor levels, including those required for mechanical and electrical work. Transfer survey markings and elevations for use with control lines and levels. Level foundations and piers from two or more locations.
- E. Record Log: Maintain a log of layout control work. Record deviations from required lines and levels. Include beginning and ending dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make the log available for reference by Architect.

3.4 FIELD ENGINEERING

- A. Reference Points: Locate existing permanent benchmarks, control points, and similar reference points before beginning the Work. Preserve and protect permanent benchmarks and control points during construction operations.
- B. Benchmarks: Establish and maintain a minimum of two permanent benchmarks on Project site, referenced to data established by survey control points. Comply with authorities having jurisdiction for type and size of benchmark.
 - 1. Record benchmark locations, with horizontal and vertical data, on Project Record Documents.
- C. Certified Survey: On completion of foundation walls, major site improvements, and other work requiring field-engineering services, prepare a certified survey showing dimensions, locations, angles, and elevations of construction and sitework.
- D. Final Property Survey: Prepare a final property survey showing significant features (real property) for Project. Include on the survey a certification, signed by professional engineer, that principal metes, bounds, lines, and levels of Project are accurately positioned as shown on the survey.

1. Recording: At Substantial Completion, have the final property survey recorded by or with authorities having jurisdiction as the official "property survey."

3.5 INSTALLATION

- A. General: Locate the Work and components of the Work accurately, in correct alignment and elevation, as indicated.
 1. Make vertical work plumb and make horizontal work level.
 2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
 3. Conceal pipes, ducts, and wiring in finished areas, unless otherwise indicated.
- B. Comply with manufacturer's written instructions and recommendations for installing products in applications indicated.
- C. Install products at the time and under conditions that will ensure the best possible results. Maintain conditions required for product performance until Substantial Completion.
- D. Conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.
- E. Tools and Equipment: Do not use tools or equipment that produce harmful noise levels.
- F. Templates: Obtain and distribute to the parties involved templates for work specified to be factory prepared and field installed. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing products to comply with indicated requirements.
- G. Anchors and Fasteners: Provide anchors and fasteners as required to anchor each component securely in place, accurately located and aligned with other portions of the Work.
 1. Mounting Heights: Where mounting heights are not indicated, mount components at heights directed by Architect.
 2. Allow for building movement, including thermal expansion and contraction.
 3. Coordinate installation of anchorages. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.
- H. Joints: Make joints of uniform width. Where joint locations in exposed work are not indicated, arrange joints for the best visual effect. Fit exposed connections together to form hairline joints.
- I. Hazardous Materials: Use products, cleaners, and installation materials that are not considered hazardous.

3.6 PROGRESS CLEANING

- A. General: Clean Project site and work areas daily, including common areas. Coordinate progress cleaning for joint-use areas where more than one installer has worked. Enforce requirements strictly. Dispose of materials lawfully.
 1. Comply with requirements in NFPA 241 for removal of combustible waste materials and debris.

2. Do not hold materials more than 7 days during normal weather or 3 days if the temperature is expected to rise above 80 deg F.
 3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, according to regulations.
- B. Site: Maintain Project site free of waste materials and debris.
- C. Work Areas: Clean areas where work is in progress to the level of cleanliness necessary for proper execution of the Work.
1. Remove liquid spills promptly.
 2. Where dust would impair proper execution of the Work, broom-clean or vacuum the entire work area, as appropriate.
- D. Installed Work: Keep installed work clean. Clean installed surfaces according to written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials that are not hazardous to health or property and that will not damage exposed surfaces.
- E. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.
- F. Exposed Surfaces in Finished Areas: Clean exposed surfaces and protect as necessary to ensure freedom from damage and deterioration at time of Substantial Completion.
- G. Waste Disposal: Burying or burning waste materials on-site will not be permitted. Washing waste materials down sewers or into waterways will not be permitted.
- H. During handling and installation, clean and protect construction in progress and adjoining materials already in place. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.
- I. Clean and provide maintenance on completed construction as frequently as necessary through the remainder of the construction period. Adjust and lubricate operable components to ensure operability without damaging effects.
- J. Limiting Exposures: Supervise construction operations to assure that no part of the construction, completed or in progress, is subject to harmful, dangerous, damaging, or otherwise deleterious exposure during the construction period.

3.7 STARTING AND ADJUSTING

- A. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.
- B. Adjust operating components for proper operation without binding. Adjust equipment for proper operation.
- C. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- D. Manufacturer's Field Service: If a factory-authorized service representative is required to inspect field-assembled components and equipment installation, comply with qualification requirements in Division 01 Section "Quality Requirements."

3.8 PROTECTION OF INSTALLED CONSTRUCTION

- A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.
- B. Comply with manufacturer's written instructions for temperature and relative humidity.

3.9 CORRECTION OF THE WORK

- A. Repair or remove and replace defective construction. Restore damaged substrates and finishes. Comply with requirements in Division 01 Section "Cutting and Patching."
 - 1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.
- B. Restore permanent facilities used during construction to their specified condition.
- C. Remove and replace damaged surfaces that are exposed to view if surfaces cannot be repaired without visible evidence of repair.
- D. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.
- E. Remove and replace chipped, scratched, and broken glass or reflective surfaces.

END OF SECTION 017300

SECTION 017329 - CUTTING AND PATCHING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes procedural requirements for cutting and patching.
- B. See applicable Divisions 02 through 33 Sections for specific requirements and limitations applicable to cutting and patching individual parts of the Work.
- C. See Division 07 Section "Penetration Firestopping" for patching fire-rated construction.

1.3 SUBMITTALS

- A. Cutting and Patching Plan: Proposal describing procedures at least 10 days before the time cutting and patching will be performed, requesting approval to proceed. Include the following information:
 - 1. Extent: Describe cutting and patching, show how they will be performed, and indicate why they cannot be avoided.
 - 2. Changes to In-Place Construction: Describe anticipated results. Include changes to structural elements and operating components as well as changes in building's appearance and other significant visual elements.
 - 3. Products: List products to be used and firms or entities that will perform the Work.
 - 4. Dates: Indicate when cutting and patching will be performed.
 - 5. Utility Services and Mechanical/Electrical Systems: List services/systems that cutting and patching procedures will disturb or affect. List services/systems that will be relocated and those that will be temporarily out of service. Indicate how long services/systems will be disrupted.
 - 6. Structural Elements only with Architect's Approval: Any defect item must be removed and replaced. No patch or cutting will be allowed on site.
 - 7. Architect's Approval: Obtain approval of cutting and patching proposal before cutting and patching. Approval does not waive right to later require removal and replacement of unsatisfactory work.

1.4 QUALITY ASSURANCE

- A. Structural Elements: Do not cut and patch structural elements in a manner that could change their load-carrying capacity or load-deflection ratio.
- B. Operational Elements: Do not cut and patch operating elements and related components in a manner that results in reducing their capacity to perform as intended or that results in increased maintenance or decreased operational life or safety. Operating elements include the following:
 - 1. Heating, ventilating and air conditioning systems.
 - 2. Plumbing systems.
 - 3. Fire protection systems.
 - 4. Fire alarm systems.

- 5. Communication systems.
- 6. Vertical transportation systems.

- C. Miscellaneous Elements: Do not cut and patch miscellaneous elements or related components in a manner that could change their load-carrying capacity, that results in reducing their capacity to perform as intended, or that results in increased maintenance or decreased operational life or safety.
- D. Visual Requirements: Do not cut and patch construction in a manner that results in visual evidence of cutting and patching. Do not cut and patch construction exposed on the exterior or in occupied spaces in a manner that would, in Architect's opinion, reduce the building's aesthetic qualities. Remove and replace construction that has been cut and patched in a visually unsatisfactory manner.

1.5 WARRANTY

- A. Existing Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during cutting and patching operations, by methods and with materials so as not to void existing warranties.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. General: Comply with requirements specified in other Sections.
- B. In-Place Materials: Use materials identical to in-place materials. For exposed surfaces, use materials that visually match in-place adjacent surfaces to the fullest extent possible.
 - 1. If identical materials are unavailable or cannot be used, use materials that, when installed, will match the visual and functional performance of in-place materials.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine surfaces to be cut and patched and conditions under which cutting and patching are to be performed.
 - 1. Compatibility: Before patching, verify compatibility with and suitability of substrates, including compatibility with in-place finishes or primers.
 - 2. Proceed with installation only after unsafe or unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Temporary Support: Provide temporary support of Work to be cut.
- B. Protection: Protect in-place construction during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portions of Project that might be exposed during cutting and patching operations.
- C. Adjoining Areas: Avoid interference with use of adjoining areas or interruption of free passage to adjoining areas.

- D. Existing Utility Services and Mechanical/Electrical Systems: Where existing services/systems are required to be removed, relocated, or abandoned, bypass such services/systems before cutting to prevent interruption to occupied areas.

3.3 PERFORMANCE

- A. General: Employ skilled workers to perform cutting and patching. Proceed with cutting and patching at the earliest feasible time, and complete without delay.
 - 1. Cut in-place construction to provide for installation of other components or performance of other construction, and subsequently patch as required to restore surfaces to their original condition.
- B. Cutting: Cut in-place construction by sawing, drilling, breaking, chipping, grinding, and similar operations, including excavation, using methods least likely to damage elements retained or adjoining construction. If possible, review proposed procedures with original Installer; comply with original Installer's written recommendations.
 - 1. In general, use hand or small power tools designed for sawing and grinding, not hammering and chopping. Cut holes and slots as small as possible, neatly to size required, and with minimum disturbance of adjacent surfaces. Temporarily cover openings when not in use.
 - 2. Finished Surfaces: Cut or drill from the exposed or finished side into concealed surfaces.
 - 3. Concrete and masonry: Cut using a cutting machine, such as an abrasive saw or a diamond-core drill.
 - 4. Excavating and Backfilling: Comply with requirements in applicable Division 31 Sections where required by cutting and patching operations.
 - 5. Mechanical and Electrical Services: Cut off pipe or conduit in walls or partitions to be removed. Cap, valve, or plug and seal remaining portion of pipe or conduit to prevent entrance of moisture or other foreign matter after cutting.
 - 6. Proceed with patching after construction operations requiring cutting are complete.
- C. Patching: Patch construction by filling, repairing, refinishing, closing up, and similar operations following performance of other Work. Patch with durable seams that are as invisible as possible. Provide materials and comply with installation requirements specified in other Sections.
 - 1. Inspection: Where feasible, test and inspect patched areas after completion to demonstrate integrity of installation.
 - 2. Exposed Finishes: Restore exposed finishes of patched areas and extend finish restoration into retained adjoining construction in a manner that will eliminate evidence of patching and refinishing.
 - 3. Floors and Walls: Where walls or partitions that are removed extend one finished area into another, patch and repair floor and wall surfaces in the new space. Provide an even surface of uniform finish, color, texture, and appearance. Remove in-place floor and wall coverings and replace with new materials, if necessary, to achieve uniform color and appearance.
 - 4. Ceilings: Patch, repair, or rehang in-place ceilings as necessary to provide an even-plane surface of uniform appearance.
 - 5. Exterior Building Enclosure: Patch components in a manner that restores enclosure to a weathertight condition.
- D. Cleaning: Clean areas and spaces where cutting and patching are performed. Completely remove paint, mortar, oils, putty, and similar materials.

END OF SECTION 017329

SECTION 017419 - CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes administrative and procedural requirements for the following:
 - 1. Recycling of nonhazardous construction waste
 - 2. Disposing of nonhazardous demolition and construction waste.
- B. Related Sections:
 - 1. Division 04 Sections "Unit Masonry" and "Cast Stone Masonry" for disposal requirements for masonry waste.

1.3 DEFINITIONS

- A. Construction Waste: Building and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.
- B. Demolition Waste: Building and site improvement materials resulting from demolition or selective demolition operations.
- C. Disposal: Removal off-site of demolition and construction waste and subsequent sale, recycling, reuse, or deposit in landfill or incinerator acceptable to authorities having jurisdiction.
- D. Recycle: Recovery of demolition or construction waste for subsequent processing in preparation for reuse.
- E. Salvage: Recovery of demolition or construction waste and subsequent sale or reuse in another facility.
- F. Salvage and Reuse: Recovery of demolition or construction waste and subsequent incorporation into the Work.

1.4 PERFORMANCE REQUIREMENTS

- A. General: Practice efficient waste management in the use of materials in the course of the Work. Use all reasonable means to divert construction and demolition waste from landfills and incinerators. Facilitate recycling and salvage of materials. Achieve End-of-Project rates for recycling acceptable to authorities having jurisdiction of total non-Hazardous solid waste generated by the Work.

1.5 ACTION SUBMITTALS

- A. Waste Management Plan: Submit plan within 15 days of date established for the Notice to Proceed.

1.6 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For waste management coordinator.
- B. Waste Reduction Progress Reports: Concurrent with each Application for Payment, submit report. Include the following information:
 - 1. Material category.
 - 2. Generation point of waste.
 - 3. Total quantity of waste in tons (tonnes).
 - 4. Quantity of waste salvaged, both estimated and actual in tons (tonnes).
 - 5. Quantity of waste recycled, both estimated and actual in tons (tonnes).
 - 6. Total quantity of waste recovered (salvaged plus recycled) in tons (tonnes).
 - 7. Total quantity of waste recovered (salvaged plus recycled) as a percentage of total waste.
- C. Waste Reduction Calculations: Before request for Substantial Completion, submit calculated end-of-Project rates for salvage, recycling, and disposal as a percentage of total waste generated by the Work.

1.7 CLOSEOUT SUBMITTALS

- A. Records of Donations: Indicate receipt and acceptance of salvageable waste donated to individuals and organizations. Indicate whether organization is tax exempt.
- B. Records of Sales: Indicate receipt and acceptance of salvageable waste sold to individuals and organizations. Indicate whether organization is tax exempt.
- C. Recycling and Processing Facility Records: Indicate receipt and acceptance of recyclable waste by recycling and processing facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.
- D. Landfill and Incinerator Disposal Records: Indicate receipt and acceptance of waste by landfills and incinerator facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.

1.8 QUALITY ASSURANCE

- A. Waste Management Coordinator Qualifications: Experienced firm, with a record of successful waste management coordination of projects with similar requirements.
- B. Waste Management Conference: Conduct conference at Project site to comply with requirements in Section 013100 "Project Management and Coordination."
- C. Regulatory Requirements: Comply with hauling and disposal regulations of authorities having jurisdiction.

1.9 WASTE MANAGEMENT PLAN

- A. General: Develop a waste management plan according to ASTM E 1609 and requirements in this Section. Plan shall consist of waste identification, waste reduction work plan, and cost/revenue analysis. Indicate quantities by weight or volume but use same units of measure throughout waste management plan.

- B. Waste Identification: Indicate anticipated types and quantities of site-clearing and construction waste generated by the Work. Include estimated quantities and assumptions for estimates.
- C. Waste Reduction Work Plan: List each type of waste and whether it will be salvaged, recycled, or disposed of in landfill or incinerator. Include points of waste generation, total quantity of each type of waste, quantity for each means of recovery, and handling and transportation procedures.
 - 1. Salvaged Materials for Reuse: For materials that will be salvaged and reused in this Project, describe methods for preparing salvaged materials before incorporation into the Work.
 - 2. Salvaged Materials for Sale: For materials that will be sold to individuals and organizations, include list of their names, addresses, and telephone numbers.
 - 3. Salvaged Materials for Donation: For materials that will be donated to individuals and organizations, include list of their names, addresses, and telephone numbers.
 - 4. Recycled Materials: Include list of local receivers and processors and type of recycled materials each will accept. Include names, addresses, and telephone numbers.
 - 5. Disposed Materials: Indicate how and where materials will be disposed of. Include name, address, and telephone number of each landfill and incinerator facility.
 - 6. Handling and Transportation Procedures: Include method that will be used for separating recyclable waste including sizes of containers, container labeling, and designated location where materials separation will be performed.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 PLAN IMPLEMENTATION

- A. General: Implement approved waste management plan. Provide handling, containers, storage, signage, transportation, and other items as required to implement waste management plan during the entire duration of the Contract.
 - 1. Comply with Division 01 Section "Temporary Facilities and Controls" for operation, termination, and removal requirements.
- B. Training: Train workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work occurring at Project site.
 - 1. Distribute waste management plan to entities when they first begin work on-site. Review plan procedures and locations established for salvage, recycling, and disposal.
- C. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.
 - 1. Designate and label specific areas on Project site necessary for separating materials that are to be salvaged, recycled, reused and sold.
 - 2. Comply with Division 01 Section "Temporary Facilities and Controls" for controlling dust and dirt, environmental protection, and noise control.

3.2 RECYCLING CONSTRUCTION WASTE, GENERAL

- A. General: Recycle paper and beverage containers used by on-site workers.

- B. Preparation of Waste: Prepare and maintain recyclable waste materials according to recycling or reuse facility requirements. Maintain materials free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process.
- C. Procedures: Separate recyclable waste from other waste materials, trash, and debris. Separate recyclable waste by type at Project site to the maximum extent practical according to approved construction waste management plan.
 - 1. Provide appropriately marked containers or bins for controlling recyclable waste until they are removed from Project site. Include list of acceptable and unacceptable materials at each container and bin.
 - a. Inspect containers and bins for contamination and remove contaminated materials if found.
 - 2. Remove recyclable waste off Owner's property and transport to recycling receiver or processor.

3.3 RECYCLING CONSTRUCTION WASTE

- A. Packaging:
 - 1. Cardboard and Boxes: Break down packaging into flat sheets. Bundle and store in a dry location.
 - 2. Polystyrene Packaging: Separate and bag materials.
 - 3. Pallets: As much as possible, require deliveries using pallets to remove pallets from Project site. For pallets that remain on-site, break down pallets into component wood pieces and comply with requirements for recycling wood.
 - 4. Crates: Break down crates into component wood pieces and comply with requirements for recycling wood.

3.4 DISPOSAL OF WASTE

- A. General: Except for items or materials to be salvaged, recycled, or otherwise reused, remove waste materials from Project site and legally dispose of them in a landfill or incinerator acceptable to authorities having jurisdiction.
 - 1. Except as otherwise specified, do not allow waste materials that are to be disposed of accumulate on-site.
 - 2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
- B. Burning: Do not burn waste materials.
- C. Disposal: Transport waste materials off Owner's property and legally dispose of them.

END OF SECTION 017419

SECTION 017700 - CLOSEOUT PROCEDURES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for contract closeout, including, but not limited to, the following:
 - 1. Inspection procedures.
 - 2. Warranties.
 - 3. Final cleaning.
- B. All Closeout procedure must refer to Contract Document.

1.3 SUBSTANTIAL COMPLETION

- A. Preliminary Procedures: Before requesting inspection for determining date of Substantial Completion, complete the following. List items below that are incomplete in request.
 - 1. Prepare a list of items to be completed and corrected (punch list), the value of items on the list, and reasons why the Work is not complete.
 - 2. Advise Owner of pending insurance changeover requirements.
 - 3. Submit specific warranties, workmanship bonds, maintenance service agreements, final certifications, and similar documents.
 - 4. Obtain and submit releases permitting Owner unrestricted use of the Work and access to services and utilities. Include occupancy permits, operating certificates, and similar releases.
 - 5. Prepare and submit Project Record Documents, test reports, certificates, manufacturer's instructions, manufacturer's field reports, operation and maintenance manuals, closeout submittals, Final Completion construction photographs, damage or settlement surveys, property surveys, and similar final record information.
 - 6. Deliver tools, spare parts, extra materials, and similar items to location designated by Owner. Label with manufacturer's name and model number where applicable.
 - 7. Install permanent locks and deliver keys to Owner. Advise Owner's personnel of changeover in security provisions.
 - 8. Complete startup testing of systems.
 - 9. Submit test/adjust/balance records.
 - 10. Terminate and remove temporary facilities from Project site, along with mockups, construction tools, and similar elements.
 - 11. Advise Owner of changeover in heat and other utilities.
 - 12. Submit changeover information related to Owner's occupancy, use, operation, and maintenance.
 - 13. Complete final cleaning requirements, including touchup painting.
 - 14. Touch up and otherwise repair and restore marred exposed finishes to eliminate visual defects.
 - 15. For additional requirements see contract documents A101 and A102.

- B. Inspection: Submit a written request for inspection for Substantial Completion. On receipt of request, Architect will either proceed with inspection or notify Contractor of unfulfilled requirements. Architect will prepare the Certificate of Substantial Completion after inspection or will notify Contractor of items, either on Contractor's list or additional items identified by Architect, that must be completed or corrected before certificate will be issued.
 - 1. Reinspection: Request reinspection when the Work identified in previous inspections as incomplete is completed or corrected.
 - 2. Results of completed inspection will form the basis of requirements for Final Completion.

1.4 FINAL COMPLETION

- A. As used in the Contract Documents, "Final Completion" (and/or "Finally Complete") shall mean, and "Final Payment" shall not be due to the Contractor until, the proper and full completion of all of the Work, including but not limited to the completion of the "Final Closeout Documents", satisfactory operation of all equipment and systems, completion or correction of all punchlist items, delivery of all maintenance and operations manuals and all warranties and guarantees (and assignments thereof), assurance of all required approvals and certificates by any authorities with jurisdiction over the Work (including final and unconditional certificates of occupancy), removal of all rubbish, tools, scaffolding and surplus materials from the Project site and correction of all property damage that is the responsibility of the Contractor pursuant to the Contract Documents. Within five (5) days following receipt of the Contractor's written notice that the Work is ready for final inspection and acceptance and upon receipt of a final Application for Payment, the Architect and the Owner will promptly make such inspection and, when the Owner finds the Work acceptable under the Contract Documents and the Contract fully performed, the Owner will remit final payment pursuant to Final Payment section of the Owner's Form of Contract and subject to fulfillment of all prerequisites to Final Payment as provided in the Contract Documents.

1.5 LIST OF INCOMPLETE ITEMS (PUNCH LIST)

- A. Refer to contract document for punch list process.

1.6 WARRANTIES

- A. Submittal Time: Submit written and digital warranties on request of Architect for designated portions of the Work where commencement of warranties other than date of Substantial Completion is indicated.
- B. Organize warranty documents into an orderly sequence based on the table of contents of the Project Manual.
 - 1. Bind warranties and bonds in heavy-duty, 3-ring, vinyl-covered, loose-leaf binders, thickness as necessary to accommodate contents, and sized to receive 8-1/2-by-11-inch paper.
 - 2. Provide heavy paper dividers with plastic-covered tabs for each separate warranty. Mark tab to identify the product or installation. Provide a typed description of the product or installation, including the name of the product and the name, address, and telephone number of Installer.
 - 3. Identify each binder on the front and spine with the typed or printed title "WARRANTIES," Project name, and name of Contractor.
- C. Warranty Electronic File: Provide warranties and bonds in PDF format. Assemble complete warranty and bond submittal package into a single electronic PDF file with bookmarks enabling navigation to each item. Provide bookmarked table of contents at beginning of document.

1. Submit by uploading to web-based project software site.

- D. Provide additional copies of each warranty to include in operation and maintenance manuals.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Cleaning Agents: Use cleaning materials and agents recommended by manufacturer or fabricator of the surface to be cleaned. Do not use cleaning agents that are potentially hazardous to health or property or that might damage finished surfaces.

PART 3 - EXECUTION

3.1 FINAL CLEANING

- A. General: Provide final cleaning in two stages as follow:
 1. Substantial completion (before handover to FM)
 2. Before opening
- B. Conduct cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.
- C. Substantial Completion Cleaning (before handover to FM): Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.
 1. The Contractor shall keep the premises and surrounding area free from accumulation of waste materials or rubbish caused by operations under the Contract. At completion of the Work, the Contractor shall remove waste materials, rubbish, the Contractor's tools, construction equipment, machinery and surplus materials from and about the Project. The Contractor shall maintain streets and sidewalks around the site in a clean condition and shall comply with all erosion control and storm water runoff ordinances and regulations. The Contractor shall remove all spillage and tracking arising from the performance of the Work from such areas and shall establish a regular maintenance program of sweeping and hosing to minimize accumulation of dirt and dust upon such areas. The Contractor shall take all steps necessary to minimize emissions and noise from equipment used by Contractor or any Subcontractors while at the Project site. If the Contractor fails to clean up as provided in the Contract Documents, the Owner may do so and Owner shall be entitled to reimbursement from the Contractor.
 2. Complete the following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a portion of Project:
 - a. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
 - b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
 - c. Rake grounds that are neither planted nor paved to a smooth, even-textured surface.
 - d. Remove tools, construction equipment, machinery, and Contractor's surplus materials from Project site no later than three (3) weeks prior to Opening.
 - e. Remove snow and ice to provide safe access to building.

- f. Clean exposed exterior and interior hard-surfaced finishes to a dirt-free condition, free of stains, films, and similar foreign substances. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to their original condition.
 - g. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.
 - h. Sweep concrete floors broom clean in unoccupied spaces.
 - i. Vacuum carpet and similar soft surfaces, removing debris and excess nap; shampoo if visible soil or stains remain.
 - j. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and other noticeable, vision-obscuring materials. Replace chipped or broken glass and other damaged transparent materials. Polish mirrors and glass, taking care not to scratch surfaces.
 - k. Remove labels that are not permanent.
 - l. Touch up and otherwise repair and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.
 - 1) Do not paint over "UL" and similar labels, including mechanical and electrical nameplates.
 - m. Wipe surfaces of mechanical and electrical equipment, elevator equipment, and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.
 - n. Replace parts subject to unusual operating conditions.
 - o. Clean plumbing fixtures to a sanitary condition, free of stains, including stains resulting from water exposure.
 - p. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of diffusers, registers, and grills.
 - q. Clean light fixtures, lamps, globes, and reflectors to function with full efficiency. Replace burned-out bulbs, and those noticeably dimmed by hours of use, and defective and noisy starters in fluorescent and mercury vapor fixtures to comply with requirements for new fixtures.
 - r. Leave Project clean and ready for occupancy.
 - s. Remove field office no later than three (3) weeks prior to Opening. See Section 015000 "Temporary Facilities and Controls".
 - t. Pressure wash building exterior, hardscape and paved surfaces (including parking lot and curbing) three (3) days prior to Opening. See Sections 042000 "Unit Masonry", 047200 "Cast Stone Masonry" and 072419 "Exterior Insulation and Finish System with Moisture Drainage".
 - u. Provide full-service lawn and landscape maintenance two (2) days prior to Opening. Replace dead and defected plant materials and provide tree staking as specified in Civil Notes and Details two (2) days prior to Opening. Stakes shall be level, consistent and cut clean. Notify Owner of Maintenance Service Agreement(s) no later than six (6) weeks prior to Opening.
- D. Cleaning Before Opening: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions
- 1. General Cleaning:
 - a. Remove marks and fingerprints on walls
 - b. Clean baseboard
 - c. Clean window sills, tracks, in/outside glass

- d. Clean closet shelves and rods
 - e. Remove fingerprints around knobs; dust on louvered door.
 - f. Professionally clean floors
2. Light Fixture:
- a. Dust light fixture. Remove cover; wash; put back in place
 - b. Replace burnt-out light bulbs
 - c. Clean switch plates. Replace missing or broken plates
 - d. Clean outlet plates. Replace missing or broken plates
3. Smoke Alarm:
- a. Test alarms; retest to insure working order.
4. Outside:
- a. Remove cobwebs from eaves and doorway
 - b. Clean exterior light fixtures; replace burnt out bulbs
 - c. Edge and weed all flower beds
 - d. Mow lawn
 - e. Remove all debris from grounds and storage area
 - f. Broom clean parking, cart corral, storage area and walkways
 - g. Remove all garbage, recyclable and yard waste from cans
5. Bakery (Working Area):
- a. Clean in/outside cupboard doors; remove fingerprints; polish
 - b. Clean drawer faces; polish
 - c. Clean cupboard shelves and inside drawers
 - d. Clean cupboard under sink
 - e. Clean counters; remove stains
 - f. Clean sink; remove stains; polish fixture
 - g. Multiple-Compartment Sink: Before using and at the end of every work period, clean the sink, scrap baskets and splash backs as follows:
 - 1) Fill all sinks to a low level with a hot solution of Liquid Detergent.
 - 2) Brush-wash inside and outside of all sinks, drain table and splash back.
 - 3) Drain and rinse sinks. Spray sanitize all compartments
 - h. Clear the Decks:
 - 1) Empty all trash containers into designated receptacles.
 - 2) Take empty boxes to the recycling area.
 - 3) Remove any packaging supplies and other non-essential items from the Bakery.
 - i. Clean wall above sink
 - j. Clean walls around and above ovens
 - k. Clean outside of hood
 - l. Remove filter; clean; replace if necessary
 - m. Clean outside panels and door handles
 - n. Clean control panel
 - o. Clean oven and racks

- p. Clean storage drawer
 - q. Clean oven tops
 - r. Clean drip pans or replace
6. Refrigerator:
- a. Clean side, tops, door and handle
 - b. Defrost and clean freezer if necessary
 - c. Remove and clean shelves and crispers
 - d. Clean inside; replace shelves and crispers if necessary
 - e. Clean door shelves and trays
 - f. Clean refrigerators shades
 - g. Leave refrigerator plugged and on.
7. Floor:
- a. Wash floors, dry and seal per SSI Maintenance specifications
8. Office Quarters:
- a. Clean all ceilings, walls and floors
 - b. Clean all side, tops, door and handle
 - c. Remove and clean shelves
 - d. Clean any stains on ceiling, walls and doors
 - e. Clean above any unit or cabinets installed
9. Back of House:
- a. Clean all walls and floors
 - b. Clean all side, tops, door and handle
 - c. Remove and clean shelves
 - d. Clean any stains on walls, doors
 - e. Clean above freezer and cooler boxes.
10. Back of House Freezer and Cooler:
- a. Clean all side, tops, door and handle
 - b. Wash outside surface and handles with hot Liquid Detergent. Special attention should be given to the back of the handles
 - c. Scrape down with metal scraper, including sides. Sweep with bench brush, then wash with a solution of hot water and Liquid Detergent, rinse and sanitize with Quat Rinse.
 - d. Defrost and clean freezer if necessary
 - e. Remove and clean shelves
 - f. Clean inside; replace shelves if necessary
 - g. Clean any stains on exterior and interior
 - h. Clean above freezer and cooler boxes.
11. Bathrooms:
- a. Bathrooms
 - b. Clean sink and soap holder
 - c. Clean mirror
 - d. Clean in/outside of toilet; remove any dye dispenser/tablet; disinfect.

- e. Wash floor and walls; remove dirt along corners and toilet.
- f. Clean top of tiles
- g. Clean baby changing station.

12. Leave Project clean and ready to open.

- E. Pest Control: Engage an experienced, licensed exterminator to make a final inspection and rid Project of rodents, insects, and other pests. Prepare a report.
- F. Comply with safety standards for cleaning. Do not burn waste materials. Do not bury debris or excess materials on Owner's property. Do not discharge volatile, harmful, or dangerous materials into drainage systems. Remove waste materials from Project site and dispose of lawfully.

END OF SECTION 017700

SECTION 017823 - OPERATION AND MAINTENANCE DATA**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for preparing operation and maintenance manuals, including the following:
 - 1. Emergency manuals.
 - 2. Operation manuals for systems, subsystems, and equipment.
 - 3. Maintenance manuals for the care and maintenance of products, materials, and finishes, systems and equipment.
- B. See Divisions 02 through 32 Sections for specific operation and maintenance manual requirements for the Work in those Sections.

1.3 SUBMITTALS

- A. Manuals: Submit each manual in final form at least 15 days before final inspection. Architect will provide comments within 15 days after final inspection.
 - 1. Correct or modify each manual to comply with Architect's comments. Re-submit each corrected manual within 15 days of receipt of Architect's comments.
 - 2. Submit by uploading to web-based project software site. Enable reviewer comments on draft submittals.

PART 2 - PRODUCTS**2.1 MANUALS, GENERAL**

- A. Organization: Unless otherwise indicated, organize each manual into a separate section for each system and subsystem, and a separate section for each piece of equipment not part of a system. Each manual shall contain a title page, table of contents, and manual contents.
- B. Title Page: Enclose title page in transparent plastic sleeve. Include the following information:
 - 1. Subject matter included in manual.
 - 2. Name and address of Project.
 - 3. Name and address of Owner.
 - 4. Date of submittal.
 - 5. Name, address, and telephone number of Contractor.
 - 6. Name and address of Architect.
 - 7. Cross-reference to related systems in other operation and maintenance manuals.
- C. Table of Contents: List each product included in manual, identified by product name, indexed to the content of the volume and cross-referenced to Specification Section number in Project Manual.

- D. Manual Contents: Organize into sets of manageable size. Arrange contents alphabetically by system, subsystem, and equipment. If possible, assemble instructions for subsystems, equipment, and components of one system into a single binder.
1. Binders: Heavy-duty, 3 D-ring, vinyl-covered, loose-leaf binders, in thickness necessary to accommodate contents, sized to hold 8-1/2-by-11-inch paper; with clear plastic sleeve on spine to hold label describing contents and with pockets inside covers to hold folded oversize sheets.
 - a. Identify each binder on front and spine, with printed title "OPERATION AND MAINTENANCE MANUAL," Project title or name, and subject matter of contents. Indicate volume number for multiple-volume sets.
 2. Dividers: Heavy-paper dividers with plastic-covered tabs for each section. Mark each tab to indicate contents. Include typed list of products and major components of equipment included in the section on each divider, cross-referenced to Specification Section number and title of Project Manual.
 3. Protective Plastic Sleeves: Transparent plastic sleeves designed to enclose diagnostic software diskettes for computerized electronic equipment.
 4. Drawings: Attach reinforced, punched binder tabs on drawings and bind with text.
 - a. If oversize drawings are necessary, fold drawings to same size as text pages and use as foldouts.
 - b. If drawings are too large to be used as foldouts, fold and place drawings in labeled envelopes and bind envelopes in rear of manual. At appropriate locations in manual, insert typewritten pages indicating drawing titles, descriptions of contents, and drawing locations.
 5. Provide a digital copy of the manual.
- E. Manuals, Electronic Files: Submit manuals in the form of a multiple file composite electronic PDF file for each manual type required.
1. Electronic Files: Use electronic files prepared by manufacturer where available. Where scanning of paper documents is required, configure scanned file for minimum readable file size.
 2. File Names and Bookmarks: Bookmark individual documents based on file names. Name document files to correspond to system, subsystem, and equipment names used in manual directory and table of contents. Group documents for each system and subsystem into individual composite bookmarked files, then create composite manual, so that resulting bookmarks reflect the system, subsystem, and equipment names in a readily navigated file tree. Configure electronic manual to display bookmark panel on opening file.

2.2 EMERGENCY MANUALS

- A. Content: Organize manual into a separate section for type of emergency, emergency instructions, and emergency procedures.
- B. Type of Emergency: Where applicable for each type of emergency indicated below, include instructions and procedures for each system, subsystem, piece of equipment, and component for fire, flood, gas leak, water leak, power failure, water outage, equipment failure and chemical release or spill.

- C. Emergency Instructions: Describe and explain warnings, trouble indications, error messages, and similar codes and signals. Include responsibilities of Owner's operating personnel for notification of Installer, supplier, and manufacturer to maintain warranties.
- D. Emergency Procedures: Include instructions on stopping, shutdown instructions for each type of emergency, operating instructions for conditions outside normal operating limits, and required sequences for electric or electronic systems.

2.3 OPERATION MANUALS

- A. Content: In addition to requirements in this Section, include operation data required in individual Specification Sections and equipment descriptions, operating standards, operating procedures, operating logs, wiring and control diagrams, and license requirements.
- B. Descriptions: Include the following:
 - 1. Product name and model number.
 - 2. Manufacturer's name.
 - 3. Equipment identification with serial number of each component.
 - 4. Equipment function.
 - 5. Operating characteristics.
 - 6. Limiting conditions.
 - 7. Performance curves.
 - 8. Engineering data and tests.
 - 9. Complete nomenclature and number of replacement parts.
- C. Operating Procedures: Include start-up, break-in, and control procedures; stopping and normal shutdown instructions; routine, normal, seasonal, and weekend operating instructions; and required sequences for electric or electronic systems.
- D. Systems and Equipment Controls: Describe the sequence of operation, and diagram controls as installed.
- E. Piped Systems: Diagram piping as installed and identify color-coding where required for identification.

2.4 PRODUCT MAINTENANCE MANUAL

- A. Content: Organize manual into a separate section for each product, material, and finish. Include source information, product information, maintenance procedures, repair materials and sources, and warranties and bonds, as described below.
- B. Source Information: List each product included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.
- C. Product Information: Include the following, as applicable:
 - 1. Product name and model number.
 - 2. Manufacturer's name.
 - 3. Color, pattern, and texture.
 - 4. Material and chemical composition.
 - 5. Reordering information for specially manufactured products.

- D. Maintenance Procedures: Include manufacturer's written recommendations and inspection procedures, types of cleaning agents, methods of cleaning, schedule for cleaning and maintenance, and repair instructions.
- E. Repair Materials and Sources: Include lists of materials and local sources of materials and related services.
- F. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that would affect validity of warranties or bonds.

2.5 SYSTEMS AND EQUIPMENT MAINTENANCE MANUAL

- A. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranty and bond information, as described below.
- B. Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.
- C. Manufacturers' Maintenance Documentation: Manufacturers' maintenance documentation including maintenance instructions, drawings and diagrams for maintenance, nomenclature of parts and components, and recommended spare parts for each component part or piece of equipment:
- D. Maintenance Procedures: Include test and inspection instructions, troubleshooting guide, disassembly instructions, and adjusting instructions, and demonstration and training videotape, that detail essential maintenance procedures:
- E. Maintenance and Service Schedules: Include service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.
- F. Spare Parts List and Source Information: Include lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.
- G. Maintenance Service Contracts: Include copies of maintenance agreements with name and telephone number of service agent.
- H. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that would affect validity of warranties or bonds.

PART 3 - EXECUTION

3.1 MANUAL PREPARATION

- A. Emergency Manual: Assemble a complete set of emergency information indicating procedures for use by emergency personnel and by Owner's operating personnel for types of emergencies indicated.

- B. Product Maintenance Manual: Assemble a complete set of maintenance data indicating care and maintenance of each product, material, and finish incorporated into the Work.
- C. Operation and Maintenance Manuals: Assemble a complete set of operation and maintenance data indicating operation and maintenance of each system, subsystem, and piece of equipment not part of a system.
- D. Manufacturers' Data: Where manuals contain manufacturers' standard printed data, include only sheets pertinent to product or component installed. Mark each sheet to identify each product or component incorporated into the Work. If data include more than one item in a tabular format, identify each item using appropriate references from the Contract Documents. Identify data applicable to the Work and delete references to information not applicable.
- E. Drawings: Prepare drawings supplementing manufacturers' printed data to illustrate the relationship of component parts of equipment and systems and to illustrate control sequence and flow diagrams. Coordinate these drawings with information contained in Record Drawings to ensure correct illustration of completed installation.
 - 1. Do not use original Project Record Documents as part of operation and maintenance manuals.
- F. Comply with Division 01 Section "Closeout Procedures" for schedule for submitting operation and maintenance documentation.

END OF SECTION 017823

SECTION 017839 - PROJECT RECORD DOCUMENTS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for Project Record Documents, including the following:
 - 1. Record Drawings.
 - 2. Record Specifications.
 - 3. Record Product Data.
- B. See Division 01 Section "Operation and Maintenance Data" for operation and maintenance manual requirements.
- C. See Divisions 02 through 32 Sections for specific requirements for Project Record Documents of the Work in those Sections.

1.3 SUBMITTALS

- A. Record Drawings: Comply with the following:
 - 1. Number of Copies: Submit one set of marked-up Record Prints.
 - 2. Number of Copies: Submit copies of Record Drawings as follows:
 - a. Initial Submittal: Submit one set of plots from corrected Record CAD Drawings and one set of marked-up Record Prints. Owner will initial and date each plot and mark whether general scope of changes, additional information recorded, and quality of drafting are acceptable. Owner will return plots and prints for organizing into sets, printing, binding, and final submittal.
 - b. Final Submittal: Submit one set of marked-up Record Prints, and the following:
 - 1) Record PDF Drawing Files: One set.
 - 2) Copies printed from Record CAD Drawing Plots: One copy. Print each Drawing, whether or not changes and additional information were recorded.
- B. Record Specifications: Submit one PDF electronic copy of Project's Specifications, including addenda and contract modifications.
- C. Record Product Data: Submit one PDF electronic copy of each Product Data submittal.

PART 2 - PRODUCTS**2.1 RECORD DRAWINGS**

- A. Record Prints: Maintain one set of blue or black& white prints of the Contract Drawings and Shop Drawings.

1. Preparation: Mark Record Prints to show the actual installation where installation varies from that shown originally. Require individual or entity who obtained record data, whether individual or entity is Installer, subcontractor, or similar entity, to prepare the marked-up Record Prints.
 - a. Give particular attention to information on concealed elements that would be difficult to identify or measure and record later.
 - b. Record data as soon as possible after obtaining it. Record and check the markup before enclosing concealed installations.
 2. Mark the Contract Drawings or Shop Drawings, whichever is most capable of showing actual physical conditions, completely and accurately. If Shop Drawings are marked, show cross-reference on the Contract Drawings.
 3. Mark record sets with erasable, red-colored pencil. Use other colors to distinguish between changes for different categories of the Work at same location.
 4. Note Construction Change Directive numbers, alternate numbers, Change Order numbers, and similar identification, where applicable.
- B. Format: Identify and date each Record Drawing; include the designation "PROJECT RECORD DRAWING" in a prominent location.
1. Record Prints: Organize Record Prints and newly prepared Record Drawings into manageable sets. Bind each set with durable paper cover sheets. Include identification on cover sheets.
 2. Record Transparencies: Organize into unbound sets matching Record Prints. Place transparencies in durable tube-type drawing containers with end caps. Mark end cap of each container with identification. If container does not include a complete set, identify Drawings included.
 3. Record CAD Drawings: Organize CAD information into separate electronic files that correspond to each sheet of the Contract Drawings. Name each file with the sheet identification. Include identification in each CAD file.
 4. Identification: As follows:
 - a. Project name.
 - b. Date.
 - c. Designation "PROJECT RECORD DRAWINGS."
 - d. Name of Architect.
 - e. Name of Contractor.

2.2 RECORD SPECIFICATIONS

- A. Preparation: Mark Specifications to indicate the actual product installation where installation varies from that indicated in Specifications, addenda, and contract modifications.
1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
 2. Mark copy with the proprietary name and model number of products, materials, and equipment furnished, including substitutions and product options selected.
 3. Record the name of manufacturer, supplier, Installer, and other information necessary to provide a record of selections made.
 4. Note related Change Orders, Record Product Data, and Record Drawings where applicable.

2.3 RECORD PRODUCT DATA

- A. Preparation: Mark Product Data to indicate the actual product installation where installation varies substantially from that indicated in Product Data submittal.
 - 1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
 - 2. Include significant changes in the product delivered to Project site and changes in manufacturer's written instructions for installation.
 - 3. Note related Change Orders, Record Specifications, and Record Drawings where applicable.

2.4 MISCELLANEOUS RECORD SUBMITTALS

- A. Assemble miscellaneous records required by other Specification Sections for miscellaneous record keeping and submittal in connection with actual performance of the Work. Bind or file miscellaneous records and identify each, ready for continued use and reference.

PART 3 - EXECUTION

3.1 RECORDING AND MAINTENANCE

- A. Recording: Maintain one copy of each submittal during the construction period for Project Record Document purposes. Post changes and modifications to Project Record Documents as they occur; do not wait until the end of Project.
- B. Maintenance of Record Documents and Samples: Store Record Documents and Samples in the field office apart from the Contract Documents used for construction. Do not use Project Record Documents for construction purposes. Maintain Record Documents in good order and in a clean, dry, legible condition, protected from deterioration and loss. Provide access to Project Record Documents for Architect's reference during normal working hours.

END OF SECTION 017839

SECTION 017900 - DEMONSTRATION AND TRAINING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for instructing Owner's personnel, including the following:
 - 1. Demonstration of operation of systems, subsystems, and equipment.
 - 2. Training in operation and maintenance of systems, subsystems, and equipment.
- B. See applicable Divisions 02 through 32 Sections for specific requirements for demonstration and training for products in those Sections.

1.3 SUBMITTALS

- A. Instruction Program: Submit two copies of outline of instructional program for demonstration and training, including a schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Include learning objective and outline for each training module. Include manufacturer's instructions.

1.4 QUALITY ASSURANCE

- A. Facilitator Qualifications: A firm or individual experienced in training or educating maintenance personnel in a training program similar in content and extent to that indicated for this Project, and whose work has resulted in training or education with a record of successful learning performance.
- B. Instructor Qualifications: A factory-authorized service representative, complying with requirements in Division 01 Section "Quality Requirements," experienced in operation and maintenance procedures and training.
- C. Preinstallation Conference: Conduct conference at Project site. Review methods and procedures related to demonstration and training.
- D. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals. Do not submit instruction program until operation and maintenance data has been reviewed and approved by Architect.

PART 2 - PRODUCTS**2.1 INSTRUCTION PROGRAM**

- A. Program Structure: Develop an instruction program that includes individual training modules for each system and equipment not part of a system, as required by individual Specification Sections.

- B. Training Modules: Develop a learning objective and teaching outline for each module. Include a description of specific skills and knowledge that participant is expected to master. For each module, include instruction for the following:
1. Basis of System Design, Operational Requirements, and Criteria: Include system and equipment descriptions, operating standards, regulatory requirements, equipment function, operating characteristics, limiting conditions, and performance curves.
 2. Documentation: Review emergency, operations, and maintenance manuals; Project Record Documents; identification systems; warranties and bonds; and maintenance service agreements.
 3. Emergencies: Include instructions on stopping; shutdown instructions; operating instructions for conditions outside normal operating limits; instructions on meaning of warnings, trouble indications, and error messages; and required sequences for electric or electronic systems.
 4. Operations: Include startup, break-in, control, and safety procedures; stopping and normal shutdown instructions; routine, normal, seasonal, and weekend operating instructions; operating procedures for emergencies and equipment failure; and required sequences for electric or electronic systems.
 5. Adjustments: Include alignments and checking, noise, vibration, economy, and efficiency adjustments.
 6. Troubleshooting: Include diagnostic instructions and test and inspection procedures.
 7. Maintenance: Include inspection procedures, types of cleaning agents, methods of cleaning, procedures for preventive and routine maintenance, and instruction on use of special tools.
 8. Repairs: Include diagnosis, repair, and disassembly instructions; instructions for identifying parts; and review of spare parts needed for operation and maintenance.

PART 3 - EXECUTION

3.1 INSTRUCTION

- A. Facilitator: Engage a qualified facilitator to prepare instruction program and training modules, to coordinate instructors, and to coordinate between Contractor and Owner for number of participants, instruction times, and location.
- B. Engage qualified instructors to instruct Owner's personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.
1. Owner will furnish an instructor to describe Owner's operational philosophy.
- C. Scheduling: Provide instruction at mutually agreed on times. For equipment that requires seasonal operation, provide similar instruction at start of each season.
1. Schedule training with Owner with at least seven days' advanced notice.

END OF SECTION 017900

SECTION 019113 - GENERAL COMMISSIONING REQUIREMENTS**PART 1 – GENERAL****1.01 SUMMARY**

- A. Work Included:
 - 1. Scope of systems and equipment to be commissioned.
 - 2. Commissioning duties and procedures at the site.

1.02 RELATED SECTIONS

- A. Division 01, General Requirements applies to this Section.
- B. Contents of Division 22, 23, and 26 apply to this Section.
- C. In addition, reference the following:
 - 1. 22 08 00, Commissioning of Plumbing
 - 2. 23 08 00, Commissioning of HVAC
 - 3. 26 08 00, Commissioning of Electrical

1.03 REFERENCES AND STANDARDS

- A. References and Standards as required by:
 - 1. Division 01, General Requirements.
 - 2. 22 08 00, Commissioning of Plumbing
 - 3. 23 08 00, Commissioning of HVAC
 - 4. 26 08 00, Commissioning of Electrical
- B. In addition, meet the following:
 - 1. Current edition of the ASHRAE Guideline 0, The Commissioning Process.

1.04 SUBMITTALS

- A. Submittals as required by:
 - 1. Division 01, General Requirements.
 - 2. 22 08 00, Commissioning of Plumbing
 - 3. 23 08 00, Commissioning of HVAC
 - 4. 26 08 00, Commissioning of Electrical
- B. In addition, provide:
 - 1. Use the following procedure to ensure quick and effective turnaround of submittals for systems to be commissioned.
 - a. The Architect forwards one set of submittals for systems to be commissioned to the Commissioning Authority at the same time as the design team.

- b. The Commissioning Authority forwards comments to the design team for consideration in their submittal response.
- c. The design team sends a consolidated response to the submittals and copies the Commissioning Authority.

1.05 QUALITY ASSURANCE

- A. Quality assurance as required by:
 - 1. Division 01, General Requirements.
 - 2. 22 08 00, Commissioning of Plumbing
 - 3. 23 08 00, Commissioning of HVAC
 - 4. 26 08 00, Commissioning of Electrical

1.06 WARRANTY

- A. Warranty of materials and workmanship as required by Division 01, General Requirements.
 - 1. Division 01, General Requirements.
 - 2. 22 08 00, Commissioning of Plumbing
 - 3. 23 08 00, Commissioning of HVAC
 - 4. 26 08 00, Commissioning of Electrical

1.07 DEFINITIONS

- A. Commissioning Authority: The Commissioning Authority is the person or entity referred to throughout the Contract Documents as if singular in number who works with the Owner's Authorized Representative under a separate Contract.
- B. Commissioning:
 - 1. Commissioning is a process for achieving, verifying, and documenting that performance of a building and its various energy consuming systems meets the Design Engineer's design intent and the Owner's operational needs.
 - 2. Commissioning includes tests for the operation of equipment and building systems to ensure that they operate as designed by the Design Engineer, and meet the needs of the building throughout the entire range of operating conditions.
 - 3. Commissioning is a cooperative effort that requires participation by the Owner's Authorized Representative, General Contractor, system and equipment installers, building automation system installer, Testing and Balancing Agency, equipment manufacturers' representatives, Architect, Architect's design engineers, and Commissioning Authority.
- C. Owner's Project Requirements (OPR): Document that details the functional requirements and expectations of how the building will be used and operated. This may include project location, goals, cost considerations, equipment manufacturers, and environmental control requirements.
- D. Basis of Design (BoD): A document that records concepts, calculations, decisions, and product selections used to meet the OPR and to satisfy applicable regulatory requirements, standards, and guidelines.

E. Commissioning Procedures:

1. Inspection and testing procedures that are written by the Commissioning Authority for equipment and systems within the scope of commissioning.
2. Inspection checklists typically address items of installation compliance with design intent and approved submittals.
3. Functional performance test procedures typically address all sequences for normal and emergency equipment and system operation. These procedures consist of a mix of One-Time Tests and Continuous Measurement.
4. One-Time Tests: Functional performance tests of equipment and systems that are performed by forcing specific conditions that are intended to trigger specific responses, per the design intent.

F. Continuous Measurements:

1. Functional performance tests of equipment and systems that are performed by observing parameters of normal operation over an extended period. This is typically accomplished by means of the BAS trend logging capabilities, by monitoring with stand-alone data logging equipment, or by some combination of both.
2. Temperature conditions in occupied spaces, control stability, and lighting levels in areas with daylighting controls are three typical subjects of continuous measurement.

G. Commissioning Plan: The document, provided by the Commissioning Authority, that states the required tests for all equipment and systems within the scope of commissioning.

H. Commissioning Meetings: Issues related to commissioning will be discussed as required during regularly scheduled progress meetings.

1.08 PERFORMANCE REQUIREMENTS

- A. Testing, inspecting and performance monitoring tasks specified in this Section and in Sections 22 08 00, 23 08 00, and 26 08 00 are the responsibility of the Commissioning Authority, unless specifically indicated otherwise, and not part of the General Construction Contract. These tasks are included in these Sections for the Contractor's information, so the Contractor can understand the standards of system performance that are required and more effectively coordinate with the process of commissioning.
- B. The Commissioning Authority will verify for the Owner's Authorized Representative that commissioned mechanical, plumbing, electrical, and controls system function interactively and in compliance with the Project design intent, and to facilitate orderly and efficient transfer of building operating systems to the Owner.
- C. Commissioning does not relieve the Contractor of Contract obligations.

1.09 EQUIPMENT AND SYSTEMS TO BE COMMISSIONED

A. Systems:

1. HVAC Equipment
2. HVAC Controls.
3. Domestic Hot Water Equipment

4. Automatic Lighting Controls (LCP, Daylighting, Occupancy Sensors)

1.10 COMMISSIONING DUTIES

- A. Duties of Owner: Provide the OPR to the Architect/Engineer and Commissioning Authority prior to design development.
- B. Duties of Architect:
 - 1. Attend commissioning portion of Progress Meetings as necessary, minimum two meetings.
 - 2. Lead the design team in assisting the resolution of deficiencies.
- C. Duties of Architect's Mechanical Engineer:
 - 1. Attend commissioning portion of Progress Meetings as necessary, minimum two meetings.
 - 2. At the request of either the Owner's Authorized Representative or the Commissioning Authority, review Commissioning Procedures and submit comments to Owner's Authorized Representative.
 - 3. Develop and provide the Basis of Design to Owner and Commissioning Authority prior to 50 percent CD.
 - 4. Assist in resolution of problems and deficiencies that are discovered during commissioning.
 - 5. Participate and respond to commissioning related issues by using the Commissioning Authority's web based commissioning software Facility Grid.
- D. Duties of Architect's Electrical Engineer:
 - 1. Attend commissioning portion of Project Meetings as necessary.
 - 2. At request of either the Owner's Authorized Representative or the Commissioning Authority, review Commissioning Procedures and submit comments to Owner's Authorized Representative.
 - 3. Develop and provide the Basis of Design to Owner and Commissioning Authority prior to 50 percent CD.
 - 4. Assist in resolution of problems and deficiencies that are discovered during commissioning.
 - 5. Participate and respond to commissioning related issues by using the Commissioning Authority's web based commissioning software Facility Grid.
- E. Duties of Commissioning Authority:
 - 1. Attend commissioning portion of Project Meetings as necessary, minimum two meetings.
 - 2. Provide plan to Owner's Authorized Representative for review and comment.
 - 3. Utilize web based Commissioning software to manage all commissioning related checklists, tests, issues, and observation reports.

4. Prepare commissioning procedures for each commissioned system based on actual system configuration.
 5. Commissioning Procedures written by Commissioning Authority will include, in field data collection format, the detailed test procedures, test conditions, and criteria for acceptance of test results.
 6. Submit any commissioning procedures that are written by Commissioning Authority to the Owner's Authorized Representative for review and approval at least 1 week prior to scheduled field testing.
 7. Provide personnel experienced in technical aspects of each system to be commissioned for execution of tests.
 8. BAS Sequence Demonstration:
 - a. Witness the Control Contractor's demonstration of their sequence tests.
 - b. If any of the demonstrated sequences fails to operate per the controls submittal, witness the repeat demonstration after corrective action has been taken.
 9. Execute the Commissioning Procedures.
 10. Prepare and submit Observation Reports and Deficiency Reports as required, but within 3 days of noting any deficiency.
 11. Submit to Owner's Authorized Representative a weekly written report of commissioning progress, unresolved deficiencies, and projected inspection, and test schedule during field testing.
 12. Take the lead in timely evaluation of deficiencies, and advise Owner's Authorized Representative on resolution.
 13. Assist in resolving commissioned system disputes by performing research to determine the scope of the dispute, and informing the involved parties on possible solutions to disputes.
 14. During the systems warranty period(s) CxA to retest any systems that had their full testing deferred during the initial functional testing due to the lack of peak season conditions. This testing must ensure that all system sequences of operations and capacity have been verified.
 15. Prepare a Commissioning Report that includes a summary of overall commissioning process, including deficiencies found, deficiency corrections, unresolved deficiencies, approved equipment and systems, discrepancies between final design intent and as-built systems, completed commissioning checklists, test documentation, and other commissioning documentation.
- F. Duties of General Contractor:
1. Attend commissioning portion of Project Meetings as necessary, minimum four meetings.
 2. Participate and respond to commissioning related issues by using the Commissioning Authority's web based commissioning software Facility Grid.
 3. Coordinate and direct system installers in executing their commissioning tasks.

4. Direct subcontractors to participate and respond to commissioning related issues by using the Commissioning Authority's web based commissioning software Facility Grid. A desktop, laptop, tablet or iPad will be required.
 5. Coordinate with Commissioning Authority on integration of construction and commissioning schedules.
 6. Oversee and perform documentation requirements for all Pre-Functional Checklists.
 7. Notify Commissioning Authority when all the following has been achieved. It is permissible, with prior approval by Commissioning Authority, to provide notification for individual systems as the following are all completed for each system.
 - a. All controls point-to-point and sequence checkout is complete.
 - b. All test and balancing is complete.
 - c. Normal equipment schedules have been activated.
 - d. All control overrides and temporary valves have been returned to normal automatic control.
 - e. All manual isolation valves have been left open.
 - f. Piping and duct systems have been cleaned and tested.
 - g. Heating water system is fully operational under normal automatic operation.
 - h. Luminaires are installed with operational daylighting controls and occupancy sensors.
 - i. Distribution boards, including overcurrent devices, containing breakers over 600 amps, are installed.
 - j. Building inspector acceptance of emergency lighting system following their site inspection.
 8. Provide all startup, flushing, pressure testing, etc results/reports for commissioned systems.
- G. Duties of Installer's and Manufacturer's Representatives:
1. Attend commissioning portion of Project Meetings as necessary, minimum two meetings.
 2. Participate and respond to commissioning related issues by using the Commissioning Authority's web based commissioning software Facility Grid.
 3. Within three months of the award of the Contract, as part of the required submittals for the contract, Contractor submits manufacturer's startup and installation procedures as well as controls point-to-point and sequence checkout and provides in checkset format for each piece of equipment and controls.
 4. Demonstrate proper system operation in the presence of the Commissioning Authority.
 5. Commissioning does not relieve installers from obligations to complete Work as required by Contract Documents.
- H. Duties of BAS Installer:
1. Attend commissioning portion of project meetings as necessary, minimum two meetings.

2. Participate and respond to commissioning related issues by using the Commissioning Authority's web based commissioning software Facility Grid.
3. Review and approve Commissioning Procedures as relevant to controls work.
4. Point-to-Point Checkout:
 - a. Perform point-to-point checkout and calibration of all energy management system points.
 - b. Document checkout and calibration on forms as approved by mechanical designer, and/or Commissioning Authority.
 - c. Submit three copies of the completed point-to-point checkout forms to the Owner's Authorized Representative within five working days of completion of field checkout. Distribute copies to the Commissioning Authority and the designer.
5. Control Sequence Testing:
 - a. Prepare control sequence test procedure forms to a degree of rigor comparable to the Commissioning Authority's Commissioning Procedures.
 - b. Submit test procedure forms to the Commissioning Authority for approval at least two weeks prior to intended sequence testing. At the contractor's option, it is acceptable to use the Commissioning Authority's Commissioning Procedures, substituting one-time tests for continuous measurement wherever applicable. However, it is still necessary to submit any edited Commissioning Authority Commissioning Procedures as least two weeks prior to intended sequence testing.
 - c. Submit the completed sequence testing forms to the Owner's Authorized Representative. The Owner's Authorized Representative distributes copies to the Commissioning Authority and the designer.
6. Submit to Commissioning Authority, prior to Sequence Demonstration, two copies of installed control Drawings, sequence narratives, control wiring diagrams, and program code or block diagrams.
7. Sequence Demonstration:
 - a. After completing and documenting all required sequence tests with own staff, demonstrate sequence tests to the Commissioning Authority. Demonstration is to be performed by the BAS installer's programmer who programmed the control system for this specific project.
 - b. If any of the demonstrated sequences fails to operate per the controls submittal, take corrective action and demonstrate the failed sequence tests to the Commissioning Authority a second time.
 - c. If the Control Contractor fails to demonstrate proper sequence operation in any of the second round of sequence tests, the Commissioning Authority's costs for witnessing all further demonstration of that sequence may be assigned to the Control Contractor by the Owner as a deduct to their contracted price. The Control Contractor will not be responsible for costs related to failure due to design or to other factors beyond their control, though it is expected to call any design concerns (and other factors beyond their control that might cause failure) to the attention of the Commissioning Authority and the Owner's Authorized Representative.

8. Assist Commissioning Authority with programming of the energy management system for trend logs to support functional performance testing during field testing.
 9. Assist Commissioning Authority with execution of the Commissioning Procedures. Commissioning Authority will present test schedule at Progress Meeting at least one week ahead of scheduled tests.
 10. The Commissioning Authority, acting with Owner authority, may request the Control Contractor to assist with or perform minor loop tuning adjustments, set point and schedule changes, and other similar minor field corrections.
 11. Recommended changes to the controls sequences, program code, and recommendations for additional points must go through the Owner's Authorized Representative and the designer. The designer is the final authority on all recommended sequence changes, and will submit such changes to the Owner's Authorized Representative for implementation.
 12. Submit to Owner's Authorized Representative, at least two weeks prior to Final Completion, two copies of as-built version of points list, including I/O and virtual points, controls Drawings, program printout, and sequence narratives.
 13. Participate in resolution of problems and deficiencies that are discovered during commissioning.
- I. Duties of Balancer:
1. Attend commissioning portion of Project Meetings as necessary, minimum two meetings.
 2. Participate in resolution of problems and deficiencies that are discovered during commissioning.
 3. Assist Commissioning Authority with execution of commissioning procedures.
 4. Demonstrate accuracy of final balance report in the presence of the Commissioning Authority. This will be a 10 percent spot check.

PART 2 - PRODUCTS (NOT USED)

PART 3 – EXECUTION

3.01 COMMISSIONING PROCEDURES AT THE SITE

A. Testing Techniques:

1. Each testing procedure may use a variety of techniques. Generally it is preferred to observe new and existing equipment and systems during normal operation.
2. When functional and emergency modes of operation occur rarely or seasonally, if possible, simulate the conditions that trigger these operational modes.
3. Simulation of conditions may involve changing set points, changing schedules, simulating pneumatic system pressures or energy management system voltages and currents, disconnecting power, jumpering contacts, or other such procedures.
4. Whenever temporary adjustments are made, restore the system to its original condition once tests are completed.

5. When testing requires observing equipment operation over an extended period, use the building energy management system's trend logging capabilities or independent monitoring equipment.
 6. Do not use the building automation system trend logging in the commissioning process prior to point-to-point checkout by Controls Contractor and approval of point-to-point checkout by Commissioning Authority.
 7. Measurement of room lighting levels during evening hours with only artificial lighting, during mid-morning, around noon and mid-afternoon with only natural lighting and with both natural and artificial lighting. Repeat same measurements following calibration of room daylighting sensor.
- B. Commissioning Documentation:
1. The Contractors are required to perform startup and checkout of their systems (prefunctional testing) and document the results in Facility Grid. The Commissioning Authority will provide electronic forms that may be used by the Contractors. The Contractors may use their own forms if they contain all the required information on the Commissioning Authority's forms, but prior approval must be obtained.
 - a. Where numeric data is required, a narrative entry or simple check-off is not acceptable.
 - b. Annotate trend logs and monitored data as necessary to clarify meaning, and attach to relevant test reports.
 - c. Do not attach irrelevant data to test reports.
 2. The Contractor sends the startup and checkout forms to the Commissioning Authority when they are complete and functional. The Contractor sends a "Certificate of Readiness" with the forms which will signal that functional testing can begin.
 3. The technician who performed the pretesting and checkout of the system completes the Pre-Functional Checklists using the web based commissioning software Facility Grid.
 4. E-mail an "issues log" weekly to inform the design and construction team of issues that need resolution. The "issues log" will open and close items as they are discovered and resolved until all items are closed.
 5. The Commissioning Authority will assemble all the information from the Commissioning Plan (test forms, trend logs, issues log, and basis of design) into a final Commissioning Report.
- C. Coordination of Commissioning and Equipment Startup: Do not initiate functional performance testing for equipment or systems in advance of their startup and checkout by affected equipment or system installers and manufacturers' representatives.
- D. Test Acceptance Criteria:
1. Acceptance Criteria are the test results that are required before the mode of performance or inspection item in question will be considered acceptable.
 2. Any procedures in Specification Sections 22 08 00, 23 08 00, or 26 08 00 that begin with "Verify that...." have an implied acceptance criterion that the sequence as stated is proven

to occur and is documented with visual observation notes, measurements, trend logs, and/or monitored data.

3. Acceptance criteria for other functional modes and checklist items are as stated in each section of the Commissioning Plan.
4. Input will be sought when necessary from the Architect's Engineer to determine if test results indicate compliance with Design Intent.
5. The Commissioning Authority will recommend acceptance or rejection of commissioned system work based on test results.

E. Resolution of Deficiencies:

1. Adjust, repair, or replace defective equipment and systems to meet Commissioning Procedure Acceptance Criteria as directed by Owner's Authorized Representative.
2. Inform the Owner's Authorized Representative and Commissioning Authority of the date for completion of corrective activities.
3. If the date for completion of corrective work passes without resolution of deficiencies, Owner's Authorized Representative reserves the right to obtain supplementary services and equipment to correct the problem as indicated in General Conditions.

F. Rechecking and Retesting Charges:

1. In the event of a second failure of a specific commissioning procedure item or test, the responsible party may be assessed charges by Owner's Authorized Representative.
2. Charges will be based on each party's actual expenses, including normal hourly billing rates for preparation, testing, and travel time, and materials, equipment rental, and travel expenses as applicable.

G. Construction and Acceptance Milestones for Tasks Related to Commissioning:

1. Equipment, ductwork, and piping installation.
2. Equipment startup.
3. Pre-functional checklists.
4. Substantial completion.
5. Point-to-point checkout and sequence testing of controls.
6. Test and balance.
7. Commissioning field testing.
8. Owner training.
9. Occupant move-in.
10. Final completion.
11. Seasonal testing.
12. Commissioning report submittal.

END OF SECTION 019113

SECTION 024113 - SELECTIVE SITE DEMOLITION**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes the following:

1. Demolition and removal of existing asphalt and/or concrete pavement, concrete and/or asphalt walks, curbs and gutters, and other exterior site items indicated or not indicated which interfere with the Work.
2. Removal and/or relocation of existing underground utilities.
3. Removal and disposal of existing sanitary sewer pipe, water pipe, storm drainage pipe and appurtenances indicated. Filling of existing pipes to be abandoned in place.
4. Removal and replacement of fencing and playground equipment.
5. Removal and relocation of existing light poles.
6. Permanent abandonment of existing wells.
7. Removal and disposal of material from dump areas.

1.3 DEFINITIONS

- A. Remove: Remove and legally dispose of items except those indicated to be reinstalled, salvaged, or to remain the Owner's property.
- B. Remove and Salvage: Items indicated to be removed and salvaged remain the Owner's property. Remove, clean, and pack or crate items to protect against damage. Identify contents of containers and deliver to Owner's designated storage area.
- C. Remove and Reinstall: Remove items indicated; clean, service, and otherwise prepare them for reuse; store and protect against damage. Reinstall items in the same locations or in locations indicated.
- D. Existing to Remain: Protect items indicated to remain against damage and soiling. When permitted by the Architect and/or Engineer, items may be removed to a suitable, protected storage location and then cleaned and reinstalled in their original locations.

1.4 MATERIALS OWNERSHIP

- A. Except for items or materials indicated to be reused, salvaged, reinstalled, or otherwise indicated to remain the Owner's property, remove demolished materials from the site with further disposition at the Contractor's option.
- B. Storage or sale of removed items or materials on-site will not be permitted.
- C. Historical items, relics, and similar objects including, but not limited to, cornerstones and their contents, commemorative plaques and tablets, antiques, and other items of interest or value to the Owner, which may be encountered, remain the Owner's property. Carefully remove and salvage each item or object in a manner to prevent damage and deliver promptly to the Owner.

1.5 SUBMITTALS

- A. Photographs or videotape, sufficiently detailed, of existing conditions of adjoining construction and site improvements that might be misconstrued as damage caused by the Work.
- B. Record drawings at Project closeout.

1. Identify and accurately locate capped utilities and other subsurface structural, electrical, or mechanical conditions.
- C. Proposed Erosion and Sediment control measures.
- D. Schedule of selective demolition activities indicating the following:
 1. Detailed sequence of selective demolition and removal work, with starting and ending dates for each activity.
 2. Interruption of utility services.
 3. Coordination for shutoff, capping, and continuation of utility services.
 4. Detailed sequence of selective demolition and removal work to ensure uninterrupted progress of Owner's on-site operations.
 5. Coordination of Owner's continuing occupancy of portions of existing building and of Owner's partial occupancy of completed Work.
 6. Locations of temporary partitions and means of egress.
- E. Inventory of items to be removed and salvaged or turned over to Owner.
- F. Landfill records indicating receipt and acceptance of hazardous wastes by a landfill facility licensed to accept hazardous wastes.

1.6 QUALITY ASSURANCE

- A. Regulatory Requirements: All work shall comply with Federal, State and Local laws and regulations concerning hauling and disposal of demolition debris.
- B. Notify the proper agencies prior to the start of work and obtain all necessary permits for this work.

1.7 PROJECT CONDITIONS

- A. Owner assumes no responsibility for actual condition of items or structures to be demolished. Conditions existing at the time of inspection for bidding purposes will be maintained by the Owner to the extent practical. However, minor variations may occur due to Owner's removal and salvage operations prior to the start of demolition work.
- B. The location of existing underground utilities indicated is approximate only. Field locate all existing underground utilities in the area of work, regardless of whether or not they are indicated. Call "Miss Utility" prior to the start of demolition work for assistance in the location of existing underground utilities.
- C. Should charted, uncharted or incorrectly charted utilities be encountered during demolition, contact the Architect and/or Engineer immediately for instructions. Cooperate with Owner and utility companies to keep services and facilities in operation.
- D. Do not interrupt existing utilities serving facilities occupied and used by the Owner and others, except when permitted in writing by the Owner. Provide acceptable temporary utility service as required to maintain Owner's operations.

1.8 SCHEDULING

- A. Owner will occupy portions of the building immediately adjacent to the Work. Conduct selective demolition so that the Owner's operations will not be disrupted. Provide not less than 72 hours notice to Owner of activities that will affect Owner's operations.
- B. Arrange selective demolition schedule so as not to interfere with Owner's on-site operations.
- C. Notify and coordinate any required relocation and/or removal of existing underground utilities, poles, meters or other above ground appurtenances with the appropriate utility company (i.e. power, telephone, cable and natural gas/propane) prior to the start of selective demolition work.

1.9 PAYMENT FOR UTILITY REMOVAL / RELOCATIONS

- A. Electric Service – The Owner will pay for the relocation of the electrical line.
- B. Phone Service
- C. Cable Television
- D. Gas
- E. Fiber Optic Lines
- F. Petroleum Pipelines

1.10 USE OF EXPLOSIVES

- A. Do not use explosives to perform selective site demolition work.

PART 2 - PRODUCTS

(Not Applicable)

2.1 CEMENT GROUT FOR WELL CAPPING

- A. Cement Grout: Cement grout shall be API Spec. 10, Class G cement or Class B similar to ASTI C 150 Type II and water in ratio of not more than six gallons per 94 pound sack of cement.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Call "Miss Utility" prior to the start of demolition work for assistance in the location of existing underground utilities. Field locate all existing underground utilities in the area of work, regardless of whether or not they are indicated.
- B. Should uncharted or incorrectly charted existing utilities be identified, contact the Architect and/or Engineer immediately for instructions. Provide a scale drawing with the location of the uncharted or incorrectly charted utilities for use by the Architect in preparing additional direction.
- C. Verify that utilities indicated as removed, abandoned and/or relocated have been disconnected and capped.
- D. Survey existing conditions and correlate with requirements indicated to determine extent of selective demolition required.
- E. Inventory and record the condition of items to be removed and reinstalled and items to be removed and salvaged and turned over to the Owner.

3.2 PROTECTION OF PERSONS AND PROPERTY

- A. Drain, purge, or otherwise remove, collect, and dispose of chemicals, gases, explosives, acids, flammables, or other dangerous materials before proceeding with selective demolition operations.
- B. Conduct demolition operations and remove debris to ensure minimum interference with roads, streets, walks and other adjacent occupied and used facilities.
 - 1. Do not close or obstruct streets, walks, or other adjacent occupied or used facilities without permission from Owner and authorities having jurisdiction. Provide alternate routes around closed or obstructed traffic ways if required by governing regulations.
- C. Conduct demolition operations to prevent injury to people and damage to adjacent buildings and facilities to remain. Ensure safe passage of people around selective demolition area.
 - 1. Erect temporary protection, such as walks, fences, railings, canopies, and covered passageways, where required by authorities having jurisdiction.
 - 2. Protect existing site improvements, appurtenances, and landscaping to remain.

- D. Barricade areas of demolition occurring as part of this work, and post with warning lights as required by authorities having jurisdiction.
- E. Protect structures, buildings, utilities, walks, pavements, existing vegetation and other facilities to remain from damage caused by settlement, lateral movement, undermining, washout and other hazards created by demolition operations.

3.3 POLLUTION CONTROLS

- A. Perform all work in accordance with the requirements of the State and Local Jurisdiction and those of the local Erosion Control official.
- B. Clean adjacent structures and improvements of dust, dirt, and debris caused by the Work. Return adjacent areas to condition existing before start of selective demolition.

3.4 DEMOLITION OF EXISTING FACILITIES

- A. Electric Service
 - 1. Coordinate the removal and/or relocation of existing utilities with Power Company.
 - 2. Contact Power Company Representative at XXX-XXX-XXXX to arrange for required removal and/or relocation of existing service.
- B. Phone Service
 - 1. Coordinate the removal and/or relocation of existing utilities with Phone Company.
 - 2. Contact Phone Company Representative at XXX-XXX-XXXX to arrange for required removal and/or relocation of existing service.
- C. Cable Television
 - 1. Coordinate the removal and/or relocation of existing utilities with Cable Company.
 - 2. Contact Cable Company Representative at XXX-XXX-XXXX to arrange for required removal and/or relocation of existing service.
- D. Gas
 - 1. Coordinate the removal and/or relocation of existing utilities with Gas Company.
 - 2. Contact Gas Company Representative at XXX-XXX-XXXX to arrange for required removal and/or relocation of existing service.
- E. Fiber Optic Lines
 - 1. Coordinate the removal and/or relocation of existing utilities with Fiber Optic Company.
 - 2. Contact Fiber Optic Company Representative at XXX-XXX-XXXX to arrange for required removal and/or relocation of existing service.
- F. Petroleum Pipelines
 - 1. Coordinate the removal and/or relocation of existing utilities with Petroleum Pipeline Company.
 - 2. Contact Petroleum Pipeline Company Representative at XXX-XXX-XXXX to arrange for required removal and/or relocation of existing service.
- G. Utilities
 - 1. Coordinate the removal and/or relocation of existing utilities with the appropriate utility companies.
 - 2. Remove existing utilities as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and to local jurisdictional codes.

3. Provide adequate means of support and protection during demolition and other construction operations for existing utilities that are to remain in place. Repair utilities damaged by construction operations to the satisfaction of the utility owner.

H. Asphalt Pavement

1. Remove asphalt concrete pavement by sawcutting to the full depth of the pavement. Provide neat sawcuts at the limits of pavement removal indicated.

I. Concrete Pavement, Walks and Curbs

1. Remove concrete pavement and walks to the nearest joint. Sawcut concrete if joints are not present adjacent to the area of demolition.
2. Sawcut concrete along straight lines to a depth of not less than 2 inches. Break out remainder of concrete, provided that the broken area is concealed in the finished work, and the remaining concrete is sound. At locations where the broken face cannot be concealed, grind smooth or sawcut entirely through concrete.

J. Light Poles

1. Remove and relocate light poles as indicated. If light poles are owned by a public utility, coordinate the relocation with them.

K. Fencing

1. Remove existing chain-link fencing as indicated on the drawings.
2. Turn fencing materials removed over to the Owner.

L. Playground Equipment

1. Remove, store and protect existing playground equipment interfering with proposed construction.
2. Turn play equipment over to the Owner for reinstallation following the completion of construction in the area.

M. Well Capping

1. Remove all well pumps, riser piping and electrical cable from wells prior to plugging operations. Clear well of all obstructions.
2. Thoroughly chlorinate the well prior to plugging. Chlorinate the well using 2 gallons of liquid chlorine bleach. Chlorine bleach shall remain in well for 24 hours before plugging operations begin.
3. Provide cement grout or clay slurry from the bottom of the well to six (6) feet below the finished ground surface. Provide sealing materials from the bottom upwards utilizing steel or plastic tremie pipe. Raise the tremie pipe slowly as the grout material is introduced and keep full continuously from start to finish. Continuously submerge the discharge end of the tremie in the grout until the well is completely filled.
4. After the well has been plugged, trim the well casing at a depth that closely matches the top of the grout. Fill in the remainder of the well opening with dirt prior to final grading of the site.
5. Notify the proper authorities having jurisdiction prior to grouting of well.
6. Comply with all regulations of local authorities related to capping the existing well.

3.5 DISPOSAL OF DEMOLISHED MATERIALS

- A. Promptly dispose of demolished materials. Do not allow demolished materials to accumulate on-site.
- B. Do not burn demolished materials or debris.

- C. Transport and legally dispose of demolished materials off of Owner's property.

3.6 CLEANUP AND REPAIR

- A. Upon completion of demolition work remove all tools, equipment and demolition materials from site. Remove demolition work area protection and leave areas clean.
- B. Repair any demolition performed in excess of that required. Return elements of construction and surfaces to remain to the condition existing prior to the start of construction. Repair adjacent construction or surfaces soiled or damaged by demolition work.

END OF SECTION 024113

SECTION 031500 – CONCRETE ACCESSORIES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Armor-Edge joint assembly for formed construction joints.
- B. Related Sections:
 - 1. Section 033000 Cast-In-Place Concrete.
 - 2. Section 033010 Cast-In-Place Interior Concrete Slabs on Ground.

1.3 REFERENCES

- A. American Concrete Institute (ACI)
 - 1. ACI Detailing Manual SP-66.
 - 2. ACI 302.1R-04 Guide for Concrete Floor and Slab Construction.
 - 3. ACI 360R-06 Guide for Design of Slabs-on-Ground
- B. American Society for Testing and Materials (ASTM)
 - 1. ASTM A 36 Standard Specifications for Carbon Structural Steel.
 - 2. ASTM A 108 Standard Specifications for Steel Bar, Carbon and Alloy, Cold Finished.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: Manufacturer's product data with application and installation instructions for proprietary materials and items, including but not limited to joint protection systems.
 - 2. Design data.
 - 3. Shop Drawings: Indicate placement of Armor-Edge® joint assemblies.
 - a. Indicate dimensions and spacings.
 - b. Comply with ACI 302.1R-04, ACI 360R-06, ACI Detailing Manual (SP-66) and PNA installation guides indicating arrangement of armored joint accessories.

1.5 QUALITY ASSURANCE

- A. Pre-Installation Meeting:

1. Convene pre-installation meeting two weeks before installation of PNA products. Require attendance of parties affecting the work of this section, including Contractor, Engineer, and installers.
2. Review installation procedures and coordinate with other work.

1.6 DELIVERY, STORAGE AND HANDLING

- A. Deliver Armor-Edge joint assemblies in manufacturer's packaging.
- B. Deliver and handle materials to prevent damage or weakening to assemblies.
- C. Prevent accumulation of rust, debris or deleterious materials on assemblies.

PART 2 - PRODUCTS

2.1 MANUFACTURER

- A. Acceptable manufacturer: PNA Construction Technologies, www.pna-inc.com; phone 800-542-0214.

2.2 ARMORED JOINT ASSEMBLIES FOR FORMED CONSTRUCTION JOINTS

- A. Acceptable Product: PNA Armor-Edge Joint Assembly.
- B. Material:
 1. Bars: Saw cut from cold rolled steel bar meeting ASTM A 108-03 grade 1018 to within 3/16" of specified length.
 2. Mounting brackets shall be manufactured from steel as provided by manufacturer.
 3. Shipping nut assembly to ensure alignment of armored joint assembly shall contain a nylon nut, bolt and ferrule.
- C. Fabrication:
 1. Bolt 2 - 2" by 3/8" steel bars back-to-back with a 4" offset and drill holes to hold two plates in perfect alignment together at 2' spacing with shipping nut assembly.
 2. Use mounting bracket for installation only.
 3. Weld 4" by 3/8" headed studs at 12" on center to steel bars.
 4. Assembly length: 12'-0" with 24" offset

2.3 DIAMOND-SHAPED LOAD PLATE FOR CONSTRUCTION JOINTS

- A. Acceptable Product: PNA Diamond Dowel System.
- B. Material:
 1. Diamond-Shaped Load Plate: 1/4" and 3/8" saw cut from hot rolled steel plate meeting ASTM A 36. 3/4" saw cut from cold rolled steel plate for acceptable tolerances meeting ASTM 108-03 grade 1018.
 2. Pocket Former: High density plastic with internal collapsible fins and spacer that hold diamond shaped load plate in correct position and creates a void to its vertical faces. This void, in addition to its tapered shape, shall allow for differential movement and shall prevent horizontal stress accumulation at joint, thus reducing likelihood of random cracking.

- a. REFER TO ACI 302.1R-04 FOR SELECTION OF PLATE SIZE AND SPACING.

3. Dimensions of Plate: 3/4" by 4-1/2" by 4-1/2"

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Notify Engineer at least 72 hours prior to placing armor joint assemblies to allow time for examination of accessories. Place no concrete in forms without Engineer's acceptance of accessories.

3.2 INSTALLATION

- A. Installation of armored edge joint assembly:

1. Set wood form 2" below top of slab. Stake form.
2. Mark chalk-line on wood forms 3/4" above slab center or minimally at top of form so center line of the Diamond Dowel plate will be a minimum of 2-1/4" below top of slab.
3. Attach Diamond Dowel pocket former onto wood form with pre-assembled nails. Align center line of dowel at mid-slab depth.
4. Secure the joint assembly to form at top and back of each assembly bracket with two 8d duplex nails per bracket. For long runs, join joint assembly at overlap using nylon nuts and bolts provided.
5. Check assembly for correct line and elevation and adjust if required. If the Armor-Edge joint assembly will butt up to a saw-cut contraction joint, cut through the full depth (2") of the abutting bar to allow activation of the joint.
6. Place and finish first slab. Use internal vibration to consolidate concrete under Diamond Dowel pocket former in accordance with industry guidelines. Remove concrete paste from top of bars during finishing.
7. Remove mounting bracket from joint assembly. Then strip form and bend nails that are protruding from pocket former so they are flush with joint face.
8. Insert Diamond Dowel plate into slot created by Diamond Dowel pocket former. Center the corner of the plate in the middle of the label and push straight through the label into the pocket former. Do not hammer, or use excessive pounding force, to insert Diamond Dowel plate. Diamond Dowel plate should be inserted within two weeks of concrete placement.
9. Place and finish second slab. Use internal vibration to consolidate concrete under plate in accordance with industry guidelines. Remove concrete paste from top of bars during finishing. Threads of nylon nut will strip when concrete contracts.

3.3 PROTECTION

- A. Protect Armor-Edge joint assemblies from displacement and from damage until concrete casting is complete.

END OF SECTION 031500

SECTION 033000 - CAST-IN-PLACE CONCRETE**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY**A. Section Includes:**

- 1. Cast-in-place concrete, including concrete materials, mixture design, placement procedures, and finishes.
- 2. For interior slabs on ground, follow Section 033010

B. Related Requirements:

- 1. Section 033010 "Cast-in-Place Interior Slabs on Ground".
- 2. Section 033543 "Polished Concrete Finish and Joint Filler."
- 3. Section 071113 "Bituminous Dampproofing"

1.3 DEFINITIONS

- A. Cementitious Materials: Portland cement alone or in combination with one or more of the following: blended hydraulic cement, fly ash, slag cement, and other pozzolans materials subject to compliance with requirements.
- B. Water/Cement Ratio (w/cm): The ratio by weight of water to cementitious materials.

1.4 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.5 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each of the following:
 - a. Portland cement.
 - b. Fly ash.
 - c. Slag cement.
 - d. Blended hydraulic cement.
 - e. Aggregates.
 - f. Admixtures:
 - 1) Include limitations of use, including restrictions on cementitious materials, supplementary cementitious materials, air entrainment, aggregates,

temperature at time of concrete placement, relative humidity at time of concrete placement, curing conditions, and use of other admixtures.

- g. Vapor retarders.
 - h. Liquid floor treatments.
 - i. Curing materials.
 - j. Joint fillers.
 - k. Reinforcement.
2. Design Data: For each of the following:
- a. Concrete.
 - b. Reinforcement.
 - c. Foundation and footings.
 - d. Vapor retarders.
3. Design Mixtures: For each concrete mixture, include the following:
- a. Mixture identification.
 - b. Minimum 28-day compressive strength.
 - c. Durability exposure class.
 - d. Maximum w/cm.
 - e. Calculated equilibrium unit weight, for lightweight concrete.
 - f. Slump limit.
 - g. Air content.
 - h. Nominal maximum aggregate size.
 - i. Indicate amounts of mixing water to be withheld for later addition at Project site if permitted.
 - j. Intended placement method.
 - k. Submit alternate design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.
4. Shop Drawings:
- a. Steel Reinforcement Shop Drawings: Placing Drawings that detail fabrication, bending, and placement.
 - b. Foundation and footings.
 - c. Vapor retarders.
 - d. Construction Joint Layout: Indicate proposed construction joints required to construct the structure.
 - 1) Location of construction joints is subject to approval of the Architect.
5. Concrete Schedule: For each location of each Class of concrete indicated in "Concrete Mixtures" Article, including the following:
- a. Concrete Class designation.
 - b. Location within Project.
 - c. Exposure Class designation.
 - d. Formed Surface Finish designation and final finish.
 - e. Final finish for floors.
 - f. Curing process.

1.6 INFORMATIONAL SUBMITTALS

A. Material Certificates: For each of the following, signed by manufacturers:

1. Cementitious materials.
2. Admixtures.
3. Curing compounds.
4. Vapor retarders.
5. Joint-filler strips.

B. Material Test Reports: For the following, from a qualified testing agency:

1. Portland cement.
2. Fly ash.
3. Slag cement.
4. Blended hydraulic cement.
5. Aggregates.
6. Admixtures

C. Preconstruction Test Reports: For each mix design.

D. Field quality-control reports.

E. Minutes of preinstallation conference.

1.7 QUALITY ASSURANCE

A. Ready-Mixed Concrete Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C 94/C 94M requirements for production facilities and equipment.

1. Manufacturer certified according to NRMCA's "Certification of Ready Mixed Concrete Production Facilities."

B. Testing Agency Qualifications: An independent agency acceptable to authorities having jurisdiction qualified according to ASTM C 1077 and ASTM E 329 for testing indicated.

1.8 PRECONSTRUCTION TESTING

A. Preconstruction Testing Service: Engage a qualified testing agency to perform preconstruction testing on each concrete mixture.

1. Include the following information in each test report:
 - a. Admixture dosage rates.
 - b. Slump.
 - c. Air content.
 - d. Seven-day compressive strength.
 - e. 28-day compressive strength.

1.9 DELIVERY, STORAGE, AND HANDLING

A. Comply with ASTM C94/C94M and ACI 301 (ACI 301M).

1.10 FIELD CONDITIONS

A. Cold-Weather Placement: Comply with ACI 306.1.

1. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators unless otherwise specified and approved in mixture designs.

B. Hot-Weather Placement: Comply with ACI 305.

PART 2 - PRODUCTS

2.1 CONCRETE, GENERAL

- #### A. ACI Publications: Comply with ACI 301 (ACI 301M) unless modified by requirements in the Contract Documents.

2.2 FORM-FACING MATERIALS

- #### A. Smooth-Formed Finished Concrete: Form-facing panels that provide continuous, true, and smooth concrete surfaces. Furnish in largest practicable sizes to minimize number of joints.

1. Exposed concrete at loading dock. See Sheet A200, Detail 5.

- #### B. Rough-Formed Finished Concrete: Plywood, lumber, metal, or another approved material. Provide lumber dressed on at least two edges and one side for tight fit.

2.3 STEEL REINFORCEMENT

- #### A. Reinforcing Bars: ASTM A 615/A 615M, Grade 60 (Grade 420), deformed.

- #### B. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded-wire reinforcement in place. Manufacture bar supports from steel wire, plastic, or precast concrete according to CRSI's "Manual of Standard Practice."

2.4 CONCRETE MATERIALS

A. Cementitious Materials:

1. Portland Cement: ASTM C 150/C 150M, Type I or Type II gray.
2. Fly Ash: ASTM C 618, Class F or C.
3. Slag Cement: ASTM C 989/C 989M, Grade 100 or 120.
4. Blended Hydraulic Cement: ASTM C 595/C 595M, Type IS, portland blast-furnace slag cement.

- #### B. Normal-Weight Aggregates: ASTM C 33/C 33M, graded. Provide aggregates from a single source.

1. Alkali-Silica Reaction: Comply with one of the following:

- a. Expansion Result of Aggregate: Not more than 0.04 percent at one-year when tested in accordance with ASTM C1293.
- b. Expansion Results of Aggregate and Cementitious Materials in Combination: Not more than 0.10 percent at an age of 16 days when tested in accordance with ASTM C1567.

- c. Alkali Content in Concrete: Not more than 4 lb./cu. yd. (2.37 kg/cu. m) for moderately reactive aggregate or 3 lb./cu. yd. (1.78 kg/cu. m) for highly reactive aggregate, when tested in accordance with ASTM C1293 and categorized in accordance with ASTM C1778, based on alkali content being calculated in accordance with ACI 301 (ACI 301M).
- 2. Maximum Coarse-Aggregate Size: 3/4 inches (19 mm) nominal.
- 3. Fine Aggregate: Free of materials with deleterious reactivity to alkali in cement.
- C. Lightweight Aggregate (if indicated on Structural Drawings): ASTM C 330/C 330M, 1-inch (25-mm) nominal maximum aggregate size.
- D. Air-Entraining Admixture: ASTM C 260/C 260M.
- E. Chemical Admixtures: Certified by manufacturer to be compatible with other admixtures and that do not contribute water-soluble chloride ions exceeding those permitted in hardened concrete. Do not use calcium chloride or admixtures containing calcium chloride.
 - 1. Water-Reducing Admixture: ASTM C 494/C 494M, Type A.
 - 2. Retarding Admixture: ASTM C 494/C 494M, Type B.
 - 3. Water-Reducing and Retarding Admixture: ASTM C 494/C 494M, Type D.
 - 4. High-Range, Water-Reducing Admixture: ASTM C 494/C 494M, Type F.
 - 5. High-Range, Water-Reducing and Retarding Admixture: ASTM C 494/C 494M, Type G.
 - 6. Plasticizing and Retarding Admixture: ASTM C 1017/C 1017M, Type II.
- F. Water and Water Used to Make Ice: ASTM C94/C94M, potable.

2.5 WATERSTOPS

- A. Chemically Resistant Flexible Waterstops: Thermoplastic elastomer rubber waterstops with factory-installed metal eyelets, for embedding in concrete to prevent passage of fluids through joints; resistant to oils, solvents, and chemicals. Factory fabricate corners, intersections, and directional changes.

2.6 CURING MATERIALS

- A. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. (305 g/sq. m) when dry.
- B. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.
- C. Water: Potable.
- D. Clear, Waterborne, Membrane-Forming Curing Compound: ASTM C 309, Type 1, Class B, dissipating.

2.7 RELATED MATERIALS

- A. Expansion- and Isolation-Joint-Filler Strips: ASTM D 1751, asphalt-saturated cellulosic fiber.

2.8 CONCRETE MIXTURES, GENERAL

- A. Prepare design mixtures for each type and strength of concrete, proportioned on the basis of laboratory trial mixture or field test data, or both, according to ACI 301 (ACI 301M).

1. Use a qualified testing agency for preparing and reporting proposed mixture designs, based on laboratory trial mixtures.
- B. Cementitious Materials: Limit percentage, by weight, of cementitious materials other than portland cement in concrete as follows:
 1. Fly Ash or Other Pozzolans: 25 percent by mass.
 2. Slag Cement: 35 percent by mass.
 3. Total of Fly Ash or Other Pozzolans, Slag Cement: 35 percent by mass, with fly ash or pozzolans not exceeding 25 percent by mass.
 4. Total of Fly Ash or Other Pozzolans: 35 percent by mass with fly ash or pozzolans not exceeding 25 percent by mass.
- C. Admixtures: Use admixtures according to manufacturer's written instructions.
 1. Use water-reducing, high-range water-reducing or plasticizing admixture in concrete, as required, for placement and workability.
 2. Use water-reducing and -retarding admixture when required by high temperatures, low humidity, or other adverse placement conditions.
 3. Use water-reducing admixture in pumped concrete.

2.9 CONCRETE MIXTURES

- A. Normal-weight concrete used for footings.
 1. Exposure Class: ACI 318 (ACI 318M) F0.
 2. Minimum Compressive Strength: 3000 psi (20.7 MPa) at 28 days.
 3. Maximum Water to Cement Ratio (w/cm): 0.55
 4. Slump Limit: 4 inches (100 mm), plus or minus 1 inch (25 mm) for concrete with verified slump of 3 inches (75 mm) plus or minus 1 inch (25 mm) before adding high-range water-reducing admixture or plasticizing admixture at Project site.
 5. Air Content: As indicated on Structural Drawings.
- B. Normal-weight concrete used for foundation walls.
 1. Exposure Class: ACI 318 (ACI 318M) F2.
 2. Minimum Compressive Strength: 4500 psi (31 MPa) at 28 days.
 3. Maximum Water to Cement Ratio (w/cm): 0.45
 4. Slump Limit: 4 inches (100 mm), plus or minus 1 inch (25 mm) for concrete with verified slump of 3 inches (75 mm) plus or minus 1 inch (25 mm) before adding high-range water-reducing admixture or plasticizing admixture at Project site.
 5. Air Content: As indicated on Structural Drawings.

2.10 FABRICATING REINFORCEMENT

- A. Fabricate steel reinforcement according to CRSI's "Manual of Standard Practice."

2.11 CONCRETE MIXING

- A. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete according to ASTM C 94/C 94M and ASTM C 1116/C 1116M and furnish batch ticket information.

1. When air temperature is between 85 and 90 deg F (30 and 32 deg C), reduce mixing and delivery time from 1-1/2 hours to 75 minutes; when air temperature is above 90 deg F (32 deg C), reduce mixing and delivery time to 60 minutes.

PART 3 - EXECUTION

3.1 CONCRETE FINISH REQUIREMENTS

- A. Based on ACI 347.3R-13 following quality should be provided for all surfaces:

1. CSC3 (formed concrete surface category)
2. T4 (texture)
3. SVR4 (Surface void ratio)
4. CU2 (Color uniformity)
5. SI4 (Surface irregularities)
6. CJ4 (Construction and facing joints)
7. FC3 (Form facing categories)

3.2 INSTALLATION OF FORMWORK

- A. Limit deflection of form-facing panels to not exceed ACI 303.1 requirements.
- B. In addition to ACI 303.1 limits on form-facing panel deflection, limit cast-in-place architectural concrete surface irregularities, designated by ACI 347 as abrupt or gradual, as follows:
1. Class A, 1/8 inch.
- C. Construct forms to result in cast-in-place architectural concrete that complies with ACI 117 (ASI 117M).
- D. Chamfer exterior corners and edges of cast-in-place architectural concrete.
- E. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.
- F. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.

3.3 INSTALLATION OF EMBEDDED ITEMS

- A. Place and secure anchorage devices and other embedded items required for adjoining Work that is attached to or supported by cast-in-place concrete.
1. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
 2. Install anchor rods, accurately located, to elevations required and complying with tolerances in Section 7.5 of ANSI/AISC 303.
 3. Install reglets to receive waterproofing and to receive through-wall flashings in outer face of concrete frame at exterior walls, where flashing is shown at lintels, shelf angles, and other conditions.

3.4 INSTALLATION OF VAPOR-RETARDER

- A. Sheet Vapor Retarders: Place, protect, and repair sheet vapor retarder according to ASTM E 1643 and manufacturer's written instructions.
 - 1. Install vapor retarder with longest dimension parallel with direction of concrete pour.
 - 2. Face laps away from exposed direction of concrete pour.
 - 3. Lap vapor retarder over footings and grade beams not less than 6 inches (150 mm), sealing vapor retarder to concrete.
 - 4. Lap joints 6 inches (150 mm) and seal with manufacturer's recommended tape.
 - 5. Terminate vapor retarder at the top of floor slabs, grade beams, and pile caps, sealing entire perimeter to floor slabs, grade beams, foundation walls, or pile caps.
 - 6. Seal penetrations in accordance with vapor retarder manufacturer's instructions.
 - 7. Protect vapor retarder during placement of reinforcement and concrete.
 - a. Repair damaged areas by patching with vapor retarder material, overlapping damages area by 6 inches (150 mm) on all sides, and sealing to vapor retarder.

3.5 STEEL REINFORCEMENT INSTALLATION

- A. General: Comply with CRSI's "Manual of Standard Practice" for fabricating, placing, and supporting reinforcement.
 - 1. Do not cut or puncture vapor retarder. Repair damage and reseal vapor retarder before placing concrete.

3.6 JOINTS

- A. Refer to Section 033543 "Polished Concrete Finish and Joint Filler" specifications.
- B. Refer to Section 033010 "Cast-In-Place Concrete Slab-On-Grade" for isolation joints and control joints.
- C. General: Construct joints true to line with faces perpendicular to surface plane of concrete.
- D. Construction Joints: Coordinate with concrete placement sequence.
 - 1. Install so strength and appearance of concrete are not impaired, at locations indicated on Drawings or as approved by Architect.
 - 2. Form keyed joints as indicated. Embed keys at least 1-1/2 inches (38 mm) into concrete.
 - 3. Locate horizontal joints in walls and columns at underside of slabs and at the top of footings or floor slabs.
 - 4. Space vertical joints in walls as indicated on Drawings. Unless otherwise indicated on Drawings, locate vertical joints beside piers integral with walls, near corners, and in concealed locations where possible.

3.7 CONCRETE PLACEMENT

- A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections are completed.
- B. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete is placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.

1. Consolidate placed concrete with mechanical vibrating equipment according to ACI 301 (ACI 301M).
- C. Notify Architect and testing and inspection agencies 24 hours prior to commencement of concrete placement.
- D. Do not add water to concrete during delivery, at Project site, or during placement unless approved by Architect in writing, but not to exceed the amount indicated on the concrete delivery ticket.
 1. Do not add water to concrete after adding high-range water-reducing admixtures to mixture.
- E. Before test sampling and placing concrete, water may be added at Project site, subject to limitations of ACI 301 (ACI 301M), but not to exceed the amount indicated on the concrete delivery ticket.
 1. Do not add water to concrete after adding high-range water-reducing admixtures to mixture.
- F. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete is placed on concrete that has hardened enough to cause seams or planes of weakness.
 1. If a section cannot be placed continuously, provide construction joints as indicated.
 2. Deposit concrete to avoid segregation.
 3. Deposit concrete in horizontal layers of depth not to exceed formwork design pressures and in a manner to avoid inclined construction joints.
 4. Consolidate placed concrete with mechanical vibrating equipment in accordance with ACI 301 (ACI 301M).
 - a. Do not use vibrators to transport concrete inside forms.
 - b. Insert and withdraw vibrators vertically at uniformly spaced locations to rapidly penetrate placed layer and at least 6 inches (150 mm) into preceding layer.
 - c. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity.
 - d. At each insertion, limit duration of vibration to time necessary to consolidate concrete, and complete embedment of reinforcement and other embedded items without causing mixture constituents to segregate.

3.8 FINISHING FORMED SURFACES

A. As-Cast Surface Finishes:

1. ACI 301 (ACI 301M) Surface Finish SF-1.0: As-cast concrete texture imparted by form-facing material.
 - a. Patch voids larger than 1-1/2 inches (38 mm) wide or 1/2 inch (13 mm) deep.
 - b. Remove projections larger than 1 inch (25 mm).
 - c. Tie holes do not require patching.
 - d. Surface Tolerance: ACI 117 (ACI 117M) Class D.
 - e. Apply to concrete surfaces not exposed to public view.

2. ACI 301 (ACI 301M) Surface Finish SF-2.0: As-cast concrete texture imparted by form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams.
 - a. Patch voids larger than 3/4 inch (19 mm) wide or 1/2 inch (13 mm) deep.
 - b. Remove projections larger than 1/4 inch (6 mm).
 - c. Patch tie holes.
 - d. Surface Tolerance: ACI 117 (ACI 117M) Class B.
 - e. Locations: Apply to concrete surfaces to receive a rubbed finish, or to be covered with a coating or covering material applied directly to concrete.
 - B. Rough-Formed Finish: As-cast concrete texture imparted by form-facing material with tie holes and defects repaired and patched. Remove fins and other projections that exceed specified limits on formed-surface irregularities.
 1. Apply to concrete surfaces not exposed to public view.
 - C. Smooth-Formed Finish: As-cast concrete texture imparted by form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams. Repair and patch tie holes and defects. Remove fins and other projections that exceed specified limits on formed-surface irregularities.
 1. Apply to concrete surfaces to receive a rubbed finish, or to be covered with a coating or covering material applied directly to concrete.
 - D. Related Unformed Surfaces:
 1. At tops of walls, horizontal offsets, and similar unformed surfaces adjacent to formed surfaces, strike off smooth and finish with a color and texture matching adjacent formed surfaces.
 2. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces unless otherwise indicated.
- 3.9 INSTALLATION OF MISCELLANEOUS CONCRETE ITEMS
- A. Filling In:
 1. Fill in holes and openings left in concrete structures after Work of other trades is in place unless otherwise indicated.
 2. Mix, place, and cure concrete, as specified, to blend with in-place construction.
 3. Provide other miscellaneous concrete filling indicated or required to complete the Work.
 - B. Curbs: Provide monolithic finish to interior curbs by stripping forms while concrete is still green and by steel-troweling surfaces to a hard, dense finish with corners, intersections, and terminations slightly rounded.
 - C. Equipment Bases and Foundations:
 1. Coordinate sizes and locations of concrete bases with actual equipment provided.
 2. Construct concrete bases 4 inches (100 mm) high unless otherwise indicated on Drawings and extend base not less than 6 inches (150 mm) in each direction beyond the maximum dimensions of supported equipment unless otherwise indicated on Drawings, or unless required for seismic anchor support.
 3. Minimum Compressive Strength: As indicated on Drawings.

4. For supported equipment, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete substrate.
5. Prior to pouring concrete, place and secure anchorage devices.
 - a. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
 - b. Cast anchor-bolt insert into bases.
 - c. Install anchor bolts to elevations required for proper attachment to supported equipment.

3.10 CONCRETE CURING

- A. Protect freshly placed concrete from premature drying and excessive cold or hot temperatures.
 1. Comply with ACI 301 (ACI 301M) and ACI 306.1 for cold weather protection during curing.
 2. Comply with ACI 301 (ACI 301M) and ACI 305.1 (ACI 305.1M) for hot-weather protection during curing.
 3. Maintain moisture loss no more than 0.2 lb/sq. ft. x h (1 kg/sq. m x h), calculated in accordance with ACI 305.1,) before and during finishing operations.
- B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 lb/sq. ft. x h (1 kg/sq. m x h) before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.
- C. Curing Formed Surfaces: Comply with ACI 308.1 (ACI 308.1M) as follows:
 1. Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces.
 2. Cure concrete containing color pigments in accordance with color pigment manufacturer's instructions.
 3. If forms remain during curing period, moist cure after loosening forms.
 4. If removing forms before end of curing period, continue curing for remainder of curing period, as follows:
 - a. Continuous Sprinkling: Maintain concrete surface continuously wet.
 - b. Absorptive Cover: Pre-dampen absorptive material before application; apply additional water to absorptive material to maintain concrete surface continuously wet.
 - c. Water-Retention Sheeting Materials: Cover exposed concrete surfaces with sheeting material, taping, or lapping seams.
 - d. Membrane-Forming Curing Compound: Apply uniformly in continuous operation by power spray or roller in accordance with manufacturer's written instructions.
 - 1) Recoat areas subject to heavy rainfall within three hours after initial application.
 - 2) Maintain continuity of coating and repair damage during curing period.

3.11 TOLERANCES

- A. Conform to ACI 117 (ACI 117M).

3.12 CONCRETE SURFACE REPAIRS

- A. Defective Concrete: Repair and patch defective areas when approved by Architect. Remove and replace concrete that cannot be repaired and patched to Architect's approval.

3.13 FIELD QUALITY CONTROL

- A. Special Inspections: Owner will engage a special inspector to perform field tests and inspections and prepare testing and inspection reports.
- B. Testing Agency: Owner will engage a qualified testing and inspecting agency to perform tests and inspections and to submit reports.
 - 1. Testing agency shall be responsible for providing curing container for composite samples on Site and verifying that field-cured composite samples are cured in accordance with ASTM C31/C31M.
 - 2. Testing agency shall immediately report to Architect, Contractor, and concrete manufacturer any failure of Work to comply with Contract Documents.
 - 3. Testing agency shall report results of tests and inspections, in writing, to Owner, Architect, Contractor, and concrete manufacturer within 48 hours of inspections and tests.
 - a. Test reports shall include reporting requirements of ASTM C31/C31M, ASTM C39/C39M, and ACI 301, including the following as applicable to each test and inspection:
 - 1) Project name.
 - 2) Name of testing agency.
 - 3) Names and certification numbers of field and laboratory technicians performing inspections and testing.
 - 4) Name of concrete manufacturer.
 - 5) Date and time of inspection, sampling, and field testing.
 - 6) Date and time of concrete placement.
 - 7) Location in Work of concrete represented by samples.
 - 8) Date and time sample was obtained.
 - 9) Truck and batch ticket numbers.
 - 10) Design compressive strength at 28 days.
 - 11) Concrete mixture designation, proportions, and materials.
 - 12) Field test results.
 - 13) Information on storage and curing of samples before testing, including curing method and maximum and minimum temperatures during initial curing period.
 - 14) Type of fracture and compressive break strengths at seven days and 28 days.
- C. Batch Tickets: For each load delivered, submit three copies of batch delivery ticket to testing agency, indicating quantity, mix identification, admixtures, design strength, aggregate size, design air content, design slump at time of batching, and amount of water that can be added at Project site.
- D. Inspections:
 - 1. Steel reinforcement placement.
 - 2. Headed bolts and studs.
 - 3. Verification of use of required design mixture.
 - 4. Concrete placement, including conveying and depositing.

5. Curing procedures and maintenance of curing temperature.
 6. Verification of concrete strength before removal of shores and forms from beams and slabs.
 7. Batch Plant Inspections: On a random basis, as determined by Architect.
- E. Concrete Tests: Testing of composite samples of fresh concrete obtained according to ASTM C 172/C 172M shall be performed according to the following requirements:
1. Testing Frequency: Obtain one composite sample for each day's pour of each concrete mixture exceeding 5 cu. yd. (4 cu. m), but less than 25 cu. yd. (19 cu. m), plus one set for each additional 50 cu. yd. (38 cu. m) or fraction thereof.
 - a. When frequency of testing provides fewer than five compressive-strength tests for each concrete mixture, testing shall be conducted from at least five randomly selected batches or from each batch if fewer than five are used.
 2. Slump: ASTM C143/C143M:
 - a. One test at point of placement for each composite sample, but not less than one test for each day's pour of each concrete mixture.
 - b. Perform additional tests when concrete consistency appears to change.
 3. Air Content: ASTM C231/C231M pressure method, for normal-weight concrete;
 - a. One test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
 4. Concrete Temperature: ASTM C1064/C1064M:
 - a. One test hourly when air temperature is 40 deg F (4.4 deg C) and below or 80 deg F (27 deg C) and above, and one test for each composite sample.
 5. Unit Weight: ASTM C567/C567M fresh unit weight of structural lightweight concrete.
 - a. One test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
 6. Compression Test Specimens: ASTM C31/C31M:
 - a. Cast and laboratory cure two sets of two of 6-inch (150 mm) by 12-inch (300 mm) or three sets of two 4-inch (100 mm) by 8-inch (200 mm) cylinder specimens for each composite sample.
 - b. Cast, initial cure, and field cure two sets of two standard cylinder specimens for each composite sample.
 7. Compressive-Strength Tests: ASTM C39/C39M.
 - a. Test one set of two laboratory-cured specimens at seven days and one set of two specimens at 28 days.
 - b. Test one set of two field-cured specimens at seven days and one set of two specimens at 28 days.
 - c. A compressive-strength test shall be the average compressive strength from a set of two specimens obtained from same composite sample and tested at age indicated.

8. When strength of field-cured cylinders is less than 85 percent of companion laboratory-cured cylinders, Contractor shall evaluate operations and provide corrective procedures for protecting and curing in-place concrete.
9. Strength of each concrete mixture will be satisfactory if every average of any three consecutive compressive-strength tests equals or exceeds specified compressive strength, and no compressive-strength test value falls below specified compressive strength by more than 500 psi (3.4 MPa) if specified compressive strength is 5000 psi (34.5 MPa), or no compressive strength test value is less than 10 percent of specified compressive strength if specified compressive strength is greater than 5000 psi (34.5 MPa).
10. Nondestructive Testing: Impact hammer, sonoscope, or other nondestructive device may be permitted by Architect but will not be used as sole basis for approval or rejection of concrete.
11. Additional Tests:
 - a. Testing and inspecting agency shall make additional tests of concrete when test results indicate that slump, air entrainment, compressive strengths, or other requirements have not been met, as directed by Architect.
 - b. Testing and inspecting agency may conduct tests to determine adequacy of concrete by cored cylinders complying with ASTM C42/C42M or by other methods as directed by Architect.
 - 1) Acceptance criteria for concrete strength shall be in accordance with ACI 301 (ACI 301M), section 1.6.6.3.
12. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.
13. Correct deficiencies in the Work that test reports and inspections indicate do not comply with the Contract Documents.

3.14 PROTECTION

A. Protect concrete surfaces as follows:

1. Protect from petroleum stains.
2. Diaper hydraulic equipment used over concrete surfaces.
3. Prohibit vehicles from interior concrete slabs.
4. Prohibit use of pipe-cutting machinery over concrete surfaces.
5. Prohibit placement of steel items on concrete surfaces.
6. Prohibit use of acids or acidic detergents over concrete surfaces.
7. Protect liquid floor treatment from damage and wear during the remainder of construction period. Use protective methods and materials, including temporary covering, recommended in writing by liquid floor treatments installer.
8. Protect concrete surfaces scheduled to receive surface hardener or polished concrete finish using Floor Slab Protective Covering.

END OF SECTION 033000

SECTION 033010 - CAST-IN-PLACE CONCRETE SLAB-ON-GRADE**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes:

- 1. Interior concrete slab-on-grade.
- 2. Reactive surface densification of non-polished, exposed slab surfaces.
- 3. For non-interior slabs on ground, follow Section 033000.
- 4. Concrete requirements, as applicable to interior slabs, in Section 033000 apply to this Section.

- B. Related Requirements:

- 1. Section 013300 – Submittal Procedures.
- 2. Section 014000 – Quality Requirements.
- 3. Section 016000 – Product Requirements.
- 4. Section 033000 – Cast-In-Place Concrete
- 5. Section 033543 – Polished Concrete Floor Finish and Joint Filler.
- 6. Section 079200 – Joint Sealants.

1.3 REFERENCE STANDARDS

- A. American Concrete Institute (ACI)

- 1. ACI 117 - Standard Specifications for Tolerances for Concrete Construction and Materials
- 2. ACI 211.1 - Standard Practice for Selecting Proportions for Normal, Heavyweight, and Mass Concrete
- 3. ACI 301 - Specifications for Structural Concrete for Buildings
- 4. ACI 305.1 - Hot Weather Concreting
- 5. ACI 306.1 - Cold Weather Concreting; American Concrete Institute International.
- 6. ACI 308.1 - Guide to Curing Concrete
- 7. ACI 318 - Building Code Requirements for Structural Concrete and Commentary
- 8. ACI 347 - Guide to Formwork for Concrete

- B. American Society for Testing and Materials (ASTM)

- 1. ASTM A 185 / A 185M - Standard Specification for Steel Welded Wire Reinforcement, Plain, for Concrete.
- 2. ASTM A 615 / A 615M - Standard Specification for Deformed and Plain Billet-Steel Bars for Concrete Reinforcement.
- 3. ASTM C 33 / C 33M - Standard Specification for Concrete Aggregates.
- 4. ASTM C 39 / C 39M - Standard Test Method for Compressive Strength of Cylindrical Concrete.
- 5. ASTM C 94 / C 94M - Standard Specification for Ready-Mixed Concrete.
- 6. ASTM C 143 / C 143M - Standard Test Method for Slump of Hydraulic-Cement Concrete.

7. ASTM C 150 / C 150M - Standard Specification for Portland Cement.
8. ASTM C171 - Standard Specification for Sheet Materials for Curing Concrete.
9. ASTM C173 - Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method.
10. ASTM C 231 - Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method.
11. ASTM C 309 - Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete.
12. ASTM C 494 / C 494M - Standard Specification for Chemical Admixtures for Concrete.
13. ASTM C 1064 - Standard Test Method for Temperature of Freshly Mixed Hydraulic-Cement Concrete.
14. ASTM C 1602 / C 1602M - Standard Specification for Mixing Water Used in the Production of Hydraulic Cement Concrete.
15. ASTM C 685 / C 685M - Standard Specification for Concrete Made by Volumetric Batching and Continuous Mixing.
16. ASTM C 779 / C 779M - Standard Test Method for Abrasion Resistance of Horizontal Concrete Surfaces.
17. ASTM E 1643 - Standard Practice for Installation of Water Vapor Retarders Used in Contact with Earth or Granular Fill Under Concrete Slabs.
18. ASTM E 1745 - Standard Specification for Plastic Water Vapor Retarders Used in Contact with Soil or Granular Fill under Concrete Slabs.

1.4 ADMINISTRATIVE REQUIREMENTS

- A. Pre-installation Meeting: Conduct a concrete slab pre-installation meeting prior to the start of the work of this section.
1. Schedule meeting at least two weeks prior to first interior concrete slab placement.
 2. Require responsible representatives of each party involved with the concrete slab work to attend the meeting.
 3. Notify all required attendees in writing of scheduled time and place at least two weeks in advance of meeting. Include copy of agenda with invitation.
 4. The meeting shall convene only when all parties are present.
 5. Attendees shall include all key parties executing, but not be limited to the following:
 - a. Owner
 - b. Consultant(s) Construction Testing Laboratory (individual present during and managing all slab placements)
 - c. General Contractor:
 - 1) Project Manager
 - 2) Superintendent
 - d. Concrete Finishing Subcontractor:
 - 1) Project Manager
 - 2) Finish Foreman (individual present during and managing all slab placements)
 - e. Concrete Supplier:
 - 1) Quality Assurance Representative (individual present during and managing all slab placements)

f. Polishing / Joint Filling Subcontractor:

- 1) Project Manager
- 2) Foreman (individual present during all polish work)

B. Agenda shall include, but not be limited to the following:

1. Definition of Owner's expectations and objectives.
2. Installation scheduling and coordination; scheduling of mock-up construction and review.
3. Confirm placement environment and necessary construction adaptations.
4. Review concrete mix design.
5. Review complete slab construction details; base, reinforcement, jointing, placing, finishing, and curing.
6. Site Testing: Base installation, concrete materials, surface profiling, surface gloss and moisture emissions.
7. Joint filler and sealant installation.
8. Concrete surface treatments, including colorant, densification and polishing
9. Concrete slab protection.

C. Changes to the contract documents from recommendations or discussions at the Pre-installation meeting must be approved in writing by the Owner or Architect prior to implementation. No other party is authorized to allow changes to any specified requirement or to approve execution of any portion of the work.

D. Minutes of the pre-installation meeting shall be recorded, typed and printed by the Contractor and distributed by him to all Parties in attendance and to those as directed by Sprouts' Representative, within five (5) days of the meeting.

1.5 CONFORMANCE AND ACTION SUBMITTALS

A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.

B. Submit the following Submittals via Owner's Project Management software platform.

1. Product Data: For each of the following:

- a. Portland cement.
- b. Fly ash.
- c. Aggregates.
- d. Admixtures:
 - 1) Include limitations of use, including restrictions on cementitious materials, supplementary cementitious materials, air entrainment, aggregates, temperature at time of concrete placement, relative humidity at time of concrete placement, curing conditions, and use of other admixtures.
- e. Vapor retarders.
- f. Liquid floor treatments.
- g. Curing materials.
- h. Joint fillers.
- i. Reinforcement.

2. Slab Joint Layout and Placement Plan: Submit slab joint layout and placement plan on a copy of the drawing with defined floor joint locations, identifying the following:
 - a. Truck access locations and direction of concrete installation.
 - b. Extent of pours including width, length, slab placement area and volume.
 - c. Construction joints. Install Sales Area concrete in one continuous monolithic placement.
 - d. Location of construction joints. Configure placements to minimize visible construction joints in exposed polished sales areas.
 - e. Review all joint locations and spacing defined in contract documents. Submission of slab joint and placement plan confirms contractor's acceptance at these locations and spacing will provide best opportunity to minimize out of joint crack mitigation.
3. Shop Drawings:
 - a. Include bar schedules, shapes of bent bars, spacing of bars, and location of splices.
 - b. Construction Joint Layout: Indicate proposed construction joints required to construct the structure.
 - 1) Location of construction joints is subject to approval of the Architect.
4. Mix Designs: Submit proposed slab mix design (.pdf format) to Architect for review a minimum of 30 days prior to commencement of slab concrete installations.
 - a. Fill out and submit Concrete Slab on Ground Mix Design Submittal Form included at the end of this section, including all requested information on last page of form.
 - b. Provide completed mix designs submittal electronically in .pdf, un-secured format (maximum size 5mb to the Architect). Do not combine other submittal items with the slab on ground mix document.
 - c. Attachments to Concrete Mix Design: Submit the following as attachments to be included with the Concrete Mix Design.
 - 1) Portland cement mill test reports.
 - 2) Designation type, quality, and source (natural or manufactured) of coarse and fine aggregate materials.
 - 3) Sieve Analysis Reports: Provide separate sieve analysis of percentages passing for coarse and fine aggregate, including data that produced values. Show values for each sieve size shown on the slab mix design form. Do not leave any line blank. Sieve analysis sampling and testing for each aggregate source shall be conducted within 60 days of concrete submittal date.
 - 4) Aggregate Supplier Statement - stating if aggregates used can potentially cause pop-outs or other disruptions from aggregate reactivity or other mechanisms, based on test or past service.
 - 5) Product data for the following concrete materials admixtures:
 - a) Water reducing.
 - b) Set retarding.
 - c) Set accelerating.
 - 6) Concrete compressive strength data as required by ACI 318.

- d. It is the responsibility of the contractor to ensure that the concrete materials supplied will produce the desired finished product as determined by the Architect and Owner. Review of the mix design by any designer of record or consultant, does not relieve the contractor of this responsibility.

1.6 INFORMATIONAL SUBMITTALS

- A. Concrete Batch Plant Certifications: Name and address of the concrete supplier's batch plant and plant certification(s) by National Ready-Mix Concrete Association or State Department of Transportation.
- B. Concrete Installer's Qualifications
 - 1. Fill out and submit Concrete Finishing Subcontractor's Qualification Statement included at the end of this section. If the place and finish work is subcontracted, all references shall be that of the actual place and finish contractor, if subcontracted by the concrete contractor.
 - 2. Included with the qualification statement ACI Certifications.
- C. Manufacturer's Certificate: Certify that reinforcing, steel and accessories meet or exceed specified requirements.
- D. Manufacturer's Installation Instructions: For concrete accessories, indicate installation procedures and interface required with adjacent construction.
- E. Project Record Documents: Accurately record actual locations of embedded utilities and components that will be concealed from view upon completion of concrete work.
- F. Slab Protection Plan.
- G. Floor surface flatness and levelness measurements indicating compliance with specified tolerances.
- H. Field quality-control test and inspection reports.
- I. Minutes of preinstallation conference.

1.7 QUALITY ASSURANCE

- A. Perform work of this section in accordance with ACI 301 and ACI 318.
 - 1. Maintain one copy of each document on site.

1.8 CONCRETING IN HOT, DRY OR WINDY WEATHER

- A. Follow ACI 305.1 specifications when concreting during hot weather, with the following superseding additions or clarifications:
 - 1. Employ methods to avoid cracking when the evaporation rate of water from the surface exceeds 0.2 pounds per square foot per hour, or when any combination of concrete materials and weather conditions are favorable for the formation of plastic shrinkage cracks.

- a. Whenever surface evaporation rate is, or anticipated to be greater than 0.2 lbs./sf./hr.:
 - 1) A specified evaporation retarder shall be used after initial channel floating.
2. Cool ingredients before mixing to reduce concrete temperature at time of placement. Mixing water may be chilled or chopped ice may be used to control the concrete temperature provided the water equivalent of the ice is calculated to the total amount of mixing water.
3. Maximum concrete temperature as measured at the point of discharge shall be 90 degrees F.
4. Reject concrete if more than one slump adjustment, as defined in ASTM C 94, is required.
5. Do not place concrete when forms, subgrade, aggregate base (granular fill) or reinforcing bars are more than 120 degrees F or the temperature differential between the forms, aggregate base (granular fill), or reinforcing bars and concrete will create conditions favorable for settlement cracks or thermal cracking.
6. Cover reinforcing steel with water-soaked burlap so the steel temperature will not exceed the ambient air temperature immediately before embedment in concrete.

1.9 COLD WEATHER CONCRETE PLACEMENTS

- A. Follow ACI 306.1 specifications when concreting during cold weather, with the following superseding additions or clarifications:
 1. Minimum base surface temperature and ambient building air temperature shall be 55 degrees F during placement and throughout five-day minimum curing period.
 2. Do not place slabs on subgrade, or base that is more than 20 degrees F cooler than concrete. Warm subgrade, or base to decrease temperature differential to 20 degrees F or less.
 3. Minimum concrete temperature as measured at the point of discharge shall be 60 degrees F.
 4. Do not use unvented combustion heaters during concrete placement so as to prevent exposure of concrete to excessive exhaust gases containing carbon dioxide (CO₂) or carbon monoxide (CO). During slab placement and curing periods, maximum CO₂ levels shall be 4,500 parts per million and maximum CO levels shall be 50 parts per million at concrete surface within 5 feet of any source of exhaust gases to minimize potential damage to concrete.

1.10 FIELD CONDITIONS

- A. Enclosed Structure:
 1. Sales floor concrete shall be placed within a completely enclosed structure after the roof membrane is completely installed and watertight.
 2. Roof construction, skylight installation, overhead painting, and roof drainage system shall be complete and weather tight prior to placement.
 3. Non-sales floor areas, and minimum concrete slab installations required for stud wall framing, shall only require compliance with the requirements of both ACI 305.1 and 306.1, with additional focus on protection from wind and precipitation during and after placement to prevent damage.
- B. Lighting:

1. Illuminate slab placement areas using fixtures suspended from the structure above, positioned and in quantity as necessary to achieve uniform lighting without distinct shadows.
 2. Maintain a lighting level of minimum 30 foot-candles on concrete surfaces during placement and initial finishing.
- C. Ventilation: Provide adequate continuous ventilation during all slab placements. Contractor is responsible for monitoring CO and CO2 levels during concrete placement, while insuring a safe working environment.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Subject to compliance with project requirements, provide products as manufactured by the following to the extent as specified hereinafter for the specific product:
1. American Decorative Concrete (800) 592-9320
 2. BASF Admixtures, Inc., (800) 628-9990.
 3. BASF Building Systems, (800) 433-9517.
 4. BW Manufacturing, Inc., (616) 447-9076.
 5. Dustless Technologies (800) 568-3949.
 6. Fortifiber Building Systems Group, (800) 773-4777.
 7. Goldblatt Industries, (877) 876.7562
 8. Greenstreak Group, Inc., (800) 325-9504.
 9. Kraft Tool Company, (800) 422-2448
 10. L&M Construction Chemicals, (800) 362-3331.
 11. PNA Construction Technologies, (800) 542-0214.
 12. Prosoco, (800) 255-4255
 13. Reliable Concrete Accessories, (855) 259-2662
 14. Raven Industries, Inc., (800) 635-3456.
 15. Sika Corporation, (800) 933-7452
 16. Soff-Cut, (951) 272-2330
 17. SpecChem (866) 791-8700.
 18. Stego Industries (877) 464-7834.
 19. Unitex, (800) 821-5846.
 20. Vexcon Chemicals, Inc., (888) 839-2661.
 21. W.R. Grace and Co., (713) 223-8353.
 22. W.R. Meadows, Inc., (847) 214-2100.
- B. Substitutions: Comply with the requirements of Section 016000 unless specified otherwise herein.

2.2 AGGREGATE BASE (GRANULAR FILL)

- A. Aggregate base (granular fill) material shall comply with local State DOT Standard Specifications for Dense Graded Aggregate material designated for road base application as follows:
1. 100% passing the 1-1/2-inch sieve
 2. 15% to 55% passing the No. 4 sieve
 3. 5% to 12% passing the No. 200 sieve.
- B. The base material shall be free of any asphalt or volatile organic materials.

2.3 FORMWORK

- A. Formwork Design and Construction: Comply with guidelines of ACI 347 to provide formwork that will produce concrete complying with tolerances of ACI 117.
- B. Furnish wood forms in largest practicable sizes to minimize number of joints and to conform to joint system shown on Drawings.
 - 1. Forms shall be dressed on at least two edges and one side for tight fit.

2.4 REINFORCEMENT

- A. Reinforcing Steel: ASTM A615/A615M Grade 60 (420).
 - 1. Type: Deformed billet-steel bars.
 - 2. Finish: Unfinished, unless otherwise indicated.
- B. Reinforcement Accessories:
 - 1. Tie Wire: Annealed, minimum 16 gage.
 - 2. Chairs, Bolsters, Bar Supports, and Spacers: Sized and shaped for adequate support of reinforcement during concrete placement.
 - 3. Provide stainless steel, galvanized, plastic, or plastic-coated steel components for placement within 1-1/2 inches of weathering surfaces.

2.5 CONCRETE MATERIALS

- A. Cement: ASTM C 150, Type I or II.
- B. Fine and Coarse Aggregates: ASTM C 33, with following additional requirements:
 - 1. Use only aggregates that the supplier does not have any knowledge of pop-outs or other disruptions from aggregate reactivity or other mechanisms, based on test or past service when used for similar interior slabs on ground applications.
 - 2. No coal, lignite or mica is acceptable in concrete slab mix.
 - 3. Fine aggregate grading requirements as defined in section 6.1 of ASTM C33 shall be strictly met without deviation unless approved by the Structural Engineer.
 - 4. Use only natural (not manufactured) sand materials in polished slab areas, unless otherwise approved.
 - 5. Maximum Coarse-Aggregate Size: 1 inch (25.4 mm) nominal.
 - 6. Fine Aggregate: Free of materials with deleterious reactivity to alkali in cement.
 - 7. Adjust proportions of coarse, intermediate, and fine aggregate to provide the following material proportioning characteristics unless otherwise approved:
 - a. Coarseness Factor of 55 to 75%.
 - 1) The Coarseness Factor (CF) is the percent of combined aggregate retained on the #8 sieve that is also retained on the 3/8 inch sieve.
 - 2) The Coarseness Factor (CF) is calculated as follows:
$$CF = \frac{\text{Aggregate retained on 3/8 inch sieve}}{\text{Aggregate retained on the \#8 sieve}}$$
 - b. Adjusted Workability Factor.

- 1) The Workability Factor (WF) is the percent of combined aggregate that passes the #8 sieve.
- 2) The Adjusted Workability Factor (Adj-WF) is calculated as follows:
 - a) $\text{Adj-WF} = \text{WF} + [(\text{Cementitious Material} - 564\text{lbs}) \div 37.6]$.
- 3) The range of accepted Adj-WF for a given CF is as follows:
 - a) $\text{Adj-WF} = [(11.25 - 15 \text{ CF}) + 36] \pm 3$.
8. The total amount of intermediate aggregate shall not exceed 300 lbs/cy (1.85 cubic feet), unless necessary to meet the maximum specified Coarseness Factor.
9. Particle distribution requirements of ASTM C33 may be waived in order to meet ranges specified.
- C. Water and Water Used to Make Ice: ASTM C94/C94M, potable.
- D. Chemical Admixtures: ASTM C 494/C 494M.
- E. Supplementary Cementitious Materials:
 1. For troweled interior slabs, fly ash is allowed.
 2. Weight of the supplementary cementitious materials divided by the weight of total cement materials shall not exceed the following:
 - a. Fly Ash: 20%.
 3. Maintain specified minimum Portland cement content.

2.6 CONCRETE MIX DESIGN

- A. Exposure Class: ACI 318 (ACI 318M) F0.
- B. Proportioning Normal Weight Concrete: Comply with ACI 211.1 recommendations.
- C. Concrete Strength:
 1. Establish required average strength for each type of concrete on the basis of field experience or trial mixtures, as specified in ACI 301.
- D. Portland Cement:
 1. Minimum concrete compressive strength shall be 4,000 psi at 28 days.
 2. Minimum required content is 520 pounds per cubic yard.
 3. Use only one cement brand throughout project.
- E. Maximum water/cement ratio of 0.55.
- F. Slump Limit: 4 inches (100 mm), plus or minus 1 inch (25 mm) for concrete with verified slump of 3 inches (75 mm) plus or minus 1 inch (25 mm) before adding high-range water-reducing admixture or plasticizing admixture.

- G. Concrete mix shall be designed with intent to provide optimized placing and finishing characteristics for the given placement environment and available materials, that will produce a uniformly finished surface acceptable by owner's representative.

2.7 CHEMICAL ADMIXTURES

- A. Add acceptable admixtures as recommended in ACI 211.1 and at rates recommended by manufacturer
- B. Do not use Type "F" water reducing admixture, or all range water reducing admixture at Type F dosage rate.
- C. Use only one water reducing admixture in any given mix.
- D. Water content required to produce indicated slump without admixture, should be targeted. Maximum total water content shall be 36 gallons per cubic yard.
- E. Do not use air entrainment admixtures in concrete materials that receive a trowel finish.
- F. Do not use chemicals that will result in total mix soluble chloride ion content in excess of 0.1 percent by weight of cement.

2.8 MIXING

- A. Measure and mix ingredients in accordance with most stringent requirements of ASTM C 94.
- B. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete in accordance with ASTM C94/C94M and furnish batch ticket information.

2.9 UNDERSLAB VAPOR BARRIER

- A. Sheet Vapor Barrier: 15 mils thick, ASTM E 1745, Class A.
 - 1. Two layers required.
- B. Water Vapor Permeance: 0.01 perm, maximum, when tested in accordance with ACI 302.2R-06 recommendations.
- C. Accessory Products: Vapor barrier manufacturer's recommended tape, adhesive, mastic, prefabricated boots, etc., for sealing seams and penetrations in vapor membrane.
- D. Products:
 - 1. Perminator; W.R. Meadows
 - 2. Stego Wrap; Stego Industries
 - 3. Vapor Block; Raven Industries
 - 4. Vaporguard; Reef Industries

2.10 REACTIVE SURFACE DENSIFIER

- A. Where drawings specify sealer at trowel finished exposed concrete floors that are not polished. See Section 033543 for sealer to be used at polished concrete floors.
- B. Applicator qualifications: Approved in writing by manufacturer.

C. Manufacturer's Warranty.

1. Furnish warranty that treated floor will remain water-repellant, dustproof, hardened and abrasion resistant 10-years from the date of densifier application.

D. Products:

1. 3D Densifier; American Decorative Concrete
2. Ashford Formula; Curecrete Chemical
3. Prosoco; Blended Densifier
4. Seal Hard; L & M Construction Chemicals
5. SpecHard; SpecChem
6. Starseal PS; Vexcon

2.11 JOINTING PRODUCTS

A. Preformed Isolation Joint: Flexible closed-cell synthetic foam expansion joint strips, non-extruding, for full depth of concrete.

1. Products:

- a. Ceramar Flexible Foam Expansion Joint; WR Meadows
- b. Deck-O-Foam Expansion Joint Filler; WR Meadow
- c. Expansion Joint Filler; BASF

B. Joint Sealant and Primer: As specified in Section 079200.

2.12 EVAPORATION RETARDER

A. Water-based polymer, sprayable (during slab placement):

1. Confilm; BASF
2. E-Con; L&M
3. SpecFilm; SpecChem.

2.13 CURING MATERIALS

A. Moisture Retaining Cure Cover: White natural cellulose fiber matting securely attached to plastic sheet backing.

1. UltraCure; McTech Group
2. ReliableCure SOG; Reliable Concrete Accessories
3. Substitutions: Not Permitted.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify lines, levels, and dimensions before proceeding with work of this section.
- B. Verify that forms are clean before applying release agent.
- C. Verify that underslab conduits and pipes are fully embedded within the aggregate base or subgrade. Do not allow conduits or pipes to be embedded in any section of the floor slab.

1. Contractor is responsible for work required to repair and correct cracks and damages due to conduits or pipes embedded within the floor slab.

3.2 AGGREGATE BASE (GRANULAR FILL)

- A. Unless otherwise indicated on the Drawings, place aggregate base (granular fill) as specified herein.
- B. Aggregate Base (granular fill):
 1. Install aggregate base (granular fill) under slab areas in thickness shown on Drawings.
 2. Compact to final thickness shown on Drawings in layers not exceeding 4 inches with minimum of 2 passes per layer with vibratory compactor.
 3. Compact lifts to at least 95 percent of ASTM D 1557 maximum dry density. Proof-roll the entire area under slabs.
- C. Aggregate Base (granular fill) Fine Grading:
 1. Compact to final thickness noted with 2 passes minimum vibratory compactor to produce smooth, flat, dense surface.
 2. Use laser guided equipment to establish specified base elevation.
 3. Do not allow excess moisture or soft soil beneath vapor barrier at time of placing concrete.
 4. Provide dry, smooth, flat, dense surface. Take corrective action if rutting or pumping is evident during the preparation of the soil support system.
 5. Proof-roll 48 hours maximum prior to concrete placement and prior to installation of vapor barrier. Depression under a fully loaded concrete truck shall not exceed 1/4 inch.
- D. Slab on Ground Aggregate Base (granular fill) Fine Grade Tolerance: +0 inch, -3/4 inch with transition no greater than 3/4 inch vertically to 8 inches horizontally for level slab.
- E. Under freezer and cooler slabs, install a base consisting of free-draining, natural, fine, screened and washed, silica-based (no limestone, white or gray color) sand. No silt allowed. Manufactured sand or stone dust is not acceptable

3.3 VAPOR BARRIER INSTALLATION

- A. Install vapor barrier in accordance with ASTM E 1643 except that Paragraph 10 – Slab Moisture Content provisions shall not apply.
- B. Install two layers of vapor barrier, with staggered seams, directly under slabs, after acceptable proof rolling of base materials.
 1. At freezer and cooler slabs, install vapor barrier directly below insulation board.
- C. Place, protect, and repair sheet vapor barrier in accordance with manufacturer's written instructions.
 1. Install vapor retarder with longest dimension parallel with direction of concrete pour.
 2. Face laps away from exposed direction of concrete pour.
 3. Lap vapor retarder over footings and grade beams not less than 6 inches (150 mm), sealing vapor retarder to concrete.
 4. Lap joints 6 inches (150 mm) and seal with manufacturer's recommended tape.

5. Terminate vapor retarder at the top of floor slabs, grade beams, and pile caps, sealing entire perimeter to floor slabs, grade beams, foundation walls, or pile caps.
6. Seal penetrations in accordance with vapor retarder manufacturer's instructions.
7. Protect vapor retarder during placement of reinforcement and concrete.
 - a. Repair damaged areas by patching with vapor retarder material, overlapping damages area by 6 inches (150 mm) on all sides, and sealing to vapor retarder.

3.4 INSTALLING REINFORCEMENT AND OTHER EMBEDDED ITEMS

- A. Comply with requirements of ACI 301.
- B. Clean reinforcement of loose rust and mill scale, and accurately position, support, and secure in place to achieve not less than minimum concrete coverage required for protection.
- C. Install welded wire reinforcement in maximum possible lengths, and offset end laps in both directions. Splice laps with tie wire.
- D. Verify that anchors, seats, plates, reinforcement and other items to be cast into concrete are accurately placed, positioned securely, and will not interfere with concrete placement.
- E. Ensure reinforcement, inserts, embedded parts, and formed construction joint devices will not be disturbed during concrete placement.
- F. The concrete finisher shall verify that all plumbing fixtures are installed and secured at finish floor elevation prior concrete placement. They shall be installed in a manner that allows for continuous finishing operations over these fixtures with riding trowel machines without buildup of concrete materials, while producing a uniform finish around the fixtures.

3.5 CONVEYING

- A. Handle concrete from mixer to place of final deposit as rapidly as practicable and in manner which will assure obtaining specified quality of concrete.
- B. Re-tempering: Discard concrete which has already begun to set. Do not re-temper with water.
- C. All concrete materials for interior slabs on ground shall be deposited from the concrete ready-mix truck by pump to point of placement, in the most expeditious, continuous manner that insures seamless consolidation without differential set or construction joints.

3.6 PLACING CONCRETE

- A. Notify Owner's Consultant not less than five (5) days prior to commencement of placement operations. A representative from this consulting firm must be on site for the entire installation of the sales area slab placement.
- B. Unless otherwise specified, place concrete in accordance with the requirements of ACI 301.
- C. The following personnel must be present during all slab placements:
 1. General Contractor's Superintendent.
 2. Concrete Subcontractor's Finish Foreman
 3. A minimum of three ACI Flatwork Finisher & Technician (or Tradesman) Certifications.
 4. Concrete Supplier's Quality Control Representative.

5. Owner's Testing Laboratory.
 - D. Do not place concrete over standing water, mud, frost, ice or snow.
 - E. Before placing concrete, verify that installation of formwork, reinforcement, embedded items, and vapor retarder is complete and that required inspections are completed.
 1. Immediately prior to concrete placement, inspect vapor retarder for damage and deficient installation, and repair defective areas.
 2. Provide continuous inspection of vapor retarder during concrete placement and make necessary repairs to damaged areas as Work progresses.
 - F. All concrete shall be delivered from only one concrete batch for any given days placement. The same concrete plant shall be used for the entire interior slab placements, unless approved in writing prior to change.
 - G. Installation of the concrete shall be completed within 90 minutes after the introduction of the mixing water to the cement and aggregates or the introduction of the cement to the aggregates.
 - H. Manage the installation process, including materials delivery, manpower and conveying methods to insure a continuous monolithic deposit without placing fresh plastic concrete adjacent to hardened concrete, creating cold seams or joints.
 - I. Deposit and consolidate concrete for floors and slabs in a continuous operation, within limits of construction joints, until placement of a panel or section is complete.
 1. Do not place concrete floors and slabs in a checkerboard sequence.
 2. Consolidate concrete during placement operations, so concrete is thoroughly worked around reinforcement and other embedded items and into corners.
 3. Maintain reinforcement in position on chairs during concrete placement.
 4. Screed slab surfaces with a straightedge and strike off to correct elevations.
 5. Level concrete, cut high areas, and fill low areas.
 6. Slope surfaces uniformly to drains where required.
 7. Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane, before excess bleedwater appears on the surface.
 8. Do not further disturb slab surfaces before starting finishing operations.
 - J. Consolidate and screed concrete slabs-on-ground using a ride-on, vibrating laser screed.
 1. Laser screed must have a minimum strike-off head width of 12' and boom reach of 20'.
 2. Check laser screed level head a minimum of 3 times during each pour. Use feeler gauge to measure deviations. If deviations are present, replace head with a new straight one.
 3. Other vibratory screeding methods are acceptable only in areas where laser screed is not accessible and approved in writing.
 - K. Use internal vibration along construction joints at both formed and slab abutments.
- 3.7 CONCRETE FINISHING
- A. Comply with ACI 302.1R recommendations for screeding, restraighening, and finishing operations for concrete surfaces. Do not wet concrete surfaces.
 - B. General:

1. Provide a dense, smooth trowel finished surface, uniform in color and profile.
2. Finish surface to produce maximum sheen free of scratches and visible or raised trowel marks.
3. At formed construction joints, finishing methods utilized shall produce the same density and final surface color across the joint. Riding finishing equipment shall extend over the formed edge for the first slab placement, with the same finishing techniques utilized up to the abutting joint for the second slab placement, minimizing hand finishing. All measures shall be taken to avoid scratching the originally placed slab surface while installing the second slab placement.
4. Do not ride trowel machines on existing hardened concrete slabs. Trowels shall be carried off of slab surfaces.
5. Do not add water to any slab surface during finishing operations.
6. Do not add cement to any slab surface during finishing operations.
7. Do not perform finishing operation while water is present on slab surface.

C. Equipment:

1. Provide a minimum of five ride-on finishing machines, eight (8) foot or larger, fully operational, clean of any debris, hardened concrete splatters, dripping oil or grease, on site and available for all slab placements as follows:
 - a. 2 machines equipped with floating pans.
 - b. 2 machines equipped with steel trowel or combination blades
 - c. 1 machine equipped with steel reinforced, plastic finish blades.
 - 1) "Poly Pro" plastic blades; Wagman Metal Products.
2. Use new finishing blades and pans that are free of rust, dirt or hardened concrete, for each installation.
3. Provide at least one operating, 24" or 30" walk-behind edge finishing machine with rotating guard ring.
4. The contractor shall provide necessary backup equipment, to ensure continuous depositing and installation of materials that prevents delays and the potential for placing of fresh concrete adjacent to previously hardened materials.

D. Initial Leveling and Aggregate Consolidation:

1. For slabs to receive a polished finish, immediately after Laser Screed strike-off, use a double roller tamp (minimum 36" wide) to consolidate the coarse aggregate below the concrete surface, as manufactured by:
 - a. Goldblatt Industries
 - b. Kraft Tool Company
 - c. Approved equal.
2. Follow roller tamping, use a 6 to 12 foot wide minimum channel float or check rod (not a bull float) in sales floor areas.
3. After initial leveling of slab surface, all further floating shall be completed with pans, with the exception of "boarding" edges at construction joints. Bump cutters or highway straight edges shall not be used to cut high points at any time during this placing and finishing process.

E. Hand and Power Floating:

1. Power float floor with pans before troweling.
2. Do not start floating until following conditions are met:
 - a. Bleeding is complete and water is gone, including water sheen on slab surface.
 - b. Mortar is not thrown by rotating blades of power float.
3. The finisher shall determine the proper time to start finishing procedures for interior slab placements on the basis the above conditions.
 - a. It is noted for advisory purposes, however, that typical setting characteristics of concrete materials will allow for initial power floating to begin 3-1/2 hour's $\pm$ 1 hour after batching, at which time the concrete should support a finisher on foot without more than approximately a 1/8 inch indentation in the slab surface when using ride-on power trowels and 1/4 inch indentation when using walk-behind edge machines along walls, pipes and columns.
 - b. Variations in concrete materials, nature, and concrete temperature will cause setting behavior to vary. It is the responsibility of the purchaser of the concrete materials to validate that the placing, finishing and setting characteristics of the materials will produce desired finish product.
4. Do not use following tools for floating:
 - a. Power troweling machines with trowel blades.
 - b. Fresno or other type of wide metal trowel.
5. Float three times minimum with each floating at right angles and 45 degrees to previous floating, and final pass at 90 degrees to previous pass.

F. Troweling for Interior Slabs:

1. For first troweling, keep blades as flat as possible and use low speed, minimizing surface "twisting", "washboard", "chatter marks" or "pitting".
2. Trowel two times minimum with first two trowelings at right angles. Cease troweling before trowel blades scratch surface.
3. Allow time between trowelings for concrete to stiffen and water sheen to disappear but ensure adequate continuous finishing of the slab surface to avoid early sealing or crusting.
4. Keep trowel machines continuously moving during trowel finishing to avoid permanent surface discoloration.
5. Darkened spots, smeared or blotchy finish, typically from troweled over knots or mortar deposits thrown from finishing machine, are not acceptable.
6. Ensure kneeboard impressions, trowel marks, footprints, or chattered areas are not evident after floor is finished.
7. All exposed in-service concrete slab surfaces, beyond 21 days after placement, shall have a minimum Mohs' hardness of 5 when measured using the following test method, or approved equivalent:
 - a. "Deluxe Hardness Picks" as manufactured by Mineralab, LLC, (800) 749-3766.

G. Surfaces to Receive Thick Floor Coverings:

1. Includes quarry tile, ceramic tile, and terrazzo with full bed setting
2. ACI 302 Class 1 - "Float" finish.

H. Surfaces to Receive Thin Floor Coverings:

1. Includes carpeting, resilient flooring, seamless flooring, thin set quarry tile, and thin set ceramic tile.
2. ACI 302 Class 2 - "Light-troweled" finish. Uniform "haze" finish without burn marks, meeting the surface tolerance requirements.

I. Surfaces to receive Polished Finish:

1. ACI 302 Class 3 - "Light-troweled" finish with the final pass using steel reinforced plastic finishing blades. Uniform tight finish without burn marks, meeting the surface tolerance requirements.

J. Other Exposed Concrete Wearing Surfaces:

1. ACI 302 Class 5 - "Steel-troweled" burnished finish.

K. Broom Finish: Apply a broom finish to exterior concrete platforms, steps, ramps, and locations indicated on Drawings.

1. Immediately after float finishing, slightly roughen trafficked surface by brooming with fiber-bristle broom perpendicular to main traffic route.
2. Coordinate required final finish with Architect before application.

3.8 SLAB JOINTING

A. General:

1. Provide construction, isolation, and control joints as indicated on Drawings and as noted below.
2. Make joints straight; perpendicular or parallel to building lines, column lines and slab edges, as appropriate.
3. Joints shall form rectangular panels with the long side less than 1-1/2 times the length of the short side.
4. Locate contraction joints at reentrant corners, when possible.
5. Coordinate with placement of joints in tile surfaces.

B. Construction Joints:

1. Shall be 1-1/2 inch minimum form lumber with top outside edge beveled 30 degrees minimum, leaving 3/8 inch to 1/2 inch level surface.
2. Provide bulkheads full depth of member.
3. After removal of bulkhead (form), lightly stone the top slab edge to remove any protrusions that may prevent a seamless abutment when placing adjacent concrete.
4. After concrete is fully cured, and prior to joint filling, saw cut joint to provide ledge to receive filler material. Maintain sawed joint in accurate alignment with originally formed construction joint, allowing time for joint to open minimum 1/16 inch width at time of cutting. Cut to a minimum depth of 1 inch, unless otherwise indicated.

C. Isolation Joints: After removing formwork, install joint-filler strips at slab junctions with vertical surfaces, such as walls, column pedestals, foundation walls, grade beams, and other locations where indicated.

1. Terminate full-width joint-filler strips not less than 1/2 inch (13 mm) or more than 1 inch (25 mm) below finished concrete surface, where joint sealant or joint filler is indicated or where aesthetic is important.
 2. Extend joint-filler strips full width and depth of joint, terminating flush with finished concrete surface in other locations not specified.
 3. Install joint-filler strips in lengths as long as practicable. Where more than one length is required, lace or clip sections together.
- D. Control (Shrinkage) Joints: Form weakened-plane control joints, sectioning concrete into areas as indicated.
1. As soon as practical after slab has set, provide Soff-Cut slab joints as indicated on drawings. Avoid raveling of the saw joint edge caused from early cutting.
 2. Non-staining chalk shall be used to mark all locations of saw cut lines.
 3. Employ sufficient number of saws and workers to complete cutting saw joints before shrinkage produces cracking. Provide at least two Soff-Cut saws with specified vacuums on site with blades capable of achieving the required depth of saw cut with operating vacuums for each saw.
 4. Special attention shall be paid by Contractor to joints at re-entrant corners, or at other locations which may require early, or deeper, cutting to prevent cracking.
 5. Saw cut to maximum width of 1/8 inch by 1/4 the slab depth, unless noted otherwise on drawings.
 6. Cutting joints into "T" intersections is not acceptable.
 7. Extend sawed joint to the slab boundaries and abutments, including columns, drains, and other penetrations in the path of a defined joint. Implement methods and timing of the saw cut beyond the limits of the Soff-Cut saw reach to provide a consistent depth of cut with minimal raveling of joint edges.
- E. Saw-Cut Joint Dust Collection (for exposed concrete slabs):
1. Take measures to prevent all sawing residue from scratching or collecting on slab surface.
 2. Apply new 2" wide Velcro tape across the bottom of the saw plate, fabric side only, for each day of sawing.
 3. Connect a specified dust collection system directly to each saw being used.
 - a. Dust Collection System:
 - 1) Pulse Vac; BW Manufacturing.
 - 2) DustDroid; Dustless Technologies.
 - 3) Soff-Vac; Husqvarna USA.
 - 4) Substitutions: Not permitted.
 4. Immediately clean any remaining residue after cutting, prior to cure.

3.9 TOLERANCES

- A. Conform to most stringent requirements of ACI 117 and ACI 301 except as specified herein.
- B. Floor Flatness F(F) and Floor Levelness F(L) tolerances for trowel finished slabs shall be in accordance with ACI 117 as follows:
1. The minimum local area shall be bound on each side by column lines.

2. Interior Floor Slabs on Ground: Specified Overall Value (SOV) F(F) 50 / F(L) 35 and Minimum Local Value (MLV) F(F) 34 / F(L) 24.

3.10 CURING AND TRAFFIC RESTRICTIONS

- A. Comply with requirements of ACI 308.1. Immediately after placement, protect concrete from premature drying, excessively hot or cold temperatures, and mechanical injury.
- B. Slab Surfaces (including exposed polish mock-up area):
 1. Immediately after finishing is complete, saturate exposed surfaces of concrete for 30 minutes for preconditioning.
 2. Apply Moisture Retaining Cure Cover with edges lapped 3 to 4 inches, sealed and secured in such manner as to prevent moisture from escaping from concrete from laps or edges in accordance with manufacturer's recommendations.
 3. After laying cure cover, use manufacturer furnished roller squeegee or push broom to ensure uniform contact between curing cover and slab eliminating all air pockets and creases.
 4. Slab surface shall remain saturated for the duration of the seven (7) day curing period after placement.
 5. After curing period, remove cure cover in sections immediately prior to scrubbing saturated slab surface. Remove only enough sheets in a given area that can be scrubbed before slab surface dries. Do not allow slab surface to dry prior to initial cleaning.
 6. Scrub floor surface immediately and as often as necessary to remove all laitance and residue in a soluble state, prior hardening of materials that can discolor the slab surface.
 7. Properly dispose of used cure covers. Do not reuse.
- C. Traffic Limitations after Slabs on Ground Placement:
 1. Barricade concrete surfaces immediately after placing and finishing.
 2. Storage of any materials on, or processing over, any slab surface that will not be covered with flooring is prohibited during the construction process.
 3. No traffic (foot or vehicular) is permitted until removal of cure covers and complete cleaning of the slab surface seven (7) days after placement.

3.11 REACTIVE SURFACE DENSIFIER

- A. Area to be treated: Apply densifier to all interior concrete floors where concrete is not polished.
- B. Examination and Preparation:
 1. The area of application shall have a minimum cure time of 14 days after placement of concrete, prior to application of densifier, and 10 days prior to racking and other in-store accessory installation.
 2. Examine surfaces receiving densifier. Verify that surfaces conform to product manufacturer's requirements for substrate conditions.
 3. Scrub floor with pre-densifier floor cleaner to remove latent salts.
 4. Do not proceed until unsatisfactory conditions have been corrected.
- C. Application:
 1. Application shall be performed by certified applicator in accordance with manufacturers published instructions.

2. Employ methods to ensure concrete surface is not damaged during application, including discoloration.
3. Apply densifier to point of rejection, minimum two (2) applications.
4. Apply with low pressure sprayer with enough coverage to keep concrete surface wet for minimum 20 minute period.
5. When treated surface gels and becomes slippery under foot, lightly sprinkle surface with water and agitate with broom to redistribute special concrete finish material evenly across surface.
6. After surface again becomes slippery, using garden hose with garden-type spray nozzle, flush entire surface with water removing excess material, alkali, or impurities. Squeegee or wet-vac surface dry to remove excess material to avoid whitening of concrete during curing. Whitening of concrete by over-application of hardener/densifier may be cause for rejection.

D. Disposal:

1. Upon completion of each densifier treatment, dispose of excess densifier material as required by local agency having jurisdiction.
2. Certified applicator shall remove all densifier product containers from job site immediately upon completion of work.

3.12 FIELD QUALITY CONTROL

- A. An independent testing agency will perform field quality control tests, as specified in Section 014000.
- B. Provide free access to concrete operations at project site and cooperate with appointed firm.
- C. Maintain records of concrete placement. Record date, location, quantity, air temperature, and test samples taken.
- D. Test concrete materials as follows:
 1. For every 50 cubic yards of concrete placed:
 - a. Perform one slump test - ASTM C143/C143M.
 - b. Measure concrete materials temperature - ASTM C1064.
 2. For every 100 cubic yards of concrete placed:
 - a. Compressive Strength Tests - ASTM C39/C39M.
 - b. Measure air content - ASTM C173/231.
 3. Complete all above tests for the first concrete truck delivered.
- E. Flatness and Levelness Testing:
 1. Test interior floor slabs for flatness and levelness in accordance with ASTM E 1155 within 24 hours of completion of floor finishing, as specified herein:
 - a. Conduct surface profile testing prior to saw cutting for each day's placement.
 - b. F(F)/F(L) minimum overall for composite of measured values (Specified Overall Value - SOV) for entire day's concrete placement and F(F)/F(L) minimum for any individual floor section (Minimum Local Values - MLV) as specified.

- c. Bound individual floor sections for testing purposes by the following that provide the smallest sections: construction joints, control joints or column and half-column lines.

- F. Test Results: The testing agency shall report test results in writing to Architect and Contractor within 24 hours of test.

3.13 DEFECTIVE CONCRETE

- A. Defective Concrete: Concrete not conforming to required lines, details, dimensions, tolerances or specified requirements for uniformity of finish, surface scratches and gouges.
- B. The contractor may be directed to remove and replace concrete for one of the following (but not limited to) conditions at the discretion of the Owner's Representative:
 - 1. Slab surfaces measuring below either of specified minimum local $F(F) / F(L)$ values.
 - 2. Any cracked panel in slabs to receive polished finish.
 - 3. Visible construction joints from delays in deposit and consolidation of the concrete materials.
 - 4. Excessive differential coloration of the slab surface, due to finishing, curing or improper protection.
 - 5. Misaligned, or defective saw cuts.
 - 6. Poorly finished, or damaged slab surface around plumbing or electrical fixtures.
- C. All costs associated with remedies for defective concrete, including extra testing as required, shall be borne by Contractor. Owner may choose to take credit for value of corrective work, in lieu of further delays to project schedule.
- D. Do not patch, fill, touch-up, repair, or replace exposed concrete except upon express direction of Architect for each individual area.

3.14 PROTECTION

- A. General Contractor to provide a written "Slab Protection Plan" prior to pre-installation meeting for approval including, but not limited to, the following items:
 - 1. All exposed polished slab areas shall be barricaded with cones. Sub-Contractors and Vendors will not be allowed in these areas unless authorized and required to complete contracted work. No product storage will be allowed in these areas.
 - 2. Dry mop slab surface as necessary to remove dust and debris daily.
 - 3. Clean slab surface as necessary to remove dust and debris daily, without damaging the slab surface.
 - 4. If an automated scrubber is used to clean the slab, use only pad type scrubber with clean red 3M pads. Do not use a drum type scrubber.
 - 5. Develop spill cleanup kit and identify locations where found in store.
 - 6. All equipment allowed on the slab surface will require:
 - a. Diapering
 - b. Non-marking tires.
 - c. Inspection by the operator twice daily for leaks and any dragging items that may damage the concrete surface.
 - d. Inspect tires for embedded nails, screws or rocks prior to each use.
 - 7. Protect concrete slabs from discoloration, staining, gouges and scratching.

8. All scissors lift battery charging will only be conducted in non-polished, back storage areas, with concrete surface protected from any potential spills.
9. Protect from all paint overspray using only breathable materials.
 - a. Temporarily use breathable material such as EZ-Cover, Ramboard or Econocover for protection in high-traffic locations.
 - b. Do not store building materials, soil, chemicals or equipment on permanently exposed slab surface.
 - c. Do not allow pipe cutting machine on slab.
- B. The slab protection plan shall be distributed and reviewed by all sub-contractors and vendors working around or near the interior concrete slabs. A poster size copy of this plan shall be clearly visible at a minimum of four locations inside the store after concrete slab installation and prior to store turnover.

END OF SECTION 033010

CONCRETE FINISHING SUBCONTRACTOR'S QUALIFICATION STATEMENT

Project Location: _____

Date: _____

Project Number: _____

Store Number: _____

By signing below Contractor confirms that the concrete finishing subcontractor meets the following requirements:

The concrete finishing subcontractor lead finisher and minimum two additional members of the finishing crew have Flatwork Finisher & Technician (or Tradesman) Certifications as granted by the American Concrete Institute. These certified finishers will be present for all sales area slab placements, with proof of certification available at all times.

The concrete floor finishing subcontractor's crews have experience in finishing exposed retail concrete floor slabs of similar size and scope in at least 5 previous projects as listed below (provide area completely installed and finished by finishing subcontractor).

Project Description (Owner, Store #)	Location (City, ST)	Area (s.f.)
1. _____	_____	_____
2. _____	_____	_____
3. _____	_____	_____
4. _____	_____	_____
5. _____	_____	_____

Concrete Finishing Subcontractor Company Name and Address:

Signature of Responsible Officer: _____

Typed Name and Title of Officer: _____

Telephone Number: _____

Contractor Company Name and Address:

Attach certificates from the ACI Flatwork Finisher & Technician (or Tradesman) Certifications for lead finisher and two additional members of the finishing crew.

Submit one copy of this Statement to the Architect for review.

CONCRETE SLAB ON GROUND MIX DESIGN SUBMITTAL FORM**STORE INFORMATION**

STORE #	_____
ADDRESS	_____
CITY, ST	_____
CONTRACTORS	_____

GENERAL	_____
CONCRETE	_____

A. CONCRETE INFORMATION

Supplier Mix Design #	_____
Design Strength (f'c)	_____ Psi
Water / Cementitious Ratio	_____
Total Air Content	_____ %
Mix Developed From:	
☐ Trial Mix Test Data (<i>attach test data</i>)	
☐ Field Experience	
Density	
Wet	_____ pcf
Dry	_____ Pcf
Slump	
" _____ (± 1") WITHOUT WR Admixture	
" _____ (± 1") WITH WR Admixture	

B. ADMIXTURE INFORMATION

	ASTM Designation	Product (Manufacturer/Brand)	Dosage (ounces)	
			oz / cy	oz / cwt
Water Reducing				
Accelerating				
Retarding				

C. MIX DESIGN

Mix Proportions (per cubic yard)

	Identification (Type, size, source, etc.)	Weight (pounds)	Density (SSD)	Volume (cu ft)	% Aggregate Absorption
Cement					
Fly Ash					
Slag					
Coarse Agg. #1					
#2					
#3					
Fine Agg. #1					
#2					
Water					
Air Content					
	TOTALS				

Coarse & Fine Aggregate Gradation Information

Sieve Size	% Passing Each Sieve (All Sieve Sizes must be entered)					Combined % Passing	Combined % Retained	
	Coarse Agg. # 1	Coarse Agg. # 2	Coarse Agg. # 3	Fine Agg. # 1	Fine Agg. # 2		Cumulative	Individual
1-1/2"								
3/4"								
3/8"								
1/2"								
3/16"								
# 4								
# 10								
# 20								
# 40								
# 60								
# 100								
# 200								
% of Vol								

Aggregate Ratios

Coarseness Factor =	$\frac{\text{Combined \% cumulative retained } 3/8" \text{ sieve}}{\text{Combined \% cumulative retained } \#8 \text{ sieve}}$		=
Workability Factor =	Combined % passing #8 sieve		=
Adj-Workability Factor =	$WF + [(Cementitious \text{ Material} - 564) \div 37.6]$		=
Allowable Adj-WF=	$Adj-WF = [(11.25 - .15 \text{ CF}) + 36] \pm 3$		= Low High

D. ATTACHMENTS: Include the following with this Mix Design Report.

- ☐ Portland Cement mill test reports
- ☐ Separate aggregate gradation reports including all required sieve sizes
 - All gradation sieve report tests dated within 60 days of this report
 - Report for each coarse and fine aggregate material in mix
- ☐ Statement of potential pop-outs or other disruptions from aggregate reactivity or other mechanisms.
- ☐ Product data for the following admixtures:
Water reducing, set retarding, set accelerating, etc.
- ☐ Measured water-soluble chloride ion content in concrete (percent by weight of cement).
- ☐ Concrete compressive strength data used for standard deviation calculations

E. CONCRETE SUPPLIER INFORMATION

Company Name _____	Tel. # (____) _____
Address _____	
City, ST Zip _____	
Technical Contact _____	Cell # (____) _____
	e-mail _____
Sales Contact _____	Cell # (____) _____
<u>PRIMARY PLANT</u>	<u>SECONDARY PLANT</u>
Plant Location: _____	_____
Miles from Site: _____	_____
Travel Time to Site: _____	_____
NRMCA Certified: ☐ YES ☐ NO	☐ YES ☐ NO
State DOT Certified: ☐ YES ☐ NO	☐ YES ☐ NO
Batch Mixing Type: ☐ DRY ☐ CENTRAL MIX	☐ DRY ☐ CENTRAL MIX

SECTION 033543 – POLISHED CONCRETE FINISH AND JOINT FILLER**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes surface treatments for concrete floors and slabs as follows:

1. Polished Concrete Flooring System.
2. Reactive Surface Densifier and Protective Surface Treatment.
3. Filler Installation for All Interior Slabs on Ground Joints.
4. Liquid Glass Top Coat.

- B. Related Requirements:

1. Section 013300 – Submittal Procedures.
2. Section 014000 – Quality Requirements.
3. Section 016000 – Product Requirements.
4. Section 017000 – Closeout Procedures.
5. Section 033010 – Cast-in-Place Interior Concrete Slabs on Ground.
6. Section 079200 – Joint Sealants.

1.3 REFERENCE STANDARDS

- A. American Society for Testing and Materials (ASTM)

1. ASTM D 4039 - Standard Test Method for Reflection Haze of High-Gloss Surfaces
2. ASTM D 523 - Standard Test Method for Specular Gloss
3. ASTM D 5767 - Standard Test Methods for Instrumental Measurement of Distinctness-of-Image Gloss of Coating Surfaces
4. ASTM F 2170 - Standard Test Method for Relative Humidity in Concrete Floors
5. ASTM F 1869 - Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride

1.4 ADMINISTRATIVE REQUIREMENTS

- A. Coordinate with Section 033010 for scheduling and coordinating work of this Section, including troweling of concrete finish and curing of concrete slab. Do not permit use of curing compounds on slabs to receive work of this Section.
- B. Coordinate where possible to finish work of this section prior to construction of vertical obstructions due to difficulty of bringing floor finish flush to walls and obstructed areas.
- C. Pre-installation Meeting: See Section 033010 for requirements.

1.5 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: Manufacturer's published data on each finishing product, including information on compatibility of different products and limitations.
 - a. Manufacturer and model of equipment to be used, including all types of grinding heads, saws, and dust extraction systems.
 - b. Diamond tooling manufacturer's grinding guide chart, detailing the appropriate diamond selection for various concrete surface hardness.
 - c. Reactive surface densification system including protective surface treatment.
 - d. Reactive grind and fill material for filling pits and voids.
 - e. Floor joint filler.
 - 2. Shop Drawings"
 - a. For layout and installation of floor joint filler.
 - 3. Samples:
 - a. For joint filler material and color.
 - 4. Furnish one complete submittal document that includes datasheet information for all items requested in checklist found at the end of this section, for review.
 - 5. Furnish written dust/water/slurry control and disposal plan to ensure compliance with all local, state and federal guidelines and regulations, for review. A copy of this plan shall be maintained at the jobsite with compliance of execution verified by the General Contractor.

1.6 INFORMATIONAL SUBMITTALS

- A. Material certificates.
- B. Field-quality control reports.

1.7 QUALITY ASSURANCE

- A. Applicator Qualifications: Company specializing in performing the work of this Section and approved by manufacturer.
 - 1. Complete Polishing Contractor's Qualification Statement (included at the end of this Section).
 - 2. Experienced artisan specializing in application of ground and polished concrete flooring systems comparable in type, scope, and quality as specified by this Section.
 - 3. Able to document minimum five years' experience, and minimum five successful commercial quality projects comparable in scope and size as specified project within the last year.
 - 4. Provide letter(s) of certification from manufacturer's representative as qualified to perform work of this Section, prior to pre-installation meeting, and accepted by Architect.

1.8 MOCKUPS

- A. In a location designated by the Owner, provide minimum of 2 each, 200 square foot mock-up slabs for polished concrete surface and joint filling.
- B. Concrete in mock-up area to have a minimum of 14 days cure time.
- C. Mock-up installation shall be completed with the same equipment and crew that will perform all polish work. Provide all equipment necessary to do a complete system including edging equipment.
- D. Prior to starting mock-up installation, verify immediate surface hardness with a Moh's testing kit.
- E. Mock-up will also be located next to a wall and contractor shall demonstrate a seamless grind and polish transitioning from the floor to the wall. Mock-up area shall include complete filling of the control joint as part of the polish processing over one or more floor joints.
- F. Provide mock-up in area with similar lighting type and wattage as store will use when in operation.
- G. Where necessary for comparison of finishes, or to obtain approval, provide up to two additional mock-ups as directed.
- H. Contractor shall use pre-conditioned (previously used) diamonds, having a complete range of tooling available for a variety of concrete surface harnesses.
- I. Maintain approved mock-up as standard for work of this Section.

1.9 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials in manufacturer's sealed packaging, including application instructions.
- B. Storage:
 - 1. Store materials in clean, dry area indoors in accordance with manufacturer's instructions, and MSDS sheets.
 - 2. Keep materials from freezing.
- C. Handling: Protect materials during handling and application to prevent contamination or damage.

1.10 FIELD CONDITIONS

- A. Maintain a lighting level of minimum 30 foot-candles at 8 feet above the floor surface over each 20 foot square area of floor being finished.
- B. All necessary electrical supply to be provided within 100 feet of the work area for all polishing equipment, including three-phase 480-volt service as needed. Electrical supplied by generator may be necessary.
- C. Maintain ambient temperature in conformance with manufacturer's instructions.
- D. Roof construction, skylight installation, overhead painting, and roof drainage system shall be complete and weather tight prior to placement of concrete and maintained.

1.11 WARRANTY

- A. See Section 017000 - Closeout Procedures, for additional warranty requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with project requirements, provide products and equipment by the following:

1. 3M (888) 364-3577
2. American Decorative Concrete (800) 592-9320
3. Convergent Concrete Technologies (866) 375-2280
4. DRC Industries (866) 365-1001
5. Eagle Solutions (800) 633-0519
6. Gorilla Concrete Tools (440) 236-5112
7. Hi-Tech Systems (800) 454-5530
8. HTC (877) 482-8700
9. Husqvarna (800) 288-5040
10. Joe Due (608) 847-6627
11. McTech Group (866) 913-8363
12. Metzger/McGuire; (800) 223-6680
13. Minuteman (800) 323-9420
14. Norton (800) 331-3604.
15. Onyx (800) 858-3533
16. Pioneer Eclipse (800) 367-3550
17. Prosoco (800) 255-4255
18. Pullman-Ermator (800) 232-2635
19. Ram Board (818) 848-0400
20. Rubbermaid Commercial Products (800) 347-9800
21. SASE (800) 522-2606
22. Scanmaskin USA (800) 400-9473
23. Skudo (888) 785-3611
24. U.S. Saws (866) 987-7297
25. VersaFlex (800) 321-0906

2.2 CONCRETE FLOOR FINISH APPLICATIONS

- A. Installation of polished concrete floor system for new interior concrete floors by grinding and polishing with various size grit metal-, resin- and/or ceramic-bonded diamonds and application of concrete densifier and surface protectant.
- B. Reactive densifier and surface protectant shall be products from the same manufacturer.

2.3 REACTIVE DENSIFIER

- A. Subject to compliance with project requirements, provide one of the following products:

1. 3D HS Densifier; American Decorative Concrete
2. Blended Densifier; Prosoco
3. D2; SASE
4. Pentra-Sil (IH); Convergent Concrete Technologies

2.4 PROTECTIVE SURFACE TREATMENT

A. Subject to compliance with project requirements, provide one of the following products:

1. Consolideck Concrete Protector SB; Prosoco
2. Pentra-Pel (SI); Convergent Concrete Technologies
3. SPR3; SASE
4. SR2; American Decorative Concrete

2.5 PIT GROUT COAT MATERIALS

A. Subject to compliance with project requirements, provide one of the following products:

1. Pentra (PCF); Convergent Concrete Technologies
2. Quick Mender XO, VersaFlex
3. Rapid ReFloor Pit Grout, Metzger/McGuire
4. Spall TX3, Hi-Tech Systems

2.6 INTERIOR FLOOR JOINT FILLER

A. Rapid setting polyurea polymer, 100% solids, multi-component, semi-rigid.

B. Color: To be selected by Architect from manufacturer's standard range.

C. Applications: Use for:

1. All interior joints in exposed concrete slabs, including polished surfaced.

D. Products:

1. PE85; Hi-Tech
2. SL/85; VersaFlex
3. Edge-Pro 80; Metzger/McGuire

2.7 JOINT FILLER PREPARATION SAW (DUSTLESS, DRY-CUT)

A. Subject to compliance with project requirements, provide one of the following products:

1. Dust Buggy; U.S. Saws
2. GCT Tank 10" Dustless Concrete Joint Saw; Gorilla Concrete Tools
3. Humpback Cutter; Joe Due
4. Perfect Trac; Metzger/McGuire

2.8 POLISHING EQUIPMENT

A. Field Grinding and Polishing Equipment:

1. Variable speed, containing 3-heads minimum (planetary, counter rotating) that contour to the floor surface.
2. Must produce a minimum of 600 pounds of head pressure.
3. Provide a minimum of four machines, with a minimum head diameter of 30 inches, fully operational for the duration of work.
4. If dry grinding, honing, or polishing, use dust extraction equipment with flow rate suitable for dust generated, with squeegee attachments.

B. Dust Extraction System:

1. Heavy-duty industrial HEPA filtration vacuum system, suitable for extracting and containing large quantities of fine concrete dust (minimum 350CFM air flow) in conjunction with manufacturer recommended pre-separator. Pre-separator is mandatory for all metal bond cuts when grinding dry.
 2. Provide one of the following:
 - a. 86D; HTC
 - b. T8600; Pullman-Ermator
 - c. Bull 1250; SASE
 - d. Approved Equal
- C. Diamond Grinding Segments:
1. Provide diamond grinding segments from one of the following:
 - a. HTC
 - b. Husqvarna
 - c. SASE
 - d. Approved Equal
 2. Allowable Grinding/Polishing Diamonds (or manufacturer's equivalent designations).
 - a. Metal Bonded Diamond Grit Sizes: 25, 40, 80, and 150.
 - b. Hybrid Style Diamond: 30, 50, 100, 200.
 - c. Resin Bonded, Phenolic Diamond Grit Sizes: 100, 200, 400, 800, 1500, and 3000 as needed to achieve specified gloss.
 3. Grinding/Polishing Pads for Edges.
 - a. Exposed floor surface, requiring handwork, where larger machines cannot reach.
 - b. Provide same progression of diamond grit sizes used for larger machine grinding/polishing.
 4. Progression of all polishing steps shall include sequential grit sizes that are not greater than twice the previous grit used.
- D. Burnishing Equipment: High speed walk-behind or ride-on machines.
1. Propane type.
 2. 22 pound head weight (40 pound maximum).
 3. Dust skirt assembly.
 4. The weight and speed of the burnisher shall generate adequate heat and friction at the concrete floor surface to transform the protective surface treatment application to the specified gloss level.
 5. Provide burnisher by one of the following:
 - a. Eagle Solutions
 - b. Minuteman
 - c. Onyx
 - d. Pioneer Eclipse
 - e. SASE
- E. Burnishing Pads:

1. Diamond Impregnated Pads (1500 grit and/or 3000 grit):
 - a. Trifecta; SASE
 - b. Scotch-Brite Plus; 3M
 - c. Twister; HTC
 2. Natural Hair Pads:
 - a. 3M; Niagra Super Hogs Hair Floor Pad 3700N
 - b. Norton; Ultra Grizzly Bear
 - c. Prosoco; Consolideck Heat Pad
 - d. SureShine Heat; SASE
- F. Applicator Pad:
1. 24" High-Performance Damp Microfiber Mop (Blue); Rubbermaid
- G. Breathable protective floor cover materials for protection of high traffic areas.
1. Econo- or Primo- Cover; DRC Industries
 2. EZ-Cover; McTech Group
 3. Ram Board; Ram Board
 4. Skudo; MT Tack-Mat

PART 3 - EXECUTION

3.1 FLOOR JOINT FILLER INSTALLATION

- A. General:
1. Installation shall be by Concrete Polishing Contractor who is approved in writing by the manufacturer's corporate office for this project.
 2. Do not use rigid filler under flooring, carpet materials or coatings.
 3. Commence placing floor joint filler no sooner than 14 days after first placement of concrete.
 4. If joint is wet or damp, vacuum completely and allow to dry prior to filling.
 5. Install test sample of the polyurea joint filler to determine acceptance of dry color by the Architect.
 6. If only resin polishing, consider the use of stain preventative film to eliminate ghosting or staining from the overage.
- B. Cleaning:
1. Immediately prior to filling, clean and prepare joint bottom and sidewalls to full depth of saw cut, in accordance with manufacturer's recommendation, using specified preparation saw with vacuum attachments.
 2. Remove any residual edge fracturing or spalling, created from initial concrete sawing, up to 1/4" in total width. Minimum saw blade width for initial joint filler preparation cut will be 3/16". Joint preparation width must be continuous to nearest joint intersection or vertical abutment.
 3. Remove dirt, debris, saw laitance, and other foreign material from the joint cavity, and the top portion of the slab. Clean inner joint walls mechanically using specified dustless dry cut saw, to the full depth of saw cuts or two-inch minimum depth in control joints that does not have a sawed bottom shelf ledge.

4. Remove all form release agent, curing compound, or other components.

C. Installation:

1. Once climatized, mix and install filler in accordance with manufacturer's recommendations. Prepare only that which can be installed before material begins to set.
2. Fill with polyurea joint filler material, preferably in two lifts from the bottom up, taking care not to entrap large air bubbles per manufacturer's recommendation. Slightly overfill and shave flush to the surface prior to polishing process.
3. Ensure that after grinding, the joint is cut smooth and flush with the finish floor surface, without concave or intermittent, darkened profile.
4. Do not allow filler to extend over joint edges in finished condition.
 - a. Do not use joint back-up material (i.e. backer rod, sand, etc.) except below bottom of saw cut in control joints. Provide a minimum joint filler depth of 2 inches for backer rod material.
 - b. Add extra filler prior to filler set if needed to prevent depressed areas. If concave filler is already set, abrade with wire wheel or similar tool to minimum depth of 1/4" below surface prior to refilling.
 - c. Razor off crowned filler flush with floor surface after filler has sufficiently set.

D. Joint refill repair and warranty requirements:

1. One week prior to store turnover and one year following store Owner turnover, the contractor shall return and walk the floor with the owner's representative to evaluate, remove and refill all joints where joint filler:
 - a. Has separated from sidewall or is splitting.
 - b. Surface profile is concave, chattered or has noticeable voids.
2. Joint refill and repair will be required by the contractor, as stated above, regardless of the conditions of environment or time after concrete placement when the joint filler is originally installed.

3.2 SURFACE EXAMINATION

A. Polish Contractor is to verify that floor surfaces are acceptable to receive the work of this section.

B. Verify the Following for New Concrete Floors Finish:

1. Floor Surface Profile: Flatness and Levelness as specified in Section 033010.
2. Concrete Curing: Minimum 5 days; fully cured.
3. All exposed concrete slab surfaces to be polished, shall have a minimum Mohs' hardness of 5 when measured using the following test method, or approved equivalent.
 - a. "Deluxe Hardness Picks" as manufactured by Mineralab, LLC, (800) 749-3766.
4. Concrete Adjacent to walls, columns and floor penetrations and at construction joints are troweled flat, with uniform color and density similar to surrounding concrete.

- C. If any surface treatments used are moisture sensitive, it will be the responsibility of the polishing contractor to provide any testing required to ensure compliance with the manufacturer's requirements for concrete moisture levels at time of installation.
- D. Notify Architect of conditions that would adversely affect aesthetics or serviceable use of the finished product. Do not begin surface preparation or installation until unacceptable conditions are either corrected or accepted by the Owner's representative.

3.3 SURFACE PREPARATION

- A. Clean concrete surfaces of dirt and other particulates and remove oil, stains, grease, adhesives, water repellents, and other substances that may be detrimental to work of this Section.
 - 1. All dust from saw cutting control joints shall be vacuumed and contained with no residual dust left on the surface.
- B. Apply materials in accordance with manufacturer's instructions.
- C. Protection: Protect surrounding areas and adjacent surfaces from the following:
 - 1. Minimal accumulation of dust from grinding and polishing.
 - 2. Prior to wet grinding, seal and protect all walls and vertical abutments a minimum four feet above finished floor from water and slurry.
 - 3. Contact with overspray of concrete densifier and protective surface treatment.
- D. Verify that surface is free of previous coatings, sealers, curing compounds, water repellents, laitance, efflorescence, fats, oils, grease, wax, soluble salts, and residues from cleaning agents, and other impediments to adhesion.
- E. Protect adjacent non-coated areas from drips, overflow, and overspray; immediately remove excess material.

3.4 CONCRETE POLISHING (F-1)

- A. Execute using materials, equipment, and procedures specified by manufacturer, using manufacturer approved installer.
- B. Complete all joint filling work, allowing to cure within manufacturer's minimum time requirements, prior to commencement of any grinding or polishing. Polyurea joint filling to be concurrent to grinding and polishing process.
- C. Aggregate Exposure: Fine Aggregate - mottled salt-and-pepper aggregate exposure.
 - 1. Class B, Level 4, as defined by the Concrete Polishing Council (CPC).
 - 2. Where low areas still have a remaining cream layer, contractor shall remove the layer of paste, exposing the fine aggregates. Aggregate Exposure shall be consistent throughout Work.
 - 3. Finished floor aesthetics should uniformly match jobsite mock up, previously approved by the Owner's Representative.
- D. Begin wet or dry grinding with initial grit necessary to produce a uniform refinement, appearance, and a surface free of pits without exposure of coarse aggregate (typically 150 metals or 50 to 100 grit resin hybrids). In the event the floor is exhibiting imperfections as a

result of non-uniform finishing more aggressive grit diamond tooling may be required. The use of metal bonded tooling below 150 will require approval from the Owners polished concrete consultant and at no added cost to the Owner.

1. Include all work necessary for proper initial grinding and removals.
 2. Overlap adjacent polishing passes by a minimum of 40 percent.
 3. Between grinding passes, and applications of densifiers, thoroughly scrub and rinse slab surface with clean water and vacuum up with auto scrubber.
 4. Dispose of waste solution in compliance with all local, state and federal guidelines and regulations for disposal.
 5. Perform each pass perpendicular to the previous.
- E. If during the initial grinding process, crazing, pits or divots are observed, a specified pit grout coat material shall be used in accordance with the manufacturer's recommendations to fill and mask these voids.
- F. Progressively hone and polish slab with 100-, 200-, 400- and polish with 800-, and 1,500- grit resin-bonded, phenolic diamonds.
1. Progressively diamond polish edges along and around all abutments. Use same progression of polishing diamonds as noted above.
 2. Sequential progression shall be required and shall be limited to no more than double the grit value of the previous diamonds used.
 3. Between polishing passes, thoroughly scrub and rinse slab surface with clean water and vacuum up with auto scrubber.
 4. Perform each pass perpendicular to the previous.
- G. Reactive Sealer-Densifier:
1. Ensure slab surface is clean and dry at time of application of sealer-densifier.
 2. A minimum of two coats of sealer-densifier shall be applied with the first application being installed after the initial grind.
- H. Apply reactive surface sealer-densifier in accordance with manufacturer's instructions, ensuring that application is applied to point of rejection. Continue progressively polishing with appropriate resin tooling, as stated above.
- I. Burnish slab with diamond impregnated pad to remove swirl marks from resin-bonded, phenolic diamonds.
- J. Concrete Sealer-Densifier for F-1 and F-3:
1. Apply two coats in accordance with manufacturer's published instructions.
 - a. Apply using specified sprayer and tip. Use new sprayer tip for each night's work.
 - b. Use new applicator pad for each coat, pre-wetted with sealer, to pull material out to create a thin film prior to drying.
 2. Application of sealer-densifier in excess of manufacturer's published instructions may result in additional cost to Contractor without an increase in Contract time. Remove product completely from areas of over application, as evidenced by surface streaking, and replace with unused sealer.
- K. High Speed Burnish for F-1 and F-3:

1. Burnish surface per manufacturer's instructions and when sealer application has had the proper dry time.
 2. Burnish at a slow movement pace using high speed machine with 1500 and/or 3000 grit diamond impregnated or natural hair burnishing pads.
 3. Burnish with several passes. Make each progressive pass at 90 degrees from previous pass.
- L. Provide final burnishing 72 hours prior to Owner turnover.
- M. Polish concrete enhancer:
1. Basis of Design Manufacturers:
 - a. AtmosphereX, LLC; 3275 Corporate View Vista, CA 92081; phone: 760-600-5188, mail: info@atmospherex.com, web: www.atmospherex.com.
 - b. Ameripolish, Inc.; 120 Commercial Ave., Lowell, AR 72745; phone: 479-725-0033; web: www.ameripolish.com.
- N. Provide a surface free from scratches, swirl marks, and has a consistent surface sheen prior to the application of guard products.

3.5 SURFACE AESTHETIC MEASUREMENTS (AVERAGES)

- A. Gloss - ASTM D-523: Specified Overall Gloss Value (SOGV) of not less than 60 and a Minimum Local Gloss Value (MLGV) of 40 as measured at a 60 degree angle. An average minimum gloss value of 50 shall be validated as achieved prior application any surface protective treatment, in the initial mock up.
- B. DOI (Distinction of Image) – ASTM D-5767: Specified Overall DOI Value (SODV) of not less than 60 and a Minimum Local DOI Value (MLDV) of 40 as at 20 degree angle.
- C. Haze – ASTM D4039: overall average shall be less than 12 at 20 degree angle.
- D. Minimum Local Values are the measurement averages within a column bay (area bound by column lines, or walls).

3.6 FIELD QUALITY CONTROL

- A. See Section 014000 - Quality Requirements, for additional requirements.
- B. Inspect completed polished concrete floor system with Owner's representative, Contractor and Installer.
- C. Testing will be validated by the Owner's Representative. A test report of all measurements and a plan and table of the measurements taken will be prepared by for submission.
- D. Final Surface Aesthetic Measurements:
1. Will be validated two days prior to Owner turnover. Contractor shall notify Owner when floors are ready for testing.
 2. Contractor shall furnish and assist in operating proper meter to measure Specular Gloss, DOI and Haze, in accordance with specified ASTM procedures. The Owner's Representative shall observe proper random measurements.
 3. Measurement and Recording:

- a. Measure surface aesthetic values at three evenly but randomly distributed locations within each column bay within the floor areas to receive a polished finish.
 - b. Record all measurements on the slab plan from the contract documents that designate floor joint location.
4. If overall or local average measured values fail to meet the specified minimums:
 - a. Gloss or Haze values, additional refinement shall be completed until the specified requirements are achieved.
 - b. DOI values, review procedures with Owner and Consultant to correct unacceptable areas of the completed polished concrete floor system.
5. Contractor shall furnish final report of all aesthetic measurements, including sample pictures of actual meter readings and long view pictures of the overall finished floor.

3.7 PROTECTION

- A. Continue to protect completed polished concrete floor system from damage until Substantial Completion in accordance with "Slab Protection Plan" required in Section 033010. This work is the responsibility of the General Contractor.

END OF SECTION 033543

CONCRETE POLISHING CONTRACTOR'S QUALIFICATION STATEMENT**Project Name:** _____ **Date:** _____**Project Location:** _____

By signing below as approved, the General Contractor hereby confirms that the qualifications of the Concrete Polishing subcontractor conform to the qualifications as follows:

The concrete polishing subcontractor has experience in polishing interior floors (including joint filling work) of similar size and scope in at least 5 previous projects within the last year. Projects name and location are identified as follows:

Project Description (Owner, Store #)	Location (City, ST)	Area (s.f.)
1.		
2.		
3.		
4.		
5.		

Concrete Polishing Subcontractor Company Name and Address:**Company Name:** _____**Address, City ST** _____**Signature of Responsible Officer:** _____**Typed Name and Title of Officer:** _____**Date:** _____**General Contractor Company Name and Address:****Company Name:** _____**Address, City ST** _____**Signature of Responsible Officer:** _____**Typed Name and Title of Officer:** _____**Date:** _____

Submit one copy of this Statement to the Architect of Record for review.

CONCRETE POLISHING SUBMITTAL CHECKLIST**1) Concrete Polish Contractor's Qualification Statement****2) Manufacturer's Letter of Contractor Certification for the following:**

- ☐ Joint filler
- ☐ Reactive surface densifier / protective surface treatments
- ☐ Polishing equipment and diamond tooling system(s)

3) Product data and MSDS for the following:

- ☐ Polyurea joint filler
- ☐ Reactive surface densifier
- ☐ Protective surface treatment
- ☐ Pit grout coat material product for filling pits and voids

4) Manufacturer and model of all equipment used to complete work, including:

- ☐ Joint filler prep saw
- ☐ Joint filler pumping equipment
- ☐ Polishing machines
- ☐ Dust extraction system
- ☐ Diamond tooling
- ☐ Diamond tooling manufacturer's grinding guide chart, detailing the appropriate diamond selection for various concrete surface hardness
- ☐ Auto scrubber
- ☐ High speed burnishing equipment
- ☐ Burnishing pads
- ☐ Sprayers and sprayer tips

SECTION 042000 - UNIT MASONRY

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

- 1. Concrete masonry units.
- 2. Decorative Concrete Masonry Units
- 3. Mortar and grout.
- 4. Steel reinforcing bars.
- 5. Masonry-joint reinforcement.

B. Products Installed but not Furnished under This Section:

- 1. Cast-stone trim in unit masonry.
- 2. Steel lintels in unit masonry.
- 3. Cavity wall insulation.

C. Related Requirements:

- 1. Section 042516 "Thin Brick Panel System" for thin brick units on veneer panel support system.
- 2. Section 042600 "Adhered Masonry Veneer" for adhered stone veneer system.
- 3. Section 072100 "Thermal Insulation" for cavity wall insulation.
- 4. Section 076200 "Sheet Metal Flashing and Trim" for exposed sheet metal flashing and for furnishing manufactured reglets installed in masonry joints.
- 5. Section 079200 "Joint Sealants" for sealants and backer rods.

1.3 DEFINITIONS

- A. CMU(s): Concrete masonry unit(s).
- B. Reinforced Masonry: Masonry containing reinforcing steel in grouted cells.

1.4 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.5 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data:

- a. For each type of concrete masonry unit.
 - b. For each type of mortar and grout.
 - c. For each type of masonry accessories.
2. Samples:
 - a. For each type, texture and color of exposed concrete masonry unit.
 - b. For each type of colored mortar.
 - c. For each type of masonry accessories.
3. Shop Drawings:
 - a. Concrete Masonry Units: Show sizes, profiles, coursing, and locations of special shapes.
 - b. Reinforcing Steel: Detail bending, lap lengths, and placement of unit masonry reinforcing bars. Comply with ACI 315. Show elevations of reinforced walls.
 - c. Masonry Accessories: Show locations and installation details. Coordinate with adjacent and adjoining construction.
4. Mix Designs: For each type of mortar and grout. Include description of type and proportions of ingredients.
 - a. Include test reports for mortar mixes required to comply with property specification. Test according to ASTM C 109/C 109M for compressive strength, ASTM C 1506 for water retention, and ASTM C 91/C 91M for air content.
 - b. Include test reports, according to ASTM C 1019, for grout mixes required to comply with compressive strength requirement.
5. Material Certificates: For each type and size of the following:
 - a. Masonry units.
 - 1) For masonry units used in structural masonry, include data and calculations establishing average net-area compressive strength of units.
 - b. Cementitious materials. Include name of manufacturer, brand name, and type.
 - c. Mortar admixtures.
 - d. Preblended, dry mortar mixes. Include description of type and proportions of ingredients.
 - e. Grout mixes. Include description of type and proportions of ingredients.
 - f. Reinforcing bars.
 - g. Joint reinforcement.
6. Statement of Compressive Strength of Masonry: For each combination of masonry unit type and mortar type, provide statement of average net-area compressive strength of masonry units, mortar type, and resulting net-area compressive strength of masonry determined according to TMS 602/ACI 530.1/ASCE 6.
7. Cold-Weather and Hot-Weather Procedures: Detailed description of methods, materials, and equipment to be used to comply with requirements.

1.6 QUALITY ASSURANCE

- A. Testing Agency Qualifications: Qualified according to ASTM C1093 for testing indicated.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store masonry units on elevated platforms in a dry location. If units are not stored in an enclosed location, cover tops and sides of stacks with waterproof sheeting, securely tied. If units become wet, do not install until they are dry.
- B. Store cementitious materials on elevated platforms, under cover, and in a dry location. Do not use cementitious materials that have become damp.
- C. Store aggregates where grading and other required characteristics can be maintained and contamination avoided.
- D. Deliver preblended, dry mortar mix in moisture-resistant containers. Store preblended, dry mortar mix in delivery containers on elevated platforms in a dry location or in covered weatherproof dispensing silos.
- E. Store masonry accessories, including metal items, to prevent corrosion and accumulation of dirt and oil.

1.8 FIELD CONDITIONS

- A. Protection of Masonry: During construction, cover tops of walls, projections, and sills with waterproof sheeting at end of each day's work. Cover partially completed masonry when construction is not in progress.
 - 1. Extend cover a minimum of 24 inches (600 mm) down both sides of walls, and hold cover securely in place.
 - 2. Where one wythe of multiwythe masonry walls is completed in advance of other wythes, secure cover a minimum of 24 inches (600 mm) down face next to unconstructed wythe, and hold cover in place.
- B. Do not apply uniform floor or roof loads for at least 12 hours and concentrated loads for at least three days after building masonry walls or columns.
- C. Stain Prevention: Prevent grout, mortar, and soil from staining the face of masonry to be left exposed or painted. Immediately remove grout, mortar, and soil that come in contact with such masonry.
 - 1. Protect base of walls from rain-splashed mud and from mortar splatter by spreading coverings on ground and over wall surface.
 - 2. Protect sills, ledges, and projections from mortar droppings.
 - 3. Protect surfaces of window and door frames, as well as similar products with painted and integral finishes, from mortar droppings.
 - 4. Turn scaffold boards near the wall on edge at the end of each day to prevent rain from splashing mortar and dirt onto completed masonry.
- D. Cold-Weather Requirements: Do not use frozen materials or materials mixed or coated with ice or frost. Do not build on frozen substrates. Remove and replace unit masonry damaged by frost or by freezing conditions. Comply with cold-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.
 - 1. Cold-Weather Cleaning: Use liquid cleaning methods only when air temperature is 40 deg F (4 deg C) and higher and will remain so until masonry has dried, but not less than seven days after completing cleaning.

- E. Hot-Weather Requirements: Comply with hot-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Source Limitations for Masonry Units: Obtain exposed masonry units of a uniform texture and color, or a uniform blend within the ranges accepted for these characteristics, from single source from single manufacturer for each product required.
- B. Source Limitations for Mortar Materials: Obtain mortar ingredients of a uniform quality, including color for exposed masonry, from single manufacturer for each cementitious component and from single source or producer for each aggregate.

2.2 PERFORMANCE REQUIREMENTS

- A. Provide structural unit masonry that develops indicated net-area compressive strengths at 28 days.
 - 1. Determine net-area compressive strength of masonry from average net-area compressive strengths of masonry units and mortar types (unit-strength method) according to TMS 602/ACI 530.1/ASCE 6.

2.3 UNIT MASONRY, GENERAL

- A. Masonry Standard: Comply with TMS 602/ACI 530.1/ASCE 6, except as modified by requirements in the Contract Documents.
- B. Defective Units: Referenced masonry unit standards may allow a certain percentage of units to contain chips, cracks, or other defects exceeding limits stated. Do not use units where such defects are exposed in the completed Work.
- C. Fire-Resistance Ratings: Comply with requirements for fire-resistance-rated assembly designs indicated.
 - 1. Where fire-resistance-rated construction is indicated, units shall be listed and labeled by a qualified testing agency acceptable to authorities having jurisdiction.

2.4 CONCRETE MASONRY UNITS

- A. Shapes: Provide shapes indicated and as follows, with exposed surfaces matching exposed faces of adjacent units unless otherwise indicated.
 - 1. Provide special shapes for lintels, corners, jambs, sashes, movement joints, headers, bonding, and other special conditions.
 - 2. Provide bullnose units as indicated and at all external corners.
- B. Integral Water Repellent: Provide units made with integral water repellent admixture for exposed exterior units.
 - 1. Integral Water Repellent: Liquid polymeric, integral water-repellent admixture that does not reduce flexural bond strength. Units made with integral water repellent, when tested according to ASTM E514/E514M as a wall assembly made with mortar containing

integral water-repellent manufacturer's mortar additive, with test period extended to 24 hours, shall show no visible water or leaks on the back of test specimen.

C. CMUs: ASTM C 90.

1. Contacts: See LIDL National Account List.
2. Manufacturers:
 - a. Oldcastle, Inc. (Basis of Design)
 - b. Nitterhouse Masonry Products, LLC
 - c. Westbrook Block Co. Inc.
 - d. York Building Products
 - e. Or Approved Alternate
3. Unit Compressive Strength: Provide units with minimum average net-area compressive strength of 2800 psi (19.3 MPa).
4. Density Classification: Normal weight.
5. Size (Width): Manufactured to dimensions 3/8 inch (10 mm) less than nominal dimensions.
6. CMU Dimensions: See Drawings.
7. Textures:
 - a. Exposed Units: [Ground-faced] [Split-faced] finish.
 - b. Concealed Units: Smooth-faced finish.
8. Colors:
 - a. Exposed Units:
 - 1) CMU Color 'A':
 - a) Oldcastle, Inc. (Pearl Mist (SF) – Betco Supreme Series)
 - b) Nitterhouse Masonry Products, LLC (A-13 SF)
 - c) Westbrook Block Co. Inc. (SF-213A)
 - d) York Building Products (Split Face – Glacier)
 - e) Or Approved Alternate Color
 - 2) CMU Color 'B':
 - a) Oldcastle, Inc. (Slate Gray (SF) – Betco Supreme Series)
 - b) Nitterhouse Masonry Products, LLC (NM-183)
 - c) Westbrook Block Co. Inc. (SF-335)
 - d) York Building Products (Split Face – Flint)
 - e) Or Approved Alternate Color
 - b. Concealed Units:
 - 1) Gray (natural)

2.5 MORTAR AND GROUT MATERIALS

- A. Portland Cement: ASTM C 150/C 150M, Type I or II, except Type III may be used for cold-weather construction. Provide natural color or white cement as required to produce mortar color indicated.

1. Alkali content shall not be more than 0.1 percent when tested according to ASTM C114.
 - B. Hydrated Lime: ASTM C 207, Type S.
 - C. Portland Cement-Lime Mix: Packaged blend of portland cement and hydrated lime containing no other ingredients.
 - D. Masonry Cement: ASTM C 91/C 91M.
 - E. Mortar Pigments: Natural and synthetic iron oxides and chromium oxides, compounded for use in mortar mixes and complying with ASTM C 979/C 979M. Use only pigments with a record of satisfactory performance in masonry mortar.
 - F. Colored Cement Products: Packaged blend made from portland cement, ground limestone, proprietary admixtures and pigment, all complying with specified requirements and containing no other ingredients.
 1. Formulate blend as required to produce color indicated or, if not indicated, as selected from manufacturer's standard colors.
 2. Pigments shall not exceed 10 percent of portland cement by weight.
 3. Pigments shall not exceed 5 percent of masonry cement by weight.
 - G. Aggregate for Mortar: ASTM C 144.
 1. For mortar that is exposed to view, use washed aggregate consisting of natural sand or crushed stone.
 2. For joints less than 1/4 inch (6 mm) thick, use aggregate graded with 100 percent passing the No. 16 (1.18-mm) sieve.
 3. White-Mortar Aggregates: Natural white sand or crushed white stone.
 4. Colored-Mortar Aggregates: Natural sand or crushed stone of color necessary to produce required mortar color.
 - H. Aggregate for Grout: ASTM C 404.
 - I. Cold-Weather Admixture: Nonchloride, noncorrosive, accelerating admixture complying with ASTM C 494/C 494M, Type C, and recommended by manufacturer for use in masonry mortar of composition indicated.
 - J. Water-Repellent Admixture: Liquid water-repellent mortar admixture intended for use with CMUs and face bricks exposed to exterior conditions.
 - K. Water: Potable.
- 2.6 REINFORCEMENT
- A. Uncoated-Steel Reinforcing Bars: ASTM A 615/A 615M or ASTM A 996/A 996M, Grade 60 (Grade 420).
 - B. Masonry-Joint Reinforcement, General: ASTM A 951/A 951M.
 1. Interior Walls: Hot-dip galvanized carbon steel.
 2. Exterior Walls: Hot-dip galvanized carbon steel.
 3. Wire Size for Side Rods: 0.148-inch (3.77-mm) (9 gauge) diameter.
 4. Wire Size for Cross Rods: 0.148-inch (3.77-mm) (9 gauge) diameter.

5. Wire Size for Veneer Ties: 0.187-inch (4.76-mm) (7 gauge) diameter.
6. Spacing of Cross Rods, Tabs, and Cross Ties: Not more than 16 inches (407 mm) o.c.
7. Provide in lengths of not less than 10 feet (3 m), with prefabricated corner and tee units.

- C. Masonry-Joint Reinforcement for Single-Wythe Masonry: Ladder or truss type with single pair of side rods.

2.7 EMBEDDED FLASHING MATERIALS

- A. Metal Flashing: Provide metal flashing complying with Section 076200 "Sheet Metal Flashing and Trim" and as follows:

1. Fabricate metal drip edges from stainless steel. Extend at least 3 inches (76 mm) into wall and 1/4 inch (6.5 mm) out from wall, with outer edge bent down 30 degrees and hemmed.
2. Fabricate metal sealant stops from stainless steel. Extend at least 3 inches (76 mm) into wall and out to exterior face of wall. At exterior face of wall, bend metal back on itself for 3/4 inch (19 mm) and down into joint 1/4 inch (6 mm) to form a stop for retaining sealant backer rod.

- B. Flexible Flashing: Use the following unless otherwise indicated:

1. Flexible Stainless Steel Flashing: Type 304 stainless steel with polymer fabric laminated to the bottom side with non-asphalt adhesive. The top face (exposed side) must not be covered with a polymer fabric.
 - a. Products: Subject to compliance with requirements, provide one of the following:
 - 1) York Flashings; Multi-Flash SS.
 - 2) Illinois Products; IPCO Stainless Steel Fabric Flashing
 - 3) Prosoco; R-Guard SS ThruWall
 - 4) STS Coatings; Wall Guardian Stainless Steel TWF
 - 5) TK Products; TK TWF
 - b. Self-Adhesive Sheet with Drip Edge at masonry openings: Elastomeric thermoplastic flashing, 0.025 inch (0.64 mm) thick, with a 0.015-inch- (0.38-mm-) thick coating of rubberized-asphalt adhesive. Where flashing extends to face of masonry, rubberized-asphalt coating is held back approximately 1-1/2 inches (38 mm) from edge.
 - c. Accessories: Provide preformed corners, end dams, other special shapes, and seaming materials produced by flashing manufacturer where flashing occurs at masonry openings. Flashing termination bars: Hohmann & Barnard, Inc, T-2 Aluminum Termination Bar.
2. Accessories: Unless otherwise indicated, use the following:
 - a. Mastic/Sealant:
 - 1) Products: York Flashing UniverSeal US100 or other compatible sealant recommended by flashing manufacturer.
 - b. Outside corner and inside corner material; manufacturer's standard available units using:

- 1) Stainless: 26 gauge stainless steel
 - c. End dam: Product may be folded in line with the flashing material or utilize preformed end dams by manufacturer using:
 - 1) Stainless: 26 gauge stainless steel
 - d. Splice material: Product standard of quality is York304 SS by York. Manufacturer's standard self-adhered metal material; material matching system material or use Multi-Flash Stainless Steel 6" lap piece and polyether sealant as a splice
 - e. Fasteners: Domestic manufactured fastener types and sizes recommended by flashing manufacturer for intended use.
3. Application: Unless otherwise indicated, use the following:
- a. Where flashing is indicated to receive counterflashing, use metal flashing.
 - b. Where flashing is indicated to be turned down at or beyond the wall face, use metal flashing.
 - c. Where flashing is partly exposed and is indicated to terminate at the wall face, use metal flashing or flexible flashing with a metal drip edge.
 - d. Where flashing is fully concealed, use flexible flashing.
- C. Adhesives, Primers, and Seam Tapes for Flashings: Flashing manufacturer's standard products or products recommended by flashing manufacturer for bonding flashing sheets to each other and to substrates.

2.8 MISCELLANEOUS MASONRY ACCESSORIES

- A. Compressible Filler: Premolded filler strips complying with ASTM D 1056, Grade 2A1; compressible up to 35 percent; of width and thickness indicated; formulated from neoprene urethane or PVC.
- B. Preformed Control-Joint Gaskets: Made from styrene-butadiene-rubber compound, complying with ASTM D 2000, Designation M2AA-805 or PVC, complying with ASTM D 2287, Type PVC-65406 and designed to fit standard sash block and to maintain lateral stability in masonry wall; size and configuration as indicated.
- C. Bond-Breaker Strips: Asphalt-saturated felt complying with ASTM D 226/D 226M, Type I (No. 15 asphalt felt).

2.9 MASONRY CLEANERS

- A. Proprietary Acidic Cleaner: Manufacturer's standard-strength cleaner designed for removing mortar/grout stains, efflorescence, and other new construction stains from new masonry without discoloring or damaging masonry surfaces. Use product expressly approved for intended use by cleaner manufacturer and manufacturer of masonry units being cleaned.
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
- a. Diedrich Technologies, Inc.; a division of Sandell Construction Solutions.
 - b. PROSOCO, Inc.

2.10 MORTAR AND GROUT MIXES

- A. General: Do not use admixtures, including pigments, air-entraining agents, accelerators, retarders, water-repellent agents, antifreeze compounds, or other admixtures unless otherwise indicated.
 - 1. Do not use calcium chloride in mortar or grout.
 - 2. Use portland cement-lime or masonry cement mortar unless otherwise indicated.
 - 3. For exterior masonry, use portland cement-lime or masonry cement mortar.
 - 4. For reinforced masonry, use portland cement-lime or masonry cement mortar.
 - 5. Add cold-weather admixture (if used) at same rate for all mortar that will be exposed to view, regardless of weather conditions, to ensure that mortar color is consistent.
- B. Preblended, Dry Mortar Mix: Furnish dry mortar ingredients in form of a preblended mix. Measure quantities by weight to ensure accurate proportions, and thoroughly blend ingredients before delivering to Project site.
- C. Mortar for Unit Masonry: Comply with ASTM C 270, Proportion Specification. Provide the following types of mortar for applications stated unless another type is indicated or needed to provide required compressive strength of masonry.
 - 1. For masonry below grade or in contact with earth, use Type M.
 - 2. For reinforced masonry, use Type S.
 - 3. For mortar parge coats, use Type S or Type N.
 - 4. For exterior, above-grade, load-bearing and nonload-bearing walls and parapet walls; for interior load-bearing walls; for interior nonload-bearing partitions; and for other applications where another type is not indicated, use Type S.
- D. Pigmented Mortar: Use colored cement product. Do not add pigments to colored cement product.
 - 1. Pigments shall not exceed 10 percent of portland cement by weight.
 - 2. Pigments shall not exceed 5 percent of masonry cement by weight.
 - 3. Application: Use pigmented mortar for exposed mortar joints in clay face brick.
- E. Colored-Aggregate Mortar: Produce required mortar color by using colored aggregates and natural color or white cement as necessary to produce required mortar color.
 - 1. Mix to match Architect's sample.
- F. Grout for Unit Masonry: Comply with ASTM C 476.
 - 1. Use grout of type indicated or, if not otherwise indicated, of type (fine or coarse) that will comply with TMS 602/ACI 530.1/ASCE 6 for dimensions of grout spaces and pour height.
 - 2. Proportion grout in accordance with ASTM C 476, Table 1 or paragraph 4.2.2 for specified 28-day compressive strength indicated, but not less than 2000 psi (14 MPa).
 - 3. Provide grout with a slump of 8 to 11 inches (200 to 280 mm) as measured according to ASTM C 143/C 143M.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
 - 1. For the record, prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.
 - 2. Verify that foundations are within tolerances specified.
 - 3. Verify that reinforcing dowels are properly placed.
 - 4. Verify that substrates are free of substances that impair mortar bond.
- B. Before installation, examine rough-in and built-in construction for piping systems to verify actual locations of piping connections.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION, GENERAL

- A. Thickness: Build cavity and composite walls and other masonry construction to full thickness shown. Build single-wythe walls to actual widths of masonry units, using units of widths indicated.
- B. Build chases and recesses to accommodate items specified in this and other Sections.
- C. Leave openings for equipment to be installed before completing masonry. After installing equipment, complete masonry to match construction immediately adjacent to opening.
- D. Use full-size units without cutting if possible. If cutting is required to provide a continuous pattern or to fit adjoining construction, cut units with motor-driven saws; provide clean, sharp, unchipped edges. Allow units to dry before laying unless wetting of units is specified. Install cut units with cut surfaces and, where possible, cut edges concealed.
- E. Select and arrange units for exposed unit masonry to produce a uniform blend of colors and textures. Mix units from several pallets or cubes as they are placed.

3.3 TOLERANCES

- A. Dimensions and Locations of Elements:
 - 1. For dimensions in cross section or elevation, do not vary by more than plus 1/2 inch (12 mm) or minus 1/4 inch (6 mm).
 - 2. For location of elements in plan, do not vary from that indicated by more than plus or minus 1/2 inch (12 mm).
 - 3. For location of elements in elevation, do not vary from that indicated by more than plus or minus 1/4 inch (6 mm) in a story height or 1/2 inch (12 mm) total.
- B. Lines and Levels:
 - 1. For bed joints and top surfaces of bearing walls, do not vary from level by more than 1/4 inch in 10 feet (6 mm in 3 m), or 1/2-inch (12-mm) maximum.
 - 2. For conspicuous horizontal lines, such as lintels, sills, parapets, and reveals, do not vary from level by more than 1/8 inch in 10 feet (3 mm in 3 m), 1/4 inch in 20 feet (6 mm in 6 m), or 1/2-inch (12-mm) maximum.

3. For vertical lines and surfaces, do not vary from plumb by more than 1/4 inch in 10 feet (6 mm in 3 m), 3/8 inch in 20 feet (9 mm in 6 m), or 1/2-inch (12-mm) maximum.
4. For conspicuous vertical lines, such as external corners, door jambs, reveals, and expansion and control joints, do not vary from plumb by more than 1/8 inch in 10 feet (3 mm in 3 m), 1/4 inch in 20 feet (6 mm in 6 m), or 1/2-inch (12-mm) maximum.
5. For lines and surfaces, do not vary from straight by more than 1/4 inch in 10 feet (6 mm in 3 m), 3/8 inch in 20 feet (9 mm in 6 m), or 1/2-inch (12-mm) maximum.

C. Joints:

1. For bed joints, do not vary from thickness indicated by more than plus or minus 1/8 inch (3 mm), with a maximum thickness limited to 1/2 inch (12 mm).
2. For head and collar joints, do not vary from thickness indicated by more than plus 3/8 inch (9 mm) or minus 1/4 inch (6 mm).
3. For exposed head joints, do not vary from thickness indicated by more than plus or minus 1/8 inch (3 mm).

3.4 LAYING MASONRY WALLS

- A. Lay out walls in advance for accurate spacing of surface bond patterns with uniform joint thicknesses and for accurate location of openings, movement-type joints, returns, and offsets. Avoid using less-than-half-size units, particularly at corners, jambs, and, where possible, at other locations.
- B. Bond Pattern for Exposed Masonry: Unless otherwise indicated, lay exposed masonry in running bond; do not use units with less-than-nominal 4-inch (100-mm) horizontal face dimensions at corners or jambs.
- C. Stopping and Resuming Work: Stop work by stepping back units in each course from those in course below; do not tooth. When resuming work, clean masonry surfaces that are to receive mortar, remove loose masonry units and mortar, and wet brick if required before laying fresh masonry.
- D. Built-in Work: As construction progresses, build in items specified in this and other Sections. Fill in solidly with masonry around built-in items.
- E. Fill space between steel frames and masonry solidly with mortar unless otherwise indicated.
- F. Where built-in items are to be embedded in cores of hollow masonry units, place a layer of metal lath, wire mesh, or plastic mesh in the joint below, and rod mortar or grout into core.
- G. Fill cores in hollow CMUs with grout where indicated on Structural Drawings, including but not limited to the following locations:
 1. 24 inches (600 mm) under bearing plates, beams, lintels, posts, and similar items unless otherwise indicated.
 2. All below grade CMUs.
 3. In wall cavity only at collar joint.

3.5 MORTAR BEDDING AND JOINTING

- A. Lay CMUs as follows:
 1. Bed face shells in mortar and make head joints of depth equal to bed joints.

2. Bed webs in mortar in all courses of piers, columns, and pilasters.
 3. Bed webs in mortar in grouted masonry, including starting course on footings.
 4. Fully bed entire units, including areas under cells, at starting course on footings where cells are not grouted.
- B. Lay solid masonry units with completely filled bed and head joints; butter ends with sufficient mortar to fill head joints and shove into place. Do not deeply furrow bed joints or slush head joints.
- C. Tool exposed joints slightly concave when thumbprint hard, using a jointer larger than joint thickness, except at the following locations.
1. Cut joints flush for masonry walls to receive plaster, EIFS, board insulation, adhered stone, or other direct-applied finishes (other than paint) unless otherwise indicated.
 2. Cut joints flush for masonry walls below grade.

3.6 CAVITY WALLS

- A. Bond wythes of cavity walls together using bonding system indicated on Drawings.
- B. Keep cavities clean of mortar droppings and other materials during construction. Bevel beds away from cavity, to minimize mortar protrusions into cavity. Do not attempt to trowel or remove mortar fins protruding into cavity.
- C. Installing Cavity Wall Insulation: Place small dabs of adhesive, spaced approximately 12 inches (300 mm) o.c. both ways, on inside face of insulation boards, or attach with plastic fasteners designed for this purpose. Fit courses of insulation between wall ties and other confining obstructions in cavity, with edges butted tightly both ways. Press units firmly against inside wythe of masonry or other construction as shown.

3.7 MASONRY-JOINT REINFORCEMENT

- A. General: Install entire length of longitudinal side rods in mortar with a minimum cover of 5/8 inch (16 mm) on exterior side of walls, 1/2 inch (13 mm) elsewhere. Lap reinforcement a minimum of 6 inches (150 mm).
1. Space reinforcement not more than 16 inches (406 mm) o.c.
 2. Space reinforcement not more than 8 inches (203 mm) o.c. in foundation walls and parapet walls.
 3. Provide reinforcement not more than 8 inches (203 mm) above and below wall openings and extending 12 inches (305 mm) beyond openings in addition to continuous reinforcement.
- B. Interrupt joint reinforcement at control and expansion joints unless otherwise indicated.
- C. Provide continuity at wall intersections by using prefabricated T-shaped units.
- D. Provide continuity at corners by using prefabricated L-shaped units.

3.8 ANCHORING MASONRY TO STRUCTURAL STEEL AND CONCRETE

- A. Anchor masonry to structural steel and concrete, where masonry abuts or faces structural steel or concrete, to comply with the following:

1. Provide an open space not less than 1 inch (25 mm) wide between masonry and structural steel or concrete unless otherwise indicated. Keep open space free of mortar and other rigid materials.
2. Anchor masonry with anchors embedded in masonry joints and attached to structure.
3. Space anchors as indicated, but not more than 24 inches (610 mm) o.c. vertically and 36 inches (915 mm) o.c. horizontally.

3.9 CONTROL AND EXPANSION JOINTS

- A. General: Install control- and expansion-joint materials in unit masonry as masonry progresses. Do not allow materials to span control and expansion joints without provision to allow for in-plane wall or partition movement.
- B. Form control joints in concrete masonry as follows:
 1. Fit bond-breaker strips into hollow contour in ends of CMUs on one side of control joint. Fill resultant core with grout and rake out joints in exposed faces for application of sealant.
 2. Install preformed control-joint gaskets designed to fit standard sash block.
 3. Install interlocking units designed for control joints. Install bond-breaker strips at joint. Keep head joints free and clear of mortar or rake out joint for application of sealant.
 4. Install temporary foam-plastic filler in head joints and remove filler when unit masonry is complete for application of sealant.

3.10 LINTELS

- A. Install steel lintels where indicated.
- B. Provide minimum bearing of 8 inches (200 mm) at each jamb unless otherwise indicated.

3.11 FLASHING

- A. General: Install embedded flashing and weep holes in masonry at shelf angles, lintels, ledges, other obstructions to downward flow of water in wall, and where indicated. Install cavity vents at shelf angles, ledges, and other obstructions to upward flow of air in cavities, and where indicated.
- B. Install flashing as follows unless otherwise indicated:
 1. Prepare masonry surfaces so they are smooth and free from projections that could puncture flashing. Where flashing is within mortar joint, place through-wall flashing on sloping bed of mortar and cover with mortar. Before covering with mortar, seal penetrations in flashing with adhesive, sealant, or tape as recommended by flashing manufacturer.
 2. At multiwythe masonry walls, including cavity walls, extend flashing through outer wythe, turned up a minimum of 4 inches (100 mm), and through inner wythe to within 1/2 inch (13 mm) of the interior face of wall in exposed masonry. Where interior face of wall is to receive furring or framing, carry flashing completely through inner wythe and turn flashing up approximately 2 inches (50 mm) on interior face.
 3. At masonry end/corner wall (adjacent to Generator space), wrap masonry end wall as indicated on Drawings to provide continuity of air/water barrier protection. Extend flashing at least 12 inches beyond inside corner of wall.

4. At lintels and shelf angles, extend flashing a minimum of 6 inches (150 mm) into masonry at each end. At heads and sills, extend flashing 6 inches (150 mm) at ends and turn up not less than 2 inches (50 mm) to form end dams.
5. Install metal drip edges beneath flexible flashing at exterior face of wall. Stop flexible flashing 1/2 inch (13 mm) back from outside face of wall and adhere flexible flashing to top of metal drip edge.
6. Install metal flashing termination beneath flexible flashing at exterior face of wall. Stop flexible flashing 1/2 inch (13 mm) back from outside face of wall and adhere flexible flashing to top of metal flashing termination.

3.12 REINFORCED UNIT MASONRY

- A. Temporary Formwork and Shores: Construct formwork and shores as needed to support reinforced masonry elements during construction.
 1. Construct formwork to provide shape, line, and dimensions of completed masonry as indicated. Make forms sufficiently tight to prevent leakage of mortar and grout. Brace, tie, and support forms to maintain position and shape during construction and curing of reinforced masonry.
 2. Do not remove forms and shores until reinforced masonry members have hardened sufficiently to carry their own weight and that of other loads that may be placed on them during construction.
- B. Placing Reinforcement: Comply with requirements in TMS 602/ACI 530.1/ASCE 6.
- C. Grouting: Do not place grout until entire height of masonry to be grouted has attained enough strength to resist grout pressure.
 1. Comply with requirements in TMS 602/ACI 530.1/ASCE 6 for cleanouts and for grout placement, including minimum grout space and maximum pour height.
 2. Limit height of vertical grout pours to not more than 48 inches (1219.2 mm).

3.13 FIELD QUALITY CONTROL

- A. Testing and Inspecting: Owner will engage special inspectors to perform tests and inspections and prepare reports. Allow inspectors access to scaffolding and work areas as needed to perform tests and inspections. Retesting of materials that fail to comply with specified requirements shall be done at Contractor's expense.
- B. Inspections: Special inspections according to Level B typically with Level C at shearwalls in TMS 402/ACI 530/ASCE 5.
 1. Begin masonry construction only after inspectors have verified proportions of site-prepared mortar.
 2. Place grout only after inspectors have verified compliance of grout spaces and of grades, sizes, and locations of reinforcement.
 3. Place grout only after inspectors have verified proportions of site-prepared grout.
- C. Testing Prior to Construction: One set of tests.
- D. Testing Frequency: One set of tests for each 5000 sq. ft. (464 sq. m) of wall area or portion thereof.

- E. Concrete Masonry Unit Test: For each type of unit provided, according to ASTM C 140 for compressive strength.
- F. Mortar Aggregate Ratio Test (Proportion Specification): For each mix provided, according to ASTM C 780.
- G. Mortar Test (Property Specification): For each mix provided, according to ASTM C 780. Test mortar for compressive strength.
- H. Grout Test (Compressive Strength): For each mix provided, according to ASTM C 1019.

3.14 PARGING

- A. Parge exterior faces of below-grade masonry walls, where indicated, in two uniform coats to a total thickness of 3/4 inch (19 mm). Dampen wall before applying first coat and scarify first coat to ensure full bond to subsequent coat.
- B. Use a steel-trowel finish to produce a smooth, flat, dense surface with a maximum surface variation of 1/8 inch per foot (3 mm per 300 mm). Form a wash at top of parging and a cove at bottom.
- C. Damp-cure parging for at least 24 hours and protect parging until cured.

3.15 REPAIRING, POINTING, AND CLEANING

- A. Remove and replace masonry units that are loose, chipped, broken, stained, or otherwise damaged or that do not match adjoining units. Install new units to match adjoining units; install in fresh mortar, pointed to eliminate evidence of replacement.
- B. Pointing: During the tooling of joints, enlarge voids and holes, except weep holes, and completely fill with mortar. Point up joints, including corners, openings, and adjacent construction, to provide a neat, uniform appearance. Prepare joints for sealant application, where indicated.
- C. In-Progress Cleaning: Clean unit masonry as work progresses.
 - 1. Keep masonry faces clean during construction whenever possible, i.e. remove all mortar tags and stains before they cure, a light brushing with an appropriate brush upon initial mortar set, minimize mortar run-down with wet masonry units, minimize water entry into constructed walls, remove mortar build up from scaffold, protect all wall projections from mortar splashes, turn over planks to avoid mortar splashes when not working, protect base of wall from all mortar and mud splashes and remove and clean grout spills immediately.
- D. Final Cleaning: After mortar is thoroughly set and cured, clean exposed masonry as follows:
 - 1. Remove large mortar particles by hand with wooden paddles and nonmetallic scrape hoes or chisels.
 - 2. Test cleaning methods on sample wall panel; leave one-half of panel uncleaned for comparison purposes. Obtain Architect's approval of sample cleaning before proceeding with cleaning of masonry.
 - 3. Protect adjacent stone and non-masonry surfaces from contact with cleaner by covering them with liquid strippable masking agent or polyethylene film and waterproof masking tape.

4. Wet wall surfaces with water before applying cleaners; remove cleaners promptly by rinsing surfaces thoroughly with clear water.
5. Clean brick by bucket-and-brush hand-cleaning method described in BIA Technical Notes 20.
6. Clean concrete masonry by applicable cleaning methods indicated in NCMA TEK 8-4A.
7. Clean masonry with a proprietary acidic cleaner applied according to manufacturer's written instructions.
8. Clean stone trim to comply with stone supplier's written instructions.

3.16 MASONRY WASTE DISPOSAL

- A. Salvageable Materials: Unless otherwise indicated, excess masonry materials are Contractor's property. At completion of unit masonry work, remove from Project site.
- B. Masonry Waste Recycling: Return broken CMUs not used as fill to manufacturer for recycling.
- C. Excess Masonry Waste: Remove excess clean masonry waste that cannot be used as fill, as described above or recycled, and other masonry waste, and legally dispose of off Owner's property.

END OF SECTION 042000

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SECTION 042516 – THIN BRICK PANEL SYSTEM**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Thin masonry veneer support system.
2. Thin brick units, mortar and accessories.

B. Related Requirements:

1. Section 042000 "Unit Masonry" CMU substrate and self-adhered flashing.
2. Section 042600 "Adhered Masonry Veneer" for adhered stone veneer system.
3. Section 072100 "Thermal Insulation" for cavity wall insulation.
4. Section 072726 "Fluid-Applied Membrane Air Barriers".
5. Section 076200 "Sheet Metal Flashing and Trim" for exposed sheet metal flashing and for furnishing manufactured reglets installed in masonry joints.
6. Section 079200 "Joint Sealants" for sealants and backer rods.

1.2 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.

- B. Submit the following Submittals via Owner's Project Management software platform.

1. Product Data: For each type of product, including manufacturer's installation instructions and details.
2. Samples:
 - a. For each color and texture of brick units.
 - b. For colored mortar.

1.4 INFORMATIONAL SUBMITTALS

- A. Product test reports.
- B. Material certificates.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.6 QUALITY ASSURANCE

- A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials in manufacturer's unopened packaging.
- B. Inspect all materials upon arrival and notify supplier if any damage is observed. Store products in manufacturer's packaging or according to manufacturer's recommendations until ready for installation.
- C. Store masonry units, mortar, and accessories off the ground to prevent contamination by mud, dust or other materials likely to cause staining or other defects.
- D. Protect materials from contamination, moisture, freezing, overheating or other damage in accordance with manufacturer's instructions. Cover all materials with a non-staining waterproof membrane material when necessary to protect from elements.
- E. Store different types of materials separately.
- F. Store adhesive and mortar additive above 32-deg Fahrenheit and below 86-deg Fahrenheit temperatures.
- G. Store and dispose of solvent-based materials and materials used with solvent-based materials, in accordance with requirements of local authorities having jurisdiction.
- H. Panels must be stored to prevent permanent distortion and kept dry before installation. Panels and accessories should be covered at all times prior to installation.
- I. Dry panels exposed to water or condensation prior to installation or application of thin brick. The presence of moisture on the brick or metal panels may adversely affect adhesive performance.

1.8 FIELD CONDITIONS

- A. Comply with requirements of referenced standards and recommendations of material manufacturers for environmental conditions before, during and after installation.
- B. Protection of Work:
 - 1. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's limits.
- C. Stain Prevention: Prevent adhesive, mortar, and soil from staining the face of masonry to be left exposed or painted. To avoid smearing of adhesive or mortar on the face of masonry, allow adhesive and mortar on face of installed masonry to become firm before trying to remove.
 - 1. Protect base of walls from rain-splashed mud and from mortar splatter.
 - 2. Protect sills, ledges, and projections from droppings of adhesive or mortar.
 - 3. Protect surfaces of window and door frames, as well as similar products with painted and integral finishes, from mortar droppings.
 - 4. Turn scaffold boards near the wall on edge at the end of each day to prevent rain from splashing mortar and dirt onto completed masonry.
- D. Cold-Weather Requirements: Do not use frozen materials or materials mixed or coated with ice or frost. Do not build on frozen substrates. Remove and replace unit masonry damaged by frost

or by freezing conditions. Comply with cold-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.

- E. Hot-Weather Requirements: Comply with hot-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6. Protect mortar from uneven and excessive evaporation.
1. The face of the installed thin brick may be dampened with water prior to mortar installation to reduce the absorption of moisture from the mortar joint and increase bond.
 2. Veneer may be fogged with water to prevent drying prior to proper hydration of mortar. Apply only enough moisture to consistently dampen the wall without allowing water to run down the face of the masonry.

1.9 WARRANTY

- A. Manufacturer's Warranty: Provide a 25-year warranty against material defects in Thin Tech Elite panels and accessories.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

- A. Obtain thin brick panel system components from single source from single manufacturer for each product required.

2.2 THIN MASONRY VENEER SUPPORT SYSTEM

- A. Manufacturer: Glen-Gery Corporation; Wyomissing, PA; phone: 610-562-3076; www.glengery.com.
- B. Thin Masonry Veneer Support Panel: Metal support panel consisting of 26 gauge (16 mil), stucco embossed, textured steel with angled ties providing structural mechanical support for exterior thin masonry veneer, installed by fastening to concrete, masonry, or metal frame construction.
1. Metal Panels for Thin Brick Support: Glen-Gery Thin Tech Elite Panel.
 2. Exterior Finish: G90 Galvanized with thermal set coating.
 3. Support Ties: 3/8 inch (10 mm)
 4. Support Panels: 4 ft. x 4 ft. nominal (16 sq. ft., 1.44 sq. m.) masonry support panels for flat wall areas, shall be for unit heights as follows:
 - a. Spacing for Modular Units: 2-5/8 in. (2.667 in., 68 mm); three courses equal 8 inches (203 mm).
- C. Fasteners (for Masonry Support Panel): Supplied by thin veneer support panel manufacturer.
1. Screw Fasteners: Pan head fasteners with a minimum #10, (0.190 in., 4.8 mm) thread diameter and corrosion resistance provided by tri-layered ceramic surface coating with a minimum protection of 1000 hrs. when tested according to ASTM B117.
 - a. Fasteners for Attachment to Concrete and Masonry: Self-tapping, point type 17.
 - b. Fasteners for Attachment to Light Gauge Steel Studs: Self-drilling, with a drilling capacity of 0.035 to 0.176 in. and minimum torsional strength of 75 lb/in.
 2. Fastener Length:

- a. Metal Framing/Support Member Fasteners: Length necessary to allow fasteners to penetrate not less than member thickness plus three threads (3/16 in., 5 mm).
 - b. Masonry or Concrete Fasteners: Length necessary to allow fasteners to penetrate substrate a minimum of 1 inch (25 mm).
 - D. Starter Angle Flashing:
 - 1. Stainless steel conforming to ASTM A240/A240M, Type 304, 0.018 in. (0.45 mm) (26 gauge) pre-bent in 10 ft. (3.05 m) lengths.
 - 2. Depth: As required to accommodate veneer units, wall cavity and continuous insulation as applicable.
 - 3. Profile: 45-degree drip.
 - E. Water-Resistive Barrier:
 - 1. See Section 072726 "Fluid-Applied Membrane Air Barriers".
 - F. Adhesive: Supplied by thin veneer support panel manufacturer.
 - 1. Heavy Duty Construction Adhesive (for thin brick maximum 1 inch (25 mm) thick): High-strength mastic exceeding ASTM D3498 and ASTM C557 specifications with less than 70 grams of VOC per liter with a shear value between the thin veneer and the panel greater than 50 PSI (10.5 kg/cm²).
 - G. Installation Hardware:
 - 1. Support Channels: 8 ft. (2.44 m) long, manufactured from 26 gauge (18 mil) stainless steel with 1/2-inch (13 mm) support leg, 3/4-inch (19 mm) wide by 1/2-in. (13 mm) high clip.
 - a. 2-3/8 inch (60 mm) Type 5 Classic Channel for 2-1/4 inch (57 mm) high units.
 - H. Universal Corner Support: 6 inches (152 mm) x 6 inches (152 mm) x 4 ft. (1.22 m) long, pre-bent 90 degrees, manufactured from 26 gauge (16 mil) , stucco embossed, textured steel.
 - 1. Exterior Finish: G90 Galvanized with thermal set coating.
 - I. Weep Holes / Vents:
 - 1. Glen-Gery Thin Tech® Air Vent: Impact resistant polypropylene copolymer.
 - a. Density: 2,000 grams/sq. m.
 - b. Size: 3/8 inch (10 mm) x 1/2 inch (13 mm) x 48 inches (1.22 m).
- 2.3 MASONRY UNITS, GENERAL
- A. Masonry Unit Weight: Not to exceed 15 psf.
 - B. Defective Units: Referenced masonry unit standards may allow a certain percentage of units to contain chips, cracks, or other imperfections exceeding limits stated in the standard. Do not use units where such defects will be exposed in the completed work.

2.4 CLAY UNIT MASONRY

- A. Manufacturer: Glen-Gery Corporation; Wyomissing, PA; phone: 610-562-3076; www.glengery.com.
- B. General: Provide shapes indicated and as follows:
 - 1. Provide special shapes for applications where flats (stretcher units) cannot accommodate special conditions, including those at corners and openings.
 - 2. Provide special shapes for applications requiring thin brick of size, form, color and texture on exposed surfaces that cannot be produced by sawing.
 - 3. Provide special shapes for applications where shapes produced by sawing would result in sawed surfaces being exposed to view.
- C. Thin Brick: ASTM C 216, Grade SW, Type FBS.
 - 1. Series: Emporium+ Series
 - 2. Style: Extruded
 - 3. Texture/Finish: Wirecut
 - 4. Modular Size(actual): 2-1/4 inches (57 mm) high by 7-5/8 inches (194 mm) long.
 - 5. Thickness: 1/2 inch (13 mm)
 - 6. Colors:
 - a. Aspen White
 - b. Black Pearl

2.5 MORTAR

- A. Cold Weather Additives (including accelerators) shall not be used in mortar.
- B. Mortar for Thin Brick: ASTM C 270, Type S
 - 1. Glen-Gery Color Mortar Blend.
 - 2. See colors on Exterior Finish Schedule on Drawings.

2.6 ANCILLARY MATERIALS

- A. Self-Adhering Flashing: See Section 042000 "Unit Masonry".
- B. Sealants: See Section 079200 "Joint Sealants".
 - 1. Bond Breaker Tape: Polyethylene tape, width to match width of movement joint.
- C. Rigid Insulation: See Section 072100 "Thermal Insulation".

2.7 MASONRY CLEANERS

- A. Proprietary Acidic Cleaner: Cleaner as recommended by thin brick manufacturer for removing mortar/grout stains, efflorescence, and other new construction stains from new masonry without discoloring or damaging masonry surfaces. Use product expressly approved for intended use by cleaner manufacturer and manufacturer of masonry units being cleaned.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Do not begin thin brick panel system installation until substrates, water-resistive barrier, foundations, rough-ins and built-in construction have been installed according to project specifications and building code requirements.
 - 1. Walls must be structurally sound and substrate system designed with a wall deflection not greater than $L/360$.
 - a. Maximum wall frame spacing for stud walls: 24 in. (610 mm) on center.
 - b. Maximum wall frame spacing for girts: 24 in. (610 mm) on center.
 - c. Minimum 0.043 in. (18 gauge; 1.09 mm) metal framing for exterior walls.
 - 2. Substrate shall provide continuous support and be flat, with surface variation not exceeding 1/8 inch (3 mm) within any 4-foot (1.2 m) square area and have no planar irregularities greater than 1/4 inch (6 mm) in 10 feet (3.05 m).
 - 3. Substrate shall be concrete, masonry or one of the following as deemed suitable for specific project conditions:
 - a. Glass fiber mat-faced sheathing not less than 1/2 inch (13 mm) in thickness.
 - b. Closed-cell insulating rigid foam not less than 1/2 inch (13 mm) thick conforming to ASTM C578 (minimum 15 psi compressive strength and minimum 1.30 lb./cu. ft. (21 kg/cu. m) density) or ASTM C1289.
 - c. Exterior grade plywood not less than 3/8 inch (10 mm) in thickness.
- B. Verify walls are plumb and corners are braced.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Clean surfaces thoroughly prior to installation. All surfaces including, but not limited to metal panel and back face of thin brick must be free of water, snow, dirt, mud, oil and other foreign materials prior to application.
- B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.
- C. Trim or flash in place per manufacturer's details and/or BIA Technical Note 28C on Thin Brick Veneer.
- D. Protect the tops of all uncompleted walls to prevent water entry.

3.3 INSTALLATION, GENERAL

- A. Install materials in accordance with manufacturer's instructions.
- B. Select and arrange exposed masonry units to produce a uniform blend of color and texture.
 - 1. Install masonry units from several pallets or cubes as they are placed.
- C. Comply with tolerance requirements in TMS 602 (ACI 530.1/ASCE 6).

3.4 FLASHING

- A. Prior to metal panel installation, install starter angle and flashing in accordance with veneer panel manufacturer's installation instructions or ASTM E2112. Prime substrate prior to installation when required by flashing manufacture.
- B. Seal penetrations in flashing with adhesive, sealant, or tape as recommended by flashing manufacturer.
- C. Extend horizontal through wall flashing vertically up the backing a minimum of 3 inches (76 mm).
- D. Lap flashing ends a minimum of 3 inches (76 mm).
- E. Seal vertical and horizontal legs of all flashing laps with compatible lap sealant.
- F. Lap water-resistive barrier over top of flashing.
- G. Turn up the ends of discontinuous (e.g., head, sill and stepped) flashings to form end dams or extend not less than 4 inches (102 mm) beyond edges of openings.
- H. Extend flashing through wall with at least 1/4 inch (6 mm) exposed to form a drip.

3.5 THIN MASONRY SUPPORT PANEL INSTALLATION

- A. Install in accordance with manufacturer's written instructions as applicable to each type of substrate required.
- B. Trim, starter angle and flashing shall be installed prior to panel installation.
- C. Panels shall be clean, dry and free of dirt, oil or any other foreign contaminant.
- D. Attach panels flat to the substrate in true and level rows with support ties aligned and level to each other, including at corners.
- E. Stagger vertical support panel joints over sheathing joints and at least 16 inches (406 mm) horizontally from panels in rows above or below.
- F. Install full-size uncut panels when possible. Cut panels using a method resulting in clean, unbent edges when necessary to provide vertical staggered panel joints, to ensure panels are fastened to studs, or to fit specific conditions.
- G. Install panels with 1/16 inch to 1/8 inch (2 to 3 mm) space between the vertical edges of panels.
- H. Do not allow panels to bridge vertical or horizontal movement joints in substrate. The space between metal panels at substrate movement joint locations shall be equal to the thickness of the substrate movement joint.
- I. Stop panel at least 3/8 inch (10 mm) from inside corners, openings and other materials to allow for movement.
- J. Start panels at least 4 inches (102 mm) above earth or 2 inches (51 mm) above paved areas. Do not install veneer support panels below grade.
 - 1. A minimum 1/2 in inch (13 mm) clearance is adequate above paving supported by the same foundation as the stud wall. Where veneer support panel is installed over exterior

concrete or CMU walls, maintain a 2-inch (51 mm) clearance from grade or 1/2-inch (13 mm) clearance from paving.

2. Coordinate clearances with drawings and manufacturer's recommendations. Obtain Architect's approval prior to installation to ensure that uniform appearance at completion.

3.6 FASTENERS FOR THIN MASONRY SUPPORT PANELS

- A. Attach fasteners to framing/support members for framed applications. Do not attach veneer support panel to the sheathing alone.
- B. Fasteners for steel studs or other steel supports shall penetrate a minimum of three threads beyond steel member thickness.
- C. Fasteners for masonry or concrete shall penetrate the substrate a minimum of 1 inch (25 mm).
- D. Fastener Installation: Mechanically attach veneer support panels with a minimum of one fastener per sq. ft. (929 cm²), reducing fastener spacing along the top and bottom of the wall and around openings, as indicated below.
 1. Spacing of pre-punched fastener holes results in the typical recommended fastener spacing of 8 inches (203 mm) vertically and 16 inches (406 mm) horizontally.
 2. Horizontal fastener spacing shall not exceed 24 inches (610 mm); vertical fastener spacing shall not exceed 16 inches (406 mm).
 3. Fasteners should not be installed behind masonry units, except where they are placed within the channels of veneer support panels.
 4. Provide additional fasteners around the perimeter of walls and around openings larger than 24 inches (610 mm) in either dimension, as well as building corners not utilizing corner panels as follows:
 - a. Fasteners shall be placed within 8 inches (203 mm) of the top and bottom of the building walls, at a maximum spacing of 12 inches (305 mm) horizontally.
 - b. At vertical ends of walls and openings, fasteners shall be spaced a maximum of 8 inches (203 mm) vertically within 4 inches (102 mm) of the vertical edge of the panel.
 - c. Where the vertical edge of veneer support panel is more than 2 inches (51 mm) from a channel, install veneer support panel shim and fasten for additional support.

3.7 SETTING OF MASONRY UNITS

- A. Arrange and trim masonry veneer units for accurate fit with uniform joint widths.
- B. Install veneer units in accordance with mortar manufacturer's written recommendations, as appropriate for substrate condition.
- C. Back face of thin brick and surface of veneer support panel must be dry and clean; free of dirt, oil or and other contaminants.
- D. Prior to installing veneer units with adhesive, establish unit layout to ensure bond patterns with uniform joint thicknesses and to reduce cutting at openings, movement joints, returns, and offsets.
 1. Avoid using less-than-half-size units, particularly at corners and jambs.
 2. Avoid using cut units at heads and sills of opening.
 3. Ensure unfinished or cut faces are not exposed to view upon completion.

- E. Lay masonry in bond pattern as indicated on drawings.
- F. Select and arrange units for exposed unit masonry to produce a uniform blend of color and texture.
- G. Leave a uniform 3/8 inch to 1/2 inch (10 to 13 mm) space around the perimeter of openings to allow for movement joint installation.
- H. Apply 1 inch (25 mm) diameter dabs of specified adhesive, not more than 6 inches (152 mm) apart to the back of thin brick and adhere units to panel. No less than two dabs/beads shall be applied to each full-length unit. Corner units shall have at least one dab/bead of adhesive applied to the head/return.
 - 1. Do not apply adhesive in horizontal streams.
 - 2. Do not use excessive adhesive as this may cause veneer units to tilt away from wall prior to adhesive set.
- I. Masonry units shall be adhered to dry panel within 5 to 10 minutes after adhesive has been applied and before film begins to form on the adhesive.
- J. When adjustment is necessary to be made after adhesive begins to harden, remove hardened adhesive and replace with fresh adhesive.
- K. Keep areas intended to receive sealant clear of thin brick, adhesive and other materials during construction.
- L. Do not allow masonry units to bridge movement joints in substrate.

3.8 MORTAR INSTALLATION AND JOINTING

- A. After adhesive has set a minimum of 6 hours, completely fill head and bed joints intended to receive mortar.
- B. Discard mortar after two hours or when too stiff to work. Retempering within two hours is permitted, but may contribute to mortar color variation.
- C. Do not fill movement joints to receive sealant.
- D. Do not fill joints to receive veneer support panel air vent.
- E. Form weep holes (head joints free of mortar) as required, immediately above starter angles and flashings.
- F. Tool exposed joints when thumbprint hard to joint concave profile.
- G. Flush cut all joints not tooled.
- H. When repointing, completely remove mortar, and refill solidly with mortar and tool joints.

3.9 WEEP HOLES / VENTS

- A. Vents for Veneer Support Panels: Omit mortar from head joints and install vents cut to 2-1/4 inches (57 mm) vertically in head joints.

1. Spacing of weep vents shall not exceed a maximum of 24 inches (610 mm) on center horizontally for units 12 inches (304.8 mm) or less in length and a maximum of 32 inches (813 mm) on center for larger units, in the joint between the flashing and masonry units above or in the lower third of the head joints immediately above the starter angles and flashings, including the base of the wall, at horizontal expansion joints and above all openings.
2. Keep vents free of mortar or other obstructions.

3.10 MOVEMENT JOINTS

- A. Locate movement joints where indicated on drawings or where recommended by veneer panel manufacturer.
- B. Provide vertical and horizontal movement joints where indicated by leaving a continuous space no less than 3/8 inch (10 mm) wide between veneer support panels, installing bond breaker tape or backer rod as specified and installing sealant as specified in Section 079200 "Joint Sealants."
 1. Provide and install bond breaker tape at movement joints prior to installing masonry units 1 inch (25 mm) or less in thickness. Install backer rod where thickness of masonry unit exceeds 1 inch (25 mm).
- C. Keep entire length of movement joint clear of mortar, veneer adhesive and debris.
- D. Install movement joints between veneer panel assemblies and other materials, including around windows and doors.
- E. Install movement joints at changes in substrate and where movement joints occur in substrate or foundation.
- F. Install movement joints at changes in wall height or thickness.
- G. Spacing of vertical movement joints must not exceed 24 feet (7.3 m) on center in walls without openings.
- H. Install movement joints at inside corners and within 2 to 4 feet (0.6 to 1.2 m) of outside corners where intersecting walls are longer than 4 feet (1.2 m).
- I. Install no less than one horizontal movement joint per story, with height between horizontal movement joints not exceeding 20 feet (6.1 m).
- J. Install horizontal movement joints below all starter angles above the base of the wall.

3.11 CLEANING AND PROTECTION

- A. Follow Manufacturer's written recommendations specific to the products installed.
- B. In-Progress Cleaning: Clean unit masonry as work progresses with a dry, non-metallic brush to remove adhesive as well as mortar fins and smears before tooling joints.
- C. Final Cleaning: After mortar is thoroughly set and cured, clean exposed masonry as follows:
 1. Cut out all defective mortar joints and holes in exposed masonry and fill with new mortar.

2. Clean preselected sample wall area with specified cleaning product as per brick manufacturer's recommendations. Do not use muriatic or hydrofluoric acid. Do not proceed with cleaning until approved by Owner.
3. Clean thin brick in accordance with cleaning product manufacturer's written instructions.
4. Protect adjacent stone and non-masonry surfaces from contact with cleaner.

END OF SECTION 042516

SECTION 042600 - ADHERED MASONRY VENEER**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Stone veneer units.
2. Air and water barrier.
3. Setting mortar and accessory materials.

B. Related Requirements:

1. Section 042000 "Unit Masonry" CMU substrate and self-adhered flashing.
2. Section 042516 "Thin Brick Panel System" for thin brick units on veneer panel support system.
3. Section 072100 "Thermal Insulation" for cavity wall insulation.
4. Section 072726 "Fluid-Applied Membrane Air Barriers".
5. Section 076200 "Sheet Metal Flashing and Trim" for exposed sheet metal flashing and for furnishing manufactured reglets installed in masonry joints.
6. Section 079200 "Joint Sealants" for sealants and backer rods.

1.2 CONFORMANCE AND ACTION SUBMITTALS**A. Conformance Submittals** are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.**B. Submit the following Submittals** via Owner's Project Management software platform.

1. **Product Data:** For each type of product.
 - a. For stone veneer units, include dimensions and finishes.
2. **Shop Drawings:** Show fabrication and installation details for brick and stone veneer units. Include dimensions, details of reinforcement and anchorages if any, and indication of finished faces.
3. **Samples:**
 - a. For each color and texture of stone veneer required.
 - b. For colored mortar.

1.3 FIELD CONDITIONS

- A. Cold-Weather Requirements:** Do not use frozen materials or materials mixed or coated with ice or frost. Do not build on frozen substrates. Remove and replace unit masonry damaged by frost or by freezing conditions. Comply with cold-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.
- B. Hot-Weather Requirements:** Comply with hot-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.

- C. Masonry units and mortar shall be delivered and stored covered and protected from the weather in such a manner as to eliminate contact with moisture. Store masonry units above ground on level platform, which permits air circulation under stack. Protect anchors, ties, and reinforcement from weather exposure and construction activity.

PART 2 - PRODUCTS

2.1 MANUFACTURED STONE VENEER

- A. Stone Veneer: Portland-cement-based manufactured stone veneer and trim units.

1. Supplier Contact: See **LIDL National Account List**.
2. Manufacturer: Glen-Gery
3. Style: Landmark
4. Profile: Glen Ridge
5. Sizes:
 - a. Length: 5 to 22 inches
 - b. Height: 1-3/4 to 7-1/2 inches
 - c. Thickness: 1 to 1-5/8 inches
6. Textures: Sand and smooth faces.
7. Colors: See Exterior Finish Schedule on Drawings.

2.2 SETTING AND GROUTING MATERIALS

- A. Materials, General: Comply with ASTM C1780 and Masonry Veneer Manufacturers Association (MVMA).
- B. Skim-Coat Mortar (if required): Polymer-fortified, non-sag mortar to skim coat or smooth out irregular surfaces.
1. Acceptable Product: Laticrete MVIS Veneer Mortar.
- C. Air and Water Barrier Membrane: Single-component, fluid-applied air and water barrier membrane.
1. Acceptable Product: Laticrete MVIS Air & Water Barrier.
- D. Cement Mortar Bed: Polymer-fortified, non-sag mortar for use as mortar bed for direct-adhered masonry veneer application.
1. Acceptable Product: Laticrete MVIS Veneer Mortar.
- E. Pointing Mortar: Factory-prepared and designed to be mixed with water; a formulation from a blend of Portland cement, graded aggregates, and color-fast pigments.
1. Colors: See Exterior Material/Finish Schedule on Drawings.
 2. Joint Width: As directed by Architect.
- F. Water: Potable.

2.3 RELATED MATERIALS

- A. Weep Screed and Weep Screed Deflector at Base of Stone Veneer Wall:
1. Manufacturer: Masonry Technology Inc. (MTI); phone 800-879-3348; www.mtidry.com.
 2. Products:
 - a. L & R Weep Screed (LR 3501); 70-deg bent angle with 3-1/2-inch vertical attachment flange, 1-1/32-inch angled short leg, pre-punched drain slots and fastener holes.
 - b. Weep Screed Deflector (WSD 1309); 120-deg bent angle with 3-1/2-inch vertical attachment flange, 51/64-inch angled short leg, and pre-punched fastener holes.
 3. Material: Zinc-coated (galvanized) steel sheet.
 - a. Thickness: 26 gauge
 - b. Length: 8 feet
 - c. Finish: Mill
- B. Cementitious Backer Board: ANSI A118.9 or ASTM C 1325, Type A; with core consisting of cement, polystyrene beads and aggregates and embedded fiberglass mesh on both faces; approved for exterior use.
1. Product: National Gypsum Co.; PermaBase Cement Board.
 2. Panel Size: Maximum lengths available to minimize end-to-end butt joints.
 3. Panel Thickness: As indicated on Drawings.
 4. Accessories:
 - a. Tape: PermaBase Tape; 4-inch (102 mm) wide polymer-coated (alkali resistant) mesh tape.
 - b. Fasteners: PermaBase Screws; drill point screws (No. 8) wafer head, corrosion-resistant, Type S-12 screws or equivalent; minimum, 1-5/8-inch (41 mm) long.
- C. Joint Sealant: One-component, on-staining, neutral cure, exterior grade 100% silicone sealant.
1. Acceptable Product: Laticrete MVIS Silicone Sealant.
 2. Use primers, backer rods, and sealant accessories recommended by sealant manufacturer.
 3. Color: As selected by Architect.
- D. Masonry Sealer: Use penetrating, non-film-forming, water-based, and VOC-compliant, of type appropriate to the masonry veneer and grout material, as recommended by masonry veneer and mortar manufacturer.
1. Acceptable Product: Laticrete L&M Aquapel.
- E. Masonry and Mortar/Grout Cleaner: A neutral, non-acidic cleaner capable of removing soil and residue, mortar/grout stains, efflorescence, and other new construction stains without discoloring or damaging masonry veneer and grout surfaces. Use product expressly approved for intended use by brick and stone veneer manufacturer and expressly approved by cleaner manufacturer for use on brick and stone veneer and adjacent masonry materials.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine surfaces indicated to receive masonry veneer, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Remove releasing agent (form oil) from new masonry surfaces. If necessary, etch surface with muriatic acid, rinse thoroughly and/or score with a wire brush, or use high pressure water or sandblasting to remove releasing agent.
- B. Protect adjacent work from contact with mortar.
- C. Remove all soil, stain, grease, oil, paint, sealer, curing compound and foreign materials that may inhibit bonding of masonry veneer. Clean masonry substrates by thoroughly scrubbing with fiber brushes and then drenching with clear water. Use only mild cleaning compounds that contain no caustic or harsh materials or abrasives.

3.3 INSTALLATION, GENERAL

- A. Comply with ASTM C1780 and Masonry Veneer Manufacturers Association (MVMA).

3.4 CEMENTITIOUS BACKER BOARD INSTALLATION

- A. Install backer board and treat joints according to ANSI A108.11 and manufacturer's written instructions for type of application indicated.

3.5 SKIM COAT INSTALLATION

- A. Where cement backer board surfaces are rough or uneven, apply a skim coat of specified mortar material to provide for a smooth and uniform surface, within tolerance level recommended by manufacturer for adhering veneer directly to substrate. Allow to cure as recommended by manufacturer prior to installing air & water barrier.
- B. Prepare substrate in accordance with manufacturer's installation instructions for the type of substrate being covered.

3.6 WEEP SCREED AND WEEP SCREED DEFLECTOR INSTALLATION

- A. Install weep screed and weep screed deflector at base of stone veneer assembly per manufacturer's installation instructions.
- B. Install weep screed in front of weep screed deflector, with base of stone veneer not less than 2 inches above paved areas.
- C. Fasten weep screed and weep screed deflector into metal framing.
- D. Lap air and moisture barrier membrane over weep screed attachment flange.

3.7 AIR AND WATER BARRIER MEMBRANE INSTALLATION

- A. Prior to installation of air and water barrier membrane, verify that all flashing, transitions, penetrations and openings have been properly detailed in accordance to manufacturer's

installation instructions. Ensure proper transition between air and water barrier membrane and dampproofing for continuity of protection against moisture migration into the building.

- B. Apply air and water barrier membrane to uniformly and fully cover skim-coated cement backer board substrate. Follow manufacturer's recommendations for coverage rate. Allow to cure before installing veneer units.
- C. Apply air and water barrier membrane in two coats for a total dry coat thickness of 20 to 30 mils (0.02 to 0.03 inch) (0.5 to 0.8 mm).

3.8 SETTING OF STONE VENEER UNITS

- A. Arrange and trim masonry veneer units for accurate fit with uniform joint widths.
- B. Install veneer units in accordance with mortar manufacturer's written recommendations, as appropriate for substrate condition.
- C. Wipe backs of veneer units with a damp cloth to remove dirt, dust and loose materials before units are installed.
- D. Spread mortar over prepared substrate using the appropriate notched trowel for the unit being installed. Back butter all veneer units to provide full bedding of veneer unit. Firmly press the veneer units into the freshly applied mortar bed, allowing excess mortar to extrude on all sides around the veneer unit. Slide and twist the veneer units to ensure good bond. Tap units into place, completely filling space between units and mortar bed to achieve 100% coverage between substrate and unit. Clean excess extruded mortar. Allow installation to set until firm and cured.
 - 1. Spread only enough mortar that can be covered with veneer units within 15 to 20 minutes. Adjust as necessary. Check mortar for complete coverage by periodically removing veneer unit and inspecting the transfer onto the back of the veneer unit.
 - 2. After the initial set has begun, do not move the stone units. If moved, remove the units, scrape off the setting mortar on the back of the units as well as on the mortar bed, and then reinstall units as specified.
- E. Install outside corner return units with short and long lengths alternated.
- F. Cutting: Perform necessary field cutting as masonry veneer unit is set. Use proper tools to provide uniform edges; take care to prevent breaking unit corners or edges. Cut lines straight and true, with edges eased slightly to prevent snipping.
- G. Joints: Place units with uniform joints, not exceeding 1/2 inch in width. When additional mortar is required, use a grout bag to fill in joints. When installing pre-fitted stone textures, fit units tight against each other with no allowance for mortar joints.
- H. Take care to avoid smearing mortar on face of stone. Remove accidental smears or mortar dropping only after mortar has become crumbly using a whisk broom or dry bristle brush. Do not use a wet brush or wire brush.
- I. Movement Joints: Locate sealant-filled joints, including expansion, control, contraction, and isolation joints as identified below, during installation of setting materials, mortar beds, and veneer units. Provide joints in width and depth as located and as recommended by manufacturer. Do not saw-cut joints after installing veneer units.

1. Substrate joints must carry through, full width, to surface of masonry veneer.
2. Locate joints in masonry veneer over construction/cold joints or construction joints in substrates.
3. Locate joints in masonry veneer where change occurs in substrate materials.
4. Locate joints where masonry veneer abuts restraining surfaces (such as perimeter walls, curbs, columns, and similar), changes in plane and corners.
5. Locate joints at maximum 10 feet in both directions.

- J. Elastomeric Joint Sealants: Prepare joints and apply elastomeric sealants to comply with grouting material manufacturer's recommendations.

3.9 ADJUSTING, CLEANING AND PROTECTION

- A. Follow Manufacturer's written recommendations specific to the products installed.
- B. In-Progress Cleaning: Clean masonry veneer as work progresses. Remove mortar fins and smears before tooling joints.
- C. Final Cleaning: After mortar is thoroughly set and cured, clean masonry veneer units.
1. Remove large mortar particles by hand with wooden paddles and nonmetallic scrape hoes or chisels.
 2. Protect adjacent masonry and non-masonry surfaces from contact with cleaner.
 3. Remove temporary coverings and protection of adjacent work areas. Remove construction debris from project site and legally dispose of debris.
 4. Cleaning: Use a strong solution of granulated soap or detergent and water with a soft bristle brush. Do not use a wire brush as it will cause damage to the surface. Rinse immediately with clean water. Do not attempt to clean using acid or acid based products. Do not clean with high pressure power washer.
 5. Salt and De-icing Chemicals: Do not use de-icing chemicals on areas immediately adjacent to a masonry veneer application.
 6. Scuffing: Remove scuff marks by cleaning as specified.
 7. Efflorescence: Allow masonry veneer to dry thoroughly, then scrub vigorously with a soft bristle brush and clean water. Rinse thoroughly with clean water and allow to dry. Do not use a wire brush. For difficult efflorescence problems, scrub thoroughly with a solution of 1 part white household vinegar to 5 parts water. Rinse thoroughly with clean water.
- D. Sealer: Apply sealer in multiple coats to full saturation, in accordance to manufacturer's installation instructions.
- E. Protection: Protect installed product and finish surfaces from damage during construction and from direct exposure to rain, direct intense sunlight and high temperature. Follow manufacturer's recommendations.

END OF SECTION 042600

SECTION 047200 - CAST STONE MASONRY**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Cast-stone water table (sill).

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - a. For cast-stone units, include dimensions and finishes.
 - 2. Shop Drawings: Show fabrication and installation details for cast-stone units. Include dimensions, details of reinforcement and anchorages if any, and indication of finished faces.
 - 3. Samples:
 - a. For each color and texture of cast stone required.
 - b. For colored mortar.

1.4 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For manufacturer.
- B. Material Test Reports: For each mix required to produce cast stone, based on testing according to ASTM C 1364, including test for resistance to freezing and thawing.

PART 2 - PRODUCTS**2.1 CAST-STONE UNITS**

- A. Provide product by the following:
 - 1. Contact: See **LIDL National Account List**.
 - 2. Manufacturers and Products:
 - a. Echelon Cordova stone.
 - 3. See arch drawings for shapes and sizes. All shapes and sizes shall be at 2'-0" length.

4. Equivalent product by other manufacturers will be considered due to availability.
- B. Cast-Stone Units: Comply with ASTM C 1364.
 1. Units shall be manufactured using the vibrant dry tamp or wet-cast method.
 2. Units shall be resistant to freezing and thawing as determined by laboratory testing according to ASTM C 666/C 666M, Procedure A, as modified by ASTM C 1364.
- C. Fabricate shaped cast stone with sharp arris and accurately reproduced details, with indicated texture on all exposed surfaces.
 1. Slope exposed horizontal surfaces 1:12 to drain unless otherwise indicated.
 2. Provide raised fillets at backs of sills and at ends indicated to be built into jambs.
 3. Provide drips on projecting elements unless otherwise indicated.
- D. Cure Units as Follows:
 1. Cure units in enclosed, moist curing room at 95 to 100 percent relative humidity and temperature of 100 deg F (38 deg C) for 12 hours or 70 deg F (21 deg C) for 16 hours.
 2. Keep units damp and continue curing to comply with one of the following:
 - a. No fewer than five days at mean daily temperature of 70 deg F (21 deg C) or above prior to shipping
- E. Acid etch units after curing to remove cement film from surfaces to be exposed to view.
- F. Stone Colors and Textures:
 1. Texture: See LIDL National Account List.
 2. Color: See LIDL National Account List.

2.2 ACCESSORIES

- A. Anchors: Type and size indicated, fabricated from Type 304 stainless steel complying with ASTM A 240/A 240M, ASTM A 276, or ASTM A 666.
- B. Dowels: 1/2-inch- (12-mm-) diameter round bars, fabricated from Type 304 stainless steel complying with ASTM A 240/A 240M, ASTM A 276, or ASTM A 666.
- C. Proprietary Acidic Cleaner: Manufacturer's standard-strength cleaner designed for removing mortar/grout stains, efflorescence, and other new construction stains from new masonry without discoloring or damaging masonry surfaces. Use product expressly approved for intended use by cast-stone manufacturer and expressly approved by cleaner manufacturer for use on cast stone and adjacent masonry materials.

2.3 MORTAR

- A. Comply with requirements in Section 042000 "Unit Masonry" for mortar mixes.
 1. For setting mortar, use Type N.
 2. For pointing mortar, use Type N.

- B. Colored Cement Products: Packaged blend made from portland cement and hydrated lime or masonry cement and mortar pigments, all complying with specified requirements, and containing no other ingredients.
 - 1. Basis of Design Product:
 - a. Manufacturer: Essroc Italcementi Group.
 - b. Product: Flamingo-Brixment Colored Masonry Cement
 - c. Colors: See Exterior Material/Finish Schedule on Drawings.
 - 2. Joint Width: As directed by Architect.

PART 3 - EXECUTION

3.1 SETTING CAST STONE IN MORTAR

- A. Install cast-stone units to comply with requirements in Section 042000 "Unit Masonry."
- B. Set units in full bed of mortar with full head joints unless otherwise indicated.
 - 1. Fill dowel holes and anchor slots with mortar.
 - 2. Fill collar joints solid as units are set.
 - 3. Build concealed flashing into mortar joints as units are set.
 - 4. Keep joints at shelf angles open to receive sealant.
- C. Rake out joints for pointing with mortar to depths of not less than 3/4 inch (19 mm). Rake joints to uniform depths with square bottoms and clean sides. Scrub faces of units to remove excess mortar as joints are raked.
- D. Point mortar joints by placing and compacting mortar in layers not greater than 3/8 inch (10 mm). Compact each layer thoroughly and allow it to become thumbprint hard before applying next layer.
- E. Tool exposed joints slightly concave when thumbprint hard. Use a smooth plastic jointer larger than joint thickness.

3.2 INSTALLATION TOLERANCES

- A. Variation from Plumb: Do not exceed 1/8 inch in 10 feet (3 mm in 3 m), 1/4 inch in 20 feet (6 mm in 6 m), or 1/2 inch (12 mm) maximum.
- B. Variation from Level: Do not exceed 1/8 inch in 10 feet (3 mm in 3 m), 1/4 inch in 20 feet (6 mm in 6 m), or 1/2 inch (12 mm) maximum.
- C. Variation in Joint Width: Do not vary joint thickness more than 1/8 inch in 36 inches (3 mm in 900 mm) or one-fourth of nominal joint width, whichever is less.
- D. Variation in Plane between Adjacent Surfaces (Lipping): Do not vary from flush alignment with adjacent units or adjacent surfaces indicated to be flush with units by more than 1/16 inch (1.5 mm), except where variation is due to warpage of units within tolerances specified.

3.3 ADJUSTING AND CLEANING

- A. Remove and replace stained and otherwise damaged units and units not matching approved Samples. Cast stone may be repaired if methods and results are approved by Architect.
- B. Replace units in a manner that results in cast stone matching approved Samples, complying with other requirements, and showing no evidence of replacement.
- C. In-Progress Cleaning: Clean cast stone as work progresses.
 - 1. Remove mortar fins and smears before tooling joints.
 - 2. Remove excess sealant immediately, including spills, smears, and spatter.
- D. Final Cleaning: After mortar is thoroughly set and cured, clean exposed cast stone as follows:
 - 1. Remove large mortar particles by hand with wooden paddles and nonmetallic scrape hoes or chisels.
 - 2. Test cleaning methods on sample; leave one sample uncleaned for comparison purposes. Obtain Architect's approval of sample cleaning before proceeding with cleaning of cast stone.
 - 3. Protect adjacent surfaces from contact with cleaner by covering them with liquid strippable masking agent or polyethylene film and waterproof masking tape.
 - 4. Wet surfaces with water before applying cleaners; remove cleaners promptly by rinsing thoroughly with clear water.
 - 5. Clean cast stone with proprietary acidic cleaner applied according to manufacturer's written instructions.

END OF SECTION 047200

SECTION 051200 - STRUCTURAL STEEL FRAMING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Structural steel.
 - 2. Shrinkage-resistant grout.
- B. Related Requirements:
 - 1. Section 052100 "Steel Joist Framing".
 - 2. Section 053100 "Steel Decking".
 - 3. Section 055000 "Metal Fabrications".
 - 4. Section 099123 "Interior Painting"

1.3 DEFINITIONS

- A. Structural Steel: Elements of the structural frame indicated on Drawings and as described in ANSI/AISC 303.

1.4 COORDINATION

- A. Coordinate selection of shop primers with topcoats to be applied over them. Comply with paint and coating manufacturers' written recommendations to ensure that shop primers and topcoats are compatible with one another.
- B. Coordinate installation of anchorage items to be embedded in or attached to other construction without delaying the Work. Provide setting diagrams, sheet metal templates, instructions, and directions for installation.

1.5 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.6 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - a. Structural-steel materials.
 - b. High-strength, bolt-nut-washer assemblies.

- c. Anchor rods.
 - d. Threaded rods.
 - e. Shop primer.
 - f. Galvanized-steel primer.
 - g. Galvanized repair paint.
 - h. Shrinkage-resistant grout.
- 2. Shop Drawings: Show fabrication of structural-steel components.
 - a. Include details of cuts, connections, splices, camber, holes, and other pertinent data.
 - b. Include embedment Drawings.
 - c. Indicate welds by standard AWS symbols, distinguishing between shop and field welds, and show size, length, and type of each weld. Show backing bars that are to be removed and supplemental fillet welds where backing bars are to remain.
 - d. Indicate type, size, and length of bolts, distinguishing between shop and field bolts.
- 3. Delegated-Design Submittal: For structural-steel connections indicated on Drawings to comply with design loads, include analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1.7 INFORMATIONAL SUBMITTALS

- A. Welding Procedure Specifications (WPSs) and Procedure Qualification Records (PQRs): Provide in accordance with AWS D1.1/D1.1M for each welded joint whether prequalified or qualified by testing, including the following:
 - 1. Power source (constant current or constant voltage).
- B. Qualification Data: For Installer, fabricator and testing agency.
- C. Welding certificates.
- D. Paint Compatibility Certificates: From manufacturers of topcoats applied over shop primers, certifying that shop primers are compatible with topcoats.
- E. Mill test reports for structural steel, including chemical and physical properties.
- F. Source quality-control reports.
- G. Field quality-control and special inspection reports.

1.8 QUALITY ASSURANCE

- A. Fabricator Qualifications: A qualified fabricator that participates in the AISC Quality Certification Program and is designated an AISC-Certified Plant, Category STD, or is accredited by the IAS Fabricator Inspection Program for Structural Steel (AC 172).
- B. Installer Qualifications: A qualified Installer who participates in the AISC Quality Certification Program and is designated an AISC-Certified Erector, Category ACSE.
- C. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

1.9 DELIVERY, STORAGE, AND HANDLING

- A. Store materials to permit easy access for inspection and identification. Keep steel members off ground and spaced by using pallets, dunnage, or other supports and spacers. Protect steel members and packaged materials from corrosion and deterioration.
 - 1. Do not store materials on structure in a manner that might cause distortion, damage, or overload to members or supporting structures. Repair or replace damaged materials or structures as directed.
- B. Store fasteners in a protected place in sealed containers with manufacturer's labels intact.
 - 1. Fasteners may be repackaged provided Owner's testing and inspecting agency observes repackaging and seals containers.
 - 2. Clean and re-lubricate bolts and nuts that become dry or rusty before use.
 - 3. Comply with manufacturers' written recommendations for cleaning and lubricating ASTM F3125/F3125M, Grade F1852 bolt assemblies and for retesting bolt assemblies after lubrication.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Comply with applicable provisions of the following specifications and documents:
 - 1. ANSI/AISC 360.
 - 2. RCSC's "Specification for Structural Joints Using High-Strength Bolts."
- B. Connection Design Information:
 - 1. Option 1: Connection designs have been completed and connections indicated on the Drawings.
- C. Construction: Shear wall system.

2.2 STRUCTURAL-STEEL MATERIALS

- A. W-Shapes: ASTM A 992/A 992M
- B. Channels, Angles, Shapes: ASTM A 36/A 36M.
- C. Plate and Bar: ASTM A 36/A 36M.
- D. Cold-Formed Hollow Structural Sections: ASTM A 500/A 500M, Grade B, structural tubing.
- E. Steel Pipe: ASTM A 53/A 53M, Type E or Type S, Grade B.
- F. Welding Electrodes: Comply with AWS requirements.

2.3 BOLTS AND CONNECTORS

- A. High-Strength A325 Bolts, Nuts, and Washers: ASTM F3125/F3125M, Grade A325 (Grade A325M), Type 1, heavy-hex steel structural bolts; ASTM A563, Grade DH

(ASTM A563M, Class 10S), heavy-hex carbon-steel nuts; and ASTM F436/F436M, Type 1, hardened carbon-steel washers; all with plain finish.

1. Direct-Tension Indicators: ASTM F959/F959M, Type 325-1 (Type 8.8-1), compressible-washer type with plain finish.

- B. Shear Stud Connectors: ASTM A108, AISI C-1015 through C-1020, headed-stud type, cold-finished carbon steel; AWS D1.1/D1.1M, Type B.

2.4 RODS

- A. Headed Anchor Rods: ASTM F1554, Grade 36, straight.

1. Nuts: ASTM A563 (ASTM A563M) heavy-hex carbon steel.
2. Plate Washers: ASTM A36/A36M carbon steel.
3. Washers: ASTM F436 (ASTM F436M), Type 1, hardened carbon steel.
4. Finish: Hot-dip zinc coating, ASTM A153/A153M, Class C.

- B. Threaded Rods: ASTM A36/A36M.

1. Nuts: ASTM A63 (ASTM A563M) heavy-hex carbon steel.
2. Washers: ASTM F436 (ASTM F436M), Type 1, hardened.
3. Finish: Hot-dip zinc coating, ASTM A153/A153M, Class C.

2.5 PRIMER

- A. Shop Primer: Shop primed grey and field painted. Fabricator's standard lead- and chromate-free, non-asphaltic, rust-inhibiting primer complying with MPI#79 and compatible with topcoat.

- B. Galvanizing Repair Paint: MPI#18, MPI#19, or SSPC-Paint 20.

- C. Painting: Apply one coat of shop primer to structural steel frame and accessories. Apply two coats of paint at ceiling sales area on all exposed structural members.

1. Color: See Drawings.

2.6 SHRINKAGE-RESISTANT GROUT

- A. Nonmetallic, Shrinkage-Resistant Grout: ASTM C 1107/C 1107M, factory-packaged, nonmetallic aggregate grout, noncorrosive and nonstaining, mixed with water to consistency suitable for application and a 30-minute working time.

2.7 FABRICATION

- A. Structural Steel: Fabricate and assemble in shop to greatest extent possible. Fabricate in accordance with ANSI/AISC 303 and to ANSI/AISC 360.

1. Fabricate beams with rolling camber up.
2. Identify high-strength structural steel in accordance with ASTM A6/A6M and maintain markings until structural-steel framing has been erected.
3. Mark and match-mark materials for field assembly.
4. Complete structural-steel assemblies, including welding of units, before starting shop-priming operations.

- B. Thermal Cutting: Perform thermal cutting by machine to greatest extent possible.
 - 1. Plane thermally cut edges to be welded to comply with requirements in AWS D1.1/D1.1M.
- C. Bolt Holes: Cut, drill, mechanically thermal cut, or punch standard bolt holes perpendicular to metal surfaces.
- D. Finishing: Accurately finish ends of columns and other members transmitting bearing loads.
- E. Cleaning: Clean and prepare steel surfaces that are to remain unpainted in accordance with SSPC-SP 3.
- F. Holes: Provide holes required for securing other work to structural steel and for other work to pass through steel members.
 - 1. Cut, drill, or punch holes perpendicular to steel surfaces.
 - 2. Baseplate Holes: Cut, drill, mechanically thermal cut, or punch holes perpendicular to steel surfaces.
 - 3. Weld threaded nuts to framing and other specialty items indicated to receive other work.

2.8 SHOP CONNECTIONS

- A. High-Strength Bolts: Shop install high-strength bolts in accordance with RCSC's "Specification for Structural Joints Using High-Strength Bolts" for type of bolt and type of joint specified.
 - 1. Joint Type: Snug tightened.
- B. Weld Connections: Comply with AWS D1.1/D1.1M and AWS D1.8/D1.8M for tolerances, appearances, welding procedure specifications, weld quality, and methods used in correcting welding work.
 - 1. Assemble and weld built-up sections by methods that maintain true alignment of axes without exceeding tolerances in ANSI/AISC 303 for mill material.

2.9 GALVANIZING

- A. Hot-Dip Galvanized Finish: Apply zinc coating by the hot-dip process to all exterior structural steel according to ASTM A123/A123M.
 - 1. Fill vent and drain holes that are exposed in the finished Work unless they function as weep holes, by plugging with zinc solder and filing off smooth.
 - 2. Galvanize lintels and shelf angles attached to structural-steel frame and located in exterior walls.

2.10 SHOP PRIMING

- A. Shop prime interior steel surfaces except the following:
 - 1. Surfaces embedded in concrete or mortar. Extend priming of partially embedded members to a depth of 2 inches (50 mm).
 - 2. Surfaces to be field welded.
 - 3. Surfaces of high-strength bolted, slip-critical connections.
 - 4. Surfaces to receive sprayed fire-resistive materials (applied fireproofing).

5. Galvanized surfaces unless indicated to be painted.
 6. Surfaces enclosed in interior construction.
- B. Surface Preparation: Clean surfaces to be painted. Remove loose rust and mill scale and spatter, slag, or flux deposits. Prepare surfaces according to the following specifications and standards:
1. SSPC-SP 3, "Power Tool Cleaning."
- C. Surface Preparation of Galvanized Steel: Prepare galvanized-steel surfaces for shop priming by thoroughly cleaning steel of grease, dirt, oil, flux, and other foreign matter, and treating with etching cleaner or in accordance with SSPC-SP 16.
- D. Priming: Immediately after surface preparation, apply primer according to manufacturer's written instructions and at rate recommended by SSPC to provide a minimum dry film thickness of 1.5 mils (0.038 mm). Use priming methods that result in full coverage of joints, corners, edges, and exposed surfaces.
1. Stripe paint corners, crevices, bolts, welds, and sharp edges.
 2. Apply two coats of shop paint to surfaces that are inaccessible after assembly or erection. Change color of second coat to distinguish it from first.

2.11 SOURCE QUALITY CONTROL

- A. Testing Agency: Owner will engage a qualified testing agency to perform shop tests and inspections.
1. Provide testing agency with access to places where structural-steel work is being fabricated or produced to perform tests and inspections.
 2. Bolted Connections: Inspect shop-bolted connections in accordance with RCSC's "Specification for Structural Joints Using High-Strength Bolts."
 3. Welded Connections: Visually inspect shop-welded connections in accordance with AWS D1.1/D1.1M and the following inspection procedures, at testing agency's option:
 - a. Liquid Penetrant Inspection: ASTM E165/E165M.
 - b. Magnetic Particle Inspection: ASTM E709; performed on root pass and on finished weld. Cracks or zones of incomplete fusion or penetration are not accepted.
 - c. Ultrasonic Inspection: ASTM E164.
 - d. Radiographic Inspection: ASTM E94/E94M.
 4. Prepare test and inspection reports.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify elevations of concrete- and masonry-bearing surfaces and locations of anchor rods, bearing plates, and other embedments for compliance with requirements.
1. Prepare a certified survey of existing conditions. Include bearing surfaces, anchor rods, bearing plates, and other embedments showing dimensions, locations, angles, and elevations.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Provide temporary shores, guys, braces, and other supports during erection to keep structural steel secure, plumb, and in alignment against temporary construction loads and loads equal in intensity to design loads. Remove temporary supports when permanent structural steel, connections, and bracing are in place unless otherwise indicated on Drawings.
 - 1. Do not remove temporary shoring supporting composite deck construction and structural-steel framing until cast-in-place concrete has attained its design compressive strength.

3.3 ERECTION

- A. Set structural steel accurately in locations and to elevations indicated and in accordance with ANSI/AISC 303 and ANSI/AISC 360.
- B. Baseplates, Bearing Plates and Leveling Plates: Clean concrete- and masonry-bearing surfaces of bond-reducing materials, and roughen surfaces prior to setting plates. Clean bottom surface of plates.
 - 1. Set plates for structural members on wedges, shims, or setting nuts as required.
 - 2. Weld plate washers to top of baseplate.
 - 3. Snug-tighten anchor rods after supported members have been positioned and plumbed. Do not remove wedges or shims but, if protruding, cut off flush with edge of plate before packing with grout.
 - 4. Promptly pack shrinkage-resistant grout solidly between bearing surfaces and plates, so no voids remain. Neatly finish exposed surfaces; protect grout and allow to cure. Comply with manufacturer's written installation instructions for grouting.
- C. Maintain erection tolerances of structural steel within ANSI/AISC 303.
- D. Align and adjust various members that form part of complete frame or structure before permanently fastening. Before assembly, clean bearing surfaces and other surfaces that are in permanent contact with members. Perform necessary adjustments to compensate for discrepancies in elevations and alignment.
 - 1. Level and plumb individual members of structure. Slope roof framing members to slopes indicated on Drawings.
 - 2. Make allowances for difference between temperature at time of erection and mean temperature when structure is completed and in service.
- E. Splice members only where indicated.
- F. Do not use thermal cutting during erection.
- G. Do not enlarge unfair holes in members by burning or using drift pins. Ream holes that must be enlarged to admit bolts.

3.4 FIELD CONNECTIONS

- A. High-Strength Bolts: Install high-strength bolts in accordance with RCSC's "Specification for Structural Joints Using High-Strength Bolts" for bolt and joint type specified.
 - 1. Joint Type: Snug tightened.

- B. Weld Connections: Comply with AWS D1.1/D1.1M and AWS D1.8/D1.8M for tolerances, appearances, welding procedure specifications, weld quality, and methods used in correcting welding work.
 - 1. Comply with ANSI/AISC 303 and ANSI/AISC 360 for bearing, alignment, adequacy of temporary connections, and removal of paint on surfaces adjacent to field welds.
 - 2. Remove backing bars or runoff tabs where indicated, back gouge, and grind steel smooth.
 - 3. Assemble and weld built-up sections by methods that maintain true alignment of axes without exceeding tolerances in ANSI/AISC 303 for mill material.

3.5 REPAIR

- A. Galvanized Surfaces: Clean areas where galvanizing is damaged or missing, and repair galvanizing to comply with ASTM A780/A780M.
- B. Touchup Painting:
 - 1. Immediately after erection, clean exposed areas where primer is damaged or missing, and paint with the same material as used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.
 - a. Clean and prepare surfaces by SSPC-SP 2 hand-tool cleaning or SSPC-SP 3 power-tool cleaning.

3.6 FIELD QUALITY CONTROL

- A. Special Inspections: Owner will engage a qualified special inspector to perform the following special inspections:
 - 1. Verify structural-steel materials and inspect steel frame joint details.
 - 2. Verify weld materials and inspect welds.
 - 3. Verify connection materials and inspect high-strength bolted connections.
- B. Testing Agency: Owner will engage a qualified testing agency to perform tests and inspections.
 - 1. Bolted Connections: Inspect bolted connections in accordance with RCSC's "Specification for Structural Joints Using High-Strength Bolts."
 - 2. Welded Connections: Visually inspect field welds in accordance with AWS D1.1/D1.1M.
 - a. In addition to visual inspection, test and inspect field welds in accordance with AWS D1.1/D1.1M and the following inspection procedures, at testing agency's option:
 - 1) Liquid Penetrant Inspection: ASTM E165/E165M.
 - 2) Magnetic Particle Inspection: ASTM E709; performed on root pass and on finished weld. Cracks or zones of incomplete fusion or penetration are not accepted.
 - 3) Ultrasonic Inspection: ASTM E164.
 - 4) Radiographic Inspection: ASTM E94/E94M.

END OF SECTION 051200

SECTION 052100 - STEEL JOIST FRAMING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY**A. Section Includes:**

1. K-series steel joists.
2. Steel joist girders.
3. Steel joist accessories.

B. Related Requirements:

1. Section 042000 "Unit Masonry" for installing bearing plates in unit masonry.
2. Section 051200 "Structural Steel Framing" for field-welded shear connectors.

1.3 DEFINITIONS

- A. SJI's "Specifications": Steel Joist Institute's "Standard Specifications, Load Tables and Weight Tables for Steel Joists and Joist Girders."
- B. Special Joists: Steel joists or joist girders requiring modification by manufacturer to support nonuniform, unequal, or special loading conditions that invalidate load tables in SJI's "Specifications."

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 1. Product Data: For each type of joist, accessory, and product.
- C. Shop Drawings:
 1. Include layout, designation, number, type, location, and spacing of joists.
 2. Include joining and anchorage details; bracing, bridging, and joist accessories; splice and connection locations and details; and attachments to other construction.
 3. Indicate locations and details of bearing plates to be embedded in other construction.
 4. Include design calculations with the appropriate shop drawings sealed by a local licensed structural engineer for review by the architect and engineer.

1.5 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For manufacturer.

- B. Welding certificates.
- C. Manufacturer certificates.
- D. Mill Certificates: For each type of bolt.
- E. Comprehensive engineering analysis of special joists signed and sealed by the qualified professional engineer responsible for its preparation.
- F. Field quality-control reports.

1.6 QUALITY ASSURANCE

- A. Manufacturer Qualifications: A manufacturer certified by SJI to manufacture joists complying with applicable standard specifications and load tables in SJI's "Specifications."
 - 1. Manufacturer's responsibilities include providing professional engineering services for designing special joists to comply with performance requirements.
- B. Welding Qualifications: Qualify field-welding procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle joists as recommended in SJI's "Specifications."
- B. Protect joists from corrosion, deformation, and other damage during delivery, storage, and handling.

1.8 SEQUENCING

- A. Deliver steel bearing plates to be built into masonry construction.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Structural Performance: Provide special joists and connections capable of withstanding design loads indicated on Drawings.
 - 1. Use ASD; data are given at service-load level.
 - 2. Design special joists to withstand design loads with live-load deflections no greater than the following:
 - a. Roof Joists: Vertical deflection of $L/240$ of the span.

2.2 MANUFACTURER

- A. Manufacturer: Vulcraft.
- B. Contact: See **LIDL National Account List**.

2.3 STEEL JOISTS

- A. K-Series Steel Joist: Manufactured steel joists of type indicated according to "Standard Specification for Open Web Steel Joists, K-Series" in SJI's "Specifications," with steel-angle top- and bottom-chord members, underslung ends, and parallel top chord.
 - 1. Joist Type: K-series steel joists.
 - 2. Provide holes in chord members for connecting and securing other construction to joists.
 - 3. Extended Ends: Extend bearing ends of joists with SJI's Type R extended ends where indicated on Drawings, complying with SJI's "Specifications."
 - 4. Camber joists according to SJI's "Specifications."
 - 5. Equip bearing ends of joists with manufacturer's standard beveled ends or sloped shoes if joist slope exceeds 1/4 inch per 12 inches (1:48).

2.4 PRIMERS

- A. Primer:
 - 1. SSPC-Paint 15, or manufacturer's standard shop primer complying with performance requirements in SSPC-Paint 15.

2.5 JOIST ACCESSORIES

- A. Bridging:
 - 1. Provide bridging anchors and number of rows of horizontal or diagonal bridging of material, size, and type required by SJI's "Specifications" for type of joist, chord size, spacing, and span. Furnish additional erection bridging if required for stability.
 - 2. Schematically indicated. Detail and fabricate according to SJI's "Specifications." Furnish additional erection bridging if required for stability.
- B. Fabricate steel bearing plates from ASTM A36/A36M steel with integral anchorages of sizes and thicknesses indicated on Drawings. Hot-dip zinc coat according to ASTM A123/A123M.
- C. Furnish ceiling extensions, either extended bottom-chord elements or a separate extension unit of enough strength to support ceiling construction.
 - 1. Extend ends to within 1/2 inch (13 mm) of finished wall surface unless otherwise indicated on Drawings.
 - 2. Finish: Plain, uncoated.
- D. High-Strength Bolts, Nuts, and Washers: ASTM F3125/F3125M, Grade A325 (Grade A325M), Type 1, heavy-hex steel structural bolts; ASTM A563, Grade DH, (ASTM A563M, Class 10S) heavy-hex carbon-steel nuts; and ASTM F436/F436M, Type 1, hardened carbon-steel washers.
 - 1. Finish: Plain.
- E. Welding Electrodes: Comply with AWS standards.
- F. Furnish miscellaneous accessories including splice plates and bolts required by joist manufacturer to complete joist assembly.

2.6 CLEANING AND SHOP PAINTING

- A. Clean and remove loose scale, heavy rust, and other foreign materials from fabricated joists and accessories by hand-tool cleaning, SSPC-SP 2.

- B. Do not prime paint joists and accessories to receive sprayed fire-resistive materials.
- C. Apply one coat of shop primer to joists and joist accessories to be primed to provide a continuous, dry paint film not less than 1 mil (0.025 mm) thick.
- D. Apply two coats of paint at ceiling sales area on all exposed structural members as specified in Section 099123 "Interior Painting".
 - 1. Color: See Drawings.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine supporting substrates, embedded bearing plates, and abutting structural framing for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Do not install joists until supporting construction is in place and secured.
- B. Install joists and accessories plumb, square, and true to line; securely fasten to supporting construction according to SJI's "Specifications," joist manufacturer's written instructions, and requirements in this Section.
 - 1. Before installation, splice joists delivered to Project site in more than one piece.
 - 2. Space, adjust, and align joists accurately in location before permanently fastening.
 - 3. Install temporary bracing and erection bridging, connections, and anchors to ensure that joists are stabilized during construction.
 - 4. Delay rigidly connecting bottom-chord extensions to columns or supports until dead loads are applied.
- C. Field weld joists to supporting steel bearing plates and framework. Coordinate welding sequence and procedure with placement of joists. Comply with AWS requirements and procedures for welding, appearance and quality of welds, and methods used in correcting welding work.
- D. Bolt joists to supporting steel framework using carbon-steel bolts.
- E. Bolt joists to supporting steel framework using high-strength structural bolts. Comply with RCSC's "Specification for Structural Joints Using ASTM A 325 or ASTM A 490 Bolts" for high-strength structural bolt installation and tightening requirements.
- F. Install and connect bridging concurrently with joist erection, before construction loads are applied. Anchor ends of bridging lines at top and bottom chords if terminating at walls or beams.

3.3 REPAIRS

- A. Repair damaged galvanized coatings on galvanized items with galvanized repair paint according to ASTM A780/A780M and manufacturer's written instructions.

B. Touchup Painting:

1. Immediately after installation, clean, prepare, and prime or reprime field connections, rust spots, and abraded surfaces of prime-painted joists, bearing plates, abutting structural steel, and accessories.
 - a. Clean and prepare surfaces by SSPC-SP 2 hand-tool cleaning or SSPC-SP 3 power-tool cleaning.
 - b. Apply a compatible primer of same type as primer used on adjacent surfaces.

3.4 FIELD QUALITY CONTROL

- A. Testing Agency: Owner will engage a qualified testing agency to perform tests and inspections.
- B. Visually inspect field welds according to AWS D1.1/D1.1M.
 1. In addition to visual inspection, test field welds according to AWS D1.1/D1.1M and the following procedures, at testing agency's option:
 - a. Liquid Penetrant Inspection: ASTM E165/E165M.
 - b. Magnetic Particle Inspection: ASTM E709.
 - c. Ultrasonic Testing: ASTM E164.
 - d. Radiographic Testing: ASTM E94.
- C. Visually inspect bolted connections.
- D. Prepare test and inspection reports.

END OF SECTION 052100

SECTION 053100 - STEEL DECKING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Roof deck.

- B. Related Requirements:

- 1. Section 055000 "Metal Fabrications" for framing deck openings with miscellaneous steel shapes.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.

- B. Submit the following Submittals via Owner's Project Management software platform.

- 1. Product Data: For each type of deck, accessory, and product indicated.
 - 2. Shop Drawings:

- a. Include layout and types of deck panels, anchorage details, reinforcing channels, pans, cut deck openings, special jointing, accessories, and attachments to other construction.

1.4 INFORMATIONAL SUBMITTALS

- A. Welding certificates.

- B. Product Certificates: For each type of steel deck.

- C. Product Test Reports: For tests performed by a qualified testing agency, indicating that each of the following complies with requirements:

- 1. Power-actuated mechanical fasteners.

- D. Field quality-control reports.

1.5 QUALITY ASSURANCE

- A. Testing Agency Qualifications: Qualified according to ASTM E 329 for testing indicated.

- B. Welding Qualifications: Qualify procedures and personnel according to AWS D1.3/D1.3M, "Structural Welding Code - Sheet Steel."

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Protect steel deck from corrosion, deformation, and other damage during delivery, storage, and handling.
- B. Stack steel deck on platforms or pallets and slope to provide drainage. Protect with a waterproof covering and ventilate to avoid condensation.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. AISI Specifications: Comply with calculated structural characteristics of steel deck according to AISI's "North American Specification for the Design of Cold-Formed Steel Structural Members."

2.2 MANUFACTURER

- A. Manufacturer: Vulcraft.
- B. Contact: See **LIDL National Account List**.

2.3 ROOF DECK

- A. Roof Deck: Fabricate panels, without top-flange stiffening grooves, to comply with "SDI Specifications and Commentary for Steel Roof Deck," in SDI Publication No. 31, and with the following:
 - 1. Galvanized and Shop-Primed Steel Sheet: ASTM A 653/A 653M, Structural Steel (SS), Grade 80 (550), G60 (Z180) zinc coating; cleaned, pretreated, and primed with manufacturer's standard baked-on, rust-inhibitive primer.
 - 2. Deck Profile: As indicated per structural drawings.
 - 3. Profile Depth: As indicated, per structural drawings.
 - 4. Design Uncoated-Steel Thickness: As indicated per structural drawings.
 - 5. Span Condition: Double span minimum.
 - 6. Side Laps: Overlapped.
- B. Paint: Field-apply two coats of paint at ceiling sales area on all exposed steel deck as specified in Division 9, Painting Sections.
 - 1. Color: See Drawings.

2.4 ACCESSORIES

- A. Provide manufacturer's standard accessory materials for deck that comply with requirements indicated.
- B. Mechanical Fasteners: Corrosion-resistant, low-velocity, power-actuated or pneumatically driven carbon-steel fasteners; or self-drilling, self-threading screws.
- C. Side-Lap Fasteners: Corrosion-resistant, hexagonal washer head; self-drilling, carbon-steel screws, No. 10 (4.8-mm) minimum diameter.
- D. Flexible Closure Strips: Vulcanized, closed-cell, synthetic rubber.

- E. Miscellaneous Sheet Metal Deck Accessories: Steel sheet, minimum yield strength of 33,000 psi (230 MPa), not less than 0.0359-inch (0.91-mm) design uncoated thickness, of same material and finish as deck; of profile indicated or required for application.
- F. Column Closures, End Closures, Z-Closures, and Cover Plates: Steel sheet, of same material, finish, and thickness as deck unless otherwise indicated.
- G. Primer: Fabricator's standard lead- and chromate-free, non-asphaltic, rust-inhibiting primer complying with MPI#79 and compatible with topcoat.
- H. Galvanizing Repair Paint: ASTM A780/A780M.
- I. Repair Paint: Manufacturer's standard rust-inhibitive primer and match the field color.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine supporting frame and field conditions for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION, GENERAL

- A. Install deck panels and accessories according to applicable specifications and commentary in SDI Publication No. 31, manufacturer's written instructions, and requirements in this Section.
- B. Locate deck bundles to prevent overloading of supporting members.
- C. Place deck panels on supporting frame and adjust to final position with ends accurately aligned and bearing on supporting frame before being permanently fastened. Do not stretch or contract side-lap interlocks.
- D. Place deck panels flat and square and fasten to supporting frame without warp or deflection.
- E. Cut and neatly fit deck panels and accessories around openings and other work projecting through or adjacent to deck.
- F. Provide additional reinforcement and closure pieces at openings as required for strength, continuity of deck, and support of other work.
- G. Comply with AWS requirements and procedures for manual shielded metal arc welding, appearance and quality of welds, and methods used for correcting welding work.

3.3 INSTALLATION OF ROOF DECK

- A. Fasten roof-deck panels to steel supporting members by arc spot (puddle) welds of the surface diameter indicated or arc seam welds with an equal perimeter that is not less than 1-1/2 inches (38 mm) long, and as follows:
 - 1. Weld Diameter: 5/8 inch (16 mm), nominal.
 - 2. Weld Spacing: As indicated on drawings.

- B. Side-Lap and Perimeter Edge Fastening: Fasten side laps and perimeter edges of panels between supports, at 12 inches, and as follows:
 - 1. Mechanically fasten with self-drilling, No. 10 (4.8-mm-) diameter or larger, carbon-steel screws.
 - 2. Fasten with a minimum of 1-1/2-inch- (38-mm-) long welds.
- C. End Bearing: Install deck ends over supporting frame with a minimum end bearing of 1-1/2 inches (38 mm), with end joints as follows:
 - 1. End Joints: Lapped 2 inches (51 mm) minimum
- D. Miscellaneous Roof-Deck Accessories: Install ridge and valley plates, finish strips, end closures, and reinforcing channels according to deck manufacturer's written instructions. Weld or mechanically fasten to substrate to provide a complete deck installation.
 - 1. Weld cover plates at changes in direction of roof-deck panels unless otherwise indicated.
- E. All roof and deck penetrations shall be temporary covered and watertight sealed until the roof units are installed.

3.4 REPAIR

- A. Galvanizing Repairs: Prepare and repair damaged galvanized coatings on both surfaces of deck with galvanized repair paint according to ASTM A780/A780M and manufacturer's written instructions.
- B. Repair Painting:
 - 1. Wire brush and clean rust spots, welds, and abraded areas on both surfaces of prime-painted deck immediately after installation and apply repair paint.

3.5 FIELD QUALITY CONTROL

- A. Testing Agency: Owner will engage a qualified testing agency to perform tests and inspections.
- B. Field welds will be subject to inspection.
- C. Prepare test and inspection reports.

END OF SECTION 053100

SECTION 054000 - COLD-FORMED METAL FRAMING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Exterior non-load-bearing wall framing.
 - 2. Interior non-load-bearing wall framing exceeding height limitations of standard, nonstructural metal framing.
- B. Related Requirements:
 - 1. Section 055000 "Metal Fabrications" for miscellaneous steel shapes, masonry shelf angles, and connections used with cold-formed metal framing.
 - 2. Section 092216 "Non-Structural Metal Framing" for standard, interior non-load-bearing, metal-stud framing, with height limitations and ceiling-suspension assemblies.

1.3 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at project site.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings:
 - a. Provide shop drawings and calculations signed and sealed by a professional engineer registered in the project's jurisdiction.
 - b. Include layout, spacings, sizes, thicknesses, and types of cold-formed steel framing; fabrication; and fastening and anchorage details, including mechanical fasteners.
 - c. Indicate reinforcing channels, opening framing, supplemental framing, strapping, bracing, bridging, splices, accessories, connection details, and attachment to adjoining work.
 - 3. Delegated-Design Submittal: For cold-formed metal framing.

1.5 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For testing agency.

- B. Product Certificates: For each type of code-compliance certification for studs and tracks.
- C. Product Test Reports.

1.6 QUALITY ASSURANCE

- A. Testing Agency Qualifications: Qualified according to ASTM E 329 for testing indicated.
- B. Product Tests: Mill certificates or data from a qualified independent testing agency, indicating steel sheet complies with requirements, including base-metal thickness, yield strength, tensile strength, total elongation, chemical requirements, and metallic-coating thickness.
- C. Welding Qualifications: Qualify procedures and personnel according to the following:
 - 1. AWS D1.1/D1.1M, "Structural Welding Code - Steel."
 - 2. AWS D1.3/D1.3M, "Structural Welding Code - Sheet Steel."

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design cold-formed steel framing.
- B. Structural Performance: Provide cold-formed steel framing capable of withstanding design loads within limits and under conditions indicated.
 - 1. Design Loads: As indicated.
 - 2. Deflection Limits: Design framing systems to withstand design loads without deflections greater than the following:
 - a. Exterior Non-Load-Bearing Framing: Horizontal deflection of L/600 of the wall height.
 - b. Interior Non-Load-Bearing Framing: Horizontal deflection of L/240 of the wall height under a horizontal load of 5 lbf/sq. ft. (239 Pa).
 - 3. Design framing systems to provide for movement of framing members located outside the insulated building envelope without damage or overstressing, sheathing failure, connection failure, undue strain on fasteners and anchors, or other detrimental effects when subject to a maximum ambient temperature change of 120 deg F (67 deg C).
 - 4. Design framing system to maintain clearances at openings, to allow for construction tolerances, and to accommodate live load deflection of primary building structure as follows:
 - a. Upward and downward movement of 3/4 inch (19 mm).
 - 5. Design exterior non-load-bearing wall framing to accommodate horizontal deflection without regard for contribution of sheathing materials.
- C. Cold-Formed Steel Framing Standards: Unless more stringent requirements are indicated, framing shall comply with AISI S100, AISI S200, and the following:
 - 1. Wall Studs: AISI S211.
 - 2. Headers: AISI S212.

3. Lateral Design: AISI S213.

D. Fire-Resistance Ratings: Comply with ASTM E 119; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

1. Indicate design designations from UL's "Fire Resistance Directory" or from the listings of another qualified testing agency.

2.2 COLD-FORMED STEEL FRAMING MATERIALS

A. Steel Sheet: ASTM A 1003/A 1003M, Structural Grade, Type H, metallic coated, of grade and coating designation as follows:

1. Grade: As required by structural performance.
2. Coating: G60 (Z180).

B. Steel Sheet for Clips: ASTM A 653/A 653M, structural steel, zinc coated, of grade and coating as follows:

1. Grade: As required by structural performance.
2. Coating: G60 (Z180).

2.3 EXTERIOR NON-LOAD-BEARING WALL FRAMING

A. Steel Studs: Manufacturer's standard C-shaped steel studs, of web depths indicated, punched, with stiffened flanges, and as follows:

1. Minimum Base-Metal Thickness: 0.0428 inch (1.09 mm).
2. Minimum Flange Width: 1-3/8 inches (35 mm).
3. Maximum Stud Spacing: 16 inches (406 mm).
4. Section Properties: As noted on drawings.

B. Steel Track: Manufacturer's standard U-shaped steel track, of web depths indicated, unpunched, with unstiffened flanges, and matching minimum base-metal thickness of steel studs.

C. Vertical Deflection Clips: Manufacturer's standard head clips, capable of accommodating upward and downward vertical displacement of primary structure through positive mechanical attachment to stud web.

D. Drift Clips: Manufacturer's standard bypass or head clips, capable of isolating wall stud from upward and downward vertical displacement and lateral drift of primary structure through positive mechanical attachment to stud web and structure.

2.4 INTERIOR NON-LOAD-BEARING WALL FRAMING

A. Steel Studs: Manufacturer's standard C-shaped steel studs, of web depths indicated, punched, with stiffened flanges, and as follows:

1. Minimum Base-Metal Thickness: 0.0329 inch (0.84 mm).
2. Flange Width: 1-3/8 inches (35 mm).
3. Maximum Stud Spacing: 16 inches (406 mm).

- B. Steel Track: Manufacturer's standard U-shaped steel track, of web depths indicated, unpunched, with unstiffened flanges, and matching minimum base-metal thickness of steel studs.
- C. Vertical Deflection Clips: Manufacturer's standard bypass or head clips, capable of accommodating upward and downward vertical displacement of primary structure through positive mechanical attachment to stud web.
- D. Drift Clips: Manufacturer's standard bypass or head clips, capable of isolating wall stud from upward and downward vertical displacement and lateral drift of primary structure through positive mechanical attachment to stud web and structure.

2.5 FRAMING ACCESSORIES

- A. Fabricate steel-framing accessories from ASTM A 1003/A 1003M, Structural Grade, Type H, metallic coated steel sheet, of same grade and coating designation used for framing members.
- B. Provide accessories of manufacturer's standard thickness and configuration, unless otherwise indicated.

2.6 ANCHORS, CLIPS, AND FASTENERS

- A. Steel Shapes and Clips: ASTM A 36/A 36M, zinc coated by hot-dip process according to ASTM A 123/A 123M.
- B. Anchor Bolts: ASTM F 1554, Grade 36, threaded carbon-steel headless, hooked bolts, carbon-steel nuts, and flat, hardened-steel washers; zinc coated by hot-dip process according to ASTM A 153/A 153M, Class C.
- C. Post-Installed Anchors: Fastener systems with bolts of same basic metal as fastened metal, if visible, unless otherwise indicated; with working capacity greater than or equal to the design load, according to an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC01, ICC-ES AC193, ICC-ES AC58, or ICC-ES AC308 as appropriate for the substrate.
 - 1. Uses: Securing cold-formed steel framing to structure.
 - 2. Type: adhesive anchor.
 - 3. Material for Interior Locations: Carbon-steel components zinc plated to comply with ASTM B 633 or ASTM F 1941 (ASTM F 1941M), Class Fe/Zn 5, unless otherwise indicated.
 - 4. Material for Exterior or Interior Locations and Where Stainless Steel Is Indicated: Alloy Group 1 (A1) stainless-steel bolts, ASTM F 593 (ASTM F 738M), and nuts, ASTM F 594 (ASTM F 836M).
- D. Power-Actuated Anchors: Fastener systems with working capacity greater than or equal to the design load, according to an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC70.
- E. Mechanical Fasteners: ASTM C 1513, corrosion-resistant-coated, self-drilling, self-tapping, steel drill screws.
 - 1. Head Type: Low-profile head beneath sheathing; manufacturer's standard elsewhere.

2.7 MISCELLANEOUS MATERIALS

- A. Galvanizing Repair Paint: ASTM A 780/A 780M MIL-P-21035B or SSPC-Paint 20.
- B. Cement Grout: Portland cement, ASTM C 150/C 150M, Type I; and clean, natural sand, ASTM C 404. Mix at ratio of 1 part cement to 2-1/2 parts sand, by volume, with minimum water required for placement and hydration.
- C. Nonmetallic, Nonshrink Grout: Factory-packaged, nonmetallic, noncorrosive, nonstaining grout, complying with ASTM C 1107/C 1107M, and with a fluid consistency and 30-minute working time.
- D. Shims: Load-bearing, high-density, multimonomer, nonleaching plastic; or cold-formed steel of same grade and metallic coating as framing members supported by shims.
- E. Sill Sealer Gaskets: Closed-cell neoprene foam, 1/4 inch (6 mm) thick, selected from manufacturer's standard widths to match width of bottom track or rim track members as required.

2.8 FABRICATION

- A. Fabricate cold-formed steel framing and accessories plumb, square, and true to line, and with connections securely fastened, according to referenced AISI's specifications and standards, manufacturer's written instructions, and requirements in this Section.
 - 1. Fabricate framing assemblies using jigs or templates.
 - 2. Cut framing members by sawing or shearing; do not torch cut.
 - 3. Fasten cold-formed steel framing members by welding, screw fastening, clinch fastening, pneumatic pin fastening, or riveting as standard with fabricator. Wire tying of framing members is not permitted.
 - a. Comply with AWS D1.3/D1.3M requirements and procedures for welding, appearance and quality of welds, and methods used in correcting welding work.
 - b. Locate mechanical fasteners and install according to Shop Drawings, with screws penetrating joined members by no fewer than three exposed screw threads.
 - 4. Fasten other materials to cold-formed steel framing by welding, bolting, pneumatic pin fastening, or screw fastening, according to Shop Drawings.
- B. Reinforce, stiffen, and brace framing assemblies to withstand handling, delivery, and erection stresses. Lift fabricated assemblies by means that prevent damage or permanent distortion.
- C. Tolerances: Fabricate assemblies level, plumb, and true to line to a maximum allowable variation of 1/8 inch in 10 feet (1:960) and as follows:
 - 1. Spacing: Space individual framing members no more than plus or minus 1/8 inch (3 mm) from plan location. Cumulative error shall not exceed minimum fastening requirements of sheathing or other finishing materials.
 - 2. Squareness: Fabricate each cold-formed steel framing assembly to a maximum out-of-square tolerance of 1/8 inch (3 mm).

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, conditions, and abutting structural framing for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Before sprayed fire-resistive materials are applied, attach continuous angles, supplementary framing, or tracks to structural members indicated to receive sprayed fire-resistive materials.
- B. After applying sprayed fire-resistive materials, remove only as much of these materials as needed to complete installation of cold-formed framing without reducing thickness of fire-resistive materials below that required to obtain fire-resistance ratings indicated. Protect remaining fire-resistive materials from damage.
- C. Install load-bearing shims or grout between the underside of load-bearing wall bottom track and the top of foundation wall or slab at locations with a gap larger than 1/4 inch (6 mm) to ensure a uniform bearing surface on supporting concrete or masonry construction.
- D. Install sill sealer gaskets at the underside of wall bottom track or rim track and at the top of foundation wall or slab at stud or joist locations.
- E. Install sill sealer gasket/termite barrier in accordance with manufacturer's written instructions at the underside of wall bottom track or rim track and at the top of foundation wall or slab at stud or joist locations.

3.3 INSTALLATION, GENERAL

- A. Cold-formed steel framing may be shop or field fabricated for installation, or it may be field assembled.
- B. Install cold-formed steel framing according to AISI S200, AISI S202, and manufacturer's written instructions unless more stringent requirements are indicated.
- C. Install cold-formed steel framing and accessories plumb, square, and true to line, and with connections securely fastened.
 - 1. Cut framing members by sawing or shearing; do not torch cut.
 - 2. Fasten cold-formed steel framing members by, screw fastening, clinch fastening, or riveting. Wire tying of framing members is not permitted.
 - a. Locate mechanical fasteners, install according to Shop Drawings, and comply with requirements for spacing, edge distances, and screw penetration.
- D. Install framing members in one-piece lengths unless splice connections are indicated for track or tension members.
- E. Install temporary bracing and supports to secure framing and support loads equal to those for which structure was designed. Maintain braces and supports in place, undisturbed, until entire integrated supporting structure has been completed and permanent connections to framing are secured.

- F. Do not bridge building expansion joints with cold-formed steel framing. Independently frame both sides of joints.
- G. Install insulation, specified in Section 072100 "Thermal Insulation," and Section 072119 "Foamed-In-Place Insulation," in framing-assembly members, such as headers, sills, boxed joists, and multiple studs at openings, that are inaccessible on completion of framing work.
- H. Fasten hole-reinforcing plate over web penetrations that exceed size of manufacturer's approved or standard punched openings.

3.4 INSTALLATION OF EXTERIOR NON-LOAD-BEARING WALL FRAMING

- A. Install continuous tracks sized to match studs. Align tracks accurately and securely anchor to supporting structure.
- B. Fasten both flanges of studs to bottom track unless otherwise indicated. Space studs as follows:
 - 1. Stud Spacing: As indicated on Drawings.
- C. Set studs plumb, except as needed for diagonal bracing or required for nonplumb walls or warped surfaces and similar requirements.
- D. Isolate non-load-bearing steel framing from building structure to prevent transfer of vertical loads while providing lateral support.
 - 1. Connect vertical deflection clips to bypassing or infill studs and anchor to building structure.
 - 2. Connect drift clips to cold-formed steel framing and anchor to building structure.
- E. Install horizontal bridging in wall studs, spaced vertically in rows indicated on Delegated Design Drawings but not more than 48 inches (1220 mm) apart. Fasten at each stud intersection.
 - 1. Channel Bridging: Cold-rolled steel channel, welded or mechanically fastened to webs of punched studs.
 - 2. Strap Bridging: Combination of flat, taut, steel sheet straps of width and thickness indicated and stud-track solid blocking of width and thickness to match studs. Fasten flat straps to stud flanges and secure solid blocking to stud webs or flanges.
 - 3. Bar Bridging: Proprietary bridging bars installed according to manufacturer's written instructions.
- F. Install miscellaneous framing and connections, including stud kickers, web stiffeners, clip angles, continuous angles, anchors, and fasteners, to provide a complete and stable wall-framing system.

3.5 INSTALLATION OF INTERIOR NON-LOAD-BEARING WALL FRAMING

- A. Install continuous tracks sized to match studs. Align tracks accurately and securely anchor to supporting structure.
- B. Fasten both flanges of studs to bottom track unless otherwise indicated. Space studs as follows:
 - 1. Stud Spacing: As indicated on Drawings.

- C. Set studs plumb, except as needed for diagonal bracing or required for nonplumb walls or warped surfaces and similar requirements.
- D. Isolate non-load-bearing steel framing from building structure to prevent transfer of vertical loads while providing lateral support.
 - 1. Connect vertical deflection clips to studs and anchor to building structure.
 - 2. Connect drift clips to cold-formed steel metal framing and anchor to building structure.
- E. Install horizontal bridging in wall studs, spaced vertically in rows indicated on Delegated Design Drawings but not more than 48 inches (1220 mm) apart. Fasten at each stud intersection.
 - 1. Channel Bridging: Cold-rolled steel channel, welded or mechanically fastened to webs of punched studs.
 - 2. Strap Bridging: Combination of flat, taut, steel sheet straps of width and thickness indicated and stud-track solid blocking of width and thickness to match studs. Fasten flat straps to stud flanges and secure solid blocking to stud webs or flanges.
 - 3. Bar Bridging: Proprietary bridging bars installed according to manufacturer's written instructions.
- F. Install miscellaneous framing and connections, including stud kickers, web stiffeners, clip angles, continuous angles, anchors, and fasteners, to provide a complete and stable wall-framing system.

3.6 INSTALLATION TOLERANCES

- A. Install cold-formed steel framing level, plumb, and true to line to a maximum allowable tolerance variation of 1/8 inch in 10 feet (1:960) and as follows:
 - 1. Space individual framing members no more than plus or minus 1/8 inch (3 mm) from plan location. Cumulative error shall not exceed minimum fastening requirements of sheathing or other finishing materials.

3.7 REPAIR

- A. Galvanizing Repairs: Prepare and repair damaged galvanized coatings on fabricated and installed cold-formed steel framing with galvanized repair paint according to ASTM A 780/A 780M and manufacturer's written instructions. Touch up and weld in field not acceptable.

3.8 FIELD QUALITY CONTROL

- A. Testing: Owner will engage a qualified independent testing and inspecting agency to perform field tests and inspections and prepare test reports.
- B. Testing agency will report test results promptly and in writing to Contractor and Architect.
- C. Cold-formed steel framing will be considered defective if it does not pass tests and inspections.
- D. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.

3.9 PROTECTION

- A. Provide final protection and maintain conditions, in a manner acceptable to manufacturer and Installer, that ensure that cold-formed steel framing is without damage or deterioration at time of Substantial Completion.

END OF SECTION 054000

SECTION 055000 - METAL FABRICATIONS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY**A. Section Includes:**

1. Miscellaneous steel framing and supports.
2. Shelf angles.
3. Metal ladders.
4. Roof access ladder.
5. Metal floor plate.
6. Metal bollards.
7. Flexible metal bollards.
8. Corner guards
9. Loose bearing and leveling plates.
10. Ram protection Type 2.

B. Products furnished, but not installed, under this Section include the following:

1. Loose steel lintels.
2. Anchor bolts, steel pipe sleeves, slotted-channel inserts, and wedge-type inserts indicated to be cast into concrete or built into unit masonry.
3. Steel weld plates and angles for casting into concrete for applications where they are not specified in other Sections.

C. Related Sections:

1. Section 055213 "Pipe and Tube Railings" for railings at Cart Corrals and Retaining Walls.

1.3 ACTION SUBMITTALS**A. Submit the following Submittals via Owner's Project Management software platform.****1. Product Data: For the following:**

- a. Paint products.
- b. Grout.

2. Shop Drawings: Show fabrication and installation details. Show anchorage and accessory items.

1.4 INFORMATIONAL SUBMITTALS

- A. Delegated-Design Submittal: For ladders, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design ladders.
- B. Structural Performance of Aluminum Ladders: Aluminum ladders shall withstand the effects of loads and stresses within limits and under conditions specified in ANSI A14.3.
- C. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes acting on exterior metal fabrications by preventing buckling, opening of joints, overstressing of components, failure of connections, and other detrimental effects.
 - 1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

2.2 METALS

- A. Metal Surfaces, General: Provide materials with smooth, flat surfaces unless otherwise indicated. For metal fabrications exposed to view in the completed Work, provide materials without seam marks, roller marks, rolled trade names, or blemishes.
- B. Stainless-Steel Bars and Shapes: ASTM A 276, Type 316L.
- C. Stainless Steel Pipe: ASTM A 312/A 312M, Grade TP 316L.
- D. Stainless Steel Sheet, Strip, Plate, and Flat Bar: ASTM A 666, Type 316L.
- E. Castings: ASTM A 743/A 743M, Grade CF 8M or CF 3M.
- F. Steel Plates, Shapes, and Bars: ASTM A 36/A 36M.
- G. Rolled-Steel Galvanized Floor Checker-Plate: ASTM A 786/A 786M, rolled from plate complying with ASTM A 36/A 36M or ASTM A 283/A 283M, Grade C or D with G90 (Z275) coating.
- H. Steel Tubing: ASTM A 500/A 500M, cold-formed steel tubing.
- I. Steel Pipe: ASTM A 53/A 53M, Standard Weight (Schedule 40) unless otherwise indicated.
- J. Cast Iron: Either gray iron, ASTM A 48/A 48M, or malleable iron, ASTM A 47/A 47M, unless otherwise indicated.
- K. Aluminum Extrusions: ASTM B 221 (ASTM B 221M), Alloy 6063-T6.
- L. Aluminum-Alloy Rolled Tread Plate: ASTM B 632/B 632M, Alloy 6061-T6.
- M. Aluminum Castings: ASTM B 26/B 26M, Alloy 443.0-F.

2.3 FASTENERS

- A. General: Unless otherwise indicated, provide Type 304 stainless-steel fasteners for exterior use and zinc-plated fasteners with coating complying with ASTM B 633 or ASTM F 1941 (ASTM F 1941M), Class Fe/Zn 5, at exterior walls. Select fasteners for type, grade, and class required.

1. Provide stainless-steel fasteners for fastening aluminum.
 2. Provide stainless-steel fasteners for fastening stainless steel.
- B. Cast-in-Place Anchors in Concrete: Either threaded type or wedge type unless otherwise indicated; galvanized ferrous castings, either ASTM A 47/A 47M malleable iron or ASTM A 27/A 27M cast steel. Provide bolts, washers, and shims as needed, all hot-dip galvanized per ASTM F 2329.
- C. Post-Installed Anchors: Torque-controlled expansion anchors.
1. Material for Interior Locations: Carbon-steel components zinc plated to comply with ASTM B 633 or ASTM F 1941 (ASTM F 1941M), Class Fe/Zn 5, unless otherwise indicated.
 2. Material for Exterior Locations and Where Stainless Steel Is Indicated: Alloy Group 1 (A1) stainless-steel bolts, ASTM F 593 (ASTM F 738M), and nuts, ASTM F 594 (ASTM F 836M).

2.4 MISCELLANEOUS MATERIALS

- A. Shop Primers: Provide primers that comply with the following sections:
1. Section 099113 "Exterior Painting"
 2. Section 099123 "Interior Painting"
 3. Section 099600 "High-Performance Coatings"
- B. Galvanizing Repair Paint: High-zinc-dust-content paint complying with SSPC-Paint 20 and compatible with paints specified to be used over it.
- C. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D 1187/D 1187M.
- D. Nonshrink, Nonmetallic Grout: Factory-packaged, nonstaining, noncorrosive, nongaseous grout complying with ASTM C 1107/C 1107M. Provide grout specifically recommended by manufacturer for interior and exterior applications.
- E. Concrete: Comply with requirements in Section 033000 "Cast-in-Place Concrete" for normal-weight, air-entrained, concrete with a minimum 28-day compressive strength of 3000 psi (20 MPa).
- F. Sustainable Design Requirements
1. VOC Content:
 - a. Flat coatings and primers applied inside the weatherproofing system of the building shall have a VOC content of 50g/L or less when calculated according to 40 DFR 59, Subpart D (EPA Method 24).
 - b. Rust-inhibiting primers applied inside the weatherproofing system of the building shall have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

2.5 FABRICATION, GENERAL

- A. Shop Assembly: Preassemble items in the shop to greatest extent possible. Use connections that maintain structural value of joined pieces.

- B. Cut, drill, and punch metals cleanly and accurately. Remove burrs and ease edges. Remove sharp or rough areas on exposed surfaces.
- C. Weld corners and seams continuously to comply with the following:
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
 - 4. At exposed connections, finish exposed welds and surfaces smooth and blended.
- D. Form exposed connections with hairline joints, flush and smooth, using concealed fasteners or welds where possible. Locate joints where least conspicuous.
- E. Fabricate seams and other connections that are exposed to weather in a manner to exclude water. Provide weep holes where water may accumulate.
- F. Where units are indicated to be cast into concrete or built into masonry, equip with integrally welded steel strap anchors not less than 8 inches (200 mm) from ends and corners of units and 24 inches (600 mm) o.c.

2.6 MISCELLANEOUS FRAMING AND SUPPORTS

- A. General: Provide steel framing and supports not specified in other Sections as needed to complete the Work.
- B. Fabricate units from steel shapes, plates, and bars of welded construction unless otherwise indicated. Fabricate to sizes, shapes, and profiles indicated and as necessary to receive adjacent construction.

2.7 SHELF ANGLES

- A. Fabricate shelf angles from steel angles of sizes indicated and for attachment to concrete framing. Provide horizontally slotted holes to receive 3/4-inch (19-mm) bolts, spaced not more than 6 inches (150 mm) from ends and 24 inches (600 mm) o.c., unless otherwise indicated.
- B. For cavity walls, provide vertical channel brackets to support angles from backup masonry and concrete.
- C. Galvanize shelf angles located in exterior walls.
- D. Prime shelf angles located in exterior walls with primer specified in Section 099600 "High-Performance Coatings."
- E. Furnish wedge-type concrete inserts, complete with fasteners, to attach shelf angles to cast-in-place concrete.

2.8 METAL LADDERS

- A. General:
 - 1. Comply with ANSI A14.3.

2.9 ROOF ACCESS LADDERS

- A. Fabricate roof access ladder to comply with ANSI A14.3. Assemble by welding or with stainless-steel fasteners. At ladder from grade, provide locking security door.
- B. Aluminum Roof Access Ladder and Locking Hatch:
 - 1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. O'Keeffe's Inc., Horizon, Model 503A Access Ladder with Platform Only.
 - b. ALACO Ladder Company
 - c. Precision Ladders, LLC
 - 2. Ladder Fabrication:
 - a. Rungs: Not less than 1-1/4 inches (32 mm) in section and 18-3/8 inches (467mm) long, formed from tubular aluminum extrusions. Squared and deeply serrated on all sides.
 - b. Channel Side Rails: Not less than 1/8 inch (3 mm) wall thickness by 3 inches (76 mm) wide.
 - c. Heavy Duty Tubular Side Rails: Assembled from two interlocking aluminum extrusions no less than 1/8 inch (3 mm) wall thickness by 3 inches (76 mm) wide. Construction shall be self-locking stainless steel fasteners, full penetration TIG welds and clean, smooth and burr-free surfaces.
 - d. Walk-Through Rail and Roof Rail Extension: Not less than 3 feet 6 inches (1067 mm) above the landing and shall be fitted with deeply serrated, square, tubular grab rails.
 - e. Landing Platform: 1-1/2 inches (38 mm) or greater diameter, tubular aluminum guardrails and decks of serrated aluminum treads.
 - f. Security Doors: Formed 1/8 inch (3 mm) thick aluminum sheet. Security panels shall extend on both sides, perpendicular to the door face, to within 2 inches (51 mm) of the wall. Security door shall be furnished with continuous aluminum piano hinge and heavy duty forged steel locking hasps.
 - 3. Provide locking security door with Combination Padlock; 5 Passwords Security.
 - 4. Aluminum Finish: Clear anodized coating.

2.10 METAL FLOOR PLATE

- A. Fabricate from rolled-steel floor plate of thickness indicated below:
 - 1. Thickness: 1/8 inch (3.2 mm), 3/16 inch (4.8 mm) and 1/4 inch (6.4 mm) as required for loading.
- B. Provide steel angle supports as indicated.

2.11 CORNER GUARDS

- A. Galvanized, checker-plate steel channel shape, minimum 10-gauge thick, with 4 inch (96 mm) legs by full width of wall and full height of opening. Anchor corner guards at 16 inches on center along each leg with countersunk holes in bent steel checker-plate and using rawl plugs for masonry or toggle bolts for drywall attachment. Provide continuous bead of silicone sealant at all edges.

2.12 MISCELLANEOUS STEEL TRIM

- A. Unless otherwise indicated, fabricate units from steel shapes, plates, and bars of profiles shown with continuously welded joints and smooth exposed edges. Miter corners and use concealed field splices where possible.
- B. Provide cutouts, fittings, and anchorages as needed to coordinate assembly and installation with other work.
- C. Prime exterior miscellaneous steel trim with primer specified in Section 099600 "High-Performance Coatings."

2.13 METAL BOLLARDS

- A. Fabricate metal bollards from ASTM A53 B steel pipe. Fabricate bollards as shown on drawings.
 - 1. Cap bollards with 1/4-inch- (6.4-mm-) thick steel plate.
- B. Prime bollards with zinc-rich primer.
- C. Fabricate stainless steel sleeves for bollards in Sales Floor and Back of House Areas.

2.14 FLEXIBLE METAL BOLLARDS

- A. Manufacturer: McCue Corporation; www.mccue.com.
- B. Product: FlexCore Post 6; Model No. CSFP-55-14-GS-KIT
 - 1. Post: Zinc-electroplated steel post with mounting base.
 - 2. Post Diameter: 5.56 inches (141 mm) diameter
 - 3. Post Height: 14 inches (355.6 mm)
 - 4. Hardware and Accessories: Self-threading concrete anchors, washer, impact absorbing core, damper, flat head socket head cap screw, and other components necessary to complete installation.

2.15 LOOSE BEARING AND LEVELING PLATES

- A. Provide loose bearing and leveling plates for steel items bearing on masonry or concrete construction. Drill plates to receive anchor bolts and for grouting.

2.16 LOOSE STEEL LINTELS

- A. Fabricate loose steel lintels from steel angles and shapes of size indicated for openings and recesses in masonry walls and partitions at locations indicated.
- B. Galvanize loose steel lintels located in exterior walls.

2.17 STEEL WELD PLATES AND ANGLES

- A. Provide steel weld plates and angles not specified in other Sections, for items supported from concrete construction as needed to complete the Work. Provide each unit with no fewer than two integrally welded steel strap anchors for embedding in concrete.

2.18 RAM PROTECTION AT EQUIPMENT (BACKSTOCK AND TECHNICAL AREA)

- A. Manufacturer: McCue Corporation; www.mccue.com.
- B. Product: Systems Protector.
 - 1. Uprights (End, Middle, Corner): 3 inches by 36 inches; heavy-duty steel with powder coated finish; Safety Yellow color.
 - a. Include mounting base flange welded to uprights and concrete anchors.
 - b. Include rail brackets welded to uprights and locking pins.
 - 2. Cross Members: Cold-rolled steel; electroplated finish; 3 inches by 48 inches.
 - 3. Concrete Anchors: 5/8-inch by 3 inches; self-threading and with anti-rotation serrations to prevent back-out.

2.19 RAM PROTECTION TYPE 1 (CRASH RAIL)

- A. Manufacturer: McCue Corporation; www.mccue.com.
- B. Product: CartStop Stainless Steel Rail Assembly; Model No. CSSSR-KIT
 - 1. Post: Stainless steel legs, Type 201.
 - 2. Rails: Stainless steel tubes, Type 201.
 - 3. Height: 7 inches (slab to bottom of rail)
 - 4. Accessories: Include connectors, end caps, elbows and other parts necessary for a complete installation.
 - 5. Flange Anchor Kit: CSFA-10-KIT; stainless steel mounting sleeve and plastic anchor.

2.20 RAM PROTECTION TYPE 2

- A. Provide galvanized steel angles and concrete base along the base of cooler/freezer walls and security screen/fence.
 - 1. Profile and Sizes: As indicated on Drawings.
 - 2. Apply clear sealer to exposed concrete surfaces.
 - 3. Apply two coats of high-solid epoxy paint to exposed steel surfaces as specified in Section 099123.

2.21 FINISHES, GENERAL

- A. Finish metal fabrications after assembly.

2.22 STEEL AND IRON FINISHES

- A. Galvanizing: Hot-dip galvanize all exterior steel items and other items as indicated to comply with ASTM A 153/A 153M for steel and iron hardware and with ASTM A 123/A 123M for other steel and iron products.
- B. Shop prime iron and steel items not indicated to be galvanized unless they are to be embedded in concrete, or masonry, or unless otherwise indicated.
 - 1. Shop prime with universal shop primer unless primers specified in Section 099600 "High-Performance Coatings" are indicated.
- C. Preparation for Shop Priming: Prepare surfaces to comply with requirements indicated below:

1. Exterior Items: SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
 2. Items Indicated to Receive Zinc-Rich Primer: SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
 3. Items Indicated to Receive Primers Specified in Section 099600 "High-Performance Coatings": SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
 4. Other Items: SSPC-SP 3, "Power Tool Cleaning."
- D. Shop Priming: Apply shop primer to comply with SSPC-PA 1, "Paint Application Specification No. 1: Shop, Field, and Maintenance Painting of Steel," for shop painting.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing metal fabrications. Set metal fabrications accurately in location, alignment, and elevation; with edges and surfaces level, plumb, true, and free of rack; and measured from established lines and levels.
- B. Fit exposed connections accurately together to form hairline joints. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Do not weld, cut, or abrade surfaces of exterior units that have been hot-dip galvanized after fabrication and are for bolted or screwed field connections.
- C. Field Welding, only for emergency use, comply with the following requirements:
1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 2. Obtain fusion without undercut or overlap.
 3. Remove welding flux immediately.
 4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.
- D. Fastening to In-Place Construction: Provide anchorage devices and fasteners where metal fabrications are required to be fastened to in-place construction.
- E. Provide temporary bracing or anchors in formwork for items that are to be built into concrete, masonry, or similar construction.

3.2 INSTALLING METAL BOLLARDS

- A. Fill metal-capped bollards solidly with concrete and allow concrete to cure seven days before installing.
- B. Anchor bollards in place with concrete footings. Place concrete and vibrate or tamp for consolidation. Support and brace bollards in position until concrete has cured.
- C. Fill bollards solidly with concrete, mounding top surface to shed water.

3.3 INSTALLING FLEXIBLE METAL BOLLARDS

- A. Securely anchor bollard mounting base to concrete slab, following manufacturer's installation instructions.

- B. Slide post onto mounting base until flush with the floor. Secure post to mounting base.

3.4 INSTALLING BEARING AND LEVELING PLATES

- A. Clean concrete and masonry bearing surfaces of bond-reducing materials and roughen to improve bond to surfaces. Clean bottom surface of plates.
- B. Set bearing and leveling plates on wedges, shims, or leveling nuts. After bearing members have been positioned and plumbed, tighten anchor bolts. Do not remove wedges or shims but, if protruding, cut off flush with edge of bearing plate before packing with nonshrink grout. Pack grout solidly between bearing surfaces and plates to ensure that no voids remain.

3.5 ADJUSTING AND CLEANING

- A. Touchup Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas. Paint uncoated and abraded areas with the same material as used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.
- B. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A 780/A 780M.

END OF SECTION 055000

SECTION 055119 - METAL GRATING STAIRS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Industrial-type, straight-run stair with steel-grating treads and platforms attached steel stringers and supports.
 - 2. Steel railings attached to metal stairs.

1.3 COORDINATION

- A. Coordinate installation of anchorages for metal stairs. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For galvanized metal grating stairs and the following:
 - a. Galvanized products.
 - b. Grout.
 - 2. Shop Drawings: Include plans, elevations, sections, details, and attachments.
- C. Delegated-Design Submittal: For stairs and railings, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1.5 INFORMATIONAL SUBMITTALS

- A. Welding certificates.

1.6 QUALITY ASSURANCE

- A. Installer Qualifications: Fabricator of products.
- B. Shop Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Structural Performance of Stairs: Galvanized Metal stairs shall withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated:
 - 1. Uniform Load: 100 lbf/sq. ft.
 - 2. Concentrated Load: 300 lbf applied on an area of 4 sq. in.
 - 3. Uniform and concentrated loads need not be assumed to act concurrently.
 - 4. Stair Framing: Capable of withstanding stresses resulting from railing loads in addition to loads specified above.
 - 5. Limit deflection of treads, platforms, and framing members to $L/360$.
- B. Seismic Performance of Stairs: Galvanized Metal stairs shall withstand the effects of earthquake motions determined according to ASCE/SEI 7-10.
 - 1. Component Importance Factor: 1.5.
- C. Structural Performance of Railings: Railings, including attachment to building construction, shall withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated:
 - 1. Handrails and Top Rails of Guards:
 - a. Uniform load of 50 lbf/ft. (0.73 kN/m) applied in any direction.
 - b. Concentrated load of 200 lbf (0.89 kN) applied in any direction.
 - c. Uniform and concentrated loads need not be assumed to act concurrently.
 - 2. Infill of Guards:
 - a. Concentrated load of 50 lbf (0.22 kN) applied horizontally on an area of 1 sq. ft. (0.093 sq. m).
 - b. Infill load and other loads need not be assumed to act concurrently.

2.2 METALS

- A. Metal Surfaces, General: Provide materials with smooth, flat surfaces unless otherwise indicated. For components exposed to view in the completed Work, provide materials without seam marks, roller marks, rolled trade names, or blemishes.
- B. Steel Plates, Shapes, and Bars: ASTM A 36/A 36M.
- C. Steel Bars for Grating Treads: ASTM A 36 or steel strip, ASTM A 1011 and ASTM A 1018
- D. Wire Rod for Grating Crossbars: ASTM A 510/A 510M.
- E. Steel Pipe for Railings: ASTM A53/A53M, Type F or Type S, Grade A, Standard Weight (Schedule 40), unless another grade and weight are required by structural loads.
- F. Provide galvanized finish for exterior installations.

2.3 FASTENERS

- A. General: Provide ASTM F 1941, Class Fe/Zn 12 for exterior use. Select fasteners for type, grade, and class required.

- B. Bolts and Nuts: Regular hexagon-head bolts, ASTM A 307, Grade A; with hex nuts, ASTM A 563; and, where indicated, flat washers.
- C. Anchor Bolts: ASTM F 1554, Grade 36, of dimensions indicated; with nuts, ASTM A 563; and, where indicated, flat washers.
 - 1. Provide hot-dip, zinc-coated anchor bolts for exterior stairs.
- D. Post-Installed Anchors: Torque-controlled expansion anchors or chemical anchors capable of sustaining, without failure, a load equal to six times the load imposed when installed in unit masonry and four times the load imposed when installed in concrete, as determined by testing according to ASTM E 488, conducted by a qualified independent testing agency.
 - 1. Material for Exterior Locations: Alloy Group 1 stainless-steel bolts, ASTM F 593, and nuts, ASTM F 594.

2.4 MISCELLANEOUS MATERIALS

- A. Welding Electrodes: Comply with AWS requirements.
- B. Galvanizing Repair Paint: High-zinc-dust-content paint complying with ASTM A780/A780M and compatible with paints specified to be used over it.
- C. Nonmetallic, Shrinkage-Resistant Grout: ASTM C1107/C1107M, factory-packaged, nonmetallic aggregate grout; recommended by manufacturer for exterior use; noncorrosive and nonstaining; mixed with water to consistency suitable for application and a 30-minute working time.

2.5 FABRICATION, GENERAL

- A. Provide complete stair assemblies, including metal framing, hangers, clips, brackets, bearing plates, and other components necessary to support and anchor stairs and platforms on supporting structure.
 - 1. Join components by shop welding only unless otherwise noted.
 - 2. Use connections that maintain structural value of joined pieces.
- B. Assemble stairs and railings in shop to greatest extent possible.
 - 1. Disassemble units only as necessary for shipping and handling limitations.
 - 2. Clearly mark units for reassembly and coordinated installation.
- C. Cut, drill, and punch metals cleanly and accurately.
 - 1. Remove burrs and ease edges to a radius of approximately 1/32 inch (1 mm) unless otherwise indicated.
 - 2. Remove sharp or rough areas on exposed surfaces.
- D. Form exposed work with accurate angles and surfaces and straight edges.
- E. Shop Weld connections to comply with the following:
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.

3. Remove welding flux immediately.
4. Weld exposed corners and seams continuously unless otherwise indicated.
5. At exposed connections, finish exposed welds to comply with NOMMA's "Voluntary Joint Finish Standards" for Type 4 welds: good quality, uniform undressed weld with minimal splatter.

F. Form exposed connections with hairline joints, flush and smooth, using concealed fasteners where possible.

1. Where exposed fasteners are required, use Phillips flat-head (countersunk) screws or bolts unless otherwise indicated.
2. Locate joints where least conspicuous.
3. Fabricate joints that are exposed to weather in a manner to exclude water.
4. Provide weep holes where water may accumulate internally.

2.6 GALVANIZED STEEL-FRAMED STAIRS

A. NAAMM Stair Standard: Comply with NAAMM AMP 510, "Metal Stairs Manual," for Industrial Class, unless more stringent requirements are indicated.

B. Galvanized Metal Bar-Grating Stairs: Form treads and platforms to configurations shown from metal bar grating; fabricate to comply with NAAMM MBG 531, "Metal Bar Grating Manual."

1. Fabricate treads and platforms from welded steel grating with openings in gratings no more than 1/2 inch in least dimension.
 - a. Surface: Serrated.
 - b. Finish: Galvanized for exterior locations.
2. Secure grating to platform framing by welding.
3. Provide toe plates at open-sided edges of grating platforms.
4. Risers: Open.

2.7 STAIR RAILINGS

A. Comply with applicable requirements in Section 055213 "Pipe and Tube Railings."

B. Steel Tube Railings: Fabricate railings to comply with requirements indicated for design, dimensions, details, finish, and member sizes, including wall thickness of tube, post spacing, and anchorage, but not less than that needed to withstand indicated loads.

1. Rails and Posts: 1-1/2-inch-square top and bottom rails and 1-1/2-inch-square posts.
2. Picket Infill: 1/2-inch-square pickets spaced less than 4 inches clear.

C. Welded Connections: Fabricate railings with welded connections.

1. Fabricate connections that are exposed to weather in a manner that excludes water.
 - a. Provide weep holes where water may accumulate internally.
2. Cope components at connections to provide close fit, or use fittings designed for this purpose.
3. Weld all around at connections, including at fittings.

4. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
5. Obtain fusion without undercut or overlap.
6. Remove flux immediately.
7. Finish welds to comply with NOMMA's "Voluntary Joint Finish Standards" for Finish #4 - Good quality, uniform undressed weld with minimal splatter] as shown in NAAMM AMP 521.

D. Form changes in direction of railings as follows:

1. As detailed.

E. For changes in direction made by bending, use jigs to produce uniform curvature for each repetitive configuration required. Maintain cross section of member throughout entire bend without buckling, twisting, cracking, or otherwise deforming exposed surfaces of components. F. Close exposed ends of railing members with prefabricated end fittings.

F. Close exposed ends of railing members with prefabricated end fittings.

G. Provide wall returns at ends of wall-mounted handrails unless otherwise indicated. Close ends of returns unless clearance between end of rail and wall is 1/4 inch or less.

H. Connect posts to stair framing by mechanically fastened unless otherwise indicated.

I. Brackets, Flanges, Fittings, and Anchors: Provide wall brackets, end closures, flanges, miscellaneous fittings, and anchors for interconnecting components and for attaching to other work. Furnish inserts and other anchorage devices for connecting to concrete or masonry work.

1. Galvanized railings, provide galvanized fittings, brackets, fasteners, sleeves, and other ferrous-metal components.
2. Provide type of bracket with predrilled hole for exposed bolt anchorage and that provides 1-1/2-inch clearance from inside face of handrail to finished wall surface.

J. Fillers: Provide fillers made from steel plate, or other suitably crush-resistant material, where needed to transfer wall bracket loads through wall finishes to structural supports. Size fillers to suit wall finish thicknesses and to produce adequate bearing area to prevent bracket rotation and overstressing of substrate.

2.8 FINISHES

A. Finish metal stairs after assembly.

B. Galvanizing: Hot-dip galvanize items as indicated to comply with ASTM A 153 for steel and iron hardware and with ASTM A 123 for other steel and iron products.

1. Do not quench or apply post galvanizing treatments that might interfere with paint adhesion.
2. Fill vent and drain holes that are exposed in the finished Work, unless indicated to remain as weep holes, by plugging with zinc solder and filing off smooth.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Fastening to In-Place Construction: Provide anchorage devices and fasteners where necessary for securing metal stairs to in-place construction. Include threaded fasteners for concrete and masonry inserts, through-bolts, lag bolts, and other connectors.
- B. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing metal stairs. Set units accurately in location, alignment, and elevation, measured from established lines and levels and free of rack.

3.2 INSTALLING METAL STAIRS WITH GROUTED BASEPLATES

- A. Clean concrete and masonry bearing surfaces of bond-reducing materials and roughen to improve bond to surfaces. Clean bottom surface of baseplates.
- B. Set steel-stair baseplates on wedges, shims, or leveling nuts. After stairs have been positioned and aligned, tighten anchor bolts. Do not remove wedges or shims, but if protruding, cut off flush with edge of bearing plate before packing with grout.
 - 1. Use nonmetallic, nonshrink grout unless otherwise indicated.
 - 2. Pack grout solidly between bearing surfaces and plates to ensure that no voids remain.

3.3 INSTALLING RAILINGS

- A. Adjust railing systems before anchoring to ensure matching alignment at abutting joints. Space posts at spacing indicated or, if not indicated, as required by design loads. Plumb posts in each direction. Secure posts and rails to stair construction as follows:
 - 1. Anchor posts to steel by welding to steel supporting members.
 - 2. Anchor handrail ends to concrete and masonry with steel round flanges welded to rail ends and anchored with post-installed anchors and bolts.
- B. Attach handrails to stair as indicated. Locate and secure brackets as indicated or, if not indicated, at spacing required to support structural loads.

3.4 ADJUSTING AND CLEANING

- A. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A780/A780M.

END OF SECTION 055119

SECTION 055213 - PIPE AND TUBE RAILINGS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Stainless-steel tube railings at Cart Corrals.
 - 2. Galvanized steel tube railings at Retaining Wall.
- B. Related Section:
 - 1. Section 055119 "Metal Grating Stairs" for railings at Loading Dock stair.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For the following:
 - a. Manufacturer's product lines of mechanically connected railings.
 - b. Railing brackets.
 - c. Grout, anchoring cement, and paint products.
 - 2. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.
 - 3. Samples: For each type of exposed finish required.

1.4 INFORMATIONAL SUBMITTALS

- A. Delegated-Design Submittal: For railings, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
- B. Product Test Reports: For pipe and tube railings, for tests performed by a qualified testing agency, according to ASTM E 894 and ASTM E 935.

PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design railings, including attachment to building construction.

- B. Structural Performance: Railings, including attachment to building construction, shall withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated:

1. Handrails and Top Rails of Guards:

- a. Uniform load of 50 lbf/ ft. (0.73 kN/m) applied in any direction.
- b. Concentrated load of 200 lbf (0.89 kN) applied in any direction.
- c. Uniform and concentrated loads need not be assumed to act concurrently.

2. Infill of Guards:

- a. Concentrated load of 50 lbf (0.22 kN) applied horizontally on an area of 1 sq. ft. (0.093 sq. m).
- b. Infill load and other loads need not be assumed to act concurrently.

2.2 METALS, GENERAL

- A. Brackets, Flanges, and Anchors: Cast or formed metal of same type of material and finish as supported rails unless otherwise indicated.

1. Provide type of bracket with predrilled hole for exposed bolt anchorage and that provides 1-1/2-inch (38-mm) clearance from inside face of handrail to finished wall surface.

2.3 STEEL

- A. Tubing: ASTM A500 (cold formed) or ASTM A513.
- B. Pipe: ASTM A53/A53M, Type F or Type S, Grade A, Standard Weight (Schedule 40), unless another grade and weight are required by structural loads.
- C. Plates, Shapes, and Bars: ASTM A36/A36M.

2.4 STAINLESS STEEL

- A. Tubing: ASTM A554, Grade MT 304.
- B. Pipe: ASTM A312/A312M, Grade TP 304.
- C. Plate and Sheet: ASTM A240/A240M or ASTM A666, Type 304.

2.5 FASTENERS

- A. General: Provide the following:

1. Hot-Dip Galvanized Railings: Type 304 stainless-steel or hot-dip zinc-coated steel fasteners complying with ASTM A 153/A 153M or ASTM F 2329 for zinc coating.
2. Stainless-Steel Railings: Type 304 stainless-steel fasteners.
3. Provide exposed fasteners with finish matching appearance, including color and texture, of railings.

- B. Fasteners for Anchoring Railings to Other Construction: Select fasteners of type, grade, and class required to produce connections suitable for anchoring railings to other types of construction indicated and capable of withstanding design loads.

C. Fasteners for Interconnecting Railing Components:

1. Provide concealed fasteners for interconnecting railing components and for attaching them to other work, unless exposed fasteners are unavoidable or are the standard fastening method for railings indicated.

D. Post-Installed Anchor: Torque-controlled expansion anchors capable of sustaining, without failure, a load equal to 6 times the load imposed when installed in unit masonry and 4 times the load imposed when installed in concrete, as determined by testing according to ASTM E 488/E 488M, conducted by a qualified independent testing agency.

1. Material for Interior Locations: Carbon-steel components zinc-plated to comply with ASTM B 633 or ASTM F 1941 (ASTM F 1941M), Class Fe/Zn 5, unless otherwise indicated.
2. Material for Exterior Locations: Alloy Group 1 (A1) stainless-steel bolts, ASTM F 593 (ASTM F 738M), and nuts, ASTM F 594 (ASTM F 836M).

2.6 MISCELLANEOUS MATERIALS

A. Welding Rods and Bare Electrodes: Select according to AWS specifications for metal alloy welded.

B. Etching Cleaner for Galvanized Metal: Complying with MPI#25.

C. Galvanizing Repair Paint: High-zinc-dust-content paint complying with SSPC-Paint 20 and compatible with paints specified to be used over it.

D. Nonshrink, Nonmetallic Grout: Factory-packaged, nonstaining, noncorrosive, nongaseous grout complying with ASTM C 1107/C 1107M. Provide grout specifically recommended by manufacturer for interior and exterior applications.

E. Sustainable Design Requirements

1. VOC Content:

- a. Flat coatings and primers applied inside the weatherproofing system of the building shall have a VOC content of 50g/L or less when calculated according to 40 DFR 59, Subpart D (EPA Method 24).
- b. Rust-inhibiting primers applied inside the weatherproofing system of the building shall have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

2.7 FABRICATION

A. General: Fabricate railings to comply with requirements indicated for design, configurations, dimensions, member sizes and spacing, details, finish, and anchorage, but not less than that required to support structural loads.

B. Shop assemble railings to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation. Use connections that maintain structural value of joined pieces.

- C. Cut, drill, and punch metals cleanly and accurately. Remove burrs and ease edges to a radius of approximately 1/32 inch (1 mm) unless otherwise indicated. Remove sharp or rough areas on exposed surfaces.
- D. Form work true to line and level with accurate angles and surfaces.
- E. Fabricate connections that are exposed to weather in a manner that excludes water. Provide weep holes where water may accumulate.
- F. Cut, reinforce, drill, and tap as indicated to receive finish hardware, screws, and similar items.
- G. Connections: Fabricate railings with welded connections unless otherwise indicated.
- H. Welded Connections: Cope components at connections to provide close fit, or use fittings designed for this purpose. Weld all around at connections, including at fittings.
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove flux immediately.
 - 4. At exposed connections, finish exposed surfaces smooth and blended so no roughness shows after finishing and welded surface matches contours of adjoining surfaces.
- I. Nonwelded Connections: Connect members with concealed mechanical fasteners and fittings. Fabricate members and fittings to produce flush, smooth, rigid, hairline joints.
- J. Form changes in direction by bending.
- K. For changes in direction made by bending, use jigs to produce uniform curvature for each repetitive configuration required. Maintain cross section of member throughout entire bend without buckling, twisting, cracking, or otherwise deforming exposed surfaces of components.
- L. Close exposed ends of railing members with prefabricated end fittings.
- M. Provide wall returns at ends of wall-mounted handrails unless otherwise indicated.
- N. Brackets, Flanges, Fittings, and Anchors: Provide wall brackets, flanges, miscellaneous fittings, and anchors to interconnect railing members to other work unless otherwise indicated.
 - 1. At brackets and fittings fastened to plaster or gypsum board partitions, provide crush-resistant fillers or other means to transfer loads through wall finishes to structural supports and prevent bracket or fitting rotation and crushing of substrate.
- O. Provide inserts and other anchorage devices for connecting railings to concrete or masonry work. Fabricate anchorage devices capable of withstanding loads imposed by railings. Coordinate anchorage devices with supporting structure.

2.8 STEEL AND IRON FINISHES

- A. Galvanized Railings:
 - 1. Hot-dip galvanize exterior steel railings, including hardware, after fabrication.
 - 2. Comply with ASTM A123/A123M for hot-dip galvanized railings.
 - 3. Comply with ASTM A153/A153M for hot-dip galvanized hardware.

4. Do not quench or apply post galvanizing treatments that might interfere with paint adhesion.
 5. Fill vent and drain holes that are exposed in the finished Work, unless indicated to remain as weep holes, by plugging with zinc solder and filing off smooth.
- B. For galvanized railings, provide hot-dip galvanized fittings, brackets, fasteners, sleeves, and other ferrous components.

2.9 STAINLESS-STEEL FINISHES

- A. Remove tool and die marks and stretch lines, or blend into finish.
- B. Grind and polish surfaces to produce uniform, directionally textured, polished finish indicated, free of cross scratches.
- C. Stainless Steel Tubing Finishes:
1. 180-Grit Polished Finish: Uniform, directionally textured finish.
- D. Stainless Steel Sheet and Plate Finishes:
1. Directional Satin Finish: ASTM A489/A480, No. 4.
- E. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Perform cutting, drilling, and fitting required for installing railings.
1. Fit exposed connections together to form tight, hairline joints.
 2. Install railings level, plumb, square, true to line; without distortion, warp, or rack.
 3. Set railings accurately in location, alignment, and elevation; measured from established lines and levels.
 4. Do not weld, cut, or abrade surfaces of railing components that are coated or finished after fabrication and that are intended for field connection by mechanical or other means without further cutting or fitting.
 5. Set posts plumb within a tolerance of 1/16 inch in 3 feet (2 mm in 1 m).
 6. Align rails so variations from level for horizontal members and variations from parallel with rake of steps and ramps for sloping members do not exceed 1/4 inch in 12 feet (6 mm in 3.5 m).
- B. Control of Corrosion: Prevent galvanic action and other forms of corrosion by insulating metals and other materials from direct contact with incompatible materials.
1. Coat, with a heavy coat of bituminous paint, concealed surfaces of aluminum that are in contact with grout, concrete, masonry, wood, or dissimilar metals.
- C. Adjust railings before anchoring to ensure matching alignment at abutting joints.
- D. Fastening to In-Place Construction: Use anchorage devices and fasteners where necessary for securing railings and for properly transferring loads to in-place construction.

3.2 RAILING CONNECTIONS

- A. Welded Connections: Use fully welded joints for permanently connecting railing components. Comply with requirements for welded connections, whether welding is performed in the shop or in the field.

3.3 ANCHORING POSTS

- A. Form or core-drill holes not less than 5 inches (125 mm) deep and 3/4 inch (20 mm) larger than OD of post for installing posts in concrete. Clean holes of loose material, insert posts, and fill annular space between post and concrete with nonshrink, nonmetallic grout, mixed and placed to comply with anchoring material manufacturer's written instructions.
- B. Cart corral guards shall be installed by core drilling to accommodate 18" of guard legs.
- C. Cart corral guards shall be set in cored holes using quick setting expansive concrete (provided by installer).

3.4 ADJUSTING AND CLEANING

- A. Clean stainless steel by washing thoroughly with clean water and soap and rinsing with clean water.
- B. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas, and repair galvanizing to comply with ASTM A780/A780M.

END OF SECTION 055213

SECTION 055313 - BAR GRATINGS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes metal bar gratings and metal frames at scrubber pit.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For the following:
 - a. Clips and anchorage devices for gratings.
 - b. Paint products.
 - 2. Shop Drawings: Include plans, sections, details, and attachments to other work.
- C. Delegated-Design Submittal: For gratings, including manufacturers' published load tables or analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design gratings.
- B. Structural Performance: Gratings shall withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated:
 - 1. Floors: For uniform and concentrated loading and depth refer to General Notes on Structural Drawings.
 - 2. Limit deflection to $L/360$ or 1/4 inch (6.4 mm), whichever is less.
- C. Seismic Performance: Gratings shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
 - 1. Component Importance Factor: 1.5.

2.2 METAL BAR GRATINGS

- A. Manufacturer: Metals, Inc.; www.metals-inc.com.

- B. Metal Bar Grating Standards: Comply with NAAMM MBG 531, "Metal Bar Grating Manual."
- C. Welded Steel Grating:
 - 1. Bearing Bar Spacing: Not less than 1/2 inch (13 mm) o.c.
 - 2. Bearing Bar Depth: As required to comply with structural performance requirements.
 - 3. Bearing Bar Thickness: As required to comply with structural performance requirements.
 - 4. Crossbar Spacing: 2 inches (51 mm) o.c.
 - 5. Traffic Surface: Plain.
 - 6. Steel Finish: Hot-dip galvanized with a coating weight of not less than 1.8 oz./sq. ft. (550 g/sq. m) of coated surface.

2.3 FERROUS METALS

- A. Steel Plates, Shapes, and Bars: ASTM A 36/A 36M.
- B. Steel Bars for Bar Gratings: ASTM A 36/A 36M or steel strip, ASTM A 1011/A 1011M or ASTM A 1018/A 1018M.
- C. Wire Rod for Bar Grating Crossbars: ASTM A 510 (ASTM A 510M).
- D. Galvanizing of Bar Grating: ASTM A123 with G90 (Z275) coating.

2.4 FASTENERS

- A. General: Unless otherwise indicated, provide zinc-plated fasteners with coating complying with ASTM B 633 or ASTM F 1941 (ASTM F 1941M), Class Fe/Zn 5. Select fasteners for type, grade, and class required.
 - 1. Provide stainless-steel fasteners for fastening to stainless steel elements.
- B. Post-Installed Anchors: Torque-controlled expansion anchors capable of sustaining, without failure, a load equal to six times the load imposed when installed in unit masonry and four times the load imposed when installed in concrete, as determined by testing according to ASTM E 488/E 488M, conducted by a qualified independent testing agency.

2.5 MISCELLANEOUS MATERIALS

- A. Galvanizing Repair Paint: High-zinc-dust-content paint complying with SSPC-Paint 20 and compatible with paints specified to be used over it.
- B. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D 1187/D 1187M.

2.6 FABRICATION

- A. Cut, drill, and punch material cleanly and accurately. Remove burrs and ease edges to a radius of approximately 1/32 inch (1 mm) unless otherwise indicated. Remove sharp or rough areas on exposed surfaces.
- B. Fit exposed connections accurately together to form hairline joints.

2.7 GRATING FRAMES AND SUPPORTS

- A. Fabricate from metal shapes, plates, and bars of welded construction to sizes, shapes, and profiles indicated and as necessary to receive gratings. Miter and weld connections for perimeter angle frames. Cut, drill, and tap units to receive hardware and similar items.
 - 1. Unless otherwise indicated, fabricate from same basic metal as gratings.
 - 2. Equip units indicated to be cast into concrete or built into masonry with integrally welded anchors. Unless otherwise indicated, space anchors 24 inches (600 mm) o.c. and provide minimum anchor units in the form of steel straps 1-1/4 inches (32 mm) wide by 1/4 inch (6 mm) thick by 8 inches (200 mm) long.
 - 3. Provide steel posts with bearing plates for post-installed anchorage to concrete floor and supporting level grating system at Scrubbing Station. Bar grating is to be removable.
 - 4. Galvanize all steel frames and supports.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing gratings. Set units accurately in location, alignment, and elevation; measured from established lines and levels and free of rack.
- B. Fit exposed connections accurately together to form hairline joints.
 - 1. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Do not weld, cut, or abrade the surfaces of exterior units that have been hot-dip galvanized after fabrication and are for bolted or screwed field connections.
- C. Attach toe plates to gratings by welding at locations indicated.

3.2 INSTALLING METAL BAR GRATINGS

- A. General: Install gratings to comply with recommendations of referenced metal bar grating standards that apply to grating types and bar sizes indicated, including installation clearances and standard anchoring details.
- B. Attach removable units to supporting members with type and size of clips and fasteners indicated or, if not indicated, as recommended by grating manufacturer for type of installation conditions shown.
- C. Attach nonremovable units to supporting members by welding where both materials are same; otherwise, fasten by bolting as indicated above.

END OF SECTION 055313

SECTION 061000 - ROUGH CARPENTRY**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Wood blocking and nailers.
 - 2. Wood furring and grounds.
 - 3. Plywood backing panels.
 - 4. Wood sheeting products

PART 2 - PRODUCTS**2.1 WOOD PRODUCTS, GENERAL**

- A. All wood products and adhesives shall be made with no added urea formaldehyde.
- B. Lumber: DOC PS 20 and applicable rules of grading agencies indicated. If no grading agency is indicated, comply with the applicable rules of any rules-writing agency certified by the ALSC Board of Review. Grade lumber by an agency certified by the ALSC Board of Review to inspect and grade lumber under the rules indicated.
 - 1. Factory mark each piece of lumber with grade stamp of grading agency.
 - 2. For exposed lumber indicated to receive a stained or natural finish, mark grade stamp on end or back of each piece or omit grade stamp and provide certificates of grade compliance issued by grading agency.
 - 3. Dress lumber, S4S, unless otherwise indicated.
- C. Maximum Moisture Content of Lumber: 15 percent for 2-inch nominal (38-mm actual) thickness or less; 19 percent for more than 2-inch nominal (38-mm actual) thickness unless otherwise indicated.

2.2 WOOD-PRESERVATIVE-TREATED LUMBER

- A. Preservative Treatment by Pressure Process: AWWA U1; Use Category UC3b for exterior construction not in contact with ground and Use Category UC4a for items in contact with ground.
 - 1. Preservative Chemicals: Acceptable to authorities having jurisdiction and containing no arsenic or chromium. Do not use inorganic boron (SBX) for sill plates.
- B. Kiln-dry lumber after treatment to a maximum moisture content of 19 percent. Do not use material that is warped or that does not comply with requirements for untreated material.
- C. Mark lumber with treatment quality mark of an inspection agency approved by the ALSC Board of Review.

D. Application: Treat items indicated on Drawings, and the following:

1. Wood sills, sleepers, blocking, furring, and similar concealed members in contact with masonry or concrete.
2. Wood framing and furring attached directly to the interior of below-grade exterior masonry or concrete walls.

2.3 FIRE-RETARDANT-TREATED MATERIALS

A. General: Where fire-retardant-treated materials are indicated, materials shall comply with requirements in this article, that are acceptable to authorities having jurisdiction, and with fire-test-response characteristics specified as determined by testing identical products per test method indicated by a qualified testing agency.

B. Fire-Retardant-Treated Lumber and Plywood by Pressure Process: Products with a flame-spread index of 25 or less when tested according to ASTM E 84, and with no evidence of significant progressive combustion when the test is extended an additional 20 minutes, and with the flame front not extending more than 10.5 feet (3.2 m) beyond the centerline of the burners at any time during the test.

1. Interior Type A: Treated materials shall have a moisture content of 28 percent or less when tested according to ASTM D 3201 at 92 percent relative humidity. Use where exterior type is not indicated.

C. Kiln-dry lumber after treatment to maximum moisture content of 19 percent. Kiln-dry plywood after treatment to maximum moisture content of 15 percent.

D. Identify fire-retardant-treated wood with appropriate classification marking of qualified testing agency.

E. Application: Treat items indicated on Drawings, and the following:

1. Concealed blocking and sleepers.
2. Plywood backing panels.
3. Oriented strand board (OSB)

2.4 MISCELLANEOUS LUMBER

A. General: Provide miscellaneous lumber indicated and lumber for support or attachment of other construction, including the following:

1. Blocking.
2. Nailers.
3. Furring.
4. Grounds.

B. Dimension Lumber Items: Construction or No. 2 grade lumber of any species.

C. Concealed Boards: 15 percent maximum moisture content and any of the following species and grades:

1. Mixed southern pine or southern pine; No. 2 grade; SPIB.
2. Eastern softwoods; No. 2 Common grade; NeLMA.
3. Northern species; No. 2 Common grade; NLGA.

4. Western woods; Construction or No. 2 Common grade; WCLIB or WWPA.

2.5 PLYWOOD BACKING PANELS

- A. Plywood Subtop at Counters: Plywood, DOC PS 1, Exterior, A-C, fire-retardant treated, in thickness indicated or, if not indicated, not less than 3/4-inch (19-mm) nominal thickness.
- B. Equipment Backing Panels: Plywood, DOC PS 1, Exterior, C-C Plugged, fire-retardant treated, in thickness indicated or, if not indicated, not less than 3/4-inch (19-mm) nominal thickness.

2.6 ORIENTED-STRAND-BOARD (OSB) PANELS

- A. Oriented-Strand-Board Sheathing: DOC PS 2, Exposure 1, Structural I sheathing, fire-retardant-treated.

1. Thickness: As indicated on Drawings.

2.7 FASTENERS

- A. General: Fasteners shall be of size and type indicated and shall comply with requirements specified in this article for material and manufacture.
 1. Where rough carpentry is exposed to weather, in ground contact, pressure-preservative treated, or in area of high relative humidity, provide fasteners of Type 304 stainless steel.
- B. Power-Driven Fasteners: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC70.
- C. Post-Installed Anchors: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC01, ICC-ES AC58, ICC-ES AC193 or ICC-ES AC308 as appropriate for the substrate.

2.8 MISCELLANEOUS MATERIALS

- A. Sill-Sealer Gaskets: Glass-fiber-resilient insulation, fabricated in strip form, for use as a sill sealer; 1-inch (25-mm) nominal thickness, compressible to 1/32 inch (0.8 mm); selected from manufacturer's standard widths to suit width of sill members indicated.
- B. Flexible Flashing: Composite, self-adhesive, flashing product consisting of a pliable, rubberized-asphalt compound, bonded to a high-density polyethylene film, aluminum foil, or spunbonded polyolefin to produce an overall thickness of not less than 0.025 inch (0.6 mm).
- C. Adhesives for Gluing Furring and Sleepers to Concrete or Masonry: Formulation complying with ASTM D 3498 that is approved for use indicated by adhesive manufacturer.
 1. Adhesives shall have a VOC content of 70 g/L or less.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Set rough carpentry to required levels and lines, with members plumb, true to line, cut, and fitted. Fit rough carpentry accurately to other construction. Locate furring, nailers, blocking,

grounds, sleepers and similar supports to comply with requirements for attaching other construction.

- B. Comply with AWPAC M4 for applying field treatment to cut surfaces of preservative-treated lumber.
- C. Where wood-preservative-treated lumber is installed adjacent to metal decking, install continuous flexible flashing separator between wood and metal decking.
- D. Securely attach rough carpentry work to substrate by anchoring and fastening as indicated, complying with the following:
 - 1. Table 2304.9.1, "Fastening Schedule," in ICC's International Building Code (IBC).
 - 2. ICC-ES evaluation report for fastener.

3.2 PLYWOOD/OSB PANEL INSTALLATION

- A. Install plywood/OSB panel with long dimension perpendicular to supports and continuous over at least one support.
 - 1. Center joints accurately over supports and securely fasten to framing.
 - 2. Allow 1/8 inch (3 mm) minimum gap at panel edges and ends. Stagger end joints.
 - 3. Provide continuous solid blocking at all panel edges.

3.3 PROTECTION

- A. Protect wood that has been treated with inorganic boron (SBX) from weather. If, despite protection, inorganic boron-treated wood becomes wet, apply EPA-registered borate treatment. Apply borate solution by spraying to comply with EPA-registered label.
- B. Protect rough carpentry from weather. If, despite protection, rough carpentry becomes wet, apply EPA-registered borate treatment. Apply borate solution by spraying to comply with EPA-registered label.

END OF SECTION 061000

SECTION 061600 – SHEATHING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Wall sheathing.
 - 2. Parapet sheathing.
 - 3. Sheathing joint and penetration treatment.

- B. Related Section:

- 1. Section 075423 "Thermoplastic Polyolefin (TPO) Roofing" for gypsum roof board on roof of entry canopy.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.

- B. Submit the following Submittals via Owner's Project Management software platform.

- 1. Product Data: For each type of process and factory-fabricated product.

1.4 INFORMATIONAL SUBMITTALS

- A. Product Test Reports: For each water-resistant glass-mat gypsum sheathing assembly, indicating compliance with specified requirements, for tests performed by a qualified testing agency.

1.5 QUALITY ASSURANCE

- A. Refer to Sections 044213 and 075423 for free-standing mock-up requirements.

PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. Fire-Resistance Ratings: As tested according to ASTM E 119; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

- 1. Fire-Resistance Ratings: Indicated by design designations from UL's "Fire Resistance Directory" or from the listings of another qualified testing agency.

2.2 WALL SHEATHING

A. Glass-Mat Gypsum Sheathing: ASTM C 1177/1177M.

1. Georgia-Pacific Gypsum LLC, DensGlass Fireguard Sheathing or approved equivalent.
2. Type and Thickness: Type X, 5/8 inch (15.9 mm) thick.

2.3 SHEATHING ON BACK OF PARAPET WALL

A. Glass-Mat Gypsum Sheathing: ASTM C 1177/1177M.

1. Georgia-Pacific Gypsum LLC, DensDeck Prime Roof Boards or approved equivalent.
2. Type and Thickness: 1/2 inch (12.7 mm) thick.

2.4 FASTENERS

A. General: Provide fasteners of size and type indicated that comply with requirements specified in this article for material and manufacture.

1. For parapet and wall sheathing, provide fasteners of Type 304 stainless steel or approve equal.

2.5 SHEATHING JOINT-AND-PENETRATION TREATMENT MATERIALS

A. Sealant for Glass-Mat Gypsum Sheathing: Silicone emulsion sealant complying with ASTM C 834, compatible with sheathing tape and sheathing and recommended by tape and sheathing manufacturers for use with glass-fiber sheathing tape and for covering exposed fasteners.

1. Sheathing Tape: Self-adhering glass-fiber tape, minimum 2 inches (50 mm) wide, 10 by 10 or 10 by 20 threads/inch (390 by 390 or 390 by 780 threads/m), of type recommended by sheathing and tape manufacturers for use with silicone emulsion sealant in sealing joints in glass-mat gypsum sheathing and with a history of successful in-service use.

B. Trim Accessories: Provide manufacturer's standard plastic or metal trim accessories for gypsum board work, per ASTM C 1047. Provide with either knurled or perforated expanded flanges for nailing or stapling, and beaded for concealment of flanges, in joint compound. Provide corner beads, L-type edge trim-beads, U-type edge trim-beads, special L-kerf-type edge trim-beads, and one-piece control joint beads.

1. Subject to compliance with requirements, provide drywall trims and accessories by Vinyl Corp.; a division of ClarkDietrich Building Systems or equivalent.

C. Interior Trim Accessories: Provide corner beads, L-type edge trim-beads, U-type edge trim-beads, special L-kerf-type edge trim-beads, and one-piece control joint beads complying with the following requirements:

1. Materials: Formed plastic with one the following requirements:
 - a. Rigid PVC or CPVC plastic.
2. Subject to compliance with requirements, provide drywall trims and accessories by Vinyl Corp.; a division of ClarkDietrich Building Systems.

- D. Form control joints and expansion joints at locations indicated on Drawings, and as recommended by Gypsum Association, with space between edges of boards prepared to receive trim accessories.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Do not use materials with defects that impair quality of sheathing or pieces that are too small to use with minimum number of joints or optimum joint arrangement. Arrange joints so that pieces do not span between fewer than three support members.
- B. Cut panels at penetrations, edges, and other obstructions of work; fit tightly against abutting construction unless otherwise indicated.
- C. Securely attach to substrate by fastening as indicated, complying with the following:
 - 1. Table 2304.9.1, "Fastening Schedule," in the ICC's International Building Code.
 - 2. ICC-ES evaluation report for fastener.
- D. Coordinate wall and parapet sheathing installation with flashing and joint-sealant installation so these materials are installed in sequence and manner that prevent exterior moisture from passing through completed assembly.
- E. Do not bridge building expansion joints; cut and space edges of panels to match spacing of structural support elements.

3.2 GYPSUM SHEATHING INSTALLATION

- A. Comply with GA-253 and with manufacturer's written instructions.
 - 1. Fasten gypsum sheathing to cold-formed metal framing with screws.
 - 2. Install panels with a 3/8-inch (9.5-mm) gap where non-load-bearing construction abuts structural elements.
 - 3. Install panels with a 1/4-inch (6.4-mm) gap where they abut masonry or similar materials that might retain moisture, to prevent wicking.
- B. Apply fasteners so heads bear tightly against face of sheathing, but do not cut into facing.
- C. Vertical Installation: Install vertical edges centered over studs. Abut ends and edges with those of adjacent panels. Attach at perimeter and within field of panel to each stud.
 - 1. Space fasteners approximately 8 inches (200 mm) o.c. and set back a minimum of 3/8 inch (9.5 mm) from edges and ends of panels.
- D. Seal sheathing joints according to sheathing manufacturer's written instructions.
 - 1. Apply elastomeric sealant to joints and fasteners and trowel flat. Apply sufficient amount of sealant to completely cover joints and fasteners after troweling. Seal other penetrations and openings.
 - 2. Apply glass-fiber sheathing tape to glass-mat gypsum sheathing joints and apply and trowel sealant to embed entire face of tape in sealant. Apply sealant to exposed fasteners with a trowel so fasteners are completely covered. Seal other penetrations and openings.

END OF SECTION 061600

SECTION 064116 - PLASTIC-LAMINATE-FACED ARCHITECTURAL CABINETS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Plastic-laminate-faced architectural cabinets and countertops.
 - 2. Wood furring, blocking, shims, and hanging strips for installing plastic-laminate-faced architectural cabinets unless concealed within other construction before cabinet installation.
- B. Related Requirements:
 - 1. Section 123661.16 - Solid surfacing countertops
- C. Coordinate sizes and locations of framing, blocking, furring, reinforcements, and other related units of Work specified in other Sections to support loads imposed by installed and fully loaded cabinets.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product, including high-pressure decorative laminate and cabinet hardware and accessories.
 - 2. Shop Drawings: Show location of each item, dimensioned plans and elevations, large-scale details, attachment devices, and other components. Apply AWI Quality Certification Program label to Shop Drawings.
 - 3. Samples:
 - a. Plastic laminates, for each color, pattern, and surface finish.
 - b. Thermoset decorative panels, for each color, pattern, and surface finish.
 - c. Complete sample base and wall cabinets with all specified cabinet hardware attached. Base cabinet will contain both a drawer and door.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications: Fabricator of products.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Do not deliver cabinets until painting and similar finish operations that might damage architectural cabinets have been completed in installation areas. Store cabinets in installation

areas or in areas where environmental conditions comply with requirements specified in "Field Conditions" Article.

1.6 FIELD CONDITIONS

- A. Environmental Limitations: Do not deliver or install cabinets until building is enclosed, wet work is complete, and HVAC system is operating and maintaining temperature and relative humidity at occupancy levels during the remainder of the construction period.
- B. Field Measurements: Where cabinets are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication, and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.
 - 1. Locate concealed framing, blocking, and reinforcements that support cabinets by field measurements before being enclosed/concealed by construction, and indicate measurements on Shop Drawings.

PART 2 - PRODUCTS

2.1 PLASTIC-LAMINATE-FACED ARCHITECTURAL CABINETS

- A. Quality Standard: Unless otherwise indicated, comply with the "Architectural Woodwork Standards" for grades of architectural plastic-laminate cabinets indicated for construction, finishes, installation, and other requirements.
 - 1. Provide labels from AWI certification program indicating that woodwork, including installation, complies with requirements of grades specified.
- B. Grade: Premium.
- C. Type of Construction: Frameless.
- D. Cabinet, Door, and Drawer Front Interface Style: Flush overlay.
- E. High-Pressure Decorative Laminate: NEMA LD 3, grades as indicated or if not indicated, as required by woodwork quality standard.
 - 1. Refer to Interior Design Finish Schedule for products, finishes and colors.
- F. Laminate Cladding for Exposed Surfaces:
 - 1. Horizontal Surfaces: Grade HGS.
 - 2. Postformed Surfaces: Grade HGP.
 - 3. Vertical Surfaces: Grade VGS.
- G. Materials for Semiexposed Surfaces:
 - 1. Surfaces Other Than Drawer Bodies: Thermoset decorative panels.
 - 2. Drawer Sides and Backs: Solid-hardwood lumber.
 - 3. Drawer Bottoms: Hardwood plywood.
- H. Dust Panels: 1/4-inch (6.4-mm) plywood or tempered hardboard above compartments and drawers unless located directly under tops.

- I. Colors, Patterns, and Finishes: Provide materials and products that result in colors and textures of exposed laminate surfaces complying with the following requirements:

- 1. As indicated by laminate manufacturer's designations as indicated on Interior Design Finish Schedule, Sheet F1.3.

2.2 PLASTIC-LAMINATE COUNTERTOPS AND BACKSPLASHES

- A. Grade: Premium.
- B. High-Pressure Decorative Laminate Grade: HGS.
 - 1. Refer to Interior Design Finish Schedule for products, finishes and colors.
- C. Edge Treatment: Same as laminate cladding on horizontal surfaces.
- D. Core Material: As indicated on Drawings.
- E. Backer Sheet: Provide plastic-laminate backer sheet, Grade BKL, on underside of countertop substrate.

2.3 WOOD MATERIALS

- A. Wood Products: Provide materials that comply with requirements of referenced quality standard for each type of woodwork and quality grade specified unless otherwise indicated.
 - 1. Wood Moisture Content: 5 to 10 percent.
- B. Composite Wood and Agrifiber Products: Provide materials that comply with requirements of referenced quality standard for each type of woodwork and quality grade specified unless otherwise indicated.
 - 1. Particleboard: ANSI A208.1, Grade M-2, made with binder containing no urea formaldehyde.
 - 2. Softwood Plywood: DOC PS 1.
 - 3. Veneer-Faced Panel Products (Hardwood Plywood): HPVA HP-1, made with adhesive containing no urea formaldehyde.
 - 4. Thermoset Decorative Panels: Particleboard or medium-density fiberboard finished with thermally fused, melamine-impregnated decorative paper and complying with requirements of NEMA LD 3, Grade VGL, for test methods 3.3, 3.4, 3.6, 3.8, and 3.10.

2.4 CABINET HARDWARE AND ACCESSORIES

- A. General: Provide cabinet hardware and accessory materials associated with architectural cabinets except for items specified in Section 087100 "Door Hardware."
- B. Frameless Concealed Hinges (European Type): BHMA A156.9, B01602, 100 degrees of opening, self-closing.
- C. Wire Pulls: Back mounted, solid metal, 4 inches (100 mm) long, 5/16 inch (8 mm) in diameter unless shown otherwise on drawings.
- D. Catches: Magnetic catches, BHMA A156.9, B03141 or Roller catches, BHMA A156.9, B03071.

- E. Adjustable Shelf Standards and Supports: BHMA A156.9, B04071; with shelf rests, B04081.
- F. Shelf Rests: BHMA A156.9, B04013; metal, two-pin type with shelf hold-down clip.
- G. Drawer Slides: BHMA A156.9.
 - 1. Grade 1 and Grade 2: Side mounted and extending under bottom edge of drawer; full-extension type; zinc-plated steel with polymer rollers.
 - 2. Grade 1HD-100 and Grade 1HD-200: Side mounted; [full-extension] [full-overtravel-extension] type; zinc-plated-steel ball-bearing slides.
 - 3. For drawers more than 3 inches (75 mm) high but not more than 6 inches (150 mm) high and not more than 24 inches (600 mm) wide, provide Grade 1HD-100.
 - 4. For drawers more than 6 inches (150 mm) high or more than 24 inches (600 mm) wide, provide Grade 1HD-200.
- H. Door Locks: BHMA A156.11, E07121.
- I. Drawer Locks: BHMA A156.11, E07041.
- J. Door and Drawer Silencers: BHMA A156.16, L03021 with soft close feature.
- K. Exposed Hardware Finishes: For exposed hardware, provide finish that complies with BHMA A156.18 for BHMA finish number indicated.
 - 1. Satin Stainless Steel: BHMA 630.

2.5 MISCELLANEOUS MATERIALS

- A. Furring, Blocking, Shims, and Hanging Strips: Softwood or hardwood lumber, kiln dried to less than 15 percent moisture content.
- B. Anchors: Select material, type, size, and finish required for each substrate for secure anchorage. Provide metal expansion sleeves or expansion bolts for post-installed anchors. Use nonferrous-metal or hot-dip galvanized anchors and inserts at inside face of exterior walls and at floors.
- C. Adhesives: Do not use adhesives that contain urea formaldehyde.
 - 1. VOC Content: Contact adhesives applied inside the weatherproofing system of the building shall have a VOC of 80 g/l or less when calculating according to 40 CFR 59, subpart D (EPA Method 24)
- D. Adhesive for Bonding Plastic Laminate: Unpigmented contact cement.
 - 1. VOC Content: Contact adhesives applied inside the weatherproofing system of the building shall have a VOC of 80 g/l or less when calculating according to 40 CFR 59, subpart D (EPA Method 24)
 - 2. Adhesive for Bonding Edges: Hot-melt adhesive or adhesive specified above for faces.

2.6 FABRICATION

- A. Complete fabrication, including assembly and hardware application, to maximum extent possible before shipment to Project site. Disassemble components only as necessary for

shipment and installation. Where necessary for fitting at site, provide ample allowance for scribing, trimming, and fitting.

- B. Shop-cut openings to maximum extent possible to receive hardware, appliances, electrical work, and similar items. Locate openings accurately and use templates or roughing-in diagrams to produce accurately sized and shaped openings. Sand edges of cutouts to remove splinters and burrs.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Before installation, condition cabinets to average prevailing humidity conditions in installation areas.

3.2 INSTALLATION

- A. Grade: Install cabinets to comply with same grade as item to be installed.
- B. Install cabinets level, plumb, true, and straight. Shim as required with concealed shims. Install level and plumb to a tolerance of 1/8 inch in 96 inches (3 mm in 2400 mm).
- C. Scribe and cut cabinets to fit adjoining work, refinish cut surfaces, and repair damaged finish at cuts.
- D. Anchor cabinets to anchors or blocking built in or directly attached to substrates. Secure with countersunk, concealed fasteners and blind nailing. Use fine finishing nails or finishing screws for exposed fastening, countersunk and filled flush with woodwork.
- E. Cabinets: Install without distortion so doors and drawers fit openings properly and are accurately aligned. Adjust hardware to center doors and drawers in openings and to provide unencumbered operation. Complete installation of hardware and accessory items as indicated.
 - 1. Install cabinets with no more than 1/8 inch in 96-inch (3 mm in 2400-mm) sag, bow, or other variation from a straight line.
 - 2. Fasten wall cabinets through back, near top and bottom, and at ends not more than 16 inches (400 mm) o.c. with No. 10 wafer-head screws sized for not less than 1-1/2-inch (38-mm) penetration into wood blocking or No. 10 wafer-head sheet metal screws through metal backing or metal framing behind wall finish.
- F. Countertops: Anchor securely to supports on underside of countertop. Pre-drill holes for screws as recommended by manufacturer. Align adjacent surfaces and, using adhesive in color to match countertop, form seams to comply with manufacturer's written instructions. Carefully dress joints smooth, remove surface scratches, and clean entire surface.
 - 1. Install countertops with no more than 1/8 inch in 96-inch sag, bow, or other variation from a straight line.
 - 2. Secure backsplashes to tops with concealed metal brackets at 16 inches (400 mm) o.c. and to walls with adhesive.
 - 3. Caulk space between backsplash and wall with sealant specified in Section 079200 "Joint Sealants."

- G. Touch up finishing work specified in this Section after installation of woodwork. Fill nail holes with matching filler where exposed.

3.3 ADJUSTING AND CLEANING

- A. Repair damaged and defective woodwork, where possible, to eliminate functional and visual defects; where not possible to repair, replace woodwork. Adjust joinery for uniform appearance.
- B. Clean, lubricate, and adjust hardware.
- C. Clean woodwork on exposed and semi-exposed surfaces. Touch up shop-applied finishes to restore damaged or soiled areas.

END OF SECTION 064116

SECTION 071113 - BITUMINOUS DAMPPROOFING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes cold-applied, emulsified-asphalt dampproofing.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.

1.4 PROJECT CONDITIONS

- A. Ventilation: Provide adequate ventilation during application of dampproofing in enclosed spaces and confined areas such as foundation trenches. Maintain ventilation until dampproofing has cured.

PART 2 - PRODUCTS**2.1 MATERIALS, GENERAL**

- A. VOC Content: Products shall comply with VOC content limits of authorities having jurisdiction unless otherwise required.

2.2 COLD-APPLIED, EMULSIFIED-ASPHALT DAMPPROOFING

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. BASF Construction Chemicals - Building Systems; Sonneborn Brand Products.
 - 2. ChemMasters, Inc.
 - 3. Euclid Chemical Company (The); an RPM company.
 - 4. Henry Company.
 - 5. Karnak Corporation.
 - 6. Koppers Inc.
 - 7. Malarkey Roofing Products.
 - 8. Meadows, W. R., Inc.
- B. Trowel Coats: ASTM D 1227, Type II, Class 1.
- C. Fibered Brush and Spray Coats: ASTM D 1227, Type II, Class 1.

- D. Brush and Spray Coats: ASTM D 1227, Type III, Class 1.
- E. VOC Content: 30 g/L or less.

2.3 AUXILIARY MATERIALS

- A. General: Furnish auxiliary materials recommended in writing by dampproofing manufacturer for intended use and compatible with bituminous dampproofing.
- B. Emulsified-Asphalt Primer: ASTM D 1227, Type III, Class 1, except diluted with water as recommended in writing by manufacturer.
- C. Asphalt-Coated Glass Fabric: ASTM D 1668, Type I.

PART 3 - EXECUTION

3.1 APPLICATION, GENERAL

- A. Comply with manufacturer's written instructions for substrate preparation, dampproofing application, cure time between coats, and drying time before backfilling unless more stringent requirements are indicated.
 - 1. Apply dampproofing to provide continuous plane of protection to below-grade, foundation walls where habitable space is not on the interior side of such walls.
 - 2. Apply additional coats if recommended in writing by manufacturer or to achieve a smooth surface and uninterrupted coverage.
- B. Where dampproofing footings and foundation walls, apply from finished-grade line to top of footing; extend over top of footing and down a minimum of 6 inches (150 mm) over outside face of footing.
 - 1. Extend dampproofing 12 inches (300 mm) onto intersecting walls and footings, but do not extend onto surfaces exposed to view when Project is completed.
 - 2. Install flashings and corner protection stripping at internal and external corners, changes in plane, construction joints, cracks, and where shown as "reinforced," by embedding an 8-inch- (200-mm-) wide strip of asphalt-coated glass fabric in a heavy coat of dampproofing. Dampproofing coat for embedding fabric is in addition to other coats required.

3.2 COLD-APPLIED, EMULSIFIED-ASPHALT DAMPPROOFING

- A. Concrete Foundations and Parged Masonry Foundation Walls: Apply two brush or spray coats at not less than 1.5 gal./100 sq. ft. (0.6 L/sq. m) for first coat and 1 gal./100 sq. ft. (0.4 L/sq. m) for second coat, one fibered brush or spray coat at not less than 3 gal./100 sq. ft. (1.2 L/sq. m), or one trowel coat at not less than 4 gal./100 sq. ft. (1.6 L/sq. m).
- B. Unparged Masonry Foundation Walls: Apply primer and two brush or spray coats at not less than 1.5 gal./100 sq. ft. (0.6 L/sq. m) for first coat and 1 gal./100 sq. ft. (0.4 L/sq. m) for second coat, primer and one fibered brush or spray coat at not less than 3 gal./100 sq. ft. (1.2 L/sq. m), or primer and one trowel coat at not less than 5 gal./100 sq. ft. (2 L/sq. m).

END OF SECTION 071113

SECTION 071900 - WATER REPELLENTS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes water repellent treatment for the following:
 - 1. Exterior exposed CMU and brick wall surfaces.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each product specified. Submit manufacturer's specifications, application instructions, and recommendations for water repellents. Include data substantiating that materials are recommended by manufacturer for applications indicated.
 - 2. Water Penetration Test Reports.

1.4 INFORMATIONAL SUBMITTALS

- A. Product Certificates: For each type of water repellent, from manufacturer.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance manual.

1.6 QUALITY ASSURANCE

- A. Applicator Qualifications: An employer of workers trained and approved by manufacturer.

1.7 DELIVERY, STORAGE AND HANDLING

- A. Storage and Protection: Comply with manufacturer's recommendations.
- B. Deliver materials in original sealed containers, clearly marked with manufacturer's brand name, grade of material, date of manufacture and batch number.
- C. Store materials in areas where temperatures are not less than 50 deg F (10 deg C), unless authorized by the manufacturer.

1.8 PROJECT CONDITIONS

- A. Limitations: Proceed with application only when the following existing and forecasted weather and substrate conditions permit water repellent to be applied according to manufacturers' written instructions and warranty requirements:
1. Cementitious surfaces have cured for not less than 28 days.
 2. Building has been closed in for not less than 30 days before treating wall assemblies.
 3. Ambient temperature is above 40 deg F (4.4 deg C) and below 90 deg F (32 deg C) and will remain so for 24 hours.
 4. Substrate is not frozen and substrate-surface temperature is above 40 deg F (4.4 deg C) and below 100 deg F (37.8 deg C).
 5. Rain or snow is not predicted within 24 hours.
 6. Not less than 24 hours have passed since surfaces were last wet.
 7. Windy conditions do not exist that might cause water repellent to be blown onto vegetation or surfaces not intended to be treated.

PART 2 - PRODUCTS

2.1 WATER REPELLENTS

- A. General: Provide clear, deep penetrating, non-film-forming, non-silicone, non-yellowing, heavy-duty water repellent solutions onto exterior vertical surfaces.
- B. Basis of Design for CMU and Brick:
1. Manufacturer: Evonik Industries; www.protectosil.com/product/protectosil
 2. Product: Protectosil Aqua-Trete Concentrate

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Applicator present, for compliance with requirements and conditions affecting performance of the Work.
1. Verify that surfaces are clean and dry according to water repellent manufacturer's requirements. Check moisture content in three representative locations by method recommended by manufacturer.
 2. Inspect for previously applied treatments that may inhibit penetration or performance of water repellent.
 3. Verify that there is no efflorescence or other removable residues that would be trapped beneath the application of water repellent.
 4. Verify that required repairs are complete, cured, and dry before applying water repellent.
- B. Test pH level according to water repellent manufacturer's written instructions to ensure chemical bond to silica-containing or siliceous minerals.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Allow cementitious materials to age before application of water repellent, according to repellent manufacturer's written instructions.

- B. Cleaning: Before application of water repellent, clean substrate of substances that could impair penetration or performance of product according to water-repellent manufacturer's written instructions and as follows:
 - 1. Surfaces to be treated with water repellent shall be cleaned of dust, dirt, oil, grease, laitance, curing compounds, sealer, bond breaker, efflorescence, salt, asphalt, laitance, paints, coatings, and other contaminants detrimental to proper installation of penetrating water repellent.
 - 2. It is the responsibility of the Contractor to prepare the surfaces to a condition acceptable to the water repellent manufacturer.
- C. Where application occurs in the field, protect adjacent surfaces from overrun and overspray with protective covering. Mask as necessary, and cover plants and shrubs, asphalt, surfaces to be painted, windows, and all other surfaces not scheduled to receive water repellent.
- D. Provide controlled area at plant or site for application of material. Area shall prevent exposure to environmental conditions which would adversely affect the successful application of the material, such as rain, excess sun and wind exposure.
- E. Coordination with Sealant Joints: Do not apply water repellent until sealants for joints adjacent to surfaces receiving water-repellent treatment have been installed and cured.
- F. Prior to application of repellent, protect all adjacent surfaces. Repellent treatment will leave streaks and other unsightly appearances on surfaces, if left unprotected. The cost of repair work to correct these unsightly appearances will be the responsibilities of the Contractor.

3.3 APPLICATION

- A. Apply water repellent to all exposed vertical CMU and brick surfaces in accordance with manufacturer's instructions and applicable recommendations.
- B. Apply a heavy saturation coating on surfaces indicated for treatment using high-volume low-pressure spray equipment with a fan spray coarse nozzle, flooding the surface to obtain uniform coverage unless otherwise recommended by the manufacturer.
 - 1. Apply at a rate determined by the field test and as necessary to meet the performance requirements.
 - 2. Apply at temperature and weather conditions recommended by the manufacturer's written instructions.
- C. Do not allow material to run onto roofs, asphalt pavement or landscape. Use movable metal catcher or an absorbent material to intercept repellent.
- D. Remove excess material; do not allow material to puddle beyond saturation.

3.4 CLEANING

- A. Immediately clean water repellent from adjoining surfaces and surfaces soiled or damaged by repellent application as work progresses. Correct damage to work of other trades caused by repellent application, as approved by Architect.
- B. Comply with manufacturer's written cleaning instructions.

END OF SECTION 071900

SECTION 072100 - THERMAL INSULATION**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Extruded polystyrene foam-plastic board (XPS)
 - 2. Mineral-wool blanket

- B. Related Requirements:

- 1. Section 042000 "Unit Masonry".
 - 2. Section 061600 "Sheathing" for sheathing installed directly on steel framing.
 - 3. Section 072119 "Foamed-in-Place Insulation" for spray-applied polyurethane foam insulation.
 - 4. Section 092900 "Gypsum Board" for sound attenuation blanket used as acoustic insulation.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.

- B. Submit the following Submittals via Owner's Project Management software platform.

- 1. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

- A. Product test reports.

- B. Research reports.

PART 2 - PRODUCTS**2.1 EXTRUDED POLYSTYRENE FOAM-PLASTIC BOARD (XPS)**

- A. Manufacturers:

- 1. Dow; Styrofoam Brand XPS Insulation.
 - 2. Owens Corning; Foamular XPS Insulation.
 - 3. Kingspan; GreenGuard XPS Insulation.

- B. Extruded polystyrene boards in this article are also called "XPS boards."

- C. Extruded Polystyrene Board: ASTM C 578, unfaced; maximum flame-spread and smoke-developed indexes of 25 and 450, respectively, per ASTM E 84.
 - 1. Fire Propagation Characteristics: Passes NFPA 285 testing as part of an approved assembly.
 - 2. Thickness: As indicated on Drawings.
 - 3. Compressive Strength:
 - a. Type IV, 25-psi (173-kPa), for the following locations:
 - 1) Exterior wall cavity.
 - 2) Interior side of single wythe exterior CMU walls.
 - 3) Underside of slab edge at building perimeter.
 - 4) Foundation walls.
 - b. Type VI, 40-psi (276-kPa), for the following locations:
 - 1) Underside of slab at Freezers and Coolers.

2.2 MINERAL-WOOL BLANKETS

- A. Mineral-Wool Blanket, Unfaced: ASTM C 665, Type I (blankets without membrane facing); consisting of fibers; with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively, per ASTM E 84; passing ASTM E 136 for combustion characteristics.
 - 1. Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. MinWool Sound Attenuation Fire Batts by Johns Manville.

2.3 ACCESSORIES

- B. Insulation for Miscellaneous Voids:
 - 1. Spray Polyurethane Foam Insulation: ASTM C 1029, Type II, closed cell, with maximum flame-spread and smoke-developed indexes of 75 and 450, respectively, per ASTM E 84.
- C. Adhesive for Bonding Insulation: Product compatible with insulation and air and water barrier materials, and with demonstrated capability to bond insulation securely to substrates without damaging insulation and substrates.
 - 1. VOC Content: Contact adhesives applied inside the weatherproofing system of the building shall have a VOC of 80 g/l or less when calculating according to 40 CFR 59, subpart D (EPA Method 24)

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean substrates of substances that are harmful to insulation, including removing projections capable of puncturing insulation, or that interfere with insulation attachment.

3.2 INSTALLATION, GENERAL

- A. Comply with insulation manufacturer's written instructions applicable to products and applications.
- B. Install insulation that is undamaged, dry, and unsoiled and that has not been left exposed to ice, rain, or snow at any time.
- C. Extend insulation to envelop entire area to be insulated. Fit tightly and fill voids with insulation. Remove projections that interfere with placement.
- D. Provide sizes to fit applications and selected from manufacturer's standard thicknesses, widths, and lengths. Apply single layer of insulation units unless multiple layers are otherwise shown or required to make up total thickness or to achieve R-value.

3.3 INSTALLATION OF FOUNDATION AND SLAB INSULATION

- A. On vertical wall surfaces, slab edge, and foundation surfaces, set XPS insulation units using manufacturer's recommended adhesive according to manufacturer's written instructions.
 - 1. If not otherwise indicated, extend 25-psi (173-kPa) minimum compressive strength insulation a minimum of 24 inches (610 mm) below exterior grade line.
- B. On horizontal surfaces below slabs at building perimeter, loosely lay XPS insulation units according to manufacturer's written instructions. Stagger end joints and tightly abut insulation units.
 - 1. Install 25-psi (173-kPa) minimum compressive strength insulation below slab a minimum of 24 inches (610 mm) in from building perimeter walls.
- C. On horizontal surfaces below freezer and cooler slabs, loosely lay XPS insulation units according to manufacturer's written instructions. Stagger end joints and tightly abut insulation units.
 - 1. Prior to installing insulation board, verify that sheet vapor barrier is installed over compacted sand base.
 - 2. Place 40-psi (276-kPa) minimum compressive strength insulation over sheet vapor barrier. Install insulation in three layers, with staggered joints in both directions.
 - 3. Extend sheet vapor barrier up to fully encapsulate edges of insulation and slab. Seal all penetrations and repair damaged areas with manufacturer's recommended tape.

3.4 INSTALLATION OF CAVITY-WALL INSULATION

- A. Install pads of adhesive spaced approximately 24 inches (610 mm) o.c. both ways on inside face and as recommended by manufacturer. Fit courses of insulation between wall ties, z-furring and other obstructions, with edges butted tightly in both directions. Press units firmly against inside substrates.

3.5 INSTALLATION OF INSULATION IN MISCELLANEOUS VOIDS

- A. Miscellaneous Voids: Install insulation in miscellaneous voids and cavity spaces where required to prevent gaps in insulation.
 - 1. Spray Polyurethane Insulation: Apply according to manufacturer's written instructions.

3.6 THERMAL BARRIER

- A. Where exposed to interior space, foam insulation must be fully concealed behind an approved 15-minute thermal barrier or other thermal barrier materials approved by authority having jurisdiction.

3.7 PROTECTION

- A. Protect installed insulation from damage due to harmful weather exposures, physical abuse, and other causes. Provide temporary coverings or enclosures where insulation is subject to abuse and cannot be concealed and protected by permanent construction immediately after installation.

END OF SECTION 072100

SECTION 072726 - FLUID-APPLIED MEMBRANE AIR BARRIERS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes fluid-applied, vapor-permeable air barrier.
- B. Related Requirements:
 - 1. Section 061600 "Sheathing" for wall sheathings and wall sheathing joint-and-penetration treatments.
 - 2. Section 042000 "Unit masonry"

1.3 DEFINITIONS

- A. Air-Barrier Material: A primary element that provides a continuous barrier to the movement of air.
- B. Air-Barrier Accessory: A transitional component of the air barrier that provides continuity.
- C. Air-Barrier Assembly: The collection of air-barrier materials and accessories applied to an opaque wall, including joints and junctions to abutting construction, to control air movement through the wall.

1.4 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.5 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - a. Include manufacturer's written instructions for evaluating, preparing, and treating each substrate; technical data; dry film thickness; and tested physical and performance properties of products.
 - 2. Shop Drawings: For air-barrier assemblies.
 - a. Include details for substrate joints and cracks, counterflashing strips, penetrations, inside and outside corners, terminations, and tie-ins with adjoining construction.

1.6 INFORMATIONAL SUBMITTALS

- A. Product certificates.
- B. Field quality control reports.

1.7 QUALITY ASSURANCE

- A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Remove and replace liquid materials that cannot be applied within their stated shelf life.
- B. Protect stored materials from direct sunlight.

1.9 FIELD CONDITIONS

- A. Environmental Limitations: Apply air barrier within the range of ambient and substrate temperatures recommended in writing by air-barrier manufacturer.
 - 1. Protect substrates from environmental conditions that affect air-barrier performance.
 - 2. Do not apply air barrier to a damp or wet substrate or during snow, rain, fog, or mist.

PART 2 - PRODUCTS

2.1 MATERIALS, GENERAL

- A. Source Limitations: Obtain primary air-barrier materials and air-barrier accessories from single source from single manufacturer.
- B. VOC Content: 100 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24) and complying with VOC content limits of authorities having jurisdiction.

2.2 PERFORMANCE REQUIREMENTS

- A. Air-Barrier Performance: Air-barrier assembly and seals with adjacent construction shall be capable of performing as a continuous air barrier and as a liquid-water drainage plane flashed to discharge to the exterior incidental condensation or water penetration. Air-barrier assemblies shall be capable of accommodating substrate movement and of sealing substrate expansion and control joints, construction material changes, penetrations, and transitions at perimeter conditions without deterioration and air leakage exceeding specified limits.
- B. Membrane Air Permeance (ASTM E 2178): Not to exceed 0.004 cfm/sq. ft. of surface area at 1.57 lbf/sq. ft. (0.2 L/s x sq. m of surface area at 75 Pa), when tested according to ASTM E 2178.
- C. Air-Barrier Assembly Air Leakage: Not to exceed 0.04 cfm/sq. ft. of surface area at 1.57 lbf/sq. ft. (0.2 L/s x sq. m of surface area at 75 Pa), when tested according to ASTM E 2357.

2.3 FLUID-APPLIED VAPOR-PERMEABLE MEMBRANE AIR-BARRIER

- A. Vapor-Permeable Air Barrier: Synthetic polymer material with an installed dry film thickness, according to manufacturer's written instructions, over smooth, void-free substrates.

B. Manufacturers and Products: Behind rainscreen cladding without open joints.

1. Hohmann & Barnard; Enviro-Barrier VP; www.h-b.com.
2. Tremco; EXOAIR 230; www.tremcosealants.com.
3. Carlisle; Barritech VP; www.carlisleccw.com

C. Manufacturers and Products: Behind rainscreen cladding with open joints.

1. Tremco; EXOAIR 230 (black, 40 mils DFT); www.tremcosealants.com.
2. Sto Corp.; Sto AirSeal (black, 40 mils DFT); www.stocorp.com.

2.4 ACCESSORY MATERIALS

A. Requirement: Provide primers, transition strips, termination strips, joint reinforcing fabric and strips, joint sealants, counterflashing strips, flashing sheets and metal termination bars, termination mastic, substrate patching materials, adhesives, tapes, foam sealants, lap sealants, and other accessory materials that are recommended in writing by air-barrier manufacturer to produce a complete air-barrier assembly and that are compatible with primary air-barrier material and adjacent construction to which they may seal.

B. Stainless-Steel Sheet: ASTM A240/A240M, Type 304 and Series 300 stainless-steel fasteners.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

1. Verify that substrates are sound and free of oil, grease, dirt, excess mortar, or other contaminants.
2. Verify that substrates have cured and aged for minimum time recommended in writing by air-barrier manufacturer.
3. Verify that substrates are visibly dry and free of moisture.
4. Verify that masonry joints are flush and completely filled with mortar.
5. Verify sealants used in sheathing are compatible with membrane air barrier. Perform field peel-adhesion test on materials to which sealants are adhered.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 SURFACE PREPARATION

A. Clean, prepare, treat, fill, and seal substrate and joints and cracks in substrate according to manufacturer's written instructions and details. Provide clean, dust-free, and dry substrate for air-barrier application.

B. Mask off adjoining surfaces not covered by air barrier to prevent spillage and overspray affecting other construction.

C. Remove grease, oil, bitumen, form-release agents, paints, curing compounds, and other penetrating contaminants or film-forming coatings from concrete.

D. Remove fins, ridges, mortar, and other projections and fill honeycomb, aggregate pockets, holes, and other voids in concrete with substrate-patching material.

- E. Remove excess mortar from masonry ties, shelf angles, and other obstructions.
- F. At changes in substrate plane, apply sealant or termination mastic beads at sharp corners and edges to form a smooth transition from one plane to another.
- G. Cover gaps in substrate plane and form a smooth transition from one substrate plane to another with stainless-steel sheet mechanically fastened to structural framing to provide continuous support for air barrier.
- H. Substrate Preparation:
 - 1. CMU: Strike mortar joints flush and full to face of CMU. Fill voids and holes, particularly at mortar joints.
 - 2. Concrete: Cure for a minimum of 48 hours before application and surface free of large voids and spalled areas.
 - 3. Exterior Sheathing: Treat joints with mastic/reinforced mesh or self-adhesive tape depending on size of gap. Fill larger voids with mortar and allow to cure.

3.3 ACCESSORIES INSTALLATION

- A. Install accessory materials according to air-barrier manufacturer's written instructions and details to form a seal with adjacent construction and ensure continuity of air and water barrier.
 - 1. Coordinate the installation of air barrier with installation of roofing membrane and base flashing to ensure continuity of air barrier with roofing membrane.
 - 2. Install transition strip on roofing membrane or base flashing so that a minimum of 3 inches (75 mm) of coverage is achieved over each substrate.
 - 3. Unless manufacturer recommends in writing against priming, apply primer to substrates at required rate and allow it to dry.
 - 4. Apply primer to substrates at required rate and allow it to dry. Limit priming to areas that will be covered by air-barrier material on same day. Reprime areas exposed for more than 24 hours.
- B. Connect and seal exterior wall air-barrier material continuously to roofing-membrane air barrier, concrete below-grade structures, floor-to-floor construction, exterior glazing and window systems, glazed storefront systems, exterior louvers, exterior door framing, and other construction used in exterior wall openings, using accessory materials.
- C. At end of each working day, seal top edge of strips and transition strips to substrate with termination mastic.
- D. Apply joint sealants forming part of air-barrier assembly within manufacturer's recommended application temperature ranges. Consult manufacturer when sealant cannot be applied within these temperature ranges.
- E. Wall Openings: Prime concealed, perimeter frame surfaces of windows, curtain walls, storefronts, and doors. Apply transition strip so that a minimum of 3 inches (75 mm) of coverage is achieved over each substrate. Maintain 3 inches (75 mm) of full contact over firm bearing to perimeter frames, with not less than 1 inch (25 mm) of full contact.
 - 1. Transition Strip: Roll firmly to enhance adhesion.
- F. Fill gaps in perimeter frame surfaces of windows, storefronts, and doors, and miscellaneous penetrations of air-barrier material with foam sealant.

- G. Seal strips and transition strips around masonry reinforcing or ties and penetrations with termination mastic.
- H. Seal top of through-wall flashings to air barrier with an additional 6-inch- (150-mm-) wide, transition strip.
- I. Seal exposed edges of strips at seams, cuts, penetrations, and terminations not concealed by metal counterflashings or ending in reglets with termination mastic.
- J. Follow manufacturer's instructions at transition and tie-in between different materials to ensure compatibility and watertight installation.
- K. Repair punctures, voids, and deficient lapped seams in strips and transition strips. Slit and flatten fishmouths and blisters. Patch with transition strips extending 6 inches (150 mm) beyond repaired areas in strip direction.

3.4 AIR-BARRIER MATERIAL INSTALLATION

- A. Apply air-barrier material to form a seal with strips and transition strips and to achieve a continuous air barrier according to air-barrier manufacturer's written instructions and details. Apply air-barrier material within manufacturer's recommended application temperature ranges.
 - 1. Unless manufacturer recommends in writing against priming, apply primer to substrates at required rate and allow it to dry.
 - 2. Limit priming to areas that will be covered by air-barrier material on same day. Reprime areas exposed for more than 24 hours.
 - 3. Where multiple prime coats are needed to achieve required bond, allow adequate drying time between coats.
- B. Apply continuous unbroken air-barrier material to substrates in thickness recommended by air barrier manufacturer. Apply an increased thickness of air-barrier material in full contact around protrusions such as masonry ties.
- C. Do not cover air barrier until it has been tested and inspected by Owner's testing agency.
- D. Correct deficiencies in or remove air barrier that does not comply with requirements; repair substrates and reapply air-barrier components.

3.5 FIELD QUALITY CONTROL

- A. Testing Agency: Owner may engage a qualified testing agency to perform tests and inspections.
- B. Inspections: Air-barrier materials, accessories, and installation are subject to inspection for compliance with requirements.
- C. Tests: As determined by Owner's testing agency from among the following tests:
 - 1. Qualitative Air-Leakage Testing: Air-barrier assemblies will be tested for evidence of air leakage according to ASTM E 1186, smoke pencil with pressurization or depressurization or ASTM E 1186, chamber pressurization or depressurization with smoke tracers.
- D. Air barriers will be considered defective if they do not pass tests and inspections.

1. Apply additional air-barrier material, according to manufacturer's written instructions, where inspection results indicate insufficient thickness.
2. Remove and replace deficient air-barrier components for retesting as specified above.

E. Repair damage to air barriers caused by testing; follow manufacturer's written instructions.

3.6 CLEANING AND PROTECTION

A. Protect air-barrier system from damage during application and remainder of construction period, according to manufacturer's written instructions.

1. Protect air barrier from exposure to UV light and harmful weather exposure as recommended in writing by manufacturer. If exposed to these conditions for longer than recommended, remove and replace air barrier or install additional, full-thickness, air-barrier application after repairing and preparing the overexposed materials according to air-barrier manufacturer's written instructions.
2. Protect air barrier from contact with incompatible materials and sealants not approved by air-barrier manufacturer.

B. Clean spills, stains, and soiling from construction that would be exposed in the completed work using cleaning agents and procedures recommended in writing by manufacturer of affected construction.

C. Remove masking materials after installation.

END OF SECTION 072726

SECTION 074213.23 - METAL COMPOSITE MATERIAL WALL PANELS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes metal composite material wall panels.

1.3 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings: Include fabrication and installation layouts of metal composite material panels; details of edge conditions, joints, panel profiles, corners, anchorages, attachment assembly, trim, flashings, closures, and accessories; and special details.
 - 3. Samples: For each type of metal composite material panel indicated.

1.5 INFORMATIONAL SUBMITTALS

- A. Product test reports.
 - 1. Physical and Performance characteristics.
 - 2. Finishes
 - 3. Storage and handling requirements.
- B. Warranties: Samples of special warranties.

1.6 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials in Manufacturer's original, unopened, undamaged packaging.
- B. Examine delivered panel materials upon receipt to ensure that no damage has occurred during shipment.
- C. Exercise care in unloading, storing and installing panels to prevent bending, warping and twisting.

- D. Store materials at temperature and in humidity conditions recommended by manufacturer and protect from exposure to harmful weather conditions and damage.

1.8 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section, with not less than 10 years documented experience.
 - 1. Manufacturer certified to ISO 9001:2015 with design.
- B. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.
- C. Source Limitation: Obtain components for Composite wall system from or approved by wall system manufacturer.

1.9 FIELD CONDITIONS

- A. Field Measurements: Verify actual dimensions by field measurement prior to starting work.

1.10 WARRANTY

- A. Product Warranty: Manufacturer's standard warranty covering failure in materials caused by manufacturing within specified warranty period.
 - 1. Finish Warranty Period: 20 years from date of Substantial Completion.
- B. Finish Warranty: Manufacturer's special warranty agreeing to repair finish or replace roof specialties that show evidence of deterioration of factory-applied finishes, including significant fading, chalking, cracking, or peeling, within specified warranty period.
 - 1. Finish Warranty Period: 30 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Structural Performance: Provide metal composite material panel systems capable of withstanding the effects of the following loads, based on testing according to ASTM E 330:
 - 1. Wind Loads: As indicated on Drawings.
 - 2. Other Design Loads including Seismic: As indicated on Drawings.
 - 3. Deflection Limits: For wind loads, no greater than 1/180 of the span.
- B. Water Penetration under Static Pressure: No water penetration when tested according to ASTM E 331 at the following test-pressure difference:
 - 1. Test-Pressure Difference: 2.86 lbf/sq. ft. (137 Pa).
- C. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes by preventing buckling, opening of joints, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

1. Temperature Change (Range): 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

- D. Fire Propagation Characteristics: Metal composite material wall panel system passes NFPA 285 testing.

2.2 MANUFACTURER

- A. Manufacturer: ATAS International; www.atas.com
- B. Contact: See **LIDL National Account List**.

2.3 METAL COMPOSITE MATERIAL WALL PANELS

- A. Metal Composite Material Wall Panel Systems: Provide factory-formed and -assembled, metal composite material wall panels fabricated from two metal facings that are bonded to a corrugated polypropylene core; formed into profile for installation method indicated. Include attachment components and accessories required for weathertight system.
- B. Aluminum-Faced Composite Wall Panels: Factory-formed from a core with coil-coated aluminum sheet facings on each side; 0.032-inch (0.81-mm) thick on front/top side and 0.015-inch (0.38 mm) thick on back/bottom side.
 1. Product: ATAS SterraCore Aluminum Composite Material Panel.
 2. Panel Thickness: 1/4 inch (6 mm)
 3. Panel Size: Manufacturer's standard size for application indicated on Drawings.
 4. Core: Corrugated polypropylene center laminated between two finished aluminum sheets.
 5. Joint Type: 1-Piece, Tight-Fit Molding Trim with PVDF painted finish matching panel.
 6. Exposed Finish: Two-coat fluoropolymer.
 7. Color: See Exterior Material/Finish Schedule on Drawings.
- C. Attachment Assembly Components: Formed from extruded aluminum with PVDF painted finish matching panel.
- D. Substitutions: See Section 016000 Product Requirements.
 1. Substitutions will be considered only if materials meet the requirements of Basis of Design and are approved by the Architect in writing 10 day prior to bid.

2.4 MISCELLANEOUS MATERIALS

- A. Panel Accessories, Flashing and Trim: Provide components, flashing and trim formed from same material as metal composite material panels as required to seal against weather and to provide a weathertight panel system and finished appearance. Components include, but are not limited to J-moldings, H-moldings, drips, corners, rakes and fillers. Match material and finish of adjacent metal composite material panels.
 1. Factory-fabricate trims and flashings in a minimum 12-foot lengths, or as dictated by project conditions and approved by panel manufacturer.
- B. Panel Accessories Fasteners: Concealed, self-tapping screws designed to withstand design loads.

- C. Panel Sealants: ASTM C 920; silicone sealant of types recommended by metal composite material panel manufacturer that are compatible with panel materials. Provide sealant types recommended by manufacturer that are compatible with panel materials, are non-staining, and do not damage panel finish.

- 1. Sealant Color: Match color of adjacent panel.

2.5 FABRICATION

- A. General: Fabricate and finish metal composite material panels and accessories at the factory, by manufacturer's standard procedures and processes, as necessary to fulfill indicated performance requirements demonstrated by laboratory testing. Comply with indicated profiles and with dimensional and structural requirements.
- B. Sheet Metal Flashing and Trim: Fabricate flashing and trim to comply with manufacturer's recommendations and recommendations in SMACNA's "Architectural Sheet Metal Manual" that apply to design, dimensions, metal, and other characteristics of item indicated.

2.6 ALUMINUM FINISH

- A. Two-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, metal composite material panel supports, and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Miscellaneous Supports: Install subframing, furring, and other miscellaneous panel support members and anchorages according to ASTM C754 and metal composite material panel manufacturer's written recommendations.

3.3 METAL COMPOSITE MATERIAL PANEL INSTALLATION

- A. General: Install metal composite material panels according to manufacturer's written instructions in orientation, sizes, and locations indicated on Drawings. Install panels perpendicular to supports unless otherwise indicated. Anchor metal composite material panels and other components of the Work securely in place, with provisions for thermal and structural movement.
 - 1. Shim or otherwise plumb substrates receiving metal composite material panels.
 - 2. Flash and seal metal composite material panels at perimeter of all openings. Fasten with self-tapping screws. Do not begin installation until air- or water-resistive barriers and flashings that will be concealed by metal composite material panels are installed.
 - 3. Install screw fasteners in predrilled holes.
 - 4. Locate and space fastenings in uniform vertical and horizontal alignment.

5. Install flashing and trim as metal composite material panel work proceeds.
6. Locate panel splices over, but not attached to, structural supports. Stagger panel splices and end laps to avoid a four-panel lap splice condition.
7. Align bottoms of metal composite material panels and fasten with blind rivets, bolts, or self-tapping screws. Fasten flashings and trim around openings and similar elements with self-tapping screws.
8. Provide weathertight escutcheons for pipe- and conduit-penetrating panels.

B. Fasteners:

1. Aluminum Panels: Use stainless-steel fasteners.

C. Metal Protection: Where dissimilar metals contact each other or corrosive substrates, protect against galvanic action as recommended in writing by metal composite material panel manufacturer.

D. Attachment Assembly, General: Install attachment elements required by panel manufacturer to support metal composite material wall panels and to provide a complete weathertight wall system, including, perimeter extrusions, drainage trim, molding extrusions, corner angles, panel adhesive, and foam tape.

1. Include attachment to supports, panel-to-panel joinery, panel-to-dissimilar-material joinery, and panel-system joint seals.

E. Installation: Attach metal composite material wall panels to supports at locations, spacings, and with fasteners required by manufacturer.

F. Accessory Installation: Install accessories with positive anchorage to building and weathertight mounting and provide for thermal expansion. Coordinate installation with flashings and other components.

G. Flashing and Trim: Comply with performance requirements, manufacturer's written installation instructions, and SMACNA's "Architectural Sheet Metal Manual." Provide concealed fasteners where possible and set units true to line and level as indicated. Install work with laps, joints, and seams that are permanently watertight.

3.4 ERECTION TOLERANCES

- A. Installation Tolerances: Shim and align metal composite material wall panel units within installed tolerance of 1/4 inch in 20 feet (6 mm in 6 m), non-accumulative, on level, plumb, and location lines as indicated, and within 1/8-inch (3-mm) offset of adjoining faces and of alignment of matching profiles.

3.5 CLEANING AND PROTECTION

- A. Remove temporary protective coverings and strippable films, if any, as metal composite material panels are installed, unless otherwise indicated in manufacturer's written installation instructions. On completion of metal composite material panel installation, clean finished surfaces as recommended by metal composite material panel manufacturer. Maintain in a clean condition during construction.
- B. After metal composite material panel installation, clear weep holes and drainage channels of obstructions, dirt, and sealant.

- C. Replace metal composite material panels that have been damaged or have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.

END OF SECTION 074213.23

SECTION 074243 - COMPOSITE WALL PANELS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes phenolic rainscreen wall panels and aluminum supporting sub structure.

1.3 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product. Include manufacturer's installation instructions and details.
 - 2. Shop Drawings: Include fabrication and installation layouts of composite wall panels; joints, corners, anchorages, attachment assembly, trim, flashings, and accessories; and special details.
 - 3. Samples: For each type of composite wall panel indicated.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Delivery: Deliver materials in Manufacturer's original, unopened, undamaged packaging.
- B. Storage: Store materials at temperature and in humidity conditions recommended by manufacturer and protect from exposure to harmful weather conditions and damage. Store materials inside on a flat, dry surface in original packaging.

1.7 QUALITY ASSURANCE

- A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer and that have a minimum of 3 years' experience installing rainscreen systems.
- B. Source Limitations: Obtain all components of the phenolic rainscreen system, aluminum sub structure and stainless-steel color matched fasteners from a single manufacturer.

1.8 WARRANTY

- A. Manufacturer's Warranty: For material defects and manufacturing tolerances.
 - 1. Warranty Period: 10 years from date of Substantial Completion.
- B. Installer's Warranty: For manufacturing defects. Repair or replace materials.
 - 1. Warranty Period: Two years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURER

- A. Manufacturer: Fiberesin Industries; phone 262-560-4460; www.stonewoodpanels.com.
- B. Contact: See **LIDL National Account List**.
- C. No substitution allowed.

2.2 MATERIALS

- A. Phenolic Exterior Wall Cladding: Solid phenolic laminate panel with UV protective clear coat.
 - 1. Style: Stonewood Select Woodgrain
 - 2. Color: See Drawings.
 - 3. Finish: Factory #60 matte
 - 4. Plank Size: 5-5/8 by 96 inches
 - 5. Thickness: 5/16 inch
 - 6. Panel Core: Phenolic resin treated layer, black and natural brown kraft paper
 - 7. Decorative Layer: Melamine resin, proprietary pigmented and treated
 - 8. Weather and UV Resistant Layer: 2-mil, proprietary layer combining pigment protection, Ultraviolet light and weather resistant layer.
 - 9. ASTM E84: Flame Development <60; Smoke Development <200
- B. Physical Properties
 - 1. Flexural Strength per ASTM D-790: >12,000
 - 2. Flexural Modulus per ASTM D-790: > 1.1 x 10⁶
 - 3. Tensile Modulus per ASTM D-638: 10,000
 - 4. Panels should have a UV and Weather resistance performance with a grey scale rating of 4 per ISO4892-2, 4892-3
 - 5. Wear Resistance NEMA 3.13: >500 cycles
 - 6. Ball Drop Impact resistance NEMA 3.8: >80"
 - 7. Cleanability NEMA 3.4: <= 10
 - 8. Dimensional Change NEMA 3.11 0.3% Length; 0.7% Width

2.3 ACCESSORIES

- A. Fasteners: Manufacturer approved austenitic stainless-steel fastener with bi-metal welded carbon steel point.
 - 1. Exposed fasteners SX3-D12 or equal, in color to match plank.

- B. Attachment System: Approved sub-frame system to support a cladding weight of up to 8 pounds per square foot, fabricated of 0.09-inches 6063 T5 extruded aluminum or 16-gauge G90 galvanized steel.

- 1. Stonewood EF Exposed Fastener Mounting.

2.4 FABRICATION

- A. Field cuts are allowed as per manufacturer's guidelines.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine materials, installation instructions, and review manufacturer's instructions on site.
- B. Verify panel style, color, and fasteners are approved by Architect.
- C. Verify substrates and adjacent surfaces are level and plumb for installation.
 - 1. Do not begin work until construction has progressed to allow installation of materials.
 - 2. Verify manufacturers fastener spacing requirements.
- D. Proceed with work when construction has progressed to allow a warranted installation.
 - 1. Installation deems acceptance of work for a warranted installation.

3.2 INSTALLATION

- A. Install in accordance with manufacturers written instructions and Shop Drawings, maintaining required 1-inch ventilation spacing requirement.
- B. Fasten panels to an approved attachment system structurally supported by aluminum, galvanized steel or wood stud supported wall.
- C. Fasten panels to an approved attachment system structurally supported by aluminum, galvanized steel or wood stud supported wall.
 - 1. Use manufacturer approved fasteners to affix phenolic panels to the appropriate sub framing system.
- D. Panels should be installed plumb and level.
- E. Do not caulk joints, apply adhesive or sealants to phenolic panels.
- F. Install panel square, edges clean and true to size.
 - 1. Cut panels to fit at perimeter and around penetrations with minimum 3/8-inch gap.
 - 2. Field cuts are allowed.
- G. Do not install damaged, irregular and defective panels.

3.3 FIELD QUALITY CONTROL

- A. Inspect panel ventilation at top and bottom of wall, and manufacturers fixed and floating holes instructions have been followed.
- B. Verify installation, fasteners and connections with adjacent materials, and transitions have been completed in accordance with shop drawings.
- C. Verify that all field cut edges have been chamfered per manufacturer's instructions.

3.4 ADJUSTING

- A. Modify, adjust and replace panels not within manufacturers tolerances and as required by Architect.

3.5 PROTECTION

- A. Protect surface, corners and components from damage prior to Owner occupancy using temporary protection.

END OF SECTION 074243

SECTION 074293 – METAL SOFFIT SYSTEMS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes metal soffit panel and brake metal.

1.2 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings: Include fabrication and installation layouts of metal panels; details of edge conditions, joints, panel profiles, corners, anchorages, attachment system, trim, flashings, closures, and accessories; and special details.
 - 3. Samples: For each type of metal panel indicated.

1.4 INFORMATIONAL SUBMITTALS

- A. Product test reports.
 - 1. Physical and Performance characteristics.
 - 2. Finishes
 - 3. Storage and handling requirements.
- B. Warranties: Samples of special warranties.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.6 QUALITY ASSURANCE

- A. Manufacturer's Qualification: Company specializing in manufacturing product specified in this section with not less than 10 years documented experience.
 - 1. Manufacturer certified ISO 9001: 2015 with design.
- B. Installer's Qualification: Company specializing in installation of products of the type specified with a minimum 3 years documented experience.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials in Manufacturer's original, unopened, undamaged packaging.

- B. Examine delivered material upon receipt to ensure that no damage has occurred during transportation.
- C. Exercise care in unloading, storing and installing panels to prevent bending, twisting and warping.
- D. Store materials at temperature and in humidity conditions recommended by manufacturer and protect from exposure to harmful weather conditions and damage.

1.8 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.9 WARRANTY

- A. Product Warranty: Manufacturer's standard warranty covering failure in materials caused by manufacturing within specified warranty period.
 - 1. Finish Warranty Period: 20 years from date of Substantial Completion.
- B. Finish Warranty: Manufacturer's special warranty agreeing to repair finish or replace roof specialties that show evidence of deterioration of factory-applied finishes, including significant fading, chalking, cracking, or peeling, within specified warranty period.
 - 1. Finish Warranty Period: 30 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Structural Performance: Provide metal panel systems capable of withstanding the effects of the following loads, based on testing according to ASTM E 1592:
 - 1. Wind Loads: As indicated on Structural Drawings.
 - 2. Other Design Loads: As indicated on Structural Drawings.
 - 3. Deflection Limits: For wind loads, no greater than 1/240 of the span.
- B. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes by preventing buckling, opening of joints, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
 - 1. Temperature Change (Range): 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

2.2 MANUFACTURER

- A. Manufacturer: ATAS International; www.atas.com
- B. Contact: **See LIDL National Account List.**
- C. Source Limitations: Obtain all components for a complete soffit assembly from or approved by the soffit system manufacturer.

2.3 METAL SOFFIT PANELS

- A. General: Provide metal soffit panels designed to be installed by lapping and interconnecting side edges of adjacent panels and mechanically attaching through panel to supports using concealed fasteners in side laps. Include accessories required for weathertight installation.
- B. Flush-Profile Metal Soffit Panels: Solid panels formed with vertical panel edges and a flat pan between panel edges; with flush joint between panels.
 - 1. Product: ATAS Opaline OPF 040 Metal Soffit
 - 2. Aluminum Sheet: Coil-coated sheet, ASTM B 209 (ASTM B 209M), alloy as standard with manufacturer, with temper as required to suit forming operations and structural performance required.
 - 3. Thickness: 0.032 inch (0.81 mm)
 - 4. Panel Net Coverage: 4 inches (102 mm)
 - 5. Panel Depth: 0.75 inch (19 mm)
 - 6. Surface: Smooth, flat finish
 - 7. Exposed Finish: Two-coat fluoropolymer.
 - 8. Color: See Exterior Material/Finish Schedule on Drawings.
- C. Edge Trim / Venting Strip: 5-inch wide, custom Z-shaped edge trim, perforated for venting allowing 9.77 square inch per linear foot of open air.
- D. Substitutions: See Section 016000 Product Requirements.
 - 1. Substitutions will be considered only if materials meet the requirements of Basis of Design and are approved by the Architect in writing 10 day prior to bid.

2.4 MISCELLANEOUS MATERIALS

- A. Miscellaneous Metal Subframing and Furring: ASTM C 645, cold-formed, metallic-coated steel sheet, ASTM A 653/A 653M, G90 (Z275 hot-dip galvanized) coating designation or ASTM A 792/A 792M, Class AZ50 (Class AZM150) aluminum-zinc-alloy coating. Provide manufacturer's standard sections as required for support and alignment of metal panel system.
- B. Panel Accessories: Provide components required for a complete, weathertight panel system including trim, clips, flashings, sealants, gaskets, fillers, closure strips, and similar items. Match material and finish of metal soffit panels.
- C. Flashing and Trim: Provide flashing and trim formed from same material as metal panels as required to seal against weather and to provide finished appearance. Finish flashing and trim with same finish system as adjacent metal panels.
- D. Panel Fasteners: Self-tapping screws designed to withstand design loads. Provide exposed fasteners with heads matching color of metal panels by means of plastic caps or factory-applied coating. Provide EPDM or PVC sealing washers for exposed fasteners.
- E. Panel Sealants: ASTM C 920; silicone sealant of types recommended by manufacturer that are compatible with panel materials, are non-staining, and do not damage panel finish.
 - 1. Sealant Tape: Pressure-sensitive, 100 percent solids, gray polyisobutylene compound sealant tape with release-paper backing; 1/8 inch (3 mm) thick.
 - 2. Sealant Color: Match color of adjacent panel.

2.5 FABRICATION

- A. General: Fabricate and finish metal panels and accessories at the factory, by manufacturer's standard procedures and processes, as necessary to fulfill indicated performance requirements demonstrated by laboratory testing. Comply with indicated profiles and with dimensional and structural requirements.
- B. Provide panel profile, including major ribs and intermediate stiffening ribs, if any, for full length of panel.
- C. Fabricate metal panel joints with factory-installed captive gaskets or separator strips that provide a weathertight seal and prevent metal-to-metal contact, and that minimize noise from movements.
- D. Sheet Metal Flashing and Trim: Fabricate flashing and trim to comply with manufacturer's recommendations and recommendations in SMACNA's "Architectural Sheet Metal Manual" that apply to design, dimensions, metal, and other characteristics of item indicated.

2.6 ALUMINUM FINISHES

- A. Two-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, metal panel supports, and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Miscellaneous Supports: Install subframing, furring, and other miscellaneous panel support members and anchorages according to ASTM C 754 and metal panel manufacturer's written recommendations.

3.3 METAL PANEL INSTALLATION

- A. Install metal panels according to manufacturer's written instructions in orientation, sizes, and locations indicated. Install panels perpendicular to supports unless otherwise indicated. Anchor metal panels and other components of the Work securely in place, with provisions for thermal and structural movement.
 - 1. Shim or otherwise plumb substrates receiving metal panels.
 - 2. Flash and seal metal panels at perimeter of all openings. Fasten with self-tapping screws. Do not begin installation until air- or water-resistive barriers and flashings that will be concealed by metal panels are installed.
 - 3. Install screw fasteners in predrilled holes.
 - 4. Locate and space fastenings in uniform vertical and horizontal alignment.
 - 5. Install flashing and trim as metal panel work proceeds.

6. Locate panel splices over, but not attached to, structural supports. Stagger panel splices and end laps to avoid a four-panel lap splice condition.
7. Provide weathertight escutcheons for pipe- and conduit-penetrating panels.

B. Fasteners:

1. Aluminum Panels: Use stainless steel fasteners.

C. Metal Soffit Panels: Fasten metal panels to supports with fasteners at each lapped joint at location and spacing recommended by manufacturer.

1. Apply panels and associated items true to line for neat and weathertight enclosure.
2. Provide metal-backed washers under heads of exposed fasteners bearing on weather side of metal panels.
3. Locate and space exposed fasteners in uniform vertical and horizontal alignment. Use proper tools to obtain controlled uniform compression for positive seal without rupture of washer.
4. Install screw fasteners with power tools having controlled torque adjusted to compress washer tightly without damage to washer, screw threads, or panels. Install screws in predrilled holes.

D. Accessory Installation: Install accessories with positive anchorage to building and weathertight mounting and provide for thermal expansion. Coordinate installation with flashings and other components.

E. Flashing and Trim: Comply with performance requirements, manufacturer's written installation instructions, and SMACNA's "Architectural Sheet Metal Manual." Provide concealed fasteners where possible and set units true to line and level as indicated. Install work with laps, joints, and seams that are permanently watertight.

3.4 CLEANING AND PROTECTION

- A. Remove temporary protective coverings and strippable films, if any, as metal panels are installed unless otherwise indicated in manufacturer's written installation instructions. On completion of metal panel installation, clean finished surfaces as recommended by metal panel manufacturer. Maintain in a clean condition during construction.
- B. After metal panel installation, clear weep holes and drainage channels of obstructions, dirt, and sealant.
- C. Replace metal panels that have been damaged or have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.

END OF SECTION 074293

SECTION 075419 - POLYVINYL-CHLORIDE (PVC) ROOFING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Mechanically fastened polyvinyl chloride (PVC) roofing system.
 - 2. Roof insulation.
- B. Related Section:
 - 1. Section 061600 "Sheathing" for sheathing on back of parapet walls.

1.3 DEFINITIONS

- A. Roofing Terminology: Definitions in ASTM D 1079 and glossary in NRCA's "The NRCA Roofing and Waterproofing Manual" apply to work of this Section.

1.4 PREINSTALLATION MEETINGS

- A. Preinstallation Roofing Conference: Conduct conference at Project site.

1.5 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings: For roofing system. Include plans, elevations, sections, details, and attachments to other work.
 - 3. Samples: For the following products:
 - a. Fabricated inside corner of sheet roofing with insulation, in color required, demonstrating seams in roofing and size as necessary to include a complete inside corner at parapet.
 - b. Walkway pads, of color required.

1.6 INFORMATIONAL SUBMITTALS

- A. Research/Evaluation Reports: For components of roofing system, from ICC-ES.
- B. Product Certificates.
- C. Sample Warranties: For manufacturer's special warranties.

1.7 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For roofing system to include in maintenance manuals.

1.8 QUALITY ASSURANCE

- A. Installer Qualifications: A qualified firm that is approved, authorized, or licensed by roofing system manufacturer to install manufacturer's product and that is eligible to receive manufacturer's special warranty.

1.9 DELIVERY, STORAGE, AND HANDLING

- A. Ship membrane rolls horizontally on pallets, stacked pyramid style and banded.
- B. Store rolls horizontally on pallets or shelves and maintain in a dry location.

1.10 WARRANTY

- A. Special Warranty: Manufacturer agrees to repair or replace components of roofing system that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period: 20 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Carlisle SynTec Incorporated.
 - 2. Firestone Building Products.
 - 3. GAF Materials Corporation, Basis of Design.
 - 4. GenFlex Roofing Systems.
- B. Source Limitations: Obtain components including roof insulation fasteners for roofing system from manufacturer approved by membrane roofing manufacturer.

2.2 PERFORMANCE REQUIREMENTS

- A. Accelerated Weathering: Roofing system shall withstand 2000 hours of exposure when tested according to ASTM G 152, ASTM G 154, or ASTM G 155.
- B. Impact Resistance: Roofing system shall resist impact damage when tested according to ASTM D 3746 or ASTM D 4272.
- C. Roofing System Design: Tested by a qualified testing agency to resist the uplift pressures in accordance with FM I-90 and ASCE/SEI 7 requirements for wind pressures as indicated on Structural Drawings.
- D. Solar Reflectance Index: Not less than 78 when calculated according to ASTM E 1980, based on testing identical products by a qualified testing agency.

- E. Energy Star Listing: Roofing system shall be listed on the DOE's ENERGY STAR "Roof Products Qualified Product List" for low-slope roof products.
- F. Energy Performance: Roofing system shall have an initial solar reflectance index of not less than 0.70 and an emissivity of not less than 0.75 when tested according to CRRC-1.
- G. Exterior Fire-Test Exposure: ASTM E 108 or UL 790, Class A; for application and roof slopes indicated; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
- H. Fire-Resistance Ratings: Comply with fire-resistance-rated assembly designs indicated. Identify products with appropriate markings of applicable testing agency.

2.3 POLYVINYL CHLORIDE (PVC) ROOFING

- A. PVC Sheet: ASTM D4434/D4434M, Type III, fabric reinforced.
 - 1. Basis of Design Product: GAF EverGuard PVC Membrane.
 - 2. Thickness: 60 mils (1.5 mm), nominal.
 - 3. Exposed Face Color: Gray

2.4 AUXILIARY ROOFING MATERIALS

- A. General: Auxiliary materials recommended by roofing system manufacturer for intended use and compatible with roofing.
 - 1. Liquid-type auxiliary materials shall comply with VOC limits of authorities having jurisdiction.
 - 2. Adhesives and sealants that are not on the exterior side of weather barrier shall comply with the following limits for VOC content:
 - a. Plastic Foam Adhesives: 50 g/L.
 - b. Multipurpose Construction Adhesives: 70 g/L.
 - c. Fiberglass Adhesives: 80 g/L.
 - d. Single-Ply Roof Membrane Adhesives: 250 g/L.
 - e. Single-Ply Roof Membrane Sealants: 450 g/L.
 - f. Nonmembrane Roof Sealants: 300 g/L.
 - g. Sealant Primers for Nonporous Substrates: 250 g/L.
 - h. Sealant Primers for Porous Substrates: 775 g/L.
 - i. Other Adhesives and Sealants: 250 g/L.
- B. Sheet Flashing: Manufacturer's standard sheet flashing of same material, type, reinforcement, thickness and color as PVC sheet.
- C. Bonding Adhesive: Manufacturer's standard, water based.
- D. Slip Sheet: Manufacturer's standard, of thickness required for application.
- E. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Global 4470, designed for fastening roofing to substrate, and acceptable to roofing system manufacturer.

- F. Miscellaneous Accessories: Provide metal termination bars, metal battens, pourable sealers, preformed cone and vent sheet flashings, preformed inside and outside corner sheet flashings, T-joint covers, lap sealants, termination reglets, and other accessories.

2.5 SUBSTRATE BOARDS ON BACK OF PARAPET WALLS

- A. Substrate Board: ASTM C1177/C1177M, glass-mat, water-resistant gypsum board.
 - 1. Georgia-Pacific Gypsum LLC, DensDeck Prime Roof Board with EONIC Technology or approved equivalent.
 - 2. Type and Thickness: 1/2 inch (12.7 mm) thick.
- B. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Approvals 4470, designed for fastening substrate board to roof deck.

2.6 ROOF INSULATION

- A. Polyisocyanurate Board Insulation: ASTM C 1289, Type II, Class 1, Grade 2, felt or glass-fiber mat facer on both major surfaces.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Carlisle SynTec Incorporated.
 - b. Firestone Building Products.
 - c. GAF Materials Corporation.
 - d. Johns Manville.
 - e. Rmax, Inc.
- B. Tapered Insulation: Provide factory-tapered insulation boards where required for sloping to drain and fabricated to slope of 1/4 inch per 12 inches (1:48) or greater.
- C. Provide preformed saddles, crickets, tapered edge strips, and other insulation shapes where indicated for sloping to drain. Fabricate to slopes indicated.
- D. Insulation Fastening Pattern: Install fasteners at equal spacing on X and Y direction from one panel to the other. Each 4X8 panel will have the spacing as following:
 - 1. 8'-0" side shall be divided in 1'-4", 2'-8", 2'-8" and 1'-4"
 - 2. 4'-0" side shall be divided in 1'-0", 2'-0" and 1'-0"

2.7 INSULATION ACCESSORIES

- A. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Global 4470, designed for fastening roof insulation to substrate, and acceptable to roofing system manufacturer.

2.8 VAPOR RETARDER

- A. Self-Adhering-Sheet Vapor Retarder: ASTM D1970/D1970M, polyethylene film laminated to layer of rubberized asphalt adhesive, minimum 30-mil (.76-mm) total thickness; maximum permeance rating of 0.1 perm (6 ng/Pa x s x sq. m); cold applied, with slip-resisting surface and

release paper backing. Provide vapor retarder of type recommended for adhering directly to steel decking. Provide primer when recommended by vapor retarder manufacturer.

2.9 PAD-TYPE SNOW GUARDS

A. Flat Surface Mounted Plastic Snow Guard Pads:

1. Snow Guard as manufactured by Zaleski Snow Guards for Roofs, Inc., Model No. 15 clear plastic.
2. Material: Clear UV-stabilized polycarbonate.
3. Adhesive Sealant: Neutral cure 100% silicone, single component as manufactured by Hi-Tec Industries, Project 1, 6700 Series, clear.

2.10 WALKWAYS

- ### A. Flexible Walkways:
- Factory-formed, nonporous, heavy-duty, slip-resisting, surface-textured walkway pads, approximately 3/16 inch (5 mm) thick and acceptable to roofing system manufacturer.

PART 3 - EXECUTION

3.1 ROOFING INSTALLATION, GENERAL

- #### A. Install roofing system according to roofing system manufacturer's written instructions.
- #### B. Complete terminations and base flashings and provide temporary seals to prevent water from entering completed sections of roofing system at the end of the workday or when rain is forecast. Remove and discard temporary seals before beginning work on adjoining roofing.

3.2 VAPOR RETARDER INSTALLATION

- #### A. Self-Adhering-Sheet Vapor Retarder: Prime substrate if required by manufacturer. Install self-adhering-sheet vapor retarder over roof decking, side and end lapping each sheet a minimum of 3-1/2 and 6 inches (90 and 150 mm), respectively.
1. Extend vertically up parapet walls and projections to a minimum height equal to height of insulation.
 2. Seal laps by rolling.
 3. Coordinate with air barrier on walls for compatibility and lapping of material for proper drainage of moisture.

3.3 SUBSTRATE BOARD INSTALLATION ON BACK OF PARAPET WALLS

- #### A. Install substrate board with long joints in continuous straight lines, with end joints staggered not less than 24 inches (610 mm) in adjacent rows.
1. At steel roof decks, install substrate board at right angle to flutes of deck.
 - a. Locate end joints over crests of steel roof deck.
 2. Tightly butt substrate boards together.
 3. Cut substrate board to fit tight around penetrations and projections, and to fit tight to intersecting sloping roof decks.

4. Fasten substrate board to top flanges of steel deck to resist uplift pressure at corners, perimeter, and field of roof according to roofing system manufacturers' written instructions.

3.4 INSULATION INSTALLATION

- A. Coordinate installing roofing system components so insulation is not exposed to precipitation or left exposed at the end of the workday.
- B. Install tapered insulation under area of roofing for crickets and to conform to slopes indicated. Refer to structural drawings for roof steel slopes.
- C. Install insulation under area of roofing to achieve required thickness. Provide a total minimum LTTR-Value of 30.0. Provide greater LTTR-Value where required by the Energy Code in-force at jurisdiction in which project is located. Where overall insulation thickness is greater than 5.00 inches (127 mm) or greater, install two or more layers with joints of each succeeding layer staggered from joints of previous layer a minimum of 6 inches (150 mm) in each direction.
- D. Fasteners: Provide factory-coated steel fasteners and metal or plastic plates meeting corrosion-resistance provisions in FMG 4470, designed for fastening membrane and insulation to substrate in conformance to specified design requirements.

3.5 MECHANICALLY ROOFING INSTALLATION

- A. Mechanically fastened roofing over main building area to receive roofing according to roofing system manufacturer's written instructions. Unroll roofing and allow to relax before retaining.
- B. Mechanically Fastened Membrane:
 1. Mechanically fastened membrane to the structural deck utilizing fasteners and fastening patterns that in accordance with roof membrane manufacturer's requirements.
 2. Cut membrane to fit neatly around all penetrations and roof projections.
 3. Unroll roofing membrane and positioned with a minimum 6 inch overlap.
- C. Seaming:
 1. Weld overlapping sheets together using hot air. Minimum weld width is 1-1/2 inches (38 mm).
 2. Check welded seams for continuity and integrity by the end of each work day. Repair all imperfections.
- D. Membrane Termination/Securement: All membrane terminations shall be completed in accordance with roof membrane manufacturer's requirements.
 1. Provide securement at all membrane terminations at the perimeter of each roof level, roof section, curb flashing, skylight, expansion joint, interior wall, penthouse, and other similar condition.
 2. Provide securement at any angle change where the slope or combined slopes exceeds two inches in one horizontal foot.
 3. In addition to adhering, mechanically fasten roofing securely at terminations, penetrations, and perimeter of roofing.
- E. Flashings: Complete all flashings and terminations as indicated on the Drawings and fully adhere to substrates in accordance with roof membrane manufacturer's requirements.

1. Base and Wall Flashings:

- a. Do not apply flashing over existing thru-wall flashings or weep holes.
- b. Secure flashing on vertical surfaces before the seam between the flashing and the main roof sheet is completed.
- c. Extend flashing membrane a minimum of 6 inches (152 mm) onto the main roof sheet beyond the mechanical securement.
- d. Use care to ensure that the flashing does not bridge locations where there is a change in direction (e.g. where the parapet meets the roof deck).

2. Penetrations:

- a. Flash all pipes, supports, soil stacks, cold vents, and other penetrations passing through the roofing membrane as indicated on the Drawings and in accordance with roof membrane manufacturer's requirements.
- b. Utilize custom prefabricated flashings supplied by roof membrane manufacture.

3. Pipe Clusters and Unusual Shapes:

- a. Clusters of pipes or other penetrations which cannot be sealed with prefabricated flashings or membrane shall be sealed by surrounding them with pourable sealer supplied by roof membrane manufacturer.
- b. Penetration pans shall be installed, flashed and filled with pourable sealer in accordance with roof membrane manufacturer's requirements.
- c. Penetration pans shall not be used where prefabricated or field fabricated flashings are possible.

F. Roof Drains:

1. Coordinate installation of roof drains and vents specified in Division 22.
2. Provide a smooth clean surface on the mating surface between the clamping ring and the drain base.
3. Taper insulation around the drain to prevent the membrane from bridging and to provide a smooth transition from the roof surface to the drain clamping ring.
4. Use sealant to seal between the membrane and the drain base using constant, even compression from the drain clamping ring.

G. Fascia/Drip Edge:

1. Provide fascia bar and cover, drip edge and gravel stop as indicated on the Drawings. Install in accordance with roof membrane manufacturer's requirements.
2. Seal joints between individual sections in accordance with roof membrane manufacturer's requirements.
3. Coordinate installation of metal flashing and counter flashing specified in Section 07 62 00 - Sheet Metal Flashing and Trim.
4. Manufactured Roof Specialties: Coordinate installation of copings, counter flashing systems, gutters, downspouts, and roof expansion assemblies specified in Section 077100 – Roof Specialties.

3.6 ADHERED ROOFING INSTALLATION

- A. Adhere roofing to back of parapet walls, curbs, and similar vertical surfaces, and over roof area indicated to receive adhered roofing according to roofing system manufacturer's written

instructions. Unroll roofing and allow to relax before retaining. Apply recommended adhesive for vertical installations.

- B. Accurately align roofing and maintain uniform side and end laps of minimum dimensions required by manufacturer. Stagger end laps.
- C. Bonding Adhesive: Apply to substrate and underside of roofing at rate required by manufacturer and allow to partially dry before installing roofing. Do not apply to splice area of roofing.
- D. In addition to adhering, mechanically fasten roofing securely at terminations, penetrations, and perimeter of roofing.
- E. Seams: Clean seam areas, overlap roofing, and hot-air weld side and end laps of roofing and sheet flashings according to manufacturer's written instructions, to ensure a watertight seam installation.
 - 1. Test lap edges with probe to verify seam weld continuity. Apply lap sealant to seal cut edges of sheet.
 - 2. Verify field strength of seams a minimum of twice daily, and repair seam sample areas.
 - 3. Repair tears, voids, and lapped seams in roofing that do not comply with requirements.
- F. Spread sealant bed over deck-drain flange at roof drains, and securely seal roofing in place with clamping ring.

3.7 PARAPET WALLS MEMBRANE INSTALLATION

- A. Prior to installing roof membrane to back of parapet walls, verify compatibility of fluid-applied air barrier. Obtain a written confirmation from membrane roofing manufacturer.
- B. Install membrane over the entire back of parapet walls, starting at 4 inches below the fasteners at roof membrane termination and extending the membrane up, over the top, and down the outside edge of parapet walls a minimum of 1/2 inch below bottom edge of wood nailer.
- C. Apply membrane sheet horizontally, in shingle fashion starting with bottom course first; fully adhere membrane to substrate following manufacturer's written recommendations. Mechanically fasten sheet where required by manufacturer.
 - 1. Overlap membrane seams as recommended by manufacturer, ensuring a watertight installation.
 - 2. Repair tears, voids, and lapped seams in membrane that does not meet requirements.

3.8 SNOW GUARD INSTALLATION

- A. Install snow guards according to manufacturer's written instructions. Space rows as recommended by manufacturer and as indicated.
 - 1. Surface-Mounted, Snow Guard Pads: Securely attach to membrane roofing with sealant adhesive compatible with roofing and plastic snow guard as recommended by manufacturer.

3.9 WALKWAY INSTALLATION

- A. Flexible Walkways: Install walkway products in locations indicated. Heat weld to substrate or adhere walkway products to substrate with compatible adhesive according to roofing system manufacturer's written instructions.

3.10 PROTECTING AND CLEANING

- A. Protect roofing system from damage and wear during remainder of construction period. When remaining construction does not affect or endanger roofing, inspect roofing for deterioration and damage, describing its nature and extent in a written report, with copies to Architect and Owner.
- B. Correct deficiencies in or remove roofing system that does not comply with requirements, repair substrates, and repair or reinstall roofing system to a condition free of damage and deterioration at time of Substantial Completion and according to warranty requirements.
- C. Clean overspray and spillage from adjacent construction using cleaning agents and procedures recommended by manufacturer of affected construction.

END OF SECTION 075423

SECTION 075423 - THERMOPLASTIC POLYOLEFIN (TPO) ROOFING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Mechanically fastened thermoplastic polyolefin (TPO) roofing system at main roof.
 - 2. Fully adhered thermoplastic polyolefin (TPO) roofing system at entry canopy roof and vertical surfaces.
 - 3. Roof insulation.
- B. Related Section:
 - 1. Section 061600 "Sheathing" for sheathing on back of parapet walls.

1.3 DEFINITIONS

- A. Roofing Terminology: Definitions in ASTM D 1079 and glossary in NRCA's "The NRCA Roofing and Waterproofing Manual" apply to work of this Section.

1.4 PREINSTALLATION MEETINGS

- A. Preinstallation Roofing Conference: Conduct conference at Project site.

1.5 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings: For roofing system. Include plans, elevations, sections, details, and attachments to other work.
 - 3. Samples: For the following products:
 - a. Fabricated inside corner of sheet roofing with insulation, in color required, demonstrating seams in roofing and size as necessary to include a complete inside corner at parapet.
 - b. Walkway pads, of color required.

1.6 INFORMATIONAL SUBMITTALS

- A. Research/Evaluation Reports: For components of roofing system, from ICC-ES.
- B. Product Certificates.

- C. Sample Warranties: For manufacturer's special warranties.

1.7 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For roofing system to include in maintenance manuals.

1.8 QUALITY ASSURANCE

- A. Installer Qualifications: A qualified firm that is approved, authorized, or licensed by roofing system manufacturer to install manufacturer's product and that is eligible to receive manufacturer's special warranty.

1.9 DELIVERY, STORAGE, AND HANDLING

- A. Ship membrane rolls horizontally on pallets, stacked pyramid style and banded.
- B. Store rolls horizontally on pallets of shelves and maintain in a dry location.

1.10 WARRANTY

- A. Special Warranty: Manufacturer agrees to repair or replace components of roofing system that fail in materials or workmanship within specified warranty period.

1. Warranty Period: 20 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 1. Carlisle SynTec Incorporated.
 2. Firestone Building Products.
 3. GAF Materials Corporation, Basis of Design.
 4. GenFlex Roofing Systems.
- B. Source Limitations: Obtain components including roof insulation fasteners for roofing system from manufacturer approved by membrane roofing manufacturer.

2.2 PERFORMANCE REQUIREMENTS

- A. Accelerated Weathering: Roofing system shall withstand 2000 hours of exposure when tested according to ASTM G 152, ASTM G 154, or ASTM G 155.
- B. Impact Resistance: Roofing system shall resist impact damage when tested according to ASTM D 3746 or ASTM D 4272.
- C. Roofing System Design: Tested by a qualified testing agency to resist the uplift pressures in accordance with FM I-90 and ASCE/SEI 7 requirements for wind pressures as indicated on Structural Drawings.
- D. Solar Reflectance Index: Not less than 78 when calculated according to ASTM E 1980, based on testing identical products by a qualified testing agency.

- E. Energy Star Listing: Roofing system shall be listed on the DOE's ENERGY STAR "Roof Products Qualified Product List" for low-slope roof products.
- F. Energy Performance: Roofing system shall have an initial solar reflectance index of not less than 0.70 and an emissivity of not less than 0.75 when tested according to CRRC-1.
- G. Exterior Fire-Test Exposure: ASTM E 108 or UL 790, Class A; for application and roof slopes indicated; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
- H. Fire-Resistance Ratings: Comply with fire-resistance-rated assembly designs indicated. Identify products with appropriate markings of applicable testing agency.

2.3 TPO ROOFING

- A. Fabric-Reinforced TPO Sheet: ASTM D 6878, internally fabric or scrim-reinforced, uniform, flexible TPO sheet.
 - 1. Thickness: 60 mils (1.5 mm), nominal, GAF EverGuard TPO.
 - 2. Exposed Face Color: Energy Gray #322.

2.4 AUXILIARY ROOFING MATERIALS

- A. General: Auxiliary materials recommended by roofing system manufacturer for intended use and compatible with roofing.
 - 1. Liquid-type auxiliary materials shall comply with VOC limits of authorities having jurisdiction.
 - 2. Adhesives and sealants that are not on the exterior side of weather barrier shall comply with the following limits for VOC content:
 - a. Plastic Foam Adhesives: 50 g/L.
 - b. Multipurpose Construction Adhesives: 70 g/L.
 - c. Fiberglass Adhesives: 80 g/L.
 - d. Single-Ply Roof Membrane Adhesives: 250 g/L.
 - e. Single-Ply Roof Membrane Sealants: 450 g/L.
 - f. Nonmembrane Roof Sealants: 300 g/L.
 - g. Sealant Primers for Nonporous Substrates: 250 g/L.
 - h. Sealant Primers for Porous Substrates: 775 g/L.
 - i. Other Adhesives and Sealants: 250 g/L.
- B. Sheet Flashing: Manufacturer's standard unreinforced TPO sheet flashing, not less than 55 mils (1.4 mm) thick, minimum, of same color as TPO sheet.
- C. Bonding Adhesive: Manufacturer's standard, water based.
- D. Slip Sheet: Manufacturer's standard, of thickness required for application.
- E. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Global 4470, designed for fastening roofing to substrate, and acceptable to roofing system manufacturer.

- F. Miscellaneous Accessories: Provide metal termination bars, metal battens, pourable sealers, preformed cone and vent sheet flashings, preformed inside and outside corner sheet flashings, T-joint covers, lap sealants, termination reglets, and other accessories.

2.5 SUBSTRATE BOARDS ON ROOF OF ENTRY CANOPY AND BACK OF PARAPET WALLS

- A. Substrate Board: ASTM C1177/C1177M, glass-mat, water-resistant gypsum board.
 - 1. Georgia-Pacific Gypsum LLC, DensDeck Prime Roof Board with EONIC Technology or approved equivalent.
 - 2. Type and Thickness: 1/2 inch (12.7 mm) thick.
- B. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Approvals 4470, designed for fastening substrate board to roof deck.

2.6 ROOF INSULATION

- A. Polyisocyanurate Board Insulation: ASTM C 1289, Type II, Class 1, Grade 2, felt or glass-fiber mat facer on both major surfaces.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Carlisle SynTec Incorporated.
 - b. Firestone Building Products.
 - c. GAF Materials Corporation.
 - d. Johns Manville.
 - e. Rmax, Inc.
- B. Tapered Insulation: Provide factory-tapered insulation boards where required for sloping to drain and fabricated to slope of 1/4 inch per 12 inches (1:48) or greater.
- C. Provide preformed saddles, crickets, tapered edge strips, and other insulation shapes where indicated for sloping to drain. Fabricate to slopes indicated.
- D. Insulation Fastening Pattern: Install fasteners at equal spacing on X and Y direction from one panel to the other. Each 4X8 panel will have the spacing as following:
 - 1. 8'-0" side shall be divided in 1'-4", 2'-8", 2'-8" and 1'-4"
 - 2. 4'-0" side shall be divided in 1'-0", 2'-0" and 1'-0"

2.7 INSULATION ACCESSORIES

- A. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Global 4470, designed for fastening roof insulation to substrate, and acceptable to roofing system manufacturer.

2.8 VAPOR RETARDER

- A. Self-Adhering-Sheet Vapor Retarder: ASTM D1970/D1970M, polyethylene film laminated to layer of rubberized asphalt adhesive, minimum 30-mil (.76-mm) total thickness; maximum permeance rating of 0.1 perm (6 ng/Pa x s x sq. m); cold applied, with slip-resisting surface and

release paper backing. Provide vapor retarder of type recommended for adhering directly to steel decking. Provide primer when recommended by vapor retarder manufacturer.

2.9 PAD-TYPE SNOW GUARDS

A. Flat Surface Mounted Plastic Snow Guard Pads:

1. Snow Guard as manufactured by Zaleski Snow Guards for Roofs, Inc., Model No. 15 clear plastic.
2. Material: Clear UV-stabilized polycarbonate.
3. Adhesive Sealant: Neutral cure 100% silicone, single component as manufactured by Hi-Tec Industries, Project 1, 6700 Series, clear.

2.10 WALKWAYS

- ### A. Flexible Walkways:
- Factory-formed, nonporous, heavy-duty, slip-resisting, surface-textured walkway pads, approximately 3/16 inch (5 mm) thick and acceptable to roofing system manufacturer.

PART 3 - EXECUTION

3.1 ROOFING INSTALLATION, GENERAL

- #### A. Install roofing system according to roofing system manufacturer's written instructions.
- #### B. Complete terminations and base flashings and provide temporary seals to prevent water from entering completed sections of roofing system at the end of the workday or when rain is forecast. Remove and discard temporary seals before beginning work on adjoining roofing.

3.2 VAPOR RETARDER INSTALLATION

- #### A. Self-Adhering-Sheet Vapor Retarder: Prime substrate if required by manufacturer. Install self-adhering-sheet vapor retarder over roof decking, side and end lapping each sheet a minimum of 3-1/2 and 6 inches (90 and 150 mm), respectively.
1. Extend vertically up parapet walls and projections to a minimum height equal to height of insulation.
 2. Seal laps by rolling.
 3. Coordinate with air barrier on walls for compatibility and lapping of material for proper drainage of moisture.

3.3 SUBSTRATE BOARD INSTALLATION ON ROOF OF ENTRY CANOPY AND BACK OF PARAPET WALLS

- #### A. Install substrate board with long joints in continuous straight lines, with end joints staggered not less than 24 inches (610 mm) in adjacent rows.
1. At steel roof decks, install substrate board with edges over crests of steel roof deck.
 2. Tightly butt substrate boards together.
 3. Cut substrate board to fit tight around penetrations and projections, and to fit tight to intersecting sloping roof decks.
 4. Fasten substrate board to top flanges of steel deck to resist uplift pressure at corners, perimeter, and field of roof according to roofing system manufacturers' written instructions.

3.4 INSULATION INSTALLATION

- A. Coordinate installing roofing system components so insulation is not exposed to precipitation or left exposed at the end of the workday.
- B. Install tapered insulation under area of roofing for crickets and to conform to slopes indicated. Refer to structural drawings for roof steel slopes.
- C. Install insulation under area of roofing to achieve required thickness. Provide a total minimum LTTR-Value of 30.0. Provide greater LTTR-Value where required by the Energy Code in-force at jurisdiction in which project is located. Where overall insulation thickness is greater than 5.00 inches (127 mm) or greater, install two or more layers with joints of each succeeding layer staggered from joints of previous layer a minimum of 6 inches (150 mm) in each direction.
- D. Fasteners: Provide factory-coated steel fasteners and metal or plastic plates meeting corrosion-resistance provisions in FMG 4470, designed for fastening membrane and insulation to substrate in conformance to specified design requirements.

3.5 MECHANICALLY ROOFING INSTALLATION

- A. Mechanically fastened roofing over main building area to receive roofing according to roofing system manufacturer's written instructions. Unroll roofing and allow to relax before retaining.
- B. Mechanically Fastened Membrane:
 - 1. Mechanically fastened membrane to the structural deck utilizing fasteners and fastening patterns that in accordance with roof membrane manufacturer's requirements.
 - 2. Cut membrane to fit neatly around all penetrations and roof projections.
 - 3. Unroll roofing membrane and positioned with a minimum 6 inch overlap.
- C. Seaming:
 - 1. Weld overlapping sheets together using hot air. Minimum weld width is 1-1/2 inches (38 mm).
 - 2. Check welded seams for continuity and integrity by the end of each work day. Repair all imperfections.
- D. Membrane Termination/Securement: All membrane terminations shall be completed in accordance with roof membrane manufacturer's requirements.
 - 1. Provide securement at all membrane terminations at the perimeter of each roof level, roof section, curb flashing, skylight, expansion joint, interior wall, penthouse, and other similar condition.
 - 2. Provide securement at any angle change where the slope or combined slopes exceeds two inches in one horizontal foot.
- E. Flashings: Complete all flashings and terminations as indicated on the Drawings and fully adhere to substrates in accordance with roof membrane manufacturer's requirements.
 - 1. Base and Wall Flashings:
 - a. Do not apply flashing over existing thru-wall flashings or weep holes.
 - b. Secure flashing on vertical surfaces before the seam between the flashing and the main roof sheet is completed.

- c. Extend flashing membrane a minimum of 6 inches (152 mm) onto the main roof sheet beyond the mechanical securement.
 - d. Use care to ensure that the flashing does not bridge locations where there is a change in direction (e.g. where the parapet meets the roof deck).
- 2. Penetrations:
 - a. Flash all pipes, supports, soil stacks, cold vents, and other penetrations passing through the roofing membrane as indicated on the Drawings and in accordance with roof membrane manufacturer's requirements.
 - b. Utilize custom prefabricated flashings supplied by roof membrane manufacture.
- 3. Pipe Clusters and Unusual Shapes:
 - a. Clusters of pipes or other penetrations which cannot be sealed with prefabricated flashings or membrane shall be sealed by surrounding them with pourable sealer supplied by roof membrane manufacturer.
 - b. Penetration pans shall be installed, flashed and filled with pourable sealer in accordance with roof membrane manufacturer's requirements.
 - c. Penetration pans shall not be used where prefabricated or field fabricated flashings are possible.
- F. Roof Drains:
 - 1. Coordinate installation of roof drains and vents specified in Division 22.
 - 2. Provide a smooth clean surface on the mating surface between the clamping ring and the drain base.
 - 3. Taper insulation around the drain to prevent the membrane from bridging and to provide a smooth transition from the roof surface to the drain clamping ring.
 - 4. Use sealant to seal between the membrane and the drain base using constant, even compression from the drain clamping ring.
- G. Fascia/Drip Edge:
 - 1. Provide fascia bar and cover, drip edge and gravel stop as indicated on the Drawings. Install in accordance with roof membrane manufacturer's requirements.
 - 2. Seal joints between individual sections in accordance with roof membrane manufacturer's requirements.
 - 3. Coordinate installation of metal flashing and counter flashing specified in Section 07 62 00 - Sheet Metal Flashing and Trim.
 - 4. Manufactured Roof Specialties: Coordinate installation of copings, counter flashing systems, gutters, downspouts, and roof expansion assemblies specified in Section 077100 – Roof Specialties.

3.6 ADHERED ROOFING INSTALLATION

- A. Adhere roofing over entry canopy roof area, to back of parapet walls, curbs, and similar vertical surfaces, and over roof area indicated to receive adhered roofing according to roofing system manufacturer's written instructions. Unroll roofing and allow to relax before retaining. Apply recommended adhesive for vertical installations.
- B. Accurately align roofing and maintain uniform side and end laps of minimum dimensions required by manufacturer. Stagger end laps.

- C. Bonding Adhesive: Apply to substrate and underside of roofing at rate required by manufacturer and allow to partially dry before installing roofing. Do not apply to splice area of roofing.
- D. In addition to adhering, mechanically fasten roofing securely at terminations, penetrations, and perimeter of roofing.
- E. Seams: Clean seam areas, overlap roofing, and hot-air weld side and end laps of roofing and sheet flashings according to manufacturer's written instructions, to ensure a watertight seam installation.
 - 1. Test lap edges with probe to verify seam weld continuity. Apply lap sealant to seal cut edges of sheet.
 - 2. Verify field strength of seams a minimum of twice daily, and repair seam sample areas.
 - 3. Repair tears, voids, and lapped seams in roofing that do not comply with requirements.
- F. Spread sealant bed over deck-drain flange at roof drains, and securely seal roofing in place with clamping ring.

3.7 PARAPET WALLS MEMBRANE INSTALLATION

- A. Prior to installing roof membrane to back of parapet walls, verify compatibility of fluid-applied air barrier. Obtain a written confirmation from membrane roofing manufacturer.
- B. Install membrane over the entire back of parapet walls, starting at 8 inches beyond the fasteners at field membrane termination. Extend the membrane up the back side, over the top, and down the outside (front) edge of parapet walls a minimum of two inches.
- C. Apply membrane sheet horizontally, in shingle fashion starting with bottom course first; fully adhere membrane to substrate following manufacturer's written recommendations. Mechanically fasten sheet where required by manufacturer.
 - 1. Overlap membrane seams as recommended by manufacturer, ensuring a watertight installation.
 - 2. Repair tears, voids, and lapped seams in membrane that does not meet requirements.
 - 3. Provide intermediate fastening of adhered membrane at 5-foot intervals maximum and of non-adhered membrane at 18-inch intervals maximum.

3.8 SNOW GUARD INSTALLATION

- A. Install snow guards according to manufacturer's written instructions. Space rows as recommended by manufacturer and as indicated.
 - 1. Surface-Mounted, Snow Guard Pads: Securely attach to membrane roofing with sealant adhesive compatible with roofing and plastic snow guard as recommended by manufacturer.

3.9 WALKWAY INSTALLATION

- A. Flexible Walkways: Install walkway products in locations indicated. Heat weld to substrate or adhere walkway products to substrate with compatible adhesive according to roofing system manufacturer's written instructions.

3.10 PROTECTING AND CLEANING

- A. Protect roofing system from damage and wear during remainder of construction period. When remaining construction does not affect or endanger roofing, inspect roofing for deterioration and damage, describing its nature and extent in a written report, with copies to Architect and Owner.
- B. Correct deficiencies in or remove roofing system that does not comply with requirements, repair substrates, and repair or reinstall roofing system to a condition free of damage and deterioration at time of Substantial Completion and according to warranty requirements.
- C. Clean overspray and spillage from adjacent construction using cleaning agents and procedures recommended by manufacturer of affected construction.

END OF SECTION 075423

SECTION 076200 - SHEET METAL FLASHING AND TRIM**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Formed low-slope roof sheet metal fabrications.
 - 2. Formed wall sheet metal fabrications.

1.3 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings: For sheet metal flashing and trim.
 - a. Include plans, elevations, sections, and attachment details.
 - b. Distinguish between shop- and field-assembled work.
 - c. Include identification of finish for each item.
 - d. Include pattern of seams and details of termination points, expansion joints and expansion-joint covers, direction of expansion, roof-penetration flashing, and connections to adjoining work.
 - 3. Samples: For each exposed product and for each color and texture specified.

1.5 INFORMATIONAL SUBMITTALS

- A. Product Certificates.
- B. Sample warranty.

1.6 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.7 QUALITY ASSURANCE

- A. Fabricator Qualifications: Employs skilled workers who custom fabricate sheet metal flashing and trim similar to that required for this Project and whose products have a record of successful in-service performance.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Do not store sheet metal flashing and trim materials in contact with other materials that might cause staining, denting, or other surface damage. Store sheet metal flashing and trim materials away from uncured concrete and masonry.
- B. Protect strippable protective covering on sheet metal flashing and trim from exposure to sunlight and high humidity, except to extent necessary for period of sheet metal flashing and trim installation.

1.9 WARRANTY

- A. Special Warranty on Finishes: Manufacturer agrees to repair finish or replace sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within specified warranty period.

1. Finish Warranty Period: 20 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. General: Sheet metal flashing and trim assemblies shall withstand wind loads, structural movement, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation, or other defects in construction. Completed sheet metal flashing and trim shall not rattle, leak, or loosen, and shall remain watertight.
- B. Sheet Metal Standard for Flashing and Trim: Comply with NRCA's "The NRCA Roofing Manual" and SMACNA's "Architectural Sheet Metal Manual" requirements for dimensions and profiles shown unless more stringent requirements are indicated.
- C. SPRI Wind Design Standard: Manufacture and install copings tested according to SPRI ES-1 and capable of resisting the following design pressure:
 1. Design Pressure: Equivalent to 90 mph for 3-second gust wind speed unless local jurisdiction authorities require greater wind design pressure.
- D. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.
 1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

2.2 SHEET METALS

- A. General: Protect mechanical and other finishes on exposed surfaces from damage by applying strippable, temporary protective film before shipping.
- B. Aluminum Sheet: ASTM B 209 (ASTM B 209M), alloy as standard with manufacturer for finish required, with temper as required to suit forming operations and performance required.

1. As-Milled Finish: Mill.
2. Exposed Coil-Coated Finish:
 - a. Two-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
 - b. Color: See Exterior Finish Schedule.
- C. Stainless-Steel Sheet: ASTM A 240/A 240M, Type 304, dead soft, fully annealed; 2D (dull, cold rolled) finish.

2.3 MISCELLANEOUS MATERIALS

- A. General: Provide materials and types of fasteners, solder, protective coatings, sealants, and other miscellaneous items as required for complete sheet metal flashing and trim installation and as recommended by manufacturer of primary sheet metal or manufactured item unless otherwise indicated.
- B. Fasteners: Wood screws, annular threaded nails, self-tapping screws, self-locking rivets and bolts, and other suitable fasteners designed to withstand design loads and recommended by manufacturer of primary sheet metal or manufactured item.
 1. General: Blind fasteners or self-drilling screws, gasketed, with hex-washer head.
 - a. Exposed Fasteners: Heads matching color of sheet metal using plastic caps or factory-applied coating. Provide metal-backed EPDM or PVC sealing washers under heads of exposed fasteners bearing on weather side of metal.
 - b. Blind Fasteners: High-strength aluminum or stainless-steel rivets suitable for metal being fastened.
 2. Fasteners for Aluminum Sheet: Aluminum or Series 300 stainless steel.
 3. Fasteners for Stainless-Steel Sheet: Series 300 stainless steel.
- C. Solder:
 1. For Stainless Steel: ASTM B 32, Grade Sn60, with acid flux of type recommended by stainless-steel sheet manufacturer.
- D. Sealant Tape: Pressure-sensitive, 100 percent solids, polyisobutylene compound sealant tape with release-paper backing. Provide permanently elastic, nonsag, nontoxic, nonstaining tape 1/2 inch (13 mm) wide and 1/8 inch (3 mm) thick.
- E. Elastomeric Sealant: ASTM C 920, elastomeric polyurethane polymer sealant; of type, grade, class, and use classifications required to seal joints in sheet metal flashing and trim and remain watertight.

2.4 FABRICATION, GENERAL

- A. General: Custom fabricate sheet metal flashing and trim to comply with details shown and recommendations in cited sheet metal standard that apply to design, dimensions, geometry, metal thickness, and other characteristics of item required. Fabricate sheet metal flashing and trim in shop to greatest extent possible.

1. Fabricate sheet metal flashing and trim in thickness or weight needed to comply with performance requirements, but not less than that specified for each application and metal.
 2. Obtain field measurements for accurate fit before shop fabrication.
 3. Form sheet metal flashing and trim to fit substrates without excessive oil canning, buckling, and tool marks; true to line, levels, and slopes; and with exposed edges folded back to form hems.
 4. Conceal fasteners and expansion provisions where possible. Do not use exposed fasteners on faces exposed to view.
- B. Fabrication Tolerances: Fabricate sheet metal flashing and trim that is capable of installation to a tolerance of 1/4 inch in 20 feet (6 mm in 6 m) on slope and location lines indicated on Drawings and within 1/8-inch (3-mm) offset of adjoining faces and of alignment of matching profiles.
- C. Expansion Provisions: Form metal for thermal expansion of exposed flashing and trim.
1. Form expansion joints of intermeshing hooked flanges, not less than 1 inch (25 mm) deep, filled with butyl sealant concealed within joints.
 2. Use lapped expansion joints only where indicated on Drawings.
- D. Sealant Joints: Where movable, nonexpansion-type joints are required, form metal to provide for proper installation of elastomeric sealant according to cited sheet metal standard.
- E. Fabricate cleats and attachment devices from same material as accessory being anchored or from compatible, noncorrosive metal.
- F. Fabricate cleats and attachment devices of sizes as recommended by cited sheet metal standard for application, but not less than thickness of metal being secured.
- G. Seams: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with elastomeric sealant unless otherwise recommended by sealant manufacturer for intended use. Rivet joints where necessary for strength.

2.5 LOW-SLOPE ROOF SHEET METAL FABRICATIONS

- A. Base Flashing: Shop fabricate interior and exterior corners. Fabricate from the following materials:
1. Aluminum: 0.040 inch (1.02 mm) thick.
- B. Counterflashing: Fabricate from the following materials:
1. Aluminum: 0.032 inch (0.81 mm) thick.
- C. Roof-Penetration Flashing: Fabricate from the following materials:
1. Stainless Steel: 0.019 inch (0.48 mm) thick.
- D. Roof-Drain Flashing: Fabricate from the following materials:
1. Stainless Steel: 0.016 inch (0.40 mm) thick.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, substrate, and other conditions affecting performance of the Work.
 - 1. Verify compliance with requirements for installation tolerances of substrates.
 - 2. Verify that substrate is sound, dry, smooth, clean, sloped for drainage, and securely anchored.
 - 3. Verify that air- or water-resistant barriers have been installed over sheathing or backing substrate to prevent air infiltration or water penetration.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION, GENERAL

- A. General: Anchor sheet metal flashing and trim and other components of the Work securely in place, with provisions for thermal and structural movement. Use fasteners, solder, protective coatings, separators, sealants, and other miscellaneous items as required to complete sheet metal flashing and trim system.
 - 1. Install sheet metal flashing and trim true to line, levels, and slopes. Provide uniform, neat seams with minimum exposure of solder, welds, and sealant.
 - 2. Install sheet metal flashing and trim to fit substrates and to result in watertight performance. Verify shapes and dimensions of surfaces to be covered before fabricating sheet metal.
 - 3. Space cleats not more than 12 inches (300 mm) apart. Attach each cleat with at least two fasteners. Bend tabs over fasteners.
 - 4. Install exposed sheet metal flashing and trim with limited oil canning, and free of buckling and tool marks.
 - 5. Torch cutting of sheet metal flashing and trim is not permitted.
- B. Metal Protection: Where dissimilar metals contact each other, or where metal contacts pressure-treated wood or other corrosive substrates, protect against galvanic action or corrosion by painting contact surfaces with bituminous coating or by other permanent separation as recommended by sheet metal manufacturer or cited sheet metal standard.
 - 1. Coat concealed side of uncoated-aluminum and stainless-steel sheet metal flashing and trim with bituminous coating where flashing and trim contact wood, ferrous metal, or cementitious construction.
 - 2. Underlayment: Where installing sheet metal flashing and trim directly on cementitious or wood substrates, install underlayment and cover with slip sheet.
- C. Expansion Provisions: Provide for thermal expansion of exposed flashing and trim. Space movement joints at maximum of 10 feet (3 m) with no joints within 24 inches (600 mm) of corner or intersection.
 - 1. Form expansion joints of intermeshing hooked flanges, not less than 1 inch (25 mm) deep, filled with sealant concealed within joints.
 - 2. Use lapped expansion joints only where indicated on Drawings.
- D. Fasteners: Use fastener sizes that penetrate substrate not less than recommended by fastener manufacturer to achieve maximum pull-out resistance.

- E. Conceal fasteners and expansion provisions where possible in exposed work and locate to minimize possibility of leakage. Cover and seal fasteners and anchors as required for a tight installation.
- F. Seal joints as required for watertight construction. Prepare joints and apply sealants to comply with requirements in Section 079200 "Joint Sealants."
- G. Soldered Joints: Clean surfaces to be soldered, removing oils and foreign matter. Pre-tin edges of sheets with solder to width of 1-1/2 inches (38 mm); however, reduce pre-tinning where pre-tinned surface would show in completed Work.
 - 1. Do not solder aluminum sheet.
 - 2. Do not use torches for soldering.
 - 3. Heat surfaces to receive solder, and flow solder into joint. Fill joint completely. Completely remove flux and spatter from exposed surfaces.
 - 4. Stainless-Steel Soldering: Tin edges of uncoated sheets, using solder for stainless steel and acid flux. Promptly remove acid flux residue from metal after tinning and soldering. Comply with solder manufacturer's recommended methods for cleaning and neutralization.
- H. Rivets: Rivet joints in uncoated aluminum where necessary for strength.

3.3 ROOF FLASHING INSTALLATION

- A. General: Install sheet metal flashing and trim to comply with performance requirements and cited sheet metal standard. Provide concealed fasteners where possible, and set units true to line, levels, and slopes. Install work with laps, joints, and seams that are permanently watertight and weather resistant.
- B. Pipe or Post Counterflashing: Install counterflashing umbrella with close-fitting collar with top edge flared for elastomeric sealant, extending minimum of 4 inches (100 mm) over base flashing. Install stainless-steel draw band and tighten.
- C. Counterflashing: Coordinate installation of counterflashing with installation of base flashing. Extend counterflashing 4 inches (100 mm) over base flashing. Lap counterflashing joints minimum of 4 inches (100 mm).
- D. Roof-Penetration Flashing: Coordinate installation of roof-penetration flashing with installation of roofing and other items penetrating roof. Seal with elastomeric sealant and clamp flashing to pipes that penetrate roof.

3.4 WALL FLASHING INSTALLATION

- A. General: Install sheet metal wall flashing to intercept and exclude penetrating moisture according to cited sheet metal standard unless otherwise indicated. Coordinate installation of wall flashing with installation of wall-opening components such as windows, doors, and louvers.
- B. Through-Wall Flashing: Installation of through-wall flashing is specified in Section 042000 "Unit Masonry."

3.5 CLEANING AND PROTECTION

- A. Clean exposed metal surfaces of substances that interfere with uniform oxidation and weathering.

- B. Clean and neutralize flux materials. Clean off excess solder.
- C. Clean off excess sealants.
- D. Remove temporary protective coverings and strippable films as sheet metal flashing and trim are installed unless otherwise indicated in manufacturer's written installation instructions.

END OF SECTION 076200

SECTION 077100 - ROOF SPECIALTIES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Coping
 - 2. Roof-edge specialties.
 - 3. Roof-edge drainage systems.
 - 4. Reglets and counterflashings.

1.3 PREINSTALLATION CONFERENCE

- A. Preinstallation Conference: Conduct conference at Project site.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings: For roof specialties.
 - a. Include plans, elevations, expansion-joint locations, keyed details, and attachments to other work. Distinguish between plant- and field-assembled work.
 - 3. Samples: For each type of roof specialty and for each color specified.

1.5 INFORMATIONAL SUBMITTALS

- A. Product Certificates: For each type of roof specialty.

1.6 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For roofing specialties to include in maintenance manuals.

1.7 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing product specified in this section with not less than 10 years documented experience.
 - 1. Manufacturer certified ISO 9001:2015 with design.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Do not store roof specialties in contact with other materials that might cause staining, denting, or other surface damage. Store roof specialties away from uncured concrete and masonry.
- B. Protect strippable protective covering on roof specialties from exposure to sunlight and high humidity, except to extent necessary for the period of roof-specialty installation.

1.9 WARRANTY

- A. Product Warranty: Manufacturer's standard warranty covering failure in materials caused by manufacturing within specified warranty period.
 - 1. Product Warranty Period: 20 years from date of Substantial Completion.
- B. Finish Warranty: Manufacturer's special warranty agreeing to repair finish or replace roof specialties that show evidence of deterioration of factory-applied finishes, including significant fading, chalking, cracking, or peeling, within specified warranty period.
 - 1. Finish Warranty Period: 30 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Independent third party tested per ANSI/SPRI/FM4435/ES-1 as required by code.
- B. FM I-49 Loss Data Prevention Sheet Guidelines.
- C. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes to prevent buckling, opening of joints, hole elongation, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Provide clips that resist rotation and avoid shear stress as a result of thermal movements. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
 - 1. Temperature Change (Range): 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

2.2 COPINGS

- A. Metal Copings: Manufactured coping system consisting of metal coping cap in section lengths not exceeding 12 feet (3.6 m), concealed anchorage; with corner units, end cap units, and concealed splice plates with finish matching coping caps.
 - 1. Manufacturer: ATAS International; www.atas.com
 - 2. Contact: See **LIDL National Account List**.
 - 3. Product: ATAS Rapid Lok Coping.
 - 4. Snap-On Cover: 0.040 inch (1.02 mm) thick sheet aluminum.
 - 5. Cleats: 20-gauge 12-inch galvanized steel, pre-punched with slotted fastener holes.
 - a. Spacing: 48 inches for coping under 12 inches wide; 36 inches for coping 12 inches and over.
 - 6. Height: See Drawings.

7. Exposed Finish: Two-coat fluoropolymer.
8. Color: See Drawings.
9. Corners: Factory mitered and mechanically clinched and sealed watertight.
10. Splice Plates: Concealed, of same material, finish, and shape as fascia cover.

2.3 ROOF-EDGE SPECIALTIES

- A. Roof-Edge Fascia: Manufactured, two-piece, roof-edge fascia consisting of snap-on metal fascia cover in section lengths not exceeding 12 feet (3.6 m) and a continuous metal receiver with integral drip-edge cleat to engage fascia cover. Provide matching corner units.

1. Manufacturer: ATAS International; www.atas.com
2. Contact: See **LIDL National Account List**.
3. Product: ATAS Rapid Lok Fascia.
4. Snap-On Cover: 0.040 inch (1.02 mm) thick sheet aluminum.
5. Receiver: 20-gauge galvanized steel, continuous, pre-punched with slotted fastener holes.
6. Exposed Finish: Two-coat fluoropolymer.
7. Color: See Drawings.

- B. One-Piece Drip Edge: Manufactured, one-piece, metal drip edge in section lengths not exceeding 12 feet (3.6 m), with 4-inch (101.6 mm) a horizontal flange and vertical leg fascia terminating in a drip edge, and concealed splice plates of same material, finish, and shape as gravel stop. Provide matching corner units.

1. Manufacturer: ATAS International; www.atas.com
2. Contact: See **LIDL National Account List**.
3. Product: ATAS Drip Edge.
4. Drip Edge: 0.040 inch (1.02 mm) thick sheet aluminum.
5. Cleat: 22-gauge galvanized steel, continuous, pre-punched with slotted fastener holes.
6. Height: 6-1/4 inches (158.75 mm) (5-1/2 inches coverage)
7. Exposed Finish: Two-coat fluoropolymer.
8. Color: See Drawings.

2.4 ROOF-EDGE DRAINAGE SYSTEMS

- A. Gutters: Manufactured in uniform section lengths of 20 feet (6.1 m) minimum and 40 feet (12.2 m) maximum between expansion joints, with matching corner units, ends, outlet tubes, and other accessories. Elevate back edge at least 1 inch (25 mm) above front edge. Furnish high performance gutter brackets, hangers, straps, expansion joints, and expansion-joint covers fabricated from same metal as gutters.

1. Manufacturer: ATAS International; www.atas.com
2. Contact: See **LIDL National Account List**.
3. Product: ATAS Ultra HP Gutter
4. Aluminum Sheet: 0.063-inch (1.60 mm) thick
5. Custom Profile: 6 inches (152 mm) high by 7 inches (178 mm) wide
6. Corners: Factory mitered and mechanically clinched and sealed watertight.
7. Gutter Supports: 0.080-inch thick aluminum internal support brackets at 24 inches on center.
8. Hangers: 0.125-inch thick by 1-inch long aluminum at 24 inches on center, matching in color.
9. Gutter Accessories: 0.032-inch thick aluminum sheet, A24 perforated, continuous screened leaf guard with metal frame and finish matching gutter.
10. Exposed Finish: Two-coat fluoropolymer.

11. Color: See Drawings.
- B. Downspouts: Plain rectangular shape manufactured in uniform section lengths of 12 feet (3.66 m) minimum and 40 feet (12.2 m) maximum between expansion joints, complete with mitered elbows. Furnish with metal straps, from same material as downspouts.
 1. Manufacturer: ATAS International; www.atas.com
 2. Contact: See **LIDL National Account List**.
 3. Product: ATAS Box Downspout with Straps
 4. Formed Aluminum: 0.063 inch (1.60 mm) thick.
 5. Profile: Rectangular, 4 by 6 inches (102 by 152 mm).
 6. Exposed Finish: Two-coat fluoropolymer.
 7. Color: See Drawings.
- C. Downspout Boots: Cast Iron transition piece connecting exposed downspout to underground cast iron sewer piping.
 1. Manufacturer: Subject to compliance with requirements, provide product by J.R. Hoe and Sons; Middlesboro, KY, (800) 245-5521; www.downspoutboots.com; Model # A-4648C 6-inch angular downspout boot with a 3 by 4 inch side cleanout.
 2. Size: 4 by 6 inches (102 by 152 mm) angular inlet, 48-inch (1219 mm) length, and 5-7/8 inches outside diameter outlet.
 3. Color: Shop-prime with manufacturer's standard color; field-paint in color to match gutter and downspout.
 4. Accessories: Include appropriate stainless-steel anchors for securing boots to wall and rubber adapters for connection to drainage pipe.

2.5 REGLETS AND COUNTERFLASHINGS

- A. Reglets: Manufactured units formed to provide secure interlocking of separate reglet and counterflashing pieces, from the following exposed metal:
 1. Stainless Steel: 0.025 inch (0.64 mm) thick.
 2. Corners: Factory mitered and soldered and sealed watertight.
 3. Surface-Mounted Type: Provide reglets with slotted holes for fastening to substrate, with neoprene or other suitable weatherproofing washers, and with channel for sealant at top edge.
 4. Masonry Type, Embedded: Provide reglets with offset top flange for embedment in masonry mortar joint.
- B. Counter Flashings: Manufactured units of heights to overlap top edges of base flashings by 4 inches (100 mm) and in lengths not exceeding 12 feet (3.6 m) designed to snap into reglets and compress against base flashings with joints lapped, from the following exposed metal:
 1. Manufacturer: ATAS International; www.atas.com
 2. Contact: See **LIDL National Account List**.
 3. Exposed to View:
 - a. Formed Aluminum: 0.040 inch (1.02 mm) thick sheet aluminum.
 - b. Exposed Finish: Two-coat fluoropolymer.
 - c. Color: Match adjacent wall color.
 4. Not Exposed to View:

- a. Stainless Steel: 0.025 inch (0.64 mm) thick.

C. Accessories

- 1. Flexible-Flashing Retainer: Provide resilient plastic or rubber accessory to secure flexible flashing in reglet where clearance does not permit use of standard metal counterflashing or where reglet is provided separate from metal counterflashing.
- 2. Counterflashing Wind-Restraint Clips: Provide clips to be installed before counterflashing to prevent wind uplift of counterflashing lower edge.

2.6 MATERIALS

- A. Aluminum Sheet: ASTM B209 (ASTM B209M), alloy as standard with manufacturer for finish required, with temper to suit forming operations and performance required.
- B. Aluminum Extrusions: ASTM B221 (ASTM B221M), alloy and temper recommended by manufacturer for type of use and finish indicated, finished as follows:
- C. Stainless Steel Sheet: ASTM A240/A240M or ASTM A666, Type 304.

2.7 MISCELLANEOUS MATERIALS

- A. Fasteners: Manufacturer's recommended fasteners, suitable for application and designed to meet performance requirements. Furnish the following unless otherwise indicated:
 - 1. Exposed Penetrating Fasteners: Gasketed screws with hex washer heads matching color of sheet metal.
 - 2. Fasteners for Aluminum: Aluminum or Series 300 stainless steel.
 - 3. Fasteners for Stainless-Steel Sheet: Series 300 stainless steel.
- B. Elastomeric Sealant: ASTM C 920, elastomeric polyurethane polymer sealant of type, grade, class, and use classifications required by roofing-specialty manufacturer for each application.

2.8 FINISHES

- A. Aluminum Finish: AAMA 2605, two-coat fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
- B. Stainless-Steel Finish: No. 2B (mill finish).

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. General: Install roof specialties according to manufacturer's written instructions. Anchor roof specialties securely in place, with provisions for thermal and structural movement. Use fasteners, welding, solder, protective coatings, separators, underlayments, sealants, and other miscellaneous items as required to complete roof-specialty systems.
 - 1. Install roof specialties level, plumb, true to line and elevation; with limited oil-canning and without warping, jogs in alignment, buckling, or tool marks.
 - 2. Provide uniform, neat seams with minimum exposure of solder and sealant.

3. Install roof specialties to fit substrates and to result in weathertight performance. Verify shapes and dimensions of surfaces to be covered before manufacture.
 4. Torch cutting of roof specialties is not permitted.
 5. Do not use graphite pencils to mark metal surfaces.
- B. Metal Protection: Protect metals against galvanic action by separating dissimilar metals from contact with each other as recommended by manufacturer.
1. Coat concealed side of uncoated aluminum and stainless-steel roof specialties with bituminous coating where in contact with wood, ferrous metal, or cementitious construction.
- C. Expansion Provisions: Allow for thermal expansion of exposed roof specialties.
1. Space movement joints at a maximum of 12 feet (3.6 m) with no joints within 18 inches (450 mm) of corners or intersections unless otherwise indicated on Drawings.
 2. When ambient temperature at time of installation is between 40 and 70 deg F (4 and 21 deg C), set joint members for 50 percent movement each way. Adjust setting proportionately for installation at higher ambient temperatures.
- D. Fastener Sizes: Use fasteners of sizes that penetrate wood blocking or substrate not less than 1-1/4 inches (32 mm) for nails and not less than 3/4 inch (19 mm) for wood screws and not less than recommended by fastener manufacturer to achieve maximum pull-out resistance.
- E. Seal joints as required for weathertight construction. Place sealant to be completely concealed in joint. Do not install sealants at temperatures below 40 deg F (4 deg C).
- F. Soldered Joints: Clean stainless steel surfaces to be soldered, removing oils and foreign matter. Pre-tin edges of sheets to be soldered to a width of 1-1/2 inches (38 mm); however, reduce pre-tinning where pre-tinned surface would show in completed Work. Do not use torches for soldering. Heat surfaces to receive solder and flow solder into joint. Fill joint completely. Completely remove flux and spatter from exposed surfaces.

3.2 REGLET AND COUNTERFLASHING INSTALLATION

- A. Embedded Reglets: See Section 042000 "Unit Masonry" for installation of reglets.
- B. Surface-Mounted Reglets: Install reglets to receive flashings where flashing without embedded reglets is indicated on Drawings. Install at height so that inserted counterflashings overlap 4 inches (100 mm) over top edge of base flashings.
- C. Counterflashings: Insert counterflashings into reglets or other indicated receivers; ensure that counterflashings overlap 4 inches (100 mm) over top edge of base flashings. Lap counterflashing joints a minimum of 4 inches (100 mm) and bed with butyl sealant. Fit counterflashings tightly to base flashings.

3.3 CLEANING AND PROTECTION

- A. Clean exposed metal surfaces of substances that interfere with uniform oxidation and weathering.
- B. Clean and neutralize flux materials. Clean off excess solder and sealants.
- C. Remove temporary protective coverings and strippable films as roof specialties are installed.

END OF SECTION 077100

SECTION 077200 - ROOF ACCESSORIES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Roof curbs.
 - 2. Equipment supports.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of roof accessory indicated.

1.4 INFORMATIONAL SUBMITTALS

- A. Product certificates.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.6 WARRANTY

- A. Special Warranty on Painted Finishes: Manufacturer's standard form in which manufacturer agrees to repair finishes or replace roof accessories that show evidence of deterioration of factory-applied finishes within 10 years from date of Substantial Completion.

PART 2 - PRODUCTS**2.1 METAL MATERIALS**

- A. Zinc-Coated (Galvanized) Steel Sheet: ASTM A 653/A 653M, G90 (Z275) coating designation.
 - 1. Exposed Coil-Coated Finish: Two-coat fluoropolymer finish; AAMA 621; system consisting of primer and fluoropolymer color topcoat containing not less than 70 percent PVDF resin by weight.
 - 2. Baked-Enamel or Powder-Coat Finish: Manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat, with a minimum dry film thickness of 1 mil (0.025 mm) for topcoat.
- B. Aluminum-Zinc Alloy-Coated Steel Sheet: ASTM A 792/A 792M, AZ50 (AZM150) coated.

1. Exposed Coil-Coated Finish: Two-coat fluoropolymer finish; AAMA 621; system consisting of primer and fluoropolymer color topcoat containing not less than 70 percent PVDF resin by weight.
 2. Baked-Enamel or Powder-Coat Finish: Manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat, with a minimum dry film thickness of 1 mil (0.025 mm) for topcoat.
- C. Steel Shapes: ASTM A 36/A 36M, hot-dip galvanized according to ASTM A 123/A 123M unless otherwise indicated.

2.2 MISCELLANEOUS MATERIALS

- A. General: Provide materials and types of fasteners, protective coatings, sealants, and other miscellaneous items required by manufacturer for a complete installation.
- B. Fasteners: Roof accessory manufacturer's recommended fasteners suitable for application and metals being fastened. Match finish of exposed fasteners with finish of material being fastened. Provide nonremovable fastener heads to exterior exposed fasteners.
- C. Sealants: As recommended by roof accessory manufacturer for installation indicated.

2.3 ROOF CURBS

- A. Roof Curbs: Internally reinforced roof-curb units capable of supporting superimposed live and dead loads, including equipment loads and other construction indicated on Drawings; with welded or mechanically fastened and sealed corner joints, stepped integral metal cant raised the thickness of roof insulation, and integrally formed deck-mounting flange at perimeter bottom.
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. AES Industries, Inc.
 - b. Greenheck Fan Corporation.
 - c. LM Curbs.
 - d. Milcor Inc.; Commercial Products Group of Hart & Cooley, Inc.
 - e. Pate Company (The).
 - f. Thybar Corporation.
- B. Material: Zinc-coated (galvanized) or Aluminum-zinc alloy-coated steel sheet, 0.079 inch (2.01 mm) thick.
1. Finish: Two-coat fluoropolymer where exposed to view and Baked enamel or powder coat where concealed from view.
 2. Color: As selected by Architect from manufacturer's full range where concealed from view.
- C. Construction:
1. Insulation: Factory insulated with 1-1/2-inch (38-mm) thick glass-fiber board insulation.
 2. Liner: Same material as curb, of manufacturer's standard thickness and finish.
 3. Factory-installed wood nailer at top of curb, continuous around curb perimeter.
 4. Fabricate curbs to minimum height of 12 inches (300 mm) above roof membrane unless otherwise indicated.

5. Top Surface: Level around perimeter with roof slope accommodated by sloping the deck-mounting flange.

2.4 EQUIPMENT SUPPORTS

- A. Equipment Supports: Internally reinforced metal equipment supports capable of supporting superimposed live and dead loads, including equipment loads and other construction indicated on Drawings; with welded or mechanically fastened and sealed corner joints, stepped integral metal cant raised the thickness of roof insulation, and integrally formed deck-mounting flange at perimeter bottom.
 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. AES Industries, Inc.
 - b. Greenheck Fan Corporation.
 - c. LM Curbs.
 - d. Milcor Inc.; Commercial Products Group of Hart & Cooley, Inc.
 - e. Pate Company (The).
 - f. Thybar Corporation.
- B. Material: Zinc-coated (galvanized) or aluminum-zinc alloy-coated steel sheet, 0.079 inch (2.01 mm) thick.
 1. Finish: Two-coat fluoropolymer where exposed to view and baked enamel or powder coat where concealed from view.
 2. Color: To be selected by Architect
- C. Construction:
 1. Insulation: Factory insulated with 1-1/2-inch (38-mm) thick glass-fiber board insulation.
 2. Liner: Same material as equipment support, of manufacturer's standard thickness and finish.
 3. Factory-installed continuous wood nailers 3-1/2 inches (90 mm) wide at tops of equipment supports.
 4. Metal Counterflashing: Manufacturer's standard, removable, fabricated of same metal and finish as equipment support.
 5. Fabricate equipment supports to minimum height of 12 inches (300 mm) above roof membrane unless otherwise indicated.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. General: Verify dimensions of roof openings for roof accessories. Install roof accessories according to manufacturer's written instructions.
 1. Install roof accessories level, plumb, true to line and elevation, and without warping, jogs in alignment, excessive oil canning, buckling, or tool marks.
 2. Anchor roof accessories securely in place so they are capable of resisting indicated loads.
 3. Use fasteners, separators, sealants, and other miscellaneous items as required to complete installation of roof accessories and fit them to substrates.

- 4. Install roof accessories to resist exposure to weather without failing, rattling, leaking, or loosening of fasteners and seals.
 - B. Metal Protection: Protect metals against galvanic action by separating dissimilar metals from contact with each other or with corrosive as recommended by manufacturer.
 - C. Roof Curb Installation: Install each roof curb so top surface is level.
 - D. Equipment Support Installation: Install equipment supports so top surfaces are level with each other.
 - E. Seal joints with sealant as required by roof accessory manufacturer.
- 3.2 REPAIR AND CLEANING
- A. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing according to ASTM A 780.
 - B. Touch up factory-finished surfaces.
 - C. Replace roof accessories that have been damaged or that cannot be successfully repaired by finish touchup or similar minor repair procedures.

END OF SECTION 077200

SECTION 078413 - PENETRATION FIRESTOPPING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Penetrations in fire-resistance-rated walls.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product indicated.
 - 2. Product Schedule: For each penetration firestopping system. Include location and design designation of qualified testing and inspecting agency.
 - a. Where Project conditions require modification to a qualified testing and inspecting agency's illustration for a particular penetration firestopping condition, submit illustration, with modifications marked, approved by penetration firestopping manufacturer's fire-protection engineer as an engineering judgment or equivalent fire-resistance-rated assembly.

1.4 INFORMATIONAL SUBMITTALS

- A. Installer Certificates: From Installer indicating penetration firestopping has been installed in compliance with requirements and manufacturer's written recommendations.
- B. Product test reports.

1.5 QUALITY ASSURANCE

- A. Fire-Test-Response Characteristics: Penetration firestopping shall comply with the following requirements:
 - 1. Penetration firestopping is identical to those tested per testing standard referenced in "Penetration Firestopping" Article. Provide rated systems bearing marking of qualified testing and inspection agency.
- B. Where UL-classified systems are indicated, they refer to system numbers in UL's "Fire Resistance Directory" under product Category XHEZ.
- C. Pre-installation Conference: Conduct conference at Project site.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
1. A/D Fire Protection Systems Inc.
 2. Grace Construction Products.
 3. Hilti, Inc.
 4. Johns Manville.
 5. Specified Technologies Inc.
 6. 3M Fire Protection Products.
 7. Tremco, Inc.; Tremco Fire Protection Systems Group.
 8. USG Corporation.

2.2 PENETRATION FIRESTOPPING

- A. Provide penetration firestopping that is produced and installed to resist spread of fire according to requirements indicated, resist passage of smoke and other gases, and maintain original fire-resistance rating of construction penetrated. Penetration firestopping systems shall be compatible with one another, with the substrates forming openings, and with penetrating items if any.
- B. Penetrations in Fire-Resistance-Rated Walls: Ratings determined per ASTM E 814 or UL 1479, based on testing at a positive pressure differential of 0.01-inch wg (2.49 Pa).
1. F-Rating: Not less than the fire-resistance rating of constructions penetrated.
- C. Penetrations in Horizontal Assemblies: Ratings determined per ASTM E 814 or UL 1479, based on testing at a positive pressure differential of 0.01-inch wg (2.49 Pa).
1. F-Rating: At least 1 hour, but not less than the fire-resistance rating of constructions penetrated.
 2. T-Rating: At least 1 hour, but not less than the fire-resistance rating of constructions penetrated except for floor penetrations within the cavity of a wall.
- D. Exposed Penetration Firestopping Systems: Provide products with flame-spread and smoke-developed indexes of less than 25 and 450, respectively, as determined per ASTM E 84.
- E. VOC Content: Penetration firestopping sealants and sealant primers shall comply with the following limits for VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24):
1. Sealants: 250 g/L.
 2. Sealant Primers for Nonporous Substrates: 250 g/L.
 3. Sealant Primers for Porous Substrates: 775 g/L.
- F. Accessories: Provide components for each penetration firestopping system that are needed to install fill materials and to maintain ratings required. Use only those components specified by penetration firestopping manufacturer and approved by qualified testing and inspecting agency for firestopping indicated.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Examine substrates and conditions, with Installer present, for compliance with requirements for opening configurations, penetrating items, substrates, and other conditions affecting performance of the Work.
- B. Install penetration firestopping to comply with manufacturer's written installation instructions and published drawings for products and applications indicated.
- C. Install forming materials and other accessories of types required to support fill materials during their application and in the position needed to produce cross-sectional shapes and depths required to achieve fire ratings indicated.
 - 1. After installing fill materials and allowing them to fully cure, remove combustible forming materials and other accessories not indicated as permanent components of firestopping.
- D. Install fill materials for firestopping by proven techniques to produce the following results:
 - 1. Fill voids and cavities formed by openings, forming materials, accessories, and penetrating items as required to achieve fire-resistance ratings indicated.
 - 2. Apply materials so they contact and adhere to substrates formed by openings and penetrating items.
 - 3. For fill materials that will remain exposed after completing the Work, finish to produce smooth, uniform surfaces that are flush with adjoining finishes.

3.2 IDENTIFICATION

- A. Identify penetration firestopping with preprinted metal or plastic labels. Attach labels permanently to surfaces adjacent to and within 6 inches (150 mm) of firestopping edge so labels will be visible to anyone seeking to remove penetrating items or firestopping. Use mechanical fasteners or self-adhering-type labels with adhesives capable of permanently bonding labels to surfaces on which labels are placed. Include the following information on labels:
 - 1. The words "Warning - Penetration Firestopping - Do Not Disturb. Notify Building Management of Any Damage."
 - 2. Contractor's name, address, and phone number.
 - 3. Designation of applicable testing and inspecting agency.
 - 4. Date of installation.
 - 5. Manufacturer's name.

END OF SECTION 078413

SECTION 079200 - JOINT SEALANTS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Silicone joint sealants.
 - 2. Urethane joint sealants.
 - 3. Latex joint sealants.
 - 4. Acoustical joint sealants.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each joint-sealant product indicated.
 - 2. Samples for Color Selection: For each kind and color of joint sealant required.
 - 3. Samples for Color Verification: For each joint sealant type and applied to joint in selected color.

1.4 INFORMATIONAL SUBMITTALS

- A. Joint-Sealant Schedule: Include the following information:
 - 1. Joint-sealant application, joint location, and designation.
 - 2. Joint-sealant manufacturer and product name.
 - 3. Joint-sealant color.
- B. Product test reports: Based on evaluation of comprehensive tests performed by a qualified testing agency, indicating that sealants comply with requirements.
- C. Installer certificates.
- D. Warranties.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Manufacturer's authorized representative who is trained and approved for installation of units required for this Project.
- B. Source Limitations: Obtain each kind of joint sealant from single source from single manufacturer.

- C. Pre-installation Conference: Conduct conference at Project site.

1.6 DELIVERY, STORAGE AND HANDLING

- A. Accept materials on site in manufacturer's unopened original packaging.
- B. Store primers and sealants in dry location with ambient temperature range of 60 to 80 deg. F (15 to 27deg. C).

1.7 PROJECT CONDITIONS

- A. Do not proceed with installation of joint sealants under the following conditions:
 - 1. When ambient and substrate temperature conditions are outside limits permitted by joint-sealant manufacturer or are below 40 deg F (5 deg C).
 - 2. When joint substrates are wet.
 - 3. Where joint widths are less than those allowed by joint-sealant manufacturer for applications indicated.
 - 4. Where contaminants capable of interfering with adhesion have not yet been removed from joint substrates.

1.8 WARRANTY

- A. Special Installer's Warranty: Manufacturer's standard form in which Installer agrees to repair or replace joint sealants that do not comply with performance and other requirements specified in this Section within specified warranty period.
 - 1. Warranty Period: Two years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MATERIALS, GENERAL

- A. Compatibility: Provide joint sealants, backings, and other related materials that are compatible with one another and with joint substrates under conditions of service and application, as demonstrated by joint-sealant manufacturer, based on testing and field experience.
- B. VOC Content of Interior Sealants: Sealants and sealant primers used inside the weatherproofing system shall comply with the following limits for VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24):
 - 1. Architectural Sealants: 250 g/L.
 - 2. Sealant Primers for Nonporous Substrates: 250 g/L.
 - 3. Sealant Primers for Porous Substrates: 775 g/L.
- C. Liquid-Applied Joint Sealants: Comply with ASTM C 920 and other requirements indicated for each liquid-applied joint sealant specified, including those referencing ASTM C 920 classifications for type, grade, class, and uses related to exposure and joint substrates.
- D. Stain-Test-Response Characteristics: Where sealants are specified to be nonstaining to porous substrates, provide products that have undergone testing according to ASTM C 1248 and have not stained porous joint substrates indicated for Project.

2.2 SILICONE JOINT SEALANTS

- A. Mildew-Resistant, Neutral-Curing Silicone Joint Sealant: ASTM C 920.
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Dow Corning Corporation, 786 Sealant M White (Gray Color).
 - b. GE Advanced Materials - Silicones.
 - c. Pecora Corporation, 898NST (Special Order, Aluminum Stone Color).
 - d. Tremco Incorporated.
 2. Type: Single component (S).
 3. Grade: Nonsag (NS).
 4. Class: Not less than 25.
 5. Uses Related to Exposure: Nontraffic (NT).
 6. Service Temperature Range (minimums): Environments from minus 50 degrees F to plus 300 degrees F.
- B. Structural Glazing Sealant: ASTM C 1184, chemically curing silicone formulation that is compatible with system components with which it comes in contact, specifically formulated and tested for use as structural sealant and approved by structural-sealant manufacturer for use in glazing at steel entry canopy assemblies.

1. Color: Clear

2.3 URETHANE JOINT SEALANTS

- A. Urethane Joint Sealant: ASTM C 920, horizontal use (outside of building)
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. BASF Building Systems.
 - b. Bostik, Inc.
 - c. Pecora Corporation.
 - d. Tremco Incorporated.
 2. Type: Single component (S).
 3. Grade: Pourable (P).
 4. Class: 50.
 5. Uses Related to Exposure: Traffic (T).
- B. Urethane Joint Sealant: ASTM C 920, vertical use
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. BASF Building Systems.
 - b. Bostik, Inc.
 - c. Pecora Corporation.
 - d. Tremco Incorporated.
 2. Type: Single component (S).
 3. Grade: Nonsag (NS).
 4. Class: 25.

5. Uses Related to Exposure: Nontraffic (NT).

2.4 HYBRID URETHANE JOINT SEALANTS

A. Hybrid Urethane Joint Sealant: ASTM C 920, vertical use

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. BASF Building Systems.
 - b. Pecora Corporation.
 - c. Tremco Incorporated.
2. Type: Single component (S).
3. Grade: Nonsag (NS).
4. Class: 50
5. Uses Related to Exposure: Nontraffic (NT).

2.5 LATEX JOINT SEALANTS

A. Latex Joint Sealant: Acrylic latex or siliconized acrylic latex, ASTM C 834, Type OP, Grade NF.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. BASF Building Systems.
 - b. Bostik, Inc.
 - c. Pecora Corporation.
 - d. Tremco Incorporated.

2.6 ACOUSTICAL JOINT SEALANTS

A. Acoustical Joint Sealant: Manufacturer's standard nonsag, paintable, nonstaining latex sealant complying with ASTM C 834. Product effectively reduces airborne sound transmission through perimeter joints and openings in building construction as demonstrated by testing representative assemblies according to ASTM E 90.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Pecora Corporation
 - b. USG Corporation

2.7 JOINT SEALANT BACKING

- A. Cylindrical Sealant Backings: ASTM C 1330, Type C (closed-cell material with a surface skin), Type O (open-cell material) or any of the preceding types, as approved in writing by joint-sealant manufacturer for joint application indicated, and of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance.
- B. Bond-Breaker Tape: Polyethylene tape or other plastic tape recommended by sealant manufacturer.

2.8 MISCELLANEOUS MATERIALS

- A. Primer: Material recommended by joint-sealant manufacturer where required for adhesion of sealant to joint substrates indicated, as determined from preconstruction joint-sealant-substrate tests and field tests.
- B. Cleaners for Nonporous Surfaces: Chemical cleaners acceptable to manufacturers of sealants and sealant backing materials.
- C. Masking Tape: Nonstaining, nonabsorbent material compatible with joint sealants and surfaces adjacent to joints.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Surface Cleaning of Joints: Clean out joints immediately before installing joint sealants to comply with joint-sealant manufacturer's written instructions.
 - 1. Remove laitance and form-release agents from concrete.
 - 2. Clean nonporous joint substrate surfaces with chemical cleaners or other means that do not stain, harm substrates, or leave residues capable of interfering with adhesion of joint sealants.
- B. Joint Priming: Prime joint substrates where recommended by joint-sealant manufacturer or as indicated by preconstruction joint-sealant-substrate tests or prior experience. Apply primer to comply with joint-sealant manufacturer's written instructions. Confine primers to areas of joint-sealant bond; do not allow spillage or migration onto adjoining surfaces.
- C. Masking Tape: Use masking tape where required to prevent contact of sealant or primer with adjoining surfaces that otherwise would be permanently stained or damaged by such contact or by cleaning methods required to remove sealant smears. Remove tape immediately after tooling without disturbing joint seal.

3.2 INSTALLATION

- A. Sealant Installation Standard: Comply with recommendations in ASTM C 1193 for use of joint sealants as applicable to materials, applications, and conditions indicated.
- B. Install sealant backings of kind indicated to support sealants during application and at position required to produce cross-sectional shapes and depths of installed sealants relative to joint widths that allow optimum sealant movement capability.
 - 1. Do not leave gaps between ends of sealant backings.
 - 2. Do not stretch, twist, puncture, or tear sealant backings.
 - 3. Remove absorbent sealant backings that have become wet before sealant application and replace them with dry materials.
- C. Install bond-breaker tape behind sealants where sealant backings are not used between sealants and backs of joints.
- D. Install sealants using proven techniques that comply with the following and at the same time backings are installed:
 - 1. Place sealants so they directly contact and fully wet joint substrates.
 - 2. Completely fill recesses in each joint configuration.

3. Produce uniform, cross-sectional shapes and depths relative to joint widths that allow optimum sealant movement capability.
- E. Tooling of Nonsag Sealants: Immediately after sealant application and before skinning or curing begins, tool sealants according to requirements specified in subparagraphs below to form smooth, uniform beads of configuration indicated; to eliminate air pockets; and to ensure contact and adhesion of sealant with sides of joint.
 1. Remove excess sealant from surfaces adjacent to joints.
 2. Use tooling agents that are approved in writing by sealant manufacturer and that do not discolor sealants or adjacent surfaces.
 3. Provide concave joint profile per Figure 8A in ASTM C 1193, unless otherwise indicated.
- F. Acoustical Sealant Installation: Comply with ASTM C 919 and with manufacturer's written recommendations.
- G. Clean off excess sealant or sealant smears adjacent to joints as the Work progresses by methods and with cleaning materials approved in writing by manufacturers of joint sealants and of products in which joints occur.

3.3 JOINT-SEALANT SCHEDULE

- A. Joint-Sealant Application: Exterior joints in horizontal traffic surfaces.
 1. Joint Locations:
 - a. Isolation and contraction joints in cast-in-place concrete slabs (outside of building).
 - b. Joints in paving units.
 - c. Joints between different materials listed above.
 2. Joint Sealant: Urethane, horizontal use.
 3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors to match adjacent surface color.
- B. Joint-Sealant Application: Exterior joints in vertical surfaces.
 1. Joint Locations:
 - a. Control joints in concrete and unit masonry including brickwork and architectural masonry.
 - b. Joints between different materials listed above.
 - c. Perimeter joints between materials listed above and frames of storefronts, entrances, doors and windows.
 2. Joint Sealant: Urethane, vertical use.
 3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors to match adjacent surfaces color.
- C. Joint-Sealant Application: Exterior joints in vertical surfaces.
 1. Joint Locations:
 - a. Control joints in EIFS.
 - b. Joints between EIFS and masonry.

- c. Perimeter joints between EIFS and frames of storefronts, entrances, doors and windows.
 - 2. Joint Sealant: Hybrid urethane, vertical use.
 - 3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors to match adjacent surfaces color.
- D. Joint-Sealant Application: Interior joints in horizontal traffic surfaces.
- 1. Joint Locations:
 - a. Control joints in interior floor slabs.
 - 2. Joint Filler Material: See Section 033543 "Polished Concrete Finish and Joint Filler".
- E. Joint-Sealant Application: Interior joints in vertical surfaces and horizontal nontraffic surfaces.
- 1. Joint Locations:
 - a. Control and expansion joints on exposed interior surfaces of exterior walls.
 - b. Perimeter joints of exterior openings where indicated.
 - c. Vertical joints on exposed surfaces of walls and partitions.
 - d. Perimeter joints between interior wall surfaces and frames of interior doors, windows and elevator entrances.
 - 2. Joint Sealant: Latex.
 - 3. Joint-Sealant Color: As selected by Architect from manufacturer's full range of colors.
- F. Joint-Sealant Application: Mildew-resistant interior joints in vertical surfaces and horizontal nontraffic surfaces.
- 1. Joint Sealant Location:
 - a. Joints between plumbing fixtures and adjoining walls, floors, and counters, and at interior and exterior of walk-in freezers and refrigerator.
 - b. Joints between plumbing fixtures and adjoining walls, floors and counters at bakery area.
 - c. Joints in tile work.
 - 2. Joint Sealant: Silicone, mildew resistant.
 - 3. Joint-Sealant Color: Match color tile where occurs; Gray in all other locations.
- G. Joint-Sealant Application: Interior acoustical joints in vertical surfaces and horizontal nontraffic surfaces.
- 1. Joint Location:
 - a. Acoustical joints at gypsum board partitions with acoustical drywall.
 - 2. Joint Sealant: Acoustical.
 - 3. Joint-Sealant Color: As selected by Architect from manufacturer's full range.

END OF SECTION 079200

SECTION 081113 - HOLLOW METAL DOORS AND FRAMES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Hollow metal doors and frames.
- B. Related Requirements:
 - 1. Section 087100 "Door Hardware" for door hardware for hollow-metal doors.

1.3 DEFINITIONS

- A. Minimum Thickness: Minimum thickness of base metal without coatings according to NAAMM-HMMA 803 or SDI A250.8.

1.4 COORDINATION

- A. Coordinate anchorage installation for hollow-metal frames. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors. Deliver such items to Project site in time for installation.

1.5 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product indicated. Include construction details, material descriptions, core descriptions, hardware reinforcements, profiles, anchors, fire-resistance rating, and finishes.
 - 2. Door hardware supplier is to furnish templates, template reference number and/or physical hardware to the steel door and frame supplier in order to prepare the doors and frames to receive the finish hardware items.
 - 3. Shop Drawings: Include elevations, door edge details, frame profiles, metal thicknesses, preparations for hardware, and other details.
 - 4. Samples: For verification of selection of exposed finishes.
 - 5. Schedule: Prepared by or under the supervision of supplier, using same reference numbers for details and openings as those on Drawings.

1.6 QUALITY ASSURANCE

- A. Source Limitations: Obtain hollow metal doors and frames through one source from a single manufacturer wherever possible.

- B. Quality Standard: In addition to requirements specified, comply with ANSI/SDI A250.8, latest edition, "Recommended Specifications for Standard Steel Doors and Frames".
- C. Fire-Rated Door Assemblies: Assemblies complying with NFPA 80 that are listed and labeled by a qualified testing agency, for fire-protection ratings indicated, based on testing at positive pressure according to UL 10C.
 - 1. Temperature-Rise Limit: At vertical exit enclosures and exit passageways, provide doors that have a maximum transmitted temperature end point of not more than 450 deg F above ambient after 30 minutes of standard fire-test exposure.
- D. Fire-Rated, Borrowed-Light Frame Assemblies: Assemblies complying with NFPA 80 that are listed and labeled, by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire-protection ratings indicated, based on testing according to NFPA 257 or UL 9. Label each individual glazed lite.
- E. Smoke-Control Door Assemblies: Comply with NFPA 105 or UL 1784.
- F. Energy Efficient Exterior Openings: Comply with minimum thermal ratings, based on ASTM C1363. Openings to be fabricated and tested as fully operable, thermal insulating door and frame assemblies.
 - 1. Thermal Performance (Exterior Openings): Independent testing laboratory certification for exterior door assemblies being tested in accordance with ASTM C1363 and meet or exceed the following requirements:
 - a. Door Operable U-Factor and R-Value Ratings: U-Factor 0.38, R-Value 2.64, including insulated door.
 - 2. Air Infiltration (Exterior Openings): Independent testing laboratory certification for exterior door assemblies being tested in accordance with ASTM E283 to meet or exceed the following requirements:
 - a. Rate of leakage of the door assembly shall not exceed 0.25 cfm per square foot of static differential air pressure of 1.567 psf (equivalent to 25 mph wind velocity).

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver hollow-metal work palletized, packaged, wrapped and crated to provide protection during transit and project-site storage. Do not use nonvented plastic.
 - 1. Provide additional protection to prevent damage to factory-finished units.
- B. Deliver welded frames with removable spreader bars across bottom of frames, tack welded to jambs and mullions.
- C. Store hollow-metal work vertically under cover at Project site with head up. Place on minimum 4-inch- (102-mm-) high wood blocking. Provide minimum 1/4-inch (6-mm) space between each stacked door to permit air circulation. Door and frames to be stacked in a vertical upright position.

PART 2 - PRODUCTS

2.1 SUPPLIER OF DOORS AND FRAMES

- A. Contact: See **LIDL National Account List**.

2.2 MATERIALS

- A. Cold-Rolled Steel Sheet: ASTM A 1008/A 1008M, CS, Type B; suitable for exposed applications.
- B. Hot-Rolled Steel Sheet: ASTM A 1011/A 1011M, CS, Type B.
- C. Metallic-Coated Steel Sheet: ASTM A 653/A 653M, Commercial Steel (CS), Type B; with minimum G60 or A60 metallic coating.
- D. Frame Anchors: ASTM A 591/A 591M, Commercial Steel (CS), 40Z coating designation; mill phosphatized.
1. For anchors built into exterior walls, steel sheet complying with ASTM A 1008/A 1008M or ASTM A 1011/A 1011M, hot-dip galvanized according to ASTM A 153/A 153M, Class B.
- E. Inserts, Bolts, and Fasteners: Hot-dip galvanized according to ASTM A 153/A 153M.
- F. Mineral-Fiber Insulation: ASTM C 665, Type I.
- G. Glazing: Division 08 Section "Glazing."

2.3 MANUFACTURERS

- A. Basis of Design: Steelcraft, an Allegion brand.
- B. Alternate Design: Ceco Door, an Assa Abloy brand.

2.4 HOLLOW METAL DOORS

- A. Basis of Design Product: Steelcraft L18 Series Flush Doors
- B. General: Comply with ANSI/ NAAMM-HMMA 861.
1. Design: Flush panel.
 2. Core Construction: Manufacturer's standard polystyrene, polyurethane, polyisocyanurate, mineral-board or kraft-paper honeycomb.
 - a. Non-Fire-Rated Interior Doors: Kraft-paper honeycomb.
 - b. Fire Door Core: As required to provide fire-protection and temperature-rise ratings indicated.
 - c. Thermal-Rated (Insulated) Exterior Doors: Insulation type to achieve R-value of not less than 2.64 deg F x h x sq. ft./Btu when tested according to ASTM C 1363.
 3. Vertical Edges for Single-Acting Doors: Manufacturer's standard beveled or square edges.
 4. Top and Bottom Edges: Closed with flush or inverted 0.042-inch- thick, end closures or channels of same material as face sheets. Top of exterior doors shall be inverted channel.
 5. Tolerances: HMMA 840-1999: "Installation and Storage of Hollow Metal Doors and Frames."

- C. Exterior Doors: Face sheets fabricated from 0.0478-inch (18 gauge) metallic-coated steel sheet. Comply with ANSI/ NAAMM-HMMA 861 for level and model and for physical performance level:

1. Level 2 per SDI A250.8 and Physical Performance Level A (Maximum Duty) per SDI A250.4, Model 1 (Full Flush).

- D. Interior Doors: Face sheets fabricated from 0.0478-inch (18 gauge) cold-rolled steel sheet. Provide doors complying with requirements indicated below by referencing ANSI/ NAAMM-HMMA 861 for level and model and for physical performance level:

1. Level 2 per SDI A250.8 and Physical Performance Level A (Maximum Duty) per SDI A250.4, Model 1 (Full Flush).

- E. Hardware Reinforcement: ANSI/ NAAMM-HMMA 861.

- F. Door Finish: Factory finished.

2.5 HOLLOW METAL FRAMES

- A. Basis of Design Product: Steelcraft F16 Series 3-Sided Flush Frames

- B. General: Comply with ANSI/ NAAMM-HMMA 861.

- C. Exterior Frames: Fabricated from metallic-coated steel sheet.

1. Fabricate frames with mitered corners.
2. Fabricate frames as face welded unless otherwise indicated. Knocked-down frames are not allowed.
3. Frames for Level 2 Steel Doors: 0.053-inch (1.34 mm) (16 gauge) thick galvanized steel sheet.

- D. Interior Frames: Fabricated from cold-rolled steel sheet.

1. Fabricate frames with mitered corners.
2. Fabricate frames as face welded or knock down compressions type.
3. Frames for Level 2 Steel Doors: 0.053-inch (1.34 mm) (16 gauge) thick steel sheet.
4. Frames for Borrowed Lights: 0.053-inch (1.34 mm) (16 gauge) thick steel sheet.

- E. Hardware Reinforcement: ANSI/NAAMM-HMMA 861.

- F. Frame Finish: Shop-primed (for field-painted finish).

2.6 FRAME ANCHORS

- A. Jamb Anchors:

1. Masonry Type: Adjustable strap-and-stirrup or T-shaped anchors to suit frame size, not less than 0.042 inch thick, with corrugated or perforated straps not less than 2 inches wide by 10 inches long; or wire anchors not less than 0.177 inch thick.
2. Stud-Wall Type: Designed to engage stud; not less than 0.042 inch thick.

- B. Floor Anchors: Formed from same material as frames, not less than 0.042 inch thick, and as follows:

1. Monolithic Concrete Slabs: Clip-type anchors, with two holes to receive fasteners.
2. Separate Topping Concrete Slabs: Adjustable-type anchors with extension clips, allowing not less than 2-inch height adjustment. Terminate bottom of frames at finish floor surface.

2.7 STOPS AND MOLDINGS

- A. Moldings for Glazed Lites in Doors: Minimum 0.032-inch thick, same material as door face sheet.
- B. Fixed Frame Moldings: Formed integral with hollow metal frames, a minimum of 5/8 inch high unless otherwise indicated.
- C. Loose Stops for Glazed Lites in Frames: Minimum 0.032-inch thick, same material as frames.

2.8 FABRICATION

- A. Tolerances: Fabricate hollow metal work to tolerances indicated in HMMA 840-1999.
- B. Hollow Metal Doors:
 1. Exterior Doors: Provide weep-hole openings in bottom of exterior doors. Seal joints in top edges of doors against water penetration.
 2. Glazed Lites: Factory cut openings in doors.
 3. Astragals: Provide overlapping astragal on one leaf of pairs of doors where required by NFPA 80 for fire-performance rating or where indicated.
- C. Hollow Metal Frames: Where frames are fabricated in sections, provide alignment plates or angles at each joint, fabricated of same thickness metal as frames.
 1. Welded Frames: Weld flush face joints continuously; grind, fill, dress, and make smooth, flush, and invisible.
 2. Sidelight Frames: Provide closed tubular members with no visible face seams or joints, fabricated from same material as door frame. Fasten members at crossings and to jambs by butt welding.
 3. Provide countersunk, flat- or oval-head exposed screws and bolts for exposed fasteners unless otherwise indicated.
 4. Floor Anchors: Apply anchors to bottom of jambs and mullions with at least four spot welds per anchor.
 5. Jamb Anchors: Provide number and spacing of anchors as follows:
 - a. Masonry Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches o.c. and as follows:
 - 1) Two anchors per jamb up to 60 inches high.
 - 2) Three anchors per jamb from 60 to 90 inches high.
 - 3) Four anchors per jamb from 90 to 120 inches high.
 - 4) Four anchors per jamb plus 1 additional anchor per jamb for each 24 inches or fraction thereof above 120 inches high.
 - b. Stud-Wall Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches o.c. and as follows:
 - 1) Three anchors per jamb up to 60 inches high.

- 2) Four anchors per jamb from 60 to 90 inches high.
 - 3) Five anchors per jamb from 90 to 96 inches high.
 - 4) Five anchors per jamb plus 1 additional anchor per jamb for each 24 inches or fraction thereof above 96 inches high.
 - 5) Two anchors per head for frames more than 42 inches wide and mounted in metal-stud partitions.
- c. Postinstalled Expansion Type: Locate anchors not more than 6 inches from top and bottom of frame. Space anchors not more than 26 inches o.c.
6. Door Silencers: Except on weather-stripped doors, drill stops to receive door silencers.
 - a. Single-Door Frames: Three door silencers.
 - b. Double-Door Frames: Two door silencers.
- D. Hardware Preparation: Factory prepare hollow metal work to receive templated mortised hardware according to the Door Hardware Schedule and templates furnished as specified in Division 08 Section "Door Hardware."
 1. Locate hardware as indicated, or if not indicated, according to ANSI/HMMA 840-1999.
 2. Reinforce doors and frames to receive nontemplated, mortised and surface-mounted door hardware.
 3. Comply with applicable requirements in ANSI/HMMA 840-1999 Series specifications for preparation of hollow metal work for hardware.
 4. Coordinate locations of conduit and wiring boxes for electrical connections with Division 26 electrical Sections.
- E. Stops and Moldings: Provide stops and moldings around glazed lites where indicated. Form corners of stops and moldings with butted or mitered hairline joints.
 1. Single Glazed Lites: Provide fixed stops and moldings welded on secure side of hollow metal work.
 2. Multiple Glazed Lites: Provide fixed and removable stops and moldings so that each glazed lite is capable of being removed independently.
 3. Provide fixed frame moldings on outside of exterior and on secure side of interior doors and frames.
 4. Provide loose stops and moldings on inside of hollow metal work.
 5. Coordinate rabbet width between fixed and removable stops with type of glazing and type of installation indicated.

2.9 STEEL FINISHES

- A. Factory Finish (Doors): Clean, pretreat, and apply manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat, complying with SDI A250.3.
 1. Color: See Drawings.
- B. Prime Finish (Frames): Clean and chemically treat frames to ensure maximum finish paint adhesion. Surfaces of the doors and frames to receive a factory-applied coat of rust inhibiting shop primer.
 1. Shop Primer: Manufacturer's standard, fast-curing, lead and chromate free primer complying with ANSI/SDI A250.10 acceptance criteria; recommended by primer manufacturer for substrate; and compatible with substrate and field-applied coatings.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Examine roughing-in for embedded and built-in anchors to verify actual locations before frame installation.
- C. Prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.
- D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Remove welded-in shipping spreaders installed at factory. Restore exposed finish by grinding, filling, and dressing, as required to make repaired area smooth, flush, and invisible on exposed faces.
- B. Drill and tap doors and frames to receive nontemplated, mortised, and surface-mounted door hardware.

3.3 INSTALLATION

- A. Hollow Metal Frames: Comply with ANSI/HMMA 840-1999.
 - 1. Set frames accurately in position, plumbed, aligned, and braced securely until permanent anchors are set. After wall construction is complete, remove temporary braces, leaving surfaces smooth and undamaged.
 - a. At fire-protection-rated openings, install frames according to NFPA 80.
 - b. Where frames are fabricated in sections because of shipping or handling limitations, field splice at approved locations by welding face joint continuously; grind, fill, dress, and make splice smooth, flush, and invisible on exposed faces.
 - c. Install frames with removable glazing stops located on secure side of opening.
 - d. Remove temporary braces necessary for installation only after frames have been properly set and secured.
 - e. Check plumbness, squareness, and twist of frames as walls are constructed. Shim as necessary to comply with installation tolerances.
 - 2. Floor Anchors: Provide floor anchors for each jamb and mullion that extends to floor, and secure with postinstalled expansion anchors.
 - a. Floor anchors may be set with powder-actuated fasteners instead of postinstalled expansion anchors if so indicated and approved on Shop Drawings.
 - 3. Metal-Stud Partitions: Solidly pack mineral-fiber insulation behind frames for fire-rated wall assemblies.
 - 4. Masonry Walls: Coordinate installation of frames to allow for solidly filling space between frames and masonry with sprayed foam insulation.

5. In-Place Gypsum Board Partitions: Secure frames in place with postinstalled expansion anchors through floor anchors at each jamb. Countersink anchors, and fill and make smooth, flush, and invisible on exposed faces.
 6. Installation Tolerances: Adjust hollow metal door frames for squareness, alignment, twist, and plumb to the following tolerances:
 - a. Squareness: Plus or minus 1/16 inch, measured at door rabbet on a line 90 degrees from jamb perpendicular to frame head.
 - b. Alignment: Plus or minus 1/16 inch, measured at jambs on a horizontal line parallel to plane of wall.
 - c. Twist: Plus or minus 1/16 inch, measured at opposite face corners of jambs on parallel lines, and perpendicular to plane of wall.
 - d. Plumbness: Plus or minus 1/16 inch, measured at jambs at floor.
- B. Hollow Metal Doors: Fit hollow metal doors accurately in frames, within clearances specified below. Shim as necessary.
1. Non-Fire-Rated Standard Steel Doors:
 - a. Jambs and Head: 1/8 inch plus or minus 1/16 inch.
 - b. Between Edges of Pairs of Doors: 1/8 inch plus or minus 1/16 inch.
 - c. Between Bottom of Door and Top of Threshold: Maximum 3/8 inch.
 - d. Between Bottom of Door and Top of Finish Floor (No Threshold): Maximum 3/4 inch.
 2. Fire-Rated Doors: Install doors with clearances according to NFPA 80.
- C. Glazing: Comply with installation requirements in Division 08 Section "Glazing" and with hollow metal manufacturer's written instructions.
1. Secure stops with countersunk flat- or oval-head machine screws spaced uniformly not more than 9 inches o.c. and not more than 2 inches o.c. from each corner.

3.4 PROTECTION

- A. Protect doors and frames from damage during delivery, handling and storage and after installation.
1. Provide door frame protectors during construction.
 2. Provide lockable entrances during construction per Section 015000 "Temporary Facilities and Controls".
 3. Provide final installation of doors immediately prior to Substantial Completion.
 4. Remove and replace doors and frames damaged during construction, at no cost to the Owner. Contractor is responsible for entire cost of removal and replacement of damaged doors and frames.

3.5 ADJUSTING AND CLEANING

- A. Final Adjustments: Check and readjust operating hardware items immediately before final inspection. Leave work in complete and proper operating condition. Remove and replace defective work, including hollow metal and thermoplastic work that is warped, bowed, damaged or otherwise unacceptable.

- B. Factory-Finish Touchup: Clean abraded areas and repair with same material used for factory finish according to manufacturer's written instructions.
- C. Prime-Coat Touchup: Immediately after erection, sand smooth rusted or damaged areas of prime coat and apply touchup of compatible air drying, rust-inhibitive primer.

END OF SECTION 081113

SECTION 083213 - ALUMINUM-FRAMED SLIDING GLASS DOORS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes exterior and interior aluminum-framed sliding automatic entrances.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings: For sliding aluminum-framed glass doors.
 - 3. Samples: For each exposed product and for each color specified, 12-inch-long (300-mm-long) section with weather stripping, glazing bead, and factory-applied color finish.

1.4 INFORMATIONAL SUBMITTALS

- A. Qualification data.
- B. Product test reports.
- C. Installer certificates.
- D. Sample warranty.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data.

1.6 QUALITY ASSURANCE

- A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.
- B. Power-Operated Pedestrian Door Standard: ANSI/BHMA A156.10 (current version).
- C. Emergency Exit door requirements: Comply with requirements of authorities having jurisdiction for automatic entrance doors serving as a required means of egress.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver aluminum work palletized, packaged, or crated to provide protection during transit and Project-site storage. Do not use nonvented plastic.

1. Provide additional protection to prevent damage to factory-finished units.
- B. Store aluminum work vertically under cover at Project site with head up. Place on minimum 4-inch- (102-mm-) high wood blocking. Provide minimum 1/4-inch (6-mm) space between each stacked door to permit air circulation.

1.8 WARRANTY

- A. Special Warranty: Manufacturer agrees to repair or replace components of aluminum-framed sliding automatic entrances that do not comply with requirements or that fail in materials or workmanship within specified warranty period.
 1. Warranty Period: Five years from date of Substantial Completion.
- B. Special Finish Warranty: Standard form in which manufacturer agrees to repair finishes or replace aluminum that shows evidence of deterioration of factory-applied finishes within specified warranty period.
 1. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Compliance with the following:
 1. ANSI/BHMA A156.10 American National Standard for Power Operated Pedestrian Doors.
 2. UL 325 listed.
- B. Automatic door equipment accommodates medium to heavy pedestrian traffic.
- C. Entrapment Force Requirements:
 1. Power Operated Sliding Doors: Not more than 30 lbf (133 N) required to prevent stopped door from closing.
 2. Sliding doors provided with a breakaway device shall require no more than 50 lbf (222N) applied 1 inch (25 mm) from the leading edge of the lock stile for the breakout panel to open.
- D. Energy Code Requirements: Sliding automatic entrances that are required to meet construction energy code requirements in those districts that have adopted ASHRAE 90.1-2010/2013 shall have been evaluated based on methodology in accordance with the following National Fenestration Rating Council (NFRC) standards:
 1. NFRC 100-2010: Procedure for Determining Fenestration Product U-Factors.
 2. NFRC 200-2010: Procedure for Determining Fenestration Product Solar Heat Gain Coefficient and Visible Transmittance at Normal Incidence.
 3. NFRC 500-2010: Procedure for Determining Fenestration Product Condensation Resistance Values.
 4. ASTM 283e-2010: Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen.
- E. Design Wind Loads for Exterior Units: As indicated on the Structural Drawings.

2.2 MANUFACTURER

- A. Source Limitations: Obtain each type of door, frame, operator and sensor components from a single source, same manufacturer.
- B. Basis of Design:
 - 1. Manufacturer: ASSA ABLOY Entrance Systems; www.assaabloyentrance.us.
 - 2. Product: Besam SL500 EcoDoor U-Factor
- C. Supplier Contact: See **LiDL National Account List**.

2.3 SLIDING AUTOMATIC ENTRANCES

- A. Sliding automatic entrance system including the following:
 - 1. Sliding panels, sidelites, transom and aluminum frame.
 - 2. Overhead concealed, electro-mechanical operator.
 - 3. Operator housing, guide system and carrier assemblies.
 - 4. Controls and accessories as required for a complete installation.
- B. Unit Description: Automatic sliding entrance system with stile and rail panels and with overhead concealed electric operator for automatic operation.
 - 1. Bi-Parting, Fixed Sidelite, Door System:
 - a. Configuration: Bi-parting, four equal panel unit with two operable leaves and two fixed sidelites.
 - b. Traffic Pattern: Two-way
 - c. Emergency Breakaway Capability: Sliding leaves only.
 - d. Mounting: Overhead header installed between jambs.
 - 2. Stile and Rail Sliding Panels and Sidelites:
 - a. Material: Extruded Aluminum, Alloy 6063-T5
 - b. Vertical Stile: Narrow stile 2-1/8 inches (54 mm)
 - c. Bottom Rail: 4 inches (102 mm)
 - d. No muntin.
 - e. Weather-Stripping: Slide-in type, replaceable pile mohair seals retained by the aluminum extrusions.
 - f. EcoDoor Seals: High pile mohair weather stripping on the lock stile of the sliding doors, integrated mohair weather stripping with vinyl fin on the joining vertical stiles of the sidelite and sliding door panels, and expandable foam inserts in leading stile of sidelite panels at pockets for interlocks. Bottom rails shall be provided with a concealed adjustable nylon sweep.
 - g. U-Factor Door Package:
 - 1) U-Factor Rating: 0.64 BTU/(h °F ft²)
 - 2) Solar Heat Gain Coefficient: 0.28
 - 3) Visible Light Transmittance: 0.45
 - 4) Condensation Resistance: 22
 - 5) Air Infiltration Rating: 0.93 cuft/min/sqft 0.28x3/Mx2/min.
 - h. Glazing for Sliding Panels, Sidelites and Transom: ANSI Z97.1

- 1) Interior: 1/4 clear tempered glass
 - 2) Exterior: 1-inch insulating glass as specified in Section 088000 "Glazing".
3. Door Carriers: Manufacturer's standard carrier assembly that allows vertical adjustment.
 - a. Carriage Assembly: Carriage bar with two-wheel assemblies. Each assembly shall have tandem roller wheels.
 - b. Roller Wheels: Two heavy duty Delrin roller wheels per wheel assembly, for a total of four (4) roller wheels, 1-7/16-inch (36.51 mm) diameter, per active door leaf for operation over a replaceable aluminum track. Single journal with sealed oil impregnated bearings.
 - c. Two heavy-duty self-aligning anti-risers per leaf.
4. Framing Members: Manufacturer's standard extruded aluminum framing reinforced as required to support loads.
 - a. Vertical Jambs: 1-3/4 inches (44.5 mm) by 4-1/2 inches (114.3 mm)
5. Header: Manufacturer's standard extruded aluminum header with a replaceable aluminum track extending full width of entrance unit. Header to conceal door operators, carrier assemblies, and roller track; complete with hinged access panel for service of door operator, and controls.

2.4 HARDWARE

A. General:

1. Hardware: Provide manufacturer's standard hardware, fabricated from a corrosion-resistant material compatible with aluminum complying with AAMA 907 and designed to smoothly operate, tightly close, and securely lock sliding aluminum-framed glass doors.
 - a. Breakaway arms and bottom pivot assemblies shall be supplied by the manufacturer and shall be adjustable to comply with applicable codes.
 - b. Magnetic catch(s) to retain breakout door and sidelite panels in the closed position.
 - c. Lock: Manufacturer's fail secure electric lock with 24-hour battery back-up. No manual locking or lock preparation is required.
2. Guide Track/Threshold: Manufacturer's threshold and guide.
 - a. Fixed Sidelite Entrance Guide Track: Aluminum guide track integrated in the bottom of the sidelite portion of the sliding automatic door assembly.
 - b. Door Guides: Floor guides for operating door leaves.
 - c. Designed to drain to the exterior.
 - d. Finish: Manufacturer's standard.
3. Door Operator and Controller:
 - a. Electro-mechanical controlled unit utilizing a high-efficiency, energy efficient, DC motor requiring a maximum of 3 amp current draw, allowing 5 operators on one 20 amp circuit. Operator allowing an adjustable hold open time delay of 0 to 60 seconds and have internal software to incorporate a self-diagnostic system.
4. Microprocessor Control Box:

- a. Factory-adjusted configuration with opening and closing speeds set to comply with ANSI/BHMA A156.10 requirements and electronic dampening to reduce wear on drive train.
 - b. Mode Selector Control: Multi-position rotary knob mode selector control allowing selection of the indicated functions to be engaged when switch is turned to the appropriate setting.
 - c. Mode Selector Control Mounting: Jamb mounted.
 - d. Mode selector control to allow the following functions:
 - 1) "Off"
 - 2) "Exit Only" one-way traffic with automatic operation from the interior.
 - 3) "Two Way Traffic" allowing automatic operation from exterior and interior.
 - 4) "Partial Opening" energy saving door position allows door to automatically adjust opening width based on amount of usage, that is, full open during high use and partial open during low use. The control for this setting is programmable allowing adjustment to both the usage setting and the opening width.
 - 5) "Hold Open" doors activated and held in the full open position.
5. Activation and Safety Control Devices:
- a. Combination Activation Motion Sensor/Safety Presence Sensor.
6. Electrical:
- a. High-Efficiency DC Motor: Maximum of 3-amp current draw, allowing 5 operators to run on one 20 Amp circuit.
 - b. Power: Self-detecting line voltage capable control. 120 VAC through 240 VAC, 50/60 Hz, 3 amp minimum incoming power with solid earth ground connection for each door system.
 - c. Wiring: Separate internal channel raceway free from moving parts.
 - d. Brown out / high voltage capability: System has capability to operate at full performance well beyond brown out and high voltage line conditions (85 V – 265 V) sensing changes and adjusting automatically.
 - e. Convenience Battery: Concealed in header and capable of full operation with blackout conditions, including sensor capabilities for minimum of 100 cycles.

2.5 GLAZING

- A. Glass and Glazing: Manufacturer's standard glazing system that produces weathertight seal. Comply with requirements indicated in Section 088000 "Glazing."
- 1. Glass: ASTM C 1036, Type 1, q3, Category II safety glass complying with testing requirements in 16 CFR 1201.
 - 2. Safety Glazing Labeling: Permanently mark safety glazing with certification label of the SGCC or another certification agency acceptable to authorities having jurisdiction. Label shall indicate manufacturer's name, type of glass, thickness, and safety glazing standard with which glass complies.
- B. Glazing Gaskets: Manufacturer's standard sealed-corner pressure-glazing system of black, resilient elastomeric glazing gaskets, setting blocks, and shims or spacers. Provide 1/4-inch (6 mm) glazing stops.
- 1. Provide sample for approval.

C. Glazing Sealants: As recommended by manufacturer.

1. Exposed Sealant Color: Black.
2. Provide Sample for verification.
3. VOC Content: Glazing sealants applied inside the weatherproofing system of the building shall have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

2.6 ACCESSORIES

- A. Anchors, Clips, and Accessories: Provide anchors, clips, and accessories of aluminum, nonmagnetic stainless steel, or zinc-coated steel or iron for sliding aluminum-framed glass doors, complying with ASTM B 456 or ASTM B 633 for SC 3 severe service conditions; provide sufficient strength to withstand design pressure indicated.
1. Windborne-Debris Resistance: Provide anchors of same design used in windborne-debris resistance testing.

2.7 FABRICATION

- A. Fabricate sliding aluminum-framed glass doors in sizes indicated. Include a complete system for assembling components and anchoring doors.
- B. Weather Stripping: Provide full-perimeter weather stripping for each door panel.
- C. Weep Holes: Provide weep holes and internal drainage passages to conduct infiltrating water to exterior.
- D. Factory-Glazed Fabrication: Glaze sliding aluminum-framed glass doors in the factory for applications indicated. Comply with requirements in Section 088000 "Glazing".

2.8 ALUMINUM FINISHES

- A. High-Performance Organic Finish: Two-coat fluoropolymer finish complying with AAMA 2605 and containing not less than 70 percent PVDF resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
1. Color: See Exterior Material/Finish Schedule on Drawings.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine doors and frames, with Installer present, for compliance with requirements for installation tolerances, wall and floor construction, and other conditions affecting performance.
- B. Examine roughing-in for electrical source power to verify actual locations of wiring connections.
- C. Proceed only after such discrepancies or conflicts have been resolved.

3.2 INSTALLATION

- A. General: Install automatic entrances according to manufacturer's written instructions and cited BHMA A156.10 for direction of pedestrian travel, including signage, controls, wiring, and connection to the building's power supply.
 - 1. Do not install damaged components. Fit frame joints to produce hairline joints free of burrs and distortion. Rigidly secure nonmovement joints. Seal joints watertight.
 - 2. Where aluminum will contact dissimilar metals, protect against galvanic action by painting contact surfaces with primer or by applying sealant or tape recommended by manufacturer for this purpose.
 - 3. Where aluminum will contact concrete or masonry, protect against corrosion by painting contact surfaces with bituminous coating.
- B. Entrances: Install automatic entrances plumb and true in alignment with established lines and grades without warp or rack of framing members and doors. Anchor securely in place.
 - 1. Install surface mounted hardware using concealed fasteners to greatest extent possible.
 - 2. Set headers, carrier assemblies, tracks, operating brackets and guides level and true to location with anchorage for permanent support.
 - 3. Install components to drain water passing joints, condensation occurring within framing members, and moisture migrating within system to exterior.
 - 4. Level recesses for recessed thresholds using nonshrink grout.
- C. Door Operators: Connect door operators to electrical power distribution system.
- D. Controls: Install and adjust activation and safety devices according to manufacturer's written instructions and cited BHMA standard for direction of pedestrian travel.
- E. Guide Rails: Install rails according to BHMA A156.10, including Appendix A, and manufacturer's written instructions unless otherwise indicated.
- F. Sealants: Comply with requirements specified in Section 079200 "Joint Sealants" to provide weathertight installation.
 - 1. Set thresholds, framing members and flashings in full sealant bed.
 - 2. Seal perimeter of framing members with sealant.
- G. Signage: Apply signage on both sides of each door, as required by cited BHMA standard for direction of pedestrian travel.
- H. Wiring within Automatic Entrance Enclosures: Bundle, lace, and train conductors to terminal points with no excess and without exceeding manufacturer's written limitations on bending radii. Provide and use lacing bars and distribution spools.

3.3 ADJUSTING

- A. Adjust hardware, moving parts, door operators, and controls to function smoothly, and lubricate as recommended by manufacturer; comply with requirements of applicable BHMA standards.
 - 1. Adjust exterior doors for tight closure.
- B. Readjust door operators and controls after repeated operation of completed installation equivalent to three days' use by normal traffic (100 to 300 cycles).

3.4 CLEANING

- A. Clean glass and metal surfaces promptly after installation. Remove excess glazing and sealant compounds, dirt, and other substances. Repair damaged finish to match original finish.

END OF SECTION 083213

SECTION 083483 – TRAFFIC DOORS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. High-Impact traffic doors.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For product specified herein. Include manufacturer's preparation and installation instructions.
 - 2. Samples: For selection of exposed finishes.
 - 3. Shop Drawings: Show fabrication and installation details; include door elevations, head, jamb, and meeting stile details including full or partial gaskets.

1.4 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Storage and Protection: Comply with manufacturer's recommendations.
 - 1. Store in manufacturer's unopened packaging until ready for installation.
 - 2. Store in a cool, dry place out of direct sunlight.
 - 3. Protect from damage by the elements and construction procedures.

1.6 PROJECT CONDITION

- A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

1.7 WARRANTY

- A. Manufacturer's standard two-year warranty that products are free of defects in material and workmanship, guaranteeing to replace (exclusive of freight and labor) parts proven defective within two years after date of shipment to purchaser.

PART 2 - PRODUCTS

2.1 SUPPLIER OF TRAFFIC DOORS AND FRAMES

- A. Contact: See LIDL National Account List.

2.2 HIGH-IMPACT TRAFFIC DOORS AND FRAMES

- A. Manufacturer: Eliason Corporation; www.eliasoncorp.com.
- B. Provide traffic door and frame as a system from the same manufacturer.
- C. High Impact Traffic Doors:
 - 1. Product: Eliason Model HCP-10.
 - 2. Overall Door Thickness: 1-1/2 inches (38 mm)
 - 3. Door Panel Core: High strength polymer cell core.
 - 4. Door Construction: High-strength, structural PVC internal frame, filled with urethane foam, clad with 0.080-inch thick impact-resistant thermoplastic (ABS) face sheets on both sides.
 - a. Facing Color: See Drawings.
- D. Offset Frame: Custom-formed C-channel frame to accommodate offset location of hinge.
 - 1. Frame Depth: Per wall construction plus 2 inches over wall thickness.
 - 2. Frame (Jamb) Material: C-channel, 12-gauge mild steel.
 - 3. Hole Pattern: Custom predrilled holes to ensure positioning of door as indicated on Drawings.
 - 4. Wrap frame around door opening.
 - 5. Remove all burrs and break all sharp edges.
 - 6. Finish: Powder coating
 - 7. Color: See Drawings.
- E. Hardware and Accessories:
 - 1. Hinges: Easy Swing Hinge System, double-action, proprietary hinges, including cover.
 - a. Finish: Zinc coated.
 - 2. Window:
 - a. Window Size: 10- by 30-inch
 - b. Window Molding: Black rubber molding.
 - c. Glazing: Clear acrylic, double-glazed.
 - d. Mount window 43-inches above finish floor to bottom of window.
 - 3. Bumpers: Easy Spring Bumpers, 0.25 inch (6 mm) thick, tear-drop shaped, impact-resistant rotational molded polyethylene (HDPE) on both sides of each door panel.
 - a. Height: 18 inches (457 mm)
 - b. Color: Black
 - 4. Mounting Screws: Included.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Do not begin installation until substrates have been properly prepared. Notify Architect in writing of unsatisfactory preparation before proceeding.
- B. Verify jambs plumb and square.
- C. Verify jamb construction of double studs or equivalent per door manufacturer's recommendations.

3.2 PREPARTION

- A. Clean surfaces thoroughly prior to installation.
- B. Prepare surfaces using methods recommended by manufacturer for achieving the best result for the substrate until project conditions.

3.3 INSTALLATION

- A. Install doors according to manufacturer's instructions and approved shop drawings.
- B. Reinforce hollow metal jambs at hardware locations.
- C. Adjust hardware and leave in good operating order with doors automatically returning to full closed position.

3.4 PROTECTION

- A. Protect installed products until completion of project.
- B. Touch-up, repair or replace damaged products before Substantial Completion.

END OF SECTION 083483

SECTION 083613 – SECTIONAL DOORS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes insulated sectional doors.
- B. Related Requirements:
 - 1. Section 055000 "Metal Fabrications" for miscellaneous steel supports.

1.3 PRE-INSTALLATION CONFERENCE

- A. Pre-Installation Conference: Convene a pre-installation conference just prior to commencement of field operations, to establish procedures to maintain optimum working conditions and to coordinate this work with related and adjacent work.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: Manufacturer's data sheets on each product to be used, including:
 - a. Preparation instructions and recommendations.
 - b. Storage and handling requirements and recommendations.
 - c. Installation methods.
 - 2. Shop Drawings: Indicate plans and elevations including opening dimensions and required tolerances, connection details, anchorage spacing, hardware locations, and installation details.
 - 3. Samples: For each exposed finish.

1.5 INFORMATIONAL SUBMITTALS

- A. Manufacturer's Certificates: Certify products meet or exceed specified requirements.

1.6 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data.

1.7 QUALITY ASSURANCE

- A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer for both installation and maintenance of units required for this Project.

1.8 DELIVERY, STORAGE AND HANDLING

- A. Store products in manufacturer's unopened packaging with seals and labels intact until ready for installation.
- B. Store materials off the ground in a dry, warm, ventilated weathertight location.
- C. Protect materials from exposure to moisture until ready for installation.

1.9 WARRANTY

- A. Warranty: Manufacturer's limited door and operators System warranty for 10 years against cracking, splitting or deterioration of steel skin due to rust.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Wind Loads: Design and size components to withstand loads caused by pressure and suction of wind acting normal to plane of wall as calculated in accordance with applicable code.
 - 1. Design pressure of 24.5 lbs./sq. ft., unless greater pressure is required by local authority having jurisdiction.
- B. Seismic Performance: Overhead coiling doors shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
- C. Single-Source Responsibility: Provide doors, tracks, motors, and accessories from one manufacturer for each type of door. Provide secondary components from source acceptable to manufacturer of primary components.

2.2 MANUFACTURERS

- A. Source Limitations: Obtain overhead coiling doors from single source from single manufacturer.
- B. Basis of Design:
 - 1. Manufacturer: Wayne Dalton; www.wayne-dalton.com.
 - 2. Product: Thermospan 500 Insulated Sectional Steel Door.
- C. Supplier Contact: See **LiDL National Account List**.

2.3 INSULATED SECTIONAL DOORS

- A. Insulated Sectional Door: Steel/polyurethane/steel sandwich type construction with thermal break.
 - 1. Door Width and Height: See Drawings.
 - 2. Panel Thickness: 2 inches (51 mm)

3. Exterior Surface: Flush non-repeating random stucco texture and 1/4-inch wide pinstripping.
4. Exterior Steel: .015 inch (0.38 mm), hot-dipped galvanized.
5. Sections roll formed with two 1-3/4 inch integral struts sealed with polypropylene rib caps per section.
6. Thermal Values: R-value of 17.50; U-value of 0.057.
7. Air Infiltration: 0.07 cfm at 15 mph.
8. Sound transmission class 22 when tested in accordance with ASTM E 413.
9. Outdoor-indoor transmission class 19 when tested in accordance with ASTM E 1332.
10. Insulation: CFC-free and HCFC-free polyurethane, fully encapsulated.
 - a. Insulated sections tested in accordance with ASTM E 84 and achieve a Flame spread Index of 10 or less, and a Smoke Developed Index of 210 or less.
 - b. Insulation material tested in accordance with ASTM D 1929 and achieve a minimum Flash Ignition temperature of 734 degrees F, and a minimum Self Ignition temperature of 950 degrees F.
 - c. Insulated sections shall meet all requirements of the UBC 17-5 corner burn.
11. Ends: Hot-dipped galvanized steel, full height with end caps.
12. Spring Counterbalance: Sized to weight of the door, with a helically wound, oil tempered torsion spring mounted on a steel shaft; cable drum of die cast aluminum with high strength galvanized aircraft cable. Sized with a minimum 5 to 1 safety factor.
 - a. High cycle spring: 25,000 cycles.
13. Finish and Color:
 - a. Two coat baked-on polyester:
 - 1) Interior color, white.
 - 2) Exterior color, white.
14. Hardware: Galvanized steel hinges and fixtures. Ball bearing rollers with hardened steel races.
15. Lock: Interior mounted slide lock.
16. Weatherstripping:
 - a. Flexible bulb-type strip at bottom section.
 - b. Flexible Jamb seals.
 - c. Flexible Header seal.
17. Track: Provide track as recommended by manufacturer to suit loading required and clearances available.
 - a. Size: 2 inches (51 mm)
 - b. Type: Standard lift.
 - c. Horizontal track shall be reinforced with continuous angle of adequate length and gauge to minimize deflection.
18. Manual Operation: Chain hoist.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates areas and conditions, with Installer present, for compliance with requirements for substrate construction and other conditions affecting performance of the Work.
- B. Verify wall openings are ready to receive work and opening dimensions and tolerances are within specified limits.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Install sectional doors and operating equipment complete with necessary hardware, anchors, inserts, hangers, and equipment supports; in accordance with manufacturer's written instructions and approved shop drawings.
- B. Coordinate installation with adjacent work to ensure proper clearances and allow for maintenance.
- C. Anchor assembly to wall construction and building framing without distortion or stress.
- D. Securely brace door tracks suspended from structure. Secure tracks to structural members only.
- E. Fit and align door assembly including hardware.

3.3 ADJUSTING

- A. Adjust hardware and moving parts to function smoothly so that doors operate easily, free of warp, twist, or distortion.
- B. Lubricate bearings and sliding parts as recommended by manufacturer.
- C. Adjust doors and seals to provide weather-resistant fit around entire perimeter.
- D. Touchup Painting Galvanized Material: Immediately after welding galvanized materials, clean welds and abraded galvanized surfaces and repair galvanizing to comply with ASTM A780/A780M.

3.4 CLEANING

- A. Clean doors and tracks using non-abrasive materials and methods recommended by manufacturer.
- B. Remove labels and visible markings.
- C. Touch-up, repair or replace damaged products before Substantial Completion.

3.5 DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain overhead coiling doors.

END OF SECTION 083613

SECTION 084113 - ALUMINUM-FRAMED STOREFRONTS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Exterior and interior storefront framing.

1.3 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings: Include plans, elevations, sections, full-size details, and attachments to other work.
 - a. Show connection to and continuity with adjacent thermal, weather, air, and vapor barriers.
 - 3. Samples: For each exposed finish required.

1.5 INFORMATIONAL SUBMITTALS

- A. Energy Performance Certificates: NFRC-certified energy performance values from manufacturer.
- B. Product test reports.
- C. Field quality-control reports.
- D. Installer certificates.
- E. Sample warranties.

1.6 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.7 QUALITY ASSURANCE

- A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.
- B. Testing Agency Qualifications: Qualified according to ASTM E 699 for testing indicated and accredited by IAS or ILAC Mutual Recognition Arrangement as complying with ISO/IEC 17025.
- C. Product Options: Information on Drawings and in Specifications establishes requirements for aesthetic effects and performance characteristics of assemblies. Aesthetic effects are indicated by dimensions, arrangements, alignment, and profiles of components and assemblies as they relate to sightlines, to one another, and to adjoining construction.
 - 1. Do not change intended aesthetic effects, as judged solely by Architect, except with Architect's approval.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver aluminum work palletized, packaged, or crated to provide protection during transit and Project-site storage. Do not use non-vented plastic.
 - 1. Provide additional protection to prevent damage to factory-finished units.
- B. Store aluminum work vertically under cover at Project site with head up. Place on minimum 4-inch- (102-mm-) high wood blocking. Provide minimum 1/4-inch (6-mm) space between each stack to permit air circulation.

1.9 WARRANTY

- A. Special Warranty: Manufacturer agrees to repair or replace components of aluminum-framed storefronts that do not comply with requirements or that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period: Five years from date of Substantial Completion.
- B. Special Finish Warranty: Standard form in which manufacturer agrees to repair finishes or replace aluminum that shows evidence of deterioration of factory-applied finishes within specified warranty period.
 - 1. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. General Performance: Comply with performance requirements specified, as determined by testing of aluminum-framed entrances and storefronts representing those indicated for this Project without failure due to defective manufacture, fabrication, installation, or other defects in construction.
 - 1. Aluminum-framed entrances and storefronts shall withstand movements of supporting structure including, but not limited to, story drift, twist, column shortening, long-term creep, and deflection from uniformly distributed and concentrated live loads.
 - 2. Failure also includes the following:
 - a. Thermal stresses transferring to building structure.

- b. Glass breakage.
 - c. Loosening or weakening of fasteners, attachments, and other components.
 - d. Failure of operating units.
- B. Structural Loads:
 - 1. Wind Loads: As indicated on Drawings.
 - 2. Other Design Loads: As indicated on Drawings.
- C. Deflection of Framing Members: At design wind pressure, as follows:
 - 1. Deflection Normal to Wall Plane: Limited to edge of glass in a direction perpendicular to glass plane not exceeding 1/175 of the glass edge length for each individual glazing lite or an amount that restricts edge deflection of individual glazing lites to 3/4 inch (19.1 mm), whichever is less.
 - 2. Deflection Parallel to Glazing Plane: Limited to amount not exceeding that which reduces glazing bite to less than 75 percent of design dimension and that which reduces edge clearance between framing members and glazing or other fixed components to less than 1/8 inch (3.2 mm).
- D. Structural: Test according to ASTM E 330 as follows:
 - 1. When tested at positive and negative wind-load design pressures, assemblies do not evidence deflection exceeding specified limits.
 - 2. When tested at 150 percent of positive and negative wind-load design pressures, assemblies, including anchorage, do not evidence material failures, structural distress, or permanent deformation of main framing members exceeding 0.2 percent of span.
 - 3. Test Durations: As required by design wind velocity, but not less than 10 seconds.
- E. Air Infiltration: Test according to ASTM E 283 for infiltration as follows:
 - 1. Fixed Framing and Glass Area:
 - a. Maximum air leakage of 0.06 cfm/sq. ft. (0.30 L/s per sq. m) at a static-air-pressure differential of 6.24 lbf/sq. ft. (300 Pa).
- F. Water Penetration under Static Pressure: Test according to ASTM E 331 as follows:
 - 1. No evidence of water penetration through fixed glazing and framing areas when tested according to a minimum static-air-pressure differential of 20 percent of positive wind-load design pressure, but not less than 10 lbf/sq. ft. (500 Pa).
- G. Energy Performance: Certified and labeled by manufacturer for energy performance as follows:
 - 1. Thermal Transmittance (U-factor):
 - a. Fixed Glazing and Framing Areas: U-factor for the system of not more than 0.41 Btu/sq. ft. x h x deg F (2.33 W/sq. m x K) as determined according to NFRC 100.
 - 2. Solar Heat-Gain Coefficient (SHGC):
 - a. Fixed Glazing and Framing Areas: SHGC for the system of not more than 0.36 as determined according to NFRC 200.

3. Air Leakage:

- a. Fixed Glazing and Framing Areas: Air leakage for the system of not more than 0.06 cfm/sq. ft. (0.30 L/s per sq. m) at a static-air-pressure differential of 6.24 lbf/sq. ft. (300 Pa) when tested according to ASTM E283.

4. Condensation Resistance Factor (CRF):

- a. Fixed Glazing and Framing Areas: CRF for the system of not less than 25 as determined according to AAMA 1503.

H. Thermal Movements: Allow for thermal movements resulting from ambient and surface temperature changes:

1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

2.2 MANUFACTURERS

A. Basis of Design:

1. Manufacturer: Kawneer, an Arconic Company; www.kawneer.com.
2. Products:
 - a. Exterior: Trifab 601T Framing System (Thermal)
 - b. Interior: Trifab 601 Framing System (Non-Thermal)

B. Supplier Contact: See **Lidl National Account List**.

2.3 FRAMING

A. Framing Members: Manufacturer's extruded- or formed-aluminum framing members of thickness required and reinforced as required to support imposed loads.

1. Construction: Thermally broken for all exterior framing and nonthermal for interior framing. Sill and head members are continuous at clerestory / office windows.
2. System Dimensions: 2 by 6 inches (50.8 by 152.4 mm) nominal.
3. Glazing System: Retained mechanically with gaskets on four sides.
4. Glazing Plane: Center of frame.
5. Finish: Two-coat fluoropolymer, 70% PVDF
 - a. Color: See Exterior Material/Finish Schedule on Drawings.

6. Fabrication Method: Shop-fabricated crew spline or shear block.

B. Backer Plates: Manufacturer's standard, continuous backer plates for framing members, if not integral, where framing abuts adjacent construction.

C. Brackets and Reinforcements: Manufacturer's standard high-strength aluminum with nonstaining, nonferrous shims for aligning system components.

D. Fasteners and Accessories: Manufacturer's standard corrosion-resistant, nonstaining, nonbleeding fasteners and accessories compatible with adjacent materials. Where exposes shall be stainless steel.

- E. Perimeter Anchors: When steel anchors are used, provide insulation between steel material and aluminum material to prevent galvanic action.
- F. Materials:
 - 1. Aluminum: Alloy and temper recommended by manufacturer for type of use and finish indicated.
 - a. Sheet and Plate: ASTM B 209 (ASTM B 209M).
 - b. Extruded Bars, Rods, Profiles, and Tubes: ASTM B 221 (ASTM B 221M).
 - c. Extruded Structural Pipe and Tubes: ASTM B 429/B 429M.
 - d. Structural Profiles: ASTM B 308/B 308M.

2.4 GLAZING

- A. Glazing: Comply with Section 088000 "Glazing."
- B. Glazing Gaskets: Manufacturer's standard compression types; replaceable, extruded EPDM rubber.
- C. Spacers and Setting Blocks: Manufacturer's standard elastomeric type.
- D. Glazing Sealants: As recommended by manufacturer.
 - 1. Weatherseal Sealant: ASTM C 920 for Type S, Grade NS, Class 25, Uses NT, G, A, and O; single-component neutral-curing formulation that is compatible with structural sealant and other system components with which it comes in contact; recommended by structural-sealant, weatherseal-sealant, and aluminum-framed-system manufacturers for this use.
 - 2. VOC Content: Glazing sealants applied inside the weatherproofing system of the building shall have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

2.5 SOLAR SHADING LOUVER

- A. Basis of Design:
 - 1. Manufacturer: Levolux Ltd.
 - 2. Product: Infiniti Fin System
- B. External Solar Shading System:
 - 1. Aerofoil Fin Bracket, Locking Plate and Fixings: Stainless steel.
 - 2. Two-Part Fins (Blades), Cover Trim and End Caps: Extruded aluminum alloy.
 - a. Fin Dimensions: 7-7/8 by 2 inches
 - b. Mounting: Horizontal
 - c. Angle: 45 degrees
 - 3. Finish: Two-coat fluoropolymer, 70% PVDF
 - a. Color: See Exterior Material/Finish Schedule on Drawings.

2.6 ACCESSORY MATERIALS

- A. Joint Sealants: For installation at perimeter of aluminum-framed systems, as specified in Division 07 Section "Joint Sealants".
- B. Bituminous Paint: Cold-applied, asphalt-mastic paint complying with SSPC-Paint 12 requirements except containing no asbestos; formulated for 30 mil (0.762 mm) thickness per coat.

2.7 FABRICATION

- A. Form or extrude aluminum shapes before finishing.
- B. Weld in concealed locations to greatest extent possible to minimize distortion or discoloration of finish. Remove weld spatter and welding oxides from exposed surfaces by descaling or grinding.
- C. Fabricate components that, when assembled, have the following characteristics:
 - 1. Profiles that are sharp, straight, and free of defects or deformations.
 - 2. Accurately fitted joints with ends coped or mitered.
 - 3. Physical and thermal isolation of glazing from framing members.
 - 4. Accommodations for thermal and mechanical movements of glazing and framing to maintain required glazing edge clearances.
 - 5. Provisions for field replacement of glazing from interior.
 - 6. Fasteners, anchors, and connection devices that are concealed from view to greatest extent possible.
- D. Mechanically Glazed Framing Members: Fabricate for flush glazing without projecting stops.
- E. Storefront Framing: Fabricate components for assembly using manufacturer's standard installation instructions.
- F. Trim: Fabricate trim, sill extension, stool, column covers, beam cover, and other brake metal profiles as necessary to complete storefront assembly.
 - 1. Aluminum Thickness: As required to produce sheet aluminum profiles without distortion or oil canning.
 - 2. Aluminum Finish and Color: Match adjacent storefronts and entrances.
- G. After fabrication, clearly mark components to identify their locations in Project according to Shop Drawings.

2.8 ALUMINUM FINISHES

- A. High-Performance Organic Finish: Two-coat fluoropolymer finish complying with AAMA 2605 and containing not less than 70 percent PVDF resin by weight in color coat.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine openings, substrates, structural support, anchorage, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of work. Verify rough opening dimensions, levelness of sill plate and operational clearances. Examine wall flashings, vapor retarders, water and weather barriers,

and other built-in components to ensure a coordinated, weather tight framed aluminum-storefront system installation.

- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. General:

1. Comply with manufacturer's written instructions.
2. Do not install damaged components.
3. Fit joints to produce hairline joints free of burrs and distortion.
4. Rigidly secure nonmovement joints.
5. Install anchors with separators and isolators to prevent metal corrosion and electrolytic deterioration and to prevent impeding movement of moving joints.
6. Seal perimeter and other joints watertight unless otherwise indicated.

B. Metal Protection:

1. Where aluminum is in contact with dissimilar metals, protect against galvanic action by painting contact surfaces with materials recommended by manufacturer for this purpose or by installing nonconductive spacers.
2. Where aluminum is in contact with concrete or masonry, protect against corrosion by painting contact surfaces with bituminous paint.

- C. Set continuous sill members and flashing in full sealant bed as specified in Section 079200 "Joint Sealants" to produce weathertight installation.

- D. Install aluminum-framed storefront system level, plumb, square, true to line, without distortion or impeding thermal movement, anchored securely in place to structural support, and in proper relation to wall flashing and other adjacent construction.

- E. Install aluminum-framed storefront system and components to drain condensation, water penetrating joints, and moisture migrating within aluminum-framed storefront system to the exterior.

- F. Where trim pieces are joined, accurately cut and fit to result in a tightly closed joint.

- G. Install glazing as specified in Section 088000 "Glazing."

3.3 ADJUSTING, CLEANING AND PROTECTION

- A. Clean aluminum surfaces immediately after installing aluminum-framed storefronts. Avoid damaging protective coatings and finishes. Remove excess sealants, glazing materials, dirt, and other substances.

- B. Clean glass immediately after installation. Comply with glass manufacturer's written recommendations for final cleaning and maintenance. Remove nonpermanent labels, and clean surfaces.

- C. Remove and replace glass that has been broken, chipped, cracked, abraded, or damaged during construction period.

END OF SECTION 084113

SECTION 087100 – DOOR HARDWARE**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Mechanical door hardware for the following:
 - a. Swinging doors.
 - b. Sliding doors.
 - 2. Cylinders for door hardware specified in other Sections.
 - 3. Electrified door hardware.
- B. Related Sections:
 - 1. Division 08 Section "Hollow Metal Doors and Frames".

1.3 REFERENCES

- A. Codes and References: Comply with the version year adopted by the Authority Having Jurisdiction.
 - 1. ANSI A117.1 - Accessible and Usable Buildings and Facilities.
 - 2. ICC/IBC - International Building Code.
 - 3. NFPA 70 - National Electrical Code.
 - 4. NFPA 80 - Fire Doors and Windows.
 - 5. NFPA 101 - Life Safety Code.
 - 6. NFPA 105 - Installation of Smoke Door Assemblies.
 - 7. State Building Codes, Local Amendments.
- B. Standards: All hardware specified herein shall comply with the following industry standards:
 - 1. ANSI/BHMA Certified Product Standards - A156 Series
 - 2. UL10C – Positive Pressure Fire Tests of Door Assemblies

1.4 ACTION SUBMITTALS

- A. See Section 013300 "Submittal Procedures" for requirements of Action Submittals.
- B. Submit Submittals via Owner's Project Management software platform.
- C. Product Data: Manufacturer's product data sheets including installation details, material descriptions, dimensions of individual components and profiles, operational descriptions and finishes.

- D. Shop Drawings: Door Hardware Schedule prepared by or under the supervision of supplier, detailing fabrication and assembly of door hardware, as well as procedures and diagrams. Coordinate the final Door Hardware Schedule with doors, frames, and related work to ensure proper size, thickness, hand, function, and finish of door hardware.
1. Format: Comply with scheduling sequence and vertical format in DHI's "Sequence and Format for the Hardware Schedule."
 2. Organization: Organize the Door Hardware Schedule into door hardware sets indicating complete designations of every item required for each door or opening. Organize door hardware sets in same order as in the Door Hardware Sets at the end of Part 3. Submittals that do not follow the same format and order as the Door Hardware Sets will be rejected and subject to resubmission.
 3. Content: Include the following information:
 - a. Type, style, function, size, label, hand, and finish of each door hardware item.
 - b. Manufacturer of each item.
 - c. Fastenings and other pertinent information.
 - d. Location of door hardware set, cross-referenced to Drawings, both on floor plans and in door and frame schedule.
 - e. Explanation of abbreviations, symbols, and codes contained in schedule.
 - f. Mounting locations for door hardware.
 - g. Door and frame sizes and materials.
 - h. Warranty information for each product.
 4. Submittal Sequence: Submit the final Door Hardware Schedule at earliest possible date, particularly where approval of the Door Hardware Schedule must precede fabrication of other work that is critical in the Project construction schedule. Include Product Data, Samples, Shop Drawings of other work affected by door hardware, and other information essential to the coordinated review of the Door Hardware Schedule.
- E. Samples: For verification of selection of exposed finishes.
- F. Keying Schedule: After a keying meeting with the owner has taken place prepare a separate keying schedule detailing final instructions. Submit the keying schedule in electronic format. Include keying system explanation, door numbers, key set symbols, hardware set numbers and special instructions. Owner must approve submitted keying schedule prior to the ordering of permanent cylinders/cores.

1.5 INFORMATIONAL SUBMITTALS

- A. Product Test Reports: Indicating compliance with cycle testing requirements, based on evaluation of comprehensive tests performed by manufacturer and witnessed by a qualified independent testing agency.

1.6 CLOSEOUT SUBMITTALS

- A. Operating and Maintenance Manuals: Provide manufacturers operating and maintenance manuals for each item comprising the complete door hardware installation in quantity as required in Division 01, Closeout Submittals.

1.7 QUALITY ASSURANCE

- A. Manufacturers Qualifications: Engage qualified manufacturers with a minimum 5 years of documented experience in producing hardware and equipment similar to that indicated for this Project and that have a proven record of successful in-service performance.
- B. Installer Qualifications: A minimum 3 years documented experience installing both standard and electrified door hardware similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.
- C. Door Hardware Supplier Qualifications: Experienced commercial door hardware distributors with a minimum 5 years documented experience supplying both mechanical and electromechanical hardware installations comparable in material, design, and extent to that indicated for this Project. Supplier recognized as a factory direct distributor by the manufacturers of the primary materials. Supplier to have on staff a certified Architectural Hardware Consultant (AHC) available during the course of the Work to consult with Contractor, Architect, and Owner concerning both standard and electromechanical door hardware and keying.
- D. Source Limitations: Obtain each type and variety of door hardware specified in this section from a single source unless otherwise indicated.
 - 1. Electrified modifications or enhancements made to a source manufacturer's product line by a secondary or third party source will not be accepted.
 - 2. Provide electromechanical door hardware from the same manufacturer as mechanical door hardware, unless otherwise indicated.
- E. Each unit to bear third party permanent label demonstrating compliance with the referenced standards.
- F. Keying Conference: Conduct conference to comply with requirements in Division 01 Section "Project Meetings." Keying conference to incorporate the following criteria into the final keying schedule document:
 - 1. Function of building, purpose of each area and degree of security required.
 - 2. Plans for existing and future key system expansion.
 - 3. Requirements for key control storage and software.
 - 4. Installation of permanent keys, cylinder cores and software.
 - 5. Address and requirements for delivery of keys.
- G. At completion of installation, provide written documentation that components were applied to manufacturer's instructions and recommendations and according to approved schedule.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Inventory door hardware on receipt and provide secure lock-up and shelving for door hardware delivered to Project site. Do not store electronic access control hardware, software or accessories at Project site without prior authorization.
- B. Tag each item or package separately with identification related to the final Door Hardware Schedule and include basic installation instructions with each item or package.
- C. Deliver, as applicable, permanent keys, cylinders, cores, access control credentials, software and related accessories directly to Owner via registered mail or overnight package service. Instructions for delivery to the Owner shall be established at the "Keying Conference".

1.9 COORDINATION

- A. Templates: Obtain and distribute to the parties involved templates for doors, frames, and other work specified to be factory prepared for installing standard and electrified hardware. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing hardware to comply with indicated requirements.
- B. Door Hardware and Electrical Connections: Coordinate the layout and installation of scheduled electrified door hardware and related access control equipment with required connections to source power junction boxes, low voltage power supplies, detection and monitoring hardware, and fire and detection alarm systems.
- C. Door and Frame Preparation: Doors and corresponding frames are to be prepared, reinforced and pre-wired (if applicable) to receive the installation of the specified electrified, monitoring, signaling and access control system hardware without additional in-field modifications.

1.10 WARRANTY

- A. General Warranty: Reference Division 01, General Requirements. Special warranties specified in this Article shall not deprive Owner of other rights Owner may have under other provisions of the Contract Documents and shall be in addition to, and run concurrent with, other warranties made by Contractor under requirements of the Contract Documents.
- B. Warranty Period: Written warranty, executed by manufacturer(s), agreeing to repair or replace components of standard and electrified door hardware that fails in materials or workmanship within specified warranty period after final acceptance by the Owner. Failures include, but are not limited to, the following:
 - 1. Structural failures including excessive deflection, cracking, or breakage.
 - 2. Faulty operation of the hardware.
 - 3. Deterioration of metals, metal finishes, and other materials beyond normal weathering.
 - 4. Electrical component defects and failures within the systems operation.
- C. Standard Warranty Period: One year from date of Substantial Completion, unless otherwise indicated.
- D. Special Warranty Periods:
 - 1. Ten years for mechanical heavy duty cylindrical (bored) locks and latches.
 - 2. Ten years for exit hardware.
 - 3. Twenty-five years for manual surface door closer bodies.
 - 4. One years for electromechanical door hardware.

1.11 MAINTENANCE SERVICE

- A. Maintenance Tools and Instructions: Furnish a complete set of specialized tools and maintenance instructions as needed for Owner's continued adjustment, maintenance, and removal and replacement of door hardware.

PART 2 - PRODUCTS

2.1 SUPPLIER

- A. Supplier Contact: See **LiDL National Account List**.

2.2 SCHEDULED DOOR HARDWARE

- A. General: Provide door hardware for each door to comply with requirements in Door Hardware Sets and each referenced section that products are to be supplied under.
- B. Designations: Requirements for quantity, item, size, finish or color, grade, function, and other distinctive qualities of each type of door hardware are indicated in the Door Hardware Sets at the end of Part 3. Products are identified by using door hardware designations.
- C. Named Manufacturer's Products: Product designation and manufacturer are listed for each door hardware type required for the purpose of establishing requirements. Manufacturers' names are abbreviated in the Door Hardware Schedule.

2.3 HANGING DEVICES

- A. Hinges: ANSI/BHMA A156.1 certified butt hinges with number of hinge knuckles as specified in the Door Hardware Sets.
 - 1. Quantity: Provide the following hinge quantity, unless otherwise indicated:
 - a. Two Hinges: For doors with heights up to 60 inches.
 - b. Three Hinges: For doors with heights 61 to 90 inches.
 - c. Four Hinges: For doors with heights 91 to 120 inches.
 - d. For doors with heights more than 120 inches, provide 4 hinges, plus 1 hinge for every 30 inches of door height greater than 120 inches.
 - 2. Hinge Size: Provide the following, unless otherwise indicated, with hinge widths sized for door thickness and clearances required:
 - a. Widths up to 3'-0": 4-1/2" heavy weight
 - b. Widths from 3'-1" to 4'-0": 4-1/2" heavy weight
 - 3. Hinge Weight and Base Material: Unless otherwise indicated, provide the following:
 - a. Exterior Doors: Heavy weight, non-ferrous, ball bearing or oil impregnated bearing hinges unless Hardware Sets indicate standard weight.
 - b. Interior Doors: Heavy weight, steel, ball bearing or oil impregnated bearing hinges unless Hardware Sets indicate heavy weight.
 - 4. Hinge Options: Comply with the following where indicated in the Hardware Sets or on Drawings:
 - a. Non-removable Pins: Provide set screw in hinge barrel that, when tightened into a groove in hinge pin, prevents removal of pin while door is closed; for the all out-swinging lockable doors.
- B. Continuous Geared Hinges: ANSI/BHMA A156.26 Grade 1-600 certified continuous geared hinge. with minimum 0.120-inch thick extruded 6060 T6 aluminum alloy hinge leaves and a minimum overall width of 4 inches. Hinges are non-handed, reversible and fabricated to template screw locations. Factory trim hinges to suit door height and prepare for electrical cut-outs.

2.4 DOOR OPERATING TRIM

- A. Flush Bolts and Surface Bolts: ANSI/BHMA A156.3 and A156.16, Grade 1, certified.

1. Manual flush bolts to be furnished with top rod of sufficient length to allow bolt location approximately six feet from the floor.
 2. Furnish dust proof strikes for bottom bolts.
 3. Surface bolts to be minimum 8" in length and U.L. listed for labeled fire doors and U.L. listed for windstorm components where applicable.
 4. Provide related accessories (mounting brackets, strikes, coordinators, etc.) as required for appropriate installation and operation.
- B. Door Push Plates and Pulls: ANSI/BHMA A156.6 certified door pushes and pulls of type and design specified in the Hardware Sets. Coordinate and provide proper width and height as required where conflicting hardware dictates.
1. Push/Pull Plates: Minimum .050 inch thick, size as indicated in hardware sets, with beveled edges, secured with exposed screws unless otherwise indicated.
 2. Door Pull and Push Bar Design: Size, shape, and material as indicated in the hardware sets. Minimum clearance of 2 1/2-inches from face of door unless otherwise indicated.
 3. Offset Pull Design: Size, shape, and material as indicated in the hardware sets. Minimum clearance of 2 1/2-inches from face of door and offset of 90 degrees unless otherwise indicated.
 4. Fasteners: Provide manufacturer's designated fastener type as indicated in Hardware Sets.

2.5 CYLINDERS AND KEYING

- A. General: Cylinder manufacturer to have minimum (10) years experience designing secured master key systems and have on record a published security keying system policy.
- B. Cylinders: Original manufacturer cylinders complying with the following:
1. Mortise Type: Threaded cylinders with rings and cams to suit hardware application.
 2. Rim Type: Cylinders with back plate, flat-type vertical or horizontal tailpiece, and raised trim ring.
 3. Bored-Lock Type: Cylinders with tailpieces to suit locks.
 4. Mortise and rim cylinder collars to be solid and recessed to allow the cylinder face to be flush and be free spinning with matching finishes.
- C. Permanent Cores: Manufacturer's standard; finish face to match lockset; complying with the following:
1. Interchangeable Cores: Core insert, removable by use of a special key; usable with other manufacturers' cylinders.
- D. Keying System: Each type of lock and cylinders to be factory keyed.
1. Conduct specified "Keying Conference" to define and document keying system instructions and requirements.
 2. Furnish factory cut, nickel-silver large bow permanently inscribed with a visual key control number as directed by Owner.
 3. New System: Key locks to a new key system as directed by the Owner.
- E. Key Quantity: Provide the following minimum number of keys:
1. Change Keys per Cylinder: Two (2)
 2. Master Keys (per Master Key Level/Group): Five (5).

- a. One Master Key at Security Monitoring Vendor
 - b. One Master Key at RDC Facilities
 - c. One Master Key at Lockbox
 - d. Five Master Keys at Store Management
 3. Construction Keys (where required): Ten (10).
 4. Construction Control Keys (where required): Two (2).
 5. Permanent Control Keys (where required): Two (2).
 - F. Construction Keying: Provide temporary keyed construction cores.
 - G. Key Registration List (Bitting List):
 1. Provide keying transcript list to Owner's representative in the proper format for importing into key control software.
 2. Provide transcript list in writing or electronic file as directed by the Owner.
- 2.6 MECHANICAL LOCKS AND LATCHING DEVICES
- A. Cylindrical Locksets, Grade 1 (Heavy Duty): ANSI/BHMA A156.2, Series 4000, Grade 1 certified.
 1. Furnish with solid cast levers, standard 2 3/4" backset, and 1/2" (3/4" at rated paired openings) throw brass or stainless steel latchbolt.
 2. Locks are to be non-handed and fully field reversible.
- 2.7 AUXILIARY LOCKS
- A. Deadbolts: ANSI/BHMA A156, Grade 1. Provide deadbolts with standard 2-3/4 inches (70 mm) backset. Provide 2-3/8 inches (60 mm) where noted or if door or frame detail requires. Provide deadbolt with full 1 inch (25 mm) throw, constructed of steel alloy.
 - B. Deadbolts to be products of the same source manufacturer and keyway as other specified locksets.
- 2.8 LOCK AND LATCH STRIKES
- A. Strikes: Provide manufacturer's standard strike with strike box for each latch or lock bolt, with curved lip extended to protect frame, finished to match door hardware set, unless otherwise indicated, and as follows:
 1. Flat-Lip Strikes: For locks with three-piece antifriction latchbolts, as recommended by manufacturer.
 2. Aluminum-Frame Strike Box: Provide manufacturer's special strike box fabricated for aluminum framing.
 3. Double-lipped strikes: For locks at double acting doors. Furnish with retractable stop for rescue hardware applications.
 - B. Standards: Comply with the following:
 1. Strikes for Mortise Locks and Latches: BHMA A156.13.
 2. Strikes for Bored Locks and Latches: BHMA A156.2.
 3. Strikes for Auxiliary Deadlocks: BHMA A156.5.
 4. Dustproof Strikes: BHMA A156.16.

2.9 CONVENTIONAL EXIT DEVICES

- A. General Requirements: All exit devices specified herein shall meet or exceed the following criteria:
1. Provide exit devices tested to ANSI/BHMA A156.3 Grade 1 and UL listed for Panic Exit or Fire Exit Hardware.
 2. Cylinders: Refer to "KEYING" article, herein.
 3. Provide touchpad type exit devices, fabricated of brass, bronze, stainless steel, or aluminum, plated to standard architectural finishes to match balance of door hardware.
 4. Touchpad must extend a minimum of one half of door width. No plastic inserts are allowed in touchpads.
 5. Provide exit devices with deadlatching feature for security and for future addition of alarm kits and/or other electrified requirements.
 6. Provide flush end caps for exit devices.
 7. Provide exit devices with manufacturer's approved strikes.
 8. Provide exit devices cut to door width and height. Install exit devices at height recommended by exit device manufacturer, allowable by governing building codes, and approved by Architect.
 9. Mount mechanism case flush on face of doors or provide spacers to fill gaps behind devices. Where glass trim or molding projects off face of door, provide glass bead kits.
 10. Provide cylinder or hex-key dogging as specified at non fire-rated openings.
 11. Provide factory drilled weep holes for exit devices used in full exterior application, highly corrosive areas, and where noted in hardware sets.
 12. Provide electrified options as scheduled.
 13. Provide exit devices with optional trim designs to match other lever and pull designs used on the project.
- B. Conventional Push Rail Exit Devices (Commercial Duty): ANSI/BHMA A156.3, Grade 1 certified panic and fire exit hardware devices furnished in the functions specified in the Hardware Sets. Fabricate latchbolts from cast stainless steel, Pullman type, incorporating a deadlocking feature.

2.10 DOOR CLOSERS

- A. All door closers specified herein shall meet or exceed the following criteria:
1. General: Door closers to be from one manufacturer, matching in design and style, with the same type door preparations and templates regardless of application or spring size. Closers to be non-handed with full sized covers including installation and adjusting information on inside of cover.
 2. Standards: Closers to comply with UL-10C for Positive Pressure Fire Test and be U.L. listed for use of fire rated doors.
 3. Cycle Testing: Provide closers which have surpassed 15 million cycles in a test witnessed and verified by UL.
 4. Size of Units: Comply with manufacturer's written recommendations for sizing of door closers depending on size of door, exposure to weather, and anticipated frequency of use. Where closers are indicated for doors required to be accessible to the physically handicapped, provide units complying with ANSI ICC/A117.1.
 5. Closer Arms: Provide heavy duty, forged steel closer arms unless otherwise indicated in Hardware Sets.
 6. Closers shall not be installed on exterior or corridor side of doors; where possible install closers on door for optimum aesthetics.
 7. Closer Accessories: Provide door closer accessories including custom templates, special mounting brackets, spacers and drop plates, and through-bolt and security type fasteners as required for proper installation.

- B. Door Closers, Surface Mounted (Commercial Duty): ANSI/BHMA 156.4, Grade 1 certified surface mounted, institutional grade door closers with complete spring power adjustment, sizes 1 thru 6; and fully operational adjustable according to door size, frequency of use, and opening force. Closers to be rack and pinion type, one piece cast iron or aluminum alloy body construction, with adjustable backcheck, closing sweep, and latch speed control valves. Provide non-handed units standard.

2.11 ARCHITECTURAL TRIM

A. Door Protective Trim

1. General: Door protective trim units to be of type and design as specified below or in the Hardware Sets.
2. Size: Fabricate protection plates (kick, armor, or mop) not more than 2" less than door width (LDW) on stop side of single doors and 1" LDW on stop side of pairs of doors, and not more than 1" less than door width on pull side. Coordinate and provide proper width and height as required where conflicting hardware dictates. Height to be as specified in the Hardware Sets.
3. Protection Plates: ANSI/BHMA A156.6 certified protection plates (kick, armor, or mop), fabricated from the following:
 - a. Stainless Steel: 300 grade, 050-inch thick.
4. Options and fasteners: Provide manufacturer's designated fastener type as specified in the Hardware Sets. Provide countersunk screw holes.

2.12 DOOR STOPS AND HOLDERS

- A. General: Door stops and holders to be of type and design as specified below or in the Hardware Sets.
- B. Door Stops and Bumpers: ANSI/BHMA A156.16, Grade 1 certified door stops and wall bumpers. Provide wall bumpers, either convex or concave types with anchorage as indicated, unless floor or other types of door stops are specified in Hardware Sets. Do not mount floor stops where they will impede traffic. Where floor or wall bumpers are not appropriate, provide overhead type stops and holders.

2.13 ARCHITECTURAL SEALS

- A. General: Thresholds, weatherstripping, and gasket seals to be of type and design as specified below or in the Hardware Sets. Provide continuous weatherstrip gasketing on exterior doors and provide smoke, light, or sound gasketing on interior doors where indicated. At exterior applications provide non-corrosive fasteners and elsewhere where indicated.
- B. Smoke Labeled Gasketing: Assemblies complying with NFPA 105 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for smoke control ratings indicated, based on testing according to UL 1784.
 1. Provide smoke labeled perimeter gasketing at all smoke labeled openings.
- C. Fire Labeled Gasketing: Assemblies complying with NFPA 80 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire ratings indicated, based on testing according to UL-10C.

1. Provide intumescent seals as indicated to meet UL10C Standard for Positive Pressure Fire Tests of Door Assemblies, and UBC 7-2, Fire Tests of Door Assemblies.

- D. Replaceable Seal Strips: Provide only those units where resilient or flexible seal strips are easily replaceable and readily available from stocks maintained by manufacturer.

2.14 ELECTRONIC ACCESSORIES

- A. Door Position Switches: Door position magnetic reed contact switches specifically designed for use in commercial door applications. On recessed models the contact and magnetic housing snap-lock into a 1" diameter hole. Surface mounted models include wide gap distance design complete with armored flex cabling. Provide SPDT, N/O switches with optional Rare Earth Magnet installation on steel doors with flush top channels.

2.15 FABRICATION

- A. Fasteners: Provide door hardware manufactured to comply with published templates generally prepared for machine, wood, and sheet metal screws. Provide screws according to manufacturers recognized installation standards for application intended.

2.16 FINISHES

- A. Standard: Designations used in the Hardware Sets and elsewhere indicate hardware finishes complying with ANSI/BHMA A156.18, including coordination with traditional U.S. finishes indicated by certain manufacturers for their products.
- B. Provide quality of finish, including thickness of plating or coating (if any), composition, hardness, and other qualities complying with manufacturer's standards, but in no case less than specified by referenced standards for the applicable units of hardware.
- C. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine scheduled openings, with Installer present, for compliance with requirements for installation tolerances, labeled fire door assembly construction, wall and floor construction, and other conditions affecting performance.
- B. Notify architect of any discrepancies or conflicts between the door schedule, door types, drawings and scheduled hardware. Proceed only after such discrepancies or conflicts have been resolved in writing.

3.2 PREPARATION

- A. Hollow Metal Doors and Frames: Comply with ANSI/DHI A115 series.

3.3 INSTALLATION

- A. Install each item of mechanical and electromechanical hardware and access control equipment to comply with manufacturer's written instructions and according to specifications.

1. Installers are to be trained and certified by the manufacturer on the proper installation and adjustment of fire, life safety, and security products including: hanging devices; locking devices; closing devices; and seals.
 - B. Mounting Heights: Mount door hardware units at heights indicated in following applicable publications, unless specifically indicated or required to comply with governing regulations:
 1. Standard Steel Doors and Frames: DHI's "Recommended Locations for Architectural Hardware for Standard Steel Doors and Frames."
 2. Where indicated to comply with accessibility requirements, comply with ANSI A117.1 "Accessibility Guidelines for Buildings and Facilities."
 3. Provide blocking in drywall partitions where wall stops or other wall mounted hardware is located.
 - C. Retrofitting: Install door hardware to comply with manufacturer's published templates and written instructions. Where cutting and fitting are required to install door hardware onto or into surfaces that are later to be painted or finished in another way, coordinate removal, storage, and reinstallation of surface protective trim units with finishing work specified in Division 9 Sections. Do not install surface-mounted items until finishes have been completed on substrates involved.
 - D. Thresholds: Set thresholds for exterior and acoustical doors in full bed of sealant complying with requirements specified in Division 7 Section "Joint Sealants."
 - E. Storage: Provide a secure lock up for hardware delivered to the project but not yet installed. Control the handling and installation of hardware items so that the completion of the work will not be delayed by hardware losses before and after installation.
- 3.4 ADJUSTING
- A. Initial Adjustment: Adjust and check each operating item of door hardware and each door to ensure proper operation or function of every unit. Replace units that cannot be adjusted to operate as intended. Adjust door control devices to compensate for final operation of heating and ventilating equipment and to comply with referenced accessibility requirements.
- 3.5 CLEANING AND PROTECTION
- A. Protect all hardware stored on construction site in a covered and dry place. Protect exposed hardware installed on doors during the construction phase. Install any and all hardware at the latest possible time frame.
 - B. Clean adjacent surfaces soiled by door hardware installation.
 - C. Clean operating items as necessary to restore proper finish. Provide final protection and maintain conditions that ensure door hardware is without damage or deterioration at time of owner occupancy.
- 3.6 DOOR HARDWARE SETS
- A. The hardware sets represent the design intent and direction of the owner and architect. They are a guideline only and should not be considered a detailed hardware schedule. Discrepancies, conflicting hardware and missing items should be brought to the attention of the architect with corrections made prior to the bidding process. Omitted items not included in a

hardware set should be scheduled with the appropriate additional hardware required for proper application and functionality.

B. Manufacturer's Abbreviations:

1. IVE – Ives
2. FAL – Falcon
3. LCN – LCN
4. SCE – Schlage Electronics
5. SCH – Schlage
6. VON – Von Duprin
7. ZER - Zero

3.7 HARDWARE SCHEDULE

HARDWARE GROUP NO. 01

For use on Door #(s):

100 Vestibule 100.1 Vestibule 100.2 Vestibule

ALL AUTOMATIC SLIDING DOOR HARDWARE BY THE DOOR MANUFACTURER FOR A COMPLETE AND PROPER INSTALLATION.

HARDWARE GROUP NO. 02

For use on Door #(s):

116 Cooler 116.1 Cooler 117 Freezer 118 Freezer 118.1 Freezer

ALL REFRIGERATION DOOR HARDWARE BY THE DOOR MANUFACTURER FOR A COMPLETE AND PROPER INSTALLATION.

HARDWARE GROUP NO. 03

For use on Door #(s):

103 Restroom 104 Restroom 120 Changing Room

Provide each SGL door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5	626	IVE
1	EA	SURFACE CLOSER	1450 RW/PA X SRT	689	LCN
1	EA	KICK PLATE	8400 10" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407-CCV	630	IVE
3	EA	SILENCERS	SR64	GRY	IVE
1	EA	COAT HOOK	574	626	IVE
1	EA	CYLINDRICAL DEAD LOCK W/ INDICATOR	B571	626	SCH
1	EA	PUSH PLATE	8200 X 4" X 16"	630	IVE
1	EA	PULL PLATE	8302 10" X 4" X 16"	630	IVE

HARDWARE GROUP NO. 04

For use on Door #(s):

107 Cash Office

Provide each SGL door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5	626	IVE
1	EA	ELECTRIC POWER TRANSFER	EPT10	689	VON
1	EA	ELECTRIC STOREROOM LOCK	ND80BDEU RHO RX 12V/24V DC	626	SCH
1	EA	DOOR CONTACT	679-05	BLK	SCE
1	EA	SURFACE CLOSER	1450 X RW/PA X SRT	689	LCN
1	EA	KICK PLATE	8400 10" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407 CCV	630	IVE
3	EA	SILENCERS	SR64	GRY	IVE
2	EA	DOOR VIEWERS	U698	626	IVE
1	EA	SFIC CORE	80-036 X EVEREST 29 R	626	SCH

DOOR VIEWERS AT 43" AND 60" AFF.

OUTSIDE CARD READER (WALL MOUNTED AND FURNISHED BY SECURITY) SIGNALS POWER SUPPLY TO RELEASE ELECTRIFIED LOCK ALLOWING ENTRY. FREE EGRESS AT ALL TIMES FROM INSIDE. REX INSIDE LOCK SHUNTS DOOR CONTACT WHEN EXITING. POWER SUPPLY PROVIDED BY SECURITY.

HARDWARE GROUP NO. 06

For use on Door #(s):

108 Manager's Office

Provide each SGL door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5	626	IVE
1	EA	ELECTRIC POWER TRANSFER	EPT10	689	VON
1	EA	ELECTRIC STOREROOM LOCK	ND80BDEU RHO RX 12V/24V DC	626	SCH
1	EA	CYLINDRICAL DEAD LOCK W/ INDICATOR	B571	626	SCH
1	EA	DOOR CONTACT	679-05	BLK	SCE
1	EA	SURFACE CLOSER	1450 RW/PA X SRT	689	LCN
1	EA	KICK PLATE	8400 10" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407 CCV	630	IVE
3	EA	SILENCERS	SR64	GRY	IVE
1	EA	SFIC CORE	80-036 X EVEREST 29 R	626	SCH

OUTSIDE CARD READER (WALL MOUNTED AND FURNISHED BY SECURITY) SIGNALS POWER SUPPLY TO RELEASE ELECTRIFIED LOCK ALLOWING ENTRY. FREE EGRESS AT ALL TIMES FROM INSIDE. REX INSIDE LOCK SHUNTS DOOR CONTACT WHEN EXITING. POWER SUPPLY PROVIDED BY SECURITY. LITE BY OTHERS.

HARDWARE GROUP NO. 07

For use on Door #(s):

110 IT Room

Provide each SGL door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5	626	IVE
1	EA	ELECTRIC POWER TRANSFER	EPT10	689	VON
1	EA	ELECTRIC STOREROOM LOCK	ND80BDEU RHO RX 12V/24V DC	626	SCH
1	EA	DOOR CONTACT	679-05	BLK	SCE
1	EA	SURFACE CLOSER	1450 RW/PA X SRT	689	LCN
1	EA	KICK PLATE	8400 10" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407 CCV	630	IVE
2	EA	DOOR VIEWERS	U698	626	IVE
1	EA	CYLINDER CORE	80-036 X EVEREST 29 R	626	SCH
1	EA	GASKETING	188SBK PSA	BK	ZER
1	EA	DOOR SWEEP	39A	A	ZER

DOOR VIEWERS AT 43" AND 60" AFF

OUTSIDE CARD READER (WALL MOUNTED AND FURNISHED BY SECURITY) SIGNALS POWER SUPPLY TO RELEASE ELECTRIFIED LOCK ALLOWING ENTRY. FREE EGRESS AT ALL TIMES FROM INSIDE. REX INSIDE LOCK SHUNTS DOOR CONTACT WHEN EXITING. POWER SUPPLY PROVIDED BY SECURITY.

HARDWARE GROUP NO. 10

For use on Door #(s):

109 Breakroom to Backstock

109.1 Hallway

Provide each SGL door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP SEC	626	IVE
1	EA	ELECTRIC POWER TRANSFER	EPT10	689	VON
1	EA	ELECTRIC STOREROOM LOCK	ND80BDEU RHO RX 12V/24V DC	626	SCH
1	EA	DOOR CONTACT	679-05	BLK	SCE
1	EA	SURFACE CLOSER	1450 RW/PA X SRT	689	LCN
1	EA	KICK PLATE	8400 10" X 2" LDW B-CS	630	IVE
1	EA	WALL STOP	WS406/407 CCV	630	IVE
1	EA	GASKETING	188SBK PSA	BK	ZER
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	SFIC CORE	80-036 X EVEREST 29 R	626	SCH

OUTSIDE CARD READER (WALL MOUNTED AND FURNISHED BY SECURITY) SIGNALS POWER SUPPLY TO RELEASE FAIL SECURE ELECTRIFIED LOCK ALLOWING ENTRY. INSIDE, FREE EGRESS AT ALL TIMES BY OPERATING INSIDE LEVER, WHICH SHUNTS DOOR CONTACT. POWER SUPPLY PROVIDED BY SECURITY. LITE BY OTHERS.

HARDWARE GROUP NO. 12

For use on Door #(s):

112 Staging 112.1 Backstock 115 Loading Dock 123 Fire Pump

Provide each SGL door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP	630	IVE
1	EA	ELECTRIC POWER TRANSFER	EPT10	689	VON
1	EA	ELEC RIM EXIT DEVICE	RX-FSE-25-R-L-DANE 24 VDC	626	FAL
1	EA	SURFACE CLOSER	1450 X SCUSH X SRT	689	LCN
1	EA	KICK PLATE	8400 10" X 2" LDW B-CS	630	IVE
1	EA	DOOR CONTACT	679-05	BLK	SCE
1	EA	DOOR SWEEP	8197AA	AA	ZER
1	EA	GASKETING	8878AA-S	AA	ZER
1	EA	RAIN DRIP	142AA	AA	ZER
2	EA	DOOR VIEWERS	U698	626	IVE
1	EA	THRESHOLD	566A-223	A	ZER
1	EA	MORTISE CYLINDER	C987 5622-IC	626	FAL
1	EA	SFIC CORE	80-036 X EVEREST 29 R	626	SCH

VIEWERS AT 43" AND 60" AFF

OUTSIDE CARD READER (WALL MOUNTED AND FURNISHED BY SECURITY) SIGNALS POWER SUPPLY TO RELEASE FAIL SECURE ELECTRIC TRIM ALLOWING ENTRY BY LEVER. KEY OVERRIDE. INSIDE, FREE EGRESS AT ALL TIMES BY EXIT DEVICE. POWER SUPPLY PROVIDED BY SECURITY.

HARDWARE GROUP NO. 14

For use on Door #(s):

101.2 Sales to Exterior

Provide each SGL door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER	FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP	630	IVE
1	EA	RIM EXIT DEVICE	25-R-EO	626	FAL
1	EA	SURFACE CLOSER	1450 SCUSH X SRT	689	LCN
1	EA	KICK PLATE	8400 10" X 2" LDW B-CS	630	IVE
1	EA	DOOR CONTACT	679-05	BLK	SCE
1	EA	GASKETING	8878AA-S	AA	ZER
1	EA	RAIN DRIP	142AA	AA	ZER
1	EA	DOOR SWEEP	8197AA	AA	ZER
1	EA	THRESHOLD	566A-223	A	ZER
2	EA	DOOR VIEWERS	U698	626	IVE

VIEWERS AT 43" AND 60" AFF

EXTERIOR, NO INGRESS AT ANY TIME. INSIDE, FREE EGRESS AT ALL TIMES BY EXIT DEVICE. POWER SUPPLY AND REX PROVIDED BY SECURITY.

HARDWARE GROUP NO. 15

For use on Door #(s):

114 Backstock

ALL HARDWARE BY CHAIN-LINK GATE SUPPLIER/INSTALLER FOR A COMPLETE AND PROPER
INSTALLATION. SEE SECTION 323113.

END OF SECTION 087100

SECTION 088000 - GLAZING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes:
 - 1. Glass for curtain wall, doors and windows.
 - 2. One-way mirror glass for interior door lite.
 - 3. Glazing sealants and accessories.

1.3 COORDINATION

- A. Coordinate glazing channel dimensions to provide necessary bite on glass, minimum edge and face clearances, and adequate sealant thicknesses, with reasonable tolerances.

1.4 ACTION SUBMITTALS

- A. See Section 013300 "Submittal Procedures" for requirements of Action Submittals.
- B. Submit Submittals via Owner's Project Management software platform.
- C. Product Data: For each type of product.
- D. Glass Samples: For each type of glass product other than clear monolithic vision glass; 12 inches (300 mm) square.
- E. Shop Drawings: Glazing Schedule of glass types and thicknesses for each size opening and location. Use same designations indicated on Drawings.

1.5 INFORMATIONAL SUBMITTALS

- A. Installer certificates.
- B. Product test reports.

1.6 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.7 QUALITY ASSURANCE

- A. Sealant Testing Agency Qualifications: An independent testing agency qualified according to ASTM C 1021 to conduct the testing indicated.

1.8 PRECONSTRUCTION TESTING

- A. Preconstruction Adhesion and Compatibility Testing: Test each glass product, tape sealant, gasket, glazing accessory, and glass-framing member for adhesion to and compatibility with elastomeric glazing sealants.
 - 1. Testing is not required if data are submitted based on previous testing of current sealant products and glazing materials matching those submitted.

1.9 WARRANTY

- A. Manufacturer's Special Warranty for Insulating Glass: Manufacturer agrees to replace insulating-glass units that deteriorate within specified warranty period. Deterioration of insulating glass is defined as failure of hermetic seal under normal use that is not attributed to glass breakage or to maintaining and cleaning insulating glass contrary to manufacturer's written instructions. Evidence of failure is the obstruction of vision by dust, moisture, or film on interior surfaces of glass.
 - 1. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Structural Performance: Glazing shall withstand the following design loads within limits and under conditions indicated determined according to the International Building Code and ASTM E 1300.
 - 1. Design Wind Pressures: As indicated on Drawings.
 - 2. Differential Shading: Design glass to resist thermal stresses induced by differential shading within individual glass lite.
- B. Safety Glazing: Where safety glazing is indicated, provide glazing that complies with 16 CFR 1201, Category II.
- C. Thermal and Optical Performance Properties: Provide glass with performance properties specified, as indicated in manufacturer's published test data, based on procedures indicated below:
 - 1. U-Factors: Center-of-glazing values, according to NFRC 100 and based on LBL's WINDOW 5.2 computer program, expressed as Btu/sq. ft. x h x deg F (W/sq. m x K).
 - 2. Solar Heat-Gain Coefficient and Visible Transmittance: Center-of-glazing values, according to NFRC 200 and based on LBL's WINDOW 5.2 computer program.
 - 3. Visible Reflectance: Center-of-glazing values, according to NFRC 300.

2.2 GLASS PRODUCTS, GENERAL

- A. Glazing Publications: Comply with published recommendations of glass product manufacturers and organizations below unless more stringent requirements are indicated. See these publications for glazing terms not otherwise defined in this Section or in referenced standards.
 - 1. GANA Publications: "Glazing Manual."
 - 2. IGMA Publication for Insulating Glass: SIGMA TM-3000, "North American Glazing Guidelines for Sealed Insulating Glass Units for Commercial and Residential Use."

- B. Safety Glazing Labeling: Where safety glazing is indicated, permanently mark glazing with certification label of the SGCC or another certification agency acceptable to authorities having jurisdiction. Label shall indicate manufacturer's name, type of glass, thickness, and safety glazing standard with which glass complies.
- C. Insulating-Glass Certification Program: Permanently marked either on spacers or on at least one component lite of units with appropriate certification label of IGCC.
- D. Thickness: Where glass thickness is indicated, it is a minimum. Provide glass that complies with performance requirements and is not less than the thickness indicated.
- E. Strength: Where annealed float glass is indicated, provide annealed float glass, heat-strengthened float glass, or fully tempered float glass as needed to comply with "Performance Requirements" Article. Where heat-strengthened float glass is indicated, provide heat-strengthened float glass or fully tempered float glass as needed to comply with "Performance Requirements" Article. Where fully tempered float glass is indicated, provide fully tempered float glass.

2.3 GLASS PRODUCTS

- A. Clear Annealed Float Glass: ASTM C 1036, Type I, Class 1 (clear), Quality-Q3.
- B. Fully Tempered Float Glass: ASTM C 1048, Kind FT (fully tempered), Condition A (uncoated) unless otherwise indicated, Type I, Class 1 (clear) or Class 2 (tinted) as indicated, Quality-Q3.
- C. Heat-Strengthened Float Glass: ASTM C1048, Kind HS (heat strengthened), Type I, Condition A (uncoated) unless otherwise indicated, Type I, Class 1 (clear) or Class 2 (tinted) as indicated, Quality-Q3.

2.4 LAMINATED GLASS

- A. Laminated Glass: ASTM C1172. Use materials that have a proven record of no tendency to bubble, discolor, or lose physical and mechanical properties after fabrication and installation.
 - 1. Construction: Laminate glass with polyvinyl butyral interlayer to comply with interlayer manufacturer's written instructions.
 - 2. Interlayer Thickness: Provide thickness not less than that indicated and as needed to comply with requirements.

2.5 INSULATING GLASS

- A. Insulating-Glass Units: Factory-assembled units consisting of sealed lites of glass separated by a dehydrated interspace, qualified according to ASTM E 2190.
 - 1. Sealing System: Dual seals.
 - 2. Perimeter Spacer: Polypropylene-covered stainless steel in manufacturer's standard Dark Grey color.
 - a. Manufacturer: Technoform Glass Insulation, Twinsburg, OH; Technoform TGI Spacer.

2.6 ONE-WAY MIRROR GLASS

- A. One-Way Mirror Glass: Transparent, clear, tempered, conforming to ASTM C 1036. Glass shall be coated on one face with a hard-adherent film of chromium or other approved coating of proven equivalent durability. Glass shall transmit not less than 3 percent nor more than 11 percent of total incident light in visible region and shall reflect from front surface of coating not less than 45 percent of total incident light in visible region.

1. Products:

- a. Saint-Gobain Glass; SGG Mirastar.
- b. Pilkington; Mirropane TM (Transparent Mirror).

2.7 GLAZING SEALANTS

A. General:

- 1. Compatibility: Compatible with one another and with other materials they contact, including glass products, seals of insulating-glass units, and glazing channel substrates, under conditions of service and application, as demonstrated by sealant manufacturer based on testing and field experience.
- 2. Suitability: Comply with sealant and glass manufacturers' written instructions for selecting glazing sealants suitable for applications indicated and for conditions existing at time of installation.
- 3. Colors of Exposed Glazing Sealants: Dark gray as selected by Architect from manufacturer's full range.

- B. Glazing Sealant: Neutral-curing silicone glazing sealant complying with ASTM C 920, Type S, Grade NS, Class 100/50, Use NT.

- C. VOC Content: All glazing sealants applied inside the weatherproofing system of the building shall have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

2.8 GLAZING TAPES

- A. Back-Bedding Mastic Glazing Tapes: Preformed, butyl-based, 100 percent solids elastomeric tape; nonstaining and nonmigrating in contact with nonporous surfaces; with or without spacer rod as recommended in writing by tape and glass manufacturers for application indicated; and complying with ASTM C 1281 and AAMA 800 for products indicated below:

- 1. AAMA 806.3 tape, for glazing applications in which tape is subject to continuous pressure.
- 2. AAMA 807.3 tape, for glazing applications in which tape is not subject to continuous pressure.

- B. Expanded Cellular Glazing Tapes: Closed-cell, PVC foam tapes; factory coated with adhesive on both surfaces; and complying with AAMA 800 for the following types:

- 1. AAMA 810.1, Type 1, for glazing applications in which tape acts as the primary sealant.
- 2. AAMA 810.1, Type 2, for glazing applications in which tape is used in combination with a full bead of liquid sealant.

2.9 MISCELLANEOUS GLAZING MATERIALS

- A. Cleaners, Primers, and Sealers: Types recommended by sealant or gasket manufacturer.

- B. Setting Blocks: Elastomeric material with a Shore, Type A durometer hardness of 85, plus or minus 5.
- C. Spacers: Elastomeric blocks or continuous extrusions of hardness required by glass manufacturer to maintain glass lites in place for installation indicated.
- D. Edge Blocks: Elastomeric material of hardness needed to limit glass lateral movement (side walking).
- E. Cylindrical Glazing Sealant Backing: ASTM C 1330, Type O (open-cell material), of size and density to control glazing sealant depth and otherwise produce optimum glazing sealant performance.

PART 3 - EXECUTION

3.1 GLAZING, GENERAL

- A. Comply with combined written instructions of manufacturers of glass, sealants, gaskets, and other glazing materials, unless more stringent requirements are indicated, including those in referenced glazing publications.
- B. Protect glass edges from damage during handling and installation. Remove damaged glass from Project site and legally dispose of off Project site. Damaged glass includes glass with edge damage or other imperfections that, when installed, could weaken glass, impair performance, or impair appearance.
- C. Apply primers to joint surfaces where required for adhesion of sealants, as determined by preconstruction testing.
- D. Install setting blocks in sill rabbets, sized and located to comply with referenced glazing publications, unless otherwise required by glass manufacturer. Set blocks in thin course of compatible sealant suitable for heel bead.
- E. Do not exceed edge pressures stipulated by glass manufacturers for installing glass lites.
- F. Provide spacers for glass lites where length plus width is larger than 50 inches (1270 mm).
- G. Provide edge blocking where indicated or needed to prevent glass lites from moving sideways in glazing channel, as recommended in writing by glass manufacturer and according to requirements in referenced glazing publications.

3.2 TAPE GLAZING

- A. Position tapes on fixed stops so that, when compressed by glass, their exposed edges are flush with or protrude slightly above sightline of stops.
- B. Install tapes continuously, but not necessarily in one continuous length. Do not stretch tapes to make them fit opening.
- C. Cover vertical framing joints by applying tapes to heads and sills first, then to jambs. Cover horizontal framing joints by applying tapes to jambs, then to heads and sills.
- D. Place joints in tapes at corners of opening with adjoining lengths butted together, not lapped. Seal joints in tapes with compatible sealant approved by tape manufacturer.

- E. Apply heel bead of elastomeric sealant.
- F. Center glass lites in openings on setting blocks and press firmly against tape by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings.
- G. Apply cap bead of elastomeric sealant over exposed edge of tape.

3.3 GASKET GLAZING (DRY)

- A. Cut compression gaskets to lengths recommended by gasket manufacturer to fit openings exactly, with allowance for stretch during installation.
- B. Insert soft compression gasket between glass and frame or fixed stop so it is securely in place with joints miter cut and bonded together at corners.
- C. Installation with Drive-in Wedge Gaskets: Center glass lites in openings on setting blocks and press firmly against soft compression gasket by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings. Compress gaskets to produce a weathertight seal without developing bending stresses in glass. Seal gasket joints with sealant recommended by gasket manufacturer.
- D. Installation with Pressure-Glazing Stops: Center glass lites in openings on setting blocks and press firmly against soft compression gasket. Install dense compression gaskets and pressure-glazing stops, applying pressure uniformly to compression gaskets. Compress gaskets to produce a weathertight seal without developing bending stresses in glass. Seal gasket joints with sealant recommended by gasket manufacturer.
- E. Install gaskets so they protrude past face of glazing stops.

3.4 SEALANT GLAZING (WET)

- A. Install continuous spacers, or spacers combined with cylindrical sealant backing, between glass lites and glazing stops to maintain glass face clearances and to prevent sealant from extruding into glass channel and blocking weep systems until sealants cure. Secure spacers or spacers and backings in place and in position to control depth of installed sealant relative to edge clearance for optimum sealant performance.
- B. Force sealants into glazing channels to eliminate voids and to ensure complete wetting or bond of sealant to glass and channel surfaces.
- C. Tool exposed surfaces of sealants to provide a substantial wash away from glass.

3.5 CLEANING AND PROTECTION

- A. Immediately after installation remove nonpermanent labels and clean surfaces.
- B. Protect glass from contact with contaminating substances resulting from construction operations. Examine glass surfaces adjacent to or below exterior concrete and other masonry surfaces at frequent intervals during construction, but not less than once a month, for buildup of dirt, scum, alkaline deposits, or stains.

1. If, despite such protection, contaminating substances do come into contact with glass, remove substances immediately as recommended in writing by glass manufacturer. Remove and replace glass that cannot be cleaned without damage to coatings.

C. Remove and replace glass that is damaged during construction period.

3.6 INSULATING GLASS SCHEDULE

A. Glass Type 1: Low-E-Coated, Clear Insulating Glass, Tempered:

1. Product: Vitro Solarban 70 (2) + Clear
2. Overall Unit Thickness: 1 inch (25 mm)
3. Minimum Thickness of Each Glass Lite: 6 mm
4. Outdoor Lite: Fully tempered clear glass
5. Interspace Content: 10% Air; 90% Argon
6. Indoor Lite: Fully tempered clear glass
7. Low-E Coating: Sputtered on second surface
8. U-Factor: 0.24 maximum (winter)
9. Visible Light Transmittance: 64%
10. Solar Heat Gain Coefficient: 0.27 maximum
11. Locations: Exterior curtain walls, storefronts and sliding entrances.

B. Glass Type 2: Low-E-Coated, Clear Insulating Glass, Heat-Strengthened:

1. Product: Vitro Solarban 70 (2) + Clear
2. Overall Unit Thickness: 1 inch (25 mm)
3. Minimum Thickness of Each Glass Lite: 6 mm.
4. Outdoor Lite: Heat-strengthened clear glass
5. Interspace Content: 10% Air; 90% Argon
6. Indoor Lite: Heat-strengthened clear glass
7. Low-E Coating: Sputtered on second surface
8. U-Factor: 0.24 maximum (winter)
9. Visible Light Transmittance: 64%
10. Solar Heat Gain Coefficient: 0.27 maximum
11. Locations: Exterior curtain walls and storefronts.

C. Glass Type 5: Low-E-Coated, Clear Insulating Glass, Laminated:

1. Product: Vitro Solarban 70 (2) + Clear
2. Overall Unit Thickness: 1 inch (25 mm)
3. Outdoor Lite: 1/4 inch (6 mm) heat-strengthened clear glass (fully tempered where required)
4. Air Space: 7/16 inch (12 mm)
 - a. Interspace Content: 10% Air; 90% Argon
5. Indoor Lite: 5/16 inch (8 mm) laminated glass consisting of:
 - a. Glass: Two layers of heat-strengthened clear glass (fully tempered where required)
 - b. Interlayer: 0.060 inch PVB in color to match storefront finish.
6. Low-E Coating: Sputtered on second surface

7. Locations: Exterior curtain walls and storefronts at Vestibule corner (if shown on Drawings)

3.7 MONOLITHIC GLASS SCHEDULE

A. Glass Type 3: Clear Float Glass, Fully Tempered:

1. Thickness: 1/4 inch (6 mm) minimum.
2. Safety glazing label required.
3. Locations:
 - a. Interior doors, windows and relites.
 - b. Storefronts and sliding entrances on the interior side of Entry Vestibule.

B. Glass Type 4: Clear Float Glass:

1. Thickness: 1/4 inch (6 mm) minimum.
2. Locations:
 - a. Interior windows.
 - b. Storefronts on the interior side of Entry Vestibule.

C. Glass Type: One-way glass, full tempered, in door lite.

1. Product: Pilkington; Mirropane TM (Transparent Mirror).
2. Overall Thickness: 1/4 inch (6 mm) nominal
3. Fully tempered glass where required.
4. Provide safety glazing label where required.
5. Install with reflective surface (coated side) towards customer side.

END OF SECTION 088000

SECTION 089119 – LOUVERS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes:
 - 1. Fixed, extruded-aluminum louvers.
- B. Related Sections:
 - 1. Section 079200 "Joint Sealants" for sealants installed in perimeter joints between louver frames and adjoining construction.
 - 2. Section 099113 "Painting" for field painting louvers.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - a. For louvers specified to bear AMCA seal, include printed catalog pages showing specified models with appropriate AMCA Certified Ratings Seals.
 - 2. Shop Drawings: For louvers and accessories. Include plans, elevations, sections, details, and attachments to other work. Show frame profiles and blade profiles, angles, and spacing.
 - a. Show weep paths, gaskets, flashings, sealants, and other means of preventing water intrusion.
 - b. Show mullion profiles and locations.
 - 3. Samples: For each type of metal finish required.
 - 4. Delegated-Design Submittal: For louvers indicated to comply with structural and seismic performance requirements and design criteria, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1.4 INFORMATIONAL SUBMITTALS

- A. Product Test Reports: Based on evaluation of comprehensive tests performed according to AMCA 500-L by a qualified testing agency or by manufacturer and witnessed by a qualified testing agency, for each type of louver and showing compliance with performance requirements specified.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.6 QUALITY ASSURANCE

- A. Welding: Qualify procedures and personnel according to the following:

1. AWS D1.2/D1.2M, "Structural Welding Code - Aluminum."

1.7 FIELD CONDITIONS

- A. Field Measurements: Verify actual dimensions of openings by field measurements before fabrication.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Source Limitations: Obtain fixed louvers from single source from a single manufacturer where indicated to be of same type, design, or factory-applied color finish.

2.2 PERFORMANCE REQUIREMENTS

- A. Delegated Design: Design louvers, including comprehensive engineering analysis by a qualified professional engineer, using structural and seismic performance requirements and design criteria indicated.
- B. Structural Performance: Louvers shall withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated without permanent deformation of louver components, noise or metal fatigue caused by louver-blade rattle or flutter, or permanent damage to fasteners and anchors. Wind pressures shall be considered to act normal to the face of the building.
 1. Wind Loads: Determine loads based on pressures as indicated on Drawings.
- C. Louver Performance Ratings: Provide louvers complying with requirements specified, as demonstrated by testing manufacturer's stock units identical to those provided, except for length and width according to AMCA 500-L.
- D. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.
 1. Temperature Change (Range): 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.
- E. SMACNA Standard: Comply with recommendations in SMACNA's "Architectural Sheet Metal Manual" for fabrication, construction details, and installation procedures.

2.3 FIXED LOUVERS

- A. Horizontal-Drainable-Blade, Sightproof Extruded Aluminum Louvers:
 1. Basis of Design Product: Construction Specialties (CS); Model B5157

2. Louver Depth: 5 inches nominal
3. Frame: Extruded 6063-T6 aluminum, 0.081-inch (2.06 mm) nominal wall thickness.
4. Fixed Blades: Drainable and sightproof design, extruded 6063-T6 aluminum, 0.060-inch nominal wall thickness.
5. Construction: Mechanically fastened
6. Vertical Mullion: None
7. Louver Performance:
 - a. Free Area: 52.6 percent (8.42 sq. ft. (0.782 sq. m.)), for 48-inch wide by 48-inch high louver.
 - b. Beginning Point of Water Penetration: 1043 fpm (5.30 m/s) free area velocity.
8. AMCA Seal: Mark units with AMCA Certified Ratings Seal.

2.4 LOUVER SCREENS

- A. General: Provide screen at each exterior louver.
 1. Screen Location for Fixed Louvers: Interior face.
 2. Screening Type: Bird screening.
- B. Secure screens to louver frames with stainless-steel machine screws, spaced a maximum of 6 inches from each corner and at 12 inches o.c.
- C. Louver Screen Frames: Fabricate with mitered corners to louver sizes indicated.
 1. Metal: Same kind and form of metal as indicated for louver to which screens are attached. Reinforce extruded-aluminum screen frames at corners with clips.
 2. Finish: Same finish as louver frames to which louver screens are attached.
 3. Type: Removable, rewirable frames with a driven spline or insert for securing screen mesh.
- D. Louver Screening for Aluminum Louvers:
 1. Bird Screening: Flattened, expanded aluminum, 5/8 inch (15.87 mm) by 0.055 inch (1.4 mm) thick.

2.5 MATERIALS

- A. Aluminum Extrusions: ASTM B 221 (ASTM B 221M), Alloy 6063-T5, T-52, or T6.
- B. Aluminum Sheet: ASTM B 209 (ASTM B 209M), Alloy 3003 or 5005 with temper as required for forming, or as otherwise recommended by metal producer for required finish.
- C. Fasteners: Use types and sizes to suit unit installation conditions.
 1. For fastening aluminum, use aluminum or 300 series stainless-steel fasteners.
 2. For color-finished louvers, use fasteners with heads that match color of louvers.
- D. Postinstalled Fasteners for Concrete and Masonry: Torque-controlled expansion anchors, fabricated from stainless-steel components, with allowable load or strength design capacities calculated according to ICC-ES AC193 and ACI 318 greater than or equal to the design load, as determined by testing according to ASTM E488/E488M conducted by a qualified testing agency.
- E. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D1187/D1187M.

2.6 FABRICATION

- A. Factory assemble louvers to minimize field splicing and assembly. Disassemble units as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation.
- B. Maintain equal louver blade spacing, including separation between blades and frames at head and sill, to produce uniform appearance.
- C. Fabricate frames, including integral sills, to fit in openings of sizes indicated, with allowances made for fabrication and installation tolerances, adjoining material tolerances, and perimeter sealant joints.
 - 1. Frame Type: Channel unless otherwise indicated.
- D. Include supports, anchorages, and accessories required for complete assembly.
- E. Provide vertical mullions of type and at spacings indicated, but not more than is recommended by manufacturer, or 72 inches (1830 mm) o.c., whichever is less.
- F. Join frame members to each other and to fixed louver blades with fillet welds concealed from view, threaded fasteners, or both, as standard with louver manufacturer unless otherwise indicated or size of louver assembly makes bolted connections between frame members necessary.

2.7 FINISHES, GENERAL

- A. Finish louvers after assembly.
- B. High-Performance Organic Finish: Two-coat fluoropolymer finish complying with AAMA 2605 and containing not less than 70 percent PVDF resin by weight in both color coat and clear topcoat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
 - 1. Color: See Exterior Material/Finish Schedule.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and openings, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Coordinate setting drawings, diagrams, templates, instructions, and directions for installation of anchorages that are to be embedded in concrete or masonry construction. Coordinate delivery of such items to Project site.

3.3 INSTALLATION

- A. Locate and place louvers and vents level, plumb, and at indicated alignment with adjacent work.

- B. Use concealed anchorages where possible. Provide brass or lead washers fitted to screws where required to protect metal surfaces and to make a weathertight connection.
- C. Form closely fitted joints with exposed connections accurately located and secured.
- D. Provide perimeter reveals and openings of uniform width for sealants and joint fillers, as indicated.
- E. Repair finishes damaged by cutting, welding, soldering, and grinding. Restore finishes so no evidence remains of corrective work. Return items that cannot be refinished in the field to the factory, make required alterations, and refinish entire unit or provide new units.
- F. Protect galvanized and nonferrous-metal surfaces from corrosion or galvanic action by applying a heavy coating of bituminous paint on surfaces that will be in contact with concrete, masonry, or dissimilar metals.
- G. Install concealed gaskets, flashings, joint fillers, and insulation as louver installation progresses, where weathertight louver joints are required. Comply with Division 7 Section "Joint Sealants" for sealants applied during louver installation.

3.4 ADJUSTING AND CLEANING

- A. Clean exposed surfaces of louvers and vents that are not protected by temporary covering, to remove fingerprints and soil during construction period. Do not let soil accumulate until final cleaning.
- B. Before final inspection, clean exposed surfaces with water and a mild soap or detergent not harmful to finishes. Thoroughly rinse surfaces and dry.
- C. Restore louvers and vents damaged during installation and construction so no evidence remains of corrective work. If results of restoration are unsuccessful, as determined by Architect, remove damaged units and replace with new units.
 - 1. Touch up minor abrasions in finishes with air-dried coating that matches color and gloss of, and is compatible with, factory-applied finish coating.

END OF SECTION 089119

SECTION 092216 - NON-STRUCTURAL METAL FRAMING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Non-load-bearing steel framing systems for interior partitions.
 - 2. Suspension systems for interior ceilings and soffits.
 - 3. Hat channel.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

- A. Product Certificates: For each type of code-compliance certification for studs and tracks.

1.5 QUALITY ASSURANCE

- A. Code-Compliance Certification of Studs and Tracks: Provide documentation that framing members are certified according to the product-certification program of the Certified Steel Stud Association, the Steel Framing Industry Association or the Steel Stud Manufacturers Association.

PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. Horizontal Deflection: For composite wall assemblies, limited deflection indicated based on horizontal loading of minimum 5 lbf/sq. ft. (239 Pa).
 - 1. Fabricate and install systems as indicated but not less than that required to comply with ASTM C754 and Steel Stud Manufacturers Association (SSMA) "Interior Wall Heights – Composite" under the following conditions:
 - a. Metal Stud Wall with Gypsum Board Partitions (Interior):
 - 1) Standard System (up to 12 feet): Maximum deflection of L/240 of partition height.

- 2) System Supporting Thin-Bed Ceramic Tile: Maximum deflection of L/360 of partition height.
- b. Interior Suspended Ceilings and Soffits: Maximum deflection of L/360 of distance between supports.
- B. Fire-Test-Response Characteristics: For fire-resistance-rated assemblies that incorporate non-load-bearing steel framing, provide materials and construction identical to those tested in assembly indicated, according to ASTM E119 by an independent testing agency.
- C. STC-Rated Assemblies: For STC-rated assemblies, provide materials and construction identical to those tested in assembly indicated on Drawings, according to ASTM E 90 and classified according to ASTM E 413 by an independent testing agency.

2.2 FRAMING SYSTEMS

- A. Framing Members, General: Comply with ASTM C 754 for conditions indicated.
 - 1. Steel Sheet Components: Comply with ASTM C 645 requirements for metal unless otherwise indicated.
 - 2. Protective Coating: Coating with equivalent corrosion resistance of ASTM A 653/A 653M, G40 (Z120), hot-dip galvanized unless otherwise indicated.
- B. Studs and Tracks: ASTM C 645.
 - 1. Steel Studs and Tracks:
 - a. Minimum Base-Metal Thickness: As required by performance requirements for horizontal deflection but not less than 0.0179 inch (0.455 mm).
 - b. Depth: 2-1/2 inches (64mm), 3-5/8 inches (92 mm), 4 inches (101.5 mm), 6 inches (152 mm) and 1-5/8 inches (41 mm) unless shown otherwise.
 - c. Spacing: Maximum of 16 inches (406 mm) o.c.
- C. Slip-Type Head Joints: Where indicated, provide one of the following:
 - 1. Double-Track System: ASTM C 645 top outer tracks, inside track with 2-inch- (51-mm-) deep flanges in thickness not less than indicated for studs and fastened to studs, and outer track sized to friction-fit over inner track.
 - 2. Deflection Track: Steel sheet top track manufactured to prevent cracking of finishes applied to interior partition framing resulting from deflection of structure above; in thickness not less than indicated for studs and in width to accommodate depth of studs.
- D. Firestop Tracks: Top track manufactured to allow partition heads to expand and contract with movement of structure while maintaining continuity of fire-resistance-rated assembly indicated; in thickness not less than indicated for studs and in width to accommodate depth of studs.
- E. Flat Strap and Backing Plate: Minimum 4-inch wide steel sheet for blocking and bracing in length and width indicated.
 - 1. Minimum Base-Metal Thickness: Not less than 0.0329 inch (0.836 mm).
- F. Cold-Rolled Channel Bridging: Steel, 0.0538-inch (1.367-mm) minimum base-metal thickness, with minimum 1/2-inch- (13-mm-) wide flanges.

1. Depth: Not less than 1-1/2 inches (38 mm).
2. Clip Angle: Not less than 1-1/2 by 1-1/2 inches (38 by 38 mm), 0.068-inch- (1.72-mm-) thick, galvanized steel.

G. Hat-Shaped, Rigid Furring Channels: ASTM C 645.

1. Minimum Base-Metal Thickness: 0.0179 inch (0.455 mm).
2. Depth: 1-1/2 inches (38 mm).

2.3 SUSPENDED DRYWALL SYSTEM

A. DROP CEILING ASSEMBLY – Location: Offices

1. Tie Wire: ASTM A 641/A 641M, Class 1 zinc coating, soft temper, 0.062-inch- (1.59-mm-) diameter wire, or double strand of 0.048-inch- (1.21-mm-) diameter wire.
2. Wire Hangers: ASTM A 641/A 641M, Class 1 zinc coating, soft temper, 0.16 inch (4.12 mm) in diameter.
3. Carrying Channels (Main Runners): Cold-rolled, commercial-steel sheet with a base-metal thickness of 0.0538 inch (1.367 mm) and minimum 1/2-inch (13-mm-) wide flanges.
4. Depth: Not less than 1-1/2 inches (38 mm).
 - a. Hat-Shaped, Rigid Furring Channels: ASTM C 645, 7/8 inch (22 mm) deep.
 - b. Minimum Base-Metal Thickness: 0.0179 inch (0.455 mm).

2.4 AUXILIARY MATERIALS

A. General: Provide auxiliary materials that comply with referenced installation standards.

1. Fasteners for Steel Framing: Of type, material, size, corrosion resistance, holding power, and other properties required to fasten steel members to substrates.

B. Isolation Strip at Exterior Walls: Provide one of the following:

1. Asphalt-Saturated Organic Felt: ASTM D 226/D 226M, Type I (No. 15 asphalt felt), nonperforated.
2. Foam Gasket: Adhesive-backed, closed-cell vinyl foam strips that allow fastener penetration without foam displacement, 1/8 inch (3.2 mm) thick, in width to suit steel stud size.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Installation Standard: ASTM C 754.

1. Gypsum Board Assemblies: Also comply with requirements in ASTM C 840 that apply to framing installation.

B. Install framing and accessories plumb, square, and true to line, with connections securely fastened.

C. Install supplementary framing, and blocking to support fixtures, equipment services, heavy trim, grab bars, toilet accessories, furnishings, or similar construction.

- D. Install bracing at terminations in assemblies.
- E. Do not bridge building control and expansion joints with non-load-bearing steel framing members. Frame both sides of joints independently.

3.2 INSTALLING FRAMED ASSEMBLIES

- A. Install framing system components according to spacings indicated, but not greater than spacings required by referenced installation standards for assembly types.
- B. Where studs are installed directly against exterior masonry walls or dissimilar metals at exterior walls, install isolation strip between studs and exterior wall.
- C. Install studs so flanges within framing system point in same direction.
- D. Install tracks at floors and overhead supports. Extend framing full height to structural supports or substrates above suspended ceilings except where partitions are indicated to terminate at suspended ceilings. Continue framing around ducts that penetrate partitions above ceiling.
 - 1. Slip-Type Head Joints: Where framing extends to overhead structural supports, install to produce joints at tops of framing systems that prevent axial loading of finished assemblies.
 - 2. Door Openings: Screw vertical studs at jambs to jamb anchor clips on door frames; install track section (for cripple studs) at head and secure to jamb studs.
 - a. Install two studs at each jamb unless otherwise indicated.
 - b. Install cripple studs at head adjacent to each jamb stud, with a minimum 1/2-inch (13-mm) clearance from jamb stud to allow for installation of control joint in finished assembly.
 - c. Extend jamb studs through suspended ceilings and attach to underside of overhead structure.
 - 3. Other Framed Openings: Frame openings other than door openings the same as required for door openings unless otherwise indicated. Install framing below sills of openings to match framing required above door heads.
 - 4. Fire-Resistance-Rated Partitions: Install framing to comply with fire-resistance-rated assembly indicated and support closures and to make partitions continuous from floor to underside of solid structure.
 - a. Firestop Track: Where indicated, install to maintain continuity of fire-resistance-rated assembly indicated.
 - 5. Sound-Rated Partitions: Install framing to comply with sound-rated assembly indicated.
- E. Direct Furring:
 - 1. Attach to concrete or masonry with stub nails, screws designed for masonry attachment, or powder-driven fasteners spaced 24 inches (610 mm) o.c.
- F. Installation Tolerance: Install each framing member so fastening surfaces vary not more than 1/8 inch (3 mm) from the plane formed by faces of adjacent framing.

3.3 INSTALLING CEILING SUSPENSION SYSTEMS

- A. Install suspension system components according to spacings indicated, but not greater than spacings required by referenced installation standards for assembly types.
- B. Isolate suspension systems from building structure where they abut or are penetrated by building structure to prevent transfer of loading imposed by structural movement.
- C. Suspend hangers from building structure as follows:
 - 1. Install hangers plumb and free from contact with insulation or other objects within ceiling plenum that are not part of supporting structural or suspension system.
 - a. Splay hangers only where required to miss obstructions and offset resulting horizontal forces by bracing, countersplaying, or other equally effective means.
 - 2. Where width of ducts and other construction within ceiling plenum produces hanger spacings that interfere with locations of hangers required to support standard suspension system members, install supplemental suspension members and hangers in the form of trapezes or equivalent devices.
 - a. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by referenced installation standards.
 - 3. Wire Hangers: Secure by looping and wire tying, either directly to structures or to inserts, eye screws, or other devices and fasteners that are secure and appropriate for substrate, and in a manner that will not cause hangers to deteriorate or otherwise fail.
 - 4. Do not attach hangers to steel roof deck.
 - 5. Do not connect or suspend steel framing from ducts, pipes, or conduit.
- D. Fire-Resistance-Rated Assemblies: Wire tie furring channels to supports.
- E. Seismic Bracing: Sway-brace suspension systems with hangers used for support where required by jurisdiction authorities.
- F. Installation Tolerances: Install suspension systems that are level to within 1/8 inch in 12 feet (3 mm in 3.6 m) measured lengthwise on each member that will receive finishes and transversely between parallel members that will receive finishes.

END OF SECTION 092216

SECTION 092900 - GYPSUM BOARD**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Interior gypsum board.
- 2. Tile backing panels.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.

1.4 DELIVERY, STORAGE AND HANDLING

- A. Store materials inside under cover and keep them dry and protected against weather, condensation, direct sunlight, construction traffic, and other potential causes of damage. Stack panels flat and supported on risers on a flat platform to prevent sagging.

1.5 FIELD CONDITIONS

- A. Environmental Limitations: Comply with ASTM C 840 requirements or gypsum board manufacturer's written recommendations, whichever are more stringent.
- B. Do not install paper-faced gypsum panels until installation areas are enclosed and conditioned.
- C. Do not install panels that are wet, those that are moisture damaged, and those that are mold damaged.
 - 1. Indications that panels are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
 - 2. Indications that panels are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. Fire-Resistance-Rated Assemblies: For fire-resistance-rated assemblies, provide materials and construction identical to those tested in assembly indicated according to ASTM E119 by an independent testing agency.

- B. STC-Rated Assemblies: For STC-rated assemblies, provide materials and construction identical to those tested in assembly indicated according to ASTM E 90 and classified according to ASTM E 413 by an independent testing agency.
- C. Low-Emitting Materials: For ceiling and wall assemblies, provide materials of construction identical to those tested in assembly and complying with the testing and product requirements of the California Department of Health Services "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers".

2.2 GYPSUM BOARD, GENERAL

- A. Size: Provide maximum lengths and widths available that will minimize joints in each area and that correspond with support system indicated.

2.3 INTERIOR GYPSUM BOARD

- A. Mold-Resistant Walls Gypsum Board: ASTM C 1396/C 1396M. With moisture- and mold-resistant core and paper surfaces.

- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. CertainTeed Corporation.
 - b. Georgia-Pacific Building Products.
 - c. National Gypsum Company.
 - d. United States Gypsum Company.
- 2. Thickness and Core: As indicated on Drawings.
 - a. 1/2 inch (12.7 mm), Type C (where indicated)
 - b. 5/8 inch (15.9 mm)
 - c. 5/8 inch (15.9 mm), Type X (where indicated)
- 3. Long Edges: Tapered.
- 4. Mold Resistance: ASTM D 3273, score of 10 as rated according to ASTM D 3274.

- B. Gypsum Ceiling Board: ASTM C 1396/C 1396M.

- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. CertainTeed Corporation.
 - b. Georgia-Pacific Building Products.
 - c. National Gypsum Company.
 - d. United States Gypsum Company.
- 2. Thickness: As indicated on Drawings.
 - a. 1/2 inch (12.7 mm)
- 3. Long Edges: Tapered.

2.4 BACKING PANELS BEHIND TILE AND FRP

A. Glass-Mat, Mold-and Water-Resistant Tile Backing Board: ASTM C1178/C1178M.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. CertainTeed Corporation.
 - b. Georgia-Pacific Building Products.
 - c. National Gypsum Company.
 - d. United States Gypsum Company.
2. Thickness and Core: As indicated on Drawings.
 - a. 5/8 inch (15.9 mm) throughout
3. Edges: Manufacturer's standard.
4. Mold Resistance: ASTM D 3273, score of 10 as rated according to ASTM D 3274.
5. Locations: Substrate for tile and FRP installation.

2.5 CEMENT BOARD BEHIND COVE BASE COATING

A. Cementitious Backer Board: ANSI A118.9, ASTM C 1288 or ASTM C 1325.

1. Thickness: 5/8 inch (15.9 mm)
2. Edges: Manufacturer's standard.
3. Locations: Cove base in Bakery, Janitor Room, and Restrooms (including Vestibule Area)

2.6 TRIM ACCESSORIES

A. Interior Trim: ASTM C 1047.

1. Material: PVC.
2. Shapes:
 - a. Cornerbead.
 - b. LC-Bead: J-shaped; exposed long flange receives joint compound.
 - c. L-Bead: L-shaped; exposed long flange receives joint compound.
 - d. Expansion (control) joint.

2.7 JOINT TREATMENT MATERIALS

A. General: Comply with ASTM C 475/C 475M.

B. Joint Tape:

1. Interior Gypsum Board: Paper.
2. Tile Backing Panels: As recommended by panel manufacturer.
3. Cement Board: As recommended by panel manufacturer.

C. Joint Compound for Interior Gypsum Board: For each coat, use formulation that is compatible with other compounds applied on previous or for successive coats.

1. Prefilling: At open joints, rounded or beveled panel edges, and damaged surface areas, use setting-type taping compound.

2. Embedding and First Coat: For embedding tape and first coat on joints, fasteners, and trim flanges, use setting-type taping compound.
 - a. Use setting-type compound for installing paper-faced metal trim accessories.
3. Fill Coat: For second coat, use drying-type, all-purpose compound.
4. Finish Coats: For third and fourth coats, use drying-type, all-purpose compound.
5. VOC Content: Joint compounds shall have a VOC content of 50g/L or less when calculated according to 40 CFR59, Subpart D (EPA Method D).

D. Joint Compound for Tile Backing Panels:

1. As recommended by backing panel manufacturer.
2. VOC Content: Joint compounds and laminating adhesives shall have a VOC content of 50g/L or less when calculated according to 40 CFR59, Subpart D (EPA Method D).

E. Joint Compound for Cement Board:

1. As recommended by cement board manufacturer.

2.8 AUXILIARY MATERIALS

A. General: Provide auxiliary materials that comply with referenced installation standards and manufacturer's written instructions.

1. Use screws complying with ASTM C 954 for fastening panels to steel members from 0.033 to 0.112 inch (0.84 to 2.84 mm) thick.
2. For fastening Glass-Mat backer units, use screws of type and size recommended by panel manufacturer.

B. Sound-Attenuation Blankets: ASTM C 665, Type I (blankets without membrane facing) produced by combining thermosetting resins with mineral fibers manufactured from glass, slag wool, or rock wool.

1. Fire-Resistance-Rated Assemblies: Comply with mineral-fiber requirements of assembly.

C. Acoustical Sealant: Manufacturer's standard nonsag, paintable, nonstaining latex sealant complying with ASTM C 834. Product effectively reduces airborne sound transmission through perimeter joints and openings in building construction as demonstrated by testing representative assemblies according to ASTM E 90.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Hilti, Inc.
 - b. Pecora Corporation.
 - c. Specified Technologies, Inc.
 - d. United States Gypsum Company.
2. VOC Content: Joint compounds shall have a VOC content of 250g/L or less when calculated according to 40 CFR59, Subpart D (EPA Method D).

D. Thermal Insulation: As specified in Section 072100 "Thermal Insulation."

PART 3 - EXECUTION

3.1 APPLYING AND FINISHING PANELS

- A. Examine panels before installation. Reject panels that are wet, moisture damaged, and mold damaged.
- B. Gypsum Board: Comply with ASTM C 840 and GA-216.
 - 1. Space screws a maximum of 12 inches (300 mm) on center for vertical applications.
 - 2. Space fasteners in panels that are tile substrates a maximum of 8 inches (200 mm) on center.
 - 3. On partitions / walls, apply gypsum panels either vertically (parallel to framing) or horizontally (perpendicular to framing), unless otherwise indicated or required by fire-resistance-rated assembly, and minimize end joints.
 - a. Stagger abutting end joints not less than one framing member in alternate courses of board. Arrange for joints on opposite sides of a partition to occur on different studs.
 - b. Space control joints not more than 24 feet (7.3 m) on center, in specific locations approved by Architect for visual effect.
 - c. Install control joints in furred assemblies where control joints occur in base exterior wall.
 - 4. On Z-furring members, apply gypsum panels vertically (parallel to framing) with no end joints. Locate edge joints over furring members.
 - 5. Use metal trim where board abuts dissimilar materials.
- C. Cement Board: Install cement board behind cove base coating, complying with ANSI A108.11 and the manufacturer's recommendations. Extend cement board from floor to 2 inches above top of cove base. Install cement boards with 1/4-inch (6 mm) gap where abut other construction or penetrations.
- D. Isolate perimeter of gypsum board applied to non-load-bearing partitions at structural abutments. Provide 1/4- to 1/2-inch- (6.4- to 12.7-mm-) wide spaces at these locations and trim edges with edge trim where edges of panels are exposed. Seal joints between edges and abutting structural surfaces with acoustical sealant.
- E. Horizontal joints must be staggered. Avoid vertical joints directly above or below a window, door or other openings for best decorating results.
- F. For trim with back flanges intended for fasteners, attach to framing with same fasteners used for panels. Otherwise, attach trim according to manufacturer's written instructions.
- G. Prefill open joints, rounded or beveled edges, and damaged surface areas.
- H. Apply joint tape over gypsum board joints, except for trim products specifically indicated as not intended to receive tape.
- I. Gypsum Board Finish Levels: Finish panels to levels indicated below and according to ASTM C 840:
 - 1. Level 1: Ceiling plenum areas, concealed areas.
 - 2. Level 2: Panels that are substrate for tile.

- 3. Level 4: At panel surfaces that will be exposed to view unless otherwise indicated.
 - a. Primer and its application to surfaces are specified in Section 099123 "Interior Painting."
- J. Glass-Mat Faced Tile Backing Board: Finish according to manufacturer's written instructions.
- 3.2 INSTALLATION OF ACOUSTIC INSULATION IN FRAMED CONSTRUCTION AND ABOVE CEILINGS
 - A. Acoustic Blanket Insulation: Install in cavities formed by framing members and above ceiling system according to the following requirements:
 - 1. Use insulation widths and lengths that fill the cavities formed by framing members. If more than one length is required to fill the cavities, provide lengths that will produce a snug fit between ends.
 - 2. Place insulation in cavities formed by framing members to produce a friction fit between edges of insulation and adjoining framing members.
 - 3. For above ceiling installation, maintain 3-inch (76-mm) clearance of insulation around recessed lighting fixtures not rated for or protected from contact with insulation.
 - 4. For metal-framed wall cavities where cavity heights exceed 96 inches (2438 mm), support unfaced blankets mechanically and support faced blankets by taping flanges of insulation to flanges of metal studs.
- 3.3 PROTECTION
 - A. Protect installed products from damage from weather, condensation, direct sunlight, construction, and other causes during remainder of the construction period.
 - B. Remove and replace panels that are wet, moisture damaged, and mold damaged.

END OF SECTION 092900

SECTION 095113 - ACOUSTICAL PANEL CEILINGS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes acoustical panels and exposed suspension systems for interior ceilings.

1.3 PREINSTALLATION MEETINGS

- A. Pre-installation Conference: Conduct conference at Project site.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Samples: For each exposed product and for each color and texture specified.
 - 3. Coordination Drawings: Reflected ceiling plans, drawn to scale, and coordinated with each other, using input from installers of the items involved.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.6 QUALITY ASSURANCE

- A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer for both installation and maintenance of units required for this Project.

PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. Low-Emitting Materials: For ceiling and wall assemblies, provide materials of construction identical to those tested in assembly and complying with the testing and product requirements of the California Department of Health Services "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers".
- B. Seismic Performance: Suspended ceilings shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
- C. Surface-Burning Characteristics: Comply with ASTM E 84; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

1. Flame-Spread Index: Class B according to ASTM E 1264.
2. Smoke-Developed Index: 450 or less.

2.2 ACOUSTICAL PANELS

- A. Manufacturers: Subject to compliance with requirements, provide products by the following:

1. Armstrong World Industries, Inc.

- B. Acoustical Panel Standard: Manufacturer's standard panels according to ASTM E 1264.

- C. Acoustic Ceiling Tile ACT-1: Armstrong Optima 9/16" Square Tegular #3251.

1. Classification: Type XII, Form 2, Pattern E, Fire Class A
2. Color: White
3. Light Reflectance (LR): 0.90
4. Noise Reduction Coefficient (NRC): 0.95
5. Ceiling Attenuation Class (CAC): 190
6. Edge/Joint Detail: Square tegular reveal with Interlude XL 9/16" suspension system.
7. Thickness: 1 inch (25 mm)
8. Modular Size: 24 by 24 inches (610 by 610 mm)

- D. Acoustic Ceiling Tile ACT-2: Armstrong Kitchen Zone 15/16" Square Lay-in Smooth Texture #673.

1. Classification: Type IX, Form 2, Pattern G, Fire Class A
2. Finish and Color: White
3. Light Reflectance (LR): 0.89
4. Ceiling Attenuation Class (CAC): 33
5. Edge/Joint Detail: 15/16" square lay-in with Prelude XL 15/16" suspension system.
6. Thickness: 5/8 inch (15 mm)
7. Modular Size: 24 by 24 inches (610 by 610 mm)

2.3 METAL SUSPENSION SYSTEM

- A. Manufacturers: Subject to compliance with requirements, provide products by the following:

1. Armstrong World Industries, Inc.

- B. Metal Suspension-System Standard: Manufacturer's standard, direct-hung, metal suspension system and accessories according to ASTM C 635/C 635M.

- C. Narrow-Face, Capped, Double-Web, Steel Suspension System for ACT-1: Main and cross runners roll formed from cold-rolled steel sheet; prepainted, electrolytically zinc coated, or hot-dip galvanized, G30 (Z90) coating designation; with prefinished 9/16-inch- (15-mm-) wide metal caps on flanges.

1. Product: Armstrong Interlude XL HRC
2. Structural Classification: Intermediate-duty system.
3. End Condition of Cross Runners: Override (stepped) type.
4. Face Design: Flat, flush.
5. Cap Material: Cold-rolled steel.
6. Cap Finish: Powder-coated finish, manufacturer's standard White.

- D. Wide-Face, Capped, Double-Web, Steel Suspension System for ACT-2: Main and cross runners roll formed from cold-rolled steel sheet; pre-painted, electrolytically zinc coated, or hot-dip galvanized, G30 (Z90) coating designation; with prefinished 15/16-inch (24-mm) wide metal caps on flanges and integral gaskets.

1. Product: Armstrong Prelude XL 15/16"
2. Structural Classification: Heavy-duty system.
3. End Condition of Cross Runners: Override (stepped) type.
4. Face Design: Flat, flush.
5. Cap Material: Cold-rolled steel.
6. Cap Finish: White.

2.4 ACCESSORIES

- A. Attachment Devices: Size for five times the design load indicated in ASTM C 635/C 635M, Table 1, "Direct Hung," unless otherwise indicated. Comply with seismic design requirements.
- B. Seismic Clips: Manufacturer's standard seismic clips designed to secure acoustical panels in place during a seismic event.

2.5 METAL EDGE MOLDINGS AND TRIM

- A. Manufacturers: Subject to compliance with requirements, provide products by the following:
1. Armstrong World Industries, Inc.
- B. Roll-Formed, Sheet-Metal Edge Moldings and Trim: Type and profile indicated or, if not indicated, manufacturer's standard hemmed angle molding for edges and penetrations that comply with seismic design requirements; formed from sheet metal of same material, finish, and color as that used for exposed flanges of suspension-system runners.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Measure each ceiling area and establish layout of acoustical panels to balance border widths at opposite edges of each ceiling. Avoid using less-than-half-width panels at borders unless otherwise indicated.
- B. Layout openings for penetrations centered on the penetrating items.

3.2 INSTALLATION

- A. Install acoustical panel ceilings according to ASTM C 636/C 636M, seismic design requirements where required by jurisdiction authorities, and manufacturer's written instructions.
- B. Install edge moldings and trim of type indicated at perimeter of acoustical ceiling area and where necessary to conceal edges of acoustical panels.
1. Apply acoustical sealant in a continuous ribbon concealed on back of vertical legs of moldings before they are installed.
 2. Do not use exposed fasteners, including pop rivets, on moldings and trim.
 3. Where required by code, install seismic clips in areas indicated; space according to panel manufacturer's written instructions unless otherwise indicated.

- C. Install tiles with extreme care, do not damage or stain any new installed tile. All tiles shall be protected and carefully removed and replaced. Any trade requiring access beyond ceiling for further installation or any other use must wear white gloves to protect the ceiling tiles already installed. Any stained or damaged tiles and ceiling grid must be replaced immediately.

END OF SECTION 095113

SECTION 096513 - RESILIENT BASE AND ACCESSORIES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Resilient base.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Samples: For each exposed product and for each color and texture specified, not less than 12 inches (300 mm) long.

PART 2 - PRODUCTS**2.1 THERMOPLASTIC-RUBBER BASE**

- A. Manufacturers: Subject to compliance with requirements, provide products by:
 - 1. Johnsonite; A Tarkett Company.
- B. Product Standard: ASTM F 1861, Type TP (rubber, thermoplastic).
 - 1. Group: I (solid, homogeneous).
 - 2. Style: Cove
- C. Thickness: 0.125 inch (3.2 mm).
- D. Height: 4 inches (101.6 mm).
- E. Lengths: Continuous.
- F. Outside Corners: Job formed.
- G. Inside Corners: Job formed.
- H. Color: Johnsonite, #30, Charcoal EG.

2.2 INSTALLATION MATERIALS

- A. Adhesives: Water-resistant type recommended by resilient-product manufacturer for resilient products and substrate conditions indicated.
 - 1. VOC Content: Joint compounds shall have a VOC content of 50g/L or less when calculated according to 40 CFR59, Subpart D (EPA Method D).

PART 3 - EXECUTION

3.1 PREPARATION

- A. Prepare substrates according to manufacturer's written instructions to ensure adhesion of resilient products. Verify that substrates are dry. Remove substrate substances that are incompatible with adhesives using mechanical methods recommended by manufacturer.
- B. Fill cracks, holes, and depressions in substrates with patching compound; remove bumps and ridges to produce a uniform and smooth substrate.
- C. Do not install resilient products until they are the same temperature as the space where they are to be installed.
- D. Do not install Johnsonite Resilient Wall Base over non-porous surfaces with Tarkett 960 Wall Base Adhesive. Utilize Tarkett 946 Premium Contact Adhesive following the non-porous application instructions for all non-porous surface installations.
- E. Steel wall and steel column surfaces must be mechanically abraded to assist with the adhesive bond. The wall must be cleaned to remove all dirt, rust and other contaminants. When applying adhesive the non-porous installation instructions must be followed.
- F. Immediately before installation, clean substrates to be covered by resilient products. Do not use liquid solvents or adhesive removers.

3.2 RESILIENT BASE INSTALLATION

- A. Allow coiled wall base to lay flat for at least 24 hours, between 65° and 85°F (18.3° and 29.4°C) with HVAC system operating, prior to installation.
- B. For installations on porous wall surfaces, apply Tarkett 960 Wall Base Adhesive to the ribbed surface (back) of the wall base with a 1/8" square-notched trowel. The adhesive should cover 80% of the back surface. Leave a 1/4" (6.35mm) uncovered space at the top of the wall base to prevent the adhesive from oozing onto the wall above the base when installed. Once contact is made to the wall surface, the wall base cannot be moved.
- C. For installations on non-porous wall surfaces (i.e.: metal, epoxy paint, ceramics, etc.) apply Tarkett 946 Premium Contact Adhesive to both the wall surface and the back of the wall base. Allow adhesive to thoroughly dry to the touch.
- D. Comply with manufacturer's written instructions for installing resilient base.
- E. Apply resilient base to walls, columns, pilasters, casework and cabinets in toe spaces, and other permanent fixtures in rooms and areas where base is required.
- F. Install resilient base in lengths as long as practical without gaps at seams and with tops of adjacent pieces aligned.

- G. Tightly adhere resilient base to substrate throughout length of each piece, with base in continuous contact with horizontal and vertical substrates.
- H. Do not stretch resilient base during installation.
- I. On masonry surfaces or other similar irregular substrates, fill voids along top edge of resilient base with manufacturer's recommended adhesive filler material.
- J. Job-Formed Corners:
 - 1. Outside Corners: Use straight pieces of maximum lengths possible and form with returns not less than 4 inches (101.6 mm) in length.
 - a. Form without producing discoloration (whitening) at bends.
 - 2. Inside Corners: Use straight pieces of maximum lengths possible and form with returns not less than 4 inches (101.6 mm) in length. Miter or cope corners to minimize open joints.

3.3 CLEANING AND PROTECTION

- A. Comply with manufacturer's written instructions for cleaning and protecting resilient products.
- B. Cover resilient products subject to wear and foot traffic until Substantial Completion.

END OF SECTION 096513

SECTION 097720 –FIBERGLASS REINFORCED WALL PANELS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Prefinished polyester glass reinforced plastic sheets and adhered to unfinished gypsum wallboard with PVC trim.

1.3 REFERENCES

- A. American Society for Testing and Materials (ASTM)
 - 1. ASTM D 256 Izod Impact Strengths (ft #/in)
 - 2. ASTM D 570 Water Absorption (%)
 - 3. ASTM D 638 Tensile Strengths (psi) & Tensile Modulus (psi)
 - 4. ASTM D 790 Flexural Strengths (psi) & Flexural Modulus (psi)
 - 5. ASTM D 2583 Barcol Hardness
 - 6. ASTM D 5319 Standard Specification for Glass-Fiber Reinforced Polyester Wall and Ceiling Panels.
 - 7. ASTM E 84 Standard Test Method for Surface Burning Characteristics of Building Materials.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: Submit sufficient manufacturer's data to indicate compliance with these specifications, including:
 - a. Preparation instructions and recommendations.
 - b. Storage and handling requirements and recommendations.
 - c. Installation methods.
 - 2. Samples: Submit appropriate section of panel for specified finish indicating the color, texture, and pattern required.
 - a. Exposed Molding and Trim: Provide samples of each type, finish, and color.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.6 QUALITY ASSURANCE

- A. Conform to building code requirements for interior finish for smoke and flame spread requirements as tested in accordance with:
 - 1. ASTM E 84 – Class C for wall rating.
- B. Sanitary Standards: System components and finishes to comply with:
 - 1. United States Department of Agriculture (USDA) requirements for food preparation facilities, incidental contact.
 - 2. Food and Drug Administration (FDA) 1999 Food Code 6-101.11.

1.7 DELIVERY, STORAGE AND HANDLING

- A. Deliver materials factory packaged on strong pallets.
- B. Store panels and trim lying flat, under cover and protected from the elements and damage. Allow panels to acclimate to room temperature (range of 60 to 75°F) for 48 hours prior to installation.

1.8 PROJECT CONDITIONS

- A. Environmental Limitations: Building are to be fully enclosed prior to installation with sufficient heat (70°) and ventilation consistent with good working conditions for finish work
- B. During installation and for not less than 48 hours before, maintain an ambient temperature and relative humidity within limits required by type of adhesive used and recommendation of adhesive manufacturer. Provide ventilation to disperse fumes during application of adhesive as recommended by the adhesive manufacturer.

PART 2 - PRODUCTS

2.1 MANUFACTURER

- A. Basis of Design:
 - 1. Manufacturer: Marlite; 1 Marlite Drive, Dover, OH 44622. 800-377-1221 FAX (330) 343-4668 Email: info@marlite.com www.marlite.com.

2.2 PANELS

- A. Fiberglass reinforced thermosetting polyester resin panel sheets complying with ASTM D 5319.
 - 1. Coating: Multi-layer print, primer and finish coats or applied over-layer.
 - 2. Panel Dimensions:
 - a. Thickness: 0.090 inch (2.29 mm) nominal
 - b. Width: 4'-0" (1.22 m) nominal
 - c. Length: 8'-0" (2.4 m) nominal
 - 3. Tolerance:
 - a. Length and Width: +/-1/8 inch (3.175 mm)

- b. Square: Not to exceed 1/8 inch for 8-foot (2.4 m) panel or 5/32 inch (3.96 mm) for 10-foot (2.4m) panel
 - B. Properties: Resistant to rot, corrosion, staining, denting, peeling, and splintering.
 - 1. Flexural Strength - 1.0×10^4 psi per ASTM D 790. (7.0 kilogram-force/square millimeter)
 - 2. Flexural Modulus - 3.1×10^5 psi per ASTM D 790. (217.9 kilogram-force/square millimeter)
 - 3. Tensile Strength - 7.0×10^3 psi per ASTM D 638. (4.9 kilogram-force/square millimeter)
 - 4. Tensile Modulus - 1.6×10^5 psi per ASTM D 638. (112.5 kilogram-force/square millimeter)
 - 5. Water Absorption - 0.72% per ASTM D 570.
 - 6. Barcol Hardness (scratch resistance) of 35 55 as per ASTM D 2583.
 - 7. Izod Impact Strength of 72 ft. lbs./in ASTM D 256
 - C. Back Surface: Smooth. Imperfections which do not affect functional properties are not cause for rejection.
 - D. FRP-1:
 - 1. Symmetrix with Sani-Coat Panels.
 - 2. Color: C100-G63 White.
 - 3. Pattern: Subway tile
 - E. FRP-2:
 - 1. Standard FRP.
 - 2. Color: S 100G White.
 - 3. Surface Finish: Smooth
- 2.3 MOLDINGS
- A. Aluminum Trim at Restrooms:
 - 1. F551 Inside Corner
 - 2. F560 Outside Corner, 7/16 inch
 - 3. F570 Edge (top, bottom, vertical), 3/8 inch
 - 4. Finish: Clear anodized coating, No. 4 satin finish.
 - B. SaniSeal Trim at All Other Locations: Co-extruded, dual-durometer polypropylene/monprene profiles with Dual-Seal Technology and high-performance pressure sensitive adhesive.
 - 1. S650 Inside Corner
 - 2. S660 Outside Corner
 - 3. S665 Division
 - 4. S670 Edge
 - 5. Color: White
- 2.4 ACCESSORIES
- A. Fasteners: Non-staining nylon drive rivets.
 - 1. Match panel colors.
 - 2. Length to suit project conditions.

B. Adhesive: Either of the following construction adhesives complying with ASTM C 557.

1. Marlite C-551 FRP Adhesive - Water- resistant, non-flammable adhesive.
2. Titebond Advanced Polymer Panel Adhesive – VOC compliant, non-flammable, environmentally safe adhesive.

C. Sealant:

1. Marlite Brand MS-251 White Silicone Sealant.

PART 3 - EXECUTION

3.1 PREPARATION

A. Examine backup surfaces to determine that corners are plumb and straight, surfaces are smooth, uniform, clean and free from foreign matter, nails countersunk, joints and cracks filled flush and smooth with the adjoining surface.

1. Verify that stud spacing does not exceed 24 inches (61cm) on-center.

B. Repair defects prior to installation.

1. Level wall surfaces to panel manufacturer's requirements. Remove protrusions and fill indentations.

3.2 INSTALLATION

A. Comply with manufacturer's recommended procedures and installation sequence.

B. Cut sheets to meet supports allowing 1/8-inch (3 mm) clearance for every 8 feet (2.4m) of panel.

1. Cut and drill with carbide tipped saw blades or drill bits or cut with shears.
2. Pre-drill fastener holes 1/8 inch (3mm) oversize with high speed drill bit.
 - a. Space at 8 inches (200mm) maximum on center at perimeter, approximately 1 inch from panel edge.
 - b. Space at in field in rows 16 feet (40.64cm) on center, with fasteners spaced at 12 inches (30.48 cm) maximum on center.

C. Apply panels to board substrate, above base, vertically oriented with seams plumb and pattern aligned with adjoining panels.

1. Install panels with manufacturer's recommended gap for panel field and corner joints.
 - a. Adhesive trowel and application method to conform to adhesive manufacturer's recommendations.
 - b. Drive fasteners for snug fit. Do not over-tighten.

D. Apply panel moldings to all panel edges using silicone sealant providing for required clearances.

1. All moldings must provide for a minimum 1/8 inch (3mm) of panel expansion at joints and edges, to insure proper installation.

2. Apply sealant to all moldings, channels and joints between the system and different materials to assure watertight installation.

3.3 CLEANING

- A. Remove excess sealant from panels and moldings. Wipe panel down using a damp cloth and mild soap solution or cleaner.
- B. Refer to manufacturer's specific cleaning recommendations Do not use abrasive cleaners.
- C. Cover panels to protect from construction damage until work in Bakery is complete and ready for Substantial Completion.

END OF SECTION 097720

SECTION 099113 - EXTERIOR PAINTING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes surface preparation and the application of paint systems including, but not limited to, the following:
 - 1. Cart Corral Columns.
 - 2. Exterior columns and beams.
 - 3. Exterior steel framing at canopies and awnings.
 - 4. Exterior frame (outside and inside surfaces).
 - 5. Exterior bollards.
 - 6. Exterior railings.
 - 7. Steel deck, beams, columns, and miscellaneous surfaces inside Entry Vestibule.
- B. Related Requirements:
 - 1. Section 055000 "Metal Fabrications" for shop priming metal fabrications.
 - 2. Section 055213 "Pipe and Tube Railings" for shop priming pipe and tube railings.

1.3 DEFINITIONS

- A. MPI Gloss Level 1: Not more than five units at 60 degrees and 10 units at 85 degrees, according to ASTM D 523.
- B. MPI Gloss Level 3: 10 to 25 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.
- C. MPI Gloss Level 4: 20 to 35 units at 60 degrees and not less than 35 units at 85 degrees, according to ASTM D 523.
- D. MPI Gloss Level 5: 35 to 70 units at 60 degrees, according to ASTM D 523.
- E. MPI Gloss Level 6: 70 to 85 units at 60 degrees, according to ASTM D 523.
- F. MPI Gloss Level 7: More than 85 units at 60 degrees, according to ASTM D 523.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product. Include preparation requirements and application instructions.

2. Samples: For each type of paint system and each color and gloss of topcoat.
3. Shop Drawings: Plan showing locations of each paint system, including color and gloss.

1.5 MAINTENANCE MATERIAL SUBMITTALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 1. Paint: Not less than 2 gal. (7.6 L) of each material and color applied.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F (7 deg C).
 1. Maintain containers in clean condition, free of foreign materials and residue.
 2. Remove rags and waste from storage areas daily.

1.7 FIELD CONDITIONS

- A. Apply paints only when temperature of surfaces to be painted and ambient air temperatures are between 50 and 95 deg F (10 and 35 deg C).
- B. Do not apply paints in snow, rain, fog, or mist; when relative humidity exceeds 85 percent; at temperatures less than 5 deg F (3 deg C) above the dew point; or to damp or wet surfaces.

1.8 QUALITY ASSURANCE

- A. Mockups: Apply mockups of each paint system indicated and each color and finish selected to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
 1. Architect will select one surface to represent surfaces and conditions for application of each paint system.
 - a. Exterior door frame and column surfaces: Provide samples of at least 25 sq. ft. (2.3 sq. m).
 2. Final approval of color selections will be based on mockups and may be incorporated in the work.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers:
 1. Sherwin-Williams Company; National Account contact Dan Giddens; phone 703-929-6487; email dan.giddens@sherwin.com.
- B. Products: Subject to compliance with requirements, provide product listed in the Exterior Painting Schedule for the paint category indicated. See architectural drawings for locations.

2.2 PAINT, GENERAL

- A. MPI Standards: Products shall comply with MPI standards indicated and shall be listed in its "MPI Approved Products Lists."
- B. Material Compatibility:
 - 1. Materials for use within each paint system shall be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.
 - 2. For each coat in a paint system, products shall be recommended in writing by topcoat manufacturers for use in paint system and on substrate indicated.
- C. VOC Content: Exterior paints applied at the project site shall have the following VOC limits, exclusive of colorants added to a tint base, when calculated according to 40 CFR59, Subpart D (EPA Method D).
 - 1. Flat paints and coatings: 50 g/L.
 - 2. Non-flat paints and coatings: 150 g/L.
 - 3. Primers, sealers and undercoaters: 200 g/L.
 - 4. Comply with regional or state/city VOC regulations when lower than the above, such as: SCAQMD, CARB, OTC and OTC Phase II.
- D. Colors: See Drawings.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.
- B. Verify suitability of substrates, including surface conditions and compatibility with existing finishes and primers.
- C. Proceed with coating application only after unsatisfactory conditions have been corrected.
 - 1. Application of coating indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Comply with manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual" applicable to substrates and paint systems indicated.
- B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.
 - 1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection.
- C. Clean substrates of substances that could impair bond of coatings, including dust, dirt, oil, grease, loose mill scales, rust, stains, and incompatible paints and encapsulants.
 - 1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce coating systems indicated.

- D. Surface Preparation for Non-Galvanized, Primed and Painted Steel Surfaces:
 - 1. SSPC-SP3, "Power Tool Cleaning".
- E. Surface Preparation for Galvanized Steel:
 - 1. SSPC-SP1 "Solvent Cleaning".
 - a. Use Clean 'n Etch by Great Lakes Laboratories or approved equivalent.
- F. Surface Preparation for Shop-Primed Door Frames: As recommended by door frame manufacturer.
- G. Surface Preparation for Brick: Allow new brick work to cure at least 30 days before painting. If painting before 30 days, see manufacturer's recommendation. Repair cracks, voids, and other holes with elastomeric patch or sealant.
- H. Spot/Stripe prime bare areas, seams, joints, connections, corners and damaged coating. Use material similar in quality to finished coating.
- I. Priming: Immediately after surface preparation, apply primer according to manufacturer's written instructions and at rate recommended by SSPC to provide a minimum dry film thickness of 1.5 mils (0.038 mm). Use priming methods that result in full coverage of joints, corners, edges, and exposed surfaces.
- J. Test Patch: Prior to starting the painting work, apply a test patch for each of the painting systems specified. Allow paints to dry at least one week before testing adhesion and compatibility. If the coating system is incompatible or adhesion fails, use surface preparation methods recommended by paint manufacturer and apply a new test patch. Remove, repeat surface preparation, and reapply if adhesion is determined to be poor and inadequate. Re-test until proper adhesion and compatibility are determined.

3.3 APPLICATION

- A. Apply paints according to manufacturer's written instructions and recommendations in "MPI Manual."
- B. Apply paints to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and color breaks.
- C. Spray and back roll on porous and rough surfaces to ensure continuous coverage in uniform thickness.

3.4 CLEANING AND PROTECTION

- A. Protect work of other trades against damage from paint application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Architect, and leave in an undamaged condition.
- B. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

3.5 EXTERIOR COATING SCHEDULE

A. Steel (Galvanized and Non-Galvanized): Existing and New Steel.

1. Epoxy Zinc-Rich Primer: Complying with MPI#20 and compatible with topcoat.
 - a. Product: Sherwin Williams; Protective & Marine Zinc Clad IV B69A8/B69V8 (VOC <340 g/L) (3 to 5 mils dry)
 - 1) Surface Prep: SSPC SP6, cleaned new or repaired ferrous metal
2. Epoxy Intermediate Coat: Complying with MPI #77 and compatible with primer and topcoat.
 - a. Product: Sherwin Williams; Protective & Marine Tile-Clad HS Epoxy B62WZ111/B60VZ70 (VOC <378 g/L) (2.2 to 3.9 mils dry)
3. Topcoat: Urethane aliphatic, two-component, pigmented, semi-gloss (MPI Gloss Level 5), complying with MPI #72 and compatible with undercoat.
 - a. Product: Sherwin Williams; Protective & Marine Acrolon 218 HS B65W611/B65V600 (VOC <300 g/L) (3 to 6 mils dry)

END OF SECTION 099113

SECTION 099123 - INTERIOR PAINTING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes surface preparation and the application of paint systems on the following interior substrates:
 - 1. Concrete masonry units (CMUs).
 - 2. Steel and iron.
 - 3. Steel decking.
 - 4. Gypsum board.
 - 5. ASJ insulation covering.
 - 6. Floor striping (if applicable).
 - 7. All ceiling components (underside of roof): steel beams, deck, joists, conduits, junction boxes, sprinklers, miscellaneous steel, duct work, and wire mesh return grilles.

1.3 DEFINITIONS

- A. MPI Gloss Level 1: Not more than five units at 60 degrees and 10 units at 85 degrees, according to ASTM D 523.
- B. MPI Gloss Level 2: Not more than 10 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.
- C. MPI Gloss Level 3: 10 to 25 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.
- D. MPI Gloss Level 4: 20 to 35 units at 60 degrees and not less than 35 units at 85 degrees, according to ASTM D 523.
- E. MPI Gloss Level 5: 35 to 70 units at 60 degrees, according to ASTM D 523.
- F. MPI Gloss Level 6: 70 to 85 units at 60 degrees, according to ASTM D 523.
- G. MPI Gloss Level 7: More than 85 units at 60 degrees, according to ASTM D 523.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product. Include preparation requirements and application instructions.

- a. Include Printout of current "MPI Approved Products List" for each product category specified, with the proposed product highlighted.
2. Samples: For each type of paint system and in each color and gloss of topcoat.
3. Shop Drawings: Plan showing locations of each paint system, including color and gloss.

1.5 MAINTENANCE MATERIAL SUBMITTALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Paint: Not less than 2 gal. (7.6 L) of each material and color applied.

1.6 QUALITY ASSURANCE

- A. Mockups: Apply mockups of each paint system indicated and each color and finish selected to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

1. Architect will select one surface to represent surfaces and conditions for application of each paint system.

- a. Gypsum Board and CMU Surfaces: Provide samples of at least 50 sq. ft. (4.5 sq. m).

2. Final approval of color selections will be based on mockups and may be incorporated in the work.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F (7 deg C).

1. Maintain containers in clean condition, free of foreign materials and residue.
 2. Remove rags and waste from storage areas daily.

1.8 FIELD CONDITIONS

- A. Apply paints only when temperature of surfaces to be painted and ambient air temperatures are between 50 and 95 deg F (10 and 35 deg C).

- B. Do not apply paints in snow, rain, fog, or mist; relative humidity exceeds 50 percent; at temperatures less than 5 deg F (3 deg C) above the dew point; or to damp or wet surfaces.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers:

1. Sherwin-Williams Company; National Account contact Dan Giddens; phone 703-929-6487; email dan.giddens@sherwin.com.

- B. Products: Subject to compliance with requirements, provide products listed in the Interior Painting Schedule for the paint category indicated.

2.2 PAINT, GENERAL

- A. MPI Standards: Products shall comply with MPI standards indicated and shall be listed in its "MPI Approved Products Lists."
- B. Material Compatibility:
 - 1. Materials for use within each paint system shall be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.
 - 2. For each coat in a paint system, products shall be recommended in writing by topcoat manufacturers for use in paint system and on substrate indicated.
- C. VOC Content: Interior paints and coatings applied at the project site shall have the following VOC limits, exclusive of colorants added to a tint base, when calculated according to 40 CFR59, Subpart D (EPA Method D).
 - 1. Flat paints and coatings: 50 g/L.
 - 2. Non-flat paints and coatings: 150 g/L.
 - 3. Primers, sealers and undercoaters: 200 g/L.
 - 4. Anticorrosive and anti-rust paints applied to ferrous metals: 250 g/L.
 - 5. Zinc-rich industrial maintenance primers: 340 g/L.
 - 6. Pretreatment wash primers: 420 g/L.
 - 7. Floor coatings: 100 g/L.
 - 8. Dry-fall coatings: 400 g/L.
 - 9. Comply with regional or state/city VOC regulations when lower than the above, such as: SCAQMD, CARB, OTC and OTC Phase II
- D. Colors: See Drawings.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.
- B. Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:
 - 1. Concrete and CMU: 12 percent.
 - 2. Gypsum Board: 12 percent.
- C. Verify suitability of substrates, including surface conditions and compatibility with existing finishes and primers.
- D. Proceed with coating application only after unsatisfactory conditions have been corrected.
 - 1. Application of coating indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Comply with manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual" applicable to substrates and paint systems indicated.
- B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.
 - 1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection if any.
- C. Clean substrates of substances that could impair bond of coatings, including dust, dirt, oil, grease, loose mill scales, rust, stains, and incompatible paints and encapsulants.
 - 1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce coating systems indicated.
- D. Surface Preparation for Non-Galvanized, Primed and Painted Steel Surfaces:
 - 1. SSPC-SP3, "Power Tool Cleaning".
- E. Surface Preparation for Galvanized Steel:
 - 1. SSPC-SP1 "Solvent Cleaning".
 - a. Use Clean 'n Etch by Great Lakes Laboratories or approved equivalent.
- F. Surface Preparation for Shop-Primed Doors and Frames: As recommended by door and frame manufacturer.
- G. Surface Preparation for Concrete/CMU:
 - 1. Remove all loose mortar and foreign material. Fill all bug holes, air pockets, and all other voids with a cement patching compound compatible with the paint coating.
 - 2. The masonry must be cured for at least 30 days at 75 degrees F (23 degrees C); pH of the surface between 6 and 9; and moisture content less than 12 percent.
 - 3. Surfaces must be free of laitance, dust, dirt, loose cement at time of paint application.
- H. Surface Preparation for Gypsum Board:
 - 1. Set and spackle all nail and screw heads. Spackle exterior surfaces with exterior grade compounds.
 - 2. Sand all taped joints and spackled nail heads smooth.
 - 3. Drywall must be clean and dry and all dust thoroughly removed at time of paint application.
- I. Spot/Stripe prime bare areas, seams, joints, connections, corners and damaged coating. Use material similar in quality to finished coating.
- J. Priming: Immediately after surface preparation, apply primer according to manufacturer's written instructions and at rate recommended by SSPC to provide a minimum dry film thickness of 1.5 mils (0.038 mm). Use priming methods that result in full coverage of joints, corners, edges, and exposed surfaces.

3.3 APPLICATION

- A. Apply paints according to manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual."
- B. Apply paints to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and color breaks.

3.4 CLEANING AND PROTECTION

- A. Protect work of other trades against damage from paint application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Architect, and leave in an undamaged condition.
- B. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

3.5 INTERIOR PAINTING SCHEDULE

A. CMU Substrates:

1. Latex System:

a. Block Filler:

- 1) Product: Sherwin Williams; PrepRite Interior-Exterior Latex Block Filler B25W00025.

b. Two Topcoats: Latex, interior (MPI Gloss Level 3).

- 1) Product: Sherwin Williams; ProMar 200 Zero VOC Interior Latex Eg-Shel B20W12651.

B. Steel Substrates (except steel base, dock steel):

1. Latex System:

a. Primer Coat: Fast dry, for shop and field application.

- 1) Product: Sherwin Williams; Pro Industrial Pro-Cryl Universal Primer B66W01310.

b. Two Topcoats: Latex, interior (MPI Gloss Level 5).

- 1) Product: Sherwin Williams; Pro Industrial DTM Acrylic Semi-Gloss B66W01151.

2. Dry-Fall System: All ceiling components (underside of roof).

a. Primer Coat: Fast dry, alkyd metal primer.

- 1) Product: Sherwin Williams; Kem Bond HS Universal Metal Primer B50WZ4.

- b. Topcoat: Dry fall, acrylic, flat (MPI Gloss Level 1).
 - 1) Product: Sherwin Williams; Pro Industrial Waterborne Acrylic Dryfall Flat W42W00181 (White).
- C. Gypsum Board Substrates:
 - 1. Latex System:
 - a. Primer Coat: Primer sealer, latex, interior.
 - 1) Product: Sherwin Williams; ProMar 200 Zero VOC Interior Latex Primer B28W02600.
 - b. Two Topcoats: Latex, interior (MPI Gloss Level 3).
 - 1) Product: Sherwin Williams; ProMar 200 Zero VOC Interior Latex Eg-Shel B20W12651.
- D. Cotton or Canvas and ASJ Insulation-Covering Substrates: Including pipe and duct coverings.
 - 1. Latex System:
 - a. Primer Coat: Primer sealer, latex, interior.
 - 1) Product: Sherwin Williams; ProMar 200 Zero VOC Interior Latex Primer B28W02600.
 - b. Two Topcoats: Latex, interior (MPI Gloss Level 3).
 - 1) Product: Sherwin Williams; ProMar 200 Zero VOC Interior Latex Eg-Shel B20W12651.
- E. Steel Angles at Ram Protection Type 2, Dock Steel Angles, Plates and Channels, Floor Striping, and Similar Areas Subject to Heavy Traffic Impact.
 - 1. Epoxy Paint:
 - a. Two Coats: Sherwin Williams; Macropoxy 646-100 Fast Cure Epoxy.
 - b. Colors: As indicated on Drawings.
 - 2. Floor Striping (if applicable):
 - a. Mask to protect areas around the stripes. Grind or shot blast surfaces to be striped to a profile of CSP 2 to 3. Remove all dirt, dust and loose materials. Surfaces must be clean and dry before painting.
 - b. Use machine equipped with dust collection to minimize dusting in warehouse.
 - c. Apply to pathway and caution striping in colors and at locations indicated on Drawings or as required by authority having jurisdiction.

END OF SECTION 099123

SECTION 099600 - HIGH-PERFORMANCE COATINGS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes surface preparation and the application of high-performance coating systems on the following substrates:

- 1. Interior Substrates:

- a. Concrete floor surface at Bakery
 - b. Concrete floor surface at Restrooms.
 - c. Concrete floor at IT Room.
 - d. CMU walls in IT Room.
 - e. Drywall in Bakery.
 - f. Concrete pit at Scrubbing Station.
 - g. Topside of drywall lid over IT Room.

- B. Related Section:

- 1. Section 092900 "Gypsum Board" for cement board behind cove base.

1.3 DEFINITIONS

- A. MPI Gloss Level 3: 10 to 25 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.
- B. MPI Gloss Level 4: 20 to 35 units at 60 degrees and not less than 35 units at 85 degrees, according to ASTM D 523.
- C. MPI Gloss Level 5: 35 to 70 units at 60 degrees, according to ASTM D 523.
- D. MPI Gloss Level 6: 70 to 85 units at 60 degrees, according to ASTM D 523.
- E. MPI Gloss Level 7: More than 85 units at 60 degrees, according to ASTM D 523.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product. Include preparation requirements and application instructions.

- a. Include printout of current "MPI Approved Products List" for each product category specified, with the proposed product highlighted.
 2. Samples: For each type of coating system and in each color and gloss of topcoat indicated.
 3. Shop Drawings: Plan showing locations of each paint system, including color and gloss.
- 1.5 CLOSEOUT SUBMITTALS
- A. Maintenance data.
- 1.6 QUALITY ASSURANCE
- A. Mockups: Apply mockups of each coating system indicated to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
 1. Architect will select one surface to represent surfaces and conditions for application of each coating system.
 2. Final approval of color selections will be based on mockups.
- 1.7 DELIVERY, STORAGE, AND HANDLING
- A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F (7 deg C).
 1. Maintain containers in clean condition, free of foreign materials and residue.
 2. Remove rags and waste from storage areas daily.
- 1.8 FIELD CONDITIONS
- A. Apply coatings only when temperature of surfaces to be coated and ambient air temperatures are between 50 and 95 deg F (10 and 35 deg C).
 - B. Do not apply coatings when relative humidity exceeds 85 percent; at temperatures less than 5 deg F (3 deg C) above the dew point; or to damp or wet surfaces.
 - C. Do not apply exterior coatings in snow, rain, fog, or mist.
 - D. General Contractor shall be responsible for providing lighting level during installation that simulates permanent lighting conditions.

PART 2 - PRODUCTS

2.1 MANUFACTURERS AND PRODUCTS

- A. Manufacturers and Products:
 1. PPG Architectural Coatings; Coraflon ADS Intermix Metallic for exterior ferrous.
 2. Sherwin-Williams Company, General Polymers; Static Dissipative coating system #18 SD for interior concrete floor.
 3. Antistatic Industries, A Division of ADM Tronics Unlimited; StaticPaint, one-part static dissipative paint for CMU wall substrates.
 4. Sherwin-Williams Company; Epoxy-Modified Latex System for drywall substrates.

5. Laticrete International; Spartacoat, Sparta-Flex Pure, two-part polyaspartic aliphatic polyurea finish with traction additive for concrete floor.
6. Silikal America; Silikal 368 CQ Decorative Quartz, for concrete floor.
7. Sherwin-Williams Company; Multi-Purpose Epoxy for floor and walls inside concrete scrubber pit.
8. Sherwin-Williams Company; General Polymers; 3552W EPO-Flex Flexible Wall Epoxy

2.2 HIGH-PERFORMANCE COATINGS, GENERAL

- A. MPI Standards: Products shall comply with MPI standards indicated and shall be listed in its "MPI Approved Products Lists."
- B. Conductivity Resistance: Provide static dissipative floor coatings in the range of $10^7 - 10^9$ OHMS.
- C. Material Compatibility:
 1. Materials for use within each paint system shall be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.
 2. For each coat in a paint system, products shall be recommended in writing by topcoat manufacturers for use in paint system and on substrate indicated.
 3. Products shall be of same manufacturer for each coat in a coating system.
- D. VOC Content: Interior paints and coatings applied at the project site shall have the following VOC limits, exclusive of colorants added to a tint base, when calculated according to 40 CFR59, Subpart D (EPA Method D).
 1. Primers, sealers and undercoaters: 200 g/L
 2. Anticorrosive and anti-rust paints applied to ferrous metals: 250 g/L
 3. Zinc-rich industrial maintenance primers: 340 g/L
 4. Pretreatment wash primers: 420 g/L
 5. Floor coatings: 100 g/L
 6. Comply with regional or state/city VOC regulations when lower than the above, such as: SCAQMD, CARB, OTC and OTC Phase II
- E. Colors: See Drawings.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.
- B. Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:
 1. Concrete and CMU: 12 percent.
 2. Gypsum Board: 12 percent.
- C. Verify suitability of substrates, including surface conditions and compatibility with existing finishes and primers.

- D. Proceed with coating application only after unsatisfactory conditions have been corrected.

- 1. Application of coating indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Comply with manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual" applicable to substrates and coating systems indicated.

- B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.

- 1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection if any.

- C. Clean substrates of substances that could impair bond of coatings, including dust, dirt, oil, grease, and incompatible paints and encapsulants.

- 1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce coating systems indicated.

- D. Surface Preparation for Nongalvanized Steel:

- 1. SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."

- E. Surface Preparation for Concrete/CMU Walls:

- 1. Remove all loose mortar and foreign material. Fill all bug holes, air pockets, and all other voids with a cement patching compound compatible with the paint coating.
 - 2. The masonry must be cured for at least 30 days at 75 degrees F (23 degrees C); pH of the surface between 6 and 9; and moisture content less than 12 percent.
 - 3. Surfaces must be free of laitance, dust, dirt, loose cement at time of paint application.

- F. Surface Preparation for Gypsum Board:

- 1. Set and spackle all nail and screw heads. Spackle exterior surfaces with exterior grade compounds.
 - 2. Sand all taped joints and spackled nail heads smooth.
 - 3. Drywall must be clean and dry and all dust thoroughly removed at time of paint application.

- G. Surface Preparation for Concrete Floors:

- 1. Concrete to be moisture cured for a minimum of 7 days at 70-deg F and to be fully cured for a minimum of 28 days prior to application of the coating system.
 - 2. Remove all surface contaminants such as curing agents, membranes, paints/coatings, and other bond breakers.
 - 3. Perform moisture testing to verify moisture emission rate of no more than 5 lbs per 1000 sq. ft. per 24-hour period as determined by proper Calcium Chloride Testing. Concrete relative humidity must be 85 percent or less as measured by protimeter. Consult with coating manufacturer for treatment recommendations, if readings are greater than allowed.

- H. Spot/Stripe prime bare areas, seams, joints, connections, corners and damaged coating. Use material similar in quality to finished coating.
- I. Priming: Immediately after surface preparation, apply primer according to manufacturer's written instructions and at rate recommended by SSPC to provide a minimum dry film thickness of 1.5 mils (0.038 mm). Use priming methods that result in full coverage of joints, corners, edges, and exposed surfaces.

3.3 APPLICATION

- A. Apply high-performance coatings according to manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual." High-Performance Coatings are moisture sensitive and moisture contamination is to be avoided.
 - 1. Pre-mix two component static control coatings to ensure dispersion of conductive elements evenly throughout the resin.
- B. Apply coatings to produce surface films without cloudiness, spotting, holidays, laps, brush marks, runs, sags, ropiness, or other surface imperfections. Produce sharp glass lines and color breaks.
- C. Test Static Dissipative floor coating and Static Control wall coating in accordance with NFPA 99 for static resistance. If deviations from acceptable range of ohms resistance occurs, re-coat surfaces and re-test until acceptable range of ohms resistance is achieved.
- D. Cove Base Installation: Install cove base according to coating manufacturer's installation instructions
 - 1. Verify that cement board substrate is installed behind cove base.
 - 2. Apply primer. Two coats may be necessary to fully coat cement board surfaces.
 - 3. Apply surface material to wall and floor in one step using a coving tool.
 - 4. Apply top coat.
- E. Floor Transition Installation: Install flooring system flush with adjacent floor finish according to coating manufacturer's installation instructions
 - 1. Saw-cut 6 to 12 inches from termination. Remove 1/2 to 3/4 inch of concrete at edge, staying inside area to receive flooring system. Demolish and remove concrete to saw-cut edge. Maintain 15 to 20 degrees slope throughout.
 - 2. Saw-cut all lines 3/4 to 1 inch below concrete slab at termination. Saw-cuts should be two times total thickness of flooring system.
 - 3. Trowel Silikal R17 mortar to within 1/8 inch of termination, at rate recommended by manufacturer, to match level of surrounding slab.
 - 4. Apply Silikal R61 resin binder, at rate recommended by manufacturer.
 - 5. Apply flooring finish materials as specified.

3.4 INTERIOR HIGH-PERFORMANCE COATING SCHEDULE

- A. Concrete Substrates, Horizontal Surfaces.
 - 1. Static Dissipative Floor Coating System in IT Room (F-2):
 - a. Prime Coat: General Polymers; Conductive Epoxy Primer, #3524.
 - b. Intermediate Coat: Matching topcoat.

- c. Topcoat: General Polymers; Static Dissipative Aliphatic Urethane #4620E (MPI Gloss Level 4).
- 2. Bakery, and Restrooms (F-4):
 - a. Moisture Vapor Treatment: Silikal RE40 (if required per Manufacturer Instructions)
 - b. Prime Coat: Silikal R41, at rate recommended by manufacturer
 - c. Roll Coat: Silikal R62 FS, at rate recommended by manufacturer.
 - 1) Color: MicroFlake #3, broadcast to saturation.
 - d. First Topcoat: Silikal R71 RE, at rate recommended by manufacturer.
 - e. Second Topcoat: Silikal R71 RE, at rate recommended by manufacturer.
 - f. Cove Base: Silikal CQ-61 with integral cove base. Trowel base to bottom of L-strip.
- B. CMU Substrates:
 - 1. Static Dissipative Wall Paint in I.T. Room:
 - a. Block Filler: Block filler, latex, interior/exterior, MPI #4, and as recommended by paint manufacturer.
 - 1) Product: Sherwin Williams; PreRite Block Filler Int./Ext. B25W00025.
 - b. Intermediate Coat: Matching topcoat.
 - c. Topcoat: Antistatic Industries; StaticPaint, one-part coating (Gloss Level 3).
- C. Drywall Substrates:
 - 1. Epoxy-Modified Latex System in Bakery:
 - a. Prime Coat: Sherwin Williams; ProMar 200 Zero VOC Interior Latex Primer B28W02600 (VOC <50 g/L) (1.0 mil dry)
 - b. Intermediate Coat: Sherwin Williams; Pro Industrial Water Based Catalyzed Epoxy Gloss, B73-300 Series (VOC <50 g/L) (2 to 5 mils dry, per coat).
 - c. Topcoat: Sherwin Williams; Pro Industrial Water Based Catalyzed Epoxy Gloss, B73-300 Series (VOC <50 g/L) (2 to 5 mils dry, per coat).
- D. Concrete Substrates, Horizontal and Vertical Surfaces of Pit at Scrubbing Station.
 - 1. Epoxy Coating System for Concrete Pit at Scrubbing Station:
 - a. Prime Coat: Sherwin Williams, Kem Cati-Coat HS Epoxy Filler/Sealer B42W400 (VOC 250 g/L) (10 to 20 mils dry per coat), if required to fill holes, cracks and voids in concrete substrate.
 - b. Intermediate Coat: Sherwin Williams, Dura-Plate 235 Multi-Purpose Epoxy (VOC 280 g/L) (4 to 8 mils dry per coat).
 - c. Topcoat: Sherwin Williams, Dura-Plate 235 Multi-Purpose Epoxy (VOC 280 g/L) (4 to 8 mils dry per coat).
- E. Topside of Drywall Lid:
 - 1. Epoxy Coating System for Topside of Drywall Lid over IT Room.

- a. Prime Coat: Sherwin Williams, General Polymers GP 3479 Epoxy Primer.
- b. Intermediate Coat: Sherwin Williams, General Polymers GP 3552W EPO-Flex Flexible Wall Epoxy (VOC 100 g/L) (8 to 10 mils wet per coat).
- c. Topcoat: Sherwin Williams, General Polymers GP 3552W EPO-Flex Flexible Wall Epoxy (VOC 100 g/L) (8 to 10 mils wet per coat).

END OF SECTION 099600

SECTION 102600 – WALL SURFACE PROTECTION**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Wall covering.
- B. Related Section:
 - 1. Section 102600.01 "Wall and Door Protection" for handrails, corner guards, bumper guards, crash rails, accent rails, wall panels, door protection.

1.3 REFERENCES

- A. National codes (IBC, UBC, SBCCI, BOCA and Life Safety)
- B. American Society for Testing and Materials (ASTM)

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each system component and installation accessory required, including installation methods for each type of substrate.
 - 2. Shop Drawings: Show locations, extent and installation details of wall covering products
 - 3. Samples: For each product specified.

1.5 INFORMATIONAL SUBMITTALS

- A. Product test reports from a qualified independent testing laboratory showing compliance of each component with requirements indicated.

1.6 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.7 QUALITY ASSURANCE

- A. Installer qualifications: Engage an installer who has no less than 3 years experience in installation of systems similar in complexity to those required for this project.

- B. Manufacturer's qualifications: Not less than 5 years experience in the production of specified products and a record of successful in-service performance.
- C. Code Compliance: Assemblies should conform to all applicable codes including IBC, UBC, SBCCI, BOCA and Life Safety.
- D. Fire Performance Characteristics: Provide stainless steel components tested in accordance with ASTM E84 for Class A/1 fire characteristics.
- E. Single Source Responsibility: Provide all components of the wall protection system manufactured by the same company to ensure compatibility of color, texture and physical properties.

1.8 DELIVERY, STORAGE AND HANDLING

- A. Deliver materials to the project site in unopened original factory packaging clearly labeled to show manufacturer.
- B. Store materials flat in an enclosed and weatherproofed storage area.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Stainless Steel Wall Protection: Material to be 16 gauge Type 304 alloy with #4 satin finish.
- B. Stainless Steel Column Protection: Material to be 22 gauge Type 304 alloy with #4 satin finish.

2.1 WALL COVERING

- A. Stainless Steel Sheet: Nominal .0625" (1.59mm) thick, 16 gauge stainless steel sheet
 - 1. Textures:
 - a. Standard smooth texture at Bakery and where indicated.
 - b. Diamond Plate texture at overhead door jambs and where indicated.
 - 2. Sheet Size: 4' x 8' or 10' (1.2m x 2.4m or 3.0m)
 - 3. Fasteners: Corrosion-resistant screws for mechanical fastening.

2.2 FABRICATION

- A. Fabricate wall protection products to comply with requirements indicated for design, dimensions, detail, finish and sizes.

2.3 FINISHES

- A. Comply with NAAMM "Metal Finishes Manual" for recommendations relative to applications and designations of finishes.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and conditions under which work is to be performed and identify conditions detrimental to proper or timely completion.
- B. Do not proceed until unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Prior to installation, clean substrate to remove dirt, debris and loose particles. Perform additional preparation procedures as required by manufacturer's instructions.
- B. Take all necessary steps to prevent damage to material during installation as required in manufacturer's installation instructions.

3.3 INSTALLATION

- A. Install the work of this section in strict accordance with the manufacturer's recommendations using only approved hardware.
- B. Install all components firmly into position, level and plumb.

3.4 CLEANING

- A. Immediately upon completion of installation, clean wall protection products and accessories in accordance with manufacturer's recommended cleaning method.
- B. Remove surplus materials, rubbish and debris resulting from installation as work progresses and upon completion of work.

3.5 PROTECTION

- A. Protect installed materials to prevent damage by other trades. Use materials that may be easily removed without leaving residue or permanent stains.

END OF SECTION 102600

SECTION 102600.01 – WALL AND DOOR PROTECTION**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Corner guards.
 - 2. Wall bumpers.
 - 3. Flexible metal bollards.
 - 4. Crash rail (Ram Protection Type 1).

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product data: Submit manufacturer's product data.
 - 2. Shop drawing: Submit shop drawings showing components, dimensions, anchorage details.
 - 3. Samples for selection and verification of product pattern and finish.

1.4 QUALITY ASSURANCE

- A. Single Source Responsibility: Provide all product systems components from a single source.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Delivery: Deliver materials to site in manufacturer's original, unopened containers and packaging with labels clearly indicating manufacturer and material.
- B. Storage: Store materials indoors in a clean, dry area protected from damage where temperature and humidity remain stable and within 40 degrees F to 100 degrees F. Materials must be stored flat.
- C. Handling: Protect materials during handling and installation to prevent damage.

1.6 PROJECT CONDITIONS

- A. Environmental conditions: Do NOT install wall surface protection systems until the installation area is enclosed and weatherproof, and until the ambient temperature within the building is maintained at not less than 70 degrees F for not less than 72 hours prior to installation.
- B. Installation locations must be enclosed weatherproofed, and climate controlled prior to commencing installation.

1.7 WARRANTY

- A. Submit supplier's written warranty against defects in material and workmanship.

PART 2 - PRODUCTS

2.1 CORNER GUARDS

- A. Manufacturer: Babcock-Davis, Brooklyn Park, MN; phone 888-412-3726; www.Babcock-Davis.com
- B. Stainless Steel Corner Guards: Model BCGS Surface mount fabricated from one-piece, formed metal with formed edge, 90-degree match wall.
 - 1. Material: Type 304 Stainless Steel, 16 Gauge with sharp edges, Satin No. 4 finish
 - 2. Wing Size; as specified between 2 inches
 - 3. Corner Radius: 1/8 inch.
 - 4. Mounting: Adhesive
 - 5. Option: embossed patterns available in Type 304
- C. Stainless Steel Corner End Wall Guards: Model BCGS-EW surface mount fabricated from one-piece, formed metal with formed edge.
 - 1. Material: Type 304 Stainless Steel, 16 gauge with sharp edges, Satin No. 4 finish
 - 2. Wing Size; as specified between 2 inches
 - 3. End wall size: as specified
 - 4. Corner Radius: 1/8 inch
 - 5. Mounting: Adhesive

2.2 WALL BUMPERS

- A. Manufacturer: Roechling Polystone; www.roechling-industrial.com.
- B. Product: Polystone Wall Protection; G HD
 - 1. Materials: High quality plastics polyethylene (PE) and polypropylene (PP).
 - 2. Size and Color: As indicated on Drawings
 - 3. Surface Finish: Smooth
 - 4. Exposed Edges: 0.197-inch radius on front face only.
 - 5. Countersunk Drilled Holes: Pre-drilled 32/125-inch (0.256 inch) diameter thru-hole spaced at 15.27 inches on center (typical).
 - 6. Fasteners: 1/4-inch diameter concrete lag bolts for CMU; 1/4-inch joint connector bolt with flat head and hexagon socket for CMU and plywood. Length as required to securely attach bumper to wall.
 - 7. Sealant: Silicone type in color matching floor slab finish.

2.3 FLEXIBLE METAL BOLLARDS AT TRAFFIC DOORS

- A. Manufacturer: McCue Corporation; www.mccue.com.
- B. Product: FlexCore Post 6; Model No. CSFP-55-14-GS-KIT
 - 1. Post: Zinc-electroplated steel post with mounting base.
 - 2. Post Diameter: 5.56 inches (141 mm) diameter

3. Post Height: 14 inches (355.6 mm)
4. Hardware and Accessories: Self-threading concrete anchors, washer, impact absorbing core, damper, flat head socket head cap screw, and other components necessary to complete installation.

2.4 CRASH RAIL (RAM PROTECTION TYPE 1) AT STOREFRONTS

A. Manufacturer: McCue Corporation; www.mccue.com.

B. Product: Model No. CSSSR-12A-7 / CSSSR-10A-7.

1. Provide stainless steel rail assembly to include the following:
 - a. Type 201 stainless steel tubular rails, polished.
 - b. Stainless steel legs.
 - c. Plastic end caps.
 - d. Elbows (if needed).
 - e. Connectors.
 - f. Stainless steel flange anchor kit for floor mounting.
 - g. Accessories: Provide other components as necessary to complete installation.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine the substrates and conditions under which the work is to be performed and notify the Contractor in writing of conditions detrimental to the proper and timely completion of the work.
- B. Do not proceed until unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Surface preparation: Prior to installation, clean surface to remove dirt, debris and loose particles. Perform additional preparation procedures as required per the manufacturer's instructions.

3.3 INSTALLATION

- A. Install the work of this section in strict accordance with manufacture's written instructions, level, plumb, and true to line without distortions.
- B. Do not use materials with chips, cracks, voids, stains, or other defects that might be visible in the finished work.
- C. Corner Guard Installation:
 1. Subsurface must be dry, clean, dust-free, and properly sealed.
 2. Apply adhesive as recommended by manufacturer for heavy duty usage.
- D. Wall Bumper Installation:
 1. Install wall bumpers with tight butt joints and secure to suitable substrate. Countersink fastener heads to set flush with face of wall bumper.
 2. Where field cuts remain exposed, use a deburring tool to smooth out the rough edges.

3. Apply backer rod and sealant in joint along floor slab; ensure that sealant is continuous, smooth and flush with face of wall bumper.

E. Flexible Metal Bollard Installation:

1. Securely anchor bollard mounting base to concrete slab, following manufacturer's installation instructions.
2. Slide post onto mounting base until flush with the floor. Secure post to mounting base.

F. Crash Rail Installation:

1. Position crash rail to location indicated on Drawings.
2. Anchor flange to concrete slab following manufacturer's installation instructions. Ensure top of stainless steel sleeve is fully seated on top of plastic flange anchor.
3. Insert legs into flange assemblies until fully seated into the flange sleeves.

3.4 CLEANING

- A. Upon completion of the work, remove excess materials, debris and rubbish resulting from the installation and leave the area in a clean and orderly condition.

END OF SECTION 102600.01

SECTION 102800 - TOILET, BATH, AND LAUNDRY ACCESSORIES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Public-use washroom accessories.
 - 2. Warm-air dryers.
 - 3. Childcare accessories.
 - 4. Underlavatory guards.
 - 5. Custodial accessories.

1.3 ACTION SUBMITTALS

- A. See Section 013300 "Submittal Procedures" for requirements of Action Submittals.
- B. Submit Submittals via Owner's Project Management software platform.
- C. Product Data: For each type of product.
- D. Samples: Full size, for each exposed product and for each finish specified.

1.4 INFORMATIONAL SUBMITTALS

- A. Product certificates.
- B. Sample warranty.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance data.

1.6 WARRANTY

- A. Manufacturer's Special Warranty for Mirrors: Manufacturer agrees to repair or replace mirrors that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period: 15 years from date of Substantial Completion.

PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2.2 PUBLIC-USE WASHROOM ACCESSORIES

A. Grab Bar #TA-1:

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. Bobrick Washroom Equipment, Inc., Model #B-6806 x 18, 36, 42 Series.
 - b. Kingston Brass. Model # GB1236CS 36, 42 Series
 - c. Bradley Washroom Accessories Model 812 Series
2. Mounting: Flanges with concealed fasteners.
3. Material: Stainless steel, 0.05 inch (1.2 mm) thick.
 - a. Finish: Smooth, No. 4 finish (satin).
4. Outside Diameter: 1-1/2 inches (38 mm).
5. Configuration and Length: Straight, as indicated on Drawings.

B. Mirror Unit #TA-3:

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. Bobrick Washroom Equipment, Inc., Model #B-165 Series.
2. Frame: Stainless-steel channel.
 - a. Corners: Manufacturer's standard.
3. Hangers: Produce rigid, tamper- and theft-resistant installation, using method indicated below.
 - a. One-piece, galvanized-steel, wall-hanger device with spring-action, snap locking design to hold mirror unit in position with no exposed screws or bolts.
4. Size: As indicated on Drawings.

C. Toilet Tissue (Roll) Dispenser #TA-4:

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. American Specialties; 7305-2S
2. Description: Double-roll dispenser.
3. Mounting: Surface mounted.
4. Operation: Noncontrol delivery with standard spindle
5. Capacity: Designed for 5-1/2-inch (140-mm-) diameter tissue rolls.
6. Material and Finish: 304 stainless steel, satin finish.

D. Surface Mounted Sanitary Napkin Disposal #TA-5

1. Manufacturers: Subject to compliance with requirements, provide products by the following:

- a. Bobrick washroom equipment, Inc. Model #B-270.
 - b. Bradley BX-Towel Dispenser, 250-15, Multi-Fold Or C-Fold Towels.
 - c. Palmer Fixture Multifold/C-Fold Towel Dispenser - Metal, XC, TD0170-12
2. Description: Sanitary Napkin Disposal
 3. Mounting: Surface mounted.
 4. Capacity: 1 Gallon.
 5. Material and Finish: Stainless steel.

E. Hat and Coat Hook #TA-8:

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. American Specialties; 7382-S
2. Description: Double hook.
3. Mounting: Surface mounted.
4. Material and Finish: 304 stainless steel, satin finish.

2.1 CHILDCARE ACCESSORIES

A. Diaper-Changing Station #TA-2:

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. Koala Kare Products, Model #KB200-01 (Grey), injection molded polypropylene.
 - b. World Dryer ABC-300H (Light Grey) Dry Baby Changing Station, injection molded polypropylene, Surface Mounted.
2. Description: ADA compliant, horizontally mounted unit that opens by folding down from recessed stored position and with child-protection strap.
 - a. Engineered to support minimum of 250-lb (113-kg) static load when opened.
3. Mounting: Surface.
4. Operation: By pneumatic shock-absorbing mechanism.
5. Material and Finish: Injection molded polypropylene, grey color, and HDPE interior in manufacturer's standard color selected.
6. Liner Dispenser: Built in.

2.2 WARM-AIR DRYERS

A. High-Speed Warm-Air Dryer #TA-6:

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. Excel Dryer Inc.; ThinAir Hand Dryer; TA-SB-110-120V 7-7.8A-750-920W
2. Description: High-speed air hand dryer with HEPA filter.
3. Mounting: Surface mounted.
4. Operation: Electronic-sensor activated with operation time of 14 seconds.

5. Cover Material and Finish: Brushed stainless steel finish.
6. Electrical Requirements: 110-120 V, 7.0-7.8A, 750-920W.

2.3 UNDERLAVATORY GUARDS

A. Underlavatory Guard:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Plumberex Specialty Products, Inc.
 - b. Truebro by IPS Corporation.
2. Description: Insulating pipe covering for supply and drain piping assemblies that prevents direct contact with and burns from piping; allow service access without removing coverings.
3. Material and Finish: Antimicrobial, molded plastic, white.

2.4 CUSTODIAL ACCESSORIES

A. Surface Mounted Toilet Brush and Holder #TA-7

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. American Specialties, Inc. Model #7387.
2. Description: Wall Mounted Toilet Brush and Holder.
3. Material and Finish: Stainless steel, Satin.
4. Brush: PVC with polypropylene shaft.

B. Mop and Broom Holder #TA-9

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. Bobrick Washroom Equipment, Inc, Model #B-239 x 34.
 - b. ASI 1308 Combination Shelf, Utility Hook, Mop Holder
 - c. Bradley Corp Model 9933, 9934
2. Description: Unit with shelf, hooks and holders.
3. Length: 34 inches.
4. Hooks: Four.
5. Mop/Broom Holders: Three, spring-loaded, rubber hat, cam type.
6. Material and Finish: Stainless steel, No. 4 finish (satin).
 - a. Shelf: Not less than nominal 0.05-inch thick stainless steel.

2.5 FABRICATION

- ### A. Keys: Provide universal keys for internal access to accessories for servicing and resupplying. Provide minimum of six keys to Owner's representative.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install accessories according to manufacturers' written instructions, using fasteners appropriate to substrate indicated and recommended by unit manufacturer. Install units level, plumb, and firmly anchored in locations and at heights indicated.
- B. Grab Bars: Install to withstand a downward load of at least 250 lbf (1112 N), when tested according to ASTM F 446.

END OF SECTION 102800

SECTION 104416 - FIRE EXTINGUISHERS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes portable, hand-carried fire extinguishers and mounting brackets for fire extinguishers.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

- A. Product certificates.
- B. Warranty: Sample of special warranty.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.6 COORDINATION

- A. Coordinate type and capacity of fire extinguishers with fire-protection cabinets to ensure fit and function.

1.7 WARRANTY

- A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace fire extinguishers that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. NFPA Compliance: Fabricate and label fire extinguishers to comply with NFPA 10, "Portable Fire Extinguishers."

- B. Fire Extinguishers: Listed and labeled for type, rating, and classification by an independent testing agency acceptable to authorities having jurisdiction.

2.2 PORTABLE, HAND-CARRIED FIRE EXTINGUISHERS

- A. Fire Extinguishers for Bakery: Type, size, and capacity for each mounting bracket location indicated.

- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Guardian Fire Equipment, Inc.
 - b. JL Industries, Inc.; a division of the Activar Construction Products Group.
 - c. Larsens Manufacturing Company; WC-6L.
 - d. Nystrom Building Products.
 - e. Potter Roemer LLC.
- 2. Instruction Labels: Include pictorial marking system complying with NFPA 10, Appendix B, and bar coding for documenting fire-extinguisher location, inspections, maintenance, and recharging.
- 3. Wet Chemical Type Fire Extinguishers: UL-rated 2-A:1-B:C:K, 1.6-gal. (6-L), nominal capacity and contain a blend of Potassium Acetate and Potassium Citrate in manufacturer's standard stainless steel container, with pressure-indicating gage.

- B. Fire Extinguishers non-kitchen areas and in locations subject to public view: Type, size, and capacity for each mounting bracket location indicated.

- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Guardian Fire Equipment, Inc.
 - b. JL Industries, Inc.; a division of the Activar Construction Products Group.
 - c. Larsens Manufacturing Company; MP10.
 - d. Nystrom Building Products.
 - e. Potter Roemer LLC.
- 2. Instruction Labels: Include pictorial marking system complying with NFPA 10, Appendix B, and bar coding for documenting fire-extinguisher location, inspections, maintenance, and recharging.
- 3. Multipurpose Dry-Chemical Type Fire Extinguishers: UL-rated 4-A:80-B:C, 10-lb (4.5-kg) nominal capacity, with monoammonium phosphate-based dry chemical in manufacturer's standard stainless steel container.

- C. Fire Extinguishers in IT room: Type, size, and capacity for each mounting bracket location indicated. Halotron 1

- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Guardian Fire Equipment, Inc.
 - b. JL Industries, Inc.; a division of the Activar Construction Products Group.
 - c. Larsens Manufacturing Company; Halotron I Series, Model HT11.
 - d. Nystrom Building Products.
 - e. Potter Roemer LLC.

2. Instruction Labels: Include pictorial marking system complying with NFPA 10, Appendix B, and bar coding for documenting fire-extinguisher location, inspections, maintenance, and recharging.
3. Clean-Agent Type Fire Extinguishers: UL-rated 1-A:10-B:C, 11-lb (5-kg), nominal capacity, with HydroChloroFluoroCarbon (HCFC) Blend B agent and inert material.

2.3 MOUNTING BRACKETS

- A. Mounting Brackets: Manufacturer's standard steel, designed to secure fire extinguisher to wall or structure, of sizes required for types and capacities of fire extinguishers indicated, with plated red baked-enamel finish.
 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Guardian Fire Equipment, Inc.
 - b. JL Industries, Inc.; a division of the Activar Construction Products Group.
 - c. Larsens Manufacturing Company.
 - d. Nystrom Building Products.
 - e. Potter Roemer LLC.
- B. Identification: Lettering complying with authorities having jurisdiction for letter style, size, spacing, color and location. Locate directly above wall mounted fire extinguishers only.
 1. Identify bracket-mounted fire extinguishers with the words "**FIRE EXTINGUISHER**" in black letter decals applied to mounting surface.
 - a. Orientation: Horizontal.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Examine fire extinguishers for proper charging and tagging.
 1. Remove and replace damaged, defective, or undercharged fire extinguishers.
- B. Install fire extinguishers and mounting brackets in locations indicated and in compliance with requirements of authorities having jurisdiction.
 1. Mounting Brackets: 48 inches (1220 mm) above finished floor to top of fire extinguisher unless otherwise stipulated by jurisdiction authorities.
- C. Mounting Brackets: Fasten mounting brackets to surfaces, square and plumb, at locations indicated.

END OF SECTION 104416

SECTION 107313 - METAL CANOPIES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Exterior metal canopy at Employees' Entrance
 - 2. Exterior metal canopy at Building Entrance

1.3 ACTION SUBMITTALS

- A. See Section 013300 "Submittal Procedures" for requirements of Action Submittals.
- B. Submit Submittals via Owner's Project Management software platform.
- C. Product Data: For each type of product.
 - 1. Include material descriptions, construction details, fabrication details, dimensions of individual components and profiles, hardware, fittings, mounting accessories, features, and finishes.
- D. Shop Drawings:
 - 1. Include plans, elevations, sections, mounting heights, and attachment details.
 - 2. Detail fabrication and assembly of metal canopy.
 - 3. Show locations for blocking, reinforcement, and supplementary structural support.
- E. Coordination Drawings: Submit coordination drawings showing structural component locations/positions, material dimensions and details of construction and assembly.
- F. Samples: For each exposed product.
- G. Delegated-Design Submittal: For metal canopy.

1.4 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data: For metal canopy to include in operation and maintenance manuals.

1.5 QUALITY ASSURANCE

- A. Regulatory Requirements: Comply with all applicable building code requirements. Provide structural calculations, signed and sealed by Engineer licensed in State where project is located.

- B. Fabricator Qualifications: Shop that employs skilled workers who custom-fabricate products similar to those required for this Project and whose products have a record of successful in-service performance.
 - 1. Fabricator's responsibilities include fabricating and installing canopies and providing professional engineering services needed to assume engineering responsibility.
- C. Installer Qualifications: Fabricator of products.
- D. Welding Qualifications: Qualify procedures and personnel according to the following:
 - 1. AWS D1.2/D1.2M, "Structural Welding Code - Aluminum."

1.6 DELIVERY, STORAGE AND HANDLING

- A. Deliver products in manufacturer's original, unopened, undamaged containers with identification labels intact.
 - 1. Protect finishes by applying heavy duty removable plastic film during production.
 - 2. Package for protection against transportation damage.
 - 3. Provide markings to identify components consistently with drawings.
 - 4. Exercise care in unloading, storing and installing metal canopy to prevent bending, warping, twisting and surface damage.
- B. Upon receipt of materials, thoroughly examine units for damage. Promptly report any observed damage manufacturer in writing. Include digital photographs of any observed damage as well as a copy of the Bill of Lading disclosing the damage.
- C. Store products protected from exposure to harmful weather conditions and at temperature conditions recommended by manufacturer.

1.7 FIELD CONDITIONS

- A. Field Measurements: Where metal canopy installation is indicated to fit to other work, verify dimensions of other work by field measurements before fabrication and indicate measurements on Shop Drawings. Allow clearances for fenestration operation throughout the entire operating range. Notify Architect of discrepancies. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Design system to withstand project requirements of wind pressure, snow load and drifting snow load in accordance to applicable building codes and authorities having jurisdiction.

2.2 EXTERIOR METAL CANOPY AT EMPLOYEES' ENTRANCE

- A. Basis-of-Design Product: Echoshade manufactured by MASA Architectural Canopies; Avenel, NJ, (800) 761-7446, www.architecturalcanopies.com.
- B. Canopy System: Fixed, building-supported, pre-engineered canopy system including framing, louvers, outriggers and attachment hardware.

- C. Framing: Extruded aluminum rectangular tube, 2 inches x 8 inches x 1/8 inch thick.
- D. Outrigger Shape: Extruded aluminum rectangular tube.
- E. Infill Louvers:
 - 1. Type: Rectangular tubing.
 - 2. Size and Angle: To be determined by Architect.
- F. Mounting: Hanger rods with square wall plates.
- G. Hardware: As recommended by manufacturer for specific installation conditions and as detailed on approved shop drawings.
- H. Other Components: As indicated or as required for system attachment and performance.

2.3 MATERIALS

- A. Aluminum Extrusions: ASTM B221 and ASTM B 429; 6061-T6 alloy and temper.
- B. Anchors and Fasteners: Stainless steel bolts, studs, sheet metal screws, and other types of size and spacing as recommended by manufacturer for specific installation conditions and as detailed on approved shop drawings.

2.4 FABRICATION

- A. Fabricate metal canopies in accordance with approved Shop Drawings.
- B. Factory-assemble louvers with plate outriggers using stainless steel, thread-cutting screws through internal screw slots in blades. Infill louvers must be removable in case of damage; welding is not acceptable.
- C. Mechanically-assemble canopies with a minimum shear stress strength of 350 lbs. Pre-welding is not acceptable.

2.5 ALUMINUM FINISHES

- A. Finish: AAMA 2605, fluoropolymer coating containing minimum 70 percent PVDF resins.
 - 1. Color: See Exterior Finish Schedule

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for supporting members, blocking, inserts, installation tolerances, and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Prior to fabrication, field-verify required dimensions.

- B. Coordinate system installation with provision of exterior wall system to ensure proper structural support is provided, attachment of metal canopy is compatible with substrate, and weather tightness of exterior envelop is maintained.

3.3 INSTALLATION

- A. Install in accordance with manufacturer's written instructions and recommendations and approved Shop Drawings.
- B. Verify dimensions of supporting structure at the site by accurate field measurements so that the work will be accurately designed, fabricated and fitted to the structure.
- C. Anchor system to building substrate as indicated on drawings, and as specified. Provide adequate clearance for movement caused by thermal expansion and contraction and wind loads.
- D. Separate dissimilar metals; use gasket fasteners, isolation shims, or isolation tape where needed to eliminate possibility of galvanic action between metals.
- E. Do not erect warped, bowed, deformed or otherwise damaged or defaced members. Remove and replace any members damaged in the erection process as directed.
- F. Set units level, plumb and true to line, with uniform joints and free of warp and twist.
- G. Erection Tolerances:
 - 1. Variation from level: +/- 1/8 inch maximum in 20 ft. runs, non-cumulative.
 - 2. Maximum Offset from True Alignment Between Adjacent Members Butting or In-Line: 1/16 inch.
- H. Do not field cut or trim units. Cut and trim component parts during erection only with the approval of the manufacturer, and in accordance with his recommendations. Restore finish completely. Remove and replace members where cutting and trimming has impaired the strength or appearance of the assembly as directed.
- I. Corrosion Protection: Coat concealed surfaces of aluminum that come in contact with grout, concrete, masonry, wood, or dissimilar metals with a heavy coat of bituminous paint.

3.4 CLEANING AND PROTECTION

- A. Remove site cuttings from finish surfaces.
- B. Clean and wash prefinished surfaces with mild soap and water; rinse with clean water.
- C. Clean aluminum surfaces in accordance with recommendations found in AAMA 609 and 610. Do not use aggressive alkaline, TSP, acid cleaners, or abrasive cleaners on aluminum surfaces.
- D. Touchup Painting: Immediately after erection, clean connections and abraded areas. Paint uncoated and abraded areas with same or compatible material as used for shop-applied finish painting, as recommended by canopy manufacturer.

END OF SECTION 107313

SECTION 108213 – ROOF SCREENS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Roof equipment screens and steel-braced framing.

1.3 ACTION SUBMITTALS

- A. Product Data: Manufacturer's data sheets on each product to be used, including:
 - 1. Preparation instructions and recommendations.
 - 2. Storage and handling requirements and recommendations.
 - 3. Installation methods.
- B. Shop Drawings: Shows the layout of the screen indicating typical dimensions of components and specific placement of frames. Welds shall be shown using standard AWS A2.4 Welding Symbols. Include weld lengths. If applicable, include elevations.
- C. Samples: For each finish product specified, minimum size 6 inches (150 mm) square, representing actual product, shape, and patterns.

1.4 QUALITY ASSURANCE

- A. Manufacturer/Installer Qualifications: Manufacturer with a minimum five years documented experience in producing/installing equipment roof screen.
- B. Design Calculations: Provide design calculations for louvers, framing and associated components resisting wind loads with seal and signature of professional engineer licensed in the state in which this project is located.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Deliver materials to the project site clearly marked for proper identification. Receive, handle and store materials in conformance with the manufacturers printed instructions.
- B. Store materials in a dry, warm, ventilated weathertight location. Store products under cover, in manufacturer's unopened packaging until ready for installation.
- C. Protect materials from exposure to moisture. Protect finishes from damage.
- D. Do not lift loads by packaging bands or straps. Use safety guidelines and good practices while transporting or handling material. When loading to roof, disperse the weight so as not to overload the building framing.

1.6 COORDINATION

- A. Coordinate Work with other operations and installation of roofing materials to avoid damage to installed insulation and membrane materials.

1.7 PROJECT CONDITIONS

- A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.
- B. Field Measurement: Prior to fabrication, verify all conditions pertaining to the installation on site. Take measurements to verify dimensions of equipment screen and specify on Shop Drawings.

1.8 WARRANTY

- A. Screen Frames: Provide manufacturer's standard which guarantees protection from structural failure due to tearing, buckling, bending, corrosion or cracking when used in normal environmental conditions for the applicable period of the warranty.
- B. Louver: Provide manufacturer's standard warranty for factory finish.
 - 1. Warranty Period: 25 years.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Design Loads: Comply with Building Code requirements for site location and building height.
 - 1. Design to resist ASCE 7 Minimum Design Loads for Buildings and Other Structures.
 - 2. Design all materials, assembly and attachments to resist snow, wind, suction and uplift loading at any point without damage or permanent set.
- B. Structural Design: Calculations shall be prepared for the equipment screen frames and attachment building.
- C. All welds to be performed by an AWS certified welder.

2.2 MANUFACTURER

- A. Basis of Design:
 - 1. Manufacturer: Cascade Rooftop Screens, Inc.; 688 Walsh Avenue, Santa Clara, CA 95050; Phone: 408-827-1700; www.cascaderooftopscreens.com.
 - 2. Roof Screen System:
 - a. Framing: Prefabricated Simple Frame Tubular Steel Framing.
 - b. Panels: Louver style.

2.3 MATERIALS

- A. Frames: Carbon steel structural tubing and plates in manufacturer's standard sizes, conforming to ASTM A500 with manufacturers galvanized coating conforming to ASTM B117 salt spray testing.

1. Wall and Plate Thickness: As determined by structural calculations.
2. Welds: Conform to AWS D1.1.
- B. Assembly Bolts and Fasteners: Hot-dip galvanized coating conforming to ASTM B117 salt spray testing.
- C. Flashing:
 1. Flashing Type: Provide flashing to match material of roof membrane.
 - a. Round TPO Roof Flashing: Fabricated from 60 mil, white, single ply TPO sheet conforming to ASTM D 6878.
 - b. Round PVC Roof Flashing: Fabricated from 60 mil, white, single ply PVC sheet conforming to ASTM D 4434.
 2. Provide with base flange that extends a minimum of 5 inches (127 mm) onto the roof surface.
 3. Hot weld all seams for water tightness.
- D. Steel Hat Channel: ASTM A 653, Class SS; G90 hot-dip galvanized coating.
- E. Bolts, Nuts, Washers, Screws and Other Hardware: Hot-dip galvanized coating.
- F. Provide anchorage and assembly parts for fabrications in compliance with engineering calculations.
- G. Welding Materials: AWS D1.1; type required for materials being welded.
- H. Louvers:
 1. Profile: Cascade; Louver Style A
 2. Material: Galvanized steel sheet.
 - a. Attachment Base: 16 gauge.
 - b. Facing Clip: 24 gauge; 5 inches high by 2 inches deep by 10 feet long.
 - c. Fasteners: Hot-dip galvanized with color-coated cap to match slats. Provide screws of type, size, and length recommended by louver manufacturer.
 3. Finish (both sides): 70% PVDF fluoropolymer coating (Kynar).
 4. Color: Charcoal.
 5. Installation: Horizontal louver slats, attached directly to frames with self-drilling screws.

2.4 FABRICATION

- A. Fabricate components in largest practical sections possible for delivery and field assembly.
- B. Fabricate components so that joints are tightly fit and flush, with no exposed jagged edges.
- C. Supply components required for anchorage of fabrications. Fabricate anchors and related components of same material and finish as fabrication, except where specifically noted otherwise.
- D. Trim: Fabricate from flat sheet metal, in thickness and finish matching facing panel. Metal to be hemmed per SMACNA guidelines.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine structural supports, substrates and roof areas where roof screens are to be installed, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of work.
- B. Verify all dimensions of in-place and subsequent construction.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Install roof screen in accordance with manufacturer's installation instructions and approved Shop Drawings.
- B. Set all items in their correct locations as shown on the final reviewed Shop Drawings. Install components level, square, plumb, accurately fitted, and at proper elevations and in alignment with other work, free from distortion or defects.
- C. Install all roof flashings and boots per the roof membrane manufacturer's specifications to ensure a watertight connection.
- D. Separate dissimilar metals and use gasketed fasteners, isolation shim, or isolation tape to eliminate possibility of corrosive or electrolytic action between metals.
- E. Exercise care when installing components so as not to damage finish surfaces. Touch up as required to repair damaged finishes.
- F. Remove all protective masking from material immediately after installation.

3.3 CLEANING AND PROTECTION

- A. Upon completion of the work remove all unused materials, debris, containers and equipment from the project site. Clean and repair finished surfaces that have been marked or otherwise damaged by work under this Section.
- B. Protect installed products until completion of project. If minor damage to finishes occurs, repair damage in accordance with manufacturer's recommendations; provide replacement components if repaired finishes are unacceptable to Architect.
- C. Remove and replace damaged roof screens at no additional cost to the Owner.

END OF SECTION 108213

SECTION 111316 - LOADING DOCK SEALS AND SHELTERS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Frame-type dock shelters.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings:
 - a. Include plans, elevations, sections, details, and attachments to other work.
 - 3. Samples: For each exposed product and for each color and texture specified.

1.4 INFORMATIONAL SUBMITTALS

- A. Product certificates.
- B. Field quality-control reports.

1.5 WARRANTY

- A. Manufacturer's Special Warranty: Manufacturer agrees to repair or replace dock levelers that fail in materials or workmanship within specified warranty period of manufacturer.

PART 2 - PRODUCTS**2.1 MANUFACTURERS**

- A. Basis of Design:
 - 1. Manufacturer: McGuire; www.wbm McGuire.com.
 - 2. Product: TC-920 Soft-Sided Dock Shelter
- B. Supplier Contact: See **LiDL National Account List**.

2.2 FRAME-TYPE DOCK SHELTERS

- A. General: Dock shelters designed to form a seal with sides and top of truck body while leaving entire width and height of truck's rear opening unobstructed.
- B. Flexible-Frame Type: Fabricated from fabric-covered side and top panels attached to retractable supporting framework with independent spring-tension extension arms. Provide head and side curtains with built-in flexible stays, pleated protectors on head curtain, and a yellow aim patch on side curtains. Provide replaceable, fabric-covered, tapered, foam-bottom pads of size and type required for application shown.
- C. Head-Pad Height: 24 inches (610 mm).
- D. Construction: Fabricate to sizes and shapes indicated or required to fit door-opening sizes shown and allow for not less than 18 inches (457 mm) of truck-body penetration when truck is docked.
 - 1. Provide pivoting canopy header.
 - 2. Provide detachable side curtains.
- E. Materials:
 - 1. Cover Fabric: Manufacturer's proprietary 40 oz vinyl cover fabric.
 - 2. Fabric Color: Black.
 - 3. Pleated Protectors: Overlapping layers of same fabric as cover.
- F. Steel Finish: Hot-dip galvanize components to comply with the following:
 - 1. ASTM A 123/A 123M for iron and steel support framing.
 - 2. ASTM A 153/A 153M or ASTM F 2329 for iron and steel hardware and anchors.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Dock Shelters: Attach dock shelters securely to building structure in proper relation to openings, dock bumpers, and dock levelers to ensure an effective seal of dock-shelter curtains with sides and top of truck body when trucks are positioned against dock bumpers.
- B. After completing installation, inspect exposed factory finishes and repair damaged finishes. Protect Dock Shelter from damage with weatherproof materials until substantial completion.

END OF SECTION 111316

SECTION 111319 - STATIONARY LOADING DOCK EQUIPMENT**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Edge-of-dock levelers.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.

- B. Submit the following Submittals via Owner's Project Management software platform.

- 1. Product Data: For each type of product.
 - 2. Shop Drawings: For stationary loading dock equipment.
 - a. Include plans, elevations, sections, details, and attachments to other work.

1.4 INFORMATIONAL SUBMITTALS

- A. Product test reports.
- B. Sample warranty.
- C. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.6 WARRANTY

- A. Manufacturer's Special Warranty: Manufacturer agrees to repair or replace dock levelers that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period for Structural Assembly: 10 years from date of Substantial Completion.
 - 2. Warranty shall be for unlimited usage of leveler for the specified rated capacity over the term of the warranty.

PART 2 - PRODUCTS**2.1 MANUFACTURERS**

- A. Basis of Design:

1. Manufacturer: McGuire; www.wbm McGuire.com.
2. Product: MEDLF-6630-15 Mechanical EOD Leveler

B. Supplier Contact: See **LIDL National Account List**.

2.2 EDGE-OF-DOCK LEVELERS

- A. General: Surface-mounted, hinged-lip-type, edge-of-dock levelers designed for permanent installation on face of loading dock platform.
- B. Standard: Comply with MH 30.1, except for structural testing to establish rated capacity.
- C. Rated Capacity: Capable of supporting total gross load of 30,000 lbs. (13,600 kg.) without permanent deflection or distortion.
- D. Platform Ramp Width: 66 inches (1676 mm).
- E. Deck and Hinged Beveled Lip: Not less than 7/16-inch (11-mm) thick, nonskid steel tread plate.
- F. Function: Dock levelers shall compensate for differences in height between truck bed and loading platform.
 1. Vertical Travel: Operating range above platform level of sufficient height to enable lip to extend and clear truck bed before contact with the following minimum working range:
 - a. Above Adjoining Platform: 3 inches (76 mm)
 - b. Below Adjoining Platform: 3 inches (76 mm)
 2. Lip Operation: Manufacturer's standard mechanism, which push/pull of the level handle extends hinged lip on ramp edge with lip resting on truck bed over dock leveler's working range, allows lip to yield under impact of incoming truck and retracts lip when truck departs.
 - a. Length of Lip Extension: 15 inches (381 mm)
- G. Bumpers: Provide two steel-faced bumpers for each dock leveler, attached to face of loading dock with bolts.
- H. Materials:
 1. Steel Plates, Shapes, and Bars: ASTM 36/A 36M.
 2. Rolled-Steel Floor Plate: ASTM A 786/A 786M, rolled from steel plate complying with ASTM A 572/A 572M, Grade 55 (380).
 3. Welding Rods and Bare Electrodes: Select according to AWS specifications for metal alloy welded.
- I. Dock-Leveler Finish: **Factory-finished in** manufacturer's standard **color**.

2.3 FINISH REQUIREMENTS

- A. Factory Finish: Clean, pretreat and apply manufacturer's standard two-coat finish consisting of prime coat and topcoat.
 1. Color: Manufacturer's standard Gray.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Edge-of-Dock Levelers: Attach dock levelers to loading dock platform in a manner that complies with requirements indicated for arrangement and position relative to top of platform.
 - 1. Bolt anchor holes in contact with continuous embedded loading dock edge, steel channel. Weld or bolt bumper blocks to face of loading dock.
- B. Adjust loading dock equipment to function smoothly and safely and lubricate as recommended by manufacturer.
- C. Test dock levelers for vertical travel within operating range indicated.
- D. After completing installation of exposed, factory-finished loading dock equipment, inspect exposed finishes and repair damaged finishes.

3.2 DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain loading dock equipment.

END OF SECTION 111319

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SECTION 113100 - RESIDENTIAL APPLIANCES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Refrigeration appliances.
 - 2. Microwave appliances.
 - 3. Dishwasher

1.3 CONFORMANCE SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section. See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform, as directed by Owner; however, the following Submittals will not be reviewed by the Architect or Engineer:
 - 1. Product Data: For each type of product.
 - 2. Samples: For each exposed product and for each color and texture specified.

1.4 INFORMATIONAL SUBMITTALS

- A. Product certificates.
- B. Field quality-control reports.
- C. Sample warranties.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.6 WARRANTY

- A. Special Warranties: Manufacturer agrees to repair or replace residential appliances or components that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period: Two years from date of Substantial Completion.

PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. Electrical Appliances: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2.1 REFRIGERATOR

A. Refrigerator #E-1: Bottom-Freezer Refrigerator, CSA Certified.

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. Whirlpool: Model #WRT138FZDM.
2. Type: Freestanding.
3. Storage Capacity: 18.2 cu. ft.
4. ENERGY STAR: Provide appliances that qualify for the EPA/DOE ENERGY STAR product-labeling program.
5. General Features:
 - a. Interior light and electronic temperature control.
 - b. Flexi-Slide Bin.
 - c. Quiet Cooling.
6. Front Panel: Stainless steel.

2.2 MICROWAVE OVENS

A. Microwave Oven #E-3:

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. Farberware, Model #FMO11AHTBKB
2. Capacity: 1.1 cu. ft.
3. Microwave Power Rating: 1000 W.
4. Dimensions: 12.03 (H) by 20.2 (W) by 16.54 inches deep.
5. Material: Stainless steel/Black.

2.3 DISHWASHERS

A. Dishwasher #E-8.

1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. Whirlpool, Model #DWF550SAH.
2. Type: Built-in undercounter.
3. Dimensions: 34.5 (H) by 23.5 (W) by 24 inches deep.
4. ENERGY STAR: Provide appliances that qualify for the EPA/DOE ENERGY STAR product-labeling program.
5. Front Panel: Silver.

B. Dishwasher #E-8: ADA Compliant.

1. Manufacturers: Subject to compliance with requirements, provide products by the following:

- a. Summit Appliances, Model #DW2435SSADA.
2. Type: Built-in undercounter.
3. Size: 32 (H) by 23.5 (W) by 22.38 inches deep.
4. ENERGY STAR: Provide appliances that qualify for the EPA/DOE ENERGY STAR product-labeling program.
5. Front Panel: Stainless Steel.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Built-in Equipment: Securely anchor units to supporting cabinets or countertops with concealed fasteners. Verify that clearances are adequate for proper functioning and that rough openings are completely concealed.
- B. Freestanding Equipment: Place units in final locations after finishes have been completed in each area. Verify that clearances are adequate to properly operate equipment.

3.2 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections with the assistance of a factory-authorized service representative:
 1. Perform visual, mechanical, and electrical inspection and testing for each appliance according to manufacturers' written recommendations. Certify compliance with each manufacturer's appliance-performance parameters.
 2. Operational Test: After installation, start units to confirm proper operation.
 3. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and components.
 4. Prepare test and inspection reports
- B. An appliance will be considered defective if it does not pass tests and inspections.

END OF SECTION 113100

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SECTION 122413 – ROLLER WINDOW SHADES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Motorized solar screen shades.

- B. Related Requirements:

- 1. Electrical wiring and hookup of motorized shade units to be responsibility of electrical contractor.

1.3 ACTION SUBMITTALS

- A. See Section 013300 "Submittal Procedures" for requirements of Action Submittals.

- B. Submit Submittals via Owner's Project Management software platform.

- C. Product Data: For each type of product.

- 1. Include styles, material descriptions, construction details, dimensions of individual components and profiles, features, finishes, and operating instructions for shades.

- D. Shop Drawings: Show fabrication and installation details for roller shades including materials and their operation.

- 1. Motorized Solar Screen Shades: Include details of installation and diagrams for power, signal, and control wiring.

- E. Samples: For each exposed product and for each color and texture specified, 10 inches (250 mm) long.

- 1. Include Samples of accessories involving color selection.

- F. Samples for Verification: For each type of shade.

- 1. Shade Material: Not less than 10 inches (250 mm) square

1.4 INFORMATIONAL SUBMITTALS

- A. Product test reports.

- B. Product certificates.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.6 QUALITY ASSURANCE

- A. Manufacturer Qualifications:

1. A firm with at least 20 years of demonstrated experience in the manufacture of motorized solar screen shades.
2. A firm that provides turnkey detailing, measuring, installation and warranty service for National Account Installations.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Deliver shades in factory packages, marked with manufacturer, product name, and location of installation using same designations indicated on Drawings.

1.8 FIELD CONDITIONS

- A. Environmental Limitations: Do not install solar screen shades until adjacent construction and finish work is complete.
- B. Field Measurements: Where roller sun shades are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication and indicate measurements on Shop Drawings. Allow clearances for operating hardware of operable glazed units through entire operating range. Notify Architect of installation conditions that vary from Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Source Limitations: Obtain roller sun shades from single source from single manufacturer.
- B. Manufacturer: Provided on **LIDL National Account List**.

2.2 MOTOR-OPERATED, SINGLE ROLLER SOLAR SCREEN SHADES

- A. Motorized Operating System: Provide factory-assembled solar shade system of size and capacity and with features, characteristics, and accessories suitable for conditions indicated.
 1. Only a UL Listed system with Certification of Compliance will be accepted. Systems manufactured with UL Recognized Components, but without a UL Listing will not be accepted.
 2. Electrical Components: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
 3. Electric Motor: Manufacturer's standard tubular, enclosed in roller.
 - a. Electrical Characteristics: 120V/60Hz AC with built-in radio receiver.
 4. Remote Control: Provide the following for remote-control activation of shades:
 - a. Provide Single Channel wireless wall switch for group control of shades to open and close shades simultaneously and to stop shade movement.

5. Limit Switches: Adjustable switches interlocked with motor controls and set to stop shades automatically at fully raised and fully lowered positions.
 6. Operating features: Group switching with integrated switch control, single faceplate for multiple switch cutouts.
 7. Color: White.
- B. Rollers: Corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of fabric panels indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of fabric panels for service.
1. Roller Drive-End Location: As indicated on Drawings or installer recommendation.
 2. Direction of Fabric Panel Roll: Regular Roll, standard shade operation where shade fabric will roll down from shade mount at top of sill
 - a. Fabric Panel-to-Roller Attachment: Adhesive strip
 - b. Provide fabric panels not less than 12 inches (300 mm) longer than desired shade height to assure solid attachment to roller tube and ability to adjust panels in field without removing mounting brackets.
- C. Mounting Hardware: Custom Colored Brackets or end caps, corrosion resistant and compatible with roller assembly, operating mechanism, installation accessories, and mounting location and conditions indicated.
- D. Fabric Panels:
1. Fabric Panel Material: RAS Sheerweave 2000
 2. Openness Factor: 5%
 3. Color: Charcoal/Grey
 4. Fabric Panel Bottom (Hem) Bar: Enclosed in hem pocket of fabric panel material, thermally sealed, not sewn.
- E. Installation Accessories:
1. Front Fascia: Custom Colored Aluminum extrusion that conceals front and underside of roller and operating mechanism and attaches to roller end caps without exposed fasteners.
 - a. Shape: L-shaped
 - b. Height: Manufacturer's standard height required to conceal roller and fabric panel when shade is fully open, but not less than 4 inches (102 mm). Color options below apply to L-shaped fascia and brackets.
 - c. Color: Standard grey.

2.3 BLIND AND SHADE FABRICATION

- A. Product Safety Standard: Fabricate shades to comply with WCMA A 100.1.
1. Between (Inside) Jamb Installation: Width equal to jamb-to-jamb dimension of opening in which shade is installed less 1/4 inch (6 mm) per side or 1/2-inch (13-mm) total, plus or minus 1/8 inch (3.1 mm). Length equal to head-to-sill or -floor dimension of opening in which shade is installed less 1/4 inch (6 mm), plus or minus 1/8 inch (3.1 mm).

- B. Fabric Panel Fabrication: Fabricate fabric panels without battens or seams to extent possible except as follows: when indicated as a need per manufacturer capabilities.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, operational clearances, accurate locations of connections to building electrical system, and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 BLIND AND SHADE INSTALLATION

- A. Install shades level, plumb, and aligned with adjacent units according to manufacturer's written instructions.
- B. Electrical Connections: Electrical Wiring and Hookup by other. Shade manufacturer/installer is not responsible for wiring and hookup.
- C. Adjust and balance shades to operate smoothly, easily, safely, and free from binding or malfunction throughout entire operational range.

3.3 CLEANING AND PROTECTION

- A. Clean shade surfaces after installation, according to manufacturer's written instructions.
- B. Provide final protection and maintain conditions, in a manner acceptable to manufacturer and installer that ensure shades are without damage or deterioration at time of Substantial Completion.
- C. Replace damaged shades that cannot be repaired, in a manner approved by Architect, before time of Substantial Completion.

3.4 DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain motor-operated shades.

END OF SECTION 122413

SECTION 123661.16 - SOLID SURFACING COUNTERTOPS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Solid surface material countertops.
 - 2. Solid surface material backsplashes.
 - 3. Solid surface material end splashes.

1.3 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For countertop materials.
 - 2. Shop Drawings: For countertops. Show materials, finishes, edge and backsplash profiles, methods of joining, and cutouts for plumbing fixtures.
 - 3. Samples: For each type of material exposed to view.

1.4 INFORMATIONAL SUBMITTALS

- A. Product test reports.
- B. Product certificates.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance data.

PART 2 - PRODUCTS**2.1 SOLID SURFACE COUNTERTOP MATERIALS**

- A. Solid Surface Material: Homogeneous-filled plastic resin complying with ICPA SS-1.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by the following:
 - a. DuPont de Nemours Inc.; Corian Solid Surface.
 - b. Aristech Surfaces; Avonite Surfaces
 - 2. Type: Provide Standard type and Veneer type made from material complying with requirements for Standard type.

3. Colors and Patterns: As indicated on Schedule – Finish Materials on drawings.

B. Plywood: Exterior softwood plywood complying with DOC PS 1, Grade C-C Plugged, touch sanded.

2.2 COUNTERTOP FABRICATION

A. Fabricate countertops according to solid surface material manufacturer's written instructions and to the AWI/AWMAC/WI's "Architectural Woodwork Standards."

1. Grade: Premium.

B. Configuration:

1. Front: Straight, slightly eased at top.
2. Backsplash: Straight, slightly eased at corner.
3. End splash: Matching backsplash.

C. Countertops: 1-1/4 inches (30-mm) thick, solid surface with front edge built up with same material.

D. Backsplashes and end splashes: 1-1/4 inches (30-mm) thick, solid surface.

E. Joints: Fabricate countertops without joints.

F. Cutouts and Holes:

1. Undercounter Plumbing Fixtures: Make cutouts for fixtures in shop using template or pattern furnished by fixture manufacturer. Form cutouts to smooth, even curves.

2.3 INSTALLATION MATERIALS

A. Adhesive/mastic: Product recommended by solid surface material manufacturer.

1. VOC Content: Multipurpose construction adhesives shall have a VOC content of 70g/L or less when calculated according to 40 CFR59, Subpart D (EPA Method D).

B. Sealant for Countertops: Comply with applicable requirements in Section 079200 "Joint Sealants."

1. VOC Content: Sealants shall have a VOC content of 250g/L or less when calculated according to 40 CFR59, Subpart D (EPA Method D).

PART 3 - EXECUTION

3.1 INSTALLATION

A. Fasten countertops by screwing through corner blocks of base units into underside of countertop. Predrill holes for screws as recommended by manufacturer.

B. Fasten subtops to cabinets by screwing through subtops into cornerblocks of base cabinets. Shim as needed to align subtops in a level plane.

- C. Secure countertops to subtops with adhesive according to solid surface material manufacturer's written instructions.
- D. Bond joints with adhesive and draw tight as countertops are set. Mask areas of countertops adjacent to joints to prevent adhesive smears.
- E. Install backsplashes and end splashes by adhering to wall and countertops with adhesive.
- F. Complete cutouts not finished in shop. Mask areas of countertops adjacent to cutouts to prevent damage while cutting. Make cutouts to accurately fit items to be installed, and at right angles to finished surfaces unless beveling is required for clearance. Ease edges slightly to prevent snipping.
- G. Apply sealant to joints between solid surface and walls; comply with Section 079200 "Joint Sealants."

END OF SECTION 123661.16

SECTION 210517 - SLEEVES AND SLEEVE SEALS FOR FIRE-SUPPRESSION PIPING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Sleeves.
 - 2. Stack-sleeve fittings.
 - 3. Sleeve-seal systems.
 - 4. Sleeve-seal fittings.
 - 5. Grout.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

PART 2 - PRODUCTS**2.1 SLEEVES**

- A. Cast-Iron Wall Pipes: Cast or fabricated of cast or ductile iron and equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop unless otherwise indicated.
- B. Galvanized-Steel Wall Pipes: ASTM A 53/A 53M, Schedule 40, with plain ends and welded steel collar; zinc coated.
- C. Galvanized-Steel-Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, zinc coated, with plain ends.
- D. PVC-Pipe Sleeves: ASTM D 1785, Schedule 40.
- E. Galvanized-Steel-Sheet Sleeves: 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint.
- F. Molded-PE or -PP Sleeves: Removable, tapered-cup shaped, and smooth outer surface with nailing flange for attaching to wooden forms.
- G. Molded-PVC Sleeves: With nailing flange for attaching to wooden forms.

2.2 STACK-SLEEVE FITTINGS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. Smith, Jay R. Mfg. Co.
 - 2. Zurn Specification Drainage Operation; Zurn Plumbing Products Group.

- B. Description: Manufactured, cast-iron sleeve with integral clamping flange. Include clamping ring, bolts, and nuts for membrane flashing.

- 1. Underdeck Clamp: Clamping ring with setscrews.

2.3 SLEEVE-SEAL SYSTEMS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

- 1. Advance Products & Systems, Inc.
 - 2. CALPICO, Inc.
 - 3. Metraflex Company (The).
 - 4. Pipeline Seal and Insulator, Inc.
 - 5. Proco Products, Inc.

- B. Description: Modular sealing-element unit, designed for field assembly, for filling annular space between piping and sleeve.

- 1. Sealing Elements: EPDM-rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
 - 2. Pressure Plates: Carbon steel.
 - 3. Connecting Bolts and Nuts: Carbon steel, with corrosion-resistant coating, of length required to secure pressure plates to sealing elements.

2.4 SLEEVE-SEAL FITTINGS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

- 1. Presealed Systems.

- B. Description: Manufactured plastic, sleeve-type, waterstop assembly made for imbedding in concrete slab or wall. Unit has plastic or rubber waterstop collar with center opening to match piping OD.

2.5 GROUT

- A. Standard: ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.
- B. Characteristics: Nonshrink; recommended for interior and exterior applications.
- C. Design Mix: 5000-psi, 28-day compressive strength.
- D. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION

- A. Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.

- B. For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide 1-inch annular clear space between piping and concrete slabs and walls.
 - 1. Sleeves are not required for core-drilled holes.
- C. Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.
 - 1. Permanent sleeves are not required for holes in slabs formed by molded-PE or -PP sleeves.
 - 2. Cut sleeves to length for mounting flush with both surfaces.
 - a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level.
 - 3. Using grout, seal the space outside of sleeves in slabs and walls without sleeve-seal system.
- D. Install sleeves for pipes passing through interior partitions.
 - 1. Cut sleeves to length for mounting flush with both surfaces.
 - 2. Install sleeves that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
 - 3. Seal annular space between sleeve and piping or piping insulation; use joint sealants appropriate for size, depth, and location of joint. Comply with requirements for sealants specified in Section 079200 "Joint Sealants."
- E. Fire-Barrier Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with firestop materials. Comply with requirements for firestopping specified in Section 078413 "Penetration Firestopping."

3.2 STACK-SLEEVE-FITTING INSTALLATION

- A. Install stack-sleeve fittings in new slabs as slabs are constructed.
 - 1. Install fittings that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
 - 2. Secure flashing between clamping flanges for pipes penetrating floors with membrane waterproofing. Comply with requirements for flashing specified in Section 076200 "Sheet Metal Flashing and Trim."
 - 3. Install section of cast-iron soil pipe to extend sleeve to 2 inches above finished floor level.
 - 4. Extend cast-iron sleeve fittings below floor slab as required to secure clamping ring if ring is specified.
 - 5. Using grout, seal the space around outside of stack-sleeve fittings.
- B. Fire-Barrier Penetrations: Maintain indicated fire rating of floors at pipe penetrations. Seal pipe penetrations with firestop materials. Comply with requirements for firestopping specified in Section 078413 "Penetration Firestopping."

3.3 SLEEVE-SEAL-SYSTEM INSTALLATION

- A. Select type, size, and number of sealing elements required for piping material and size and for sleeve ID or hole size. Position piping in center of sleeve. Center piping in penetration, assemble sleeve-seal system components, and install in annular space between piping and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make a watertight seal.

3.4 SLEEVE-SEAL-FITTING INSTALLATION

- A. Install sleeve-seal fittings in new walls and slabs as they are constructed.
- B. Assemble fitting components of length to be flush with both surfaces of concrete slabs and walls. Position waterstop flange to be centered in concrete slab or wall.
- C. Secure nailing flanges to concrete forms.
- D. Using grout, seal the space around outside of sleeve-seal fittings.

3.5 SLEEVE AND SLEEVE-SEAL SCHEDULE

- A. Use sleeves and sleeve seals for the following piping-penetration applications:
 - 1. Concrete Slabs above Grade:
 - a. Piping Smaller Than NPS 6: Galvanized-steel-pipe sleeves.
 - 2. Interior Partitions:
 - a. Piping Smaller Than NPS 6: Galvanized-steel-pipe sleeves.

END OF SECTION 210517

SECTION 210518 - ESCUTCHEONS FOR FIRE-SUPPRESSION PIPING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Escutcheons.
 - 2. Floor plates.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

PART 2 - PRODUCTS**2.1 ESCUTCHEONS**

- A. One-Piece, Cast-Brass Type: White finish and setscrew fastener.
- B. One-Piece, Deep-Pattern Type: Deep-drawn, box-shaped brass with white finish and spring-clip fasteners.
- C. One-Piece, Stamped-Steel Type: White finish and spring-clip fasteners.
- D. Split-Casting Brass Type: White finish and with concealed hinge and setscrew.

2.2 FLOOR PLATES

- A. One-Piece Floor Plates: Cast-iron flange with holes for fasteners.
- B. Split-Casting Floor Plates: Chrome plated brass with concealed hinge.

PART 3 - EXECUTION**3.1 INSTALLATION**

- A. Install escutcheons for piping penetrations of walls, ceilings, and finished floors.
- B. Install escutcheons with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.
 - 1. Escutcheons for New Piping:
 - a. Piping with Fitting or Sleeve Protruding from Wall: One-piece, deep-pattern type.

- b. Painted Piping: One-piece, cast-brass type with white painted finish.
 - c. Bare Piping at Wall and Floor Penetrations in Finished Spaces: One-piece, cast- brass or split-casting brass type with white finish.
 - d. Bare Piping at Ceiling Penetrations in Finished Spaces: One-piece, cast-brass type with white finish.
 - e. Bare Piping in Unfinished Service Spaces: One-piece, cast-brass or split-casting brass type with white finish.
 - f. Bare Piping in Equipment Rooms: One-piece, cast-brass or split-casting brass type with white finish.
- C. Install floor plates for piping penetrations of equipment-room floors.
- D. Install floor plates with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.
1. New Piping: One-piece, floor-plate type.

3.2 FIELD QUALITY CONTROL

- A. Replace broken and damaged escutcheons and floor plates using new materials.

END OF SECTION 210518

SECTION 210523 - GENERAL-DUTY VALVES FOR WATER-BASED FIRE-SUPPRESSION PIPING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Two-piece ball valves with indicators.
 - 2. Bronze butterfly valves with indicators.
 - 3. Iron butterfly valves with indicators.
 - 4. Check valves.
 - 5. Bronze OS&Y gate valves.
 - 6. Iron OS&Y gate valves.
 - 7. NRS gate valves.
 - 8. Trim and drain valves.

1.3 DEFINITIONS

- A. NBR: Acrylonitrile-butadiene, Buna-N, or nitrile rubber.
- B. NRS: Nonrising stem.
- C. OS&Y: Outside screw and yoke.
- D. SBR: Styrene-butadiene rubber.

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of valve.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Prepare valves for shipping as follows:
 - 1. Protect internal parts against rust and corrosion.
 - 2. Protect threads, flange faces, and weld ends.
 - 3. Set valves open to minimize exposure of functional surfaces.
- B. Use the following precautions during storage:
 - 1. Maintain valve end protection.
 - 2. Store valves indoors and maintain at higher than ambient dew point temperature. If outdoor storage is necessary, store valves off the ground in watertight enclosures.

- C. Use sling to handle large valves; rig sling to avoid damage to exposed parts. Do not use operating handles or stems as lifting or rigging points.
- D. Protect flanges and specialties from moisture and dirt.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR VALVES

- A. UL Listed: Valves shall be listed in UL's "Online Certifications Directory" under the headings listed below and shall bear UL mark:
 - 1. Main Level: HAMV - Fire Main Equipment.
 - a. Level 1: HCBZ - Indicator Posts, Gate Valve.
 - b. Level 1: HLOT - Valves.
 - 1) Level 3: HLUG - Ball Valves, System Control.
 - 2) Level 3: HLXS - Butterfly Valves.
 - 3) Level 3: HMER - Check Valves.
 - 4) Level 3: HMRZ - Gate Valves.
 - 2. Main Level: VDGT - Sprinkler System & Water Spray System Devices.
 - a. Level 1: VQGU - Valves, Trim and Drain.
- B. FM Global Approved: Valves shall be listed in its "Approval Guide," under the headings listed below:
 - 1. Automated Sprinkler Systems:
 - a. Valves.
 - 1) Gate valves.
 - 2) Check valves.
 - a) Single check valves.
 - 3) Miscellaneous valves.
- C. Source Limitations for Valves: Obtain valves for each valve type from single manufacturer.
- D. ASME Compliance:
 - 1. ASME B16.1 for flanges on iron valves.
 - 2. ASME B1.20.1 for threads for threaded-end valves.
 - 3. ASME B31.9 for building services piping valves.
- E. AWWA Compliance: Comply with AWWA C606 for grooved-end connections.
- F. NFPA Compliance: Comply with NFPA 24 for valves.

- G. Valve Pressure Ratings: Not less than the minimum pressure rating indicated or higher as required by system pressures.
- H. Valve Sizes: Same as upstream piping unless otherwise indicated.
- I. Valve Actuator Types:
 - 1. Worm-gear actuator with handwheel for quarter-turn valves, except for trim and drain valves.
 - 2. Handwheel: For other than quarter-turn trim and drain valves.
 - 3. Handlever: For quarter-turn trim and drain valves NPS 2 and smaller.

2.2 TWO-PIECE BALL VALVES WITH INDICATORS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. NIBCO INC.
 - 2. Victaulic Company.
- B. Description:
 - 1. UL 1091, except with ball instead of disc and FM Global standard for indicating valves (butterfly or ball type), Class Number 1112.
 - 2. Minimum Pressure Rating: 175 psig.
 - 3. Body Design: Two piece.
 - 4. Body Material: Forged brass or bronze.
 - 5. Port Size: Full or standard.
 - 6. Seats: PTFE.
 - 7. Stem: Bronze or stainless steel.
 - 8. Ball: Chrome-plated brass.
 - 9. Actuator: Worm gear or traveling nut.
 - 10. Supervisory Switch: Internal or external.
 - 11. End Connections for Valves NPS 1 through NPS 2: Threaded ends.
 - 12. End Connections for Valves NPS 2-1/2: Grooved ends.

2.3 BRONZE BUTTERFLY VALVES WITH INDICATORS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. Fivalco Inc.
 - 2. Globe Fire Sprinkler Corporation.
 - 3. Milwaukee Valve Company.
- B. Description:
 - 1. Standard: UL 1091 and FM Global standard for indicating valves, (butterfly or ball type), Class Number 1112.
 - 2. Minimum: Pressure rating: 175 psig.

3. Body Material: Bronze.
4. Seat Material: EPDM.
5. Stem Material: Bronze or stainless steel.
6. Disc: Bronze with EPDM coating.
7. Actuator: Worm gear or traveling nut.
8. Supervisory Switch: Internal or external.
9. Ends Connections for Valves NPS 1 through NPS 2: Threaded ends.
10. Ends Connections for Valves NPS 2-1/2: Grooved ends.

2.4 IRON BUTTERFLY VALVES WITH INDICATORS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Anvil International, Inc.
2. Fivalco Inc.
3. Globe Fire Sprinkler Corporation.
4. Kennedy Valve; a division of McWane, Inc.
5. NIBCO INC.
6. Tyco Fire & Building Products LP.
7. Victaulic Company.

- B. Description:

1. Standard: UL 1091 and FM Global standard for indicating valves, (butterfly or ball type), Class Number 112.
2. Minimum Pressure Rating: 175 psig.
3. Body Material: Cast or ductile iron with nylon, EPDM, epoxy, or polyamide coating.
4. Seat Material: EPDM.
5. Stem: Stainless steel.
6. Disc: Ductile iron, nickel plated and EPDM or SBR coated.
7. Actuator: Worm gear or traveling nut.
8. Supervisory Switch: Internal or external.
9. Body Design: Grooved-end connections.

2.5 CHECK VALVES

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Anvil International, Inc.
2. Fire Protection Products, Inc.
3. Fivalco Inc.
4. Globe Fire Sprinkler Corporation.
5. Kennedy Valve; a division of McWane, Inc.
6. Matco-Norca.
7. Mueller Co.; Water Products Division.
8. NIBCO INC.
9. Reliable Automatic Sprinkler Co., Inc.

10. Shurjoint Piping Products.
11. Tyco Fire & Building Products LP.
12. United Brass Works, Inc.
13. Venus Fire Protection Ltd.
14. Victaulic Company.
15. Viking Corporation.
16. Watts Water Technologies, Inc.
17. Wilson & Cousins Inc.

B. Description:

1. Standard: UL 312 and FM Global standard for swing check valves, Class Number 1210.
2. Minimum Pressure Rating: 175 psig.
3. Type: Single swing check.
4. Body Material: Cast iron, ductile iron, or bronze.
5. Clapper: Bronze, ductile iron, or stainless steel with elastomeric seal.
6. Clapper Seat: Brass, bronze, or stainless steel.
7. Hinge Shaft: Bronze or stainless steel.
8. Hinge Spring: Stainless steel.
9. End Connections: Flanged, grooved, or threaded.

2.6 BRONZE OS&Y GATE VALVES

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Milwaukee Valve Company.
2. NIBCO INC.
3. United Brass Works, Inc.

B. Description:

1. Standard: UL 262 and FM Global standard for fire-service water control valves (OS&Y- and NRS-type gate valves).
2. Minimum Pressure Rating: 175 psig.
3. Body and Bonnet Material: Bronze or brass.
4. Wedge: One-piece bronze or brass.
5. Wedge Seat: Bronze.
6. Stem: Bronze or brass.
7. Packing: Non-asbestos PTFE.
8. Supervisory Switch: External.
9. End Connections: Threaded.

2.7 IRON OS&Y GATE VALVES

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. American Cast Iron Pipe Company; Waterous Company subsidiary.
2. Clow Valve Company; a division of McWane, Inc.
3. Hammond Valve.
4. Kennedy Valve; a division of McWane, Inc.
5. Mueller Co.; Water Products Division.
6. NIBCO INC.
7. Victaulic Company.
8. Watts Water Technologies, Inc.

B. Description:

1. Standard: UL 262 and FM Global standard for fire-service water control valves (OS&Y- and NRS-type gate valves).
2. Minimum Pressure Rating: 175 psig.
3. Body and Bonnet Material: Cast or ductile iron.
4. Wedge: Cast or ductile iron, or bronze with elastomeric coating.
5. Wedge Seat: Cast or ductile iron, or bronze with elastomeric coating.
6. Stem: Brass or bronze.
7. Packing: Non-asbestos PTFE.
8. Supervisory Switch: External.
9. End Connections: Grooved.

2.8 NRS GATE VALVES

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. American Cast Iron Pipe Company; Waterous Company subsidiary.
2. Clow Valve Company; a division of McWane, Inc.
3. Kennedy Valve; a division of McWane, Inc.
4. Mueller Co.; Water Products Division.
5. NIBCO INC.
6. Victaulic Company.

B. Description:

1. Standard: UL 262 and FM Global standard for fire-service water control valves (OS&Y- and NRS-type gate valves).
2. Minimum Pressure Rating: 175 psig.
3. Body and Bonnet Material: Cast or ductile iron.
4. Wedge: Cast or ductile iron with elastomeric coating.
5. Wedge Seat: Cast or ductile iron, or bronze with elastomeric coating.
6. Stem: Brass or bronze.
7. Packing: Non-asbestos PTFE.
8. Supervisory Switch: External.
9. End Connections: Grooved.

2.9 TRIM AND DRAIN VALVES

A. Ball Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Conbraco Industries, Inc.; Apollo Valves.
 - b. Fire-End & Croker Corporation.
 - c. Fire Protection Products, Inc.
 - d. Flowserve.
 - e. FNW; Ferguson Enterprises, Inc.
 - f. Jomar International, LTD.
 - g. Kitz Corporation.
 - h. Legend Valve.
 - i. Metso Automation USA Inc.
 - j. Milwaukee Valve Company.
 - k. NIBCO INC.
 - l. Potter Roemer.
 - m. Red-White Valve Corporation.
 - n. Tyco Fire & Building Products LP.
 - o. Victaulic Company.
 - p. Watts Water Technologies, Inc.
2. Description:
 - a. Pressure Rating: 175 psig.
 - b. Body Design: Two piece.
 - c. Body Material: Forged brass or bronze.
 - d. Port size: Full or standard.
 - e. Seats: PTFE.
 - f. Stem: Bronze or stainless steel.
 - g. Ball: Chrome-plated brass.
 - h. Actuator: Handlever.
 - i. End Connections for Valves NPS 1 through NPS 2-1/2: Threaded ends.
 - j. End Connections for Valves NPS 1-1/4 and NPS 2-1/2: Grooved ends.

B. Angle Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Fire Protection Products, Inc.
 - b. NIBCO INC.
 - c. United Brass Works, Inc.
2. Description:
 - a. Pressure Rating: 175 psig.
 - b. Body Material: Brass or bronze.

- c. Ends: Threaded.
- d. Stem: Bronze.
- e. Disc: Bronze.
- f. Packing: Asbestos free.
- g. Handwheel: Malleable iron, bronze, or aluminum.

C. Globe Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

- a. NIBCO INC.
- b. United Brass Works, Inc.

2. Description:

- a. Pressure Rating: 175 psig.
- b. Body Material: Bronze with integral seat and screw-in bonnet.
- c. Ends: Threaded.
- d. Stem: Bronze.
- e. Disc Holder and Nut: Bronze.
- f. Disc Seat: Nitrile.
- g. Packing: Asbestos free.
- h. Handwheel: Malleable iron, bronze, or aluminum.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine valve interior for cleanliness, freedom from foreign matter, and corrosion. Remove special packing materials, such as blocks, used to prevent disc movement during shipping and handling.
- B. Operate valves in positions from fully open to fully closed. Examine guides and seats made accessible by such operations.
- C. Examine threads on valve and mating pipe for form and cleanliness.
- D. Examine mating flange faces for conditions that might cause leakage. Check bolting for proper size, length, and material. Verify that gasket is of proper size, that its material composition is suitable for service, and that it is free from defects and damage.
- E. Do not attempt to repair defective valves; replace with new valves.

3.2 GENERAL REQUIREMENTS FOR VALVE INSTALLATION

- A. Comply with requirements in the following Sections for specific valve installation requirements and applications:

1. Section 211313 "Wet-Pipe Sprinkler Systems" for application of valves in wet-pipe, fire- suppression sprinkler systems.
 2. Section 211316 "Dry-Pipe Sprinkler Systems" for application of valves in dry-pipe, fire- suppression sprinkler systems.
-
- B. Install listed fire-protection shutoff valves supervised-open, located to control sources of water supply except from fire-department connections. Install permanent identification signs indicating portion of system controlled by each valve.
 - C. Install check valve in each water-supply connection. Install backflow preventers instead of check valves in potable-water-supply sources.
 - D. Install valves having threaded connections with unions at each piece of equipment arranged to allow easy access, service, maintenance, and equipment removal without system shutdown. Provide separate support where necessary.
 - E. Install valves in horizontal piping with stem at or above the pipe center.
 - F. Install valves in position to allow full stem movement.
 - G. Install valve tags. Comply with requirements in Section 210553 "Identification for Fire- Suppression Piping and Equipment" for valve tags and schedules and signs on surfaces concealing valves; and the NFPA standard applying to the piping system in which valves are installed. Install permanent identification signs indicating the portion of system controlled by each valve.
 - H. Install listed fire-protection shutoff valves supervised-open, located to control sources of water supply except from fire-department connections.
 - I. Install check valve in each water-supply connection. Install backflow preventers instead of check valves in potable-water-supply sources.

END OF SECTION 210523

SECTION 210548 - VIBRATION AND SEISMIC CONTROLS FOR FIRE-SUPPRESSION PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY**A. Section Includes:**

1. Elastomeric isolation pads.
2. Elastomeric isolation mounts.
3. Restrained elastomeric isolation mounts.
4. Pipe-riser resilient supports.
5. Resilient pipe guides.
6. Elastomeric hangers.
7. Snubbers.
8. Restraint channel bracings.
9. Seismic-restraint accessories.
10. Mechanical anchor bolts.
11. Adhesive anchor bolts.

B. Related Requirements:

1. Section 220548 "Vibration and Seismic Controls for Plumbing Piping and Equipment" for devices for plumbing equipment and systems.

1.3 DEFINITIONS

- A. IBC: International Building Code.
- B. ICC-ES: ICC-Evaluation Service.
- C. OSHPD: Office of Statewide Health Planning & Development (for the State of California).

1.4 ACTION SUBMITTALS**A. Product Data: For each type of product.**

1. Include rated load, rated deflection, and overload capacity for each vibration isolation device.
2. Illustrate and indicate style, material, strength, fastening provision, and finish for each type and size of vibration isolation device and seismic-restraint component required.
 - a. Tabulate types and sizes of seismic restraints, complete with report numbers and rated strength in tension and shear as evaluated by an agency acceptable to authorities having jurisdiction.

- b. Annotate to indicate application of each product submitted and compliance with requirements.
 - 3. Interlocking Snubbers: Include ratings for horizontal, vertical, and combined loads.
- B. Delegated-Design Submittal: For each vibration isolation and seismic-restraint device.
 - 1. Include design calculations and details for selecting vibration isolators and seismic restraints complying with performance requirements, design criteria, and analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
 - 2. Design Calculations: Calculate static and dynamic loading due to equipment weight and operation, due to seismic forces required to select vibration isolators, and due to seismic restraints.
 - 3. Riser Supports: Include riser diagrams and calculations showing anticipated expansion and contraction at each support point, initial and final loads on building structure, spring deflection changes, and seismic loads. Include certification that riser system was examined for excessive stress and that none exists.
 - 4. Seismic-Restraint Details:
 - a. Design Analysis: To support selection and arrangement of seismic restraints. Include calculations of combined tensile and shear loads.
 - b. Details: Indicate fabrication and arrangement. Detail attachments of restraints to the restrained items and to the structure. Show attachment locations, methods, and spacings. Identify components, list their strengths, and indicate directions and values of forces transmitted to the structure during seismic events. Indicate association with vibration isolation devices.
 - c. Coordinate seismic-restraint and vibration isolation details with wind-restraint details required for equipment mounted outdoors. Comply with requirements in other Sections for equipment mounted outdoors.
 - d. Preapproval and Evaluation Documentation: By an agency acceptable to authorities having jurisdiction, showing maximum ratings of restraint items and the basis for approval (tests or calculations).

1.5 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Show coordination of vibration isolation device installation and seismic bracing for fire-suppression piping and equipment with other systems and equipment in the vicinity, including other supports and restraints, if any.
- B. Qualification Data: For testing agency.
- C. Welding certificates.
- D. Field quality-control reports.

1.6 QUALITY ASSURANCE

- A. Testing Agency Qualifications: An independent agency, with the experience and capability to conduct the testing indicated, that is an NRTL as defined by OSHA in 29 CFR 1910.7 and that is acceptable to authorities having jurisdiction.
- B. Comply with seismic-restraint requirements in the IBC unless requirements in this Section

are more stringent.

- C. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."
- D. Seismic-restraint devices shall have horizontal and vertical load testing and analysis and shall bear anchorage preapproval OPA number from OSHPD, preapproval by ICC-ES, or preapproval by another agency acceptable to authorities having jurisdiction, showing maximum seismic-restraint ratings. Ratings based on independent testing are preferred to ratings based on calculations. If preapproved ratings are unavailable, submittals based on independent testing are preferred. Calculations (including combining shear and tensile loads) to support seismic-restraint designs must be signed and sealed by a qualified professional engineer.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Seismic-Restraint Loading:
 - 1. Site Class as Defined in the IBC: B.
 - 2. Assigned Seismic Use Group or Building Category as Defined in the IBC: [I] [II] [III].
 - a. Component Importance Factor: 1.0.
 - b. Component Response Modification Factor: 1.
 - c. Component Amplification Factor: 1.0.
 - 3. Design Spectral Response Acceleration at Short Periods (0.2 Second):.
 - 4. Design Spectral Response Acceleration at 1.0-Second Period:.
 - 5. Rated strengths, features, and applications shall be as defined in reports by an agency acceptable to authorities having jurisdiction.
 - a. Structural Safety Factor: Allowable strength in tension, shear, and pullout force of components shall be at least four times the maximum seismic forces to which they are subjected.

2.2 ELASTOMERIC ISOLATION PADS

- A. Elastomeric Isolation Pads:
 - 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Ace Mountings Co., Inc.
 - b. California Dynamics Corporation.
 - c. Isolation Technology, Inc.
 - d. Kinetics Noise Control, Inc.
 - e. Mason Industries, Inc.
 - f. Vibration Eliminator Co., Inc.
 - g. Vibration Isolation.
 - h. Vibration Mountings & Controls, Inc.

2. Fabrication: Single or multiple layers of sufficient durometer stiffness for uniform loading over pad area.
3. Size: Factory or field cut to match requirements of supported equipment.
4. Pad Material: Oil and water resistant with elastomeric properties.
5. Surface Pattern: Ribbed pattern.
6. Infused nonwoven cotton or synthetic fibers.
7. Load-bearing metal plates adhered to pads.
8. Sandwich-Core Material: Resilient and elastomeric.
 - a. Surface Pattern: Ribbed pattern.
 - b. Infused nonwoven cotton or synthetic fibers.

2.3 ELASTOMERIC ISOLATION MOUNTS

A. Double-Deflection, Elastomeric Isolation Mounts: .

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Ace Mountings Co., Inc.
 - b. California Dynamics Corporation.
 - c. Isolation Technology, Inc.
 - d. Kinetics Noise Control, Inc.
 - e. Mason Industries, Inc.
 - f. Vibration Eliminator Co., Inc.
 - g. Vibration Isolation.
 - h. Vibration Mountings & Controls, Inc.
2. Mounting Plates:
 - a. Top Plate: Encapsulated steel load transfer top plates, factory drilled and threaded with threaded studs or bolts.
 - b. Baseplate: Encapsulated steel bottom plates with holes provided for anchoring to support structure.
3. Elastomeric Material: Molded, oil-resistant rubber, neoprene, or other elastomeric material.

2.4 RESTRAINED ELASTOMERIC ISOLATION MOUNTS

A. Restrained Elastomeric Isolation Mounts: .

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Ace Mountings Co., Inc.
 - b. California Dynamics Corporation.
 - c. Isolation Technology, Inc.

- d. Kinetics Noise Control, Inc.
 - e. Mason Industries, Inc.
 - f. Vibration Eliminator Co., Inc.
 - g. Vibration Isolation.
 - h. Vibration Mountings & Controls, Inc.
2. Description: All-directional isolator with seismic restraints containing two separate and opposing elastomeric elements that prevent central threaded element and attachment hardware from contacting the housing during normal operation.
- a. Housing: Cast-ductile iron or welded steel.
 - b. Elastomeric Material: Molded, oil-resistant rubber, neoprene or other elastomeric material.

2.5 PIPE-RISER RESILIENT SUPPORT

- A. Description: All-directional, acoustical pipe anchor consisting of two steel tubes separated by a minimum 1/2-inch- thick neoprene.
- 1. Vertical-Limit Stops: Steel and neoprene vertical-limit stops arranged to prevent vertical travel in both directions.
 - 2. Maximum Load Per Support: 500 psig on isolation material providing equal isolation in all directions.

2.6 RESILIENT PIPE GUIDES

- A. Description: Telescopic arrangement of two steel tubes or post-and-sleeve arrangement separated by a minimum 1/2-inch- thick neoprene.
- 1. Factory-Set Height Guide with Shear Pin: Shear pin shall be removable and reinsertable to allow for selection of pipe movement. Guides shall be capable of motion to meet location requirements.

2.7 ELASTOMERIC HANGERS

- A. Elastomeric Mount in a Steel Frame with Upper and Lower Steel Hanger Rods: .
- 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Ace Mountings Co., Inc.
 - b. California Dynamics Corporation.
 - c. Isolation Technology, Inc.
 - d. Kinetics Noise Control, Inc.
 - e. Mason Industries, Inc.
 - f. Vibration Eliminator Co., Inc.
 - g. Vibration Mountings & Controls, Inc.
 - 2. Frame: Steel, fabricated with a connection for an upper threaded hanger rod and an opening on the underside to allow for a maximum of 30 degrees of angular lower

- hanger- rod misalignment without binding or reducing isolation efficiency.
3. Dampening Element: Molded, oil-resistant rubber, neoprene, or other elastomeric material with a projecting bushing for the underside opening preventing steel to steel contact.

2.8 SNUBBERS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 1. Kinetics Noise Control, Inc.
 2. Mason Industries, Inc.
 3. Vibration Mountings & Controls, Inc.
- B. Description: Factory fabricated using welded structural-steel shapes and plates, anchor bolts, and replaceable resilient isolation washers and bushings.
 1. Anchor bolts for attaching to concrete shall be seismic-rated, drill-in, and stud-wedge or female-wedge type.
 2. Resilient Isolation Washers and Bushings: Oil- and water-resistant neoprene.
 3. Maximum 1/4-inch air gap, and minimum 1/4-inch- thick resilient cushion.

2.9 RESTRAINT CHANNEL BRACINGS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 1. Cooper B-Line, Inc.
 2. Hilti, Inc.
 3. Mason Industries, Inc.
 4. Unistrut.
- B. Description: MFMA-4, shop- or field-fabricated bracing assembly made of slotted steel channels with accessories for attachment to braced component at one end and to building structure at the other end and other matching components and with corrosion-resistant coating; rated in tension, compression, and torsion forces.

2.10 SEISMIC-RESTRAINT ACCESSORIES

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 1. Cooper B-Line, Inc.
 2. Kinetics Noise Control, Inc.
 3. Mason Industries, Inc.
 4. TOLCO.
- B. Hanger-Rod Stiffener: Reinforcing steel angle clamped to hanger rod.

- C. Hinged and Swivel Brace Attachments: Multifunctional steel connectors for attaching hangers to rigid channel bracings.
- D. Bushings for Floor-Mounted Equipment Anchor Bolts: Neoprene bushings designed for rigid equipment mountings, and matched to type and size of anchor bolts and studs.
- E. Bushing Assemblies for Wall-Mounted Equipment Anchorage: Assemblies of neoprene elements and steel sleeves designed for rigid equipment mountings, and matched to type and size of attachment devices used.
- F. Resilient Isolation Washers and Bushings: One-piece, molded, oil- and water-resistant neoprene, with a flat washer face.

2.11 MECHANICAL ANCHOR BOLTS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. Cooper B-Line, Inc.
 - 2. Hilti, Inc.
 - 3. Kinetics Noise Control, Inc.
 - 4. Mason Industries, Inc.
- B. Mechanical Anchor Bolts: Drilled-in and stud-wedge or female-wedge type in zinc-coated steel for interior applications and stainless steel for exterior applications. Select anchor bolts with strength required for anchor and as tested according to ASTM E 488.

2.12 ADHESIVE ANCHOR BOLTS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. Hilti, Inc.
 - 2. Kinetics Noise Control, Inc.
 - 3. Mason Industries, Inc.
- B. Adhesive Anchor Bolts: Drilled-in and capsule anchor system containing PVC or urethane methacrylate-based resin and accelerator, or injected polymer or hybrid mortar adhesive. Provide anchor bolts and hardware with zinc-coated steel for interior applications and stainless steel for exterior applications. Select anchor bolts with strength required for anchor and as tested according to ASTM E 488.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and equipment to receive vibration isolation and seismic-control devices for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Examine roughing-in of reinforcement and cast-in-place anchors to verify actual locations before installation.

- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 APPLICATIONS

- A. Multiple Pipe Supports: Secure pipes to trapeze member with clamps approved for application by an agency acceptable to authorities having jurisdiction.
- B. Hanger-Rod Stiffeners: Install hanger-rod stiffeners where indicated or scheduled on Drawings to receive them and where required to prevent buckling of hanger rods due to seismic forces.
- C. Strength of Support and Seismic-Restraint Assemblies: Where not indicated, select sizes of components so strength is adequate to carry present and future static and seismic loads within specified loading limits.

3.3 VIBRATION CONTROL AND SEISMIC-RESTRAINT DEVICE INSTALLATION

- A. Coordinate the location of embedded connection hardware with supported equipment attachment and mounting points and with requirements for concrete reinforcement.
- B. Installation of vibration isolators must not cause any change of position of equipment, piping, or ductwork resulting in stresses or misalignment.
- C. Equipment Restraints:
 - 1. Install seismic snubbers on fire-suppression equipment mounted on vibration isolators. Locate snubbers as close as possible to vibration isolators and bolt to equipment base and supporting structure.
 - 2. Install resilient bolt isolation washers on equipment anchor bolts where clearance between anchor and adjacent surface exceeds 0.125 inch.
 - 3. Install seismic-restraint devices using methods approved by an agency acceptable to authorities having jurisdiction that provides required submittals for component.
- D. Piping Restraints:
 - 1. Comply with requirements in MSS SP-127.
 - 2. Space lateral supports a maximum of 40 feet o.c., and longitudinal supports a maximum of 80 feet o.c.
 - 3. Brace a change of direction longer than 12 feet.
- E. Install seismic-restraint devices using methods approved by an agency acceptable to authorities having jurisdiction that provides required submittals for component.
- F. Install bushing assemblies for anchor bolts for floor-mounted equipment, arranged to provide resilient media between anchor bolt and mounting hole in concrete base.
- G. Install bushing assemblies for mounting bolts for wall-mounted equipment, arranged to provide resilient media where equipment or equipment-mounting channels are attached to wall.
- H. Attachment to Structure: If specific attachment is not indicated, anchor bracing to structure

at flanges of beams, at upper truss chords of bar joists, or at concrete members.

I. Drilled-in Anchors:

1. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Do not damage existing reinforcing or embedded items during coring or drilling. Notify the structural engineer if reinforcing steel or other embedded items are encountered during drilling. Locate and avoid prestressed tendons, electrical and telecommunications conduit, and gas lines.
2. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength.
3. Wedge Anchors: Protect threads from damage during anchor installation. Heavy-duty sleeve anchors shall be installed with sleeve fully engaged in the structural element to which anchor is to be fastened.
4. Adhesive Anchors: Clean holes to remove loose material and drilling dust prior to installation of adhesive. Place adhesive in holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the adhesive.
5. Set anchors to manufacturer's recommended torque, using a torque wrench.
6. Install zinc-coated steel anchors for interior and stainless-steel anchors for exterior applications.

3.4 ACCOMMODATION OF DIFFERENTIAL SEISMIC MOTION

- A. Install flexible connections in piping where they cross seismic joints, where adjacent sections or branches are supported by different structural elements, and where the connections terminate with connection to equipment that is anchored to a different structural element from the one supporting the connections as they approach equipment. Comply with requirements in Section 211313 "Wet-Pipe Sprinkler Systems," and Section 211316 "Dry-Pipe Sprinkler Systems" for piping flexible connections.

3.5 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.
- B. Perform tests and inspections.
- C. Tests and Inspections:
1. Provide evidence of recent calibration of test equipment by a testing agency acceptable to authorities having jurisdiction.
 2. Schedule test with Owner, through Architect, before connecting anchorage device to restrained component (unless postconnection testing has been approved), and with at least seven days' advance notice.
 3. Obtain Architect's approval before transmitting test loads to structure. Provide temporary load-spreading members.
 4. Test at least four of each type and size of installed anchors and fasteners selected by Architect.
 5. Test to 90 percent of rated proof load of device.
 6. Measure isolator restraint clearance.
 7. Measure isolator deflection.
 8. Verify snubber minimum clearances.

D. Remove and replace malfunctioning units and retest as specified above.

E. Prepare test and inspection reports.

END OF SECTION 210548

SECTION 210553 - IDENTIFICATION FOR FIRE-SUPPRESSION PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Equipment labels.
 - 2. Warning signs and labels.
 - 3. Pipe labels.
 - 4. Stencils.
 - 5. Valve tags.
 - 6. Warning tags.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples: For color, letter style, and graphic representation required for each identification material and device.
- C. Equipment-Label Schedule: Include a listing of all equipment to be labeled and the proposed content for each label.
- D. Valve Schedules: Valve numbering scheme.

PART 2 - PRODUCTS**2.1 EQUIPMENT LABELS**

- A. Metal Labels for Equipment:
 - 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Brady Corporation.
 - b. Brimar Industries, Inc.
 - c. Carlton Industries, LP.
 - d. Champion America.
 - e. Craftmark.
 - f. emedco.
 - g. Kolbi Pipe Marker Co.

- h. LEM Products Inc.
 - i. Marking Services Inc.
 - j. Seton Identification Products.
 - 2. Material and Thickness: Brass, 0.032 inch stainless steel, 0.025 inch aluminum, 0.032 inch or anodized aluminum, 0.032 inch thick, with predrilled holes for attachment hardware.
 - 3. Letter Color: White.
 - 4. Background Color: Red.
 - 5. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.
 - 6. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
 - 7. Fasteners: Stainless-steel rivets or self-tapping screws.
 - 8. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.
- B. Plastic Labels for Equipment:
- 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Brady Corporation.
 - b. Brimar Industries, Inc.
 - c. Carlton Industries, LP.
 - d. Champion America.
 - e. Craftmark.
 - f. emedco.
 - g. Kolbi Pipe Marker Co.
 - h. LEM Products Inc.
 - i. Marking Services Inc.
 - j. Seton Identification Products.
 - 2. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, with predrilled holes for attachment hardware.
 - 3. Letter Color: White.
 - 4. Background Color: Red.
 - 5. Maximum Temperature: Able to withstand temperatures up to 160 deg F.
 - 6. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.
 - 7. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
 - 8. Fasteners: Stainless-steel rivets or self-tapping screws.
 - 9. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

- C. Label Content: Include equipment's Drawing designation or unique equipment number, Drawing numbers where equipment is indicated (plans, details, and schedules), and the Specification Section number and title where equipment is specified.
- D. Equipment-Label Schedule: For each item of equipment to be labeled, on 8-1/2-by-11-inch bond paper. Tabulate equipment identification number and identify Drawing numbers where equipment is indicated (plans, details, and schedules) and the Specification Section number and title where equipment is specified. Equipment schedule shall be included in operation and maintenance data.

2.2 WARNING SIGNS AND LABELS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. Brady Corporation.
 - 2. Brimar Industries, Inc.
 - 3. Carlton Industries, LP.
 - 4. Champion America.
 - 5. Craftmark.
 - 6. emedco.
 - 7. LEM Products Inc.
 - 8. Marking Services Inc.
 - 9. National Marker Company.
 - 10. Seton Identification Products.
 - 11. Stranco, Inc.
- B. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, with predrilled holes for attachment hardware.
- C. Letter Color: White.
- D. Background Color: Red.
- E. Maximum Temperature: Able to withstand temperatures up to 160 deg F.
- F. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.
- G. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
- H. Fasteners: Stainless-steel rivets or self-tapping screws.
- I. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.
- J. Label Content: Include caution and warning information, plus emergency

notification instructions.

2.3 PIPE LABELS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. ActionCraft Products, Inc.; a division of Industrial Test Equipment Co., Inc.
 2. Brady Corporation.
 3. Brimar Industries, Inc.
 4. Carlton Industries, LP.
 5. Champion America.
 6. Craftmark.
 7. emedco.
 8. Kolbi Pipe Marker Co.
 9. LEM Products Inc.
 10. Marking Services Inc.
 11. Seton Identification Products.
- B. General Requirements for Manufactured Pipe Labels: Preprinted, color-coded, with lettering indicating service and showing flow direction according to ASME A13.1.
- C. Pretensioned Pipe Labels: Precoiled, semirigid plastic formed to cover full circumference of pipe and to attach to pipe without fasteners or adhesive.
- D. Self-adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.
- E. Pipe-Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings, pipe size, and an arrow indicating flow direction.
1. Flow-Direction Arrows: Integral with piping-system service lettering to accommodate both directions or as separate unit on each pipe label to indicate flow direction.
 2. Lettering Size: Size letters according to ASME A13.1 for piping.
- F. Pipe-Label Colors:
1. Background Color: Safety Red.
 2. Letter Color: White.

2.4 STENCILS

- A. Stencils for Piping:
1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

- a. Brimar Industries, Inc.
- b. Carlton Industries, LP.
- c. Champion America.
- d. Craftmark.
- e. Kolbi Pipe Marker Co.
- f. Marking Services Inc.

2. Lettering Size: Size letters according to ASME A13.1 for piping.
3. Stencil Material: Fiberboard.
4. Stencil Paint: Safety Red, exterior, gloss, acrylic enamel. Paint may be in pressurized spray-can form.
5. Identification Paint: White, exterior, acrylic enamel. Paint may be in pressurized spray- can form.

2.5 VALVE TAGS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. ActionCraft Products, Inc.
2. Brady Corporation.
3. Brimar Industries, Inc.
4. Carlton Industries, LP.
5. Champion America.
6. Craftmark.
7. emedco.
8. Kolbi Pipe Marker Co.
9. LEM Products Inc.
10. Marking Services Inc.
11. Seton Identification Products.

- B. Description: Stamped or engraved with 1/4-inch letters for piping-system abbreviation and 1/2- inch numbers.

1. Tag Material: Brass, 0.032 inch stainless steel, 0.025 inch aluminum, 0.032 inch or anodized aluminum, 0.032 inch thick, with predrilled holes for attachment hardware.
2. Fasteners: Brass wire-link chain beaded chain or S-hook.
3. Valve-Tag Color: Safety Red.
4. Letter Color: White.

- C. Valve Schedules: For each piping system, on 8-1/2-by-11-inch bond paper. Tabulate valve number, piping system, system abbreviation (as shown on valve tag), location of valve (room or space), normal-operating position (open, closed, or modulating), and variations for identification. Mark valves for emergency shutoff and similar special uses.

1. Valve-tag schedule shall be included in operation and maintenance data.

2.6 WARNING TAGS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. Brady Corporation.
 2. Brimar Industries, Inc.
 3. Carlton Industries, LP.
 4. Champion America.
 5. Craftmark.
 6. emedco.
 7. Kolbi Pipe Marker Co.
 8. LEM Products Inc.
 9. Marking Services Inc.
 10. Seton Identification Products.
- B. Description: Preprinted or partially preprinted, accident-prevention tags, of plasticized card stock with matte finish suitable for writing.
1. Size: Approximately 4 by 7 inches.
 2. Fasteners: Brass grommet and wire.
 3. Nomenclature: Large-size primary caption such as "DANGER," "CAUTION," or "DO NOT OPERATE."
 4. Color: Safety Yellow background with black lettering.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean piping and equipment surfaces of incompatible primers, paints, and encapsulants, as well as dirt, oil, grease, release agents, and other substances that could impair bond of identification devices.

3.2 GENERAL INSTALLATION REQUIREMENTS

- A. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be installed.
- B. Coordinate installation of identifying devices with locations of access panels and doors.
- C. Install identifying devices before installing acoustical ceilings and similar concealment.

3.3 EQUIPMENT LABEL INSTALLATION

- A. Install or permanently fasten labels on each major item of mechanical equipment.
- B. Locate equipment labels where accessible and visible.

3.4 PIPE LABEL INSTALLATION

- A. Piping: Painting of piping is specified in Section 099123 "Interior Painting."

- B. Stenciled Pipe-Label Option: Stenciled labels may be provided instead of manufactured pipe labels, at Installer's option. Install stenciled pipe labels, complying with ASME A13.1, on each piping system.
 - 1. Identification Paint: Use for contrasting background.
 - 2. Stencil Paint: Use for pipe marking.
- C. Pipe-Label Locations: Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:
 - 1. Near each valve and control device.
 - 2. Near each branch connection excluding short takeoffs. Where flow pattern is not obvious, mark each pipe at branch.
 - 3. Near penetrations and on both sides of through walls, floors, ceilings, and inaccessible enclosures.
 - 4. At access doors, manholes, and similar access points that permit a view of concealed piping.
 - 5. Near major equipment items and other points of origination and termination.
 - 6. Spaced at maximum intervals of 50 feet along each run. Reduce intervals to 25 feet in areas of congested piping and equipment.
 - 7. On piping above removable acoustical ceilings. Omit intermediately spaced labels.
- D. Directional Flow Arrows: Arrows shall be used to indicate direction of flow in pipes including pipes where flow is allowed in both directions.

3.5 VALVE-TAG INSTALLATION

- A. Install tags on valves and control devices in fire-suppression piping systems. List tagged valves in a valve-tag schedule.
- B. Valve-Tag Application Schedule: Tag valves according to size, shape, and with captions similar to those indicated in "Valve-Tag Size and Shape" Subparagraph below:
 - 1. Valve-Tag Size and Shape:
 - a. Wet-Pipe Sprinkler System: 2 inches, square.
 - b. Dry-Pipe Sprinkler System: 2 inches, square.

3.6 WARNING-TAG INSTALLATION

- A. Write required message on, and attach warning tags to, equipment and other items where required.

END OF SECTION 210553

SECTION 211119 - FIRE-DEPARTMENT CONNECTIONS**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Flush-type fire-department connections.

1.2 ACTION SUBMITTALS**A. Product Data:** For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for each fire-department connection.

PART 2 - PRODUCTS**2.1 FLUSH-TYPE FIRE-DEPARTMENT CONNECTION****A. Manufacturers:** Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. AFAC Inc.
2. Central Sprinkler Corp.
3. Elkhart Brass Mfg. Co., Inc.
4. Fire-End and Croker Corp.
5. Fire Protection Products, Inc.
6. GMR International Equipment Corporation.
7. Guardian Fire Equipment Incorporated.
8. Potter-Roemer; Fire-Protection Div.
9. Reliable Automatic Sprinkler Co., Inc.
10. United Brass Works, Inc.

B. Standard: UL 405.**C. Type:** Flush, for wall mounting.**D. Pressure Rating:** 175 psig minimum.**E. Body Material:** Corrosion-resistant metal.**F. Inlets:** Brass with threads according to NFPA 1963 and matching local fire-department sizes and threads. Include extension pipe nipples, brass lugged swivel connections, and check devices or clappers.**G. Caps:** Chrome Plated Brass, lugged type, with gasket and chain.**H. Escutcheon Plate:** Rectangular, chrome plated brass, wall type.

- I. Outlet: With pipe threads.
- J. Body Style: Horizontal.
- K. Number of Inlets: Two.
- L. Outlet Location: Top.
- M. Escutcheon Plate Marking: Similar to "AUTO SPKR."
- N. Finish: Chrome.
- O. Outlet Size: NPS 6.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install wall-type fire-department connections.
- B. Install automatic (ball-drip) drain valve at each check valve for fire-department connection.

END OF SECTION 211119

SECTION 211313 - WET-PIPE SPRINKLER SYSTEM**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY**A. Section Includes:**

- 1. Pipes, fittings, and specialties.
- 2. Fire-protection valves.
- 3. Sprinklers.
- 4. Alarm devices.
- 5. Manual control stations.
- 6. Control panels.
- 7. Pressure gages.

B. Related Sections:

- 1. Section 211316 "Dry-Pipe Sprinkler Systems" for dry-pipe sprinkler piping.

1.3 DEFINITIONS

- A. High-Pressure Sprinkler Piping: Wet-pipe sprinkler system piping designed to operate at working pressure higher than standard 175 psig, but not higher than 250 psig.
- B. Standard-Pressure Sprinkler Piping: Wet-pipe sprinkler system piping designed to operate at working pressure of 175 psig maximum.

1.4 SYSTEM DESCRIPTIONS

- A. Wet-Pipe Sprinkler System: Automatic sprinklers are attached to piping containing water and that is connected to water supply through alarm valve. Water discharges immediately from sprinklers when they are opened. Sprinklers open when heat melts fusible link or destroys frangible device. Hose connections are included if indicated.

1.5 PERFORMANCE REQUIREMENTS

- A. Standard-Pressure Piping System Component: Listed for 175-psig minimum working pressure.
- B. Delegated Design: Design sprinkler system(s), including comprehensive engineering analysis by a qualified professional engineer, using performance requirements and design criteria indicated.
- C. Sprinkler system design shall be approved by authorities having jurisdiction.
 - 1. Margin of Safety for Available Water Flow and Pressure: 10 percent, including losses through water-service piping, valves, and backflow preventers.

2. Sprinkler Occupancy Hazard Classifications:
 - a. Building Service Areas: Ordinary Hazard, Group 1.
 - b. Electrical Equipment Rooms: Ordinary Hazard, Group 1.
 - c. General Storage Areas: Ordinary Hazard, Group 1.
 - d. Mechanical Equipment Rooms: Ordinary Hazard, Group 1.
 - e. Office and Public Areas: Light Hazard.
 3. Minimum Density for Automatic-Sprinkler Piping Design:
 - a. Light-Hazard Occupancy: 0.10 gpm over 1500-sq. ft. area.
 - b. Ordinary-Hazard, Group 1 Occupancy: 0.15 gpm over 1500-sq. ft. area.
 4. Maximum Protection Area per Sprinkler: Per UL listing.
 5. Maximum Protection Area per Sprinkler:
 - a. Office Spaces: 225 sq. ft..
 - b. Storage Areas: 130 sq. ft..
 - c. Mechanical Equipment Rooms: 130 sq. ft..
 - d. Electrical Equipment Rooms: 130 sq. ft..
 - e. Other Areas: According to NFPA 13 recommendations unless otherwise indicated.
- D. Seismic Performance: Sprinkler piping shall withstand the effects of earthquake motions determined according to NFPA 13 and ASCE/SEI 7.

1.6 SUBMITTALS

- A. Product Data: For each type of product indicated. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.
- B. Shop Drawings: For wet-pipe sprinkler systems. Include plans, elevations, sections, details, and attachments to other work.
 1. Wiring Diagrams: For power, signal, and control wiring.
- C. Delegated-Design Submittal: For sprinkler systems indicated to comply with performance requirements and design criteria, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
- D. Coordination Drawings: Sprinkler systems, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved:
 1. Domestic water piping.
 2. HVAC hydronic piping.
 3. Items penetrating finished ceiling include the following:
 - a. Lighting fixtures.
 - b. Air outlets and inlets.
- E. Qualification Data: For qualified Installer and professional engineer.

- F. Approved Sprinkler Piping Drawings: Working plans, prepared according to NFPA 13, that have been approved by authorities having jurisdiction, including hydraulic calculations if applicable.
- G. Welding certificates.
- H. Fire-hydrant flow test report.
- I. Field Test Reports and Certificates: Indicate and interpret test results for compliance with performance requirements and as described in NFPA 13. Include "Contractor's Material and Test Certificate for Aboveground Piping."
- J. Field quality-control reports.
- K. Operation and Maintenance Data: For sprinkler specialties to include in emergency, operation, and maintenance manuals.

1.7 QUALITY ASSURANCE

- A. Installer Qualifications:
 - 1. Installer's responsibilities include designing, fabricating, and installing sprinkler systems and providing professional engineering services needed to assume engineering responsibility. Base calculations on results of fire-hydrant flow test.
 - a. Engineering Responsibility: Preparation of working plans, calculations, and field test reports by a qualified professional engineer.
- B. Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code.
- C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- D. NFPA Standards: Sprinkler system equipment, specialties, accessories, installation, and testing shall comply with the following:
 - 1. NFPA 13, "Installation of Sprinkler Systems."

1.8 PROJECT CONDITIONS

- A. Interruption of Existing Sprinkler Service: Do not interrupt sprinkler service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary sprinkler service according to requirements indicated:
 - 1. Notify Owner no fewer than five days in advance of proposed interruption of sprinkler service.
 - 2. Do not proceed with interruption of sprinkler service without Owner's written permission.

1.9 COORDINATION

- A. Coordinate layout and installation of sprinklers with other construction that penetrates ceilings,

including light fixtures, HVAC equipment, console locations, rack equipment in equipment rooms and partition assemblies.

PART 2 - PRODUCTS

2.1 PIPING MATERIALS

- A. Comply with requirements in "Piping Schedule" Article for applications of pipe, tube, and fitting materials, and for joining methods for specific services, service locations, and pipe sizes.

2.2 STEEL PIPE AND FITTINGS

- A. Standard Weight, Black-Steel Pipe: ASTM A 53/A 53M. Pipe ends may be factory or field formed to match joining method.
- B. Schedule 40, Steel Pipe: ASTM A 135; ASTM A 795/A 795M, or ASME B36.10M, wrought steel; with wall thickness not less than Schedule 40. Pipe ends may be factory or field formed to match joining method.
- C. Schedule 10, Black-Steel Pipe: ASTM A 135 or ASTM A 795/A 795M, Schedule 10 in NPS 5 and smaller; and NFPA 13-specified wall thickness in NPS 6 to NPS 10, plain end.
- D. Black-Steel Pipe Nipples: ASTM A 733, made of ASTM A 53/A 53M, standard-weight, seamless steel pipe with threaded ends.
- E. Steel Couplings: ASTM A 865, threaded.
- F. Gray-Iron Threaded Fittings: ASME B16.4, Class 125, standard pattern.
- G. Malleable- or Ductile-Iron Unions: UL 860.
- H. Cast-Iron Flanges: ASME 16.1, Class 125.
- I. Steel Flanges and Flanged Fittings: ASME B16.5, Class 150.
- J. Steel Welding Fittings: ASTM A 234/A 234M and ASME B16.9.
- K. Grooved-Joint, Steel-Pipe Appurtenances:
 - 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Anvil International, Inc.
 - b. Corcoran Piping System Co.
 - c. National Fittings, Inc.
 - d. Shurjoint Piping Products.
 - e. Tyco Fire & Building Products LP.
 - f. Victaulic Company.
 - 2. Pressure Rating: 175 psig minimum.

3. Grooved-End Fittings for Steel Piping: ASTM A 47/A 47M, malleable-iron casting or ASTM A 536, ductile-iron casting; with dimensions matching steel pipe.
 4. Grooved-End-Pipe Couplings for Steel Piping: AWWA C606 and UL 213, rigid pattern, unless otherwise indicated, for steel-pipe dimensions. Include ferrous housing sections, EPDM-rubber gasket, and bolts and nuts.
- L. Steel Pressure-Seal Fittings: UL 213, FM-approved, 175-psig pressure rating with steel housing, rubber O-rings, and pipe stop; for use with fitting manufacturers' pressure-seal tools.
1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Victaulic Company.

2.3 COPPER TUBE AND FITTINGS

- A. Hard Copper Tube: ASTM B 88, Type L water tube, drawn temper.
- B. Cast-Copper, Solder-Joint Fittings: ASME B16.18, pressure fittings.
- C. Wrought-Copper, Solder-Joint Fittings: ASME B16.22, pressure fittings.
- D. Bronze Flanges: ASME B16.24, Class 150, with solder-joint ends.
- E. Copper Unions: MSS SP-123, cast-copper-alloy, hexagonal-stock body, with ball-and-socket, metal-to-metal seating surfaces, and solder-joint or threaded ends.
- F. Copper Pressure-Seal Fittings:
 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following]:
 - a. Viega; Plumbing & Heating Systems.
 2. Standard: UL 213.
 3. NPS 2 and Smaller: Wrought-copper fitting with EPDM-rubber O-ring seal in each end.
 4. NPS 2-1/2 to NPS 4: Cast-bronze fitting with EPDM-rubber O-ring seal in each end.

2.4 PIPING JOINING MATERIALS

- A. Pipe-Flange Gasket Materials: AWWA C110, rubber, flat face, 1/8 inch thick or ASME B16.21, nonmetallic and asbestos free.
 1. Class 125, Cast-Iron Flanges and Class 150, Bronze Flat-Face Flanges: Full-face gaskets.
 2. Class 250, Cast-Iron Flanges and Class 300, Steel Raised-Face Flanges: Ring-type gaskets.
- B. Metal, Pipe-Flange Bolts and Nuts: ASME B18.2.1, carbon steel unless otherwise indicated.

- C. Brazing Filler Metals: AWS A5.8/A5.8M, BCuP Series, copper-phosphorus alloys for general- duty brazing unless otherwise indicated.
- D. Welding Filler Metals: Comply with AWS D10.12M/D10.12 for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.

2.5 LISTED FIRE-PROTECTION VALVES

A. General Requirements:

- 1. Valves shall be UL listed or FM approved.
- 2. Minimum Pressure Rating for Standard-Pressure Piping: 175 psig.

B. Ball Valves:

- 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Anvil International, Inc.
 - b. Victaulic Company.
- 2. Standard: UL 1091 except with ball instead of disc.
- 3. Valves NPS 1-1/2 and Smaller: Bronze body with threaded ends.
- 4. Valves NPS 2 and NPS 2-1/2: Bronze body with threaded ends or ductile-iron body with grooved ends.
- 5. Valves NPS 3: Ductile-iron body with grooved ends.

C. Bronze Butterfly Valves:

- 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Fivalco Inc.
 - b. Global Safety Products, Inc.
 - c. Milwaukee Valve Company.
- 2. Standard: UL 1091.
- 3. Pressure Rating: 175 psig.
- 4. Body Material: Bronze.
- 5. End Connections: Threaded.

D. Iron Butterfly Valves:

- 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Anvil International, Inc.

- b. Fivalco Inc.
 - c. Global Safety Products, Inc.
 - d. Kennedy Valve; a division of McWane, Inc.
 - e. Milwaukee Valve Company.
 - f. NIBCO INC.
 - g. Pratt, Henry Company.
 - h. Shurjoint Piping Products.
 - i. Tyco Fire & Building Products LP.
 - j. Victaulic Company.
- 2. Standard: UL 1091.
 - 3. Pressure Rating: 175 psig.
 - 4. Body Material: Cast or ductile iron.
 - 5. Style: Lug or wafer.
 - 6. End Connections: Grooved.
- E. Check Valves:
- 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Anvil International, Inc.
 - b. Crane Co.; Crane Valve Group; Crane Valves.
 - c. Crane Co.; Crane Valve Group; Jenkins Valves.
 - d. Crane Co.; Crane Valve Group; Stockham Division.
 - e. Fire-End & Croker Corporation.
 - f. Fivalco Inc.
 - g. Globe Fire Sprinkler Corporation.
 - h. Kennedy Valve; a division of McWane, Inc.
 - i. Milwaukee Valve Company.
 - j. Mueller Co.; Water Products Division.
 - k. NIBCO INC.
 - l. Potter Roemer.
 - m. Reliable Automatic Sprinkler Co., Inc.
 - n. Shurjoint Piping Products.
 - o. Tyco Fire & Building Products LP.
 - p. United Brass Works, Inc.
 - q. Venus Fire Protection Ltd.
 - r. Victaulic Company.
 - s. Viking Corporation.
 - aa. Watts Water Technologies, Inc.
 - 2. Standard: UL 312.
 - 3. Pressure Rating: 250 psig minimum.
 - 4. Type: Swing check.
 - 5. Body Material: Cast iron.
 - 6. End Connections: Flanged or grooved.
- F. Bronze OS&Y Gate Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Crane Co.; Crane Valve Group; Crane Valves.
 - b. Crane Co.; Crane Valve Group; Stockham Division.
 - c. Milwaukee Valve Company.
 - d. NIBCO INC.
 - e. United Brass Works, Inc.
2. Standard: UL 262.
3. Pressure Rating: 175 psig.
4. Body Material: Bronze.
5. End Connections: Threaded.

G. Iron OS&Y Gate Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. American Cast Iron Pipe Company; Waterous Company Subsidiary.
 - b. American Valve, Inc.
 - c. Clow Valve Company; a division of McWane, Inc.
 - d. Crane Co.; Crane Valve Group; Crane Valves.
 - e. Crane Co.; Crane Valve Group; Jenkins Valves.
 - f. Crane Co.; Crane Valve Group; Stockham Division.
 - g. Hammond Valve.
 - h. Milwaukee Valve Company.
 - i. Mueller Co.; Water Products Division.
 - j. NIBCO INC.
 - k. Shurjoint Piping Products.
 - l. Tyco Fire & Building Products LP.
 - m. United Brass Works, Inc.
 - n. Watts Water Technologies, Inc.
2. Standard: UL 262.
3. Pressure Rating: 250 psig minimum.
4. Body Material: Cast or ductile iron.
5. End Connections: Flanged or grooved.

H. Indicating-Type Butterfly Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Anvil International, Inc.
 - b. Fivalco Inc.
 - c. Global Safety Products, Inc.

- d. Kennedy Valve; a division of McWane, Inc.
 - e. Milwaukee Valve Company.
 - f. NIBCO INC.
 - g. Shurjoint Piping Products.
 - h. Tyco Fire & Building Products LP.
 - i. Victaulic Company.
- 2. Standard: UL 1091.
 - 3. Pressure Rating: 175 psig minimum.
 - 4. Valves NPS 2 and Smaller:
 - a. Valve Type: Ball or butterfly.
 - b. Body Material: Bronze.
 - c. End Connections: Threaded.
 - 5. Valves NPS 2-1/2 and Larger:
 - a. Valve Type: Butterfly.
 - b. Body Material: Cast or ductile iron.
 - c. End Connections: Flanged, grooved, or wafer.
 - 6. Valve Operation: Integral electrical, 115-V ac, prewired, single-circuit, supervisory switch visual indicating device.
- I. NRS Gate Valves:
- 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. American Cast Iron Pipe Company; Waterous Company Subsidiary.
 - b. American Valve, Inc.
 - c. Clow Valve Company; a division of McWane, Inc.
 - d. Crane Co.; Crane Valve Group; Stockham Division.
 - e. Kennedy Valve; a division of McWane, Inc.
 - f. Mueller Co.; Water Products Division.
 - g. NIBCO INC.
 - h. Tyco Fire & Building Products LP.
 - 2. Standard: UL 262.
 - 3. Pressure Rating: 250 psig minimum.
 - 4. Body Material: Cast iron with indicator post flange.
 - 5. Stem: Nonrising.
 - 6. End Connections: Flanged or grooved.
- 2.6 TRIM AND DRAIN VALVES
- A. General Requirements:
- 1. Standard: UL's "Fire Protection Equipment Directory" listing or "Approval Guide," published by FM Global, listing.

2. Pressure Rating: 175 psig minimum.

B. Angle Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Fire Protection Products, Inc.
 - b. United Brass Works, Inc.

C. Ball Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Anvil International, Inc.
 - b. Conbraco Industries, Inc.; Apollo Valves.
 - c. Fire-End & Croker Corporation.
 - d. Fire Protection Products, Inc.
 - e. Flowserve.
 - f. FNW.
 - g. Kennedy Valve; a division of McWane, Inc.
 - h. Legend Valve.
 - i. Milwaukee Valve Company.
 - j. NIBCO INC.
 - k. Potter Roemer.
 - l. Tyco Fire & Building Products LP.
 - m. Victaulic Company.
 - n. Watts Water Technologies, Inc.

D. Globe Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Fire Protection Products, Inc.
 - b. United Brass Works, Inc.

2.7 SPECIALTY VALVES

A. General Requirements:

1. Standard: UL's "Fire Protection Equipment Directory" listing or "Approval Guide," published by FM Global, listing.
2. Pressure Rating:
 - a. Standard-Pressure Piping Specialty Valves: 175 psig minimum.

3. Body Material: Cast or ductile iron.
4. Size: Same as connected piping.
5. End Connections: Flanged or grooved.

B. Alarm Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. AFAC Inc.
 - b. Globe Fire Sprinkler Corporation.
 - c. Reliable Automatic Sprinkler Co., Inc.
 - d. Tyco Fire & Building Products LP.
 - e. Venus Fire Protection Ltd.
 - f. Victaulic Company.
 - g. Viking Corporation.
2. Standard: UL 193.
3. Design: For horizontal or vertical installation.
4. Include trim sets for bypass, drain, electrical sprinkler alarm switch, pressure gages, retarding chamber, and fill-line attachment with strainer.
5. Drip Cup Assembly: Pipe drain without valves and separate from main drain piping.
6. Drip Cup Assembly: Pipe drain with check valve to main drain piping.

C. Automatic (Ball Drip) Drain Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. AFAC Inc.
 - b. Reliable Automatic Sprinkler Co., Inc.
 - c. Tyco Fire & Building Products LP.
2. Standard: UL 1726.
3. Pressure Rating: 175 psig minimum.
4. Type: Automatic draining, ball check.
5. Size: NPS 3/4.
6. End Connections: Threaded.

2.8 SPRINKLER SPECIALTY PIPE FITTINGS

A. Branch Outlet Fittings:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

- a. Anvil International, Inc.
 - b. National Fittings, Inc.
 - c. Shurjoint Piping Products.
 - d. Tyco Fire & Building Products LP.
 - e. Victaulic Company.
 2. Standard: UL 213.
 3. Pressure Rating: 175 psig minimum.
 4. Body Material: Ductile-iron housing with EPDM seals and bolts and nuts.
 5. Type: Mechanical-T and -cross fittings.
 6. Configurations: Snap-on and strapless, ductile-iron housing with branch outlets.
 7. Size: Of dimension to fit onto sprinkler main and with outlet connections as required to match connected branch piping.
 8. Branch Outlets: Grooved, plain-end pipe, or threaded.
- B. Flow Detection and Test Assemblies:
1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. AGF Manufacturing Inc.
 - b. Reliable Automatic Sprinkler Co., Inc.
 - c. Tyco Fire & Building Products LP.
 - d. Victaulic Company.
 2. Standard: UL's "Fire Protection Equipment Directory" listing or "Approval Guide," published by FM Global, listing.
 3. Pressure Rating: 175 psig minimum.
 4. Body Material: Cast- or ductile-iron housing with orifice, sight glass, and integral test valve.
 5. Size: Same as connected piping.
 6. Inlet and Outlet: Threaded.
- C. Branch Line Testers:
1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Elkhart Brass Mfg. Company, Inc.
 - b. Fire-End & Croker Corporation.
 - c. Potter Roemer.
 2. Standard: UL 199.
 3. Pressure Rating: 175 psig.
 4. Body Material: Brass.
 5. Size: Same as connected piping.
 6. Inlet: Threaded.
 7. Drain Outlet: Threaded and capped.
 8. Branch Outlet: Threaded, for sprinkler.

D. Sprinkler Inspector's Test Fittings:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. AGF Manufacturing Inc.
 - b. Triple R Specialty.
 - c. Tyco Fire & Building Products LP.
 - d. Victaulic Company.
 - e. Viking Corporation.
2. Standard: UL's "Fire Protection Equipment Directory" listing or "Approval Guide," published by FM Global, listing.
3. Pressure Rating: 175 psig minimum.
4. Body Material: Cast- or ductile-iron housing with sight glass.
5. Size: Same as connected piping.
6. Inlet and Outlet: Threaded.

E. Adjustable Drop Nipples:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. CECA, LLC.
 - b. Corcoran Piping System Co.
 - c. Merit Manufacturing; a division of Anvil International, Inc.
2. Standard: UL 1474.
3. Pressure Rating: 250 psig minimum.
4. Body Material: Steel pipe with EPDM-rubber O-ring seals.
5. Size: Same as connected piping.
6. Length: Adjustable.
7. Inlet and Outlet: Threaded.

2.9 SPRINKLERS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Globe Fire Sprinkler Corporation.
2. Reliable Automatic Sprinkler Co., Inc.
3. Tyco Fire & Building Products LP.
4. Venus Fire Protection Ltd.
5. Victaulic Company.
6. Viking Corporation.

B. General Requirements:

1. Standard: UL's "Fire Protection Equipment Directory" listing or "Approval Guide," published by FM Global, listing.
 2. Pressure Rating for Residential Sprinklers: 175 psig maximum.
 3. Pressure Rating for Automatic Sprinklers: 175 psig minimum.
- C. Automatic Sprinklers with Heat-Responsive Element:
1. Early-Suppression, Fast-Response Applications: UL 1767.
 2. Nonresidential Applications: UL.
 3. Characteristics: Nominal 1/2-inch orifice with Discharge Coefficient K of 5.6, and for "Ordinary" temperature classification rating unless otherwise indicated or required by application.
- D. Open Sprinklers with Heat-Responsive Element Removed: UL 199.
1. Characteristics:
 - a. Nominal 1/2-inch Orifice: With Discharge Coefficient K between 5.3 and 5.8.
- E. Sprinkler Finishes:
1. Chrome plated.
 2. Bronze.
 3. Painted.
- F. Special Coatings:
1. Wax.
 2. Corrosion-resistant paint.
- G. Sprinkler Escutcheons: Materials, types, and finishes for the following sprinkler mounting applications. Escutcheons for concealed, flush, and recessed-type sprinklers are specified with sprinklers.
1. Ceiling Mounting: Plastic, concealed.
 2. Sidewall Mounting: Plastic, concealed.
 3. Mechanical and electrical equipment spaces, provide protective cages.
- H. Sprinkler Guards:
1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Reliable Automatic Sprinkler Co., Inc.
 - b. Tyco Fire & Building Products LP.
 - c. Victaulic Company.
 - d. Viking Corporation.
 2. Standard: UL 199.
 3. Type: Wire cage with fastening device for attaching to sprinkler.

2.10 ALARM DEVICES

A. Alarm-device types shall match piping and equipment connections.

B. Water-Flow Indicators:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. ADT Security Services, Inc.
 - b. McDonnell & Miller; ITT Industries.
 - c. Potter Electric Signal Company.
 - d. System Sensor; a Honeywell company.
 - e. Viking Corporation.
 - f. Watts Industries (Canada) Inc.
2. Standard: UL 346.
3. Water-Flow Detector: Electrically supervised.
4. Components: Two single-pole, double-throw circuit switches for isolated alarm and auxiliary contacts, 7 A, 125-V ac and 0.25 A, 24-V dc; complete with factory-set, field-adjustable retard element to prevent false signals and tamperproof cover that sends signal if removed.
5. Type: Paddle operated.
6. Pressure Rating: 250 psig.
7. Design Installation: Horizontal or vertical.

C. Pressure Switches:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. AFAC Inc.
 - b. Barksdale, Inc.
 - c. Detroit Switch, Inc.
 - d. Potter Electric Signal Company.
 - e. System Sensor; a Honeywell company.
 - f. Tyco Fire & Building Products LP.
 - g. United Electric Controls Co.
 - h. Viking Corporation.
2. Standard: UL 346.
3. Type: Electrically supervised water-flow switch with retard feature.
4. Components: Single-pole, double-throw switch with normally closed contacts.
5. Design Operation: Rising pressure signals water flow.

D. Valve Supervisory Switches:

1. Manufacturers: Subject to compliance with requirements, available manufacturers

offering products that may be incorporated into the Work include, but are not limited to, the following:

- a. Fire-Lite Alarms, Inc.; a Honeywell company.
- b. Kennedy Valve; a division of McWane, Inc.
- c. Potter Electric Signal Company.
- d. System Sensor; a Honeywell company.

2. Standard: UL 346.
3. Type: Electrically supervised.
4. Components: Single-pole, double-throw switch with normally closed contacts.
5. Design: Signals that controlled valve is in other than fully open position.

2.11 MANUAL CONTROL STATIONS

- A. Description: UL listed or FM approved, hydraulic operation, with union, NPS 1/2 pipe nipple, and bronze ball valve. Include metal enclosure labeled "MANUAL CONTROL STATION" with operating instructions and cover held closed by breakable strut to prevent accidental opening.

2.12 CONTROL PANELS

- A. Description: Single-area, two-area, or single-area cross-zoned control panel as indicated, including NEMA ICS 6, Type 1 enclosure, detector, alarm. Panels contain power supply; battery charger; standby batteries; field-wiring terminal strip; electrically supervised solenoid valves and polarized fire-alarm bell; lamp test facility; single-pole, double-throw auxiliary alarm contacts; and rectifier.
 1. Panels: UL listed and FM approved when used with thermal detectors and Class A detector circuit wiring. Electrical characteristics are 120-V ac, 60 Hz, with 24-V dc rechargeable batteries.
 2. Manual Control Stations: Electric operation, metal enclosure, labeled "MANUAL CONTROL STATION" with operating instructions and cover held closed by breakable strut to prevent accidental opening.

2.13 PRESSURE GAGES

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 1. AMETEK; U.S. Gauge Division.
 2. Ashcroft, Inc.
 3. Brecco Corporation.
 4. WIKA Instrument Corporation.
- B. Standard: UL 393.
- C. Dial Size: 3-1/2- to 4-1/2-inch diameter.
- D. Pressure Gage Range: 0 to 250 psig minimum.

- E. Water System Piping Gage: Include "WATER" or "AIR/WATER" label on dial face.
- F. Air System Piping Gage: Include retard feature and "AIR" or "AIR/WATER" label on dial face.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Perform fire-hydrant flow test according to NFPA 13 and NFPA 291. Use results for system design calculations required in "Quality Assurance" Article.
- B. Report test results promptly and in writing.

3.2 PIPING INSTALLATION

- A. Locations and Arrangements: Drawing plans, schematics, and diagrams indicate general location and arrangement of piping. Install piping as indicated, as far as practical.
 - 1. Deviations from approved working plans for piping require written approval from authorities having jurisdiction. File written approval with Architect before deviating from approved working plans.
- B. Piping Standard: Comply with requirements for installation of sprinkler piping in NFPA 13.
- C. Install seismic restraints on piping. Comply with requirements for seismic-restraint device materials and installation in NFPA 13.
- D. Use listed fittings to make changes in direction, branch takeoffs from mains, and reductions in pipe sizes.
- E. Install unions adjacent to each valve in pipes NPS 2 and smaller.
- F. Install flanges, flange adapters, or couplings for grooved-end piping on valves, apparatus, and equipment having NPS 2-1/2 and larger end connections.
- G. Install "Inspector's Test Connections" in sprinkler system piping, complete with shutoff valve, and sized and located according to NFPA 13.
- H. Install sprinkler piping with drains for complete system drainage.
- I. Install sprinkler control valves, test assemblies, and drain risers adjacent to standpipes when sprinkler piping is connected to standpipes.
- J. Install alarm devices in piping systems.
- K. Install hangers and supports for sprinkler system piping according to NFPA 13. Comply with requirements for hanger materials in NFPA 13.
- L. Install pressure gages on riser or feed main, at each sprinkler test connection, and at top of each standpipe. Include pressure gages with connection not less than NPS 1/4 and with soft metal seated globe valve, arranged for draining pipe between gage and valve. Install gages to permit removal, and install where they will not be subject to freezing.

- M. Fill sprinkler system piping with water.
- N. Install sleeves for piping penetrations of walls, ceilings, and floors. Comply with requirements for sleeves specified in Section 210517 "Sleeves and Sleeve Seals for Fire-Suppression Piping."
- O. Install sleeve seals for piping penetrations of concrete walls and slabs. Comply with requirements for sleeve seals specified in Section 210517 "Sleeves and Sleeve Seals for Fire- Suppression Piping."
- P. Install escutcheons for piping penetrations of walls, ceilings, and floors. Comply with requirements for escutcheons specified in Section 210518 "Escutcheons for Fire-Suppression Piping."

3.3 JOINT CONSTRUCTION

- A. Install couplings, flanges, flanged fittings, unions, nipples, and transition and special fittings that have finish and pressure ratings same as or higher than system's pressure rating for aboveground applications unless otherwise indicated.
- B. Install unions adjacent to each valve in pipes NPS 2 and smaller.
- C. Install flanges, flange adapters, or couplings for grooved-end piping on valves, apparatus, and equipment having NPS 2-1/2 and larger end connections.
- D. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
- E. Remove scale, slag, dirt, and debris from inside and outside of pipes, tubes, and fittings before assembly.
- F. Flanged Joints: Select appropriate gasket material in size, type, and thickness suitable for water service. Join flanges with gasket and bolts according to ASME B31.9.
- G. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
 - 1. Apply appropriate tape or thread compound to external pipe threads.
 - 2. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged.
- H. Steel-Piping, Pressure-Sealed Joints: Join lightwall steel pipe and steel pressure-seal fittings with tools recommended by fitting manufacturer.
- I. Welded Joints: Construct joints according to AWS D10.12M/D10.12, using qualified processes and welding operators according to "Quality Assurance" Article.
 - 1. Shop weld pipe joints where welded piping is indicated. Do not use welded joints for galvanized-steel pipe.
- J. Steel-Piping, Cut-Grooved Joints: Cut square-edge groove in end of pipe according to

AWWA C606. Assemble coupling with housing, gasket, lubricant, and bolts. Join steel pipe and grooved-end fittings according to AWWA C606 for steel-pipe joints.

- K. Steel-Piping, Roll-Grooved Joints: Roll rounded-edge groove in end of pipe according to AWWA C606. Assemble coupling with housing, gasket, lubricant, and bolts. Join steel pipe and grooved-end fittings according to AWWA C606 for steel-pipe grooved joints.
- L. Brazed Joints: Join copper tube and fittings according to CDA's "Copper Tube Handbook," "Braze Joints" Chapter.
- M. Grooved-end fittings according to AWWA C606 for steel-pipe grooved joints.
- N. Copper-Tubing, Pressure-Sealed Joints: Join copper tube and copper pressure-seal fittings with tools recommended by fitting manufacturer.
- O. Dissimilar-Material Piping Joints: Make joints using adapters compatible with materials of both piping systems.

3.4 VALVE AND SPECIALTIES INSTALLATION

- A. Install listed fire-protection valves, trim and drain valves, specialty valves and trim, controls, and specialties according to NFPA 13 and authorities having jurisdiction.
- B. Install listed fire-protection shutoff valves supervised open, located to control sources of water supply except from fire-department connections. Install permanent identification signs indicating portion of system controlled by each valve.
- C. Install check valve in each water-supply connection. Install backflow preventers instead of check valves in potable-water-supply sources.
- D. Specialty Valves:
 - 1. General Requirements: Install in vertical position for proper direction of flow, in main supply to system.
 - 2. Alarm Valves: Include bypass check valve and retarding chamber drain-line connection.

3.5 SPRINKLER INSTALLATION

- A. Install sprinklers in suspended ceilings in center of narrow dimension of acoustical ceiling panels.
- B. Install dry-type sprinklers with water supply from heated space. Do not install pendent or sidewall, wet-type sprinklers in areas subject to freezing.

3.6 IDENTIFICATION

- A. Install labeling and pipe markers on equipment and piping according to requirements in NFPA 13.

3.7 FIELD QUALITY CONTROL

- A. Perform tests and inspections.

B. Tests and Inspections:

1. Leak Test: After installation, charge systems and test for leaks. Repair leaks and retest until no leaks exist.
2. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
3. Flush, test, and inspect sprinkler systems according to NFPA 13, "Systems Acceptance" Chapter.
4. Energize circuits to electrical equipment and devices.
5. Coordinate with fire-alarm tests. Operate as required.
6. Coordinate with fire-pump tests. Operate as required.
7. Verify that equipment hose threads are same as local fire-department equipment.

C. Sprinkler piping system will be considered defective if it does not pass tests and inspections.

D. Prepare test and inspection reports.

3.8 CLEANING

- A. Clean dirt and debris from sprinklers.
- B. Remove and replace sprinklers with paint other than factory finish.

3.9 DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain specialty valves.

3.10 PIPING SCHEDULE

- A. Sprinkler specialty fittings may be used, downstream of control valves, instead of specified fittings.
- B. Standard-pressure, wet-pipe sprinkler system, NPS 2 and smaller, shall be one of the following:
 1. Schedule 40, black-steel pipe with threaded ends; uncoated, gray-iron threaded fittings; and threaded joints.
 2. Schedule 40, black-steel pipe with cut- or roll-grooved ends; uncoated, grooved-end fittings for steel piping; grooved-end-pipe couplings for steel piping; and grooved joints.
 3. Schedule 40, black-steel pipe with plain ends; steel welding fittings; and welded joints.
- C. Standard-pressure, wet-pipe sprinkler system, NPS 2-1/2 to NPS 4, shall be one of the following:
 1. Standard-weight or Schedule 30, black-steel pipe with threaded ends; uncoated, gray-iron threaded fittings; and threaded joints.
 2. Standard-weight or Schedule 30, black-steel pipe with cut- or roll-grooved ends; uncoated, grooved-end fittings for steel piping; grooved-end-pipe couplings for steel piping; and grooved joints.
 3. Standard-weight or Schedule 30, black-steel pipe with plain ends; steel welding fittings;

- and welded joints.
- 4. Thinwall Schedule 10, black-steel pipe with roll-grooved ends; uncoated, grooved-end fittings for steel piping; grooved-end-pipe couplings for steel piping; and grooved joints.
- 5. Thinwall Schedule 10, black-steel pipe with plain ends; welding fittings; and welded joints.

3.11 SPRINKLER SCHEDULE

A. Use sprinkler types in subparagraphs below for the following applications:

- 1. Rooms without Ceilings: Upright or pendant sprinklers.
- 2. Rooms with Suspended Ceilings: Concealed sprinklers.
- 3. Wall Mounting: Sidewall sprinklers.
- 4. Spaces Subject to Freezing Upright, pendent, dry sprinklers; and sidewall, dry sprinklers as necessary.

B. Provide sprinkler types in subparagraphs below with finishes indicated.

- 1. Concealed Sprinklers: Rough brass, with factory-painted white cover plate.
- 2. Upright, Pendent and Sidewall Sprinklers: Chrome plated in finished spaces exposed to view; rough bronze in unfinished spaces not exposed to view; wax coated where exposed to acids, chemicals, or other corrosive fumes.

END OF SECTION 211313

SECTION 213113 - ELECTRIC-DRIVE, CENTRIFUGAL FIRE PUMPS**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. In-line fire pumps.
2. Fire-pump accessories and specialties.

1.2 REQUIREMENT

- A. Pressure booster system requirements subject to engineer of record recommendations based on site specific conditions and requirements.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.

- B. Shop Drawings: For fire pumps, motor drivers, and fire-pump accessories and specialties.

1. Include plans, elevations, sections, and mounting and attachment details.
2. Include details of equipment assemblies. Indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
3. Include diagrams for power, signal, and control wiring.

1.4 INFORMATIONAL SUBMITTALS

- A. Seismic Qualification Certificates: For fire pumps, accessories, and components, from manufacturer.

- B. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.6 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. NFPA Compliance: Comply with NFPA 20.

- B. Seismic Performance: Fire pumps shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified."
 2. Component Importance Factor: 1.5.
- C. Pump Equipment, Accessory, and Specialty Pressure Rating: 175 psig minimum unless higher pressure rating is indicated.
- D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2.2 GENERAL REQUIREMENTS FOR CENTRIFUGAL FIRE PUMPS

- A. Description: Factory-assembled and -tested fire-pump and driver unit.
- B. Base: Fabricated and attached to fire-pump and driver unit, with reinforcement to resist movement of pump during seismic events when base is anchored to building substrate.
- C. Finish: Red paint applied to factory-assembled and -tested unit before shipping.

2.3 IN-LINE FIRE PUMPS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. A-C Pump; ITT Industries.
 2. Armstrong Darling, Inc.
 3. Aurora Pump; Pentair Pump Group.
 4. Patterson Pump Company.
- B. Pump:
1. Standard: UL 448, for in-line pumps for fire service.
 2. Casing: Radially split case, cast iron, with ASME B16.1 pipe-flange connections.
 3. Impeller: Cast bronze, statically and dynamically balanced, and keyed to shaft.
 4. Wear Rings: Replaceable bronze.
 5. Shaft and Sleeve: Steel shaft with bronze sleeve.
 - a. Shaft Bearings: Grease-lubricated ball bearings in cast-iron housing.
 - b. Seals: Stuffing box with minimum of four rings of graphite-impregnated braided yarn and bronze packing gland.
 6. Mounting: Pump and driver shaft is vertical, with motor above pump and pump on base. Motor and pump rotating assembly shall be removable from top without removing the pump casing from the piping.
- C. Coupling: None or rigid.
- D. Driver:

1. Standard: UL 1004A.
2. Type: Electric motor; NEMA MG 1, polyphase Design B.

E. Capacities and Characteristics (See Fire Pump Schedule)

2.4 FIRE-PUMP ACCESSORIES AND SPECIALTIES

- A. Automatic Air-Release Valves: Comply with NFPA 20 for installation in fire-pump casing.
- B. Circulation Relief Valves: UL 1478, brass, spring loaded; for installation in pump discharge piping.
- C. Relief Valves:
 1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Hammond Valve.
 - b. Mueller Co.; Water Products Division.
 - c. NIBCO INC.
 2. Description: UL 1478, bronze or cast iron, spring loaded; for installation in fire- suppression water-supply piping.
- D. Inlet Fitting: Eccentric tapered reducer at pump suction inlet.
- E. Outlet Fitting: Concentric tapered reducer at pump discharge outlet.
- F. Discharge Cone: Closed or open type.

2.5 GROUT

- A. Standard: ASTM C 1107, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.
- B. Characteristics: Nonshrink and recommended for interior and exterior applications.
- C. Design Mix: 5000-psi, 28-day compressive strength.
- D. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Fire-Pump Installation Standard: Comply with NFPA 20 for installation of fire pumps, relief valves, and related components.
- B. Equipment Mounting:
 1. Install fire pumps on cast-in-place concrete equipment bases. Comply with

- requirements for equipment bases and foundations specified in Section 033000 "Cast-in-Place Concrete."
2. Comply with requirements for vibration isolation and seismic-control devices specified in Section 210548 "Vibration and Seismic Controls for Fire-Suppression Piping and Equipment."
 3. Comply with requirements for vibration isolation devices specified in Section 210548.13 "Vibration Controls for Fire-Suppression Piping and Equipment."
- C. Install fire-pump suction and discharge piping equal to or larger than sizes required by NFPA 20.
- D. Support piping and pumps separately, so weight of piping does not rest on pumps.
- E. Install valves that are same size as connecting piping. Comply with requirements for fire-protection valves specified in Section 211313 "Wet-Pipe Sprinkler Systems."
- F. Install pressure gages on fire-pump suction and discharge flange pressure-gage tappings. Comply with requirements for pressure gages specified in Section 211313 "Wet-Pipe Sprinkler Systems."
- G. Install piping hangers and supports, anchors, valves, gages, and equipment supports according to NFPA 20.
- H. Electrical Wiring: Install electrical devices furnished by equipment manufacturers but not factory mounted. Furnish copies of manufacturers' wiring diagram submittals to electrical Installer.
- I. Wiring Method: Conceal conductors and cables in accessible ceilings, walls, and floors where possible.
- 3.2 ALIGNMENT
- A. After alignment is correct, tighten anchor bolts evenly. Fill baseplate completely with grout, with metal blocks and shims or wedges in place. Tighten anchor bolts after grout has hardened. Check alignment and make required corrections.
 - B. Align piping connections.
 - C. Align pump and driver shafts for angular and parallel alignment according to HI 1.4 and to tolerances specified by manufacturer.
- 3.3 CONNECTIONS
- A. Comply with requirements for piping and valves specified in Section 211313 "Wet-Pipe Sprinkler Systems." Drawings indicate general arrangement of piping, fittings, and specialties.
 - B. Install piping adjacent to pumps and equipment to allow service and maintenance.
 - C. Connect relief-valve discharge to drainage piping or point of discharge.
 - D. Connect flowmeter-system meters, sensors, and valves to tubing.

- E. Connect fire pumps to their controllers.

3.4 IDENTIFICATION

- A. Identify system components. Comply with requirements for fire-pump marking according to NFPA 20.

3.5 FIELD QUALITY CONTROL

- A. Test each fire pump with its controller as a unit. Comply with requirements for electric-motor-driver fire-pump controllers specified in Section 2623933 "Controllers for Fire-Pump Drivers."
- B. Perform the following tests and inspections with the assistance of a factory-authorized service representative.
 - 1. After installing components, assemblies, and equipment, including controller, test for compliance with requirements.
 - 2. Test according to NFPA 20 for acceptance and performance testing.
 - 3. Leak Test: After installation, charge system and test for leaks. Repair leaks and retest until no leaks exist.
 - 4. Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
 - 5. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- C. Components, assemblies, and equipment will be considered defective if they do not pass tests and inspections.
- D. Prepare test and inspection reports.
- E. Furnish fire hoses in number, size, and length required to reach storm drain or other acceptable location to dispose of fire-pump test water. Hoses are for tests only and do not convey to Owner.

3.6 STARTUP SERVICE

- A. Perform startup service.
 - 1. Complete installation and startup checks according to manufacturer's written instructions.

END OF SECTION 213113

SECTION 213400 - PRESSURE-MAINTENANCE PUMPS**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Regenerative-turbine, pressure-maintenance pumps.
2. Vertical-turbine, pressure-maintenance pumps.

B. Related Requirements:

1. Section 262933 "Controllers for Fire-Pump Drivers" for pressure-maintenance-pump controllers.

1.2 REQUIREMENT

- A. Pressure booster system requirements subject to engineer of record recommendations based on site specific conditions and requirements.

1.3 ACTION SUBMITTALS**A. Product Data:** For each type of product.**B. Shop Drawings:** For pumps, accessories, and specialties.

1. Include plans, elevations, sections, and attachment details.
2. Include details of equipment assemblies. Indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
3. Include diagrams for power, signal, and control wiring.

1.4 INFORMATIONAL SUBMITTALS

- A. Field quality-control reports.

1.5 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

1.6 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

PART 2 - PRODUCTS**2.1 REGENERATIVE-TURBINE, PRESSURE-MAINTENANCE PUMPS**

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. A-C Pump; ITT Industries.
 2. Armstrong Darling, Inc.
 3. Aurora Pump; Pentair Pump Group.
 4. Patterson Pump Company.
- B. Description: Factory-assembled and -tested, close-coupled, single-stage, regenerative-turbine centrifugal pump as defined in HI 1.1-1.2 and HI 1.3; with pump and motor mounted horizontally.
- C. Pump Construction:
1. Casing: Radially split, cast iron, with threaded inlet and outlet.
 2. Impeller: Bronze, balanced, and keyed to shaft.
 3. Pump Shaft: Stainless steel[**or steel**] with deflector.
 4. Shaft Sleeve: Bronze.
 5. Seal: Mechanical type with spring-loaded rotating head.
- D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- E. Motor: Single speed with permanently lubricated ball bearings. Comply with requirements in Section 210513 "Common Motor Requirements for Fire Suppression Equipment."
1. Power Cord: Factory-connected to motor for field connection to controller and at least 10 feet long.
- F. Nameplate: Permanently attached to pump and indicating capacity and characteristics.
- G. Capacities and Characteristics (See Fire Pump Schedule):

2.2 VERTICAL-TURBINE, PRESSURE-MAINTENANCE PUMPS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. A-C Pump; ITT Industries.
 2. Armstrong Darling, Inc.
 3. Aurora Pump; Pentair Pump Group.
 4. Patterson Pump Company.
- B. Description: Factory-assembled and -tested, vertical, multistage, open-line-shaft turbine pump as defined in HI 2.1-2.2 and HI 2.3; with pump motor mounted above pump head.
- C. Pump Construction:
1. Pump Head: Cast iron, for surface discharge, with flange except connections may be threaded in sizes in which flanges are not available.
 2. Pump Head Seal: Stuffing box and stuffing.
 3. Line Shaft: Stainless steel or steel, with corrosion-resistant shaft sleeves.

4. Line Shaft Bearings: Rubber sleeve, water lubricated.
 5. Line Shaft: Steel.
 6. Line Shaft Bearings: Corrosion resistant, oil lubricated.
 7. Impeller Shaft: Monel metal or stainless steel.
 8. Bowl Section: Multiple cast-iron bowls with closed-type bronze or stainless-steel impellers.
 9. Column Pipe: ASTM A 53/A 53M, Schedule 40, galvanized-steel pipe with threaded ends and cast-iron or steel fittings, in sections 10 feet or less, with strainer of cast or fabricated bronze or stainless steel at bottom.
- D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- E. Motor: Single speed with permanently lubricated ball bearings. Comply with requirements in Section 210513 "Common Motor Requirements for Fire Suppression Equipment."
1. Power Cord: Factory-connected to motor for field connection to controller and at least 10 feet long.
- F. Base: Cast iron or steel with hole for electrical cable.
- G. Nameplate: Permanently attached to pump and indicating capacity and characteristics.

2.3 MOTORS

- A. Comply with NEMA designation, temperature rating, service factor, enclosure type, and efficiency requirements for motors specified in Section 210513 "Common Motor Requirements for Fire Suppression Equipment."
1. Motor Sizes: Minimum size as indicated; if not indicated, large enough so driven load will not require motor to operate in service factor range above 1.0.

PART 3 - EXECUTION

3.1 EQUIPMENT INSTALLATION

- A. NFPA Standard: Comply with NFPA 20 for installation of pressure-maintenance pumps.
- B. Equipment Mounting:
1. Install regenerative-turbine, pressure-maintenance pumps according to HI 1.4.
 2. Install vertical-turbine, pressure-maintenance pumps according to HI 2.4.
 3. Install base-mounted pumps on cast-in-place concrete equipment base(s). Comply with requirements for equipment bases and foundations specified in Section 033000 "Cast-in-Place Concrete."
 - a. Comply with requirements for vibration isolation and seismic control devices specified in Section 210548 "Vibration and Seismic Controls for Fire-Suppression Piping and Equipment."
 - b. Comply with requirements for vibration isolation devices specified in Section 210548.13 "Vibration Controls for Fire-Suppression Piping and Equipment."
 - c. For supported equipment, install epoxy-coated anchor bolts that extend through

- concrete base and anchor into structural concrete floor.
 - d. Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
 - e. Install anchor bolts to elevations required for proper attachment to supported equipment.
 - f. Attach pumps to equipment base using anchor bolts.
 - g. Shim pumps as needed to make them level.
4. Install isolation valves in both inlet and outlet pipes near the pump. Comply with requirements for valves specified in Section 211313 "Wet-Pipe Sprinkler Systems."

3.2 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections with the assistance of a factory-authorized service representative:
- 1. Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
 - 2. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
 - 3. Pressure-maintenance pumps will be considered defective if they do not pass tests and inspections.
- B. Prepare test and inspection reports.

3.3 ADJUSTING

- A. Lubricate pumps as recommended by manufacturer.
- B. Set field-adjustable pressure-switch ranges as indicated.

END OF SECTION 213400

SECTION 220517 - SLEEVES AND SLEEVE SEALS FOR PLUMBING PIPING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Sleeves.
- 2. Stack-sleeve fittings.
- 3. Sleeve-seal systems.
- 4. Sleeve-seal fittings.
- 5. Grout.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

PART 2 - PRODUCTS**2.1 SLEEVES**

- A. Cast-Iron Wall Pipes: Cast or fabricated of cast or ductile iron and equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop unless otherwise indicated.
- B. Galvanized-Steel Wall Pipes: ASTM A 53/A 53M, Schedule 40, with plain ends and welded steel collar; zinc coated.
- C. Galvanized-Steel-Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, zinc coated, with plain ends.
- D. PVC-Pipe Sleeves: ASTM D 1785, Schedule 40.
- E. Galvanized-Steel-Sheet Sleeves: 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint.
- F. Molded-PE or -PP Sleeves: Removable, tapered-cup shaped, and smooth outer surface with nailing flange for attaching to wooden forms.
- G. Molded-PVC Sleeves: With nailing flange for attaching to wooden forms.

2.2 STACK-SLEEVE FITTINGS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include,

but are not limited to, the following:

1. Smith, Jay R. Mfg. Co.
2. Zurn Specification Drainage Operation: Zurn Plumbing Products Group.

B. Description: Manufactured, cast-iron sleeve with integral clamping flange. Include clamping ring, bolts, and nuts for membrane flashing.

1. Underdeck Clamp: Clamping ring with setscrews.

2.3 SLEEVE-SEAL SYSTEMS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Advance Products & Systems, Inc.
2. CALPICO, Inc.
3. Metraflex Company (The).
4. Pipeline Seal and Insulator, Inc.
5. Proco Products, Inc.

B. Description: Modular sealing-element unit, designed for field assembly, for filling annular space between piping and sleeve.

1. Sealing Elements: EPDM-rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
2. Pressure Plates: Carbon steel.
3. Connecting Bolts and Nuts: Stainless steel of length required to secure pressure plates to sealing elements.

2.4 SLEEVE-SEAL FITTINGS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Presealed Systems.

B. Description: Manufactured plastic, sleeve-type, waterstop assembly made for imbedding in concrete slab or wall. Unit has plastic or rubber waterstop collar with center opening to match piping OD.

2.5 GROUT

A. Standard: ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.

B. Characteristics: Nonshrink; recommended for interior and exterior applications.

C. Design Mix: 5000-psi, 28-day compressive strength.

- D. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION

- A. Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.
- B. For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide 1-inch annular clear space between piping and concrete slabs and walls.
1. Sleeves are not required for core-drilled holes.
- C. Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.
1. Permanent sleeves are not required for holes in slabs formed by molded-PE or -PP sleeves.
 2. Cut sleeves to length for mounting flush with both surfaces.
 - a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level.
 3. Using grout, seal the space outside of sleeves in slabs and walls without sleeve-seal system.
- D. Install sleeves for pipes passing through interior partitions.
1. Cut sleeves to length for mounting flush with both surfaces.
 2. Install sleeves that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
 3. Seal annular space between sleeve and piping or piping insulation; use joint sealants appropriate for size, depth, and location of joint. Comply with requirements for sealants specified in Section 079200 "Joint Sealants."
- E. Fire-Barrier Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with firestop materials. Comply with requirements for firestopping specified in Section 078413 "Penetration Firestopping."

3.2 STACK-SLEEVE-FITTING INSTALLATION

- A. Install stack-sleeve fittings in new slabs as slabs are constructed.
1. Install fittings that are large enough to provide 1/4-inch (6.4-mm) annular clear space between sleeve and pipe or pipe insulation.
 2. Secure flashing between clamping flanges for pipes penetrating floors with membrane waterproofing. Comply with requirements for flashing specified in Section 076200 "Sheet Metal Flashing and Trim."
 3. Install section of cast-iron soil pipe to extend sleeve to 2 inches above

- finished floor level.
- 4. Extend cast-iron sleeve fittings below floor slab as required to secure clamping ring if ring is specified.
- 5. Using grout, seal the space around outside of stack-sleeve fittings.
- B. Fire-Barrier Penetrations: Maintain indicated fire rating of floors at pipe penetrations. Seal pipe penetrations with firestop materials. Comply with requirements for firestopping specified in Section 078413 "Penetration Firestopping."

3.3 SLEEVE-SEAL-SYSTEM INSTALLATION

- A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at service piping entries into building.
- B. Select type, size, and number of sealing elements required for piping material and size and for sleeve ID or hole size. Position piping in center of sleeve. Center piping in penetration, assemble sleeve-seal system components, and install in annular space between piping and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make a watertight seal.

3.4 SLEEVE-SEAL-FITTING INSTALLATION

- A. Install sleeve-seal fittings in new walls and slabs as they are constructed.
- B. Assemble fitting components of length to be flush with both surfaces of concrete slabs and walls. Position waterstop flange to be centered in concrete slab or wall.
- C. Secure nailing flanges to concrete forms.
- D. Using grout, seal the space around outside of sleeve-seal fittings.

3.5 SLEEVE AND SLEEVE-SEAL SCHEDULE

- A. Use sleeves and sleeve seals for the following piping-penetration applications:
 - 1. Exterior Concrete Walls above Grade:
 - a. Piping Smaller Than NPS 6 Galvanized-steel wall sleeves
 - 2. Exterior Concrete Walls below Grade:
 - a. Piping Smaller Than NPS 6 Galvanized-steel wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.
 - 3. Concrete Slabs above Grade:
 - a. Piping Smaller Than NPS 6 : Galvanized-steel-pipe sleeves.

4. Interior Partitions:

- a. Piping Smaller Than NPS 6: PVC-pipe sleeves.

END OF SECTION 220517

SECTION 220518 - ESCUTCHEONS FOR PLUMBING PIPING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Escutcheons.
 - 2. Floor plates.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

PART 2 - PRODUCTS**2.1 ESCUTCHEONS**

- A. One-Piece, Cast-Brass Type: With white plated finish and setscrew fastener.
- B. One-Piece, Deep-Pattern Type: Deep-drawn, box-shaped brass with white finish and spring-clip fasteners.
- C. Split-Casting Brass Type: With polished, white finish and with concealed hinge and setscrew.

2.2 FLOOR PLATES

- A. One-Piece Floor Plates: Cast-iron flange with holes for fasteners. Split-Casting Floor Plates: White Plated Cast brass with concealed hinge.

PART 3 - EXECUTION**3.1 INSTALLATION**

- A. Install escutcheons for piping penetrations of walls, ceilings, and finished floors.
- B. Install escutcheons with ID to closely fit around pipe, tube, and insulation of insulated piping and with OD that completely covers opening.
 - 1. Escutcheons for New Piping:
 - a. Piping with Fitting or Sleeve Protruding from Wall: One-piece, deep-pattern type.
 - b. White Piping: One-piece, cast-brass or split-casting brass type with polished, white finish.
 - c. Bare Piping at Wall and Floor Penetrations in Finished Spaces: One-piece,

- cast- brass or split-casting brass type with polished, white finish.
 - d. Bare Piping at Ceiling Penetrations in Finished Spaces: One-piece, cast-brass or split-casting brass type with polished, white finish.
 - e. Bare Piping in Unfinished Service Spaces: One-piece, cast-brass or split-casting brass type with polished, white finish.
 - f. Bare Piping in Equipment Rooms: One-piece, cast-brass or split-casting brass type with rough-brass finish.
- C. Install floor plates for piping penetrations of equipment-room floors.
- D. Install floor plates with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.
- 1. New Piping: One-piece, floor-plate type.
 - 2. Existing Piping: Split-casting, floor-plate type.

2.3 FIELD QUALITY CONTROL

- A. Replace broken and damaged escutcheons and floor plates using new materials.

END OF SECTION 220518

SECTION 220523 - GENERAL-DUTY VALVES FOR PLUMBING PIPING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY**A. Section Includes:**

1. Bronze ball valves.
2. Bronze lift check valves.
3. Bronze swing check valves.
4. Iron swing check valves.
5. Iron center guided check valves.
6. Iron gate valves.
7. Balancing Valves.
8. Drain Valves
- 9.

B. Related Sections:

1. Division 22 plumbing piping Sections for specialty valves applicable to those Sections only.
2. Division 22 Section "Identification for Plumbing Piping and Equipment" for valve tags and schedules.
3. Division 33 water distribution piping Sections for general-duty and specialty valves for site construction piping.

1.3 DEFINITIONS

- A. CWP: Cold working pressure.
- B. EPDM: Ethylene propylene copolymer rubber.
- C. NBR: Acrylonitrile-butadiene, Buna-N, or nitrile rubber.
- D. NRS: Nonrising stem.
- E. OS&Y: Outside screw and yoke.
- F. RS: Rising stem.
- G. SWP: Steam working pressure.

1.4 SUBMITTALS

- A. Product Data: For each type of valve indicated.

1.5 QUALITY ASSURANCE

- A. Source Limitations for Valves: Obtain each type of valve from single source from single manufacturer.
- B. ASME Compliance:
 - 1. ASME B16.10 and ASME B16.34 for ferrous valve dimensions and design criteria.
 - 2. ASME B31.1 for power piping valves.
 - 3. ASME B31.9 for building services piping valves.
- C. NSF Compliance: NSF 61 for valve materials for potable-water service.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Prepare valves for shipping as follows:
 - 1. Protect internal parts against rust and corrosion.
 - 2. Protect threads, flange faces, grooves, and weld ends.
 - 3. Set angle, gate, and globe valves closed to prevent rattling.
 - 4. Set ball and plug valves open to minimize exposure of functional surfaces.
 - 5. Block check valves in either closed or open position.
- B. Use the following precautions during storage:
 - 1. Maintain valve end protection.
 - 2. Store valves indoors and maintain at higher than ambient dew point temperature. If outdoor storage is necessary, store valves off the ground in watertight enclosures.
- C. Use sling to handle large valves; rig sling to avoid damage to exposed parts. Do not use hand wheels or stems as lifting or rigging points.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR VALVES

- A. Refer to valve schedule articles for applications of valves.
- B. Valve Pressure and Temperature Ratings: Not less than indicated and as required for system pressures and temperatures.
- C. Valve Sizes: Same as upstream piping unless otherwise indicated.
- D. Valve Actuator Types:
 - 1. Handwheel: For valves other than quarter-turn types.
 - 2. Handlever: For quarter-turn valves NPS 6 and smaller.
- E. Valves in Insulated Piping: With valve stem extensions and the following features:
 - 1. Gate Valves: With rising stem.
 - 2. Ball Valves: With extended operating handle of non-thermal-conductive material, and protective sleeve that allows operation of valve without breaking the vapor seal or disturbing insulation.
- F. Valve-End Connections: Match pipe joining method

1. Flanged: With flanges according to ASME B16.1 for iron valves.
2. Grooved: With grooves according to AWWA C606.
3. Solder Joint: With sockets according to ASME B16.18.
4. Threaded: With threads according to ASME B1.20.1.

G. Valve Bypass and Drain Connections: MSS SP-45.

2.2 BRONZE BALL VALVES

A. Two-Piece, Full-Port, Bronze Ball Valves with Stainless-Steel Trim:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Beeco.
 - b. Crane Co.; Crane Valve Group; Crane Valves.
 - c. Conbraco Industries, Inc.; Apollo Valves.
 - d. Hammond Valve.
 - e. Milwaukee Valve Company.
 - f. NIBCO INC.
 - g. Watts Regulator Co.; a division of Watts Water Technologies, Inc.
2. Description:
 - a. Standard: MSS SP-110.
 - b. CWP Rating: 200 psig minimum, match plans for greater pressures.
 - c. Body Design: Two piece.
 - d. Body Material: Bronze.
 - e. Ends: SOLDER
 - f. Seats: PTFE or TFE.
 - g. Stem: Stainless steel.
 - h. Ball: Stainless steel, vented.
 - i. Port: Full.

2.3 BRONZE LIFT CHECK VALVES

A. Lift Check Valves with Nonmetallic Disc:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Conbraco Industries, Inc.; Apollo Valves.
 - b. Crane Co.; Crane Valve Group; Crane Valves.
 - c. Hammond Valve.
 - d. Milwaukee Valve Company.
 - e. NIBCO INC.
 - f. Watts Regulator Co.; a division of Watts Water Technologies, Inc.
2. Description:
 - a. Standard: MSS SP-80, Type 2.
 - b. CWP Rating: psig minimum, match plans for greater pressures.
 - c. Body Design: Vertical flow.
 - d. Body Material: ASTM B 61 or ASTM B 62, bronze.

- e. Ends: SOLDER
- f. Disc: NBR, PTFE, or TFE.

2.4 BRONZE SWING CHECK VALVES

A. Bronze Swing Check Valves with Bronze or non-metallic Disc:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Crane Co.
 - b. Hammond Valve.
 - c. Kitz Corporation.
 - d. Milwaukee Valve Company.
 - e. NIBCO INC.
 - f. Watts Regulator Co.; a division of Watts Water Technologies, Inc.
2. Description:
 - a. Standard: MSS SP-80
 - b. CWP Rating: 200 minimum, match plans for greater pressures.
 - c. Body Design: Horizontal flow.
 - d. Body Material: ASTM B 62, bronze.
 - e. Ends: Threaded.
 - f. Disc: Bronze.

2.5 IRON SWING CHECK VALVES

A. Iron Swing Check Valves with Metal Seats:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Crane Co
 - b. Hammond Valve.
 - c. Kitz Corporation.
 - d. Milwaukee Valve Company.
 - e. NIBCO INC.
 - f. Watts Regulator Co.; a division of Watts Water Technologies, Inc.
2. Description:
 - a. Standard: MSS SP-71
 - b. CWP Rating: 200 psig. minimum, match plans for greater pressures.
 - c. Body Design: Clear or full waterway.
 - d. Body Material: ASTM A 126, gray iron with bolted bonnet.
 - e. Ends: Match pipe system
 - f. Trim: Bronze.

2.6 IRON, CENTER-GUIDED CHECK VALVES

A. Class 125, Iron, Compact-Wafer, Center-Guided Check Valves with Metal Seat:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- a. Crane Co
- b. Hammond Valve.
- c. Kitz Corporation.
- d. Milwaukee Valve Company.
- e. NIBCO INC.
- f. Watts Regulator Co.; a division of Watts Water Technologies, Inc.

2. Description:

- a. Standard: MSS SP-125.
- b. CWP Rating: 200 psig. minimum, match plans for greater pressures.
- c. Body Material: ASTM A 126, gray iron.
- d. Style: Compact wafer.
- e. Seat: Bronze.

2.7 IRON GATE VALVES

A. OS&Y, Iron Gate Valves:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- a. Beeco.
- b. Hammond Valve.
- c. Crane Co.
- d. Kitz Corporation.
- e. Milwaukee Valve Company.
- f. NIBCO INC.
- g. Watts Regulator Co.; a division of Watts Water Technologies, Inc.

2. Description:

- a. Standard: MSS SP-70
- b. CWP Rating: 200 psig minimum, match plans for greater pressures.
- c. Body Material: ASTM A 126, gray iron with bolted bonnet.
- d. Ends: Flanged.
- e. Trim: Bronze.
- f. Disc: Solid wedge.
- g. Packing and Gasket: Asbestos free.

2.8 BALANCING VALVES

A. Copper-Alloy Calibrated Balancing Valves

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- a. ITT Industries; Bell & Gossett Div.
- b. NIBCO INC.
- c. Taco, Inc.
- d. Watts Industries, Inc.; Water Products Div.

2. Type: Ball or globe valve with two readout ports and memory setting indicator.
3. Body: bronze.
4. Size: Same as connected piping

5. Accessories: Meter hoses, fittings, valves, differential pressure meter, and carrying case.

B. Accessories: Meter hoses, fittings, valves, differential pressure meter, and carrying case.

2.9 DRAIN VALVES

A. Ball-Valve-Type, Hose-End Drain Valves

1. Standard: MSS SP-110 for standard-port, two-piece ball valves.
2. Pressure Rating: 400-psig minimum CWP.
3. Size: NPS 3/4.
4. Body: Copper alloy.
5. Ball: Chrome-plated brass.
6. Seats and Seals: Replaceable.
7. Handle: Vinyl-covered steel.
8. Inlet: Threaded or solder joint.
9. Outlet: Threaded, short nipple with garden-hose thread complying with ASME B1.20.7 and cap with brass chain.

B. Gate-Valve-Type, Hose-End Drain Valves

1. Standard: MSS SP-80 for gate valves.
2. Pressure Rating: Class 125.
3. Size: NPS 3/4.
4. Body: ASTM B 62 bronze.
5. Inlet: NPS 3/4 threaded or solder joint.
6. Outlet: Garden-hose thread complying with ASME B1.20.7 and cap with brass chain.

C. Stop-and-Waste Drain Valves

1. Standard: MSS SP-110 for ball valves or MSS SP-80 for gate valves.
2. Pressure Rating: 200-psig minimum CWP or Class 125.
3. Size: NPS 3/4.
4. Body: Copper alloy or ASTM B 62 bronze.
5. Drain: NPS 1/8 side outlet with cap.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine valve interior for cleanliness, freedom from foreign matter, and corrosion. Remove special packing materials, such as blocks, used to prevent disc movement during shipping and handling.
- B. Operate valves in positions from fully open to fully closed. Examine guides and seats made accessible by such operations.
- C. Examine threads on valve and mating pipe for form and cleanliness.
- D. Examine mating flange faces for conditions that might cause leakage. Check bolting for proper size, length, and material. Verify that gasket is of proper size, that its material composition is suitable for service, and that it is free from defects and damage.
- E. Do not attempt to repair defective valves; replace with new valves.

3.2 VALVE INSTALLATION

- A. Install valves with unions or flanges at each piece of equipment arranged to allow service, maintenance, and equipment removal without system shutdown.
- B. Locate valves for easy access and provide separate support where necessary.
- C. Install valves in horizontal piping with stem at or above center of pipe.
- D. Install valves in position to allow full stem movement.
- E. Install balancing valves in locations where they can easily be adjusted.
- F. Install chainwheels on operators for gate valves NPS 4 and larger and more than 96 inches above floor. Extend chains to 60 inches above finished floor.
- G. Install check valves for proper direction of flow and as follows:
 - 1. Swing Check Valves: In horizontal position with hinge pin level.
 - 2. Center-Guided and Plate-Type Check Valves: In horizontal or vertical position, between flanges.
 - 3. Lift Check Valves: With stem upright and plumb.

3.3 ADJUSTING

- A. Adjust or replace valve packing after piping systems have been tested and put into service but before final adjusting and balancing. Replace valves if persistent leaking occurs.
- B. Set field-adjustable flow set points of balancing valves.

3.4 GENERAL REQUIREMENTS FOR VALVE APPLICATIONS

- A. If valve applications are not indicated, use the following:
 - 1. Shutoff Service:
 - a. NPS 3 and smaller: Ball valve.
 - b. NPS 4 and larger: OS&Y gate.
 - 2. Throttling Service: Ball.
 - 3. Sewage and sump basin discharge to have swing check valves with optional lever and weight or lever and spring, for quiet operation.
- B. If valves with specified SWP classes or CWP ratings are not available, the same types of valves with higher SWP classes or CWP ratings may be substituted.
- C. Select valves with the following end connections:
 - 1. NPS 3 and smaller: Match the piping system joining method.
 - 2. NPS 4 and larger: Flanged.

3.5 DOMESTIC, HOT- AND COLD-WATER VALVE SCHEDULE

- A. Pipe NPS 3 and Smaller: Bronze ball valves

- B. Pipe NPS 4 and larger: OS&Y gate valves

3.6 SANITARY-WASTE AND STORM-DRAINAGE VALVE SCHEDULE

- A. Pipe NPS 2 and Smaller: Bronze with threaded ends

- B. Pipe NPS 3 and larger: Iron with flanged ends.

END OF SECTION 220523

SECTION 220529 - HANGERS AND SUPPORTS FOR PLUMBING PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY**A. Section Includes:**

1. Metal pipe hangers and supports.
2. Trapeze pipe hangers.
3. Metal framing systems.
4. Thermal-hanger shield inserts.
5. Fastener systems.
6. Pipe positioning systems.
7. Equipment supports.

B. Related Sections:

1. Section 055000 "Metal Fabrications" for structural-steel shapes and plates for trapeze hangers for pipe and equipment supports.
2. Section 220516 "Expansion Fittings and Loops for Plumbing Piping" for pipe guides and anchors.
3. Section 220548 "Vibration and Seismic Controls for Plumbing Piping and Equipment" for vibration isolation devices.

1.3 DEFINITIONS

- A. MSS: Manufacturers Standardization Society of The Valve and Fittings Industry Inc.

1.4 PERFORMANCE REQUIREMENTS

- A. Delegated Design: Design trapeze pipe hangers and equipment supports, including comprehensive engineering analysis by a qualified professional engineer, using performance requirements and design criteria indicated.

- B. Structural Performance: Hangers and supports for plumbing piping and equipment shall withstand the effects of gravity loads and stresses within limits and under conditions indicated according to ASCE/SEI 7.

1. Design supports for multiple pipes, including pipe stands, capable of supporting combined weight of supported systems, system contents, and test water.
2. Design equipment supports capable of supporting combined operating weight of supported equipment and connected systems and components.
3. Design seismic-restraint hangers and supports for piping and equipment and obtain approval from authorities having jurisdiction.

1.5 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Shop Drawings: Show fabrication and installation details and include calculations for the following; include Product Data for components:
 - 1. Trapeze pipe hangers.
 - 2. Metal framing systems.
 - 3. Fiberglass strut systems.
 - 4. Pipe stands.
 - 5. Equipment supports.
- C. Delegated-Design Submittal: For trapeze hangers indicated to comply with performance requirements and design criteria, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
 - 1. Detail fabrication and assembly of trapeze hangers.
 - 2. Design Calculations: Calculate requirements for designing trapeze hangers.

1.6 INFORMATIONAL SUBMITTALS

- A. Welding certificates.

1.7 QUALITY ASSURANCE

- A. Structural Steel Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."
- B. Pipe Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code.

PART 2 - PRODUCTS

2.1 METAL PIPE HANGERS AND SUPPORTS

- A. Carbon-Steel Pipe Hangers and Supports:
 - 1. Description: MSS SP-58, Types 1 through 58, factory-fabricated components.
 - 2. Galvanized Metallic Coatings: Pregalvanized or hot dipped.
 - 3. Nonmetallic Coatings: Plastic coating, jacket, or liner.
 - 4. Padded Hangers: Hanger with fiberglass or other pipe insulation pad or cushion to support bearing surface of piping.
 - 5. Hanger Rods: Continuous-thread rod, nuts, and washer made of stainless steel.
- B. Stainless-Steel Pipe Hangers and Supports:
 - 1. Description: MSS SP-58, Types 1 through 58, factory-fabricated components.
 - 2. Padded Hangers: Hanger with fiberglass or other pipe insulation pad or cushion to support bearing surface of piping.

3. Hanger Rods: Continuous-thread rod, nuts, and washer made of stainless steel.

C. Copper Pipe Hangers:

1. Description: MSS SP-58, Types 1 through 58, copper-coated-steel, factory-fabricated components.
2. Hanger Rods: Continuous-thread rod, nuts, and washer made of stainless steel.

2.2 TRAPEZE PIPE HANGERS

- A. Description: MSS SP-69, Type 59, shop- or field-fabricated pipe-support assembly made from structural carbon-steel shapes with MSS SP-58 carbon-steel hanger rods, nuts, saddles, and U- bolts.

2.3 METAL FRAMING SYSTEMS

A. MFMA Manufacturer Metal Framing Systems:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Allied Tube & Conduit.
 - b. Cooper B-Line, Inc.
 - c. Flex-Strut Inc.
 - d. GS Metals Corp.
 - e. Thomas & Betts Corporation.
 - f. Unistrut Corporation; Tyco International, Ltd.
 - g. Wesanco, Inc.
2. Description: Shop- or field-fabricated pipe-support assembly for supporting multiple parallel pipes.
3. Standard: MFMA-4.
4. Channels: Continuous slotted steel channel with intumed lips.
5. Channel Nuts: Formed or stamped steel nuts or other devices designed to fit into channel slot and, when tightened, prevent slipping along channel.
6. Hanger Rods: Continuous-thread rod, nuts, and washer made of stainless steel.
7. Metallic Coating: Electroplated zinc, Hot-dipped galvanized, Mill galvanized or Mechanically-deposited zinc.
8. Paint Coating: Vinyl, Epoxy or Polyester.
9. Plastic Coating: PVC, Epoxy or Polyester.

B. Non-MFMA Manufacturer Metal Framing Systems:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Anvil International; a subsidiary of Mueller Water Products Inc.
 - b. Empire Industries, Inc.
 - c. ERICO International Corporation.
 - d. Haydon Corporation; H-Strut Division.
 - e. NIBCO INC.

- f. PHD Manufacturing, Inc.
 - g. PHS Industries, Inc.
- 2. Description: Shop- or field-fabricated pipe-support assembly made of steel channels, accessories, fittings, and other components for supporting multiple parallel pipes.
 - 3. Standard: Comply with MFMA-4.
 - 4. Channels: Continuous slotted steel channel with inturned lips.
 - 5. Channel Nuts: Formed or stamped steel nuts or other devices designed to fit into channel slot and, when tightened, prevent slipping along channel.
 - 6. Hanger Rods: Continuous-thread rod, nuts, and washer made of stainless steel.
 - 7. Coating: Zinc, Paint or PVC.

2.4 THERMAL-HANGER SHIELD INSERTS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Carpenter & Paterson, Inc.
 - 2. Clement Support Services.
 - 3. ERICO International Corporation.
 - 4. National Pipe Hanger Corporation.
 - 5. PHS Industries, Inc.
 - 6. Pipe Shields, Inc.; a subsidiary of Piping Technology & Products, Inc.
 - 7. Piping Technology & Products, Inc.
 - 8. Rilco Manufacturing Co., Inc.
 - 9. Value Engineered Products, Inc.
- B. Insulation-Insert Material for Cold Piping: ASTM C 552, Type II cellular glass with 100-psig or ASTM C 591, Type VI, Grade 1 polyisocyanurate with 125-psig minimum compressive strength and vapor barrier.
- C. Insulation-Insert Material for Hot Piping: ASTM C 552, Type II cellular glass with 100-psig or ASTM C 591, Type VI, Grade 1 polyisocyanurate with 125-psig minimum compressive strength.
- D. For Trapeze or Clamped Systems: Insert and shield shall cover entire circumference of pipe.
- E. For Clevis or Band Hangers: Insert and shield shall cover lower 180 degrees of pipe.
- F. Insert Length: Extend 2 inches beyond sheet metal shield for piping operating below ambient air temperature.

2.5 FASTENER SYSTEMS

- A. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.
- B. Mechanical-Expansion Anchors: Insert-wedge-type, zinc-coated or stainless- steel anchors, for use in hardened portland cement concrete; with pull-out, tension,

and shear capacities appropriate for supported loads and building materials where used.

2.6 PIPE POSITIONING SYSTEMS

- A. Description: IAPMO PS 42, positioning system of metal brackets, clips, and straps for positioning piping in pipe spaces; for plumbing fixtures in commercial applications.

2.7 EQUIPMENT SUPPORTS

- A. Description: Welded, shop- or field-fabricated equipment support made from structural carbon- steel shapes.

2.8 MISCELLANEOUS MATERIALS

- A. Structural Steel: ASTM A 36/A 36M, carbon-steel plates, shapes, and bars; black and galvanized.
- B. Grout: ASTM C 1107, factory-mixed and -packaged, dry, hydraulic-cement, nonshrink and nonmetallic grout; suitable for interior and exterior applications.
 - 1. Properties: Nonstaining, noncorrosive, and nongaseous.
 - 2. Design Mix: 5000-psi, 28-day compressive strength.

PART 3 - EXECUTION

3.1 HANGER AND SUPPORT INSTALLATION

- A. Metal Pipe-Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Install hangers, supports, clamps, and attachments as required to properly support piping from the building structure.
- B. Metal Trapeze Pipe-Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Arrange for grouping of parallel runs of horizontal piping, and support together on field-fabricated trapeze pipe hangers.
 - 1. Pipes of Various Sizes: Support together and space trapezes for smallest pipe size or install intermediate supports for smaller diameter pipes as specified for individual pipe hangers.
 - 2. Field fabricate from ASTM A 36/A 36M, carbon-steel shapes selected for loads being supported. Weld steel according to AWS D1.1/D1.1M.
- C. Fiberglass Pipe-Hanger Installation: Comply with applicable portions of MSS SP-69 and MSS SP-89. Install hangers and attachments as required to properly support piping from building structure.
- D. Metal Framing System Installation: Arrange for grouping of parallel runs of piping, and support together on field-assembled metal framing systems.
- E. Fiberglass Strut System Installation: Arrange for grouping of parallel runs of piping, and support together on field-assembled fiberglass struts.

- F. Thermal-Hanger Shield Installation: Install in pipe hanger or shield for insulated piping.
- G. Fastener System Installation:
 - 1. Install powder-actuated fasteners for use in lightweight concrete or concrete slabs less than 4 inches thick in concrete after concrete is placed and completely cured. Use operators that are licensed by powder-actuated tool manufacturer. Install fasteners according to powder-actuated tool manufacturer's operating manual.
 - 2. Install mechanical-expansion anchors in concrete after concrete is placed and completely cured. Install fasteners according to manufacturer's written instructions.
- H. Pipe Stand Installation:
 - 1. Pipe Stand Types except Curb-Mounted Type: Assemble components and mount on smooth roof surface. Do not penetrate roof membrane.
 - 2. Curb-Mounted-Type Pipe Stands: Assemble components or fabricate pipe stand and mount on permanent, stationary roof curb. See Section 077200 "Roof Accessories" for curbs.
- I. Pipe Positioning-System Installation: Install support devices to make rigid supply and waste piping connections to each plumbing fixture.
- J. Install hangers and supports complete with necessary attachments, inserts, bolts, rods, nuts, washers, and other accessories.
- K. Equipment Support Installation: Fabricate from welded-structural-steel shapes.
- L. Install hangers and supports to allow controlled thermal and seismic movement of piping systems, to permit freedom of movement between pipe anchors, and to facilitate action of expansion joints, expansion loops, expansion bends, and similar units.
- M. Install lateral bracing with pipe hangers and supports to prevent swaying.
- N. Load Distribution: Install hangers and supports so that piping live and dead loads and stresses from movement will not be transmitted to connected equipment.
- O. Pipe Slopes: Install hangers and supports to provide indicated pipe slopes and to not exceed maximum pipe deflections allowed by ASME B31.9 for building services piping.
- P. Insulated Piping:
 - 1. Attach clamps and spacers to piping.
 - a. Piping Operating above Ambient Air Temperature: Clamp may project through insulation.
 - b. Piping Operating below Ambient Air Temperature: Use thermal-hanger shield insert with clamp sized to match OD of insert.

- c. Do not exceed pipe stress limits allowed by ASME B31.9 for building services piping.
- 2. Install MSS SP-58, Type 39, protection saddles if insulation without vapor barrier is indicated. Fill interior voids with insulation that matches adjoining insulation.
 - a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.
- 3. Install MSS SP-58, Type 40, protective shields on cold piping with vapor barrier. Shields shall span an arc of 180 degrees.
 - a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.
- 4. Shield Dimensions for Pipe: Not less than the following:
 - a. NPS 1/4 to NPS 3-1/2: 12 inches long and 0.048 inch thick.
- 5. Thermal-Hanger Shields: Install with insulation same thickness as piping insulation.

3.2 EQUIPMENT SUPPORTS

- A. Fabricate structural-steel stands to suspend equipment from structure overhead or to support equipment above floor.
- B. Grouting: Place grout under supports for equipment and make bearing surface smooth.
- C. Provide lateral bracing, to prevent swaying, for equipment supports.

3.3 METAL FABRICATIONS

- A. Cut, drill, and fit miscellaneous metal fabrications for trapeze pipe hangers and equipment supports.
- B. Fit exposed connections together to form hairline joints. Field weld connections that cannot be shop welded because of shipping size limitations.
- C. Field Welding: Comply with AWS D1.1/D1.1M procedures for shielded, metal arc welding; appearance and quality of welds; and methods used in correcting welding work; and with the following:
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
 - 4. Finish welds at exposed connections so no roughness shows after finishing and so contours of welded surfaces match adjacent contours.

3.4 ADJUSTING

- A. Hanger Adjustments: Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe.
- B. Trim excess length of continuous-thread hanger and support rods to 1-1/2 inches.

3.5 PAINTING

- A. Touchup: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.
 - 1. Apply paint by brush or spray to provide a minimum dry film thickness of 2.0 mils.
- B. Touchup: Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal are specified in Section 099123 "Interior Painting."
- C. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

3.6 HANGER AND SUPPORT SCHEDULE

- A. Specific hanger and support requirements are in Sections specifying piping systems and equipment.
- B. Comply with MSS SP-69 for pipe-hanger selections and applications that are not specified in piping system Sections.
- C. Use hangers and supports with galvanized metallic coatings for piping and equipment that will not have field-applied finish.
- D. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.
- E. Use carbon-steel pipe hangers and supports metal trapeze pipe hangers and metal framing systems and attachments for general service applications.
- F. Use corrosion-resistant attachments for hostile environment applications.
- G. Use copper-plated pipe hangers and copper or stainless-steel attachments for copper piping and tubing.
- H. Use padded hangers for piping that is subject to scratching.
- I. Use thermal-hanger shield inserts for insulated piping and tubing.
- J. Horizontal-Piping Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
 - 1. Adjustable, Steel Clevis Hangers (MSS Type 1): For suspension of

- noninsulated or insulated, stationary pipes NPS 1/2 to NPS 30.
2. Carbon- or Alloy-Steel, Double-Bolt Pipe Clamps (MSS Type 3): For suspension of pipes NPS 3/4 to NPS 36, requiring clamp flexibility and up to 4 inches of insulation.
 3. Steel Pipe Clamps (MSS Type 4): For suspension of cold and hot pipes NPS 1/2 to NPS 24 if little or no insulation is required.
 4. Pipe Hangers (MSS Type 5): For suspension of pipes NPS 1/2 to NPS 4, to allow off- center closure for hanger installation before pipe erection.
 5. Adjustable, Swivel Split- or Solid-Ring Hangers (MSS Type 6): For suspension of noninsulated, stationary pipes NPS 3/4 to NPS 8.
 6. Adjustable, Steel Band Hangers (MSS Type 7): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
 7. Adjustable Band Hangers (MSS Type 9): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
 8. Adjustable, Swivel-Ring Band Hangers (MSS Type 10): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
 9. Split Pipe Ring with or without Turnbuckle Hangers (MSS Type 11): For suspension of noninsulated, stationary pipes NPS 3/8 to NPS 8.
 10. Extension Hinged or Two-Bolt Split Pipe Clamps (MSS Type 12): For suspension of noninsulated, stationary pipes NPS 3/8 to NPS 3.
 11. U-Bolts (MSS Type 24): For support of heavy pipes NPS 1/2 to NPS 30.
 12. Clips (MSS Type 26): For support of insulated pipes not subject to expansion or contraction.
 13. Pipe Saddle Supports (MSS Type 36): For support of pipes NPS 4 to NPS 36, with steel- pipe base stanchion support and cast-iron floor flange or carbon-steel plate.
 14. Pipe Stanchion Saddles (MSS Type 37): For support of pipes NPS 4 to NPS 36, with steel-pipe base stanchion support and cast-iron floor flange or carbon-steel plate, and with U-bolt to retain pipe.
 15. Adjustable Pipe Saddle Supports (MSS Type 38): For stanchion-type support for pipes NPS 2-1/2 to NPS 36 if vertical adjustment is required, with steel-pipe base stanchion support and cast-iron floor flange.
 16. Single-Pipe Rolls (MSS Type 41): For suspension of pipes NPS 1 to NPS 30, from two rods if longitudinal movement caused by expansion and contraction might occur.
 17. Complete Pipe Rolls (MSS Type 44): For support of pipes NPS 2 to NPS 42 if longitudinal movement caused by expansion and contraction might occur but vertical adjustment is not necessary.
 18. Pipe Roll and Plate Units (MSS Type 45): For support of pipes NPS 2 to NPS 24 if small horizontal movement caused by expansion and contraction might occur and vertical adjustment is not necessary.
 19. Adjustable Pipe Roll and Base Units (MSS Type 46): For support of pipes NPS 2 to NPS 30 if vertical and lateral adjustment during installation might be required in addition to expansion and contraction.
- K. Vertical-Piping Clamps: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Extension Pipe or Riser Clamps (MSS Type 8): For support of pipe risers NPS 3/4 to NPS 24.
 2. Carbon- or Alloy-Steel Riser Clamps (MSS Type 42): For support of pipe risers NPS 3/4 to NPS 24 if longer ends are required for riser clamps.

- L. Hanger-Rod Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Steel Turnbuckles (MSS Type 13): For adjustment up to 6 inches for heavy loads.
 2. Steel Clevises (MSS Type 14): For 120 to 450 deg F piping installations.
 3. Swivel Turnbuckles (MSS Type 15): For use with MSS Type 11, split pipe rings.
 4. Malleable-Iron Sockets (MSS Type 16): For attaching hanger rods to various types of building attachments.
 5. Steel Weldless Eye Nuts (MSS Type 17): For 120 to 450 deg F piping installations.
- M. Building Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Steel or Malleable Concrete Inserts (MSS Type 18): For upper attachment to suspend pipe hangers from concrete ceiling.
 2. Top-Beam C-Clamps (MSS Type 19): For use under roof installations with bar-joist construction, to attach to top flange of structural shape.
 3. Side-Beam or Channel Clamps (MSS Type 20): For attaching to bottom flange of beams, channels, or angles.
 4. Center-Beam Clamps (MSS Type 21): For attaching to center of bottom flange of beams.
 5. Welded Beam Attachments (MSS Type 22): For attaching to bottom of beams if loads are considerable and rod sizes are large.
 6. C-Clamps (MSS Type 23): For structural shapes.
 7. Top-Beam Clamps (MSS Type 25): For top of beams if hanger rod is required tangent to flange edge.
 8. Side-Beam Clamps (MSS Type 27): For bottom of steel I-beams.
 9. Steel-Beam Clamps with Eye Nuts (MSS Type 28): For attaching to bottom of steel I-beams for heavy loads.
 10. Linked-Steel Clamps with Eye Nuts (MSS Type 29): For attaching to bottom of steel I-beams for heavy loads, with link extensions.
 11. Malleable-Beam Clamps with Extension Pieces (MSS Type 30): For attaching to structural steel.
 12. Welded-Steel Brackets: For support of pipes from below or for suspending from above by using clip and rod. Use one of the following for indicated loads:
 - a. Light (MSS Type 31): 750 lb.
 - b. Medium (MSS Type 32): 1500 lb.
 - c. Heavy (MSS Type 33): 3000 lb.
 13. Plate Lugs (MSS Type 57): For attaching to steel beams if flexibility at beam is required.
 14. Horizontal Travelers (MSS Type 58): For supporting piping systems subject to linear horizontal movement where headroom is limited.
- N. Saddles and Shields: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Protection Shields (MSS Type 40): Of length recommended in writing by manufacturer to prevent crushing insulation.
 2. Thermal-Hanger Shield Inserts: For supporting insulated pipe.
- O. Spring Hangers and Supports: Unless otherwise indicated and except as

specified in piping system Sections, install the following types:

1. Restraint-Control Devices (MSS Type 47): Where indicated to control piping movement.
 2. Spring Cushions (MSS Type 48): For light loads if vertical movement does not exceed 1- 1/4 inches.
 3. Spring-Cushion Roll Hangers (MSS Type 49): For equipping Type 41, roll hanger with springs.
 4. Spring Sway Braces (MSS Type 50): To retard sway, shock, vibration, or thermal expansion in piping systems.
 5. Variable-Spring Hangers (MSS Type 51): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from hanger.
 6. Variable-Spring Base Supports (MSS Type 52): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from base support.
 7. Variable-Spring Trapeze Hangers (MSS Type 53): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from trapeze support.
 8. Constant Supports: For critical piping stress and if necessary to avoid transfer of stress from one support to another support, critical terminal, or connected equipment. Include auxiliary stops for erection, hydrostatic test, and load-adjustment capability. These supports include the following types:
 - a. Horizontal (MSS Type 54): Mounted horizontally.
 - b. Vertical (MSS Type 55): Mounted vertically.
 - c. Trapeze (MSS Type 56): Two vertical-type supports and one trapeze member.
- P. Comply with MSS SP-69 for trapeze pipe-hanger selections and applications that are not specified in piping system Sections.
- Q. Comply with MFMA-103 for metal framing system selections and applications that are not specified in piping system Sections.
- R. Use powder-actuated fasteners or mechanical-expansion anchors instead of building attachments where required in concrete construction.
- S. Use pipe positioning systems in pipe spaces behind plumbing fixtures to support supply and waste piping for plumbing fixtures.

END OF SECTION 220529

SECTION 220548 - VIBRATION AND SEISMIC CONTROLS FOR PLUMBING PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY**A. Section Includes:**

1. Elastomeric isolation pads.
2. Elastomeric isolation mounts.
3. Restrained elastomeric isolation mounts.
4. Pipe-riser resilient supports.
5. Resilient pipe guides.
6. Elastomeric hangers.
7. Spring hangers.
8. Snubbers.
9. Restraint channel bracings.
10. Restraint cables.
11. Seismic-restraint accessories.
12. Mechanical anchor bolts.
13. Adhesive anchor bolts.

B. Related Requirements:

1. Section 210548 "Vibration and Seismic Controls for Fire-Suppression Piping and Equipment" for devices for fire-suppression equipment and systems.
2. Section 230548 "Vibration and Seismic Controls for HVAC" for devices for HVAC equipment and systems.

1.3 DEFINITIONS

- A. IBC: International Building Code.
- B. ICC-ES: ICC-Evaluation Service.
- C. OSHPD: Office of Statewide Health Planning & Development (for the State of California).

1.4 ACTION SUBMITTALS**A. Product Data: For each type of product.**

1. Include rated load, rated deflection, and overload capacity for each vibration isolation device.
2. Illustrate and indicate style, material, strength, fastening provision, and

finish for each type and size of vibration isolation device and seismic-restraint component required.

- a. Tabulate types and sizes of seismic restraints, complete with report numbers and rated strength in tension and shear as evaluated by an agency acceptable to authorities having jurisdiction.
 - b. Annotate to indicate application of each product submitted and compliance with requirements.
 3. Interlocking Snubbers: Include ratings for horizontal, vertical, and combined loads.
- B. Shop Drawings:
1. Detail fabrication and assembly of equipment bases. Detail fabrication including anchorages and attachments to structure and to supported equipment.
- C. Delegated-Design Submittal: For each vibration isolation and seismic-restraint device.
1. Include design calculations and details for selecting vibration isolators and seismic restraints complying with performance requirements, design criteria, and analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
 2. Design Calculations: Calculate static and dynamic loading due to equipment weight and operation, due to seismic forces required to select vibration isolators, and due to seismic restraints.
 3. Riser Supports: Include riser diagrams and calculations showing anticipated expansion and contraction at each support point, initial and final loads on building structure, spring deflection changes, and seismic loads. Include certification that riser system was examined for excessive stress and that none exists.
 4. Seismic-Restraint Details:
 - a. Design Analysis: To support selection and arrangement of seismic restraints. Include calculations of combined tensile and shear loads.
 - b. Details: Indicate fabrication and arrangement. Detail attachments of restraints to the restrained items and to the structure. Show attachment locations, methods, and spacings. Identify components, list their strengths, and indicate directions and values of forces transmitted to the structure during seismic events. Indicate association with vibration isolation devices.
 - c. Coordinate seismic-restraint and vibration isolation details with wind-restraint details required for equipment mounted outdoors. Comply with requirements in other Sections for equipment mounted outdoors.
 - d. Preapproval and Evaluation Documentation: By an agency acceptable to authorities having jurisdiction, showing maximum ratings of restraint items and the basis for approval (tests or calculations).

1.5 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Show coordination of vibration isolation device installation and seismic bracing for plumbing piping and equipment with other systems and equipment in the vicinity, including other supports and restraints, if any.
- B. Qualification Data: For testing agency.
- C. Welding certificates.
- D. Field quality-control reports.

1.6 QUALITY ASSURANCE

- A. Testing Agency Qualifications: An independent agency, with the experience and capability to conduct the testing indicated, that is an NRTL as defined by OSHA in 29 CFR 1910.7 and that is acceptable to authorities having jurisdiction.
- B. Comply with seismic-restraint requirements in the IBC unless requirements in this Section are more stringent.
- C. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."
- D. Seismic-restraint devices shall have horizontal and vertical load testing and analysis and shall bear anchorage preapproval OPA number from OSHPD, preapproval by ICC-ES, or preapproval by another agency acceptable to authorities having jurisdiction, showing maximum seismic-restraint ratings. Ratings based on independent testing are preferred to ratings based on calculations. If preapproved ratings are unavailable, submittals based on independent testing are preferred. Calculations (including combining shear and tensile loads) to support seismic-restraint designs must be signed and sealed by a qualified professional engineer.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Seismic-Restraint Loading:
 - 1. Site Class as Defined in the IBC: B.
 - 2. Assigned Seismic Use Group or Building Category as Defined in the IBC: [I] [II] [III].
 - a. Component Importance Factor: 1.0.
 - b. Component Response Modification Factor: 1.5.
 - c. Component Amplification Factor: 1.0.
 - 3. Design Spectral Response Acceleration at Short Periods (0.2 Second).
 - 4. Design Spectral Response Acceleration at 1.0-Second Period.
 - 5. Rated strengths, features, and applications shall be as defined in reports by an agency acceptable to authorities having jurisdiction.

- a. Structural Safety Factor: Allowable strength in tension, shear, and pullout force of components shall be at least four times the maximum seismic forces to which they are subjected.

2.2 ELASTOMERIC ISOLATION PADS

A. Elastomeric Isolation Pads:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Ace Mountings Co., Inc.
 - b. California Dynamics Corporation.
 - c. Isolation Technology, Inc.
 - d. Kinetics Noise Control, Inc.
 - e. Mason Industries, Inc.
 - f. Vibration Eliminator Co., Inc.
 - g. Vibration Isolation.
 - h. Vibration Mountings & Controls, Inc.
2. Fabrication: Single or multiple layers of sufficient durometer stiffness for uniform loading over pad area.
3. Size: Factory or field cut to match requirements of supported equipment.
4. Pad Material: Oil and water resistant with elastomeric properties.
5. Surface Pattern: Ribbed pattern.
6. Infused nonwoven cotton or synthetic fibers.
7. Load-bearing metal plates adhered to pads.
8. Sandwich-Core Material: Resilient and elastomeric.
 - a. Surface Pattern: Ribbed pattern.
 - b. Infused nonwoven cotton or synthetic fibers.

2.3 ELASTOMERIC ISOLATION MOUNTS

A. Double-Deflection, Elastomeric Isolation Mounts: .

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Ace Mountings Co., Inc.
 - b. California Dynamics Corporation.
 - c. Isolation Technology, Inc.
 - d. Kinetics Noise Control, Inc.
 - e. Mason Industries, Inc.
 - f. Vibration Eliminator Co., Inc.
 - g. Vibration Isolation.
 - h. Vibration Mountings & Controls, Inc.
2. Mounting Plates:
 - a. Top Plate: Encapsulated steel load transfer top plates, factory

- drilled and threaded with threaded studs or bolts.
- b. Baseplate: Encapsulated steel bottom plates with holes provided for anchoring to support structure.

- 3. Elastomeric Material: Molded, oil-resistant rubber, neoprene, or other elastomeric material.

2.4 RESTRAINED ELASTOMERIC ISOLATION MOUNTS

A. Restrained Elastomeric Isolation Mounts:

- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Ace Mountings Co., Inc.
 - b. California Dynamics Corporation.
 - c. Isolation Technology, Inc.
 - d. Kinetics Noise Control, Inc.
 - e. Mason Industries, Inc.
 - f. Vibration Eliminator Co., Inc.
 - g. Vibration Isolation.
 - h. Vibration Mountings & Controls, Inc.
- 2. Description: All-directional isolator with seismic restraints containing two separate and opposing elastomeric elements that prevent central threaded element and attachment hardware from contacting the housing during normal operation.
 - a. Housing: Cast-ductile iron or welded steel.
 - b. Elastomeric Material: Molded, oil-resistant rubber, neoprene, or other elastomeric material.

2.5 PIPE-RISER RESILIENT SUPPORT

- A. Description: All-directional, acoustical pipe anchor consisting of two steel tubes separated by a minimum 1/2-inch- thick neoprene.
 - 1. Vertical-Limit Stops: Steel and neoprene vertical-limit stops arranged to prevent vertical travel in both directions.
 - 2. Maximum Load Per Support: 500 psig on isolation material providing equal isolation in all directions.

2.6 RESILIENT PIPE GUIDES

- A. Description: Telescopic arrangement of two steel tubes or post and sleeve arrangement separated by a minimum 1/2-inch- thick neoprene.
 - 1. Factory-Set Height Guide with Shear Pin: Shear pin shall be removable and reinsertable to allow for selection of pipe movement. Guides shall be capable of motion to meet location requirements.

2.7 ELASTOMERIC HANGERS

A. Elastomeric Mount in a Steel Frame with Upper and Lower Steel Hanger Rods: .

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Ace Mountings Co., Inc.
 - b. California Dynamics Corporation.
 - c. Isolation Technology, Inc.
 - d. Kinetics Noise Control, Inc.
 - e. Mason Industries, Inc.
 - f. Vibration Eliminator Co., Inc.
 - g. Vibration Mountings & Controls, Inc.
2. Frame: Steel, fabricated with a connection for an upper threaded hanger rod and an opening on the underside to allow for a maximum of 30 degrees of angular lower hanger- rod misalignment without binding or reducing isolation efficiency.
3. Dampening Element: Molded, oil-resistant rubber, neoprene, or other elastomeric material with a projecting bushing for the underside opening preventing steel to steel contact.

2.8 SNUBBERS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Kinetics Noise Control, Inc.
2. Mason Industries, Inc.
3. Vibration Mountings & Controls, Inc.

B. Description: Factory fabricated using welded structural-steel shapes and plates, anchor bolts, and replaceable resilient isolation washers and bushings.

1. Anchor bolts for attaching to concrete shall be seismic-rated, drill-in, and stud-wedge or female-wedge type.
2. Resilient Isolation Washers and Bushings: Oil- and water-resistant neoprene.
3. Maximum 1/4-inch air gap, and minimum 1/4-inch- thick resilient cushion.

2.9 RESTRAINT CHANNEL BRACINGS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Cooper B-Line, Inc.
2. Hilti, Inc.
3. Mason Industries, Inc.
4. Unistrut.

B. Description: MFMA-4, shop- or field-fabricated bracing assembly made of

slotted steel channels with accessories for attachment to braced component at one end and to building structure at the other end and other matching components and with corrosion-resistant coating; rated in tension, compression, and torsion forces.

2.10 RESTRAINT CABLES

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
1. Kinetics Noise Control, Inc.
 2. Loos & Co., Inc.
 3. Vibration Mountings & Controls, Inc.
- B. Restraint Cables: ASTM A 492 stainless-steel cables. End connections made of steel assemblies with thimbles, brackets, swivel, and bolts designed for restraining cable service; with a minimum of two clamping bolts for cable engagement.

2.11 SEISMIC-RESTRAINT ACCESSORIES

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
1. Cooper B-Line, Inc.
 2. Kinetics Noise Control, Inc.
 3. Mason Industries, Inc.
 4. TOLCO.
- B. Hanger-Rod Stiffener: Reinforcing steel angle clamped to hanger rod.
- C. Hinged and Swivel Brace Attachments: Multifunctional steel connectors for attaching hangers to rigid channel bracings and restraint cables.
- D. Bushings for Floor-Mounted Equipment Anchor Bolts: Neoprene bushings designed for rigid equipment mountings, and matched to type and size of anchor bolts and studs.
- E. Bushing Assemblies for Wall-Mounted Equipment Anchorage: Assemblies of neoprene elements and steel sleeves designed for rigid equipment mountings, and matched to type and size of attachment devices used.
- F. Resilient Isolation Washers and Bushings: One-piece, molded, oil- and water-resistant neoprene, with a flat washer face.

2.12 MECHANICAL ANCHOR BOLTS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
1. Cooper B-Line, Inc.
 2. Hilti, Inc.

3. Kinetics Noise Control, Inc.
4. Mason Industries, Inc.

- B. Mechanical Anchor Bolts: Drilled-in and stud-wedge or female-wedge type in zinc-coated steel for interior applications and stainless steel for exterior applications. Select anchor bolts with strength required for anchor and as tested according to ASTM E 488.

2.13 ADHESIVE ANCHOR BOLTS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Hilti, Inc.
2. Kinetics Noise Control, Inc.
3. Mason Industries, Inc.

- B. Adhesive Anchor Bolts: Drilled-in and capsule anchor system containing PVC or urethane methacrylate-based resin and accelerator, or injected polymer or hybrid mortar adhesive. Provide anchor bolts and hardware with zinc-coated steel for interior applications and stainless steel for exterior applications. Select anchor bolts with strength required for anchor and as tested according to ASTM E 488.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and equipment to receive vibration isolation and seismic-control devices for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Examine roughing-in of reinforcement and cast-in-place anchors to verify actual locations before installation.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 APPLICATIONS

- A. Multiple Pipe Supports: Secure pipes to trapeze member with clamps approved for application by an agency acceptable to authorities having jurisdiction.
- B. Hanger-Rod Stiffeners: Install hanger-rod stiffeners where indicated or scheduled on Drawings to receive them and where required to prevent buckling of hanger rods due to seismic forces.
- C. Strength of Support and Seismic-Restraint Assemblies: Where not indicated, select sizes of components so strength is adequate to carry present and future static and seismic loads within specified loading limits.

3.3 VIBRATION CONTROL AND SEISMIC-RESTRAINT DEVICE INSTALLATION

- A. Coordinate the location of embedded connection hardware with supported equipment attachment and mounting points and with requirements for concrete reinforcement and formwork specified in Section 033000 "Cast-in-Place Concrete." or Section 033053 "Miscellaneous Cast-in-Place Concrete."
- B. Installation of vibration isolators must not cause any change of position of equipment, piping, or ductwork resulting in stresses or misalignment.
- C. Equipment Restraints:
 - 1. Install seismic snubbers on plumbing equipment mounted on vibration isolators. Locate snubbers as close as possible to vibration isolators and bolt to equipment base and supporting structure.
 - 2. Install resilient bolt isolation washers on equipment anchor bolts where clearance between anchor and adjacent surface exceeds 0.125 inch.
 - 3. Install seismic-restraint devices using methods approved by an agency acceptable to authorities having jurisdiction that provides required submittals for component.
- D. Piping Restraints:
 - 1. Comply with requirements in MSS SP-127.
 - 2. Space lateral supports a maximum of 40 feet o.c., and longitudinal supports a maximum of 80 feet o.c.
 - 3. Brace a change of direction longer than 12 feet.
- E. Install cables so they do not bend across edges of adjacent equipment or building structure.
- F. Install seismic-restraint devices using methods approved by an agency acceptable to authorities having jurisdiction that provides required submittals for component.
- G. Install bushing assemblies for anchor bolts for floor-mounted equipment, arranged to provide resilient media between anchor bolt and mounting hole in concrete base.
- H. Install bushing assemblies for mounting bolts for wall-mounted equipment, arranged to provide resilient media where equipment or equipment-mounting channels are attached to wall.
- I. Attachment to Structure: If specific attachment is not indicated, anchor bracing to structure at flanges of beams, at upper truss chords of bar joists, or at concrete members.
- J. Drilled-in Anchors:
 - 1. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Do not damage existing reinforcing or embedded items during coring or drilling. Notify the structural engineer if reinforcing steel or other embedded items are encountered during drilling. Locate and avoid prestressed tendons, electrical and telecommunications conduit, and gas lines.
 - 2. Do not drill holes in concrete or masonry until concrete, mortar, or grout

has achieved full design strength.

3. Wedge Anchors: Protect threads from damage during anchor installation. Heavy-duty sleeve anchors shall be installed with sleeve fully engaged in the structural element to which anchor is to be fastened.
4. Adhesive Anchors: Clean holes to remove loose material and drilling dust prior to installation of adhesive. Place adhesive in holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the adhesive.
5. Set anchors to manufacturer's recommended torque, using a torque wrench.
6. Install zinc-coated steel anchors for interior and stainless-steel anchors for exterior applications.

3.4 ACCOMMODATION OF DIFFERENTIAL SEISMIC MOTION

- A. Install flexible connections in piping where they cross seismic joints, where adjacent sections or branches are supported by different structural elements, and where the connections terminate with connection to equipment that is anchored to a different structural element from the one supporting the connections as they approach equipment. Comply with requirements in Section 221116 "Domestic Water Piping" for piping flexible connections.

3.5 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.
- B. Perform tests and inspections.
- C. Tests and Inspections:
 1. Provide evidence of recent calibration of test equipment by a testing agency acceptable to authorities having jurisdiction.
 2. Schedule test with Owner, through Architect, before connecting anchorage device to restrained component (unless post connection testing has been approved), and with at least seven days' advance notice.
 3. Obtain Architect's approval before transmitting test loads to structure. Provide temporary load-spreading members.
 4. Test at least four of each type and size of installed anchors and fasteners selected by Architect.
 5. Test to 90 percent of rated proof load of device.
 6. Measure isolator restraint clearance.
 7. Measure isolator deflection.
 8. Verify snubber minimum clearances.

D. Remove and replace malfunctioning units and retest as specified above.

E. Prepare test and inspection reports.

3.6 ADJUSTING

- A. Adjust isolators after piping system is at operating weight.

- B. Adjust limit stops on restrained-spring isolators to mount equipment at normal operating height. After equipment installation is complete, adjust limit stops so they are out of contact during normal operation.

END OF SECTION 220548

SECTION 220553 - IDENTIFICATION FOR PLUMBING PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Equipment labels.
 - 2. Warning signs and labels.
 - 3. Pipe labels.
 - 4. Stencils.
 - 5. Valve tags.
 - 6. Warning tags.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Samples: For color, letter style, and graphic representation required for each identification material and device.
- C. Equipment Label Schedule: Include a listing of all equipment to be labeled with the proposed content for each label.
- D. Valve numbering scheme.
- E. Valve Schedules: For each piping system to include in maintenance manuals.

1.4 COORDINATION

- A. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.
- B. Coordinate installation of identifying devices with locations of access panels and doors.
- C. Install identifying devices before installing acoustical ceilings and similar concealment.

PART 2 - PRODUCTS**2.1 EQUIPMENT LABELS**

- A. Metal Labels for Equipment:
 - 1. Material and Thickness: Brass, 0.032-inch Stainless steel, 0.025-inch Aluminum, 0.032- inch or anodized aluminum, 0.032-inch minimum

- thickness, and having predrilled or stamped holes for attachment hardware.
2. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.
 3. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
 4. Fasteners: Stainless-steel rivets or self-tapping screws.
 5. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.
- B. Plastic Labels for Equipment:
1. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/16 inch thick, and having predrilled holes for attachment hardware.
 2. Letter Color: Black.
 3. Background Color: White.
 4. Maximum Temperature: Able to withstand temperatures up to 160 deg F.
 5. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.
 6. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
 7. Fasteners: Stainless-steel rivets or self-tapping screws.
 8. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.
- C. Label Content: Include equipment's Drawing designation or unique equipment number, Drawing numbers where equipment is indicated (plans, details, and schedules), plus the Specification Section number and title where equipment is specified.
- D. Equipment Label Schedule: For each item of equipment to be labeled, on 8-1/2-by-11-inch bond paper. Tabulate equipment identification number and identify Drawing numbers where equipment is indicated (plans, details, and schedules), plus the Specification Section number and title where equipment is specified. Equipment schedule shall be included in operation and maintenance data.

2.2 WARNING SIGNS AND LABELS

- A. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/16 inch thick, and having predrilled holes for attachment hardware.
- B. Letter Color: Red.
- C. Background Color: White.
- D. Maximum Temperature: Able to withstand temperatures up to 160 deg F.
- E. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

- F. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
- G. Fasteners: Stainless-steel rivets or self-tapping screws.
- H. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.
- I. Label Content: Include caution and warning information, plus emergency notification instructions.

2.3 PIPE LABELS

- A. General Requirements for Manufactured Pipe Labels: Preprinted, color-coded, with lettering indicating service, and showing flow direction.
- B. Pretensioned Pipe Labels: Precoiled, semirigid plastic formed to cover full circumference of pipe and to attach to pipe without fasteners or adhesive.
- C. Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.
- D. Pipe Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings, pipe size, and an arrow indicating flow direction.
 - 1. Flow-Direction Arrows: Integral with piping system service lettering to accommodate both directions, or as separate unit on each pipe label to indicate flow direction.
 - 2. Lettering Size: At least 1-1/2 inches high.

2.4 STENCILS

- A. Stencils: Prepared with letter sizes according to ASME A13.1 for piping; and minimum letter height of 3/4 inch for access panel and door labels, equipment labels, and similar operational instructions.
 - 1. Stencil Material: Aluminum.
 - 2. Stencil Paint: Exterior, gloss, acrylic enamel black unless otherwise indicated. Paint may be in pressurized spray-can form.
 - 3. Identification Paint: Exterior, acrylic enamel in colors according to ASME A13.1 unless otherwise indicated.

2.5 VALVE TAGS

- A. Valve Tags: Stamped or engraved with 1/4-inch letters for piping system abbreviation and 1/2- inch numbers.
 - 1. Tag Material: Brass, 0.032-inch Stainless steel, 0.025-inch Aluminum, 0.032-inch or anodized aluminum, 0.032-inch minimum thickness, and

- having predrilled or stamped holes for attachment hardware.
- 2. Fasteners: Brass wire-link or beaded chain; or S-hook.
- B. Valve Schedules: For each piping system, on 8-1/2-by-11-inch bond paper. Tabulate valve number, piping system, system abbreviation (as shown on valve tag), location of valve (room or space), normal-operating position (open, closed, or modulating), and variations for identification. Mark valves for emergency shutoff and similar special uses.
 - 1. Valve-tag schedule shall be included in operation and maintenance data.

2.6 WARNING TAGS

- A. Warning Tags: Preprinted or partially preprinted, accident-prevention tags, of plasticized card stock with matte finish suitable for writing.
 - 1. Size: Approximately 4 by 7 inches.
 - 2. Fasteners: Brass grommet and wire.
 - 3. Nomenclature: Large-size primary caption such as "DANGER," "CAUTION," or "DO NOT OPERATE."
 - 4. Color: Yellow background with black lettering.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean piping and equipment surfaces of substances that could impair bond of identification devices, including dirt, oil, grease, release agents, and incompatible primers, paints, and encapsulants.

3.2 EQUIPMENT LABEL INSTALLATION

- A. Install or permanently fasten labels on each major item of mechanical equipment.
- B. Locate equipment labels where accessible and visible.

3.3 PIPE LABEL INSTALLATION

- A. Piping Color-Coding: Painting of piping is specified in Section 099123 "Interior Painting."
- B. Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:
 - 1. Near each valve and control device.
 - 2. Near each branch connection, excluding short takeoffs for fixtures and terminal units. Where flow pattern is not obvious, mark each pipe at branch.
 - 3. Near penetrations through walls, floors, ceilings, and inaccessible enclosures.
 - 4. At access doors, manholes, and similar access points that permit view of concealed piping.
 - 5. Near major equipment items and other points of origination and termination.

6. Spaced at maximum intervals of 50 feet along each run. Reduce intervals to 25 feet in areas of congested piping and equipment.
7. On piping above removable acoustical ceilings. Omit intermediately spaced labels.

C. Pipe Label Color Schedule:

1. Domestic Water Piping:
 - a. Background Color: Yellow.
 - b. Letter Color: Black.
2. Sanitary Waste and Storm Drainage Piping:
 - a. Background Color: Yellow.
 - b. Letter Color: Black.

3.4 VALVE-TAG INSTALLATION

- A. Install tags on valves and control devices in piping systems, except check valves; valves within factory-fabricated equipment units; shutoff valves; faucets; convenience and lawn-watering hose connections; and similar roughing-in connections of end-use fixtures and units. List tagged valves in a valve schedule.
- B. Valve-Tag Application Schedule: Tag valves according to size, shape, and color scheme and with captions similar to those indicated in the following subparagraphs:
 1. Valve-Tag Size and Shape:
 - a. Cold Water: 1-1/2 inches, round.
 - b. Hot Water: 1-1/2 inches, round.
 2. Valve-Tag Color:
 - a. Cold Water: Natural.
 - b. Hot Water: Natural.
 3. Letter Color:
 - a. Cold Water: Black.
 - b. Hot Water: Black.

3.5 WARNING-TAG INSTALLATION

- A. Write required message on, and attach warning tags to, equipment and other items where required.

END OF SECTION 220553

SECTION 220719 - PLUMBING PIPING INSULATION**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes insulating the following plumbing piping services:
 - 1. Domestic cold-water piping.
 - 2. Domestic hot-water piping.
 - 3. Supplies and drains for handicap-accessible lavatories and sinks.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated. Include thermal conductivity, water-vapor permeance thickness, and jackets (both factory- and field-applied, if any).
- B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.
 - 1. Detail application of protective shields, saddles, and inserts at hangers for each type of insulation and hanger.
 - 2. Detail attachment and covering of heat tracing inside insulation.
 - 3. Detail insulation application at pipe expansion joints for each type of insulation.
 - 4. Detail insulation application at elbows, fittings, flanges, valves, and specialties for each type of insulation.
 - 5. Detail removable insulation at piping specialties, equipment connections, and access panels.
 - 6. Detail application of field-applied jackets.
 - 7. Detail application at linkages of control devices.
- C. Samples: For each type of insulation and jacket indicated. Identify each Sample, describing product and intended use. Sample sizes are as follows:
 - 1. Preformed Pipe Insulation Materials: 12 inches long by NPS 2.
 - 2. Jacket Materials for Pipe: 12 inches long by NPS 2.
 - 3. Sheet Jacket Materials: 12 inches square.
 - 4. Manufacturer's Color Charts: For products where color is specified, show the full range of colors available for each type of finish material.

1.4 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For qualified Installer.
- B. Material Test Reports: From a qualified testing agency acceptable to authorities having jurisdiction indicating, interpreting, and certifying test results for compliance of insulation materials, sealers, attachments, cements, and jackets, with requirements indicated. Include dates of tests and test methods employed.

- C. Field quality-control reports.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or another craft training program certified by the Department of Labor, Bureau of Apprenticeship and Training.
- B. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products according to ASTM E 84 by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.
 - 1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.
- C. Mockups: Before installing insulation, build mockups for each type of insulation and finish listed below to demonstrate quality of insulation application and finishes. Build mockups in the location indicated or, if not indicated, as directed by Architect. Use materials indicated for the completed Work.
 - 1. Piping Mockups:
 - a. One 10-foot section of NPS 2 straight pipe.
 - b. One each of a 90-degree threaded, welded, and flanged elbow.
 - c. One each of a threaded, welded, and flanged tee fitting.
 - d. One NPS 2 or smaller valve, and one NPS 2-1/2 or larger valve.
 - e. Four support hangers including hanger shield and insert.
 - f. One threaded strainer and one flanged strainer with removable portion of insulation.
 - g. One threaded reducer and one welded reducer.
 - h. One pressure temperature tap.
 - i. One mechanical coupling.
 - 2. For each mockup, fabricate cutaway sections to allow observation of application details for insulation materials, adhesives, mastics, attachments, and jackets.
 - 3. Notify Architect seven days in advance of dates and times when mockups will be constructed.
 - 4. Obtain Architect's approval of mockups before starting insulation application.
 - 5. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
 - 6. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
 - 7. Demolish and remove mockups when directed.
- D. Comply with the following applicable standards and other requirements specified for miscellaneous components:
 - 1. Supply and Drain Protective Shielding Guards: ICC A117.1.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Packaging: Insulation material containers shall be marked by manufacturer with appropriate ASTM standard designation, type and grade, and maximum use temperature.

1.7 COORDINATION

- A. Coordinate sizes and locations of supports, hangers, and insulation shields specified in Section 220529 "Hangers and Supports for Plumbing Piping and Equipment."
- B. Coordinate clearance requirements with piping Installer for piping insulation application. Before preparing piping Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.
- C. Coordinate installation and testing of heat tracing.

1.8 SCHEDULING

- A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results.
- B. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

PART 2 - PRODUCTS

2.1 INSULATION MATERIALS

- A. Comply with requirements in "Piping Insulation Schedule, General," "Indoor Piping Insulation Schedule," articles for where insulating materials shall be applied.
- B. Products shall not contain asbestos, lead, mercury, or mercury compounds.
- C. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C 871.
- D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C 795.
- E. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.
- F. Cellular Glass: Inorganic, incombustible, foamed or cellulated glass with annealed, rigid, hermetically sealed cells. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.

- 1. Products: Subject to compliance with requirements, provide one of the following:

- a. Pittsburgh Corning Corporation; Foamglas.

2. Block Insulation: ASTM C 552, Type I.
3. Special-Shaped Insulation: ASTM C 552, Type III.
4. Preformed Pipe Insulation without Jacket: Comply with ASTM C 552, Type II, Class 1.
5. Preformed Pipe Insulation with Factory-Applied ASJ: Comply with ASTM C 552, Type II, Class 2.
6. Factory fabricate shapes according to ASTM C 450 and ASTM C 585.

G. Mineral-Fiber, Preformed Pipe Insulation:

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Fibrex Insulations Inc.; Coreplus 1200.
 - b. Johns Manville; Micro-Lok.
 - c. Knauf Insulation; 1000-Degree Pipe Insulation.
 - d. Manson Insulation Inc.; Alley-K.
 - e. Owens Corning; Fiberglas Pipe Insulation.
2. Type I, 850 Deg F Materials: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 547, Type I, Grade A, with factory-applied ASJ. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.

2.2 INSULATING CEMENTS

A. Mineral-Fiber Insulating Cement: Comply with ASTM C 195.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Ramco Insulation, Inc.; Super-Stik.

B. Expanded or Exfoliated Vermiculite Insulating Cement: Comply with ASTM C 196.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Ramco Insulation, Inc.; Thermokote V.

2.3 ADHESIVES

A. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated, unless otherwise indicated.

B. Cellular-Glass Adhesive: Two-component, thermosetting urethane adhesive containing no flammable solvents, with a service temperature range of minus 100 to plus 200 deg F.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 81-84.
2. For indoor applications, adhesive shall have a VOC content of 50 g/L or less

when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

C. Mineral-Fiber Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-127.
 - b. Eagle Bridges - Marathon Industries; 225.
 - c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 85-60/85-70.
 - d. Mon-Eco Industries, Inc.; 22-25.
2. For indoor applications, adhesive shall have a VOC content of 80 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
3. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

D. ASJ Adhesive, and FSK Jacket Adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-82.
 - b. Eagle Bridges - Marathon Industries; 225.
 - c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 85-20.
 - d. Mon-Eco Industries, Inc.; 22-25.
2. For indoor applications, adhesive shall have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
3. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

E. PVC Jacket Adhesive: Compatible with PVC jacket.

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Dow Corning Corporation; 739, Dow Silicone.
 - b. Johns Manville; Zeston Perma-Weld, CEEL-TITE Solvent Welding Adhesive.
 - c. P.I.C. Plastics, Inc.; Welding Adhesive.
 - d. Speedline Corporation; Polyco VP Adhesive.
2. For indoor applications, adhesive shall have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
3. Adhesive shall comply with the testing and product requirements of the California

Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

2.4 MASTICS

- A. Materials shall be compatible with insulation materials, jackets, and substrates; comply with MIL-PRF-19565C, Type II.
 - 1. For indoor applications, use mastics that have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
- B. Vapor-Barrier Mastic: Water based; suitable for indoor use on below-ambient services.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 30-80/30-90.
 - b. Vimasco Corporation; 749.
 - 2. Water-Vapor Permeance: ASTM E 96/E 96M, Procedure B, 0.013 perm at 43-mil dry film thickness.
 - 3. Service Temperature Range: Minus 20 to plus 180 deg F.
 - 4. Solids Content: ASTM D 1644, 58 percent by volume and 70 percent by weight.
 - 5. Color: White.
- C. Vapor-Barrier Mastic: Solvent based; suitable for indoor use on below-ambient services.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-30.
 - b. Eagle Bridges - Marathon Industries; 501.
 - c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 30-35.
 - d. Mon-Eco Industries, Inc.; 55-10.
 - 2. Water-Vapor Permeance: ASTM F 1249, 0.05 perm at 35-mil dry film thickness.
 - 3. Service Temperature Range: 0 to 180 deg F.
 - 4. Solids Content: ASTM D 1644, 44 percent by volume and 62 percent by weight.
 - 5. Color: White.

2.5 LAGGING ADHESIVES

- A. Description: Comply with MIL-A-3316C, Class I, Grade A, and shall be compatible with insulation materials, jackets, and substrates.
 - 1. For indoor applications, use lagging adhesives that have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 - 2. Products: Subject to compliance with requirements, provide one of the following:

- a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-50 AHV2.
 - b. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 30-36.
 - c. Vimasco Corporation; 713 and 714.
3. Fire-resistant, water-based lagging adhesive and coating for use indoors to adhere fire-resistant lagging cloths over pipe insulation.
 4. Service Temperature Range: 0 to plus 180 deg F.
 5. Color: White.

2.6 SEALANTS

A. Joint Sealants:

1. Joint Sealants for Cellular-Glass and Phenolic Products: Subject to compliance with requirements, provide one of the following:
 - a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-76.
 - b. Eagle Bridges - Marathon Industries; 405.
 - c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 30-45.
 - d. Mon-Eco Industries, Inc.; 44-05.
 - e. Pittsburgh Corning Corporation; Pittseal 444.
2. Materials shall be compatible with insulation materials, jackets, and substrates.
3. Permanently flexible, elastomeric sealant.
4. Service Temperature Range: Minus 100 to plus 300 deg F.
5. Color: White or gray.
6. For indoor applications, sealants shall have a VOC content of 420 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
7. Sealants shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

2.7 FACTORY-APPLIED JACKETS

A. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

1. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C 1136, Type I.
2. ASJ-SSL: ASJ with self-sealing, pressure-sensitive, acrylic-based adhesive covered by a removable protective strip; complying with ASTM C 1136, Type I.
3. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C 1136, Type II.

2.8 FIELD-APPLIED JACKETS

- A. Field-applied jackets shall comply with ASTM C 921, Type I, unless otherwise indicated.
- B. PVC Jacket: High-impact-resistant, UV-resistant PVC complying with ASTM D 1784, Class 16354-C; thickness as scheduled; roll stock ready for shop or field cutting and forming. Thickness is indicated in field-applied jacket schedules.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. Johns Manville; Zeston.
 - b. P.I.C. Plastics, Inc.; FG Series.
 - c. Proto Corporation; LoSmoke.
 - d. Speedline Corporation; SmokeSafe.
 - 2. Adhesive: As recommended by jacket material manufacturer.
 - 3. Color: White.
 - 4. Factory-fabricated fitting covers to match jacket if available; otherwise, field fabricate.
 - a. Shapes: 45- and 90-degree, short- and long-radius elbows, tees, valves, flanges, unions, reducers, end caps, soil-pipe hubs, traps, mechanical joints, and P-trap and supply covers for lavatories.

2.9 TAPES

- A. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C 1136.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. ABI, Ideal Tape Division; 428 AWF ASJ.
 - b. Avery Dennison Corporation, Specialty Tapes Division; Fasson 0836.
 - c. Compac Corporation; 104 and 105.
 - d. Venture Tape; 1540 CW Plus, 1542 CW Plus, and 1542 CW Plus/SQ.
 - 2. Width: 3 inches.
 - 3. Thickness: 11.5 mils.
 - 4. Adhesion: 90 ounces force/inch in width.
 - 5. Elongation: 2 percent.
 - 6. Tensile Strength: 40 lbf/inch in width.
 - 7. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.
- B. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor and outdoor applications.
 - 1. Products: Subject to compliance with requirements, provide one of the following:
 - a. ABI, Ideal Tape Division; 370 White PVC tape.
 - b. Compac Corporation; 130.
 - c. Venture Tape; 1506 CW NS.
 - 2. Width: 2 inches.

3. Thickness: 6 mils.
4. Adhesion: 64 ounces force/inch in width.
5. Elongation: 500 percent.
6. Tensile Strength: 18 lbf/inch in width.

2.10 SECUREMENTS

A. Bands:

1. Products: Subject to compliance with requirements, provide one of the following:
 - a. ITW Insulation Systems; Gerrard Strapping and Seals.
 - b. RPR Products, Inc.; Insul-Mate Strapping and Seals.
2. Stainless Steel: ASTM A 167 or ASTM A 240/A 240M, Type 304; 0.015 inch thick, 3/4 inch wide with closed seal.

B. Staples: Outward-clinching insulation staples, nominal 3/4-inch- wide, stainless steel or Monel.

C. Protective Shielding Piping Enclosures,:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Truebro; a brand of IPS Corporation.
 - b. Zurn Industries, LLC; Tubular Brass Plumbing Products Operation.
2. Description: Manufactured plastic enclosure for covering plumbing fixture hot- and cold- water supplies and trap and drain piping. Comply with ADA requirements.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions for compliance with requirements for installation tolerances and other conditions affecting performance of insulation application.
 1. Verify that systems to be insulated have been tested and are free of defects.
 2. Verify that surfaces to be insulated are clean and dry.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.
- B. Mix insulating cements with clean potable water; if insulating cements are to be in contact with stainless-steel surfaces, use demineralized water.

3.3 GENERAL INSTALLATION REQUIREMENTS

- A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of piping including fittings, valves, and specialties.
- B. Install insulation materials, forms, vapor barriers or retarders, jackets, and thicknesses required for each item of pipe system as specified in insulation system schedules.
- C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.
- D. Install insulation with longitudinal seams at top and bottom of horizontal runs.
- E. Install multiple layers of insulation with longitudinal and end seams staggered.
- F. Do not weld brackets, clips, or other attachment devices to piping, fittings, and specialties.
- G. Keep insulation materials dry during application and finishing.
- H. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.
- I. Install insulation with least number of joints practical.
- J. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.
 - 1. Install insulation continuously through hangers and around anchor attachments.
 - 2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.
 - 3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.
 - 4. Cover inserts with jacket material matching adjacent pipe insulation. Install shields over jacket, arranged to protect jacket from tear or puncture by hanger, support, and shield.
- K. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.
- L. Install insulation with factory-applied jackets as follows:
 - 1. Draw jacket tight and smooth.
 - 2. Cover circumferential joints with 3-inch- wide strips, of same material as insulation jacket. Secure strips with adhesive and outward clinching staples along both edges of strip, spaced 4 inches o.c.
 - 3. Overlap jacket longitudinal seams at least 1-1/2 inches. Install insulation with longitudinal seams at bottom of pipe. Clean and dry surface to receive self-sealing lap. Staple laps with outward clinching staples along edge at 4 inches o.c.
 - a. For below-ambient services, apply vapor-barrier mastic over staples.

4. Cover joints and seams with tape, according to insulation material manufacturer's written instructions, to maintain vapor seal.
 5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints and at ends adjacent to pipe flanges and fittings.
- M. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.
- N. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.
- O. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.
- P. For above-ambient services, do not install insulation to the following:
1. Vibration-control devices.
 2. Testing agency labels and stamps.
 3. Nameplates and data plates.
 4. Cleanouts.

3.4 PENETRATIONS

- A. Insulation Installation at Interior Wall and Partition Penetrations (That Are Not Fire Rated): Install insulation continuously through walls and partitions.
- B. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Install insulation continuously through penetrations of fire-rated walls and partitions.
1. Comply with requirements in Section 078413 "Penetration Firestopping" for firestopping and fire-resistive joint sealers.
- C. Insulation Installation at Floor Penetrations:
1. Pipe: Install insulation continuously through floor penetrations.
 2. Seal penetrations through fire-rated assemblies. Comply with requirements in Section 078413 "Penetration Firestopping."

3.5 GENERAL PIPE INSULATION INSTALLATION

- A. Requirements in this article generally apply to all insulation materials except where more specific requirements are specified in various pipe insulation material installation articles.
- B. Insulation Installation on Fittings, Valves, Strainers, Flanges, and Unions:
1. Install insulation over fittings, valves, strainers, flanges, unions, and other specialties with continuous thermal and vapor-retarder integrity unless otherwise indicated.
 2. Insulate pipe elbows using preformed fitting insulation or mitered fittings made from same material and density as adjacent pipe insulation. Each piece shall be butted

- tightly against adjoining piece and bonded with adhesive. Fill joints, seams, voids, and irregular surfaces with insulating cement finished to a smooth, hard, and uniform contour that is uniform with adjoining pipe insulation.
3. Insulate tee fittings with preformed fitting insulation or sectional pipe insulation of same material and thickness as used for adjacent pipe. Cut sectional pipe insulation to fit. Butt each section closely to the next and hold in place with tie wire. Bond pieces with adhesive.
 4. Insulate valves using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. For valves, insulate up to and including the bonnets, valve stuffing-box studs, bolts, and nuts. Fill joints, seams, and irregular surfaces with insulating cement.
 5. Insulate strainers using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Fill joints, seams, and irregular surfaces with insulating cement. Insulate strainers so strainer basket flange or plug can be easily removed and replaced without damaging the insulation and jacket. Provide a removable reusable insulation cover. For below-ambient services, provide a design that maintains vapor barrier.
 6. Insulate flanges and unions using a section of oversized preformed pipe insulation. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker.
 7. Cover segmented insulated surfaces with a layer of finishing cement and coat with a mastic. Install vapor-barrier mastic for below-ambient services and a breather mastic for above-ambient services. Reinforce the mastic with fabric-reinforcing mesh. Trowel the mastic to a smooth and well-shaped contour.
 8. For services not specified to receive a field-applied jacket except for flexible elastomeric and polyolefin, install fitted PVC cover over elbows, tees, strainers, valves, flanges, and unions. Terminate ends with PVC end caps. Tape PVC covers to adjoining insulation facing using PVC tape.
 9. Stencil or label the outside insulation jacket of each union with the word "union." Match size and color of pipe labels.
- C. Insulate instrument connections for thermometers, pressure gages, pressure temperature taps, test connections, flow meters, sensors, switches, and transmitters on insulated pipes. Shape insulation at these connections by tapering it to and around the connection with insulating cement and finish with finishing cement, mastic, and flashing sealant.
- D. Install removable insulation covers at locations indicated. Installation shall conform to the following:
1. Make removable flange and union insulation from sectional pipe insulation of same thickness as that on adjoining pipe. Install same insulation jacket as adjoining pipe insulation.
 2. When flange and union covers are made from sectional pipe insulation, extend insulation from flanges or union long at least two times the insulation thickness over adjacent pipe insulation on each side of flange or union. Secure flange cover in place with stainless-steel or aluminum bands. Select band material compatible with insulation and jacket.
 3. Construct removable valve insulation covers in same manner as for flanges, except divide the two-part section on the vertical center line of valve body.
 4. When covers are made from block insulation, make two halves, each consisting of

mitered blocks wired to stainless-steel fabric. Secure this wire frame, with its attached insulation, to flanges with tie wire. Extend insulation at least 2 inches over adjacent pipe insulation on each side of valve. Fill space between flange or union cover and pipe insulation with insulating cement. Finish cover assembly with insulating cement applied in two coats. After first coat is dry, apply and trowel second coat to a smooth finish.

5. Unless a PVC jacket is indicated in field-applied jacket schedules, finish exposed surfaces with a metal jacket.

3.6 INSTALLATION OF CELLULAR-GLASS INSULATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of insulation to pipe with wire or bands and tighten bands without deforming insulation materials.
2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.
3. For insulation with factory-applied jackets on above-ambient services, secure laps with outward clinched staples at 6 inches o.c.
4. For insulation with factory-applied jackets on below-ambient services, do not staple longitudinal tabs. Instead, secure tabs with additional adhesive as recommended by insulation material manufacturer and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Fittings and Elbows:

1. Install preformed sections of same material as straight segments of pipe insulation when available. Secure according to manufacturer's written instructions.
2. When preformed sections of insulation are not available, install mitered sections of cellular-glass insulation. Secure insulation materials with wire or bands.

C. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed sections of cellular-glass insulation to valve body.
2. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
3. Install insulation to flanges as specified for flange insulation application.

3.7 INSTALLATION OF MINERAL-FIBER INSULATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of preformed pipe insulation to pipe with wire or bands and tighten bands without deforming insulation materials.
2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.
3. For insulation with factory-applied jackets on above-ambient surfaces, secure laps with outward clinched staples at 6 inches o.c.
4. For insulation with factory-applied jackets on below-ambient surfaces, do not staple longitudinal tabs. Instead, secure tabs with additional adhesive as recommended by insulation material manufacturer and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Fittings and Elbows:

1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed insulation elbows and fittings are not available, install mitered sections of pipe insulation, to a thickness equal to adjoining pipe insulation. Secure insulation materials with wire or bands.

C. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed sections are not available, install mitered sections of pipe insulation to valve body.
3. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
4. Install insulation to flanges as specified for flange insulation application.

3.8 FIELD-APPLIED JACKET INSTALLATION

A. Where glass-cloth jackets are indicated, install directly over bare insulation or insulation with factory-applied jackets.

1. Draw jacket smooth and tight to surface with 2-inch overlap at seams and joints.
2. Embed glass cloth between two 0.062-inch- thick coats of lagging adhesive.
3. Completely encapsulate insulation with coating, leaving no exposed insulation.

B. Where PVC jackets are indicated, install with 1-inch overlap at longitudinal seams and end joints. Seal with manufacturer's recommended adhesive.

1. Apply two continuous beads of adhesive to seams and joints, one bead under lap and the finish bead along seam and joint edge.

3.9 FINISHES

A. Insulation with ASJ, Glass-Cloth, or Other Paintable Jacket Material: Paint jacket with paint system identified below and as specified in Section 099113 "Exterior Painting" and Section 099123 "Interior Painting."

1. Flat Acrylic Finish: Two finish coats over a primer that is compatible with jacket material and finish coat paint. Add fungicidal agent to render fabric mildew proof.
 - a. Finish Coat Material: Interior, flat, latex-emulsion size.

B. Flexible Elastomeric Thermal Insulation: After adhesive has fully cured, apply two coats of insulation manufacturer's recommended protective coating.

C. Color: Final color as selected by Architect. Vary first and second coats to allow visual inspection of the completed Work.

3.10 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.
- B. Perform tests and inspections.
- C. Tests and Inspections:
- D. Inspect pipe, fittings, strainers, and valves, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to three locations of straight pipe, three locations of threaded fittings, three locations of welded fittings, two locations of threaded strainers, two locations of welded strainers, three locations of threaded valves, and three locations of flanged valves for each pipe service defined in the "Piping Insulation Schedule, General" Article.
- E. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements.

3.11 PIPING INSULATION SCHEDULE, GENERAL

- A. Acceptable preformed pipe and tubular insulation materials and thicknesses are identified for each piping system and pipe size range. If more than one material is listed for a piping system, selection from materials listed is Contractor's option.
- B. Items Not Insulated: Unless otherwise indicated, do not install insulation on the following:
 - 1. Drainage piping located in crawl spaces.
 - 2. Underground piping.
 - 3. Chrome-plated pipes and fittings unless there is a potential for personnel injury.

3.12 INDOOR PIPING INSULATION SCHEDULE

- A. Domestic Cold Water:
 - 1. NPS 1 and Smaller: Insulation shall be one of the following:
 - a. Cellular Glass: 1-1/2 inches thick.
 - b. Mineral-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.
 - 2. NPS 1-1/4 and Larger: Insulation shall be one of the following:
 - a. Cellular Glass: 1-1/2 inches thick.
 - b. Mineral-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.
- B. Domestic Hot Water:
 - 1. NPS 1-1/4 and Smaller: Insulation shall be one of the following:
 - a. Cellular Glass: 1-1/2 inches thick.
 - b. Mineral-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.
 - 2. NPS 1-1/2 and Larger: Insulation shall be one of the following:

- a. Cellular Glass: 1-1/2 inches thick.
 - b. Mineral-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.
- C. Exposed Sanitary Drains, Domestic Water, Domestic Hot Water, and Stops for Plumbing Fixtures for People with Disabilities:
 - 1. All Pipe Sizes: Insulation shall be one of the following:
 - a. Flexible Elastomeric: 3/4 inch thick.
 - b. Mineral-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.
 - c. Polyolefin: 3/4 inch thick.

3.13 INDOOR, FIELD-APPLIED JACKET SCHEDULE

- A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.
- B. If more than one material is listed, selection from materials listed is Contractor's option.
- C. Piping, Concealed:
 - 1. None.
- D. Piping, Exposed:
 - 1. PVC: 20 mils thick.

END OF SECTION 220719

SECTION 220800 - COMMISSIONING OF PLUMBING**PART 1 – GENERAL****1.01 SUMMARY****A. Work Included:**

1. See Division 01, General Requirements for overall objectives and comply with requirements.
 - a. This section covers the Contractor's responsibilities for commissioning; installer responsible for installation of a particular system or equipment item to be commissioned is responsible for commissioning activities relating to that system or equipment item.
 - b. Pre-Functional Checklist and Functional Test requirements specified in this Section are in addition to, not a substitute for, inspection or testing specified in other sections.

1.02 RELATED SECTIONS

- A. Contents of Division 22, Plumbing and Division 01, General Requirements apply to this section.

1.03 REFERENCES AND STANDARDS

- A. References and Standards as required by Section 22 00 00, Plumbing Basic Requirements and Division 01, General Requirements.
- B. Meet requirements of ASHRAE Guideline 0, The Commissioning Process.

1.04 SUBMITTALS

- A. Submittals as required by Section 22 00 00, Plumbing Basic Requirements and Division 01, General Requirements.
- B. In addition, provide Pressure Tests, Flushing Reports, and Startup Reports. Submit for approval of Commissioning Authority.

1.05 QUALITY ASSURANCE

- A. Quality assurance as required by Section 22 00 00, Plumbing Basic Requirements and Division 01, General Requirements.

1.06 WARRANTY

- A. Warranty of materials and workmanship as required by Section 22 00 00, Plumbing Basic Requirements and Division 01, General Requirements.
- B. In addition, provide:
 1. Commissioning, inspecting, and testing will not modify terms or time periods of mechanical equipment, systems, and controls warranties including related equipment and system, and adjacent work.

2. Control system warranty period starts from date of Commissioning Agent acceptance.

1.07 COORDINATION

- A. Reference Section 01 91 13, General Commissioning Requirements for requirements pertaining to coordination during the commissioning process.

1.08 PURPOSE

- A. Purpose of commissioning process is to provide Owner assurance that systems have been installed in prescribed manner and will operate within performance guidelines. Commissioning is intended to enhance quality of system startup and aid in orderly transfer of systems to beneficial use by Owner.
- B. Commissioning procedures and results will be observed by Commissioning Authority or Owner's staff. Contractor is expected to verify functional readiness of systems to be tested prior to performing the tests in presence of Owner's witness. A high rate of test failure will indicate that Contractor has not adequately verified readiness of systems.

PART 2 – PRODUCTS

2.01 TEST EQUIPMENT

- A. Provide standard testing equipment required to perform startup and initial checkout and required functional performance testing; unless otherwise noted such testing equipment will NOT become property of Owner.
- B. Specialized tools, test equipment, and instruments required to execute Start-up, checkout, and testing of equipment are to be of sufficient quality and accuracy to test and/or measure system performance within specified tolerances. A testing laboratory must have calibrated its test equipment within the previous 12 months. Calibration to be NIST traceable. Contractor must calibrate test equipment and instruments according to manufacturer's recommended intervals and whenever the test equipment is dropped or damaged. Calibration tags must be affixed to the test equipment or certificates readily available.
- C. Equipment-Specific Tools: Where special testing equipment, tools and instruments are specific to a piece of equipment, are only available from the vendor, and are required in order to accomplish startup or Functional Testing, provide such equipment, tools, and instruments as part of the work at no extra cost to Owner; such equipment, tools, and instruments are to become property of Owner.

PART 3 – EXECUTION

3.01 PREPARATION

- A. Cooperate with Commissioning Authority in development of the Pre-Functional Checklists and Functional Test Procedures.
- B. As part of required submittals for contract, within three months of award of contract, submit for each piece of equipment and controls, manufacturer's startup and installation procedures as well as controls point-to-point and sequence checkout and provide in check list format.
- C. Furnish additional information requested by the Commissioning Authority.
- D. Prepare a preliminary schedule for plumbing pipe systems testing, flushing and cleaning, equipment start-up and testing, adjusting, and balancing start and completion for use by the Commissioning Authority; update schedule as appropriate.

- E. Notify Commissioning Authority when pipe system testing, flushing, cleaning, startup of each piece of equipment and testing, adjusting, and balancing will occur; when commissioning activities not yet performed or not yet scheduled will delay construction notify ahead of time and be proactive in seeing that Commissioning Authority has scheduling information needed to efficiently execute commissioning process.
- F. Put equipment and systems into operation and continue operation during each working day of testing, adjusting, and balancing and commissioning, as required.
- G. Provide temperature and pressure taps in accordance with Contract Documents.
- H. Provide a pressure/temperature plug at each water sensor which is an input point to control system.

3.02 CONTRACTOR'S RESPONSIBILITIES

- A. Perform commissioning tests at the direction of the Commissioning Authority.
- B. Participate in Plumbing systems, assemblies, equipment, and component maintenance orientation and inspection as directed by the Commissioning Authority.
- C. Provide information requested by the Commissioning Authority for final commissioning documentation.
- D. Include requirements for submittal data, operation and maintenance data, and training in each purchase order or sub-contract written.
- E. Prepare preliminary schedule for Plumbing system orientations and inspections, operation and maintenance manual submissions, training sessions, pipe and duct system testing, flushing and cleaning, equipment startup, testing and balancing and task completion for Owner. Distribute preliminary schedule to commissioning team members.
- F. Update schedule as required throughout the construction period.
- G. During the startup and initial checkout process, execute the related portions of the Pre-Functional Checklists for commissioned equipment.
- H. Contractor to participate and complete checklists using the Commissioning Authority's web based commissioning software Facility Grid. A desktop, laptop, tablet, or iPad will be required.
- I. Assist the Commissioning Authority in verification and Functional Performance Tests.
- J. Gather operation and maintenance literature on equipment and assemble in binders as required by the Specifications. Submit to Commissioning Authority 45 days after substantial completion.
- K. Coordinate with the Commissioning Authority to provide 48 hour advance notice so that the witnessing of equipment and system startup and testing can begin.
- L. Notify the Commissioning Authority a minimum of one week in advance of the time for the start of the balancing work.
- M. Participate in, and schedule vendors and contractors to participate in the training sessions.
- N. Provide written notification to the CM/GC and Commissioning Authority that the following work has been completed in accordance with the Contract Documents, and that the equipment, systems, and sub-system are operating as required.

1. Plumbing equipment including domestic water heaters, pumps, plumbing fixtures, and other equipment furnished under this Division.
 2. Gas piping, sanitary waste and vent piping, storm drainage piping, sump pumps and automatic sprinkler system.
- O. Provide training of the Owner's operating staff using expert qualified personnel, as specified.
- P. Reference Section 01 91 13, General Commissioning Requirements, for additional contractor responsibilities.

3.03 OWNER'S RESPONSIBILITIES

- A. Reference Section 01 91 13, General Commissioning Requirements for Owner's responsibilities.

3.04 DESIGN PROFESSIONAL'S RESPONSIBILITIES

- A. Reference Section 01 91 13, General Commissioning Requirements for the Architect, Mechanical, Electrical, and Plumbing Engineer's responsibilities.

3.05 COMMISSIONING AUTHORITY'S RESPONSIBILITIES

- A. Reference Section 01 91 13, General Commissioning Requirements for the Commissioning Authority's responsibilities.

3.06 TESTING PREPARATION

- A. Certify, in writing, to the Commissioning Authority that plumbing instrumentation and control systems have been completed and calibrated, that they are operating according to the Contract Documents, and that pre-test setpoints have been recorded.
- B. Certify, in writing, that discrepancies discovered during the test and balance process have been resolved and that testing, adjusting, and balancing is completed.
- C. Set systems, subsystems, and equipment into operating mode to be tested (e.g. normal auto position, normal manual position, unoccupied mode, emergency power and alarm conditions).
- D. Inspect and verify the position of each device and interlock identified on checklists.
- E. Check safety cutouts, alarms, and interlocks with smoke control and life-safety systems during each mode operation.

3.07 TAB COORDINATION

- A. TAB: Testing, adjusting, and balancing of Plumbing.
- B. Coordinate commissioning schedule with TAB schedule.
- C. Review the TAB plan to determine capabilities of the control system toward completing TAB.
- D. Provide necessary unique instruments and instruct TAB technicians in their use; such as handheld control system interface, etc.
- E. Have required Pre-Functional Checklists, calibrations, startup and component Functional Tests of the system completed and approved by Commissioning Authority prior to starting TAB.
- F. Provide a qualified control system technician to operate controls to assist TAB technicians or provide sufficient training for TAB technicians to operate system without assistance.

3.08 GENERAL TESTING REQUIREMENTS

- A. Provide technicians, instrumentation, and tools to perform commissioning test at the direction of Commissioning Authority.
- B. Scope of Plumbing testing to include entire Plumbing installation. Testing to include measuring capacities and effectiveness of operational and control functions.
- C. Test operating modes, interlocks, control responses, and responses to abnormal or emergency conditions, and verify proper response of building automation system controllers and sensors.
- D. The Commissioning Authority along with the Plumbing contractor, balancing subcontractor to prepare detailed testing plans, procedures, and checklists for Plumbing systems, subsystems, and equipment.
- E. Tests will be performed using design conditions whenever applicable.
- F. Simulated conditions may need to be imposed using an artificial load when it is not practical to test under design conditions. Before simulating conditions, calibrate testing instruments. Provide equipment to simulate loads. Set simulated conditions as directed by the Commissioning Authority and document simulated conditions and methods of simulation. After tests, return settings to normal operating conditions.
- G. The Commissioning Authority may direct that setpoints be altered when simulating conditions is not practical.
- H. The Commissioning Authority may direct that sensor values be altered with a signal generator when design or simulating conditions and altering setpoints are not practical.
- I. If tests cannot be completed because of a deficiency outside the scope of the Plumbing system, document the deficiency and report it to the Owner. After deficiencies are resolved, reschedule tests.

3.09 PLUMBING SYSTEMS, SUBSYSTEMS AND EQUIPMENT TESTING PROCEDURES

- A. Pipe system cleaning, flushing, hydrostatic tests, and chemical treatment: Test requirements are specified in Division 22, Plumbing Piping sections. Plumbing Contractor to prepare a pipe system cleaning, flushing, and hydrostatic testing plan. Provide cleaning, flushing, testing, and treating plan and final reports to the Commissioning Authority. Plan should include the following.
 - 1. Sequence of testing procedures for each section of pipe to be tested, identified by pipe zone or sector identifications marker. Markers to be keyed to Drawings for each pipe sector, showing the physical location of each designated pipe test section. Drawings keyed to pipe zones or sectors to be formatted to allow each section to be physically located and identified when referred to in pipe system cleaning, flushing, hydrostatic testing, and chemical treatment plan.
 - 2. Description of equipment for flushing operations.
 - 3. Minimum flushing velocity.
- B. Functional Performance Tests: Tests will be fully documented with test procedures, expected results for each procedure, and documented in either pass or fail. Tests are written by the Commissioning Authority and performed by the Contractor. The Commissioning Authority documents the results of the test.

3.10 DEFICIENCIES / NON-CONFORMANCE AND COST OF RETESTING

- A. The Commissioning Authority documents the results of the tests. Corrections of minor deficiencies identified are made during the tests at the discretion of the Commissioning Authority. The Commissioning Authority documents the testing results on the Functional Performance Testing document. Deficiencies or non-conformance issues are noted and reported to the GC and Owner via the Master Cx Issues/Resolutions Log. The Contractor with then correct deficiencies, notify the Commissioning Authority of the correction, and then schedule retesting of the issue with the GC and Commissioning Authority. For areas in dispute of the issue between the Commissioning Authority and Contractor to go directly to the A/E. A/E to provide direction of the design intent and expected result to clear up the dispute.
- B. If the Plumbing contractor fails to demonstrate proper sequence of operation in any of the second round of Functional Performance Tests, the Commissioning Authority's costs for witnessing further demonstration of that test procedure may be assigned to the Plumbing contractor by the Owner as a deduct to their contracted price. The Plumbing contractor will not be responsible for costs related to failure due to design or other factors beyond their control, though it is expected to call any design concerns (and other factors beyond their control that might cause failure) to the attention of the GC and Commissioning Authority.
- C. Reference Section 01 91 13, General Commissioning Requirements for additional contractor responsibilities

3.11 OPERATION AND MAINTENANCE MANUALS

- A. See Division 01, General Requirements for additional requirements.
- B. Submit manuals related to items that were commissioned to Commissioning Authority for review; make changes recommended by Commissioning Authority.
- C. Commissioning Authority will add commissioning records to manuals after submission to Owner.

3.12 DEMONSTRATION AND TRAINING

- A. See Division 01, General Requirements for additional requirements.
- B. Demonstrate operation and maintenance of Plumbing systems to Owner's personnel; if during any demonstration, system fails to perform in accordance with information included in Operations and Maintenance (O&M) manual, stop demonstration, repair or adjust, and repeat demonstration. Demonstrations may be combined with training sessions if appropriate.
- C. These demonstrations are in addition to, and not a substitute for, Pre-Functional Checklists and demonstrations to Commissioning Authority during Functional Testing.
- D. Training:
 - 1. Submit a written training plan to the Owner and Architect/Engineer for review and approval. Contractor's training plan to cover the following elements:
 - a. Equipment included in training.
 - 1) Intended audience.
 - 2) Location of training.
 - 3) Objectives.

- b. Subjects covered.
 - c. Duration of training on each subject.
 - d. Instructor for each subject.
 - e. Methods (classroom lecture, video, Site walk-through, actual operational demonstrations, written hand outs, etc.).
 - f. Instructors and qualifications.
2. Contractor is to have the following training responsibilities:
- a. Provide a training plan ten calendar days prior to the scheduled training, in accordance with Division 01, General Requirements.
 - b. Provide Owner personnel with comprehensive training in the understanding of the systems and the operation and maintenance of each major piece of commissioned mechanical equipment or system.
 - c. Training to start with classroom sessions, if necessary, followed by hands-on training on each piece of equipment, which will illustrate the various modes of operation, including start-up, shutdown, fire/smoke alarm, power failure, etc.
 - d. During any demonstration, should the system fail to perform in accordance with the requirements of the O&M manual or sequence of operations, the system will be repaired or adjusted as necessary and the demonstration repeated.
 - e. The appropriate trade or manufacturer's representative will provide the instructions on each major piece of equipment. This representative may be the Start-up technician for the piece of equipment, the installing contractor, or the manufacturer's representative. Practical building operating expertise, as well as in-depth knowledge of modes of operation of the specific piece of equipment, is required. More than one party may be required to execute the training.
- E. Provide the services of manufacturer representatives to assist instructors where necessary.

END OF SECTION 22 08 00

SECTION 221116 - DOMESTIC WATER PIPING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Aboveground domestic water pipes, tubes, and fittings inside buildings.

1.3 ACTION SUBMITTALS

- A. Product Data: For transition fittings and dielectric fittings.

1.4 INFORMATIONAL SUBMITTALS

- A. System purging and disinfecting activities report.
- B. Field quality-control reports.

1.5 FIELD CONDITIONS

- A. Interruption of Existing Water Service: Do not interrupt water service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary water service according to requirements indicated:
 - 1. Notify Owner no fewer than two days in advance of proposed interruption of water service.
 - 2. Do not interrupt water service without Owner's written permission.

PART 2 - PRODUCTS**2.1 PIPING MATERIALS**

- A. Comply with requirements in "Piping Schedule" Article for applications of pipe, tube, fitting materials, and joining methods for specific services, service locations, and pipe sizes.
- B. Potable-water piping and components shall comply with NSF 14 and NSF 61. Plastic piping components shall be marked with "NSF-pw."

2.2 COPPER TUBE AND FITTINGS

- A. Hard Copper Tube: ASTM B 88, Type L water tube, drawn temper.
- B. Soft Copper Tube: ASTM B 88, Type K water tube, annealed temper.
- C. Cast-Copper, Solder-Joint Fittings: ASME B16.18, pressure fittings.

- D. Wrought-Copper, Solder-Joint Fittings: ASME B16.22, wrought-copper pressure fittings.
- E. Bronze Flanges: ASME B16.24, Class 150, with solder-joint ends.
- F. Copper Unions:
 - 1. MSS SP-123.
 - 2. Cast-copper-alloy, hexagonal-stock body.
 - 3. Ball-and-socket, metal-to-metal seating surfaces.
 - 4. Solder-joint or threaded ends.
- G. Copper Pressure-Seal-Joint Fittings:
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Elkhart Products Corporation.
 - b. NIBCO Inc.
 - c. Viega.
 - 2. Fittings for NPS 2 and Smaller: Wrought-copper fitting with EPDM-rubber, O-ring seal in each end.
 - 3. Fittings for NPS 2-1/2 to NPS 4: Cast-bronze or wrought-copper fitting with EPDM-rubber, O-ring seal in each end.

2.3 CPVC PIPE AND FITTINGS

- 1. CPVC, Schedule 80 Pipe: ASTM D 2846.
 - a. CPVC, Schedule 80 Socket Fittings: ASTM D 2846
 - b. ASTM F 493 Solvent Cements for CPVC.

2.4 DIELECTRIC FITTINGS

- A. General Requirements: Assembly of copper alloy and ferrous materials with separating nonconductive insulating material. Include end connections compatible with pipes to be joined.
- B. Dielectric Unions:
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Capitol Manufacturing Company; member of the Phoenix Forge Group.
 - b. Central Plastics Company.
 - c. Hart Industries International, Inc.
 - d. Jomar International.
 - e. Matco-Norca.
 - f. McDonald, A. Y. Mfg. Co.
 - g. Watts; a division of Watts Water Technologies, Inc.
 - h. Wilkins; a Zurn company.

2. Standard: ASSE 1079.
3. Pressure Rating: 125 psig minimum at 180 deg F.
4. End Connections: Solder-joint copper alloy and threaded ferrous.

PART 3 - EXECUTION

3.1 PIPING INSTALLATION

- A. Drawing plans, schematics, and diagrams indicate general location and arrangement of domestic water piping. Indicated locations and arrangements are used to size pipe and calculate friction loss, expansion, and other design considerations. Install piping as indicated unless deviations to layout are approved on coordination drawings.
- B. Install copper tubing under building slab according to CDA's "Copper Tube Handbook."
- C. Install shutoff valve, hose-end drain valve, strainer, pressure gage, and test tee with valve inside the building at each domestic water-service entrance. Comply with requirements for drain valves and strainers in Section 221119 "Domestic Water Piping Specialties."
- D. Install shutoff valve immediately upstream of each dielectric fitting.
- E. Install water-pressure-reducing valves downstream from shutoff valves. Comply with requirements for pressure-reducing valves in Section 221119 "Domestic Water Piping Specialties."
- F. Install domestic water piping level with 0.25 percent slope downward toward drain and plumb.
- G. Install seismic restraints on piping. Comply with requirements for seismic-restraint devices in Section 220548 "Vibration and Seismic Controls for Plumbing Piping and Equipment."
- H. Install piping concealed from view and protected from physical contact by building occupants unless otherwise indicated and except in equipment rooms and service areas.
- I. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.
- J. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal, and coordinate with other services occupying that space.
- K. Install piping to permit valve servicing.
- L. Install nipples, unions, special fittings, and valves with pressure ratings the same as or higher than the system pressure rating used in applications below unless otherwise indicated.
- M. Install piping free of sags and bends.
- N. Install fittings for changes in direction and branch connections.
- O. Install unions in copper tubing at final connection to each piece of equipment, machine, and specialty.

- P. Install sleeves for piping penetrations of walls, ceilings, and floors. Comply with requirements for sleeves specified in Section 220517 "Sleeves and Sleeve Seals for Plumbing Piping."
- Q. Install sleeve seals for piping penetrations of concrete walls and slabs. Comply with requirements for sleeve seals specified in Section 220517 "Sleeves and Sleeve Seals for Plumbing Piping."
- R. Install escutcheons for piping penetrations of walls, ceilings, and floors. Comply with requirements for escutcheons specified in Section 220518 "Escutcheons for Plumbing Piping."

3.2 JOINT CONSTRUCTION

- A. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
- B. Remove scale, slag, dirt, and debris from inside and outside of pipes, tubes, and fittings before assembly.
- C. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
 - 1. Apply appropriate tape or thread compound to external pipe threads.
 - 2. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged.
- D. Brazed Joints for Copper Tubing: Comply with CDA's "Copper Tube Handbook," "Braze Joints" chapter.
- E. Soldered Joints for Copper Tubing: Apply ASTM B 813, water-flushable flux to end of tube. Join copper tube and fittings according to ASTM B 828 or CDA's "Copper Tube Handbook."
- F. Pressure-Sealed Joints for Copper Tubing: Join copper tube and pressure-seal fittings with tools recommended by fitting manufacturer.
- G. Joints for Dissimilar-Material Piping: Make joints using adapters compatible with materials of both piping systems.

3.3 TRANSITION FITTING INSTALLATION

- A. Install transition couplings at joints of dissimilar piping.
- B. Transition Fittings in Aboveground Domestic Water Piping NPS 2 and Smaller: Plastic-to-metal transition fittings or unions.

3.4 DIELECTRIC FITTING INSTALLATION

- A. Install dielectric fittings in piping at connections of dissimilar metal piping and tubing.
- B. Dielectric Fittings for NPS 2 and Smaller: Use dielectric couplings or unions.

3.5 HANGER AND SUPPORT INSTALLATION

- A. Comply with requirements for seismic-restraint devices in Section 220548 "Vibration and Seismic Controls for Plumbing Piping and Equipment."
- B. Comply with requirements for pipe hanger, support products, and installation in Section 220529 "Hangers and Supports for Plumbing Piping and Equipment."
 - 1. Vertical Piping: MSS Type 8 or 42, clamps.
 - 2. Individual, Straight, Horizontal Piping Runs:
 - a. 100 Feet and Less: MSS Type 1, adjustable, steel clevis hangers.
 - b. Longer Than 100 Feet: MSS Type 43, adjustable roller hangers.
 - c. Longer Than 100 Feet if Indicated: MSS Type 49, spring cushion rolls.
 - 3. Multiple, Straight, Horizontal Piping Runs 100 Feet or Longer: MSS Type 44, pipe rolls. Support pipe rolls on trapeze.
 - 4. Base of Vertical Piping: MSS Type 52, spring hangers.
- C. Support vertical piping and tubing at base and at each floor.
- D. Rod diameter may be reduced one size for double-rod hangers, to a minimum of 3/8 inch.
- E. Install hangers for copper tubing with the following maximum horizontal spacing and minimum rod diameters:
 - 1. NPS 3/4 and Smaller: 60 inches with 3/8-inch rod.
 - 2. NPS 1 and NPS 1-1/4: 72 inches with 3/8-inch rod.
 - 3. NPS 1-1/2 and NPS 2: 96 inches with 3/8-inch rod.
- F. Install supports for vertical copper tubing every 10 feet.

3.6 CONNECTIONS

- A. Drawings indicate general arrangement of piping, fittings, and specialties.
- B. When installing piping adjacent to equipment and machines, allow space for service and maintenance.
- C. Connect domestic water piping to exterior water-service piping. Use transition fitting to join dissimilar piping materials.
- D. Connect domestic water piping to water-service piping with shutoff valve; extend and connect to the following:
 - 1. Water Heaters: Cold-water inlet and hot-water outlet piping in sizes indicated, but not smaller than sizes of water heater connections.
 - 2. Plumbing Fixtures: Cold- and hot-water-supply piping in sizes indicated, but not smaller than that required by plumbing code.
 - 3. Equipment: Cold- and hot-water-supply piping as indicated, but not smaller than equipment connections. Provide shutoff valve and union for each connection. Use flanges

instead of unions for NPS 2-1/2 and larger.

3.7 IDENTIFICATION

- A. Identify system components. Comply with requirements for identification materials and installation in Section 220553 "Identification for Plumbing Piping and Equipment."
- B. Label pressure piping with system operating pressure.

3.8 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections:

1. Piping Inspections:

- a. Do not enclose, cover, or put piping into operation until it has been inspected and approved by authorities having jurisdiction.
- b. During installation, notify authorities having jurisdiction at least one day before inspection must be made. Perform tests specified below in presence of authorities having jurisdiction:
 - 1) Roughing-in Inspection: Arrange for inspection of piping before concealing or closing in after roughing in and before setting fixtures.
 - 2) Final Inspection: Arrange for authorities having jurisdiction to observe tests specified in "Piping Tests" Subparagraph below and to ensure compliance with requirements.
- c. Reinspection: If authorities having jurisdiction find that piping will not pass tests or inspections, make required corrections and arrange for reinspection.
- d. Reports: Prepare inspection reports and have them signed by authorities having jurisdiction.

2. Piping Tests:

- a. Fill domestic water piping. Check components to determine that they are not air bound and that piping is full of water.
- b. Test for leaks and defects in new piping and parts of existing piping that have been altered, extended, or repaired. If testing is performed in segments, submit a separate report for each test, complete with diagram of portion of piping tested.
- c. Leave new, altered, extended, or replaced domestic water piping uncovered and unconcealed until it has been tested and approved. Expose work that was covered or concealed before it was tested.
- d. Cap and subject piping to static water pressure of 50 psig above operating pressure, without exceeding pressure rating of piping system materials. Isolate test source and allow it to stand for four hours. Leaks and loss in test pressure constitute defects that must be repaired.
- e. Repair leaks and defects with new materials, and retest piping or portion thereof until satisfactory results are obtained.
- f. Prepare reports for tests and for corrective action required.

- B. Domestic water piping will be considered defective if it does not pass tests and inspections.

- C. Prepare test and inspection reports.

3.9 ADJUSTING

- A. Perform the following adjustments before operation:

1. Close drain valves, hydrants, and hose bibbs.
2. Open shutoff valves to fully open position.
3. Open throttling valves to proper setting.
4. Adjust balancing valves in hot-water-circulation return piping to provide adequate flow.
 - a. Manually adjust ball-type balancing valves in hot-water-circulation return piping to provide hot-water flow in each branch.
 - b. Adjust calibrated balancing valves to flows indicated.
5. Remove plugs used during testing of piping and for temporary sealing of piping during installation.
6. Remove and clean strainer screens. Close drain valves and replace drain plugs.
7. Remove filter cartridges from housings and verify that cartridges are as specified for application where used and are clean and ready for use.
8. Check plumbing specialties and verify proper settings, adjustments, and operation.

3.10 CLEANING

- A. Clean and disinfect potable domestic water piping as follows:

1. Purge new piping and parts of existing piping that have been altered, extended, or repaired before using.
2. Use purging and disinfecting procedures prescribed by authorities having jurisdiction; if methods are not prescribed, use procedures described in either AWWA C651 or AWWA C652 or follow procedures described below:
 - a. Flush piping system with clean, potable water until dirty water does not appear at outlets.
 - b. Fill and isolate system according to either of the following:
 - 1) Fill system or part thereof with water/chlorine solution with at least 50 ppm of chlorine. Isolate with valves and allow to stand for 24 hours.
 - 2) Fill system or part thereof with water/chlorine solution with at least 200 ppm of chlorine. Isolate and allow to stand for three hours.
 - c. Flush system with clean, potable water until no chlorine is in water coming from system after the standing time.
 - d. Repeat procedures if biological examination shows contamination.
 - e. Submit water samples in sterile bottles to authorities having jurisdiction.

- B. Clean non-potable domestic water piping as follows:

1. Purge new piping and parts of existing piping that have been altered, extended, or

- repaired before using.
2. Use purging procedures prescribed by authorities having jurisdiction or; if methods are not prescribed, follow procedures described below:
 - a. Flush piping system with clean, potable water until dirty water does not appear at outlets.
 - b. Submit water samples in sterile bottles to authorities having jurisdiction. Repeat procedures if biological examination shows contamination.
 - C. Prepare and submit reports of purging and disinfecting activities. Include copies of water-sample approvals from authorities having jurisdiction.
 - D. Clean interior of domestic water piping system. Remove dirt and debris as work progresses.

3.11 PIPING SCHEDULE

- A. Transition and special fittings with pressure ratings at least equal to piping rating may be used in applications below unless otherwise indicated.
- B. Flanges and unions may be used for aboveground piping joints unless otherwise indicated.
- C. Aboveground domestic water piping, NPS 2 and smaller, shall be the following:
 1. Hard copper tube, ASTM B 88, Type L; cast- or wrought-copper, solder-joint fittings; and soldered joints.
 2. Hard copper tube, ASTM B 88, Type L; copper pressure-seal-joint fittings; and pressure-sealed joints.

3.12 VALVE SCHEDULE

- A. Where specific valve types are not indicated, the following requirements apply:
 1. Shutoff Duty: Use ball or gate valves for piping NPS 2 and smaller.
 2. Drain Duty: Hose-end drain valves.
- B. Use check valves to maintain correct direction of domestic water flow to and from equipment.

END OF SECTION 221116

SECTION 221119 - DOMESTIC WATER PIPING SPECIALTIES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY**A. Section Includes:**

1. Vacuum breakers.
2. Backflow preventers.
3. Water pressure-reducing valves.
4. Strainers.
5. Hose bibbs.
6. Drain valves.
7. Water-hammer arresters.
8. Trap-seal primer valves.
9. Specialty valves.
10. Flexible connectors.

B. Related Requirements:

1. Section 224716 "Pressure Water Coolers" for water filters for water coolers.

1.3 ACTION SUBMITTALS**A. Product Data:** For each type of product.**B. Shop Drawings:** For domestic water piping specialties.

1. Include diagrams for power, signal, and control wiring.

1.4 INFORMATIONAL SUBMITTALS**A. Field quality-control reports.****1.5 CLOSEOUT SUBMITTALS**

- A. Operation and Maintenance Data: For domestic water piping specialties to include in emergency, operation, and maintenance manuals.

PART 2 - PRODUCTS**2.1 GENERAL REQUIREMENTS FOR PIPING SPECIALTIES**

- A. Potable-water piping and components shall comply with NSF 61.

2.2 PERFORMANCE REQUIREMENTS

- A. Minimum Working Pressure for Domestic Water Piping Specialties: 125 psig unless otherwise indicated.

2.3 VACUUM BREAKERS

- A. Pipe-Applied, Atmospheric-Type Vacuum Breakers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Conbraco Industries, Inc.
 - b. FEBCO; a division of Watts Water Technologies, Inc.
 - c. Watts; a division of Watts Water Technologies, Inc.; Watts Regulator Company.
 - d. Zurn Industries, LLC; Plumbing Products Group; Wilkins Water Control Products.
2. Standard: ASSE 1001.
3. Size: NPS 1/4 to NPS 3, as required to match connected piping.
4. Body: Bronze.
5. Inlet and Outlet Connections: Threaded.
6. Finish: Rough bronze.

- B. Hose-Connection Vacuum Breakers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Arrowhead Brass Products.
 - b. Conbraco Industries, Inc.
 - c. MIFAB, Inc.
 - d. Watts; a division of Watts Water Technologies, Inc.; Watts Regulator Company.
 - e. Woodford Manufacturing Company; a division of WCM Industries, Inc.
 - f. Zurn Industries, LLC; Plumbing Products Group; Light Commercial Products.
2. Standard: ASSE 1011.
3. Body: Bronze, nonremovable, with manual drain.
4. Outlet Connection: Garden-hose threaded complying with ASME B1.20.7.
5. Finish: Rough bronze.

- C. Pressure Vacuum Breakers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Conbraco Industries, Inc.
 - b. FEBCO; a division of Watts Water Technologies, Inc.
 - c. Flomatic Corporation.
 - d. Watts; a division of Watts Water Technologies, Inc.; Watts Regulator Company.
 - e. Zurn Industries, LLC; Plumbing Products Group; Wilkins Water Control Products.
2. Standard: ASSE 1020.

3. Operation: Continuous-pressure applications.
4. Pressure Loss: 5 psig maximum, through middle third of flow range.
 - a. Valves: Ball type, on inlet and outlet.

2.4 BACKFLOW PREVENTERS

A. Intermediate Atmospheric-Vent Backflow Preventers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Conbraco Industries, Inc.
 - b. FEBCO; a division of Watts Water Technologies, Inc.
 - c. Watts; a division of Watts Water Technologies, Inc.; Watts Regulator Company.
 - d. Zurn Industries, LLC; Plumbing Products Group; Wilkins Water Control Products.
2. Standard: ASSE 1012.
3. Operation: Continuous-pressure applications.
4. Size: NPS 1/2.
5. Body: Bronze.
6. End Connections: Solder joint.
7. Finish: Rough bronze.

B. Reduced-Pressure-Principle Backflow Preventers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Ames Fire & Waterworks; a division of Watts Water Technologies, Inc.
 - b. Conbraco Industries, Inc.
 - c. FEBCO; a division of Watts Water Technologies, Inc.
 - d. Watts; a division of Watts Water Technologies, Inc.; Watts Regulator Company.
 - e. Zurn Industries, LLC; Plumbing Products Group; Wilkins Water Control Products.
2. Standard: ASSE 1013.
3. Operation: Continuous-pressure applications.
4. Pressure Loss: 10 psig maximum, through middle third of flow range.
5. Body: Bronze for NPS 2 and smaller
6. End Connections: Threaded for NPS 2 and smaller;
7. Configuration: Designed for horizontal, straight-through flow.
8. Accessories:
 - a. Valves NPS 2 and Smaller: Ball type with threaded ends on inlet and outlet.
 - b. Air-Gap Fitting: ASME A112.1.2, matching backflow-preventer connection.

C. Beverage-Dispensing-Equipment Backflow Preventers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- a. Conbraco Industries, Inc.
 - b. Watts; a division of Watts Water Technologies, Inc.; Watts Regulator Company.
 - c. Zurn Industries, LLC; Plumbing Products Group; Wilkins Water Control Products.
2. Standard: ASSE 1022.
 3. Operation: Continuous-pressure applications.
 4. Size: NPS 1/4 or NPS 3/8.
 5. Body: Stainless steel.
 6. End Connections: Threaded.

D. Hose-Connection Backflow Preventers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Conbraco Industries, Inc.
 - b. Watts; a division of Watts Water Technologies, Inc.; Watts Regulator Company.
 - c. Woodford Manufacturing Company; a division of WCM Industries, Inc.
2. Standard: ASSE 1052.
3. Operation: Up to 10-foot head of water back pressure.
4. Inlet Size: NPS 1/2 or NPS 3/4.
5. Outlet Size: Garden-hose thread complying with ASME B1.20.7.
6. Capacity: At least 3-gpm flow.

2.5 TEMPERATURE-ACTUATED, WATER MIXING VALVES

A. Water-Temperature Limiting Devices:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Conbraco Industries, Inc.
 - b. Symmons Industries, Inc.
 - c. Watts; a division of Watts Water Technologies, Inc.; Watts Regulator Company.
 - d. Zurn Industries, LLC; Plumbing Products Group; Wilkins Water Control Products.
2. Standard: ASSE 1016.
3. Pressure Rating: 125 psig.
4. Type: Thermostatically controlled, water mixing valve.
5. Material: Bronze body with corrosion-resistant interior components.
6. Connections: Threaded inlets and outlet.
7. Accessories: Check stops on hot- and cold-water supplies, and adjustable, temperature- control handle.
8. Tempered-Water Setting: 105 deg F.
9. Tempered-Water Design Flow Rate: 0.5 gpm.
10. Valve Finish: Chrome plated.

2.6 STRAINERS FOR DOMESTIC WATER PIPING

A. Y-Pattern Strainers:

1. Pressure Rating: 125 psig minimum unless otherwise indicated.
2. Body: Bronze for NPS 2 and smaller;
3. End Connections: Threaded for NPS 2 and smaller;
4. Screen: Stainless steel with round perforations unless otherwise indicated.
5. Perforation Size:
 - a. Strainers NPS 2 and Smaller: 0.020 inch.
6. Drain: Pipe plug.

2.7 HOSE BIBBS

A. Hose Bibbs:

1. Standard: ASME A112.18.1 for sediment faucets.
2. Body Material: Bronze.
3. Seat: Bronze, replaceable.
4. Supply Connections: NPS 1/2 or NPS 3/4 threaded or solder-joint inlet.
5. Outlet Connection: Garden-hose thread complying with ASME B1.20.7.
6. Pressure Rating: 125 psig.
7. Vacuum Breaker: Integral nonremovable, drainable, hose-connection vacuum breaker complying with ASSE 1011.
8. Finish for Equipment Rooms: Rough bronze, or chrome or nickel plated.
9. Finish for Service Areas: Rough bronze.
10. Finish for Finished Rooms: Chrome or nickel plated.
11. Include operating key with each operating-key hose bibb.

2.8 WATER-HAMMER ARRESTERS

A. Water-Hammer Arresters:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. AMTROL, Inc.
 - b. Josam Company.
 - c. MIFAB, Inc.
 - d. Precision Plumbing Products, Inc.
 - e. Sioux Chief Manufacturing Company, Inc.
 - f. Smith, Jay R. Mfg. Co.; Division of Smith Industries, Inc.
 - g. Tyler Pipe; Wade Div.
 - h. Watts Drainage Products.
 - i. Zurn Industries, LLC; Plumbing Products Group; Specification Drainage Products.
2. Standard: ASSE 1010 or PDI-WH 201.

3. Type: Copper tube with piston.
4. Size: ASSE 1010, Sizes AA and A through F, or PDI-WH 201, Sizes A through F.

2.9 TRAP-SEAL PRIMER DEVICE

A. Supply-Type, Trap-Seal Primer Device:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. MIFAB, Inc.
 - b. Precision Plumbing Products, Inc.
 - c. Sioux Chief Manufacturing Company, Inc.
 - d. Smith, Jay R. Mfg. Co.; Division of Smith Industries, Inc.
 - e. Watts; a division of Watts Water Technologies, Inc.; Watts Regulator Company.
2. Standard: ASSE 1018.
3. Pressure Rating: 125 psig minimum.
4. Body: Bronze.
5. Inlet and Outlet Connections: NPS 1/2 threaded, union, or solder joint.
6. Gravity Drain Outlet Connection: NPS 1/2 threaded or solder joint.
7. Finish: Chrome plated, or rough bronze for units used with pipe or tube that is not chrome finished.

2.10 FLEXIBLE CONNECTORS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Flex-Hose Co., Inc.
2. Flexicraft Industries.
3. Flex Pression, Ltd.
4. Flex-Weld Incorporated.
5. Hyspan Precision Products, Inc.
6. Mercer Gasket & Shim, Inc.
7. Metraflex, Inc.
8. Proco Products, Inc.
9. TOZEN Corporation.
10. Unaflex.Universal Metal Hose; a Hyspan company.

B. Bronze-Hose Flexible Connectors: Corrugated-bronze tubing with bronze wire-braid covering and ends brazed to inner tubing.

1. Working-Pressure Rating: Minimum 250 psig.
2. End Connections NPS 2 and Smaller: Threaded copper pipe or plain-end copper tube.

C. Stainless-Steel-Hose Flexible Connectors: Corrugated-stainless-steel tubing with stainless-steel wire-braid covering and ends welded to inner tubing.

1. Working-Pressure Rating: Minimum 250 psig.
2. End Connections NPS 2 and Smaller: Threaded steel-pipe nipple.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install backflow preventers in each water supply to mechanical equipment and systems and to other equipment and water systems that may be sources of contamination. Comply with authorities having jurisdiction.
 1. Install drain for backflow preventers with atmospheric-vent drain connection with air-gap fitting, fixed air-gap fitting, or equivalent positive pipe separation of at least two pipe diameters in drain piping and pipe-to-floor drain. Locate air-gap device attached to or under backflow preventer. Simple air breaks are unacceptable for this application.
 2. Do not install bypass piping around backflow preventers.
- B. Install water-hammer arresters in water piping according to PDI-WH 201.
- C. Install supply-type, trap-seal primer valves with outlet piping pitched down toward drain trap a minimum of 1 percent, and connect to floor-drain body, trap, or inlet fitting. Adjust valve for proper flow.
- D. Install drainage-type, trap-seal primer valves as lavatory trap with outlet piping pitched down toward drain trap a minimum of 1 percent, and connect to floor-drain body, trap, or inlet fitting.
- E. Install trap-seal primer systems with outlet piping pitched down toward drain trap a minimum of 1 percent, and connect to floor-drain body, trap, or inlet fitting. Adjust system for proper flow.

3.2 CONNECTIONS

- A. Comply with requirements for ground equipment in Section 260526 "Grounding and Bonding for Electrical Systems."

3.3 LABELING AND IDENTIFYING

- A. Equipment Nameplates and Signs: Install engraved plastic-laminate equipment nameplate or sign on or near each of the following:
 1. Pressure vacuum breakers.
 2. Intermediate atmospheric-vent backflow preventers.
 3. Reduced-pressure-principle backflow preventers.
 4. Double-check, backflow-prevention assemblies.
 5. Dual-check-valve backflow preventers.
 6. Outlet boxes.
 7. Hose stations.
 8. Supply-type, trap-seal primer valves.
- B. Distinguish among multiple units, inform operator of operational requirements, indicate safety

and emergency precautions, and warn of hazards and improper operations, in addition to identifying unit. Nameplates and signs are specified in Section 220553 "Identification for Plumbing Piping and Equipment."

3.4 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections:
 - 1. Test each pressure vacuum breaker and reduced-pressure-principle backflow preventer according to authorities having jurisdiction and the device's reference standard.
- B. Domestic water piping specialties will be considered defective if they do not pass tests and inspections.
- C. Prepare test and inspection reports.

3.5 ADJUSTING

- A. Set field-adjustable pressure set points of water pressure-reducing valves.
- B. Set field-adjustable flow set points of balancing valves.
- C. Set field-adjustable temperature set points of temperature-actuated, water mixing valves.

END OF SECTION 221119

SECTION 221123 - DOMESTIC WATER PUMPS**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. In-line, sealless centrifugal pumps.
2. Horizontally mounted, in-line, close-coupled centrifugal pumps.

1.2 ACTION SUBMITTALS**A. Product Data:** For each type of product indicated.**1.3 CLOSEOUT SUBMITTALS****A. Operation and maintenance data.****1.4 QUALITY ASSURANCE****A. Electrical Components, Devices, and Accessories:** Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.**B. UL Compliance:** Comply with UL 778 for motor-operated water pumps.**1.5 WARRANTY****A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.****PART 2 - PRODUCTS****2.1 IN-LINE, SEALLESS CENTRIFUGAL PUMPS****A. Description:** Factory-assembled and -tested, in-line, close-coupled, canned-motor, sealless, overhung-impeller centrifugal pumps.**B. Pump Construction:**

1. Pump and Motor Assembly: Hermetically sealed, replaceable-cartridge type with motor and impeller on common shaft and designed for installation with pump and motor shaft horizontal.
2. Casing: Bronze, with threaded or companion-flange connections.
3. Impeller: Plastic.
4. Motor: Single speed, unless otherwise indicated.

C. Capacities and Characteristics:

1. Capacity: See basis of design product schedule on drawings.
2. Total Dynamic Head: See basis of design product schedule on drawings.
3. Minimum Working Pressure: 125 psig
4. Maximum Continuous Operating Temperature: 220 deg F

5. Inlet and Outlet Size: See basis of design product schedule on drawings.
6. Pump Speed: See basis of design product schedule on drawings.
7. Pump Control: Thermostat, or Timer
8. Motor Horsepower: See basis of design product schedule on drawings.
9. Electrical Characteristics: See basis of design product schedule on drawings.

2.2 MOTORS

- A. Comply with NEMA designation, temperature rating, service factor, enclosure type, and efficiency requirements for motors specified in Section 220513 "Common Motor Requirements for Plumbing Equipment."
 1. Motor Sizes: Minimum size as indicated. If not indicated, large enough so driven load will not require motor to operate in service factor range above 1.0.

2.3 CONTROLS

- A. Thermostats: Electric; adjustable for control of hot-water circulation pump.
 1. Type: Water-immersion temperature sensor, for installation in piping.
 2. Range: 50 to 125 deg F
 3. Enclosure: NEMA 250 >.
 4. Operation of Pump: On or off.
 5. Transformer: Provide if required.
 6. Power Requirement: Per approved product.
 7. Settings: Start pump at 105 deg F and stop pump at 120 deg.
- B. Timers: Electric, for control of hot-water circulation pump.
 1. Type: Programmable, seven-day clock with manual override on-off switch.
 2. Enclosure: NEMA 250, , suitable for wall mounting.
 3. Operation of Pump: On or off.
 4. Transformer: Provide if required.
 5. Power Requirement: Per approved product.
 6. Programmable Sequence of Operation: Based on code and owner requirements.

PART 3 - EXECUTION

3.1 PUMP INSTALLATION

- A. Comply with HI 1.4.
- B. Install in-line, sealless centrifugal pumps with shaft horizontal unless otherwise indicated.
- C. Install horizontally mounted, in-line, close-coupled centrifugal pumps with shaft horizontal.
- D. Install continuous-thread hanger rods and spring hangers, or spring hangers with vertical-limit stop of size required to support pump weight.
 1. Comply with requirements for vibration isolation devices specified in Section 220548.13 "Vibration Controls for Plumbing Piping and Equipment." Fabricate brackets or supports as required.
 2. Comply with requirements for hangers and supports specified in Section 220529

"Hangers and Supports for Plumbing Piping and Equipment."

- E. Install thermostats in hot-water return piping.
- F. Install timers per owner requirements.

3.2 CONNECTIONS

- A. Comply with requirements for piping specified in Section 221116 "Domestic Water Piping." Drawings indicate general arrangement of piping, fittings, and specialties.
- B. Install piping adjacent to pumps to allow service and maintenance.
- C. Connect domestic water piping to pumps. Install suction and discharge piping equal to or greater than size of pump nozzles.
 - 1. Install flexible connectors adjacent to pumps in suction and discharge piping of the following pumps:
 - a. Horizontally mounted, in-line, close-coupled centrifugal pumps.
 - b. Comply with requirements for flexible connectors specified in Section 221116 "Domestic Water Piping."
 - 2. Install shutoff valve and strainer on suction side of each pump, and check, shutoff, and throttling valves on discharge side of each pump. Install valves same size as connected piping. Comply with requirements for valves specified in Section 220523.12 "Ball Valves for Plumbing Piping," Section 220523.13 "Butterfly Valves for Plumbing Piping," Section 220523.14 "Check Valves for Plumbing Piping," and Section 220523.15 "Gate Valves for Plumbing Piping," and comply with requirements for strainers specified in Section 221119 "Domestic Water Piping Specialties."
 - 3. Install pressure gage and snubber at suction of each pump and pressure gage and snubber at discharge of each pump. Install at integral pressure-gage tappings where provided or install pressure-gage connectors in suction and discharge piping around pumps. Comply with requirements for pressure gages and snubbers specified in Section 220519 "Meters and Gages for Plumbing Piping."
- D. Connect thermostats or timers to pumps that they control.

3.3 ADJUSTING

- A. Adjust domestic water pumps to function smoothly, and lubricate as recommended by manufacturer.
- B. Adjust initial temperature set points.
- C. Set field-adjustable switches and circuit-breaker trip ranges as indicated.

END OF SECTION 221123

SECTION 221123.13 - DOMESTIC-WATER PACKAGED BOOSTER PUMPS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 REQUIREMENT

- A. Pressure booster system requirements subject to engineer of record recommendations based on site specific conditions and requirements.

1.3 SUMMARY

- A. Section Includes:
 - 1. Multiplex, variable-speed booster pumps.
- B. Related Sections:
 - 1. Division 22 Section "Domestic Water Pumps" for domestic-water circulation pumps.
 - 2. Division 22 Section "Facility Indoor Potable-Water Storage Tanks" for separate hydropneumatic domestic-water tanks for multiplex booster pumps.

1.4 DEFINITIONS

- A. VFC: Variable-frequency controller(s).

1.5 PERFORMANCE REQUIREMENTS

- A. Seismic Performance: Booster pumps shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
 - 1. The term "withstand" means "the booster pump will remain in place without separation of any parts from the booster pump when subjected to the seismic forces specified."

1.6 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Shop Drawings: For booster pumps. Include plans, elevations, sections, details, and attachments to other work.
 - 1. Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 2. Wiring Diagrams: For power, signal, and control wiring.
- C. Seismic Qualification Certificates: For booster pumps, accessories, and components, from manufacturer.

1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
 2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
 3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.
- D. Operation and Maintenance Data: For booster pumps to include in emergency, operation, and maintenance manuals.

1.7 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. ASME Compliance: Comply with ASME B31.9 for piping.
- C. UL Compliance for Packaged Pumping Systems:
1. UL 508, "Industrial Control Equipment."
 2. UL 508A, "Industrial Control Panels."
 3. UL 778, "Motor-Operated Water Pumps."
 4. UL QCZJ, "Packaged Pumping Systems."
- D. Booster pumps shall be listed and labeled as packaged pumping systems by testing agency acceptable to authorities having jurisdiction.
- E. NSF Compliance: All wetted components shall comply with NSF Standard 61.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Retain protective coatings and flange's protective covers during storage.

1.9 COORDINATION

- A. Coordinate sizes and locations of concrete bases with actual equipment provided.

1.10 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

PART 2 - PRODUCTS

2.1 DUPLEX, VARIABLE-SPEED BOOSTER PUMPS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- B. Basis-of-Design Product: See drawings.
1. Advanced Mechanical Technologies Inc.
 2. Bell & Gossett Domestic Pump; ITT Corporation.
 3. Grundfos Pumps Corporation U.S.A.
- C. Description: Factory-assembled and -tested, fluid-handling system for domestic water, with pumps, piping, valves, specialties, and controls, and mounted on base.
- D. Pumps:
1. Type: Vertical, multistage as defined in HI 1.1-1.2 and HI 1.3 for in-line, multistage, separately coupled, overhung-impeller, centrifugal pump.
 2. Casing: Cast-iron or steel base and stainless-steel chamber.
 3. Impeller: Closed, stainless steel; statically and dynamically balanced and keyed to shaft.
 4. Shaft: Stainless steel.
 5. Seal: Mechanical.
 6. Bearing: Water-lubricated sleeve type.
- E. Valves:
1. Shutoff Valves NPS 2 and Smaller: two-piece, full-port ball valve, in each pump's suction and discharge piping.
- F. Dielectric Fittings: With insulating material isolating joined dissimilar metals.
- G. Control Panel: Factory installed and connected as an integral part of booster pump; automatic for multiple-pump, variable-speed operation, with load control and protection functions.
1. Control Logic: Solid-state system with transducers, programmable microprocessor, VFC, and other devices in controller. Install VFC for pump motors larger than 25 hp in separate panel; same type as motor control panel enclosure.
 2. Motor Controller: NEMA ICS 2, variable-frequency, solid-state type.
 - a. Control Voltage: 120-V ac, with integral control-power transformer.
 3. Enclosure: NEMA 250.
 4. Motor Overload Protection: Overload relay in each phase.
 5. Starting Devices: Hand-off-automatic selector switch for each pump in cover of control panel, plus pilot device for automatic control.
 - a. Duplex, Automatic, Alternating Starter: Switches lead pump to lag main pump and to two-pump operation.
 - b. Triplex, Sequence (Lead-Lag-Lag) Starter: Switches lead pump to one lag main pump and to three-pump operation.
 6. Pump Operation and Sequencing: Pressure-sensing method.
 7. VFC: Voltage-source, pulse-width, modulating-frequency converter for each pump.
 8. Manual Bypass: Magnetic contactor arranged to transfer to constant-speed operation upon VFC failure.
 9. Instrumentation: Suction and discharge pressure gages.
 10. Lights: Running light for each pump.
 11. Alarm Signal Device: Sounds alarm when backup pumps are operating.

12. Thermal-bleed cutoff.
13. Low-suction-pressure cutout.
14. High-suction-pressure cutout.
15. Low-discharge-pressure cutout.
16. High-discharge-pressure cutout.
17. Building Automation System Interface: See drawings.
 - a. On-off status of each pump.
 - b. Alarm status.

H. Base: Stainless steel.

I. Capacities and Characteristics: See drawings.

2.2 MOTORS

- A. Comply with NEMA designation, temperature rating, service factor, enclosure type, and efficiency requirements for motors.
1. Motor Sizes: Minimum size as indicated. If not indicated, large enough so driven load will not require motor to operate in service factor range above 1.0.
 2. Controllers, Electrical Devices, and Wiring: Comply with requirements for electrical devices and connections specified in NFPA 70.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine roughing-in for booster pumps to verify actual locations of piping connections before booster-pump installation.

3.2 INSTALLATION

- A. Equipment Mounting: Install booster pumps on concrete base with isolation device.
1. Install dowel rods to connect concrete base to concrete floor. Unless otherwise indicated, install dowel rods on 18-inch centers around the full perimeter of concrete base.
 2. For supported equipment, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete floor.
 3. Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
 4. Install anchor bolts to elevations required for proper attachment to supported equipment.
- B. Support connected domestic-water piping so weight of piping is not supported by booster pumps.

3.3 CONNECTIONS

- A. Comply with requirements for piping specified in Division 22 Section "Domestic Water Piping." Drawings indicate general arrangement of piping, fittings, and specialties.
- B. Connect domestic-water piping to booster pumps. Install suction and discharge pipe equal to or greater than size of system suction and discharge headers.
 - 1. Install shutoff valves on piping connections to booster-pump suction and discharge headers. Install ball valves same size as suction and discharge piping. Comply with requirements for general-duty valves specified in Division 22 Section "General-Duty Valves for Plumbing Piping."
 - 2. Install union, flanged, or grooved-joint connections on suction and discharge headers at connection to domestic-water piping. Comply with requirements for unions and flanges specified in Division 22 Section "Domestic Water Piping."
 - 3. Install valved bypass, same size as and between piping, at connections to booster-pump suction and discharge headers. Comply with requirements for domestic-water piping specified in Division 22 Section "Domestic Water Piping."
 - 4. Install flexible connectors, same size as piping, on piping connections to booster-pump suction and discharge headers. Comply with requirements for flexible connectors specified in Division 22 Section "Domestic Water Piping."
 - 5. Install piping adjacent to booster pumps to allow service and maintenance.

3.4 IDENTIFICATION

- A. Identify system components. Comply with requirements for identification specified in Division 22 Section "Identification for Plumbing Piping and Equipment."

3.5 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust components, assemblies, and equipment installations, including connections.
- B. Perform tests and inspections.
 - 1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.
- C. Tests and Inspections:
 - 1. Perform visual and mechanical inspection.
 - 2. Leak Test: After installation, charge booster pump and test for leaks. Repair leaks and retest until no leaks exist.
 - 3. Operational Test: After electrical circuitry has been energized, start booster pumps to confirm proper motor rotation and booster-pump operation.
 - 4. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- D. Pumps and controls will be considered defective if they do not pass tests and inspections.
- E. Prepare test and inspection reports.

3.6 STARTUP SERVICE

- A. Engage a factory-authorized service representative to perform startup service.
 - 1. Complete installation and startup checks according to manufacturer's written instructions.

3.7 ADJUSTING

- A. Adjust booster pumps to function smoothly, and lubricate as recommended by manufacturer.
- B. Adjust pressure set points.

3.8 DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain booster pumps.

END OF SECTION 221123.13

SECTION 221316 - SANITARY WASTE AND VENT PIPING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Pipe, tube, and fittings.
 - 2. Specialty pipe fittings.

1.3 PERFORMANCE REQUIREMENTS

- A. Components and installation shall be capable of withstanding the following minimum working pressure unless otherwise indicated:
 - 1. Soil, Waste, and Vent Piping: 10-foot head of water.
- B. Seismic Performance: Soil, waste, and vent piping and support and installation shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

1.5 INFORMATIONAL SUBMITTALS

- A. Seismic Qualification Certificates: For waste and vent piping, accessories, and components, from manufacturer.
 - 1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
 - 2. Detailed description of piping anchorage devices on which the certification is based and their installation requirements.
- B. Field quality-control reports.

1.6 QUALITY ASSURANCE

- A. Piping materials shall bear label, stamp, or other markings of specified testing agency.

1.7 PROJECT CONDITIONS

- A. Interruption of Existing Sanitary Waste Service: Do not interrupt service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary service according to requirements indicated:
 - 1. Notify Owner no fewer than five days in advance of proposed interruption of sanitary

- waste service.
2. Do not proceed with interruption of sanitary waste service without Owner's written permission.

PART 2 - PRODUCTS

2.1 PIPING MATERIALS

- A. Comply with requirements in "Piping Schedule" Article for applications of pipe, tube, fitting materials, and joining methods for specific services, service locations, and pipe sizes.

2.2 HUB-AND-SPIGOT, CAST-IRON SOIL PIPE AND FITTINGS

- A. Pipe and Fittings: ASTM A 74, Service class.
- B. Gaskets: ASTM C 564, rubber.
- C. Calking Materials: ASTM B 29, pure lead and oakum or hemp fiber.

2.3 HUBLESS, CAST-IRON SOIL PIPE AND FITTINGS

- A. Pipe and Fittings: ASTM A 888 or CISPI 301.
- B. Sovent Stack Fittings: ASME B16.45 or ASSE 1043, hubless, cast-iron aerator and deaerator drainage fittings.
- C. CISPI, Hubless-Piping Couplings:
 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Fernco Inc.
 - b. MIFAB, Inc.
 - c. Mission Rubber Company; a division of MCP Industries, Inc.
 - d. Tyler Pipe.
 2. Standards: ASTM C 1277 and CISPI 310.
 3. Description: Stainless-steel corrugated shield with stainless-steel bands and tightening devices; and ASTM C 564, rubber sleeve with integral, center pipe stop.
- D. Cast-Iron, Hubless-Piping Couplings:
 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. MG Piping Products Company.
 2. Standard: ASTM C 1277.
 3. Description: Two-piece ASTM A 48/A 48M, cast-iron housing; stainless-steel bolts and nuts; and ASTM C 564, rubber sleeve with integral, center pipe stop.

2.4 DUCTILE-IRON PIPE AND FITTINGS

A. Ductile-Iron, Mechanical-Joint Piping:

1. Ductile-Iron Pipe: AWWA C151/A21.51, with mechanical-joint bell and plain spigot end unless grooved or flanged ends are indicated.
2. Ductile-Iron Fittings: AWWA C110/A21.10, mechanical-joint, ductile- or gray-iron standard pattern or AWWA C153/A21.53, ductile-iron compact pattern.

2.5 COPPER TUBE AND FITTINGS

A. Copper DWV Tube: ASTM B 306, drainage tube, drawn temper.

B. Copper Drainage Fittings: ASME B16.23, cast copper or ASME B16.29, wrought copper, solder-joint fittings.

C. Hard Copper Tube: ASTM B 88, Type L and Type M, water tube, drawn temper.

D. Soft Copper Tube: ASTM B 88, Type L, water tube, annealed temper.

E. Copper Pressure Fittings:

1. Copper Fittings: ASME B16.18, cast-copper-alloy or ASME B16.22, wrought-copper, solder-joint fittings. Furnish wrought-copper fittings if indicated.
2. Copper Unions: MSS SP-123, copper-alloy, hexagonal-stock body with ball-and-socket, metal-to-metal seating surfaces, and solder-joint or threaded ends.

F. Solder: ASTM B 32, lead free with ASTM B 813, water-flushable flux.

2.6 PVC PIPE AND FITTINGS

A. Solid-Wall PVC Pipe: ASTM D 1784 or ASTM F 2618, drain, waste, and vent.

1. PVC Socket Fittings: ASTM D 1784 or ASTM F 2618, socket type, drain, waste, and vent patterns adhering to ASTM D 3311 DWV patterns.
 - a. ASTM F 493 Solvent Cements for CPVC Pipe and fittings.

B. Prohibited locations: Areas where discharge temperatures may reach 140F; all bakery drainage to point of building exit.

2.7 SPECIALTY PIPE FITTINGS

A. Dielectric Fittings:

1. General Requirements: Assembly of copper alloy and ferrous materials with separating nonconductive insulating material. Include end connections compatible with pipes to be joined.
2. Dielectric Unions:
 - a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- 1) Watts Regulator Co.; a division of Watts Water Technologies, Inc.
- 2) Wilkins; a Zurn company.

b. Description:

- 1) Standard: ASSE 1079.
- 2) Pressure Rating: 125 psig minimum at 180 deg F.
- 3) End Connections: Solder-joint copper alloy and threaded ferrous.

PART 3 - EXECUTION

3.1 PIPING INSTALLATION

- A. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated unless deviations to layout are approved on coordination drawings.
- B. Install piping in concealed locations unless otherwise indicated and except in equipment rooms and service areas.
- C. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.
- D. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.
- E. Install piping to permit valve servicing.
- F. Install piping at indicated slopes.
- G. Install piping free of sags and bends.
- H. Install fittings for changes in direction and branch connections.
- I. Install piping to allow application of insulation.
- J. Install seismic restraints on piping. Comply with requirements for seismic-restraint devices specified in Section 220548 "Vibration and Seismic Controls for Plumbing Piping and Equipment."
- K. Make changes in direction for soil and waste drainage and vent piping using appropriate branches, bends, and long-sweep bends. Sanitary tees and short-sweep 1/4 bends may be used on vertical stacks if change in direction of flow is from horizontal to vertical. Use long-turn, double Y-branch and 1/8-bend fittings if two fixtures are installed back to back or side by side with common drain pipe. Straight tees, elbows, and crosses may be used on vent lines. Do not change direction of flow more than 90 degrees. Use proper size of standard increasers and reducers if pipes of different sizes are connected. Reducing size of drainage piping in direction of flow is prohibited.
- L. Lay buried building drainage piping beginning at low point of each system. Install true to grades and alignment indicated, with unbroken continuity of invert. Place hub ends of piping upstream. Install required gaskets according to manufacturer's written instructions for use of lubricants,

cements, and other installation requirements. Maintain swab in piping and pull past each joint as completed.

M. Install soil and waste drainage and vent piping at the following minimum slopes unless otherwise indicated:

1. Building Sanitary Drain: 2 percent downward in direction of flow for piping NPS 3 and smaller; 1 percent downward in direction of flow for piping NPS 4 and larger.
2. Horizontal Sanitary Drainage Piping: 2 percent downward in direction of flow.
3. Vent Piping: 1 percent down toward vertical fixture vent or toward vent stack.

N. Install cast-iron soil piping according to CISPI's "Cast Iron Soil Pipe and Fittings Handbook," Chapter IV, "Installation of Cast Iron Soil Pipe and Fittings."

O. Install steel piping according to applicable plumbing code.

P. Install aboveground copper tubing according to CDA's "Copper Tube Handbook."

Q. Install engineered soil and waste drainage and vent piping systems as follows:

1. Combination Waste and Vent: Comply with standards of authorities having jurisdiction.
2. Solvent Drainage System: Comply with ASSE 1043 and solvent fitting manufacturer's written installation instructions.
3. Reduced-Size Venting: Comply with standards of authorities having jurisdiction.

R. Plumbing Specialties:

1. Install cleanouts at grade and extend to where building sanitary drains connect to building sanitary sewers in sanitary drainage gravity-flow piping. Install cleanout fitting with closure plug inside the building in sanitary drainage force-main piping. Comply with requirements for cleanouts specified in Section 221319 "Sanitary Waste Piping Specialties."
2. Install drains in sanitary drainage gravity-flow piping. Comply with requirements for drains specified in Section 221319 "Sanitary Waste Piping Specialties."

S. Do not enclose, cover, or put piping into operation until it is inspected and approved by authorities having jurisdiction.

T. Install sleeves for piping penetrations of walls, ceilings, and floors. Comply with requirements for sleeves specified in Section 220517 "Sleeves and Sleeve Seals for Plumbing Piping."

U. Install sleeve seals for piping penetrations of concrete walls and slabs. Comply with requirements for sleeve seals specified in Section 220517 "Sleeves and Sleeve Seals for Plumbing Piping."

V. Install escutcheons for piping penetrations of walls, ceilings, and floors. Comply with requirements for escutcheons specified in Section 220518 "Escutcheons for Plumbing Piping."

3.2 JOINT CONSTRUCTION

A. Join hub-and-spigot, cast-iron soil piping with gasket joints according to CISPI's "Cast Iron Soil Pipe and Fittings Handbook" for compression joints.

- B. Join hub-and-spigot, cast-iron soil piping with calked joints according to CISPI's "Cast Iron Soil Pipe and Fittings Handbook" for lead-and-oakum calked joints.
- C. Join hubless, cast-iron soil piping according to CISPI 310 and CISPI's "Cast Iron Soil Pipe and Fittings Handbook" for hubless-piping coupling joints.
- D. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
 - 1. Apply appropriate tape or thread compound to external pipe threads unless dry seal threading is specified.
 - 2. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds.
- E. Join copper tube and fittings with soldered joints according to ASTM B 828. Use ASTM B 813, water-flushable, lead-free flux and ASTM B 32, lead-free-alloy solder.
- F. Join CPVC pipe and fittings in accordance with

3.3 SPECIALTY PIPE FITTING INSTALLATION

- A. Transition Couplings:
 - 1. Install transition couplings at joints of piping with small differences in OD's.
 - 2. In Drainage Piping: Shielded, nonpressure transition couplings.
 - 3. In Aboveground Force Main Piping: Fitting-type transition couplings.
 - 4. In Underground Force Main Piping:
 - a. NPS 1-1/2 and Smaller: Fitting-type transition couplings.
 - b. NPS 2 and Larger: Pressure transition couplings.
- B. Dielectric Fittings:
 - 1. Install dielectric fittings in piping at connections of dissimilar metal piping and tubing.
 - 2. Dielectric Fittings for NPS 2 and Smaller: Use dielectric unions.
 - 3. Dielectric Fittings for NPS 2-1/2 to NPS 4: Use dielectric nipples.

3.4 VALVE INSTALLATION

- A. General valve installation requirements are specified in Section 220523.12 "Ball Valves for Plumbing Piping," Section 220523.13 "Butterfly Valves for Plumbing Piping," Section 220523.14 "Check Valves for Plumbing Piping," and Section 220523.15 "Gate Valves for Plumbing Piping."
- B. Shutoff Valves:
 - 1. Install shutoff valve on each sewage pump discharge.
 - 2. Install gate or full-port ball valve for piping NPS 2 and smaller.
 - 3. Install gate valve for piping NPS 2-1/2 and larger.
- C. Check Valves: Install swing check valve, between pump and shutoff valve, on each sewage pump discharge.

3.5 HANGER AND SUPPORT INSTALLATION

- A. Comply with requirements for seismic-restraint devices specified in Section 220548 "Vibration and Seismic Controls for Plumbing Piping and Equipment."
- B. Comply with requirements for pipe hanger and support devices and installation specified in Section 220529 "Hangers and Supports for Plumbing Piping and Equipment."
 - 1. Install carbon-steel pipe hangers for horizontal piping in noncorrosive environments.
 - 2. Install stainless-steel pipe hangers for horizontal piping in corrosive environments.
 - 3. Install carbon-steel pipe support clamps for vertical piping in noncorrosive environments.
 - 4. Install stainless-steel pipe support clamps for vertical piping in corrosive environments.
 - 5. Vertical Piping: MSS Type 8 or Type 42, clamps.
 - 6. Install individual, straight, horizontal piping runs:
 - a. 100 Feet and Less: MSS Type 1, adjustable, steel clevis hangers.
 - b. Longer Than 100 Feet: MSS Type 43, adjustable roller hangers.
 - c. Longer Than 100 Feet if Indicated: MSS Type 49, spring cushion rolls.
 - 7. Base of Vertical Piping: MSS Type 52, spring hangers.
- C. Support horizontal piping and tubing within 12 inches of each fitting, valve, and coupling.
- D. Support vertical piping and tubing at base and at each floor.
- E. Rod diameter may be reduced one size for double-rod hangers, with 3/8-inch minimum rods.
- F. Install hangers for cast-iron soil piping with the following maximum horizontal spacing and minimum rod diameters:
 - 1. NPS 1-1/2 and NPS 2: 60 inches with 3/8-inch rod.
 - 2. NPS 3: 60 inches with 1/2-inch rod.
 - 3. NPS 4 and NPS 5: 60 inches with 5/8-inch rod.
 - 4. Spacing for 10-foot lengths may be increased to 10 feet. Spacing for fittings is limited to 60 inches.
- G. Install supports for vertical cast-iron soil piping every 15 feet.
- H. Install hangers for steel piping with the following maximum horizontal spacing and minimum rod diameters:
 - 1. NPS 1-1/4: 84 inches with 3/8-inch rod.
 - 2. NPS 1-1/2: 108 inches with 3/8-inch rod.
 - 3. NPS 2: 10 feet with 3/8-inch rod.
 - 4. NPS 2-1/2: 11 feet with 1/2-inch rod.
 - 5. NPS 3: 12 feet with 1/2-inch rod.
 - 6. NPS 4 and NPS 5: 12 feet with 5/8-inch rod.
- I. Install supports for vertical steel piping every 15 feet.
- J. Install hangers for copper tubing with the following maximum horizontal spacing and minimum rod diameters:

1. NPS 1-1/4: 72 inches with 3/8-inch rod.
2. NPS 1-1/2 and NPS 2: 96 inches with 3/8-inch rod.
3. NPS 2-1/2: 108 inches with 1/2-inch rod.
4. NPS 3 and NPS 5: 10 feet with 1/2-inch rod.
5. NPS 6: 10 feet with 5/8-inch rod.
6. NPS 8: 10 feet with 3/4-inch rod.

K. Install supports for vertical copper tubing every 10 feet.

L. Support piping and tubing not listed above according to MSS SP-69 and manufacturer's written instructions.

3.6 CONNECTIONS

A. Drawings indicate general arrangement of piping, fittings, and specialties.

B. Connect soil and waste piping to exterior sanitary sewerage piping. Use transition fitting to join dissimilar piping materials.

C. Connect drainage and vent piping to the following:

1. Plumbing Fixtures: Connect drainage piping in sizes indicated, but not smaller than required by plumbing code.
2. Plumbing Fixtures and Equipment: Connect atmospheric vent piping in sizes indicated, but not smaller than required by authorities having jurisdiction.
3. Plumbing Specialties: Connect drainage and vent piping in sizes indicated, but not smaller than required by plumbing code.
4. Install test tees (wall cleanouts) in conductors near floor and floor cleanouts with cover flush with floor.
5. Comply with requirements for cleanouts and drains specified in Section 221319 "Sanitary Waste Piping Specialties."

D. Where installing piping adjacent to equipment, allow space for service and maintenance of equipment.

3.7 IDENTIFICATION

A. Identify exposed sanitary waste and vent piping. Comply with requirements for identification specified in Section 220553 "Identification for Plumbing Piping and Equipment."

3.8 FIELD QUALITY CONTROL

A. During installation, notify authorities having jurisdiction at least 24 hours before inspection must be made. Perform tests specified below in presence of authorities having jurisdiction.

1. Roughing-in Inspection: Arrange for inspection of piping before concealing or closing-in after roughing-in and before setting fixtures.
2. Final Inspection: Arrange for final inspection by authorities having jurisdiction to observe tests specified below and to ensure compliance with requirements.

B. Reinspection: If authorities having jurisdiction find that piping will not pass test or inspection, make required corrections and arrange for reinspection.

- C. Reports: Prepare inspection reports and have them signed by authorities having jurisdiction.
- D. Test sanitary drainage and vent piping according to procedures of authorities having jurisdiction or, in absence of published procedures, as follows:
 - 1. Test for leaks and defects in new piping and parts of existing piping that have been altered, extended, or repaired. If testing is performed in segments, submit separate report for each test, complete with diagram of portion of piping tested.
 - 2. Leave uncovered and unconcealed new, altered, extended, or replaced drainage and vent piping until it has been tested and approved. Expose work that was covered or concealed before it was tested.
 - 3. Roughing-in Plumbing Test Procedure: Test drainage and vent piping except outside leaders on completion of roughing-in. Close openings in piping system and fill with water to point of overflow, but not less than 10-foot head of water. From 15 minutes before inspection starts to completion of inspection, water level must not drop. Inspect joints for leaks.
 - 4. Finished Plumbing Test Procedure: After plumbing fixtures have been set and traps filled with water, test connections and prove they are gastight and watertight. Plug vent-stack openings on roof and building drains where they leave building. Introduce air into piping system equal to pressure of 1-inch wg. Use U-tube or manometer inserted in trap of water closet to measure this pressure. Air pressure must remain constant without introducing additional air throughout period of inspection. Inspect plumbing fixture connections for gas and water leaks.
 - 5. Repair leaks and defects with new materials and retest piping, or portion thereof, until satisfactory results are obtained.
 - 6. Prepare reports for tests and required corrective action.

3.9 CLEANING AND PROTECTION

- A. Clean interior of piping. Remove dirt and debris as work progresses.
- B. Protect drains during remainder of construction period to avoid clogging with dirt and debris and to prevent damage from traffic and construction work.
- C. Place plugs in ends of uncompleted piping at end of day and when work stops.

3.10 PIPING SCHEDULE

- A. Aboveground, soil and waste piping NPS 4 and smaller shall be any of the following:
 - 1. Service class, cast-iron soil pipe and fittings; gaskets; and gasketed joints.
 - 2. Hubless, cast-iron soil pipe and fittings and solvent stack fittings; CISPI hubless-piping couplings; and coupled joints.
 - 3. Copper DWV tube, copper drainage fittings, and soldered joints.
 - 4. Dissimilar Pipe-Material Couplings: Shielded, nonpressure transition couplings.
- B. Aboveground, vent piping NPS 4 and smaller shall be any of the following:
 - 1. Service class, cast-iron soil pipe and fittings; gaskets; and gasketed joints.
 - 2. Hubless, cast-iron soil pipe and fittings; CISPI hubless-piping couplings; and coupled joints.
 - 3. Copper DWV tube, copper drainage fittings, and soldered joints.

- a. Option for Vent Piping, NPS 2-1/2 and NPS 3-1/2: Hard copper tube, Type M; copper pressure fittings; and soldered joints.
4. Dissimilar Pipe-Material Couplings: Shielded, nonpressure transition couplings.

END OF SECTION 221316

SECTION 221319 - SANITARY WASTE PIPING SPECIALTIES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Cleanouts.
- 2. Floor drains.
- 3. Miscellaneous sanitary drainage piping specialties.

1.3 INFORMATIONAL SUBMITTALS

- A. Manufacturer Seismic Qualification Certification: Submit certification that accessories, and components will withstand seismic forces defined in Section 220548 "Vibration and Seismic Controls for Plumbing Piping and Equipment." Include the following:

- 1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
 - a. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified."
 - b. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."
- 2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
- 3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.

- B. Field quality-control reports.

1.4 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data: For drainage piping specialties to include in emergency, operation, and maintenance manuals.

1.5 QUALITY ASSURANCE

- A. Drainage piping specialties shall bear label, stamp, or other markings of specified testing agency.
- B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- C. Comply with NSF 14, "Plastics Piping Components and Related Materials," for plastic sanitary piping specialty components.

PART 2 - PRODUCTS

2.1 CLEANOUTS

A. Exposed Metal Cleanouts:

1. ASME A112.36.2M, Cast-Iron Cleanouts:

- a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- 1) Josam Company.
- 2) MIFAB, Inc.
- 3) Smith, Jay R. Mfg. Co.
- 4) Tyler Pipe.
- 5) Watts Drainage Products.
- 6) Zurn Plumbing Products Group.
- 7) Sioux Chief

B. Cast-Iron Wall Cleanouts:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- a. Josam Company; Josam Div.
- b. MIFAB, Inc.
- c. Smith, Jay R. Mfg. Co.
- d. Tyler Pipe; Wade Div.
- e. Watts Drainage Products.
- f. Zurn Plumbing Products Group; Specification Drainage Operation.
- g. Sioux Chief

2. Standard: ASME A112.36.2M. Include wall access.
3. Size: Same as connected drainage piping.
4. Body: Hub-and-spigot, cast-iron soil pipe T-branch Hubless, cast-iron soil pipe test tee as required to match connected piping.
5. Closure: Countersunk or raised-head, brass plug.
6. Closure Plug Size: Same as or not more than one size smaller than cleanout size.
7. Wall Access: Round, deep, chrome-plated bronze flat, chrome-plated brass or stainless- steel cover plate with screw.
8. Wall Access: Round, nickel-bronze, copper-alloy, or stainless-steel wall-installation frame and cover.

2.2 FLOOR DRAINS

A. Cast-Iron Floor Drains:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- a. ACO Drain
- b. Blucher

- c. Sioux Chief
- d. Zurn

- 2. Standard: EN 1253-1.
- 3. Pattern: Floor drain.
- 4. Body Material: 316L Stainless Steel.
- 5. Seepage Flange: Required.
- 6. Anchor Flange: Required.
- 7. Clamping Device: Required.
- 8. Outlet: Bottom.
- 9. Backwater Valve: Not required.
- 10. Coating on Interior and Exposed Exterior Surfaces: 316L Stainless Steel.
- 11. Sediment Bucket: Standard 316L Stainless Steel.
- 12. Top or Strainer Material: 316L Stainless Steel.
- 13. Top Shape: Square.
- 14. Inlet Fitting: 316L Stainless Steel, no hub connection.
- 15. Trap Material: Cast iron.
- 16. Trap Pattern: Standard 316L Stainless Steel foul air trap.
- 17. Trap Features: Trap-seal primer valve drain connection.

2.3 Miscellaneous sanitary drainage piping specialties.

A. Mechanical Trap Seal

- 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Mifab
 - b. RectorSeal
- 2. Mechanical trap seals for drains that are not subject to intermittent drainage to protect trap seals will be provided with drop-in style, self-sealing mechanical trap seal.

PART 3 – EXECUTION

3.1 INSTALLATION

- A. Install cleanouts in aboveground piping and building drain piping according to the following, unless otherwise indicated:
 - 1. Size same as drainage piping up to NPS 4. Use NPS 4 for larger drainage piping unless larger cleanout is indicated.
 - 2. Locate at each change in direction of piping greater than 45 degrees.
 - 3. Locate at minimum intervals of 50 feet for piping NPS 4 and smaller and 100 feet for larger piping.
 - 4. Locate at base of each vertical soil and waste stack.
- B. For floor cleanouts for piping below floors, install cleanout deck plates with top flush with finished floor.
- C. For cleanouts located in concealed piping, install cleanout wall access covers, of types indicated, with frame and cover flush with finished wall.
- D. Install floor drains at low points of surface areas to be drained. Set grates of drains flush with finished floor, unless otherwise indicated.

1. Position floor drains for easy access and maintenance.
 2. Install floor-drain flashing collar or flange so no leakage occurs between drain and adjoining flooring. Maintain integrity of waterproof membranes where penetrated.
 3. Install individual traps for floor drains connected to sanitary building drain, unless otherwise indicated.
- E. Assemble open drain fittings and install with top of hub above floor.
- F. Install floor-drain, trap-seal primer fittings on inlet to floor drains that require trap-seal primer connection.
1. Size: Same as floor drain inlet.
- G. Install sleeve flashing device with each riser and stack passing through floors with waterproof membrane.

3.2 CONNECTIONS

- A. Comply with requirements in Section 221316 "Sanitary Waste and Vent Piping" for piping installation requirements. Drawings indicate general arrangement of piping, fittings, and specialties.
- B. Install piping adjacent to equipment to allow service and maintenance.
- C. Ground equipment according to Section 260526 "Grounding and Bonding for Electrical Systems."
- D. Connect wiring according to Section 260519 "Low-Voltage Electrical Power Conductors and Cables."

3.3 FLASHING INSTALLATION

- A. Fabricate flashing from single piece unless large pans, sumps, or other drainage shapes are required. Join flashing according to the following if required:
1. Lead Sheets: Burn joints of lead sheets 6.0-lb/sq. ft., 0.0938-inch thickness or thicker. Solder joints of lead sheets 4.0-lb/sq. ft., 0.0625-inch thickness or thinner.
 2. Copper Sheets: Solder joints of copper sheets.
- B. Install sheet flashing on pipes, sleeves, and specialties passing through or embedded in floors and roofs with waterproof membrane.
1. Pipe Flashing: Sleeve type, matching pipe size, with minimum length of 10 inches, and skirt or flange extending at least 8 inches around pipe.
 2. Sleeve Flashing: Flat sheet, with skirt or flange extending at least 8 inches around sleeve.
 3. Embedded Specialty Flashing: Flat sheet, with skirt or flange extending at least 8 inches around specialty.
- C. Set flashing on floors and roofs in solid coating of bituminous cement.
- D. Secure flashing into sleeve and specialty clamping ring or device.
- E. Fabricate and install flashing and pans, sumps, and other drainage shapes.

3.4 FIELD QUALITY CONTROL

A. Perform tests and inspections and prepare test reports.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect field-assembled and their installation, including piping and electrical connections, and to assist in testing.

B. Tests and Inspections:

1. Leak Test: After installation, charge system and test for leaks. Repair leaks and retest until no leaks exist.
2. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

3.5 PROTECTION

A. Protect drains during remainder of construction period to avoid clogging with dirt or debris and to prevent damage from traffic or construction work.

B. Place plugs in ends of uncompleted piping at end of each day or when work stops.

END OF SECTION 221319

SECTION 221319.13 - SANITARY DRAINS**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Drains.

1.2 DEFINITIONS

- A. ABS: Acrylonitrile-butadiene styrene.
- B. FRP: Fiberglass-reinforced plastic.
- C. HDPE: High-density polyethylene.
- D. PE: Polyethylene.
- E. PP: Polypropylene.
- F. PVC: Polyvinyl chloride.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.

PART 2 – PRODUCTS**2.1 DRAIN ASSEMBLIES**

- A. Sanitary drains shall bear label, stamp, or other markings of specified testing agency.
- B. Comply with NSF 14 for plastic sanitary piping specialty components.

2.2 MANUFACTURERS

- A. Products to be as listed on drawings or from one of the following approved manufacturers:
 1. Mifab
 2. Zurn
 3. JR Smith
 4. Sioux Chief

3.1 INSTALLATION

- A. Install trench drains at low points of surface areas to be drained.
 1. Set grates of drains flush with finished surface, unless otherwise indicated.
 2. Install on support devices, so that top will be flush with adjacent surface.

- B. Install open drain fittings with top of hub 2 inches above floor.

3.2 CONNECTIONS

- A. Comply with requirements in Section 221316 "Sanitary Waste and Vent Piping" for piping installation requirements. Drawings indicate general arrangement of piping, fittings, and specialties.
- B. Comply with requirements in Section 221319 "Sanitary Waste Piping Specialties" for backwater valves, air admittance devices and miscellaneous sanitary drainage piping specialties.
- C. Comply with requirements in Section 221323 "Sanitary Waste Interceptors" for grease interceptors, grease-removal devices, oil interceptors, sand interceptors, and solid interceptors.
- D. Install piping adjacent to equipment to allow service and maintenance.
- E. Ground equipment according to Section 260526 "Grounding and Bonding for Electrical Systems."
- F. Connect wiring according to Section 260519 "Low-Voltage Electrical Power Conductors and Cables."

3.3 LABELING AND IDENTIFYING

- A. Distinguish among multiple units, inform operator of operational requirements, indicate safety and emergency precautions, and warn of hazards and improper operations, in addition to identifying unit. Nameplates and signs are specified in Section 220553 "Identification for Plumbing Piping and Equipment."

3.4 PROTECTION

- A. Protect drains during remainder of construction period to avoid clogging with dirt or debris and to prevent damage from traffic or construction work.
- B. Place plugs in ends of uncompleted piping at end of each day or when work stops.

END OF SECTION 221319.13

SECTION 221323 - GREASE WASTE INTERCEPTORS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Grease interceptors.
 - 2. Sediment interceptors.

1.3 DEFINITIONS

- A. FRP: Fiberglass-reinforced plastic.
- B. PDI: Plumbing and Drainage Institute
- C. GREASE INTERCEPTOR: Elliptical fiberglass (FRP) tank system designed with built-in inlet piping and baffle penetration that introduces wastewater in a tangential laminar flow to reduce disruption of collected grease and solids. Tank system is designed to capture and hold grease and solids to maximize waste retention and optimize Stokes Law separation. System backed by 30 year manufacturer warranty.
 - 1. GRAVITY GREASE INTERCEPTOR: Elliptical fiberglass (FRP) interceptor that is certified to meet IAPMO/ANSI Z1001-2007 and applicable sections of the latest editions of the Uniform Plumbing Code or the International Plumbing Code. Interceptor is designed to deliver 100 PPM non-emulsified free-floating oil and grease and 350 PPM Total Suspended Solids effluent quality based on inlet peak fixture flow.
 - 2. HYDROMECHANICAL OR PDI CERTIFIED GREASE INTERCEPTOR: Elliptical fiberglass (FRP) interceptor that is certified to meet PDI G-101 Standard rated up to 100 GPM flow.
 - 3. CSA CERTIFIED GREASE INTERCEPTOR: Elliptical fiberglass (FRP) interceptor that is certified to meet CSA B481 Standard rated at either 50 GPM or 100 GPM flow.

1.4 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

1.5 ACTION SUBMITTALS

- A. Product Data: Include materials of fabrication, dimensions, rated capacities, retention capacities, operating characteristics, size and location of each pipe connection, furnished specialties, and accessories.
- B. Shop Drawings: For each type and size of interceptor indicated.
 - 1. Include materials of construction, dimensions, rated capacities, retention capacities, location and size of each pipe connection, furnished specialties, and accessories.

1.6 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Interceptors, drawn to scale, on which the following items are shown and coordinated with each other, based on input from Installers of the items involved:
1. Grease Interceptors.
 2. Piping connections. Include size, location, and elevation of each.
 3. Interface with underground structures and utility services.

PART 2 - PRODUCTS

2.1 GREASE INTERCEPTORS (CSA certified)

- A. Basis of Design: ZURN Z250H grease interceptor shall be lifetime guaranteed and made in USA of seamless, rotationally-molded polyethylene. Interceptor shall be furnished for below grade installation. Interceptor shall be certified to ASME A112.14.3 (type C) and CSA B481.1, with built-in flow control, three inlet options and three outlet options. Interceptor flow rate shall be 100 GPM. Interceptor grease capacity shall be 200 lbs. @ 100 GPM. Size, discharge and installation requirements to be confirmed with local AHJ.
1. Engineered inlet and outlet diffusers with inspection ports removable to inspect / clean piping.
 2. Include with field adjustable riser/accessway system.
 3. Cover shall provide water/gas-tight seal and have minimum 16,000 lbs. load capacity

2.2 GREASE INTERCEPTORS (Gravity Type Alternate)

- A. Alternate provided where local AHJ requires gravity type interceptor.
- B. Basis of Design: Elliptical Fiberglass (FRP) grease interceptor construction, as supplied by Green Turtle Americas Ltd. or Green Turtle Technologies Ltd., with inlet piping and baffle penetration designed to introduce wastewater in a tangential laminar flow pattern, to be appropriately sized based on anticipated usage and flow rates to meet applicable sanitary sewer discharge limits, incl. municipal by-laws.
1. Include accessways, tanks, and piping and baffle openings to retain grease and to permit tangential laminar wastewater flow.
 2. Factory installed Schedule 40 PVC cement welded type socket ports, or straight pipe, fitted into interceptor walls for each pipe connection.
 3. Accessway Extension Collar:
 - a. Fiberglass risers (EC2), 36-inch optional alternate.
 4. Accessway Frames and Covers: Round cover with non slip cover finish, gasketed and non vented top design with "Proceptor" lettering cast into cover.
 - a. Cast Iron: AASHTO M306 Traffic load rated. 24 inch-diameter cover with 0.25" gasket. Two closed pickholes. Non Bolted or Bolted option. Weight 249 lbs. ASTM A48 CL35B. 36-inch optional alternate is acceptable to match fiberglass risers.
 - b. Fiberglass: Pedestrian loading 24" diameter bolted and gasketed.
 5. Watertight Flexible Caulking: Sikaflex 255 or Sikaflex 221 or approved alternate to provide watertight seal at extension collar joints.

2.3 FIBERGLASS ACCESSWAY RISERS

- A. Fiberglass accessway extensions: Fiberglass wound pipe.
 - 1. Length: From top of underground tank to underside of access frame at grade.
 - 2. Extension Sections: 0.25-inch minimum thickness and 24-inch as a single continuous piece, without joints unless approved by the manufacturer.
 - a. Sealant: Watertight Flexible Caulking, Sikaflex 255 or Sikaflex 221 or approved alternate to provide watertight seal at extension collar joining to tank on bottom and access frame at top.

PART 3 - EXECUTION

3.1 EARTHWORK

- A. Excavating, trenching, and backfilling are specified in "Earth Moving."

3.2 INSTALLATION

- A. Install fiberglass interceptors according to manufacturer's installation instructions.

3.3 CONNECTIONS

- A. Piping installation requirements are specified in Section 221316 "Sanitary Waste and Vent Piping." Drawings indicate general arrangement of piping, fittings, and specialties.
- B. Make piping connections between interceptors and piping systems.

3.4 IDENTIFICATION

- A. Identification materials and installation are specified in "Earth Moving." Arrange for installation of green warning tapes directly over piping and at outside edges of underground interceptors.
 - 1. Use warning tapes or detectable warning tape over ferrous piping.
 - 2. Use detectable warning tape over nonferrous piping and over edges of underground structures.

END OF SECTION 221323

SECTION 223300 - ELECTRIC DOMESTIC WATER HEATERS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes the following electric water heaters:
 - 1. Commercial, storage electric water heaters.
 - 2. Compression tanks.
 - 3. Water heater accessories.

1.3 SUBMITTALS

- A. Product Data: For each type and size of water heater indicated. Include rated capacities, operating characteristics, furnished specialties, and accessories.
- B. Shop Drawings: Diagram power, signal, and control wiring.
- C. Product Certificates: For each type of commercial electric water heater, signed by product manufacturer.
- D. Manufacturer Seismic Qualification Certification: Submit certification that commercial water heaters, accessories, and components will withstand seismic forces defined in Division 22 Section "Vibration and Seismic Controls for Plumbing Piping and Equipment." Include the following:
 - 1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
 - a. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified."
 - b. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."
 - 2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
 - 3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.
- E. Source quality-control test reports.
- F. Field quality-control test reports.
- G. Operation and Maintenance Data: For electric water heaters to include in emergency, operation, and maintenance manuals.
- H. Warranty: Special warranty specified in this Section.

1.4 QUALITY ASSURANCE

- A. Source Limitations: Obtain same type of electric water heaters through one source from a single manufacturer.
- B. Product Options: Drawings indicate size, profiles, and dimensional requirements of electric water heaters and are based on the specific system indicated. Refer to Division 01 Section "Product Requirements."
- C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- D. ASME Compliance: Where indicated, fabricate and label commercial water heater storage tanks to comply with ASME Boiler and Pressure Vessel Code: Section VIII, Division 1.
- E. Comply with NSF 61, "Drinking Water System Components - Health Effects; Sections 1 through 9," for all components that will be in contact with potable water.

1.5 COORDINATION

- A. Coordinate size and location of concrete bases with Architectural and Structural Drawings.

1.6 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.
- B. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of electric water heaters that fail in materials or workmanship within specified warranty period.
 - 1. Failures include, but are not limited to, the following:
 - a. Structural failures including storage tank and supports.
 - b. Faulty operation of controls.
 - c. Deterioration of metals, metal finishes, and other materials beyond normal use.
 - 2. Warranty Period(s): From date of Substantial Completion:
 - a. Commercial Electric Water Heaters:
 - 1) Storage Tank: Five years.
 - 2) Controls and Other Components: Five years.
 - b. Compression Tanks: One year(s).

PART 2 - PRODUCTS

2.1 COMMERCIAL ELECTRIC WATER HEATERS

- A. Commercial, Storage Electric Water Heaters: Comply with UL 1453 requirements for storage-tank-type water heaters.
1. Manufacturers:
 - a. Lochinvar Corporation.
 - b. Rheem Water Heater Div.; Rheem Manufacturing Company.
 - c. Smith, A. O. Water Products Company.
 2. Storage-Tank Construction: Steel.
 - a. Tappings: ASME B1.20.1 pipe thread.
 - b. Pressure Rating: 150 psig.
 - c. Interior Finish: Comply with NSF 61 barrier materials for potable-water tank linings, including extending lining material into tappings.
 3. Factory-Installed Storage-Tank Appurtenances:
 - a. Anode Rod: Replaceable magnesium.
 - b. Drain Valve: Corrosion-resistant metal complying with ASSE 1005. Insulation: Comply with ASHRAE/IESNA 90.1. Jacket: Rectangular shaped, with stainless-steel front panel, unless otherwise indicated. Heating Elements: Electric, screw-in or bolt-on immersion type arranged in multiples of two or three. Temperature Control: Adjustable thermostat, to setting of at least 180 deg F.
 - c. Safety Controls: High-temperature-limit and low-water cutoff devices or systems.
 - d. Relief Valve: ASME rated and stamped and complying with ASME PTC 25.3, combination temperature and pressure relief valve. Include relieving capacity at least as great as heat input, and include pressure setting less than water heater working-pressure rating. Select relief valve with sensing element that extends into storage tank.
 - e. Gages: Combination temperature and pressure type or separate thermometer and pressure gage.
 4. Capacity and Characteristics:
 - a. Capacity: See schedule on drawings.
 - b. Recovery: See schedule on drawings.
 - c. Temperature Setting: 140 deg F
 - d. Electrical Characteristics:
 - 1) Power Demand: See schedule on drawings.
 - 2) Volts: See schedule on drawings.
 - 3) Phases: See schedule on drawings.
 - 4) Hertz: 60.
 - 5) Full-Load Amperes: See schedule on drawings.
 - 6) Minimum Circuit Ampacity: See schedule on drawings.
 - 7) Maximum Overcurrent Protection: See schedule on drawings.

2.2 COMPRESSION TANKS

- A. Description: Steel pressure-rated tank constructed with welded joints and factory-installed butyl-rubber diaphragm. Include air precharge to minimum system-operating pressure at tank.
1. Manufacturers:
 - a. AMTROL Inc.
 - b. Watts Regulator Co.
 - c. Wessels Co.
 2. Construction:
 - a. Tappings: Factory-fabricated steel, welded to tank before testing and labeling. Include ASME B1.20.1, pipe thread.
 - b. Interior Finish: Comply with NSF 61 barrier materials for potable-water tank linings, including extending finish into and through tank fittings and outlets.
 - c. Air-Charging Valve: Factory installed.
 3. Capacity and Characteristics:
 - a. Working-Pressure Rating: 150 psig.
 - b. Capacity Acceptable: 8 gal. minimum.
 - c. Air Precharge Pressure: Matched to system pressure.

2.3 WATER HEATER ACCESSORIES

- A. Combination Temperature and Pressure Relief Valves: ASME rated and stamped and complying with ASME PTC 25.3. Include relieving capacity at least as great as heat input, and include pressure setting less than water heater working-pressure rating. Select relief valves with sensing element that extends into storage tank.
- B. Pressure Relief Valves: ASME rated and stamped and complying with ASME PTC 25.3. Include pressure setting less than water heater working-pressure rating.
- C. Purge/Dump Valves: How water heater to be provided with a hose- thread T-port style purge/dump valve directly downstream the water heater on the domestic hot water piping for use during maintenance.
- D. Water Heater Stand and Drain-Pan Units: High-density-polyethylene-plastic, 18-inch- high, enclosed-base stand complying with IAPMO PS 103 and IAS No. 2. Include integral or separate drain pan with raised edge and NPS 1 drain outlet with ASME B1.20.1 pipe thread.
- E. Water Heater Mounting Brackets: Water heater manufacturer's factory-fabricated steel bracket for wall mounting and capable of supporting water heater and water.
- F. Drain Pans: Corrosion-resistant metal with raised edge. Include dimensions not less than base of water heater and include drain outlet not less than NPS 3/4.
- G. Piping Manifold Kits: Water heater manufacturer's factory-fabricated inlet and outlet piping arrangement for multiple-unit installation. Include piping and valves for field assembly that are capable of isolating each water heater and of providing balanced flow through each water heater.

- H. Piping-Type Heat Traps: Field-fabricated piping arrangement according to ASHRAE/IESNA 90.1 or ASHRAE 90.2.
- I. Water Regulators: ASSE 1003, water-pressure reducing valve. Set at 25-psig- maximum outlet pressure, unless otherwise indicated.
- J. Shock Absorbers: ASSE 1010 or PDI WH 201, Size A water hammer arrester.

2.4 SOURCE QUALITY CONTROL

- A. Test and inspect water heater storage tanks, specified to be ASME-code construction, according to ASME Boiler and Pressure Vessel Code.
- B. Hydrostatically test commercial water heater storage tanks before shipment to minimum of one and one-half times pressure rating.
- C. Prepare test reports.

PART 3 - EXECUTION

3.1 WATER HEATER INSTALLATION

- A. Install commercial water heater on elevated shelf structurally suitable for carrying at least 1,000 lb.
- B. Install water heaters level and plumb, according to layout drawings, original design, and referenced standards. Maintain manufacturer's recommended clearances. Arrange units so controls and devices needing service are accessible.
- C. Install seismic restraints for commercial water heaters. Anchor to substrate.
- D. Install combination temperature and pressure relief valves in top portion of storage tanks. Use relief valves with sensing elements that extend into tanks. Extend commercial-water-heater relief-valve outlet, with drain piping same as domestic water piping in continuous downward pitch, and discharge by positive air gap onto closest floor drain.
- E. Install combination temperature and pressure relief valves in water piping for water heaters without storage. Extend commercial-water-heater relief-valve outlet, with drain piping same as domestic water piping in continuous downward pitch, and discharge by positive air gap onto closest floor drain.
- F. Install water-heater drain piping as indirect waste to spill by positive air gap into open drains or over floor drains. Install hose-end drain valves at low points in water piping for water heaters that do not have tank drains. Refer to Division 22 Section "Domestic Water Piping Specialties" for hose-end drain valves.
- G. Install thermometer on outlet piping of water heaters. Refer to Division 22 Section "Meters and Gages for Plumbing Piping" for thermometers.

- H. Install thermometers on inlet and outlet piping of household, collector-to-tank, solar-electric water heaters. Refer to Division 22 Section "Meters and Gages for Plumbing Piping" for thermometers.
- I. Install pressure gage(s) on inlet and outlet of commercial electric water- heater piping. Refer to Division 22 Section "Meters and Gages for Plumbing Piping" for pressure gages.
- J. Assemble and install inlet and outlet piping manifold kits for multiple water heaters. Fabricate, modify, or arrange manifolds for balanced water flow through each water heater. Include shutoff valve, thermometer in each water heater inlet and outlet, and throttling valve in each water heater outlet. Refer to Division 22 Section "General-Duty Valves for Plumbing Piping" for general-duty valves and to Division 22 Section "Meters and Gages for Plumbing Piping" for thermometers.
- K. Install water regulator, with integral bypass relief valve, in booster-heater inlet piping and water hammer arrester in booster-heater outlet piping.
- L. Install piping-type heat traps on inlet and outlet piping of water heater storage tanks without integral or fitting-type heat traps.
- M. Fill water heaters with water.
- N. Charge compression tanks with air.

3.2 CONNECTIONS

- A. Piping installation requirements are specified in other Division 22 Sections. Drawings indicate general arrangement of piping, fittings, and specialties.
- B. Install piping adjacent to water heaters to allow service and maintenance. Arrange piping for easy removal of water heaters.
- C. Ground equipment according to Division 26 Section "Grounding and Bonding for Electrical Systems."
- D. Connect wiring according to Division 26 Section "Low-Voltage Electrical Power Conductors and Cables."

3.3 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust field-assembled components and equipment installation, including connections, and to assist in field testing. Report results in writing.
- B. Perform the following field tests and inspections and prepare test reports:
 - 1. Leak Test: After installation, test for leaks. Repair leaks and retest until no leaks exist.
 - 2. Operational Test: After electrical circuitry has been energized, confirm proper operation.
 - 3. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

- C. Remove and replace water heaters that do not pass tests and inspections and retest as specified above.

3.4 DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain commercial electric water heaters. Refer to Division 01 Section "Demonstration and Training."

END OF SECTION 223300

SECTION 224213.13 - COMMERCIAL WATER CLOSETS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Water closets.
 - 2. Flushometer valves.
 - 3. Toilet seats.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
 - 1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for water closets.
 - 2. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.
- B. Shop Drawings: Include diagrams for power, signal, and control wiring.

1.4 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data: For flushometer valves and electronic sensors to include in operation and maintenance manuals.

1.5 MAINTENANCE MATERIAL SUBMITTALS

- A. Furnish extra materials that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Flushometer-Valve Repair Kits: Equal to 10 percent of amount of each type installed, but no fewer than one of each type.

PART 2 - PRODUCTS**2.1 WALL-MOUNTED WATER CLOSETS**

- A. Water Closets: Wall mounted, top spud, accessible.

1. Required Product: See schedule on drawings.
2. Bowl:
 - a. Standards: ASME A112.19.2/CSA B45.1 and ASME A112.19.5.
 - b. Material: Vitreous china.
 - c. Type: Siphon jet.
 - d. Style: Flushometer valve.
 - e. Height: Standard.
 - f. Rim Contour: Elongated.
 - g. Water Consumption: 1.28 gal. per flush.
3. Support:
 - a. Standard: ASME A112.6.1M.
 - b. Description: Waste-fitting assembly, as required to match drainage piping material and arrangement with faceplates, couplings gaskets, and feet; bolts and hardware matching fixture. Include additional extension coupling, faceplate, and feet for installation in wide pipe space.
 - c. Water-Closet Mounting Height: See Architectural drawings.

2.2 FLUSHOMETER VALVES

- A. Solenoid-Actuator, Diaphragm Flushometer Valves:
 1. Required Product: See schedule on drawings.
 2. Standard: ASSE 1037.
 3. Minimum Pressure Rating: 125 psig.
 4. Features: Include integral check stop and backflow-prevention device.
 5. Material: Brass body with corrosion-resistant components.
 6. Exposed Flushometer-Valve Finish: Chrome plated.
 7. Panel Finish: Stainless steel.
 8. Style: Concealed.
 9. Actuator: Solenoid complying with UL 1951, and listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
 10. Trip Mechanism: Battery powered electronic sensor complying with UL 1951, and listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application and manual override button.
 11. Consumption: 1.28 gal. per flush.
 12. Minimum Inlet: NPS 1.
 13. Minimum Outlet: NPS 1-1/4.

2.3 TOILET SEATS

- A. Toilet Seats: The toilet seats should be from the same manufacturers of the toilet.
 1. Basis-of-Design Product: Subject to compliance with requirements, provide product indicated on Drawings or comparable product by one of the following:
 - a. American Standard America.
 - b. Bemis Manufacturing Company.
 - c. Church Seats.

d. Kohler Co.

2. Standard: IAPMO/ANSI Z124.5.
3. Material: Plastic.
4. Type: Commercial (Heavy duty).
5. Shape: Elongated rim, open front.
6. Hinge: Self-sustaining, check.
7. Hinge Material: Noncorroding metal.
8. Seat Cover: Not required.
9. Color: White.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine roughing-in of water supply and sanitary drainage and vent piping systems to verify actual locations of piping connections before water-closet installation.
- B. Examine walls and floors for suitable conditions where water closets will be installed.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Water-Closet Installation:
 1. Install level and plumb according to roughing-in drawings.
 2. Install floor-mounted water closets on bowl-to-drain connecting fitting attachments to piping or building substrate.
 3. Install accessible, wall-mounted water closets at mounting height for handicapped/elderly, according to ICC/ANSI A117.1.
- B. Support Installation:
 1. Install supports, affixed to building substrate, for floor-mounted, back-outlet water closets.
 2. Use carrier supports with waste-fitting assembly and seal.
 3. Install floor-mounted, back-outlet water closets attached to building floor substrate, onto waste-fitting seals; and attach to support.
 4. Install wall-mounted, back-outlet water-closet supports with waste-fitting assembly and waste-fitting seals; and affix to building substrate.
- C. Flushometer-Valve Installation:
 1. Install flushometer-valve, water-supply fitting on each supply to each water closet.
 2. Attach supply piping to supports or substrate within pipe spaces behind fixtures.
 3. Install actuators in locations that are easy for people with disabilities to reach.
- D. Install toilet seats on water closets.
- E. Joint Sealing:

1. Seal joints between water closets and walls and floors using sanitary-type, one-part, mildew-resistant silicone sealant.
2. Match sealant color to water-closet color.
3. Comply with sealant requirements specified in Section 079200 "Joint Sealants."

3.3 CONNECTIONS

- A. Connect water closets with water supplies and soil, waste, and vent piping. Use size fittings required to match water closets.
- B. Comply with water piping requirements specified in Section 221116 "Domestic Water Piping."
- C. Comply with soil and waste piping requirements specified in Section 221316 "Sanitary Waste and Vent Piping."
- D. Where installing piping adjacent to water closets, allow space for service and maintenance.

3.4 ADJUSTING

- A. Operate and adjust water closets and controls. Replace damaged and malfunctioning water closets, fittings, and controls.
- B. Adjust water pressure at flushometer valves to produce proper flow.

3.5 CLEANING AND PROTECTION

- A. Clean water closets and fittings with manufacturers' recommended cleaning methods and materials.
- B. Install protective covering for installed water closets and fittings.
- C. Do not allow use of water closets for temporary facilities unless approved in writing by Owner.

END OF SECTION 224213.13

SECTION 224216.13 - COMMERCIAL LAVATORIES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

- 1. Lavatories.
- 2. Faucets.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
 - 1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for lavatories.
 - 2. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.
- B. Shop Drawings: Include diagrams for power, signal, and control wiring of automatic faucets.

1.4 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Counter cutout templates for mounting of counter-mounted lavatories.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data: For lavatories and faucets to include in operation and maintenance manuals.
 - 1. In addition to items specified in Section 017823 "Operation and Maintenance Data," include the following:
 - a. Servicing and adjustments of automatic faucets.

1.6 MAINTENANCE MATERIAL SUBMITTALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Faucet Washers and O-Rings: Equal to 10 percent of amount of each type and size installed.
 - 2. Faucet Cartridges and O-Rings: Equal to 5 percent of amount of each type and size installed.

PART 2 - PRODUCTS

2.1 VITREOUS-CHINA, WALL-MOUNTED LAVATORIES

A. Lavatory: Vitreous china, wall mounted.

1. Required Product: See schedule on drawings.
2. Fixture:
 - a. Standard: ASME A112.19.2/CSA B45.1.
 - b. Type: For wall hanging.
 - c. Faucet-Hole Punching: One hole.
 - d. Faucet-Hole Location: Top.
 - e. Color: White.
 - f. Mounting Material: Chair carrier.
3. Support: ASME A112.6.1M, Type II, concealed-arm lavatory carrier. Include rectangular, steel uprights.

2.2 SOLID-BRASS, AUTOMATIC METERING FAUCETS

A. NSF Standard: Comply with NSF/ANSI 61, "Drinking Water System Components - Health Effects," for faucet materials that will be in contact with potable water.

B. Lavatory Faucets: Battery Powered sensor operated, single-control mixing, solid-brass valve.

1. Required Product: See schedule on drawings.:
2. Standard: ASME A112.18.1/CSA B125.1.
3. General: Include hot- and cold-water indicators; coordinate faucet inlets with supplies and fixture hole punchings; coordinate outlet with spout and fixture receptor.
4. Body Type: Single hole.
5. Body Material: Commercial, solid brass.
6. Finish: Polished chrome plate.
7. Maximum Flow Rate: 0.5 gpm.
8. Spout: Rigid type.
9. Spout Outlet: Aerator.
10. Drain: Grid.

2.3 SUPPLY FITTINGS

A. NSF Standard: Comply with NSF/ANSI 61, "Drinking Water System Components - Health Effects," for supply-fitting materials that will be in contact with potable water.

B. Standard: ASME A112.18.1/CSA B125.1.

C. Supply Piping: Chrome-plated-brass pipe or chrome-plated copper tube matching water-supply piping size. Include chrome-plated-brass or stainless-steel wall flange.

- D. Supply Stops: Chrome-plated-brass, one-quarter-turn, ball-type or compression valve with inlet connection matching supply piping.
- E. Operation: Loose key.
- F. Risers:
 - 1. NPS 1/2.
 - 2. Chrome-plated, rigid-copper-pipe and brass straight or offset tailpieces riser.

2.4 WASTE FITTINGS

- A. Standard: ASME A112.18.2/CSA B125.2.
- B. Drain: Grid type with NPS 1-1/4 offset and straight tailpiece.
- C. Trap:
 - 1. Size: NPS 1-1/2 by NPS 1-1/4.
 - 2. Material: Chrome-plated, two-piece, cast-brass trap and swivel elbow with 0.032-inch-thick brass tube to wall; and chrome-plated, brass wall flange.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine roughing-in of water supply and sanitary drainage and vent piping systems to verify actual locations of piping connections before lavatory installation.
- B. Examine counters and walls for suitable conditions where lavatories will be installed.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Install lavatories level and plumb according to roughing-in drawings.
- B. Install supports, affixed to building substrate, for wall-mounted lavatories.
- C. Install accessible wall-mounted lavatories at handicapped/elderly mounting height for people with disabilities or the elderly, according to ICC/ANSI A117.1.
- D. Install wall flanges or escutcheons at piping wall penetrations in exposed, finished locations. Use deep-pattern escutcheons if required to conceal protruding fittings. Comply with escutcheon requirements specified in Section 220518 "Escutcheons for Plumbing Piping."
- E. Seal joints between lavatories, counters, and walls using sanitary-type, one-part, mildew-resistant silicone sealant. Match sealant color to fixture color. Comply with sealant requirements specified in Section 079200 "Joint Sealants."

- F. Install protective shielding pipe covers and enclosures on exposed supplies and waste piping of accessible lavatories. Comply with requirements in Section 220719 "Plumbing Piping Insulation."

3.3 CONNECTIONS

- A. Connect fixtures with water supplies, stops, and risers, and with traps, soil, waste, and vent piping. Use size fittings required to match fixtures.
- B. Comply with water piping requirements specified in Section 221116 "Domestic Water Piping."
- C. Comply with soil and waste piping requirements specified in Section 221316 "Sanitary Waste and Vent Piping."

3.4 ADJUSTING

- A. Operate and adjust lavatories and controls. Replace damaged and malfunctioning lavatories, fittings, and controls.
- B. Adjust water pressure at faucets to produce proper flow.

3.5 CLEANING AND PROTECTION

- A. After completing installation of lavatories, inspect and repair damaged finishes.
- B. Clean lavatories, faucets, and other fittings with manufacturers' recommended cleaning methods and materials.
- C. Provide protective covering for installed lavatories and fittings.
- D. Do not allow use of lavatories for temporary facilities unless approved in writing by Owner.

END OF SECTION 224216.13

SECTION 224216.16 - COMMERCIAL SINKS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Kitchen sinks.
 - 2. Sink faucets.
 - 3. Supply fittings.
 - 4. Waste fittings.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
 - 1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for sinks.
 - 2. Include rated capacities, operating characteristics and furnished specialties and accessories.

1.4 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Counter cutout templates for mounting of counter-mounted lavatories.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For sinks to include in maintenance manuals.

1.6 MAINTENANCE MATERIAL SUBMITTALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Faucet Cartridges and O-Rings: Equal to 5 percent of amount of each type and size installed.

PART 2 - PRODUCTS**2.1 SINKS**

- A. Sinks three-bowl, stainless steel.
 - 1. Stainless-Steel utility Sinks:
 - a. Required Product: See schedule on sheet.
 - a. Standard: ASME A112.19.3/CSA B45.4 for stainless-steel sinks.
 - 1) Drain: 3-1/2-inch grid to 2-inch valve controlled outlet.
 - a) Location: Center of each compartment.

B. Sinks One bowl, under mounted, stainless steel.

1. Stainless-Steel Kitchen Sinks:

a. Required Product: See schedule on drawings.

b. Standard: ASME A112.19.3/CSA B45.4 for stainless-steel kitchen sinks.

1) Drain: 3-1/2-inch grid.

a) Location: Near back of bowl.

2.2 SINK FAUCETS

A. NSF Standard: Comply with NSF/ANSI 61, "Drinking Water System Components - Health Effects," for faucet-spout materials that will be in contact with potable water.

B. Sink Faucets: Manual type, two handle valve.

1. For 3 compartment sink:

a. Basis of Design: ELKAY SS835LR4

2. For Kitchen Sink:

a. Basis of Design: ADVANCE TABCO FC-WM-2721

3. For Sink Faucets:

a. Basis of Design: KROWNE 17-109WL

2.3 SUPPLY FITTINGS

A. NSF Standard: Comply with NSF/ANSI 61, "Drinking Water System Components - Health Effects," for supply-fitting materials that will be in contact with potable water.

B. Standard: ASME A112.18.1/CSA B125.1.

C. Supply Piping: Chrome-plated brass pipe or chrome-plated copper tube matching water-supply piping size. Include chrome-plated brass or stainless-steel wall flange.

D. Supply Stops: Chrome-plated brass, one-quarter-turn, ball-type or compression valve with inlet connection matching supply piping.

E. Operation: Loose key

F. Risers:

1. NPS 3/8

2. Chrome-plated, soft-copper flexible tube or ASME A112.18.6, braided or corrugated stainless-steel flexible hose.

2.4 WASTE FITTINGS

A. Standard: ASME A112.18.2/CSA B125.2.

B. Drain: Grid type with NPS 1-1/2 offset and straight tailpiece.

C. Trap:

1. Size: NPS 1-1/2.
2. Material: Chrome-plated, two-piece, cast-brass trap and swivel elbow with 0.032-inch-thick brass tube to wall

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine roughing-in of water supply and sanitary drainage and vent piping systems to verify actual locations of piping connections before sink installation.
- B. Examine walls, floors, and counters for suitable conditions where sinks will be installed.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Install sinks level and plumb according to roughing-in drawings.
- B. Install supports, affixed to building substrate, for wall-hung sinks.
- C. Install water-supply piping with stop on each supply to each sink faucet.
 1. Exception: Use ball or gate valves if supply stops are not specified with sink. Comply with valve requirements specified in Section 220523.12 "Ball Valves for Plumbing Piping" and Section 220523.15 "Gate Valves for Plumbing Piping."
 2. Install stops in locations where they can be easily reached for operation.
- D. Install wall flanges or escutcheons at piping wall penetrations in exposed, finished locations. Use deep-pattern escutcheons if required to conceal protruding fittings. Comply with escutcheon requirements specified in Section 220518 "Escutcheons for Plumbing Piping."
- E. Seal joints between sinks and counters, floors, and walls using sanitary-type, one-part, mildew-resistant silicone sealant. Match sealant color to fixture color. Comply with sealant requirements specified in Section 079200 "Joint Sealants."
- F. Install protective shielding pipe covers and enclosures on exposed supplies and waste piping of accessible sinks. Comply with requirements in Section 220719 "Plumbing Piping Insulation."

3.3 CONNECTIONS

- A. Connect sinks with water supplies, stops, and risers, and with traps, soil, waste, and vent piping. Use size fittings required to match fixtures.
- B. Comply with water piping requirements specified in Section 221116 "Domestic Water Piping."
- C. Comply with soil and waste piping requirements specified in Section 221316 "Sanitary Waste and Vent Piping."

3.4 ADJUSTING

- A. Operate and adjust sinks and controls. Replace damaged and malfunctioning sinks, fittings, and controls.
- B. Adjust water pressure at faucets to produce proper flow.

3.5 CLEANING AND PROTECTION

- A. After completing installation of sinks, inspect and repair damaged finishes.
- B. Clean sinks, faucets, and other fittings with manufacturers' recommended cleaning methods and materials.
- C. Provide protective covering for installed sinks and fittings.
- D. Do not allow use of sinks for temporary facilities unless approved in writing by Owner.

END OF SECTION 224216.16

SECTION 224716 – DRINKING FOUNTAINS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes drinking fountains and related components.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of drinking fountain.
 - 1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.
 - 2. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.
- B. Shop Drawings: Include diagrams for power, signal, and control wiring.

1.4 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For drinking fountains to include in maintenance manuals.

1.5 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

PART 2 - PRODUCTS**2.1 DRINKING FOUNTAINS**

- A. Drinking Fountains: Wall Mounted (Wheelchair Accessible)
 - 1. Required Product: Basis of Design: ELKAY EZSTLDDWSVRSK.
 - 2. Cabinet: Bi-level with two attached cabinets, all stainless steel.
 - 3. Bubbler: One, with adjustable stream regulator, located on each cabinet deck.
 - 4. Control: ADA Compliant.
 - 5. Drain: Grid with NPS 1-1/4 tailpiece.
 - 6. Supply: NPS 3/8 with shutoff valve.
 - 7. Waste Fitting: ASME A112.18.2/CSA B125.2, NPS 1-1/4 brass P-trap.
 - 8. Filter: One or more water filters complying with NSF 42 and NSF 53 for cyst and lead reduction to below EPA standards; with capacity sized for unit peak flow rate.
 - 9. Support: ASME A112.6.1M, Type I carrier.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine roughing-in for water-supply and sanitary drainage and vent piping systems to verify actual locations of piping connections before fixture installation.
- B. Examine walls and floors for suitable conditions where fixtures will be installed.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Install fixtures level and plumb according to roughing-in drawings. For fixtures indicated for children, install at height required by authorities having jurisdiction.
- B. Install off-the-floor carrier supports, affixed to building substrate, for wall-mounted fixtures.
- C. Install mounting frames, affixed to building construction, and attach recessed, drinking fountains to mounting frames.
- D. Install water-supply piping with shutoff valve on supply to each fixture to be connected to domestic-water distribution piping. Use ball or gate valve. Install valves in locations where they can be easily reached for operation. Valves are specified in Section 220523.12 "Ball Valves for Plumbing Piping" and Section 220523.15 "Gate Valves for Plumbing Piping."
- E. Install trap and waste piping on drain outlet of each fixture to be connected to sanitary drainage system.
- F. Install wall flanges or escutcheons at piping wall penetrations in exposed, finished locations. Use deep-pattern escutcheons where required to conceal protruding fittings. Comply with escutcheon requirements specified in Section 220518 "Escutcheons for Plumbing Piping."
- G. Seal joints between fixtures and walls using sanitary-type, one-part, mildew-resistant, silicone sealant. Match sealant color to fixture color. Comply with sealant requirements specified in Section 079200 "Joint Sealants."

3.3 CONNECTIONS

- A. Connect fixtures with water supplies, stops, and risers, and with traps, soil, waste, and vent piping. Use size fittings required to match fixtures.
- B. Comply with water piping requirements specified in Section 221116 "Domestic Water Piping."
- C. Install ball or gate shutoff valve on water supply to each fixture. Install valve upstream from filter for water cooler. Comply with valve requirements specified in Section 220523.12 "Ball Valves for Plumbing Piping" and Section 220523.15 "Gate Valves for Plumbing Piping."
- D. Comply with soil and waste piping requirements specified in Section 221316 "Sanitary Waste and Vent Piping."

3.4 ADJUSTING

- A. Adjust fixture flow regulators for proper flow and stream height.
- B. Adjust pressure water-cooler temperature settings.

3.5 CLEANING

- A. After installing fixture, inspect unit. Remove paint splatters and other spots, dirt, and debris. Repair damaged finish to match original finish.
- B. Clean fixtures, on completion of installation, according to manufacturer's written instructions.
- C. Provide protective covering for installed fixtures.
- D. Do not allow use of fixtures for temporary facilities unless approved in writing by Owner.

END OF SECTION 224716

SECTION 230503 - GENERAL PROVISIONS FOR HVAC WORK**PART 1 - GENERAL****1.1 SUMMARY**

- A. The work under Division 23 "Heating, Ventilating, and Air Conditioning (HVAC)" shall include all labor, services, materials and equipment and performance of all work required for the installation of all mechanical work as shown on the Drawings and herein specified in the following Sections.
- B. Should there be any discrepancies or a question of intent, refer the matter to the Architect/Engineer for decision before ordering any equipment or materials or before starting any related work.
- C. Where work connects to that of another trade, or to piping or equipment in place, take measurements in the field to make connecting work come true and line up with the item being connected.
- D. Where work specified under other Divisions of the Specifications connects to equipment which is a part of Division 23, provide proper connection(s) to such equipment.
- E. Minor items and accessories or devices reasonably inferable as necessary, to the complete and proper installation and operation of any system, shall be provided by the Trade Contractor for such system whether or not they are specifically called for by the Specifications or Drawings.
- F. The Drawings and Specifications are to be taken together. Work specified and not shown or work shown and not specified shall be performed or furnished as though mentioned in both Specifications and Drawings. If there is a discrepancy between the Drawings and Specifications as to the quantity or quality to be provided, the greater quantity or the better quality shall be provided.

1.2 DEFINITIONS

- A. "Piping" includes, in addition to pipe, all fittings, valves, hangers, and other supports and accessories related to such piping.
- B. "Ductwork" includes, in addition to ducts, all fittings, transitions, dampers, hangers and other supports and accessories related to such ductwork.
- C. "Concealed" means hidden from sight in chases, furred spaces, shafts, hung ceilings, embedded in construction, in crawl spaces or buried.
- D. "Exposed" means not installed underground or "concealed" as defined above.
- E. "Invert Elevations" means the elevation of the inside bottom of pipe or duct.
- F. "HVAC Work" is all of the work in Division 23.

1.3 QUALITY ASSURANCE

- A. Each major component of equipment to have the manufacturer's name, address, model number and rating on a plate securely affixed in a conspicuous place.

- B. Code Ratings, labels or other data which are die-stamped or otherwise affixed to the surface of the equipment shall be in visible location.
- C. All equipment provided under Division 23 to perform with the least possible noise and vibration consistent with its duty. Quietness of operation of all equipment is a requirement. Any equipment, as determined by the Owner's Representative or Architect/Engineer to be producing objectionable noise or transmitting noise or vibration to the building to be repaired or removed and replaced.
- D. All workmanship shall be first class in every respect and shall be performed only by skilled mechanics.
- E. Shutdown and Notifications:
 - 1. It is imperative that service interruptions on the various existing utilities be held to an absolute minimum. Wherever possible provide suitable temporary services or connections, where continuity of service for essential systems can be maintained by this means. It will be the Owner's final prerogative to decide which systems are to be considered as essential, and to establish the maximum allowable shutdown time, if any, for each system.
 - 2. Owner will require not less than 72 hours advance notice, in writing, that an interruption of service in any system is desired. Such notice shall identify the system or systems involved, and shall be submitted in duplicate, one copy of which will be signed and returned by the Owner's authorized representative stating whether the requested shutdown will be permitted or not.
- F. Existing Utilities:
 - 1. Location of utilities as shown on the drawings has been determined from the best available information and is given for convenience; however, Owner does not assume responsibility in the event that during construction, utilities other than those shown may be encountered, and that the actual location of those which are shown may be different from the location as shown on the plans.
 - 2. Assume responsibility for interference with or damage to any existing utilities, and repair or replace same with the least possible delay.
- G. Notify Architect of broken or open pipes discovered during construction.
- H. Layout and establish the lines and levels necessary for work.
- I. The following Standards shall be used where referenced by the following abbreviations:
 - 1. AABC: Associated Air Balance Council
 - 2. ADC: Air Diffusion Council
 - 3. AGA: American Gas Association
 - 4. AIA: American Institute of Architects
 - 5. AMCA: Air Moving and Conditioning Association
 - 6. ANSI: American National Standards Institute
 - 7. ARI: Air Conditioning and Refrigeration Institute
 - 8. ASE: Association of Safety Engineers
 - 9. ASHRAE: American Society of Heating, Refrigeration and Air Conditioning Engineers
 - 10. ASME: American Society of Mechanical Engineers
 - 11. ASPE: American Society of Plumbing Engineers
 - 12. ASTM: American Society of Testing and Materials
 - 13. AWPB: American Wood Preserves Bureau
 - 14. AWS: American Welding Society

- 15. AWWA: American Water Works Association
- 16. CSA: Canadian Standards Association
- 17. CISPI: Cast Iron Soil Pipe Institute
- 18. EIA: Electronic Industries Association
- 19. EPA: Environmental Protection Agency
- 20. FDA: Food and Drug Administration
- 21. FM: Factory Mutual Insurance Association
- 22. HIS: Hydraulic Institute Standards
- 23. IRI: Industrial Risk Insurers
- 24. IBR: Institute of Boiler and Radiator Manufacturers
- 25. IEEE: Institute of Electrical and Electronics Engineers
- 26. MCAA: Mechanical Contractors' Association of America
- 27. NIST: National Institute of Standards and Testing
- 28. NEBB: National Environmental Balancing Bureau
- 29. NEC: National Electric Code
- 30. NECA: National Electric Contractors Association
- 31. NEMA: National Electrical Manufacturers Association
- 32. NFPA: National Fire Protection Association
- 33. NSC: National Safety Council
- 34. NSF: National Sanitation Foundation
- 35. OSHA: Occupational Safety & Health Administration
- 36. SAE: Society of Automotive Engineers
- 37. SBI: Steel Boiler Institute Industry
- 38. SMACNA: Sheet Metal and Air Conditioning Contractors National Association
- 39. TIMA: Thermal Insulation Manufacturers Association
- 40. UL: Underwriters' Laboratories
- 41. USDA: United States Department of Agriculture

J. Project Certification:

- 1. Each trade shall submit a project certification, guaranteeing that this project was constructed and will operate in accordance with the performance requirements of the Drawings and Specifications. This certification shall be signed by a principal of the firm and shall be delivered to the Architect/Engineer prior to final payment.

K. Drawings:

- 1. The Drawings are essentially diagrammatic in nature and show general arrangement of the equipment, piping, ductwork, accessories, etc. Because of the small scale of the Drawings, it is not possible to show each offsets, fittings, and accessories, which may be required. Carefully investigate the structural conditions, Architectural Drawings, Equipment Drawings, and the finished conditions of the work and arrange such work accordingly, furnish any fittings, pipe accessories that may be required to meet such conditions.
- 2. Any changes from the plans necessary to make the work conform to building as constructed and to fit work of other trades, or to conform to rules of the governing authorities and regulations, shall be met without extra cost to the Owner.
- 3. The layout of the piping, ductwork, equipment, etc., as shown on the Drawings shall be checked and exact locations shall be determined by the dimensions of equipment approved and Contractor shall obtain the Architect's approval for revised layout before the apparatus is installed. Consult the Architectural, Structural, and Equipment Drawings for the dimensions, locations of partitions, locations and sizes of structural supports, foundations, etc.
- 4. Refer to the Architectural Plans for details and large scale Drawings and to approved Shop Drawing of equipment furnished under other Contracts or Sections of the Specifications for exact location of service connections. The equipment Shop Drawings

will be furnished to the Contractor before roughing in. Contractor shall not install any piping or ductwork for said equipment until they have received approved Coordination Drawings for same.

L. Minor Deviations:

1. The dimensions of equipment hereinafter specified or indicated on the Drawings are intended to establish the outlines and characteristics of such equipment in general. Minor deviations in dimensions will be permitted to allow the manufacturers specified to bid on their nearest stock equipment, provided the specified ratings are met or exceeded.
2. Where manufacturers' catalog numbers or types are mentioned in the Specifications or indicated on the Drawings, they are intended to be used as a guide only and shall not be interpreted as taking precedence over the basic rating and duty specified. In all cases, manufacturers shall verify the duty specified with particular characteristics of the equipment they intend to offer for approval and shall also pay the additional charges as may be required under other Divisions.

M. Interferences:

1. Before making any installation, the work of the trades must be coordinated and the necessary changes shall be made to avoid interferences or improper effect on work to be performed by any other Section. In the event that interferences develop, the Architect's decision will be final and no additional compensation will be allowed for moving of misplaced piping, ducts, conduit and/or equipment.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. Unless otherwise specified, all material and equipment incorporated in the work under the contract shall be new.
- B. Material and equipment specified by one or more manufacturer's name, trade name and/or model number does not limit a bidder from bidding on other equipment providing the procedure set forth in the Conditions of the Contract and hereinafter specified is followed.
- C. The various mechanical systems have been engineered and designed on equipment name and catalog numbers specified or designated on the Drawings.
- D. A Contractor who intends to furnish equipment listed as approved equal shall proceed as follows:
 1. Obtain Architect/Engineer's approval of said equipment.
 2. Be fully responsible for said equipment.
 3. Include in the Base Bid, all cost for any changes that may be required in his work and/or work of other trades for the proper installation and functioning of said equipment.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Each Trade Contractor shall be responsible for all of his work fitting into place in a satisfactory and neat workmanlike manner acceptable to the Architect/Engineer.

- B. Confer with other Trade Contractors regarding the location and size of pipes, equipment, fixtures, conduit, ducts, openings, switches, outlets, etc., in order that there may be no interference between the installation of the progress of the work of any Trade Contractor on the project. The Architectural Drawings shall take precedence over the Mechanical and Electrical Drawings.
- C. The Mechanical Drawings are diagrammatic and shall be followed as closely as actual construction of the building and the work of other trade contractors will permit. All changes from Drawings necessary to make the work of each Contractor conform to the building construction and the work of other trade contractors shall be done at the appropriate Trade Contractor's expense.
- D. Unless explicitly stated to the contrary, each Trade Contractor shall furnish and install each item of equipment or material hereinafter specified, complete with all necessary fittings, supports, trim, piping, insulation, etc., as required for a complete and operating installation.
- E. All equipment and materials shall be installed according to the manufacturer's instructions unless otherwise specifically directed by the Trade Contract Documents. All piping, valves, connections, and other like items recommended by the manufacturer or required for proper operation shall be provided without additional cost to the Owner.
- F. All references to Contractors in Specifications and Drawings shall refer to the respective Trade Contractor performing that portion of the work.
- G. In general, all piping, ductwork and similar items shall be installed concealed from view above the ceiling, in partitions, shafts, chases, unless otherwise indicated.
- H. Locations of items not definitely fixed by dimensions are approximate only and exact locations necessary to secure the best conditions and results shall be determined at the site, subject to review.
- I. Where pipes are in partitions, furred out spaces and chases, obtain information as to their exact location and size and install work so as to be entirely concealed in the allotted space. If conflicts arise making this impossible, obtain instructions from the Architect/Engineer before proceeding with the work.
- J. Wherever two or more pipes are to be installed in parallel, or parallel to the piping of other trades, the piping shall be installed with sufficient space between pipes to allow for the proper application of pipe covering, painting and servicing.
- K. Furnish advance information on locations and sizes of frames, boxes, sleeves and openings needed for the work, and also furnish information and shop drawings necessary to permit installation of other work without delay.
- L. Where there is evidence that parts of the Mechanical Work will interfere with other work, assist in working out space conditions and/or the structure, make necessary adjustments to accommodate the work.
- M. Mechanical Work installed before coordinating with other work so as to cause interference with other work shall be changed to correct such condition without additional cost to the Owner.
- N. In no case shall any pipe, conduit, duct, or item of equipment be installed where it is supported on or suspended from another pipe, conduit, duct or item of equipment.

- O. Where an item or task is specified to be provided "under this Section," it shall be understood that, that item or task is the responsibility of the trade responsible for that Section, but the work must be performed by qualified workmen of the appropriate trade.
- P. Accessibility:
 - 1. Install Mechanical work to permit removal (without damage to other parts) of coils, heat exchangers, pumps, fan shafts and wheels, belt guards, sheaves and drives, and other parts requiring periodic replacement or maintenance.
 - 2. Arrange pipes, ducts, and equipment to permit ready access to valves, cocks, traps, starters, motors, dampers, control components, and to clear the openings of swinging and overhead doors and of access panels.
 - 3. Change dimensions of ductwork when required to meet job conditions but maintain the same equivalent cross-sectional area. Based on equal friction to airflow.
 - 4. Provide access panels in equipment, ducts, and like items for inspection of interiors and proper maintenance.

END OF SECTION 230503

SECTION 230505 - BASIC HVAC MATERIALS AND METHODS**PART 1 - GENERAL****1.1 SUMMARY****A. Section includes the following:**

1. Piping materials and installation instructions common to most piping systems.
2. Concrete base construction requirements.
3. Escutcheons.
4. Mechanical sleeve seals.
5. Equipment nameplate data requirements.
6. Nonshrink grout for equipment installations.
7. Field-fabricated metal and wood equipment supports.
8. Installation requirements common to equipment specification sections.
9. Cutting and patching.
10. Touchup painting and finishing.
11. Access Panels
12. Flashings
13. Cleaning

B. Pipe and pipe fitting materials are specified in Division 22.**1.2 DEFINITIONS**

- A. **Finished Spaces:** Spaces other than mechanical and electrical equipment rooms, furred spaces, pipe and duct shafts, unheated spaces immediately below roof, spaces above ceilings, unexcavated spaces, crawl spaces, and tunnels.
- B. **Exposed, Interior Installations:** Exposed to view indoors. Examples include finished occupied spaces and mechanical equipment rooms.
- C. **Exposed, Exterior Installations:** Exposed to view outdoors, or subject to outdoor ambient temperatures and weather conditions. Examples include rooftop locations.
- D. **Concealed, Interior Installations:** Concealed from view and protected from physical contact by building occupants. Examples include above ceilings and in duct shafts.
- E. **Concealed, Exterior Installations:** Concealed from view and protected from weather conditions and physical contact by building occupants, but subject to outdoor ambient temperatures. Examples include installations within unheated shelters.
- F. **The following are industry abbreviations for rubber materials:**
 1. BS: Acrylonitrile-butadiene-styrene plastic.
 2. CPVC: Chlorinated polyvinyl chloride plastic.
 3. CR: Chlorosulfonated polyethylene synthetic rubber.
 4. EPDM: Ethylene-propylene-diene terpolymer rubber.
 5. NBR: Acrylonitrile-butadiene rubber.
 6. PE: Polyethylene plastic.
 7. PVC: Polyvinyl chloride plastic.

1.3 SUBMITTALS

- A. Refer to Division 01 Section "Submittal Procedures" for administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other miscellaneous submittals.
- B. Product Data: For dielectric fittings, flexible connectors, mechanical sleeve seals, access panels and identification materials and devices.
- C. Coordination Drawings: For access panel and door locations.
- D. Coordination Drawings: Detail major elements, components, and systems of mechanical equipment and materials in relationship with other systems, installations, and building components. Show space requirements for installation and access. Indicate if sequence and coordination of installations are important to efficient flow of the Work. Use BIM software for coordination. Include the following:
 - 1. Planned piping layout, including valve and specialty locations and valve-stem movement.
 - 2. Planned duct systems layout, including elbow radii and duct accessories.
 - 3. Clearances for installing and maintaining insulation.
 - 4. Clearances for servicing and maintaining equipment, accessories, and specialties, including space for disassembly required for periodic maintenance.
 - 5. Equipment and accessory service connections and support details.
 - 6. Exterior wall and foundation penetrations.
 - 7. Fire-rated wall and floor penetrations.
 - 8. Sizes and location of required concrete pads and bases.
 - 9. Scheduling, sequencing, movement, and positioning of large equipment into building during construction.
 - 10. Floor plans, elevations, and details to indicate penetrations in floors, walls, and ceilings and their relationship to other penetrations and installations.
 - 11. Reflected ceiling plans to coordinate and integrate installation of air outlets and inlets, light fixtures, communication system components, sprinklers, and other ceiling-mounted items.
 - 12. Access panel locations in ceilings/walls/floors.

1.4 QUALITY ASSURANCE

- A. Requirements of Regulatory Agencies:
 - 1. American Society for Testing and Materials
 - a. ASTM A 53-98: Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
 - b. ASTM B 32-96: Specification for Solder Metal
 - c. ASTM B 813-93: Specification for Liquid and Paste Fluxes for Soldering Applications of Copper and Copper Alloy Tube
 - d. ASTM B 828-98: Practice for Making Capillary Joints by Soldering of Copper and Copper Alloy Tube and Fittings
 - e. ASTM C 1107-97: Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
 - f. ASTM C 1173-97: Specification for Flexible Transition Couplings for Underground Piping Systems
 - g. ASTM D 1785-96b: Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120

- h. ASTM D 2235-96a: Specification for Solvent Cement for Acrylonitrile-Butadiene-Styrene (ABS) Plastic Pipe and Fittings
 - i. ASTM D 2564-96a: Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems
 - j. ASTM D 2672-96a: Specification for Joints for IPS PVC Pipe Using Solvent Cement
 - k. ASTM D 2855-96: Practice for Making Solvent-Cemented Joints with Poly(Vinyl Chloride) (PVC) Pipe and Fittings
 - l. ASTM D 3139-98: Specification for Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals
 - m. ASTM F 402-93: Practice for Safe Handling of Solvent Cements, Primers, and Cleaners Used for Joining Thermostatic Pipe and Fittings
 - n. ASTM F 493-97: Specification for Solvent Cements for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe and Fittings
 - o. ASTM F 656-96a: Specification for Primers for Use in Solvent Cement Joints of Poly(Vinyl Chloride) (PVC) Plastic Pipe and Fittings
2. American Water Works Association
- a. AWWA C110-98: Ductile-Iron and Gray-Iron Fittings, 3 In. through 48 In. (76 mm through 1219 mm), for Water and Other Liquids
 - b. AWWA C219-97: Bolted, Sleeve-Type Couplings for Plain-End Pipe
3. American Welding Society
- a. AWS A5.8-92: Specification for Filler Metals for Brazing and Braze Welding
 - b. AWS D1.1-98: Structural Welding Code--Steel
 - c. AWS D10.12-89: Recommended Practices and Procedures for Welding Low Carbon Steel Pipe
 - d. Brazing Handbook. 1991.
4. ASME International
- a. ASME B1.20.1-83 (Reaffirmed 1992): Pipe Threads, General Purpose (Inch)
 - b. ASME B16.21-92: Nonmetallic Flat Gaskets for Pipe Flanges
 - c. ASME B18.2.1-96: Square and Hex Bolts and Screws--Inch Series
 - d. ASME B31 Series: Code for Pressure Piping
 - e. 1998 ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications"
5. Copper Development Association Inc.
- a. Copper Tube Handbook. 1995.
6. Manufacturers Standardization Society of the Valve and Fittings Industry, Inc.
- a. MSS SP-107-91: Transition Union Fittings for Joining Metal and Plastic Products
- B. Steel Support Welding: Qualify processes and operators according to AWS D1.1, "Structural Welding Code--Steel."
- C. Steel Pipe Welding: Qualify processes and operators according to ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."

1. Comply with provisions in ASME B31 Series, "Code for Pressure Piping."
2. Certify that each welder has passed AWS qualification tests for welding processes involved and that certification is current.

- D. Equipment of higher electrical characteristics, physical dimensions, capacities, and ratings may be furnished provided such proposed equipment is approved in writing and connecting mechanical and electrical services, circuit breakers, conduit, motors, bases, and equipment spaces are increased. Additional costs shall be approved in advance by appropriate Contract Modification for these increases. If minimum energy ratings or efficiencies of equipment are specified, equipment must meet design and commissioning requirements.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver pipes and tubes with factory-applied end caps. Maintain end caps through shipping, storage, and handling to prevent pipe-end damage and prevent entrance of dirt, debris, and moisture.
- B. Protect stored pipes and tubes from moisture and dirt. Elevate above grade. Do not exceed structural capacity of floor, if stored inside.
- C. Protect flanges, fittings, and piping specialties from moisture and dirt.

1.6 SEQUENCING AND SCHEDULING

- A. Coordinate phasing and sequencing of all work with the Owner and Building Engineer.
- B. Coordinate mechanical equipment installation with other building components.
- C. Arrange for pipe spaces, chases, slots, and openings in building structure during progress of construction to allow for mechanical installations.
- D. Coordinate installation of required supporting devices and sleeves in poured-in-place concrete and other structural components, as they are constructed.
- E. Sequence, coordinate, and integrate installations of mechanical materials and equipment for efficient flow of the Work. Coordinate installation of large equipment requiring positioning before closing in building.
- F. Coordinate connection of mechanical systems with exterior underground and overhead utilities and services. Comply with requirements of governing regulations, franchised service companies, and controlling agencies.
- G. Coordinate requirements for access panels and doors if mechanical items requiring access are concealed behind finished surfaces. Access panels and doors are specified in Division 08 Section "Access Doors and Frames."
- H. Coordinate installation of identifying devices after completing covering and painting, if devices are applied to surfaces. Install identifying devices before installing acoustical ceilings and similar concealment.
- I. Coordinate connection of electrical services.

1.7 WARRANTY

- A. Provide warranty on materials and labor for 30 months starting from date of delivery, or two years from date of substantial completion, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following manufacturers:

- 1. Mechanical Sleeve Seals:

- a. Calpico, Inc.
 - b. Metraflex Co.
 - c. Thunderline/Link-Seal.

- 2. Metal, Flexible Connectors:

- a. Grinnell Corp.; Grinnell Supply Sales Co..
 - b. Mercer Rubber Co.
 - c. Metraflex Co.

- 3. Rubber, Flexible Connectors:

- a. General Rubber Corp.
 - b. Metraflex Co.
 - c. Red Valve Co., Inc.

2.2 PIPE, TUBE, AND FITTINGS

- A. Refer to individual Division 22 piping Sections for pipe, tube, and fitting materials and joining methods.
- B. Pipe Threads: ASME B1.20.1 for factory-threaded pipe and pipe fittings.

2.3 JOINING MATERIALS

- A. Refer to individual Division 22 piping Sections for special joining materials not listed below.
- B. Pipe-Flange Gasket Materials: Suitable for chemical and thermal conditions of piping system contents.
 - 1. ASME B16.21, nonmetallic, flat, 1/8-inch maximum thickness unless thickness or specific material is indicated.
 - a. Full-Face Type: For flat-face, Class 125, cast-iron and cast-bronze flanges.
 - b. Narrow-Face Type: For raised-face, Class 250, cast-iron and steel flanges.
 - 2. AWWA C110, rubber, flat face, 1/8 inch thick, unless otherwise indicated; and full-face or ring type, unless otherwise indicated.

- C. Flange Bolts and Nuts: ASME B18.2.1, carbon steel, unless otherwise indicated.
- D. Plastic, Pipe-Flange Gasket, Bolts, and Nuts: Type and material recommended by piping system manufacturer, unless otherwise indicated.
- E. Solder Filler Metals: ASTM B 32, lead-free alloys. Include water-flushable flux according to ASTM B 813.
- F. Brazing Filler Metals: AWS A5.8, BCuP Series, copper-phosphorus alloys for general-duty brazing, unless otherwise indicated; and AWS A5.8, BAg1, silver alloy for refrigerant piping, unless otherwise indicated.
- G. Welding Filler Metals: Comply with AWS D10.12 for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.
- H. Solvent Cements for Joining Plastic Piping:
 - 1. PVC Piping: ASTM D 2564. Include primer according to ASTM F 656.

2.4 MECHANICAL SLEEVE SEALS

- A. Description: Modular sealing element unit, designed for field assembly, to fill annular space between pipe and sleeve.
 - 1. Sealing Elements: EPDM interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
 - 2. Pressure Plates: Carbon steel. Include two for each sealing element.
 - 3. Connecting Bolts and Nuts: Carbon steel with corrosion-resistant coating of length required to secure pressure plates to sealing elements. Include one for each sealing element.

2.5 FLEXIBLE CONNECTORS

- A. General: Fabricated from materials suitable for system fluid and that will provide flexible pipe connections. Include 125-psig minimum working-pressure rating, unless higher working pressure is indicated, and ends according to the following:
 - 1. 2-Inch NPS and Smaller: Threaded.
 - 2. 2-1/2-Inch NPS and Larger: Flanged.
 - 3. Option for 2-1/2-Inch NPS and Larger: Grooved for use with keyed couplings.

2.6 SLEEVES

- A. Galvanized-Steel Sheet: 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint.
- B. Steel Pipe: ASTM A 53, Type E, Grade B, Schedule 40, galvanized, plain ends.
- C. Cast Iron: Cast or fabricated "wall pipe" equivalent to ductile-iron pressure pipe, with plain ends and integral water-stop, unless otherwise indicated.
- D. Stack Sleeve Fittings: Manufactured, cast-iron sleeve with integral clamping flange. Include clamping ring and bolts and nuts for membrane flashing.

1. Under-deck Clamp: Clamping ring with set screws.

2.7 ESCUTCHEONS

- A. Description: Manufactured wall and ceiling escutcheons and floor plates, with an ID to closely fit around pipe, tube, and insulation of insulated piping and an OD that completely covers opening.
- B. One-Piece, Deep-Pattern Type: Deep-drawn, box-shaped brass with polished chrome-plated finish.
- C. One-Piece, Cast-Brass Type: With set screw.
 1. Finish: Rough brass.
- D. Split-Casting, Cast-Brass Type: With concealed hinge and set screw.
 1. Finish: Rough brass.
- E. One-Piece, Stamped-Steel Type: With set screw and chrome-plated finish.
- F. Split-Plate, Stamped-Steel Type: With concealed hinge, set screw, and chrome-plated finish.
- G. One-Piece, Floor-Plate Type: Cast-iron floor plate.
- H. Split-Casting, Floor-Plate Type: Cast brass with concealed hinge and set screw.

2.8 ACCESS PANELS

- A. General: Refer to Division 08 Section, "Access Doors and Frames," for access panel manufacturers and other requirements.

2.9 GROUT

- A. Description: ASTM C 1107, Grade B, nonshrink and nonmetallic, dry hydraulic-cement grout.
 1. Characteristics: Post-hardening, volume-adjusting, nonstaining, noncorrosive, nongaseous, and recommended for interior and exterior applications.
 2. Design Mix: 5000-psi, 28-day compressive strength.
 3. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 PIPING SYSTEMS - COMMON REQUIREMENTS

- A. Install piping according to the following requirements and Division 22 Sections specifying piping systems.
- B. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated unless deviations to layout are approved on Coordination Drawings.

- C. Install piping in concealed locations, unless otherwise indicated and except in equipment rooms and service areas.
- D. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.
- E. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.
- F. Install piping to permit valve servicing.
- G. Install piping at indicated slopes.
- H. Install piping free of sags and bends.
- I. Install fittings for changes in direction and branch connections.
- J. Install piping to allow application of insulation.
- K. Select system components with pressure rating equal to or greater than system operating pressure.
- L. Install escutcheons for penetrations of walls, ceilings, and floors according to the following:
 - 1. New Piping:
 - a. Piping with Fitting or Sleeve Protruding from Wall: One-piece, deep-pattern type.
 - b. Chrome-Plated Piping: One-piece, cast-brass type with polished chrome-plated finish.
 - c. Insulated Piping: One-piece, stamped-steel type with spring clips.
 - d. Bare Piping at Wall and Floor Penetrations in Finished Spaces: One-piece, stamped-steel type.
 - e. Bare Piping at Ceiling Penetrations in Finished Spaces: One-piece, stamped-steel type and set screw.
 - f. Bare Piping in Unfinished Service Spaces: One-piece, stamped-steel type with concealed hinge and set screw.
 - g. Bare Piping in Equipment Rooms: One-piece, stamped-steel type with set screw.
 - h. Bare Piping at Floor Penetrations in Equipment Rooms: One-piece, floor-plate type.
- M. Sleeves are not required for core-drilled holes.
- N. Permanent sleeves are not required for holes formed by removable PE sleeves.
- O. Install sleeves for pipes passing through concrete and masonry walls, gypsum-board partitions, and concrete floor and roof slabs.
 - 1. Cut sleeves to length for mounting flush with both surfaces.
 - a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level. Extend cast-iron sleeve fittings below floor slab as required to secure clamping ring if ring is specified.
 - 2. Install sleeves in new walls and slabs as new walls and slabs are constructed.

3. Install sleeves that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation. Use the following sleeve materials:
 - a. Steel Pipe Sleeves: For pipes smaller than NPS 6.
 - b. Steel Sheet Sleeves: For pipes NPS 6 and larger, penetrating gypsum-board partitions.
 - c. Stack Sleeve Fittings: For pipes penetrating floors with membrane waterproofing. Secure flashing between clamping flanges. Install section of cast-iron soil pipe to extend sleeve to 2 inches above finished floor level. Refer to Division 07 Section "Sheet Metal Flashing and Trim" for flashing.
 - 1) Seal space outside of sleeve fittings with grout.
 4. Except for underground wall penetrations, seal annular space between sleeve and pipe or pipe insulation, using joint sealants appropriate for size, depth, and location of joint. Refer to Division 07 Section "Joint Sealants" for materials and installation.
 - P. Aboveground, Exterior-Wall Pipe Penetrations: Seal penetrations using sleeves and mechanical sleeve seals. Select sleeve size to allow for 1-inch (25-mm) annular clear space between pipe and sleeve for installing mechanical sleeve seals.
 1. Install steel pipe for sleeves smaller than 6 inches in diameter.
 2. Install cast-iron "wall pipes" for sleeves 6 inches and larger in diameter.
 3. Mechanical Sleeve Seal Installation: Select type and number of sealing elements required for pipe material and size. Position pipe in center of sleeve. Assemble mechanical sleeve seals and install in annular space between pipe and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.
 - Q. Underground, Exterior-Wall Pipe Penetrations: Install cast-iron "wall pipes" for sleeves. Seal pipe penetrations using mechanical sleeve seals. Select sleeve size to allow for 1-inch annular clear space between pipe and sleeve for installing mechanical sleeve seals.
 1. Mechanical Sleeve Seal Installation: Select type and number of sealing elements required for pipe material and size. Position pipe in center of sleeve. Assemble mechanical sleeve seals and install in annular space between pipe and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.
 - R. Fire-Barrier Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with firestop materials. Refer to Division 07 Section "Penetration Firestopping" for materials.
 - S. Verify final equipment locations for roughing-in.
- 3.2 PIPING JOINT CONSTRUCTION
- A. Join pipe and fittings according to the following requirements and Division 22 Sections specifying piping systems.
 - B. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
 - C. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.

- D. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
1. Note internal length of threads in fittings or valve ends, and proximity of internal seat or wall, to determine how far pipe should be threaded into joint.
 2. Apply appropriate tape or thread compound to external pipe threads unless dry seal threading is specified.
 3. Align threads at point of assembly.
 4. Tighten joint with wrench. Apply wrench to valve end into which pipe is being threaded.
 5. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds.
- E. Welded Joints: Construct joints according to AWS D10.12, using qualified processes and welding operators according to Part 1 "Quality Assurance" Article.
- F. Plastic Piping Solvent-Cement Joints: Clean and dry joining surfaces. Join pipe and fittings according to the following:
1. Comply with ASTM F 402 for safe-handling practice of cleaners, primers, and solvent cements.
 2. PVC Pressure Piping: Join schedule number ASTM D 1785, PVC pipe and PVC socket fittings according to ASTM D 2672. Join other-than-schedule-number PVC pipe and socket fittings according to ASTM D 2855.
 3. PVC Non-pressure Piping: Join according to ASTM D 2855.
- G. Plastic Pressure Piping Gasketed Joints: Join according to ASTM D 3139.
- H. Plastic Non-pressure Piping Gasketed Joints: Join according to ASTM D 3212.
- I. PE Piping Heat-Fusion Joints: Clean and dry joining surfaces by wiping with clean cloth or paper towels. Join according to ASTM D 2657.
1. Plain-End Pipe and Fittings: Use butt fusion.
 2. Plain-End Pipe and Socket Fittings: Use socket fusion.

3.3 PIPING CONNECTIONS

- A. Make connections according to the following, unless otherwise indicated:
1. Install unions, in piping NPS 2 and smaller, adjacent to each valve and at final connection to each piece of equipment.
 2. Install flanges, in piping NPS 2-1/2 and larger, adjacent to flanged valves and at final connection to each piece of equipment.
 3. Dry Piping Systems: Install dielectric unions and flanges to connect piping materials of dissimilar metals.
 4. Wet Piping Systems: Install dielectric coupling and nipple fittings to connect piping materials of dissimilar metals

3.4 EQUIPMENT INSTALLATION - COMMON REQUIREMENTS

- A. Install equipment to provide maximum possible headroom, if mounting heights are not indicated.

- B. Install equipment according to approved submittal data. Portions of the Work are shown only in diagrammatic form. Refer conflicts to Architect.
- C. Install equipment level and plumb, parallel and perpendicular to other building systems and components in exposed interior spaces, unless otherwise indicated.
- D. Install mechanical equipment to facilitate service, maintenance, and repair or replacement of components. Connect equipment for ease of disconnecting, with minimum interference to other installations. Extend grease fittings to accessible locations.
- E. Install equipment giving right of way to piping installed at required slope.
- F. Install flexible connectors on equipment side of shutoff valves, horizontally and parallel to equipment shafts if possible.

3.5 PAINTING AND FINISHING

- A. Refer to Division 09 Section "Painting" for paint materials, surface preparation, and application of paint.
- B. Do not paint piping specialties with factory-applied finish.
- C. Damage and Touchup: Repair marred and damaged factory-painted finishes with materials and procedures to match original factory finish.

3.6 CUTTING AND PATCHING

- A. Cut, channel, chase, and drill floors, walls, partitions, ceilings, and other surfaces necessary for mechanical installations. Perform cutting by skilled mechanics of trades involved.
- B. Repair cut surfaces to match adjacent surfaces.

3.7 GROUTING

- A. Install nonmetallic, nonshrink, grout for mechanical equipment base bearing surfaces, pump and other equipment base plates, and anchors. Mix grout according to manufacturer's written instructions.
- B. Clean surfaces that will come into contact with grout.
- C. Provide forms as required for placement of grout.
- D. Avoid air entrapment during placing of grout.
- E. Place grout, completely filling equipment bases.
- F. Place grout on concrete bases to provide smooth bearing surface for equipment.
- G. Place grout around anchors.
- H. Cure placed grout according to manufacturer's written instructions.

3.8 CLEANING

- A. Coordinate general cleanup with the work as specified in Division 1.

3.9 ACCESS PANELS

- A. Where control valves, shut-off valves, drip traps, heating coils, dampers, pull boxes or other specialties, which require service or adjustment, are installed above inaccessible type furred ceilings or within furred walls, the Trade Contractor whose equipment is involved shall furnish and install access panels as required.
- B. Each Trade shall confer with other trades with respect to access panel locations, and shall wherever practical group valves, traps, dampers, etc. in such a way as to be accessible from a single panel and eliminate as many access panels as possible.

3.10 ERECTION

- A. Provide all necessary rigging, scaffolding, tools, tackle, labor and other like items necessary for the complete installation of the equipment.
- B. Adapt his work to job conditions and install his work to clear beams, joists and light fixtures, adjusting risers, avoiding interferences with windows and openings, raising or lowering work to permit the passing of ductwork or the work of other trades, all as required or as job conditions dictate, without additional costs to the Owner.
- C. Trade Contractor shall not rig, tie to, or rest weight upon any part of the building or make use of any stairway until specific permission is obtained.
 - 1. Permission to rig to or make use of any part of the building premises shall not relieve the contractor of responsibility for any damage.
- D. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor mechanical materials and equipment.
- E. Field Welding: Comply with AWS D1.1, "Structural Welding Code--Steel."

3.11 ERECTION OF WOOD SUPPORTS AND ANCHORAGE

- A. Cut, fit, and place wood grounds, nailers, blocking, and anchorage to support and anchor mechanical materials and equipment.
- B. Select fastener sizes that will not penetrate members if opposite side will be exposed to view or will receive finish materials. Tighten connections between members. Install fasteners without splitting wood members.
- C. Attach to substrates as required to support applied loads.

3.12 FLASHINGS

- A. Openings in roof shall be flashed with 40 lb. sheet lead, in one piece, extending 24" under roofing in all directions from drains, pipes, or ducts, with upper edge at least 8" above the roof and turned over and down into increaser or sleeve or over curb for at least 1" and tightly bound with nonferrous wire.

1. All lead flashings shall be entirely painted with a good coat of black asphaltum before installation.
2. Roof flashing must be approved in writing by the Roofing Trade contractor to the effect that such flashing will not void guarantees or bonds pertaining to the roofing contract.
3. Coordinate with the work under Division 07.

3.13 OPENINGS

- A. Where temporary openings are necessary thru walls and partitions of the building for the entry or installation of tanks, fans, or other machinery or apparatus, or for driveways and other facilities, the permanent work of the mechanical trades at said openings shall be temporarily omitted and installed after equipment is brought into the building or after temporary facilities are removed.
- B. Refer to other Sections of the Specifications for framing of openings for ducts, grilles, registers, etc., in walls, partitions, floors, roofs, etc. The trade for each service shall be responsible for locating and providing the proper dimensions for all required openings.
 1. Space between ducts and wall or floor openings shall be sealed as specified in Division 07 Section "Penetration Firestopping."
- C. No cutting or drilling of any building structural members will be permitted, unless the specific extent and limits are approved, in writing, by the Architect.
- D. All openings in the existing structure shall be core drilled with a diamond drill. The use of jackhammers will not be permitted.
- E. The Contractor shall notify the Architect if any existing openings are uncovered adjacent to location of a new opening. The new opening shall not be drilled if the existing can be used unless directed by the Architect.

END OF SECTION 230505

SECTION 230513 - COMMON MOTOR REQUIREMENTS FOR HVAC EQUIPMENT**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes basic requirements for factory- and field-installed motors.

1.2 DEFINITIONS

- A. **Factory-Installed Motor:** A motor installed by motorized-equipment manufacturer as a component of equipment.

1.3 SUBMITTALS

- A. **Product Data for Field-Installed Motors:** For each type and size of motor, provide nameplate data and ratings; shipping, installed, and operating weights; enclosure type and mounting arrangements; size, type, and location of winding terminations; conduit entry and ground lug locations; and information on coatings or finishes.
- B. **Qualification Data:** For testing agency.
- C. **Source quality-control test reports.**
- D. **Training Reports:** Submit reports on training documenting dates and attendance.

1.4 QUALITY ASSURANCE

- A. **Testing Agency Qualifications:** An independent agency, with the experience and capability to conduct the testing indicated, that is a member company of the InterNational Electrical Testing Association or is a nationally recognized testing laboratory (NRTL) as defined by OSHA in 29 CFR 1910.7, and that is acceptable to authorities having jurisdiction.
 - 1. **Testing Agency's Field Supervisor:** Person currently certified by the InterNational Electrical Testing Association or the National Institute for Certification in Engineering Technologies to supervise on-site testing specified in PART 3.
- B. **Electrical Components, Devices, and Accessories:** Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- C. **Comply with NFPA 70.**

1.5 DELIVERY, STORAGE AND HANDLING

- A. Follow manufacturer's instructions for storage and handling of motors.

1.6 COORDINATION

- A. **Coordinate features of motors, installed units, and accessory devices and features that comply with the following:**
 - 1. **Compatible with the following:**
 - a. **Magnetic controllers.**

- b. Multispeed controllers.
 - c. Reduced-voltage controllers.
 - 2. Designed and labeled for use with variable frequency controllers, and suitable for use throughout speed range without overheating.
 - 3. Matched to torque and horsepower requirements of the load.
 - 4. Matched to ratings and characteristics of supply circuit and required control sequence.
 - B. Coordinate motor support with requirements for driven load; access for maintenance and motor replacement; installation of accessories, belts, belt guards; and adjustment of sliding rails for belt tensioning.
 - C. Coordinate size and location of concrete bases. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified in Division 03.
- 1.7 WARRANTY
- A. Written manufacturer's warranty covering parts and labor for a period of one year from substantial completion, or eighteen months from shipment, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide products by one of the following manufacturers:
 - 1. Motors:
 - a. Baldor.
 - b. General Electric.
 - c. Lincoln Motors.
 - d. Marathon.
 - e. Reliance.
 - f. U.S. Motors.

2.2 MOTOR REQUIREMENTS

- A. Motor requirements apply to factory- and field-installed motors except as follows:
 - 1. Different ratings, performance, or characteristics for motor are specified in another Section.
 - 2. Motorized-equipment manufacturer requires ratings, performance, or characteristics, other than those specified in this Section, to meet performance specified.

2.3 MOTOR CHARACTERISTICS

- A. Motors 1/2 HP and Larger: Three phase.
- B. Motors Smaller Than 1/2 HP: Single phase.
- C. Frequency Rating: 60 Hz.

- D. Voltage Rating: NEMA standard voltage selected to operate on nominal circuit voltage to which motor is connected. 208V motors shall be rated for continuous operation at 200V.
- E. Service Factor: 1.15 for open dripproof motors; 1.0 for totally enclosed motors.
- F. Duty: Continuous duty at ambient temperature of 105 deg F and at altitude of 3300 feet above sea level.
- G. Capacity and Torque Characteristics: Sufficient to start, accelerate, and operate connected loads at designated speeds, at installed altitude and environment, with indicated operating sequence, and without exceeding nameplate ratings or considering service factor.
- H. Enclosure: Open dripproof.

2.4 POLYPHASE MOTORS

- A. Description: NEMA MG 1, Design B, medium induction motor.
- B. Efficiency: Premium, as defined in NEMA MG 1.
- C. Stator: Copper windings, unless otherwise indicated.
 - 1. Multispeed motors shall have separate winding for each speed.
- D. Rotor: Squirrel cage, unless otherwise indicated.
- E. Bearings: Double-shielded, pre-lubricated ball bearings suitable for radial and thrust loading.
- F. Temperature Rise: Match insulation rating, unless otherwise indicated.
- G. Insulation: Class F, unless otherwise indicated.
- H. Code Letter Designation:
 - 1. Motors 15 HP and Larger: NEMA starting Code F or Code G.
 - 2. Motors Smaller Than 15 HP: Manufacturer's standard starting characteristic.
- I. Enclosure: Cast iron for motors 7.5 hp and larger; rolled steel for motors smaller than 7.5 hp.
 - 1. Finish: Gray enamel.

2.5 POLYPHASE MOTORS WITH ADDITIONAL REQUIREMENTS

- A. Motors Used with Reduced-Inrush Controllers: Match wiring connection requirements for controller with required motor leads. Provide terminals in motor terminal box, suited to control method.
- B. Motors Used with Variable Frequency Controllers: Motors shall be inverter-duty. Ratings, characteristics, and features coordinated with and approved by controller manufacturer.
 - 1. Designed with critical vibration frequencies outside operating range of controller output.
 - 2. Temperature Rise: Matched to rating for Class B insulation.
 - 3. Insulation: Class F.

4. Thermal Protection: Comply with NEMA MG 1 requirements for thermally protected motors.
 5. Provide Aegis shaft grounding rings.
- C. Rugged-Duty Motors: Totally enclosed, with 1.25 minimum service factor, greased bearings, integral condensate drains, and capped relief vents. Windings insulated with non-hygroscopic material.
1. Finish: Chemical-resistant paint over corrosion-resistant primer.
- D. Source Quality Control for Field-Installed Motors: Perform the following tests on each motor according to NEMA MG 1:
1. Measure winding resistance.
 2. Read no-load current and speed at rated voltage and frequency.
 3. Measure locked rotor current at rated frequency.
 4. Perform high-potential test.

2.6 SINGLE-PHASE MOTORS

- A. Type: One of the following, to suit starting torque and requirements of specific motor application:
1. Permanent-split capacitor.
 2. Split-phase start, capacitor run.
 3. Capacitor start, capacitor run.
- B. Shaded-Pole Motors: For motors 1/20 hp and smaller only.
- C. Thermal Protection: Internal protection to automatically open power supply circuit to motor when winding temperature exceeds a safe value calibrated to temperature rating of motor insulation. Thermal-protection device shall automatically reset when motor temperature returns to normal range.
- D. Bearings: Ball type for belt-connected motors and other motors with high radial forces on motor shaft; sealed, pre-lubricated-sleeve type for other single-phase motors.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine roughing-in for conduit systems to verify actual locations of conduit connections before motor installation.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.
- C. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust field-assembled components and equipment installation, including connections, and to assist in field testing. Report results in writing.
- D. Perform the following field tests and inspections and prepare test reports:

1. Perform electrical tests and visual and mechanical inspections including optional tests and inspections stated in NETA ATS on factory- and field-installed motors. Certify compliance with test parameters.
2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.

3.2 CLEANING

- A. Comply with applicable requirements in Division 23 Section "HVAC Equipment Cleaning."

3.3 CONTRACTOR STARTUP AND REPORTING

- A. Prepare for acceptance tests.

1. Align motors, bases, shafts, pulleys, and belts. Tension belts according to manufacturer's written instructions.
2. Verify bearing lubrication.
3. Run each motor with its controller. Demonstrate correct rotation, alignment, and speed at motor design load.
4. Test interlocks and control and safety features for proper operation.
5. Verify that current and voltage for each phase comply with nameplate rating and NEMA MG 1 tolerances.

- B. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust field-assembled components and equipment installation, including connections, and to assist in field testing. Report results in writing.

- C. Perform the following field tests and inspections and prepare test reports:

1. Perform electrical tests and visual and mechanical inspections including optional tests and inspections stated in NETA ATS on factory- and field-installed motors. Certify compliance with test parameters.
2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.

3.4 DEMONSTRATION AND COMMISSIONING

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain field-installed motors. Refer to Division 01 Section "Demonstration and Training."

1. Train Owner's maintenance personnel on procedures and schedules for starting up and shutting down, troubleshooting, servicing, and maintaining chillers. The training will occur after the startup report has been provided to the owner and the trainer will provide two (2) Installation and Operations manuals for the use of the owners personnel during training.
2. Review data in maintenance manuals. Refer to Division 01 Section "Operation and Maintenance Data." All required and recommended maintenance will be reviewed as well as operational trouble shooting. If the IOM does not include a written trouble shooting guide one will be provided.
3. Schedule training with Owner, through Architect, with at least seven days' advance notice.

- B. Demonstrate proper operation of equipment to commissioning agent or designated owners personnel. The scope of the demonstration will include functional performance requirements

under both local and building automation control, as well as any commissioning requirements in Division 01 or 23.

END OF SECTION 230513

SECTION 230517 - SLEEVES AND SLEEVE SEALS FOR HVAC PIPING**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Sleeves.
2. Sleeve-seal systems.
3. Grout.

1.2 ACTION SUBMITTALS**A. Product Data:** For each type of product indicated.**PART 2 - PRODUCTS****2.1 SLEEVES**

- A. Cast-Iron Wall Pipes: Cast or fabricated of cast or ductile iron and equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop unless otherwise indicated.
- B. Galvanized-Steel Wall Pipes: ASTM A 53/A 53M, Schedule 40, with plain ends and welded steel collar; zinc coated.
- C. Galvanized-Steel-Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, zinc coated, with plain ends.
- D. PVC-Pipe Sleeves: ASTM D 1785, Schedule 40.
- E. Galvanized-Steel-Sheet Sleeves: 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint.

2.2 SLEEVE-SEAL SYSTEMS

- A. Description: Modular sealing-element unit, designed for field assembly, for filling annular space between piping and sleeve.
 1. Sealing Elements: EPDM-rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
 2. Pressure Plates: Plastic.
 3. Connecting Bolts and Nuts: Carbon steel, with corrosion-resistant coating, of length required to secure pressure plates to sealing elements.

2.3 GROUT

- A. Standard: ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.

- B. Characteristics: Nonshrink; recommended for interior and exterior applications.
- C. Design Mix: 5000-psi, 28-day compressive strength.
- D. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION

- A. Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.
- B. For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide 1-inch annular clear space between piping and concrete slabs and walls.
 - 1. Sleeves are not required for core-drilled holes.
- C. Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.
 - 1. Cut sleeves to length for mounting flush with both surfaces.
 - a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level.
 - 2. Using grout, seal the space outside of sleeves in slabs and walls without sleeve-seal system.
- D. Install sleeves for pipes passing through interior partitions.
 - 1. Cut sleeves to length for mounting flush with both surfaces.
 - 2. Install sleeves that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
 - 3. Seal annular space between sleeve and piping or piping insulation; use joint sealants appropriate for size, depth, and location of joint. Comply with requirements for sealants specified in Section 079200 "Joint Sealants."
- E. Fire-Barrier Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with firestop materials. Comply with requirements for firestopping specified in Section 078413 "Penetration Firestopping."

3.2 SLEEVE-SEAL-SYSTEM INSTALLATION

- A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at service piping entries into building.
- B. Select type, size, and number of sealing elements required for piping material and size and for sleeve ID or hole size. Position piping in center of sleeve. Center piping in penetration, assemble sleeve-seal system components, and install in annular space between piping and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make a watertight seal.

3.3 SLEEVE AND SLEEVE-SEAL SCHEDULE

A. Use sleeves and sleeve seals for the following piping-penetration applications:

1. Exterior Concrete Walls above Grade:
 - a. Piping Smaller Than NPS 6: Galvanized-steel wall sleeves.
 - b. Piping NPS 6 and Larger: Galvanized-steel wall sleeves.
2. Exterior Concrete Walls below Grade:
 - a. Piping Smaller Than NPS 6: Galvanized-steel wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.
 - b. Piping NPS 6 and Larger: Galvanized-steel wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.
3. Concrete Slabs-on-Grade:
 - a. Piping Smaller Than NPS 6: Galvanized-steel wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.
 - b. Piping NPS 6 and Larger: Galvanized-steel wall sleeves with sleeve-seal system.
 - 1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.
4. Concrete Slabs above Grade:
 - a. Piping Smaller Than NPS 6: Galvanized-steel-pipe sleeves.
 - b. Piping NPS 6 and Larger: Galvanized-steel-pipe sleeves.
5. Interior Partitions:
 - a. Piping Smaller Than NPS 6: Galvanized-steel-pipe sleeves.
 - b. Piping NPS 6 and Larger: Galvanized-steel-sheet sleeves.

END OF SECTION 230517

SECTION 230518 - ESCUTCHEONS FOR HVAC PIPING**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Escutcheons.
2. Floor plates.

1.2 ACTION SUBMITTALS**A. Product Data:** For each type of product indicated.**PART 2 - PRODUCTS****2.1 ESCUTCHEONS**

- A. One-Piece, Cast-Brass Type: With polished, chrome-plated finish and setscrew fastener.
- B. One-Piece, Deep-Pattern Type: Deep-drawn, box-shaped brass with chrome-plated finish and spring-clip fasteners.
- C. One-Piece, Stamped-Steel Type: With chrome-plated finish and spring-clip fasteners.

2.2 FLOOR PLATES

- A. One-Piece Floor Plates: Cast-iron flange.

PART 3 - EXECUTION**3.2 INSTALLATION**

- A. Install escutcheons for piping penetrations of walls, ceilings, and finished floors.
- B. Install escutcheons with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.
 1. Escutcheons for New Piping:
 - a. Piping with Fitting or Sleeve Protruding from Wall: One-piece, deep-pattern type.

- b. Chrome-Plated Piping: One-piece, cast-brass type with polished, chrome-plated finish.
 - c. Insulated Piping: One-piece, stamped-steel type.
 - d. Bare Piping at Wall and Floor Penetrations in Finished Spaces: One-piece, cast- brass type with polished, chrome-plated finish.
 - e. Bare Piping at Wall and Floor Penetrations in Finished Spaces: One-piece, stamped-steel type.
 - f. Bare Piping at Ceiling Penetrations in Finished Spaces: One-piece, cast-brass type with polished, chrome-plated finish.
 - g. Bare Piping at Ceiling Penetrations in Finished Spaces: One-piece, stamped-steel type.
 - h. Bare Piping in Unfinished Service Spaces: One-piece, cast-brass type with polished, chrome-plated finish.
 - i. Bare Piping in Unfinished Service Spaces: One-piece, stamped-steel type.
 - j. Bare Piping in Equipment Rooms: One-piece, cast-brass type with polished, chrome-plated finish.
 - k. Bare Piping in Equipment Rooms: One-piece, stamped-steel type.
- C. Install floor plates for piping penetrations of equipment-room floors.
- D. Install floor plates with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.
1. New Piping: One-piece, floor-plate type.

3.3 FIELD QUALITY CONTROL

- A. Replace broken and damaged escutcheons and floor plates using new materials.

END OF SECTION 230518

SECTION 230529 - HANGERS AND SUPPORTS FOR PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the following hangers and supports for mechanical system piping and equipment:
 - 1. Steel pipe hangers and supports.
 - 2. Trapeze pipe hangers.
 - 3. Metal framing systems.
 - 4. Thermal-hanger shield inserts.
 - 5. Fastener systems.
 - 6. Rooftop Pipe Supports.
 - 7. Pipe positioning systems.

1.2 DEFINITIONS

- A. MSS: Manufacturers Standardization Society for the Valve and Fittings Industry Inc.
- B. Terminology: As defined in MSS SP-90, "Guidelines on Terminology for Pipe Hangers and Supports."

1.3 SUBMITTALS

- A. Product Data: For the following:
 - 1. Steel pipe hangers and supports.
 - 2. Thermal-hanger shield inserts.
 - 3. Powder-actuated fastener systems.
 - 4. Pipe positioning systems.
- B. Shop Drawings: Signed and sealed by a qualified professional engineer. Show fabrication and installation details and include calculations for the following:
 - 1. Trapeze pipe hangers. Include Product Data for components.
 - 2. Metal framing systems. Include Product Data for components.
 - 3. Rooftop Pipe Supports. Include Product Data for components.
 - 4. Equipment supports.
- C. Welding certificates.

1.4 QUALITY ASSURANCE

- A. Welding: Qualify procedures and personnel according to AWS D1.1, "Structural Welding Code-Steel." and ASME Boiler and Pressure Vessel Code: Section IX.

1.5 PERFORMANCE REQUIREMENTS

- A. Design supports for multiple pipes, including pipe stands, capable of supporting combined weight of supported systems, system contents, and test water.

1.6 WARRANTY

- A. Provide warranty on materials and labor for 18 months starting from date of delivery, or one year from date of substantial completion, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following manufacturers:

1. Steel Pipe Hangers and Supports:
 - a. Anvil
 - b. B-Line Systems, Inc.; a division of Cooper Industries.
 - c. Carpenter & Paterson, Inc.
2. Powder-Actuated Fasteners:
 - a. Hilti, Inc.
 - b. ITW Ramset/Red Head.
3. Mechanical-Expansion Anchors:
 - a. B-Line Systems, Inc.; a division of Cooper Industries.
 - b. Hilti, Inc.
 - c. ITW Ramset/Red Head.
4. Rooftop Pipe Supports:
 - a. Pate
 - b. RPS
 - c. Thybar
5. Thermal-Hanger Shield Inserts:
 - a. Carpenter & Paterson, Inc.
 - b. PHS Industries, Inc.
 - c. Pipe Shields, Inc.
6. Pipe Positioning Systems:
 - a. C & S Mfg. Corp.
 - b. HOLDRITE Corp.; Hubbard Enterprises.
 - c. Samco Stamping, Inc.

2.2 STEEL PIPE HANGERS AND SUPPORTS

- A. Description: MSS SP-58, Types 1 through 58, factory-fabricated components. Refer to PART 3 "Hanger and Support Applications" Article for where to use specific hanger and support types.

- B. Corrosion Protection: Hangers and components shall be galvanized or painted with carbo-zinc #11.
- C. Threads: All threads shall be UNC unless otherwise specified.
- D. Heat Transmission: Supports, guides and anchors shall limit the amount of heat transmitted to the structural steel. Temperature of supporting parts shall be based on a 100°F per inch temperature gradient from the outside pipe surface.

2.3 TRAPEZE PIPE HANGERS

- A. Description: MSS SP-69, Type 59, shop- or field-fabricated pipe-support assembly made from structural-steel shapes with MSS SP-58 hanger rods, nuts, saddles, and U-bolts.

2.4 FASTENER SYSTEMS

- A. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.
- B. Mechanical-Expansion Anchors: Insert-wedge-type zinc-coated steel, for use in hardened portland cement concrete with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

2.5 ROOFTOP PIPE SUPPORTS

- A. Description: Provide factory-fabricated steel sheet structural mounting supports. Construction shall include a welded 18-gauge galvanized steel shell, base plate and removable counterflashing. Support shall include a factory-installed wood nailer and internal bulkhead reinforcement. End sections shall be fully mitered. Support shall be compatible with insulated roof decks and include a 3 inch cant and variable step to match deck insulation thickness. The pipe roller assembly shall have galvanized 18 inches long continuous threaded rods to permit 12 inch vertical adjustments and a galvanized removable pipe retainer bracket.

2.6 THERMAL-HANGER SHIELD INSERTS

- A. Description: 100 psig minimum, compressive-strength insulation insert encased in sheet metal shield.
- B. Insulation-Insert Material for Cold Piping: Water-repellent treated, ASTM C 533, Type I calcium silicate with vapor barrier.
- C. Insulation-Insert Material for Hot Piping: Water-repellent treated, ASTM C 533, Type I calcium silicate.
- D. For Trapeze or Clamped Systems: Insert and shield shall cover entire circumference of pipe.
- E. For Clevis or Band Hangers: Insert and shield shall cover lower 180 degrees of pipe.
- F. Insert Length: Extend 2 inches beyond sheet metal shield for piping operating below ambient air temperature.

2.7 PIPE POSITIONING SYSTEMS

- A. Description: IAPMO PS 42, system of metal brackets, clips, and straps for positioning piping in pipe spaces for plumbing fixtures for commercial applications.

2.8 MISCELLANEOUS MATERIALS

- A. Structural Steel: ASTM A 36/A 36M, steel plates, shapes, and bars; black and galvanized.
- B. Grout: ASTM C 1107, factory-mixed and -packaged, dry, hydraulic-cement, nonshrink and nonmetallic grout; suitable for interior and exterior applications.
 - 1. Properties: Nonstaining, noncorrosive, and nongaseous.
 - 2. Design Mix: 5000-psi, 28-day compressive strength.

PART 3 - EXECUTION

3.1 HANGER AND SUPPORT APPLICATIONS

- A. Specific hanger and support requirements are specified in Sections specifying piping systems and equipment.
- B. Comply with MSS SP-69 for pipe hanger selections and applications that are not specified in piping system Sections.
- C. Use hangers and supports with galvanized, metallic coatings for piping and equipment that will not have field-applied finish. Hangers installed outdoors shall have two coats of rust inhibitor paint after installation and adjustment.
- D. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.
- E. Use padded hangers for piping that is subject to scratching.
- F. Horizontal-Piping Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
 - 1. Adjustable, Steel Clevis Hangers (MSS Type 1): For suspension of non-insulated or insulated stationary pipes, NPS 1/2 to NPS 30.
 - 2. U-Bolts (MSS Type 24): For support of heavy pipes, NPS 1/2 to NPS 30.
 - 3. Single Pipe Rolls (MSS Type 41): For suspension of pipes, NPS 1 to NPS 30, from 2 rods if longitudinal movement caused by expansion and contraction might occur.
 - 4. Adjustable Roller Hangers (MSS Type 43): For suspension of pipes, NPS 2-1/2 to NPS 20, from single rod if horizontal movement caused by expansion and contraction might occur.
- G. Vertical-Piping Clamps: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
 - 1. Extension Pipe or Riser Clamps (MSS Type 8): For support of pipe risers, NPS 3/4 to NPS 20.

- H. Hanger-Rod Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Steel Turnbuckles (MSS Type 13): For adjustment up to 6 inches for heavy loads.
 2. Steel Clevises (MSS Type 14): For 120 to 450 deg F piping installations.
 3. Swivel Turnbuckles (MSS Type 15): For use with MSS Type 11, split pipe rings.
 4. Malleable-Iron Sockets (MSS Type 16): For attaching hanger rods to various types of building attachments.
 5. Steel Weldless Eye Nuts (MSS Type 17): For 120 to 450 deg F piping installations.
- I. Building Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Steel or Malleable Concrete Inserts (MSS Type 18): For upper attachment to suspend pipe hangers from concrete ceiling.
 2. Side-Beam or Channel Clamps (MSS Type 20): For attaching to bottom flange of beams, channels, or angles.
 3. Center-Beam Clamps (MSS Type 21): For attaching to center of bottom flange of beams.
 4. Welded Beam Attachments (MSS Type 22): For attaching to bottom of beams if loads are considerable and rod sizes are large.
 5. Top-Beam Clamps (MSS Type 25): For top of beams if hanger rod is required tangent to flange edge.
 6. Side-Beam Clamps (MSS Type 27): For bottom of steel I-beams.
 7. Steel-Beam Clamps with Eye Nuts (MSS Type 28): For attaching to bottom of steel I-beams for heavy loads.
 8. Linked-Steel Clamps with Eye Nuts (MSS Type 29): For attaching to bottom of steel I-beams for heavy loads, with link extensions.
 9. Malleable Beam Clamps with Extension Pieces (MSS Type 30): For attaching to structural steel.
 10. Welded-Steel Brackets: For support of pipes from below, or for suspending from above by using clip and rod. Use one of the following for indicated loads:
 - a. Light (MSS Type 31): 750 lb (340 kg).
 - b. Medium (MSS Type 32): 1500 lb (680 kg).
 - c. Heavy (MSS Type 33): 3000 lb (1360 kg).
 11. Side-Beam Brackets (MSS Type 34): For sides of steel or wooden beams.
 12. Plate Lugs (MSS Type 57): For attaching to steel beams if flexibility at beam is required.
 13. Horizontal Travelers (MSS Type 58): For supporting piping systems subject to linear horizontal movement where headroom is limited.
- J. Saddles and Shields: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Steel Pipe-Covering Protection Saddles (MSS Type 39): To fill interior voids with insulation that matches adjoining insulation.
 2. Protection Shields (MSS Type 40): Of length recommended in writing by manufacturer to prevent crushing insulation.
 3. Thermal-Hanger Shield Inserts: For supporting insulated pipe.
- K. Comply with MSS SP-69 for trapeze pipe hanger selections and applications that are not specified in piping system Sections.

- L. Comply with MFMA-102 for metal framing system selections and applications that are not specified in piping system Sections.
- M. Use powder-actuated fasteners or mechanical-expansion anchors instead of building attachments where required in concrete construction.
- N. Use pipe positioning systems in pipe spaces behind plumbing fixtures to support supply and waste piping for plumbing fixtures.

3.2 HANGER AND SUPPORT INSTALLATION

- A. Steel Pipe Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Install hangers, supports, clamps, and attachments as required to properly support piping from building structure.
- B. Trapeze Pipe Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Arrange for grouping of parallel runs of horizontal piping and support together on field-fabricated trapeze pipe hangers.
 - 1. Pipes of Various Sizes: Support together and space trapezes for smallest pipe size or install intermediate supports for smaller diameter pipes as specified above for individual pipe hangers.
 - 2. Field fabricate from ASTM A 36/A 36M, steel shapes selected for loads being supported. Weld steel according to AWS D1.1.
 - 3. Trapeze bars shall be tightly secured to structural members at two points with bolts or other similar mechanical fasteners. Hangers from bar joist and fabricated truss members shall be located at the panel points of the structural members. C-clamp type hangers attached to one side of double-angle bottom members are not allowed. Point loads shall not exceed the lesser of:
 - a. Manufacturer's certified recommendation for the component parts.
 - b. The following maximum point loads, and maximum hanger spacings as herein specified, for structural elements in any direction; except as specifically approved by the Structural Engineer of Record:

Structural Element Type	Maximum Hanger Point Load (lb)
Metal deck without concrete topping	50
Composite metal deck slab with concrete topping	50
Steel Beams:	
All channels, W4 through W8	100
W10 through W14	200
W16 through W24	400
W27 through W36	750
Built-up structural steel trusses	250

Reinforced post-tensioned concrete elements

Slabs up to 6 inches thick	150
Slabs over 6 inches thick	250
Joists 8 inches wide	250
Beam/girders 8 inches wide	500

C. Fastener System Installation:

1. Drive screws, pins, studs, etc., of the type which are secured in place by means of explosive force may be used as a means of securing any of the hangers subject to the following:
 - a. The stud, pin or fastener shall be caused to have a velocity not in excess of 300 feet per second when measured 6-1/2 feet from the muzzle of the tool by accepted ballistic test methods.
 - b. Only workmen qualified by instructions of the manufacturers representative and/or licensed by the state and local authorities shall be assigned to use a powder actuated fastening tool.
 - c. Where practical, tools of only one manufacturer shall be used on a project.
 - d. Only cartridges and fasteners supplied by the manufacturer of the tool shall be used to operate that tool.
 - e. Powder actuated fastening tools shall be handled with the same care as firearms.
 - f. All safety devices incorporated in the tool by the manufacturer shall be used at all times.
 - g. Acceptable types of powder actuated fastening tools are:
 - 1) Piston Tool - Low Velocity Type - is a tool utilizing a piston, activated by the power of a blank cartridge furnished by the manufacturer for use with it, to drive a stud, pin or fastener into a work surface.
 - 2) Powder Assisted Hammer Drive Tool - Low Velocity Type - is a tool utilizing a captive piston, activated by a blow from a 4 lb. hammer supplemented by the power of a blank cartridge furnished by the manufacturer for use with the tool, to drive a stud, pin or fastener into a work surface.
2. Install mechanical-expansion anchors in concrete after concrete is placed and completely cured. Install fasteners according to manufacturer's written instructions.

- D. Pipe Positioning System Installation: Install support devices to make rigid supply and waste piping connections to each plumbing fixture. Refer to Division 22 Section "Plumbing Fixtures" for plumbing fixtures.
- E. Install hangers and supports complete with necessary inserts, bolts, rods, nuts, washers, and other accessories.
- F. Equipment Support Installation: Fabricate from welded-structural-steel shapes.

- G. Install hangers and supports to allow controlled thermal and seismic movement of piping systems, to permit freedom of movement between pipe anchors, and to facilitate action of expansion joints, expansion loops, expansion bends, and similar units.
- H. Install lateral bracing with pipe hangers and supports to prevent swaying.
- I. Install building attachments within concrete slabs or attach to structural steel. Install additional attachments at concentrated loads, including valves, flanges, and strainers, NPS 2-1/2 and larger and at changes in direction of piping. Install concrete inserts before concrete is placed; fasten inserts to forms and install reinforcing bars through openings at top of inserts.
- J. Load Distribution: Install hangers and supports so piping live and dead loads and stresses from movement will not be transmitted to connected equipment.
- K. Pipe Slopes: Install hangers and supports to provide indicated pipe slopes and so maximum pipe deflections allowed by ASME B31.1 (for power piping) and ASME B31.9 (for building services piping) are not exceeded.
- L. Insulated Piping: Comply with the following:
 - 1. Attach clamps and spacers to piping.
 - a. Piping Operating above Ambient Air Temperature: Clamp may project through insulation.
 - b. Piping Operating below Ambient Air Temperature: Use thermal-hanger shield insert with clamp sized to match OD of insert.
 - c. Do not exceed pipe stress limits according to ASME B31.1 for power piping and ASME B31.9 for building services piping.
 - 2. Install MSS SP-58, Type 39, protection saddles if insulation without vapor barrier is indicated. Fill interior voids with insulation that matches adjoining insulation.
 - a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.
 - 3. Install MSS SP-58, Type 40, protective shields on cold piping with vapor barrier. Shields shall span an arc of 180 degrees.
 - a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.
 - 4. Shield Dimensions for Pipe: Not less than the following:
 - a. NPS 1/4 to NPS 3-1/2: 12 inches long and 0.048 inch thick.
 - b. NPS 4: 12 inches long and 0.06 inch thick.
 - c. NPS 5 and NPS 6: 18 inches long and 0.06 inch thick.
 - d. NPS 8 to NPS 14: 24 inches long and 0.075 inch thick.
 - 5. Pipes NPS 8 and Larger: Include wood inserts.
 - 6. Insert Material: Length at least as long as protective shield.
 - 7. Thermal-Hanger Shields: Install with insulation same thickness as piping insulation.

3.3 EQUIPMENT SUPPORTS

- A. Fabricate structural-steel stands to suspend equipment from structure overhead or to support equipment above floor.
- B. Grouting: Place grout under supports for equipment and make smooth bearing surface.
- C. Provide lateral bracing, to prevent swaying, for equipment supports.

3.4 METAL FABRICATIONS

- A. Cut, drill, and fit miscellaneous metal fabrications for trapeze pipe hangers and equipment supports.
- B. Fit exposed connections together to form hairline joints. Field weld connections that cannot be shop welded because of shipping size limitations.
- C. Field Welding: Comply with AWS D1.1 procedures for shielded metal arc welding, appearance and quality of welds, and methods used in correcting welding work, and with the following:
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
 - 4. Finish welds at exposed connections so no roughness shows after finishing and contours of welded surfaces match adjacent contours.

3.5 ADJUSTING

- A. Hanger Adjustments: Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe.
- B. Trim excess length of continuous-thread hanger and support rods to 1-1/2 inches.

3.6 PAINTING

- A. Touch Up: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.
 - 1. Apply paint by brush or spray to provide minimum dry film thickness of 2.0 mils.
 - 2. Use hangers and supports with galvanized, metallic coatings for piping and equipment that will not have field-applied finish. Hangers installed outdoors shall have two coats of rust inhibitor paint after installation and adjustment.
 - 3. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.
- B. Touch Up: Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal are specified in Division 09 Section "Painting."
- C. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

END OF SECTION 230529

SECTION 230548 - VIBRATION CONTROLS FOR HVAC PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 SUMMARY**

- A. This Section includes the following:
 - 1. Vibration isolation for piping, duct and equipment.
 - 2. Equipment isolation bases.
 - 3. Flexible piping connections.
 - 4. Resilient pipe anchors and guides.

1.2 SUBMITTALS

- A. Catalog cut sheets that include rated load, rated deflection, and overload capacity for each vibration isolation device.
- B. Coordination Drawings: Show coordination of seismic bracing for HVAC piping and equipment with other systems and equipment in the vicinity, including other supports.
- C. Submit details of equipment bases including dimensions, structural member sizes and support point locations.
- D. Submit details of isolation hangers for ceiling hung equipment, piping and ductwork.
- E. Submit details of mountings for floor-supported equipment, piping and ductwork.
- F. All hanger, mounting or pad drawings shall indicate deflections and model numbers as well as any other requirements in the specifications.
- G. Spring diameters, rated loads and deflections, heights at rated load and closed height shall be provided for all springs shown in the submittals in tabular form.
- H. Provide complete flexible connector details.

1.3 QUALITY ASSURANCE

- A. The isolation materials manufacturer shall be responsible for the proper selection of spring rates to accomplish the specified minimum static deflections for all spring and pay type isolators based on the weight distribution of equipment to be isolated.
- B. The isolation materials manufacturer shall be responsible for the structural design of steel beam bases and concrete inertia bases to support mechanical equipment scheduled or specified to receive a supplementary base.
- C. Furnish a complete set of final Shop Drawings of all mechanical equipment to receive vibration isolation devices to the vibration isolation materials manufacturer. These Drawings will be the basis upon which the selection of vibration isolators and design of supplementary bases will be completed. The Shop Drawings to be furnished shall include operating weight of the equipment to be isolated and the distribution of weight to the support points.

- D. Furnish a complete layout of piping and ductwork to be isolated, including vertical risers, showing size or weight and support points of the piping system to the vibration isolation materials manufacturer for selection and layout of isolation hangers.

1.4 DELIVERY, STORAGE AND HANDLING

- A. Store equipment in accordance with manufacturer's instructions.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide diffusers by one of the following:

1. Mason Industries.
2. Vibration Eliminator Company.
3. Kinetics Noise Control, Inc.
4. Vibration Mountings and Controls, Inc.

2.2 VIBRATION ISOLATOR SELECTION

- A. Noise and vibration isolator types, minimum operating static deflections, and supplemental bases shall be provided for individual mechanical equipment units according to selection criteria delineated in Table 1 incorporated as part of this specification or as tabulated in the equipment schedules of the project Drawings.
- B. Isolator types are scheduled to establish minimum standards. At the Contractor's option labor saving accessories can be an integral part of isolators supplied to provide initial lift of equipment to operating height, hold piping at fixed elevations during installation and initial system filling operations, and similar installation advantages, provided isolators supplied incorporate and specified isolator type, and do not degrade the noise and vibration isolation of equipment mounted.
- C. Supplemental equipment base types tabulated can be deleted for unitary packaged air handling equipment having a rigid frame and casing providing a distortion free platform for attachment of vibration isolators.
- D. Isolators exposed to the outdoors shall have weather-proof finish on all parts.

2.3 NEOPRENE MOUNTINGS

- A. Neoprene mountings shall have a minimum static deflection of 0.35". All metal surfaces shall be neoprene covered and have friction pads both top and bottom. Bolt holes shall be provided on the bottom and a tapped hole and cap screw on top. Steel rails shall be used above the mountings under equipment such as small vent sets to compensate for the overhang. Mountings shall be type ND or rails type DNR as manufactured by Mason Industries, Inc.

2.4 SPRING ISOLATORS

- A. General: Spring isolators shall be free standing and laterally stable without any housing and complete with a molded neoprene cup or 1/4"(6mm) neoprene acoustical friction pad between the baseplate and the support. All mountings shall have leveling bolts that must be rigidly bolted to the equipment. Installed and operating heights shall be equal. The ratio of the spring

diameter divided by the compressed spring height shall be no less than 0.8. Springs shall have a minimum additional travel to solid equal to 50% of the rated deflection. Submittals shall include spring diameters, deflection, compressed spring height and solid spring height. Mountings shall be type SLF, as manufactured by Mason Industries, Inc.

- B. Equipment with Significant Weight Variations: Equipment with significant variations in the operating and installed weight, and equipment exposed to the wind shall be mounted on spring mountings as described above. Provide a neoprene acoustical pad within a rigid-sided housing with vertical limit stops to prevent spring extension when weight is removed. Also provide temporary steel spacers between the upper and lower housings. Housings shall serve as blocking during erection. When the equipment is at full operating weight, the springs shall be adjusted to assume the weight and the spacers removed, without changing the installed and operating heights. All restraining bolts shall have large rubber grommets to provide cushioning in the vertical as well as horizontal modes. The hole through the bushing shall be a minimum of 0.75" larger in diameter than the restraining bolt. Horizontal clearance on the sides between the spring assembly and the housing shall be a minimum of 0.5" to avoid bumping and interfering with the spring action. Vertical limit stops shall be out of contact during normal operation. When there is no provision for direct mounting, cooling tower mounts are to be located between the supporting steel and the roof, or the grillage and dunnage as shown on the drawings. Housings and springs shall be powder coated and hardware electro-galvanized. Mountings shall be SLR as manufactured by Mason Industries, Inc.

2.5 HANGERS

- A. General: Hangers shall consist of rigid steel frames containing minimum 1-1/4" thick neoprene elements at the top, a steel spring with general characteristics as described in Paragraph 2.4, and shall be seated in a steel washer-reinforced neoprene cup on the bottom. The neoprene element and the cup shall have neoprene bushings projecting through the steel box. In order to maintain stability the boxes shall not be articulated. The neoprene element shall not be stacked on top of the spring. Spring diameters and hanger box lower hole sizes shall be large enough to permit the hanger rod to swing through a 30° arc from side to side before contacting the cup bushing. Submittals shall include a hanger drawing showing the 30° capability. Hangers shall be type 30N as manufactured by Mason Industries, Inc.
- B. Spring Locks: Hangers shall be precompressed and locked at the rated deflection by means of a resilient upstop to keep the piping or equipment at a fixed elevation during installation. The hangers shall be designed with a release mechanism to free the spring after the installation is complete and the hanger is subjected to its full load. Deflection shall be clearly indicated by means of a scale. Submittals shall include a drawing of the hanger showing the 30° capability. Hangers shall be type PC30N as manufactured by Mason Industries, Inc.
- C. Hangers shall be manufactured with minimum characteristics as described in Paragraph 2.4, but without the neoprene element. Springs shall be seated in a steel washer reinforced neoprene cup that has a neoprene bushing projecting through the bottom hole to prevent rod hanger contact. Spring diameters and the lower hole sizes shall be large enough to allow the hanger rod to swing through a 30° arc from side to side before contacting the cup bushing. For ducts suspended by flat strap iron, the hanger assembly shall be modified by the manufacturer with an eye on top of the box and on the bottom of the spring hanger rod to allow for bolting to the hanger straps. Submittals shall include a scale drawing of the hanger showing the 30° capability. Hangers for rods shall be Type 30 or for straps W30 as manufactured by Mason Industries, Inc.

2.6 HORIZONTAL THRUST RESTRAINTS

- A. When total air thrust exceeds 10% of the isolated weight, floor mounted or suspended air handling equipment shall be protected against excessive displacement by the use of horizontal thrust restraints. The restraint shall consist of a modified spring mounting. Restraint springs shall have the same deflection as the isolator springs. The assembly shall be preset at the factory and adjusted in the field to allow for a maximum of 1/4" movement from stop to maximum thrust. The assemblies shall be furnished with rod and angle brackets for attachment to both the equipment and duct work or structure. Attach restraints at the centerline of thrust and symmetrically on both sides of the unit. Horizontal thrust restraints shall be WB as manufactured by Mason Industries, Inc.

2.7 STEEL BASES

- A. General: Vibration isolation manufacturer shall furnish integral structural steel bases. Rectangular bases are standard for all equipment. Centrifugal refrigeration machines and pump bases may require T or L shapes. Pump bases for split case pumps shall be of the dimensions to support suction and discharge elbows. All perimeter members shall be steel beams with a minimum depth equal to 1/10 of the longest dimension of the base. Base depth shall not exceed 14" if the deflection and misalignment is kept within acceptable limits as determined by the manufacturer. Provide height saving brackets in all mounting locations for a base clearance of 1". Bases shall be type WF as manufactured by Mason Industries, Inc.
- B. Partial Bases: Vibration isolation manufacturer shall provide steel members welded to height saving brackets to cradle equipment having legs or bases that do not require a complete supplementary base. Members shall have sufficient rigidity to prevent distortion of equipment. Inverted saddles shall be type ICS, as manufactured by Mason Industries, Inc.
- C. Floating Bases: Manufacturer shall furnish rectangular steel concrete pouring forms for floating concrete bases. Bases for split case pumps shall be of sufficient size to provide support for suction and discharge elbows. Bases shall be a minimum of 1/12 of the longest dimension of the base but not less than 6". The base depth need not exceed 12" unless specifically recommended by the base manufacturer for mass or rigidity. Forms shall include minimum concrete reinforcing consisting of 1/2" bars welded in place on 6" centers running both ways in a layer 1-1/2" above the bottom. Furnish forms with steel templates to hold the anchor bolt sleeves and anchor bolts while pouring concrete. Employ height-saving brackets in all mounting locations to maintain a 1" clearance below the base. Wooden formed bases are not acceptable. Base shall be type BMK or K as manufactured by Mason Industries, Inc.

2.8 EXPANSION JOINTS

- A. Rubber expansion joints shall be peroxide cured EPDM throughout with Kevlar tire cord reinforcement. Substitutions must have certifiable equal or superior characteristics. The raised face rubber flanges must encase solid steel rings to prevent pull out. Flexible cable wire is not acceptable. Sizes 1-1/2" through 14" shall have a ductile iron external ring between the two spheres. Sizes 16" through 24" may be single sphere. Sizes 3/4" through 2" may have one sphere, bolted threaded flange assemblies and cable retention.
- B. Minimum ratings through 14"(350mm) shall be 250psi at 170°F and 215psi at 250°F., 16" through 24" 180psi at 170°F and 150psi at 250°F. Higher published rated connectors may be used where required.
- C. Safety factors shall be a minimum of 3/1. All expansion joints must be factory tested to 150% of maximum pressure for 12 minutes before shipment.

- D. The piping gap shall be equal to the length of the expansion joint under pressure. Control rods passing through 1/2" thick Neoprene washer bushings large enough to take the thrust at 1000 psi of surface area may be used on unanchored piping where the manufacturer determines the condition exceeds the expansion joint rating without them. Submittals shall include two test reports by independent consultants showing minimum reductions of 20 DB in vibration accelerations and 10 DB in sound pressure levels at typical blade passage frequencies on this or a similar product by the same manufacturer. All expansion joints shall be installed on the equipment side of the shut off valves. Expansion joints shall be SAFEFLEX SFDEJ, SFEJ, SFDCR or SFU and Control Rods CR as manufactured by Mason Industries, Inc.

2.9 SPLIT SEALS

- A. Split Seals consist of pipe halves with minimum 3/4" thick neoprene sponge cemented to the inner faces. The seal shall be tightened around the pipe to eliminate clearance between the inner sponge face and the piping. Concrete may be packed around the seal to make it integral with the floor, wall or ceiling if the seal is not in place prior to the construction of the building member. Seals shall project a minimum of 1" past either face of the wall. Where temperatures exceed 240°F, 10 lb. density fiberglass may be used in lieu of the sponge. Seals shall be Type SWS as manufactured by Mason Industries, Inc.

2.10 ACOUSTICAL PIPE ANCHORS

- A. All directional acoustical pipe anchors, consist of two sizes of steel tubing separated by a minimum 1/2" thickness of 60 duro or softer neoprene. Vertical restraint shall be provided by similar material arranged to prevent up or down vertical travel. Allowable loads on the isolation material shall not exceed 500 psi and the design shall be balanced for equal resistance in any direction. All directional anchors shall be type ADA as manufactured by Mason Industries, Inc.

2.11 PIPE GUIDES

- A. General: Pipe guides shall consist of a telescopic arrangement of two sizes of steel tubing separated by a minimum 1/2" thickness of 60 durometer or softer neoprene. The height of the guides shall be preset with a shear pin to allow vertical motion due to pipe expansion or contraction. Guides shall be capable of $\pm 1\frac{5}{8}$ " motion, or to meet location requirements. Pipe guides shall be type VSG as manufactured by Mason Industries, Inc.

2.12 HORIZONTAL PIPE ISOLATION

- A. The first four pipe hangers in the main lines near the mechanical equipment shall be as described Paragraph 2.5B. Hangers supporting piping 2" and larger in all other locations throughout the building shall be isolated by hangers as described in Paragraph 2.5A. Floor supported piping shall rest on isolators as described in Paragraph 2.5B. Heat exchangers and expansion tanks are considered part of the piping run. The first four isolators from the isolated equipment shall have the same static deflection as specified for the mountings under the connected equipment. If piping is connected to equipment located in basements and hangs from ceilings under occupied spaces the first four hangers shall have 0.75" deflection for pipe sizes up to and including 3", 1-1/2" deflection for pipe sizes over 3" and up to and including 6", and 2-1/2" deflection thereafter. Where piping connects to mechanical equipment install expansion joints as shown in Paragraph 2.10 or stainless hoses as shown in Paragraph 2.11. All piping passing through the equipment walls, floors or ceilings shall be protected against sound leakage by means of an acoustical seal, as described in Paragraph 2.12.

2.13 VERTICAL PIPE ISOLATION

- A. Support vertical risers by spring isolators, designed to support the riser filled with water, if it is a water line. Assigned loads must be within the building design limits at the support points. Neutral central resilient anchors close to the center of the run shall direct movement up and down. The anchors shall be capable of holding an upward force equal to the water weight when the system is drained. If one level cannot accommodate this force, anchors can be located on 2 or 3 adjacent floors. Resilient guides shall be spaced and sized properly depending on the pipe diameter. Submittals must include the initial load, initial deflection, change in deflection, final load and change in load at all spring and anchor support locations, as well as guide spacing. The initial spring deflection shall be a minimum of 0.75", or four times the thermal movement at the isolator location, whichever is greater. Calculations shall include pipe stress at end conditions and branch off locations and the manufacturer must include installation instructions. Submittal must be stamped and signed by a licensed professional engineer in the employ of the vibration vendor for at least 5 years. Proper provision shall be made for seismic protection in seismic zones. The isolator manufacturer shall be the same firm supplying the mechanical contract. Support spring mountings, anchors, and telescoping guides shall be in accordance with previous paragraphs of this section.

2.14 DUCT ISOLATION

- A. All air ducts with a cross section of 2 square feet or larger shall be isolated from the building structure by hangers as shown in Paragraph 2.5C or floor supports with a minimum deflection of 0.75". Isolators shall continue for 50' from the equipment. If air velocity exceeds 1000 fpm, hangers or supports shall continue for an additional 50' or as shown on the drawings.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and equipment to receive vibration isolation and seismic[- and wind]-control devices for compliance with requirements for installation tolerances and other conditions affecting performance.
- B. Examine roughing-in of reinforcement and cast-in-place anchors to verify actual locations before installation.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 GENERAL

- A. All vibration isolators must be installed in strict accordance with the manufacturers written instructions and all certified submittal data.
- B. Installation of vibration isolators must not cause any change of position of equipment, piping or duct work resulting in stresses or misalignment.
- C. No rigid connections between equipment and the building structure shall be made that degrades the noise and vibration control system herein specified.
- D. The contractor shall not install any equipment, piping, duct or conduit which makes rigid connections with the building. This includes, but is not limited to, slabs, beams, columns, studs and walls.

- E. Coordinate work with other trades to avoid rigid contact with the building.
- F. Any conflicts with other trades which will result in rigid contact with equipment or piping due to inadequate space or other unforeseen conditions should be brought to the architects/engineers attention prior to installation. Corrective work necessitated by conflicts after installation shall be at the responsible contractors expense.
- G. Bring to the architects/engineers attention any discrepancies between the specifications and the field conditions or changes required due to specific equipment selection, prior to installation. Corrective work necessitated by discrepancies after installation shall be at the responsible contractors expense.
- H. Correct, at no additional cost, all installations which are deemed defective in workmanship and materials at the contractors expense.
- I. Hand built elastomeric expansion joints may be used when pipe sizes exceed 24" or specified movements exceed specified capabilities.
- J. Where piping passes through walls, floors or ceilings the vibration isolation manufacturer shall provide seals as described in this specification.
- K. Locate isolation hangers as near to the overhead support structure as possible.
- L. Air handling equipment and centrifugal fans shall be protected against excessive displacement which results from high air thrust when thrust forces exceed 10% of the equipment weight. Horizontal thrust restraints shall be in accordance with this specification.
- M. Rooftop equipment isolators must be bolted to the equipment and structure. Mountings must be designed to resist 100 mph wind loads.

3.3 VIBRATION ISOLATION OF PIPING

- A. Horizontal pipe shall be installed in accordance with 2.15 Horizontal Pipe Isolation.
- B. Risers shall be installed in accordance with 2.16 Riser Isolation.

3.4 VIBRATION ISOLATION OF DUCTWORK

- A. All duct runs shall be installed in accordance with 2.17 Duct Isolation.

3.5 ISOLATOR SCHEDULE

Equipment	Basement Slab Or Slab On Grade		Upper Floors	
	<i>Paragraph</i>	<i>Min Static Deflect</i>	<i>Paragraph</i>	<i>Min Static Deflect</i>
Cabinet Fan				
Section High				
Pressure				
(5" & Above)				
Low Pressure				
(Less than 5")				
Boilers				

	Basement Slab Or Slab On Grade		Upper Floors	
Pumps				
Close Coupled				
Thru 5HP				
7-1/2HP & Over				
Rooftop Units		-	2.4B	0.75
Suspended				
Centrifugal	2.4A, 2.7C,	2.5	2.4A, 2.7C,	2.5
Blowers	2.11		2.11	
15HP & Less	2.4A, 2.7C,	1.5	2.4A, 2.7C,	2.5
	2.11		2.11	
Condensing	2.4A, 2.7C	1.0	2.4A, 2.7C	3.0
Units				
Power Roof Exhausters Roof				
Mounted				
	2.5A, 2.6	1.0	2.5A, 2.6	1.0
	2.3, 2.6	0.35	2.4A, 2.6	1.0
	2.3, 2.7C, 2.6	0.35	2.4A, 2.7C, 2.6	1.0
			2.4B	2.0
			2.8	1.0

END OF SECTION 230548

SECTION 230553 - IDENTIFICATION FOR HVAC PIPING AND EQUIPMENT**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Equipment labels.
2. Warning signs and labels.
3. Pipe labels.
4. Valve tags.
5. Warning tags.
6. Sample Schedules.

1.2 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Samples: For color, letter style, and graphic representation required for each identification material and device.
- C. Equipment Label Schedule: Include a listing of all equipment to be labeled with the proposed content for each label. An electronic copy and hard copy will be provided to THE OWNER before preliminary acceptance. Furnish extra copies (in addition to mounted copies) for Maintenance.
- D. Valve numbering scheme.
- E. Valve Schedules: For each piping system to include in maintenance manuals. An electronic copy and hard copy will be provided to the Owner before preliminary acceptance. Furnish extra copies (in addition to mounted copies) for Maintenance.

1.3 QUALITY ASSURANCE

- A. Follow manufacturer's recommended installation procedures.
- B. As applicable, comply with ASME A13.1 "Scheme for the Identification of Piping System"

1.4 DELIVERY, STORAGE AND HANDLING

- A. Store materials in a dry and secure area on-site and protect against dirt and moisture damage
- B. Do not apply or install damaged materials.

1.5 COORDINATION

- A. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.
- B. Coordinate installation of identifying devices with locations of access panels and doors.
- C. Install identifying devices before installing acoustical ceilings and similar concealment.

1.6 WARRANTY

- A. Provide warranty on materials and labor for 18 months starting from date of delivery, or one year from date of substantial completion, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide products by one of the following manufacturers:
 - 1. Seton
 - 2. Brady
 - 3. Best
 - 4. Milwaukee

2.2 EQUIPMENT LABELS

- A. Metal Labels for Equipment:
 - 1. Material and Thickness: Brass, 0.032-inch or Stainless steel, 0.025-inch minimum thickness, and having predrilled or stamped holes for attachment hardware.
 - 2. Minimum Letter Size: 4 by 2 inch. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
 - 3. Fasteners: Stainless-steel self-tapping screws.
 - 4. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.
- B. Label Content: Include equipment's Drawing designation or unique equipment number, Drawing numbers where equipment is indicated (plans, details, and schedules), plus the Specification Section number and title where equipment is specified.
- C. Equipment Label Schedule: For each item of equipment to be labeled, on 8-1/2-by-11-inch bond paper. Tabulate equipment identification number and identify Drawing numbers where equipment is indicated (plans, details, and schedules), plus the Specification Section number and title where equipment is specified. Equipment schedule shall be included in operation and maintenance data. See Sample Equipment Schedule at end of this Specification.

2.3 WARNING SIGNS AND LABELS

- A. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, and having predrilled holes for attachment hardware.
- B. Letter Color: Black.
- C. Background Color: Yellow.
- D. Maximum Temperature: Able to withstand temperatures up to 160 deg F.
- E. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

- F. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
- G. Fasteners: Stainless-steel self-tapping screws.
- H. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.
- I. Label Content: Include caution and warning information, plus emergency notification instructions. Coordinate with Architect and Owner for specific warning label information to be included.

2.4 PIPE LABELS

- A. General Requirements for Manufactured Pipe Labels: Preprinted, color-coded, with lettering indicating service, and showing flow direction. Do not provide labels for bare pipes conveying fluids at temperatures of 125 °F or higher.
- B. Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.
- C. Pipe Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings, pipe size, and an arrow indicating flow direction.
 - 1. Flow-Direction Arrows: Integral with piping system service lettering to accommodate both directions, or as separate unit on each pipe label to indicate flow direction.
 - 2. Lettering Size: At least 1-1/2 inches high.

2.5 VALVE TAGS

- A. Valve Tags: Stamped or engraved with 1/4-inch letters for piping system abbreviation and 1/2-inch numbers.
 - 1. Tag Material: Brass, 0.032-inch or Stainless steel, 0.025-inch minimum thickness, and having predrilled or stamped holes for attachment hardware.
 - 2. Fasteners: Brass beaded chain.
- B. Valve Schedules: For each piping system, on 8-1/2-by-11-inch bond paper. Tabulate valve number, piping system, system abbreviation (as shown on valve tag), location of valve (room or space), normal-operating position (open, closed, or modulating), and variations for identification. Mark valves for emergency shutoff and similar special uses.
 - 1. Valve-tag schedule shall be included in operation and maintenance data. See Sample Valve Tag Schedule at the end of this Specification.

2.6 WARNING TAGS

- A. Warning Tags: Preprinted or partially preprinted, accident-prevention tags, of plasticized card stock with matte finish suitable for writing.
 - 1. Size: 4 by 7 inches.
 - 2. Fasteners: Reinforced grommet and wire.
 - 3. Nomenclature: Large-size primary caption such as "DANGER," "CAUTION," or "DO NOT OPERATE."

4. Color: Yellow background with black lettering.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean piping and equipment surfaces of substances that could impair bond of identification devices, including dirt, oil, grease, release agents, and incompatible primers, paints, and encapsulants.
- B. Identification for piping, duct or equipment in sales floor should be installed on surfaces away from direct view of customers.

3.2 EQUIPMENT LABEL INSTALLATION

- A. Install or permanently fasten labels on each major item of mechanical equipment.
- B. Locate equipment labels where accessible and visible.

3.3 PIPE LABEL INSTALLATION

- A. Piping Color-Coding: Painting of piping is specified in Division 09 Section " Painting"
- B. Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:
 1. Near each valve and control device.
 2. Near each branch connection, excluding short takeoffs for fixtures and terminal units. Where flow pattern is not obvious, mark each pipe at branch.
 3. Near penetrations through walls, floors, ceilings, and inaccessible enclosures.
 4. At access doors, manholes, and similar access points that permit view of concealed piping.
 5. Near major equipment items and other points of origination and termination.
 6. Spaced at maximum intervals of 25 feet along each run. Reduce intervals to 15 feet in areas of congested piping and equipment.
 7. On piping above removable acoustical ceilings. Omit intermediately spaced labels.
- C. Pipe Label Color Schedule:
 1. Refrigerant Piping:
 - a. Background Color: White.
 - b. Letter Color: Black.

3.4 DUCT LABEL INSTALLATION

- A. Install self-adhesive duct labels with permanent adhesive on air ducts in the following color codes:
 1. Blue: For cold-air supply ducts.
 2. Yellow: For hot-air supply ducts.
 3. Green: For exhaust-, outside-, relief-, return-, and mixed-air ducts.

- B. Locate labels near points where ducts enter into concealed spaces and at maximum intervals of 50 feet in each space where ducts are exposed or concealed by removable ceiling system.

3.5 VALVE-TAG INSTALLATION

INSTALL TAGS ON VALVES AND CONTROL DEVICES IN PIPING SYSTEMS, EXCEPT CHECK VALVES; VALVES WITHIN FACTORY-FABRICATED EQUIPMENT UNITS; SHUTOFF VALVES; FAUCETS; CONVENIENCE AND LAWN-WATERING HOSE CONNECTIONS; AND HVAC TERMINAL DEVICES AND SIMILAR ROUGHING-IN CONNECTIONS OF END-USE FIXTURES AND UNITS. LIST TAGGED VALVES IN A VALVE SCHEDULE. SEE SAMPLE VALVE TAG SCHEDULE AT END OF THIS SPECIFICATION.

- A. Valve-Tag Application Schedule: Tag valves according to size, shape, and color scheme and with captions similar to those indicated in the following subparagraphs:
1. Valve-Tag Size and Shape:
 - a. Refrigerant: 2 inches, round..
 - b. Gas: 2 inches, round..
 2. Valve-Tag Color:
 - a. Refrigerant: Natural.
 - b. Gas: Natural.
 3. Letter Color:
 - a. Refrigerant: Black.
 - b. Gas: Black.

3.6 WARNING-TAG INSTALLATION

- A. Write required message on, and attach warning tags to, equipment and other items where required.

3.7 SAMPLE SCHEDULES

- A. The following tables are examples of schedules required to be submitted by the Contractor. Examples of the kinds of information to be included are in the first row in these tables.

EQUIPMENT LABEL SCHEDULE				
EQUIPMENT IDENTIFICATION	EQUIPMENT LOCATION	SPECIFICATION SECTION		REMARKS
		NUMBER	TITLE	
AHU-xx	MECHANICAL ROOM M-xxx	23 xx xx		MODULAR INDOOR AIR-HANDLING UNITS

3.8 CLEANING

- A. Follow manufacturer's instructions for surface preparation prior to application or installation.

- B. Follow Manufacturer's instructions for cleaning labels and tags.

END OF SECTION 230553

SECTION 230593 - TESTING, ADJUSTING, AND BALANCING FOR HVAC**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes procedural requirements for testing, adjusting, and balancing (TAB) of new and existing systems to achieve the required flows within the limits of the fan and the motor HP. The testing, adjusting, and balancing work includes producing design objectives for the following:
 - 1. Air Systems:
 - a. Constant-volume air systems.
 - b. VRF systems
 - c. Roof Top Units
 - 2. Hydronic Piping Systems:
 - a. Condensing Units.
 - 3. HVAC equipment quantitative-performance settings.

1.2 DEFINITIONS

- A. AABC: Associated Air Balance Council.
- B. NEBB: National Environmental Balancing Bureau.
- C. TAB: Testing, Adjusting, and Balancing.
- D. TABB: Testing, Adjusting, and Balancing Bureau.
- E. TAB Firm: Entity responsible for performing and reporting TAB procedures.
- F. TAB Specialist: Entity engaged by TAB Firm to perform TAB work.

1.3 SUBMITTALS

- A. Qualification Data: Within **30** days of Notice to Proceed, submit documentation for the TAB contractor and the Project's TAB team members.
- B. Contract Documents Examination Report: Within **30** days of Notice to Proceed, submit the Contract Documents Examination Report.
- C. TAB Plan: Within **30** days of Notice to Proceed, submit TAB strategies and step-by-step procedures as specified in "Preparation" Article.
- D. Certified TAB reports. Submit two copies of the TAB reports prepared as specified in this specification on approved forms certified by the TAB contractor.
- E. Sample report forms. Submit two copies of the sample TAB report forms
- F. Instrument calibration reports, including the following:

1. Instrument type and make.
2. Serial number.
3. Application.
4. Dates of use.
5. Dates of calibration.

1.4 QUALITY ASSURANCE

- A. TAB Contractor Qualifications: Engage a TAB entity certified by AABC, NEBB, or TABB.
 1. TAB Field Supervisor: Employee of the TAB contractor and certified by AABC, NEBB or TABB.
 2. TAB Technician: Employee of the TAB contractor and who is certified by AABC, NEBB or TABB as a TAB technician.
 3. Agenda Items:
 - a. The Contract Documents examination report.
 - b. The TAB plan.
 - c. Coordination and cooperation of trades and subcontractors.
 - d. Coordination of documentation and communication flow.
 - e. Submittal distribution requirements.
 - f. Work Schedule and Project-Site requirements.
- B. Certification of TAB Reports: Certify TAB field data reports and perform the following:
 1. Review field data reports to validate accuracy of data and to prepare certified TAB reports.
 2. Certify that the TAB team complied with the approved TAB plan and the procedures specified and referenced in this Specification.
- C. TAB Report Forms: Use standard TAB forms from NEBB, AABC or TABB as well as providing any additional information required by this specification.
- D. Instrumentation Type, Quantity, Accuracy, and Calibration: As described in ASHRAE 111, Section 5, "Instrumentation."
- E. Owner reserves the right to select at random 10% of the TAB report data for field verification witnessed by the commission agent. The TAB contractor will be given sufficient notice of the date of field verification. The same instruments that were used when the original test was recorded shall be used. A failure of more than 10% of the selected field verification items shall result in a repeat of the testing of the entire system at the TAB contractor's expense. The repeated work is also subject to field verification.

1.5 COORDINATION

- A. Coordinate the efforts of factory-authorized service representatives for systems and equipment, HVAC controls installers, and other mechanics to operate HVAC systems and equipment to support and assist TAB activities.
- B. Notice: Provide seven days' advance notice for each test. Include scheduled test dates and times.
- C. Perform TAB after leakage and pressure tests on air and water distribution systems have been satisfactorily completed.

1.6 WARRANTY

A. Provide warranty in accordance with AABC, NEBB or TABB standards:

1. National Project Performance Guarantee: Provide a guarantee on AABC's "National Standards for Testing and Balancing Heating, Ventilating, and Air Conditioning Systems" forms stating that AABC will assist in completing requirements of the Contract Documents if TAB firm fails to comply with the Contract Documents.
2. Special Guarantee: Provide a guarantee on NEBB forms stating that NEBB will assist in completing requirements of the Contract Documents if TAB firm fails to comply with the Contract Documents.
3. Guarantee shall include the following provisions:
 - a. The certified TAB firm has tested and balanced systems according to the Contract Documents.
 - b. Systems are balanced to optimum performance capabilities within design and installation limits.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine the Contract Documents and field conditions to become familiar with Project requirements and to discover conditions that may preclude proper TAB of systems and equipment.
- B. Examine the approved submittals for HVAC systems and equipment.
- C. Examine design data including HVAC system descriptions, statements of design assumptions for environmental conditions and systems' output, and statements of philosophies and assumptions about HVAC system and equipment controls.
- D. Examine ceiling plenums used for return air to verify that they meet the leakage class of connected ducts as specified in Division 23 Section "Metal Ducts," and are properly separated from adjacent areas. Verify that penetrations in plenum walls are properly sealed or, as required, fire-stopped.
- E. Examine system and equipment installations and verify that field quality-control testing, cleaning, and adjusting specified in individual Sections have been performed.
- F. Examine test reports specified in individual system and equipment Sections.
- G. Examine HVAC equipment and filters and verify that bearings are greased, belts are aligned and tight, and equipment with functioning controls is ready for operation.
- H. Examine heat-transfer coils for correct piping connections and for clean and straight fins.
- I. Examine equipment for installation and properly operating safety interlocks and controls.

- J. Report deficiencies discovered before and during performance of TAB procedures. Observe and record system reactions to changes in conditions. Record default set points if different from indicated values.

3.2 PREPARATION

- A. Prepare a TAB Plan that includes strategies and step-by-step procedures.
- B. Complete system-readiness checks and prepare reports. Verify the following:
 - 1. Permanent electrical-power wiring is complete.
 - 2. Automatic temperature-control systems are operational.
 - 3. Equipment and duct access doors are securely closed.
 - 4. Balance, smoke, and fire dampers are open.
 - 5. Isolating and balancing valves are open and control valves are operational.
 - 6. Ceilings are installed in critical areas where air-pattern adjustments are required and access to balancing devices is provided.
 - 7. Windows and doors can be closed so indicated conditions for system operations can be met.

3.3 GENERAL PROCEDURES FOR TESTING AND BALANCING

- A. Perform testing and balancing procedures on each system according to the procedures contained in AABC's "National Standards for Total System Balance," ASHRAE 111, NEBB's "Procedural Standards for Testing, Adjusting, and Balancing of Environmental Systems," or SMACNA's "HVAC Systems - Testing, Adjusting, and Balancing," and this Section.
 - 1. Comply with requirements in ASHRAE 62.1-2007, Section 7.2.2, "Air Balancing."
- B. Cut insulation, ducts, pipes, and equipment cabinets for installation of test probes to the minimum extent necessary for TAB procedures.
 - 1. After testing and balancing, patch probe holes in ducts with same material and thickness as used to construct ducts.
- C. Mark equipment and balancing devices, including damper-control positions, valve position indicators, fan-speed-control levers, and similar controls and devices, with paint or other suitable, permanent identification material to show final settings.
- D. Take and report testing and balancing measurements in inch-pound (IP) units.
- E. HVAC system lineup. The contractor will test the system in the normal system lineups.
 - 1. A hydronic system with two chillers designed to run at the same time, the system will be balanced with pumps flowing to both chillers. All pump data will also be collected when the pump is flowing to only one chiller.
 - 2. Air systems will be balanced while aligned for minimum outside air flow and the position of the outside air dampers will be recorded for the minimum position.
 - 3. It is not acceptable to balance air systems when the normal boundaries, doors, walls, or ductwork are not in their design configuration.

3.4 GENERAL PROCEDURES FOR BALANCING AIR SYSTEMS

- A. Prepare test reports for both fans and outlets. Obtain manufacturer's outlet factors and recommended testing procedures. Crosscheck the summation of required outlet volumes with required fan volumes.
- B. Prepare schematic diagrams of systems' "as-built" duct layouts.
- C. For variable-air-volume systems, develop a plan to simulate diversity.
- D. Determine the best locations in main and branch ducts for accurate duct-airflow measurements.
- E. Check airflow patterns from the outdoor-air louvers and dampers and the return- and exhaust-air dampers through the supply-fan discharge and mixing dampers.
- F. Locate start-stop and disconnect switches, electrical interlocks, and motor starters.
- G. Verify that motor starters are equipped with properly sized thermal protection.
- H. Check dampers for proper position to achieve desired airflow path.
- I. Check for airflow blockages.
- J. Check condensate drains for proper connections and functioning.
- K. Check for proper sealing of air-handling-unit components.
- L. Check for proper sealing of air duct system.

3.5 PROCEDURES FOR CONSTANT-VOLUME AIR SYSTEMS

- A. Adjust fans to deliver total indicated airflows within the maximum allowable fan speed listed by fan manufacturer and within the limits of the motor HP. All fans shall deliver the design air flow capacity at actual static pressure up to design static. Fan sheaves shall be replaced as necessary to obtain desired results.
 - 1. Measure total airflow.
 - a. Where sufficient space in ducts is unavailable for Pitot-tube-traverse measurements, measure airflow at terminal outlets and inlets and calculate the total airflow.
 - 2. Measure fan static pressures as follows to determine actual static pressure:
 - a. Measure outlet static pressure as far downstream from the fan as practical and upstream from restrictions in ducts such as elbows and transitions.
 - b. Measure static pressure directly at the fan outlet or through the flexible connection.
 - c. Measure inlet static pressure of single-inlet fans in the inlet duct as near the fan as possible, upstream from the flexible connection, and downstream from duct restrictions.
 - d. Measure inlet static pressure of double-inlet fans through the wall of the plenum that houses the fan.

3. Measure static pressure across each component that makes up an air-handling unit, rooftop unit, and other air-handling and -treating equipment.
 - a. Report the cleanliness status of filters and the time static pressures are measured.
 4. Measure static pressures entering and leaving other devices, such as sound traps, heat-recovery equipment, and air washers, under final balanced conditions.
 5. Review Record Documents to determine variations in design static pressures versus actual static pressures. Calculate actual system-effect factors. Recommend adjustments to accommodate actual conditions.
 6. Do not make fan-speed adjustments that result in motor overload. Consult equipment manufacturers about fan-speed safety factors. Modulate dampers and measure fan-motor amperage to ensure that no overload will occur. Measure amperage in full-cooling, full-heating, economizer, and any other operating mode to determine the maximum required brake horsepower.
- B. Adjust volume dampers for main duct, sub-main ducts, and major branch ducts to indicated airflows within specified tolerances.
1. Measure airflow of sub-main and branch ducts.
 - a. Where sufficient space in sub-main and branch ducts is unavailable for Pitot-tube-traverse measurements, measure airflow at terminal outlets and inlets and calculate the total airflow for that zone.
 2. Measure static pressure at a point downstream from the balancing damper, and adjust volume dampers until the proper static pressure is achieved.
 3. Re-measure each sub-main and branch duct after all have been adjusted. Continue to adjust sub-main and branch ducts to indicated airflows within specified tolerances.
- C. Measure air outlets and inlets without making adjustments.
1. Measure terminal outlets using a direct-reading hood or outlet manufacturer's written instructions and calculating factors.
- D. Adjust air outlets and inlets for each space to indicated airflows within specified tolerances of indicated values. Make adjustments using branch volume dampers rather than extractors and the dampers at air terminals.
1. Adjust each outlet in same room or space to within specified tolerances of indicated quantities without generating noise levels above the limitations prescribed by the Contract Documents.
 2. Adjust patterns of adjustable outlets for proper distribution without drafts.
- 3.6 PROCEDURES FOR VARIABLE-AIR-VOLUME SYSTEMS
- A. Compensating for Diversity: When the total airflow of all terminal units is more than the indicated airflow of the fan, place a selected number of terminal units at a minimum set-point airflow with the remainder at maximum-airflow condition until the total airflow of the terminal units equals the indicated airflow of the fan. Select the reduced-airflow terminal units so they are distributed evenly among the branch ducts.
1. Minimum outside air flow rate must be verified under three conditions, with the OA damper position recorded in each configuration.

- a. System operating with all boxes at minimum primary air flow.
 - b. System at maximum supply flow (diversity may not allow all boxes to be open).
 - c. System flow halfway between box minimum and maximum.
- B. Pressure-Independent, Variable-Air-Volume Systems: After the fan systems have been adjusted, adjust the variable-air-volume systems as follows:
1. Set outdoor-air dampers at minimum, and set return- and exhaust-air dampers at a position that simulates full-cooling load.
 2. Select the terminal unit that is most critical to the supply-fan airflow and static pressure. Measure static pressure. Adjust system static pressure so the entering static pressure for the critical terminal unit is not less than the sum of the terminal-unit manufacturer's recommended minimum inlet static pressure plus the static pressure needed to overcome terminal-unit discharge system losses.
 3. Measure total system airflow. Adjust to within indicated airflow.
 4. Set terminal units at maximum airflow and adjust controller or regulator to deliver the designed maximum airflow. Use terminal-unit manufacturer's written instructions to make this adjustment. When total airflow is correct, balance the air outlets downstream from terminal units the same as described for constant-volume air systems.
 5. Set terminal units at minimum airflow and adjust controller or regulator to deliver the designed minimum airflow. Check air outlets for a proportional reduction in airflow the same as described for constant-volume air systems.
 - a. If air outlets are out of balance at minimum airflow, report the condition but leave outlets balanced for maximum airflow.
 6. Re-measure the return airflow to the fan while operating at maximum return airflow and minimum outdoor airflow.
 - a. Adjust the fan and balance the return-air ducts and inlets the same as described for constant-volume air systems.
 7. Measure static pressure at the most critical terminal unit and adjust the static-pressure controller at the main supply-air sensing station to ensure that adequate static pressure is maintained at the most critical unit.
 8. Record final fan-performance data.

3.7 PROCEDURES FOR MOTORS

- A. Motors, 1/2 HP and Larger: Test at final balanced conditions and record the following data:
1. Manufacturer's name, model number, and serial number.
 2. Motor horsepower rating.
 3. Motor rpm.
 4. Efficiency rating.
 5. Nameplate and measured voltage, each phase.
 6. Nameplate and measured amperage, each phase.
 7. Starter thermal-protection-element rating.
- B. Motors Driven by Variable-Frequency Controllers: Test for proper operation at speeds varying from minimum to maximum. Test the manual bypass of the controller to prove proper operation. Record observations including name of controller manufacturer, model number, serial number, and nameplate data.

3.8 TEMPERATURE-CONTROL VERIFICATION

- A. Verify that controllers are calibrated and commissioned.
- B. Check transmitter and controller locations and note conditions that would adversely affect control functions.
- C. Record controller settings and note variances between set points and actual measurements.
- D. Check the operation of limiting controllers (i.e., high- and low-temperature controllers).
- E. Check free travel and proper operation of control devices such as damper and valve operators.
- F. Check the sequence of operation of control devices. Note air pressures and device positions and correlate with airflow and water flow measurements. Note the speed of response to input changes.
- G. Check the interaction of electrically operated switch transducers.
- H. Check the interaction of interlock and lockout systems.
- I. Check main control supply-air pressure and observe compressor and dryer operations.
- J. Record voltages of power supply and controller output. Determine whether the system operates on a grounded or nongrounded power supply.
- K. Note operation of electric actuators using spring return for proper fail-safe operations.

3.9 TOLERANCES

- A. Set HVAC system air flow rates and water flow rates within the following tolerances:
 - 1. Supply, Return, and Exhaust Fans and Equipment with Fans: Plus or minus 10 percent.
 - 2. Air Outlets and Inlets: Plus or minus 10 percent.
 - 3. Heating-Water Flow Rate: Plus or minus 10 percent.
 - 4. Cooling-Water Flow Rate: Plus or minus 5 percent.

3.10 REPORTING

- A. Initial Construction-Phase Report: Based on examination of the Contract Documents as specified in "Examination" Article, prepare a report on the adequacy of design for systems' balancing devices. Recommend changes and additions to systems' balancing devices to facilitate proper performance measuring and balancing. Recommend changes and additions to HVAC systems and general construction to allow access for performance measuring and balancing devices.

3.11 FINAL REPORT

- A. General: Prepare a certified written report; tabulate and divide the report into separate sections for tested systems and balanced systems.
 - 1. Include a certification sheet at the front of the report's binder, signed and sealed by the certified testing and balancing engineer.
 - 2. Include a list of instruments used for procedures, along with proof of calibration.

- B. Final Report Contents: In addition to certified field-report data, include the following:
1. Fan curves.
 2. Manufacturers' test data.
 3. Field test reports prepared by system and equipment installers.
 4. Other information relative to equipment performance; do not include Shop Drawings and product data.
- C. General Report Data: In addition to form titles and entries, include the following data:
1. Title page.
 2. Name and address of the TAB contractor.
 3. Project name.
 4. Project location.
 5. Architect's name and address.
 6. Engineer's name and address.
 7. Contractor's name and address.
 8. Report date.
 9. Signature of TAB supervisor who certifies the report.
 10. Table of Contents with the total number of pages defined for each section of the report. Number each page in the report.
 11. Summary of contents including the following:
 - a. Indicated versus final performance.
 - b. Notable characteristics of systems.
 - c. Description of system operation sequence if it varies from the Contract Documents.
 12. Nomenclature sheets for each item of equipment.
 13. Data for terminal units, including manufacturer's name, type, size, and fittings.
 14. Notes to explain why certain final data in the body of reports vary from indicated values.
 15. Test conditions for fans and pump performance forms including the following:
 - a. Settings for outdoor-, return-, and exhaust-air dampers.
 - b. Conditions of filters.
 - c. Cooling coil, wet- and dry-bulb conditions.
 - d. Face and bypass damper settings at coils.
 - e. Fan drive settings including settings and percentage of maximum pitch diameter.
 - f. Inlet vane settings for variable-air-volume systems.
 - g. Settings for supply-air, static-pressure controller.
 - h. Other system operating conditions that affect performance.
- D. System Diagrams: Include schematic layouts of air and hydronic distribution systems. Present each system with single-line diagram and include the following:
1. Quantities of outdoor, supply, return, and exhaust airflows.
 2. Duct, outlet, and inlet sizes.
 3. Pipe and valve sizes and locations.
 4. Terminal units.
 5. Balancing stations.
 6. Position of balancing devices.
- E. Roof Top Unit Test Reports: For air-handling units with coils, include the following:

1. Unit Data:
 - a. Unit identification.
 - b. Location.
 - c. Make and type.
 - d. Model number and unit size.
 - e. Manufacturer's serial number.
 - f. Unit arrangement and class.
 - g. Discharge arrangement.
 - h. Sheave make, size in inches, and bore.
 - i. Center-to-center dimensions of sheave, and amount of adjustments in inches.
 - j. Number, make, and size of belts.
 - k. Number, type, and size of filters.
2. Motor Data:
 - a. Motor make, and frame type and size.
 - b. Horsepower and rpm.
 - c. Volts, phase, and hertz.
 - d. Full-load amperage and service factor.
 - e. Sheave make, size in inches, and bore.
 - f. Center-to-center dimensions of sheave, and amount of adjustments in inches.
3. Test Data (Indicated and Actual Values):
 - a. Total air flow rate in cfm.
 - b. Total system static pressure in inches wg.
 - c. Fan rpm.
 - d. Discharge static pressure in inches wg.
 - e. Filter static-pressure differential in inches wg.
 - f. Cooling coil static-pressure differential in inches wg.
 - g. Heating coil static-pressure differential in inches wg.
 - h. Outdoor airflow in cfm.
 - i. Return airflow in cfm.
 - j. Outdoor-air damper position.
 - k. Return-air damper position.

F. Apparatus-Coil Test Reports:

1. Test Data (Indicated and Actual Values):
 - a. Air flow rate in cfm.
 - b. Average face velocity in fpm.
 - c. Air pressure drop in inches wg.
 - d. Outdoor-air, wet- and dry-bulb temperatures in deg F.
 - e. Return-air, wet- and dry-bulb temperatures in deg F.
 - f. Entering-air, wet- and dry-bulb temperatures in deg F.
 - g. Leaving-air, wet- and dry-bulb temperatures in deg F.
 - h. Refrigerant expansion valve and refrigerant types.
 - i. Refrigerant suction pressure in psig.
 - j. Refrigerant suction temperature in deg F.

G. Gas- and Oil-Fired Heat Apparatus Test Reports: In addition to manufacturer's factory startup equipment reports, include the following:

1. Unit Data:
 - a. System identification.
 - b. Location.
 - c. Make and type.
 - d. Model number and unit size.
 - e. Manufacturer's serial number.
 - f. Fuel type in input data.
 - g. Output capacity in Btu/h.
 - h. Ignition type.
 - i. Burner-control types.
 - j. Motor horsepower and rpm.
 - k. Motor volts, phase, and hertz.
 - l. Motor full-load amperage and service factor.
 - m. Sheave make, size in inches, and bore.
 - n. Center-to-center dimensions of sheave, and amount of adjustments in inches.

2. Test Data (Indicated and Actual Values):
 - a. Total air flow rate in cfm.
 - b. Entering-air temperature in deg F.
 - c. Leaving-air temperature in deg F.
 - d. Air temperature differential in deg F.
 - e. Entering-air static pressure in inches wg.
 - f. Leaving-air static pressure in inches wg.
 - g. Air static-pressure differential in inches wg.
 - h. Low-fire fuel input in Btu/h.
 - i. High-fire fuel input in Btu/h.
 - j. Manifold pressure in psig.
 - k. High-temperature-limit setting in deg F.
 - l. Operating set point in Btu/h.
 - m. Motor voltage at each connection.
 - n. Motor amperage for each phase.
 - o. Heating value of fuel in Btu/h.

H. Electric-Coil Test Reports: For electric furnaces, duct coils, and electric coils installed in central-station air-handling units, include the following:

1. Unit Data:
 - a. System identification.
 - b. Location.
 - c. Coil identification.
 - d. Capacity in kW
 - e. Number of stages.
 - f. Connected volts, phase, and hertz.
 - g. Rated amperage.
 - h. Air flow rate in cfm.
 - i. Face area in sq. ft.
 - j. Minimum face velocity in fpm.
2. Test Data (Indicated and Actual Values):
 - a. Heat output in kW
 - b. Air flow rate in cfm.

- c. Air velocity in fpm.
 - d. Entering-air temperature in deg F.
 - e. Leaving-air temperature in deg F.
 - f. Voltage at each connection.
 - g. Amperage for each phase.
- I. Fan Test Reports: For supply, return, and exhaust fans, include the following:
 - 1. Fan Data:
 - a. System identification.
 - b. Location.
 - c. Make and type.
 - d. Model number and size.
 - e. Manufacturer's serial number.
 - f. Arrangement and class.
 - g. Sheave make, size in inches, and bore.
 - h. Center-to-center dimensions of sheave, and amount of adjustments in inches.
 - 2. Motor Data:
 - a. Motor make, and frame type and size.
 - b. Horsepower and rpm.
 - c. Volts, phase, and hertz.
 - d. Full-load amperage and service factor.
 - e. Sheave make, size in inches, and bore.
 - f. Center-to-center dimensions of sheave, and amount of adjustments in inches.
 - g. Number, make, and size of belts.
 - 3. Test Data (Indicated and Actual Values):
 - a. Total airflow rate in cfm.
 - b. Total system static pressure in inches wg.
 - c. Fan rpm.
 - d. Discharge static pressure in inches wg.
 - e. Suction static pressure in inches wg.
- J. Round, Flat-Oval, and Rectangular Duct Traverse Reports: Include a diagram with a grid representing the duct cross-section and record the following:
 - 1. Report Data:
 - a. System and air-handling-unit number.
 - b. Location and zone.
 - c. Traverse air temperature in deg F.
 - d. Duct static pressure in inches wg.
 - e. Duct size in inches.
 - f. Duct area in sq. ft.
 - g. Indicated air flow rate in cfm.
 - h. Indicated velocity in fpm.
 - i. Actual air flow rate in cfm.
 - j. Actual average velocity in fpm.
 - k. Barometric pressure in psig.

- K. Compressor and Condenser Reports: For refrigerant side of unitary systems, stand-alone refrigerant compressors, air-cooled condensing units, or water-cooled condensing units, include the following:

1. Unit Data:
 - a. Unit identification.
 - b. Location.
 - c. Unit make and model number.
 - d. Compressor make.
 - e. Compressor model and serial numbers.
 - f. Refrigerant weight in lb.
 - g. Low ambient temperature cutoff in deg F.
2. Test Data (Indicated and Actual Values):
 - a. Inlet-duct static pressure in inches wg.
 - b. Outlet-duct static pressure in inches wg.
 - c. Entering-air, dry-bulb temperature in deg F.
 - d. Leaving-air, dry-bulb temperature in deg F.
 - e. Condenser entering-water temperature in deg F.
 - f. Condenser leaving-water temperature in deg F.
 - g. Condenser-water temperature differential in deg F.
 - h. Condenser entering-water pressure in feet of head or psig.
 - i. Condenser leaving-water pressure in feet of head or psig.
 - j. Condenser-water pressure differential in feet of head or psig.
 - k. Control settings.
 - l. Unloader set points.
 - m. Low-pressure-cutout set point in psig.
 - n. High-pressure-cutout set point in psig.
 - o. Suction pressure in psig.
 - p. Suction temperature in deg F.
 - q. Condenser refrigerant pressure in psig.
 - r. Condenser refrigerant temperature in deg F.
 - s. Oil pressure in psig.
 - t. Oil temperature in deg F.
 - u. Voltage at each connection.
 - v. Amperage for each phase.
 - w. Kilowatt input.
 - x. Crankcase heater kilowatt.
 - y. Number of fans.
 - z. Condenser fan rpm.
 - aa. Condenser fan airflow rate in cfm.
 - bb. Condenser fan motor make, frame size, rpm, and horsepower.
 - cc. Condenser fan motor voltage at each connection.
 - dd. Condenser fan motor amperage for each phase.

- L. Condenser Test Reports: For condensers, include the following:

1. Unit Data:
 - a. Unit identification.
 - b. Make and type.
 - c. Model and serial numbers.
 - d. Nominal cooling capacity in tons.

- e. Refrigerant type and weight in lb.
- f. Water-treatment chemical feeder and chemical.
- g. Number and type of fans.
- h. Fan motor make, frame size, rpm, and horsepower.
- i. Fan motor voltage at each connection.
- j. Sheave make, size in inches, and bore.
- k. Sheave dimensions, center-to-center, and amount of adjustments in inches.
- l. Number of belts, make, and size.
- m. Pump make and model number.
- n. Pump manufacturer's serial number.
- o. Pump motor make and frame size.
- p. Pump motor horsepower and rpm.

M. Instrument Calibration Reports:

1. Report Data:

- a. Instrument type and make.
- b. Serial number.
- c. Application.
- d. Dates of use.
- e. Dates of calibration.

3.12 INSPECTIONS

A. Initial Inspection:

- 1. After testing and balancing are complete, operate each system and randomly check measurements to verify that the system is operating according to the final test and balance readings documented in the final report.
- 2. Check the following for each system:
 - a. Measure airflow of at least 10 percent of air outlets.
 - b. Measure water flow of at least 5 percent of terminals.
 - c. Measure room temperature at each thermostat/temperature sensor. Compare the reading to the set point.
 - d. Verify that balancing devices are marked with final balance position.
 - e. Note deviations from the Contract Documents in the final report.

B. Final Inspection:

- 1. After initial inspection is complete and documentation by random checks verifies that testing and balancing are complete and accurately documented in the final report, request that a final inspection be made by Owner's Representative.
- 2. The TAB contractor's test and balance engineer shall conduct the inspection in the presence of the Owner Authorized Representative.
- 3. The Owner shall randomly select measurements, documented in the final report, to be rechecked. Rechecking shall be limited to either 10 percent of the total measurements recorded or the extent of measurements that can be accomplished in a normal 8-hour business day.
- 4. If rechecks yield measurements that differ from the measurements documented in the final report by more than the tolerances allowed, the measurements shall be noted as "FAILED."

5. If the number of "FAILED" measurements is greater than 10 percent of the total measurements checked during the final inspection, the testing and balancing shall be considered incomplete and shall be rejected.
 - C. TAB Work will be considered defective if it does not pass final inspections. If TAB Work fails, proceed as follows:
 1. Recheck all measurements and make adjustments. Revise the final report and balancing device settings to include all changes; resubmit the final report and request a second final inspection.
 2. If the second final inspection also fails, Owner may contract the services of another TAB contractor to complete TAB Work according to the Contract Documents and deduct the cost of the services from the original TAB contractor's final payment.
 - D. Prepare test and inspection reports.
- 3.13 ADDITIONAL TESTS
- A. Within 90 days of completing TAB, perform additional TAB to verify that balanced conditions are being maintained throughout and to correct unusual conditions.
 - B. Seasonal Periods: If initial TAB procedures were not performed during near-peak summer and winter conditions, perform additional TAB during near-peak summer and winter conditions.
 - C. Include 80 hours of additional TAB time for use at the direction of the engineer.

END OF SECTION 230593

SECTION 230700 - HVAC INSULATION**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Insulation Materials:
 - a. Flexible elastomeric.
 - b. Mineral fiber.
 - c. Polyisocyanurate.
2. Fire-rated insulation systems.
3. Insulating cements.
4. Adhesives.
5. Mastics.
6. Lagging adhesives.
7. Sealants.
8. Factory-applied jackets.
9. Field-applied jackets.
10. Identification
11. Tapes.
12. Securements.
13. Corner angles.

1.2 DEFINITIONS

- A. Hot Surfaces: Normal operating temperatures of 100 deg F or higher.
- B. Dual-Temperature Surfaces: Normal operating temperatures that vary from hot to cold.
- C. Cold Surfaces: Normal operating temperatures less than 75 deg F.
- D. Thermal Resistivity: "R-values" represent the reciprocal of thermal conductivity (K-value). Thermal conductivity is the rate of heat flow through a homogeneous material exactly 1 inch thick. Thermal resistivities are expressed by the temperature difference in degrees F between two exposed faces required to cause one Btu to flow through one square foot of material, in one hour, at a given mean temperature.
- E. Density: Is expressed in lb/cu. ft.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated. Include thermal conductivity, thermal resistivity (R-value), thickness, and jackets (both factory and field applied, if any).
- B. Shop Drawings:
 1. Detail application of protective shields, saddles, and inserts at hangers for each type of insulation and hanger.
 2. Detail attachment and covering of heat tracing inside insulation.
 3. Detail insulation application at pipe expansion joints for each type of insulation.

4. Detail insulation application at elbows, fittings, flanges, valves, and specialties for each type of insulation.
 5. Detail removable insulation at piping specialties, equipment connections, and access panels.
 6. Detail application of field-applied jackets.
 7. Detail application of Identification
 8. Detail application at linkages of control devices.
 9. Detail field application for each equipment type.
 10. Detail outdoor duct insulation installation.
- C. Samples: For each type of insulation jacket, and identification indicated. Identify each Sample, describing product and intended use.
1. Sample Sizes:
 - a. Preformed Pipe Insulation Materials: 12 inches long by NPS 2.
 - b. Sheet Form Insulation Materials: 12 inches square.
 - c. Jacket Materials for Pipe: 12 inches long by NPS 2.
 - d. Sheet Jacket Materials: 12 inches square.
 - e. Manufacturer's Color Charts: For products where color is specified, show the full range of colors available for each type of finish material.
- D. Material Test Reports: From a qualified testing agency acceptable to the authorities having jurisdiction indicating, interpreting, and certifying test results for compliance of insulation materials, sealers, attachments, cements, and jackets, with requirements indicated. Include dates of tests and test methods employed.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or another craft training program certified by the Department of Labor, Bureau of Apprenticeship and Training.
- B. Fire-Test-Response Characteristics: Except where indicated otherwise, insulation and related materials shall have fire-test-response characteristics indicated, as determined by testing identical products per ASTM E 84, by UL or another testing and inspecting agency acceptable to the authorities having jurisdiction. Identify materials with appropriate markings of applicable testing and inspecting agency.
1. Insulation Installed Indoors:
 - a. Flame-Spread Index: 25 or less.
 - b. Smoke-Developed Index: 50 or less.
 2. Insulation Installed Outdoors:
 - a. Flame-Spread Index: 75 or less.
 - b. Smoke-Developed Index: 150 or less.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials to Project site in original packages and containers with seals unbroken and bearing manufacturer's original labels, including manufacturer's name, product name, and directions for storing, handling, and use.

- B. Store materials in a clean, dry, fully enclosed, conditioned space where they will be protected against damage from moisture, humidity, temperature extremes, direct sunlight, surface contamination, and other causes.

- 1. Remove damaged or wet insulation from Project site.

1.6 COORDINATION

- A. Coordinate size and location of supports, hangers, and insulation shields specified in Division 23 Section "Hangers and Supports for Piping and Equipment."
- B. Coordinate clearance requirements with piping Installer for piping insulation application, duct Installer for duct insulation application, and equipment Installer for equipment insulation application. Before preparing piping and ductwork Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.
- C. Coordinate installation and testing of heat tracing.

1.7 SCHEDULING

- A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results.
- B. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

1.8 WARRANTY

- A. Provide warranty on materials and labor for 60 months starting from date of delivery, or one year from date of Preliminary Acceptance or Substantial Completion, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers and Products: Subject to compliance with requirements, provide one of the following products for each of the products identified:
 - 1. Flexible Elastomeric Insulation:
 - a. Aeroflex USA Inc.; Aerocel.
 - b. Armacell LLC; AP Armaflex.
 - c. RBX Corporation; Insul-Sheet 1800 and Insul-Tube 180.
 - 2. Mineral-Fiber Blanket Insulation:
 - a. CertainTeed Corp.; Duct Wrap.
 - b. Johns Manville; Microlite.
 - c. Owens Corning; All-Service Duct Wrap.
 - 3. Mineral-Fiber Board Insulation:

- a. CertainTeed Corp.; Commercial Board.
 - b. Fibrex Insulations Inc.; FBX.
 - c. Johns Manville; 800 Series Spin-Glas.
 - d. Knauf Insulation; Insulation Board.
 - e. Manson Insulation Inc.; AK Board.
 - f. Owens Corning; Fiberglas 700 Series.
4. Fire-Rated Blanket Insulation:
 - a. CertainTeed Corp.; FlameChek.
 - b. Johns Manville; Firetemp Wrap.
 - c. Nelson Firestop Products; Nelson FSB Flameshield Blanket.
 - d. Thermal Ceramics; FireMaster Duct Wrap.
 - e. 3M; Fire Barrier Wrap Products.
 - f. Unifrax Corporation; FyreWrap.
 - g. Vesuvius; PYROSCAT FP FASTR Duct Wrap.
5. Mineral-Fiber, Preformed Pipe Insulation:
 - a. Fibrex Insulations Inc.; Coreplus 1200.
 - b. Johns Manville; Micro-Lok.
 - c. Knauf Insulation; 1000 Pipe Insulation.
 - d. Owens Corning; Fiberglas Pipe Insulation.
6. Mineral-Fiber, Pipe and Tank Insulation:
 - a. CertainTeed Corp.; CrimpWrap.
 - b. Johns Manville; MicroFlex.
 - c. Knauf Insulation; Pipe and Tank Insulation.
 - d. Manson Insulation Inc.; AK Flex.
 - e. Owens Corning; Fiberglas Pipe and Tank Insulation.
7. Polyisocyanurate Insulation:
 - a. Apache Products Company; ISO-25.
 - b. Dow Chemical Company (The); Trymer.
 - c. Duna USA Inc.; Corafoam.
 - d. Elliott Company; Elfoam.
8. Mineral-Fiber Insulating Cement:
 - a. Insulco, Division of MFS, Inc.; Triple I.
 - b. P. K. Insulation Mfg. Co., Inc.; Super-Stik.
9. Mineral-Fiber, Hydraulic-Setting Insulating and Finishing Cement:
 - a. Insulco, Division of MFS, Inc.; SmoothKote.
 - b. P. K. Insulation Mfg. Co., Inc.; PK No. 127, and Quik-Cote.
 - c. Rock Wool Manufacturing Company; Delta One Shot.
10. Polyisocyanurate, Adhesive:
 - a. Childers Products, Division of ITW; CP-96.
 - b. Foster Products Corporation, H. B. Fuller Company; 81-33.

11. Flexible Elastomeric and Polyolefin Adhesive:
 - a. Aeroflex USA Inc.; Aero seal.
 - b. Armacell LCC; 520 Adhesive.
 - c. Foster Products Corporation, H. B. Fuller Company; 85-75.
 - d. RBX Corporation; Rubatex Contact Adhesive.
12. Mineral-Fiber Adhesive:
 - a. Childers Products, Division of ITW; CP-82.
 - b. Foster Products Corporation, H. B. Fuller Company; 85-20.
 - c. ITW TACC, Division of Illinois Tool Works; S-90/80.
13. ASJ Adhesive, and FSK Adhesive:
 - a. Childers Products, Division of ITW; CP-82.
 - b. Foster Products Corporation, H. B. Fuller Company; 85-20.
 - c. ITW TACC, Division of Illinois Tool Works; S-90/80.
14. PVC Jacket Adhesive:
 - a. Dow Chemical Company (The); 739, Dow Silicone.
 - b. Johns-Manville; Zeston Perma-Weld, CEEL-TITE Solvent Welding Adhesive.
 - c. Speedline Corporation; Speedline Vinyl Adhesive.
15. Vapor-Barrier Mastic:
 - a. Childers Products, Division of ITW; CP-35.
 - b. Foster Products Corporation, H. B. Fuller Company; 30-90.
 - c. ITW TACC, Division of Illinois Tool Works; CB-50.
 - d. Vimasco Corporation; 749.
16. Lagging Adhesives:
 - a. Childers Products, Division of ITW; CP-52.
 - b. Foster Products Corporation, H. B. Fuller Company; 81-42.
 - c. Vimasco Corporation; 136.
17. Joint Sealants for Cellular-Glass, Phenolic, and Polyisocyanurate Products:
 - a. Childers Products, Division of ITW; CP-76.
 - b. Foster Products Corporation, H. B. Fuller Company; 30-45.
 - c. Pittsburgh Corning Corporation; Pittseal 444.
 - d. Vimasco Corporation; 750.
18. Metal Jacket Flashing Sealants:
 - a. Childers Products, Division of ITW; CP-76-8.
 - b. Foster Products Corporation, H. B. Fuller Company; 95-44.
 - c. Vimasco Corporation; 750.
19. ASJ Flashing Sealants and PVC Jacket Flashing Sealants:
 - a. Childers Products, Division of ITW; CP-76.

- 20. PVC Jacket:
 - a. Johns Manville; Zeston.
 - b. P.I.C. Plastics, Inc.; FG Series.
 - c. Proto PVC Corporation; LoSmoke.
 - d. Speedline Corporation; SmokeSafe.
- 21. Metal Jacket:
 - a. Childers Products, Division of ITW; Metal Jacketing Systems.
 - b. IDC Corporation.
 - c. RPR Products, Inc.; Insul-Mate.
- 22. Identification:
 - a. Seton.
 - b. Brady.
 - c. Best.
 - d. Milwaukee.
- 23. ASJ Tape:
 - a. Avery Dennison Corporation, Specialty Tapes Division; Fasson 0835.
 - b. Compac Corp.; 104 and 105.
 - c. Ideal Tape Co., Inc., an American Biltrite Company; 428 AWF ASJ.
 - d. Venture Tape; 1540 CW Plus, 1542 CW Plus, and 1542 CW Plus/SQ.
- 24. FSK Tape:
 - a. Avery Dennison Corporation, Specialty Tapes Division; Fasson 0827.
 - b. Compac Corp.; 110 and 111.
 - c. Ideal Tape Co., Inc., an American Biltrite Company; 491 AWF FSK.
 - d. Venture Tape; 1525 CW, 1528 CW, and 1528 CW/SQ.
- 25. PVC Tape:
 - a. Avery Dennison Corporation, Specialty Tapes Division; Fasson 0555.
 - b. Compac Corp.; 130.
 - c. Ideal Tape Co., Inc., an American Biltrite Company; 370 White PVC tape.
 - d. Venture Tape; 1506 CW NS.
- 26. Bands:
 - a. Childers Products; Bands.
 - b. IDC Corporation.
 - c. RPR Products, Inc.; Bands.
- 27. Insulation Pins and Hangers:
 - a. AGM Industries, Inc.; CWP-1.
 - b. GEMCO; Cupped Head Weld Pin.
 - c. Midwest Fasteners, Inc.; Cupped Head.
 - d. Nelson Stud Welding; CHP

28. Wire:

- a. C & F Wire.
- b. Childers Products.
- c. PABCO Metals Corporation.
- d. RPR Products, Inc.

2.2 INSULATION MATERIALS

- A. Comply with requirements in PART 3 schedule articles for where insulating materials shall be applied.
- B. Products shall not contain asbestos, lead, mercury, or mercury compounds.
- C. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C 871.
- D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C 795.
- E. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.
- F. Flexible Elastomeric: Closed-cell, sponge- or expanded-rubber materials. Comply with ASTM C 534, Type I for tubular materials and Type II for sheet materials.
- G. Mineral-Fiber Blanket Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 553, Type II and ASTM C 1290, Type III with factory-applied FSK jacket. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.
- H. Mineral-Fiber Board Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 612, Type IB. For duct and plenum applications, provide insulation with factory-applied FSK jacket. For equipment applications, provide insulation with factory-applied ASJ. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.
- I. Fire-Rated Blanket: High-temperature, flexible, blanket insulation with FSK jacket that is tested by UL, or another testing and inspecting agency acceptable to the authorities having jurisdiction, and certified to provide a 2-hour fire rating.
- J. Mineral-Fiber, Preformed Pipe Insulation:
- K. Mineral-Fiber, Pipe and Tank Insulation: Mineral or glass fibers bonded with a thermosetting resin. Semirigid board material with factory-applied FSK jacket complying with ASTM C 1393, Type II or Type IIIA Category 2. Nominal density is 2.5 lb/cu. ft. or more. Thermal conductivity (k-value) at 100 deg F is 0.29 Btu x in./h x sq. ft. x deg F or less. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.
- L. Polyisocyanurate: Unfaced, preformed, rigid cellular polyisocyanurate material intended for use as thermal insulation.
 1. Comply with ASTM C 591, Type I or Type IV, except thermal conductivity (k-value) shall not exceed 0.19 Btu x in./h x sq. ft. x deg F at 75 deg F after 180 days of aging.
 2. Flame-spread index shall be 25 or less and smoke-developed index shall be 50 or less for thickness up to 1-1/2 inches as tested by ASTM E 84.

3. Fabricate shapes according to ASTM C 450 and ASTM C 585.
4. Factory-Applied Jacket: Requirements are specified in "Factory-Applied Jackets" Article.
 - a. Pipe Applications: ASJ.
 - b. Equipment Applications: ASJ.

2.3 FIRE-RATED INSULATION SYSTEMS

- A. Fire-Rated Blanket: High-temperature, flexible, blanket insulation with FSK jacket that is tested and certified to provide a 2-hour fire rating by UL or another testing and inspecting agency acceptable to the authorities having jurisdiction.

2.4 INSULATING CEMENTS

- A. Mineral-Fiber Insulating Cement: Comply with ASTM C 195.
- B. Mineral-Fiber, Hydraulic-Setting Insulating and Finishing Cement: Comply with ASTM C 449/C 449M.

2.5 ADHESIVES

- A. Adhesives, General: All adhesives and mastics installed within the building envelope shall be shown to comply either with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers," including 2004 Addenda, or with the VOC limits established in South Coast Air Quality Management District (SCAQMD) Rule #1168.
 1. Aerosol adhesives shall comply with Green Seal Standard for Commercial Adhesives GS-36 (2000).
- B. Compatibility: Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated, unless otherwise indicated.
- C. Polyisocyanurate Adhesive: Solvent-based resin adhesive, with a service temperature range of minus 75 to plus 300 deg F.
- D. Flexible Elastomeric Adhesive: Comply with MIL-A-24179A, Type II, Class I.
- E. Mineral-Fiber Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.
- F. ASJ Adhesive, and FSK Adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints.
- G. PVC Jacket Adhesive: Compatible with PVC jacket.
- H. Mastics: Comply with MIL-C-19565C, Type II. Materials shall be compatible with insulation materials, jackets, and substrates.
 1. Vapor-Barrier Mastic: Water based; suitable for indoor and outdoor use on below ambient services.
 - a. Water-Vapor Permeance: ASTM E 96, Procedure B, 0.013 perm at 43-mildry film thickness.

- b. Service Temperature Range: Minus 20 to plus 180 deg F
- c. Solids Content: ASTM D 1644, 59 percent by volume and 71 percent by weight.
- d. Color: White.

2.6 Lagging Adhesives: Comply with MIL-A-3316C Class I, Grade A. Materials shall be compatible with insulation materials, jackets, and substrates.

- a. Fire-resistant, water-based lagging adhesive and coating for use indoors to adhere fire-resistant lagging cloths over duct, equipment, and pipe insulation.
- b. Service Temperature Range: Minus 50 to plus 180 deg F.
- c. Color: White.

2.7 SEALANTS

A. Sealants, General: All sealants installed within the building envelope shall be shown to comply either with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers," including 2004 Addenda, or with the VOC limits established in South Coast Air Quality Management District (SCAQMD) Rule #1168.

B. Joint Sealants:

- 1. Materials shall be compatible with insulation materials, jackets, and substrates.
- 2. Permanently flexible, elastomeric sealant.
- 3. Service Temperature Range: Minus 100 to plus 300 deg F.
- 4. Color: White or gray.

C. Metal Jacket Flashing Sealants:

- 1. Materials shall be compatible with insulation materials, jackets, and substrates.
- 2. Fire- and water-resistant, flexible, elastomeric sealant.
- 3. Service Temperature Range: Minus 40 to plus 250 deg F
- 4. Color: Aluminum.

D. ASJ Flashing Sealants and PVC Jacket Flashing Sealants:

- 1. Materials shall be compatible with insulation materials, jackets, and substrates.
- 2. Fire- and water-resistant, flexible, elastomeric sealant.
- 3. Service Temperature Range: Minus 40 to plus 250 deg F.
- 4. Color: White.

2.8 JACKETS

A. Factory-Applied Jackets: Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

- 1. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C 1136, Type I..
- 2. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C 1136, Type II.

B. Field-Applied Jackets: Comply with ASTM C 921, Type I, unless otherwise indicated. Field-applied jackets shall be provided on all piping below 8 ft. height.

1. PVC Jackets: High-impact-resistant, UV-resistant PVC complying with ASTM D 1784, Class 16354-C; thickness as scheduled; roll stock ready for shop or field cutting and forming. Thickness is indicated in field-applied jacket schedules.
 - a. Adhesive: As recommended by jacket material manufacturer.
 - b. Color: White.
 - c. Factory-fabricated fitting covers to match jacket if available; otherwise, field fabricate.
 - 1) Shapes: 45- and 90-degree, short- and long-radius elbows, tees, valves, flanges, unions, reducers, end caps, soil-pipe hubs, traps, mechanical joints, and P-trap and supply covers for lavatories.
 - d. Factory-fabricated tank heads and tank side panels.
2. Metal Jackets:
 - a. Stainless-Steel Jackets: ASTM A 167 or ASTM A 240.
 - 1) Sheet and roll stock ready for shop or field sizing.
 - 2) Material, finish, and thickness are indicated in field-applied jacket schedules.
 - 3) Moisture Barrier for Indoor Applications: 1-mil-thick, heat-bonded polyethylene and kraft paper.
 - 4) Moisture Barrier for Outdoor Applications: 2.5-mil-thick Polysurlyn.
 - 5) Factory-Fabricated Fitting Covers: Fabricated from the same material, finish, and thickness as jacket. Provide factory-fabricated covers for preformed 2-piece or gore, 45- and 90-degree, short- and long-radius elbows; tee covers; flange and union covers; end caps; beveled collars; and valve covers. Field fabricate fitting covers only if factory-fabricated fitting covers are not available.
 - b. Aluminum Jackets: Comply with ASTM B 209, Alloy 3003, Temper H-14.
 - 1) Sheet and roll stock ready for shop or field sizing.
 - 2) Finish and thickness are indicated in field-applied jacket schedule.
 - 3) Moisture Barrier for Indoor Applications: 3-mil thick Polysurlyn.
 - 4) Moisture Barrier for Outdoor Applications: 3-mil thick Polysurlyn.
 - 5) Factory-Fabricated Fitting Covers: Fabricated from the same material, finish, and thickness as jacket. Provide factory-fabricated covers for preformed 2-piece or gore, 45- and 90-degree, short- and long-radius elbows; tee covers; flange and union covers; end caps; beveled collars; and valve covers. Field fabricate fitting covers only if factory-fabricated fitting covers are not available.

2.9 IDENTIFICATION

- A. General: Products specified are manufacturer's standard products of categories and types required for each application as referenced in other Division 23 Sections. Where more than single type is specified for listed application, selection is Installer's option but provide single selection for each product category.
- B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Seton.
2. Brady.
3. Best.
4. Milwaukee.

C. Snap-On Plastic Pipe Markers: Manufacturer's standard pre-printed, semi-rigid snap-on, color-coded pipe markers.

1. Lettering: Use piping system terms as indicated and abbreviate only as necessary for each application length.
2. Arrows: Either integrally with piping system service lettering (to accommodate both directions), or as separate unit, on each pipe marker to indicate direction of flow.

D. Lettering and Graphics: Coordinate names, abbreviations, and other designations used in mechanical identification, with corresponding designations indicated. Use numbers, letters, and terms indicated for proper identification, operation, and maintenance of mechanical systems and equipment.

1. Multiple Systems: Where multiple systems of same name are indicated, identify individual system number as well as service.

E. Color Coding:	<u>Background</u>	<u>Lettering</u>
1. Domestic Cold Water	White	Black
2. Domestic Hot Water/ Hot Water Return	Blue	White
3. Non Potable Water/ Makeup Water	Purple	White
4. Storm Water	Grey	White
5. Air Conditioning Condensate	Grey	White
6. Refrigerant Suction/ Hot Gas	Clear	Black
7. Natural Gas	Yellow	Black
8. Fire Protection	Red	White

2.10 TAPES

A. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C 1136.

1. Width: 3 inches.
2. Thickness: 11.5 mils.
3. Adhesion: 90 ounces force/inch in width.
4. Elongation: 2 percent.
5. Tensile Strength: 40 lbf/inch in width.
6. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.

B. FSK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C 1136.

1. Width: 3 inches.
2. Thickness: 6.5 mils.
3. Adhesion: 90 ounces force/inch in width.
4. Elongation: 2 percent.
5. Tensile Strength: 40 lbf/inch in width.
6. FSK Tape Disks and Squares: Precut disks or squares of FSK tape.

- C. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive. Suitable for indoor and outdoor applications.

1. Width: 2 inches.
2. Thickness: 6 mils
3. Adhesion: 64 ounces force/inch in width.
4. Elongation: 500 percent.
5. Tensile Strength: 18 lbf/inch in width.

2.11 SECUREMENTS

A. Bands:

1. Stainless Steel: ASTM A 167 or ASTM A 240/A 240M, Type 304; 0.020-inch-thick, 3/4-inch-wide with wing or closed seal.
2. Springs: Twin spring set constructed of stainless steel with ends flat and slotted to accept metal bands. Spring size determined by manufacturer for application.

B. Insulation Pins and Hangers:

1. Capacitor-Discharge-Weld Pins: Copper-coated steel pin, fully annealed for capacitor-discharge welding, 0.106-inch-diameter shank, length to suit depth of insulation indicated.
2. Cupped-Head, Capacitor-Discharge-Weld Pins: Copper- or zinc-coated steel pin, fully annealed for capacitor-discharge welding, 0.106-inch-diameter shank, length to suit depth of insulation indicated with integral 1-1/2-inch galvanized carbon-steel washer.
3. Metal, Adhesively Attached, Perforated-Base Insulation Hangers: Baseplate welded to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place. Comply with the following:
 - a. Baseplate: Perforated, galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.
 - b. Spindle: Copper- or zinc-coated, low carbon steel, fully annealed, 0.106-inch-diameter shank, length to suit depth of insulation indicated.
 - c. Adhesive: Recommended by hanger manufacturer. Product with demonstrated capability to bond insulation hanger securely to substrates indicated without damaging insulation, hangers, and substrates.
4. Nonmetal, Adhesively Attached, Perforated-Base Insulation Hangers: Baseplate fastened to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place. Comply with the following:
 - a. Baseplate: Perforated, nylon sheet, 0.030 inch thick by 1-1/2 inches in diameter.
 - b. Spindle: Nylon, 0.106-inch-diameter shank, length to suit depth of insulation indicated, up to 2-1/2 inches.
 - c. Adhesive: Recommended by hanger manufacturer. Product with demonstrated capability to bond insulation hanger securely to substrates indicated without damaging insulation, hangers, and substrates.
5. Self-Sticking-Base Insulation Hangers: Baseplate welded to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place. Comply with the following requirements:

- a. Baseplate: Galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.
 - b. Spindle: Copper- or zinc-coated, low carbon steel, fully annealed, 0.106-inch-diameter shank, length to suit depth of insulation indicated.
 - c. Adhesive-backed base with a peel-off protective cover.
 - 6. Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch- thick, galvanized-steel sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.
 - a. Protect ends with capped self-locking washers incorporating a spring steel insert to ensure permanent retention of cap in exposed locations.
 - 7. Nonmetal Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch-thick nylon sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.
 - C. Staples: Outward-clinching insulation staples, nominal 3/4-inch- wide, stainless steel or Monel.
 - D. Wire: 0.062-inch soft-annealed, stainless steel.
- 2.12 CORNER ANGLES
- A. PVC Corner Angles: 30 mils thick, minimum 1 by 1 inch, PVC according to ASTM D 1784, Class 16354-C. White or color-coded to match adjacent surface.
 - B. Stainless-Steel Corner Angles: 0.024 inch thick, minimum 1 by 1 inch, stainless steel according to ASTM A 167 or ASTM A 240, Type 304 or Type 316.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions for compliance with requirements for installation and other conditions affecting performance of insulation application.
 - 1. Verify that systems and equipment to be insulated have been tested and are free of defects.
 - 2. Verify that surfaces to be insulated are clean and dry.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.
- B. Coordinate insulation installation with the trade installing heat tracing, if any. Comply with requirements for heat tracing that apply to insulation.
- C. Mix insulating cements with clean potable water; if insulating cements are to be in contact with stainless-steel surfaces, use demineralized water.

3.3 INSTALLATION – GENERAL

- A. Install insulation materials, accessories, and finishes according to the manufacturer's written instructions with smooth, straight, and even surfaces; free of voids throughout the length of equipment, ducts and fittings, and piping including fittings, valves, and specialties.
- B. Install insulation materials, forms, vapor barriers or retarders, jackets, and thicknesses required for each item of equipment, duct system, and pipe system as specified in insulation system schedules.
- C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.
- D. Keep insulation materials dry during application and finishing.
- E. Install insulation with least number of joints practical.
- F. Install insulation with longitudinal seams at top and bottom of horizontal runs.
- G. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.
- H. Install multiple layers of insulation with longitudinal and end seams staggered.
- I. Install vapor barriers on insulated pipes, ductwork, and equipment having surface operating temperatures below 60 deg F.
- J. Do not weld brackets, clips, or other attachment devices to piping, fittings, and specialties.
- K. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.
 - 1. Install insulation continuously through hangers and around anchor attachments.
 - 2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.
 - 3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.
 - 4. Cover inserts with jacket material matching adjacent pipe insulation. Install shields over jacket, arranged to protect jacket from tear or puncture by hanger, support, and shield.
- L. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.
- M. Install insulation with factory-applied jackets as follows:
 - 1. Draw jacket tight and smooth.
 - 2. Cover circumferential joints with 3-inch- wide strips, of same material as insulation jacket. Secure strips with adhesive and outward clinching staples along both edges of strip, spaced 4 inches o.c.

3. Overlap jacket longitudinal seams at least 1-1/2 inches. Install insulation with longitudinal seams at bottom of pipe. Clean and dry surface to receive self-sealing lap. Staple laps with outward clinching staples along edge at 2 inches o.c.
 - a. For below ambient services, apply vapor-barrier mastic over staples.
 - b. Exception: Do not staple longitudinal laps on insulation applied to piping systems with a surface temperature at or below 35 deg F.
 4. Cover joints and seams with tape as recommended by insulation material manufacturer to maintain vapor seal.
 5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints and at ends adjacent to duct and pipe flanges and fittings.
 6. At penetrations in jackets for thermometers, and pressure gages, fill and seal voids with vapor barrier coating.
- N. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.
- O. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.
- P. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.
- Q. For above ambient services, do not install insulation to the following:
1. Vibration-control devices.
 2. Testing agency labels and stamps.
 3. Nameplates and data plates.
 4. Manholes.
 5. Handholes.
 6. Cleanouts.

3.4 PENETRATIONS

- A. Insulation Installation at Roof Penetrations: Install insulation continuously through roof penetrations.
1. Seal penetrations with flashing sealant.
 2. For applications requiring only indoor insulation, terminate insulation above roof surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
 3. Extend jacket of outdoor insulation outside roof flashing at least 2 inches below top of roof flashing.
 4. Seal jacket to roof flashing with flashing sealant.
- B. Insulation Installation at Underground Exterior Wall Penetrations: Terminate insulation flush with sleeve seal. Seal terminations with flashing sealant.
- C. Insulation Installation at Aboveground Exterior Wall Penetrations: Install insulation continuously through wall penetrations.

1. Seal penetrations with flashing sealant.
 2. For applications requiring only indoor insulation, terminate insulation inside wall surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
 3. Extend jacket of outdoor insulation outside wall flashing and overlap wall flashing at least 2 inches.
 4. Seal jacket to wall flashing with flashing sealant.
- D. Insulation Installation at Interior Wall and Partition Penetrations (That Are Not Fire Rated): Install insulation continuously through walls and partitions.
- E. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Install insulation continuously through penetrations of fire-rated walls and partitions in accordance with requirements of approved and tested assembly indicated. Terminate insulation at fire damper sleeves for fire-rated wall and partition penetrations. Externally insulate damper sleeves to match adjacent insulation and overlap duct insulation at least 2 inches.
1. Comply with requirements in Division 07 Section "Penetration Firestopping" for firestopping and fire-resistive joint sealers.
- F. Insulation Installation at Floor Penetrations:
1. Duct: Install insulation continuously through floor penetrations that are not fire rated.
 - a. For penetrations through fire-rated assemblies, terminate insulation at fire damper sleeves and externally insulate damper sleeve beyond floor to match adjacent duct insulation. Overlap damper sleeve and duct insulation at least 2 inches.
 - b. Comply with requirements in Division 07 Section "Penetration Firestopping" for firestopping and fire-resistive joint sealers.
 2. Pipe: Install insulation continuously through floor penetrations.

3.5 PIPE INSULATION INSTALLATION – GENERAL

- A. General: Requirements in this article apply to all insulation materials, unless more specific requirements are indicated for individual pipe insulation types.
- B. Insulation Installation on Fittings, Valves, Strainers, Flanges, and Unions:
1. Install insulation over fittings, valves, strainers, flanges, unions, and other specialties with continuous thermal and vapor-retarder integrity, unless otherwise indicated.
 2. Pipe Elbows: Insulate pipe elbows using preformed fitting insulation or mitered fittings made from same material and density as adjacent pipe insulation. Each piece shall be butted tightly against adjoining piece and bonded with adhesive. Fill joints, seams, voids, and irregular surfaces with insulating cement finished to a smooth, hard, and uniform contour that is uniform with adjoining pipe insulation.
 3. Tee Fittings: Insulate tee fittings with preformed fitting insulation or sectional pipe insulation of same material and thickness as used for adjacent pipe. Cut sectional pipe insulation to fit. Butt each section closely to the next and hold in place with tie wire. Bond pieces with adhesive.
 4. Valves: Insulate valves using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe

- diameter, whichever is thicker. For valves, insulate up to and including the bonnets, valve stuffing-box studs, bolts, and nuts. Fill joints, seams, and irregular surfaces with insulating cement.
5. Strainers: Insulate strainers using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Fill joints, seams, and irregular surfaces with insulating cement. Insulate strainers so strainer basket flange or plug can be easily removed and replaced without damaging the insulation and jacket. Provide a removable reusable insulation cover. For below ambient services, provide a design that maintains vapor barrier.
 6. Fittings and Unions: Insulate flanges and unions using a section of oversized preformed pipe insulation. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker.
 7. Cover segmented insulated surfaces with a layer of finishing cement and coat with a mastic. Install vapor-barrier mastic for below ambient services and a breather mastic for above ambient services. Reinforce the mastic with fabric-reinforcing mesh. Trowel the mastic to a smooth and well-shaped contour.
 8. For services not specified to receive a field-applied jacket except for flexible elastomeric, install fitted PVC cover over elbows, tees, strainers, valves, flanges, and unions. Terminate ends with PVC end caps. Tape PVC covers to adjoining insulation facing using PVC tape.
 9. Stencil or label the outside insulation jacket of each union with the word "UNION." Match size and color of pipe labels.
- C. Insulate instrument connections for thermometers, pressure gages, pressure temperature taps, test connections, flow meters, sensors, switches, and transmitters on insulated pipes, vessels, and equipment. Shape insulation at these connections by tapering it to and around the connection with insulating cement and finish with finishing cement, mastic, and flashing sealant.
- D. Install removable insulation covers at locations indicated on Drawings. Installation shall conform to the following:
1. Make removable flange and union insulation from sectional pipe insulation of same thickness as that on adjoining pipe. Install same insulation jacket as adjoining pipe insulation.
 2. When flange and union covers are made from sectional pipe insulation, extend insulation from flanges or union long at least two times the insulation thickness over adjacent pipe insulation on each side of flange or union. Secure flange cover in place with stainless-steel. Select band material compatible with insulation and jacket.
 3. Construct removable valve insulation covers in same manner as for flanges except divide the two-part section on the vertical center line of valve body.
 4. When covers are made from block insulation, make two halves, each consisting of mitered blocks wired to stainless-steel fabric. Secure this wire frame, with its attached insulation, to flanges with tie wire. Extend insulation at least 2 inches over adjacent pipe insulation on each side of valve. Fill space between flange or union cover and pipe insulation with insulating cement. Finish cover assembly with insulating cement applied in two coats. After first coat is dry, apply and trowel second coat to a smooth finish.
 5. Unless a PVC jacket is indicated in field-applied jacket schedules, finish exposed surfaces with a metal jacket.

3.6 FLEXIBLE ELASTOMERIC INSULATION INSTALLATION

- A. Seal longitudinal seams and end joints with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

B. Insulation Installation on Pipe Flanges:

1. Install pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.
3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with cut sections of sheet insulation of same thickness as pipe insulation.
4. Secure insulation to flanges and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

C. Insulation Installation on Pipe Fittings and Elbows:

1. Install mitered sections of pipe insulation.
2. Secure insulation materials and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

D. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed valve covers manufactured of same material as pipe insulation when available.
2. When preformed valve covers are not available, install cut sections of pipe and sheet insulation to valve body. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
3. Install insulation to flanges as specified for flange insulation application.
4. Secure insulation to valves and specialties and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

3.7 MINERAL-FIBER INSULATION INSTALLATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of preformed pipe insulation to pipe with wire or bands and tighten bands without deforming insulation materials.
2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.
3. For insulation with factory-applied jackets on above ambient surfaces, secure laps with outward clinched staples at 6 inches o.c.
4. For insulation with factory-applied jackets on below ambient surfaces, do not staple longitudinal tabs but secure tabs with additional adhesive as recommended by insulation material manufacturer and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Flanges:

1. Install preformed pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.
3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with mineral-fiber blanket insulation.
4. Install jacket material with manufacturer's recommended adhesive, overlap seams at least 1 inch, and seal joints with flashing sealant.

C. Insulation Installation on Pipe Fittings and Elbows:

1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed insulation elbows and fittings are not available, install mitered sections of pipe insulation, to a thickness equal to adjoining pipe insulation. Secure insulation materials with wire or bands.

D. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed sections are not available, install mitered sections of pipe insulation to valve body.
3. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
4. Install insulation to flanges as specified for flange insulation application.

E. Blanket Insulation Installation on Ducts and Plenums: Secure with adhesive and insulation pins.

1. Apply adhesives according to manufacturer's recommended coverage rates per unit area, for 100 percent coverage of duct and plenum surfaces.
2. Apply adhesive to entire circumference of ducts and to all surfaces of fittings and transitions.
3. Install either capacitor-discharge-weld pins and speed washers or cupped-head, capacitor-discharge-weld pins on sides and bottom of horizontal ducts and sides of vertical ducts as follows:
 - a. On duct sides with dimensions 18 inches and smaller, place pins along longitudinal centerline of duct. Space 3 inches maximum from insulation end joints, and 16 inches o.c.
 - b. On duct sides with dimensions larger than 18 inches place pins 16 inches o.c. each way, and 3 inches maximum from insulation joints. Install additional pins to hold insulation tightly against surface at cross bracing.
 - c. Pins may be omitted from top surface of horizontal, rectangular ducts and plenums.
 - d. Do not over compress insulation during installation.
 - e. Impale insulation over pins and attach speed washers.
 - f. Cut excess portion of pins extending beyond speed washers or bend parallel with insulation surface. Cover exposed pins and washers with tape matching insulation facing.
4. For ducts and plenums with surface temperatures below ambient, install a continuous unbroken vapor barrier. Create a facing lap for longitudinal seams and end joints with insulation by removing 2 inches from 1 edge and 1 end of insulation segment. Secure laps to adjacent insulation section with 1/2-inch outward-clinching staples, 1 inch o.c. Install vapor barrier consisting of factory- or field-applied jacket, adhesive, vapor-barrier mastic, and sealant at joints, seams, and protrusions.
 - a. Repair punctures, tears, and penetrations with tape or mastic to maintain vapor-barrier seal.
 - b. Install vapor stops for ductwork and plenums operating below 50 deg F at 18-foot intervals. Vapor stops shall consist of vapor-barrier mastic applied in a Z-shaped pattern over insulation face, along butt end of insulation, and over the surface. Cover insulation face and surface to be insulated a width equal to 2 times the insulation thickness but not less than 3 inches

5. Overlap unfaced blankets a minimum of 2 inches on longitudinal seams and end joints. At end joints, secure with steel bands spaced a maximum of 18 inches o.c.
 6. Install insulation on rectangular duct elbows and transitions with a full insulation section for each surface. Install insulation on round and flat-oval duct elbows with individually mitered gores cut to fit the elbow.
 7. Insulate duct stiffeners, hangers, and flanges that protrude beyond insulation surface with 6-inch-wide strips of same material used to insulate duct. Secure on alternating sides of stiffener, hanger, and flange with pins spaced 6 inches o.c.
- F. Board Insulation Installation on Ducts and Plenums: Secure with adhesive and insulation pins.
1. Apply adhesives according to manufacturer's recommended coverage rates per unit area, for 100 percent coverage of duct and plenum surfaces.
 2. Apply adhesive to entire circumference of ducts and to all surfaces of fittings and transitions.
 3. Install either capacitor-discharge-weld pins and speed washers or cupped-head, capacitor-discharge-weld pins on sides and bottom of horizontal ducts and sides of vertical ducts as follows:
 - a. On duct sides with dimensions 18 inches and smaller, place pins along longitudinal centerline of duct. Space 3 inches maximum from insulation end joints, and 16 inches o.c.
 - b. On duct sides with dimensions larger than 18 inches space pins 16 inches o.c. each way, and 3 inches maximum from insulation joints. Install additional pins to hold insulation tightly against surface at cross bracing.
 - c. Pins may be omitted from top surface of horizontal, rectangular ducts and plenums.
 - d. Do not over compress insulation during installation.
 - e. Cut excess portion of pins extending beyond speed washers or bend parallel with insulation surface. Cover exposed pins and washers with tape matching insulation facing.
 4. For ducts and plenums with surface temperatures below ambient, install a continuous unbroken vapor barrier. Create a facing lap for longitudinal seams and end joints with insulation by removing 2 inches from 1 edge and 1 end of insulation segment. Secure laps to adjacent insulation section with 1/2-inch outward-clinching staples, 1 inch o.c. Install vapor barrier consisting of factory- or field-applied jacket, adhesive, vapor-barrier mastic, and sealant at joints, seams, and protrusions.
 - a. Repair punctures, tears, and penetrations with tape or mastic to maintain vapor-barrier seal.
 - b. Install vapor stops for ductwork and plenums operating below 50 deg F at 18-foot intervals. Vapor stops shall consist of vapor-barrier mastic applied in a Z-shaped pattern over insulation face, along butt end of insulation, and over the surface. Cover insulation face and surface to be insulated a width equal to 2 times the insulation thickness but not less than 3 inches.
 5. Install insulation on rectangular duct elbows and transitions with a full insulation section for each surface. Groove and score insulation to fit as closely as possible to outside and inside radius of elbows. Install insulation on round and flat-oval duct elbows with individually mitered gores cut to fit the elbow.
 6. Insulate duct stiffeners, hangers, and flanges that protrude beyond insulation surface with 6-inch-wide strips of same material used to insulate duct. Secure on alternating sides of stiffener, hanger, and flange with pins spaced 6 inches o.c.

3.8 POLYISOCYANURATE INSULATION INSTALLATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of insulation to pipe with tape or bands and tighten without deforming insulation materials. Orient longitudinal joints between half sections in 3 o'clock and 9 o'clock positions on the pipe.
2. For insulation with factory-applied jackets with vapor barriers, do not staple longitudinal tabs but secure tabs with additional adhesive or tape as recommended by insulation material manufacturer and seal with vapor-barrier mastic.
3. All insulation shall be tightly butted and free of voids and gaps at all joints. Vapor barrier must be continuous. Before installing jacket material, install vapor-barrier system.

B. Insulation Installation on Pipe Flanges:

1. Install preformed pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, same thickness of adjacent pipe insulation, not to exceed 1-1/2-inch thickness.
3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with cut sections of polyisocyanurate block insulation of same thickness as pipe insulation.

C. Insulation Installation on Fittings and Elbows:

1. Install preformed sections of same material as straight segments of pipe insulation. Secure according to manufacturer's written instructions.

D. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed sections of polyisocyanurate insulation to valve body.
2. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
3. Install insulation to flanges as specified for flange insulation application.

3.9 WEATHERPROOF DUCT INSULATION INSTALLATION

A. Provide thickness of polyisocyanurate insulation as indicated in the Schedule at the end of PART 3. Provide on the exterior of all supply, and return, ducts exposed to the weather.

B. Cover insulation with a field-applied jacket.

C. Top of duct insulation shall overlap the side of duct insulation. Side of duct insulation shall overlap the bottom of duct insulation. Provide corner angles at each corner.

D. Top of duct insulation shall be pitched for drainage. Pitch shall be a minimum of 1/2 inch per foot from the high point at the duct horizontal longitudinal centerline to the low points at the outsides of the duct. Insulation thickness at the low points shall be the minimum indicated in the Schedule at the end of PART 3. Duct pitch shall be achieved by either of the following:

1. Tapered insulation.
2. Insulation with tapered fesco board to achieve the pitch.
3. For ducts over 6 ft. wide, tapered continuous wood shims (1 in. wide) running laterally from the duct centerline high point to the low points at the outsides. Shims shall be located at 12 inches o.c. along the length of the duct.

- E. Vapor retarder type mastic, or joint sealer, shall be applied on longitudinal and butt joints to prevent moisture and moisture vapor infiltration. Vapor retarder butt joints shall be sealed with 3 in. wide vapor retarder tape.
- F. Apply insulation in a solid bed of waterproof adhesive.
- G. Tightly wrap the insulation circumferentially with saran film. Overlap the seams by a minimum of 2 inches. Seal the overlapped seams with vapor retarder tape.
- H. Secure the insulation with stainless steel bands on 12-inch centers.
- I. Jacketing shall be secured with 1/2-inch wide stainless steel bands on 12 inch centers. Rivets, screws, staples, or any other fastener capable of penetrating the underlying vapor retarder shall not be used.

3.10 FIELD-APPLIED JACKET INSTALLATION

- A. Where PVC jackets are indicated, install with 1-inch overlap at longitudinal seams and end joints; for horizontal applications, install with longitudinal seams along top and bottom of tanks and vessels. Seal with manufacturer's recommended adhesive.
 - 1. Apply two continuous beads of adhesive to seams and joints, one bead under lap and the finish bead along seam and joint edge.
- B. Where Stainless Steel, or Aluminum, jackets are indicated, install with 2-inch overlap at longitudinal seams and end joints. Overlap longitudinal seams arranged to shed water. Seal end joints with weatherproof sealant recommended by insulation manufacturer. Secure jacket with stainless-steel bands 12 inches o.c. and at end joints.

3.11 LABELING AND IDENTIFYING INSTALLATION

- A. Piping Systems: Install pipe markers on each system. Include arrows showing normal direction of flow.
 - 1. Plastic markers with application systems. Install on pipe insulation segment where required for hot non-insulated pipes.
 - a. Fasten markers on pipes smaller than 6-inch diameter by one of following methods:
 - 1) Snap-on application of pre-tensioned semi-rigid plastic pipe marker.
 - 2) Adhesive lap joint in pipe marker overlap.
 - 3) Laminated or bonded application of pipe marker to pipe (or insulation).
 - 4) Taped to pipe (or insulation) with color-coded plastic adhesive tape, not less than 3/4-inch wide, lapped 1-1/2 inches minimum at both ends of pipe marker, and covering full circumference of pipe.
 - b. Fasten markers on pipes 6-inch and larger diameter by one of following methods:
 - 1) Laminated or bonded application of pipe marker to pipe (or insulation).
 - 2) Taped to pipe (or insulation) with color-coded plastic adhesive tape, not less than 1-1/2-inches wide, lapped 3 inches minimum at both ends of pipe marker, and covering full circumference of pipe.

- 3) Strapped to pipe (or insulation) with manufacturer's standard stainless steel bands.
2. Locate pipe markers and color bands as follows wherever piping is exposed in finished spaces, machine rooms, accessible maintenance spaces (shafts, tunnels, plenums) and exterior non-concealed locations.
 - a. Near each valve and control device.
 - b. Near each branch connection, excluding short take-offs for fixtures and terminal units. Mark each pipe at branch, where flow pattern is not obvious.
 - c. Near penetrations through walls, floors, ceilings, or enter non-accessible enclosures.
 - d. At access doors, manholes, and similar access points that permit view of concealed piping.
 - e. Near major equipment items and other points of origination and termination.
 - f. Spaced at a maximum of 50-feet o.c. along each run. Reduce intervals to 25 feet in congested areas of piping and equipment.
 - g. On piping above removable acoustical ceilings, except omit intermediately spaced markers.

3.12 FIRE-RATED INSULATION SYSTEM INSTALLATION

- A. Where fire-rated insulation system is indicated, secure system to ducts and duct hangers and supports to maintain a continuous fire rating.
- B. Insulate duct access panels and doors to achieve same fire rating as duct.
- C. Install firestopping at penetrations through fire-rated assemblies in accordance with Division 07 Section "Penetration Firestopping."

3.13 FIELD QUALITY CONTROL

- A. Testing Agency: Contractor shall engage a qualified testing agency to perform tests and inspections.
- B. Tests and Inspections:
 1. Inspect ductwork, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to one location for each duct system defined in the Article "Duct Insulation Schedule, General".
 2. Inspect field-insulated equipment, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to one location for each type of equipment requiring insulation for this project. For large equipment, remove only a portion adequate to determine compliance.
 3. Inspect pipe, fittings, strainers, and valves, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to three locations of straight pipe, three locations of threaded fittings, three locations of welded fittings, two locations of threaded strainers, two locations of welded strainers, three locations of threaded valves, and three locations of flanged valves for each pipe service defined in the Article "Piping Insulation Schedule, General".

- C. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements. Architect may reject all work if sample work is found to be defective.

3.14 DUCT INSULATION PERFORMANCE , GENERAL

A. Plenums and Ducts Requiring Insulation:

1. Indoor, supply and outdoor air.
2. Indoor, concealed return located in non-conditioned space.
3. Indoor, return located in non-conditioned space.
4. Indoor, kitchen hood exhaust.
5. Indoor, exhaust between isolation damper and penetration of building exterior.
6. Outdoor, supply, return and outdoor air.

B. Items Not Insulated:

1. Factory-insulated flexible ducts.
2. Factory-insulated plenums and casings.
3. Flexible connectors.
4. Vibration-control devices.
5. Factory-insulated access panels and doors.

3.15 DUCT AND PLENUM INSULATION SCHEDULE

A. INSULATION APPLICATION SCHEDULE

1. General: Abbreviations used in the following schedules include:
 - a. Field-Applied Jackets: P - PVC, K - Foil and Paper, A - Aluminum, SS - Stainless Steel.

- B. Indoor, Concealed duct insulation shall be mineral fiber blanket unless indicated otherwise.

C. DUCT SYSTEMS INSULATION SCHEDULE

DUCT LOCATION	OUTSIDE AIR, SUPPLY AIR, EXHAUST AIR (DOWNSTREAM OF DAMPER) INSULATION INSTALLED R-VALUE (H·Ft ² ·deg F/BTU)	RETURN AIR INSULATION INSTALLED R-VALUE (H·Ft ² ·deg F/BTU)
Exterior of Building (Notes 5, 6, 8)	8.0	8.0
Ventilated Attic (Notes 5, 7)	8.0	8.0
Unvented Attic Above Insulated Ceiling (Notes 5, 7)	8.0	8.0
Unvented Attic with Roof Insulation (Notes 1, 5, 7)	6.0	N/A
Unconditioned Space (Notes 2, 5, 6, 7)	8.0	8.0
Indirectly Conditioned	N/A	N/A

Space (Notes 3, 7)		
Ceiling Cavity / Shafts / Soffits / Me- chanical Spaces and Rooms (Notes 4, 5, 6, 7)	6.0	N/A
Exposed Locations within Conditioned Space (Note 9)	N/A	N/A

NOTE 1: INSULATION R-VALUES, MEASURED IN (H X FT² X F)/BTU, ARE FOR THE INSULATION AS INSTALLED AND DO NOT INCLUDE FILM RESISTANCE. WHERE EXTERIOR WALLS ARE USED AS PLENUM WALLS, WALL INSULATION SHALL BE AS REQUIRED BY THE MOST RESTRICTIVE CONDITION OF ASHRAE 90.1-2004 SECTION 5 OR 6.4.4.2. INSULATION RESISTANCE MEASURED ON A HORIZONTAL PLANE IN ACCORDANCE WITH ASTM C518 AT A MEAN TEMPERATURE OF 75F AT THE INSTALLED THICKNESS.

NOTE 2: INCLUDING CRAWL SPACES (BOTH VENTILATED/NON-VENTILATED), FRAMED CAVITIES IN WALLS, FLOOR AND CEILING ASSEMBLIES WHICH (A) SEPARATE CONDITIONED SPACE FROM UNCONDITIONED SPACE OR OUTSIDE AIR, AND (B) ARE UNINSULATED ON THE SIDE FACING AWAY FROM CONDITIONED SPACE.

NOTE 3: RETURN AIR PLENUMS WITH OR WITHOUT EXPOSED ROOFS ABOVE.

NOTE 4: CAVITY CONTAINED WITHIN THE INSULATED BUILDING ENVELOPE.

NOTE 5: VAPOR BARRIER REQUIRED.

NOTE 6: FIELD APPLIED JACKET (STAINLESS STEEL, OR ALUMINUM, FOR EXTERIOR APPLICATIONS, PVC FOR INTERIOR EXPOSED LOCATIONS).

NOTE 7: PROVIDE MINERAL FIBER BOARD WITH FIELD APPLIED JACKET (SS, OR A, EXTERIOR, ALL SERVICE INTERIOR) IN EXPOSED LOCATIONS IN LIEU OF MINERAL FIBER BLANKET.

NOTE 8: PROVIDE POLYISOCYANURATE BOARD INSULATION FOR OUTDOOR DUCTWORK

NOTE 9: UNLESS NOTED OTHERWISE IN THE CONTRACT DOCUMENTS.

D. KITCHEN EXHAUST DUCTS

MATERIAL	FORM	THICKNESS (INCHES)	VAPOR BARRIER REQUIRED	FIELD-APPLIED JACKET
Fire-Rated Blanket	Blanket	Note 1	No	Aluminum

NOTE 1: AS REQUIRED TO PROVIDE A 2-HOUR FIRE RATING.

3.16 EQUIPMENT INSULATION SCHEDULE

- A. General: Insulation materials and thicknesses are identified below. If more than one material is listed for a type of equipment, selection from materials listed is Contractor's option.
- B. Insulate indoor and outdoor equipment in paragraphs below that are not factory insulated.

3.17 PIPING INSULATION SCHEDULE, GENERAL

- A. Acceptable preformed pipe and tubular insulation materials and thicknesses are identified for each piping system and pipe size range. If more than one material is listed for a piping system, selection from materials listed is Contractor's option.
- B. Items Not Insulated: Unless otherwise indicated, do not install insulation on the following:
1. Drainage piping located in crawl spaces.
 2. Underground piping.
 3. Chrome-plated pipes and fittings unless there is a potential for personnel injury.

3.18 PIPING INSULATION SCHEDULE

- A. Abbreviations used in the following schedule include:
1. Field-Applied Jackets: P - PVC, K - Foil and Paper, A - Aluminum, SS - Stainless Steel.
 2. Pipe Sizes: NPS - Nominal Pipe Size.
- B. Minimum HVAC Pipe Insulation Thickness Table :

HEATING SYSTEMS (Steam and Hot Water) (Note 1)	FLUID TEMPERATURE RANGE (deg F)	INSULATION TYPE / FIELD-APPLIED JACKET / IDENTIFICATION			PIPE SIZE AND INSULATION THICKNESS (INCHES) (Note 5, 6)				
		Glass Fiber	Polyiso-cyanurate	Flexible Elastomeric	<1" (Note 4)	1" to <1-1/2"	1-1/2" to <4"	4" to 6"	8" and Larger
COOLING SYSTEMS									
Refrigerant	40 – 60	X (Note 2, 7)	X (Note 2, 3, 7)	X (Note 7)	1/2	3/4	1	1	1
	Below 40	X (Note 2, 7)	X (Note 2, 3, 7)	X (Note 7)	1	1-1/2	1-1/2	1-1/2	1-1/2
Indoor Air-Conditioning Condensate Drains	40 – 60	X		X	1	1	1	1	1

NOTE 1: GLASS-FIBER INSULATION ONLY FOR HYDRONIC PIPING.

NOTE 2: (P), (A), OR (SS) FIELD-APPLIED JACKET ON OUTDOOR INSTALLATIONS, EXPOSED AND CONCEALED.

NOTE 3: FOR OUTDOOR USE ONLY.

NOTE 4: PIPING INSULATION IS NOT REQUIRED BETWEEN THE CONTROL VALVE AND COIL ON RUNOUTS, WHEN THE CONTROL VALVE IS WITHIN 4-FEET OF THE COIL AND THE PIPE SIZE IS 1-INCH OR LESS.

NOTE 5: FOR PIPING EXPOSED TO OUTDOOR AIR, INCREASE INSULATION THICKNESS BY 1-INCH.

NOTE 6: INSULATION THICKNESS IS BASED ON INSULATION HAVING A THERMAL CONDUCTIVITY OF 0.22 – 0.25 BTU·INCH/(H·FT²·deg F)

NOTE 7: VAPOR BARRIER REQUIRED.

3.19 INDOOR, FIELD-APPLIED JACKET SCHEDULE

- A. Install jacket over insulation material for all piping below 8 ft. height. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.
- B. Provide identification for all piping.
- C. Piping, Exposed:
 - 1. PVC, White – below 8 ft. height: 20 mils thick.

3.20 OUTDOOR, FIELD-APPLIED JACKET SCHEDULE

- A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.
- B. Ducts and Plenums, Exposed, up to 48 Inches in Diameter or with Flat Surfaces up to 72 Inches
 - 1. Stainless Steel, Type 304, Smooth 2B Finish: 0.020 inch thick.
 - 2. Aluminum, Alloy 3003, Temper H-14, Smooth Finish: 0.020 inch thick.
- C. Ducts and Plenums, Exposed, Larger Than 48 Inches in Diameter or with Flat Surfaces Larger Than 72 Inches:
 - 1. Stainless Steel, Type 304, Smooth Finish: 0.020 inch thick.
 - 2. Aluminum, Alloy 3003, Temper H-14, Smooth Finish: 0.020 inch thick.
- D. Piping, Exposed
 - 1. Stainless Steel, Type 304 Smooth 2B Finish 0.016 inch thick.

END OF SECTION 230700

SECTION 23 08 00 - COMMISSIONING OF HVAC**PART 1 – GENERAL****1.01 SUMMARY****A. Work Included:**

1. Definitions, warranties, test equipment requirements, and mechanical commissioning requirements.

1.02 RELATED SECTIONS

- A. Contents of Division 23, HVAC and Division 01, General Requirements apply to this Section.
- B. In addition, reference the following:
 1. Section 01 91 13, General Commissioning Requirements.

1.03 REFERENCES AND STANDARDS

- A. References and Standards as required by Section 23 00 00, HVAC Basic Requirements and Division 01, General Requirements.
- B. In addition, meet the following:
 1. Current edition of ASHRAE Guideline 0, The Commissioning Process.

1.04 SUBMITTALS

- A. Submittals as required by Section 23 00 00, HVAC Basic Requirements and Division 01, General Requirements.
- B. In addition, provide:
 1. Certificates of readiness.
 2. Certificates of completion of installation, prestart, and startup activities.
 3. Operation and Maintenance Manuals.
 4. Test reports.
 5. Control Drawings Submittal
 - a. Provide a key to abbreviations.
 - b. Provide graphic schematic depictions of the systems and each component.
 - c. Include the system and component layout of any equipment that the control system monitors, enables or controls, even if the equipment is primarily controlled by packaged or integral controls.
 - d. Provide a full points list with at least the following included for each point:
 - 1) Controlled system
 - 2) Point abbreviation

- 3) Point description
 - 4) Display unit
 - 5) Control point or set point (Yes / No)
 - 6) Monitoring point (Yes / No)
 - 7) Intermediate point (Yes / No)
 - 8) Calculated point (Yes / No)
6. Architect forwards one set of submittals for systems to be commissioned to Commissioning Agent at same time as design team.
 7. Commissioning Agent forwards comments to design team for consideration in their submittal response.
 8. Design team sends consolidated response to submittals and copies to Commissioning Agent.

1.05 QUALITY ASSURANCE

- A. Quality assurance as required by Section 23 00 00, HVAC Basic Requirements and Division 01, General Requirements.
- B. Test Equipment Calibration Requirements: Contractors will comply with test manufacturer's calibration procedures and intervals. Recalibrate test instruments immediately after instruments have been repaired resulting from being dropped or damaged. Affix calibration tags to test instruments. Furnish calibration records to Commissioning Authority upon request.

1.06 WARRANTY

- A. Warranty of materials and workmanship as required by Section 23 00 00, HVAC Basic Requirements and Division 01, General Requirements.
- B. In addition, provide:
 1. Commissioning, inspecting, and testing will not modify terms or time periods of mechanical equipment, systems, and controls warranties including related equipment and systems, and adjacent work.
 2. Control system warranty period starts from date of Commissioning Agent acceptance.

1.07 COORDINATION

- A. Reference Section 01 91 13, General Commissioning Requirements, for requirements pertaining to coordination during the commissioning process.

1.08 PURPOSE

- A. Purpose of commissioning process is to provide Owner assurance that systems have been installed in prescribed manner and will operate within performance guidelines. Commissioning is intended to enhance quality of system startup and aid in orderly transfer of systems to beneficial use by Owner.
- B. Commissioning procedures and results will be observed by Commissioning Authority or Owner's staff. Contractor is expected to verify functional readiness of systems to be tested prior to performing the tests in presence of Owner's witness. A high rate of test failure will indicate that Contractor has not adequately verified readiness of systems.

PART 2 – PRODUCTS

2.01 TEST EQUIPMENT

- A. Provide standard testing equipment required to perform startup, initial checkout and functional performance testing for the equipment being tested. For example, the mechanical contractor of Division 23, HVAC will ultimately be responsible for standard testing equipment for the HVAC&R system and controls system in Division 23, HVAC, except for the equipment specific to and used by TAB in their commissioning responsibilities. Provide a sufficient quantity of two-way radios by each subcontractor.
- B. Include special equipment, tools and instruments (specific to a piece of equipment and only available from vendor) required for testing in the base bid price to the Owner and leave on site, except for stand-alone data logging equipment that may be used by the Commissioning Authority.
- C. Manufacturer of equipment to provide proprietary test equipment and software required for programming and/or start-up, whether specified or not. Manufacturer provides the test equipment, demonstrates its use, and assists in the commissioning process as needed. Proprietary test equipment (and software) become the property of the Owner upon completion of the commissioning process.
- D. Data logging equipment and software required to test equipment will be provided by the Commissioning Authority, and will not become the property of the Owner.
- E. Use only testing equipment of sufficient quality and accuracy to test and/or measure system performance with the tolerances specified in the specifications. If not otherwise noted, the following minimum requirements apply: Temperature sensors and digital thermometers have a certified calibration within the past year to an accuracy of 0.5 degree F and a resolution of plus or minus 0.1 degree F. Pressure sensors have an accuracy of plus or minus 2.0 percent of the value range being measured (not full range of meter) and have been calibrated within the last year.

PART 3 – EXECUTION

3.01 GENERAL DOCUMENTATION REQUIREMENTS

- A. With assistance from the installing contractors, the Commissioning Authority will prepare prefunctional checklists for commissioned components, equipment, and systems

B. Red-Lined Drawings:

1. Verify equipment, systems, instrumentation, wiring and components are shown correctly on red-lined drawings.
2. Preliminary red-lined drawings must be made available to the Commissioning Team for use prior to the start of Functional Performance Testing.
3. Changes, as a result of Functional Testing, must be incorporated into the final as-built drawings, which will be created from the red-lined drawings.
4. The contracted party, as defined in the Contract Documents will create the as-built drawings.

C. Operation and Maintenance (O&M) Data:

1. Contractor will provide a copy of O&M literature within 45 days of each submittal acceptance for use during the commissioning process for commissioned equipment and systems.
2. The Commissioning Authority will review the O&M literature once for conformance to project requirements.
3. The Commissioning Authority will receive a copy of the final approved O&M literature once corrections have been made by the Contractor.

D. Demonstration and Training:

1. Contractor will provide demonstration and training as required by the specifications.
2. A complete training plan and schedule must be submitted by the contractor to the Commissioning Authority four weeks prior to any training.
3. A training agenda for each training session must be submitted to the Commissioning Authority one week prior the training session.
4. Notify the Commissioning Authority at least 72 hours in advance of scheduled tests so that testing may be observed by the Commissioning Authority and Owner's Authorized Representative. Provide a copy of the test record to the Commissioning Authority, Owner, and Architect.
5. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain specific equipment.
6. Train Owner's maintenance personnel on procedures and schedules for starting and stopping, trouble shooting, servicing, and maintaining equipment.
7. Review data in O&M Manuals.

E. Systems Manual Requirements:

1. The Systems Manual is intended to be a usable information resource containing the information related to the systems, assemblies, and Commissioning Process in one place with indexes and cross references.
2. Include final approved versions of the following information for the Systems Manual:
 - a. Facility Description.

- b. Basis of Design (A/E).
 - c. A list of contractors, subcontractors, suppliers, architects, and engineers involved in the project along with their contact information.
 - d. Overview of each commissioned system including operational schedules and sequence of operations.
 - e. General maintenance recommendations and procedures.
 - f. Recommendations for recalibration frequency of sensors and actuators
 - g. Recommended best practices for keeping the system running efficiently.
 - h. Blank Functional Performance Tests so the Owner can recommission the facility at a later date.
3. Organize and arrange information by building system, such as fire alarm, chilled water, heating hot water, etc.
 4. Provide Information in an electronic version to the extent possible. Legible, scanned images are acceptable for non-electronic documentation to facilitate this deliverable.

3.02 CONTRACTOR'S RESPONSIBILITIES

- A. Mechanical, Controls and TAB Contractors. The commissioning responsibilities applicable to each of the mechanical, controls and TAB contractors of Division 23, HVAC are as follows (references apply to commissioned equipment only):
 1. Perform commissioning tests at the direction of the Commissioning ~~Authority~~Agent.
 2. Attend construction phase controls coordination meetings.
 3. Attend testing, adjusting, and balancing review and coordination meetings.
 4. Participate in HVAC&R systems, assemblies, equipment, and component maintenance orientation and inspection as directed by the Commissioning Authority.
 5. Provide information requested by the Commissioning Authority for final commissioning documentation.
 6. Include requirements for submittal data, operation and maintenance data, and training in each purchase order or subcontract written.
 7. Prepare preliminary schedule for mechanical system orientations and inspections, operation and maintenance manual submissions, training sessions, pipe and duct system testing, flushing and cleaning, equipment start-up, testing and balancing and task completion for owner. Distribute preliminary schedule to commissioning team members.
 8. Update schedule as required throughout the construction period.
 9. During the startup and initial checkout process, execute the related portions of the prefunctional checklists for commissioned equipment.

10. Contractor to participate and complete checklists using the Commissioning Authority's web based commissioning software ~~Facility Grid~~Checklisten App. A desktop, laptop, tablet, or iPad will be required.
 11. Assist the Commissioning Authority in verification and functional performance tests.
 12. Gather operation and maintenance literature on equipment, and assemble in binders as required by the specifications. Submit to Commissioning Authority 45 days after submittal acceptance.
- B. Coordinate with the Commissioning Authority to provide 48 hour advance notice so that the witnessing of equipment and system start-up and testing can begin.
- C. Notify the Commissioning Authority a minimum of two weeks in advance of the time for start of the testing and balancing work. Attend the initial testing and balancing meeting for review of the official testing and balancing procedures.
- D. Participate in, and schedule vendors and contractors to participate in the training sessions.
- E. Provide written notification to the Construction Manager/General Contractor (CM/GC) and Commissioning Authority that the following work has been completed in accordance with the Contract Documents, and that the equipment, systems, and sub-system are operating as required.
1. HVAC&R equipment including fans, air handling units, ductwork, dampers, terminals, and other equipment furnished under this Division.
 2. Fire stopping in the fire rated construction, including fire and smoke damper installation, caulking, gasketing and sealing of smoke barriers.
 3. Fire detection and smoke detection devices furnished under other divisions of the specification.
- F. Equipment supplier to document the performance of his equipment.
- G. Test, Adjust and Balance Contractor:
- ~~1. Attend initial commissioning coordination meeting scheduled by the Commissioning Authority.~~
 2. Participate in verification of the testing and balancing report, which will consist of repeating measurements contained in the testing and balancing reports. Assist in diagnostic purposes when directed.
- H. Provide training of the Owner's operating staff using expert qualified personnel, as specified.
- I. Equipment Suppliers:
1. Provide requested submittal data, including detailed start-up procedures and specific responsibilities of the Owner, to keep warranties in force.
 2. Assist in equipment testing per agreements with contractors.
 3. Provide information requested by Commissioning Authority regarding equipment sequence of operation and testing procedures.
- J. Reference Section 01 91 13, General Commissioning Requirements for additional contractor responsibilities.

3.03 OWNER'S RESPONSIBILITIES

- A. Reference Section 01 91 13, General Commissioning Requirements for Owner's Responsibilities.

3.04 DESIGN PROFESSIONAL'S RESPONSIBILITIES

- A. Reference Section 01 91 13, General Commissioning Requirements for Design Professional's Responsibilities.

3.05 COMMISSIONING AUTHORITY'S RESPONSIBILITIES

- A. Reference Section 01 91 13, General Commissioning Requirements for Commissioning Authority's Responsibilities.

3.06 TESTING PREPARATION

- A. Certify in writing to the Commissioning Authority that HVAC&R systems, subsystems, and equipment have been installed, calibrated, and started and are operating according to the Contract Documents.
- B. Certify in writing to the Commissioning Authority that HVAC&R instrumentation and control systems have been completed and calibrated, that they are operating according to the Contract Documents, and that pretest set points have been recorded.
- C. Certify in writing that testing, adjusting, and balancing procedures have been completed and that testing, adjusting, and balancing reports have been submitted, discrepancies corrected, and corrective work approved.
- D. Place systems, subsystems, and equipment into operating mode to be tested (e.g., normal shutdown, normal auto position, normal manual position, unoccupied cycle, emergency power, and alarm conditions).
- E. Inspect and verify the position of each device and interlock identified on checklists.
- F. Check safety cutouts, alarms, and interlocks with smoke control and life-safety systems during each mode of operation.
- G. Testing Instrumentation: Install measuring instruments and logging devices to record test data as directed by the Commissioning Authority.

3.07 TESTING, ADJUSTING AND BALANCING VERIFICATION

- A. Prior to performance of Testing, Adjusting and Balancing work, provide copies of reports, sample forms, checklists, and certificates to the Commissioning Authority.
- B. Notify the Commissioning Authority at least 10 days in advance of testing and balancing Work, and provide access for the Commissioning Authority to witness testing and balancing Work.

- C. Provide technicians, instrumentation, and tools to verify testing and balancing of HVAC&R systems at the direction of the Commissioning Authority.
 - 1. The Commissioning Authority will notify testing and balancing subcontractor 10 days in advance of the date of field verification. Notice will not include data points to be verified.
 - 2. Testing and balancing subcontractor to use the same instruments (by model and serial number) that were used when original data were collected.
 - 3. Failure of an item includes, other than sound, a deviation of more than 10 percent. Failure of more than 10 percent of selected items to result in rejection of final testing, adjusting, and balancing report. For sound pressure readings, a deviation of 3 dB to result in rejection of final testing. Variations in background noise must be considered.
 - 4. Remedy the deficiency and notify the Commissioning Authority so verification of failed portions can be performed.

3.08 GENERAL TESTING REQUIREMENTS

- A. Provide technicians, instrumentation, and tools to perform commissioning test at the direction of the Commissioning Authority.
- B. Scope of HVAC&R testing to include entire HVAC&R installation, from central equipment for heat generation and refrigeration through distribution systems to each conditioned space. Testing to include measuring capacities and effectiveness of operational and control functions.
- C. Test operating modes, interlocks, control responses, and responses to abnormal or emergency conditions, and verify proper response of building automation system controllers and sensors.
- D. The Commissioning Authority along with the HVAC&R contractor, testing and balancing Subcontractor, and HVAC&R Instrumentation and Control Subcontractor to prepare detailed testing plans, procedures, and checklists for HVAC&R systems, subsystems, and equipment.
- E. Tests will be performed using design conditions whenever possible.
- F. Simulated conditions may need to be imposed using an artificial load when it is not practical to test under design conditions. Before simulating conditions, calibrate testing instruments. Provide equipment to simulate loads. Set simulated conditions as directed by the Commissioning Authority and document simulated conditions and methods of simulation. After tests, return settings to normal operating conditions.
- G. The Commissioning Authority may direct that set points be altered when simulating conditions is not practical.
- H. The Commissioning Authority may direct that sensor values be altered with a signal generator when design or simulating conditions and altering set points are not practical.
- I. If tests cannot be completed because of a deficiency outside the scope of the HVAC&R system, document the deficiency and report it to the Owner. After deficiencies are resolved, reschedule tests.
- J. If the testing plan indicates specific seasonal testing, complete appropriate initial performance tests and documentation and schedule seasonal tests.

3.09 HVAC&R SYSTEMS, SUBSYSTEMS, AND EQUIPMENT TESTING PROCEDURES

- A. Equipment Testing and Acceptance Procedures: Testing requirements are specified in individual Division 23, HVAC Sections. Provide submittals, test data, inspector record, and certifications to the Commissioning Authority.
- B. HVAC&R Instrumentation and Control System Testing: Field testing plans and testing requirements are specified in Division 23, HVAC Sections "Instrumentation and Control for HVAC" and "Sequence of Operations for HVAC Controls." Assist the Commissioning Authority with preparation of testing plans.
- C. Pipe System Cleaning, Flushing, Hydrostatic Tests, and Chemical Treatment: Test requirements are specified in Division 23, HVAC Piping Sections. HVAC&R Contractor to prepare a pipe system cleaning, flushing, and hydrostatic testing plan. Provide cleaning, flushing, testing, and treating plan and final reports to the Commissioning Authority. Plan to include the following:
 - 1. Sequence of testing and testing procedures for each section of pipe to be tested, identified by pipe zone or sector identification marker. Markers keyed to Drawings for each pipe sector, showing the physical location of each designated pipe test section. Provide drawings keyed to pipe zones or sectors formatted to allow each section of piping to be physically located and identified when referred to in pipe system cleaning, flushing, hydrostatic testing, and chemical treatment plan.
 - 2. Description of equipment for flushing operations.
 - 3. Minimum flushing water velocity.
 - 4. Tracking checklist for managing and ensuring that pipe sections have been cleaned, flushed, hydrostatically tested, and chemically treated.
- D. The work included in the commissioning process involves a complete and thorough evaluation of the operation and performance of components, systems and sub-systems. Evaluate the following equipment and systems:
 - 1. HVAC Equipment and Systems (all)
 - 3. Building Automation System
 - 5. Computer Room AC
 - 6. Domestic Hot Water Systems
 - 8. Pumps

3.10 DEFICIENCIES/NONCONFORMANCE, COST OF RETESTING, FAILURE DUE TO MANUFACTURER DEFECT

- A. Reference Division 01, General Requirements for requirements pertaining to deficiencies/nonconformance, cost of retesting, or failure due to manufacturer defect.

3.11 OPERATION AND MAINTENANCE MANUALS

- A. The Operation and Maintenance Manuals to conform to Contract Documents requirements as stated in Division 23, HVAC.
- B. Provide an updated as-built version of the control drawings and sequences of operation in the final controls O&M manual submittal.

END OF SECTION 23 08 00

SECTION 23 09 20 – BUILDING AUTOMATION SYSTEM (BAS) - GENERAL**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the installation of a Building Automation/Management System (BAS/BMS) for control and monitoring of systems (HVAC, energy, and lighting) as set forth with Lidl National (NAT) & International (INT).

1.2 APPLICATION OF BUILDING AUTOMATION SYSTEM (BAS)

- A. The distributed digital control (DDC) and Building Automation System (BAS/BMS) defined herein shall provide a complete BMS for all mechanical systems and other facility systems as included in the project documents. The contractor shall provide a complete and operational system to perform all sequences of operations stated in Section 23 or as shown on the control drawings.
- B. The BMS Head Controller/Building Level Controller shall communicate with Lidl server in Germany via an approved "L-Box" or equal configuration.
- C. The BMS shall utilize electronic sensing, microprocessor-based digital control, and electronic actuation of dampers and valves to perform the control sequences and functions specified.
- D. The BMS for this project shall consist of monitoring and control of the systems indicated.
- E. All interlock wiring for mechanical system equipment shall be by this contractor unless specifically stated otherwise. This shall include, but not be limited to, items such as thermostats for unit heaters, interlock wiring to rooftop factory controllers, flow switches, and duct smoke detectors.
- F. System to be a fully integrated BMS incorporating direct digital control (DDC) for energy management, equipment monitoring and control, and subsystems with open communications capabilities as herein specified.
- G. All wiring, conduit, panels, and accessories for a complete operational system.
- H. BMS Contractor shall be responsible for all electrical work associated with the BMS:
 - 1. Perform all wiring in accordance with all local and national codes.
 - 2. All line voltage and low voltage control wiring, concealed or exposed, shall be in conduit in accordance with the division 26 specifications, NEC and local building code.
 - 3. Provide a maximum of 50 feet extension of 120 volt, 20 amp circuits and circuit breakers from Emergency power panels for all BMS equipment power. All control panels shall be wired to the UPS systems.
 - 4. Surge transient protection shall be incorporated in design of system to protect electrical components in all DDC Controllers and the operator interface.
 - 5. Provide installation kits for fluid flow meters, which allow service and removal under pressure.

1.3 APPLICATION OF ENERGY CONSERVATION INITIATIVES

- A. Night set back schedules for space unoccupied times shall be incorporated into design.

- B. Demand control ventilation shall be incorporated as a function of rooftop factory controls.
- C. Design systems for spaces that need 24 hours per day operations separate from spaces that can utilize night set back. Example, data centers shall be design separate from the building HVAC system.
- D. Incoming electrical and gas utilities shall be metered.

1.4 APPLICATION OF OPEN PROTOCOLS

- A. Provide open communications system. The system shall be an open architecture with the capabilities to support a multi-vendor environment. The BMS shall be capable of utilizing industry-standard protocols and as well as with integrate third-party systems as follows:
 - 1. Capable of Modbus RTU communication
 - 2. Capable of BACnet-MSTP communication according to ANSI/ASHRAE 135-2004.
 - 3. Capable of BACnet-IP communication according to ANSI/ASHRAE 135-2008.
 - 4. All supervisory (BC) and field-level controllers (AAC, ASC) shall be certified and listed under the appropriate device profile.
- B. Software Licensing:
 - 1. Include open/non-proprietary licensing for all software packages at the required Operator Interface located on the BMS panel.
 - 2. All operator interface, programming environment, networking, database management and any other software used by the Contractor to install the system or needed to operate the system to its full capabilities shall be open-licensed and provided to the Owner.
 - 3. Open/non-proprietary licensing shall allow access to all aspects of the system including system access, thin-client workstations, points, programming, database management, graphics etc. No restrictions shall be placed on the licensing.

1.5 APPLICATION OF HARDWARE

- A. Provide hardware, software, and wiring to provide communication and/or interfaces with each of the applicable major systems listed below:
 - 1. Rooftop Units
 - 2. AC Units
 - 3. Exhaust Fan Units
 - 4. Lighting Control System
 - 5. UPS System
 - 6. Generator System
 - 7. Gas Meter
 - 8. Electrical Meter
 - 9. Miscellaneous Monitoring
- B. Operator interface hardware and standard peripherals located on the BMS panel shall be furnished by the BMS Contractor and it shall match Lidl's IT Department standards.

1.6 APPLICATION OF GRAPHICS AND WEB-BASED SYSTEM

- A. Provide system graphics for each controlled device and/or integrated systems as required by the owner.

- B. Origin of information shall be transparent to the operator and shall be controlled, displayed, trended, etc. as if the points were hardwired to the BMS.
- C. System should be web based with the user interface software – including graphics - imbedded in each Building Controller.
- D. Any owner provided BMS computer(s) connected physically or virtually to the store's local BMS network shall have access to the web based user interface.

1.7 APPROVED CONTROL SYSTEM AND INTEGRATION CONTRACTORS

- A. The BMS system shall be furnished, engineered, installed and subject to compliance with requirements, provided on **LiDL National Account List**:
 - a. Emerson Climate Technologies, Inc.
 - b. No Substitutions/Exceptions without owner approval
- B. The BMS system shall be designed and installed, commissioned, and serviced by factory trained personnel. BMS contractor shall have an in-place support facility within 100 miles of the site with technical staff, spare parts inventory and necessary test and diagnostic equipment. The BMS contractor shall provide full time, on site, experienced project manager for this work, responsible for direct supervision of the design, installation, start up and commissioning of the BMS. The Bidder shall be regularly engaged in the installation and maintenance of BMS systems and shall have a minimum of ten (10) years of demonstrated technical expertise and experience in the installation and maintenance of BMS systems similar in size and complexity to this project.
- C. The BMS contractor shall maintain a service organization consisting of factory trained service personnel and provide a list of 10 projects, similar in size and scope to this project, completed within the last five years.

1.8 CODES, STANDARDS, AND REFERENCES

- A. The BMS design shall meet the most stringent of the Local Code, ASHRAE, or Lidl's Building Design Criteria Matrix conditions and shall meet latest the most stringent ASHRAE's standard 55 Thermal Environmental Conditions for occupant comfort zone or Lidl's operating temperatures.
- B. Work, materials, and equipment shall comply with the most restrictive of local, state, and federal authorities' codes and ordinances or these plans and specifications. As a minimum, the design shall comply with current editions of the following codes:
- C. National Electric Code (NEC), latest edition
- D. ANSI Standards, latest edition
- E. Local Building Code, latest edition (including mechanical, plumbing, fire protection, and energy codes)
- F. FM Global Property Loss Prevention Data Sheets and approved products as applicable.
- G. American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE) Standards, Latest edition.

1. 55: Thermal Environmental Conditions for Human Occupancy.
2. 62.1: Ventilation for Acceptable Indoor Air Quality.
3. 90.1: Energy Standard for Buildings.
4. 135: 2010: BACnet-A Data Communication Protocol for Building Automation and Control Networks.

H. Electronics Industries Alliance:

1. EIA-709.1-A-99: Control Network Protocol Specification.
2. EIA-709.3-99: Free-Topology Twisted-Pair Channel Specification.
3. EIA-232: Interface between Data Terminal Equipment and Data Circuit Terminating Equipment Employing Serial Binary Data Interchange.
4. EIA-458: Standard Optical Fiber Material Classes and Preferred Sizes.
5. EIA-485: Standard for Electrical Characteristics of Generator and Receivers for use in Balanced Digital Multipoint Systems.
6. EIA-472: General and Sectional Specifications for Fiber Optic Cable.
7. EIA-475: Generic and Sectional Specifications for Fiber Optic Connectors and all Sectional Specifications.
8. EIA-573: Generic and Sectional Specifications for Field Portable Polishing Device for Preparation Optical Fiber and all Sectional Specifications.
9. EIA-590: Standard for Physical Location and Protection of Below-Ground Fiber Optic Cable Plant and all Sectional Specifications.
10. EIA-852: Tunneling of Component Network Data over IP Channels.

I. Underwriters Laboratories:

1. UL 916: Energy Management Systems.
2. ULC C100: Smoke Control System Equipment.
3. UL 864: Control Unit and Accessories for Fire Alarm Systems.

J. NEMA Compliance:

1. NEMA 250: Enclosure for Electrical Equipment.
2. NEMA ICS 1: General Standards for Industrial Controls.

K. NFPA Compliance:

1. NFPA 90A: "Standard for the Installation of Air Conditioning and Ventilating Systems" where applicable to controls and control sequences.
2. NFPA 70: National Electrical Code (NEC).

L. Institute of Electrical and Electronics Engineers (IEEE):

1. IEEE 142: Recommended Practice for Grounding of Industrial and Commercial Power Systems.
2. IEEE 802.3: CSMA/CD (Ethernet – Based) LAN.
3. IEEE 802.4: Token Bus Working Group (ARCNET – Based) LAN.

M. European Electromagnetic Compatibility (EMC):

1. EN-50081-2: Emission Standard for Residential, Commercial, and Light-Industrial Environments.

2. EN-50082-2: Emission Standard for Industrial Environments.
 - N. Federal Communications Commission (FCC):
 1. 47-CFR-15: Regulations regarding Unlicensed Transmissions.
 - O. International Organization for Standardizations (ISO):
 1. ISO-9001:2008: Model for Quality Assurance in Production, Installation, and Servicing. Quality Management Systems.
 2. ISO-140001: Application of Well-Accepted Business Management Principles to the Environment.
 - P. Materials and equipment shall be the catalogued products of manufacturers regularly engaged in production and installation of automatic temperature control systems and shall be manufacturer's latest standard design that complies with the specification requirements.
 - Q. All BMS peer-to-peer building controllers and local user displays shall be UL Listed under Standard UL 916, category PAZX; Standard ULC C100, category UUKL7; and under Standard UL 864, categories UUKL, UDTZ, and QVAX and be so listed at the time of bid.
 - R. The BMS peer-to-peer network controllers and local user display shall also comply with the European Electromagnetic Compatibility (EMC) Framework, and bear the C-Tic Mark to show compliance. The purpose of the regulation is to minimize electromagnetic interference between electronic products, which may diminish the performance of electrical products or disrupt essential communications.
 - S. DDC peer-to-peer controllers shall be compliant with the European EMC Directive, Standards EN 50081-2 and EN 50082-2, at the Industrial Levels. Additionally the equipment shall be compliant with the European LVD Directive and bear the CE mark in order to show compliance to both directives.
 - T. All field level controllers shall comply, at a minimum, with UL Standard UL 916 category PAZX; and be so listed at the time of Bid.
 - U. All electronic equipment shall conform to the requirements of FCC Regulation, Part 15, and Governing Radio Frequency Electromagnetic Interference and be so labeled.
 - V. The manufacturer of the building automation system shall provide documentation supporting compliance with ISO-9001:2008 and ISO-140001. The intent of this specification requirement is to ensure that the products from the manufacturer are delivered through a Quality System and Framework that will assure consistency in the products delivered for this project.
- 1.9 DEFINITIONS
- A. Acknowledged: Data is broadcast repeatedly until an acknowledgement is received. Used for critical data using one to one bindings only. This type of service shall not be used for one to many bindings.
 - B. Adjustable (ADJ): A characteristic of a control logic parameter such that it can be varied by the operator without downloading the program.
 - C. Advanced Application Controller (AAC): A device with limited resources relative to the Building Controller (BC). It may support a level of programming and may also be intended for

application specific applications. A fully programmable control module. This control module shall be capable of certain advanced features found in Building Controllers (e.g. storing trends, and initiating read and write requests) but it shall not serve as a master controller. Advanced Application Controllers may reside on either the Ethernet/IP backbone or on a subnet. A BACnet device to be used as an AAC will meet the requirements of ASHRAE 135, Annex L and will be listed as an AAC by BACnet Testing Labs. A BTL listed device will carry the BTL Mark.

- D. Application Programming Tool: A vendor unique software tool used to create applications for programmable controllers.
- E. Application Specific Controller (ASC): A device with limited resources relative to the Advanced Application Controller (AAC). A pre-programmed control module, intended for use in a specific application. ASCs shall have limited configurability, allowing the user to select various pre-programmed options, but it shall not be fully customizable. A BACnet device used as an ASC will meet the requirements of ASHRAE 135, Annex L and will be listed as an ASC by BACnet Testing Labs. A BTL listed device will carry the BTL Mark.
- F. BACnet/BACnet Standard: BACnet communication requirements, as defined by ASHRAE/ANSI 135, current version including all annexes and addenda.
- G. Bandwidth Utilization: The average utilization of the network capacity. Network loading is controlled by the use of event driven broadcast based data propagation and the use of appropriate binding services.
- H. Broadcasting: The propagation of data from a device to the control network. Software objects that broadcast data to the network shall include the following parameters:
- I. Building Automation System (BAS): Same as Building Management System (BMS). The entire integrated energy management and control system.
- J. Building Controller (BC) (also "Head Controller"): A fully programmable control module capable of storing trends and schedules, serving as a router to devices on a subnet, and initiating read and write requests to other controllers. Typically this controller is located on the Ethernet/IP backbone of the BMS. A BACnet device to be used as a BC shall meet the requirements of ASHRAE 135, Annex L and will be listed as a BC by BACnet Testing Labs. A BTL listed device will carry the BTL Mark.
- K. Change of Value (COV): An event that occurs when a measured or calculated analog value changes by a predefined amount (ASHRAE/ANSI 135-1995).
- L. Client: A device that is the requestor of services from a server. A client device makes requests of, and receives responses from, a server device.
- M. Configuration Parameter: An input network variable to a controller that is written to the EEPROM.
- N. Connection: Made when a device is physically connected to a communication cable. Devices that count against the number of connections limit include adapters, any sensor, actuator, or controller, and a router or repeater. Terminators are not considered to be a connection.
- O. Continuous Monitoring: Sampling and recording of a variable based on time or change of state (e.g. trending an analog value, monitoring a binary change of state).

- P. Web Server (WS): Provides access to the control system. This device will allow access to the control system with a web browser over the Owner's network. The webserver shall have NIC cards so that it functions as the bridge between the local supervisory LAN and the Owner.
- Q. Direct Digital Control (DDC): Microprocessor-based control including Analog/Digital conversion and program logic
- R. Media Access Control address (MAC): Hardware address that uniquely identifies each node of a network. Each different type of network medium requires a different MAC layer.
- S. Managed Communication: Transmission of data from a controller to a data manager, which in turn re-broadcasts the data to a second controller.
- T. Manual Control: Where the operator takes control of an end device and forces a specific position or state. The manual mode and the desired manual position or states are parameters that are set by the operator.
- U. Network Controller: Term used to describe a BMS hardware component that serves several key functions:
 - 1. Serve as the LANID.
 - 2. Transmission of data to an owner interface on the TCP/IP network.
 - 3. Capable of collecting/integrating with different protocol data such as BACnet IP, BACnet MS/TP, Modbus, LON, OPC, etc.
 - 4. Location for time schedules to support all of the devices.
 - 5. Location for trend logs for all data to be trended from the devices.
 - 6. Location for alarm handling software.
 - 7. Shall process event broadcasted data from the devices (alarm indication) and enter the appropriate alarm information in the alarm reporting system at the TCP/IP level.
- V. Stand-Alone Controller: A stand-alone controller has provisions for all of the physical inputs and physical outputs associated with a single mechanical component such as a terminal unit, air handling unit, chiller or boiler. The controller shall also have embedded in it all of the control logic that associated the physical inputs to the physical outputs. A stand-alone controller may rely on other networked devices for time schedule inputs and trend data storage.
- W. Test Mode: A concept where the operator from the operator interface can interrupt the flow of data from a sensor to the control logic and insert a mandatory test value or test state to be used by the control logic. The test mode and the desired test value or states are parameters that are set by the operator.
- X. Unacknowledged Repeated: The data being broadcast is sent three times and an acknowledgement of receipt is not required. This type of service shall be used for most process control related data requiring timely receipt of the data.
- Y. Unacknowledged: The data being broadcast is sent one time and an acknowledgement of receipt is not required. This type of service shall be used for non-critical data where there is no significant impact should the receiving device have to wait for the next broadcast.

1.10 FUNCTIONAL INTENT

- A. Where detailed functional or performance requirements are specified, products intended for the Project, conforming to the specified requirements, must be submitted to, and approved by, Owner prior to shipment to the Project site.

1.11 SUBMITTALS

- A. Electronic Submittals: While all requirements for hard copy submittal apply, control submittals and O&M information shall also be provided in electronic format as follows.
1. Drawings and Diagrams: Shop drawings shall be provided on electronic media as an AutoCAD drawing file and/or Adobe Portable Document Format (PDF) file. All 'X-Reference' and font files must be provided with AutoCAD files.
 2. Other Submittals: All other submittals shall be provided in Adobe PDF. Provide documents, such as Cheat Sheets and Trouble Shooting Guide, in rich text format (RTF) or Microsoft Word format as required.
 3. BIM modeling for temperature controls shall be furnished by the BMS Contractor. All control panels and raceways (at $\frac{3}{4}$ " or greater) shall be modeled for clash-detection efforts with the prime contractor.
 4. The naming convention for all physical control points and software objects shall be reviewed in a meeting with the Owner's Representative, Commissioning Agent, and Engineer before generation of control drawings and sequences of operation.
 5. Where applicable, any deviations against the design intent shall be summarized with associated description of said deviation and itemized with an associated credit for final approval. Deviation(s) not listed and submitted may warrant an automatic rejection.
 6. All sequences shall be reviewed with thought and care with respect to programming needs for the specific project's design. Insufficient or generic sequence details may warrant an automatic rejection.
 7. Generic, non-compliant submittals shall not be reviewed, and shall be returned for resubmittal.
 8. BMS Contractor shall not order material, begin fabrication or field installation until receiving authorization to proceed in the form of an approved submittal. BMS Contractor shall be solely responsible for the removal and replacement of any item not approved by submittal at no cost to the Owner.
 9. It is the BMS contractors' responsibility to coordinate and allow adequate time for an approval process. Where an expedited submittal review is required, the contractor(s) shall still be responsible for any associated project delays and pay all additional services incurred by the design engineer, at the design engineer's current standard rate.
 10. The design engineer shall perform no more than two reviews of each submittal item. Should the submittals not be acceptable after the second review, the Contractor shall pay all additional expenses incurred by the design engineer in reviewing additional submittals at the design engineer's current standard rate.
 11. At the time of the 23 09 00 series submittal, the BMS Contractor shall submit a copy of the following with a response of "comply" or "non-comply" for every bullet point for:
 - a. 23 09 20, Section 1.9 "SUBMITTALS"
 - b. Entirety of the 23 09 00 series and sub-sections, upon further request
- B. Shop Drawings: Submit shop drawings for each control system, including a complete drawing for each air handling unit, system, pump, device, etc. with all point descriptors, addresses and point names indicated. Each shop drawing shall contain the following information:
1. System Architecture and System Layout:

- a. One-line diagram indicating schematic locations of all control units, owner interface, LAN interface devices, gateways, etc. Indicate network number, device ID, address, device instance, MAC address, drawing reference number, and controller type for each control unit. Indicate media, protocol, baud rate, and type of each LAN. All optical isolators, repeaters, end-of-line resistors, junctions, terminators, ground locations etc. shall be located on the diagram.
 - b. Provide floor plans locating all control units, owner interfaces, head controller, LAN interface devices, gateways, etc. Include all WAN and LAN communication wiring routing, power wiring, power originating sources, and low voltage power wiring. Indicate network number, device ID, address, device instance, MAC address, drawing reference number, and controller type for each control unit. All optical isolators, repeaters, end-of-line resistors, junctions, ground locations etc. shall be located on the floor plans. Wiring routing as-built conditions shall be maintained accurately throughout the construction period and the drawing shall be updated to accurately reflect accurate, actual installed conditions.
2. Schematic flow diagram of each air and water system showing fans, coils, dampers, valves, pumps, heat exchange equipment and control devices. Include contractor written description of sequence of operation.
3. All physical points on the schematic flow diagram shall be indicated with names, descriptors, and point addresses identified as listed in the point summary table.
4. Provide a dedicated shop drawings of all space sensors, thermostats, OA sensors, and space pressure sensors for final EOR and AOR review.
5. With each schematic, provide a point summary table listing building number and abbreviation, system type, equipment type, full point name, point description, Ethernet backbone network number, network number, device ID, object ID (object type, instance number). Refer to other Building Automation System (BAS) 23 09 00 specification series for additional requirements.
6. Label each control device with setting.
7. Label each input and output with the appropriate range.
8. Provide a Bill of Materials with each schematic. Indicate device identification to match schematic and actual field labeling, quantity, actual product ordering number, manufacturer, description, size, voltage range, pressure range, temperature range, etc. as applicable. Also identify the specification section and specification reference.
9. Indicate all required electrical wiring. Electrical wiring diagrams shall include both ladder logic type diagram for motor starter, RTU, control, and safety circuits and detailed digital interface panel point termination diagrams with all wire numbers and terminal block numbers identified. Provide panel termination drawings on separate drawings. Ladder diagrams shall appear on system schematic. Clearly differentiate between portions of wiring, factory-installed and portions to be field-installed. For all devices with safety circuits, field wiring will be labeled and all added devices will be properly mounted. Any internal wiring changes shall be approved by the manufacturer in writing. If for example a gas booster needs to be tied into the burner circuit the manufacturer shall identify the terminal points and provide an updated control diagram.
10. Sample Operator Interface Graphic Screens for each unique type of system, with final screens to be received 60 days prior to system startup.
11. Details of control panels, including controls, instruments, and labeling shown in plan or elevation indicating the installed locations.
12. Sheets shall be consecutively numbered.
13. Each sheet shall have a title indicating the type of information included and the HVAC system controlled.
14. Table of Contents listing sheet titles and sheet numbers.
15. Legend and list of abbreviations.

16. Provide an operating schedule for review. The schedule shall have a schedule for each AHU/RTU and the associated equipment. Terminal units serving offices will have a separate schedule from the labs.
- C. Product Data:
1. For each control device, panel, and accessory indicated or furnished. Include dimensions, capacities, performance and electrical characteristics, and material finishes.
 2. Include installation and start-up instructions.
 3. When manufacturer's cutsheets apply to a product series rather than a specific product, clearly indicate applicable data by highlighting or by other means general catalogs shall not be accepted as cutsheets to fulfill submittal requirements.
- D. Product Line Demonstrated History: The product line being proposed for the project must have an installed history of demonstrated satisfactory operation for a length of one (1) year since date of final completion in at least ten (10) installations of comparative size and complexity. Submittals shall document this requirement with references.
- E. Qualifications: For manufacturer, companies, Contractor and key personnel.
- F. Checkout and Testing Forms: Submit a blank copy of the forms that will be used during Point-to-Point Checkout, Prefunctional Checkout, and Functional Performance Testing. Those forms should be structured to capture the following information at a minimum during each particular testing phase.
1. Point-to-Point Checkout Form containing the following information:
 - a. Each point is addressed, labeled and that proper communication exists between the controller and the field device.
 - b. Documents that installed condition match the control drawings and that any changes or differences are noted on the drawings.
 2. Prefunctional Checkout Forms containing the following information:
 - a. Documents correct voltage and or current present as well as verifying circuits are free from grounds or faults for each control device.
 - b. Obtain and Record Test and Balance settings and incorporate into the BMS. Information from the TAB contractor shall include:
 - 1) Air system differential pressure and flow settings.
 - 2) RTU minimum outside air control point or damper setting.
 - c. Calibration data for all sensing and actuating devices recording final measured and displayed value. Record the type and model of the meter(s) that determined the measured value for analog inputs.
 - d. For analog outputs record both the displayed output as well state of the receiving device.
 - e. For digital input/outputs record the signal at the controller and the state of the sensing/control device.
 - f. For actuators:
 - 1) Check to ensure that actuated device moves smoothly and results are repeatable thru full range and seals tightly when the appropriate signal is applied to the operator.

- 2) Check for appropriate fail position, and that the stroke and range is as required.
 - 3) For sequenced electronic actuators, calibrate in accordance with manufacturer's instructions to required ranges. Record final settings.
 - g. For all actuators, verify the actual position against the Operator Interface readout.
 - h. For air flow measuring stations the data recorded shall include the independent flow measurement, area, and the independently measured output of the flow station. The BMS input from the flow station and any factors used to calculate the flow including area and any constants used in the calculation of flow. Two sets of data shall be collected. The first at design flow and the second at 50% of design flow. It is not acceptable to simply add a correction factor to address differences between the flow station and the independent reading.
 - i. For Operator Interfaces and Web accessible display:
 - 1) Verify that all elements on the graphics are functional and are properly bound to physical devices and/or virtual points, and that hot links or page jumps are functional and logical.
 - 2) Output all specified BMS reports for review and approval.
 - 3) Verify that the alarm pop ups, printing, and logging are functional and in accordance with requirements.
 - 4) Verify that all points are trended and are archiving to disk. Provide a sample to the Commissioning Authority and owner for review.
 - 5) Verify that paging/dial-out alarm annunciation is functional.
 - 6) Verify the functionality of remote Operator Interfaces and that a robust connection can be established consistently.
 - 7) Verify that required third party software applications required with the bid are installed and are functional.
 - j. For all actuating devices record final settings for device.
 - k. Document verification of point to graphics binding for all points displayed on the owner interface and that webserver display have been mapped correctly, and display the correct information.
 - l. Document that the webserver is on LAN and can be viewed from off site, that the modem is connected, and that the BMS is accessible by the contractor.
3. Functional Performance Forms shall contain:
- a. List of all sequences, modes of operation and setpoint that initiates each sequence and/or mode. For each confirm that proper sequence of operation. Document any variance between designed sequence and actual condition.
 - b. Record tuning parameters and response time for each control loop.
 - c. Document all alarm and safeties test and final results.
 - d. Results of trends including controlled points, setpoints, actual readings, and other point defined by the Boards Authorized Representative.

G. Testing Plan:

1. Submit a plan for executing all phases of testing and completion of checkout forms. This includes the following: manufacturers' normal testing, point-to-point testing, pre-functional testing, and functional performance testing. The testing plan shall show the overall milestones of the controls work and testing of the controls system.
2. Provide the schedule for completing each phase of testing for each system or set of equipment including, but not limited to, rooftop units, exhaust fans, miscellaneous

monitoring network wiring, and operator interface. Schedules shall show the time frame needed to complete the tasks.

3. The testing plan shall identify other trade milestones that impact the successful completion of during each phase of testing.
4. This plan is not meant to take precedence over any other plan but is intended to provide coordination assistance to all trades as the project is scheduled.

H. Open Protocol Information:

1. General: Provide all information necessary for review of the proposed system, including information required by the authority maintaining the protocol standard to determine if the product selected for implementation complies with the protocol standards specified.
2. BACnet Systems:
 - a. BACnet object description, object ID, and device ID, for each I/O point.
 - b. Documentation for any non-standard BACnet objects, properties, or enumerations used detailing their structure, data types, and any associated lists of enumerated values.
 - c. Documentation of all explicit messaging.
 - d. Submit PICS indicating the standardized BACnet device profile, functionality and configuration of each controller along with proof of BTL listing.

I. Framed Control Drawings: After completion of installation and check out, but prior to training, laminated control drawings including system control schematics, sequences of operation, and panel termination drawings, shall be provided in panels for major pieces of equipment. Terminal unit drawings shall be located in the central plant equipment panel or mechanical room panel.

J. Control Logic Documentation:

1. Documents are to be submitted, recieved, and approved prior to training.
2. Submit control logic program listings (for graphical programming) and logic flow charts illustrating (for line type programs) to document the control software of all control units.
3. Control logic shall be annotated to describe how it accomplishes the sequence of operation. Annotations shall be sufficient to allow an operator to relate each program component (block or line) to corresponding portions of the specified Sequence of Operation.
4. Include written description of each control sequence.
5. Include control response, settings, setpoints, throttling ranges, gains, reset schedules, adjustable parameters, and limits.
6. Sheets shall be consecutively numbered.
7. Each sheet shall have a title indicating the controller designations and the HVAC system controlled.
8. Include Table of Contents listing sheet titles and sheet numbers
9. Submit one complete set of programming and operating manuals for all digital controllers concurrently with control logic documentation.

K. Operation and Maintenance Manual:

1. Completion of project shall include Operation and Maintenance (O&M) Manuals. Consultant shall outline requirements for Contractor's submission and review with Lidl.
2. Operations and Maintenance Manuals shall be submitted during construction. Manuals shall be formatted in current CSI Master Specification format. Reviewed O&M Manuals shall be submitted in full prior to Lidl's facilities operation participation in start-up activities

- near the end of construction and to be coordinated with Information Resources standards.
3. In addition to other copies required, submit one copy of the approved materials directly for building operation staff.
 4. Submit maintenance instructions and spare parts lists for each type of control device, control unit, and accessory.
 5. Submit BMS User's Guides (Operating Manuals) for each controller type and for all operator interface hardware and software and peripherals.
 6. Submit BMS advanced Programming Manuals for each controller type and for all operator interface software.
 7. Include all as-built submittals (product data, shop drawings, control logic documentation, hardware manuals, software manuals, installation guides or manuals, maintenance instructions, and spare parts lists) in maintenance manual, in accordance with requirements of Division 01 Section "Operations and Maintenance Manual."
- L. Video Training: The BMS contractor shall adhere to the Demonstration and Training Videos requirements detailed in section 01 79 00.
- M. Demonstration of operator interface device display capabilities.
- N. Manufacturers Certificates: For all listed and/or labeled products, provide certificate of conformance.
- O. Product Warranty Certificates: Submit manufacturers product warranty certificates covering the hardware provided.

1.12 PROJECT RECORD DOCUMENTS

- A. The Project Record documents that have not already been submitted as part of the Operating and Maintenance Manual or Training Manual are to be submitted with the Record Documents. Any documents in the Operating and Maintenance Manual or Training Manual that have changed since they were submitted will need to be re-submitted as part of the Project record documents. All of these documents may be submitted electronically.
- B. Record copies of product data and control shop drawings updated to reflect the final installed condition.
- C. Record copies of approved control logic programming and database on USB drives. The USB drives shall contain all information required to reinstall the control system program. It will include actual setpoints and settings of controls, final sequence of operation, including changes to programs made after submission and approval of shop drawings and including changes to programs made during specified testing. One set of USB drives will be stored at the building in the main control panel and the second set will be provided to Owner's facilities department. A third set shall be provided to the EOR for record keeping only.
- D. Graphic Software: Record copies of approved project specific graphic software on USB drives.
- E. Provide as-built network architecture drawings showing all BACnet nodes including a description field with specific controller identification, description and location information.
- F. Include individual floor plans with controller locations with all interconnecting wiring routing including space sensors, LAN wiring, power wiring, low voltage power wiring. Indicate device instance, MAC address and drawing reference number.

- G. Provide record riser diagram showing the location of all controllers.
- H. Maintain project record documents throughout the warranty period and submit final documents at the end of the warranty period

1.13 OPERATOR INTERFACE

- A. The Operator Interface (OI) shall provide for overall system supervision, graphical user interface, management report generation, alarm annunciation, remote monitoring, and trend reporting. Refer to other Building Automation System (BAS) 23 09 00 specification series for additional requirements.

1.14 SYSTEM ARCHITECTURE

- A. Application of Open Protocols:
 - 1. Subject to the detailed requirements provided throughout the specifications, the BMS and digital control and communications components installed, as work of this contract shall be an integrated distributed processing system utilizing the following standards:
 - a. BACnet or Hybrid System: The system architecture shall consist of a BACnet IP Router, a single Local Area Network (LAN) or two-level LANs that support BCs, AACs, ASCs, Operator Interface, Smart Devices (SD), and Remote Communication Devices (RCDs) as applicable. In no event shall there be more than two levels of LAN topology within the system, excluding wiring to sensors with no control intelligence.
- B. The system provided shall incorporate hardware resources sufficient to meet the functional requirements specified. The Contractor shall include all items not specifically itemized in these Specifications that are necessary to implement, maintain, and operate the system in compliance with the functional intent of these Specifications.
- C. The system shall be configured as a distributed processing network(s) capable of expansion as specified below. Refer to the network architecture on the BMS drawings for other requirements and details.
- D. The system architecture shall consist of an Ethernet-based, wide area network (WAN), a single Local Area Network (LAN) or multi-leveled LANs that support BCs, Operator Interface, and Remote Communication Devices (RCDs) as applicable. The following indicates a functional description of the BMS structure.
 - 1. Owner WAN: Intranet-based network connecting multiple facilities with a central data warehouse and BMS interface/server, accessible via standard web-browser.
 - 2. Local BMS Supervisory LAN: The Local BMS Supervisory LAN shall be an Ethernet-based, 100 Mbps LAN connecting Primary Control LANs and OWSs. The LAN serves as the inter-BC gateway and OWS-to-BC gateway and communications path and as the connection point for the Owner WAN. LAN shall be IEEE 802.3 Ethernet over Fiber or Category 5 cable with switches and routers that support 100 Mbps throughput. Power-line carrier communication shall not be acceptable for communications. The higher level layers of this network shall be BACnet/IP of the BACnet standard, and shall share a common network number for the Ethernet backbone, as defined in BACnet.
Primary Controller LAN ('Primary LAN'): High-speed, peer-to-peer communicating LAN used to connect and BC and communicate exclusively control information. A BACnet network shall be used to connect AACs, ASCs or SDs. These can be Master Slave/

Token Passing or polling, or Modbus RTU, in addition to those allowed for Primary Controller LANs. Network speed vs. the number of controllers on the LAN shall be dictated by the response time and trending requirements. The primary network shall communicate at a minimum of 38 kbps. Each secondary network may support up to 32 communicating devices without segmentation or repeaters subject to the requirements for response time, trending and bandwidth utilization.

- E. Dynamic Data Access: Any data throughout any level of the network shall be available to and accessible by all other devices, Controllers and OWS, whether directly connected or connected remotely.
- F. Remote Data Access: The system shall support the following methods of remote access to the building data.
 - 1. Browser-based access: A remote user, connecting via the Owner WAN and using a standard browser shall be able access all control system facilities and graphics with proper password. The remote access user will require a password to enable applications to view the web pages.
- G. Network Performance: The communication speed between the controllers, control LAN interface devices, and operator interface devices shall be sufficient to ensure fast system response time under any loading condition. Contractor shall submit guaranteed response times with shop drawings including calculations to support the guarantee. Contractor shall reconfigure LAN as necessary to accomplish these performance requirements. The performance shall also include the trending of all AI, AO and DI points at 15-minute intervals.
- H. Data Storage and Operator Interface:
 - 1. The BMS contractor shall provide a Temperature Control Panel (TCP) with a DDC user LCD interface and a Building Level Controller (BLC) with Web Server (WS) functionalities. These two items shall maintain the systems configuration and programming database and is the operating platform BMS System.
 - 2. It shall hold the backup files of the information downloaded into the individual controllers and as such support uploading and downloading that information directly to/from the controllers.
 - 3. It shall be located within each facility.
 - 4. It shall also act as a control information server to non-HVAC system based controls.
 - 5. It shall allow secure multiple-access to the control information.
 - 6. Refer to Division 23 Section "Building Automation System (BAS)" sections and drawings for further requirements.
- I. The BC shall monitor, control, and provide the field interface for all points specified. Each BC shall be capable of performing all specified energy management functions, and all DDC functions, independent of other BCs and operator interface devices as more fully specified in Division 23 "Building Automation System (BAS)" sections.
- J. Systems Configuration Database: The system architecture shall support maintaining the systems configuration database on a BMS Operator Interface or BC on the Local Supervisory LAN. User tools provided to Owner shall allow configuring, updating, and maintaining current configurations and settings at the BMS Operator Interface or BC.
 - 1. Database Schema shall be published and provided to Owner to facilitate easy access to the data.

2. Database shall be ODBC compliant or a data access driver shall be provided to act as an ODBC or OLE DB data provider.
 - K. Interruptions or fault at any point on any Primary Controller LAN shall not interrupt communications between other nodes on the network. If a LAN is severed, two separate networks shall be formed and communications within each network shall continue uninterrupted.
 - L. All line drivers, repeaters, terminators, signal boosters, and signal conditioners shall be provided as necessary for proper data communication.
 - M. Anytime any controller's database or program is changed in the field, the controller shall be capable of automatically uploading the new data to the BMS Operator Interface and BC controller.
- 1.15 WARRANTY MAINTENANCE
- A. Warranty shall include materials and labor during first two (2) years of operation from substantial completion date or completion of all punch list items, whichever is later.
 - B. The Owner reserves the right to make changes to the BMS during the warranty period. Such changes do not constitute a waiver of warranty. The Contractor shall warrant parts and installation work regardless of any such changes made by owner, unless the Contractor provides clear and convincing evidence that a specific problem is the result of such changes to the BMS.
 - C. At no cost to the Owner during the warranty period, the Contractor shall provide maintenance services for software and hardware components as specified below:
 1. Maintenance services shall be provided for all devices and hardware specified in Division 23 Section "Building Automation System (BMS)" Sections. Service all equipment per the manufacturer's recommendations. All devices shall be calibrated within the last month of the warranty period.
 2. Emergency Service: Any malfunction, failure, or defect in any hardware component or failure of any control programming that would result in property damage or loss of comfort control shall be corrected and repaired following telephonic notification by Owner to the Contractor.
 - a. Response by telephone to any request for service shall be provided within two (2) hours of the initial telephone request for service.
 - b. In the event that the malfunction, failure, or defect is not corrected through the telephonic communication, at least one (1) hardware and software technician, trained in the system to be serviced, shall be dispatched to the site within eight (8) hours of the initial telephone request for such services, as specified.
 - c. Emergency service shall be available on a 24-hour, 7-day-a-week basis.
 3. Normal Service: Any malfunction, failure, or defect in any hardware component or failure of any control programming that would not result in property damage or loss of comfort control shall be corrected and repaired following telephonic notification by Owner to the Contractor.
 - a. Response by telephone to any request for service shall be provided within eight (8) working hours (contractor specified 40 hr/week normal working period) of the initial telephone request for service.

- b. In the event that the malfunction, failure, or defect is not corrected through the telephonic communication, at least one (1) hardware and software technician, trained in the system to be serviced, shall be dispatched to the site within three (3) working days of the initial telephone request for such services, as specified.
- 4. Telephonic Request for Service: Contractor shall provide up to three telephone numbers for Owner to call in the event of a need for service. At least one of the lines shall be attended 24 hours a day, 7 days a week. Alternatively, pagers can be used for technicians trained in system to be serviced. A technician shall respond to every call within 15 minutes.
- 5. Technical Support: Contractor shall provide technical support by telephone throughout the warranty period.
- 6. Preventive maintenance shall be provided throughout the warranty period in accordance with the hardware component manufacturer's requirements.

1.16 DELIVERY, STORAGE, AND HANDLING

- A. Provide factory-shipping cartons for each piece of equipment and control device. Maintain cartons during shipping, storage and handling as required to prevent equipment damage, and to eliminate dirt and moisture from equipment. Store equipment and materials inside and protect from weather.

1.17 LISTING AND LABELING

- A. The BMS and components shall be listed by Underwriters Laboratories (UL 916) as an Energy Management System.
- B. BACnet controllers, B-BC, B-AAC, B-ASC, etc. shall carry the BTL Mark for their device profile.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. Materials shall be new, the best of their respective kinds without imperfections or blemishes, and not be damaged in any way. Used equipment shall not be used in any way for the permanent installation, no exception.

2.2 UNIFORMITY

- A. To the extent practical, all equipment of the same type serving the same function shall be identical and from the same manufacturer.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Examine areas and conditions under which control systems are to be installed. Do not proceed with work until unsatisfactory conditions have been corrected.
- B. Where a discrepancy exist among details, schematic, specification, or sequence, the most stringent shall apply to complete a fully functional system.

3.2 INSTALLATION OF CONTROL SYSTEMS

- A. General: Install systems and materials in accordance with manufacturer's instructions, roughing-in drawings and details shown on drawings.

- B. Factory/Owner Acceptance Test: a test shall be conducted by this contractor with the owner and designer with a minimum of three months prior to onsite installation. The intent of this verification/approval is to determine that all custom built panels, programs, graphical user interface screens, PID loops for Rooftop Units equipment comply with specifications and construction codes. Approval by Owner and EOR are required prior to installation. All delays incurred due to non-compliance shall be at the cost of this contractor.
- C. The Contractor shall furnish and install any power supply surge protection, filters, and other equipment as necessary for proper operation and protection of all BCs, operator interfaces, routers, gateways and other hardware and interface devices. All equipment shall be capable of handling voltage variations 10% above or below measured nominal value, with no effect on hardware, software, communications, and data storage.

3.3 CONTROL POWER SOURCE AND SUPPLY

- A. Extend all power source wiring required for operation of all equipment and devices provided under Division 23 "Building Automation System (BAS)" sections.

3.4 STARTUP, COMMISSIONING, AND TRAINING

- A. Refer to Division 01 "Demonstration and Training" sections.
- B. Refer to Division 23 "Building Automation System (BAS)" sections.
- C. At the direction of the engineer during construction, testing implementation, and/or after acceptance, allow for a minimum of (24) hours of application programming modifications to the sequence of operation to accommodate tuning of controlled system devices to match intended functional performance.

3.5 SEQUENCE OF OPERATION

- A. Refer to Division 23 "Building Automation System (BAS)" sections or control drawings.

3.6 IDENTIFICATION STANDARDS

- A. Controller Identification. All controllers shall be identified by a plastic engraved nameplate securely fastened to the outside of the controller enclosure.
- B. Panel Identification. All local control panels shall be identified by a plastic engraved nameplate securely fastened to the outside of the controller enclosure.
- C. Field Devices. All field devices shall be identified by a typed (not handwritten) securely attached tag label.
- D. Panel Devices. All panel devices shall be identified by a typed label securely fastened to the backplane of the local control panel.
- E. Raceway Identification. All the covers to junction and pull boxes of the control system raceways shall be painted blue or have identification labels stating "Control System Wiring" affixed to the covers. Labels shall be typed, not hand written.
- F. Wire Identification. All low and line voltage control wiring shall be identified by a number, as referenced to the associated control diagram, at each end of the conductor or cable. Identification number shall be permanently secured to the conductor or cable and shall be typed

END OF SECTION 230920

SECTION 230921 - BUILDING AUTOMATION SYSTEM (BAS) – INTERFACE AND COMMUNICATION DEVICES**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes network integration devices.

1.2 DESCRIPTION OF WORK

- A. The BMS head controller and interface controllers (e.g. BX-300) shall be approved by LIDL.
- B. The BMS Contractor shall install a LCD Operator Interfaces and BC controller with web serving capabilities for the BMS functions specified.
- C. Contractor shall provide all interface devices and software to provide an integrated system connecting Building Controllers, Advanced Application Controllers, Application Specific Controllers, Gateways, and Web Server. The BMS System shall be connected to the Owner's WAN.
- D. The BMS Contractor shall provide all software and licenses for the BAS System to integrate with project specific equipment with factory provided BACnet MSTP, BACnet IP, Modbus RTU, or LON.
- E. Refer to Division 23 Section "Building Automation System (BAS)" for general requirements.

PART 2 - PRODUCTS**2.1 GENERAL**

- A. Materials:
 - 1. All products used in this project installation shall be new and currently manufactured and shall have been applied in similar installations. Do not use this installation as a product test site unless explicitly approved in writing by Owner or Owner's representative. Spare parts shall be available for at least five years after completion of this contract.
- B. The Owner's WAN:
 - 1. Internet-based network connecting multiple facilities with a central data warehouse and server, accessible via standard web-browser.
 - 2. This is a new infrastructure and Contractor is required to configure any components of this WAN. However, coordination with LIDL's BMS team is required for final connectivity.
 - 3. Contractor shall provide Modbus data to the Owner's WAN.

2.2 NETWORK CONNECTION AND COMMUNICATION

- A. The design of the BMS shall support networking of an owner's thin-client connectivity, a local Operator Interface, a Building Controller (BC), and field level controllers (AAC or ASC). The network architecture shall consist of two levels, an Ethernet-based primary network for the owner's connectivity and Building DDC controllers along with secondary Field Level Networks (FLN) for terminal equipment application specific controllers (ASC).

- B. Access to system data shall not be restricted by the hardware configuration of the BMS system. The hardware configuration of the BMS network shall be totally transparent to the user when accessing data or developing control programs.
- C. Operator GUI Communication:
 - 1. All color graphic shall reside on the interface or BC via the ethernet network and the consoles shall be set up in a client/server configuration.
 - 2. The web servers shall act as the central database for system graphics and databases to provide consistency throughout all system.
 - 3. The network shall allow concurrent use of multiple BAS software site licenses.
 - 4. Provide a web server and a shared hard disk drive array. The web server shall be an enterprise level fault-tolerant server.
- D. Management Level Network Communication (MLN)
 - 1. Any owner remote BMS workstations shall be simultaneously direct connect to the Ethernet Management Level Network without the use of an interposing device.
 - 2. Operator Interface and/or BC controller shall be capable of simultaneous direct connection and communication with BACnet/IP, Modbus, OPC, and TCP/IP corporate level networks without the use of interposing devices.
 - 3. The Management Level Network shall not impose a maximum constraint on the number of owner remote BMS workstations.
 - 4. Any controller residing on the Building level networks shall connect to Ethernet network without the use of a PC or a gateway with a hard drive.
 - 5. Any owner remote BMS workstations on the Management Level Network shall have transparent communication with controllers on the building level networks connected via Ethernet.
 - 6. Any break in Ethernet communication from an owner's IT network to the web server on the building level networks shall result in a notification at the web server.
 - 7. The standard client and BMS web server on the Management Level Network shall reside on industry standard Ethernet utilizing standard TCP/IP, IEEE 802.3.
 - 8. System software applications shall run as a service to allow communication with Building Controllers without the need for user log in. Closing the application or logging off shall not prevent the processing of alarms, network status, panel failures, and trend information.
 - 9. Access to the system database shall be available from any standard owner provided workstation on the Management Level Network.
 - 10. Client access to web server BC configurations over the Internet network shall be available via Web browser interface.
 - 11. Thin-client remote access to a web server BC or operator interface shall be available and provided via web browser interface. Remote access setup shall provide multiple, independent sessions without the need to install a dedicated workstation software on a local or remote hard drive.
- E. Primary Network - Panel to Panel Communication:
 - 1. Only one (1) BMS BC controller shall directly reside on the primary IP Ethernet network.
 - 2. Systems that operate via polled response or other types of protocols that rely on a central processor, file server, or similar device to manage panel- to-panel or device-to-device communications shall not be acceptable.
 - 3. All operator interfaces shall have the ability to access all point status and application report data or execute control functions for any and all other devices. Access to data shall be based upon logical identification of building equipment. No hardware or software limits shall be imposed on the number of devices with global access to the network data.

4. The primary network shall use Modbus over Ethernet. All devices must:
 - a. Auto-sense 10/100 Mbps networks.
 - b. Receive an IP Address from a Dynamic Host Configuration Protocol (DHCP) Server or be configured with a Fixed IP Address.
 - c. Resolve Name to IP Addresses for devices using a Domain Name Service (DNS) Server on the Ethernet network.
 - d. Allow MMI access to an individual Primary Network Controller using industry standard Telnet software to view and edit entire Primary Network.
 5. The primary network shall provide the following minimum performance:
 - a. Provide high-speed data transfer rates for alarm reporting, report generation from a BC controller and upload/download efficiency between network devices.
 - b. System performance shall insure that an alarm occurring at the BC is displayed at any owner's remote BMS workstations, all BCs, and other interfaces within 15 seconds.
 - c. Message and alarm buffering to prevent information from being lost.
 - d. Error detection, correction, and re-transmission to guarantee data integrity.
 - e. Synchronization of real-time clocks, including automatic daylight savings time corrections
 - f. The primary network shall allow the BC to access any data from, or send control commands and alarm reports directly to, any other controllers on the network without dependence upon a central or intermediate processing device. The network shall also allow any Building controller to access, edit, modify, add, delete, back up, restore all system point database and all programs.
 - g. The primary network controllers shall back-up and restore their own current database including programs, and points without the requirement for connection to a mass storage device.
 - h. The primary network controllers shall provide system-wide wild card point search, command, and access direct from any building controller on the network.
 - i. The primary network shall allow the BC to access on-demand display and reports regarding system-wide information including point names, point status, present value, command priority array, trend information, and field panel configuration information.
 - j. The primary network shall allow the BC to be configured system-wide by software based tools, and by direct access from any Building Controller on the network. Proprietary vendor specific software shall not be required for system configuration.
 - k. The primary network shall allow the BC to assign password access and control priorities to each point individually. The logon password (at any PC workstation or portable operator terminal) shall enable the operator to monitor, adjust and control only the points that the operator is authorized for. All other points shall not be displayed at the PC workstation or portable terminal. (e.g. all base building and all tenant points shall be accessible to any base building operators, but only certain base building and tenant points shall be accessible to tenant building operators). Passwords and priorities for every point shall be fully programmable and adjustable.
 - l. Devices containing custom programming may reside on the Primary Network.
- F. Secondary Network – Advanced Application Controller (AAC) and Application Specific Controller (ASC) Communication:
1. The BC controller shall directly connect to all pertinent sub-networks such that communications may be executed directly between BC controller and Field Level Network controllers on a peer-to-peer basis.

2. Communication over the secondary network shall be either BACnet MS/TP or Modbus RTU protocols.
3. This level communication shall support a family of AAC for pieces of equipment requiring full programmability and ASC for terminal equipment.
4. The AACs and ASCs shall communicate bi-directionally with the BC controllers for transmission of global data.
5. A maximum of 96 terminal equipment controllers may be configured on individual secondary network trunks to insure adequate global data and alarm response times.

G. Internet Based Communication:

1. Web Based Operator Interface

- a. The BMS shall provide a HTML-5 web based graphical interface that allows users to access the BMS data via the Internet, extranet, or Intranet. The interface shall use HTML based ASP pages to send and receive data from the BMS to a web browser.
- b. All information exchanged over Internet shall be optionally encrypted and secure via SSL (provided by Owner).
- c. Access to the web interface must be password protected. Users' rights and privileges to points and graphics shall be the same as those assigned at the BMS system. An option shall exist to only allow users "read" access via the web browser, while maintaining "command" privileges via any owner's remote BMS workstation.
- d. The web interface shall not require modification or creation of HTML or ASP pages using an HTML editor.
- e. The web based interface shall provide the following functionality to users, based on their access and privilege rights:
 - 1) Logon Screen - allows the user to enter their name, password and domain name for logging into the web server.
 - 2) Alarm Display - a display of current BMS System alarms to which the user has access will be displayed. Users will be able to acknowledge and erase active alarms, and link to additional alarm information including alarm messages. Any alarm acknowledgments initiated through the web interface will be recorded to the BAS System activity log.
 - 3) Graphic Display - Display of system graphics, including animated motion, shall be available for viewing over the web browser. A graphic selector list will allow users to select any graphics to which they have access. Graphics displays will automatically refresh with the latest change of values. Users will have the ability to command and override points from the graphic display as determined by their user account rights.
 - 4) Point Details - users will have access to point detail information including operational status, operational priority, physical address, and alarm limits, for point objects to which they have access.
 - 5) Point Commanding - users will be able to override and command points they have access to via the web browser interface. Any commands or overrides initiated via the web browser interface will be written to the BMS system's central workstation activity log.
- f. The web server licensing options shall allow concurrent access by a minimum of five (5) browser connections.
- g. Internet connections, ISP services, as well as necessary firewalls or proxy servers shall be provided by the owner as required to support the web access feature.

H. Remote Notification Paging System

1. The BMS web serving BC shall be configured to send out messages to numeric pagers, alphanumeric pagers, phones (via text to speech technology), SMS (Simple Messaging Service, text messaging) Devices, and email accounts based on a point's alarm condition.
2. There shall be no limit to the number of points that can be configured for remote notification of alarm conditions and no limit on the number of remote devices which can receive messages from the system.
3. On a per point basis, system shall be configurable to send messages to an individual or group and shall be configurable to send different messages to different remote devices based on alarm message priority level.
4. Remote devices may be scheduled as to when they receive messages from the system to account for operators' work schedules.
5. System must be configurable to send messages to an escalation list so that if the first device does not respond, the message is sent on to the next device after a configurable time has elapsed.
6. Message detail shall be configurable on a per user basis.
7. During a "flood" of alarms, remote notification messages shall have the ability to optimize several alarms into an individual remote notification message.
8. The BMS web serving BC shall have the ability to send manual messages allowing an operator to type in a message to be sent immediately.
9. The BMS web serving BC shall have a feature to send a heartbeat message to periodically notify users that they have communication with the system.

2.3 COMMUNICATION AND CONTROL WIRING

A. General:

1. Provide copper wiring, plenum cable, and raceways as specified in the applicable sections of Division 16 unless otherwise noted herein.
2. All insulated wire to be copper conductors, UL labeled for 90°C minimum service.

B. Wire Sizing and Insulation

1. Wiring shall comply with minimum wire size and insulation based on services listed below:

a.	Service	Minimum Gage/Type	Insulation Class
b.	AC 24V Power	12 Ga Solid, Shielded	600 Volt
c.	DC 24V Power	10 Ga Solid, Shielded	600 Volt
d.	Class 1	14 Ga Stranded, Shielded	600 Volt
e.	Class 2	18 Ga Stranded, Shielded	300 Volt
f.	Class 3	18 Ga Stranded, Shielded	300 Volt
2. Provide plenum-rated cable when open cable is permitted in supply or return air plenum where allowed per execution specifications defined in Paragraph 3.07

C. Power Wiring:

1. 115V power circuit wiring above 100 feet distance shall use minimum 10 gage.
2. 24V control power wiring above 200 feet distance shall use minimum 12 gage.

D. Control Wiring:

1. Digital Input/output wiring shall use Class 2 twisted pair, insulated and jacketed and require a grounded shield.
2. Analog inputs shall use Class 2 twisted shielded pair, insulated, and jacketed and require a grounded shield.
3. Actuators with tri-state control shall use 3 conductor, twisted shielded, insulated, and jacketed and require a grounded shield.

E. Communication Wiring

1. Ethernet Cable shall be minimum CAT5
2. Secondary level network shall be 18-24 gage, TSP, low capacitance cable

F. Approved Cable Manufacturers:

1. Wiring from the following manufacturers which meet the above criteria shall be acceptable:
 - a. Anixter
 - b. Belden

G. Data Information/Connection Outlet

1. Technology contractor is responsible for furnishing and installing the Category 6A Ethernet cable, network switches and routers, and network data "drops" required for the BMS system. Follow all LIDL Facilities Maintenance Management IT services and standards for network hardware requirements. The BMS Contractor shall follow the installation requirements as stated below:
 - a. Ethernet cable shall be Commscope Uniprise Category 6A, non-plenum, 4-pair cable supporting 10GB/s Ethernet requirements.
 - b. At the IDF end, the cable will be terminated at a Category 6A RJ45 Uniprise patch panel.
 - c. Each data outlet shall have two Category 6A cables installed. One white cable and one gray cable.
 - d. At the data outlet end, the cable will be terminated at Category 6A 8- position 568B/RJ45 jacks, gray in color, and terminated in a 2-port face plate
 - e. Horizontal face plate configuration:
 - 1) In a 2-port face plate, the "C1" data jack (first, white cable) shall be in the bottom position and the "C2" data jack (second, gray cable) shall be in the top position.
 - 2) In a 4-port face plate, the "C1" data jack shall be in the bottom left position, the "C2" data jack shall be in the top left position, the "C3" data jack shall be in the bottom right position, and the "C4" data jack shall be in the top right position.
 - f. Vertical face plate configuration:
 - 1) In a 2-port face plate, the "C1" data jack (first, white cable) shall be in the left position and the "C2" data jack (second, gray cable) shall be in the right position.
 - 2) In a 4-port face plate, the "C1" data jack shall be in the top left position, the "C2" data jack shall be in the top right position, the "C3" data jack shall be in

the bottom left position, and the "C4" data jack shall be in the bottom right position.

- g. All cable will be secured and dressed with Velcro tie-wraps.
- h. Do not install compressed, kinked, scored, deformed, or abraded cable. If any cable is, or becomes damaged, replace the entire length of damaged cable.
- i. All data outlets shall be serviced with 1" conduit to the basket tray, or pull box via horizontal conduit system. Conduits shall service only one data outlet each. No daisy-chaining or looping between data outlets is allowed.
- j. Labeling:
 - 1) For vertical face plates, the location labels shall be installed on the top label holder on the face plate.
 - 2) For horizontal face plates, the location labels shall be installed on the left label holder on the face plate.
 - 3) For vertical face plates, the IDF labels shall be installed on the bottom label holder on the face plate. This label will contain the IDF room number only.
 - 4) For horizontal face plates, the location labels shall be installed on the right label holder on the face plate. This label will contain the IDF room number only.

2.4 INTEGRATION:

A. BACnet IP Node:

- 1. Equipment with factory-furnished BACnet IP communication shall be reside on a sub-network under the BMS BC/head controller. No exceptions.
- 2. The web server with BMS enterprise software shall perform information translation between the BACnet IP node and other devices on other BMS sub-networks.
- 3. The server shall communicate points as required for the project to a BC to support interrogation, full control, and/or utilize information to command all associated AACs and ASCs under the Primary Controller LAN.

B. Modbus Node:

- 1. An equipment with factory-furnished Modbus-RTU communication shall be reside on a sub-network under the BMS BC/head controller. No exceptions.
- 2. The web server with BMS enterprise software shall perform information translation between the Modbus nodes and other devices on other BMS sub-networks.
- 3. BC shall support interrogation, full control, and/or utilize information to command all associated AACs and ASCs under the Primary Controller LAN.

PART 3 - EXECUTION

3.1 INSPECTION:

- A. Examine areas and conditions under which control systems are to be installed. Do not proceed with work until unsatisfactory conditions have been corrected in manner acceptable to Installer.

3.2 INSTALLATION OF CONTROL SYSTEMS:

- A. General: Install systems and materials in accordance with manufacturer's instructions, roughing-in drawings and details shown on drawings.

- B. BMS web server shall be approved by Lidl IT group.
- C. BMS Contractor shall install and fully configure and provide all interface devices and software to provide an integrated system.
- D. Contractor shall closely coordinate with the Owner, or designated representative, to establish IP addresses and communications to assure proper operation of the building control system with the owner's WAN and owner's operator workstation.
- E. No license, software component, key, etc or any piece of information required to install, configure, operate, diagnose and maintain the system shall be withheld from the Owner.
- F. Install electronic control system Operation and Maintenance Manuals, programming guides, network configuration tools, and control shop drawings etc on the web server or as directed by the owner. Provide interface or shortcuts to guide user to the appropriate information.
- G. Install systems and materials in accordance with manufacturer's instructions.
- H. Refer to Division 23 Section "Building Automation System (BAS)" for general requirements.

END OF SECTION 230921

SECTION 230922 - BUILDING AUTOMATION SYSTEM (BAS) - BASIC MATERIALS, INTERFACE DEVICES, AND SENSORS**PART 1 - GENERAL****1.1 SUMMARY****A. Section includes the following:**

1. Specified in this section are the following hard wired input/output devices connected to a BC, AAC, or ASC.
 - a. Binary Temperature Device
 - b. Temperature Sensor
 - c. Dew Point/Humidity Sensor
 - d. Indoor Air Quality (CO2) Sensor
 - e. Differential Pressure Transmitter
 - f. Light Level Sensor
 - g. Current Switches
 - h. Relays
 - i. Voltage Transformers
 - j. Water Leak Detector
 - k. Duct Smoke Detector
 - l. Electric Control Components
 - m. Audio/Visual Alarm Units
 - n. Nameplates.
 - o. Testing Equipment.

1.2 DESCRIPTION OF WORK

- A. Refer to Division 23 Section "Building Automation System (BAS)" for general requirements.
- B. Refer to other Division 23 Sections for installation of instruments in mechanical systems; not work of this Section.
- C. Provide the following electrical work as work of this Section, complying with requirements of Division 26 Sections:
 1. Control wiring between field-installed controls, indicating devices, and unit control panels.
 2. Interlock wiring between electrically interlocked devices, sensors, and between a hand or auto position of motor starters as indicated for all mechanical and controls.
 3. Wiring associated with indicating and alarm panels (remote alarm panels) and connections to their associated field devices.
 4. All other necessary wiring for fully complete and functional control system as specified.
 5. Power wiring from spare circuits in electrical panels to BMS Field Panels.

1.3 WORK BY OTHERS

- A. Flow Switches, Flow Meters, etc. that will have wet surfaces, shall be installed under the applicable piping Section under the direction of the BMS Contractor who will be fully responsible for the proper installation and application.
- B. Controlled Equipment Power Wiring shall be furnished and installed under Division 26. Where control involves 120V control devices controlling 120V equipment, Division 26 Contractor shall extend power wiring to the equipment. Building Automation System Contractor shall extend it from the equipment to the control device.
- C. BMS Contractor shall provide adequate time to coordinate and work with Equipment Manufacturer where factory-controls are provided to achieve the design sequence of operations.
- D. BMS Contractor shall provide adequate time to coordinate and work with Test and Balancing Contractor to achieve the design sequence of operations.

PART 2 - PRODUCTS

2.1 DDC FUNCTIONALITY

- A. General: These requirements clarify the requirement for stand-alone DDC functionality relative to packaging I/O devices with a controller. Stand-alone functionality is specified with the controller and for each Application Category specified in PART 3. This item refers to acceptable paradigms for associating the points with the processor.
- B. Functional Boundary: Provide controllers so that all points associated with and common to one unit or other complete system/equipment shall reside within a single control unit. The boundaries of a standalone system shall be as dictated in the contract documents. Generally, systems specified for the Application Category will dictate the boundary of the standalone control functionality. See related restrictions below. When referring to the controller as pertains to the standalone functionality, reference is specifically made to the processor. One processor shall execute all the related I/O control logic via one operating system that uses a common programming and configuration tool.
- C. The following configurations are considered acceptable with reference to a controller's standalone DDC functionality:
 - 1. Points packaged as integral to the controller such that the point configuration is listed as an essential piece of information for ordering the controller (having a unique ordering number).
 - 2. Controllers with processors and modular back planes that allow plug in point modules as an integral part of the controller.
 - 3. I/O point expander boards, plugged directly into the main controller board to expand the point capacity of the controller.
 - 4. I/O point expansion devices connected to the main controller board via wiring and as such shall be remote from the controller and that communicate via a sub LAN protocol. These arrangements to be considered standalone shall have a sub LAN that is dedicated to that controller and include no other controller devices. All wiring to interconnect the I/O expander board shall be:
 - a. Contained in the control panel enclosure;
 - b. Or run in conduit. Wiring shall only be accessible at the terminations.

5. General purpose I/O or Smart Devices racked with a processor module shall be in the same contiguous physical enclosure. The controller shall also include its own dedicated processor module and bridge or router making the controllers LAN communication a subnet or LAN segment dedicated to that controller as specified under Application Categories below. The following are additional requirements of this configuration:
 - a. Configuration must meet the requirements for battery back up.
 - b. If processor fails, the I/O devices shall go to their fail condition.
 - c. Logic must provide for orderly sequencing of I/O during a power interruption and restart of program logic upon restoration of power.
 - d. Programming must facilitate a robust uploading scheme and limit available bandwidth during upload.
 - e. Trending shall be buffered in the processor or dedicated data logging module and uploaded to a Building or Network Controller.

D. Building Controller (BC)

E. General Requirements:

1. The BC(s) shall provide fully distributed control independent of the operational status of any workstation/server. All necessary calculations required to achieve control shall be executed within the BC independent of any other device.
2. BCs shall be connected to a controller network that qualifies as a Primary Controlling LAN. BCs shall share information with the entire network of BCs and AACs/ASCs for full global control.
3. BCs shall be programmable from an operator workstation or an operator's terminal device. Each controller shall permit multi-user operation from multiple owner interfaces connected either locally or over the Primary Controller LAN.
4. All control strategies performed by the BCs shall be both operator definable and modifiable through the Operator Interfaces. All point data, algorithms and application software within each BC shall be modifiable from an owner workstation.
5. Each BC shall be capable of standalone direct digital operation utilizing its own processor, non-volatile memory, input/output, wiring terminal strips, A/D converters, and voltage transient and lightning protection devices.
6. BCs shall perform overall system coordination, accept control programs, perform automated HVAC functions, control peripheral devices and perform all necessary mathematical and logical functions.
7. Each unit shall have its own internal RAM, non-volatile memory, microprocessor, battery backup, regulated power supply, power conditioning equipment, ports for connection of operating interface devices, and control enclosure.
8. BC shall contain sufficient memory for all specified global control strategies, user defined reports and trending, communication programs, and central alarming.
9. All BCs shall be protected from any memory loss due to a loss of power by one or a combination of the following:
 - a. Volatile RAM shall have a battery backup using a lithium battery with a rated service life of fifty (50) hours, and a rated shelf life of at least five years. Self-diagnostic routine shall report an alarm for a low battery condition.
 - b. EEPROM, EPROM, or NOVRAM non-volatile memory
10. Each BC shall provide for point mix flexibility and expandability. This requirement shall be met via either a family of expander boards, modular input/output configuration, or a combination thereof.
11. Each BC shall execute application programs, calculations, and commands via a microprocessor resident in the BC.

12. The database and all application programs for each BC shall be stored in non-volatile or battery backed volatile memory within the BC and shall be able to upload/download to/from an owner workstation.
13. BCs shall provide buffer for holding alarms, messages, trends etc.
14. Each BC shall contain software to perform full DDC/PID control loops.
15. For systems requiring end-of-line resistors those resistors shall be located in the BC, if it has I/O capability.
16. A communication port for operator interface through a terminal shall be provided in each BC. It shall be possible to perform all program and database back-up, system monitoring, control functions, and BC diagnostics through this port. Standalone BC panels shall allow temporary use of portable devices without interrupting the normal operation of permanently connected modems, printers, or workstations.
17. Each BC shall be equipped with loop tuning algorithm for precise proportional, integral, derivative (PID) control. Loop tuning tools provided with a workstation software is acceptable. Tools to support loop tuning must be provided such that P, I, and D gains are automatically calculated.
18. Slope intercepts and gain adjustments shall be available on a per-point basis.
19. Each BC shall be at least 32 bit, multi-tasking, multi-user, real-time 100 MHz digital control processors consisting of modular hardware with plug-in enclosed processors, communication controllers. Controller size shall be sufficient to fully meet the requirements of this specification and the attached point list.
20. Each BC shall have sufficient memory, a minimum of 24 megabyte, to support its own operating system and databases, including control processes, energy management applications, alarm management applications, historical/trend data for points specified, maintenance support applications, custom processes, operator I/O, and dial-up communications.
21. Each BC shall have an integral real-time clock and calendar.
22. Each BC shall support firmware upgrades without the need to change hardware.

F. Each Building Controller shall support:

1. Monitoring of industry standard analog and digital inputs, without the addition of equipment outside the Building Controller cabinet.
2. Monitoring of industry standard analog and digital outputs, without the addition of equipment outside the Building Controller cabinet.

G. Spare Point Capacity. Each Building Controller shall have a minimum of 20% spare point capacity.

1. The type of spares shall be in the same proportion as the implemented I/O functions of the panel, but in no case shall there be less than one spare of each implemented I/O type.
2. Provide all processors, power supplies, and communication controllers so that the implementation of adding a point to the spare point location only requires the addition of the appropriate:
 - a. Expansion modules
 - b. Sensor/actuator
 - c. Field wiring/tubing.

H. Serial Communication. Building Controllers shall provide at least one data communication port for operation of operator I/O devices such as industry standard printers, operator terminals, and portable laptop operator's terminals. Building Controllers shall allow temporary use of portable devices without interrupting the normal operation of permanently connected terminals. A USB port shall alternatively be available to support local HMI tools connection.

- I. Self-Diagnostics. Each Building Controller shall continuously perform self-diagnostics, communication diagnosis, and diagnosis of all panel components. The Building Controller shall provide both local and remote annunciation of any detected component failures, low battery conditions or repeated failure to establish communication for any system.
- J. Power loss:
 - 1. In the event of the loss of power, there shall be an orderly shutdown of all Building Controllers to prevent the loss of database or operating system software. Non-volatile memory shall be incorporated for all critical controller configuration data and battery backup shall be provided to support the real-time clock and all volatile memory for a minimum of 100 hours.
 - 2. Upon a loss of power to any BC, the other units on the primary controlling network shall not in any way be affected.
 - 3. Upon a loss of power to any BC, the UPS backup shall ensure that the energy management control software, the Direct Digital Control software, the database parameters, and all other programs and data stored in the RAM are retained.
 - 4. Upon restoration of power within the specified UPS backup period, the BC shall resume full operation without operator intervention. The BC shall automatically reset its clock such that proper operation of any time dependent function is possible without manual reset of the clock. All monitored functions shall be updated.
 - 5. Should the duration of a loss of power exceed the specified UPS backup period or BC panel memory be lost for any reason, the panel shall automatically report the condition (upon resumption of power) and be capable of receiving a download via the network, and connected workstation.
 - 6. The owner shall be able to upload the most current versions of all energy management control programs, DDC programs, database parameters, and all other data and programs in the memory of each BC to an owner workstation via the local area network, or via the telephone line dial-up modem where applicable, or to the laptop PC via the local RS-232C port.
- K. BC Failure:
 - 1. Building Controller LAN Data Transmission Failure: BC shall continue to operate in stand-alone mode. BC shall store loss of communication alarm along with the time of the event. All control functions shall continue with the global values programmable to either last value or a specified value. Peer BCs shall recognize the loss, report alarm and reconfigure the LAN.
 - 2. BC Hardware Failure: BC shall cease operation and terminate communication with other devices.
- L. Environment.
 - 1. Controller hardware shall be suitable for the anticipated ambient conditions for most mechanical spaces.
 - 2. Controllers used in conditioned space shall be mounted in dust-proof enclosures and shall be rated for operation at 0°C to 49°C (32°F to 120°F).
- M. Immunity to power and noise.
 - 1. Controller shall be able to operate at 90% to 110% of nominal voltage rating and shall perform an orderly shutdown below 80% nominal voltage.
 - 2. Operation shall be protected against electrical noise of 5 to 120 Hz and from keyed radios up to 5 W at 1 m (3 ft).

3. Isolation shall be provided at all primary network terminations, as well as all field point terminations to suppress induced voltage transients consistent with:
 - a. RF-Conducted Immunity (RFCI) per ENV 50141 (IEC 1000-4-6) at 3 V
 - b. Electro Static Discharge (ESD) Immunity per EN 61000-4-2 (IEC 1000-4-2) at 8 kV air discharge, 4 kV contact
 - c. Electrical Fast Transient (EFT) per EN 61000-4-4 (IEC 1000-4-4) at 500 V signal, 1 kV power
 - d. Output Circuit Transients per UL 864 (2,400V, 10A, 1.2 Joule max)
4. Isolation shall be provided at all Building Controller's AC input terminals to suppress induced voltage transients consistent with:
 - a. IEEE Standard 587 1980
 - b. UL 864 Supply Line Transients
 - c. Voltage Sags, Surge, and Dropout per EN 61000-4-11 (EN 1000- 4-11)
5. All products shall be BACnet certified, and shall be designed according to the BACnet Interoperability Guidelines. Product documentation and devices shall display the BACnet symbol, indicating conformance to the BACnet Testing Laboratory (BTL) Standards.

N. Advanced Application Controller (AAC)

1. Each Building Controller shall be able to communicate with Advanced Application Controllers (AACs) over the Secondary Network.
2. AACs shall be at least 32 bit, multi-tasking, multi-user, real-time 100 MHz digital control processors consisting of modular hardware with plug-in enclosed processors, communication controllers. Controller size shall be sufficient to fully meet the requirements of this specification and the attached point list.
3. Each AAC shall have sufficient memory, a minimum of 24 megabyte, to support its own operating system and databases, including control processes, energy management applications, alarm management applications, historical/trend data for points specified, maintenance support applications, custom processes, operator I/O, and dial-up communications.
4. AACs shall have an integral real-time clock.
5. Each AAC shall support firmware upgrades without the need to change hardware.
6. Each AAC shall support:
 - a. Monitoring of industry standard analog and digital inputs, without the addition of equipment outside the AAC cabinet.
 - b. Monitoring of industry standard analog and digital outputs, without the addition of equipment outside the AAC cabinet.
7. Spare Point Capacity. Each AAC shall have a minimum of 20% spare point capacity.
 - a. The type of spares shall be in the same proportion as the implemented I/O functions of the panel, but in no case shall there be less than one spare of each implemented I/O type.
 - b. Provide all processors, power supplies, and communication controllers so that the implementation of adding a point to the spare point location only requires the addition of the appropriate:
 - 1) Expansion modules
 - 2) Sensor/actuator

- 3) Field wiring/tubing.
8. Serial Communication. AACs shall provide at least one data communication port for operation of operator I/O devices such as industry standard printers, operator terminals, and portable laptop operator's terminals. AACs shall allow temporary use of portable devices without interrupting the normal operation of permanently connected terminals. A USB port shall alternatively be available to support local HMI tools connection.
9. Self-Diagnostics. Each AAC shall continuously perform self-diagnostics, communication diagnosis, and diagnosis of all panel components. The Building Controller shall provide both local and remote annunciation of any detected component failures, low battery conditions or repeated failure to establish communication for any system.
10. Power loss. In the event of the loss of power, there shall be an orderly shutdown of all AACs to prevent the loss of database or operating system software. Non-volatile memory shall be incorporated for all critical controller configuration data and battery backup shall be provided to support the real-time clock and all volatile memory for a minimum of 100 hours.
11. Environment.
 - a. Controller hardware shall be suitable for the anticipated ambient conditions for most mechanical spaces.
 - b. Controllers used outdoors and/or in wet ambient conditions shall be mounted within waterproof enclosures and shall be rated for operation at 0°C to 49°C (32°F to 120°F).
 - c. Controllers used in conditioned space shall be mounted in dust-proof enclosures and shall be rated for operation at 0°C to 49°C (32°F to 120°F).
12. Immunity to power and noise.
 - a. Controller shall be able to operate at 90% to 110% of nominal voltage rating and shall perform an orderly shutdown below 80% nominal voltage.
 - b. Operation shall be protected against electrical noise of 5 to 120 Hz and from keyed radios up to 5 W at 1 m (3 ft).
 - c. Isolation shall be provided at all primary network terminations, as well as all field point terminations to suppress induced voltage transients consistent with:
 - 1) RF-Conducted Immunity (RFCI) per ENV 50141 (IEC 1000-4-6) at 3 V
 - 2) Electro Static Discharge (ESD) Immunity per EN 61000-4-2 (IEC 1000-4-2) at 8 kV air discharge, 4 kV contact
 - 3) Electrical Fast Transient (EFT) per EN 61000-4-4 (IEC 1000-4-4) at 500 V signal, 1 kV power
 - 4) Output Circuit Transients per UL 864 (2,400V, 10A, 1.2 Joule max)
 - d. Isolation shall be provided at all AAC's AC input terminals to suppress induced voltage transients consistent with:
 - 1) IEEE Standard 587 1980
 - 2) UL 864 Supply Line Transients
 - 3) Voltage Sags, Surge, and Dropout per EN 61000-4-11 (EN 1000-4-1)
- O. APPLICATION SPECIFIC CONTROLLER (ASC)
 1. Each Building Controller shall be able to communicate with application specific controllers (ASCs) over the Secondary Network to control terminal equipment only.

2. The use of Secondary Network controllers with custom program applications to control RTU's, AC units, etc. is not acceptable.
3. Each ASC shall operate as a stand-alone controller capable of performing its specified control responsibilities independently of other controllers in the network. Each ASC shall be a microprocessor-based, multi-tasking, real-time digital control processor.
4. Each ASC shall include all point inputs and outputs necessary to perform the specified control sequences. The ASC shall accept input and provide output signals that comply with industry standards. Controllers utilizing proprietary control signals shall not be acceptable. Outputs utilized either for two-state, modulating floating, or proportional control, allowing for additional system flexibility.
5. Space Temperature Sensors. Each controller performing space temperature control shall be provided with a matching room temperature sensor.
 - a. Wired Sensor specifications. The sensor may be either RTD or thermistor type providing the following.

1) Accuracy:	+ .5 F
2) Operating Range:	35 to 115° F
3) Set Point Adjustment Range:	55 to 95° F
4) Calibration Adjustments:	None required
5) Installation:	Up to 100 ft. from controller
6) Auxiliary Communications Port:	as required
 - b. Space temperature sensors are to be provided with neither display nor setpoint adjustment. The owner and authorized facility operator have the option of modifying individual sensor types and/or functionality. Coordinate with the owner's staff and authorized facility operator prior to the installation of the space temperature sensors.
6. Communication. Each controller shall perform its primary control function independent of other Secondary Network communication, or if Secondary Network communication is interrupted. Reversion to a fail- safe mode of operation during Secondary Network interruption is not acceptable.
7. Control Algorithms. The controller shall receive its real-time data from the Building Controller time clock to insure Secondary Network continuity. Each controller shall include algorithms incorporating proportional, integral and derivative (PID) gains for all applications. All PID gains and biases shall be field-adjustable by the user via the operator's terminal. Controllers that incorporate proportional and integral (PI) control algorithms only shall not be acceptable.
8. Control Applications. Operating programs shall be field-selectable for specific applications. In addition, specific applications may be modified to meet the user's exact control strategy requirements, allowing for additional system flexibility. Controllers that require factory changes of all applications are not acceptable.
9. Memory:
 - a. Provide each ASC with sufficient memory to accommodate point databases, operating programs, local alarming and local trending. All databases and programs shall be stored in non-volatile EEPROM, EPROM and PROM, or minimum of 72-hour battery backup shall be provided. The controllers shall be able to return to full normal operation without user intervention after a power failure of unlimited duration.
 - b. Upon replacement, new ASCs shall recover control function and site specific defaults automatically and resume normal operation.

10. Power Supply. The ASCs shall be powered from a 24 VAC source and shall function normally under an operating range of 18 to 28 VAC, allowing for power source fluctuations and voltage drops. Power supply for the ASC must be rated at a minimum of 125% of ASC power consumption and shall be of the fused or current limiting type. The BMS contractor shall provide 24 VAC power to the terminal units by utilizing a dedicated line voltage power source and isolation transformers at a central location and installing 24VAC power trunk to supply multiple ASCs in the area.
11. Environment. The controllers shall function normally under ambient conditions of 32 to 122 F (0 to 50 C) and 10% to 95%RH (non- condensing). Provide each controller with a suitable cover or enclosure to protect the circuit board assembly.
12. Immunity to noise. Operation shall be protected against electrical noise of 5-120 Hz and from keyed radios up to 5 W at 1 m (3 ft).

P. INPUT-OUTPUT INTERFACES

1. Hardwired inputs and outputs may tie into the system through building or application specific controllers.
2. All input points and output points shall be protected such that shorting of the point to itself, to another point, or to ground will cause no damage to the controller. All input and output points shall be protected from voltage up to 24 V of any duration, such that contact with this voltage will cause no damage to the controller.
3. Binary inputs shall allow the monitoring of On/Off signals from remote devices. The binary inputs shall provide a wetting current of at least 12 mA to be compatible with commonly available control devices and shall be protected against the effects of contact bounce and noise. Binary inputs shall sense "dry contact" closure without external power (other than that provided by the controller) being applied.
4. Pulse accumulation input objects. This type of object shall conform to all the requirements of binary input objects and also accept up to 10 pulses per second for pulse accumulation.
5. Analog inputs shall allow the monitoring of low-voltage (0 to 10 VDC), current (4 to 20 mA), or resistance signals (thermistor, RTD). Analog inputs shall be compatible with— and field configurable to— commonly available sensing devices.
6. Binary outputs shall provide for On/Off operation or a pulsed low-voltage signal for pulse width modulation control. Binary outputs on building and custom application controllers shall have three-position (On/Off/Auto) override switches and status lights. Outputs shall be selectable for either normally open or normally closed operation.
7. Analog outputs shall provide a modulating signal for the control of end devices. Outputs shall provide either a 0 to 10 VDC, 4 to 20 mA or 0-20 PSI signal as required to provide proper control of the output device. Analog outputs on building or custom application controllers shall have status lights and a two-position (AUTO/MANUAL) switch and manually adjustable potentiometer for manual override. Analog outputs shall not exhibit a drift of greater than 0.4% of range per year.
8. System Object Capacity. The system size shall be expandable to at least twice the number of input/ output objects required for this project. Additional controllers (along with associated devices and wiring) shall be all that is necessary to achieve this capacity requirement. The operator interfaces installed for this project shall not require any hardware additions or software revisions in order to expand the system.

Q. Power Supplies and Line Filtering

1. Control transformers shall be UL listed. Furnish Class 2 current-limiting type or furnish over-current protection in both primary and secondary circuits for Class 2 service in accordance with NEC requirements. Limit connected loads to 80% of rated capacity.
2. DC power supply output shall match output current and voltage requirements. Unit shall be full-wave rectifier type with output ripple of 5.0 mV maximum peak-to-peak. Regulation

shall be 1.0% line and load combined, with 100-microsecond response time for 50% load changes. Unit shall have built-in over-voltage and over-current protection and shall be able to withstand a 150% current overload for at least three seconds without trip-out or failure.

3. Unit shall operate between 0°C and 50°C (32°F and 120°F). EM/RF shall meet FCC Class B and VDE 0871 for Class B and MILSTD 810C for shock and vibration.
4. Line voltage units shall be UL recognized and CSA approved.
5. Power line filtering and surge protection.
 - a. Provide transient voltage and surge suppression for network controllers and other critical panels either internally or as an external component. Surge protection shall have the following at a minimum:
 - b. Dielectric strength of 1000 volts minimum
 - c. Response time of 10 nanoseconds or less
 - d. Transverse mode noise attenuation of 65 dB or greater
 - e. Common mode noise attenuation of 150 dB or better at 40 Hz to 100 Hz.
 - f. Hot swappable batteries
 - g. Automatic voltage regulation
 - h. Automatic bypass
 - i. Energy efficient design
 - j. Remote on/off
 - k. Web/SNMP enabled

2.2 MATERIALS AND EQUIPMENT

- A. General: Provide electronic control products in sizes and capacities indicated, consisting of valves, dampers, thermostats, clocks, controllers, sensors, and other components as required for complete installation. Except as otherwise indicated, provide manufacturer's standard materials and components as published in their product information; designed and constructed as recommended by manufacturer, and as required for application indicated.
- B. Communication Wiring: All wiring shall be in accordance with National Electrical Codes and Division 26 of this specification.
 1. Contractor shall supply all communication wiring between Building Controllers (BC), Routers, Gateways, Advanced Application Controllers (AAC), Application Specific Controllers (ASC) and local and remote peripherals (e.g., workstations, printers, and modems).
 2. Local Supervisory LAN: For any portions of this network required under this Section of the specification, contractor shall use Category 5 of standard TIA/EIA 68 (10BaseT). Network shall be run with no splices and separate from any wiring over thirty (30) volts.
 3. Primary and Secondary Controller LANs: Communication wiring shall be individually 100% shielded pairs per manufacturers recommendations for distances installed, with overall PVC cover, Class 2, plenum-rated run with no splices and separate from any wiring over thirty (30) volts. Shield shall be terminated and wiring shall be grounded as recommended by BC manufacturer.
- C. Control Panels: Provide control panels with suitable brackets for wall mounting for each control system. Locate panel adjacent to systems served.

1. Fabricate panels of 16-gage furniture-grade steel, or 6063-T5 extruded aluminum alloy, totally enclosed on four sides, with hinged door and keyed lock, with manufacturer's standard shop-painted finish and color.
 2. Provide UL-listed cabinets for use with line voltage devices.
 3. All gauges and control components shall be identified by means of nameplates.
 4. All control tubing and wiring shall be run neatly and orderly in open slot wiring duct with cover.
 5. Complete wiring and tubing termination drawings shall be mounted in or adjacent to panel.
- D. Signal Wiring: BMS Contractor shall run all signal wiring in accordance with National Electric Codes and Division 26 of this Specification.
1. Signal wiring to all field devices, including, but not limited to, all sensors, transducers, transmitters, switches, etc. shall be twisted, 100% shielded pair, minimum 18-gauge wire, with PVC cover. Signal wiring shall be run with no splices and separate from any wiring above thirty (30) volts.
 2. Signal wiring shield shall be grounded at controller end only unless otherwise recommended by the controller manufacturer.
- E. Low Voltage Analog Output Wiring: Contractor shall run all low voltage control wiring in accordance with National Electric Codes and Division 26 of this Specification.
1. Low voltage control wiring shall be minimum 16-gauge, twisted pair, 100% shielded, with PVC cover, Class 2 plenum-rated. Low voltage control wiring shall be run with no splices separate from any wiring above thirty (30) volts.
- F. Instrument Pipe and Tube (Low Pressure Air Instrument Sensing Lines)
1. Connections: Use suitable bulkhead type fitting and static sensing tip for static pressure connections. Adapt tubing to instrument connection.
 2. Tubing: Virgin polyethylene non-metallic tubing type FR, ASTM D 2737, and with flame-retardant harness for multiple tubing. Use compression or push-on brass fittings.

2.3 GENERAL FIELD DEVICES

- A. Provide field devices for input and output of digital (binary) and analog signals into controllers (BCs, AACs, ASCs). Provide signal conditioning for all field devices as recommended by field device manufacturers, and as required for proper operation in the system.
- B. It shall be the Contractor's responsibility to assure that all field devices are compatible with controller hardware and software.
- C. Field devices specified herein are generally 'two-wire' type transmitters, with power for the device to be supplied from the respective controller. If the controller provided is not equipped to provide this power, or is not designed to work with 'two-wire' type transmitters, or if field device is to serve as input to more than one controller, or where the length of wire to the controller will unacceptably affect the accuracy, the Contractor shall provide 'four-wire' type equal transmitter and necessary regulated DC power supply or 120 VAC power supply, as required.

- D. For field devices specified hereinafter that require signal conditioners, signal boosters, signal repeaters, or other devices for proper interface to controllers, Contractor shall furnish and install proper device, including 120V power as required. Such devices shall have accuracy equal to, or better than, the accuracy listed for respective field devices.
- E. Accuracy: As stated in this Section, accuracy shall include combined effects of nonlinearity, non-repeatability and hysteresis.

2.4 TEMPERATURE SENSORS

- A. Provide the following instrumentation as required by the monitoring, control and optimization functions. All temperature sensor shall use platinum or nickel RTD elements only, silicon elements are not acceptable. All control signals shall be via a 4-20 mA loop. For critical system applications, only temperature transmitters will be accepted. RTDs and transmitters are acceptable for non-critical applications.

- B. Room Temperature:

- | | |
|--------------------------------------|---|
| 1. Temperature monitoring range | +40/+90 F (+40/120 F for high temp alarms) |
| 2. Output signal | 4-20 mA, 0-10VDC, 1K ohm |
| 3. Installation adjustments | none required |
| 4. Calibration adjustments | zero & span |
| 5. Factory calibration point | 70 °F |
| 6. Accuracy at calibration point | +0.5 °F |
| 7. Wall mounted Sensor Type: | Flat Plate |
| 8. Sensor Body/Cover Color: | White |
| 9. Approved Manufacturer: Subject to | compliance with requirements, provided on LiDL |
| National Account List: | BAPI, Emerson, or equal (|

- C. Combined Room Temperature and Humidity Sensor:

- | | |
|--|--|
| 1. Temperature monitoring range | +40/+90 F (+40/120 F for high temp alarms) |
| 2. Output signal | 4-20 mA, 0-10 VDC, 1K ohm |
| 3. Installation adjustments | none required |
| 4. Calibration adjustments | zero & span |
| 5. Factory calibration point | 70 (degree symbol) F |
| 6. Accuracy at calibration point | +0.5 (degree symbol) F |
| 7. Data-Column Sensor Type: | Flush Button Type |
| 8. Sensor Body/Cover Color: | White |
| 9. Approved manufacturer: Subject to | compliance with requirements, provided on LiDL |
| National Account List: Elan, Emerson, or equal | |

- D. Slab Temperature:

- | | |
|--------------------------------------|---|
| 1. Temperature monitoring range | +40/+90 F (+40/120 F for high temp alarms) |
| 2. Output signal | 4-20 mA, 0-10VDC, 1K ohm |
| 3. Installation adjustments | none required |
| 4. Calibration adjustments | zero & span |
| 5. Factory calibration point | 70 °F |
| 6. Accuracy at calibration point | +0.5 °F |
| 7. Wall mounted Sensor Type: | Flat Plate |
| 8. Data-Column Sensor Type: | Flush Button Type |
| 9. Approved Manufacturer: Subject to | compliance with requirements, provided on LiDL |
| National Account List: | BAPI, Emerson, or equal |

E. Duct (Single Point) Temperature

- | | | |
|----|-----------------------------------|--|
| 1. | Temperature monitoring range | +20/+120 F |
| 2. | Output signal | 4-20 mA, 0-10VDC, 1k ohm |
| 3. | Installation adjustments | none required |
| 4. | Calibration adjustments | zero & span |
| 5. | Factory calibration point | 70°F |
| 6. | Accuracy at calibration point | +0.5 F |
| 7. | Sensor Body/Cover Color: | White |
| 8. | Approved Manufacturer: Subject to | compliance with requirements, provided on LIDL National Account List : BAPI, RTU manufacturer, or equal |

2.5 DEW POINT/HUMIDITY SENSORS

A. Duct Relative Humidity:

- | | | |
|-----|--------------------------|---|
| 1. | Sensor Humidity range | 0 to 100% |
| 2. | Operating temperature | 15°F to +170°F |
| 3. | Accuracy | +2% RH |
| 4. | Sensing element | Capacitive sensor |
| 5. | Output signal | 4-20 mA DC |
| 6. | Installation adjustments | zero & span |
| 7. | Operating temperature | 15°F to +170°F |
| 8. | Voltage requirement | 12-36 VDC |
| 9. | Sensor Body/Cover Color: | White |
| 10. | Approved Manufacturer: | BAPI, RTU manufacturer, Emerson, or equal |

2.6 INDOOR AIR QUALITY (CO2) SENSORS

- A. Provide indoor air quality duct-mounted sensors to monitor Carbon Dioxide (CO2) levels.
- B. The sensors shall utilize single-beam, dual-wavelength, non-dispersive infrared technology.
- C. The sensor shall meet the following requirements:

- | | | |
|-----|--------------------------|----------------------------------|
| 1. | Operating voltage: | 24 VAC/DC |
| 2. | Accuracy: | ± 2% of range + 2% reading |
| 3. | Power consumption: | <2.5W |
| 4. | Measuring range: | 0 – 2000 ppm |
| 5. | Tolerance: | +/- 100 ppm |
| 6. | Output: | 4-20mA, 0 – 10 VDC |
| 7. | Calibration: | every 5 years recommended |
| 8. | Long-term Stability: | <± 5% of range/5 years |
| 9. | Response time: | 1 minute |
| 10. | Sensor Body/Cover Color: | White |
| 11. | Approved Manufacturer: | BAPI, RTU manufacturer, or equal |

2.7 DIFFERENTIAL PRESSURE TRANSMITTERS (DP)

A. General Air Pressure Sensor Requirements:

1. Pressure sensors shall be constructed to withstand 100% pressure over-range without damage and to hold calibrated accuracy when subject to a momentary 40% over-range input.

2. Pressure transmitters shall transmit a 0 to 5 VDC, 0 to 10 VDC, or 4 to 20 mA output signal.
3. Differential pressure transmitters used for flow measurement shall be sized to the flow sensing device and shall be supplied with Tee fittings and shut-off valves in the high and low sensing pick-up lines to allow the balancing contractor and Owner permanent easy-to-use connection.
4. Provide a minimum of a NEMA 1 housing for the transmitter. Locate transmitters in accessible local control panels wherever possible.
5. Provide a two year warranty for each transmitter. Replace all transmitters found to be defective at no cost to the Owner during the warranty period.
6. Operating voltage: 24 VAC/DC
7. Accuracy: ± 0.1 " w.c.
8. Measuring range: ± 0.25 " to 40" w.c.
9. Output: 4-20 mA, 0 – 10 VDC
10. Long-term Stability: $<\pm 5\%$ of range/5 years
11. Response time: 8 seconds
12. Output Update Rate: 1 second
13. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
14. Sensor Body/Cover Color: White
15. Approved Manufacturer: BAPI, RTU manufacturer, or equal

2.8 LIGHT LEVEL SENSOR

A. Photodiode Sensor:

1. Measuring Range: 0 – 10,000 FC
2. Output: 0-9 Volts
3. Response Time: Instant
4. Listings: UL 916, Title 24, RoHS compliant
5. Sensor Body/Cover Color: White
6. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
7. Approved Manufacturer: PLC Sensor or equal

2.9 CURRENT SWITCHES (CS)

A. Current-operated switches shall be self-powered, solid-state with adjustable trip current. The switches shall be selected to match the current of the application and output requirements of the DDC system.

1. Sensor Power: 24 VAC/DC
2. Minimum Trip Point: 0.5 A
3. Adjustable Trip Point: Required
4. Status LEDs: Required
5. Insulation Class: ± 0.1 " w.c.
6. Frequency Range: 50/60 Hz
7. Temperature Range: 5 to 140 °F
8. Humidity Range: 10 to 90% RH, Non-condensing
9. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
10. Approved Manufacturer: Veris, or equal

2.10 RELAYS

- A. Control relays shall be UL listed plug-in type with dust cover and LED "energized" indicator. Contact rating, configuration, and coil voltage shall be suitable for application.
- B. Time delay relays shall be UL listed solid-state plug-in type with adjustable time delay. Delay shall be adjustable $\pm 200\%$ (minimum) from set point shown on plans. Contact rating, configuration, and coil voltage shall be suitable for application. Provide NEMA 1 enclosure when not installed in local control panel.
- C. Where required, provide "delay on break" relays where the output remains energized for the pre-determined time delay after the input has been de-energized. Reclosing the input switch during timing resets the time delay. Loss of input voltage resets the time delay as well as the output.
- D. Approved Manufacturer: Idec, Functional Devices, or equal

2.11 VOLTAGE TRANSFORMER

- A. AC voltage transformers shall be UL/CSA Recognized, 600 VAC rated, complete with built-in fuse protection.
- B. Transformers shall be suitable for ambient temperatures of 4°C to 55°C (40°F to 130°F) and shall provide $\pm 0.5\%$ accuracy at 24 VAC and a 5 VA load.
- C. Windings (except for terminals) shall be completely enclosed with metal or plastic material.

2.12 WATER LEAK DETECTOR

- A. Single point leak detector (external powered models)
 - 1. Service: Water or conductive fluids
 - 2. Minimum Sensing Gap: 1/32"
 - 3. Switch Type: DPDT
 - 4. Audible Alarm: at least 85 dB @ 1' distance
 - 5. Visual Alarm: Red LED for water level, Green LED for power condition
 - 6. Temperature Limits: 32 to 122 °F
 - 7. Power Requirements: 24 VAC or 11 to 27 VDC
 - 8. Listings: CE, RoHS
 - 9. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
 - 10. Approved Manufacturer: Dwyer (WD3 series) or equal

2.13 DUCT SMOKE DETECTOR

- A. Photoelectric detector with sampling tube that spans the entire width of duct. .
- B. Velocity Rating: 100 to 4000 fpm or 500 to 4000fpm depending on the minimum velocity in the duct. Provide the 100 to 4000 fpm detector if the min duct velocity is below 550 fpm.
- C. Output Contact: Alarm, two sets form "C" rated at 10amps 115V resistive. One set of alarm contacts for BMS monitoring and fan shutdown. Trouble, one set of contacts.

D. Temperature & RH limits: 32 to 120°F and 10 to 85% relative humidity.

E. Acceptable Manufacturer:

1. Invensys FIREX model 2650
2. Sensor Systems DH100ACDCLP
3. Air Products and Controls SL-2000

2.14 ELECTRIC CONTROL COMPONENTS

A. Horn & Strobe:

1. Description: Xenon strobe light source with horn
2. Flash Rate: 60 FPM
3. Sound: 100 dB at 1 meter/90 dB at 10 ft. (measured in anechoic chamber)
4. Wiring: Terminals
5. Housing: Gray, engineered thermoplastic housing
6. Operating Temp Range: Indoor: 85% relative humidity at 86 °F, 32 °F to 120 °F variable ambient temperature
Outdoor: 95% relative humidity at 86 °F, -31 °F to 150 °F variable ambient temperature
7. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
8. Approved Manufacturer: Edwards Signal (869STR series) or equal

B. Timer with Relay:

1. Description: Multifunction timer
2. Input Voltage: 24, 120, 230 VAC, 50/60 Hz
3. Output Current: 10 A resistive, max.
4. Time Delay: Adjustable from 0.1 second to 1023 minutes
5. Repeatability: 0.1% or 20 mS, whichever is greater
6. Time Delay Accuracy: 2.5% or 50 mS, whichever is greater
7. Recycle / Start-up Time Delay: 250 mS, typ.
8. Mechanical: 2.5" x 2.5" case, Surface mount
9. Protection: Encapsulated circuitry, MOV transient protection
10. Warranty Period: same extent as the BMS system, or per the manufacturer's warranty, whichever is greater
11. Approved Manufacturer: Airotronics (MC363 series) or equal

2.15 NAMEPLATES

- A. Provide engraved phenolic or micarta nameplates for all equipment, components, and field devices furnished. Nameplates shall be 1/8 thick, black, with white center core, and shall be minimum 1" x 3", with minimum 1/4" high block lettering. Nameplates for devices smaller than 1" x 3" shall be attached to adjacent surface.
- B. Each nameplate shall identify the function for each device.

2.16 TESTING EQUIPMENT

- A. BMS Contractor shall test and calibrate all signaling circuits of all field devices to ascertain that required digital and accurate analog signals are transmitted, received, and displayed at system operator terminals, and make all repairs and recalibrations required to complete test. Contractor shall be responsible for test equipment required to perform these tests and calibrations. Test equipment used for testing and calibration of field devices shall be at least twice as accurate as respective field device (e.g., if field device is +/-0.5% accurate, test equipment shall be +/-0.25% accurate over same range).

PART 3 - EXECUTION

3.1 INSPECTION

- A. Examine areas and conditions under which control systems are to be installed. Do not proceed with work until unsatisfactory conditions have been corrected in manner acceptable to Contractor.

3.2 INSTALLATION OF CONTROL SYSTEMS

- A. General: Install systems and materials in accordance with manufacturer's instructions, roughing-in drawings and details shown on drawings. Install electrical components and use electrical products complying with requirements of National Electric Code and all local codes.
- B. Control Wiring: The term "control wiring" is defined to include providing of wire, conduit and miscellaneous materials as required for mounting and connection of electric control devices.
 - 1. Wiring System: Install complete wiring system for electric control systems. Install all control wiring external to panels in electric metallic tubing or raceway. On Renovation projects, wiring in finished areas shall be routed in wire mold. The routing of wiring in finished areas must be specifically approved by the AOR/EOR. Installation of wiring shall generally follow building lines. Install in accordance with National Electrical Code and Division 26 of this Specification. Fasten flexible conductors bridging cabinets and doors, neatly along hinge side, and protect against abrasion. Tie and support conductors neatly.
 - 2. Control Wiring Conductors: Install control wiring conductors, without splices between terminal points, color-coded. Install in neat workmanlike manner, securely fastened. Install in accordance with National Electrical Code and Division 26 of this Specification.
 - 3. Communication wiring, signal wiring and low voltage control wiring shall be installed separate from any wiring over thirty (30) volts. Signal wiring shield shall be grounded at controller end only, unless otherwise recommended by the controller manufacturer.
 - 4. All WAN and LAN Communication wiring shield shall be terminated as recommended by controller manufacturer. All WAN and LAN Communication wiring shall be labeled with a network number, device ID at each termination and shall correspond with the WAN and LAN system architecture and floor plan submittals. All WAN and LAN cabling shall comply with applicable Division 26 requirements.
 - 5. Number-code or color-code conductors appropriately for future identification and servicing of control system. Code shall be as indicated on approved installation drawings.

- C. Room Temperature Sensors: Install sensors as shown on the drawings. Seal conduit penetrations at the wall box airtight. Install batt insulation in the wall box to completely fill the box. Electrical connections shall be made using a twist-on sealant filled connectors suitable for the installation.
- D. Relative Humidity Sensors: Provide element guard as recommended by manufacturer for high velocity installations. For high limit sensors, position remote enough to allow full moisture absorption into the air stream before reaching the sensor.
- E. Current Switches for Motor Status Monitoring: Adjust so that setpoint is below minimum operating current and above motor no load current.

3.3 DDC CONTROLLERS:

A. INSPECTION:

- 1. Examine areas and conditions under which control systems are to be installed. Do not proceed with work until unsatisfactory conditions have been corrected in manner acceptable to Contractor.

B. System Access:

- 1. Provide an Ethernet connection and 4-port hub at the panel for the BC controller or controllers that provides access to the Local Supervisory LAN and to the BMS web server for all controllers, other than an Application Category-1 controllers. The user shall be able to access each controller on the system using this connection via the Control System Server database for graphics, schedules, programming, controller configuration etc.

C. INSTALLATION OF CONTROL SYSTEMS:

- 1. General: Install systems and materials in accordance with manufacturer's instructions, specifications roughing-in drawings and details shown on drawings. BMS Contractor shall install all controllers in accordance with manufacturer's installation procedures and practices.

D. HARDWARE APPLICATION REQUIREMENTS

- 1. General: The functional intent of this specification is to allow cost effective application of manufacturers standard products while maintain the integrity and reliability of the control functions. Specific requirements indicated below are required for the respective application. Manufacturer shall apply the most cost-effective unit that meets the requirement of that application.
- 2. Standalone Capability: Each Control Unit shall be capable of performing the required sequence of operation for the associated equipment. All physical point data and calculated values required to accomplish the sequence of operation shall originate within the associated control unit with only the exceptions enumerated below.
- 3. Where associated control functions involve functions from different categories identified below, the requirements for the most restrictive category shall be met.
- 4. Primary DDC panels shall be as follows:
 - a. Application Category-0: Distributed Monitoring
 - 1) Applications in this category include the following:

- a) Monitoring of variables that are not used in a control loop, sequence logic, or safety.
 - 2) Points on BCs, AACs, and ASCs may be used in these applications as well as Ds and/or general-purpose I/O modules.
 - 3) Where these points are trended, contractor shall verify and document that the network bandwidth is acceptable for such trends and is still capable of acceptable and timely control function.
 - 4) LAN Restrictions: These points may reside on any controller
- b. Application Category-1: Application Specific Controller (ASC)
 - 1) Minimum one (1) ASC for each mechanical equipment:
 - a) Rooftop Unit
 - b) Air Conditioning Unit
 - c) Unitary equipment <15 tons
 - 2) LAN Segment Restrictions: BACnet Systems: Limit the number of AAC's/ASC's servicing any one of these applications on the LAN Segment to 40.
- c. Application Category-2: Advance Application Controller (AAC)
 - 1) Minimum one (1) AAC for building control of the following:
 - a) Lighting Control
 - b) Exhaust Fan Control & Monitoring
 - c) Slab Monitoring
 - d) UPS Monitoring
 - e) Leak Detection Monitoring
 - f) Fire Alarm System Monitoring
 - g) Intrusion Controls Monitoring
 - h) Generator Monitoring
 - i) Misc. Monitoring and Meters (not directly associated with a control sequence and where trending is not critical)
 - 2) LAN Segment Restrictions: BACnet Systems: Limit the number of AAC's servicing any one of these applications on the LAN Segment to 40.
- d. Application Category-3: Building Controller (BC) with Advance Application Controller (AAC) or Expansion I/O Modules (IOM)
 - 1) Minimum one (1) BC for global operations of the following:
 - a) Occupancy Scheduling
 - b) Modbus-RTU Communications
 - c) BACnet-MS/TP Communications
 - d) BACnet-IP Communications
- 5. Motors in motor control centers shall be controlled from the DDC controller associated with HVAC system. It shall not be acceptable to control all motors in a MCC from one DDC controller dedicated to the MCC. The intent of this specification is that the loss of

any one DDC controller shall not affect the operation of other HVAC systems, only for the points connected to the DDC controller.

END OF SECTION 230922

SECTION 230923 – BUILDING AUTOMATION SYSTEM (BAS) – GENERAL SEQUENCE**PART 1 - GENERAL****1.1 SYSTEM DESCRIPTION**

- A. Refer to Division 23 Section and Control Drawings for a description of the systems to be controlled.
- B. Refer to Control Drawings for detailed sequence of operations for major HVAC equipment.
- C. This Section contains general sequence and alarming method for Illinois Institute of Technologies.

1.2 SUBMITTALS

- A. Refer to Division 23 Section "Building Automation System (BAS)" and Division 01 Sections for requirements for control shop drawings, product data, Users Manual, etc.
- B. Programming Manual: Provide DDC system programming manual as well as documentation of site specific programming prior to the start of Acceptance Phase.

1.3 PROJECT RECORD DOCUMENTS

- A. Within two weeks of the completion of commissioning, provide record documents to represent the final control configuration with actual setpoints and tuning parameters as existed at acceptance.
- B. Record documents shall be modified control drawings with the actual installed information. Drawings shall be delivered in both reproducible hard copy and electronic format in AutoCAD v13 or later. Provide all supporting files, blocks, fonts, etc. required by the drawings.
- C. Provide final points list
- D. Provide final detailed wiring diagrams with all wire numbers and termination points indicated
- E. Accurately record final sequences and control logic made after submission of shop drawings.

1.4 DEFINITIONS/ABBREVIATIONS

- A. Absolute Minimum OA: Minimum flow rate setpoint to which the OA or primary air may throttle down. This value is acceptable as long as CO₂ levels are within acceptable limits.
- B. DDC: Direct Digital Control
- C. Design Minimum OA: Minimum flow rate setpoint based on code requirements or designed system and coil capacities.
- D. BAS: Building Automation System (Same as "Building Management System (BMS)")
- E. MVR: Minimum required ventilation rate
- F. OA: Outdoor Air

- G. Physical Point: A point on the BMS that is physically connected to an I/O device such that a hardware point exists
- H. Virtual Point: A point to store values (i.e.: a setpoint) that do not represent a physical device

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 GENERAL SEQUENCE NOTES

- A. Sequences specified herein indicate the functional intent of the systems operation and may not fully detail every aspect of the programming that may be required to obtain the indicated operation. Contractor shall provide all programming necessary to obtain the sequences/system operation indicated.
- B. When a rooftop unit is not in operation, control devices shall remain in their "off" positions. "Off" positions may differ from the "normal" (meaning failed) position.
- C. Except as specified otherwise, throttling ranges, proportional bands, and cycle differentials shall be centered on the associated setpoint. All modulating feedback control loops shall include the capability of having proportional, integral, and derivative action. Unless the loop is specified "proportional only" or "P+I", Contractor shall apply appropriate elements of integral and derivative gain to each control loop which shall result in stable operation, minimum settling time, and shall maintain the primary variable within the specified maximum allowable variance.
- D. Scheduling Terminology: When rooftop units are scheduled, the following defines the terminology used:
 - 1. Occupied Period: period of time when the building is in use and occupied. Generally systems will be fully operational throughout this period and ventilation air shall be continuously introduced. Space temperature setpoints will generally be in the "normal" range of 69°-76°F.
 - 2. Unoccupied period: period of time when the building or zone is not in use and unoccupied. Ventilation air shall not be introduced.
 - 3. Preoccupancy Period: Time prior to the Occupied period when the systems are returning the space temperatures from setback to "normal" or occupied setpoints (warm-up and cool-down). Ventilation air shall not be introduced unless outside air conditions permit free-cooling. Time period shall be determined by an optimum start strategy unless otherwise specified.
 - 4. Setback Period: Setback start will typically coincide with the end of the occupied period and end with the start of the preoccupancy period, however it shall be provided with its own schedule. Generally systems will be off except to maintain a "setback" temperature.
- E. Automatic Daylight Savings Time Switchover
 - 1. The system shall provide automatic time adjustment for switching to/from Daylight Savings Time.
- F. Night Setback Control
 - 1. The system shall provide the ability to automatically adjust setpoints for night control.

G. Enthalpy Switchover (Economizer)

1. The Building Controller Software shall control the position of the rooftop unit relief, return, and outside air dampers. If the outside air enthalpy falls below changeover set point the BCS shall modulate the dampers to provide 100% outside air. The user shall be able to quickly changeover to an economizer system based on dry bulb temperature and shall be able to override the economizer cycle and return to minimum outside air operation at any time.

H. Loop Control

1. A Model-Free Adaptive Control algorithm or alternatively a PID (proportional-integral-derivative) closed-loop control algorithm with direct or reverse action and anti-windup shall be supplied. The algorithm shall calculate a time-varying analog value that is used to position an output or stage a series of outputs. The controlled variable, set point, and weighting parameters shall be user-selectable.

I. Sequencing

1. Provide application software based upon the sequences of operation specified to properly sequence equipment.

J. All setpoints shall be adjustable (adj.), they shall be modifiable, with the proper password level, from the operator interface or via a function block menu. For these points, it is unacceptable to have to modify programming statements to change the setpoint.

K. Where "prove operation" of a device (generally controlled by a digital output) is indicated in the sequence, it shall require that the BMS, after an adjustable time delay & after the device is commanded to operate (feedback delay,) confirm that the device is operational via the status input. If the status point does not confirm operation after the time delay or anytime thereafter for an adjustable time delay (debounce delay) while the device is commanded to run, an alarm shall be enunciated audibly and via an alarm message at the operator interface and print at the alarm printers. A descriptive message shall be attached to the alarm message indicating the nature of the alarm and actions to be taken. Contractor shall provide messages to meet this intent.

L. Wherever a value is indicated to be dependent on another value (i.e.: setpoint plus 5°F) the BMS shall use that equation to determine the value. Simply providing a virtual point that the operator must set is unacceptable. In this case three virtual points shall be provided. One to store the parameter (5°F), one to store the setpoint, and one to store the value which is the result of the equation.

M. Some sequences rely on integration with third-party manufacturer control equipment. See the control equipment specifications, equipment schedules and equipment specifications for more information on this integration.

3.2 GENERAL ALARMING NOTES

- A. Alarming and faults shall be a function of rooftop unit factory controller BACnet object data to a BMS System (Subject to compliance with requirements, provided on **LiDL National Account List**. basis-of-design: Emerson). The BMS system shall read all space temperature (where applicable), duct CO2 (where applicable), and duct relative humidity (where applicable), and place high and low limits for consumer comfort and product tempering.

- B. Analog Input Alarms: For each analog input, program an alarm message for reporting whenever the analog value is outside of the programmed alarm limits. Report a 'Return-to-Normal' message after the analog value returns to the normal range, using a programmed alarm differential. The alarm limits shall be individually selected by the Contractor based on the following criteria:
1. Space temperature, except as otherwise stated in sequence of operation:
 - a. Low alarm: 5°F (adj) below setpoint
 - b. High alarm: 5°F (adj) above setpoint
 2. Space humidity:
 - a. Low alarm: 2°F (adj) below setpoint
 - b. High alarm: 2°F (adj) above setpoint
- C. Maintenance Alarms: Enunciate alarms when runtime accumulation exceeds a value specified by the operator.
1. *DEVICE XXXX REQUIRES MAINTENANCE.* Runtime has exceeded specified value since last reset.

END OF SECTION 230923

SECTION 231123 - NATURAL GAS PIPING**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the following:
 - 1. Pipes, tubes, and fittings.
 - 2. Piping specialties.
 - 3. Joining materials for piping and tubing.
 - 4. Valves.
 - 5. Pressure regulators.
 - 6. Service meters.
 - 7. Mechanical sleeve seals.

1.2 DEFINITIONS

- A. Finished Spaces: Spaces other than mechanical and electrical equipment rooms, furred spaces, pipe and duct shafts, unheated spaces immediately below roofs, spaces above ceilings, unexcavated spaces, crawlspaces, and tunnels.
- B. Exposed, Interior Installations: Exposed to view indoors. Examples include finished occupied spaces and mechanical equipment rooms.
- C. Exposed, Exterior Installations: Exposed to view outdoors or subject to outdoor ambient temperatures and weather conditions. Examples include rooftop locations.
- D. Anodless Risers and Sweeps: Fittings that can be installed without additional cathodic protection.

1.3 PERFORMANCE REQUIREMENTS

- A. Natural-Gas System Pressure within Buildings: 2-psig or less.
- B. Operating Pressure Ratings:
 - 1. Piping and Valves: 100-psig minimum unless otherwise indicated.
 - 2. Service Regulators: 65-psig minimum unless otherwise indicated.
 - 3. Operating Pressure of Service Meter: 5-psig minimum unless otherwise indicated.

1.4 SUBMITTALS

- A. Product Data: For each type of the following:
 - 1. Piping specialties.
 - 2. Corrugated, stainless-steel tubing with associated components.
 - 3. Valves. Include pressure rating, capacity, settings, and electrical connection data of selected models.
 - 4. Pressure regulators. Indicate pressure ratings and capacities.
 - 5. Dielectric fittings.
 - 6. Mechanical sleeve seals.
 - 7. Escutcheons.

- B. Shop Drawings: Include plans, piping layout and elevations, sections, and details for fabrication of pipe anchors, hangers, supports for multiple pipes, alignment guides, expansion joints and loops, and attachments of the same to building structure. Detail locations of anchors, alignment guides, and expansion joints and loops. Include a site plan, on which natural-gas piping is indicated and coordinated with other services and utilities.
 - 1. Shop Drawing Scale: Not less than $1/4" = 1'-0"$.
- C. Coordination Drawings: Plans and details, drawn to scale, on which natural-gas piping is shown and coordinated with other installations, using input from installers of the items involved.
- D. Installer Qualifications.
- E. Welding Certificates.
- F. Operation and Maintenance Data: For motorized gas valves, pressure regulators, and service meters, include data in operation, and maintenance manuals.
- G. Record Drawings: Indicate, at $1/4" = 1'-0"$ scale, the actual natural-gas piping installation layout and elevations, sections, fabrication of pipe anchors, hangers, supports for multiple pipes, alignment guides, expansion joints and loops, and attachments of the same to the building structure. Detail location of anchors, alignment guides, and expansion joints and loops.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an installer that is experienced in the installation of natural-gas piping of the types required for the Project, and whose work has resulted in a record of successful in-service performance. Installer licensed to perform natural-gas piping installation and maintenance in the project jurisdiction.
- B. Steel Support Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."
- C. Pipe Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code.
- D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- E. Regulatory Requirements: Natural-gas piping installations shall comply with the International Fuel Gas Code, 2000 edition.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Handling Flammable Liquids: Remove and dispose flammable liquids from existing natural-gas piping according to requirements of authorities having jurisdiction.
- B. Deliver pipes and tubes with factory-applied end caps. Maintain end caps throughout shipping, storage, and handling to prevent pipe end damage and to prevent entrance of dirt, debris, and moisture.
- C. Store and handle pipes and tubes having factory-applied protective coatings to avoid damaging coating, and protect from direct sunlight.

- D. Protect stored polyethylene pipes and valves from direct sunlight.

1.7 PROJECT CONDITIONS

- A. Perform site survey, research public utility records, and verify existing utility locations. Contact utility-locating service for area where Project is located.

1.8 COORDINATION

- A. Coordinate sizes and locations of concrete bases with actual equipment provided.
- B. Coordinate installation of anchorages for natural-gas systems. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete. Deliver such items to Project site in time for installation and to not cause a delay in the Work.
- C. Coordinate requirements for access panels and doors for valves and controls installed concealed behind finished surfaces, and for areas requiring routine maintenance. Comply with requirements in Division 08 Section "Access Doors and Frames."

1.9 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.
- B. Special Warranty: Submit written warranty, executed by the manufacturer and Installer, agreeing to repair or replace components that develop defects in material or workmanship within the specified warranty period.
 - 1. Warranty Period for Natural-Gas Systems: One year from date of Substantial Completion, or eighteen months from date of shipment, whichever is greater.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide products by one of the manufacturers indicated for each component:
 - 1. Appliance Flexible Connector: No preference.
 - 2. Two-Piece, Full-Port, Bronze Ball Valves with Bronze Trim:
 - a. Conbraco Industries, Inc.; Apollo Div.
 - b. Lyall, R. W. & Company, Inc.
 - c. McDonald, A. Y. Mfg. Co.
 - d. Perfection Corporation; a subsidiary of American Meter Company.
 - 3. Bronze Plug Valves:
 - a. Lee Brass Company.
 - b. McDonald, A. Y. Mfg. Co.
 - 4. Cast-Iron, Non-lubricated Plug Valves:

- a. McDonald, A. Y. Mfg. Co.
 - b. Mueller Co.; Gas Products Div.
 - c. Xomox Corporation; a Crane company.
5. Cast-Iron, Lubricated Plug Valves:
- a. Flowserve.
 - b. McDonald, A. Y. Mfg. Co.
 - c. Mueller Co.; Gas Products Div.
 - d. R&M Energy Systems, a Unit of Robbins & Myers, Inc.
6. Automatic Gas Valves:
- a. ASCO Power Technologies, LP; Division of Emerson.
 - b. Eaton Corporation; Controls Div.
 - c. Eclipse Combustion, Inc.
 - d. Honeywell International Inc.
 - e. Johnson Controls.
7. Electrically Operated Valves:
- a. ASCO Power Technologies, LP; Division of Emerson.
 - b. Eclipse Combustion, Inc.
 - c. Goyen Valve Corp.; Tyco Environmental Systems.
 - d. Magnatrol Valve Corporation.
 - e. Parker Hannifin Corporation; Climate & Industrial Controls Group; Skinner Valve Div.
 - f. Watts Regulator Co.; Division of Watts Water Technologies, Inc.
8. Line Pressure Regulators:
- a. Eclipse Combustion, Inc.
 - b. Fisher Control Valves and Regulators; Division of Emerson Process Management.
 - c. Invensys.
 - d. Maxitrol Company.
 - e. Richards Industries; Jordan Valve Div.
9. Appliance Pressure Regulators:
- a. Eaton Corporation; Controls Div.
 - b. Harper Wyman Co.
 - c. Maxitrol Company.
 - d. SCP, Inc.
10. Dielectric Unions: No preference.
11. Dielectric Flanges: No preference.
12. Mechanical Sleeves: No preference.

2.2 PIPES, TUBES, AND FITTINGS

- A. Steel Pipe: ASTM A53/A53M, black steel, Schedule 40, Type E or S, Grade B.
- 1. Malleable-Iron Threaded Fittings: ASME B16.3, Class 150, standard pattern.
 - 2. Wrought-Steel Welding Fittings: ASTM A234/A234M for butt welding and socket welding.

3. Unions: ASME B16.39, Class 150, malleable iron with brass-to-iron seat, ground joint, and threaded ends.
4. Forged-Steel Flanges and Flanged Fittings: ASME B16.5, minimum Class 150, including bolts, nuts, and gaskets of the following material group, end connections, and facings:
 - a. Material Group: 1.1.
 - b. End Connections: Threaded or butt welding to match pipe.
 - c. Lapped Face: Not permitted underground.
 - d. Gasket Materials: ASME B16.20, metallic, flat, asbestos free, aluminum o-rings, and spiral-wound metal gaskets.
 - e. Bolts and Nuts: ASME B18.2.1, carbon steel aboveground and stainless steel underground.

B. Polyethylene (PE) Pipe: ASTM D2513, SDR 11 (Underground Piping Only).

1. PE Fittings: ASTM D2683, socket-fusion type or ASTM D3261, butt-fusion type with dimensions matching PE pipe.
2. PE Transition Fittings: Factory-fabricated fittings with PE pipe complying with ASTM D2513, SDR 11; and steel pipe complying with ASTM A53/A53M, black steel, Schedule 40, Type E or S, Grade B.
3. Anodeless Service-Line Risers: Factory fabricated and leak tested.
 - a. Underground Portion: PE pipe complying with ASTM D2513, SDR 11 inlet. Underground piping to receive factory-applied, three layer coating of epoxy, adhesive, and polyethylene.
 - b. Casing: Steel pipe complying with ASTM A53/A53M, Schedule 40, black steel, Type E or S, Grade B, with corrosion-protective coating covering. Vent casing aboveground.
 - c. Aboveground Portion: PE transition fitting.
 - d. Outlet shall be threaded or flanged or suitable for welded connection.
 - e. Tracer wire connection.
 - f. Ultraviolet shield.
 - g. Stake supports with factory finish to match steel pipe casing or carrier pipe.
 - h. Joint Cover Kits: Epoxy paint, adhesive, and heat-shrink polyethylene sleeves.
4. Protective Coating for Underground Piping: Factory-applied, three-layer coating of epoxy, adhesive and polyethylene.

2.3 PIPING SPECIALTIES

A. Appliance Flexible Connectors:

1. Indoor, Fixed-Appliance Flexible Connectors: Comply with ANSI Z21.24.
2. Indoor, Movable-Appliance Flexible Connectors: Comply with ANSI Z21.69.
3. Outdoor, Appliance Flexible Connectors: Comply with ANSI Z21.75.
4. Corrugated stainless-steel tubing with polymer coating.
5. Operating-Pressure Rating: 0.0.5-psig End Fittings: Zinc-coated steel.
6. Threaded Ends: Comply with ASME B1.20.1.
7. Maximum Length: 72-inches.

B. Y-Pattern Strainers:

1. Body: ASTM A126, Class B, cast iron with bolted cover and bottom drain connection.

2. End Connections: Threaded ends for NPS 2 and smaller; flanged ends for NPS 2-1/2 and larger.
3. Strainer Screen: 40-mesh startup strainer and perforated stainless-steel basket with 50-percent free area.
4. CWP Rating: 125-psig.

C. Basket Strainers:

1. Body: ASTM A126, Class B, high-tensile cast iron with bolted cover and bottom drain connection.
2. End Connections: Threaded ends for NPS 2 and smaller; flanged ends for NPS 2-1/2 and larger.
3. Strainer Screen: 40-mesh startup strainer and perforated stainless-steel basket with 50 percent free area.
4. CWP Rating: 125-psig.

D. T-Pattern Strainers:

1. Body: Ductile or malleable iron with removable access coupling and end cap for strainer maintenance.
2. End Connections: Grooved ends.
3. Strainer Screen: 40-mesh startup strainer and perforated stainless-steel basket with 57-percent free area.
4. CWP Rating: 750-psig.

- E. Weatherproof Vent Cap: Cast- or malleable-iron increaser fitting with corrosion-resistant wire screen, with free area at least equal to cross-sectional area of connecting pipe and threaded-end connection.

2.4 JOINING MATERIALS

- A. Joint Compound and Tape: Suitable for natural-gas pipe.
- B. Welding Filler Metals: Comply with AWS D10.12/D10.12M for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.

2.5 MANUAL GAS SHUTOFF VALVES

- A. Refer to Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule" for where each valve type is applied in various services.
- B. General Requirements for Metallic Valves, NPS 2 and Smaller: Comply with ASME B16.33.
1. CWP Rating: 125-psig.
 2. Threaded Ends: Comply with ASME B1.20.1.
 3. Dryseal Threads on Flare Ends: Comply with ASME B1.20.3.
 4. Tamperproof Feature: Locking feature for valves indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
 5. Listing: Listed and labeled by an NRTL acceptable to authorities having jurisdiction for valves 1 inch diameter and smaller.
 6. Service Mark: Valves NPS 1-1/4 to NPS 2 shall have initials "WOG" permanently marked on valve body.

- C. General Requirements for Metallic Valves, NPS 2-1/2 and Larger: Comply with ASME B16.38.
1. CWP Rating: 125-psig.
 2. Flanged Ends: Comply with ASME B16.5 for steel flanges.
 3. Tamperproof Feature: Locking feature for valves indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
 4. Service Mark: Initials "WOG" shall be permanently marked on valve body.
- D. Two-Piece, Full-Port, Bronze Body: Complying with ASTM B584.
1. Ball: Chrome-plated bronze.
 2. Stem: Bronze; blowout proof.
 3. Seats: Reinforced TFE; blowout proof.
 4. Packing: Threaded-body packnut design with adjustable-stem packing.
 5. Ends: Threaded, flared, or socket as indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
 6. CWP Rating: 600-psig.
 7. Listing: Valves NPS 1 and smaller shall be listed and labeled by an NRTL acceptable to authorities having jurisdiction.
 8. Service: Suitable for natural-gas service with "WOG" indicated on valve body.
- E. Bronze Plug Valves: MSS SP-78.
1. Body: Bronze, complying with ASTM B584.
 2. Plug: Bronze.
 3. Ends: Threaded, socket, or flanged as indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
 4. Operator: Square head or lug type with tamperproof feature where indicated.
 5. Pressure Class: 125-psig.
 6. Listing: Valves NPS 1 and smaller shall be listed and labeled by an NRTL acceptable to authorities having jurisdiction.
 7. Service: Suitable for natural-gas service with "WOG" indicated on valve body.
- F. Cast-Iron, Non-lubricated Plug Valves: MSS SP-78.
1. Body: Cast iron, complying with ASTM A126, Class B.
 2. Plug: Bronze or nickel-plated cast iron.
 3. Seat: Coated with thermoplastic.
 4. Stem Seal: Compatible with natural-gas.
 5. Ends: Threaded or flanged as indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
 6. Operator: Square head or lug type with tamperproof feature where indicated.
 7. Pressure Class: 125-psig.
 8. Listing: Valves NPS 1 and smaller shall be listed and labeled by an NRTL acceptable to authorities having jurisdiction.
 9. Service: Suitable for natural-gas service with "WOG" indicated on valve body.
- G. Cast-Iron, Lubricated Plug Valves: MSS SP-78.
1. Body: Cast iron, complying with ASTM A126, Class B.
 2. Plug: Bronze or nickel-plated cast iron.
 3. Seat: Coated with thermoplastic.
 4. Stem Seal: Compatible with natural-gas.

5. Ends: Threaded or flanged as indicated in Articles "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule."
6. Operator: Square head or lug type with tamperproof feature where indicated.
7. Pressure Class: 125- psig.
8. Listing: Valves NPS 1 and smaller shall be listed and labeled by an NRTL acceptable to authorities having jurisdiction.
9. Service: Suitable for natural-gas service with "WOG" indicated on valve body.

H. Valve Boxes:

1. Cast-iron, two-section box.
2. Top section with embossed cover with letters "GAS".
3. Bottom section with base to fit over valve and a minimum 5-inches diameter barrel.
4. Adjustable cast-iron extensions of length required for depth of embedment.
5. Include tee-handle, steel operating wrench with socket end fitting, either valve nut or flat head, with stem of length required to operate valve.

2.6 MOTORIZED GAS VALVES

A. Electrically Operated Valves: Comply with UL 429.

1. Pilot operated.
2. Body: Brass or aluminum.
3. Seats and Disc: Nitrile rubber.
4. Springs and Valve Trim: Stainless steel.
5. 120-V ac, 60-Hz, Class B, continuous-duty molded coil, and replaceable.
6. NEMA ICS 6, Type 4, coil enclosure.
7. Normally closed.
8. Visual position indicator.

2.7 PRESSURE REGULATORS

A. General Requirements:

1. Single stage and suitable for natural-gas.
2. Steel jacket and corrosion-resistant components.
3. Elevation compensator.
4. End Connections: Threaded for regulators NPS 2 and smaller; flanged for regulators NPS 2-1/2 and larger.

B. Line Pressure Regulators: Comply with ANSI Z21.80.

1. Body and Diaphragm Case: Cast iron or die-cast aluminum.
2. Springs: Zinc-plated steel; interchangeable.
3. Diaphragm Plate: Zinc-plated steel.
4. Seat Disc: Nitrile rubber resistant to gas impurities, abrasion, and deformation at the valve port.
5. Orifice: Aluminum; interchangeable.
6. Seal Plug: Ultraviolet-stabilized, mineral-filled nylon.
7. Single-port, self-contained regulator with orifice no larger than required at maximum pressure inlet, and no pressure sensing piping external to the regulator.
8. Pressure regulator shall maintain discharge pressure setting downstream, and not exceed 1.5 times the design discharge pressure at shutoff.
9. Overpressure Protection Device: Factory mounted on pressure regulator.

10. Atmospheric Vent: Factory- or field-installed, stainless-steel screen in opening if not connected to vent piping.
11. Maximum Inlet Pressure: 2-psig.

C. Appliance Pressure Regulators: Comply with ANSI Z21.18.

1. Body and Diaphragm Case: Die-cast aluminum.
2. Springs: Zinc-plated steel; interchangeable.
3. Diaphragm Plate: Zinc-plated steel.
4. Seat Disc: Nitrile rubber.
5. Seal Plug: Ultraviolet-stabilized, mineral-filled nylon.
6. Factory-Applied Finish: Minimum three-layer polyester and polyurethane paint finish.
7. Regulator may include vent limiting device, instead of vent connection, if approved by authorities having jurisdiction.
8. Maximum Inlet Pressure: 2-psig

2.8 DIELECTRIC FITTINGS

A. Dielectric Unions:

1. Minimum Operating-Pressure Rating: 150-psig.
2. Combination fitting of copper alloy and ferrous materials.
3. Insulating materials suitable for natural-gas.
4. Combination fitting of copper alloy and ferrous materials with threaded, brazed-joint, plain, or welded end connections that match piping system materials.

B. Dielectric Flanges:

1. Minimum Operating-Pressure Rating: 150-psig
2. Combination fitting of copper alloy and ferrous materials.
3. Insulating materials suitable for natural-gas.
4. Combination fitting of copper alloy and ferrous materials with threaded, brazed-joint, plain, or welded end connections that match piping system materials.

2.9 SLEEVES

A. Steel Pipe Sleeves: ASTM A53, Type E, Grade B, Schedule 40, galvanized steel, plain ends.

2.10 MECHANICAL SLEEVE SEALS

A. Description: Modular sealing element unit, designed for field assembly, to fill annular space between pipe and sleeve.

1. Sealing Elements: EPDM interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe and sleeve.
2. Pressure Plates: Stainless steel.
3. Connecting Bolts and Nuts: Stainless steel of length required to secure pressure plates to sealing elements. Include one nut and bolt for each sealing element.

2.11 ESCUTCHEONS

A. General Requirements for Escutcheons: Manufactured wall and ceiling escutcheons and floor plates, with ID to fit around pipe or tube, and OD that completely covers opening.

- B. One-Piece, Deep-Pattern Escutcheons: Deep-drawn, box-shaped brass with polished chrome-plated finish.
- C. One-Piece, Stamped-Steel Escutcheons: With set screw and chrome-plated finish.
- D. Split-Plate, Stamped-Steel Escutcheons: With concealed hinge, set screw, and chrome-plated finish.
- E. One-Piece, Floor-Plate Escutcheons: Cast-iron floor plate.
- F. Split-Casting, Floor-Plate Escutcheons: Cast brass with concealed hinge and set screw.

2.12 LABELING AND IDENTIFYING

- A. Detectable Warning Tape: Acid- and alkali-resistant, PE film warning tape manufactured for marking and identifying underground utilities, a minimum of 6-inches wide and 4-mils thick, continuously inscribed with a description of utility and the word "CAUTION", and colored yellow.
- B. Safety Tracer Wire: Metallic core (copper minimum AWG 14) encased in a protective insulated jacket for corrosion protection, detectable by metal detector when tape is buried up to 36-inches deep; colored yellow.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine roughing-in for natural-gas piping system to verify actual locations of piping connections before equipment installation.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Close equipment shutoff valves before turning off natural-gas to premises or piping section.
- B. Inspect natural gas piping according to the International Fuel Gas Code to determine that natural gas utilization devices are turned off in piping section(s) affected.
- C. Comply with the International Fuel Gas Code requirements for prevention of accidental ignition.

3.3 OUTDOOR PIPING INSTALLATION

- A. Comply with the International Fuel Gas Code for installation and purging of natural-gas piping.
- B. Underground Installations: Install underground natural-gas piping according to ASTM D2774.
 - 1. Locate natural-gas piping between 18-and 30-inches below finished grade, except in areas subjected to vehicular traffic, where natural-gas piping shall be located between 24- and 30-inches below finished grade. In all cases, when the backfilled trench is to be compacted, the natural-gas piping shall be placed at a depth sufficient to ensure the piping will be covered by not less than 18-inches of backfill material. Comply with requirements in Division 31 Section "Earthwork" for excavating, trenching, and backfilling.

2. At building, PE piping shall extend to a point 12-inches above finished grade with shut-off valve on either end.
 3. Install a safety tracer wire along the entire length of underground piping. Extend safety tracer wire to a point 12-inches above finished grade on the outside gas riser.
 4. Provide a continuous detectable warning tape 12-inches below finished grade, directly above - the entire length of the underground piping. Install detectable warning tape 6-inches below subgrade under paved areas and slabs.
 5. Line the trench bottom with a 3-inches thick bedding layer of clean sand, taking care to ensure the pipe will be uniformly and continuously supported over its entire length. Place PE pipe. Cover pipe with clean sand, not less than 6-inches above top of pipe, prior to backfilling the trench. Cover the sand with not less than 12-inches of excavated soil, free of rocks, frozen clods, or debris greater than 3-inches diameter, prior to compacting the backfilled trench.
- C. Steel Piping with Protective Coating (Above Ground Only):
1. Apply joint cover kits to pipe after joining to cover, seal, and protect joints.
 2. Replace pipe having damaged protective coating with new pipe.
- D. Aboveground Natural-Gas Piping:
1. Pipe Diameter – 2-inches Diameter and Smaller: Steel pipe with malleable-iron fittings and threaded joints.
 2. Pipe Diameter – Larger than 2-inches Diameter: Steel pipe with wrought-steel fittings and welded joints.
 3. Concealed Piping: All concealed natural-gas piping shall have welded joints.
- E. Install fittings for changes in direction and branch connections.
1. Anodeless risers, pieces, or sweeps shall be used when rising up from soil and when penetrating basement walls. Anodeless components shall be secured to the foundation with proper brackets when used as risers.
- F. Above-ground, Exterior Wall Pipe Penetrations: Seal penetrations using sleeves and mechanical sleeve seals. Select sleeve size to allow for 1-inch annular clear space between pipe and sleeve for installing mechanical sleeve seals.
1. Install steel pipe for sleeves smaller than 6-inches diameter.
 2. Install cast-iron "wall pipes" for sleeves 6-inches [diameter](#) and larger.
- G. Exterior Wall Pipe Penetrations: Install cast-iron "wall pipes" for sleeves. Seal pipe penetrations using mechanical sleeve seals. Select sleeve size to allow for 1-inch annular clear space between pipe and sleeve for installing mechanical sleeve seals.
- H. Mechanical Sleeve Seal Installation: Select type and number of sealing elements required for pipe material and size. Position pipe in center of sleeve. Assemble mechanical sleeve seals and install in annular space between pipe and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.
- I. Exterior Rooftop Pipe Installation:
1. Rooftop piping should meet the requirements of above ground natural gas piping in the paragraphs above.

2. Support steel piping above roof with pillow block support system equal to Miro Industries model 3- R or 3-RAH. Support system shall consist of polycarbonate base mounted on elastomeric walk pad or support pad. Support system should include pipe rollers to allow movement.
 3. Exterior Roof Pipe Penetrations: Penetration of all piping through the roof shall be through pipe curb cap assembly with sealed neoprene boots equal to Pate Company model pca-2 or equal.
- J. Install pressure gage upstream and downstream from each service regulator. Pressure gages are specified in Division 23 Section "Meters and Gages for HVAC Piping."

3.4 INDOOR PIPING INSTALLATION

- A. Comply with the International Fuel Gas Code for installation and purging of natural-gas piping.
 - B. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems. Indicated locations and arrangements are used to size pipe and calculate friction loss, expansion, and other design considerations. Install piping as indicated unless deviations to layout are approved by Architect on Coordination Drawings.
 - C. Arrange for pipe spaces, chases, slots, sleeves, and openings in building structure during progress of construction, to allow for mechanical installations.
 - D. Install piping in concealed locations unless otherwise indicated and except in equipment rooms and service areas.
 - E. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building surfaces. Diagonal runs are prohibited unless specifically indicated otherwise.
 - F. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.
 - G. Locate valves for easy access.
 - H. Install piping free of sags and bends.
 - I. Install fittings for changes in direction and branch connections.
 - J. Install escutcheons at penetrations of interior walls, ceilings, and floors.
1. New Piping:
 - a. Piping with Fitting or Sleeve Protruding from Wall: One-piece, deep-pattern type.
 - b. Piping at Wall and Floor Penetrations in Finished Spaces: One-piece, stamped-steel type.
 - c. Piping at Ceiling Penetrations in Finished Spaces: One-piece, stamped-steel type or split-plate, stamped-steel type with concealed hinge and set screw.
 - d. Piping in Unfinished Service Spaces: One-piece, stamped-steel type with concealed hinge and set screw.
 - e. Piping in Equipment Rooms: One-piece, stamped-steel type with set screw.
 - f. Piping at Floor Penetrations in Equipment Rooms: One-piece, floor-plate type.

- K. Fire-Barrier Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with fire stop materials. Comply with requirements in Division 07 Section "Penetration Firestopping."
 - L. Verify final equipment locations for roughing-in.
 - M. Comply with requirements in Sections specifying gas-fired appliances and equipment for roughing-in requirements.
 - N. Drips and Sediment Traps: Provide drips at points where condensate may collect, including service-meter outlets. Locate where accessible to permit cleaning and emptying. Do not install where condensate is subject to freezing.
 - 1. Construct drips and sediment traps using tee fitting with bottom outlet plugged or capped. Use nipple a minimum length of 3 pipe diameters, but not less than 3 inches long and same size as connected pipe. Install with space below bottom of drip to remove plug or cap.
 - O. Extend relief vent connections for service regulators, line regulators, and overpressure protection devices to outdoors and terminate with weatherproof vent cap.
 - P. Concealed natural-gas piping shall be installed according to the International Fuel Code.
 - Q. Natural-gas piping valves and regulators shall not be installed in concealed spaces.
 - 1. In Floor Channels: Install natural-gas piping in floor channels. Channels must have cover and be open to space above cover for ventilation.
 - 2. Prohibited Locations:
 - a. Do not install natural-gas piping in or through circulating air ducts, chimneys or gas vents (flues), ventilating ducts, or dumbwaiter or elevator hoistways.
 - b. Do not install natural-gas piping in solid walls or partitions.
 - c. Do not install natural-gas piping underground below building slabs.
 - R. Use eccentric reducer fittings to make reductions in pipe sizes. Install fittings with level side down.
 - S. Connect branch piping from top or side of horizontal piping.
 - T. Provide unions in pipes NPS 2 and smaller, adjacent to each valve, at final connection to each piece of equipment. Unions are not required at flanged connections.
 - U. Do not use natural-gas piping as grounding electrode.
 - V. Provide strainer on inlet of each line-pressure regulator and automatic or electrically operated valve.
 - W. Provide pressure gage upstream and downstream from each pipe regulator. Pressure gages are specified in Division 23 Section "Meters and Gages for HVAC Piping."
- 3.5 VALVE INSTALLATION
- A. Provide manual gas shutoff valve for each gas appliance ahead of appliance flexible connector. Valve shall be located in same room as equipment.

- B. Provide outdoor underground valves with valve boxes.
- C. Provide a shutoff valve, acceptable to the authority having jurisdiction, on the outside riser.
- D. Install regulators and overpressure protection devices with maintenance access space adequate for servicing and testing.
- E. Provide anode for metallic valves in underground PE piping.
- F. Provide pressure reducing valve at each equipment and appliance with relief piped outdoors.
- G. As a minimum install automatic gas shut off valve with alarm or remote operation for kitchen equipment motor hood with fire suppression system.

3.6 THERMAL MASS NATURAL GAS FLOW METER

- A. Type: Insertion Flow Meter, Outdoor Conditions.
- B. Flow Meter shall be complete with all installation hardware necessary to enable insertion and removal of the meter without system interruption.
- C. Fitment: Meter shall be installed in accordance with the manufacturer's installation guide including meter orientation and straight pipe recommendations. Applicable pipe size shall be from 0.5 to 8 inches.
- D. Maximum Operating Pressure: The flow meter shall be able to operate at up to 60 psig (standard. High pressure to 150 psig).
- E. Materials: Wetted metal components shall be 316 Stainless Steel.
- F. Electrical Connections: Enclosed terminal blocks, cable access through two ¾" NPT conduit fittings
- G. Flow Accuracy (natural gas / propane gas):
 - 1. ± 1.0% of reading from 500 – 7,000 SFPM.
 - 2. ± 2.0% of reading from 100 – 500 SFPM.
- H. Temperature Accuracy: + 1.0 °F over the range of -40 °F to 250 °F.
- I. Ambient Temperature Range: -40 °F to 158 °F.
- J. Output Signals: One (1) 4-20 mA analog output. Include Modbus RTU connectivity.
- K. Installation: Coordinate manufacturer's recommended installation requirements with piping contractor to ensure proper installation.
- L. Power: 12-28VDC, 6W minimum power.
- M. Contractor shall provide written coordination on capacity, performance, and location with trade contractor during the submittal phase.
- N. Acceptable Manufacturer: Onicon F-5500 Series or Sierra Instruments QuadraTherm-640i Series. No exception.

3.7 PIPING JOINT CONSTRUCTION

- A. Ream ends of pipes and remove burrs.
- B. Remove scale, slag, dirt, and debris from inside and outside of pipes and fittings before assembly.
- C. Threaded Joints:
 - 1. Thread pipe with tapered pipe threads according to ASME B1.20.1.
 - 2. Cut threads full and clean using sharp dies.
 - 3. Ream threaded pipe ends to remove burrs and restore full inside diameter of pipe.
 - 4. Apply appropriate tape or thread compound to external pipe threads unless dry seal threading is specified.
- D. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds. Welded Joints:
 - 1. Construct joints according to AWS D10.12/D10.12M, using qualified processes and welding operators.
 - 2. Bevel plain ends of steel pipe.
 - 3. Patch factory-applied protective coating as recommended by manufacturer at field welds and where damage to coating occurs during construction.
- E. Flanged Joints: Provide gasket material, size, type, and thickness appropriate for natural-gas service. Install gasket concentrically positioned.
- F. PE Piping Heat-Fusion Joints: Clean and dry joining surfaces by wiping with clean cloth or paper towels. Join according to ASTM D2657.
 - 1. Plain-End Pipe and Fittings: Use butt fusion.
 - 2. Plain-End Pipe and Socket Fittings: Use socket fusion.

3.8 HANGER AND SUPPORT INSTALLATION

- A. Comply with requirements for pipe hangers and supports specified in Division 23 Section "Hangers and Supports for Piping and Equipment."
- B. Provide hangers for horizontal steel piping with the following maximum spacing and minimum rod sizes:
 - 1. NPS 1 and Smaller: Maximum span, 96 inches; minimum rod size, 3/8 inch.
 - 2. NPS 1-1/2: Maximum span, 108 inches; minimum rod size, 3/8 inch.
 - 3. NPS 1-1/2 and NPS 2: Maximum span, 108 inches; minimum rod size, 3/8 inch.
 - 4. NPS 2-1/2 to NPS 3-1/2: Maximum span, 10 feet; minimum rod size, 1/2 inch.
 - 5. NPS 4 and Larger: Maximum span, 10 feet; minimum rod size, 5/8 inch.

3.9 CONNECTIONS

- A. Install natural-gas piping electrically continuous, and bonded to gas appliance equipment grounding conductor of the circuit powering the appliance according to NFPA 70.
- B. Install piping adjacent to appliances to allow service and maintenance of appliances.

- C. Connect piping to appliances using manual gas shutoff valves and unions. Install valve within 72 inches of each gas-fired appliance and equipment. Install union between valve and appliances or equipment.
- D. Sediment Traps: Provide tee fitting with capped nipple in bottom to form drip, as close as practical to inlet of each appliance.
- E. Connect wiring according to Division 26 Section "Conductors and Cables".

3.10 LABELING AND IDENTIFYING

- A. Comply with requirements in Division 23 Section "Identification for HVAC" for piping and valve identification.
- B. Below Grade: Provide detectable warning tape continuously and directly above gas piping, 12-inches below finish grade; 6-inches below sub-grade under pavements and concrete slabs.
- C. Labeling and identification materials are specified in Division 22 "Identification for Plumbing Piping and Equipment".

3.11 FIELD QUALITY CONTROL

- A. Perform tests and inspections.
- B. Tests and Inspections:
 - 1. Test, inspect, and purge natural-gas according to the International Fuel Gas Code and authorities having jurisdiction.
- C. Natural-gas piping will be considered defective if it does not pass tests and inspections.
- D. Prepare test and inspection reports.

3.12 OUTDOOR PIPING SCHEDULE

- A. Underground Natural-Gas Piping: Polyethylene (PE) pipe and fittings joined by heat fusion, or mechanical coupling. Service-line riser with tracer wire shall be terminated in an accessible location.
- B. Aboveground Natural-Gas Piping: Steel pipe, with joints as indicated in Article, "Outdoor Piping Installation."

3.13 INDOOR PIPING SCHEDULE FOR SYSTEM PRESSURES LESS THAN 0.5 PSIG

- A. Aboveground, branch piping NPS 1 and smaller shall be the following:
 - 1. Steel pipe with malleable-iron fittings and threaded joints.
- B. Aboveground, distribution piping shall be the following:
 - 1. Steel pipe with malleable-iron fittings and threaded joints NPS 2 and smaller.
 - 2. Steel pipe with wrought-steel fittings and welded joints NPS 2-1/2 and larger.
 - 3. All concealed natural-gas piping shall be welded.
 - 4. Underground, below building, piping shall not be allowed.

3.14 INDOOR PIPING SCHEDULE FOR SYSTEM PRESSURES MORE THAN 0.5 PSIG ND LESS THAN 2 PSIG

A. Aboveground piping shall be the following:

1. Steel pipe with malleable-iron fittings and threaded joints smaller or equal to NPS 2.
2. Steel pipe with steel welding fittings and welded joints NPS 2-1/2 and larger.
3. All concealed natural-gas piping shall be in a double wall configuration and vented to the outdoors.

3.15 ABOVEGROUND MANUAL GAS SHUTOFF VALVE SCHEDULE

A. Valves for pipe sizes NPS 2 and smaller at service meter shall be the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.

B. Valves for pipe sizes NPS 2-1/2 and larger at service meter shall be the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.
3. Cast-iron, non-lubricated plug valve.

C. Distribution piping valves for pipe sizes NPS 2 and smaller shall be the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.

D. Distribution piping valves for pipe sizes NPS 2-1/2 and larger shall be the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.
3. Cast-iron, lubricated plug valve.

E. Valves in branch piping for single appliance shall be the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.

END OF SECTION 231123

SECTION 232300 - REFRIGERANT PIPING**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes refrigerant piping used for air-conditioning applications.

1.2 SUBMITTALS

- A. Product Data: For each type of valve and refrigerant piping specialty indicated. Include pressure drop, based on manufacturer's test data, for the following:
 - 1. Thermostatic expansion valves.
 - 2. Solenoid valves.
 - 3. Hot-gas bypass valves.
 - 4. Filter dryers.
 - 5. Strainers.
 - 6. Pressure-regulating valves.
- B. Shop Drawings: Show layout of refrigerant piping and specialties, including pipe, tube, and fitting sizes, flow capacities, valve arrangements and locations, slopes of horizontal runs, oil traps, double risers, wall and floor penetrations, and equipment connection details. Show interface and spatial relationships between piping and equipment.
 - 1. Shop Drawing Scale: 1/4 inch equals 1 foot.
- C. Welding certificates.
- D. Field quality-control test reports.
 - 1. Submit written reports documenting the activities required to be performed in PART 3. These reports are to be submitted two weeks after the startup is completed.
- E. Operation and Maintenance Data: For refrigerant valves and piping specialties to include in maintenance manuals.

1.3 QUALITY ASSURANCE

- A. Welding: Qualify procedures and personnel according to ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."
- B. Comply with ASHRAE 15, "Safety Code for Refrigeration Systems."
- C. Comply with ASME B31.5, "Refrigeration Piping and Heat Transfer Components."

1.4 DELIVERY STORAGE AND HANDLING

- A. Store piping in a clean and protected area with end caps in place to ensure that piping interior and exterior are clean when installed.

1.5 COORDINATION

- A. Coordinate size and location of roof curbs, equipment supports, and roof penetrations. These items are specified in Division 07 Section "Roof Accessories."

1.6 PERFORMANCE REQUIREMENTS

- A. Line Test Pressure for Refrigerant R-410A:
 - 1. Suction Lines for Air-Conditioning Applications: 300 psig.
 - 2. Suction Lines for Heat-Pump Applications: 535 psig.
 - 3. Hot-Gas and Liquid Lines: 535 psig.

PART 2 - PRODUCTS

2.1 MANUFACTURERS:

- A. Subject to compliance with requirements, provide products by one of the following manufacturers:
 - 1. Refrigerants:
 - a. Atofina Chemicals, Inc.
 - b. DuPont Company; Fluorochemicals Div.
 - c. Honeywell, Inc.; Genetron Refrigerants.
 - d. INEOS Fluor Americas LLC.

2.2 COPPER TUBE AND FITTINGS

- A. Copper Tube: ASTM B 280, Type ACR or type K complying with ASTM B88 or ASTM B819.
- B. Wrought-Copper Fittings: ASME B16.22.
- C. Wrought-Copper Unions: ASME B16.22.
- D. Brazing Filler Metals: AWS A5.8.
- E. Flexible Connectors:
 - 1. Body: Tin-bronze bellows with woven, flexible, tinned-bronze-wire-reinforced protective jacket.
 - 2. End Connections: Socket ends.
 - 3. Offset Performance: Capable of minimum 3/4-inch (20-mm) misalignment in minimum 7-inch- (180-mm-) long assembly.
 - 4. Pressure Rating: Factory test at minimum 500 psig (3450 kPa).
 - 5. Maximum Operating Temperature: 250 deg F (121 deg C).

2.3 VALVES AND SPECIALTIES

- A. Diaphragm Packless Valves:
 - 1. Body and Bonnet: Forged brass or cast bronze; globe design with straight-through or angle pattern.

2. Diaphragm: Phosphor bronze and stainless steel with stainless-steel spring.
3. Operator: Rising stem and hand wheel.
4. Seat: Nylon.
5. End Connections: Socket, union, or flanged.
6. Working Pressure Rating: 500 psig.
7. Maximum Operating Temperature: 275 deg F.

B. Packed-Angle Valves:

1. Body and Bonnet: Forged brass or cast bronze.
2. Packing: Molded stem, back seating, and replaceable under pressure.
3. Operator: Rising stem.
4. Seat: Nonrotating, self-aligning polytetrafluoroethylene.
5. Seal Cap: Forged-brass or valox hex cap.
6. End Connections: Socket, union, threaded, or flanged.
7. Working Pressure Rating: 500 psig.
8. Maximum Operating Temperature: 275 deg F.

C. Check Valves:

1. Body: Ductile iron, forged brass, or cast bronze; globe pattern.
2. Bonnet: Bolted ductile iron, forged brass, or cast bronze; or brass hex plug.
3. Piston: Removable polytetrafluoroethylene seat.
4. Closing Spring: Stainless steel.
5. End Connections: Socket, union, threaded, or flanged.
6. Maximum Opening Pressure: 0.50 psig.
7. Working Pressure Rating: 500 psig.
8. Maximum Operating Temperature: 275 deg F.

D. Service Valves:

1. Body: Forged brass with brass cap including key end to remove core.
2. Core: Removable ball-type check valve with stainless-steel spring.
3. Seat: Polytetrafluoroethylene.
4. End Connections: Copper spring.
5. Working Pressure Rating: 500 psig.

E. Solenoid Valves: Comply with ARI 760 and UL 429; listed and labeled by an NRTL.

1. Body and Bonnet: Plated steel.
2. Solenoid Tube, Plunger, Closing Spring, and Seat Orifice: Stainless steel.
3. Seat: Polytetrafluoroethylene.
4. End Connections: Threaded.
5. Electrical: Molded, watertight coil in NEMA 250 enclosure of type required by location with 1/2-inch (16-GRC) conduit adapter, and 115-V ac coil.
6. Working Pressure Rating: 400 psig.
7. Maximum Operating Temperature: 240 deg F.

F. Safety Relief Valves: Comply with ASME Boiler and Pressure Vessel Code; listed and labeled by an NRTL.

1. Body and Bonnet: Ductile iron and steel, with neoprene O-ring seal.
2. Piston, Closing Spring, and Seat Insert: Stainless steel.
3. Seat Disc: Polytetrafluoroethylene.

4. End Connections: Threaded.
 5. Working Pressure Rating: 400 psig.
 6. Maximum Operating Temperature: 240 deg F.
- G. Thermostatic Expansion Valves: Comply with ARI 750.
1. Body, Bonnet, and Seal Cap: Forged brass or steel.
 2. Diaphragm, Piston, Closing Spring, and Seat Insert: Stainless steel.
 3. Packing and Gaskets: Non-asbestos
 4. Capillary and Bulb: Copper tubing filled with refrigerant charge.
 5. Suction Temperature: See Equipment Schedules
 6. Superheat: Adjustable.
 7. Reverse-flow option (for heat pump applications).
 8. End Connections; Socket, flare, or threaded union.
 9. Working Pressure Rating: 700 psig.
- H. Hot-Gas Bypass Valves: Comply with UL 429, listed and labeled by NRTL.
1. Body, Bonnet, and Seal Cap: Ductile iron or steel.
 2. Diaphragm, Piston, Closing Spring, and Seat Insert: Stainless steel.
 3. Packing and Gaskets: Non-asbestos.
 4. Solenoid Tube, Plunger, Closing Spring, and Seat Orifice: Stainless steel.
 5. Seat: Polytetrafluoroethylene.
 6. Equalizer: Internal or External.
 7. Electrical: Molded, watertight coil in NEMA 250 enclosure of type required by location with 1/2-inch conduit adapter, and [24] [115] [208]-V ac coil.
 8. End connections: Socket.
 9. Set Pressure: As required.
 10. Throttling Range: Maximum 5 psig.
 11. Working Pressure Range: 500 psig.
 12. Maximum Operating temperature: 240 deg F.
- I. Straight-Type Strainers:
1. Body: Welded steel with corrosion-resistant coating.
 2. Screen: 100-mesh stainless steel.
 3. End Connections: Socket or flare.
 4. Working Pressure Rating: 500 psig.
 5. Maximum Operating Temperature: 275 deg F.
- J. Angle-Type Strainers:
1. Body: Forged brass or cast bronze.
 2. Drain Plug: Brass hex plug.
 3. Screen: 100-mesh monel.
 4. End Connections: Socket or flare.
 5. Working Pressure Rating: 500 psig.
 6. Maximum Operating Temperature: 275 deg F.
- K. Moisture/Liquid Indicators:
1. Body: Forged brass.
 2. Window: Replaceable, clear, fused glass window with indicating element protected by filter screen.

3. Indicator: Color coded to show moisture content in ppm.
4. Minimum Moisture Indicator Sensitivity: Indicate moisture above 60 ppm.
5. End Connections: Socket or flare.
6. Working Pressure Rating: 500 psig.
7. Maximum Operating Temperature: 240 deg F.

L. Replaceable-Core Filter Dryers: Comply with ARI 730.

1. Body and Cover: Painted-steel shell with ductile-iron cover, stainless-steel screws, and neoprene gaskets.
2. Filter Media: 10 micron, pleated with integral end rings; stainless-steel support.
3. Desiccant Media: Activated charcoal.
4. End Connections: Socket.
5. Access Ports: NPS 1/4 connections at entering and leaving sides for pressure differential measurement.
6. Maximum Pressure Loss: 2 psig.
7. Rated Flow: See equipment sizing
8. Working Pressure Rating: 500 psig.
9. Maximum Operating Temperature: 240 deg F.

M. Permanent Filter Dryers: Comply with ARI 730.

1. Body and Cover: Painted-steel shell.
2. Filter Media: 10 micron, pleated with integral end rings; stainless-steel support.
3. Desiccant Media: Activated charcoal.
4. End Connections: Socket.
5. Access Ports: NPS 1/4 connections at entering and leaving sides for pressure differential measurement.
6. Maximum Pressure Loss: 2 psig.
7. Rated Flow: See equipment sizing
8. Working Pressure Rating: 500 psig.
9. Maximum Operating Temperature: 240 deg F.

N. Mufflers:

1. Body: Welded steel with corrosion-resistant coating.
2. End Connections: Socket or flare.
3. Working Pressure Rating: 500 psig.
4. Maximum Operating Temperature: 275 deg F.

O. Receivers: Comply with ARI 495.

1. Comply with ASME Boiler and Pressure Vessel Code; listed and labeled by an NRTL.
2. Comply with UL 207; listed and labeled by an NRTL.
3. Body: Welded steel with corrosion-resistant coating.
4. Tappings: Inlet, outlet, liquid level indicator, and safety relief valve.
5. End Connections: Socket or threaded.
6. Working Pressure Rating: 500 psig.
7. Maximum Operating Temperature: 275 deg F.

P. Liquid Accumulators: Comply with ARI 495.

1. Body: Welded steel with corrosion-resistant coating.
2. End Connections: Socket or threaded.

3. Working Pressure Rating: 500 psig.
4. Maximum Operating Temperature: 275 deg F.

2.4 REFRIGERANTS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 1. Atofina Chemicals, Inc.
 2. DuPont Company; Fluorochemicals Div.
 3. Honeywell, Inc.; Genetron Refrigerants.
 4. INEOS Fluor Americas LLC.
- B. ASHRAE 34, R-410A: Pentafluoroethane/Difluoromethane.

PART 3 - EXECUTION

3.1 REFRIGERANT PIPE SIZING:

- A. All refrigerant pipe sizing shall be the responsibility of the Contractor in accordance with the equipment manufacturer's recommendations.
- B. Pipe sizing shall be in accordance with the recommendations in the 2010 ASHRAE Handbook – Refrigeration, Chapter 2 – System Practices for Halocarbon Refrigerants.
- C. Refrigerant piping indicated on Drawings is schematic only. Size piping and design actual piping layout, including oil traps, double risers, specialties, and pipe and tube sizes to accommodate, as a minimum, equipment provided, elevation difference between compressor and evaporator, and length of piping to ensure proper operation and compliance with warranties of connected equipment.

3.2 PIPING APPLICATIONS FOR REFRIGERANT R-410A

- A. Suction Lines NPS 3-1/2 and Smaller for Conventional Air-Conditioning Applications: Copper, Type ACR, drawn-temper tubing and wrought-copper fittings with brazed or soldered joints.
- B. Safety-Relief-Valve Discharge Piping: Copper, Type ACR, annealed- or drawn-temper tubing and wrought-copper fittings with brazed joints.

3.3 VALVE AND SPECIALTY APPLICATIONS

- A. Install packed-angle valves in suction and discharge lines of compressor.
- B. Install service valves for gage taps at inlet and outlet of hot-gas bypass valves and strainers if they are not an integral part of valves and strainers.
- C. Install a check valve at the compressor discharge and a liquid accumulator at the compressor suction connection.
- D. Except as otherwise indicated, install packed-angle valves on inlet and outlet side of filter dryers.
- E. Install a full-sized, three-valve bypass around filter dryers.

- F. Install solenoid valves upstream from each expansion valve and hot-gas bypass valve. Install solenoid valves in horizontal lines with coil at top.
- G. Install thermostatic expansion valves as close as possible to distributors on evaporators.
 - 1. Install valve so diaphragm case is warmer than bulb.
 - 2. Secure bulb to clean, straight, horizontal section of suction line using two bulb straps. Do not mount bulb in a trap or at bottom of the line.
 - 3. If external equalizer lines are required, make connection where it will reflect suction-line pressure at bulb location.
- H. Install safety relief valves where required by ASME Boiler and Pressure Vessel Code. Pipe safety-relief-valve discharge line to outside according to ASHRAE 15.
- I. Install moisture/liquid indicators in liquid line at the inlet of the thermostatic expansion valve or at the inlet of the evaporator coil capillary tube.
- J. Install strainers upstream from and adjacent to the following unless they are furnished as an integral assembly for device being protected:
 - 1. Solenoid valves.
 - 2. Thermostatic expansion valves.
 - 3. Compressor.
- K. Install filter dryers in liquid line between compressor and thermostatic expansion valve, and in the suction line at the compressor.
- L. Install receivers sized to accommodate pump-down charge.
- M. Install flexible connectors at compressors.

3.4 PIPING INSTALLATION

- A. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems; indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated unless deviations to layout are approved on Shop Drawings.
- B. Install refrigerant piping according to ASHRAE 15.
- C. Install piping in concealed locations unless otherwise indicated and except in equipment rooms and service areas.
- D. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.
- E. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.
- F. Install piping adjacent to machines to allow service and maintenance.
- G. Install piping free of sags and bends.
- H. Install fittings for changes in direction and branch connections.

- I. Select system components with pressure rating equal to or greater than system operating pressure.
 - J. Refer to Division 23 Sections "Building Automation System (BAS)" and "Building Automation System (BAS) - Sequence of Operation" for solenoid valve controllers, control wiring, and sequence of operation.
 - K. Install piping as short and direct as possible, with a minimum number of joints, elbows, and fittings.
 - L. Arrange piping to allow inspection and service of refrigeration equipment. Install valves and specialties in accessible locations to allow for service and inspection. Install access doors or panels as specified in Division 08 Section "Access Doors and Frames" if valves or equipment requiring maintenance is concealed behind finished surfaces.
 - M. Install refrigerant piping in protective conduit where installed belowground.
 - N. Install refrigerant piping in rigid or flexible conduit in locations where exposed to mechanical injury.
 - O. Slope refrigerant piping as follows:
 - 1. Install horizontal suction lines with a uniform slope downward to compressor.
 - 2. Install traps and double risers to entrain oil in vertical runs.
 - 3. Liquid lines may be installed level.
 - P. When brazing , remove solenoid-valve coils and sight glasses; also remove valve stems, seats, and packing, and accessible internal parts of refrigerant specialties. Do not apply heat near expansion-valve bulb.
 - Q. Install pipe sleeves at penetrations in exterior walls and floor assemblies.
 - R. Seal penetrations through fire and smoke barriers according to Division 07 Section "Penetration Firestopping."
 - S. Install piping with adequate clearance between pipe and adjacent walls and hangers or between pipes for insulation installation.
 - T. Install sleeves through floors, walls, or ceilings, sized to permit installation of full-thickness insulation.
 - U. Seal pipe penetrations through exterior walls according to Division 07 Section "Joint Sealants" for materials and methods.
 - V. Identify refrigerant piping and valves according to Division 23 Section "Identification for HVAC."
- 3.5 PIPE JOINT CONSTRUCTION
- A. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
 - B. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.

- C. Brazed Joints: Construct joints according to AWS's "Brazing Handbook," Chapter "Pipe and Tube."
 - 1. Use Type BcuP, copper-phosphorus alloy for joining copper socket fittings with copper pipe.
 - 2. Use Type BAg, cadmium-free silver alloy for joining copper with bronze or steel.
- D. Threaded Joints: Thread steel pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
 - 1. Apply appropriate tape or thread compound to external pipe threads unless dry-seal threading is specified.
 - 2. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds.
- E. Steel pipe can be threaded, but threaded joints must be seal brazed or seal welded.
- F. Welded Joints: Construct joints according to AWS D10.12/D10.12M.
- G. Flanged Joints: Select appropriate gasket material, size, type, and thickness for service application. Install gasket concentrically positioned. Use suitable lubricants on bolt threads.

3.6 HANGERS AND SUPPORTS

- A. Hanger, support, and anchor products are specified in Division 23 Section "Hangers and Supports for Piping and Equipment."
- B. Install the following pipe attachments:
 - 1. Adjustable steel clevis hangers for individual horizontal runs less than 20 feet (6 m) long.
 - 2. Roller hangers and spring hangers for individual horizontal runs 20 feet (6 m) or longer.
 - 3. Pipe Roller: MSS SP-58, Type 44 for multiple horizontal piping 20 feet (6 m) or longer, supported on a trapeze.
 - 4. Spring hangers to support vertical runs.
 - 5. Copper-clad hangers and supports for hangers and supports in direct contact with copper pipe.
- C. Install hangers for copper tubing with the following maximum spacing and minimum rod sizes:
 - 1. NPS 1/2: Maximum span, 60 inches; minimum rod size, 1/4 inch.
 - 2. NPS 5/8: Maximum span, 60 inches; minimum rod size, 1/4 inch.
 - 3. NPS 1: Maximum span, 72 inches; minimum rod size, 1/4 inch.
 - 4. NPS 1-1/4: Maximum span, 96 inches; minimum rod size, 3/8 inch.
 - 5. NPS 1-1/2: Maximum span, 96 inches; minimum rod size, 3/8 inch.
 - 6. NPS 2: Maximum span, 96 inches; minimum rod size, 3/8 inch.
- D. Support multifloor vertical runs at least at each floor.

3.7 FIELD QUALITY CONTROL

- A. Perform tests and inspections and prepare test reports.
- B. Tests and Inspections:

1. Comply with ASME B31.5, Chapter VI.
2. Test refrigerant piping, specialties, and receivers. Isolate compressor, condenser, evaporator, and safety devices from test pressure if they are not rated above the test pressure.
3. Test high- and low-pressure side piping of each system separately at not less than the pressures indicated in Part 1 "Performance Requirements" Article.
 - a. Fill system with nitrogen to the required test pressure.
 - b. System shall maintain test pressure at the manifold gage throughout duration of test.
 - c. Test joints and fittings with electronic leak detector or by brushing a small amount of soap and glycerin solution over joints.
 - d. Remake leaking joints using new materials, and retest until satisfactory results are achieved.

3.8 SYSTEM CHARGING

A. Charge system using the following procedures:

1. Install core in filter dryers after leak test but before evacuation.
2. Evacuate entire refrigerant system with a vacuum pump to 500 micrometers (67 Pa). If vacuum holds for 12 hours, system is ready for charging.
3. Break vacuum with refrigerant gas, allowing pressure to build up to 2 psig (14 kPa).
4. Charge system with a new filter-dryer core in charging line.

3.9 ADJUSTING

- A. Adjust thermostatic expansion valve to obtain proper evaporator superheat.
- B. Adjust high- and low-pressure switch settings to avoid short cycling in response to fluctuating suction pressure.
- C. Adjust set-point temperature of air-conditioning or chilled-water controllers to the system design temperature.
- D. Perform the following adjustments before operating the refrigeration system, according to manufacturer's written instructions:
 1. Open shutoff valves in condenser water circuit.
 2. Verify that compressor oil level is correct.
 3. Open compressor suction and discharge valves.
 4. Open refrigerant valves except bypass valves that are used for other purposes.
 5. Check open compressor-motor alignment and verify lubrication for motors and bearings.
- E. Replace core of replaceable filter dryer after system has been adjusted and after design flow rates and pressures are established.

END OF SECTION 232300

SECTION 233113 - METAL DUCTS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes metal ducts for supply, return, outside, transfer and exhaust air-distribution systems in pressure classes from minus four (4)- inch wg to plus ten (10)-inch wg. Metal ducts include the following:
 - 1. Rectangular ducts and fittings.
 - 2. Single-wall, round, and flat-oval spiral-seam ducts and formed fittings.
 - 3. Duct Liner
- B. Sealing Requirements Definitions: For the purposes of duct systems sealing requirements specified in this Section, the following definitions apply:
 - 1. Seams: A seam is defined as joining of two longitudinally (in the direction of airflow) oriented edges of duct surface material occurring between two joints. All other duct surface connections made on the perimeter are deemed to be joints.
 - 2. Joints: Joints include girth joints; branch and sub-branch intersections; so-called duct collar tap-ins; fitting subsections; louver and air terminal connections to ducts; access door and access panel frames and jambs; duct, plenum, and casing abutments to building structures.

1.2 PERFORMANCE REQUIREMENTS

- A. Delegated Hanger and Support Design: Hangers and supports shall comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible."

1.3 SUBMITTALS

- A. Shop Drawings: Drawn at a scale of not less than 1/4" = 1'-0". Show fabrication and installation details for metal ducts.
 - 1. Fabrication, assembly, and installation, including plans, elevations, sections, components, and attachments to other work.
 - 2. Duct layout indicating sizes and pressure classes.
 - 3. Elevations of top and bottom of ducts.
 - 4. Dimensions of main duct runs from building grid lines.
 - 5. Fittings.
 - 6. Reinforcement and spacing.
 - 7. Seam and joint construction.
 - 8. Penetrations through fire-rated and other partitions.
 - 9. Equipment installation based on equipment being used on Project.
 - 10. Duct accessories, including access doors and panels.
 - 11. Hangers and supports, including methods for duct and building attachment and vibration isolation.
- B. Delegated-Design Submittal:
 - 1. Spacing of hangers and supports.

2. Design calculations: Calculations, including analysis data, signed and sealed by the qualified professional engineer responsible for their preparation for selecting hangers and supports.
 - C. Coordination Drawings: Reflected ceiling plans and BIM model, drawn to scale, on which the following items are shown and coordinated with each other, based on input from installers of the items involved:
 1. Ceiling suspension assembly members.
 2. Other systems installed in same space as ducts.
 3. Ceiling- and wall-mounting access doors and panels required to provide access to dampers and other operating devices.
 4. Ceiling-mounting items, including lighting fixtures, diffusers, grilles, speakers, sprinklers, access panels, and special moldings.
 - D. Field quality-control test reports.
- 1.4 QUALITY ASSURANCE
- A. Welding: Qualify procedures and personnel according to AWS D9.1, "Sheet Metal Welding Code," for duct joint and seam welding.
 - B. NFPA Compliance:
 1. NFPA 90A, "Installation of Air Conditioning and Ventilating Systems."
 2. NFPA 90B, "Installation of Warm Air Heating and Air Conditioning Systems."
 - C. Comply with NFPA 96, "Ventilation Control and Fire Protection of Commercial Cooking Operations," Ch. 3, "Duct System," for range hood ducts, unless otherwise indicated.
- 1.5 DELIVERY, STORAGE AND HANDLING
- A. All materials shall be stored in a designated area and protected from inclement weather.
 - B. All materials shall be secured so as not to be a hazard during the construction process.
 - C. Store ductwork with tight-fitting seals on open ends to ensure ductwork is free of all dirt, debris and moisture during the installation process.

PART 2 - PRODUCTS

2.1 SHEET METAL MATERIALS

- A. Comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for acceptable materials, material thicknesses, and duct construction methods, unless otherwise indicated. Sheet metal materials shall be free of pitting, seam marks, roller marks, stains, discolorations, and other imperfections.
- B. Galvanized Sheet Steel: Lock-forming quality; complying with ASTM A 653/A 653M and having G90 coating designation; ducts shall have mill-phosphatized finish for surfaces exposed to view.
- C. Carbon-Steel Sheets: ASTM A 366/A 366M, cold-rolled sheets; commercial quality; with oiled, matte finish for exposed ducts.

- D. Stainless Steel: ASTM A 480/A 480M, Type 304 & 316.
- E. Aluminum Sheets: ASTM B 209, alloy 3003, temper H14; with mill finish for concealed ducts and standard, 1-side bright finish for exposed ducts.
- F. Reinforcement Shapes and Plates: Galvanized-steel reinforcement where installed on galvanized sheet metal ducts.
- G. Tie Rods: Galvanized steel, 1/4-inch minimum diameter for lengths 36 inches or less; 3/8-inch minimum diameter for lengths longer than 36 inches.
- H. Insulated Flexible Ducts: Flexible ducts wrapped with flexible glass fiber insulation, enclosed by seamless aluminum pigmented plastic vapor barrier jacket; maximum 0.23 K value at 75 degrees F.

2.2 SEALANT MATERIALS

- A. Joint and Seam Sealants, General: The term "sealant" is not limited to materials of adhesive or mastic nature but includes combinations of open-weave fabric strips and mastics.
- B. Water-Based Joint and Seam Sealant: Flexible, adhesive sealant, resistant to UV light when cured, UL 723 listed, and complying with NFPA requirements for Class 1 ducts.
- C. Flanged Joint Mastic: One-part, acid-curing, silicone, elastomeric joint sealant complying with ASTM C 920, Type S, Grade NS, Class 25, Use O.
- D. Flange Gaskets: Butyl rubber or EPDM polymer with polyisobutylene plasticizer.

2.3 HANGERS AND SUPPORTS

- A. Building Attachments: Concrete inserts, powder-actuated fasteners, or structural-steel fasteners appropriate for construction materials to which hangers are being attached.
 - 1. Use powder-actuated concrete fasteners for standard-weight aggregate concretes or for slabs more than 4 inches thick.
 - 2. Exception: Do not use powder-actuated concrete fasteners for lightweight-aggregate concretes or for slabs less than 4 inches thick.
- B. Hanger Materials: Galvanized sheet steel or threaded steel rod.
 - 1. Hangers Installed in Corrosive Atmospheres: All-thread rods used in pool areas, pool equipment rooms, and pool supporting spaces shall be aluminum if the ducts are aluminum and stainless steel if the ducts are stainless steel.
 - 2. Strap and Rod Sizes: Comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for steel sheet width and thickness and for steel rod diameters.
 - 3. Galvanized-steel straps attached to aluminum ducts shall have contact surfaces painted with zinc-chromate primer.
- C. Duct Attachments: Sheet metal screws, blind rivets, or self-tapping metal screws; compatible with duct materials.
- D. Trapeze and Riser Supports: Steel shapes complying with ASTM A 36/A 36M.
 - 1. Supports for Galvanized-Steel Ducts: Galvanized-steel shapes and plates.

2. Supports for Stainless-Steel Ducts: Stainless-steel support materials.
3. Supports for Aluminum Ducts: Aluminum support materials..

2.4 RECTANGULAR DUCT FABRICATION

- A. Fabricate ducts, elbows, transitions, offsets, branch connections, and other construction according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" and complying with requirements for metal thickness, reinforcing types and intervals, tie-rod applications, and joint types and intervals.
 1. Lengths: Fabricate rectangular ducts in lengths appropriate to reinforcement and rigidity class required for pressure class.
 2. Deflection: Duct systems shall not exceed deflection limits according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible."
- B. Transverse Joints: Prefabricated slide-on joints and components constructed using manufacturer's guidelines for material thickness, reinforcement size and spacing, and joint reinforcement.
- C. Formed-On Flanges: Construct according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," Figure 1-4, using corner, bolt, cleat, and gasket details.
 1. Duct Size: Maximum 30 inches wide and up to 2-inch wg pressure class.
 2. Longitudinal Seams: Pittsburgh lock sealed with noncuring polymer sealant.
- D. Cross Breaking or Cross Beading: Cross break or cross bead duct sides 19 inches and larger and 0.0359 inch thick or less, with more than 10 sq. ft. of nonbraced panel area unless ducts are lined.

2.5 ROUND AND FLAT-OVAL DUCT AND FITTING FABRICATION

- A. Manufacturers: applicable to factory-fabricated duct and fittings:
- B. Subject to compliance with requirements, provide products by one of the following manufacturers:
 1. Ductmate Industries, Inc.
 2. Lindab.
 3. Lockformer.
 4. McGill Airflow.
 5. Nexus Inc.
 6. Semco, Inc.
 7. Ward Industries.
- C. Diameter as applied to flat-oval ducts in this Article is the diameter of a round duct with a circumference equal to the perimeter of a given size of flat-oval duct.
- D. Round, Longitudinal- and Spiral Lock-Seam Ducts: Fabricate supply ducts of galvanized steel according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible."
- E. Flat-Oval, Longitudinal- and Spiral Lock-Seam Ducts: Fabricate supply ducts according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible. Fabricate ducts larger than 72 inches in diameter with butt-welded longitudinal seams.

F. Duct Joints:

1. Ducts up to 20 Inches in Diameter: Interior, center-beaded slip coupling, sealed before and after fastening, attached with sheet metal screws.
2. Ducts 21 to 72 Inches in Diameter: Three-piece, gasketed, flanged joint consisting of two internal flanges with sealant and one external closure band with gasket.
3. Ducts Larger Than 72 Inches in Diameter: Companion angle flanged joints per SMACNA "HVAC Duct Construction Standards--Metal and Flexible," Figure 3-2.
4. Round Ducts: Prefabricated connection system consisting of double-lipped, EPDM rubber gasket. Manufacture ducts according to connection system manufacturer's tolerances.
5. Flat-Oval Ducts: Prefabricated connection system consisting of two flanges and one synthetic rubber gasket.

G. 90-Degree Tees and Laterals and Conical Tees: Fabricate to comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," with metal thicknesses specified for longitudinal-seam straight ducts.

H. Diverging-Flow Fittings: Fabricate with reduced entrance to branch taps and with no excess material projecting from fitting onto branch tap entrance.

I. Fabricate elbows using die-formed, gored, pleated, or mitered construction. Bend radius of die-formed, gored, and pleated elbows shall be 1-1/2 times duct diameter. Unless elbow construction type is indicated, fabricate elbows as follows:

1. Mitered-Elbow Radius and Number of Pieces: Welded construction complying with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," unless otherwise indicated.
2. Round Mitered Elbows: Welded construction with the following metal thickness for pressure classes from minus 2- to plus 2-inch wg:
 - a. Ducts 3 to 36 Inches in Diameter: 0.034 inch.
 - b. Ducts 37 to 50 Inches in Diameter: 0.040 inch.
 - c. Ducts 52 to 60 Inches in Diameter: 0.052 inch.
 - d. Ducts 62 to 84 Inches in Diameter: 0.064 inch.
3. Round Mitered Elbows: Welded construction with the following metal thickness for pressure classes from 2- to 10-inch wg:
 - a. Ducts 3 to 26 Inches in Diameter: 0.034 inch.
 - b. Ducts 27 to 50 Inches in Diameter: 0.040 inch.
 - c. Ducts 52 to 60 Inches in Diameter: 0.052 inch.
 - d. Ducts 62 to 84 Inches in Diameter: 0.064 inch.
4. 90-Degree, 2-Piece, Mitered Elbows: Use only for supply systems or for material-handling Class A or B exhaust systems and only where space restrictions do not permit using radius elbows. Fabricate with single-thickness turning vanes.
5. Round Elbows 8 Inches and Less in Diameter: Fabricate die-formed elbows for 45- and 90-degree elbows and pleated elbows for 30, 45, 60, and 90 degrees only. Fabricate nonstandard bend-angle configurations or nonstandard diameter elbows with gored construction.
6. Round Elbows 9 through 14 Inches in Diameter: Fabricate gored or pleated elbows for 30, 45, 60, and 90 degrees unless space restrictions require mitered elbows. Fabricate

nonstandard bend-angle configurations or nonstandard diameter elbows with gored construction.

7. Round Elbows Larger Than 14 Inches in Diameter and All Flat-Oval Elbows: Fabricate gored elbows unless space restrictions require mitered elbows.
8. Die-Formed Elbows for Sizes through 8 Inches in Diameter and All Pressures 0.040 inch thick with 2-piece welded construction.
9. Pleated Elbows for Sizes through 14 Inches in Diameter and Pressures through 10-Inch wg: 0.022 inch.

2.6 DUCTWORK LINING:

- A. Ductwork liner shall be Johns Manville Linacoustic RC. Ductwork liner as manufactured by Knauf, Owens-Corning or Certain-Teed will be acceptable if equal. The product name is to be imprinted on the air stream surface.
- B. Ductwork Liner shall be installed in accordance with the Sheet Metal and Air Conditioning Contractor's National Association Duct Liner Application Standard, Second Edition, shall meet ASTM C1071 standards and liner facing must meet EPA registration requirements for antimicrobial protection.
- C. Transfer air duct shall be provided with ½" duct lining with reinforced air side coating and anti-microbial protection.
 1. 1/2" thick liner to be 1-1/2 pcf density, shall have a minimum sound NRC of 0.55 based on a Type "A" mounting, and a "K" factor not exceeding 0.46 at 75° F mean temperature.
- D. Supply, return and exhaust air duct shall be provided with 1 ½" duct lining with reinforced air side coating and anti-microbial protection.
 1. 1 1/2" thick liner to be 1-1/2 pcf density, shall have a minimum sound NRC of 0.85 based on a Type "A" mounting, and a "K" factor not exceeding 0.16 at 75° F mean temperature.
- E. Unless otherwise indicated, the duct dimensions given on the drawings are to be clear inside dimensions after insulation is applied.
- F. Round ductwork indicated to be lined shall be lined with 1" thick "Permacote Spiracoustic Plus" insulation manufactured by Johns Manville. Liner shall be installed per manufacturer's recommendations and requirements.

PART 3 - EXECUTION

3.1 DUCT APPLICATIONS

- A. Static-Pressure Classes: Unless otherwise indicated, construct ducts according to the following:
 1. Supply Ducts (constant volume units): +2".
 2. Supply Ducts (between fan and first system fire damper): +8".
 3. Return Ducts (Negative Pressure): -2".
 4. Return Ducts (between nearest fire damper and return fan inlet): -4".
 5. Return Ducts (return fan discharge and AHU intake / exhaust damper): -4".
 6. Exhaust Ducts (Negative Pressure): -2".

B. All ducts shall be galvanized steel except as follows:

1. Range Hood Exhaust Ducts: Comply with NFPA 96.

- a. Type 430 Stainless Steel Double wall with 20 gauge interior and 24 gauge exterior
- b. Grease duct joints shall be held together by means of formed V clamps and sealed with 3M Fire Barrier 2000+.

2. Exposed Supply Ducts In Occupied Spaces (Sales areas / Lobbies / back-of-house etc.):

- a. Spiral round/oval galvanized sheet steel unless otherwise indicated as rectangular with paint grip finish.

3.2 DUCT INSTALLATION

- A. Construct and install ducts according to SMACNA's "HVAC Duct Construction Standards--Metal and Flexible," unless otherwise indicated.
- B. Install round and flat-oval ducts in lengths not less than 12 feet unless interrupted by fittings.
- C. Install ducts with fewest possible joints.
- D. Install fabricated fittings for changes in directions, size, and shape and for connections.
- E. Install couplings tight to duct wall surface with a minimum of projections into duct. Secure couplings with sheet metal screws. Install screws at intervals of 12 inches, with a minimum of 3 screws in each coupling.
- F. Install ducts, unless otherwise indicated, vertically and horizontally and parallel and perpendicular to building lines; avoid diagonal runs.
- G. Install ducts close to walls, overhead construction, columns, and other structural and permanent enclosure elements of building.
- H. Install ducts with a clearance of 1 inch, plus allowance for insulation thickness.
- I. Conceal ducts from view in finished spaces. Do not encase horizontal runs in solid partitions unless specifically indicated.
- J. Coordinate layout with suspended ceiling, fire- and control dampers, lighting layouts, and similar finished work.
- K. Seal all joints and seams. Apply sealant to male end connectors before insertion, and afterward to cover entire joint and sheet metal screws.
- L. Electrical Equipment Spaces: Route ducts to avoid passing through transformer vaults and electrical equipment spaces and enclosures.
- M. Non-Fire-Rated Partition Penetrations: Where ducts pass through interior partitions and exterior walls and are exposed to view, conceal spaces between construction openings and ducts or duct insulation with sheet metal flanges of same metal thickness as ducts. Overlap openings on 4 sides by at least 1-1/2 inches.

- N. Fire-Rated Partition Penetrations: Where ducts pass through interior partitions and exterior walls, install appropriately rated fire dampers, sleeves, and firestopping sealant. Fire and smoke dampers are specified in Division 23 Section "Air Duct Accessories." Firestopping materials and installation methods are specified in Division 07.
- O. Protect duct interiors from moisture, construction debris and dust, and other foreign materials. Comply with SMACNA's "IAQ Guidelines for Occupied Buildings Under Construction," Appendix G, "Duct Cleanliness for New Construction Guidelines."

3.3 RANGE HOOD EXHAUST DUCTS, SPECIAL INSTALLATION REQUIREMENTS

- A. Install ducts to allow for thermal expansion through 2000 deg F temperature range.
- B. Install ducts without dips or traps that may collect residues unless traps have continuous or automatic residue removal.
- C. Install access panels at each change in direction and at intervals defined by NFPA 96; locate on sides of duct a minimum of 1-1/2 inches from bottom; and fit with grease-tight covers of same material as duct.
- D. Do not penetrate fire-rated assemblies except as permitted by applicable building codes.

3.4 SEAM AND JOINT SEALING

- A. Seal all duct seams and joints to the most severe requirement between the latest Chicago Building Code and SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for duct pressure class indicated.
- B. Utilize sealant designed for outdoor use with ductwork exposed to the outdoors.
- C. Seal ducts before external insulation is applied.

3.5 HANGING AND SUPPORTING

- A. Support horizontal ducts within 24 inches of each elbow and within 48 inches of each branch intersection.
- B. Support vertical ducts at maximum intervals of 16 feet and at each floor.
- C. Install upper attachments to structures with an allowable load not exceeding one-fourth of failure (proof-test) load.
- D. For concrete structure installations: Install concrete inserts before placing concrete.
- E. For concrete structure installations: Install powder-actuated concrete fasteners after concrete is placed and completely cured.
 - 1. Do not use powder-actuated concrete fasteners for lightweight-aggregate concretes or for slabs less than 4 inches thick.

3.6 CONNECTIONS

- A. Make connections to equipment with flexible connectors according to Division 23 Section "Air Duct Accessories."

- B. Comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for branch, outlet and inlet, and terminal unit connections.

3.7 DUCTWORK LINER:

- A. All rectangular and square ductwork shall be lined single wall. Liner shall be applied to ductwork indicated on plans for lining application.
- B. Duct liner shall be cut to assure overlapped and compressed longitudinal corner joints. Cut edges shall be properly sealed with mastic to assure no erosion of insulation.
- C. Fasteners shall start within 3" of the upstream transverse edges of the duct liner and 3" from the longitudinal joints and shall be spaced at a maximum of 12" o.c. except that they shall be placed not more than 6" from a longitudinal joint of the liner nor 12" from a corner break.
- D. Duct liner shall be adhered to the sheet metal (with 100% coverage) and edges coated with one (1) of the adhesives conforming to the Standard for Adhesives for Duct Liner, ASC-A-7001C - 1972 of the Adhesive and Sealant Council, Inc.
- E. Duct liner shall be further secured with fasteners conforming to the Mechanical Fastener Standard, MF-1-1975 on Page 22 of the Duct Liner Application Standard, Second Edition, of the Sheet Metal and Air Conditioning Contractor's National Association.
- F. Duct liner in medium pressure ductwork shall be installed per manufacturer's recommendations for velocities above 2,000 fpm.
- G. The leading edge of duct liner joints (in direction of air flow) shall be provided with metal nosing or a hardening mastic to prevent lining from coming loose.

3.8 FIELD QUALITY CONTROL

- A. Perform the following field tests and inspections according to SMACNA's "HVAC Air Duct Leakage Test Manual" and prepare test reports:
 - 1. Disassemble, reassemble, and seal segments of systems to accommodate leakage testing and for compliance with test requirements.
 - 2. Conduct tests at static pressures equal to maximum design pressure of system or section being tested. If pressure classes are not indicated, test entire system at maximum system design pressure. Do not pressurize systems above maximum design operating pressure. Give seven days' advance notice for testing.
 - 3. Maximum Allowable Leakage: Comply with requirements for Leakage Class 3 for round and flat-oval ducts, Leakage Class 12 for rectangular ducts in pressure classes lower than and equal to 2-inch wg (both positive and negative pressures), and Leakage Class 6 for pressure classes from 2- to 10-inch wg.
 - 4. Remake leaking joints and retest until leakage is equal to or less than maximum allowable.

END OF SECTION 233113

SECTION 233300 - AIR DUCT ACCESSORIES**PART 1 - GENERAL****1.1 SUMMARY****A. Section includes the following:**

1. Backdraft dampers.
2. Volume dampers.
3. Fire dampers.
4. Turning vanes.
5. Duct-mounting access doors.
6. Flexible connectors.
7. Flexible ducts.
8. Duct accessory hardware.

1.2 SUBMITTALS**A. Product Data: For the following:**

1. Backdraft dampers.
2. Volume dampers.
3. Fire dampers.
4. Turning vanes.
5. Duct-mounting access doors.
6. Flexible connectors.
7. Flexible ducts.

B. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.

1. Special fittings.
2. Backdraft dampers.
3. Manual-volume damper installations.
4. Motorized-control damper installations.
5. Fire-damper and combination fire- and smoke-damper installations, including sleeves and duct-mounting access doors.
6. Duct mounted access doors.
7. Flexible connectors.
8. Flexible ducts.
9. Wiring Diagrams: Power, signal, and control wiring.

C. Coordination Drawings: Reflected ceiling plans, drawn to scale and coordinating penetrations and ceiling-mounting items. Show ceiling-mounting access panels and access doors required for access to duct accessories.**1.3 QUALITY ASSURANCE****A. Comply with NFPA 90A, "Installation of Air Conditioning and Ventilating Systems," and NFPA 90B, "Installation of Warm Air Heating and Air Conditioning Systems."**

1.4 DELIVERY, STORAGE AND HANDLING

- A. Materials delivered to the site must be coordinated with the site supervisor prior to delivery.
- B. All materials shall be stored in a designated area and protected from the environment.
- C. All materials shall be secured so as not to be a hazard during the construction process.
- D. All materials must be free of all dirt, debris and moisture during the installation process.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide products by one of the following manufacturers:
 - 1. Backdraft Dampers:
 - a. Greenheck.
 - b. Ruskin.
 - c. Vent Products Company.
 - 2. Volume Dampers
 - a. Nailor.
 - b. Ruskin.
 - c. Vent Products Company.
 - 3. Fire Dampers:
 - a. Greenheck.
 - b. Ruskin.
 - c. Vent Products Company.
 - 4. Duct-Mounting Access Doors:
 - a. CESCO Products.
 - b. Ductmate Industries.
 - c. Greenheck.
 - 5. Flexible Connectors:
 - a. Ductmate Industries.
 - b. Ventfabrics, Inc.
 - c. Ward Industries.
 - 6. Flexible Ducts:
 - a. Flexmaster USA.(MCA-TLD)
 - b. Clevaflex (DBA)
 - c. Novaflex (T/L-AM)

2.2 SHEET METAL MATERIALS

- A. Comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for acceptable materials, material thicknesses, and duct construction methods, unless otherwise indicated.
- B. Galvanized Sheet Steel: Lock-forming quality; complying with ASTM A 653/A 653M and having G90coating designation; ducts shall have mill-phosphatized finish for surfaces exposed to view.
- C. Stainless Steel: ASTM A 480/A 480M Type 304 (specify Type 314 as required by the application).
- D. Aluminum Sheets: ASTM B 209 alloy 3003, temper H14; with mill finish for concealed ducts and standard, 1-side bright finish for exposed ducts.
- E. Extruded Aluminum: ASTM B 221, alloy 6063, temper T6.
- F. Reinforcement Shapes and Plates: Galvanized-steel reinforcement where installed on galvanized sheet metal ducts; compatible materials for aluminum and stainless-steel ducts.
- G. Tie Rods: Galvanized steel, 1/4-inch minimum diameter for lengths 36 inches or less; 3/8-inch minimum diameter for lengths longer than 36 inches.

2.3 BACKDRAFT DAMPERS

- A. Description: Multiple-blade, parallel action gravity balanced, with center-pivoted blades of maximum 6-inch width, with sealed edges, assembled in rattle-free manner with 90-degree stop, steel ball bearings, and axles; adjustment device to permit setting for varying differential static pressure.
- B. Frame: 0.063-inch thick extruded aluminum, with welded corners and mounting flange.
- C. Blades: 0.050-inch thick aluminum sheet.
- D. Blade Seals: Neoprene.
- E. Blade Axles: Galvanized steel.
- F. Tie Bars and Brackets: Galvanized steel.
- G. Return Spring: Adjustable tension.

2.4 VOLUME DAMPERS

- A. General Description: Factory fabricated, with required hardware and accessories. Stiffen damper blades for stability. Include locking device to hold single-blade dampers in a fixed position without vibration. Close duct penetrations for damper components to seal duct consistent with pressure class.
 - 1. Pressure Classes of 3-Inch wg or Higher: End bearings or other seals for ducts with axles full length of damper blades and bearings at both ends of operating shaft.

- B. Standard Volume Dampers: Multiple- or single-blade, parallel- or opposed-blade design as indicated, standard leakage rating, with linkage outside airstream and suitable for horizontal or vertical applications.
1. Steel Frames (For use in steel ductwork): Hat-shaped, galvanized sheet steel channels, minimum of 0.064 inch thick, with mitered and welded corners; frames with flanges where indicated for attaching to walls and flangeless frames where indicated for installing in ducts.
 2. Roll-Formed Steel Blades (For use with steel frames): 0.064-inch- thick, galvanized sheet steel.
 3. Aluminum Frames (For use in stainless steel or aluminum ductwork): Hat-shaped, 0.10-inch- thick, aluminum sheet channels; frames with flanges where indicated for attaching to walls; and flangeless frames where indicated for installing in ducts.
 4. Roll-Formed Aluminum Blades (For use with aluminum frames): 0.10-inch- thick aluminum sheet.
 5. Extruded-Aluminum Blades (For use with aluminum frames): 0.050-inch- thick extruded aluminum.
 6. Blade Axles: Galvanized steel. Drive shaft will be the full length of the blade.
 7. Bearings: Stainless-steel sleeve.
 8. Tie Bars and Brackets: Aluminum (aluminum or stainless steel ductwork applications), Galvanized steel (galvanized steel ductwork applications).
- C. Low-Leakage Volume Dampers: Multiple- or single-blade, parallel- or opposed-blade design as indicated, low-leakage rating, with linkage outside airstream, and suitable for horizontal or vertical applications.
1. Steel Frames (For use in steel ductwork) : galvanized sheet steel channels, minimum of 0.064 inch thick, with mitered and welded corners; frames with flanges where indicated for attaching to walls and flangeless frames where indicated for installing in ducts.
 2. Roll-Formed Steel Blades (For use with steel frames): 0.064-inch- thick, galvanized sheet steel.
 3. Aluminum Frames (For use in stainless steel or aluminum ductwork) : 0.10-inch- thick, aluminum sheet channels; frames with flanges where indicated for attaching to walls and flangeless frames where indicated for installing in ducts.
 4. Roll-Formed Aluminum Blades (For use with aluminum frames): 0.10-inch- thick aluminum sheet.
 5. Extruded-Aluminum Blades (For use with aluminum frames): 0.050-inch- thick extruded aluminum.
 6. Blade Axles: Galvanized steel. Drive shaft will be the full length of the blade.
 7. Bearings: Stainless-steel sleeve thrust or ball.
 8. Blade Seals: Neoprene.
 9. Jamb Seals: Cambered stainless steel.
 10. Tie Bars and Brackets: Aluminum (aluminum or stainless steel ductwork applications), Galvanized steel (galvanized steel ductwork applications).
- D. Jackshaft: 1-inch- diameter, galvanized-steel pipe rotating within pipe-bearing assembly mounted on supports at each mullion and at each end of multiple-damper assemblies.
1. Length and Number of Mountings: Appropriate to connect linkage of each damper in multiple-damper assembly.
- E. Damper Hardware: Zinc-plated, die-cast core with dial and handle made of 3/32-inch- thick zinc-plated steel, and a 3/4-inch hexagon locking nut. Include center hole to suit damper operating-rod size. Include elevated platform for insulated duct mounting.

2.5 FIRE DAMPERS

- A. Fire dampers shall be labeled according to UL 555.
- B. Fire Rating: Insert rating as required by application hours.
- C. Frame: Curtain type with blades outside airstream fabricated with roll-formed, 0.034-inch- thick galvanized steel; with mitered and interlocking corners.
- D. Mounting Sleeve: Factory- or field-installed, galvanized sheet steel.
 - 1. Minimum Thickness: 0.052 or 0.138 inch thick as indicated and of length to suit application.
 - 2. Exceptions: Omit sleeve where damper frame width permits direct attachment of perimeter mounting angles on each side of wall or floor, and thickness of damper frame complies with sleeve requirements.
- E. Mounting Orientation: Vertical or horizontal as indicated.
- F. Blades: Roll-formed, interlocking, 0.034-inch- thick, galvanized sheet steel. In place of interlocking blades, use full-length, 0.034-inch- thick, galvanized-steel blade connectors.
- G. Horizontal Dampers: Include blade lock and stainless-steel closure spring.
- H. Fusible Links: Replaceable, 165 deg F rated.

2.6 TURNING VANES

- A. Fabricate to comply with SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for vanes and vane runners. Vane runners shall automatically align vanes.
- B. Manufactured Turning Vanes: Fabricate 1-1/2-inch- wide, single-vane, curved blades of galvanized sheet steel set 3/4 inch o.c.; support with bars perpendicular to blades set 2 inches o.c.; and set into vane runners suitable for duct mounting.
- C. Acoustic Turning Vanes: Fabricate airfoil-shaped aluminum extrusions with perforated faces and fibrous-glass fill.

2.7 DUCT-MOUNTING ACCESS DOORS

- A. General Description: Fabricate doors airtight and suitable for duct pressure class.
- B. Door: Double wall, duct mounting, and rectangular; fabricated of galvanized sheet metal with insulation fill and thickness as indicated for duct pressure class. Include vision panel where indicated. Include piano hinge and cam latches.
 - 1. Frame: Galvanized sheet steel, with bend-over tabs and foam gaskets.
 - 2. Provide number of hinges and locks as follows:
 - a. Less than 12 Inches Square: Secure with two sash locks.
 - b. Up to 18 Inches Square: Continuous hinge and two sash locks.
 - c. Larger than 18 inches square: Continuous hinge and two compression latches with outside and inside handles.

- C. Door: Double wall, duct mounting, and round; fabricated of galvanized sheet metal with insulation fill and 1-inch thickness. Include cam latches.

- 1. Frame: Galvanized sheet steel, with spin-in notched frame.

- D. Pressure Relief Access Door: Double wall and duct mounting; fabricated of galvanized sheet metal with insulation fill and thickness as indicated for duct pressure class. Include vision panel where indicated, latches, and retaining chain.

- 1. Frame: Galvanized sheet steel, with bend-over tabs and foam gaskets.

- E. Seal around frame attachment to duct and door to frame with neoprene or foam rubber.

- F. Insulation: 1-inch- thick, fibrous-glass or polystyrene-foam board.

2.8 FLEXIBLE CONNECTORS

- A. General Description: Flame-retardant or noncombustible fabrics, coatings, and adhesives complying with UL 181, Class 1.

- B. Metal-Edged Connectors: Factory fabricated with a fabric strip 5-3/4 inches wide attached to two strips of 2-3/4-inch- wide, 0.028-inch- thick, galvanized sheet steel or 0.032-inch- thick aluminum sheets. Select metal compatible with ducts.

- C. Indoor System, Flexible Connector Fabric: Glass fabric double coated with neoprene.

- 1. Minimum Weight: 26 oz./sq. yd..
 - 2. Tensile Strength: 480 lbf/inch in the warp and 360 lbf/inch in the filling.
 - 3. Service Temperature: Minus 40 to plus 200 deg F.

- D. Outdoor System, Flexible Connector Fabric: Glass fabric double coated with weatherproof, synthetic rubber resistant to UV rays and ozone.

- 1. Minimum Weight: 24 oz./sq. yd..
 - 2. Tensile Strength: 530 lbf/inch in the warp and 440 lbf/inch in the filling.
 - 3. Service Temperature: Minus 50 to plus 250 deg F.

2.9 FLEXIBLE DUCTS

- A. Insulated-Duct Connectors: UL 181, Class 1, 2-ply vinyl film supported by helically wound, spring-steel wire; fibrous-glass insulation; aluminized vapor barrier film.

- 1. Pressure Rating: 10-inch wg positive and 1.0-inch wg negative.
 - 2. Maximum Air Velocity: 4000 fpm.
 - 3. Temperature Range: Minus 10 to plus 160 deg F

- B. Flexible Duct Clamps: Stainless-steel band with cadmium-plated hex screw to tighten band with a worm-gear action, in sizes 3 through 18 inches to suit duct size.

2.10 DUCT ACCESSORY HARDWARE

- A. Instrument Test Holes: Cast iron or cast aluminum to suit duct material, including screw cap and gasket. Size to allow insertion of pitot tube and other testing instruments and of length to suit duct insulation thickness.

- B. Adhesives: High strength, quick setting, neoprene based, waterproof, and resistant to gasoline and grease.

PART 3 - EXECUTION

3.1 APPLICATION AND INSTALLATION

- A. Install duct accessories according to applicable details in SMACNA's "HVAC Duct Construction Standards--Metal and Flexible" for metal ducts and in NAIMA AH116, "Fibrous Glass Duct Construction Standards," for fibrous-glass ducts.
- B. Provide duct accessories of materials suited to duct materials; use galvanized-steel accessories in galvanized-steel and fibrous-glass ducts, stainless-steel accessories in stainless-steel ducts, and aluminum accessories in aluminum ducts.
- C. Install backdraft dampers on exhaust fans or exhaust ducts nearest to outside and where indicated.
- D. Provide balancing dampers at points on supply, return, and exhaust systems where branches lead from larger ducts as required for air balancing. Install at a minimum of two duct widths from branch takeoff.
- E. Provide test holes at fan inlets and outlets and elsewhere as indicated.
- F. Install fire dampers, with fusible links, according to manufacturer's UL-approved written instructions.
- G. Install duct silencers rigidly to ducts.
- H. Install duct access doors to allow for inspecting, adjusting, and maintaining duct accessories, control devices - sensors and terminal units as follows:
 - 1. On both sides of duct coils. On terminal units coordinate upstream coil access door with equipment supplier.
 - 2. Downstream from volume dampers, turning vanes, and duct mounted equipment.
 - 3. Adjacent to fire dampers, providing access to reset or reinstall fusible links.
 - 4. To interior of ducts for cleaning; before and after each change in direction, at maximum 50-foot spacing.
 - 5. On sides of ducts where adequate clearance is available.
 - 6. Where indicated on plans.
 - 7. Upstream and downstream of ducted fans.
- I. Label access doors according to Division 23 Section "Identification for HVAC."
- J. Install flexible connectors immediately adjacent to equipment in ducts associated with fans and motorized equipment supported by vibration isolators.
- K. For fans developing static pressures of 5-inch wg and higher, cover flexible connectors with loaded vinyl sheet held in place with metal straps.
- L. Connect terminal units to supply ducts directly or with maximum 12-inch lengths of flexible duct. Do not use flexible ducts to change directions.

- M. Connect diffusers or light troffer boots to low pressure ducts directly or with maximum 60-inch lengths of flexible duct clamped or strapped in place.
- N. Connect flexible ducts to metal ducts with adhesive plus sheet metal screws.
- O. Install duct test holes where indicated and required for testing and balancing purposes.
- P. Provide turning vanes in all short radius / square elbows (>45 degrees) and tees.

3.2 ADJUSTING

- A. Adjust duct accessories for proper settings.
- B. Adjust fire dampers for proper action.
- C. Final positioning of manual-volume dampers is specified in Division 23 Section "Testing, Adjusting, and Balancing for HVAC."

3.3 DEMONSTRATION AND COMMISSIONING

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain the duct accessories.
 - 1. Train Owner's maintenance personnel on troubleshooting, servicing, and maintaining duct accessories. The training will occur after all devices are installed including all access doors. The trainer will provide two (2) Installation and Operations manuals for the use of the owners personnel during training.
 - 2. Review data in maintenance manuals. Refer to Division 01 Section "Operation and Maintenance Data." All required and recommended maintenance will be reviewed as well as operational trouble shooting. If the IOM does not include a written trouble shooting guide one will be provided.
 - 3. Schedule training with Owner, through Architect, with at least seven days' advance notice.
 - 4. Training will occur in one (1) two (2) hour session and will include the dropping and resetting of 3 fire dampers selected by the owner. This portion of the training may not take longer than 30 minutes of the training session.
- B. Demonstrate proper operation of equipment to commissioning agent or designated owners personnel. The scope of the demonstration will include functional performance requirements under both local and building automation control as well as any commissioning requirements in Division 01 or 23.
 - 1. For all fire dampers or access doors for fire dampers installed on this project the Contractor will demonstrate that any fire dampers selected by the owner can be dropped and reset using the provided access doors.

END OF SECTION 233300

SECTION 233423 - HVAC POWER VENTILATORS**PART 1 - GENERAL****1.1 SUMMARY****A. Section includes the following:**

1. Upblast propeller roof exhaust fans.
2. In-line centrifugal fans.
3. Destratification fans.

1.2 SUBMITTALS**A. Product Data:** Include rated capacities, furnished specialties, and accessories for each type of product indicated and include the following:

1. Certified fan performance curves with system operating conditions indicated.
2. Certified fan sound-power ratings.
3. Motor ratings and electrical characteristics, plus motor and electrical accessories.
4. Material thickness and finishes, including color charts.
5. Dampers, including housings, linkages, and operators.

B. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection. **Provide wiring Diagrams:** Power, signal, and control wiring.**C. Operation and Maintenance Data:** For power ventilators to include in emergency, operation, and maintenance manuals.**D. Field Quality-Control Test Reports:** Submit reports documenting the activities required to be performed in PART 3. These reports are to be submitted two weeks after the startup is completed.**E. Training Reports:** Submit reports on training documenting dates and attendance.**1.3 QUALITY ASSURANCE****A. Electrical Components, Devices, and Accessories:** Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.**B. AMCA Compliance:** Products shall comply with performance requirements and shall be licensed to use the AMCA-Certified Ratings Seal.**C. NEMA Compliance:** Motors and electrical accessories shall comply with NEMA standards.**D. UL Standard:** Power ventilators shall comply with UL 705.**1.4 DELIVERY, STORAGE, AND HANDLING****A. Deliver fans as factory-assembled unit, to the extent allowable by shipping limitations, with protective crating and covering.**

- B. Disassemble and reassemble units, as required for moving to final location, according to manufacturer's written instructions.
- C. Lift and support units with manufacturer's designated lifting or supporting points.

1.5 PERFORMANCE REQUIREMENTS

- A. Project Altitude: Base fan-performance ratings on sea level.
- B. Operating Limits: Classify according to AMCA 99.

1.6 COORDINATION

- A. Coordinate size and location of structural-steel support members.
- B. Coordinate installation of roof curbs, equipment supports, and roof penetrations.

1.7 EXTRA MATERIALS

- A. Furnish one set of belts that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1.8 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provide diffusers by one of the following:

- 1. Upblast Centrifugal Roof Ventilators:

- a. Greenheck.
- b. Loren Cook Company.
- c. Penn Ventilation.
- d. Twin City

- 2. In-line Centrifugal Fans:

- a. Greenheck.
- b. Loren Cook Company.
- c. Penn Ventilation.
- d. Twin City

- 3. Destratification Fans:

- a. Air-Row

2.2 CENTRIFUGAL ROOF VENTILATORS

- A. Description: Direct- or belt-driven centrifugal fans consisting of housing, wheel, fan shaft, bearings, motor and disconnect switch, drive assembly, curb base, and accessories.
- B. Housing: Removable, spun-aluminum, dome top and outlet baffle; square, one-piece, aluminum base with venturi inlet cone.
 - 1. Upblast Units: Provide spun-aluminum discharge baffle to direct discharge air upward, with rain and snow drains.
- C. Fan Wheels: Aluminum hub and wheel with backward-inclined blades.
- D. Belt-Driven Drive Assembly (Provide direct drive for all units less than $\frac{1}{2}$ HP): Resiliently mounted to housing, with the following features:
 - 1. Fan Shaft: Turned, ground, and polished steel; keyed to wheel hub.
 - 2. Shaft Bearings: Permanently lubricated, permanently sealed, self-aligning ball bearings.
 - 3. Pulleys: Cast-iron, adjustable-pitch motor pulley.
 - 4. Fan and motor isolated from exhaust airstream. (UL-762 Labeled)
- E. Accessories:
 - 1. Disconnect Switch: Non-fusible type, with thermal-overload protection mounted inside fan housing, factory wired through an internal aluminum conduit.
 - 2. Specify roof curbs, if they are site fabricated, in Division 07 Section "Roof Accessories."
 - 3. Minimum 1-gallon grease trap / container to prevent grease spilling on the roof.
- F. Roof Curbs: Galvanized steel; mitered and welded corners; 2-inch- thick, rigid, fiberglass insulation adhered to inside walls; and 2-inch wood nailer. Size as required to suit roof opening and fan base.

2.3 IN-LINE CENTRIFUGAL FANS

- A. Description: In-line, belt-driven centrifugal fans consisting of housing, wheel, outlet guide vanes, fan shaft, bearings, motor and disconnect switch, drive assembly, mounting brackets, and accessories.
- B. Housing: Split, spun aluminum with aluminum straightening vanes, inlet and outlet flanges, and support bracket adaptable to floor, side wall, or ceiling mounting.
- C. Direct-Driven Units: Motor mounted in airstream, factory wired to disconnect switch located on outside of fan housing[; with wheel, inlet cone, and motor on swing-out service door].
- D. Belt-Driven Units: Motor mounted on adjustable base, with adjustable sheaves, enclosure around belts within fan housing, and lubricating tubes from fan bearings extended to outside of fan housing.
- E. Fan Wheels: Aluminum, airfoil blades welded to aluminum hub.
- F. Accessories:

1. Variable-Speed Controller: Solid-state control to reduce speed from 100 to less than 50 percent. Required for all direct drive fans.
2. Volume-Control Damper: Manually operated with quadrant lock, located in fan outlet.
3. Companion Flanges: For inlet and outlet duct connections.
4. Fan Guards: 1/2- by 1-inch mesh of galvanized steel in removable frame. Provide guard for inlet or outlet for units not connected to ductwork.

2.4 DESTRATIFICATION FANS

- A. Description: Destratification fan for open ceilings. Designed to eliminate stratified air layers from ceiling to floor and help deliver energy savings.
- B. Housing: Durable and high quality .040 aluminum. Unique square design. Light weight 1.8 lbs not including the EC motor. Color is white unless otherwise noted.
- C. Motor: 92% efficient EC (Electronically Commentated) motor. Built-in thermal overload protector. Dual customizable speed levels with standard soft start feature.
- D. Fan Wheels: PA plastic fan blade material.
- E. Accessories:
 1. Cord: white UL listed 110 V cord and plug unless otherwise noted.
 2. Blade Guard: Heavy duty plated steel powder coated black.
 3. Hanging Kits: Two 5' safety cables with easy push button mechanisms by Gripple. Each cable is UL certified with a 33 lb wight restriction and a 5:1 safety factor rating.

2.5 MOTORS

- A. Comply with requirements in Division 23 Section "Common Motor Requirements for HVAC Equipment."
- B. Enclosure Type: Totally enclosed, fan cooled.

2.6 SOURCE QUALITY CONTROL

- A. Sound-Power Level Ratings: Comply with AMCA 301, "Methods for Calculating Fan Sound Ratings from Laboratory Test Data." Factory test fans according to AMCA 300, "Reverberant Room Method for Sound Testing of Fans." Label fans with the AMCA-Certified Ratings Seal.
- B. Fan Performance Ratings: Establish flow rate, pressure, power, air density, speed of rotation, and efficiency by factory tests and ratings according to AMCA 210, "Laboratory Methods of Testing Fans for Rating."

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install power ventilators level and plumb.
- B. Support units using spring isolators having a static deflection of 1 inch.

1. Secure vibration and seismic controls to concrete bases using anchor bolts cast in concrete base.
- C. Secure roof-mounting fans to roof curbs with cadmium-plated hardware
- D. Ceiling Units: Suspend units from structure; use steel wire or metal straps.
- E. Support suspended units from structure using threaded steel rods and spring hangers having a static deflection of 1 inch.
- F. Install units with clearances for service and maintenance.
- G. Label units according to requirements specified in Division 23 Section "Identification for HVAC."

3.2 CONNECTIONS

- A. Duct installation and connection requirements are specified in other Division 23 Sections. Drawings indicate general arrangement of ducts and duct accessories. Make final duct connections with flexible connectors. Flexible connectors are specified in Division 23 Section "Air Duct Accessories."
- B. Install ducts adjacent to power ventilators to allow service and maintenance.
- C. Ground equipment according to Division 26 Section "Grounding and Bonding for Electrical Systems."
- D. Connect wiring according to Division 26 Section "Conductors and Cables for Electrical Systems."

3.3 FIELD QUALITY CONTROL

- A. Perform the following field tests and inspections and prepare test reports:
 1. Verify that shipping, blocking, and bracing are removed.
 2. Verify that unit is secure on mountings and supporting devices and that connections to ducts and electrical components are complete. Verify that proper thermal-overload protection is installed in motors, starters, and disconnect switches.
 3. Verify that cleaning and adjusting are complete.
 4. Disconnect fan drive from motor, verify proper motor rotation direction, and verify fan wheel free rotation and smooth bearing operation. Reconnect fan drive system, align and adjust belts, and install belt guards.
 5. Adjust belt tension.
 6. Adjust damper linkages for proper damper operation.
 7. Verify lubrication for bearings and other moving parts.
 8. Verify that manual and automatic volume control and fire and smoke dampers in connected ductwork systems are in fully open position.
 9. Disable automatic temperature-control operators, energize motor and adjust fan to indicated rpm, and measure and record motor voltage and amperage.
 10. Shut unit down and reconnect automatic temperature-control operators.
 11. Remove and replace malfunctioning units and retest as specified above.
- B. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

3.4 DEMONSTRATION

- A. Train Owner's maintenance personnel on procedures and schedules related to startup and shutdown, troubleshooting, servicing, and preventive maintenance.
- B. Review data in the operation and maintenance manuals. Refer to Division 01 Section "Contract Closeout."
- C. Schedule training with Owner, through Architect, with at least 7 days' advance notice.
- D. Demonstrate operation of power ventilators. Conduct walking tour of the Project. Briefly identify location and describe function, operation, and maintenance of each power ventilator.

3.5 ADJUSTING

- A. Adjust damper linkages for proper damper operation.
- B. Adjust belt tension.
- C. Refer to Division 23 Section "Testing, Adjusting, and Balancing for HVAC" for testing, adjusting, and balancing procedures.
- D. Replace fan and motor pulleys as required to achieve design airflow.
- E. Lubricate bearings.

3.6 CLEANING

- A. After completing installation, inspect exposed finish. Remove burrs, dirt, and construction debris, and repair damaged finishes including chips, scratches, and abrasions.
- B. Clean fan interiors to remove foreign material and construction debris. Vacuum clean fan wheel and cabinet.

3.7 CONTRACTOR STARTUP AND REPORTING

- A. Final Checks before Startup: Perform the following operations and checks before startup. Startup service includes the testing, inspections and startup test reports:
 - 1. Verify that shipping, blocking, and bracing are removed.
 - 2. Verify that unit is secure on mountings and supporting devices and that connections for piping, ducts, and electrical components are complete. Verify that proper thermal-overload protection is installed in motors, starters, and disconnects.
 - 3. Perform cleaning and adjusting specified in this Section.
 - 4. Disconnect fan drive from motor, verify proper motor rotation direction, and verify fan wheel free rotation and smooth bearing operation. Reconnect fan drive system, align and adjust belts, and install belt guards.
 - 5. Lubricate bearings, pulleys, belts, and other moving parts with factory-recommended lubricants.
 - 6. Verify that manual and automatic volume control and fire and smoke dampers in connected ductwork systems are in the fully open position.
 - 7. Disable automatic temperature-control operators.
- B. Starting procedures for fans are as follows:

1. Energize motor; verify proper operation of motor, drive system, and fan wheel. Adjust fan to indicated RPM.
 2. Measure and record motor voltage and amperage.
- C. Shut unit down and reconnect automatic temperature-control operators.
- D. Refer to Division 23 Section "Testing, Adjusting, and Balancing for HVAC" for procedures for air-handling-system testing, adjusting, and balancing.
- E. Replace fan and motor pulleys as required to achieve design conditions.
- 3.8 DEMONSTRATION AND COMMISSIONING
- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain the fans.
1. Train Owner's maintenance personnel on procedures and schedules for starting up and shutting down, troubleshooting, servicing, and maintaining chillers. The training will occur after the startup report has been provided to the owner and the trainer will provide two (2) Installation and Operations manuals for the use of the owner's personnel during training.
 2. Review data in maintenance manuals. Refer to Division 01 Section "Operation and Maintenance Data." All required and recommended maintenance will be reviewed as well as operational trouble shooting. If the IOM does not include a written trouble shooting guide one will be provided.
 3. Schedule training with Owner, through Architect, with at least seven days' advance notice.
- B. Demonstrate proper operation of equipment to commissioning agent or designated owners personnel. The scope of the demonstration will include functional performance requirements under both local and building automation control as well as any commissioning requirements in Division 01 and 23.

END OF SECTION 233423

SECTION 233713 - DIFFUSERS, REGISTERS, AND GRILLES**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes ceiling- and wall-mounted diffusers, registers, and grilles.

1.2 DEFINITIONS

- A. Diffuser: Circular, square, or rectangular air distribution outlet, generally located in the ceiling and comprised of deflecting members discharging supply air in various directions and planes and arranged to promote mixing of primary air with secondary room air.
- B. Grille: A louvered or perforated covering for an opening in an air passage, which can be located in a sidewall, ceiling, or floor.
- C. Register: A combination grille and damper assembly over an air opening.

1.3 SUBMITTALS

- A. Product Data: For each product indicated, include the following:
 - 1. Data Sheet: Indicate materials of construction, finish, and mounting details; and performance data including throw and drop, static-pressure drop, and noise ratings.
 - 2. Diffuser, Register, and Grille Schedule: Indicate Drawing designation, room location, quantity, model number, size, and accessories furnished.
- B. Coordination Drawings: Reflected ceiling plans, drawn to scale, on which the following items are shown and coordinated with each other, based on input from installers of the items involved:
 - 1. Ceiling suspension assembly members.
 - 2. Method of attaching hangers to building structure.
 - 3. Size and location of initial access modules for acoustical tile.
 - 4. Ceiling-mounted items including lighting fixtures, diffusers, grilles, speakers, sprinklers, access panels, and special moldings.
 - 5. Duct access panels.

1.4 QUALITY ASSURANCE

- A. ARI Compliance: Test and rate diffusers, registers, and grilles in accordance with ARI 650 "Standard for Diffusers, registers, and grilles".
- B. ASHRAE Compliance: Test and rate diffusers, registers, and grilles in accordance with ASHRAE 70 "Method of Testing for Rating the Air Flow Performance of Outlets and Inlets".
- C. ADC Compliance: Test and rate diffusers, registers, and grilles in certified laboratories under requirements of ADC 1062 "Certification, Rating and Test Manual".
- D. ADC Seal: Provide diffusers, registers, and grilles bearing ADC Certified Rating Seal.
- E. NFPA Compliance: Install diffusers, registers, and grilles in accordance with NFPA 90A "Standard for the Installation of Air Conditioning and Ventilating Systems".

1.5 DELIVERY, STORAGE AND HANDLING

- A. Deliver diffusers, registers, and grilles wrapped in factory-fabricated fiber- board type containers. Identify on outside of container type of outlet or inlet and location to be installed. Avoid crushing or bending and prevent dirt and debris from entering and settling in devices.
- B. Store diffusers, registers, and grilles in original cartons and protect from weather and construction work traffic. Where possible, store indoors; when necessary to store outdoors, store above grade and enclose with waterproof wrapping.

1.6 WARRANTY

- A. Provide warranty on materials and labor for 18 months starting from date of delivery, or one year from date of substantial completion, whichever is longer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provided on **LiDL National Account List**:

1. Ceiling Air Diffusers:

- a. Nailor.
- b. Titus Products Div.; Philips Industries, Inc.
- c. TROX.
- d. Price Industries.

2. Duct and Wall Registers and Grilles

- a. Nailor.
- b. Titus Products Div.; Philips Industries, Inc.
- c. TROX.
- d. Price Industries.

3. Plenum Box Diffusers

- a. Ruskin Rooftop Systems

2.2 CEILING AIR DIFFUSERS

- A. General: Except as otherwise indicated, provide manufacturer's standard ceiling air diffusers where shown; of size, shape, capacity and type indicated; constructed of materials and components as indicated, and as required for complete installation.
- B. Performance: Provide ceiling air diffusers that have, as minimum, temperature and velocity traverses, throw and drop, and noise criteria ratings for each size device as listed in manufacturer's current data.
- C. Ceiling Compatibility: Provide diffusers with border styles that are compatible with adjacent ceiling systems, and that are specifically manufactured to fit into ceiling module with accurate fit and adequate support. Refer to general construction drawings and specifications for types of ceiling systems which will contain each type of ceiling air diffuser.

- D. Types: Provide ceiling diffusers of type, capacity, and with accessories and finishes as listed on diffuser schedule.

2.3 WALL REGISTERS AND GRILLES

- A. General: Except as otherwise indicated, provide manufacturer's standard wall registers and grilles where shown; of size, shape, capacity and type indicated; constructed of materials and components as indicated, and as required for complete installation.
- B. Performance: Provide wall registers and grilles that have, as minimum, temperature and velocity traverses, throw and drop, and noise criteria ratings for each size device as listed in manufacturer's current data.
- C. Wall Compatibility: Provide registers and grilles with border styles that are compatible with adjacent wall systems, and that are specifically manufactured to fit into wall construction with accurate fit and adequate support. Refer to general construction drawings and specifications for types of wall construction which will contain each type of wall register and grille.
- D. Types: Provide wall registers and grilles of type, capacity, and with accessories and finishes as listed on schedule.

2.4 PLENUM BOX DIFFUSER

- A. General: Except as otherwise indicated, provide manufacturer's standard plenum box where shown; of size, shape, capacity and type indicated; construction of materials and components as indicated, and as required for complete installation.
- B. Performance: provide plenum box that have, as minimum, temperature and velocity traverses, throw and drop, and noise criteria ratings for each size device as listed in manufacturer's current data.
- C. Drum Louver diffuser shall be heavy duty construction and provide minimum 30 degree vertical adjustment with full horizontal blade adjustment to deliver single point supply air for large space applications. Manufacturers published data sheet shall include CFM rating, noise, static, and throw information. Drum louver shall be in all respects equivalent to Ruskin Rooftop Systems model 578. Critical: Drum louver shall incorporate a standard 578 precision tuned internal air deflector to ensure quite even airflow to all sides with low pressure drop.
- D. Types: Provide plenum box of type, capacity, and with accessories and finishes as listed on schedule.

2.5 SOURCE QUALITY CONTROL

- A. Verification of Performance: Rate diffusers, registers, and grilles according to ASHRAE 70, "Method of Testing for Rating the Performance of Air Outlets and Inlets."

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas where diffusers, registers, and grilles are to be installed for compliance with requirements for installation tolerances and other conditions affecting performance of equipment.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Install diffusers, registers, and grilles level and plumb.
- B. Ceiling-Mounted Outlets and Inlets: Drawings indicate general arrangement of ducts, fittings, and accessories. Air outlet and inlet locations have been indicated to achieve design requirements for air volume, noise criteria, airflow pattern, throw, and pressure drop. Make final locations where indicated, as much as practicable. For units installed in lay-in ceiling panels, locate units in the center of panel. Where architectural features or other items conflict with installation, notify Architect for a determination of final location.
- C. Install diffusers, registers, and grilles with airtight connections to ducts and to allow service and maintenance of dampers, air extractors, and fire dampers.

3.3 CLEANING

- A. After installation of diffusers, registers, and grilles, inspect exposed finish. Clean exposed surfaces to remove burrs, dirt, and smudges. Replace diffusers, registers, and grilles that have damaged finishes.

3.4 CONTRACTOR STARTUP AND REPORTING

- A. After installation, adjust diffusers, registers, and grilles to air patterns indicated, or as directed, before starting air balancing.

END OF SECTION 233713

SECTION 238119 - SELF-CONTAINED AIR-CONDITIONERS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes packaged, outdoor, central-station air-handling units (rooftop units) with the following components and accessories:
 - 1. Direct-expansion cooling.
 - 2. Gas furnace.
 - 3. Economizer outdoor- and return-air damper section.
 - 4. Integral, space temperature controls.
 - 5. Roof curbs.

1.2 DEFINITIONS

- A. DDC: Direct-digital controls.
- B. RTU: Rooftop unit. As used in this Section, this abbreviation means packaged, outdoor, central-station air-handling units. This abbreviation is used regardless of whether the unit is mounted on the roof or on a concrete base on ground.
- C. Supply-Air Fan: The fan providing supply air to conditioned space. "Supply air" is defined as the air entering a space from air-conditioning, heating, or ventilating apparatus.

1.3 SUBMITTALS.

- A. Product Data: Include refrigerant, rated capacities, operating characteristics, furnished specialties, and accessories.
 - 1. Performance at ARI standard conditions and at conditions scheduled (provide EER).
 - 2. Performance at ARI standard unloading conditions and at conditions scheduled (provide EER).
 - 3. Refrigerant type and capacity of unit.
 - 4. Characteristics of safety relief valves.
 - 5. Minimum entering condenser-air temperature
 - 6. Sound data.
 - 7. Dimensions
 - 8. Unit construction.
 - 9. Components
 - 10. Included options
 - 11. Required service and operational clearances
 - 12. Furnished specialties and accessories.
 - 13. ASHRAE/IESNA 90.1 for energy compliance statement.
- B. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 1. Wiring Diagrams: Power, signal, and control wiring.
- C. Coordination Drawings: Plans and other details, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved:

1. Structural members to which RTUs will be attached.
2. Roof openings
3. Roof curbs and flashing.

D. Field quality-control test reports.

E. Operation and Maintenance Data: For RTUs to include in emergency, operation, and maintenance manuals.

F. Submit complete warranty information as described in Article 1.7.

1.4 QUALITY ASSURANCE

A. ARI Compliance:

1. Comply with ARI 210/240 and ARI 340/360 for testing and rating energy efficiencies for RTUs.
2. Comply with ARI 270 for testing and rating sound performance for RTUs.

B. ASHRAE Compliance:

1. Comply with ASHRAE 15 for refrigeration system safety.
2. Comply with ASHRAE 33 for methods of testing cooling and heating coils.
3. Comply with ASHRAE 62.1-2004 for condensate drain pans.
4. Comply with ASHRAE/IESNA 90.1 for minimum efficiency of heating and cooling.

C. NFPA Compliance: Comply with NFPA 90A and NFPA 90B.

D. UL Compliance: Comply with UL 1995.

E. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

1.5 DELIVERY, STORAGE AND HANDLING

A. Follow manufacturer's instructions for unloading, rigging and storage of equipment.

B. Maintain manufacturer's recommended temperature and humidity limits during storage and installation. Protect equipment from dirt, dust and other jobsite contaminants and conditions detrimental to the equipment.

1.6 WARRANTY

A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

B. Manufacturer's standard form in which manufacturer agrees to replace components of RTUs that fail in materials or workmanship within specified warranty period. Provide warranty on materials and labor for 18 months starting from date of delivery, or one year from date of preliminary acceptance, whichever is longer.

1. Warranty Period for Compressors: Manufacturer's standard, but not less than five years from date of preliminary acceptance.

2. Warranty Period for Gas Furnace Heat Exchangers: Manufacturer's standard, but not less than ten years from date of preliminary acceptance.
3. Warranty Period for Solid-State Ignition Modules: Manufacturer's standard, but not less than one year from date of preliminary acceptance.
4. Warranty Period for Control Boards: Manufacturer's standard, but not less than one year from date of preliminary acceptance.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with requirements, provided on **LiDL National Account List**:

1. Lennox
2. Carrier
3. TBD

2.2 CAPACITIES AND CHARACTERISTICS:

- A. Efficiency:

1. Air conditioners shall have a EER better than ASHRAE Standard 90.1 – latest edition under ARI test procedures. When air conditioners with higher efficiencies than the Standard are scheduled on the drawings, the more efficient value shall be the minimum project requirement.

2.3 CASING

- A. General: Casings shall be formed and reinforced insulated panels, fabricated to allow removal for access to internal parts and components, with joints between sections sealed.
- B. Exterior casing material shall be galvanized steel with factory-painted finish, with pitched roof panels and knockouts with grommet seals for electrical and piping connections and lifting lugs. Cabinet shall be tested 1000 hour in salt spray test in compliance with ASTM B117.
- C. Casing Insulation shall comply with NFPA 90A or NFPA 90B and ASTM C 1071, Type I. Minimum thickness shall be 1 inch, with a hydrostatic coating on all surfaces in contact with the air stream.
- D. Condensate drain pans shall be formed sections of stainless steel, with a minimum ¾" NPT drain connection

2.4 FANS

- A. Evaporator fan shall be forward-curved, double-width, double inlet constant volume centrifugal. Fans shall be belt-driven with adjustable pulleys. Bearings shall be sealed and permanently lubricated.
1. Provide factory VFD with phase protection for variable volume fans per schedule.
- B. Condenser fans shall be propeller, mounted on shaft of permanently lubricated motor.

- C. Fan Motor: Comply with requirements in Division 23 Section "Common Motor Requirements for HVAC Equipment."
- D. Provide power exhaust fans, or standard exhaust fans per sheet schedules.
 - 1. Power exhaust fans shall be centrifugal type and be fully modulating, integrated with economizer operation.

2.5 COILS

- A. Evaporator and condenser refrigerant coils shall incorporate aluminum-plate fins mechanically bonded to seamless copper tubes. Coils shall be factory pressure tested per ARI standards and leak tested at 500 psig.
 - 1. Provide E-coating on condenser coils.

2.6 REFRIGERANT CIRCUIT COMPONENTS

- A. Compressor: Hermetic scroll, mounted on vibration isolators; with internal overcurrent and high-temperature protection. For units more than 7.5 tons, provide two compressors. Motor shall be suction gas cooled.
 - 1. For units operating with variable air volume control, provide digital scroll type compressor(s).
- B. Refrigeration Specialties:
 - 1. Refrigerant Charge: R-407 or 410A. Units shall be fully charged when delivered to the site.
 - 2. Expansion valve with replaceable thermostatic element.
 - 3. Refrigerant filter/dryer.
 - 4. Manual-reset high-pressure safety switch.
 - 5. Automatic-reset compressor motor thermal overload.
 - 6. Service ports installed in compressor suction and liquid lines.
 - 7. Crankcase heaters.
 - 8. Freezestats.

2.7 AIR FILTRATION

- A. Minimum arrestance according to ASHRAE 52.1, and a minimum efficiency reporting value (MERV) according to ASHRAE 52.2. Filters shall be glass fiber, two-inch thick (MERV 8).

2.8 GAS FURNACE

- A. Description: Factory assembled, piped, and wired; complying with ANSI Z21.47 and NFPA 54. ASHRAE 90.1 – latest edition compliant.
- B. Burners: Stainless steel with a minimum thermal efficiency of 80 percent.
 - 1. Fuel: Natural gas.
 - 2. Ignition: Electronically controlled electric spark.
- C. Heat-Exchanger: Stainless steel.

- D. Combustion system shall be induced-draft.
- E. Safeties: Provide high-temperature limit switches, flame rollout switch, flame proving controls and induced draft pressure sensor.

2.9 DAMPERS

- A. Outdoor- and Return-Air Mixing Dampers: Parallel- or opposed-blade galvanized-steel dampers mechanically fastened to cadmium plated for galvanized-steel operating rod in reinforced cabinet. Connect operating rods with common linkage and interconnect linkages so dampers operate simultaneously.
 - 1. Damper Motor: Modulating with adjustable minimum position.
 - 2. Relief-Air Damper: Gravity actuated with bird screen and hood.

2.10 ELECTRICAL POWER CONNECTION

- A. Provide for single connection of power to unit with unit-mounted disconnect switch accessible from outside unit and control-circuit transformer with built-in overcurrent protection.

2.11 CONTROLS

- A. Control equipment and sequence of operation are specified in Division 23 Section "Building Automation System (BAS)" and in the controls drawings.
- B. Provide CO2 sensor and controls where called out on drawing sheets supplied.

2.12 ACCESSORIES

- A. Duplex, 115-V, ground-fault-interrupter outlet with 15-A overcurrent protection. Include transformer if required.
- B. Filter differential pressure switch with sensor tubing on either side of filter. Set for final filter pressure loss.

2.13 ROOF CURBS

- A. Materials: Galvanized steel, insulated with corrosion-protection coating, watertight gaskets, and factory-installed wood nailer; complying with NRCA standards. Roof curb height shall be 14 inches minimum.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of RTUs.
- B. Examine roughing-in for RTUs to verify actual locations of piping and duct connections before equipment installation.
- C. Examine roofs for suitable conditions where RTUs will be installed.

- D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Roof Curb: Install on roof structure, level and secure, according to ARI Guideline B. Install RTUs on curbs and coordinate roof penetrations and flashing with roof construction specified in Division 07 Section "Roof Accessories." Secure RTUs to upper curb rail, and secure curb base to roof framing or concrete base with anchor bolts.

3.3 CONNECTIONS

- A. Install condensate drain, minimum connection size, with trap and indirect connection to nearest roof drain or area drain.
- B. Install piping adjacent to RTUs to allow service and maintenance.
 - 1. Gas Piping: Comply with applicable requirements in Division 23 Section "Natural Gas Piping." Connect gas piping to burner, full size of gas train inlet, and connect with union and shutoff valve with sufficient clearance for burner removal and service.
- C. Duct installation requirements are specified in other Division 23 Sections. Drawings indicate the general arrangement of ducts. The following are specific connection requirements:
 - 1. Install ducts to termination at top of roof curb.
 - 2. Remove roof decking only as required for passage of ducts. Do not cut out decking under entire roof curb.
 - 3. Connect supply ducts to RTUs with flexible duct connectors specified in Division 23 Section "Air Duct Accessories."
 - 4. Install return-air duct continuously through roof structure.

3.4 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust components, assemblies, and equipment installations, including connections. Report results in writing.
 - 1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing. Report results in writing.
- B. Tests and Inspections:
 - 1. After installing RTUs and after electrical circuitry has been energized, test units for compliance with requirements.
 - 2. Inspect for and remove shipping bolts, blocks, and tie-down straps.
 - 3. Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
 - 4. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- C. Remove and replace malfunctioning units and retest as specified above.

3.5 CLEANING AND ADJUSTING

- A. Occupancy Adjustments: When requested within 12 months of date of preliminary acceptance, provide on-site assistance in adjusting system to suit actual occupied conditions. Provide up to two visits to site during other-than-normal occupancy hours for this purpose.
- B. After completing system installation and testing, adjusting, and balancing RTU and air-distribution systems, clean filter housings and install new filters.

3.6 CONTRACTOR STARTUP AND REPORTING

- A. Engage a factory-authorized service representative to perform startup service.
- B. Complete installation and startup checks according to manufacturer's written instructions and do the following:
 - 1. Inspect for visible damage to unit casing.
 - 2. Inspect for visible damage to furnace combustion chamber.
 - 3. Inspect for visible damage to compressor, coils, and fans.
 - 4. Inspect internal insulation.
 - 5. Verify that labels are clearly visible.
 - 6. Verify that clearances have been provided for servicing.
 - 7. Verify that controls are connected and operable.
 - 8. Verify that filters are installed.
 - 9. Clean condenser coil and inspect for construction debris.
 - 10. Clean furnace flue and inspect for construction debris.
 - 11. Connect and purge gas line.
 - 12. Remove packing from vibration isolators.
 - 13. Verify lubrication on fan and motor bearings.
 - 14. Inspect fan-wheel rotation for movement in correct direction without vibration and binding.
 - 15. Adjust fan belts to proper alignment and tension.
 - 16. Start unit according to manufacturer's written instructions.
 - a. Start refrigeration system.
 - b. Do not operate below recommended low-ambient temperature.
 - c. Complete startup sheets and attach copy with Contractor's startup report.
 - 17. Inspect and record performance of interlocks and protective devices; verify sequences.
 - 18. Operate unit for an initial period as recommended or required by manufacturer.
 - 19. Perform the following operations for both minimum and maximum firing. Adjust burner for peak efficiency.
 - a. Measure gas pressure on manifold.
 - b. Inspect operation of power vents.
 - c. Measure combustion-air temperature at inlet to combustion chamber.
 - d. Measure flue-gas temperature at furnace discharge.
 - e. Perform flue-gas analysis. Measure and record flue-gas carbon dioxide and oxygen concentration.
 - f. Measure supply-air temperature and volume when burner is at maximum firing rate and when burner is off. Calculate useful heat to supply air.
 - 20. Calibrate thermostats.
 - 21. Adjust and inspect high-temperature limits.
 - 22. Inspect outdoor-air dampers for proper stroke and interlock with return-air dampers.

23. Start refrigeration system and measure and record the following when ambient is a minimum of 15 deg F above return-air temperature:
 - a. Coil leaving-air, dry- and wet-bulb temperatures.
 - b. Coil entering-air, dry- and wet-bulb temperatures.
 - c. Outdoor-air, dry-bulb temperature.
 - d. Outdoor-air-coil, discharge-air, dry-bulb temperature.
24. Inspect controls for correct sequencing of heating, mixing dampers, refrigeration, and normal and emergency shutdown.
25. Measure and record the following minimum and maximum airflows. Plot fan volumes on fan curve.
 - a. Supply-air volume.
 - b. Return-air volume.
 - c. Relief-air volume.
 - d. Outdoor-air intake volume.
26. Simulate maximum cooling demand and inspect the following:
 - a. Compressor refrigerant suction and hot-gas pressures.
 - b. Short circuiting of air through condenser coil or from condenser fans to outdoor-air intake.
27. Verify operation of remote panel including pilot-light operation and failure modes. Inspect the following:
 - a. High-temperature limit on gas-fired heat exchanger.
 - b. Low-temperature safety operation.
 - c. Filter high-pressure differential alarm.
 - d. Economizer to minimum outdoor-air changeover.
 - e. Relief-air fan operation.
 - f. Smoke and firestat alarms.
28. After startup and performance testing and prior to preliminary acceptance, replace existing filters with new filters.

3.7 APPENDIX

A. Lennox Startup Instructions

1. Installing contractor must complete Contractor Acknowledgement Form and submit to Lennox National Account Services (NAS) to request the specific date for the EOC and an additional Contractor Acknowledgement Form for the specific date for the Commissioning. If either/both of the dates need to be rescheduled, the installing contractor must communicate the reschedule date to NAS at a minimum of 1 week prior in order to prevent possibility of extra charges (up to \$3000).
2. Lennox EOC and Commissioning scopes of work.
3. Letter from Lennox Technical Support documenting the proper procedure to be followed in order to operate the units during construction.
4. Directions for the installing contractor to set up the Prodigy board for units to run in temporary mode until EMS controls are installed. There is no need for temporary thermostats, as all Lennox Emergence units have standard on board sensors (outside, return, and discharge air).

5. See equipment submittals for sensors being used for Lidl projects which are ordered by and shipped to the contractor ordering the RTUs and split systems from Lennox.

B. Lennox Contacts

Equipment Quotes, Ordering, Tracking, and Missing Parts/Accessories

Lennox National Account Advocate (NAA)

Royce Hamilton

972-497-6788 800-367-6285 Opt #1

LennoxNationalAccounts@LennoxInd.com

Damaged Equipment, Installation Assistance, Equipment Alarms/Not Operating Properly

Lennox Technical Support

800-367-6285 Opt #2

NationalAccountsTechnicalSupport@LennoxInd.com

EOC and/or Commissioning Questions or Scheduling

Lennox National Account Services (NAS)

Courtney Marrero

800-333-4001

LennoxServices@LennoxNAS.com

Lidl/Lennox National Account Questions

Lennox National Account Escalation

Ron Johnston

National Account Manager

(704) 621-2606 Cell

Ronald.Johnston@LennoxInd.com



Contractor Acknowledgement Form

Requested EOC/Commissioning date _____ SAP Order# _____

Minimum 3 weeks notice is required for scheduling all services.

Contact Information

Date _____ Installing Contractor _____
 Project Name _____ Site Contact _____
 Store # _____ Phone Number _____
 Project Street Address _____ Email Address _____
 Project City/State/Zip _____

Complete checklist to indicate that ALL items will be complete and ready for Commissioning services

HVAC READY CHECKLIST	YES	NO
RTU's have permanent power connected		
RTU's have gas lines connected and gas is available		
All rooftop unit GFIs have been wired and powered		
Economizer hoods are installed and caulked according to installation instructions		
Units have clean filters installed		
All diffusers, grilles and ducts are installed and dampers are in the open position		
Mechanical contractor will be onsite during start up procedure		
MSAV & VAV Commissioning (Additional to HVAC Ready Checklist)		
All VAV boxes and zone dampers must be at and stay at 100% open during the start up.		
Static pressure transducer must be installed, wired and powered past 1 st supply air elbow from unit or at least 15 feet from unit		
Discharge air sensor shipped with unit must be installed past 1 st supply air elbow from unit or at least 15 feet from unit		
L Connection Network® Zoning Control System		
Third party control systems (Field provided and field installed)		
L Connection® Controls Commissioning (Additional to HVAC Ready Checklist)		
Communication wire pulled, labeled, and verified for validity		
Devices mounted and labeled according to site plans		

Customer Signature _____

Date _____

Fax or E Mail to Lennox NAS As Soon As Possible prior to Contractor completion of work for final Commissioning date. Failure to properly complete all items before the Lennox NAS technicians arrive on site will result in additional trip charges. Return fees are paid by the Contractor based on travel requirements, not to exceed \$3,000.00

Fax To: Lennox National Account Services 214-576-3873 Att: Services
 E.Mail lennoxservices@lennoxnas.com Phone 1-800-333-4001

Lennox® Factory Equipment Operational Check: Emergence™ (EOC)\ L Series® RTU \ Variable Air Volume (VAV)\ Constant Volume (CV)\ Constant Air Volume w/Bypass (CAVB)\ Strategos™ Rooftop Units

Performance of Factory Equipment Operations Check shall provide verification and documentation of equipment condition, integrity of installation and operational performance with regard to Lennox® specifications. The following listed items shall be performed by a Lennox® factory trained service technician. The following equipment and installation integrity checks shall be performed as part of the Lennox® factory Equipment Operational Check. Any installer defects shall be noted and any factory defects shall be repaired. A report for each unit along with a summary report for the job site will be provided to the owner, installing contractor and general contractor upon completion.

Lennox representative will perform the following services during Factory Equipment Operations Check:

Unit Installation Check:

- A. Record RTU# - Unit Model # - Unit Serial # - Unit Catalog #
- B. Check curb installation for fit, air leakage, including optional vibration isolation devices and wind, or seismic restraints – verify per manufacturers installation manuals and project specifications.
- C. Check door alignment and adjust as necessary.
- D. Check insulation is secure and clean.
- E. Check installation of condensate trap and drain lines per Lennox® installation instructions.
- F. Check unit clearances and verify unit installation per Lennox® installation instructions.
- G. Check area for cleanliness and dispose of Lennox related debris.

Electrical Systems Check:

- A. Check/Record proper incoming voltage and phase – verify per Lennox® specifications for each specific unit.
- B. Verify installation and proper sizing of electrical disconnect or circuit breaker.
- C. Check electrical connections and tighten as needed.
- D. Check/Record unit's control transformer(s) for proper secondary voltage and adjust if necessary.
- E. Check/Record installation of wiring for optional Lennox® factory installed GFCI receptacle.
- F. Verify class 2 controls wiring installation to terminal board of unit to include any project specific thermostat wiring – relative humidity sensors – indoor air quality sensors – space temperature sensors.

Prodigy™ Control System Setup and Check:

- A. Record hardware and software versions of the Controller and display panel if present.
- B. Save profile and report to USB storage device for each unit (additional charge if requested by customer).

Control System/Sensor Check:

Thermostat/Thermostat Input (DDC type controls) Check:

- A. Record thermostat or DDC controller – make and model.
- B. Verify class 2 controls wiring installation to terminal board of unit.
- C. Verify Occupancy Input for Thermostat Mode of operation (OCP input).
- D. Verify Fan, and multi stage Thermostat/DDC inputs for equipment that has multi stage operation for correct termination (G/Y1/Y2/W1/W2 – Only requires G/Y1/W1 for basic discharge air control models).
- E. Verify type of Humiditrol® control and verify correct termination has been accomplished (HUM or DI4).
- F. Verify type of economizer enable control and verify correct termination has been accomplished (GLO).

Sensor Check

Zone Temperature Sensor Check – Lennox® non/communicating style

- A. Check/Verify correct termination of sensor wires.

Relative Humidity Sensor Check - Lennox® non/communicating (LOCAL) style

- A. Check/Verify correct termination of sensor wires.

Indoor Air Quality Sensor Check - Lennox® non/communicating (LOCAL) style

- A. Check/Verify correct termination of sensor wires.

Supply Static Pressure Sensor Check

- A. Check/Verify correct termination of wires/installation of remote mounted supply duct static pressure sensor.

Building Static Pressure Sensor Check

- A. Check/Verify correct termination of wires/installation of building static pressure sensor.

Discharge air Temperature Sensor – Remote / Unit Mounted –

- A. Check/Verify units utilizing discharge air control mode have a remote mounted discharge air temperature sensor.
B. Check/Verify correct termination of sensor wires.

Return Air/OA Sensors

- A. Check/Verify correct termination of sensor wires

Blower System Setup and Check:

- A. Check/Record condition and type of filters.
B. Check/Record correct blower rotation.
C. Check/Record amp draw is per Lennox® specifications.
D. For Lennox® single speed belt drive blowers: check belt tension and alignment. Check pulley set screws for proper torque. Adjust as needed.
E. For Lennox® two speed belt drive motors: set two speed operation and blower over-load settings. Check pulley set screws for proper torque. Adjust as needed.
F. For Lennox® ECM motors: check ECM motor output per Lennox® specifications.
G. For Lennox® factory installed VFD belt driven motors check belt tension & alignment. Check pulley set screws for proper torque. Adjust as needed.
H. For Lennox® Strategos™ MSAV units: verify motor set per Lennox® specifications. Check pulley set screws for proper torque. Adjust as needed.

Outdoor Air Accessory Check:

- A. Check operation of economizer or motorized outdoor air damper by driving it full open and closed.
B. Check/Record unit economizer operation mode and sensors are present.
C. Check/Record economizer setting.
D. Check operation of barometric relief damper (if installed).
E. For Lennox® single speed belt drive power exhaust fan: check belt tension & alignment. Check pulley set screws for proper torque. Adjust as needed.
F. For Lennox® factory installed VFD belt driven power exhaust fan: check belt tension & alignment. Check pulley set screws for proper torque. Adjust as needed.
a. Check and verify power exhaust operation.

Cooling System Set Up and Check

- A. Check coil integrity and cleanliness.
B. Visual leak check all circuits. Correct as needed if operation performance suggests a possible issue.
C. Confirm proper rotation for each compressor and correct phasing only if occurs within the unit as needed (We do not correct power supply phasing to unit disconnect). Record amp draw.

- D. Adjust refrigerant charge as needed.
- E. Record temperature drop across the evaporator coil in full cooling (All compressors running).

Humiditrol[®] System Set Up and Check

- A. Check number of Humiditrol[®] stages.
- B. Verify operation of Humiditrol[®] demand.

Gas Heating System Set Up and Check

- A. Record fuel type.
- B. Check installation of intake and exhaust hoods. Verify per Lennox[®] installation instructions.
- C. Check installation of combustion air inducer intake and exhaust extensions (if applicable). Verify per Lennox[®] installation instructions.
- D. Check installation of gas unions for leaks.
- E. Check/Record incoming gas pressure to unit.
- F. Check/Record manifold gas pressure from the outlet of the gas valve(s) per Lennox[®] specifications. Adjust pressure as needed.
- G. Verify/Record temperature rise across heat exchanger in full heat. Adjust airflow if necessary.

Electric Heat System Set Up and Check:

- A. Check/Record amp draw of the individual heating elements (blower on).
- B. Check/Record temperature rise across heating elements in full heat. Adjust airflow if necessary.

Necessary requirements that must be complete prior to Factory EOC:

- A. Complete installation of rooftop unit and accessories per mechanical drawings and Lennox Installation Instructions.
- B. Rooftop unit must be started and running (24 hours before scheduled start up with no heating or cooling call).
- C. Unit's return air filters must be new and at least equivalent to factory provided filters.
- D. Static pressure transducer must be installed, wired and powered past 1st supply air elbow from unit or at least 5 feet from unit, if applicable.
- E. Discharge air sensor shipped with unit must be installed past 1st supply air elbow from unit or at least 5 feet from unit, if applicable.
- F. All VAV boxes and zone dampers must be at and stay at 100% open during the start up, if applicable.

The Factory EOC does not include the following:

- A. Any repairs or adjustment not listed above.
- B. Test and Balance.
- C. Pulling of wires or installation of components, accessories or controls.
- D. Additional jobsite visits if start-up cannot be completed due to no fault of Lennox.
- E. If EOC is scheduled and cannot be completed due to site readiness as noted above, return trip charges will apply.
- F. Any custom settings or site specific setups.

Lennox® Factory Commissioning: Emergence™ RTU/Constant Air Volume (CV) W/Humiditrol

The following listed items shall be performed by a Lennox® factory trained commissioning technician. These Integrity checks and equipment setup procedures shall be performed as part of the Lennox® factory commissioning service. Any installer related defects shall be noted and any factory defects shall be repaired/scheduled for repair. A report for each unit along with a summary report for the job site will be provided to the owner, installing contractor and general contractor upon completion. Lennox® factory commissioning is performed to ensure that the Lennox® RTU controller are configured to function per the project specifications.

Unit Check:

- A. Record RTU# - Unit Model # - Unit Serial # - Unit Catalog #.
- B. Check door alignment and adjust as necessary.
- C. Check unit's factory installed accessories for proper wiring and operation.
- D. Check and note installation of any Lennox® provided field installed accessories and verify per Lennox® and project specifications.

Electrical Systems Check:

- A. Check for and record proper incoming voltage – verify per Lennox® specifications.
- B. Check and record unit's control transformer(s) for proper secondary voltage and adjust if necessary.
- C. Verify class 2 controls wiring installation to terminal board of unit to include any project specific thermostat/communication/sensor wiring.

Prodigy™ / IMC Control System – the following options are dependent upon unit and accessories:

- A. Complete guided setup of the Prodigy™ controller.
- B. Record hardware and software versions of the controller and display panel.
- C. Check/Verify/Program Lennox® controller for proper operation in the local controller (IMC/PRODIGY™) per project specifications.
- D. Check/Verify/Program Lennox® controller for proper system mode of operation in the local controller IMC/PRODIGY™ per project specifications.
- E. Save profile and report of equipment with USB storage device for each Prodigy™ controller.
 - a. Check/Verify/Program zone temperature setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - b. Check/Verify/Program relative humidity setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - c. Check/Verify/Program IAQ setpoints in the local controller (IMC/PRODIGY™) per project specifications (if applicable).
 - d. Check/Verify/Program discharge air heating/cooling setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - e. Check/Verify/Program Fresh Air Tempering heating/cooling setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - f. Check/Verify/Program economizer control mode setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - g. Check/Verify/Program motorized damper min position setpoints in the local controller (IMC/PRODIGY™) per project specifications.
 - h. Check/Verify/Program Lennox® ERS control mode of operation in the local controller (IMC/PRODIGY™) per project specifications.
 - i. Check/Verify/Program smoke shutdown mode of operation in the local controller (IMC/PRODIGY™) per project specifications.
 - j. Check/Verify/Program power exhaust mode of operation in the local controller (IMC/PRODIGY™) per project specifications.
 - k. Optional features. Only verified if called out in project specifications.

Control System/Sensor Check:**Thermostat/Thermostat Input (DDC type controls) Check:**

- A. Verify thermostat or DDC system make, model number.
- B. Verify occupancy input for thermostat mode of operation.
- C. Verify type of economizer enable control and verify correct termination has been accomplished (if applicable).

Sensor Check: (if applicable)**Zone Temperature Sensor Check – Lennox® non/communicating (LOCAL) style**

- A. Check/Verify correct termination of sensor wires.
- B. Check/Verify resistance value of temperature sensor if temperature reading is inaccurate.
- C. Check/Verify Lennox® sensor type matches job specification.

Relative Humidity Sensor Check - Lennox® non/communicating (LOCAL) style

- A. Check/Verify correct termination of sensor wires.
- B. Check/Verify DC voltage feedback of sensor data if inaccurate reading suspected.
- C. Check/Verify Lennox® sensor type matches job specification.

Indoor Air Quality Sensor Check - Lennox® non/communicating (LOCAL) style

- A. Check/Verify correct termination of sensor wires.
- B. Check/Verify DC voltage feedback of sensor data if inaccurate reading suspected.
- C. Check/Verify Lennox® sensor type matches job specification.
- D. Check/Verify IAQ Sensor is accurate.

Discharge air Temperature Sensor – Remote / Unit Mounted

- A. Check/Verify units utilizing discharge air control mode/ fresh air tempering control mode have a remote mounted discharge air temperature sensor.
- B. Check/Verify correct termination of sensor wires.
- C. Check/Verify sensor resistance values within Lennox® specification if temperature reading inaccurate.

Return Air/OA Sensors

- A. Check/Verify correct termination of sensor wires.
- B. Check/Verify sensor resistance values within Lennox® specification if temp reading inaccurate.

LON/BACnet module

- A. Check/Verify module integrity.
- B. Check/Verify control system wiring completion.

Supply Blower System Check:

- A. Check/Verify Lennox® local controller is configured for proper blower operation per project specifications.

Outdoor Air Accessory Check: (if applicable)

- A. Check operation of economizer or motorized outdoor air damper per project specifications.
- B. Check/Verify economizer sensors are present – Note deficiencies.
- C. Check/Record power exhaust type.
- D. Check damper position setting for manual OAD.

Cooling System Check:

- A. Operate stages of cooling and confirm correct staging sequence occurs based on the local controller (IMC/PRODIGY™) programmed parameters.

Heating System Check: (if applicable)

- A. Operate stages of heating and confirm correct staging sequence occurs based on the local controller (IMC/PRODIGY™) programmed parameters.

Post Operational Check:

- A. The post operational checks will be accomplished using the units primary means of control – otherwise software/local devices will be utilized to prove the units will function under normal operating conditions per project specifications.
- B. Perform cooling simulation test verify cooling stages per project specifications.
- C. Capture all report data and programming data via Lennox® USB or LConnection® software.
 - a. Perform heating simulation test verify heating stages per project specifications.
 - b. Perform ventilation operational test (if applicable) verify ventilation operation per project specifications.
 - c. Perform supply static setpoint / damper actuator control is fully functional.
 - d. Perform power exhaust operation and verify it is fully functional.
 - e. Optional features. Only verified if called out in project specifications.

Necessary requirements that must be complete prior to Factory Commissioning:

- A. Complete installation of rooftop unit and accessories per mechanical drawings and Lennox® Installation Instructions.
- B. Rooftop unit must be powered (24 hours before scheduled commissioning with no heating or cooling call).
- C. Unit's return air filters must be new and at least equivalent to factory provided filters.
 - a. Discharge air sensor shipped with unit must be installed past 1st supply air elbow from unit or at least 5 feet from unit.
 - b. Optional feature. Only verified if called out in project specifications.
- D. If gas heat units, gas supply pressure to units must be within Lennox specifications

The Factory Commissioning does not include the following and will adhere to:

- A. Any repairs or adjustment not listed above.
- B. Any sequence of operation unrelated to the Lennox® rooftop commissioning scope.
- C. Pulling of wires or installation of components/accessories/controls.
- D. Additional jobsite visits if commissioning cannot be completed due to no fault of Lennox®.
- E. If project specifications have not been presented the equipment will be configured for the Lennox® controller default parameters that will safely operate the equipment - Lennox® Commercial Controls Technical Support Team can be available to assist the contractors with technical questions throughout the project and after the commissioning by calling 1-800-308-5492 – for parts replacement and other inquiries please speak with your sales representative or call 1-800-4-Lennox.

Mechanical Contractor Equipment Startup

In regard to the startup of Lennox HVAC product, please see below excerpt from page 5 of our installations manual. This breaks down what is needed to keep from voiding the Lennox Limited Product Warranty if used during the construction phase.

Use of the unit as a construction heater or air conditioner is not recommended during any phase of construction. Very low return air temperatures, harmful vapors and operation of the unit with clogged or misplaced filters will damage the unit.

If this unit has been used for heating or cooling of buildings or structures under construction, the following conditions must be met or the warranty will be void:

- The vent hood must be installed per these installation instructions.
- A room thermostat must control the unit. The use of fixed jumpers that will provide continuous heating or cooling is not allowed. (Unit may run in Prodigy Stand Alone mode)
- A pre-filter must be installed at the entry to the return air duct.
- The return air duct must be provided and sealed to the unit.
- Return air temperature range between 55°F (13°C) and 80°F (27°C) must be maintained.
- Air filters must be replaced and pre-filters must be removed upon construction completion.
- The input rate and temperature rise must be set per the unit rating plate.
- The heat exchanger, components, duct system, air filters and evaporator coil must be thoroughly cleaned following final construction clean-up.
- The unit operating conditions (including airflow, cooling operation, ignition, input rate, temperature rise and venting) must be verified according to these installation instructions.

Q4. How can I setup the unit to operate temporarily using the factory-installed return air sensor, prior to installation of the A2 zone sensor? (How do I set the unit up for temporary heating and cooling?)

You can also refer to the HowTo video at the following link for help:

<http://www.lennoxcommercial.com/landing/prodigy/how-to-videos.asp>

In order to temporarily operate the unit with the installed return air sensors, go to

SETUP > NETWORK INTERGRATION > RTU STANDALONE press the **SAVE** button. Room sensor back up mode will appear and use the adjust value buttons until you see **= RETURN AIR BACKUP** and press the **SAVE** button. Continue through and set the desired temperature set points, until the display returns to the **SETUP** menu.

Since the unit is now running off of the factory installed return air sensor we need to make the blower run continuous so it has a good sample of the air temperature. To do this go to;

SETTINGS > RTU OPTION> EDIT PARAMETER=154

Press and hold the adjust up button until a parameter **154** appears and press **SAVE**. Press the adjust up button until **=ON-CONTINUOUS 1** is displayed and press **SAVE**. Press the back button as many times as needed to return to the unit status screen. After a five minute time delay the unit will start operation based on the temperature read at the return air sensor.

(Edit parameter not required for blower operations change. It can be done through settings/blower menu. Also, blower should not have to run continuously when run in standalone)

CAUTION: The use of this unit as a construction heater or air conditioner is not recommended during any phase of construction. Very low return air temperatures, harmful vapors and operation of the unit with clogged or misplaced filters will damage the unit. If this unit has been used for heating or cooling of buildings or structures under construction, the following conditions must be met or the warranty will be void:

- ❑ The vent hood must be installed per these installation instructions.
- ❑ A room thermostat must control the unit. The use of fixed jumpers that will provide continuous heating or cooling is not allowed.
- ❑ A pre-filter must be installed at the entry to the return air duct.
- ❑ The return air duct must be provided and sealed to the unit.
- ❑ Return air temperature range between 55°F (13°C) and 80°F (27°C) must be maintained.
- ❑ Air filters must be replaced and pre-filter must be removed upon construction completion.
- ❑ The input rate and temperature rise must be set per the unit rating plate.
- ❑ The heat exchanger, components, duct system, air filters and evaporator coil must be thoroughly cleaned following final construction clean-up.
- ❑ The unit operating conditions (including airflow, cooling operation, ignition, input rate, temperature rise and venting) must be verified according to these installation instructions.

Lennox® Emergence™ Prodigy 2.0: T&B Help – MSAV Units

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PRODIGY® 2.0**INSTALLATION AND SETUP GUIDE****Prodigy® 2.0 (M3 Unit Controller) Installation and Setup Guide**

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Commercial innovation at work



Introducing the Prodigy® Smartphone App

Intelligence in Control, now on the go!

Building upon our legacy of innovation, the Prodigy app represents the next generation of Lennox support tools. This unique tool is designed to help make technical support and troubleshooting easier for customers by putting the answers in the palm of their hands. Access to information while at a jobsite is the most critical and we now have an app for that!

Key Customer Benefits:

- Shorter service time with quick and easy access to information
- Enhanced troubleshooting with step by step guides
- Eliminate need for manuals at the jobsite

Feature Overview

- Watch "How to" videos on the go
- Access Prodigy related technical literature with one touch, with the ability to email links
- Alarm lookup tables with step by step troubleshooting guide
- Searchable list of configuration parameters with detailed description
- Frequently Asked Questions list

Availability

- Available **NOW** on iPhone and Android
- Free download from iTunes and Google Play
 - Keyword Search "Lennox Prodigy"



App Store



Google play

Notes for use by T&B Contractors:

- With the Prodigy 2.0 controller, MSAV is calibrated by target 100% CFM. The controller controls other CFM settings through internal calculations according to CFMs input in Setting menus.
- Any adjustments to Max. CFM airflow required by T&B must be performed by Blower sheave adjustment only and not in Prodigy controller settings or the VFD parameters.
Note: ECM/Direct Drive units will be adjusted by torque% on Prodigy controller.
- If adjustments are required to sheave, blower running amps should be re-verified.
- High and Low Minimum damper positions should be set by T&B on controller.
Note: Unit controller has factory defaults which may not be suitable in all applications.

To see current cfm settings follow these paths:

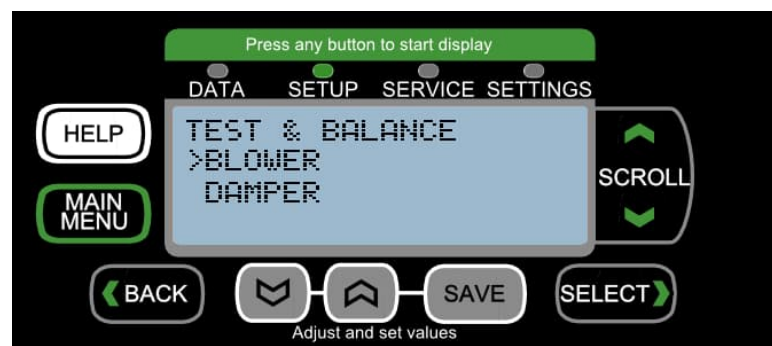
From the Main Menu, go to **SETUP** press **SELECT**.



Then **TEST & BALANCE**, press **SELECT**.



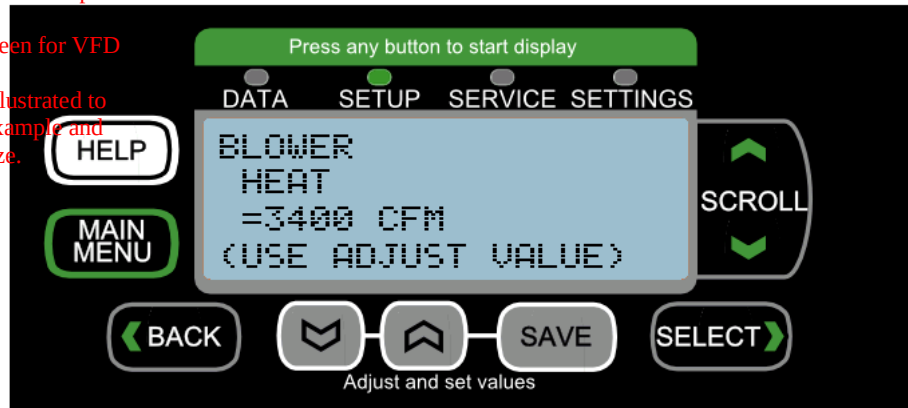
Scroll to **BLOWER** and press **SELECT**.



Heat CFM will show 1st then press SAVE.

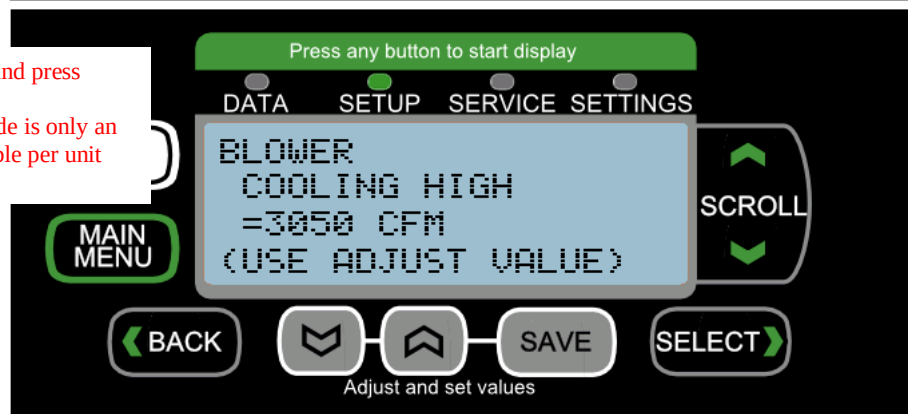
Note 1: this is the screen for VFD units.

Note 2: The setting illustrated to the right is only an example and adjustable per unit size.



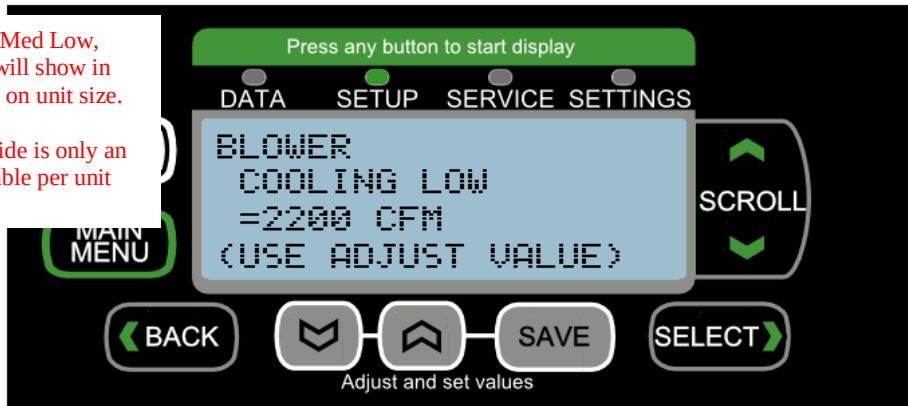
Cooling High CFM and press SAVE.

Note 1: Setting in slide is only an example and adjustable per unit size.



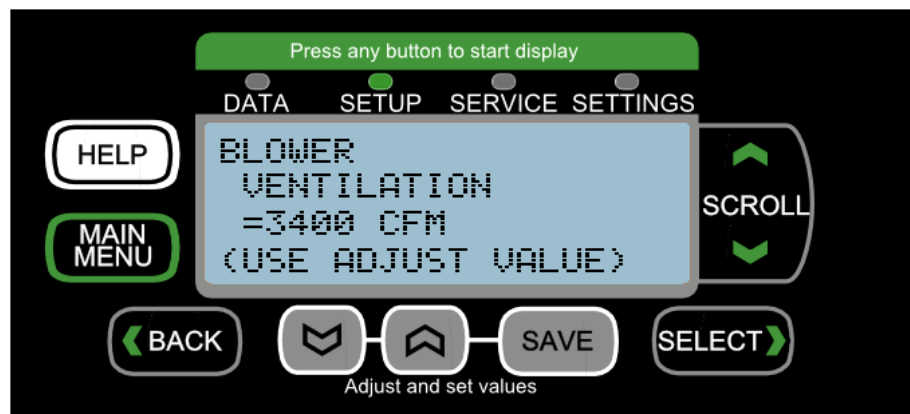
Cooling Med High, Med Low, Low CFM settings will show in sequence depending on unit size. Press SAVE again.

Note 1: Setting in slide is only an example and adjustable per unit size.



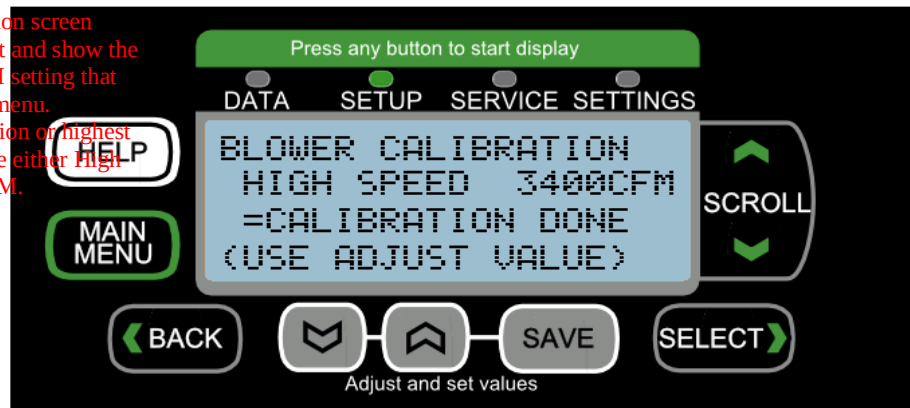
Ventilation CFM press SAVE.

Note 1: Setting in slide is only an example and adjustable per unit size.

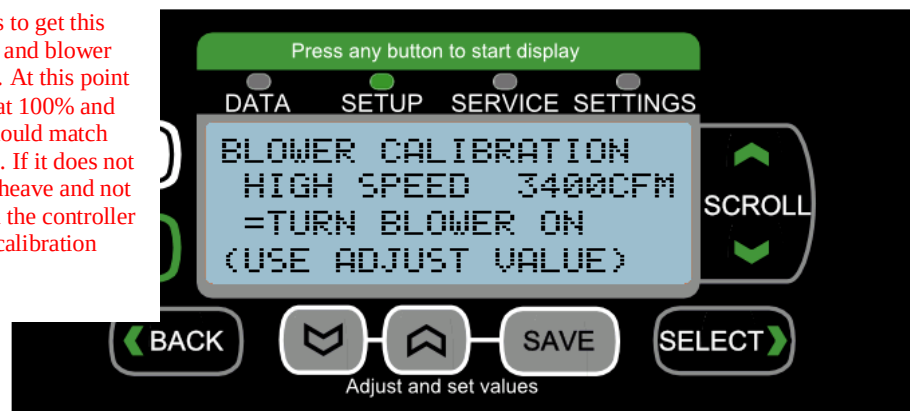


The blower calibration screen should come up next and show the highest blower CFM setting that was entered in this menu.

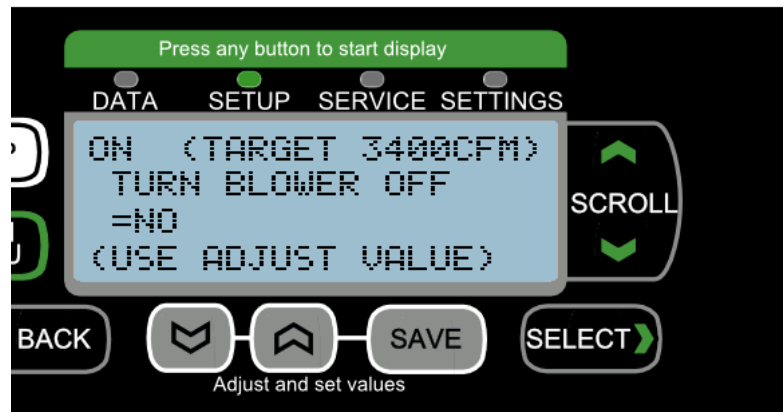
Note 1: The calibration or highest speed setting may be either High Cool or Heating CFM.



Use up/down arrows to get this screen. Press SAVE and blower will go to full speed. At this point the blower will run at 100% and measured airflow should match the calibration CFM. If it does not match, only adjust sheave and not any CFM settings in the controller until measured and calibration CFM match.



You can turn blower off after CFM measurement. If necessary, turn off unit and adjust pulley for target/calibration CFM. Retest until you have airflow set at highest CFM shown on screen. In this example, your Target would be 3400 CFM which is the highest CFM required at 100%. Once actual blower output is measured to be 3400 CFM during calibration menu, you can adjust the other CFM settings in the Prodigy directly, such as ventilation CFM.



Damper Menu Help

**All MSAV units have 2 Minimum damper positions for ventilation air.

For Damper Settings go
To SETTINGS>RTU
OPTIONS>DAMPER



*Depending on economizer type the menus will differ slightly.

Press SAVE



Press SAVE



*Depending on economizer type the menus will differ slightly.

Press SAVE.
Do not adjust.



Use UP/DOWN arrows to adjust Min. Damper For High Speed. Unit will run in high speed and damper will open to setting input, for testing, then press SAVE.



*Depending on economizer type the menus will differ slightly.

Use UP/DOWN arrows to adjust Min. Damper For Low Speed. Unit will run in low speed and damper will open to setting input, for testing, then press SAVE.

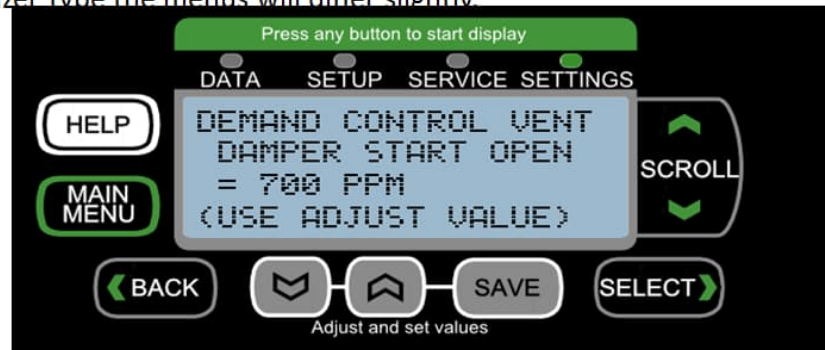


Next menus will be for DCV Options. Press SAVE.



*Depending on economizer type the menus will differ slightly.

The PPM settings should be already set, just press SAVE.



*Depending on economizer type the menus will differ slightly.

The Max Damper, FAH, and FAC options should be already set, just press SAVE until back to initial screen.



This should complete Final T&B setup required.

END OF SECTION 238119

SECTION 238126 - SPLIT-SYSTEM AIR-CONDITIONERS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes split-system air-conditioning units consisting of separate evaporator-fan and compressor-condenser components.

1.2 SUBMITTALS

- A. Product Data: For each type of product indicated. Include rated capacities, operating characteristics, and furnished specialties and accessories. Include performance data in terms of capacities, outlet velocities, static pressures, sound power characteristics, motor requirements, and electrical characteristics.
 - 1. Include documentation on refrigerants, including printed statement that refrigerants are free of HCFCs.
- B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.
 - 1. Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 2. Wiring Diagrams: For power, signal, and control wiring.
- C. Field quality-control reports.
- D. Operation and Maintenance Data: For split-system air-conditioning units to include in emergency, operation, and maintenance manuals.
- E. Recording(s) of training session(s).
- F. Warranty: Sample of special warranty.

1.3 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in Chicago Electric Code, by a qualified testing agency, and marked for intended location and application.
- B. ASHRAE Compliance:
 - 1. Fabricate and label refrigeration system to comply with ASHRAE 15, "Safety Standard for Refrigeration Systems."
 - 2. ASHRAE Compliance: Applicable requirements in ASHRAE 62.1-latest edition, Section 4 - "Outdoor Air Quality," Section 5 - "Systems and Equipment," Section 6 - "Procedures," and Section 7 - "Construction and System Start-Up."
- C. ASHRAE/IESNA Compliance: Applicable requirements in ASHRAE/IESNA 90.1-most current version.

1.4 COORDINATION

- A. Ground-Mounted Remote Condensing Units: Coordinate sizes and locations of concrete bases with actual equipment provided.
- B. Roof-Mounted Remote Condensing Units: Coordinate sizes and locations of roof curbs, equipment supports, and roof penetrations with actual equipment provided.

1.5 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.
- B. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of split-system air-conditioning units that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period:
 - a. Compressor: Six years from date of Preliminary Acceptance or Substantial Completion.
 - b. Parts: One year from date of Preliminary Acceptance or Substantial Completion.

1.6 EXTRA MATERIALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Filters: One set for each rooftop unit.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**:
 - 1. Lennox HVAC
 - 2. Carrier
 - 3. TBD

2.2 INDOOR UNITS (5 TONS OR LESS)

- A. Wall-Mounted, Evaporator-Fan Components:
 - 1. Cabinet: Enameled steel with removable panels on front and ends in color selected by Architect, and discharge drain pans with drain connection.
 - 2. Refrigerant Coil: Copper tube, with mechanically bonded aluminum fins and thermal-expansion valve. Comply with ARI 210/240.
 - 3. The unit must have low ambient cooling in temperatures as low as -22°F.
 - 4. Fan: Direct drive, centrifugal.
 - 5. Fan Motors:

- a. Comply with NEMA designation, temperature rating, service factor, enclosure type, and efficiency requirements specified in Division 23.
 - b. Multi-tapped, multi-speed with internal thermal protection and permanent lubrication.
 - c. Enclosure Type: Totally enclosed, fan cooled.
 - d. NEMA Premium (TM) efficient motors as defined in NEMA MG 1.
 - e. Controllers, Electrical Devices, and Wiring: Comply with requirements for electrical devices and connections specified in Division 26 Sections.
 - f. Mount unit-mounted disconnect switches on exterior or interior of unit.
6. Airstream Surfaces: Surfaces in contact with the airstream shall comply with requirements in ASHRAE 62.1-latest edition.
7. Condensate Drain Pans:
- a. Fabricated with one or two percent slope in at least two planes to collect condensate from cooling coils (including coil piping connections, coil headers, and return bends) and humidifiers, and to direct water toward drain connection.
 - 1) Length: Extend drain pan downstream from leaving face to comply with ASHRAE 62.1-latest edition.
 - 2) Pan Depth: 1 inch, minimum.
 - b. Single-wall, non-corrosive non-condensing with moisture tight seal.
 - c. Drain Connection: Located at lowest point of pan and sized to prevent overflow. Terminate with threaded nipple on one end of pan.
 - 1) Minimum Connection Size: NPS 1.
8. Air Filtration Section:
- a. Comply with NFPA 90A.
 - b. Minimum Arrestance: According to ASHRAE 52.1 and MERV according to ASHRAE 52.2.
 - c. Filter-Holding Frames: Arranged for flat or angular orientation, with access doors on both sides of unit. Filters shall be removable from one side or lifted out from access plenum.
 - d. Disposable Panel Filters:
 - 1) Thickness: 1 inch.
 - 2) Frame: Galvanized steel, with metal grid on outlet side, steel rod grid on inlet side, and hinged; with pull and retaining handles.

2.3 OUTDOOR UNITS (5 TONS OR LESS)

A. Air-Cooled, Compressor-Condenser Components:

- 1. Casing: Steel, finished with baked enamel, in color selected by Architect, with removable panels for access to controls, weep holes for water drainage, and mounting holes in base. Provide brass service valves, fittings, and gage ports on exterior of casing.
- 2. Compressor: Hermetically sealed with crankcase heater and mounted on vibration isolation device. Compressor motor shall have thermal- and current-sensitive overload devices, start capacitor, relay, and contactor.
 - a. Compressor Type: Scroll.

- b. Two-speed compressor motor with manual-reset high-pressure switch and automatic-reset low-pressure switch.
 - c. Refrigerant Charge: R-407C or R-410A as scheduled.
 - d. Refrigerant Coil: Copper tube, with mechanically bonded aluminum fins and liquid sub-cooler. Comply with ARI 210/240.
- 3. Fan: Aluminum-propeller type, directly connected to motor.
 - 4. Motor: Permanently lubricated, with integral thermal-overload protection.
 - 5. Low Ambient Kit: Permits operation down to -20 deg F.
 - 6. Mounting Base: Polyethylene.

2.4 ACCESSORIES

- A. Thermostat: Low voltage with sub-base to control compressor and evaporator fan.
 - 1. Compressor time delay.
 - 2. 24-hour time control of system stop and start.
 - 3. Liquid-crystal display indicating temperature, set-point temperature, time setting, operating mode, and fan speed.
 - 4. Fan-speed selection including auto setting.
- B. Automatic-reset timer to prevent rapid cycling of compressor.
- C. Refrigerant Line Kits: Soft-annealed copper suction and liquid lines factory cleaned, dried, pressurized, and sealed; factory-insulated suction line with flared fittings at both ends.
- D. Drain Hose: For condensate.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install units level and plumb.
- B. Install evaporator-fan components using manufacturer's standard mounting devices securely fastened to building structure.
- C. Install manufacturer provided condensate pump. Interlock condensate pump with operation of unit serving.
- D. Install ground-mounted, compressor-condenser components on 4-inches thick, reinforced concrete base that is 4 inches larger, on each side, than unit, with cast-in anchor-bolt inserts. Concrete, reinforcement, and formwork are specified in Division 03 Section "Cast-In-Place Concrete."
- E. Install roof-mounted, compressor-condenser components on equipment supports specified in Division 07 Section, "Roof Accessories." Anchor units to supports with removable, cadmium-plated fasteners.
- F. Install and connect pre-charged refrigerant pipes to component's quick-connect fittings. Install pipes to allow access to unit.

3.2 PIPE CONNECTIONS

- A. Pipe installation requirements are specified in other Division 22 Sections. Drawings indicate general arrangement of piping, fittings, and specialties.
- B. Pipe installations shall allow space for service and maintenance of system components.

3.3 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.
- B. Tests and Inspections:
 - 1. Leak Test: After installation, charge system and test for leaks. Repair leaks and repeat test until no leaks exist.
 - 2. Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
 - 3. Controls and Safeties: Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- C. Remove and replace malfunctioning units and retest as specified above.

3.4 STARTUP SERVICE

- A. Engage a factory-authorized service representative to perform startup service.
 - 1. Complete installation and startup checks according to manufacturer's written instructions.

3.5 TRAINING AND DEMONSTRATION

- A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain unit as specified below:
 - 1. Train Owner's maintenance personnel on procedures and schedules for starting up and shutting down, troubleshooting, servicing, and maintaining system components. The training will occur after the startup report has been provided to the Owner and the trainer will provide four Installation and Operation manuals for the use of the Owner's personnel during training.
 - 2. Review data in maintenance manuals. Refer to Division 01 Section "Operation and Maintenance Data." All required and recommended maintenance will be reviewed as well as operational troubleshooting. If the IOM does not include a written troubleshooting guide, one will be provided.
 - 3. Schedule training with Owner, through Architect, with at least seven days advance notice.
- B. Demonstrate proper operation of equipment to commissioning agent, if one, and designated Owner's personnel. The scope of the demonstration shall include functional performance requirements under local control as well as any commissioning requirements in Division 01 and 23.
- C. Video record the training session(s). The manufacturer may submit a standard training video or training CD for review as an alternate to recording of the training session. The standard video

must be reviewed and accepted by the Owner and Commissioning Authority, if one, for the alternate to be acceptable.

END OF SECTION 238126

SECTION 260503 - GENERAL REQUIREMENTS FOR ELECTRICAL SYSTEMS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes general administrative, material, and procedural requirements for electrical installations. The following administrative and procedural requirements are included in this Section to expand the requirements specified in Division 01.
 - 1. Electrical product general requirements and accesses.
 - 2. Substitutions.
 - 3. Submittals.
 - 4. Alternates.
 - 5. Coordination drawings.
 - 6. Record documents.
 - 7. Maintenance manuals.
 - 8. Rough-ins.
 - 9. Electrical installations.
 - 10. Cutting and patching.

1.2 DEFINITIONS

- A. Definitions:
 - 1. Listed: Equipment or materials included in a list published by an organization acceptable to the authority having jurisdiction and concerned with product evaluation, that maintain periodic inspection of production of listed equipment or materials, and whose listing states either that the equipment or material meets appropriate designated standards or has been tested and found suitable for use in a specified manner.
 - 2. Labeled: Equipment or materials to which has been attached a label, symbol, or other identifying mark of an organization that is acceptable to the authority having jurisdiction and concerned with product evaluation that maintains periodic inspection of production of labeled equipment or materials and by whose labeling the manufacturer indicates compliance with appropriate standards or performance in a specified manner.
 - 3. General Explanation: A substantial amount of specification language consists of definitions of terms found in other Contract Documents, including Drawings. (Drawings are recognized as being diagrammatic in nature and not completely descriptive of the requirements indicated thereon). Certain terms used in Contract Documents are defined in this article. Definitions and explanations contained in this Section are not necessarily either complete or exclusive but are general for the Work to the extent that they are not stated more explicitly in another element of the Contract Documents.
 - 4. General Requirements: The provisions or requirements of other Division 01 Sections apply to entire work of the Contract and where so indicated, to other elements which are included in the project.
 - 5. Indicated: The term "indicated" is a cross-reference to graphic representations, notes or schedules on drawings, to other paragraphs or schedules in the specifications, and to similar means of recording requirements in the Contract Documents. Where terms such as "shown," "noted," "scheduled," and "specified" are used in lieu of "indicated," it is for purpose of helping the reader locate cross- references, and no limitation of location is intended except as specifically noted.

6. Directed, Requested, etc: Where not otherwise explained, terms such as "directed", "requested", "authorized", "selected", "reviewed", "required", "accepted", and "permitted" mean "directed by Architectural/Engineer", and similar phrases. However, no such implied meaning will be interpreted to extend the Architect/Engineer's responsibility into the Contractor's area of construction supervision.
7. Furnish: Except as otherwise defined in greater detail, the term "furnish" is used to mean supply and deliver to the project site, ready for unloading, unpacking, assembly, installation, and similar operations, as applicable in each instance.
8. Install: Except as otherwise defined in greater detail, the term "install" is used to describe operations at the job site, including unloading, unpacking, assembly, erection, placing, anchoring, mounting, connecting, testing, protecting and cleaning, placing in working condition and similar operations, as applicable in each instance.
9. Provide: Except as otherwise defined in greater detail, the term "provide" means to furnish and install, complete and ready for intended use, as applicable in each instance.
10. Installer: The term "installer" is defined as the entity (person or firm) engaged by the contractor, its subcontractor or sub-subcontractor for performance of a particular unit of work at the project site. It is a general requirement that such entities (installers) be expert in the operation they are engaged to perform.
11. Connect: The term "connect" means to provide power sources, overcurrent devices, raceway, conductors, terminations, insulation and other materials required for the operation and control of the equipment noted by the term.
12. Wiring: The term wiring means all raceways, fittings, conductors, connectors, tape, junction and outlet boxes, connectors, splices, and all other items necessary and/or required in connection with such work.
13. Conduit: The term conduit means the inclusion of all fittings, hangers, supports, sleeves, etc.
14. Concealed: The term concealed means embedded in masonry or other construction, installed behind wall furring or within partitions, or installed within suspended ceilings.
15. Exposed: The term exposed means not installed underground or concealed as defined above.
16. Accessible: The term accessible means being capable of being reached without the use of ladders or without climbing or crawling under, through or over obstacles such as other mechanical or electrical equipment, building members or structure, piping, ductwork or going through doors.

1.3 SUBMITTALS

- A. General: Follow the procedures specified in Division 01 Section "Submittal Procedures." If submittals include any deviations from specified equipment/materials, these deviations must be clearly identified. The reason for the deviation must also be indicated.
- B. Prepare shop drawings and obtain approvals from inspection authorities for power utility company, fire alarm and life systems, and other electrical installations requiring specific approval.
- C. Submit coordination drawings for areas specified and those areas defined as "problem" coordination areas during construction.
- D. Submit ¼" scaled coordination drawings in electronic PDF format of electrical equipment room layouts prior to issuing electrical equipment submittals to engineer of record. Layout shall be based on equipment being submitted for approval. In addition to electrical equipment room layouts, contractor shall also include all major conduit raceway systems 2" conduit trade size and larger for both underground and overhead including all pull and junction boxes. The contractor's electrical coordination drawings shall be fully coordinated with all trade contractor's

equipment and shall bear each trade contractor's approval stamp and signature on each coordination plan drawing submittal.

- E. Electronic drawing files in AutoCad of the electrical drawings for use in preparing submittals may be purchased from the engineer. These drawings will not be provided without charge to the contractor or any of the subcontractors.
- F. Documents will not be accepted for review unless:
 - 1. They comply as to number of copies and type of paper indicated in the General Requirements.
 - 2. They include complete information pertaining to appurtenances and accessories.
 - 3. They are submitted as a package where they pertain to related items.
 - 4. Where they consist of standard catalog sheets displaying other items which are not applicable, they are properly marked with the electrical data, product identification and accessories as related to this specific project.
 - 5. They indicate the project and address along with the Contractor's name, address and phone number.
 - 6. Where they consist of standard factory assembly or field installation drawings, they are properly marked with external connection identification as related to this specific project.
- G. Any materials, fixtures, apparatus, or equipment that are not in accordance with specification requirements can and will be rejected for use in this installation and construction.
- H. Any materials, fixtures, apparatus or equipment installed without stamped or written review will be removed by the Contractor and replaced with specified equipment at the direction of the Architect/Engineer and without recourse for additional compensation.
- I. Prepare and submit all shop drawings to governmental agencies and utility companies which are required by these agencies for their approval.

1.4 QUALITY ASSURANCE

- A. Carefully examine the contract documents, visit the site, and become thoroughly familiar with the local conditions relating to the work. Failure to do so will not relieve the contractor of the obligations of the Contract.
- B. Discovery of any conflicting design information or any design intentions which are not readily interpreted shall be referred to the Architect/Engineers for further description or illustration prior to any product selection or execution of work.
- C. Discovery of any materials or equipment which are damaged, unsuitable, incompatible, or non-compliant with any applicable codes, laws, ordinances or other regulations shall be brought to the direct attention of the Architect/Engineer.
- D. Should there be any discrepancies or question of intent, refer the matter to the Architect/Engineer for a final decision before ordering any equipment or materials and before starting any relating work.
 - 1. In case of conflict between project specifications and drawings, the Contractor shall assume the more expensive method for purposes of bidding, unless the Architect/Engineer rules otherwise.

- E. Manufacturers of equipment shall be firms regularly engaged in manufacturing factory fabricated systems and equipment whose products have been in satisfactory use in similar service for not less than 5 years.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Delivery product to the project properly identified with names, model numbers, types grades, compliance labels, and other information needed for identification.

1.6 RULES AND REGULATIONS

- A. Work and materials shall conform to and be execute, inspected and tested in accordance with the latest edition of the governing rules and regulations of Federal, State and Local governmental agencies.
- B. Other codes which will apply to this installation include the current editions of:
 - 1. ANSI C2 – National Electrical Safety Code.
 - 2. ASME/ANSI A17.1 - Safety Code for Elevators and Escalators.
 - 3. ASTM - American Society for Testing and Materials.
 - 4. ICEA - Standards for Wire and Cable.
 - 5. IEEE Standards.
 - 6. IESNA Standards.
 - 7. NEMA Standards.
 - 8. NFPA 20 - Standards.
 - 9. OSHA Regulations.
 - 10. Underwriters Laboratories.
- C. Where governing codes indicate the Drawings and Specifications do not comply with the minimum requirements of applicable codes, the Contractor shall either notify the Architect/Engineer in writing during the bidding period identifying the revisions required to meet code requirements or provide an installation which will comply with the code requirements.
- D. Where regulations of electric utility and telephone companies apply, conformance with their regulations is mandatory and any costs involved shall be included in the Contract, with the exception of extra facility and other charges which are directly paid by the Owner.
- E. Where any materials, equipment or installation is not in compliance with the more stringent of the applicable codes, laws, ordinances, regulations and contract documents, they shall be entirely removed, replaced, modified or otherwise corrected at no additional cost to the Owner.

1.7 SUBSTITUTIONS

- A. The materials, products and equipment described in the Bidding Documents establish a standard of required functions, dimensions, appearance and quality to be met by any proposed substitutions.
- B. No substitution will be considered prior to receipt of Bids unless written request for approval has been received by the Architect/Engineer at least ten (10) days prior to the date for receipt of Bids. Each such request shall include the name of the material or equipment for which it is to be substituted and a complete description of the proposed substitute including drawings, cuts, performance, test data and warranties, and any other information necessary for an evaluation. A statement setting forth any changes in other materials, equipment or other work than incorporation of the substitute would require shall be included. The burden of proof of the merit

of the proposed substitute is upon the proposer. The Architect's decision of approval or disapproval of a proposed substitution shall be final.

- C. If the Architect approved any proposed substitution prior to receipt of Bids, such approval will be set forth in an Addendum. This Addendum shall then be issued to all Bidders.
- D. Requests for substitution shall be made only by a Bidder. Request for substitution received by the Architect from Sales representative, vendors, suppliers etc., are not acceptable.
- E. Refer to Division 01 Section "Substitution Procedures" for additional instructions on substitution.

1.8 RELATED WORK SPECIFIED IN OTHER SECTIONS

- A. Openings: Wall, floor, ceiling, and roof openings specifically shown and identified on the Architectural and Structural or Electrical Drawing shall be provided.
- B. Roof Sleeves: Electrical Contractor shall provide all roof penetrations and sleeves for all conduits serving HVAC equipment and shall be incorporated into the finished roofing and made watertight according to the roof construction type manufacturer's recommendations.
- C. Painting: Painting of all exposed-to-view conduit, pipes, unfinished hangers, supports, and equipment, insulated or not, in finished and unfinished areas, shall be provided. Furnish all manufactured equipment in factory-finished baked enamel, unless otherwise specified.

1.9 WIRING AND CONTROLS

- A. Wiring and controls associated with equipment shall be furnished, installed, and wired in accordance with the manufacturer's recommendations and applicable standards and codes. Provide installation instructions, locating dimensions, and wiring diagrams for the other trades. Supervise the installation and start-up and test the equipment unless otherwise specified.
- B. Equipment Furnished by Other Divisions: Equipment specified in other Divisions and requiring electrical supply shall be erected, aligned, leveled, and prepared for operation. Provided required controls and accessories along with installation instructions, diagrams, dimensions and supervision of installation and start-up. Provide the required electrical rough-ins, and confirm the electrical controls and accessories furnished under the specifications for the other Divisions. Install those controls and accessories not located in the mechanical piping and ductwork. Provide additional electrical controls, accessories, fittings, and devices not specified under the equipment but required for a finished, operating job. Make final electrical connections. Participate in the start-up and test services.

1.10 PERMIT AND INSPECTIONS

- A. Permits: Obtain and pay for all permits, bonds, licenses, tap-in fees, etc., required by the City, State, or other authority having jurisdiction over the work, as a part of the work of the affected Section.
- B. Inspections: Arrange and pay for all inspections required by the above when they become due as part of the work of the Sections affected. Conceal no work until approved by these governing authorities. Present the Contractor, Architect/Engineer with properly signed certificate of final inspection.

1.11 REVIEW OF MATERIALS

- A. Within 21 calendar days after award of this Contract, submit a typewritten list of all items of equipment and material proposed for installation on this Project to the Architect/Engineer for review for design conformance. Set forth the specification page number, manufacturer's name, model number, size, nonstandard accessories specified or required, and any other information required to identify each item.

1.12 PROJECT SITE CONDITIONS

- A. Inspect and examine the site before submitting the proposal. Note the location of any existing facilities, existing services or interference with other trades. Immediately contact Architect/Engineer indicating discrepancies. Failure to do so will not relieve the Contractor of the obligations of the Contract.
- B. Visit the site or premises in order to become familiar with job conditions. No extras will be allowed for work which could have been foreseen by an examination of the site or premises.
- C. Adjust work to meet actual conditions existing at the job.
- D. Inspect and examine the site to determine how equipment will be transported to final mounting locations. No extra charges will be allowed for moving, hoisting or otherwise transporting equipment to final mounting location.
- E. The location and routing of power and communication utility service raceway systems shown on the site plan and associated plan drawings are diagrammatic only. The contractor shall coordinate with all new and existing overhead and underground utilities (power utility service company, communication utility service company, etc.). Contractor shall verify and confirm with power and communication service utility providers the exact location and routing of proposed underground and overhead conduits prior to rough-in. Contractor shall obtain an official sign-off from such utility service providers prior to performing any work. Failure to do so shall not relieve the contractor from providing what is required by the utility service providers and shall not result in an added cost to the contract.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. All materials, unless otherwise specified, shall be new and be the standard products of the manufacturer. Seconds, rejects, or damaged materials will be rejected.
- B. The equipment to be provided under these Specifications shall be essentially the standard commercial grade product of the manufacturer. Where two or more units of the same class of equipment are required, these units shall be products of a single manufacturer.
- C. The listing of a manufacturer for certain equipment and systems does not indicate acceptance of a standard or catalogued item of equipment. All equipment and systems shall conform to the Specifications.

2.2 U.L. LISTING & LABELING

- A. All equipment shall bear the Underwriter's Laboratories (U.L.), or other approved agency, listing label.

- B. Wherein an item of equipment is specified to be U.L. Listed, the entire assembly shall be listed by Underwriters laboratories, Inc. Any modifications to suit the intent of the Specifications, shall be performed in accordance with the local codes and guidelines.

PART 3 - EXECUTION

3.1 ROUGH-IN

- A. Verify final locations for rough-ins with field measurements and with the requirements of the actual equipment to be connected.
- B. Refer to equipment specifications in Divisions 02 through 14, 22, 23, and 26 for rough-in requirements.

3.2 ELECTRICAL INSTALLATIONS

- A. General: Sequence, coordinate, and integrate the various elements of electrical systems, materials, and equipment. Comply with the following requirements:
 - 1. The Architect shall control the placement of wall and ceiling mounted electrical devices, fixtures, and outlets. The intent is to aesthetically locate fixtures/outlets by providing rough-in hardware, boxes and/or mounting plates, as required, when stud or furring may not be readily available for direct mounting. When drawing details are not available, consult with Architect's representative for actual placement.
 - 2. Coordinate electrical systems, equipment, and materials installation with other building components. Be responsible for any changes in openings and locations necessitated by the equipment installed.
 - 3. Verify all dimensions by field measurements.
 - 4. Arrange for chases, slots, and openings in other building components during progress of construction, to allow for electrical installations.
 - 5. Coordinate the installation of required supporting devices and sleeves to be set in poured-in-concrete and other structural components, as they are constructed.
 - 6. Sequence, coordinate, and integrate installations of electrical materials and equipment for efficient flow of the Work. Give particular attention to large equipment requiring positioning prior to closing in the building.
 - 7. Where mounting heights are not detailed or dimensioned, install systems, materials, and equipment to provide the maximum headroom possible.
 - 8. Install systems, materials, and equipment to conform with approved submittal data, including coordination drawings, to greatest extent possible. Conform to arrangements indicated by the Contract Documents, recognizing that portions of the Work are shown only in diagrammatic form. Where coordination requirements conflict with individual system requirements, refer conflict to the Architect.
 - 9. Switchgear/Switchboard/Motor Control Center Assembly Selection: The drawings indicate sizes, profiles, and dimensional requirements of assembly equipment. Equipment having equal performance characteristics and complying with indicated maximum dimensions and profiles may be considered, provided deviations do not change the design concept, intended performance, or code/future extension provision clearances. The burden of proof of equality is on the proposer a minimum of 10 days prior to bid.
 - 10. Protect all equipment and materials from the elements, dirt and other damage from the time it is removed from the point of storage until final acceptance.
 - 11. Equipment shall include the component parts thereof such as disconnect switches, motor starters, motors, drives, and guards necessary to the satisfactory and safe operation of the equipment.

12. Installation shall include setting equipment to accurate line and grade, leveling equipment, aligning equipment components, providing and installing couplings, bolts, guards and anchor bolts.
13. All tolerances in alignment and leveling, and the quality of workmanship for each class and stage of work shall be subject to manufacturer's installation instructions.
14. All manufacturers' finished equipment surfaces damaged during construction shall be brought to an "as new" condition by touch up or repairing. Any rust shall be completely removed and the surface primed prior to repainting.
15. Workmanship shall conform to the "Standard of Installation" published by the National Electrical Contractors Association.
16. Prior to start of utility trench work, notify the Architect of utilities that require excavations lower than the building foundations. Do not proceed with trench work without direction from the Architect to ensure the building foundations will not be compromised by the utility installation.
17. Provide all scaffolding, rigging, hoisting and services necessary for erection and delivery of equipment and apparatus furnished into the premises. These items shall be removed from the premises when no longer required.
18. No electrical equipment, raceways or other work of any kind shall be covered up or hidden from view before it has been examined and approved. Any unsatisfactory work or materials shall be removed and corrected immediately.
19. Install systems, materials, and equipment level and plumbing, parallel and perpendicular to other building systems and components.
20. Install electrical equipment to facilitate servicing, maintenance, and repair or replacement of equipment components. As much as practical, connect equipment for ease of disconnecting, with minimum of interference with other installations.
21. Install access panels or doors where units are concealed behind finished surfaces. Access panels and doors are specified in Division 26 Section "Basic Electrical Materials and Methods."
22. Install systems, materials, and equipment giving right-of-way priority to systems required to be installed at a specified slope.

3.3 MANUFACTURER'S DIRECTIONS AND SUPERVISION

- A. Where supervision by a manufacturer is specified, follow all instructions, recommended manufacturer and specified field tests, and other recommendations of the manufacturer. The manufacturer shall supervise the installation, connection, start-up, testing, and adjustment, instruction of the Owner and final tests of such equipment or system. Where two or more manufacturer's equipment are interrelated, take responsibility to coordinate their work and provide supervision.
- B. Have the manufacturer instruct the Owner in the proper operation and maintenance techniques of all equipment, systems, etc., at the time of completion of all work.
- C. Prior to final acceptance by the Owner prepare and submit to the Architect for review 3 copies of operation and maintenance (O and M) instructions in printed form for each item of equipment or system installed in the building. Complete instructions for each system shall be assembled and bound in a brochure. Detailed contents of the O and M manuals are as hereinafter specified. Refer to appropriate Division 01 Sections for general requirements affecting this work.

3.4 PAINTING

- A. Provide the prime painting of all equipment and materials furnished under Division 26 specifications, unless specifically stated otherwise. In general, all equipment except raceways

and galvanized boxes that are not provided with a factory-applied final finish shall be delivered to the job site with a shop-applied prime coat of paint.

3.5 TEST AND INSPECTION

- A. Upon completion of the work, notify the Architect in writing, that the entire electrical installation has been examined, inspected, tested, calibrated or adjusted as specified and that it is ready for final inspection. Work to be connected prior to final inspection and also to include all of the work specified for "Manufacturers' Directions and Supervision." Include documentation of specified testing and inspection.
- B. Prior to each inspection, provide a written certification that each system or piece of equipment to be operated during that test has been tested and does meet design performance criteria of the Contract Documents.
- C. On completion of work, obtain Certificates of Compliance, and approval or acceptance from all authorities having jurisdiction over the work, and deliver these certificates to the Architect. The work shall not be deemed to have reached a state of completion until the certificates have been delivered.

3.6 LOOSE EQUIPMENT

- A. Provide four keys for every different piece of electrical equipment which is equipped with a lock.
- B. Provide all other loose equipment specified/supplied for use with all systems.

3.7 SHOP DRAWINGS

- A. Refer to Division 01 for quantities and types of shop drawings.
- B. Required shop drawings shall be submitted in groups by systems. For example, all lighting fixtures, lamps, ballasts and accessories shall be submitted simultaneously in one package.
- C. Refer to individual Division 26 Sections for required shop drawings.
- D. Shop drawings submitted for other than those specifically required in the appropriate Specification Section will not be reviewed or returned.

3.8 OPERATION AND MAINTENANCE MANUALS

- A. Prepare maintenance manuals in accordance with Division 01 Section "Operating and Maintenance Data". In addition to the requirements specified in Division 01, include specific Division 26 Section requirements, and the following information for equipment items:
 - 1. Description of function, normal operating characteristics and limitations, performance curves, engineering data and tests, and complete nomenclature and commercial numbers of replacement parts.
 - 2. Manufacturer's printed operating procedures to include start-up, break-in, and routine and normal operating instructions; regulation, control, stopping, shutdown, and emergency instructions; and summer and winter operating instructions.
 - 3. Maintenance procedures for routine preventative maintenance and troubleshooting; disassembly, repair, and reassembly; aligning and adjusting instructions.
 - 4. Servicing instructions and lubrication charts and schedules.

- B. The minimum information that shall be furnished in the maintenance manual shall include the following:
1. Individual Characteristics for trouble shooting sequences for each item of:
 - a. Branch circuit panel.
 - b. Distribution panel.
 - c. Fire alarm system.
 - d. Individual motor starter.
 - e. Switchboard.
 - f. Transformer.
 - g. UPS.
 - h. Battery Inverter System.
 - i. Lighting Inverter.
 - j. Generator and associated transfer switch(es).
 2. Catalog cut sheets for every item for which a shop drawing is required.
 3. Schedule of loads served for each:
 - a. Branch circuit panel.
 - b. Distribution panel.
 - c. Switchboard.
 4. On-hand spare parts list and complete parts list for each:
 - a. Distribution panel.
 - b. Individual motor starter.
 - c. Switchboard.
 5. Tap setting schedule for each:
 - a. Transformer.
 6. Bolt tightening torques and inspection intervals on each:
 - a. Bolted bus connection.
 - b. Cable connection.
 - c. Miscellaneous bolted electrical connections.
 7. Manufacturers' recommended cleaning intervals and special procedures for each:
 - a. Cooling fins.
 - b. Dry-type transformer coil assembly.
 - c. Electrical equipment interior.
 - d. Electrical equipment ventilation opening.
 - e. Lighting fixture lenses, louvers and reflectors.
 8. Main and arcing contact adjustment and replacement for each:
 - a. Contactor.
 - b. Circuit breaker.
 - c. Fused switch.
 - d. Interrupter switch.

- e. Motor starter.
- 9. Calibration and exercise procedures and intervals for each:
 - a. Control system.
 - b. Emergency battery.
 - c. Molded case breaker.
 - d. Relay.
- 10. "As designed" and "as left" relay settings.
- 11. Testing interval and target values for ground fault protection circuit relays.
- 12. Testing and trouble shooting procedures unique to special systems.
- 13. Approved special construction details that differ from the details shown on Drawings.

3.9 COORDINATION DRAWINGS

- A. Prepare coordination drawings to a scale of $\frac{1}{4}" = 1'-0"$ or larger; detailing major elements, components, and systems of electrical equipment and materials in relationship with other systems, installations, and building components. Indicate locations where space is limited to installation and access and where sequencing and coordination of installations are of importance to the efficient flow of the Work, including (but not necessarily limited to) the following:
 - 1. Indicate the proposed locations of major raceway systems, equipment, and materials. Include the following:
 - a. Clearances for servicing equipment, including space for equipment disassembly required for periodic maintenance.
 - b. Exterior wall and foundation penetrations.
 - c. Fire-rated wall and floor penetrations.
 - d. Equipment connections and support details.
 - e. Sizes and location of required concrete pads and bases.
 - 2. Coordination drawings shall be provided by Division 26 for the following:
 - a. Access door locations.
 - b. Communication rooms.
 - c. Electrical equipment rooms.
 - d. Mechanical equipment rooms.
 - 3. Indicate scheduling, sequencing, movement, and positioning of large equipment into the building during construction.
 - 4. Prepare floor plans, elevations, and details to indicate penetrations in floors, walls, and ceilings and their relationship to other penetrations and installations.
 - 5. Advise Architect in event conflict occurs in location or connection of equipment. Bear costs resulting from failure to properly coordinate installation or failure to advise Architect of conflict.

3.10 RECORD "AS-BUILT" DOCUMENTS

- A. The Contractor shall be responsible for creating and providing accurate "As Built" drawings from field data collected during the course of the project. Field data is defined as information collected on site while constructing the project that is not available from the contract documents,

addenda, change orders, or site instructions. It is of importance that the Contractor record on the "As Builts" all field information relating to concealed conditions.

- B. The revisions from contractor markups and field inspection notes shall be transferred to the final as-built set of drawings. The final as-built drawings include modifications during construction, field requested changes, bulletins, shop drawing modifications, and contractor designs.
- C. At the completion of construction, all of the contractor's red-lined changes shall be transferred to PDF electronic drawings format, typically using the engineer of record's original design drawing files as the starting point. These PDF drawings shall then be considered the final "as-builts" and shall be submitted to the engineer and owner in PDF format as part of the project close out process. Hand marked-up as built drawings shall not be acceptable.
- D. Prepare record "As-Built" documents/drawings in accordance with the requirements in Division 01 Section "Project Closeout." In addition to the requirements specified in Division 01, indicate installed conditions for:
 - 1. Plan drawings indicating major raceway systems (100 amps and greater), conduit and wire sizes, routing and location, for both exterior and interior, locations of equipment, switchboards, panelboards, control devices; distribution and branch electrical circuitry; and fuse and circuit breaker size and arrangements.
 - 2. Major equipment locations (exposed and concealed), dimensioned from prominent building lines.
 - 3. Accurate electrical diagrams (electrical riser diagrams, wiring diagrams, etc.) reflective of all modifications during construction, field requested changes, bulletins, shop drawing modifications, and contractor designs.
 - 4. Accurate record of all switchboard and panelboard branch circuit directories/schedules which reflect actual installed circuits.
 - 5. Accurate branch circuit numbers and associated panelboard designations of all power receptacle outlet devices, electrical equipment, lighting fixtures and associated controls, etc.
 - 6. Contract Modifications and actual equipment and materials installed.
- E. Engage the services of a Professional Engineer registered in the State of Maryland to record the locations and invert elevations of underground raceway installations.

END OF SECTION 260503

SECTION 260505 - BASIC ELECTRICAL MATERIALS AND METHODS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the limited scope construction materials and methods for application with electrical installations as follows:
 - 1. Excavation for underground utilities and services, including underground raceways, vaults, and equipment.
 - 2. Miscellaneous materials for support of electrical materials and equipment.
 - 3. Concrete equipment bases.
 - 4. Cutting and patching for electrical construction.
 - 5. Touchup painting.
 - 6. Mounting heights.
 - 7. Electrical equipment coordination and installation.
 - 8. Sleeves for raceways and cables.
 - 9. Sleeve seals.
 - 10. Common electrical installation requirements.

1.2 DEFINITIONS

- A. The following definitions apply to excavation operations:
 - 1. Additional Excavation: Where excavation has reached required subgrade elevations, if unsuitable bearing materials are encountered, continue excavation until suitable bearing materials are reached. The Contract Sum may be adjusted by an appropriate Contract Modification
 - 2. Subbase: As used in this Section refers to the compacted soil layer used in pavement systems between subgrade and the pavement base course material.
 - 3. Subgrade: As used in this Section refers to the compacted soil immediately below the slab or pavement system.
 - 4. Unauthorized excavation consists of removal of materials beyond indicated subgrade elevations or dimensions without specific direction from the Architect.

1.3 SUBMITTALS

- A. Shop drawings are not required for material and equipment specified under this Section of the specifications.

1.4 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- B. Comply with local authorities having jurisdiction.
- C. Installer Qualifications: Engage an experienced installer for the installation and application of joint sealers, access panels, and doors.
- D. Qualify welding processes and welding operators in accordance with AWS D1.1 "Structural Welding Code – Steel."

1. Certify that welder has satisfactorily passed AWS qualification tests for welding processes involved and, if pertinent, has undergone recertification.

1.5 PROJECT CONDITIONS

A. Conditions Affecting Excavations: The following project conditions apply:

1. Maintain and protect existing building services which transit the area affected by selective demolition.
2. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by excavation operations.
3. Site Information: Subsurface conditions were investigated during the design of the Project. Reports of these investigations are available for information only; data in the reports are not intended as representations or warranties of accuracy or continuity of conditions. The Owner will not be responsible for interpretations or conclusions drawn from this information.
4. Existing Utilities: Locate existing underground utilities in excavation areas. If utilities are indicated to remain, support and protect services during excavation operations.
5. Remove existing underground utilities indicated to be removed.
 - a. Uncharted or Incorrectly Charted Utilities: Contact utility owner immediately for instructions.
 - b. Provide temporary utility services to affected areas. Provide minimum of 48-hour notice to Architect prior to utility interruption.
6. Use of explosives is not permitted.

B. Environmental Conditions: Apply joint sealers under temperature and humidity conditions within the limits permitted by the joint sealer manufacturer. Do not apply joint sealers to wet substrates.

1.6 COORDINATION

A. Coordinate chases, slots, inserts, sleeves, and openings with general construction work and arrange in building structure during progress of construction to facilitate the electrical installations that follow.

1. Set inserts and sleeves in poured-in-place concrete, masonry work, and other structural components as they are constructed.

B. Sequence, coordinate, and integrate installing electrical materials and equipment for efficient flow of the Work. Coordinate installing large equipment requiring positioning before closing in the building.

1. Coordinate installation and connection to exterior underground and overhead utilities and services, including provision for electricity-metering components.
2. Comply with requirements of authorities having jurisdiction and of utility company providing electrical power and other services.

C. Coordinate location of access panels and doors for electrical items that are concealed by finished surfaces. Access doors and panels are specified in Division 08 Section "Access Doors and Frames."

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following manufacturers:

1. Manufacturers:
 - a. Advance Products & Systems, Inc.
 - b. Calpico, Inc.
 - c. Metrafex Co.
 - d. Pipeline Seal and Insulator, Inc.

2.2 MISCELLANEOUS MATERIALS

- A. Material: Cold-formed steel, with corrosion-resistant coating acceptable to authorities having jurisdiction.
- B. Metal Items for Use Outdoors or in Damp Locations: Hot-dip galvanized steel.
- C. Pipe Sleeves: ASTM A 53, Type E, Grade A, Schedule 40, galvanized steel, plain ends.
- D. Expansion Anchors: Carbon-steel wedge or sleeve type.
- E. Toggle Bolts: All-steel springhead type.
- F. Powder-Driven Threaded Studs: Heat-treated steel.

2.3 CONCRETE BASES

- A. Concrete Forms and Reinforcement Materials: As specified in Division 03 Section "Cast-in-Place Concrete."
- B. Concrete: 3000 psi, 28 day compressive strength as specified in Division 03 Section "Cast-in-Place Concrete."

2.4 TOUCHUP PAINT

- A. For Equipment: Equipment manufacturer's paint selected to match installed equipment finish.
- B. Galvanized Surfaces: Zinc-rich paint recommended by item manufacturer.

2.5 SLEEVES FOR RACEWAYS

- A. Steel Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, galvanized steel, plain ends.
- B. Cast-Iron Pipe Sleeves: Cast or fabricated "wall pipe," equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop, unless otherwise indicated.
- C. Sleeves for Rectangular Openings: Galvanized sheet steel with minimum 0.052 inch or 0.138 inch thickness as indicated and of length to suit application.

- D. Coordinate sleeve selection and application with selection and application of firestopping specified in Division 07 Section "Penetration Firestopping."

2.6 SLEEVE

- A. Description: Modular sealing device, designed for field assembly, to fill annular space between sleeve and raceway.
 - 1. Sealing Elements: EPDM interlocking links shaped to fit surface of conduit. Include type and number required for material and size of raceway.
 - 2. Pressure Plates: Carbon steel. Include two for each sealing element.
 - 3. Connecting Bolts and Nuts: Carbon steel with corrosion-resistant coating of length required to secure pressure plates to sealing elements. Include one for each sealing element.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting installation and application of access panels. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.2 EXCAVATION

- A. Slope sides of excavations to comply with local codes and ordinances. Shore and brace as required for stability of excavation.
- B. Shoring and Bracing: Establish requirements for trench shoring and bracing to comply with local codes and authorities. Maintain shoring and bracing in excavations regardless of time period excavations will be open.
 - 1. Remove shoring and bracing when no longer required. Where sheeting is allowed to remain, cut top of sheeting at an elevation of 30 inches below finished grade elevation.
- C. Install sediment and erosion control measures in accordance with local codes and ordinances.
- D. Dewatering: Prevent surface water and subsurface or ground water from flowing into excavations and from flooding project site and surrounding area.
 - 1. Do not allow water to accumulate in excavations. Remove water to prevent softening of bearing materials. Provide and maintain dewatering system components necessary to convey water away from excavations.
 - 2. Establish and maintain temporary drainage ditches and other diversions outside excavation limits to convey surface water to collecting or run-off areas. Do not use trench excavations as temporary drainage ditches.
- E. Material Storage: Stockpile satisfactory excavated materials where directed, until required for backfill or fill. Place, grade, and shape stockpiles for proper drainage.
 - 1. Locate and retain soil materials away from edge of excavations. Do not store within drip-line of trees indicated to remain.

2. Remove and legally dispose of excess excavated materials and materials not acceptable for use as backfill or fill.
- F. Excavation for Underground Vaults and Electrical Structures: Conform to elevations and dimensions shown within a tolerance of plus or minus 0.10 foot; plus a sufficient distance to permit placing and removal of concrete formwork, installation of services, other construction, and for inspection.
1. Excavate, by hand, areas within dri-line of large trees. Protect the root system from damage and dry-out. Maintain moist conditions for root system and cover exposed roots with burlap. Paint root cuts of 1 inch in diameter and larger with emulsified asphalt tree paint.
 2. Take care not to disturb bottom of excavation. Excavate by hand to final grade just before concrete reinforcement is place.
- G. Trenching: Excavate trenches for electrical installations as follows:
1. Excavate trenches to the uniform width, sufficiently wide to provide ample working room and a minimum of 6 to 9 inches clearance on both sides of raceways and equipment.
 2. Excavate trenches to depth indicated or required.
 3. Limit the length of open trench to that in which installations can be made and the trench backfilled within the same day.
 4. Where rock is encountered, carry excavation below required elevation and backfill with a layer of crushed stone or gravel prior to installation of raceways and equipment. Provide a minimum of 6 inches of stone or gravel cushion between rock bearing surface and electrical installations.
- H. Cold Weather Protection: Protect excavation bottoms against freezing when atmospheric temperature is less than 35 deg F.
- I. Backfilling and Filling: Place soil materials in layers to required subgrade elevations for each area classification listed below, using materials specified in PART 2 of this Section.
1. Under walls and pavements, use a combination of subbase materials and excavated or borrowed materials.
 2. Under building slabs, use drainage fill materials.
 3. Under piping and equipment, use subbase materials where required over rock bearing surface and for correction of unauthorized excavation.
 4. For raceways less than 30 inches below surface of roadways, provide 4 inch thick concrete base slab support. After installation of raceways, provide a 4 inch thick concrete encasement (sides and top) prior to backfilling and placement of roadway subbase.
 5. Other areas, use excavated or borrowed materials.
- J. Backfill excavations as promptly as work permits, but not until completion of the following:
1. Inspection, testing, approval, and locations of underground utilities have been recorded.
 2. Removal of concrete formwork.
 3. Removal of shoring and bracing, and backfilling of voids.
 4. Removal of trash and debris.
- K. Placement and Compaction: Place backfill and fill materials in layers of not more than 8 inches in loose depth for material compacted by heavy equipment, and not more than 4 inches in loose depth for material compacted by hand-operated tampers.

- L. Before compaction, moisten or aerate each layer as necessary to provide optimum moisture content. Compact each layer to required percentage of maximum dry density or relative dry density for each area classification specified below. Do not place backfill or fill material on surfaces that are muddy, frozen, or contain frost or ice.
- M. Place backfill and fill materials evenly adjacent to structures, piping, and equipment to required elevations. Prevent displacement of raceways and equipment by carrying material uniformly around them to approximately same elevation in each lift.
- N. Compaction: Control soil compaction during construction, providing minimum percentage of density specified for each area classification indicated below.
 - 1. Percentage of Maximum Density Requirements: Company soil to not less than the following percentages of maximum density for soils which exhibit a well-defined moisture-density relationship (cohesive soils), determined in accordance with ASTM D 1557 and not less than the following percentages of relative density, determined in accordance with ASTM D 2049, for soils which will not exhibit a well-defined moisture-density relationship (cohesionless soils).
 - a. Areas Under Structures, Building Slabs and Steps, Pavements: Compact top 12 inches of subgrade and each layer of backfill or fill material to 90 percent maximum density for cohesive material, or 95 percent relative density for cohesionless material.
 - b. Areas Under Walkways: Compact top 6 inches of subgrade and each layer of backfill or fill material to 90 percent maximum density for cohesive material, or 95 percent relative density for cohesionless material.
 - c. Other Areas: Compact top 6 inches of subgrade and each layer of backfill or fill material to 85 percent maximum density for cohesive soils, and 90 percent relative density for cohesionless soils.
 - 2. Moisture Control: Where subgrade or layer of soil material must be moisture conditioned before compaction, uniformly apply water. Apply water in minimum quantity necessary to achieve required moisture content and to prevent water appearing on surface during, or subsequent to, compaction operations.
- O. Subsidence: Where subsidence occurs at electrical installation excavations during the period 12 months after Substantial Completion, remove surface treatment (i.e., pavement, lawn, or other finish), add backfill material, compact to specified conditions, and replace surface treatment. Restore appearance, quality, and condition of surface of finish to match adjacent areas.

3.3 ELECTRICAL EQUIPMENT INSTALLATION

- A. Headroom Maintenance: If mounting heights or other location criteria are not indicated, arrange and install components and equipment to provide the maximum possible headroom.
- B. Materials and Components: Install level, plumb, and parallel and perpendicular to other building systems and components, unless otherwise indicated.
- C. Equipment: Install to facilitate service, maintenance, and repair or replacement of components. Connect for ease of disconnecting, with minimum interference with other installations.
- D. Right of Way: Give to raceways and piping systems installed at a required slope.

3.4 FIRESTOPPING

- A. Apply firestopping to cable and raceway penetrations of fire-rated floor, ceiling and wall assemblies to achieve fire-resistance rating of the assembly. Provide 3M™ Fire Barrier Moldable Putty Pads. MPP+ shall be a one component, ready-to-use, intumescent elastomer capable of expanding a minimum of 3 times at 1000°F. The material shall be thixotropic and shall be applicable to overhead, vertical and horizontal firestops. Under normal conditions, 3M™ Fire Barrier Moldable Putty Pads MPP+ shall be noncorrosive to metal and compatible with synthetic cable jackets. The putty shall be listed by independent test agencies such as UL, Intertek or FM. 3M™ Fire Barrier Moldable Putty Pads MPP+ shall be tested to and pass the criteria of ASTM E 814 (UL 1479) Standard Test Method for Fire Tests of Penetration Firestop Systems and CAN/ULC S115 Standard Method of Fire Tests of Firestop Systems. 3M™ Fire Barrier Moldable Putty Pads MPP+ meets the requirements of the IBC, NFPA 5000, NEC (NFPA 70), NFPA 101.
- B. Additional firestopping materials and installation requirements are specified in Division 07 Section "Penetration Firestopping."
- C. Provide Endothermic Mat where electrical boxes, panels and equipment are installed within fire rated floors, walls and ceilings using 3M™ Instream™ Endothermic Mat products to match fire ratings.

3.5 CONCRETE BASES

- A. Construct concrete bases of dimensions indicated, but not less than 4 inches larger, in both directions, than supported unit. Follow supported equipment manufacturer's anchorage recommendations and setting templates for anchor-bolt and tie locations, unless otherwise indicated. Use 3000 psi, 28 day compressive-strength concrete and reinforcement as specified in Division 03 Section "Cast-in-Place Concrete."

3.6 SLEEVE INSTALLATION FOR ELECTRICAL PENETRATIONS

- A. Electrical penetrations occur when raceways penetrate concrete slabs, concrete or masonry walls, or fire-rated floor and wall assemblies.
- B. Coordinate sleeve selection and application with selection and application of firestopping specified in Division 07 Section "Penetration Firestopping."
- C. Concrete Slabs and Walls: Install sleeves for penetrations unless core-drilled holes or formed openings are used. Install sleeves during erection of slabs and walls.
- D. Use pipe sleeves unless penetration arrangement requires rectangular sleeved opening.
- E. Rectangular sleeve Minimum Metal Thickness:
 - 1. For sleeve cross-section rectangle perimeter less than 50 inches and no side greater than 16 inches, thickness shall be 0.052 inch.
 - 2. For sleeve cross-section rectangle perimeter equal to or greater than 50 inches, and 1 or more sides equal to or greater than 16 inches, thickness shall be 0.138 inch.
- F. Fire-Rated Assemblies: Install sleeves for penetrations of fire-rated floor and wall assemblies unless openings compatible with firestop system used are fabricated during construction of floor or wall.

- G. Cut sleeves to length for mounting flush with both surfaces of walls.
- H. Extend sleeves installed in floors 2 inches above finished floor level.
- I. Seal space outside of sleeves with grout for penetrations of concrete and masonry with approved joint compound for gypsum board assemblies.
- J. Interior Penetrations of Non-Fire-Rated Walls and Floors: Seal annular space between sleeve and raceway or cable, using joint sealant appropriate for size, depth, and location of joint. Refer to Division 07 Section "Joint Sealants" for materials and installation.
- K. Fire-Rated-Assembly Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at raceway and cable penetrations. Install sleeves and seal raceway penetration sleeves with firestop materials. Comply with Division 07 Section "Penetration Firestopping."
- L. Roof-Penetration Sleeves: Seal penetration of individual raceways with flexible boot-type flashing units applied in coordination with roofing work.
- M. Aboveground, Exterior-Wall Penetrations: Seal penetrations using sleeves and mechanical sleeve seals. Select sleeve size to allow for 1-inch annular clear space between pipe and sleeve for installing mechanical sleeve seals.
- N. Underground, Exterior-Wall Penetrations: Install cast-iron "wall pipes" for sleeves. Size sleeves to allow for 1-inch annular clear space between raceway or cable and sleeve for installing mechanical sleeve seals.

3.7 CUTTING AND PATCHING

- A. General: Perform cutting and patching in accordance with Division 01 Section "Cutting and Patching." In addition to the requirements specified in Division 01, the following requirements apply:
 - 1. Perform cutting, fitting, and patching of electrical equipment and materials required to:
 - a. Uncover Work to provide for installation of ill-timed Work.
 - b. Remove and replace defective Work.
 - c. Remove and replace Work not conforming to requirements of the Contract Documents.
 - d. Remove samples of installed Work as specified for testing.
 - e. Upon written instructions for the Architect, uncover and restore Work to provide for Architect observation of concealed work.
 - f.
 - 2. Cut, remove, and legally dispose of electrical equipment, components, and materials as indicated, including but not limited to removal of electrical items indicated to be removed and items made obsolete by the new Work.
 - 3. Protect the structure, furnishings, finishes, and adjacent materials not indicated or scheduled to be removed.
 - 4. Provide and maintain temporary partitions or dust barriers adequate to prevent the spread of dust and direct to adjacent areas.
 - 5. Protection of Installed Work: During cutting and patching operations, protect adjacent installations.
 - 6. Patch finished surfaces and building components using new materials matching materials and experienced Installers. Installers' qualifications refer to the materials and methods required for the surface and building components being patched.

3.8 FIELD QUALITY CONTROL

- A. Inspect installed components for damage and faulty work, including the following:
1. Excavation for underground utilities.
 2. Concrete bases.
 3. Cutting and patching for electrical construction.
 4. Touchup painting.
- B. Inspect installed sleeve and sleeve-seal installations and associated firestopping for damage and faulty work.

3.9 REFINISHING AND TOUCHUP PAINTING

- A. Refinish and touch up paint. Paint materials and application requirements are specified in Division 09 Section "Painting."
1. Clean damaged and disturbed areas and apply primer, intermediate, and finish coats to suit the degree of damage at each location.
 2. Follow paint manufacturer's written instructions for surface preparation and for timing and application of successive coats.
 3. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
 4. Repair damage to PVC or paint finishes with matching touchup coating recommended by manufacturer.

3.10 MOUNTING HEIGHTS

- A. Mounting heights of electrical items shall be as listed below, unless otherwise specified, or by the Architect/Engineer's field instructions. Dimensions are above finished floor, unless otherwise indicated. In areas where code requires different mount heights, as in hazardous areas, comply with code requirements.

- | | |
|---|------------------------------|
| 1. General Receptacles | -18" to C.L. |
| 2. Outdoor Receptacles | -24" to C.L. |
| 3. General Tele and Data Outlets | -18" to C.L. |
| 4. General Toggle Switches | -48" to C.L. |
| 5. Fire Alarm Pull Stations | -48" to C.L. |
| 6. Security Call Stations | -48" to C.L. |
| 7. General Television Outlets | -18" to C.L. |
| 8. Fire Alarm Audio Visual Devices | -80" to C.L. |
| 9. General Wall Mounted Lights | -90" to C.L. |
| 10. Corridor Wall Sconces (>4" deep) | -66" to C.L. |
| 11. Exit Signs | -90" to C.L. |
| 12. Individual Disconnects and Starters | -60" to C.L. |
| 13. Grouped Disconnects and Starters | >12" to C.L.
>72" to C.L. |
| 14. Panelboard Overcurrent Devices | >12" to C.L.
>72" to C.L. |
| 15. Grouped Utility Revenue Meters | >30" to C.L.
>66" to C.L. |

3.11 CLEANING AND PROTECTION

- A. On completion of installation, including outlets, fittings, and devices, inspect exposed finish. Remove burrs, dirt, paint spots, and construction debris.
- B. Protect equipment and installations and maintain conditions to ensure that coatings, finishes, and cabinets are without damage or deterioration at time of Substantial Completion.

END OF SECTION 260505

SECTION 260507 - TEMPORARY POWER AND COMMUNICATION SYSTEMS**PART 1 - GENERAL****1.1 SUMMARY**

- A. This Section includes requirements for temporary commercial AC power, telephone and security service utilities to Contractor's construction trailer and/or common construction support facilities.

1.2 USE CHARGES

- A. General: Cost or use charges for temporary facilities shall be included in the Contract Sum. Allow other entities to use temporary services and facilities without cost, including, but not limited to, Owner's construction forces, Architect, occupants of Project, testing agencies, and authorities having jurisdiction.

1.3 SUBMITTALS

- A. Site Plan: Show temporary construction trailer and/or building facilities, utility hookups, staging areas, and parking areas for construction personnel.
- B. Electrical contractor shall complete and submit a temporary electrical service meter application (load letter) to the local power utility company indicating requested electrical service size, voltage, and all associated loads. Load letter shall also include service start and completion dates

1.4 QUALITY ASSURANCE

- A. Electric Service: Comply with NECA, NEMA, and UL standards and regulations for temporary electric service. Install service to comply with NFPA 70, Local Building Code and Local Power utility service requirements.
- B. Tests and Inspections: Arrange for authorities having jurisdiction to test and inspect each temporary utility before use. Obtain required certifications and permits.

1.5 PROJECT CONDITIONS

- A. Temporary Use of Permanent Facilities: Installer of each permanent service shall assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before Owner's acceptance, regardless of previously assigned responsibilities.

PART 2 - EXECUTION

2.1 INSTALLATION, GENERAL

- A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.
- B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

2.2 TEMPORARY UTILITY INSTALLATION

- A. Temporary overhead/underground electric power utility service shall be coordinated with local power utility company. Provide all electrical equipment infrastructure, connections and extensions of utility services and distribution system of sufficient size, capacity, and power characteristics required for construction operations. Pay electric power service use charges for electricity used by all entities for construction operations.
 - 1. Provide a temporary commercial AC electrical service rated not less than 400 amps at 277/480V, 3-phase, 4-wire, 60 Hz. For construction equipment and a separate temporary commercial AC electrical service rated not less than 100 amps at 120/208V, 3-phase, 4-wire, 60 Hz. For construction trailer(s).
 - 2. Arrange with utility company, Owner, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services.
 - 3. Install electric power service overhead, unless otherwise indicated.
 - 4. Connect temporary service to Owner's existing power source, as directed by Owner.
- B. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, security, inspections, and traffic conditions.
 - 1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.
 - 2. Install lighting for Project identification sign.
- C. Telephone Service: Provide temporary telephone service in common-use facilities and/or contractor's construction trailer for use by all construction personnel. Install three T1 telephone line(s) for each field office, whether it is in the contractor's construction trailer or on-site building construction facility.
 - 1. Provide additional telephone lines for the following:
 - a. Provide a dedicated telephone line for each facsimile machine and computer in each field office.
 - b. Provide one telephone line(s) for Owner's use.
 - 2. At each telephone, post a list of important telephone numbers.
 - a. Police and fire departments.
 - b. Ambulance service.

- c. Contractor's home office.
 - d. Architect's office.
 - e. Engineers' offices.
 - f. Owner's office.
 - g. Principal subcontractors' field and home offices.
- 3. Provide superintendent with cellular telephone or portable two-way radio for use when away from field office.
- D. Electronic Communication Service: Provide temporary electronic communication service, including electronic mail, video surveillance security systems in common-use facilities and site.
 - 1. Provide one high speed DSL or T-1 line in primary field office.
 - 2. Coordinate with telephone utility company a minimum of four (4) weeks prior to construction start date.

END OF SECTION 260507

SECTION 260511 - CONDUCTORS AND CABLES FOR ELECTRICAL SYSTEMS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the following:
 - 1. Building wires and cables rated 600 V and less.
 - 2. Connectors, splices, and terminations rated 600 V and less.
 - 3. Sleeves and sleeve seals for cables.

1.2 DEFINITIONS

- A. EPDM: Ethylene-propylene-diene terpolymer rubber.
- B. NBR: Acrylonitrile-butadiene rubber.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Field quality-control test reports.

1.4 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- B. Comply with local authorities having jurisdiction.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Deliver wire and cables according to NEMA WC26.

1.6 COORDINATION

- A. Set sleeves in cast-in-place concrete, masonry walls, and other structural components as they are constructed.
- B. Coordinate layout and installation of cables with other contractors.

PART 2 - PRODUCTS**2.1 MANUFACTURERS**

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following manufacturers:
 - 1. Conductors and Cables
 - a. American Insulated Wire Corp.; a Leviton Company.
 - b. General Cable Corporation.

c. Southwire Company.

2. Connectors and Splices

- a. Hubbell Power Systems, Inc.
- b. O-Z/Gedney; EGS Electrical Group LLC.
- c. 3M; Electrical Products Division.
- d. Tyco Electronics Corp.

2.2 CONDUCTORS AND CABLES

- A. Copper Conductors: Comply with NEMA WC 70. Compact aluminum conductors compliant with NEMA WC 70 shall be acceptable for 100 amps and greater only.
- B. Conductor Insulation: Comply with NEMA WC 70 for Types THW, THHN-THWN, and XHHW.

2.3 CONNECTORS AND SPLICES

- A. Description: Factory-fabricated connectors and splices of size, ampacity rating, material, type, and class for application and service indicated.

PART 3 - EXECUTION

3.1 CONDUCTOR MATERIAL APPLICATIONS

- A. Feeders: Copper. Solid for No. 12 AWG and smaller; stranded for No. 10 AWG and larger. Conductors shall be increased if voltage drop exceeds 3% and conductor length is greater than 100 linear feet. Conduit size shall be increased as required to conform with electrical code.
- B. Branch Circuits: Copper. Solid for No. 12 AWG and smaller; stranded for No. 10 AWG and larger. Conductors shall be increased if voltage drop exceeds 3% and conductor length is greater than 100 linear feet. Conduit size shall be increased as required to conform with electrical code.

3.2 CONDUCTOR INSULATION AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

- A. Service Entrance: Type THHN-THWN, single conductors in raceway or Type XHHW, single conductors in raceway.
- B. Exposed Feeders: Type THHN-THWN, single conductors in raceway.
- C. Feeders Concealed in Ceilings, Walls, Partitions, and Crawlspace: Type THHN-THWN, single conductors in raceway.
- D. Feeders Concealed in Concrete, below Slabs-on-Grade, and Underground: Type THHN-THWN, single conductors in raceway.
- E. Exposed Branch Circuits, Including in Crawlspace: Type THHN-THWN, single conductors in raceway.
- F. Branch Circuits Concealed in Ceilings, Walls, and Partitions: Type THHN-THWN, single conductors in raceway.

- G. Branch Circuits Concealed in Concrete, below Slabs-on-Grade, and Underground: Type THHN-THWN, single conductors in raceway.
- H. Class 1 Control Circuits: Type THHN-THWN, in raceway.
- I. Class 2 Control Circuits: Type THHN-THWN in raceway.
- J. Fire Alarm Circuits: Red conductors, type THHN-THWN in raceway.

3.3 INSTALLATION OF CONDUCTORS AND CABLES

- A. Conceal cables in finished walls, ceilings, and floors, unless otherwise indicated.
- B. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.
- C. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips, that will not damage cables or raceway.
- D. Install exposed cables parallel and perpendicular to surfaces of exposed structural members and follow surface contours where possible.
- E. Identify and color-code conductors and cables according to Division 26 Section "Identification for Electrical Systems."
- F. Install wires and cables according to manufacturer's written instructions and NECA's "Standard of Care".
- G. Remove existing abandoned wires from raceway before pulling in new conductors.

3.4 CONNECTIONS

- A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A and UL 486B.
- B. Make splices and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.
- C. Wiring at Outlets: Install conductor at each outlet, with at least 6 inches of slack.
- D. Keep conductor splices to a minimum.
- E. Connect conductors to outlets and components as indicated and as instructed by manufacturers.

3.5 CONTRACTOR STARTUP AND REPORTING

- A. Perform tests and inspections and prepare test reports.
- B. Tests and Inspections:

1. After installing conductors and cables and before electrical circuitry has been energized, test service entrance and feeder conductors, and conductors feeding the following critical equipment and services for compliance with requirements.
 2. Perform each visual and mechanical inspection and electrical test.
- C. Test Reports: Prepare a written report to record the following:
1. Test procedures used.
 2. Test results that comply with requirements.
 3. Test results that do not comply with requirements and corrective action taken to achieve compliance with requirements.
- D. Remove and replace malfunctioning units and retest as specified above.

END OF SECTION 260511

SECTION 260519 - LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Copper building wire rated 600 V or less.
2. Metal-clad cable, Type MC, rated 600 V or less.
3. Connectors, splices, and terminations rated 600 V and less.

1.2 ACTION SUBMITTALS**A. Product Data:** For each type of product.**B. Sustainable Design Submittals:**

1. Product Data: For each conductor and cable indicating lead content

C. Product Schedule: Indicate type, use, location, and termination locations.**1.3 INFORMATIONAL SUBMITTALS****A. Field quality-control reports.****PART 2 - PRODUCTS****2.1 COPPER BUILDING WIRE****A. Description:** Flexible, insulated and uninsulated, drawn copper current-carrying conductor with an overall insulation layer or jacket, or both, rated 600 V or less.**B. Manufacturers:** Subject to compliance with requirements, provide products by one of the following:

1. Alpha Wire Company.
2. American Bare Conductor.
3. Belden Inc.
4. General Cable Technologies Corporation.
5. Southwire Company.

C. Standards:

1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
2. RoHS compliant.
3. Conductor and Cable Marking: Comply with wire and cable marking according to UL's "Wire and Cable Marking and Application Guide."

D. Conductors: Copper, complying with ASTM B 3 for bare annealed copper and with ASTM B 8 for stranded conductors.

E. Conductor Insulation:

1. Type THHN and Type THWN-2: Comply with UL 83.
2. Type THW and Type THW-2: Comply with NEMA WC-70/ICEA S-95-658 and UL 83.
3. Type XHHW-2: Comply with UL 44.

2.2 METAL-CLAD CABLE, TYPE MC

A. Description: A factory assembly of one or more current-carrying insulated conductors in an overall metallic sheath.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. American Bare Conductor.
2. Belden Inc.
3. General Cable Technologies Corporation.
4. Southwire Company.

C. Standards:

1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
2. Comply with UL 1569.
3. RoHS compliant.
4. Conductor and Cable Marking: Comply with wire and cable marking according to UL's "Wire and Cable Marking and Application Guide."

D. Circuits:

1. Single circuit and multicircuit with color-coded conductors.
2. Power-Limited Fire-Alarm Circuits: Comply with UL 1424.

E. Conductors: Copper, complying with ASTM B 3 for bare annealed copper and with ASTM B 8 for stranded conductors.

F. Ground Conductor: Bare.

G. Conductor Insulation:

1. Type TFN/THHN/THWN-2: Comply with UL 83.
2. Type XHHW-2: Comply with UL 44.

H. Armor: Steel, interlocked.

I. Jacket: PVC applied over armor.

2.3 CONNECTORS AND SPLICES

- A. Description: Factory-fabricated connectors and splices of size, ampacity rating, material, type, and class for application and service indicated; listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
- B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. 3M Electrical Products.
 - 2. AFC Cable Systems; a part of Atkore International.
 - 3. Hubbell Power Systems, Inc.
 - 4. Ideal Industries, Inc.
 - 5. O-Z/Gedney; a brand of Emerson Industrial Automation.
 - 6. Service Wire Co.

PART 3 - EXECUTION

3.1 CONDUCTOR MATERIAL APPLICATIONS

- A. Feeders: Copper; solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.
- B. Branch Circuits: Copper. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.
- C. Power-Limited Fire Alarm and Control: Solid for No. 12 AWG and smaller.

3.2 CONDUCTOR INSULATION AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

- A. Service Entrance: Type THHN/THWN-2, single conductors in raceway.
- B. Exposed Feeders: Type THHN/THWN-2, single conductors in raceway.
- C. Feeders Concealed in Ceilings, Walls, Partitions, and Crawlspace: Type THHN/THWN-2, single conductors in raceway.
- D. Feeders Concealed in Concrete, below Slabs-on-Grade, and Underground: Type THHN/THWN-2, single conductors in raceway.
- E. Feeders in Cable Tray: Type THHN/THWN-2, single conductors in raceway.
- F. Exposed Branch Circuits, Including in Crawlspace: Type THHN/THWN-2, single conductors in raceway.
- G. Branch Circuits Concealed in Ceilings, Walls, and Partitions: Type THHN/THWN-2, single conductors in raceway.
- H. Branch Circuits Concealed in Concrete, below Slabs-on-Grade, and

Underground: Type THHN/THWN-2, single conductors in raceway.

3.3 INSTALLATION OF CONDUCTORS AND CABLES

- A. Conceal cables in finished walls, ceilings, and floors unless otherwise indicated.
- B. Complete raceway installation between conductor and cable termination points according to Section 260533 "Raceways and Boxes for Electrical Systems" prior to pulling conductors and cables.
- C. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.
- D. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips, that will not damage cables or raceway.
- E. Install exposed cables parallel and perpendicular to surfaces of exposed structural members, and follow surface contours where possible.
- F. Support cables according to Section 260529 "Hangers and Supports for Electrical Systems."

3.4 CONNECTIONS

- A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A-486B.
- B. Make splices, terminations, and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.
 - 1. Use oxide inhibitor in each splice, termination, and tap for aluminum conductors.
- C. Wiring at Outlets: Install conductor at each outlet, with at least 6 inches of slack.

3.5 IDENTIFICATION

- A. Identify and color-code conductors and cables according to Section 260553 "Identification for Electrical Systems."
- B. Identify each spare conductor at each end with identity number and location of other end of conductor, and identify as spare conductor.

3.6 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

- A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies.

3.7 FIRESTOPPING

- A. Apply firestopping to electrical penetrations of fire-rated floor and wall assemblies to restore original fire-resistance rating of assembly.

END OF SECTION 260519

SECTION 260526 - GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes grounding and bonding systems and equipment.
- B. Section includes grounding and bonding systems and equipment, plus the following special applications:
 - 1. Underground distribution grounding.
 - 2. Foundation steel electrodes.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Sustainable Design Submittals:
 - 1. Product Data: For each conductor and cable indicating lead content.

1.3 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Plans showing dimensioned as-built locations of grounding features specified in "Field Quality Control" Article.
- B. Qualification Data: For testing agency and testing agency's field supervisor.
- C. Field quality-control reports.

1.4 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.
 - 1. Plans showing as-built, dimensioned locations of grounding features specified in "Field Quality Control" Article, including the following:
 - a. Ground rods.

PART 2 - PRODUCTS**2.1 SYSTEM DESCRIPTION**

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. Comply with UL 467 for grounding and bonding materials and equipment.

2.2 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
1. Burndy; Part of Hubbell Electrical Systems.
 2. O-Z/Gedney; a brand of Emerson Industrial Automation.
 3. Thomas & Betts Corporation; A Member of the ABB Group.

2.3 CONDUCTORS

- A. Insulated Conductors: Copper or tinned-copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction.
- B. Bare Copper Conductors:
1. Solid Conductors: ASTM B 3.
 2. Stranded Conductors: ASTM B 8.
 3. Tinned Conductors: ASTM B 33.
 4. Bonding Cable: 28 kcmil, 14 strands of No. 17 AWG conductor, 1/4 inch in diameter.
 5. Bonding Conductor: No. 4 or No. 6 AWG, stranded conductor.
 6. Bonding Jumper: Copper tape, braided conductors terminated with copper ferrules; 1-5/8 inches wide and 1/16 inch thick.
 7. Tinned Bonding Jumper: Tinned-copper tape, braided conductors terminated with copper ferrules; 1-5/8 inches wide and 1/16 inch thick.
- C. Grounding Bus: Predrilled rectangular bars of annealed copper, 1/4 by 4 inches in cross section, with 9/32-inch holes spaced 1-1/8 inches apart. Stand-off insulators for mounting shall comply

with UL 891 for use in switchboards, 600 V and shall be Lexan or PVC, impulse tested at 5000 V.

- D. Lead Content: Less than 300 parts per million.

2.4 CONNECTORS

- A. Listed and labeled by an NRTL acceptable to authorities having jurisdiction for applications in which used and for specific types, sizes, and combinations of conductors and other items connected.
- B. Welded Connectors: Exothermic-welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.
- C. Bus-Bar Connectors: Mechanical type, cast silicon bronze, solderless compression-type wire terminals, and long-barrel, two-bolt connection to ground bus bar.
- D. Bus-Bar Connectors: Compression type, copper or copper alloy, with two wire terminals.
- E. Beam Clamps: Mechanical type, terminal, ground wire access from four directions, with dual, tin-plated or silicon bronze bolts.
- F. Cable-to-Cable Connectors: Compression type, copper or copper alloy.
- G. Cable Tray Ground Clamp: Mechanical type, zinc-plated malleable iron.
- H. Conduit Hubs: Mechanical type, terminal with threaded hub.
- I. Ground Rod Clamps: Mechanical type, copper or copper alloy, terminal with hex head bolt.
- J. Lay-in Lug Connector: Mechanical type, copper rated for direct burial terminal with set screw.
- K. Service Post Connectors: Mechanical type, bronze alloy terminal, in short- and long-stud lengths, capable of single and double conductor connections.
- L. Signal Reference Grid Clamp: Mechanical type, stamped-steel terminal with hex head screw.
- M. Straps: Solid copper, cast-bronze clamp. Rated for 600 A.
- N. Tower Ground Clamps: Mechanical type, copper or copper alloy, terminal one-piece clamp.
- O. U-Bolt Clamps: Mechanical type, copper or copper alloy, terminal listed for direct burial.
- P. Water Pipe Clamps:
 - 1. Mechanical type, two pieces with zinc-plated bolts.
 - a. Material: Tin-plated aluminum.
 - b. Listed for direct burial.

2. U-bolt type with malleable-iron clamp and copper ground connector.

Q. Lead Content: Less than 300 parts per million.

2.5 GROUNDING ELECTRODES

- A. Ground Rods: Copper-clad steel; 3/4 inch by 10 feet.

PART 3 - EXECUTION

3.1 APPLICATIONS

- A. Conductors: Install solid conductor for No. 10 AWG and smaller, and stranded conductors for No. 8 AWG and larger unless otherwise indicated.
- B. Underground Grounding Conductors: Install bare tinned-copper conductor, No. ~~3/0~~ 1/0 AWG minimum.
 1. Bury at least 24 inches below grade.
- C. Grounding Bus: Install in electrical equipment rooms, in rooms housing service equipment, and elsewhere as indicated.
 1. Install bus horizontally, on insulated spacers 2 inches minimum from wall, 6 inches above finished floor unless otherwise indicated.
 2. Where indicated on both sides of doorways, route bus up to top of door frame, across top of doorway, and down; connect to horizontal bus.
- D. Conductor Terminations and Connections:
 1. Pipe and Equipment Grounding Conductor Terminations: Bolted connectors.
 2. Underground Connections: Welded connectors. except ~~at test wells~~ and as otherwise indicated.
 3. Connections to Ground Rods ~~at Test Wells~~: Bolted connectors.
 4. Connections to Structural Steel: Welded connectors.

3.2 GROUNDING AT THE SERVICE

- A. Equipment grounding conductors and grounding electrode conductors shall be connected to the ground bus. Install a main bonding jumper between the neutral and ground buses.

3.3 EQUIPMENT GROUNDING

- A. Install insulated equipment grounding conductors with all feeders and branch circuits.

- B. Install insulated equipment grounding conductors with the following items, in addition to those required by NFPA 70:
 - 1. Feeders and branch circuits.
 - 2. Lighting circuits.
 - 3. Receptacle circuits.
 - 4. Single-phase motor and appliance branch circuits.
 - 5. Three-phase motor and appliance branch circuits.
 - 6. Flexible raceway runs.
 - 7. Armored and metal-clad cable runs.
 - 8. Busway Supply Circuits: Install insulated equipment grounding conductor from grounding bus in the switchgear, switchboard, or distribution panel to equipment grounding bar terminal on busway.
- C. Air-Duct Equipment Circuits: Install insulated equipment grounding conductor to duct-mounted electrical devices operating at 120 V and more, including air cleaners, heaters, dampers, humidifiers, and other duct electrical equipment. Bond conductor to each unit and to air duct and connected metallic piping.
- D. Water Heater, Heat-Tracing, c old storage frost heave prevention system and Anti-frost Heating Cables: Install a separate insulated equipment grounding conductor to each electric water heater and heat-tracing cable. Bond conductor to heater units, piping, connected equipment, and components.
- E. Poles Supporting Outdoor Lighting Fixtures: Install grounding electrode and a separate insulated equipment grounding conductor in addition to grounding conductor installed with branch-circuit conductors.

3.4 INSTALLATION

- A. Grounding Conductors: Route along shortest and straightest paths possible unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.
- B. Ground Rods: Drive rods until tops are 2 inches below finished floor or final grade unless otherwise indicated.
 - 1. Interconnect ground rods with grounding electrode conductor below grade and as otherwise indicated. Make connections without exposing steel or damaging coating if any.
- C. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance except where routed through short lengths of conduit.

1. Bonding to Structure: Bond straps directly to basic structure, taking care not to penetrate any adjacent parts.
2. Bonding to Equipment Mounted on Vibration Isolation Hangers and Supports: Install bonding so vibration is not transmitted to rigidly mounted equipment.
3. Use exothermic-welded connectors for outdoor locations; if a disconnect-type connection is required, use a bolted clamp.

D. Grounding and Bonding for Piping:

1. Metal Water Service Pipe: Install insulated copper grounding conductors, in conduit, from building's main service equipment, or grounding bus, to main metal water service entrances to building. Connect grounding conductors to main metal water service pipes; use a bolted clamp connector or bolt a lug-type connector to a pipe flange by using one of the lug bolts of the flange. Where a dielectric main water fitting is installed, connect grounding conductor on street side of fitting. Bond metal grounding conductor conduit or sleeve to conductor at each end.
2. Water Meter Piping: Use braided-type bonding jumpers to electrically bypass water meters. Connect to pipe with a bolted connector.
3. Bond each aboveground portion of gas piping system downstream from equipment shutoff valve.

3.5 FIELD QUALITY CONTROL

A. Perform Megger Test Report tests and inspections.

B. Tests and Inspections:

1. After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements.
2. Inspect physical and mechanical condition. Verify tightness of accessible, bolted, electrical connections with a calibrated torque wrench according to manufacturer's written instructions.
3. Test completed grounding system at each location where a maximum ground-resistance level is specified, at service disconnect enclosure grounding terminal, and at individual ground rods. Make tests at ground rods before any conductors are connected.
 - a. Measure ground resistance no fewer than two full days after last trace of precipitation and without soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance.
 - b. Perform tests by fall-of-potential method according to IEEE 81.
4. Prepare dimensioned Drawings locating each ground rod and ground-rod assembly, and other grounding electrodes. Identify each by letter in alphabetical order, and key to the record of tests and observations. Include the number of rods driven and their depth at each location, and include observations of weather and other phenomena that may affect test results. Describe measures taken to improve test results.

- C. Grounding system will be considered defective if it does not pass tests and inspections.
- D. Prepare test and inspection reports.
- E. Report measured ground resistances that exceed the following values:
 - 1. Power and Lighting Equipment or System with Capacity of 500 kVA and Less: 10 ohms.
 - 2. Power and Lighting Equipment or System with Capacity of 500 to 1000 kVA: 5 ohms.
 - 3. Power and Lighting Equipment or System with Capacity More Than 1000 kVA: 3 ohms.
 - 4. Power Distribution Units or Panelboards Serving Electronic Equipment: 1 ohm(s).
- F. Excessive Ground Resistance: If resistance to ground exceeds specified values, notify Architect promptly and include recommendations to reduce ground resistance.

END OF SECTION 260526

SECTION 260529 - HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Steel slotted support systems.
2. Conduit and cable support devices.
3. Support for conductors in vertical conduit.
4. Structural steel for fabricated supports and restraints.
5. Mounting, anchoring, and attachment components, including powder-actuated fasteners, mechanical expansion anchors, concrete inserts, clamps, through bolts, toggle bolts, and hanger rods.
6. Fabricated metal equipment support assemblies.

B. Related Requirements:

1. Section 260548.16 "Seismic Controls for Electrical Systems" for products and installation requirements necessary for compliance with seismic criteria.

1.2 ACTION SUBMITTALS**A. Product Data:** For each type of product.**B. Shop Drawings:** For fabrication and installation details for electrical hangers and support systems.

1. Hangers. Include product data for components.
2. Slotted support systems.
3. Equipment supports.
4. Vibration Isolation Base Details: Detail fabrication including anchorages and attachments to structure and to supported equipment. Include adjustable motor bases, rails, and frames for equipment mounting.

C. Delegated-Design Submittal: For hangers and supports for electrical systems.

1. Include design calculations and details of hangers.
2. Include design calculations for seismic restraints.

1.3 INFORMATIONAL SUBMITTALS**A. Coordination Drawings:** Reflected ceiling plan(s) and other details, drawn to scale, and coordinated with each other, using input from installers of the items involved.

- B. Seismic Qualification Data: Certificates, for hangers and supports for electrical equipment and systems, accessories, and components, from manufacturer.
- C. Welding certificates.

1.4 QUALITY ASSURANCE

- A. Welding Qualifications: Qualify procedures and personnel.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Delegated Design: Engage a qualified professional engineer to design hanger and support system.
- B. Seismic Performance: Hangers and supports shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
 - 1. The term "withstand" means "the supported equipment and systems will remain in place without separation of any parts when subjected to the seismic forces specified and the supported equipment and systems will be fully operational after the seismic event."
 - 2. Component Importance Factor: 1.5.
- C. Surface-Burning Characteristics: Comply with ASTM E 84; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
 - 1. Flame Rating: Class 1.
 - 2. Self-extinguishing according to ASTM D 635.

2.2 SUPPORT, ANCHORAGE, AND ATTACHMENT COMPONENTS

- A. Steel Slotted Support Systems: Preformed steel channels and angles with minimum 13/32-inch- diameter holes at a maximum of 8 inches o.c. in at least one surface.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Allied Tube & Conduit; a part of Atkore International.
 - b. B-line, an Eaton business.
 - c. Flex-Strut Inc.
 - d. G-Strut.
 - e. Thomas & Betts Corporation; A Member of the ABB Group.
 - f. Unistrut; Part of Atkore International.

2. Standard: Comply with MFMA-4 factory-fabricated components for field assembly.
 3. Material for Channel, Fittings, and Accessories: Galvanized steel.
 4. Channel Width: Selected for applicable load criteria.
 5. Metallic Coatings: Hot-dip galvanized after fabrication and applied according to MFMA- 4.
 6. Nonmetallic Coatings: Manufacturer's standard PVC, polyurethane, or polyester coating applied according to MFMA-4.
 7. Painted Coatings: Manufacturer's standard painted coating applied according to MFMA- 4.
 8. Protect finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.
- B. Conduit and Cable Support Devices: Steel and malleable-iron hangers, clamps, and associated fittings, designed for types and sizes of raceway or cable to be supported.
- C. Support for Conductors in Vertical Conduit: Factory-fabricated assembly consisting of threaded body and insulating wedging plug or plugs for non-armored electrical conductors or cables in riser conduits. Plugs shall have number, size, and shape of conductor gripping pieces as required to suit individual conductors or cables supported. Body shall be made of malleable iron.
- D. Structural Steel for Fabricated Supports and Restraints: ASTM A 36/A 36M steel plates, shapes, and bars; black and galvanized.
- E. Mounting, Anchoring, and Attachment Components: Items for fastening electrical items or their supports to building surfaces include the following:
1. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete, steel, or wood, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
 - a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1) Sammys
 - 2) Hilti, Inc.
 - 3) MKT Fastening, LLC.
 2. Mechanical-Expansion Anchors: Insert-wedge-type, zinc-coated steel, for use in hardened portland cement concrete, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
 - a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1) Sammys
 - 2) B-line, an Eaton business.
 - 3) Hilti, Inc.

3. Concrete Inserts: Steel or malleable-iron, slotted support system units are similar to Sammys Press Anchor type XP-200 and comply with MFMA-4 or MSS SP-58.
4. Clamps for Attachment to Steel Structural Elements: MSS SP-58 units are suitable for attached structural element.
5. Through Bolts: Structural type, hex head, and high strength. Comply with ASTM A 325.
6. Toggle Bolts: All-steel springhead type.
7. Hanger Rods: Threaded steel.

2.3 FABRICATED METAL EQUIPMENT SUPPORT ASSEMBLIES

- A. Description: Welded or bolted structural-steel shapes, shop or field fabricated to fit dimensions of supported equipment.
- B. Materials: Comply with requirements for steel shapes and plates.

PART 3 - EXECUTION

3.1 APPLICATION

- A. Comply with the following standards for application and installation requirements of hangers and supports, except where requirements on Drawings or in this Section are stricter:
 1. NECA 1.
 2. NECA 101
 3. NECA 102.
 4. NECA 105.
 5. NECA 111.
- B. Comply with requirements in Section 078413 "Penetration Firestopping" for firestopping materials and installation for penetrations through fire-rated walls, ceilings, and assemblies.
- C. Comply with requirements for raceways and boxes specified in Section 260533 "Raceways and Boxes for Electrical Systems."
- D. Maximum Support Spacing and Minimum Hanger Rod Size for Raceways: Space supports for EMT, IMC, and RMC as required by NFPA 70. Minimum rod size shall be 1/4 inch in diameter.
- E. Multiple Raceways or Cables: Install trapeze-type supports fabricated with steel slotted or other support system, sized so capacity can be increased by at least 25 percent in future without exceeding specified design load limits.
 1. Secure raceways and cables to these supports with single-bolt conduit clamps.
- F. Spring-steel clamps designed for supporting single conduits without bolts may be used for 1- 1/2-inch and smaller raceways serving branch circuits and communication systems above suspended ceilings, and for fastening raceways to trapeze supports.

3.2 SUPPORT INSTALLATION

- A. Comply with NECA 1 and NECA 101 for installation requirements except as specified in this article.
- B. Raceway Support Methods: In addition to methods described in NECA 1, EMT IMC and RMC may be supported by openings through structure members, according to NFPA 70.
- C. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb.
- D. Mounting and Anchorage of Surface-Mounted Equipment and Components: Anchor and fasten electrical items and their supports to building structural elements by the following methods unless otherwise indicated by code:
 - 1. To Wood: Fasten with lag screws or through bolts.
 - 2. To New Concrete: Bolt to concrete inserts.
 - 3. To Masonry: Approved toggle-type bolts on hollow masonry units and expansion anchor fasteners on solid masonry units.
 - 4. To Existing Concrete: Expansion anchor fasteners.
 - 5. Instead of expansion anchors, powder-actuated driven threaded studs provided with lock washers and nuts may be used in existing standard-weight concrete 4 inches thick or greater. Do not use for anchorage to lightweight-aggregate concrete or for slabs less than 4 inches thick.
 - 6. To Steel: Beam clamps (MSS SP-58, Type 19, 21, 23, 25, or 27), complying with MSS SP-69.
 - 7. To Light Steel: Sheet metal screws.
 - 8. Items Mounted on Hollow Walls and Nonstructural Building Surfaces: Mount cabinets, panelboards, disconnect switches, control enclosures, pull and junction boxes, transformers, and other devices on slotted-channel racks attached to substrate by means that comply with seismic-restraint strength and anchorage requirements.
- E. Drill holes for expansion anchors in concrete at locations and to depths that avoid the need for reinforcing bars.

3.3 INSTALLATION OF FABRICATED METAL SUPPORTS

- A. Comply with installation requirements for site-fabricated metal supports.
- B. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor electrical materials and equipment.
- C. Field Welding: Comply with AWS D1.1/D1.1M.

END OF SECTION 260529

SECTION 260533 - RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Metal conduits, tubing, and fittings.
2. Nonmetallic conduits, tubing, and fittings.
3. Metal wireways and auxiliary gutters.
4. Surface raceways.
5. Boxes, enclosures, and cabinets.

B. Related Requirements:

1. Section 260543 "Underground Ducts and Raceways for Electrical Systems" for exterior ductbanks, manholes, and underground utility construction.
2. Section 270528 "Pathways for Communications Systems" for conduits, wireways, surface pathways, innerduct, boxes, faceplate adapters, enclosures, cabinets, and handholes serving communications systems.

1.2 ACTION SUBMITTALS**A. Product Data:** For each type of product.**B. Sustainable Design Submittals:**

1. Product Data: For solvents and adhesives, indicating VOC content.
2. Laboratory Test Reports: For solvents and adhesives, indicating compliance with requirements for low-emitting materials.

C. Shop Drawings: For custom enclosures and cabinets. Include plans, elevations, sections, and attachment details.**1.3 INFORMATIONAL SUBMITTALS****A. Coordination Drawings:** Conduit routing plans, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of items involved:

1. Structural members in paths of conduit groups with common supports.
2. HVAC and plumbing items and architectural features in paths of conduit groups with common supports.

B. Seismic Qualification Certificates: For enclosures, cabinets, and conduit racks and their mounting provisions, including those for internal components, from manufacturer.**PART 2 - PRODUCTS**

2.1 METAL CONDUITS, TUBING, AND FITTINGS

A. Manufacturers: Subject to compliance with requirements, provide products by the following:

1. Allied Tube & Conduit; a part of Atkore International.
2. Anamet Electrical, Inc.
3. O-Z/Gedney; a brand of Emerson Industrial Automation.
4. Republic Conduit.
5. Southwire Company.
6. Thomas & Betts Corporation; A Member of the ABB Group.

B. Listing and Labeling: Metal conduits, tubing, and fittings shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

C. GRC: Comply with ANSI C80.1 and UL 6.

D. ARC: Comply with ANSI C80.5 and UL 6A.

E. IMC: Comply with ANSI C80.6 and UL 1242.

F. PVC-Coated Steel Conduit: PVC-coated rigid steel conduit.

1. Comply with NEMA RN 1.
2. Coating Thickness: 0.040 inch, minimum.

G. EMT: Comply with ANSI C80.3 and UL 797.

H. FMC: Comply with UL 1; zinc-coated steel or aluminum.

I. LFMC: Flexible steel conduit with PVC jacket and complying with UL 360.

J. Fittings for Metal Conduit: Comply with NEMA FB 1 and UL 514B.

1. Conduit Fittings for Hazardous (Classified) Locations: Comply with UL 1203 and NFPA 70.
2. Fittings for EMT:
 - a. Material: Steel.
 - b. Type: Setscrew.
3. Expansion Fittings: PVC or steel to match conduit type, complying with UL 651, rated for environmental conditions where installed, and including flexible external bonding jumper.
4. Coating for Fittings for PVC-Coated Conduit: Minimum thickness of 0.040 inch, with overlapping sleeves protecting threaded joints.

K. Joint Compound for IMC, GRC, or ARC: Approved, as defined in NFPA 70, by authorities having jurisdiction for use in conduit assemblies, and compounded for use to lubricate and protect threaded conduit joints from corrosion and to enhance their conductivity.

2.2 NONMETALLIC CONDUITS AND FITTINGS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. CertainTeed Corporation.
 - 2. Condux International, Inc.
 - 3. Thomas & Betts Corporation; A Member of the ABB Group.
- B. Listing and Labeling: Nonmetallic conduits, tubing, and fittings shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- C. ENT: Comply with NEMA TC 13 and UL 1653.
- D. RNC: Type EPC-40-PVC, complying with NEMA TC 2 and UL 651 unless otherwise indicated.
- E. LFNC: Comply with UL 1660.
- F. Fittings for ENT and RNC: Comply with NEMA TC 3; match to conduit or tubing type and material.
- G. Fittings for LFNC: Comply with UL 514B.
- H. Solvents and Adhesives: As recommended by conduit manufacturer.
 - 1. VOC Content: 510 g/L or less for PVC conduit and fittings.

2.3 METAL WIREWAYS AND AUXILIARY GUTTERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Legrand, a Wiremold company.
 - 2. B-line, an Eaton business.
 - 3. Hoffman; a brand of Pentair Equipment Protection.
- B. Description: Sheet metal, complying with UL 870 and NEMA 250, Type 1 or Type 3R unless otherwise indicated, and sized according to NFPA 70.
 - 1. Metal wireways installed outdoors shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- C. Fittings and Accessories: Include covers, couplings, offsets, elbows, expansion joints, adapters, hold-down straps, end caps, and other fittings to match and mate with wireways as required for complete system.

2.4 BOXES, ENCLOSURES, AND CABINETS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Crouse-Hinds, an Eaton business.
 - 2. Hoffman; a brand of Pentair Equipment Protection.
 - 3. Hubbell Incorporated.
 - 4. Hubbell Incorporated; Wiring Device-Kellems.
 - 5. O-Z/Gedney; a brand of Emerson Industrial Automation.
 - 6. RACO; Hubbell.
 - 7. Thomas & Betts Corporation; A Member of the ABB Group.
- B. General Requirements for Boxes, Enclosures, and Cabinets: Boxes, enclosures, and cabinets installed in wet locations shall be listed for use in wet locations.
- C. Sheet Metal Outlet and Device Boxes: Comply with NEMA OS 1 and UL 514A.
- D. Cast-Metal Outlet and Device Boxes: Comply with NEMA FB 1, ferrous alloy, Type FD, with gasketed cover.
- E. Nonmetallic Outlet and Device Boxes: Comply with NEMA OS 2 and UL 514C.
- F. Metal Floor Boxes:
 - 1. Material: Cast metal or sheet metal.
 - 2. Type: Fully adjustable.
 - 3. Shape: Rectangular.
 - 4. Listing and Labeling: Metal floor boxes shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- G. Nonmetallic Floor Boxes: Nonadjustable, rectangular.
 - 1. Listing and Labeling: Nonmetallic floor boxes shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- H. Luminaire Outlet Boxes: Nonadjustable, designed for attachment of luminaire weighing 50 lb. Outlet boxes designed for attachment of luminaires weighing more than 50 lb shall be listed and marked for the maximum allowable weight.
- I. Paddle Fan Outlet Boxes: Nonadjustable, designed for attachment of paddle fan weighing 70 lb.
 - 1. Listing and labeling: Paddle fan outlet boxes shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- J. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.
- K. Cast-Metal Access, Pull, and Junction Boxes: Comply with NEMA FB 1 and UL 1773, cast aluminum with gasketed cover.

- L. Box extensions used to accommodate new building finishes shall be of same material as recessed box.
- M. Device Box Dimensions: 4 inches square by 2-1/8 inches deep.
- N. Gangable boxes are allowed.
- O. Hinged-Cover Enclosures: Comply with UL 50 and NEMA 250, Type 1 or Type 3R with continuous-hinge cover with flush latch unless otherwise indicated.
 - 1. Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
 - 2. Nonmetallic Enclosures: Plastic.
 - 3. Interior Panels: Steel; all sides finished with manufacturer's standard enamel.
- P. Cabinets:
 - 1. NEMA 250, Type 1 or Type 3R galvanized-steel box with removable interior panel and removable front, finished inside and out with manufacturer's standard enamel.
 - 2. Hinged door in front cover with flush latch and concealed hinge.
 - 3. Key latch to match panelboards.
 - 4. Metal barriers to separate wiring of different systems and voltage.
 - 5. Accessory feet where required for freestanding equipment.
 - 6. Nonmetallic cabinets shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

PART 3 - EXECUTION

3.1 RACEWAY APPLICATION

- A. Outdoors: Apply raceway products as specified below unless otherwise indicated:
 - 1. Exposed Conduit: GRC, IMC, or RNC, Type EPC-40-PVC.
 - 2. Concealed Conduit, Aboveground: GRC, IMC, EMT.
 - 3. Underground Conduit: RNC, Type EPC-40-PVC, direct buried concrete encased.
 - 4. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): LFMC.
 - 5. Boxes and Enclosures, Aboveground: NEMA 250, Type 3R.
- B. Indoors: Apply raceway products as specified below unless otherwise indicated.
 - 1. Exposed, Not Subject to Physical Damage: EMT.
 - 2. Exposed, Not Subject to Severe Physical Damage: EMT.
 - 3. Exposed and Subject to Severe Physical Damage: EMT ~~IMC~~. Raceway locations include the following:
 - a. Loading dock.
 - b. Corridors used for traffic of mechanized carts, forklifts, and pallet-handling units.
 - c. Mechanical mezzanine.

4. Concealed in Ceilings and Interior Walls and Partitions: EMT.
 5. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): FMC, except use LFMC in damp or wet locations.
 6. Damp or Wet Locations: IMC.
 7. Boxes and Enclosures: NEMA 250, Type 1, except use NEMA 250, Type 4 stainless steel in institutional and commercial kitchens and damp or wet locations.
- C. Minimum Raceway Size: 1/2-inch trade size.
- D. Raceway Fittings: Compatible with raceways and suitable for use and location.
1. Rigid and Intermediate Steel Conduit: Use threaded rigid steel conduit fittings unless otherwise indicated. Comply with NEMA FB 2.10.
 2. PVC Externally Coated, Rigid Steel Conduits: Use only fittings listed for use with this type of conduit. Patch and seal all joints, nicks, and scrapes in PVC coating after installing conduits and fittings. Use sealant recommended by fitting manufacturer and apply in thickness and number of coats recommended by manufacturer.
 3. EMT: Use ~~setscrew~~ or compression, steel fittings. Comply with NEMA FB 2.10.
 4. Flexible Conduit: Use only fittings listed for use with flexible conduit. Comply with NEMA FB 2.20.
- E. Do not install aluminum conduits, boxes, or fittings in contact with concrete or earth.
- F. Install surface raceways only where indicated on Drawings.
- G. Do not install nonmetallic conduit where ambient temperature exceeds 120 deg F.

3.2 INSTALLATION

- A. Comply with NECA 1 and NECA 101 for installation requirements except where requirements on Drawings or in this article are stricter. Comply with NECA 102 for aluminum conduits. Comply with NFPA 70 limitations for types of raceways allowed in specific occupancies and number of floors.
- B. Keep raceways at least 6 inches away from parallel runs of flues and steam or hot-water pipes. Install horizontal raceway runs above water and steam piping.
- C. Comply with requirements in Section 260529 "Hangers and Supports for Electrical Systems" for hangers and supports.
- D. Arrange stub-ups so curved portions of bends are not visible above finished slab.
- E. Install no more than the equivalent of three 90-degree bends in any conduit run except for control wiring conduits, for which fewer bends are allowed. Support within 12 inches of changes in direction.
- F. Conceal conduit and EMT within finished walls, ceilings, and floors unless otherwise indicated. Install conduits parallel or perpendicular to building lines.
- G. Support conduit within 12 inches of enclosures to which attached.

H. Raceways Embedded in Slabs:

1. Run conduit larger than 1-inch trade size, parallel or at right angles to main reinforcement. Where at right angles to reinforcement, place conduit close to slab support. Secure raceways to reinforcement at maximum 10-foot intervals.
2. Arrange raceways to cross building expansion joints at right angles with expansion fittings.
3. Arrange raceways to keep a minimum of 1 inch of concrete cover in all directions.
4. Do not embed threadless fittings in concrete unless specifically approved by Architect for each specific location.
5. Change from ENT to GRC or IMC before rising above floor.

I. Stub-ups to Above Recessed Ceilings:

1. Use EMT, IMC, or RMC for raceways.
2. Use a conduit bushing or insulated fitting to terminate stub-ups not terminated in hubs or in an enclosure.

J. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound to threads of raceway and fittings before making up joints. Follow compound manufacturer's written instructions.

K. Coat field-cut threads on PVC-coated raceway with a corrosion-preventing conductive compound prior to assembly.

L. Raceway Terminations at Locations Subject to Moisture or Vibration: Use insulating bushings to protect conductors including conductors smaller than No. 4 AWG.

M. Terminate threaded conduits into threaded hubs or with locknuts on inside and outside of boxes or cabinets. Install bushings on conduits up to 1-1/4-inch trade size and insulated throat metal bushings on 1-1/2-inch trade size and larger conduits terminated with locknuts. Install insulated throat metal grounding bushings on service conduits.

N. Install pull wires in empty raceways. Use polypropylene or monofilament plastic line with not less than 200-lb tensile strength. Leave at least 12 inches of slack at each end of pull wire. Cap underground raceways designated as spare above grade alongside raceways in use.

O. Surface Raceways:

1. Install surface raceway with a minimum 2-inch radius control at bend points.
2. Secure surface raceway with screws or other anchor-type devices at intervals not exceeding 48 inches and with no less than two supports per straight raceway section. Support surface raceway according to manufacturer's written instructions. Tape and glue are not acceptable support methods.

P. Install raceway sealing fittings at accessible locations according to NFPA 70 and fill them with listed sealing compound. For concealed raceways, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent

plates or surfaces.

- Q. Install devices to seal raceway interiors at accessible locations. Locate seals so no fittings or boxes are between the seal and the following changes of environments. Seal the interior of all raceways at the following points:
1. Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
 2. Where an underground service raceway enters a building or structure.
 3. Where otherwise required by NFPA 70.
- R. Expansion-Joint Fittings:
1. Install type and quantity of fittings that accommodate temperature change listed for each of the following locations:
 - a. Outdoor Locations Not Exposed to Direct Sunlight: 125 deg F temperature change.
 - b. Outdoor Locations Exposed to Direct Sunlight: 155 deg F temperature change.
 - c. Indoor Spaces Connected with Outdoors without Physical Separation: 125 deg F temperature change.
 2. Install fitting(s) that provide expansion and contraction for at least 0.00041 inch per foot of length of straight run per degree F of temperature change for PVC conduits.
 3. Install expansion fittings at all locations where conduits cross building or structure expansion joints.
 4. Install each expansion-joint fitting with position, mounting, and piston setting selected according to manufacturer's written instructions for conditions at specific location at time of installation. Install conduit supports to allow for expansion movement.
- S. Flexible Conduit Connections: Comply with NEMA RV 3. Use a maximum of 72 inches of flexible conduit for recessed and semi-recessed luminaires, equipment subject to vibration, noise transmission, or movement; and for transformers and motors.
1. Use LFMC in damp or wet locations subject to severe physical damage.
 2. Use LFMC in damp or wet locations not subject to severe physical damage.
- T. Mount boxes at heights indicated on Drawings. If mounting heights of boxes are not individually indicated, give priority to ADA requirements. Install boxes with height measured to center of box unless otherwise indicated.
- U. Recessed Boxes in Masonry Walls: Saw-cut opening for box in center of cell of masonry block, and install box flush with surface of wall. Prepare block surfaces to provide a flat surface for a raintight connection between the box and cover plate or the supported equipment and box.
- V. Horizontally separate boxes mounted on opposite sides of walls so they are not in the same vertical channel.
- W. Locate boxes so that cover or plate will not span different building finishes.
- X. Support boxes of three gangs or more from more than one side by spanning

two framing members or mounting on brackets specifically designed for the purpose.

- Y. Fasten junction and pull boxes to or support from building structure. Do not support boxes by conduits.
- Z. Set metal floor boxes level and flush with finished floor surface.
- AA. Set nonmetallic floor boxes level. Trim after installation to fit flush with finished floor surface.

3.3 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

- A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies. Comply with requirements in Section 260544 "Sleeves and Sleeve Seals for Electrical Raceways and Cabling."

3.4 FIRESTOPPING

- A. Install firestopping at penetrations of fire-rated floor and wall assemblies.

3.5 PROTECTION

- A. Protect coatings, finishes, and cabinets from damage and deterioration.
 - 1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
 - 2. Repair damage to PVC coatings or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION 260533

SECTION 260536 - CABLE TRAYS FOR ELECTRICAL SYSTEMS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section Includes:
 - 1. Wire-mesh cable tray.
 - 2. Cable tray accessories.
 - 3. Warning signs.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings: For cable tray.
 - 1. Show fabrication and installation details of cable trays, including plans, elevations, and sections of components and attachments to other construction elements. Designate components and accessories, including clamps, brackets, hanger rods, splice-plate connectors, expansion-joint assemblies, straight lengths, and fittings.
 - 2. Cable tray layout, showing cable tray route to scale, with relationship between the tray and adjacent structural, electrical, and mechanical elements. Include the following:
 - a. Vertical and horizontal offsets and transitions.
 - b. Clearances for access above and to sides of cable trays.
 - c. Vertical elevation of cable trays above the floor or bottom of ceiling structure.
 - d. Load calculations to show dead and live loads as not exceeding manufacturer's rating for tray and its support elements.
- C. Delegated-Design Submittal: For seismic restraints.
 - 1. Seismic-Restraint Details: Signed and sealed by a qualified professional engineer who is licensed in the state where Project is located and who is responsible for their preparation.
 - 2. Design Calculations: Calculate requirements for selecting seismic restraints.
 - 3. Detail fabrication, including anchorages and attachments to structure and to supported cable trays.

1.3 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Floor plans and sections, drawn to scale, and coordinated with each other, using input from installers of the items involved:
- B. Seismic Qualification Certificates: For cable trays, accessories, and components, from manufacturer.
- C. Field quality-control reports.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design cable tray supports and seismic bracing.
- B. Seismic Performance: Cable trays and supports shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
 - 1. The term "withstand" means "cable trays will remain in place without separation of any parts when subjected to the seismic forces specified."
 - 2. Component Importance Factor: 1.5.
- C. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes in cable tray installed outdoors.
 - 1. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

2.2 GENERAL REQUIREMENTS FOR CABLE TRAY

- A. Cable Trays and Accessories: Identified as defined in NFPA 70 and marked for intended location, application, and grounding.
- B. Sizes and Configurations: Provide tray widths 6-inches; 9-inches; 12-inches; 18-inches; 24-inches as shown on the Drawings.
- C. Structural Performance: See articles on individual cable tray types for specific values for the following parameters:
 - 1. Uniform Load Distribution: Capable of supporting a uniformly distributed load on the indicated support span when supported as a simple span and tested according to NEMA VE 1.
 - 2. Concentrated Load: A load applied at midpoint of span and centerline of tray.
 - 3. Load and Safety Factors: Applicable to both side rails and rung capacities.

2.3 WIRE-MESH AND ENCLOSED CABLE TRAY

- A. Description: steel wire mesh, complying with NEMA VE 1.
- B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. B-line, an Eaton business. FLEXTRAY GS Finish Code Series
 - 2. Cooper Industries; Cooper B-Line; GS Metals Corp.
 - 3. Hubbell Incorporated; Wiring Device-Kellems.
 - 4. Legrand US.
 - 5. MonoSystems, Inc.
 - 6. MP Husky USA Cable Tray & Cable Bus.
 - 7. Snake-tray.
- C. Capacities and Characteristics:
 - 1. Width: 24 inches unless otherwise indicated on Drawings.

2. Minimum Usable Load Depth: 4 inches.
3. Straight Section Lengths: 10 feet, except where shorter lengths are required to facilitate tray assembly.
4. Structural Performance: Capable of supporting a maximum cable load, with a safety factor of 1.5, plus a 200-lb concentrated load, when tested according to NEMA VE 1.
5. Class Designation: Comply with NEMA VE 1, Class 5AA.
6. Splicing Assemblies: Bolted type using serrated flange locknuts.
7. Splice-Plate Capacity: Splices located within support span shall not diminish rated loading capacity of cable tray.

D. Materials and Finishes:

1. Steel:
 - a. Straight Sections and Fittings: Steel complies with the minimum mechanical properties of ASTM A 1011/A 1011M, SS, Grade 33.
 - b. Steel Tray Splice Plates: ASTM A 1011/A 1011M, HSLAS, Grade 50, Class 1.
 - c. Fasteners: Steel complies with the minimum mechanical properties of ASTM A 510/A 510M, Grade 1008.
 - d. Finish: Standard pre-galvanized finish.
 - 1) Hardware: Chromium-zinc plated, ASTM F 1136.

2.4 CABLE TRAY ACCESSORIES

- A. Fittings: Tees, crosses, risers, elbows, and other fittings as indicated, of same materials and finishes as cable tray.
- B. Barrier Strips: Same materials and finishes as for cable tray.
- C. Cable tray supports and connectors, including bonding jumpers, as recommended by cable tray manufacturer.

2.5 WARNING SIGNS

- A. Lettering: 1-1/2-inch- high, black letters on yellow background, with legend "WARNING! NOT TO BE USED AS WALKWAY, LADDER, OR SUPPORT FOR LADDERS OR PERSONNEL."
- B. Comply with Section 260553 "Identification for Electrical Systems."

PART 3 - EXECUTION

3.1 CABLE TRAY INSTALLATION

- A. Install cable tray and support systems according to NEMA FG 1.
- B. Install cable tray as a complete system, including fasteners, hold-down clips, support systems, barrier strips, adjustable horizontal and vertical splice plates, elbows, reducers, tees,

crosses, cable dropouts, adapters, covers, and bonding.

- C. Install cable tray, so that the tray is accessible for cable installation and all splices are accessible for inspection and adjustment.
- D. Remove burrs and sharp edges from cable trays.
- E. Fasten cable tray supports to building structure and install seismic restraints.
- F. Design fasteners and supports to carry cable tray, cables, and a concentrated load of 200 lb. Comply with requirements in Section 260529 "Hangers and Supports for Electrical Systems." Comply with seismic-restraint details according to Section 260548.16 "Seismic Controls for Electrical Systems."
- G. Place supports, so that spans do not exceed maximum spans on schedules, and provide clearances shown on Drawings. Install intermediate supports when cable weight exceeds the load-carrying capacity of tray rungs.
- H. Construct supports from channel members, threaded rods, and other appurtenances furnished by cable tray manufacturer. Arrange supports in trapeze or wall-bracket form as required by application.
- I. Do not install more than one cable tray splice between supports.
- J. Make changes in direction and elevation using manufacturer's recommended fittings.
- K. Make cable tray connections using manufacturer's recommended fittings.
- L. Seal penetrations through fire and smoke barriers.
- M. Install capped metal sleeves for future cables through firestop-sealed cable tray penetrations of fire and smoke barriers.
- N. Install cable trays with enough workspace to permit access for installing cables.
- O. Install barriers to separate cables of different systems, such as power, communications, and data processing, or of different insulation levels, such as 600, 5000, and 15 000 V.
- P. Install permanent covers and cover clamps, if used, after installing cable.
- Q. Clamp covers on cable trays installed outdoors with heavy-duty clamps.
- R. Install warning signs in visible locations on or near cable trays after cable tray installation.

3.2 CABLE TRAY GROUNDING

- A. Ground cable trays according to NFPA 70 unless additional grounding is specified. Comply with requirements in Section 260526 "Grounding and Bonding for Electrical Systems."
- B. Cable trays with electrical power conductors shall be bonded together with splice plates listed for grounding purposes or with listed bonding jumpers.
- C. Cable trays with single-conductor power conductors shall be bonded together with a grounding

conductor run in the tray along with the power conductors and bonded to the tray at 72-inch intervals. The grounding conductor shall be sized according to NFPA 70, Article 250.122, "Size of Equipment Grounding Conductors," and Article 392, "Cable Trays."

- D. When using epoxy- or powder-coat painted cable trays as a grounding conductor, completely remove coating at all splice contact points or ground connector attachment. After completing splice-to-grounding-bolt attachment, repair the coated surfaces with coating materials recommended by cable tray manufacturer.
- E. Bond cable trays to power source for cables contained within with bonding conductors sized according to NFPA 70, Article 250.122, "Size of Equipment Grounding Conductors."

3.3 CONNECTIONS

- A. Remove paint from all connection points before making connections. Repair paint after the connections are completed.
- B. Connect raceways to cable trays according to requirements in NEMA VE 2 and NEMA FG 1.

3.4 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections:
 - 1. After installing cable trays and after electrical circuitry has been energized, survey for compliance with requirements.
 - 2. Visually inspect cable insulation for damage. Correct sharp corners, protuberances in cable trays, vibrations, and thermal expansion and contraction conditions, which may cause or have caused damage.
 - 3. Verify that the number, size, and voltage of cables in cable trays do not exceed that permitted by NFPA 70. Verify that communications or data-processing circuits are separated from power circuits by barriers or are installed in separate cable trays.
 - 4. Verify that there are no intruding items, such as pipes, hangers, or other equipment, in the cable tray.
 - 5. Remove dust deposits, industrial process materials, trash of any description, and any blockage of tray ventilation.
 - 6. Visually inspect each cable tray joint and each ground connection for mechanical continuity. Check bolted connections between sections for corrosion. Clean and re-torque in suspect areas.
 - 7. Check for improperly sized or installed bonding jumpers.
 - 8. Check for missing, incorrect, or damaged bolts, bolt heads, or nuts. When found, replace with specified hardware.
 - 9. Perform visual and mechanical checks for adequacy of cable tray grounding; verify that all takeoff raceways are bonded to cable trays. Test entire cable tray system for continuity. Maximum allowable resistance is 1 ohm.

3.5 PROTECTION

- A. Protect installed cable trays and cables.
 - 1. Install temporary protection for cables in open trays to safeguard exposed cables against falling objects or debris during construction. Temporary protection for cables and cable tray can be constructed of wood or metal materials and shall remain in place until the risk of damage is over.
 - 2. Repair damage to galvanized finishes with zinc-rich paint recommended by cable tray

3. manufacturer.
Repair damage to paint finishes with matching touchup coating recommended by cable tray manufacturer.

END OF SECTION 260536

SECTION 260543 - UNDERGROUND DUCTS AND RACEWAYS FOR ELECTRICAL SYSTEMS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.
- B. Contractor shall provide separate manholes and hand holes for both power and low voltage systems.
- C. Where a single hand hole is indicated on the drawings which serve both power and low voltage utilities, the hand hole shall be provided with a UL listed divider in order to provide physical separation of such utilities.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Conduit, ducts, and duct accessories for direct-buried and concrete-encased duct banks, and in single duct runs.
 - 2. Handholes and boxes.

1.3 DEFINITION

- A. RNC: Rigid nonmetallic conduit.
- B. PVC: Poly-Vinyl Chloride conduit.

1.4 SUBMITTALS

- A. Product Data: For the following:
 - 1. Duct-bank materials, including separators and miscellaneous components.
 - 2. Ducts and conduits and their accessories, including elbows, end bells, bends, fittings, and solvent cement.
 - 3. Accessories for manholes, handholes, boxes, and other utility structures.
 - 4. Warning tape.
 - 5. Warning planks.
- B. Duct-Bank Coordination Drawings: Show duct profiles and coordination with other utilities and underground structures.
 - 1. Include plans and sections, drawn to scale, and show bends and locations of expansion fittings.
 - 2. Drawings shall be signed and sealed by a qualified professional engineer.
- C. Product Certificates: For concrete and steel used in precast concrete manholes and handholes, as required by ASTM C 858.

- D. Qualification Data: For professional engineer and testing agency.
- E. Source quality-control test reports.
- F. Field quality-control test reports.

1.5 QUALITY ASSURANCE

- A. Testing Agency Qualifications: Qualified according to ASTM E 329 for testing indicated.
- B. Comply with ANSI C2.
- C. Comply with NFPA 70.
- D. Comply with local codes and guidelines.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver ducts to Project site with ends capped. Store nonmetallic ducts with supports to prevent bending, warping, and deforming.
- B. Store precast concrete and other factory-fabricated underground utility structures at Project site as recommended by manufacturer to prevent physical damage. Arrange so identification markings are visible.
- C. Lift and support precast concrete units only at designated lifting or supporting points.

1.7 PROJECT CONDITIONS

- A. Interruption of Existing Electrical Service: Do not interrupt electrical service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary electrical service according to requirements indicated:
 - 1. Notify Construction Manager no fewer than two days in advance of proposed interruption of electrical service.
 - 2. Do not proceed with interruption of electrical service without Construction Manager's written permission.

1.8 COORDINATION

- A. Coordinate layout and installation of ducts, manholes, handholes, and boxes with final arrangement of other utilities, site grading, and surface features as determined in the field.
- B. Coordinate elevations of ducts and duct-bank entrances into manholes, handholes, and boxes with final locations and profiles of ducts and duct banks as determined by coordination with other utilities, underground obstructions, and surface features. Revise locations and elevations from those indicated as required to suit field conditions and to ensure that duct runs drain to manholes and handholes, and as approved by Architect.

1.9 EXTRA MATERIALS

- A. Furnish extra materials described below that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
- B. Furnish cable-support stanchions, arms, insulators, and associated fasteners in quantities equal to 5 percent of quantity of each item installed.

PART 2 - PRODUCTS

2.1 CONDUIT

- A. Rigid Steel Conduit: Galvanized. Comply with ANSI C80.1.
 - 1. All ninety degree conduit elbows and conduit stub-ups shall be PVC coated Rigid Steel Conduit, no exceptions.
- B. RNC: NEMA TC 2, Type EPC-40-PVC, UL 651, with matching fittings by same manufacturer as the conduit, complying with NEMA TC 3 and UL 514B.

2.2 NONMETALLIC DUCTS AND DUCT ACCESSORIES

- A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. ARNCO Corp.
 - 2. Beck Manufacturing.
 - 3. Cantex, Inc.
 - 4. CertainTeed Corp.; Pipe & Plastics Group.
 - 5. Condux International, Inc.
 - 6. ElecSys, Inc.
 - 7. Electri-Flex Company.
 - 8. IPEX Inc.
 - 9. Lamson & Sessions; Carlon Electrical Products.
 - 10. Manhattan/CDT; a division of Cable Design Technologies.
 - 11. Spiraduct/AFC Cable Systems, Inc.
- B. Duct Accessories:
 - 1. Duct Separators: Factory-fabricated rigid PVC interlocking spacers, sized for type and sizes of ducts with which used, and selected to provide minimum duct spacings indicated while supporting ducts during concreting or backfilling (Underground Devices Incorporated (847) 205-9000, www.udevices.com).
 - a. Duct bank shall be encased in concrete with at least three inches of concrete at the top and bottom and two inches on each side.
 - b. A horizontal and vertical separation between the ducts of two inches shall be maintained by installing Underground Devices High Impact Polystyrene Wunpeece Spacers.
 - c. Spacers shall be interlocked horizontally only. Along the length of the duct run spacers shall be staggered at least 6 inches vertically and shall be placed at an interval of five spacers per 20 feet or per manufacturer's recommendations.

2. Warning Tape: Underground-line warning tape specified in Division 26 Section "Identification for Electrical Systems."
3. Concrete Warning Planks: Nominal 12 by 24 by 3 inches in size, manufactured from 6000-psi concrete.
 - a. Color: Red dye added to concrete during batching.
 - b. Mark each plank with "ELECTRIC" in 2-inch- high, 3/8-inch- deep letters.

2.3 PRECAST CONCRETE HANDHOLES AND BOXES

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 1. Carder Concrete Products.
 2. Christy Concrete Products.
 3. Oldcastle Precast Group.
 4. Riverton Concrete Products; a division of Cretex Companies, Inc.
 5. Utility Concrete Products, LLC.
 6. Utility Vault Co.
 7. Wausau Tile, Inc.
- B. Comply with ASTM C 858 for design and manufacturing processes.
- C. Description: Factory-fabricated, reinforced-concrete, monolithically poured walls and bottom unless open-bottom enclosures are indicated. Frame and cover shall form top of enclosure and shall have load rating consistent with that of handhole or box.
 1. Frame and Cover: Weatherproof cast-iron frame, with cast-iron cover with recessed cover hook eyes and tamper-resistant, captive, cover-securing bolts.
 2. Frame and Cover: Weatherproof steel frame, with steel cover with recessed cover hook eyes and tamper-resistant, captive, cover-securing bolts.
 3. Frame and Cover: Weatherproof steel frame, with hinged steel access door assembly with tamper-resistant, captive, cover-securing bolts.
 - a. Cover Hinges: Concealed, with hold-open ratchet assembly.
 - b. Cover Handle: Recessed.
 4. Frame and Cover: Weatherproof aluminum frame with hinged aluminum access door assembly with tamper-resistant, captive, cover-securing bolts.
 - a. Cover Hinges: Concealed, with hold-open ratchet assembly.
 - b. Cover Handle: Recessed.
 5. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
 6. Cover Legend: Molded lettering, "ELECTRIC.", "TELEPHONE." Or As indicated on the drawings for each service.
 7. Configuration: Units shall be designed for flush burial and have closed bottom, unless otherwise indicated.
 8. Extensions and Slabs: Designed to mate with bottom of enclosure. Same material as enclosure.
 - a. Extension shall provide increased depth of 12 inches.
 - b. Slab: Same dimensions as bottom of enclosure, and arranged to provide closure.

9. Windows: Precast openings in walls, arranged to match dimensions and elevations of approaching ducts and duct banks plus an additional 12 inches vertically and horizontally to accommodate alignment variations.
 - a. Windows shall be located no less than 6 inches from interior surfaces of walls, floors, or frames and covers of handholes, but close enough to corners to facilitate racking of cables on walls.
 - b. Window opening shall have cast-in-place, welded wire fabric reinforcement for field cutting and bending to tie in to concrete envelopes of duct banks.
 - c. Window openings shall be framed with at least two additional No. 4 steel reinforcing bars in concrete around each opening.
10. Duct Entrances in Handhole Walls: Cast end-bell or duct-terminating fitting in wall for each entering duct.
 - a. Type and size shall match fittings to duct or conduit to be terminated.
 - b. Fittings shall align with elevations of approaching ducts and be located near interior corners of handholes to facilitate racking of cable.
11. Handholes 12 inches wide by 24 inches long and larger shall have inserts for cable racks and pulling-in irons installed before concrete is poured.
12. Where power and communication service conduits are routed in parallel to a common piece of equipment or device, the handhole shall be provided with an FHR divider panel and separate access covers in order to maintain code required physical separation of conduits and associated cabling.

PART 3 - EXECUTION

3.1 UNDERGROUND DUCT APPLICATION

- A. Ducts for Electrical Feeders 600 V and Less: RNC, NEMA Type EPC-40-PVC, in direct-buried duct bank, unless otherwise indicated.
- B. Ducts for Electrical Branch Circuits: RNC, NEMA Type EPC-40-PVC, in direct-buried duct bank, unless otherwise indicated.
- C. Underground Ducts for Telephone, Communications, or Data Utility Service Cables: RNC, NEMA Type EPC-40-PVC, in concrete-encased duct bank, unless otherwise indicated.
- D. Underground Ducts for Telephone, Communications, or Data Utility Service Cables: RNC, NEMA Type EPC-40-PVC, installed in direct-buried duct bank, unless otherwise indicated.
- E. Underground Ducts for Telephone, Communications, or Data Circuits: RNC, NEMA Type EPC-40-PVC, in direct-buried duct bank, unless otherwise indicated.
- F. Underground Ducts for Telephone, Communications, or Data Circuits: RNC, NEMA Type EB-20-PVC, in concrete-encased duct bank, unless otherwise indicated.
- G. Underground Ducts Crossing Paved Paths, Walks and Driveways/Roadways: RNC, NEMA Type EPC-40-PVC, encased in reinforced concrete.

3.2 UNDERGROUND ENCLOSURE APPLICATION

- A. Handholes and Boxes for 600 V and Less, Including Telephone, Communications, and Data Wiring:
 - 1. Units in Roadways and Other Deliberate Traffic Paths: Precast concrete. AASHTO HB 17, H-10 structural load rating.
 - 2. Units in Driveway, Parking Lot, and Off-Roadway Locations, Subject to Occasional, Non-deliberate Loading by Heavy Vehicles: Precast concrete, AASHTO HB 17, H-20 structural load rating.
 - 3. Units in Sidewalk and Similar Applications with a Safety Factor for Non-deliberate Loading by Vehicles: Precast concrete, AASHTO HB 17, H-10 structural load rating.
 - 4. Units Subject to Light-Duty Pedestrian Traffic Only: Fiberglass-reinforced polyester resin, structurally tested according to SCTE 77 with 3000-lbf vertical loading.

3.3 EARTHWORK

- A. Excavation and Backfill: Comply with Division 31 Section "Earth Moving," but do not use heavy-duty, hydraulic-operated, compaction equipment.
- B. Restore surface features at areas disturbed by excavation and reestablish original grades, unless otherwise indicated. Replace removed sod immediately after backfilling is completed.
- C. Restore areas disturbed by trenching, storing of dirt, cable laying, and other work. Restore vegetation and include necessary topsoiling, fertilizing, liming, seeding, sodding, sprigging, and mulching. Comply with Division 32 Sections "Turf and Grasses" and "Plants."
- D. Cut and patch existing pavement in the path of underground ducts and utility structures according to Division 01 Section "Cutting and Patching."

3.4 DUCT INSTALLATION

- A. Slope: Pitch ducts a minimum slope of 1:300 down toward manholes and handholes and away from buildings and equipment. Slope ducts from a high point in runs between two manholes to drain in both directions.
- B. Curves and Bends: Use 5-degree angle couplings for small changes in direction. Use manufactured long sweep bends with a minimum radius of 48 inches, both horizontally and vertically, at other locations, unless otherwise indicated.
- C. Joints: Use solvent-cemented joints in ducts and fittings and make watertight according to manufacturer's written instructions. Stagger couplings so those of adjacent ducts do not lie in same plane.
- D. Duct Entrances to Manholes and Concrete and Polymer Concrete Handholes: Use end bells, spaced approximately 10 inches o.c. for 5-inch ducts, and vary proportionately for other duct sizes.
 - 1. Begin change from regular spacing to end-bell spacing 10 feet from the end bell without reducing duct line slope and without forming a trap in the line.
 - 2. Direct-Buried Duct Banks: Install an expansion and deflection fitting in each conduit in the area of disturbed earth adjacent to manhole or handhole.

3. Grout end bells into structure walls from both sides to provide watertight entrances.
- E. Building Wall Penetrations: Make a transition from underground duct to rigid steel conduit at least 10 feet outside the building wall without reducing duct line slope away from the building, and without forming a trap in the line. Use fittings manufactured for duct-to-conduit transition. Install conduit penetrations of building walls as specified in Division 26 Section "Common Work Results for Electrical."
- F. Sealing and Waterproofing: Provide temporary mechanical waterproof closure/plug at all terminations of ducts that have cables pulled. Provide mechanical waterproof plug on all spare empty ducts at each duct/conduit end. Sealing and waterproofing mechanical plugs shall withstand at least 15-psig hydrostatic pressure. Sealing and waterproofing system shall be capable of adding and removing cables. Provide the following products for sealing and waterproofing conduits and ducts:
 1. Tyco Electronics Corporation – Jackmoon Series Duct Plugs.
- G. Pulling Cord: Install 100-lbf- test nylon cord in ducts, including spares.
- H. Cleaning: Underground conduit duct shall completely dry and free from water and debris prior to installation of cables and nylon pull ropes. Clean entire length of all underground electrical conduit ducts with compressed air. Temporarily cover all ends of conduit ducts in order to prevent water and debris infiltration during the entire length of construction.
- I. Concrete-Encased Ducts: Support ducts on duct separators.
 1. Separator Installation: Space separators close enough to prevent sagging and deforming of ducts, with not less than 5 spacers per 20 feet of duct. Secure separators to earth and to ducts to prevent floating during concreting. Stagger separators approximately 6 inches between tiers. Tie entire assembly together using fabric straps; do not use tie wires or reinforcing steel that may form conductive or magnetic loops around ducts or duct groups.
 2. Concreting Sequence: Pour each run of envelope between manholes or other terminations in one continuous operation.
 - a. Start at one end and finish at the other, allowing for expansion and contraction of ducts as their temperature changes during and after the pour. Use expansion fittings installed according to manufacturer's written recommendations, or use other specific measures to prevent expansion-contraction damage.
 - b. If more than one pour is necessary, terminate each pour in a vertical plane and install 3/4-inch reinforcing rod dowels extending 18 inches into concrete on both sides of joint near corners of envelope.
 3. Pouring Concrete: Spade concrete carefully during pours to prevent voids under and between conduits and at exterior surface of envelope. Do not allow a heavy mass of concrete to fall directly onto ducts. Use a plank to direct concrete down sides of bank assembly to trench bottom. Allow concrete to flow to center of bank and rise up in middle, uniformly filling all open spaces. Do not use power-driven agitating equipment unless specifically designed for duct-bank application.
 4. Reinforcement: Reinforce concrete-encased duct banks where they cross disturbed earth and where indicated. Arrange reinforcing rods and ties without forming conductive or magnetic loops around ducts or duct groups.
 5. Forms: Use walls of trench to form side walls of duct bank where soil is self-supporting and concrete envelope can be poured without soil inclusions; otherwise, use forms.
 6. Minimum Space between Ducts: 3 inches between ducts and exterior envelope wall, 2 inches between ducts for like services, and 4 inches between power and signal ducts.

7. Depth: Install top of duct bank at least 24 inches below finished grade in areas not subject to deliberate traffic, and at least 30 inches below finished grade in deliberate traffic paths for vehicles, unless otherwise indicated.
8. Stub-Ups: Use manufactured duct elbows for stub-ups at poles and equipment and at building entrances through the floor, unless otherwise indicated. Extend concrete encasement throughout the length of the elbow.
9. Stub-Ups: Use manufactured rigid steel conduit elbows for stub-ups at poles and equipment and at building entrances through the floor.
 - a. Couple steel conduits to ducts with adapters designed for this purpose, and encase coupling with 3 inches of concrete.
 - b. Stub-Ups to Equipment: For equipment mounted on outdoor concrete bases, extend steel conduit horizontally a minimum of 60 inches from edge of base. Install insulated grounding bushings on terminations at equipment.
10. Warning Tape: Bury warning tape approximately 12 inches above all concrete-encased ducts and duct banks. Align tape parallel to and within 3 inches of the centerline of duct bank. Provide an additional warning tape for each 12-inch increment of duct-bank width over a nominal 18 inches. Space additional tapes 12 inches apart, horizontally.

J. Direct-Buried Duct Banks:

1. Support ducts on duct separators coordinated with duct size, duct spacing, and outdoor temperature.
2. Space separators close enough to prevent sagging and deforming of ducts, with not less than 5 spacers per 20 feet of duct. Secure separators to earth and to ducts to prevent displacement during backfill and yet permit linear duct movement due to expansion and contraction as temperature changes. Stagger spacers approximately 6 inches between tiers.
3. Excavate trench bottom to provide firm and uniform support for duct bank. Prepare trench bottoms as specified in Division 31 Section "Earth Moving" for pipes less than 6 inches in nominal diameter.
4. Install backfill as specified in Division 31 Section "Earth Moving."
5. After installing first tier of ducts, backfill and compact. Start at tie-in point and work toward end of duct run, leaving ducts at end of run free to move with expansion and contraction as temperature changes during this process. Repeat procedure after placing each tier. After placing last tier, hand-place backfill to 4 inches over ducts and hand tamp. Firmly tamp backfill around ducts to provide maximum supporting strength. Use hand tamper only. After placing controlled backfill over final tier, make final duct connections at end of run and complete backfilling with normal compaction as specified in Division 31 Section "Earth Moving."
6. Install ducts with a minimum of 3 inches between ducts for like services and 6 inches between power and signal ducts.
7. Depth: Install top of duct bank at least 36 inches below finished grade, unless otherwise indicated.
8. Set elevation of bottom of duct bank below the frost line.
9. Install manufactured duct elbows for stub-ups at poles and equipment and at building entrances through the floor, unless otherwise indicated. Encase elbows for stub-up ducts throughout the length of the elbow.
10. Install manufactured rigid steel conduit elbows for stub-ups at poles and equipment and at building entrances through the floor.
 - a. Couple steel conduits to ducts with adapters designed for this purpose, and encase coupling with 3 inches of concrete.

- b. For equipment mounted on outdoor concrete bases, extend steel conduit horizontally a minimum of 60 inches from edge of equipment pad or foundation. Install insulated grounding bushings on terminations at equipment.

3.5 INSTALLATION OF CONCRETE MANHOLES, HANDHOLES, AND BOXES

A. Precast Concrete Handhole Installation:

1. Comply with ASTM C 891, unless otherwise indicated.
2. Install units level and plumb and with orientation and depth coordinated with connecting ducts to minimize bends and deflections required for proper entrances.
3. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1-inch sieve to No. 4 sieve and compacted to same density as adjacent undisturbed earth.

B. Elevations:

1. Install handholes with bottom below the frost line, 36 inches below grade.
2. Handhole Covers: In paved areas and trafficways, set surface flush with finished grade. Set covers of other handholes 1 inch above finished grade.
3. Where indicated, cast handhole cover frame integrally with handhole structure.

C. Drainage: Install drains in bottom of manholes where indicated. Coordinate with drainage provisions indicated.

D. Waterproofing: Apply waterproofing to exterior surfaces of manholes and handholes after concrete has cured at least three days. Waterproofing materials and installation are specified in Division 07 Section "Elastomeric Sheet Waterproofing." After ducts have been connected and grouted, and before backfilling, waterproof joints and connections and touch up abrasions and scars. Waterproof exterior of manhole chimneys after mortar has cured at least three days.

E. Dampproofing: Apply dampproofing to exterior surfaces of manholes and handholes after concrete has cured at least three days. Dampproofing materials and installation are specified in Division 07 Section "Bituminous Dampproofing." After ducts have been connected and grouted, and before backfilling, dampproof joints and connections and touch up abrasions and scars. Dampproof exterior of manhole chimneys after mortar has cured at least three days.

F. Hardware: Install removable hardware, including pulling eyes, cable stanchions, and cable arms, and insulators, as required for installation and support of cables and conductors and as indicated.

G. Fixed Manhole Ladders: Arrange to provide for safe entry with maximum clearance from cables and other items in manholes.

H. Field-Installed Bolting Anchors in Manholes and Concrete Handholes: Do not drill deeper than 3-7/8 inches for manholes and 2 inches for handholes, for anchor bolts installed in the field. Use a minimum of two anchors for each cable stanchion.

I. Warning Sign: Install "Confined Space Hazard" warning sign on the inside surface of each manhole cover.

3.6 GROUNDING

- A. Ground underground ducts and utility structures according to Division 26 Section "Grounding and Bonding for Electrical Systems."

3.7 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections and prepare test reports:
 - 1. Demonstrate capability and compliance with requirements on completion of installation of underground ducts and utility structures.
 - 2. Pull aluminum or wood test mandrel through duct to prove joint integrity and test for out-of-round duct. Provide mandrel equal to 80 percent fill of duct. If obstructions are indicated, remove obstructions and retest.
 - 3. Test manhole and handhole grounding to ensure electrical continuity of grounding and bonding connections. Measure and report ground resistance as specified in Division 26 Section "Grounding and Bonding for Electrical Systems."
- B. Correct deficiencies and retest as specified above to demonstrate compliance.

3.8 CLEANING

- A. Pull leather-washer-type duct cleaner, with graduated washer sizes, through full length of ducts. Follow with rubber duct swab for final cleaning and to assist in spreading lubricant throughout ducts.
- B. Clean internal surfaces of handholes, including sump. Remove foreign material.

END OF SECTION 260543

SECTION 260553 - IDENTIFICATION FOR ELECTRICAL SYSTEMS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY**A. Section Includes:**

- 1. Color and legend requirements for raceways, conductors, and warning labels and signs.
- 2. Labels.
- 3. Bands and tubes.
- 4. Tapes and stencils.
- 5. Tags.
- 6. Signs.
- 7. Cable ties.
- 8. Paint for identification.
- 9. Fasteners for labels and signs.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples: For each type of label and sign to illustrate composition, size, colors, lettering style, mounting provisions, and graphic features of identification products.
- C. Delegated-Design Submittal: For arc-flash hazard

study. PART 2 - PRODUCTS**2.1 PERFORMANCE REQUIREMENTS**

- A. Comply with ASME A13.1 and IEEE C2.
- B. Comply with NFPA 70.
- C. Comply with 29 CFR 1910.144 and 29 CFR 1910.145.
- D. Comply with ANSI Z535.4 for safety signs and labels.
- E. Comply with NFPA 70E and Section 260574 "Overcurrent Protective Device Arc-Flash Study" requirements for arc-flash warning labels.
- F. Adhesive-attached labeling materials, including label stocks, laminating adhesives, and inks used by label printers, shall comply with UL 969.
- G. Thermal Movements: Allow for thermal movements from ambient and surface

temperature changes.

1. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

2.2 COLOR AND LEGEND REQUIREMENTS

A. Raceways and Cables Carrying Circuits at 600 V or Less:

1. Black letters on an orange field.
2. Legend: Indicate voltage and system or service type.

B. Color-Coding for Phase- and Voltage-Level Identification, 600 V or Less: Use colors listed below for ungrounded service feeder and branch-circuit conductors.

1. Color shall be factory applied or field applied for sizes larger than No. 8 AWG if authorities having jurisdiction permit.
2. Colors for 208/120-V Circuits:
 - a. Phase A: Black.
 - b. Phase B: Red.
 - c. Phase C: Blue.
3. Colors for 480/277-V Circuits:
 - a. Phase A: Brown.
 - b. Phase B: Orange.
 - c. Phase C: Yellow.
4. Color for Neutral: White or gray.
5. Color for Equipment Grounds: Green or Green with a yellow stripe.
6. Colors for Isolated Grounds: Green with white stripe.

C. Warning Label Colors:

1. Identify system voltage with black letters on an orange background.

D. Warning labels and signs shall include, but are not limited to, the following legends:

1. Multiple Power Source Warning: "DANGER - ELECTRICAL SHOCK HAZARD - EQUIPMENT HAS MULTIPLE POWER SOURCES."
2. Workspace Clearance Warning: "WARNING - OSHA REGULATION - AREA IN FRONT OF ELECTRICAL EQUIPMENT MUST BE KEPT CLEAR FOR 36 INCHES."

2.3 LABELS

- A. Vinyl Wraparound Labels: Preprinted, flexible labels laminated with a clear, weather- and chemical-resistant coating and matching wraparound clear adhesive tape for securing label ends.
- B. Snap-around Labels: Slit, pre-tensioned, flexible, preprinted, color-coded acrylic sleeves, with diameters sized to suit diameter and that stay in place by gripping action.

- C. Self-Adhesive Wraparound Labels: Preprinted, 3-mil-thick, vinyl flexible label with acrylic pressure-sensitive adhesive.
 - 1. Self-Lamination: Clear; UV-, weather- and chemical-resistant; self-laminating, protective shield over the legend. Labels sized such that the clear shield overlaps the entire printed legend.
 - 2. Marker for Labels: Permanent, waterproof, black ink marker recommended by tag manufacturer.
 - 3. Marker for Labels: Machine-printed, permanent, waterproof, black ink recommended by printer manufacturer.
- D. Self-Adhesive Labels: Vinyl, thermal, transfer-printed, 3-mil-thick, multicolor, weather- and UV-resistant, pressure-sensitive adhesive labels, configured for intended use and location.
 - 1. Minimum Nominal Size:
 - a. 1-1/2 by 6 inches for raceway and conductors.
 - b. 3-1/2 by 5 inches for equipment.

2.4 BANDS AND TUBES

- A. Snap-around, Color-Coding Bands: Slit, pre-tensioned, flexible, solid-colored acrylic sleeves, 2 inches long, with diameters sized to suit diameter and that stay in place by gripping action.
- B. Heat-Shrink Preprinted Tubes: Flame-retardant polyolefin tubes with machine-printed identification labels, sized to suit diameters of and shrunk to fit firmly around item being identified. Full shrink recovery occurs at a maximum of 200 deg F. Comply with UL 224.

2.5 TAPES AND STENCILS

- A. Marker Tapes: Vinyl or vinyl-cloth, self-adhesive wraparound type, with circuit identification legend machine printed by thermal transfer or equivalent process.
- B. Self-Adhesive Vinyl Tape: Colored, heavy duty, waterproof, fade resistant; not less than 3 mils thick by 1 to 2 inches wide; compounded for outdoor use.
- C. Tape and Stencil: 4-inch-wide black stripes on 10-inch centers placed diagonally over orange background and is 12 inches wide. Stop stripes at legends.
- D. Floor Marking Tape: 2-inch-wide, 5-mil pressure-sensitive vinyl tape, with yellow and black stripes and clear vinyl overlay.
- E. Underground-Line Warning Tape:
 - 1. Tape:
 - a. Recommended by manufacturer for the method of installation and suitable to identify and locate underground electrical and communications utility lines.
 - b. Printing on tape shall be permanent and shall not be damaged by burial operations.
 - c. Tape material and ink shall be chemically inert and not subject to degradation when exposed to acids, alkalis, and other destructive substances commonly

found in soils.

2. Color and Printing:

- a. Comply with ANSI Z535.1, ANSI Z535.2, ANSI Z535.3, ANSI Z535.4, and ANSI Z535.5.
- b. Inscriptions for Red-Colored Tapes: "ELECTRIC LINE, HIGH VOLTAGE".
- c. Inscriptions for Orange-Colored Tapes: "TELEPHONE CABLE, CATV CABLE, COMMUNICATIONS CABLE, OPTICAL FIBER CABLE".
- d. Foil Core Thickness: 0.35 mil.
- e. Weight: 28 lb/1000 sq. ft..
- f. Tensile according to ASTM D 882: 70 lbf and 4600 psi.

- F. Stenciled Legend: In nonfading, waterproof, black ink or paint. Minimum letter height shall be 1 inch.

2.6 TAGS

A. Write-on Tags:

1. Polyester Tags: 0.010 inch thick, with corrosion-resistant grommet and cable tie for attachment.
2. Marker for Tags: Permanent, waterproof, black ink marker recommended by tag manufacturer.
3. Marker for Tags: Machine-printed, permanent, waterproof, black ink marker recommended by printer manufacturer.

2.7 SIGNS

A. Baked-Enamel Signs:

1. Preprinted aluminum signs, high-intensity reflective, punched or drilled for fasteners, with colors, legend, and size required for application.
2. 1/4-inch grommets in corners for mounting.
3. Nominal Size: 7 by 10 inches.

B. Metal-Backed Butyrate Signs:

1. Weather-resistant, nonfading, preprinted, cellulose-acetate butyrate signs, with 0.0396- inch galvanized-steel backing, punched and drilled for fasteners, and with colors, legend, and size required for application.
2. 1/4-inch grommets in corners for mounting.
3. Nominal Size: 10 by 14 inches.

C. Laminated Acrylic or Melamine Plastic Signs:

1. Engraved legend.
2. Thickness:
 - a. For signs up to 20 sq. in., minimum 1/16 inch.
 - b. For signs larger than 20 sq. in., 1/8 inch thick.

- c. Engraved legend with black letters on white face.
- d. Punched or drilled for mechanical fasteners with 1/4-inch grommets in corners for mounting.
- e. Framed with mitered acrylic molding and arranged for attachment at applicable equipment.

2.8 CABLE TIES

- A. General-Purpose Cable Ties: Fungus inert, self-extinguishing, one piece, self-locking, and Type 6/6 nylon.
 - 1. Minimum Width: 3/16 inch.
 - 2. Tensile Strength at 73 Deg F according to ASTM D 638: 12,000 psi.
 - 3. Temperature Range: Minus 40 to plus 185 deg F.
 - 4. Color: Black, except where used for color-coding.
- B. UV-Stabilized Cable Ties: Fungus inert, designed for continuous exposure to exterior sunlight, self-extinguishing, one piece, self-locking, and Type 6/6 nylon.
 - 1. Minimum Width: 3/16 inch.
 - 2. Tensile Strength at 73 Deg F according to ASTM D 638: 12,000 psi.
 - 3. Temperature Range: Minus 40 to plus 185 deg F.
 - 4. Color: Black.

2.9 MISCELLANEOUS IDENTIFICATION PRODUCTS

- A. Paint: Comply with requirements in painting Sections for paint materials and application requirements. Retain paint system applicable for surface material and location (exterior or interior).
- B. Fasteners for Labels and Signs: Self-tapping, stainless-steel screws or stainless-steel machine screws with nuts and flat and lock washers.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Verify and coordinate identification names, abbreviations, colors, and other features with requirements in other Sections requiring identification applications, Drawings, Shop Drawings, manufacturer's wiring diagrams, and operation and maintenance manual. Use consistent designations throughout Project.
- B. Provide self-adhesive identification label on all wiring device faceplates indicating panelboard and circuit number (1/4" high black lettering on white background).
- C. Install identifying devices before installing acoustical ceilings and similar concealment.
- D. Verify identity of each item before installing identification products.

- E. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and operation and maintenance manual.
- F. Apply identification devices to surfaces that require finish after completing finish work.
- G. Install signs with approved legend to facilitate proper identification, operation, and maintenance of electrical systems and connected items.
- H. Self-Adhesive Identification Products: Before applying electrical identification products, clean substrates of substances that could impair bond, using materials and methods recommended by manufacturer of identification product.
- I. System Identification for Raceways and Cables under 600 V: Identification shall completely encircle cable or conduit. Place identification of two-color markings in contact, side by side.
 - 1. Secure tight to surface of conductor, cable, or raceway.
- J. System Identification for Raceways and Cables over 600 V: Identification shall completely encircle cable or conduit. Place adjacent identification of two-color markings in contact, side by side.
 - 1. Secure tight to surface of conductor, cable, or raceway.
- K. Auxiliary Electrical Systems Conductor Identification: Identify field-installed alarm, control, and signal connections.
- L. Emergency Operating Instruction Signs: Install instruction signs with white legend on a red background with minimum 3/8-inch-high letters for emergency instructions at equipment used for power transfer.
- M. Elevated Components: Increase sizes of labels, signs, and letters to those appropriate for viewing from the floor.
- N. Accessible Fittings for Raceways: Identify the covers of each junction and pull box of the following systems with the wiring system legend and system voltage. System legends shall be as follows:
 - 1. "EMERGENCY POWER."
 - 2. "POWER."
 - 3. "UPS."
 - 4. "FIRE ALARM."
- O. Vinyl Wraparound Labels:
 - 1. Secure tight to surface at a location with high visibility and accessibility.
 - 2. Attach labels that are not self-adhesive type with clear vinyl tape, with adhesive appropriate to the location and substrate.
- P. Snap-around Labels: Secure tight to surface at a location with high visibility and accessibility.
- Q. Self-Adhesive Wraparound Labels: Secure tight to surface of raceway or cable at a location with high visibility and accessibility.

R. Self-Adhesive Labels:

1. On each unit of equipment, install unique designation label that is consistent with wiring diagrams, schedules, and operation and maintenance manual.
2. Unless otherwise indicated, provide a single line of text with 1/2-inch-high letters on 1- 1/2-inch-high label; where two lines of text are required, use labels 2 inches high.

S. Snap-around Color-Coding Bands: Secure tight to surface at a location with high visibility and accessibility.

T. Heat-Shrink, Preprinted Tubes: Secure tight to surface at a location with high visibility and accessibility.

U. Marker Tapes: Secure tight to surface at a location with high visibility and accessibility.

V. Self-Adhesive Vinyl Tape: Secure tight to surface at a location with high visibility and accessibility.

1. Field-Applied, Color-Coding Conductor Tape: Apply in half-lapped turns for a minimum distance of 6 inches where splices or taps are made. Apply last two turns of tape with no tension to prevent possible unwinding.

W. Tape and Stencil: Comply with requirements in painting Sections for surface preparation and paint application.

X. Floor Marking Tape: Apply stripes to finished surfaces following manufacturer's written instructions.

Y. Underground Line Warning Tape:

1. During backfilling of trenches, install continuous underground-line warning tape directly above cable or raceway at 6 to 8 inches below finished grade. Use multiple tapes where width of multiple lines installed in a common trench or concrete envelope exceeds 16 inches overall.
2. Limit use of underground-line warning tape to direct-buried cables.
3. Install underground-line warning tape for direct-buried cables and cables in raceways.

Z. Write-on Tags:

1. Place in a location with high visibility and accessibility.
2. Secure using general-purpose cable ties.

AA. Baked-Enamel Signs:

1. Attach signs that are not self-adhesive type with mechanical fasteners appropriate to the location and substrate.
2. Unless otherwise indicated, provide a single line of text with 1/2-inch-high letters on minimum 1-1/2-inch-high sign; where two lines of text are required, use signs minimum 2 inches high.

AA. Metal-Backed Butyrate Signs:

1. Attach signs that are not self-adhesive type with mechanical fasteners appropriate to the location and substrate.

2. Unless otherwise indicated, provide a single line of text with 1/2-inch-high letters on minimum 1-1/2-inch-high sign; where two lines of text are required, use signs minimum 2 inches high.

BB. Laminated Acrylic or Melamine Plastic Signs:

1. Attach signs and plastic labels that are not self-adhesive type with mechanical fasteners appropriate to the location and substrate.
2. Unless otherwise indicated, provide a single line of text with 1/2-inch-high letters on minimum 1-1/2-inch-high sign; where two lines of text are required, use signs minimum 2 inches high.

CC. Cable Ties: General purpose, for attaching tags, except as listed below:

1. Outdoors: UV-stabilized nylon.
2. In Spaces Handling Environmental Air: Plenum rated.

3.2 IDENTIFICATION SCHEDULE

- A. Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment. Install access doors or panels to provide view of identifying devices.
- B. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, pull points, and locations of high visibility. Identify by system and circuit designation.
- C. Accessible Raceways and Metal-Clad Cables, 600 V or Less, for Service, Feeder, and Branch Circuits, More Than 30 A and 120 V to Ground: Identify with self-adhesive raceway labels.
 1. Locate identification at changes in direction, at penetrations of walls and floors, at 50-foot maximum intervals in straight runs, and at 25-foot maximum intervals in congested areas.
- D. Accessible Fittings for Raceways and Cables within Buildings: Identify the covers of each junction and pull box of the following systems with self-adhesive labels containing the wiring system legend and system voltage. System legends shall be as follows:
 1. "EMERGENCY POWER."
 2. "POWER."
 3. "UPS."
 4. "FIRE ALARM."
- E. Power-Circuit Conductor Identification, 600 V or Less: For conductors in vaults, pull and junction boxes, manholes, and handholes, use vinyl wraparound labels to identify the phase.
 1. Locate identification at changes in direction, at penetrations of walls and floors, at 50-foot maximum intervals in straight runs, and at 25-foot maximum intervals in congested areas.
- F. Locations of Underground Lines: Underground-line warning tape for power, lighting, communication, and control wiring and optical-fiber cable.
- G. Workspace Indication: Apply floor marking tape or tape and stencil to finished surfaces. Show

working clearances in the direction of access to live parts. Workspace shall comply with NFPA 70 and 29 CFR 1926.403 unless otherwise indicated. Do not install at flush-mounted panelboards and similar equipment in finished spaces.

- H. Instructional Signs: Self-adhesive labels, including the color code for grounded and ungrounded conductors.
- I. Warning Labels for Indoor Cabinets, Boxes, and Enclosures for Power and Lighting: Self-adhesive equipment labels.
 - 1. Apply to exterior of door, cover, or other access.
 - 2. For equipment with multiple power or control sources, apply to door or cover of equipment, including, but not limited to, the following:
 - a. Power-transfer switches.
 - b. Controls with external control power connections.
- J. Arc Flash Warning Labeling: Self-adhesive labels.
- K. Operating Instruction Signs: Self-adhesive labels.
- L. Emergency Operating Instruction Signs: Self-adhesive labels with white legend on a red background with minimum 3/8-inch-high letters for emergency instructions at equipment used for power transfer.
- M. Equipment Identification Labels:
 - 1. Indoor Equipment: Self-adhesive label.
 - 2. Outdoor Equipment:

END OF SECTION 260553

SECTION 260573 - OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes computer-based, fault-current and overcurrent protective device coordination studies. Protective devices shall be set based on results of the protective device coordination study performed by a licensed professional engineer hired by the electrical contractor.
- B. Section includes computer based arc flash hazard analysis study. Recommendations for reducing Arc Flash Incident Energy levels and enhancing worker safety shall be issued based on the results of the study.

1.2 DEFINITIONS

- A. AFIE - Arc Flash Incident Energy.

1.3 SUBMITTALS

- A. The complete fault current, arc flash and coordination study is required to be submitted seven (7) calendar days prior to the submittal of any switchgear, panelboards, overcurrent devices, motor control devices, UPS and transfer switches.
- B. Product Data: For computer software program to be used for studies.
- C. Product Certificates: For coordination-study and fault-current-study computer software programs, certifying compliance with IEEE 399, for arc flash study - compliance with IEEE 1584.
- D. Qualification Data: For coordination-study specialist. Submit names and references for at least five actual Arc-Flash Hazard Analysis performed in the past year.
- E. Other Action Submittals: The following submittals shall be made after the approval process for system protective devices has been completed. Submittals shall be in digital form.
 - 1. Coordination-study input data, including completed computer program input data sheets.
 - 2. Study and Equipment Evaluation Reports.
 - 3. Coordination-Study Report.
 - 4. Arc Flash Study Report.

1.4 QUALITY ASSURANCE

- A. Studies shall use computer programs that are distributed nationally and are in wide use. Software algorithms shall comply with requirements of standards and guides specified in this Section. Manual calculations are not acceptable.
- B. Coordination-Study and Arc Flash Study Specialist Qualifications: An entity experienced in the application of computer software used for studies, having performed successful studies of similar magnitude on electrical distribution systems using similar devices.
 - 1. Professional engineer, licensed in the state of Maryland, shall be responsible for the study. All elements of the study shall be performed under the direct supervision and control of engineer.

- C. Comply with IEEE 241 and 242 for short-circuit currents and coordination time intervals.
- D. Comply with IEEE 399 for general study procedures.
- E. Comply with IEEE 1584-2002 for arc flash calculations procedures.
- F. Comply with the National Electrical Code.
- G. Comply with National Electrical Code Article 110.

1.5 REFERENCES

- A. Institute of Electrical and Electronics Engineers, Inc. (IEEE):
 - 1. IEEE 242 – Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems.
 - 2. IEEE 241 – Recommended Practice for Electric Power Systems in Commercial Buildings.
 - 3. IEEE 1015 – Recommended Practice for Applying Low-Voltage Circuit Breakers Used in Industrial and Commercial Power Systems.
- B. American National Standards Institute (ANSI):
 - 1. C37.13 – Standard for Low Voltage AC Power Circuit Breakers Used in Enclosures
 - 2. ANSI Z535.4 – Identification of AFIE and appropriate Personal Protective Equipment Classes.

PART 2 - PRODUCTS

2.1 COMPUTER SOFTWARE DEVELOPERS

- A. Computer Software Developers: Subject to compliance with requirements, provide products by one of the following software developers:
 - 1. SKM Systems Analysis, Inc.
 - 2. CGI CYME.
 - 3. EDSA Micro Corporation.
 - 4. ESA Inc.
 - 5. Operation Technology, Inc.

2.2 COMPUTER SOFTWARE PROGRAM REQUIREMENTS

- A. Analytical features of fault-current-study computer software program shall include "mandatory," "very desirable," and "desirable" features as listed in IEEE 399.
- B. Computer software program shall be capable of plotting and diagramming time-current-characteristic curves as part of its output. Computer software program shall report device settings and ratings of all overcurrent protective devices and shall demonstrate selective coordination by computer-generated, time-current coordination plots.
 - 1. Optional Features:
 - a. Arcing faults.
 - b. Simultaneous faults.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine Project overcurrent protective device submittals for compliance with electrical distribution system coordination requirements and other conditions affecting performance. All new and existing devices and circuit breakers associated with the project shall be fully coordinated.
1. Proceed with coordination study only after relevant equipment submittals have been assembled.
 2. Proceed with arc flash analysis only after relevant equipment submittals have been assembled.
 3. Overcurrent protective devices that have not been submitted and approved prior to coordination study may not be used in study.
 4. Contractor shall set all of the new and existing device values as recommended by the coordination study results.

3.2 POWER SYSTEM DATA

- A. Gather and tabulate the following input data to support coordination study:
1. Product Data for overcurrent protective devices specified in other Division 26 Sections and involved in overcurrent protective device coordination studies. Use equipment designation tags that are consistent with electrical distribution system diagrams, overcurrent protective device submittals, input and output data, and recommended device settings.
 2. Impedance of utility service entrance.
 3. Electrical Distribution System Diagram: In hard-copy and electronic-copy formats, showing the following:
 - a. Circuit-breaker and fuse-current ratings and types.
 - b. Relays and associated power and current transformer ratings and ratios.
 - c. Transformer kilovolt amperes, primary and secondary voltages, connection type, impedance, and X/R ratios.
 - d. Generator kilovolt amperes, size, voltage, and source impedance.
 - e. Cables: Indicate conduit material, sizes of conductors, conductor material, insulation, and length.
 - f. Busway ampacity and impedance.
 - g. Motor horsepower and code letter designation according to NEMA MG 1.
 4. Data sheets to supplement electrical distribution system diagram, cross-referenced with tag numbers on diagram, showing the following:
 - a. Special load considerations, including starting inrush currents and frequent starting and stopping.
 - b. Transformer characteristics, including primary protective device, magnetic inrush current, and overload capability.
 - c. Motor full-load current, locked rotor current, service factor, starting time, type of start, and thermal-damage curve.
 - d. Generator thermal-damage curve.
 - e. Ratings, types, and settings of utility company's overcurrent protective devices.
 - f. Special overcurrent protective device settings or types stipulated by utility company.

- g. Time-current-characteristic curves of devices indicated to be coordinated.
- h. Manufacturer, frame size, interrupting rating in amperes rms symmetrical, ampere or current sensor rating, long-time adjustment range, short-time adjustment range, and instantaneous adjustment range for circuit breakers.
- i. Manufacturer and type, ampere-tap adjustment range, time-delay adjustment range, instantaneous attachment adjustment range, and current transformer ratio for overcurrent relays.
- j. Panelboards, switchboards, motor-control center ampacity, and interrupting rating in amperes rms symmetrical.

3.3 FAULT-CURRENT STUDY

- A. Calculate the maximum available short-circuit current in amperes rms symmetrical at circuit-breaker positions of the electrical power distribution system. The calculation shall be for a current immediately after initiation and for a three-phase bolted short circuit at each of the following:
 - 1. Switchgear and switchboard bus.
 - 2. Motor-control center.
 - 3. Distribution panelboard.
 - 4. Branch circuit panelboard.
- B. Study electrical distribution system from normal and alternate power sources throughout electrical distribution system for Project. Include studies of system-switching configurations and alternate operations that could result in maximum fault conditions. Obtain primary fault level data from utility company.
- C. Calculate momentary and interrupting duties on the basis of maximum available fault current.
- D. Calculations to verify interrupting ratings of overcurrent protective devices shall comply with IEEE 241 and IEEE 242.
 - 1. Transformers:
 - a. IEEE C57.96.
 - 2. Low-Voltage Circuit Breakers: IEEE 1015 and IEEE C37.20.1.
 - 3. Low-Voltage Fuses: IEEE C37.46.
- E. Study Report:
 - 1. Show calculated X/R ratios and equipment interrupting rating (1/2-cycle) fault currents on electrical distribution system diagram.
- F. Equipment Evaluation Report:
 - 1. For 600-V overcurrent protective devices, ensure that interrupting ratings are equal to or higher than calculated 1/2-cycle symmetrical fault current.
 - 2. For devices and equipment rated for asymmetrical fault current, apply multiplication factors listed in the standards to 1/2-cycle symmetrical fault current.
 - 3. Verify adequacy of phase conductors at maximum three-phase bolted fault currents; verify adequacy of equipment grounding conductors and grounding electrode conductors at maximum ground-fault currents. Ensure that short-circuit withstand ratings are equal to or higher than calculated 1/2-cycle symmetrical fault current.

3.4 COORDINATION STUDY

- A. Perform coordination study using approved computer software program. Prepare a written report using results of fault-current study. Comply with IEEE 399.
 - 1. Calculate the maximum and minimum 1/2-cycle short-circuit currents.
 - 2. Calculate the maximum and minimum ground-fault currents.
- B. Comply with IEEE 241 and IEEE 242 recommendations for fault currents and time intervals.
- C. Transformer Primary Overcurrent Protective Devices:
 - 1. Device shall not operate in response to the following:
 - a. Inrush current when first energized.
 - b. Self-cooled, full-load current or forced-air-cooled, full-load current, whichever is specified for that transformer.
 - c. Permissible transformer overloads according to IEEE C57.96 if required by unusual loading or emergency conditions.
 - 2. Device settings shall protect transformers according to IEEE C57.12.00, for fault currents.
- D. Conductor Protection: Protect cables against damage from fault currents according to ICEA P-32-382, ICEA P-45-482, and conductor melting curves in IEEE 242. Demonstrate that equipment withstands the maximum short-circuit current for a time equivalent to the tripping time of the primary relay protection or total clearing time of the fuse. To determine temperatures that damage insulation, use curves from cable manufacturers or from listed standards indicating conductor size and short-circuit current.
- E. Coordination-Study Report: Prepare a written report indicating the following results of coordination study:
 - 1. Tabular Format of Settings Selected for Overcurrent Protective Devices:
 - a. Device tag.
 - b. Relay-current transformer ratios; and tap, time-dial, and instantaneous-pickup values.
 - c. Circuit-breaker sensor rating; and long-time, short-time, and instantaneous settings.
 - d. Fuse-current rating and type.
 - e. Ground-fault relay-pickup and time-delay settings.
 - 2. Coordination Curves: Prepared to determine settings of overcurrent protective devices to achieve selective coordination. Graphically illustrate that adequate time separation exists between devices installed in series, including power utility company's upstream devices. Prepare separate sets of curves for the switching schemes and for emergency periods where the power source is local generation. Show the following information:
 - a. Device tag.
 - b. Voltage and current ratio for curves.
 - c. Three-phase and single-phase damage points for each transformer.
 - d. No damage, melting, and clearing curves for fuses.
 - e. Cable damage curves.
 - f. Transformer inrush points.

- g. Maximum fault-current cutoff point.
 - F. Prepare and include in study completed data sheets for setting of overcurrent protective devices.
 - G. Devices to coordinate down to 0.01 seconds. Coordination required for emergency systems and legally required systems.
- 3.5 ARC FLASH HAZARD ANALYSIS
- A. Perform Arc-Flash Hazard Analysis with the aid of computer software intended for this purpose in order to calculate AFIE levels and flash protection boundary distances.
 - B. Perform Arc-Flash Hazard Analysis in conjunction with a short-circuit analysis and a time-current coordination analysis.
 - C. Submit the results of the Analysis in tabular form, and include device or bus name, bolted fault and arcing fault current levels, flash protection boundary distances, personal-protective equipment classes and AFIE levels.
 - D. Perform the analysis under worst-case Arc-Flash conditions, and describe in the final report when applicable, how these conditions differ from worst-case bolted fault conditions.
 - E. Provide equipment labels in compliance with ANSI Z535.4 to identify AFIE and appropriate Personal Protective Equipment classes.

END OF SECTION 260573

SECTION 260800 – COMMISSIONING OF ELECTRICAL**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes commissioning process requirements for HVAC&R systems, assemblies, and equipment.
- B. Related Sections:
 - 1. Division 01 Section "General Commissioning Requirements" for general commissioning process requirements.
 - 2. Division 23 Sections related to Testing, Adjusting and Balancing (TAB)

1.3 COMMISSIONED SYSTEMS

- 1. Lighting Controls
- 2. Exterior Lighting Controls
- 3. Occupancy Sensors

1.4 CONTRACTOR'S RESPONSIBILITIES

- A. The contractor shall be responsible for scheduling, supervising and performing start-up, testing and commissioning activities specified in this section and necessary to demonstrate to the owner successful operation of the commissioned systems.
- B. Review submittal data for conformance with requirements, authorize the initial starting of equipment and systems in a manner to avoid damage during construction, oversee start-up and testing, and document that the operational requirements of each system have been accomplished.

1.5 SUBMITTALS

- A. Certificates of readiness.
- B. Certificates of completion of installation, prestart, and startup activities.

PART 2 - PRODUCTS

2.1 SUBMITTALS

- A. Provide means for the CxA to access, observe and visually confirm proper operation of all equipment and systems. These means shall be in compliance with all OSHA and job-site safety regulations.

2.2 TEST EQUIPMENT

- A. Provide means for the CxA to fully test the commissioned systems as defined in the functional performance test procedures provided by the CxA.
- B. Meet the following minimum requirements for test equipment:
 - 1. All test equipment shall be in good mechanical and electrical condition.
 - 2. Field test metering used to check power system meter calibration shall be more accurate than the instrument being tested.
 - 3. Accuracy of metering in test equipment shall be appropriate for the test being performed.
 - 4. Waveform and frequency of test equipment output waveforms shall be appropriate for the test and the tested equipment.
- C. Calibration
 - 1. Calibration of all test equipment shall be current.
 - 2. Calibration accuracy shall be traceable to National Institute of Standards and Technology (NIST).
 - 3. Calibrate test equipment in accordance with the following schedule.
 - a. Field instruments
 - 1) Analog – At least every 6 months
 - 2) Digital – At least every 12 months
 - b. Leased Specialty Equipment – At least every 12 months
 - 4. Dated calibration labels shall be visible on all test equipment.
 - 5. Provide calibration records for all test equipment used in the project.

PART 3 - EXECUTION

3.1 GENERAL

- A. Attend commissioning meetings as required by the CxA.
- B. Report in writing to the CxA concerning the status of activities as they affect the commissioning process and the status of deficiencies found during site visits or the performance of functional testing.
- C. Provide the CxA with data sheets and submittals for equipment to be commissioned.
- D. Prepare a preliminary schedule for commissioning activities.
- E. Notify the CxA when commissioning activities not yet performed or scheduled will delay construction

3.2 START-UP PLAN

- A. Perform start-up testing for each piece of equipment to ensure that the equipment and systems are properly installed and ready for operation, so that functional performance testing may proceed without delay.
- B. Prepare a start-up plan for each piece of equipment. This plan shall be submitted to the CxA for review and comment. The start-up plan shall consist, at a minimum of the following:
 - 1. The manufacturer's standard start-up and check out procedures copied from the installation instructions.
 - 2. Checklists and procedures with specific spaces for recording and documenting the inspection of each procedure and a summary block for deficiencies and explanations.
 - 3. Equipment Prefunctional Checklist
- C. Four (4) weeks prior to expected start-up for a piece of equipment, notify the owner and the CxA in writing. The execution of the start-up plan shall be directed and performed by the Contractor. The CxA and/or Owner may be present for the start-up of the equipment.
- D. Submit the completed start-up checklists, reports and equipment prefunctional checklists to the CxA for review. Note all noncompliance items on these checklists. Notify the CxA when outstanding items have been corrected.
- E. Complete the start-up plan and resolve or correct all issues before functional testing begins.

3.3 FUNCTIONAL TESTING PREPARATION

- A. Certify that systems, subsystems, and equipment have been installed, calibrated, and started and are operating according to the Contract Documents.

- B. Certify that instrumentation and control systems have been completed and calibrated, that they are operating according to the Contract Documents, and that pretest set points have been recorded.
- C. Certify that testing, adjusting, and balancing procedures have been completed and that testing, adjusting, and balancing reports have been submitted, discrepancies corrected, and corrective work approved.

3.4 FUNCTIONAL PERFORMANCE TESTING

- A. Provide all documentation as requested to the CxA for development of functional performance testing procedures. This documentation shall include, at a minimum, manufacturer installation, start-up, operation and maintenance procedures. The CxA may request further documentation as necessary for the development of functional performance tests.
- B. Review the functional performance test procedures developed by the CxA.
 - 1. Respond in writing to the CxA regarding the acceptability of the proposed test procedures
 - 2. Note any necessary modifications to the procedures due to the actual equipment/systems or safety concerns and submit these to the CxA for consideration.
- C. Place equipment and systems into operation and continue the operation as required during each working day of the testing activities.
- D. Accomplish the functional performance testing of equipment based on procedures developed by the CxA and as reviewed by the Contractor.
 - 1. Provide skilled technicians to operate the systems during functional performance testing.
 - 2. Correct any deficiencies identified during testing and retest equipment as required.
- E. Functional performance testing is intended to begin upon completion of a system. Functional testing may proceed prior to the completion of the system at the discretion of the CxA and the Contractor.
- F. Verify all sequences of operation defined in the Contract Documents for the commissioned equipment and systems.
 - 1. Perform testing by overriding set points or sensor readings at the DDC System or by other means mutually agreed to by the Contractor, the CxA and the owner, to initiate sequences of operation and verifying the response of the system.
 - 2. Sequences of operation shall be verified under normal power, emergency power, and fire alarm scenarios.
 - 3. Upon successful completion of all functional performance tests, perform Integrated Systems Testing. The testing shall document and verify the proper response of all Plumbing systems to all potential utility and emergency power operating and failure scenarios.

3.5 TEST AND VERIFICATION

- A. The Contractor shall complete and a short circuit study on all systems. Findings to be provided to CxA and The owner.
- B. The Contractor shall regard this verification process as a functional performance test for purposes of time allowed to correct deficiencies and requirements regarding retesting if major problems are discovered.

3.6 DEFERRED FUNCTIONAL PERFORMANCE TESTING

- A. Perform any deferred testing as required to properly demonstrate successful operation to the owner.
 - 1. Some test conditions may not be able to be simulated and thus require these actual conditions to be present to implement and test.
 - 2. A mutually convenient time to owner, CxA and Contractor will be scheduled when these test conditions will be present to conduct this deferred testing.
- B. Perform these tests as indicated in the functional performance test procedures.
- C. Correct any deficiencies or failures identified in the process of performing these tests.

END OF SECTION 260800

SECTION 260813 - TESTING OF ELECTRICAL SYSTEMS**PART 1 - GENERAL****1.1 SUMMARY**

A. Section includes requirements for Electrical Testing including, but not limited to:

1. Testing of electrical components and systems.
 - a. Insulation resistance test.
 - b. Continuity test.
 - c. Voltage test.
 - d. Phase relationship verification.
2. Correction of defective components or systems.
3. Retest of corrected components, systems.

1.2 SUBMITTALS

A. Test reports: Submit six (6) copies of all test reports to Architect/Engineer of Record (3 copies), General Contractor (1).

1. Type each test report on 8-1/2" x 11" paper. Include:
 - a. Project title and location.
 - b. Test performed.
 - c. Data performed
 - d. Test equipment used.
 - e. Contractor's name, address and telephone number.
 - f. Testing firm's name, address and telephone number.
 - g. Names and titles of persons:
 - 1) Performing the test.
 - 2) Observing test
 - h. Statement verifying each test.
 - i. Nameplate data from each motor and equipment item tested.
 - j. Test results.
 - k. Retest results after correction of defective components, systems.
 - l. Dates and time of test.
2. For each copy, assemble all test reports and bind them in a folder. Label each folder, "Electrical Test Reports" and include LiDL Project Number, title and location.

PART 2 - PRODUCTS**2.1 MATERIALS:**

A. Furnish all equipment, manpower and casual labor to perform specified testing.

PART 3 - EXECUTION**3.1 PREPARATION**

- A. Ensure that all electrical work is completed and ready for testing.
- B. Disconnect all devices or equipment that may be damaged by application of test voltages, voltage or reversed phase sequence or other procedures.

3.2 TESTING:

- A. Conduct tests and adjust equipment to verify compliance with specified performance.

3.3 INSULATION RESISTANCE TESTS

- A. Resistance measured: line-to-ground.

- B. Perform testing on the following items:

Item Tested	Min. Acceptance Voltage of Test	Resistance in Megohms
No. 2. and Larger Cables (600v)	1000v	50

3.4 CONTINUITY TESTS:

- A. Test branch circuits and control circuits to determine continuity of wiring and connection.

3.5 VOLTAGE TESTS

- A. Make and record voltage tests and record at the following listed points. Conduct tests under normal load conditions.
 - 1. Terminals of all motors.
 - 2. Terminals of all equipment, i.e., UPS, refrigeration compressors, etc.

3.6 PHASE RELATIONSHIP

- A. Examine connections to equipment for proper phase relationships. Verify proper motor rotation.

3.7 CORRECTION OF DEFECTS

- A. When tests disclose any unsatisfactory workmanship or equipment furnished under this Contract, correct defects and retest. Repeat tests until satisfactory results are obtained.
- B. When any wiring or equipment is damaged by tests, repairs or replace such wiring or equipment. Test repaired items to ensure satisfactory operation.

3.8 CONTRACTOR STARTUP AND REPORTING

- A. Contractor shall prepare and submit a complete set of test reports as outlined in this section.

END OF SECTION 260813

SECTION 260923 - LIGHTING CONTROL DEVICES**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section Includes:
 - 1. Photoelectric switches.
 - 2. Standalone daylight-harvesting switching controls.
 - 3. Indoor occupancy and vacancy sensors.
 - 4. Outdoor motion sensors.
 - 5. Switchbox-mounted occupancy and vacancy sensors
- B. Related Requirements:
 - 1. Section 262726 "Wiring Devices" for wall-box dimmers, non-networkable wall-switch occupancy sensors, and manual light switches.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings:
 - 1. Show installation details for the following:
 - a. Occupancy sensors.
 - b. Vacancy sensors.
 - 2. Interconnection diagrams showing field-installed wiring.
 - 3. Include diagrams for power, signal, and control wiring.

1.3 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Reflected ceiling plan(s) and elevations, drawn to scale and coordinated with each other, using input from installers of the items involved.
- B. Field quality-control reports.
- C. Sample warranty.

1.4 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.
- B. Software and firmware operational documentation.

1.5 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.
- B. Manufacturer's Warranty: Manufacturer and Installer agree to repair or replace lighting control devices that fail(s) in materials or workmanship within specified warranty period.
 - 1. Warranty Period: Two year(s) from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 OUTDOOR PHOTOELECTRIC SWITCHES

- A. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**:
 - 1. Cooper Industries, Inc.
 - 2. Emerson Inc.
 - 3. Leviton Manufacturing Co., Inc.
 - 4. Wattstopper / Legrand
- B. Description: Solid state, with SPST dry contacts rated for 1800 VA inductive, to operate connected relay, contactor coils, or microprocessor input; complying with UL 773A, and compatible with ballasts and LED lamps.
 - 1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
 - 2. Light-Level Monitoring Range: 1.5 to 10 fc, with an adjustment for turn-on and turn-off levels within that range, and a directional lens in front of the photocell to prevent fixed light sources from causing turn-off.
 - 3. Time Delay: Fifteen-second minimum, to prevent false operation.
 - 4. Surge Protection: Metal-oxide varistor.
 - 5. Mounting: Twist lock complies with NEMA C136.10, with base-and-stem mounting or stem-and-swivel mounting accessories as required to direct sensor to the north sky exposure.
 - 6. Failure Mode: Luminaire stays ON.
- C. Description: Solid state; one set of NO dry contacts rated for 24 V dc at 1 A, to operate connected load, complying with UL 773, and compatible with lighting control panelboard.
 - 1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
 - 2. Light-Level Monitoring Range: 1.5 to 10 fc, with an adjustment for turn-on and turn-off levels within that range.
 - 3. Time Delay: Thirty-second minimum, to prevent false operation.
 - 4. Mounting: 1/2-inch threaded male conduit.
 - 5. Failure Mode: Luminaire stays ON.
 - 6. Power Pack: Dry contacts rated for 20-A ballast or LED load at 120- and 277-V ac.
 - a. LED status lights to indicate load status.

- b. Plenum rated.
- 7. Power Pack: Digital controller capable of accepting three RJ45 inputs with one outputs rated for 20-A LED load at 120- and 277-V ac, for 16-A ballast or LED at 120- and 277-V ac. Sensor has 24-V dc, Class 2 power source, as defined by NFPA 70.
 - a. With integral current monitoring
 - b. Compatible with digital addressable lighting interface.
 - c. Plenum rated.

2.2 INDOOR OCCUPANCY AND VACANCY SENSORS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Wattstopper / Legrand Cooper Industries, Inc.
 - 2. Hubbell Building Automation, Inc.
 - 3. Leviton Manufacturing Co., Inc.
 - 4. Lutron Electronics Co., Inc.
 - 5. Lighting fixture supplied integral sensors.
- B. General Requirements for Sensors:
 - 1. Wall or Ceiling-mounted, solid-state indoor occupancy and vacancy sensors.
 - 2. Dual technology.
 - 3. Integrated power pack.
 - 4. Hardwired connection to switch and BAS; and BAS and lighting control system.
 - 5. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
 - 6. Operation:
 - a. Occupancy Sensor: Unless otherwise indicated, turn lights on when coverage area is occupied, and turn them off when unoccupied; with a time delay for turning lights off, adjustable over a minimum range of 1 to 15 minutes.
 - b. Vacancy Sensor: Unless otherwise indicated, lights are manually turned on and sensor turns lights off when the room is unoccupied; with a time delay for turning lights off, adjustable over a minimum range of 1 to 15 minutes.
 - c. Combination Sensor: Unless otherwise indicated, sensor shall be programmed to turn lights on when coverage area is occupied and turn them off when unoccupied, or to turn off lights that have been manually turned on; with a time delay for turning lights off, adjustable over a minimum range of 1 to 15 minutes.
 - 7. Sensor Output: Sensor is powered from the power pack.
 - 8. Power: Line voltage.
 - 9. Power Pack: Dry contacts rated for 20-A ballast or LED load at 120- and 277-V ac, for 13-A tungsten at 120-V ac. Sensor has 24-V dc, 150-mA, Class 2 power source, as defined by NFPA 70.
 - 10. Mounting:
 - a. Sensor: Suitable for mounting in any position on a standard outlet box.

- b. Relay: Externally mounted through a 1/2-inch knockout in a standard electrical enclosure.
 - c. Time-Delay and Sensitivity Adjustments: Recessed and concealed behind hinged door.
 - 11. Indicator: Digital display, to show when motion is detected during testing and normal operation of sensor.
 - 12. Bypass Switch: Override the "on" function in case of sensor failure.
 - 13. Automatic Light-Level Sensor: Adjustable from 2 to 200 fc; turn lights off when selected lighting level is present.
- C. Dual-Technology Type: Wall or Ceiling mounted; detect occupants in coverage area using PIR and ultrasonic detection methods. The particular technology or combination of technologies that control on-off functions is selectable in the field by operating controls on unit.
- 1. Sensitivity Adjustment: Separate for each sensing technology.
 - 2. Detector Sensitivity: Detect occurrences of 6-inch-minimum movement of any portion of a human body that presents a target of not less than 36 sq. in., and detect a person of average size and weight moving not less than 12 inches in either a horizontal or a vertical manner at an approximate speed of 12 inches/s.
 - 3. Detection Coverage (Standard Room): Detect occupancy anywhere within a circular area of 1000 sq. ft. when mounted on a 96-inch-high ceiling.
 - 4. Detection Coverage (Room, Wall Mounted): Detect occupancy anywhere within a 180-degree pattern centered on the sensor over an area of 1000 square feet when mounted 48 inches above finished floor.

2.3 SWITCHBOX-MOUNTED OCCUPANCY SENSORS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
- 1. Wattstopper / Legrand Cooper Industries, Inc.
 - 2. Hubbell Building Automation, Inc.
 - 3. Leviton Manufacturing Co., Inc.
 - 4. Lutron Electronics Co., Inc.
- B. General Requirements for Sensors: Automatic-wall-switch occupancy sensor with manual on-off switch, suitable for mounting in a single gang switchbox, with provisions for connection to BAS using hardwired connection.
- 1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
 - 2. Occupancy Sensor Operation: Unless otherwise indicated, turn lights on when coverage area is occupied, and turn lights off when unoccupied; with a time delay for turning lights off, adjustable over a minimum range of 1 to 15 minutes.
 - 3. Operating Ambient Conditions: Dry interior conditions, 32 to 120 deg F.
 - 4. Switch Rating: Not less than 800-VA ballast or LED load at 120 V, 1200-VA ballast or LED load at 277 V.

2.4 LIGHTING CONTACTORS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Schneider Electric Inc.
 - 2. Wattstopper / Legrand Cooper Industries, Inc.
 - 3. General Electric
 - 4. Hubbell Building Automation, Inc.
 - 5. Leviton Manufacturing Co., Inc.
- B. Description: Electrically operated and electrically held complying with NEMA ICS 2 and UL 508.
 - 1. Current Rating for Switching: Listing or rating consistent with type of load served, including tungsten filament, inductive, and high-inrush ballast (ballast with 15 percent or less THD of normal load current).
 - 2. Fault Current Withstand Rating: Equal to or exceeding the available fault current at the point of installation.
 - 3. Enclosure: Comply with NEMA 250.

2.5 OUTDOOR MOTION SENSORS (PIR)

- C. Performance Requirements: Suitable for operation in ambient temperatures ranging from minus 40 to plus 130 deg F, rated as rain-tight according to UL 773A.
 - 1. Operation: Turn lights on when sensing infrared energy changes between background and moving body in area of coverage; with a time delay for turning lights off, adjustable over a minimum range of 1 to 15 minutes.
 - 2. Mounting:
 - a. Sensor: Suitable for mounting in any position on a standard outdoor junction box.
 - b. Relay: Internally mounted in a standard weatherproof electrical enclosure.
 - c. Time-Delay and Sensitivity Adjustments: Recessed and concealed behind hinged door.
 - 3. Bypass Switch: Override the on function in case of sensor failure.
 - 4. Automatic Light-Level Sensor: Adjustable from 1 to 20 fc; keep lighting off during daylight hours.
- D. Detector Sensitivity: Detect occurrences of 6-inch- minimum movement of any portion of a human body that presents a target of not less than 36 sq. in.
- E. Detection Coverage: Up to 35 feet, with a field of view of 180 degrees.
- F. Retain "Interface with DDC System for HVAC" Paragraph below for interface with the DDC system for HVAC. Coordinate with Section 230923 "Direct Digital Control (DDC) System for HVAC."

2.6 EMERGENCY SHUNT RELAY

- G. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Wattstopper / Legrand Cooper Industries, Inc.
 - 2. General Electric
 - 3. Hubbell Building Automation, Inc.
 - 4. Leviton Manufacturing Co., Inc.
 - 5. Lutron Electronics Co., Inc.

- H. Description: NC, electrically held relay, arranged for wiring in parallel with automatic switching contacts; complying with UL 924.

2.7 CONDUCTORS AND CABLES

- A. Power Wiring to Supply Side of Remote-Control Power Sources: Not smaller than No. 12 AWG. Comply with requirements in Section 260519 "Low-Voltage Electrical Power Conductors and Cables."
- B. Classes 2 and 3 Control Cable: Multi-conductor cable with stranded-copper conductors not smaller than No. 18 AWG. Comply with requirements in Section 260519 "Low-Voltage Electrical Power Conductors and Cables."
- C. Class 1 Control Cable: Multi-conductor cable with stranded-copper conductors not smaller than No. 18 AWG. Comply with requirements in Section 260519 "Low-Voltage Electrical Power Conductors and Cables."

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Comply with NECA 1.
- B. Examine lighting control devices before installation. Reject lighting control devices that are wet, moisture damaged, or mold damaged.
- C. Coordinate layout and installation of ceiling-mounted devices with other construction that penetrates ceilings or is supported by them, including light fixtures, HVAC equipment, smoke detectors, fire-suppression systems, and partition assemblies.
- D. Install and aim sensors in locations to achieve not less than 90-percent coverage of areas indicated. Do not exceed coverage limits specified in manufacturer's written instructions.
- E. Mount electrically held lighting contactors with elastomeric isolator pads to eliminate structure-borne vibration unless contactors are installed in an enclosure with factory-installed vibration isolators.

3.2 WIRING INSTALLATION

- A. Wiring Method: Comply with Section 260519 "Low-Voltage Electrical Power Conductors and Cables." Minimum conduit size is 1/2 inch.
- B. Wiring within Enclosures: Separate power-limited and non-power-limited conductors according to conductor manufacturer's written instructions.
- C. Size conductors according to lighting control device manufacturer's written instructions unless otherwise indicated.
- D. Splices, Taps, and Terminations: Make connections only on numbered terminal strips in junction, pull, and outlet boxes; terminal cabinets; and equipment enclosures.

3.3 IDENTIFICATION

- A. Identify components and power and control wiring according to Section 260553 "Identification for Electrical Systems."
- B. Label time switches and contactors with a unique designation.

3.4 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections:
 - 1. Operational Test: After installing time switches and sensors, and after electrical circuitry has been energized, start units to confirm proper unit operation.
 - 2. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- B. Lighting control devices will be considered defective if they do not pass tests and inspections.
- C. Prepare test and inspection reports.

3.5 ADJUSTING

- A. Occupancy Adjustments: When requested within 12 months from date of Substantial Completion, provide on-site assistance in adjusting lighting control devices to suit actual occupied conditions. Provide up to two visits to Project during other-than-normal occupancy hours for this purpose.
 - 1. For occupancy and motion sensors, verify operation at outer limits of detector range. Set time delay to suit Owner's operations.
 - 2. For daylighting controls, adjust set points and dead band controls to suit Owner's operations.
 - 3. Align high-bay occupancy sensors using manufacturer's laser aiming tool.

3.6 DEMONSTRATION

- A. Train Owner's maintenance personnel to adjust, operate, and maintain lighting control devices.

END OF SECTION 260923

SECTION 262126 – COLD STORAGE FROST HEAVE PREVENTION SYSTEMS**PART 1 - GENERAL****1.1 SUMMARY**

- A. This Section includes a UL Listed and CSA Certified freezer frost heave prevention heat tracing system consisting of self-regulating heating cable, connection kits and electronic controller.
- B. Related Requirements:
 - 1. Section 114126 – Walk-in Freezers.
 - 2. Section 03 30 00 – Cast-In Place Concrete
 - 3. Section 25 12 16 – Direct-Protocol Integration Network Gateways
 - 4. Section 25 51 00 – Integrated Automation Control of Facility Equipment.
- C. System Description:
 - 1. System for preventing frost heave in freezers, cold rooms and ice arenas with temperature control, monitoring, integrated ground-fault circuit protection and BMS communication capabilities
 - 2. The system shall consist of all equipment and materials for a complete floor warming/space heating system to be installed.
 - 3. The entire installation of the system shall comply with National Electrical Code (NEC) for US installations and all 'AHJ' requirements.
 - 4. The entire system shall comply with electrical loads and characteristics allocated in the electrical design.

1.2 SUBMITTALS

- A. Product Data: For each type of product.
 - 1. Heating cable data sheet
 - 2. CSA or FM approval certificates for freezer frost heave prevention
 - 3. Freezer Frost Heave Prevention design guide
 - 4. System installation and operation manual
 - 5. System installation details
 - 6. Controller/Power Panel data sheet
 - 7. Controller/Power Panel wiring diagram
- B. Shop Drawings: Signed and sealed by a qualified professional engineer. For fabrication and installation details for electrical hangers and support systems.
 - 1. Hangers. Include product data for components.
 - 2. Slotted support systems.
 - 3. Equipment supports.

1.3 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Reflected ceiling plan(s) and other details, drawn to scale, and coordinated with each other, using input from installers of the items involved.

1.4 QUALITY ASSURANCE

A. Manufacturers' Qualifications

1. Manufacturer to show minimum of thirty (30) years' experience in manufacturing mineral insulated heating cables.
2. Manufacturer is ISO-9001 registered.
3. Manufacturer to provide products that comply with UL 515, CSA 22.2 No 130-03 and IEEE 515.1 requirements.

B. Installer Qualifications

1. System installer shall have a complete understanding of product and product literature from manufacturer or authorized representative prior to installation. Electrical connections shall be performed by a licensed electrician.

C. Regulatory Requirements and Approvals

1. The system (heating cable, connection kits, and controller) shall be UL Listed and CSA Certified for freezer frost heave prevention.

D. Electrical Components, Devices, and Accessories: Listed and labelled as defined in Article 100 of NFPA 70 National Electrical Code by a Nationally Recognized Testing Laboratory (NRTL), and marked for intended use.

1.5 DELIVERY, STORAGE AND HANDLING

A. General Requirements: Deliver, store and handle products to prevent their deterioration or damage due to moisture, temperature changes, contaminants or other causes.

B. Delivery and Acceptance Requirements: Deliver products to site in original, unopened containers or packages with intact and legible manufacturers' labels identifying the following:

1. Product and Manufacturer
2. Length/Quantity
3. Lot Number
4. Installation and Operation Manual
5. MSDS (if applicable) Manufacturer to show minimum of thirty (30) years' experience in manufacturing mineral.

C. Storage and Handling Requirements

1. Store the heating cable in a clean, dry location with a temperature range not below -40°F (-40°C) or exceeding 140°F (60°C).
2. Protect the heating cable from mechanical damage.

1.6 WARRANTY

A. Extended Warranty

1. Manufacturer shall make available a ten (10) year limited warranty for heating cables and components. Manufacturer shall also provide one (1) year warranty for all heat trace controllers.
2. Contractor shall submit to owner the results of all installation tests required by the manufacturer

PART 2 - PRODUCTS

2.1 MANUFACTURERS AND PRODUCTS

- A. Contract Documents are based on manufacturer and products named below to establish a standard of quality.
- B. Basis of Design:
 - 1. Basis of Design Product Selections
 - a. Manufacturer:
 - 1) Manufacturer shall have more than thirty (30) years' experience with manufacture & installation self-regulating heating cables.
 - 2) Manufacturer shall provide UL and CSA approval certificates for freezer frost heave prevention.
 - 3) Manufacturer shall be Heatizon Systems, 4137 South 500 West, Murray, Utah 84123 (888) 293-1232; www.heatizon.com – info@heatizon.com

2.2 COMPONENTS

- A. Heating element
 - 1. Cozy Heat® SR Cable and all accessories shall be supplied by Heatizon Systems (Tel: 888-239-1232, Fax: 801-293-3077)
 - 2. Shall consist of two (2) 16 AWG nickel-copper bus wires embedded in parallel in a Radiation Cross-Linked Polyolefin Core that varies its power output to respond to temperature along the entire length, allowing the heating cable to be cut to length in the field. The core is surrounded by a liner then a Radiation Cross-Linked Modified Polyolefin Jacket covers the heating cable core and liner. To provide a ground path and to enhance the heating cable's ruggedness, the heating cable shall have a Tinned Copper Braid (-C), as required by Article 427.23(A) of the NEC-2011. An ultraviolet stabilized weatherproof over-jacket composed of: Modified Polyolefin shall cover the grounding braid. The cable shall be suited for use in concrete, a mortar bed, and other cementitious materials.
 - 3. For energy efficiency and in order to provide heat output that is sufficient for floor and space heating, the heating cable shall have a nominal rating, in correlation with selection of the heating cable, of 5, 9, or 12 watts per lineal foot in Iced Water @ 32°F.
 - 4. The heating cable shall operate on line voltages of (select: 120, 208, 220, 240, or 277) volts.
 - 5. Power connection, end seal, splice kit, and tee kit components shall be supplied by Heatizon Systems and applied in the field.
 - 6. A ground-fault device for equipment protection, rated at 30-mA trip, shall protect heating cable circuit. This requirement is in accordance with Article 426 of the NEC-2011.
- B. Relay Control Panel or Contactor Panel
 - 1. Heatizon M330 Relay Panel Control Series
 - a. M330 Series Relay Panel accommodating 2, 4, 6, or 8, 30A resistive poles with manual activation switches.
 - b. M330G Series Relay Panel accommodating 2, 4, 6, or 8, 30A resistive poles with built in Ground Fault Equipment Protection and manual activation switches.
 - c. M330G-40 Series Relay Panel accommodating 2, 4, 6, or 8, 40A resistive poles with built in Ground Fault Equipment Protection and manual activation switches.
 - d. M330-50 Series Relay Panel accommodating 2, 4, 6, or 8, 50A resistive poles with manual activation switches
 - 2. Heatizon M530 Contactor Panel Series
 - a. M530-1 Contactor Panel accommodating 4, 50A resistive poles.
 - b. M530-2 Contactor Panel accommodating 8, 50A resistive poles.
 - c. M530-3 Contactor Panel accommodating 12, 50A resistive poles.
 - d. M530-4 Contactor Panel accommodating 16, 50A resistive poles.

C. Activation

1. All Cozy Heat® SR Cable circuits shall be activated by a Heatizon Systems approved activator for Cozy Heat® SR Cable:
 - a. Thermostatic Control:
 - 1) The system shall be controlled by an ambient or floor sensing thermostat that is manually adjusted or programmable. Example Models: M317 / M337 / M337G / M429 / M336
 - b. Multiple Circuits for Control, Monitoring and Power Distribution:
 - 1) Where the rating of the activator would be exceeded, it shall be used in conjunction with a Heatizon Relay Control Panel (M330 Series) or Contactor Panel (M530 Series) for activation of multiple heater circuits.

D. Control Methodology:

2. Single Circuit Local Digital Controller:

- a. Heating cable manufacturer shall provide a local digital controller with built-in GFPD compatible with selected heating cable.
- b. Digital controller shall be capable of operating with supply voltages from 100 V to 277 V.
- c. Digital controller will have a built-in self-test feature to verify proper functionality of heating cable system.
- d. Digital controller will also be able to communicate with BMS by one of the following protocols. Certain protocols will require the DigiTrace ProtoNode multi-protocol gateway.
 - 1) Modbus®
 - 2) Metasys® N2
- e. Digital controller will also supply an isolated triac alarm relay and a dry contact relay for alarm annunciation back to the BMS.
- f. The following variables will be monitored by the digital controller and reported back to the BMS
 - 1) Temperature
 - 2) Ground-fault
 - 3) Current draw
 - 4) Power consumption
 - 5) Associated alarms
- g. Digital controller shall have c-CSA-us approvals

2.3 ACCESSORIES

A. Insulation:

1. Concrete Slab and Pavers:
 - a. Provide $\frac{3}{4}$ " or 1" thick extruded polystyrene insulation below concrete slab prior to concrete pour. Insulation shall be rated at the appropriate mechanical properties for each application.

B. Reinforcing:

1. Provide welded wire fabric of the appropriate pattern and gauge to accommodate the spacing of the heating element in new pour applications. Wire fabric will hold heating element in place while concrete is poured. Concrete or Plastic chairs support the welded wire fabric and the heating element at the appropriate depth from the surface of the slab.

C. Control Joints:

1. Jumper under any and all joints and markers in asphalt and concrete using Heatizon SRJMPKIT joint/marker jumper kit.

2.2 SYSTEM LISTING

- A. The system (heating cable, connection kits, and controller) shall be UL Listed and CSA Certified for freezer frost heave prevention.
- B. The freezer frost heave prevention system shall have a design, installation and operating manual.

PART 3 - EXECUTION

3.1 ACCEPTABLE INSTALLERS

- A. Subject to compliance with requirements of Contract Documents, installer shall be familiar with installing heating cable and equipment.

3.2 INSTALLATION

- A. Comply with manufacturer's recommendations in the Installation and Operation Manual.
- B. Install and secure the heating cable in accordance with the System Installation and Operation Manual.
- C. Install electric heating cable according to the drawings and the manufacturer's instructions. The installer shall be responsible for providing a complete functional system, installed in accordance with applicable national and local requirements.
- D. Grounding of controller shall be performed according to Section 26 05 26 "Grounding and Bonding for Electrical Systems."
- E. Connection of all electrical wiring shall be according to Section 26 05 19 "Low-Voltage Electrical Power Conductors and Cables."

3.3 FIELD QUALITY CONTROL

- A. Start-up of system shall be performed by factory technician or factory representative per the owner's requirements.
- B. Field Testing and Inspections
 1. The system shall be commissioned in accordance to the Installation and Operation manual.
 2. The heating cable circuit integrity shall be tested using a 2500-Vdc megohmmeter at the following intervals. Minimum acceptable insulation resistance shall be 1000 megohms.
 - a. Before installing the heating cable
 - b. After installing the cables
 - c. Prior to initial start-up (commissioning)
 - d. As part of the regular system maintenance
 3. The technician shall verify that the controller alarm contacts are correctly connected to the BMS.

4. The technician shall verify that all communication from controller is configured correctly with the BMS.
5. All commissioning results will be recorded and presented to the owner.

3.4 MAINTENANCE

A. Maintenance Service

1. Comply with manufacturer's recommendations found in manufacturer's Installation and Operation Manual.

3.5 DEMONSTRATION

- A. Provide adequate demonstration and training to Owner in operation and maintenance of system.

END OF SECTION 262126

SECTION 262200 - LOW-VOLTAGE TRANSFORMERS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section Includes: Distribution, dry-type transformers rated 600 V and less, with capacities up to 1500 kVA.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings:
 - 1. Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 2. Vibration Isolation Base Details: Detail fabrication including anchorages and attachments to structure and to supported equipment.
 - 3. Include diagrams for power, signal, and control wiring.

1.3 INFORMATIONAL SUBMITTALS

- A. Seismic Qualification Certificates: For transformers, accessories, and components, from manufacturer.
- B. Qualification Data: For testing agency.
- C. Source quality-control reports.
- D. Field quality-control reports.

1.4 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Temporary Heating: Apply temporary heat according to manufacturer's written instructions.

1.6 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

PART 2 - PRODUCTS**2.1 MANUFACTURERS**

- A. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**: (Shall be the same manufacturer as the panelboards)
 - 1. Schneider Electric

2. Eaton.
3. General Electric Company.
4. Siemens Power Transmission & Distribution, Inc.

2.2 GENERAL TRANSFORMER REQUIREMENTS

- A. Description: Factory-assembled and -tested, air-cooled units for 60-Hz service.
- B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- C. Transformers Rated 15 kVA and Larger: Comply with NEMA TP 1 energy-efficiency levels as verified by testing according to NEMA TP 2.
 1. Coil Material: Aluminum.
- D. Encapsulation: Transformers smaller than 30 kVA shall have core and coils completely resin encapsulated.

2.3 DISTRIBUTION TRANSFORMERS

- A. Comply with NFPA 70, and list and label as complying with UL 1561.
- B. Provide transformers that are constructed to withstand seismic forces specified in Section 260548.16 "Seismic Controls for Electrical Systems."
- C. Cores: One leg per phase.
- D. Enclosure: Ventilated.
 1. NEMA 250, Type 2: Core and coil shall be encapsulated within resin compound to seal out moisture and air.
- E. Transformer Enclosure Finish: Comply with NEMA 250
 1. Finish Color: Gray.
- F. Taps for Transformers 3 kVA and Smaller: One 5 percent tap above normal full capacity.
- G. Taps for Transformers 7.5 to 24 kVA: One 5 percent tap above and one 5 percent tap below normal full capacity.
- H. Taps for Transformers 25 kVA and Larger: Two 2.5 percent taps above and two 2.5 percent taps below normal full capacity.
- I. Insulation Class, Smaller than 30 kVA: 185 deg C, UL-component-recognized insulation system with a maximum of 115-deg C rise above 40-deg C ambient temperature.
- J. Insulation Class, 30 kVA and Larger: 220 deg C, UL-component-recognized insulation system with a maximum of 150-deg C rise above 40-deg C ambient temperature.

2.4 IDENTIFICATION DEVICES

- A. Nameplates: Engraved, laminated-plastic or metal nameplate for each distribution

transformer, mounted with corrosion-resistant screws. Nameplates and label products are specified in Section 260553 "Identification for Electrical Systems."

2.5 SOURCE QUALITY CONTROL

- A. Test and inspect transformers according to IEEE C57.12.01 and IEEE C57.12.91.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Verify that ground connections are in place and requirements in Section 260526 "Grounding and Bonding for Electrical Systems" have been met. Maximum ground resistance shall be 5 ohms at location of transformer.
- B. Environment: Enclosures shall be rated for the environment in which they are located. Covers for NEMA 250, Type 4X enclosures shall not cause accessibility problems.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.
- D. Install wall-mounted transformers level and plumb with wall brackets fabricated by transformer manufacturer.
 - 1. Coordinate installation of wall-mounted and structure-hanging supports with actual transformer provided.
 - 2. Brace wall-mounted transformers as specified in Section 260548.16 "Seismic Controls for Electrical Systems."
- E. Install transformers level and plumb on a concrete base with vibration-dampening supports. Locate transformers away from corners and not parallel to adjacent wall surface.
- F. Construct concrete bases according to and anchor floor-mounted transformers according to manufacturer's written instructions, seismic codes applicable to Project, and requirements in Section 260529 "Hangers and Supports for Electrical Systems."
 - 1. Coordinate size and location of concrete bases with actual transformer provided. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified with concrete.
- G. Secure transformer to concrete base according to manufacturer's written instructions.
- H. Secure covers to enclosure and tighten all bolts to manufacturer-recommended torques to reduce noise generation.
- I. Remove shipping bolts, blocking, and wedges.

3.2 CONNECTIONS

- A. Ground equipment according to Section 260526 "Grounding and Bonding for Electrical Systems."
- B. Connect wiring according to Section 260519 "Low-Voltage Electrical Power Conductors and Cables."

- C. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A-486B.
- D. Provide flexible connections at all conduit and conductor terminations and supports to eliminate sound and vibration transmission to the building structure.

3.3 FIELD QUALITY CONTROL

- A. Perform tests and inspections and prepare test reports.
 - 1. Perform each visual and mechanical inspection and electrical test stated in NETA ATS for dry-type, air-cooled, low-voltage transformers. Certify compliance with test parameters.
- B. Remove and replace units that do not pass tests or inspections and retest as specified above.
- C. Infrared Scanning: Two months after Substantial Completion, perform an infrared scan of transformer connections.
 - 1. Use an infrared-scanning device designed to measure temperature or detect significant deviations from normal values. Provide documentation of device calibration.
 - 2. Perform two follow-up infrared scans of transformers, one at four months and the other at 11 months after Substantial Completion.
 - 3. Prepare a certified report identifying transformer checked and describing results of scanning. Include notation of deficiencies detected, remedial action taken, and scanning observations after remedial action.

3.4 ADJUSTING

- A. Record transformer secondary voltage at each unit for at least 48 hours of typical occupancy period. Adjust transformer taps to provide optimum voltage conditions at secondary terminals. Optimum is defined as not exceeding nameplate voltage plus 5 percent and not being lower than nameplate voltage minus 3 percent at maximum load conditions. Submit recording and tap settings as test results.
- B. Connect buck-boost transformers to provide nameplate voltage of equipment being served, plus or minus 5 percent, at secondary terminals.
- C. Output Settings Report: Prepare a written report recording output voltages and tap settings.

END OF SECTION 262200

SECTION 262413.70 - INTEGRATED POWER CENTER**PART 1 GENERAL****1.1 SUMMARY**

- A. Integrated Power Center – Furnish and install an integrated solution as specified herein and where shown on the associated schedules and drawings.

1. The lineups shall include:

- a. Power panel interiors
- b. Lighting and Appliance panel interiors.
- c. Lighting controls
- d. Integrated transformers
- e. Surge Protection devices
- f. Third Party Equipment

1.2 DEFINITIONS

- A. EMI: Electromagnetic interference.
- B. GFCI: Ground-fault circuit interrupter.
- C. RFI: Radio-frequency interference.
- D. RMS: Root mean square.
- E. SPDT: Single pole, double throw.

1.3 SUBMITTALS

- A. Product Data: For each assembly indicated. Include dimensions and manufacturers' technical data on features, performance, electrical characteristics, ratings, and finishes. Submit all integrated power center documentation coincidentally.

- B. Shop Drawings: For each panelboard and related equipment.

1. Dimensioned plans, elevations, sections, and details, including required clearances and service space around equipment. Show tabulations of installed devices, equipment features, and ratings. Include the following:

- a. Enclosure types and details for types other than NEMA 250, Type 1.
- b. Bus configuration, current, and voltage ratings.
- c. Short-circuit current rating of panelboards, panelboards and overcurrent protective devices.

- C. Operation and Maintenance Data: For panelboards and components to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 01 Section "Closeout Procedures," include the following:

- 1. Routine maintenance requirements for panelboards and all installed components.
- 2. Manufacturer's written instructions for testing and adjusting overcurrent protective devices.

3. Time-current curves, including selectable ranges for each type of overcurrent protective device.

1.4 QUALITY ASSURANCE

- A. Source Limitations: Obtain panelboards through one source from a single manufacturer.
- B. Product Selection for Restricted Space: Drawings indicate maximum dimensions for panelboards including clearances between panelboards and adjacent surfaces and other items. Comply with indicated maximum dimensions.
- C. Electrical Components, Devices, and Accessories: Listed and by a testing agency acceptable to authorities having jurisdiction and marked for intended use.
- D. Comply with the local codes and ordinances.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver in sections or lengths that can be moved past obstructions in delivery path.
- B. Store indoors in clean dry space with uniform temperature to prevent condensation. Protect from exposure to dirt, fumes, water, corrosive substances, and physical damage.
- C. If stored in areas subjected to weather, cover panelboards to provide protection from weather, dirt, dust, corrosive substances, and physical damage. Remove loose packing and flammable materials from inside panelboards; install electric heating (250 W per section) to prevent condensation.
- D. Handle panelboards according to NEMA PB 2.1 and NECA 400.

1.6 PROJECT CONDITIONS

- A. Installation Pathway: Remove and replace access fencing, doors, lift-out panels, and structures to provide pathway for moving panelboards into place.
- B. Verify dimensions by field measurements.
- C. Verify clearance requirements. Locate panelboard to meet installation tolerances.
- D. Service Conditions: NEMA PB 2, usual service conditions, as follows:
 1. Ambient temperatures within limits specified.
 2. Altitude not exceeding 6600 feet (2000 m).

1.7 COORDINATION

- A. Coordinate layout and installation of panelboards and components with other construction including conduit, piping, equipment, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

- B. Coordinate size and location of concrete bases. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified in Division 03.

1.8 WARRANTY

- A. Each piece of equipment shall be warranted by the equipment manufacturer to be free of defects in material and workmanship. Equipment manufacturer shall provide warranty of 30 months from the date shipment or 24 months from the date installation, including on-the-premises maintenance service during normal working hours at no cost to owner for parts or labor.

1.9 REFERENCES

- A. The equipment referenced herein are designed and manufactured according to the latest revision of the following standards:
 - 1. UL 50 – Standards for enclosures for electrical equipment
 - 2. UL 67 – Standards for panelboards
 - 3. UL 489 – Standards for molded-case circuit breakers
 - 4. UL 508A – Standards for Industrial Controls
 - 5. NEMA AB 1 – Standards for molded case circuit breakers
 - 6. NEMA PB 1 - Standards for panelboards
 - 7. NFPA 70 - National Electrical Code (NEC) Article 408-Panelboards and Panelboards
 - 8. Federal Specification W-P-115C - Type I Class 1
 - 9. Federal Specification W-C-375B/Gen - Circuit Breakers, Molded Case, Branch Circuit and Service.
 - 10. ASCE 7 – American Society of Civil Engineers, Minimum Design Loads for Buildings and Other Structures – Seismic compliance requirements
 - 11. ICC ES AC156 – International Code Council Evaluation Services Acceptance Criteria for Seismic Qualification by Shake-Table Testing of Nonstructural Components and Systems – seismic testing protocol

1.03 SUBMITTAL AND RECORD DOCUMENTATION

- A. Approval documents shall include drawings. Drawings shall contain overall panelboard enclosure dimensions. The location of the main, branches, and solid neutral shall be shown clearly. In addition, the drawing shall illustrate one line diagrams with applicable voltage systems.
- B. Record drawings shall be made available to the customer. The customer may request any of the following formats: DXF, DWG or PDF.

1.04 QUALIFICATIONS

- A. Company specializing in assembly of panelboard products with a minimum of twenty-five (25) years documented experience.
- B. Panelboards shall be manufactured in accordance with standards listed in "References".
- C. The company that assembles and provides the integrated equipment enclosure(s) shall be the same company that manufactures or purchases the panelboards, circuit breakers and dry type transformers.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Inspect and report concealed damage to carrier within their required time period.
- B. Handle carefully to avoid damage to panelboard internal components, enclosure, and finish.
- C. Store in a clean, dry environment. Maintain factory packaging and, if required, provide an additional heavy canvas or heavy plastic cover to protect enclosure(s) from dirt, water, construction debris, and traffic.

1.06 OPERATIONS AND MAINTENANCE MATERIALS

- A. Manufacturer shall provide installation instructions and NEMA Standards Publication PB 1.1 – “Instructions for Safe Installation, Operation and Maintenance of Panelboards Rated 600 Volts or Less”.

1.07 WARRANTY

- A. Manufacturer shall warrant specified equipment free from defects in materials and workmanship for thirty (30) months from the date of invoice.

PART 2 PRODUCTS

- A. MANUFACTURES: Subject to compliance with requirements, provided on LiDL National Account List:

- 1. PCX Prebuilt Power Solutions

2.1 RATINGS

- A. The assembly shall be rated to withstand mechanical forces exerted during short circuit conditions when connected to a power source having an available fault current of 65,000 amperes symmetrical at rated voltage as shown on the drawing. This value to be confirmed with the overcurrent protective device coordination study. Refer to specification section 26 05 73 OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY
- B. Voltage ratings to be as indicated on the drawings.
- C. The short circuit current rating of any section shall be limited to the lowest short circuit rating or series connected rating of any installed device in the section.
- D. UL listed series ratings shall be made available upon request.
- E. The equipment shall be suitable for and certified to meet applicable seismic requirements of the International Building Code (IBC), 2003 ICC Edition and ASCE 7-02, or later, for the site-specific location. Guidelines for the installation shall be provided by the equipment manufacture and based on testing of representative equipment. Equipment capacity shall be determined from tri-axial seismic shake table results as defined in ICC ES AC 156.

2.2 ENCLOSURES

- A. Each lineup shall consist of the required number of vertical sections bolted together to form a rigid assembly.
- B. All sections of the lineup shall be rear aligned with the depth as shown on the drawings and shall be front-accessible.
- C. NEMA Type 1 enclosures shall be constructed in accordance with UL 891 requirements.
- D. Enclosures shall have ANSI 70 gray enamel electrodeposited paint finish.
 - 1. All sections shall be of aluminum or steel construction.
- F. Enclosure Depths
 - 1. NEMA Type 1 panel, contactor, BMS, third party equipment space sections shall be 15 inches minimum, or as shown on the drawings.
 - 2. NEMA Type 1 Transformer Assembly sections shall be 25" to 40" deep depending on the size, type and rating of the transformer(s).
- G. Enclosure Height
 - 1. All panel, contactor, BMS, third party equipment space sections shall be 91.5" high maximum.
 - 2. Transformer Assembly sections shall be 91.5" high maximum as standard.
- H. Single section shipping splits shall be as shown on the drawings.
- I. Multiple section shipping splits shall be as shown on the drawings.

2.3 FRONTS

- A. Trim front shall meet strength and rigidity requirements of applicable UL standards.
- B. Each panelboard shall have a full height hinged door.
- C. The latch shall have a cylindrical tumbler type keyed lock.
- D. All lock assemblies shall be keyed alike. One (1) key shall be provided with each lock.
- E. Panelboard door shall have lift off hinges to facilitate maintenance.
- F. Directory cardholder with typed circuit directory shall be included with each panel interior.
- G. The manufacture shall provide a hinged door with a [quarter-turn latch] or [lockable latch] over the deadfront of each lighting & appliance panelboard interior.

2.4 WIRING/TERMINATION

- A. Factory installed type THHN power cables shall electrically connect the power panel

interiors, lighting and appliance panel interiors and transformers in the lineup.

- B. All control wire shall be type THHN or MTW bundled and secured with nylon ties.
- C. When shipping splits are required, power cables shall be cut to length, stripped, labeled and rolled back in one of the adjacent enclosures.

2.5 PANEL INTERIORS

- A. Panel interiors shall be flush mounted with the front of the enclosure to allow easy access to line and/or load conductor's entering/exiting top or bottom. Recessing the panel interior more than 3 inches from the front of the enclosure will not be acceptable.
- B. If two 42-circuit panel interiors are stacked in the same vertical section, a horizontal barrier that complies with UL 891 requirements shall separate the panel interiors.
- C. Power panel interiors shall be SPECTRA I-LINE circuit breaker type or equivalent, rated 600 Vac maximum.
- D. Lighting and Appliance panel interiors shall be NQ circuit breaker type, or equivalent rated 240 Vac maximum.
- E. Lighting and Appliance panel interiors shall be NF circuit breaker type rated 480Y/277 Vac maximum.
- F. All panelboard bussing shall be Aluminum.

2.6 LIGHTING CONTROLS

- A. Lighting contactors
 - 1. Lighting contactors shall be pan mounted as indicated on the associated schedules and drawings.
 - 2. Branch breakers shall be factory wired to the line side of the contactors.
 - 3. Lighting contactors shall be in accordance with specification section 26 09 23 LIGHTING CONTROL DEVICES.

2.7 SURGE SUPPRESSION DEVICE UNITS

- A. Integrated surge protective devices shall be Listed and Component Recognized in accordance with UL 1449 latest Edition to include Section 37.3 highest fault current category. SPD shall be UL 1283 listed.
- B. All units shall be provided in accordance with specification section 26 43 00 SURGE PROTECTION DEVICES

2.8 INTEGRATED TRANSFORMERS

- A. All integrated transformers shall be factory installed in the power center.

- B. Performance of the transformers will be in accordance to specification section 26
22 00 LOW-VOLTAGE TRANSFORMERS

PART 3 EXECUTION By Contractor

3.01 INSTALLATION By Contractor

- A. Install the Integrated Power Center in accordance with manufacturer's written instructions, NEMA PB 2.1 "Proper Handling, Installation, Operation and Maintenance of Deadfront Panelboards Rated 600 volts or Less" and NEC standards.

3.02 FIELD QUALITY CONTROL By Contractor

- A. Inspect complete installation for physical damage, proper alignment, anchorage, and grounding.
- B. Check tightness of bolted connections and circuit breaker connections using calibrated torque wrench or torque screwdriver per manufacturer's written specifications. Contractor shall provide a Torque Test Report.

END OF SECTION 262413.70

SECTION 262416 - PANELBOARDS**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Distribution panelboards.
2. Lighting and appliance branch-circuit panelboards.

1.2 DEFINITIONS

- A. MCCB: Molded-case circuit breaker.
- B. SPD: Surge protective device.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of panelboard.
- B. Shop Drawings: For each panelboard and related equipment.
 1. Include dimensioned plans, elevations, sections, and details.
 2. Detail enclosure types including mounting and anchorage, environmental protection, knockouts, corner treatments, covers and doors, gaskets, hinges, and locks.
 3. Detail bus configuration, current, and voltage ratings.
 4. Short-circuit current rating of panelboards and overcurrent protective devices.
 5. Include evidence of NRTL listing for SPD as installed in panelboard.
 6. Detail features, characteristics, ratings, and factory settings of individual overcurrent protective devices and auxiliary components.
 7. Include wiring diagrams for power, signal, and control wiring.
 8. Include time-current coordination curves for each type and rating of overcurrent protective device included in panelboards.

1.4 INFORMATIONAL SUBMITTALS

- A. Panelboard schedules for installation in panelboards.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.6 FIELD CONDITIONS

A. Service Conditions: NEMA PB 1, usual service conditions, as follows:

1. Ambient temperatures within limits specified.
2. Altitude not exceeding 6600 feet.

1.7 WARRANTY

A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

B. Manufacturer's Warranty: Manufacturer agrees to repair or replace panelboards that fail in materials or workmanship within specified warranty period.

1. Panelboard Warranty Period: 18 months from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PANELBOARDS COMMON REQUIREMENTS

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Comply with NEMA PB 1.

C. Comply with NFPA 70.

D. Enclosures: Surface-mounted, dead-front cabinets.

1. Rated for environmental conditions at installed location.

- a. Indoor Dry and Clean Locations: NEMA 250, Type 1.

2. Height: 86 inches maximum.

3. Front: Secured to box with concealed trim clamps. For surface-mounted fronts, match box dimensions; for flush-mounted fronts, overlap box. Trims shall cover all live parts and shall have no exposed hardware.

4. Hinged Front Cover: Entire front trim hinged to box and with standard door within hinged trim cover. Trims shall cover all live parts and shall have no exposed hardware.

E. Incoming Mains Location: Convertible between top and bottom.

F. Phase, Neutral, and Ground Buses: Hard-drawn copper, 98 percent conductivity.

G. Conductor Connectors: Suitable for use with conductor material and sizes.

1. Material: Hard-drawn copper, 98 percent conductivity.

2. Main and Neutral Lugs: Compression type, with a lug on the neutral bar for each pole

- in the panelboard.
 - 3. Ground Lugs and Bus-Configured Terminators: Compression type, with a lug on the bar for each circuit breaker in the panelboard.
 - 4. Feed-Through Lugs: Compression type, suitable for use with conductor material. Locate at opposite end of bus from incoming lugs or main device.
 - 5. Subfeed (Double) Lugs: Compression type suitable for use with conductor material. Locate at same end of bus as incoming lugs or main device.
- H. NRTL Label: Panelboards shall be labeled by an NRTL acceptable to authority having jurisdiction for use as service equipment with one or more main service disconnecting and overcurrent protective devices. Panelboards shall have meter enclosures, wiring, connections, and other provisions for utility metering. Coordinate with utility company for exact requirements.
- I. Future Devices: Panelboards shall have mounting brackets, bus connections, filler plates, and necessary appurtenances required for future installation of devices.
- J. Panelboard Short-Circuit Current Rating: Rated for series-connected system with integral or remote upstream overcurrent protective devices and labeled by an NRTL. Include label or manual with size and type of allowable upstream and branch devices listed and labeled by an NRTL for series-connected short-circuit rating.
- K. Panelboard Short-Circuit Current Rating: Fully rated to interrupt symmetrical short-circuit current available at terminals. Assembly listed by an NRTL for 100 percent interrupting capacity.

2.2 PERFORMANCE REQUIREMENTS

- A. Seismic Performance: Panelboards shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
- 1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified."
- B. Surge Suppression: Factory installed as an integral part of indicated panelboards, complying with latest edition of UL 1449 SPD Type 1.

2.3 POWER PANELBOARDS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
- 1. Square D I Line; by Schneider Electric.
 - 2. Eaton
 - 3. ABB/General Electric
 - 4. Siemens
 - 5. Or approved equivalent.
- B. Panelboards: NEMA PB 1, distribution type.
- C. Doors: Secured with vault-type latch with tumbler lock; keyed alike.

1. For doors more than 48 inches high, provide two latches, keyed alike.
- D. Mains: Circuit breaker or Lugs only as noted on drawings.
- E. Branch Overcurrent Protective Devices for Circuit-Breaker Frame Sizes 150 A and Smaller: Bolt-in circuit breakers.
- F. Branch Overcurrent Protective Devices for Circuit-Breaker Frame Sizes Larger Than 115 A: Bolt-in circuit breakers.
- G. Branch Overcurrent Protective Devices: Circuit Breakers.

2.4 LIGHTING AND APPLIANCE BRANCH-CIRCUIT PANELBOARDS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following: (Shall be the same manufacture as the power panelboards)
 1. Square D, NQ and NF; by Schneider Electric.
- B. Panelboards: NEMA PB 1, lighting and appliance branch-circuit type.
- C. Mains: Circuit breaker or lugs only.
- D. Branch Overcurrent Protective Devices: Bolt-on circuit breakers, replaceable without disturbing adjacent units.
- E. Doors: Concealed hinges; secured with flush latch with tumbler lock; keyed alike.
- F. Column-Type Panelboards: Single row of overcurrent devices with narrow gutter extension and overhead junction box equipped with ground and neutral terminal buses.

2.5 DISCONNECTING AND OVERCURRENT PROTECTIVE DEVICES

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following: (Shall be the same manufacture as the power panelboards)
 1. Square D; by Schneider Electric.
 2. Eaton
 3. ABB/General Electric
 4. Siemens
 5. Or approved equivalent.
- B. MCCB: Comply with UL 489, with interrupting capacity to meet available fault currents.
 1. Thermal-Magnetic Circuit Breakers:
 - a. Inverse time-current element for low-level overloads.
 - b. Instantaneous magnetic trip element for short circuits.
 - c. Adjustable magnetic trip setting for circuit-breaker frame sizes 250 A and larger.
 2. Adjustable Instantaneous-Trip Circuit Breakers: Magnetic trip element with

- front- mounted, field-adjustable trip setting.
3. Electronic Trip Circuit Breakers:
 - a. RMS sensing.
 - b. Field-replaceable rating plug or adjustable electronic trip.
 - c. Digital display of settings, trip targets, and indicated metering displays.
 - d. Multi-button keypad to access programmable functions and monitored data.
 - e. Ten-event, trip-history log. Each trip event shall be recorded with type, phase, and magnitude of fault that caused the trip.
 - f. Integral test jack for connection to portable test set or laptop computer.
 - g. Field-Adjustable Settings:
 - 1) Instantaneous trip.
 - 2) Long- and short-time pickup levels.
 - 3) Long and short time adjustments.
 - 4) Ground-fault pickup level, time delay, and I squared T response.
 4. Current-Limiting Circuit Breakers: Frame sizes 600 A, 400A 250A and 150A and rated 65 kA or higher; let-through ratings less than NEMA FU 1, RK-5, or UL listed as current limiting
 5. GFCI Circuit Breakers: Single- and double-pole configurations with Class A ground-fault protection (6-mA trip).
 6. GFEP Circuit Breakers: Class B ground-fault protection (30-mA trip).
 7. Arc-Fault Circuit Interrupter Circuit Breakers: Comply with UL 1699; 120/240-V, single-pole configuration.
 8. Subfeed Circuit Breakers: Vertically mounted.
 9. MCCB Features and Accessories:
 - a. Standard frame sizes, trip ratings, and number of poles.
 - b. Breaker handle indicates tripped status.
 - c. UL listed for reverse connection without restrictive line or load ratings.
 - d. Lugs: Compression style, suitable for number, size, trip ratings, and conductor materials.
 - e. Application Listing: Appropriate for application; Type SWD for switching fluorescent lighting loads; Type HID for feeding fluorescent and HID lighting circuits.
 - f. Ground-Fault Protection: Integrally mounted relay and trip unit with adjustable pickup and time-delay settings, push-to-test feature, and ground-fault indicator.
 - g. Communication Capability: Circuit-breaker-mounted communication module with Modbus RTU with functions and features compatible with power monitoring and control system specified in Section 260913 "Electrical Power Monitoring and Control."
 - h. Shunt Trip: 120-V trip coil energized from separate circuit, set to trip at 55 percent of rated voltage.

- i. Handle Padlocking Device: Fixed attachment, for locking circuit-breaker handle in on or off position.
 - j. Handle Clamp: Loose attachment, for holding circuit-breaker handle in on position.
- C. Fused Switch: NEMA KS 1, Type HD; clips to accommodate specified fuses; lockable handle.
 - 1. Fuses and Spare-Fuse Cabinet: Comply with requirements specified in Section 262813 "Fuses."

2.6 IDENTIFICATION

- A. Panelboard Label: Manufacturer's name and trademark, voltage, amperage, number of phases, and number of poles shall be located on the interior of the panelboard door.
- B. Breaker Labels: Faceplate shall list current rating, UL and IEC certification standards, and AIC rating.
- C. Circuit Directory: Directory card inside panelboard door, mounted in metal frame with transparent protective cover.

2.7 ACCESSORY COMPONENTS AND FEATURES

- A. Portable Test Set: For testing functions of solid-state trip devices without removing from panelboard. Include relay and meter test plugs suitable for testing panelboard meters and switchboard class relays.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Comply with NECA 1.
- B. Install panelboards and accessories according to NECA 407.
- C. Comply with mounting and anchoring requirements specified in Section 260548.16 "Seismic Controls for Electrical Systems."
- D. Mount top of trim 90 inches above finished floor unless otherwise indicated.
- E. Mount panelboard cabinet plumb and rigid without distortion of box.
- F. Mount recessed panelboards with fronts uniformly flush with wall finish and mating with back box.
- G. Install overcurrent protective devices and controllers not already factory installed.

1. Set field-adjustable, circuit-breaker trip ranges.
- H. Make grounding connections and bond neutral for services and separately derived systems to ground. Make connections to grounding electrodes, separate grounds for isolated ground bars, and connections to separate ground bars.
- I. Install filler plates in unused spaces.
- J. Arrange conductors in gutters into groups and bundle and wrap with wire ties.

3.2 IDENTIFICATION

- A. Identify field-installed conductors, interconnecting wiring, and components; install warning signs complying with requirements in Section 260553 "Identification for Electrical Systems."
- B. Create a directory to indicate installed circuit loads; incorporate Owner's final room designations. Obtain approval before installing. Handwritten directories are not acceptable. Install directory inside panelboard door.
- C. Panelboard Nameplates: Label each panelboard with a nameplate complying with requirements for identification specified in Section 260553 "Identification for Electrical Systems."
- D. Device Nameplates: Label each branch circuit device in power panelboards with a nameplate complying with requirements for identification specified in Section 260553 "Identification for Electrical Systems."
- E. Install warning signs complying with requirements in Section 260553 "Identification for Electrical Systems" identifying source of remote circuit.

3.3 FIELD QUALITY CONTROL

- A. Perform tests and inspections.
- B. Acceptance Testing Preparation:
 1. Test insulation resistance for each panelboard bus, component, connecting supply, feeder, and control circuit.
 2. Test continuity of each circuit.
- C. Tests and Inspections:
 1. Perform each visual and mechanical inspection and electrical test for low-voltage air circuit breakers.
 2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
- D. Panelboards will be considered defective if they do not pass tests and inspections.

- E. Prepare test and inspection reports, including a certified report that identifies panelboards included and that describes scanning results, with comparisons of the two scans. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

END OF SECTION 262416

SECTION 262713 - ELECTRICITY METERING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes equipment for utility company's electricity metering.

1.3 SUBMITTALS

- A. Product Data: Include construction details, material descriptions, dimensions of individual components and profiles, and finishes. Describe electrical characteristics, features, and operating sequences, both automatic and manual. Include the following:
 - 1. Electricity-metering equipment.
- B. Shop Drawings for Electricity-Metering Equipment:
 - 1. Dimensioned plans and sections or elevation layouts.
 - 2. Wiring Diagrams: Power, signal, and control wiring specific to this Project. Identify terminals and wiring designations and color codes to facilitate installation, operation, and maintenance. Indicate recommended types, wire sizes, and circuiting arrangements for field-installed wiring, and show circuit protection features.
 - 3. Mounting and anchoring devices recommended by manufacturer to resist seismic forces specified in Division 26 Section "Vibration and Seismic Controls for Electrical Systems."
- C. Manufacturer Seismic Qualification Certification for Electricity-Metering Equipment: Submit certification that equipment components and their mounting and anchorage provisions have been designed to remain in place without separation of any parts or loosening of factory-made connections when subjected to the seismic forces defined in Division 26 Section "Vibration and Seismic Controls for Electrical Systems." Include the following:
 - 1. Basis for Certification: Indicate whether certification is based on actual test of assembled components or on calculations.
 - 2. Detailed description of equipment mounting and anchorage devices on which the certification is based and their installation requirements.
- D. Field quality-control test reports.
- E. Operation and Maintenance Data: For electricity-metering equipment to include in emergency, operation, and maintenance manuals.

1.4 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Receive, store, and handle modular meter center as specified in NECA 400.

1.6 COORDINATION

- A. Electrical Service Connections: Coordinate with utility companies and components they furnish as follows:
 - 1. Comply with requirements of utilities providing electrical power and communication services.
 - 2. Coordinate installation and connection of utilities and services, including provision for electricity-metering components.

1.7 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. In other Part 2 articles where titles below introduce lists, the following requirements apply to product selection:
 - 1. Manufacturers: Subject to compliance with requirements by utility company, provide products by one of the manufacturers specified.
 - a. Asco

2.2 EQUIPMENT FOR ELECTRICITY METERING BY UTILITY COMPANY

- A. Current-Transformer Cabinets: Factory-coordinated assembly of a main service terminal box with lugs only. Comply with requirements of electrical power utility company.
 - 1. Available Manufacturers:
 - a. Cutler-Hammer; Eaton Corporation.
 - b. General Electric Company; Electrical Distribution & Control Div.
 - c. Siemens Energy & Automation, Inc.
 - d. Milbank.
 - e. Erickson Electrical Equipment Co.
 - f. Cooper Industries.

2. Housing: NEMA Type 4X (stainless steel) enclosure.
 - a. Structural strength of the housing, its anchorage and component attachment provisions, and anchorage devices recommended for anchoring the housing in place shall be adequate to prevent separation of equipment and its components from their installed positions during a seismic event as defined in Division 26 Section "Vibration and Seismic Controls for Electrical Systems."
3. Minimum Short-Circuit Rating: 100,000 amperes symmetrical at rated voltage.
4. Meter Socket: Comply with requirements of electrical power utility company. Type as approved by utility company, with rating coordinated with indicated tenant feeder circuit rating.
5. Provide a 3-pole fused switch (size as indicated on the drawings) on load side of utility meter within common enclosure. Provide metal barrier in accordance with power utility company requirements.
6. Metering equipment shall be compatible with Emerson's Building Automation System interface.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Comply with equipment installation requirements in NECA 1.
- B. Install equipment for utility company metering. Install raceways and equipment according to utility company's written requirements. Provide empty conduits for metering leads and extend grounding connections as required by utility company.
- C. Install modular meter center according to NECA 400 switchboard installation requirements.

3.2 FIELD QUALITY CONTROL

- A. Test Owner's electricity-metering installation for proper operation, accuracy, and usability of output data.
 1. Connect a load of known kilowatt rating, 1.5 kW minimum, to a circuit supplied by metered feeder.
 2. Turn off circuits supplied by metered feeder and secure them in off condition.
 3. Run test load continuously for eight hours, minimum, or longer to obtain a measurable meter indication. Use test load placement and setting that ensures continuous, safe operation.
 4. Check and record meter reading at end of test period and compare with actual electricity used based on test load rating, duration of test, and sample measurements of supply voltage at test load connection. Record test results.
 5. Repair or replace deficient or malfunctioning metering equipment, or correct test setup; then retest. Repeat for each meter in installation until proper operation of entire system is verified.

END OF SECTION 262713

SECTION 262716 – INDOOR SERVICE POLES**PART 1 GENERAL****1.1 SECTION INCLUDES**

- A. Indoor service poles.

1.2 RELATED SECTIONS

- A. Section 07 84 13 - Penetration Firestopping.
- B. Section 26 05 03 – General Requirements for Electrical Systems.

1.3 REFERENCES

- A. ASTM International (ASTM): ASTM E 814 - Standard Test Method for Fire Tests of Penetration Firestop Systems.
- B. National Fire Protection Association (NFPA): NFPA 70 - National Electrical Code.
- C. Underwriters Laboratories Inc. (UL):
 - 1. UL 5 - Surface Metal Raceways and Fittings.
 - 2. UL 94 - Test for Flammability of Plastic Materials for Parts in Devices and Appliances.
 - 3. UL 514A - Metallic Outlet Boxes.
 - 4. UL 514C - Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers.

1.4 SUBMITTALS

- A. Submit under provisions of Section 01 30 00 - Administrative Requirements.
- B. Product Data: Manufacturer's data sheets and descriptive literature on each product to be used, including:
 - 1. Preparation instructions and recommendations.
 - 2. Storage and handling requirements and recommendations.
 - 3. Installation methods.
- C. Shop Drawings: Indicate raceway layouts, each system component required for complete system, raceway lengths, device types, locations, elevations, details, sections and attachment to other work; identify all circuits.
- D. Samples:
 - 1. Submit three 6 inch lengths of exposed type surface raceways with required color and finish.
 - 2. Submit service pole and architectural column samples with required color and finish.
- E. Test Reports: Certified reports from independent testing laboratory supporting compliance of raceway system to specified requirements.
- F. Manufacturer's instructions: Printed installation instructions for each system component; include storage requirements.
- G. Project Record Documents: If variations from approved shop drawings occur during installation of raceway system, submit final as-built drawings indicating such variations.

1.5 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Firms regularly engaged in manufacture of raceway and box distribution products and systems of the types and sizes required, whose products have been in satisfactory use in similar service for not less than 10 years. Provide raceway and box distribution systems produced by Wiremold Company.
- B. Source Limitation: Obtain each type of round recessed raised and stage floor box through one source from a single manufacturer.
- C. Mock-Up: Provide a mock-up for evaluation of surface preparation techniques and application workmanship.
 - 1. Finish areas designated by Architect.
 - 2. Do not proceed with remaining work until workmanship, appearance and performance are approved by Architect.
 - 3. Refinish mock-up area as required to produce acceptable work.
- D. Electrical Components, Devices, and Accessories: Comply with requirements of applicable local codes, NEC, UL, and NEMA Standards pertaining to indoor service poles and architectural columns. Listed and labeled in accordance with NFPA 70, Article 100.
- E. Electrical Floor Boxes and Components: Comply with requirements of applicable local codes, NEC, UL, and NEMA Standards pertaining to floor boxes and components. Meet or exceed UL Fire Classification requirements for recessed and flush style boxes. Listed and labeled in accordance with NFPA 70, Article 100.
- F. Electrical Boxes and Fittings: Comply with requirements of applicable local codes, NEC, UL, ETL, and NEMA Standards pertaining to boxes and fittings. Listed and labeled in accordance with NFPA 70, Section 300-21.
- G. Products: Listed and classified by Underwriters Laboratories, Inc. as suitable for the purpose specified and indicated.
- H. Accessibility Compliance: Design device flange to meet ADA Accessibility Guidelines as to changes in floor and ground surface levels. Flanges shall be beveled so the slope is no greater than 1:2.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Store products in manufacturer's unopened packaging until ready for installation.
- B. Store and handle in strict compliance with manufacturer's written instructions and recommendations.
- C. Protect from damage due to weather, excessive temperature, and construction operations.

1.7 PROJECT CONDITIONS

- A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's recommended limits.

PART 2 PRODUCTS

2.1 MANUFACTURERS

- A. Acceptable Manufacturer: Legrand, a Wiremold company, which is located at: 60 Woodlawn

St. ; West Hartford, CT 06110; Toll Free Tel: 800-621-0049; Tel: 877-295-3472; Fax: 860-232-2062; Email: request info (customer.support-wm@legrand.us); Web: www.legrand.us/Wiremold.aspx

- B. Requests for substitutions will be considered in accordance with provisions of Section 01 60 00 - Product Requirements, and should include documentation of UL listings and a sample of components.

2.2 INDOOR SERVICE POLES

- A. Acceptable Product: Wiremold Tele-Power Pole System; Two separate compartments.
- B. General: Provide indoor service pole system to extend branch circuit wiring and data network, voice, video, and other communication cabling to points of use shown on the Drawings. System shall consist of pole multi-outlet assemblies, and appropriate fittings and accessories as required for a complete assembly.
- C. Classification and Use: Indoor service pole system shall be suitable for use in dry interior locations, only as covered in Articles 353 and 352 Part A of the National Electrical Code, as adopted by the National Fire Protection Association and as approved by the American National Standards Institute.
- D. The indoor service pole system is UL Listed under File Nos. E15191 Guide PVGT, E53857 Guide PVUR, E41751 Guide RJPR, and E169069 Guide ZTFR; also CUL Listed in the above files, or are CSA Certified in File LR350.
- E. Series: 25DTC-4 Series Blank Steel Poles.
 - 1. Color: Custom Color SW 7005 Pure White Factory Finish.
 - 2. Cross Section: 2.25 inches by 2.3125 inches (57 mm by 59 mm).
 - 3. First Compartment: Factory wired with two duplex styles 20A, 125V. NEMA 5-20R grounding-type specification grade receptacles and colored to match the pole finish. Receptacles shall be UL tested to meet the performance requirements of Fed. Spec. W-C695G General Specification for Electrical Power Connectors and conform to NEMA Specification WD 1-7.01 to 7.10 "Heavy Duty General Use Grounding Receptacles". Receptacles shall also be UL Listed and be in compliance with UL 498.
 - 4. Harness: Single circuit (2 conductor plus ground) with No. 12 AWG solid type THHN conductors, factory assembled to the receptacles. 6 inches (152 mm) conductor leads are to be furnished for termination to the overhead wiring system. An 8 inches (203 mm) removable cover section shall be provided at the top of the power compartment to facilitate the hard wiring of the pole wire harness.
 - 5. Second Compartment: To be for field installation of telephone or data network cabling. An 8 inches (203 mm) removable cover section at the bottom of this compartment shall be provided to assemble and mount communications connectors. This section shall be removable without dismantling or removing the Tele-Power Pole after installation. The cover section is to have six knockouts for modular voice-data jacks (RJ-type) and a 1.375 inches by 2.7 inches (35 mm by 69 mm) rectangular knockout for a modular furniture outlet. A "mouse hole" knockout with furnished grommet is to be included for straight through communication cable access.
 - 6. Fittings: Provide full compliment of fittings necessary to make system workable, including but not limited to entrance end fittings for top of each electrical channel, ceiling trim plate, pole-mounting bracket, Velcro(tm) carpet gripper pad, and adhesive pad. (If for air handling spaces, an entrance end fitting shall be furnished for the communications channel.)
 - 7. Field install power device covers to add duplex, single 1.40 inch (36 mm) and 1.59 inch (40 mm), and rectangular-type receptacles, colored to match appropriate Tele-Power Pole.

8. Add-on communication covers shall be available to mount workstation device faceplates, inserts, and specialty mounting bezels. The power pole manufacturer will provide a complete line of connectivity outlets with multi-media modular inserts for UTP fiber optic, coaxial, and other cabling types.
9. UTP inserts shall feature a unique recessed area for port labeling and shall be able to accommodate designation icon buttons or icon labels. Custom label capabilities shall be available using templates that can be downloaded from the Internet.
10. Fiber Optic/Category 5 Cable Radius: There shall be a 2 inches (51 mm) controlled radius entrance fitting, which meets or exceeds the specifications for fiber optic and Category 5 cabling and TIA/EIA-569A requirements for communications pathways, for the 25DT and AMDT Series Tele-Power Poles.
11. Modular Communications Insert Tele-Power Poles: The manufacturer shall provide a Tele-Power Pole with one compartment, factory wired with two duplex styles 20A, 125V NEMA 5-20R grounding-type specification grade receptacles, and colored to match the pole finish. Receptacles shall be UL tested to meet the performance requirements of Fed. Spec. W-C695G General Specification for Electrical Power Connectors and conform to NEMA Specification WD 1-7.01 to 7.10 "Heavy Duty General Use Grounding Receptacles". Receptacles shall also be UL Listed and be in compliance with UL 498. The harness is to be single circuit (2 conductor plus ground) with #12 AWG solid type THHN conductors, factory assembled to the receptacles. 6 inches (153) conductor leads are to be furnished for termination to the overhead wiring system. The other compartment shall have a modular communications insert bezel installed and three modular communications inserts, as well as labeling.

PART 3 EXECUTION

3.1 EXAMINATION

- A. Examine conditions under which raceways and boxes are to be installed. Do not proceed with installation until substrates have been properly prepared and deviations from Manufacturer's recommended tolerances are corrected. Commencement of installation constitutes acceptance of conditions.

3.2 PREPARATION

- A. Prepare substrates using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.
- B. If preparation is the responsibility of another installer, notify Architect in writing of deviations from Manufacturer's recommended installation tolerances and conditions.

3.3 INSTALLATION - GENERAL

- A. Install in accordance with manufacturer's instructions for system components and approved shop drawings. Coordinate installation with adjacent work to ensure proper clearances and to prevent electrical hazards.
- B. Install in accordance with complete system instruction sheets.
- C. Install enclosures to be mechanically continuous and connected to all electrical outlets, boxes, device mounting brackets, and cabinets, in accordance with manufacturer's installation sheets.
- D. Install enclosures to be electrically continuous and bonded in accordance with the National Electric Code for proper grounding.
 1. Mechanical Security: Raceway systems shall be mechanically continuous and

connected to all electrical outlets, boxes, device mounting brackets, and cabinets, in accordance with Manufacturer's installation sheets.

2. Electrical Security: Metal raceway shall be electrically continuous and bonded in accordance with the National Electric Code for proper grounding.
3. Raceway Support: Raceway shall be supported at intervals not exceeding 5 feet (1524 mm) or in accordance with Manufacturer's installation sheets.
4. Accessories: Provide accessories as required for a complete installation, including insulated bushings and inserts where required by manufacturer.
5. Unused Openings: Close unused raceway openings using manufacturer's recommended accessories.

3.4 POWER POLE INSTALLATION

- A. Install indoor power poles and architectural columns in accordance with manufacturer's suggested installation instructions, and as specified. Locate the indoor power poles and architectural columns in strict accordance with the Drawings in order to maintain the indicated pole and column spacing.
- B. Mechanical Security: Install system in such a manner that it is mechanically continuous and connected to all electrical outlets, boxes, and cabinets, in accordance with manufacturer's installation instructions.
- C. Electrical Security: Install system in such a manner that it is electrically continuous and bonded in accordance with the provisions of the National Electric Code.
- D. System Support: Securely support system at the ceiling and floor in accordance with manufacturer's installation instructions.
- E. Install system complete with insulating bushings and inserts where required by manufacturer's installation instructions. Close all unused openings where wire is exposed.

3.5 PROTECTION

- A. Clean exposed surfaces using non-abrasive materials and methods recommended by manufacturer.
- B. Protect installed products until completion of project.
- C. Touch-up, repair or replace damaged products before Substantial Completion.

END OF SECTION 262716

SECTION 262726 - WIRING DEVICES**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Straight-blade convenience receptacles.
2. GFCI receptacles.
3. Controlled receptacles.
4. Twist-locking receptacles
5. Toggle switches.
6. Decorator-style convenience.
7. Wall switch sensor light switches with dual technology sensors.
8. Wall-box dimmers.
9. Wall plates.

1.2 DEFINITIONS**A. Abbreviations of Manufacturers' Names:**

1. Cooper: Copper Wiring Devices; Division of Cooper Industries, Inc.
2. Hubbell: Hubbell Incorporated: Wiring Devices-Kellems.
3. Leviton: Leviton Mfg. Company, Inc.
4. Pass & Seymour: Pass& Seymour/Legrand.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings: List of legends and description of materials and process used for premarking wall plates.
- C. Samples: One for each type of device and wall plate specified, in each color specified.

1.4 INFORMATIONAL SUBMITTALS

- A. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

PART 2 - PRODUCTS**2.1 GENERAL WIRING-DEVICE REQUIREMENTS**

- A. Wiring Devices, Components, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. Comply with NFPA 70.
- C. Devices that are manufactured for use with modular plug-in connectors may be substituted under the following conditions:
 - 1. Connectors shall comply with UL 2459 and shall be made with stranding building wire.
 - 2. Devices shall comply with the requirements in this Section.
- D. Devices for Owner-Furnished Equipment:
 - 1. Receptacles: Match plug configurations.
- E. Source Limitations: Obtain each type of wiring device and associated wall plate from single source from single manufacturer.

2.2 STRAIGHT-BLADE RECEPTACLES

- A. Duplex Convenience Receptacles, 125 V, 20 A: Comply with NEMA WD 1, NEMA WD 6 Configuration 5-20R, UL 498, and FS W-C-596.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Eaton (Arrow Hart).
 - b. Hubbell Incorporated; Wiring Device-Kellems.
 - c. Leviton Manufacturing Co., Inc.

2.3 TWIST-LOCKING RECEPTACLES

- A. Single Convenience Receptacles, 125 V, 20 A: Comply with NEMA WD 1, NEMA WD 6 configuration L5-20R, and UL 498.

2.4 GFCI RECEPTACLES

- A. General Description:
 - 1. 125 V, 20 A, straight blade, feed-through type.
 - 2. Comply with NEMA WD 1, NEMA WD 6 Configuration 5-20R, UL 498, UL 943 Class A, and FS W-C-596.
 - 3. Include indicator light that shows when the GFCI has malfunctioned and no longer provides proper GFCI protection.
- B. Duplex GFCI Convenience Receptacles:
 - 1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - a. Eaton (Arrow Hart).
 - b. Hubbell Incorporated; Wiring Device-Kellems.

- c. Leviton Manufacturing Co., Inc.

2.5 CONTROLLED RECEPTACLES

A. Summary:

1. Automatic Receptacle Control: 50% of all 125 volt 15- and 20-ampere receptacles in private offices, conference rooms, printing/copying rooms, break rooms and individual workstations shall be automatically switched off when the space in which they are located is not occupied. Individual workstations created with modular furniture are required to have at least 50% of the receptacles in each workstation automatically controlled.
2. All controlled receptacles shall be visually distinct from non-controlled receptacles. Switched receptacles shall be identified by receptacle color, indication lights integral to the receptacle, permanently marked/engraved receptacle labeling, etc. Regardless of the means of identification, maintaining clear, uniform identification throughout a building will increase occupant understanding and reduce confusion.
3. All controlled receptacles shall be controlled via local stand-alone ceiling mounted occupancy/vacancy sensors.

2.6 TOGGLE SWITCHES

A. Comply with NEMA WD 1, UL 20, and FS W-S-896.

B. Switches, 120/277 V, 20 A:

1. Single Pole:

- a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- 1) Eaton (Arrow Hart).
- 2) Hubbell Incorporated; Wiring Device-Kellems.
- 3) Leviton Manufacturing Co., Inc.

2. Three Way:

- a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

- 1) Eaton (Arrow Hart).
- 2) Hubbell Incorporated; Wiring Device-Kellems.
- 3) Leviton Manufacturing Co., Inc.

2.7 WALL SWITCH SENSOR LIGHT SWITCH, DUAL TECHNOLOGY

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Eaton (Arrow Hart).
2. Hubbell Incorporated; Wiring Device-Kellems.
3. Leviton Manufacturing Co., Inc.

- B. Description: Switchbox-mounted, combination lighting-control sensor and conventional switch lighting-control unit using dual technology.

1. Connections: Provisions for connection to BAS.
2. Connections: Hard wired.
3. Connections: Wireless.
4. Rated 960 W at 120-V ac for tungsten lighting, 10 A at 120-V ac or 10 A at 277-V ac for fluorescent or LED lighting, and 1/4 hp at 120-V ac.
5. Integral relay for connection to BAS.
6. Adjustable time delay of 15 minutes.
7. Able to be locked to Automatic-On mode.
8. Automatic Light-Level Sensor: Adjustable from 2 to 200 fc.
9. Comply with NEMA WD 1, UL 20, and FS W-S-896.

2.8 WALL-BOX DIMMERS

- A. Dimmer Switches: Modular, full-wave, solid-state units with integral, quiet on-off switches, with audible frequency and EMI/RFI suppression filters.
- B. Control: Continuously adjustable slider; with single-pole or three-way switching. Comply with UL 1472.
- C. Fluorescent Lamp Dimmer Switches: Modular; compatible with dimmer ballasts; trim potentiometer to adjust low-end dimming; dimmer-ballast combination capable of consistent dimming with low end not greater than 20 percent of full brightness.
- D. LED Lamp Dimmer Switches: Modular; compatible with LED lamps; trim potentiometer to adjust low-end dimming; capable of consistent dimming with low end not greater than 20 percent of full brightness.

2.9 WALL PLATES

- A. Single and combination types shall match corresponding wiring devices.
1. Plate-Securing Screws: Metal with head color to match plate finish.
 2. Material for Finished Spaces: 0.035-inch-thick, satin-finished, Type 302 stainless steel.
 3. Material for Unfinished Spaces: Galvanized steel.
 4. Material for Damp Locations: Cast aluminum with spring-loaded lift cover, and listed and labeled for use in wet and damp locations.
- B. Wet-Location, Weatherproof Cover Plates: NEMA 250, complying with Type 3R, weather-resistant, die-cast aluminum with lockable cover.

2.10 FINISHES

- A. Device Color:
1. Wiring Devices Connected to Normal Power System: As selected by Architect unless otherwise indicated or required by NFPA 70 or device listing.

2. Isolated-Ground Receptacles: As specified above, with orange triangle on face.
- B. Wall Plate Color: For plastic covers, match device color.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Comply with NECA 1, including mounting heights listed in that standard, unless otherwise indicated.
- B. Coordination with Other Trades:
1. Protect installed devices and their boxes. Do not place wall finish materials over device boxes and do not cut holes for boxes with routers that are guided by riding against outside of boxes.
 2. Keep outlet boxes free of plaster, drywall joint compound, mortar, cement, concrete, dust, paint, and other material that may contaminate the raceway system, conductors, and cables.
 3. Install device boxes in brick or block walls so that the cover plate does not cross a joint unless the joint is troweled flush with the face of the wall.
 4. Install wiring devices after all wall preparation, including painting, is complete.
- C. Conductors:
1. Do not strip insulation from conductors until right before they are spliced or terminated on devices.
 2. Strip insulation evenly around the conductor using tools designed for the purpose. Avoid scoring or nicking of solid wire or cutting strands from stranded wire.
 3. The length of free conductors at outlets for devices shall meet provisions of NFPA 70, Article 300, without pigtails.
 4. Existing Conductors:
 - a. Cut back and pigtail, or replace all damaged conductors.
 - b. Straighten conductors that remain and remove corrosion and foreign matter.
 - c. Pigtailling existing conductors is permitted, provided the outlet box is large enough.
- D. Device Installation:
1. Replace devices that have been in temporary use during construction and that were installed before building finishing operations were complete.
 2. Keep each wiring device in its package or otherwise protected until it is time to connect conductors.
 3. Do not remove surface protection, such as plastic film and smudge covers, until the last possible moment.
 4. Connect devices to branch circuits using pigtails that are not less than 6 inches in length.
 5. When there is a choice, use side wiring with binding-head screw terminals. Wrap solid conductor tightly clockwise, two-thirds to three-fourths of the way around terminal screw.
 6. Use a torque screwdriver when a torque is recommended or required by manufacturer.
 7. When conductors larger than No. 12 AWG are installed on 15- or 20-A circuits,

- splice No. 12 AWG pigtails for device connections.
- 8. Tighten unused terminal screws on the device.
- 9. When mounting into metal boxes, remove the fiber or plastic washers used to hold device-mounting screws in yokes, allowing metal-to-metal contact.
- E. Receptacle Orientation:
 - 1. Install ground pin of vertically mounted receptacles down, and on horizontally mounted receptacles to the right.
- F. Device Plates: Do not use oversized or extra-deep plates. Repair wall finishes and remount outlet boxes when standard device plates do not fit flush or do not cover rough wall opening.
- G. Dimmers:
 - 1. Install dimmers within terms of their listing.
 - 2. Verify that dimmers used for fan-speed control are listed for that application.
 - 3. Install unshared neutral conductors on line and load side of dimmers according to manufacturers' device listing conditions in the written instructions.
- H. Arrangement of Devices: Unless otherwise indicated, mount flush, with long dimension vertical and with grounding terminal of receptacles on top. Group adjacent switches under single, multigang wall plates.
- I. GFCI Receptacles: Install non-feed-through-type GFCI receptacles where protection of downstream receptacles is not required.

3.2 FIELD QUALITY CONTROL

- A. Test Instruments: Use instruments that comply with UL 1436.
- B. Test Instrument for Convenience Receptacles: Digital wiring analyzer with digital readout or illuminated digital-display indicators of measurement.
- C. Perform the following tests and inspections:
 - 1. Tests for Convenience Receptacles:
 - a. Line Voltage: Acceptable range is 105 to 132 V.
 - b. Percent Voltage Drop under 15-A Load: A value of 6 percent or higher is unacceptable.
 - c. Ground Impedance: Values of up to 2 ohms are acceptable.
 - d. GFCI Trip: Test for tripping values specified in UL 1436 and UL 943.
 - e. Using the test plug, verify that the device and its outlet box are securely mounted.
 - f. Tests shall be diagnostic, indicating damaged conductors, high resistance at the circuit breaker, poor connections, inadequate fault current path, defective devices, or similar problems. Correct circuit conditions, remove malfunctioning units and replace with new ones, and retest as specified above.
- D. Wiring device will be considered defective if it does not pass tests and inspections.
- E. Prepare test and inspection reports.

END OF SECTION 262726

SECTION 262813 – FUSES**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Cartridge fuses rated 600 V ac and less for use in the following:
 - a. Control circuits.
 - b. Motor-control centers.
 - c. Panelboards.
 - d. Switchboards.
 - e. Enclosed controllers.
 - f. Enclosed switches.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.

1.3 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

PART 2 - PRODUCTS**2.1 MANUFACTURERS**

- A. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**:
 1. Schneider Electric.
 2. Bussmann, an Eaton business.
 3. Edison; a brand of Bussmann by Eaton.
 4. Littelfuse, Inc.

2.2 CARTRIDGE FUSES

- A. Characteristics: NEMA FU 1, current-limiting, nonrenewable cartridge fuses with voltage ratings consistent with circuit voltages.
 1. Type RK-1: 250 and 600-V, zero- to 600-A rating, 200 kAIC, time delay.
 2. Type RK-5: 250 and 600-V, zero- to 600-A rating, 200 kAIC, time delay.
- B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- C. Comply with NEMA FU 1 for cartridge fuses.

- D. Comply with NFPA 70.
- E. Coordinate fuse ratings with utilization equipment nameplate limitations of maximum fuse size and with system short-circuit current levels.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install fuses in fusible devices. Arrange fuses so rating information is readable without removing fuse.
- B. Install spare-fuse cabinet(s) in location shown on the Drawings or as indicated in the field by Owner.

3.2 IDENTIFICATION

- A. Install labels complying with requirements for identification specified in Section 260553 "Identification for Electrical Systems" and indicating fuse replacement information inside of door of each fused switch and adjacent to each fuse block, socket, and holder.

END OF SECTION 262813

SECTION 262816 - ENCLOSED SWITCHES AND CIRCUIT BREAKERS**PART 1 - GENERAL****1.1 SUMMARY****A. Section Includes:**

1. Fusible switches.
2. Nonfusible switches.
3. Receptacle switches.
4. Shunt trip switches.
5. Molded-case circuit breakers (MCCBs).
6. Molded-case switches.
7. Enclosures.

1.2 ACTION SUBMITTALS**A. Product Data:** For each type of enclosed switch, circuit breaker, accessory, and component indicated. Include nameplate ratings, dimensioned elevations, sections, weights, and manufacturers' technical data on features, performance, electrical characteristics, ratings, accessories, and finishes.

1. Include time-current coordination curves (average melt) for each type and rating of overcurrent protective device; include selectable ranges for each type of overcurrent protective device. Provide in PDF electronic format.

B. Shop Drawings: For enclosed switches and circuit breakers.

1. Include plans, elevations, sections, details, and attachments to other work.
2. Include wiring diagrams for power, signal, and control wiring.

1.3 INFORMATIONAL SUBMITTALS**A. Qualification Data:** For qualified testing agency.**B. Seismic Qualification Certificates:** For enclosed switches and circuit breakers, accessories, and components, from manufacturer.**C. Field quality-control reports.****1.4 CLOSEOUT SUBMITTALS****A. Operation and maintenance data.****1.5 WARRANTY****A. Manufacturer's Warranty:** Manufacturer and Installer agree to repair or replace components that fail in materials or workmanship within specified warranty period.

1. Warranty Period: One year(s) from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Seismic Performance: Enclosed switches and circuit breakers shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
 - 1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."

2.2 GENERAL REQUIREMENTS

- A. Source Limitations: Obtain enclosed switches and circuit breakers, overcurrent protective devices, components, and accessories, within same product category, from single manufacturer.
- B. Product Selection for Restricted Space: Drawings indicate maximum dimensions for enclosed switches and circuit breakers, including clearances between enclosures, and adjacent surfaces and other items. Comply with indicated maximum dimensions.
- C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.
- D. Comply with NFPA 70.

2.3 FUSIBLE SWITCHES

- A. Manufacturers: Subject to compliance with requirements, provided on **LIDL National Account List**: (Shall be same manufacturer as the panelboards)
 - 1. Schneider Electric
- B. Type HD, Heavy Duty:
 - 1. Single throw.
 - 2. Three pole.
 - 3. 240 and 600-V ac.
 - 4. 1200 A and smaller.
 - 5. UL 98 and NEMA KS 1, horsepower rated, with clips or bolt pads to accommodate indicated fuses.
 - 6. Lockable handle with capability to accept three padlocks, and interlocked with cover in closed position.
- C. Accessories:
 - 1. Equipment Ground Kit: Internally mounted and labeled for copper and aluminum ground conductors.
 - 2. Neutral Kit: Internally mounted; insulated, capable of being grounded and bonded; labeled for copper and aluminum neutral conductors.
 - 3. Isolated Ground Kit: Internally mounted; insulated, labeled for copper

- and aluminum neutral conductors.
- 4. Class R Fuse Kit: Provides rejection of other fuse types when Class R fuses are specified.
- 5. Service-Rated Switches: Labeled for use as service equipment.

2.4 NONFUSIBLE SWITCHES

- A. **Manufacturers:** Subject to compliance with requirements, provided on **LiDL National Account List**: (Shall be the same manufacturer as the panelboards)
 - 1. Schneider Electric
- B. Type HD, Heavy Duty, Three Pole, Single Throw, 240 and 600-V ac, 1200 A and Smaller: UL 98 and NEMA KS 1, horsepower rated, lockable handle with capability to accept three padlocks, and interlocked with cover in closed position.
- C. **Accessories:**
 - 1. Equipment Ground Kit: Internally mounted and labeled for copper and aluminum ground conductors.
 - 2. Neutral Kit: Internally mounted; insulated, capable of being grounded and bonded; labeled for copper and aluminum neutral conductors.
 - 3. Class R Fuse Kit: Provides rejection of other fuse types when Class R fuses are specified.
 - 4. Service-Rated Switches: Labeled for use as service equipment.

2.5 MOLDED-CASE CIRCUIT BREAKERS

- A. **Manufacturers:** Subject to compliance with requirements, provided on **LiDL National Account List**: (Shall be the same manufacturer as the panelboards)
 - 1. Schneider Electric
- B. Circuit breakers shall be constructed using glass-reinforced insulating material. Current carrying components shall be completely isolated from the handle and the accessory mounting area.
- C. Circuit breakers shall have a toggle operating mechanism with common tripping of all poles, which provides quick-make, quick-break contact action. The circuit-breaker handle shall be over center, be trip free, and reside in a tripped position between on and off to provide local trip indication. Circuit-breaker escutcheon shall be clearly marked on and off in addition to providing international I/O markings. Equip circuit breaker with a push-to-trip button, located on the face of the circuit breaker to mechanically operate the circuit-breaker tripping mechanism for maintenance and testing purposes.
- D. The maximum ampere rating and UL, IEC, or other certification standards with applicable voltage systems and corresponding interrupting ratings shall be clearly marked on face of circuit breaker. Circuit breakers shall be 100 percent rated. Circuit breaker/circuit breaker combinations for series connected interrupting ratings shall be listed by UL as recognized component combinations. Any series rated combination used shall be marked on the end-use equipment along with the statement "Caution - Series Rated System. Amps Available. Identical

Replacement Component Required."

- E. MCCBs shall be equipped with a device for locking in the isolated position.
- F. Lugs shall be suitable for 194 deg F rated wire, sized according to the 167 deg F temperature rating in NFPA 70.
- G. Standards: Comply with UL 489 and NEMA AB 3, with interrupting capacity to comply with available fault currents.
- H. Thermal-Magnetic Circuit Breakers: Inverse time-current thermal element for low-level overloads and instantaneous magnetic trip element for short circuits. Adjustable magnetic trip setting for circuit-breaker frame sizes 250 A and larger.
- I. Adjustable, Instantaneous-Trip Circuit Breakers: Magnetic trip element with front-mounted, field-adjustable trip setting.
- J. Electronic Trip Circuit Breakers: Field-replaceable rating plug, rms sensing, with the following field-adjustable settings:
 - 1. Long- and short-time pickup levels.
 - 2. Long- and short-time time adjustments.
 - 3. Ground-fault pickup level, time delay, and I-squared t response.
- K. Current-Limiting Circuit Breakers: Frame sizes 400 A and smaller, and let-through ratings less than NEMA FU 1, RK-5.
- L. Features and Accessories:
 - 1. Standard frame sizes, trip ratings, and number of poles.
 - 2. Application Listing: Appropriate for application; Type SWD for switching fluorescent lighting loads; Type HID for feeding fluorescent and high-intensity discharge lighting circuits.
 - 3. Shunt Trip: Trip coil energized from separate circuit, with coil-clearing contact.
 - 4. Auxiliary Contacts: One SPDT switch with "a" and "b" contacts; "a" contacts mimic circuit-breaker contacts, "b" contacts operate in reverse of circuit-breaker contacts.
 - 5. Alarm Switch: One NO contact that operates only when circuit breaker has tripped.

2.6 ENCLOSURES

- A. Enclosed Switches and Circuit Breakers: UL 489, NEMA KS 1, NEMA 250, and UL 50, to comply with environmental conditions at installed location.
- B. Enclosure Finish: The enclosure shall be gray baked enamel paint, electrodeposited on cleaned, phosphatized galvanized steel (NEMA 250 Types 3R, 12).
- C. Operating Mechanism: The circuit-breaker operating handle shall be directly operable through the dead front trim of the enclosure (NEMA 250 Type 3R). The cover interlock mechanism shall have an externally operated override. The override shall not permanently disable the interlock mechanism, which shall return to the locked position once the override is released. The tool used to override the cover interlock mechanism shall not be required to enter the

enclosure in order to override the interlock.

PART 3 - EXECUTION

3.1 ENCLOSURE ENVIRONMENTAL RATING APPLICATIONS

- A. Enclosed Switches and Circuit Breakers: Provide enclosures at installed locations with the following environmental ratings.
 - 1. Indoor, Dry and Clean Locations: NEMA 250, Type 1.
 - 2. Outdoor Locations: NEMA 250, Type 3R.
 - 3. Kitchen Wash-Down Areas: NEMA 250, Type 4X, stainless steel.
 - 4. Other Wet or Damp, Indoor Locations: NEMA 250, Type 4.

3.2 INSTALLATION

- A. Coordinate layout and installation of switches, circuit breakers, and components with equipment served and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.
- B. Install individual wall-mounted switches and circuit breakers with tops at uniform height unless otherwise indicated.
- C. Comply with mounting and anchoring requirements specified in Section 260548.16 "Seismic Controls for Electrical Systems."
- D. Temporary Lifting Provisions: Remove temporary lifting of eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.
- E. Install fuses in fusible devices.
- F. Comply with NFPA 70 and NECA 1.
- G. Set field-adjustable circuit-breaker trip ranges to values indicated on the Drawings.

3.3 IDENTIFICATION

- A. Comply with requirements in Section 260553 "Identification for Electrical Systems."
 - 1. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs.
 - 2. Label each enclosure with engraved metal or laminated-plastic nameplate.

3.4 FIELD QUALITY CONTROL

- A. Testing Agency: Owner will engage a qualified testing agency to perform tests and inspections.
- B. Testing Agency: Engage a qualified testing agency to perform tests and inspections.
- C. Perform tests and inspections.
- D. Tests and Inspections for Switches:

1. Visual and Mechanical Inspection:
 - a. Inspect physical and mechanical condition.
 - b. Inspect anchorage, alignment, grounding, and clearances.
 - c. Verify that the unit is clean.
 - d. Verify blade alignment, blade penetration, travel stops, and mechanical operation.
 - e. Verify that fuse sizes and types match the Specifications and Drawings.
 - f. Verify that each fuse has adequate mechanical support and contact integrity.
 - g. Inspect bolted electrical connections for high resistance using one of the two following methods:
 - 1) Use a low-resistance ohmmeter.
 - a) Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from those of similar bolted connections by more than 50 percent of the lowest value.
 - 2) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data.
 - a) Bolt-torque levels shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data.
 - h. Verify that operation and sequencing of interlocking systems is as described in the Specifications and shown on the Drawings.
 - i. Verify correct phase barrier installation.
 - j. Verify lubrication of moving current-carrying parts and moving and sliding surfaces.
2. Electrical Tests:
 - a. Perform resistance measurements through bolted connections with a low-resistance ohmmeter. Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.
 - b. Measure contact resistance across each switchblade fuseholder. Drop values shall not exceed the high level of the manufacturer's published data. If manufacturer's published data are not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.
 - c. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with switch closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, investigate values of insulation resistance less than those published in Table 100.1 or as recommended in manufacturer's published data.
 - d. Measure fuse resistance. Investigate fuse-resistance values that deviate from each other by more than 15 percent.

- e. Perform ground fault test.
- E. Tests and Inspections for Molded Case Circuit Breakers:
- 1. Visual and Mechanical Inspection:
 - a. Verify that equipment nameplate data are as described in the Specifications and shown on the Drawings.
 - b. Inspect physical and mechanical condition.
 - c. Inspect anchorage, alignment, grounding, and clearances.
 - d. Verify that the unit is clean.
 - e. Operate the circuit breaker to ensure smooth operation.
 - f. Inspect bolted electrical connections for high resistance using one of the two following methods:
 - 1) Use a low-resistance ohmmeter.
 - a) Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from those of similar bolted connections by more than 50 percent of the lowest value.
 - 2) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data.
 - a) Bolt-torque levels shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data.
 - g. Inspect operating mechanism, contacts, and chutes in unsealed units.
 - h. Perform adjustments for final protective device settings in accordance with the coordination study.
 - 2. Electrical Tests:
 - a. Perform resistance measurements through bolted connections with a low-resistance ohmmeter. Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.
 - b. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with circuit breaker closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. Investigate values of insulation resistance less than those published in Table 100.1 or as recommended in manufacturer's published data.
 - c. Perform a contact/pole resistance test. Drop values shall not exceed the high level of the manufacturer's published data. If manufacturer's published data are not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.
 - d. Perform insulation resistance tests on all control wiring with respect to ground. Applied potential shall be 500-V dc for 300-V rated cable

and 1000-V dc for 600- V rated cable. Test duration shall be one minute. For units with solid state components, follow manufacturer's recommendation. Insulation resistance values shall be no less than two megohms.

- e. Determine the following by primary current injection:
 - 1) Long-time pickup and delay. Pickup values shall be as specified. Trip characteristics shall not exceed manufacturer's published time-current characteristic tolerance band, including adjustment factors.
 - 2) Short-time pickup and delay. Short-time pickup values shall be as specified. Trip characteristics shall not exceed manufacturer's published time-current characteristic tolerance band, including adjustment factors.
 - 3) Ground-fault pickup and time delay. Ground-fault pickup values shall be as specified. Trip characteristics shall not exceed manufacturer's published time-current characteristic tolerance band, including adjustment factors.
 - 4) Instantaneous pickup. Instantaneous pickup values shall be as specified and within manufacturer's published tolerances.
 - f. Test functionality of the trip unit by means of primary current injection. Pickup values and trip characteristics shall be as specified and within manufacturer's published tolerances.
 - g. Perform minimum pickup voltage tests on shunt trip and close coils in accordance with manufacturer's published data. Minimum pickup voltage of the shunt trip and close coils shall be as indicated by manufacturer.
 - h. Verify correct operation of auxiliary features such as trip and pickup indicators; zone interlocking; electrical close and trip operation; trip-free, anti-pump function; and trip unit battery condition. Reset all trip logs and indicators. Investigate units that do not function as designed.
 - i. Verify operation of charging mechanism. Investigate units that do not function as designed.
- 3. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
 - 4. Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.
- F. Enclosed switches and circuit breakers will be considered defective if they do not pass tests and inspections.
- G. Prepare test and inspection reports.
- 1. Test procedures used.
 - 2. Include identification of each enclosed switch and circuit breaker tested and describe test results.
 - 3. List deficiencies detected, remedial action taken, and observations after remedial action.

END OF SECTION 262816

SECTION 262913 - ENCLOSED CONTROLLERS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the following enclosed controllers rated 600 V and less:

1. Full-voltage manual.
2. Full-voltage magnetic.
3. Multispeed.

1.2 DEFINITIONS

- A. CPT: Control power transformer.
- B. MCCB: Molded-case circuit breaker.
- C. MCP: Motor circuit protector.
- D. N.C.: Normally closed.
- E. N.O.: Normally open.
- F. OCPD: Overcurrent protective device.

1.3 PERFORMANCE REQUIREMENTS

- A. Seismic Performance: Enclosed controllers shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of enclosed controller.
- B. Shop Drawings: For each enclosed controller. Include dimensioned plans, elevations, sections, details, and required clearances and service spaces around controller enclosures.
1. Wiring Diagrams: For power, signal, and control wiring.

1.5 INFORMATIONAL SUBMITTALS

- A. Seismic Qualification Certificates: For enclosed controllers, accessories, and components, from manufacturer.
- B. Field quality-control reports.

1.6 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.7 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. Comply with NFPA 70.
- C. IEEE Compliance: Fabricate and test enclosed controllers according to IEEE 344 to withstand seismic forces defined in Section 260548.16 "Seismic Controls for Electrical Systems."

PART 2 - PRODUCTS

2.1 FULL-VOLTAGE CONTROLLERS

- A. General Requirements for Full-Voltage Controllers: Comply with NEMA ICS 2, general purpose, Class A.
- B. Motor-Starting Switches: "Quick-make, quick-break" toggle or push-button action; marked to show whether unit is off or on.
 - 1. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**:
 - a. Schneider Electric
 - b. Eaton.
 - c. General Electric Company.
 - d. Rockwell Automation, Inc.
 - e. Siemens Industry, Inc.
 - 2. Configuration: Non-reversing.
 - 3. Surface mounting.
 - 4. Pilot light.
- C. Fractional Horsepower Manual Controllers: "Quick-make, quick-break" toggle or push-button action; marked to show whether unit is off, on, or tripped.

1. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**:
 - a. Schneider Electric
 - b. Eaton.
 - c. General Electric Company.
 - d. Rockwell Automation, Inc.
 - e. Siemens Industry, Inc.
 2. Configuration: Non-reversing.
 3. Overload Relays: Inverse-time-current characteristics; NEMA ICS 2, Class 10 tripping characteristics; heaters matched to nameplate full-load current of actual protected motor; external reset push button; bimetallic type.
 4. Surface mounting.
 5. Pilot light.
- D. Integral Horsepower Manual Controllers: "Quick-make, quick-break" toggle or push-button action; marked to show whether unit is off, on, or tripped.
1. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**:
 - a. Schneider Electric
 - b. Eaton.
 - c. General Electric Company.
 - d. Rockwell Automation, Inc.
 - e. Siemens Industry, Inc.
 2. Configuration: Nonreversing.
 3. Overload Relays: Inverse-time-current characteristics; NEMA ICS 2, Class 10 tripping characteristics; heaters and sensors in each phase, matched to nameplate full-load current of actual protected motor and having appropriate adjustment for duty cycle; external reset push button; bimetallic type.
 4. Surface mounting.
 5. Pilot light.
- E. Magnetic Controllers: Full voltage, across the line, electrically held.
1. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**:
 - a. Schneider Electric
 - b. Eaton.
 - c. General Electric Company.
 - d. Rockwell Automation, Inc.
 - e. Siemens Industry, Inc.
 2. Configuration: Non-reversing.
 3. Contactor Coils: Pressure-encapsulated type.

- a. Operating Voltage: Depending on contactor NEMA size and line-voltage rating, manufacturer's standard matching control power or line voltage.
 4. Power Contacts: Totally enclosed, double-break, silver-cadmium oxide; assembled to allow inspection and replacement without disturbing line or load wiring.
 5. Control Circuits: 24-V ac; obtained from integral CPT, with primary and secondary fuses, with CPT of sufficient capacity to operate integral devices and remotely located pilot, indicating, and control devices.
 6. Melting Alloy Overload Relays:
 - a. Inverse-time-current characteristic.
 - b. Class 10 tripping characteristic.
 - c. Heaters in each phase matched to nameplate full-load current of actual protected motor and with appropriate adjustment for duty cycle.
 7. Bimetallic Overload Relays:
 - a. Inverse-time-current characteristic.
 - b. Class 10 tripping characteristic.
 - c. Heaters in each phase matched to nameplate full-load current of actual protected motor and with appropriate adjustment for duty cycle.
 8. Solid-State Overload Relay:
 - a. Switch or dial selectable for motor running overload protection.
 - b. Sensors in each phase.
 - c. Class 10 tripping characteristic selected to protect motor against voltage and current unbalance and single phasing.
 9. External overload reset push button.
- F. Combination Magnetic Controller: Factory-assembled combination of magnetic controller, OCPD, and disconnecting means.
1. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**:
 - a. Schneider Electric
 - b. Eaton.
 - c. General Electric Company.
 - d. Rockwell Automation, Inc.
 - e. Siemens Industry, Inc.
 2. Fusible Disconnecting Means:
 - a. NEMA KS 1, heavy-duty, horsepower-rated, fusible switch with clips or bolt pads to accommodate Class J fuses.
 - b. Lockable Handle: Accepts three padlocks and interlocks with cover in closed position.
 3. Auxiliary Contacts: N.O./N.C., arranged to activate before switch blades open.
 4. Non-fusible Disconnecting Means:

- a. NEMA KS 1, heavy-duty, horsepower-rated, non-fusible switch.
 - b. Lockable Handle: Accepts three padlocks and interlocks with cover in closed position.
 - c. Auxiliary Contacts: N.O./N.C., arranged to activate before switch blades open.
5. MCP Disconnecting Means:
- a. UL 489, NEMA AB 1, and NEMA AB 3, with interrupting capacity to comply with available fault currents, instantaneous-only circuit breaker with front-mounted, field-adjustable, short-circuit trip coordinated with motor locked-rotor amperes.
 - b. Lockable Handle: Accepts three padlocks and interlocks with cover in closed position.
 - c. Auxiliary contacts "a" and "b" arranged to activate with MCP handle.
6. MCCB Disconnecting Means:
- a. UL 489, NEMA AB 1, and NEMA AB 3, with interrupting capacity to comply with available fault currents; thermal-magnetic MCCB, with inverse time-current element for low-level overloads and instantaneous magnetic trip element for short circuits.
 - b. Front-mounted, adjustable magnetic trip setting for circuit-breaker frame sizes 250 A and larger.
 - c. Lockable Handle: Accepts three padlocks and interlocks with cover in closed position.
 - d. Auxiliary contacts "a" and "b" arranged to activate with MCCB handle.

2.2 ENCLOSURES

- A. Enclosed Controllers: NEMA ICS 6, to comply with environmental conditions at installed location.
- 1. Dry and Clean Indoor Locations: Type 1.
 - 2. Outdoor Locations: Type 3R.
 - 3. Kitchen Wash-Down Areas: Type 4X, stainless steel.
 - 4. Other Wet or Damp Indoor Locations: Type 3R.

2.3 ACCESSORIES

- A. Push Buttons, Pilot Lights, and Selector Switches: NEMA ICS 5; heavy-duty type; factory installed in controller enclosure cover unless otherwise indicated.
- B. Control Relays: Auxiliary and adjustable time-delay relays.
- C. Phase-Failure, Phase-Reversal, and Undervoltage and Overvoltage Relays: Solid-state sensing circuit with isolated output contacts for hard-wired connections. Provide adjustable undervoltage, overvoltage, and time-delay settings.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Wall-Mounted Controllers: Install enclosed controllers on walls with tops at uniform height, and with disconnect operating handles not higher than 79 inches above finished floor, unless otherwise indicated, and by bolting units to wall or mounting on lightweight structural-steel channels bolted to wall. For controllers not at walls, provide freestanding racks complying with Section 260529 "Hangers and Supports for Electrical Systems."
- B. Floor-Mounted Controllers: Install enclosed controllers on 4-inch nominal-thickness concrete base. Comply with requirements for concrete base specified in other sections of these specifications.
 - 1. Install dowel rods to connect concrete base to concrete floor. Unless otherwise indicated, install dowel rods on 18-inch centers around the full perimeter of concrete base.
 - 2. For supported equipment, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete floor.
 - 3. Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
 - 4. Install anchor bolts to elevations required for proper attachment to supported equipment.
- C. Seismic Bracing: Comply with requirements specified in Section 260548.16 "Seismic Controls for Electrical Systems."
- D. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.
- E. Install fuses in each fusible-switch enclosed controller.
- F. Install fuses in control circuits if not factory installed. Comply with requirements in Section 262813 "Fuses."
- G. Install heaters in thermal overload relays. Select heaters based on actual nameplate full-load amperes after motors have been installed.
- H. Comply with NECA 1.

3.2 IDENTIFICATION

- A. Identify enclosed controllers, components, and control wiring. Comply with requirements for identification specified in Section 260553 "Identification for Electrical Systems."
 - 1. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs.
 - 2. Label each enclosure with engraved nameplate.
 - 3. Label each enclosure-mounted control and pilot device.

3.3 CONTROL WIRING INSTALLATION

- A. Install wiring between enclosed controllers and remote devices and facility's central control system.
- B. Bundle, train, and support wiring in enclosures.
- C. Connect selector switches and other automatic-control selection devices where applicable.
 - 1. Connect selector switches to bypass only those manual- and automatic-control devices that have no safety functions when switch is in manual-control position.
 - 2. Connect selector switches with enclosed-controller circuit in both manual and automatic positions for safety-type control devices such as low- and high-pressure cutouts, high-temperature cutouts, and motor overload protectors.

3.4 FIELD QUALITY CONTROL

- A. Perform tests and inspections.
- B. Acceptance Testing Preparation:
 - 1. Test insulation resistance for each enclosed controller, component, connecting supply, feeder, and control circuit.
 - 2. Test continuity of each circuit.
- C. Tests and Inspections:
 - 1. Inspect controllers, wiring, components, connections, and equipment installation. Test and adjust controllers, components, and equipment.
 - 2. Test insulation resistance for each enclosed-controller element, component, connecting motor supply, feeder, and control circuits.
 - 3. Test continuity of each circuit.
 - 4. Verify that voltages at controller locations are within plus or minus 10 percent of motor nameplate rated voltages. If outside this range for any motor, notify Owner before starting the motor(s).
 - 5. Test each motor for proper phase rotation.
 - 6. Perform each electrical test and visual and mechanical inspection stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
 - 7. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
 - 8. Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.
- D. Enclosed controllers will be considered defective if they do not pass tests and inspections.
- E. Prepare test and inspection reports. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

3.5 ADJUSTING

- A. Set field-adjustable switches and overload-relay pickup and trip ranges.
- B. Adjust the trip settings of MCPs and thermal-magnetic circuit breakers with adjustable instantaneous trip elements. Initially adjust to six times the motor nameplate full-load ampere ratings and attempt to start motors several times, allowing for motor cooldown between starts. If tripping occurs on motor inrush, adjust settings in increments until motors start without tripping. Do not exceed eight times the motor full-load amperes (or 11 times for NEMA Premium Efficient motors if required). Where these maximum settings do not allow starting of a motor, notify Owner before increasing settings.

3.6 DEMONSTRATION

- A. Train Owner's maintenance personnel to adjust, operate, and maintain enclosed controllers.

END OF SECTION 262913

SECTION 262923 - VARIABLE-FREQUENCY MOTOR CONTROLLERS**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes separately enclosed, preassembled, combination VFCs, rated 600 V and less, for speed control of three-phase, squirrel-cage induction motors.

1.2 DEFINITIONS

- A. CE: Conformance Europeene (European Compliance).
- B. CPT: Control power transformer.
- C. DDC: Direct digital control.
- D. EMI: Electromagnetic interference.
- E. OCPD: Overcurrent protective device.
- F. PID: Control action, proportional plus integral plus derivative.
- G. RFI: Radio-frequency interference.
- H. VFC: Variable-frequency motor controller.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type and rating of VFC indicated.
- B. Shop Drawings: For each VFC indicated.
 - 1. Include details of equipment assemblies. Indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
 - 2. Include diagrams for power, signal, and control wiring.

1.4 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For testing agency.
- B. Seismic Qualification Certificates: For each VFC, accessories, and components, from manufacturer.
 - 1. Certificate of compliance.
 - 2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
 - 3. Detailed description of equipment anchorage devices on which the certification is based, and their installation requirements.
- C. Product certificates.

- D. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.6 QUALITY ASSURANCE

- A. Testing Agency Qualifications: Member company of NETA or an NRTL.

1.7 WARRANTY

- A. Special Warranty: Manufacturer agrees to repair or replace VFCs that fail in materials or workmanship within specified warranty period.

- 1. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provided on **LIDL National Account List**:

- 1. Eaton.
 - 2. Emerson Industrial Automation.
 - 3. General Electric Company.
 - 4. Rockwell Automation, Inc.
 - 5. Schneider Electric USA, Inc.
 - 6. Siemens Industry, Inc.

2.2 SYSTEM DESCRIPTION

- A. General Requirements for VFCs:

- 1. VFCs and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
 - 2. Comply with NEMA ICS 7, NEMA ICS 61800-2, and UL 508A.

- B. Application: Constant torque and variable torque.

- C. VFC Description: Variable-frequency motor controller, consisting of power converter that employs pulse-width-modulated inverter, factory built and tested in an enclosure, with integral disconnecting means and overcurrent and overload protection; listed and labeled by an NRTL as a complete unit; arranged to provide self-protection, protection, and variable-speed control of one or more three-phase induction motors by adjusting output voltage and frequency.

- 1. Units suitable for operation of NEMA MG 1 motors.
 - 2. Listed and labeled for integrated short-circuit current (withstand) rating

by an NRTL acceptable to authorities having jurisdiction.

- D. Design and Rating: Match load type, such as fans, blowers, and pumps; and type of connection used between motor and load such as direct or through a power-transmission connection.
- E. Output Rating: Three phase; 10 to 60 Hz, with voltage proportional to frequency throughout voltage range; maximum voltage equals input voltage.
- F. Unit Operating Requirements:
 - 1. Input AC Voltage Tolerance: Plus 10 and minus 10 percent of VFC input voltage rating.
 - 2. Input AC Voltage Unbalance: Not exceeding 3 percent.
 - 3. Input Frequency Tolerance: Plus or minus 3 percent of VFC frequency rating.
 - 4. Minimum Efficiency: 96 percent at 60 Hz, full load.
 - 5. Minimum Displacement Primary-Side Power Factor: 96 percent under any load or speed condition.
 - 6. Minimum Short-Circuit Current (Withstand) Rating: 22 kA.
 - 7. Ambient Temperature Rating: Not less than 32 deg F and not exceeding 104 deg F.
 - 8. Humidity Rating: Less than 95 percent (noncondensing).
 - 9. Altitude Rating: Not exceeding 3300 feet.
 - 10. Vibration Withstand: Comply with NEMA ICS 61800-2.
 - 11. Overload Capability: 1.1 times the base load current for 60 seconds; minimum of 1.8 times the base load current for three seconds.
 - 12. Starting Torque: Minimum 100 percent of rated torque from 3 to 60 Hz.
 - 13. Speed Regulation: Plus or minus 5 percent.
 - 14. Output Carrier Frequency: Selectable; 0.5 to 15 kHz.
 - 15. Stop Modes: Programmable; includes fast, free-wheel, and dc injection braking.
- G. Inverter Logic: Microprocessor based, 32 bit, isolated from all power circuits.
- H. Isolated Control Interface: Allows VFCs to follow remote-control signal over a minimum 40:1 speed range.
 - 1. Signal: Electrical.
- I. Internal Adjustability Capabilities:
 - 1. Minimum Speed: 5 to 25 percent of maximum rpm.
 - 2. Maximum Speed: 80 to 100 percent of maximum rpm.
 - 3. Acceleration: 0.1 to 999.9 seconds.
 - 4. Deceleration: 0.1 to 999.9 seconds.
 - 5. Current Limit: 30 to minimum of 150 percent of maximum rating.
- J. Self-Protection and Reliability Features:
 - 1. Surge Suppression: Factory installed as an integral part of the VFC, complying with UL 1449 SPD, Type 1 or Type 2.
 - 2. Surge Suppression: Field-mounted surge suppressors complying with Section 264313 "Surge Protection for Low-Voltage Electrical Power Circuits," UL 1449 SPD, Type 2.

3. Loss of Input Signal Protection: Selectable response strategy, including speed default to a percent of the most recent speed, a preset speed, or stop; with alarm.
 4. Under- and overvoltage trips.
 5. Inverter overcurrent trips.
 6. VFC and Motor-Overload/Overtemperature Protection: Microprocessor-based thermal protection system for monitoring VFCs and motor thermal characteristics, and for providing VFC overtemperature and motor-overload alarm and trip; settings selectable via the keypad.
 7. Critical frequency rejection, with three selectable, adjustable deadbands.
 8. Instantaneous line-to-line and line-to-ground overcurrent trips.
 9. Loss-of-phase protection.
 10. Reverse-phase protection.
 11. Short-circuit protection.
 12. Motor-overtemperature fault.
- K. Automatic Reset/Restart: Attempt three restarts after drive fault or on return of power after an interruption and before shutting down for manual reset or fault correction; adjustable delay time between restart attempts.
- L. Power-Interruption Protection: To prevent motor from re-energizing after a power interruption until motor has stopped, unless "Bidirectional Autospeed Search" feature is available and engaged.
- M. Bidirectional Autospeed Search: Capable of starting VFC into rotating loads spinning in either direction and returning motor to set speed in proper direction, without causing damage to drive, motor, or load.
- N. Torque Boost: Automatically varies starting and continuous torque to at least 1.5 times the minimum torque to ensure high-starting torque and increased torque at slow speeds.
- O. Motor Temperature Compensation at Slow Speeds: Adjustable current fall-back based on output frequency for temperature protection of self-cooled, fan-ventilated motors at slow speeds.
- P. Integral Input Disconnecting Means and OCPD: UL 489, instantaneous-trip circuit breaker with pad-lockable, door-mounted handle mechanism.
1. Disconnect Rating: Not less than 115 percent of VFC input current rating.
 2. Disconnect Rating: Not less than 115 percent of NFPA 70 motor full-load current rating or VFC input current rating, whichever is larger.
 3. Auxiliary Contacts: NO or NC, arranged to activate before switch blades open.
 4. Auxiliary contacts "a" and "b" arranged to activate with circuit-breaker handle.
 5. NC alarm contact that operates only when circuit breaker has tripped.

2.3 PERFORMANCE REQUIREMENTS

- A. Seismic Performance: VFCs shall withstand the effects of earthquake motions determined according to ASCE/SEI 7. The designated VFCs shall be tested and certified by an NRTL as meeting the ICC-ES AC 156 test procedure requirements.

1. The term "withstand" means "the unit will remain in place without separation of any parts when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."

2.4 CONTROLS AND INDICATION

- A. Status Lights: Door-mounted LED indicators displaying the following conditions:
 1. Power on.
 2. Run.
 3. Overvoltage.
 4. Line fault.
 5. Overcurrent.
 6. External fault.
- B. Panel-Mounted Operator Station: Manufacturer's standard front-accessible, sealed keypad and plain-English-language digital display; allows complete programming, program copying, operating, monitoring, and diagnostic capability.
 1. Keypad: In addition to required programming and control keys, include keys for HAND, OFF, and AUTO modes.
 2. Security Access: Provide electronic security access to controls through identification and password with at least one level of access: View only; view and operate; and view, operate, and service.
 - a. Control Authority: Supports at least four conditions: Off, local manual control at VFC, local automatic control at VFC, and automatic control through a remote source.
- C. Historical Logging Information and Displays:
 1. Real-time clock with current time and date.
 2. Running log of total power versus time.
 3. Total run time.
 4. Fault log, maintaining last four faults with time and date stamp for each.
- D. Indicating Devices: Digital display and additional readout devices as required, mounted flush in VFC door and connected to display VFC parameters including, but not limited to:
 1. Output frequency (Hz).
 2. Motor speed (rpm).
 3. Motor status (running, stop, fault).
 4. Motor current (amperes).
 5. Motor torque (percent).
 6. Fault or alarming status (code).
 7. PID feedback signal (percent).
 8. DC-link voltage (V dc).
 9. Set point frequency (Hz).
 10. Motor output voltage (V ac).

E. Control Signal Interfaces:

1. Electric Input Signal Interface:

- a. A minimum of two programmable analog inputs: 0- to 10-V dc.
- b. A minimum of six multifunction programmable digital inputs.

2. Remote Signal Inputs: Capability to accept any of the following speed-setting input signals from the DDC system for HVAC or other control systems:

- a. 0- to 10-V dc.
- b. 4- to 20-mA dc.
- c. Potentiometer using up/down digital inputs.
- d. Fixed frequencies using digital inputs.

3. Output Signal Interface: A minimum of one programmable analog output signal(s) (0- to 10-V dc), which can be configured for any of the following:

- a. Output frequency (Hz).
- b. Output current (load).
- c. DC-link voltage (V dc).
- d. Motor torque (percent).
- e. Motor speed (rpm).
- f. Set point frequency (Hz).

F. PID Control Interface: Provides closed-loop set point, differential feedback control in response to dual feedback signals. Allows for closed-loop control of fans and pumps for pressure, flow, or temperature regulation.

1. Number of Loops: One.

2.5 BYPASS SYSTEMS

A. Bypass Operation: Manually transfers motor between power converter output and bypass circuit. Unit is capable of stable operation (starting, stopping, and running) with motor completely disconnected from power converter.

B. Bypass Mode: Manual operation only; requires local operator selection at VFC. Transfer between power converter and bypass contactor, and retransfer shall only be allowed with the motor at zero speed.

C. Bypass Controller: Two-contactor-style bypass allows motor operation via the power converter or the bypass controller; with input isolating switch and barrier arranged to isolate the power converter and permit safe troubleshooting and testing, both energized and de-energized, while motor is operating in bypass mode.

- 1. Bypass Contactor: Load-break, IEC-rated contactor.
- 2. Output Isolating Contactor: Non-load-break, IEC-rated contactor.
- 3. Isolating Switch: Non-load-break switch arranged to isolate power converter and permit safe troubleshooting and testing of the power converter, both

energized and de-energized, while motor is operating in bypass mode; pad-lockable, door-mounted handle mechanism.

D. Bypass Contactor Configuration: Full-voltage (across-the-line) type.

1. NORMAL/BYPASS selector switch.
2. HAND/OFF/AUTO selector switch.
3. NORMAL/TEST Selector Switch: Allows testing and adjusting of VFC while the motor is running in the bypass mode.
4. Contactor Coils: Pressure-encapsulated type with coil transient suppressors.
 - a. Operating Voltage: Depending on contactor NEMA size and line-voltage rating, manufacturer's standard matching control power or line voltage.
 - b. Power Contacts: Totally enclosed, double break, and silver-cadmium oxide; assembled to allow inspection and replacement without disturbing line or load wiring.
5. Control Circuits: 120-V ac; obtained from integral CPT, with primary and secondary fuses, with of sufficient capacity to operate all integral devices and remotely located pilot, indicating, and control devices.
 - a. CPT Spare Capacity: 100 VA.
6. Overload Relays: NEMA ICS 2.

2.6 OPTIONAL FEATURES

- A. Damper control circuit with end-of-travel feedback capability.
- B. Communication Port: RS-232 port, USB 2.0 port, or equivalent connection capable of connecting a printer and a notebook computer.

2.7 ENCLOSURES

- A. VFC Enclosures: NEMA 250, to comply with environmental conditions at installed location.
 1. Dry and Clean Indoor Locations: Type 1.
 2. Outdoor Locations: Type 3R.
 3. Kitchen Wash-Down Areas: Type 4X, stainless steel.
 4. Other Wet or Damp Indoor Locations: Type 3R.
- B. Plenum Rating: UL 1995; NRTL certification label on enclosure, clearly identifying VFC as "Plenum Rated."

2.8 ACCESSORIES

- A. General Requirements for Control-Circuit and Pilot Devices: NEMA ICS 5; factory installed in VFC enclosure cover unless otherwise indicated.

1. Push Buttons: Covered.
 2. Pilot Lights: Push to test.
 3. Selector Switches: Rotary type.
- B. NC bypass contactor auxiliary contact(s).
- C. Control Relays: Auxiliary and adjustable solid-state time-delay relays.
- D. Phase-Failure, Phase-Reversal, and Undervoltage and Overvoltage Relays: Solid-state sensing circuit with isolated output contacts for hard-wired connections. Provide adjustable undervoltage, overvoltage, and time-delay settings.
1. Current Transformers: Continuous current rating, basic impulse insulating level (BIL) rating, burden, and accuracy class suitable for connected circuitry. Comply with IEEE C57.13.
- E. Supplemental Digital Meters:
1. Elapsed-time meter.
 2. Kilowatt meter.
 3. Kilowatt-hour meter.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Wall-Mounting Controllers: Install with tops at uniform height and with disconnect operating handles not higher than 79 inches above finished floor, unless otherwise indicated, and by bolting units to wall or mounting on lightweight structural-steel channels bolted to wall. For controllers not on walls, provide freestanding racks complying with Section 260529 "Hangers and Supports for Electrical Systems."
- B. Roof-Mounting Controllers: Install VFC on roofs with tops at uniform height and with disconnect operating handles not higher than 79 inches above finished roof surface unless otherwise indicated, and by bolting units to curbs or mounting on freestanding, lightweight, structural-steel channels bolted to curbs. Seal roof penetrations after raceways are installed.
1. Curbs and roof penetrations are specified in Section 077200 "Roof Accessories."
 2. Structural-steel channels are specified in Section 260529 "Hangers and Supports for Electrical Systems."
- C. Seismic Bracing: Comply with requirements specified in Section 260548.16 "Seismic Controls for Electrical Systems."
- D. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.
- E. Install fuses in each fusible-switch VFC.

- F. Install fuses in control circuits if not factory installed. Comply with requirements in Section 262813 "Fuses."
- G. Install heaters in thermal-overload relays. Select heaters based on actual nameplate full-load amperes after motors are installed.
- H. Install, connect, and fuse thermal-protector monitoring relays furnished with motor-driven equipment.
- I. Comply with NECA 1.

3.2 CONTROL WIRING INSTALLATION

- A. Install wiring between VFCs and remote devices and facility's central-control system.
- B. Bundle, train, and support wiring in enclosures.

3.3 IDENTIFICATION

- A. Identify VFCs, components, and control wiring. Comply with requirements for identification specified in Section 260553 "Identification for Electrical Systems."
 - 1. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs.
 - 2. Label each VFC with engraved nameplate.
 - 3. Label each enclosure-mounted control and pilot device.

3.4 FIELD QUALITY CONTROL

- A. Perform tests and inspections.
- B. Acceptance Testing Preparation:
 - 1. Test insulation resistance for each VFC element, bus, component, connecting supply, feeder, and control circuit.
 - 2. Test continuity of each circuit.
- C. Tests and Inspections:
 - 1. Inspect VFC, wiring, components, connections, and equipment installation. Test and adjust controllers, components, and equipment.
 - 2. Test insulation resistance for each VFC element, component, connecting motor supply, feeder, and control circuits.
 - 3. Test continuity of each circuit.
 - 4. Verify that voltages at VFC locations are within 10 percent of motor nameplate rated voltages. If outside this range for any motor, notify Owner before starting the motor(s).
 - 5. Test each motor for proper phase rotation.

6. Perform tests according to the Inspection and Test Procedures for Adjustable Speed Drives stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
 7. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
 8. Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.
- D. VFCs will be considered defective if they do not pass tests and inspections.
- E. Prepare test and inspection reports, including a certified report that identifies the VFC and describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations made after remedial action.

3.5 ADJUSTING

- A. Program microprocessors for required operational sequences, status indications, alarms, event recording, and display features. Clear events memory after final acceptance testing and prior to Substantial Completion.
- B. Set field-adjustable switches, auxiliary relays, time-delay relays, timers, and overload-relay pickup and trip ranges.
- C. Adjust the trip settings of instantaneous-only circuit breakers and thermal-magnetic circuit breakers with adjustable, instantaneous trip elements. Initially adjust to 6 times the motor nameplate full-load amperes and attempt to start motors several times, allowing for motor cool-down between starts. If tripping occurs on motor inrush, adjust settings in increments until motors start without tripping. Do not exceed 8 times the motor full-load amperes (or 11 times for NEMA Premium Efficient motors if required). Where these maximum settings do not allow starting of a motor, notify Owner before increasing settings.
- D. Set the taps on reduced-voltage autotransformer controllers.
- E. Set field-adjustable circuit-breaker trip ranges
- F. Set field-adjustable pressure switches.

3.6 DEMONSTRATION

- A. Train Owner's maintenance personnel to adjust, operate, reprogram, and maintain VFCs.

END OF SECTION 262923

SECTION 263353 - STATIC UNINTERRUPTIBLE POWER SUPPLY**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes three-phase, on-line, double-conversion, static-type, UPS installations complete with transient voltage surge suppression, input harmonics reduction, rectifier-charger, battery, battery disconnect device, inverter, static bypass transfer switch, and internal maintenance bypass/isolation switch.

1.2 DEFINITIONS

- A. EMI: Electromagnetic interference.
- B. LCD: Liquid-crystal display.
- C. LED: Light-emitting diode.
- D. THD: Total harmonic distortion.
- E. UPS: Uninterruptible power supply.

1.3 SUBMITTALS

- A. Product Data: Include data on features, components, ratings, and performance for each UPS component indicated.
- B. Shop Drawings: Detail assemblies of equipment indicating dimensions, weights, components, heat dissipation and location and identification of each field connection. Show access, workspace, and clearance requirements; details of control panels; and battery arrangement.
 - 1. Power, signal, and control wiring diagrams.
 - 2. Detailed layouts of customer power and control connections.
 - 3. Detailed installation drawings including all terminal locations.
- C. Qualification Data: For Installer and power quality specialist.
- D. Manufacturer Certificates: For each product, signed by manufacturers.
- E. Factory Test Reports: Comply with specified requirements.
- F. Field Quality-Control and Performance Test Reports: Indicate test results compared with specified performance requirements, and provide justification and resolution of differences if values do not agree.
- G. Operation and Maintenance Data: For UPS units to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 01 Section "Operation and Maintenance Data", include the following:
 - 1. Lists of spare parts and replacement components recommended being stored at Project site for ready access.
 - 2. Intended operation narrative with detailed operating instructions covering operation under both normal and abnormal conditions.

3. Tools Required
4. Recommended maintenance practices
5. Manufacturer service department contact information
6. Submittal Data

H. Warranties: Provide complete manufacturers warranty information on all products provided.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications: Manufacturer's authorized representative who is trained and approved for both installation and maintenance of units required for this Project.
- B. Power Quality Specialist Qualifications: A factory-trained registered professional electrical engineer or engineering technician, currently certified by the National Institute for Certification in Engineering Technologies, NICET Level 4, minimum, experienced in performance testing UPS installations and in performing power quality surveys similar to that required in Part 3 "Performance Testing" Article.
- C. Manufacturer Qualifications: A minimum of twenty years experience in the design, manufacture, and testing of solid-state UPS systems is required. The system shall be designed and manufactured according to world class quality standards. The manufacturer shall be ISO 9001 certified. Maintain, within 50 miles of Project site, a service center capable of providing training, parts, and emergency maintenance repairs with fours hours maximum on-site service response time.
- D. Manufacturer Service Qualifications:
1. The UPS manufacturer shall directly employ a nationwide service organization. The service personal shall be the manufacturers employees and consisting of factory trained field service personnel dedicated to the start-up, maintenance, and repair of UPS and power equipment. Third party or non-manufacturer employees are not allowed. The organization shall consist of regional and local manufacturer offices. A minimum of 5 dedicated manufacturer field service employees trained in three phase UPS service shall be based within 50 miles of the site.
 2. The manufacturer shall provide a fully automated national dispatch center to coordinate field service personnel schedules. One toll-free number shall reach a qualified support person 24hours/day, 7days/week, and 365 days/year. If emergency service is required, response time shall be 20 minutes or less with 4 hours on site service response.
 3. An automated procedure shall be in place to insure that the manufacturer is dedicating the appropriate technical support resources to match escalating customer needs.
 4. Parts shall be available through an extensive network to ensure around-the-clock parts availability throughout the country.
 5. Recommended spare parts shall be fully stocked by local field service personnel with back-up available from national parts center and the manufacturing location. The national parts center Customer Support Parts Coordinators shall be on-call 24 hours/day, 7 days/week, and 365 days/year for immediate parts availability. Parts from the national parts center shall be shipped within 4 hours on the next available flight out and delivered to the customer's site within 24 hours.
- E. Source Limitations: Obtain the UPS and associated components specified in this Section from a single manufacturer with responsibility for entire UPS installation.
- F. Electrical Components, Devices, and Accessories: Listed and labeled by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

- G. UL Compliance: Listed and labeled under UL 1778.
- H. Comply with NFPA 101 and local ordinances and amendments.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver equipment in fully enclosed vehicles after specified environmental conditions have been permanently established in spaces where equipment is to be placed.
- B. Store equipment in spaces with environments controlled within manufacturer's ambient temperature and humidity tolerances for non-operating equipment.

1.6 EXTRA MATERIALS

- A. Furnish extra materials described below that match products installed and that are packaged with protective covering for storage and identified with labels describing contents. Deliver extra materials to Owner.
 - 1. Fuses: One for every 10 of each type and rating, but no less than 1 of each.
 - 2. Cabinet Ventilation Filters: One complete set.

1.7 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.
- B. UPS Warranties: The UPS manufacturer shall warrant the UPS module against defects in materials and workmanship for 12-months after initial start-up or 18-months after ship date, whichever period expires first.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following manufacturers:
 - 1. UPS Module
 - a. Emerson Network Power model #GXT4 8/10 kVA with external battery cabinets
 - 2. Battery
 - a. C&D Dynasty
 - b. Enersys Data Safe
 - c. GNB Absolyte IIP

2.2 PERFORMANCE DESCRIPTION

- A. Automatic operation includes the following:

1. Normal Conditions: Supply the load with ac power flowing from the normal ac power input terminals, through the rectifier-charger and inverter, with the battery connected in parallel with the rectifier-charger output.
2. Abnormal Supply Conditions: If normal ac supply deviates from specified and adjustable voltage, voltage waveform, or frequency limits, the battery supplies energy to maintain constant, regulated inverter ac power output to the load without switching or disturbance.
3. If normal power fails, energy supplied by the battery through the inverter continues supply-regulated ac power to the load without switching or disturbance.
4. When power is restored at the normal supply terminals of the system, controls automatically synchronize the inverter with the external source before transferring the load. The rectifier-charger then supplies power to the load through the inverter and simultaneously recharges the battery.
5. If the battery becomes discharged and normal supply is available, the rectifier-charger charges the battery. On reaching full charge, the rectifier-charger automatically shifts to float-charge mode.
6. If any element of the UPS system fails and power is available at the normal supply terminals of the system, the static bypass transfer switch switches the load to the normal ac supply circuit without disturbance or interruption.
7. If a fault occurs in the system supplied by the UPS, and current flows in excess of the overload rating of the UPS system, the static bypass transfer switch operates to bypass the fault current to the normal ac supply circuit for fault clearing.
8. When the fault has cleared, the static bypass transfer switch returns the load to the UPS system.
9. If the battery is disconnected, the UPS continues to supply power to the load with no degradation of its regulation of voltage and frequency of the output bus.
10. Provide Modbus interface card for Building Automation monitoring and alarm.

B. Manual operation includes the following:

1. Turning the inverter off causes the static bypass transfer switch to transfer the load directly to the bypass source without disturbance or interruption.
2. Turning the inverter on causes the static bypass transfer switch to transfer the load to the inverter.

C. Maintenance Bypass/Isolation Switch Operation: Switch is interlocked so it cannot be operated unless the static bypass transfer switch is in the bypass mode. Device provides manual selection between the following three conditions without interrupting supply to the load during switching:

1. Full Isolation: Load is supplied, bypassing the UPS. Normal UPS ac input circuit, static bypass transfer switch, and UPS load terminals are completely disconnected from external circuits.
2. Maintenance Bypass: Load is supplied, bypassing the UPS. UPS ac supply terminals are energized to permit operational checking, but system load terminals are isolated from the load.
3. Normal: Normal UPS ac supply terminals are energized and the load is supplied through either the static bypass transfer switch and the UPS rectifier-charger and inverter, or the battery and the inverter.

2.3 SERVICE CONDITIONS

- A. Environmental Conditions: The UPS shall be capable of operating continuously in the following environmental conditions without mechanical or electrical damage or degradation of operating capability, except battery performance.

1. Operating Ambient Temperature for Electronic Components: 32 to 104 deg F.
2. Storage/Transport Ambient Temperature for Electronic Components: -4 to 158 deg F.
3. Operating Ambient Temperature for Battery: 77 +/- 9 deg F.
4. Storage/Transport Ambient Temperature for Battery: -4 to 92 deg F.
5. Relative Humidity: 0 to 95 percent, noncondensing.
6. Operating Altitude: Mean Sea level to 6,600 feet. Derated for higher altitude applications.
7. Storage/Transport Altitude: Mean Sea level to 40,000 feet.

2.4 SYSTEM DESIGN REQUIREMENTS

- A. UPS Voltage: Input/Output and bypass voltage of the UPS shall be as shown on the drawings.
- B. UPS Output Load Capacity: Specified output load capacity of the UPS shall be as shown on the drawings and shall have a 0.8 lagging power factor.
- C. Battery Cells: Sealed, lead-acid, valve-regulated.
- D. Battery Reserve Time: The reserve time shall be as shown on the drawings and based on full load, 0.8 power factor, with ambient temperature between 68 to 86 deg F.
- E. Battery Recharge Time: Recharge to 95% capacity within ten (10) times discharge time.
- F. Unbalanced Load: The percentage of continuously unbalanced load is as shown on the drawings.

2.5 PERFORMANCE REQUIREMENTS

- A. The UPS shall perform as specified in this Article while supplying rated full-load current, composed of any combination of linear and nonlinear load, up to 100 percent nonlinear load with a load crest factor of 3.0, under the following conditions or combinations of the following conditions:
 1. Inverter is switched to battery source.
 2. Steady-state ac input voltage deviates up to plus 10 or minus 20 percent from nominal voltage.
 3. Steady-state input frequency deviates up to plus or minus 5 percent from nominal frequency.
 4. THD of input voltage is 15 percent or more with a minimum crest factor of 3.0, and the largest single harmonic component is a minimum of 5 percent of the fundamental value.

- B. Overall UPS Efficiency: Equal to or greater than the efficiencies in the following table:

NOMINAL OVERALL UPS EFFICIENCY RATINGS			
SIZE RANGE OF UPS UNITS	EFFICIENCY AT 100% RATED LOAD	EFFICIENCY AT 75% RATED LOAD	EFFICIENCY AT 50% RATED LOAD
30 kVA and Smaller	86	85	84

- C. Maximum Acoustical Noise: Noise generated by the UPS under any condition of normal operation shall not exceed 65 dBA measured 3 feet from the nearest surface of the UPS.
- D. Maximum Energizing Inrush Current: Eight times the full-load current.

- E. Maximum AC Output-Voltage Regulation:
1. Plus or minus 0.5 percent three-phase RMS average for a balanced three-phase load for the combined variation effects of input voltage, connected load, battery voltage, ambient temperature, and load power factor.
 2. Plus or minus 1.0 percent three-phase RMS average for a 100% unbalanced load for the combined variation effects of input voltage, connected load, battery voltage, ambient temperature, and load power factor.
- F. Output Frequency: Nominal frequency plus or minus 0.1 percent over the full range of input voltage, load, and battery voltage.
- G. Limitation of harmonic distortion of input current to the UPS shall be as follows:
1. Description: A tuned harmonic filter shall limit THD to 10 percent, maximum, at rated full UPS load current.
- H. Maximum AC Output Voltage Distortion:
1. 1 percent total harmonic distortion (THD) for linear loads.
 2. 2.5 percent THD for 100 percent nonlinear loads (3:1) crest factor) without kVA/kW derating.
- I. Minimum Overload Capacity of UPS at Rated Voltage in all operating modes:
1. 125 percent of rated full load for 10 minutes
 2. 150 percent for one minute
 3. 200 percent for 10 cycles, pulse paralleling with the static switch
- J. Maximum Output-Voltage Transient Excursions from Rated Value: For the following instantaneous load changes, stated as percentages of rated full UPS load, voltage shall remain within stated percentages of rated value and recover to, and remain within, plus or minus 2 percent of that value within 100 ms:
1. 50 Percent: Plus or minus 5 percent.
 2. 100 Percent: Plus or minus 2.5 percent.
 3. Loss of AC Input Power: Plus or minus 1 percent.
 4. Restoration of AC Input Power: Plus or minus 1 percent.
 5. Manual transfer of 100% load: Plus or minus 3 percent
- K. Input Power Factor: Between 0.90 – 0.96 lagging minimum from full load to half load with nominal input voltage.
- L. Input Current Limit:
1. 115 percent of nominal AC input current maximum
 2. 100 percent of nominal for optimal generator operation
- M. Input Current Walk-In: 15 seconds to full rated input current maximum. Field selectable 5 through 15 seconds.
- N. Output Frequency Slew Rate: 5.0 Hertz per second maximum. Field selectable from 0.1 to 5.0 Hz per second.

- O. Output Phase Displacement:
 - 1. Plus or minus 0.5 degrees for balanced load
 - 2. Plus or minus 1.0 degrees for 100% unbalanced load
- P. Output Bypass Line Sync Range:
 - 1. Plus or minus 0.5 Hertz
 - 2. Field selectable plus or minus 0.5 to 5.0 Hertz
- Q. Output Power Rating: Rated kVA at 0.8 lagging power factor
- R. Inverter Output Voltage Adjustment: Plus or minus 5% manual adjustment
- S. Load Power Factor Range: 1.0 to 0.7 lagging without derating
- T. Transient Recovery Time: to within 1% of output voltage within one cycle.
- U. AC Output Voltage Unbalance: 100% unbalanced load plus or minus 1 percent
- V. Fault Clearing: Sub-cycle current of at least 300 percent
- W. EMI Emissions: Comply with FCC Rules and Regulations, and with 47 CFR 15 for Class A equipment.

2.6 UPS SYSTEMS

- A. General:
 - 1. All materials of the UPS shall be new, of current manufacture, high grade and free from all defects and shall not have been in prior service except as required during factory testing.
 - 2. The maximum working voltage, current, and di/dt of all solid-state power components and electronic devices shall not exceed 75% of the ratings established by their manufacturer.
 - 3. The operating temperature of solid-state component sub-assembly shall not be greater than 75% of their ratings.
 - 4. Electrolytic capacitors shall be computer grade and be operated at no more than 95% of their voltage rating at the maximum rectifier charging voltage.
 - 5. Wiring practices, materials and coding shall be in accordance with the requirements of the local authority having jurisdiction.
 - 6. All bolted connections of bus bars, lugs and cables shall be in accordance with requirements of the local authority having jurisdiction.
 - 7. All electrical power connections are to be torqued to the required value and marked with a visual indicator.
 - 8. Provision shall be made for power cables to enter or leave from the top or bottom of the UPS cabinet.
- B. Electronic Equipment: Solid-state devices using hermetically sealed, semiconductor elements. Devices include rectifier-charger, inverter, static bypass transfer switch, and system controls.
- C. Enclosures:
 - 1. The UPS unit, comprised of input transformer (if required), rectifier/charger with input filter, inverter, static transfer switch output transformer and maintenance bypass switch,

- shall be housed in a single free-standing NEMA Type 1 enclosure, unless otherwise indicated.
 - 2. Cabinet doors/covers shall require a tool for gaining access.
 - 3. Casters and stops shall be provided for ease of installation.
 - 4. Front access only shall be required for expedient servicing, adjustments, and installation.
 - 5. The UPS cabinet shall be structurally adequate and have provisions for hoisting, jacking, and forklift handling.
 - 6. The UPS cabinet shall be cleaned, primed, and painted with the manufacturer's standard color.
 - 7. The UPS shall be constructed of replaceable subassemblies. Like assemblies and like components shall be interchangeable.
- D. Control Assemblies: Mount on modular plug-ins, readily accessible for maintenance.
- E. Surge Protection: Sustains input surges without damage per criteria listed in ANSI C62.41 Category A and B.
- F. Grounding: The AC output neutral shall be electrically isolated from the UPS chassis. The UPS chassis shall have an equipment ground terminal. Provisions for local bonding shall be provided.
- G. Input Transformer: Where required, the input transformer shall be factory installed inside the UPS module cabinet without increasing the standard footprint.

2.7 RECTIFIER-CHARGER

- A. General: The rectifier/charger shall be a phase-controlled, solid-state SCR type with constant voltage/current limiting control circuitry.
- B. Capacity: Adequate to supply the inverter during rated full output load conditions and simultaneously recharge the battery from fully discharged condition to 95 percent of full charge within 10 times the rated discharge time for duration of supply under battery power at full load.
- C. AC Input Current Limiting: The rectifier/charger unit shall be provided with AC input current limiting whereby the maximum input current shall be limited to 115 percent of the full input current rating. The rectifier/charger shall operate at a reduced current limit mode whenever the critical load is powered from the UPS static bypass circuit such that the maximum UPS input current will not exceed 115 percent of full load input current. In addition, the rectifier/charger shall have a separate battery current limit, adjustable from 0 to 15 percent of the full load input current. An optional second circuit shall limit the battery recharge current to zero when activated by a customer-supplied contact closure to signal a customer function such as generator operation.
- D. Input Current Walk-In: The rectifier/charger shall contain a timed walk-in circuit that causes the unit to gradually assume the load over a 15-second time interval after input voltage is applied. Walk-in time shall be field selectable for 5 through 20 seconds. If applicable, coordinate this time for maximum compatibility with local generator-set power source.
- E. Output Ripple: The rectifier/charger shall have an output filter to minimize ripple voltage into the battery. Under no conditions shall ripple voltage into the battery exceed 1% RMS. The filter shall be adequate to insure that the DC output of the rectifier/charger will meet the input requirements of the inverter. The inverter shall be able to operate from the rectifier/charger with the battery disconnected.

- F. Fuse Failure Protection: Power semiconductors in the rectifier/charger shall be fused with fast-acting fuses, so that loss of any one power semiconductor shall not cause cascading failures.
- G. Rectifier-Charger Control Circuits: Immune to frequency variations within rated frequency ranges of normal and emergency power sources.
- H. Battery Float-Charging Conditions: Comply with battery manufacturer's written instructions for battery terminal voltage and charging current required for maximum battery life.
- I. Automatic Rectifier Restart: Upon restoration of utility AC power, after a utility AC power outage and prior to a UPS automatic end-of-discharge shutdown, the rectifier/charger shall automatically restart, walk-in, and gradually assume the inverter and battery recharge loads.
- J. Over-voltage Protection: There shall be DC over-voltage protection so that if the DC voltage rises to the pre-set limit, the UPS is to shut down automatically and initiate an uninterrupted load transfer to the static bypass line.
- K. Input Filter: The rectifier/charger shall include an input filter to reduce reflected input current distortion to 10% THD at full load with nominal input voltage. The input filter shall maintain the input power factor at 0.90 to 0.96 lagging minimum from full load to half load with nominal input voltage. The input filter shall have a disconnect that automatically disconnects the filter under low load conditions.

2.8 INVERTER

- A. Description: The term inverter shall denote the solid-state equipment and controls to convert DC power from the rectifier/charger or battery to regulated AC power for supporting the critical load. The inverter shall use Insulated Gate Bipolar Transistors (IGBTs) in a phase-controlled, pulse width modulated (PWM) design capable of providing the specified AC output.
- B. Overload Capability: The inverter shall be capable of supplying current and voltage for overloads exceeding 100 percent and up to 200 percent of full load current. A status indicator and audible alarm shall indicate overload operation. The UPS shall transfer the load to bypass when overload capacity is exceeded.
- C. Fault Clearing and Current Limit: The inverter shall be capable of supplying an overload current of 150 percent of its full-load rating for one minute. For greater currents or longer time duration, the inverter shall have electronic current-limiting protection to prevent damage to components. The critical load will be transferred to the static bypass automatically and uninterrupted. The inverter shall be self-protecting against any magnitude of connected output overload. Inverter control logic shall sense and disconnect the inverter from the critical AC load without the requirement to clear protective fuses.
- D. Step Load Response: The output voltage shall be maintained to within plus or minus 2.5 percent with a 0-to-100 percent step load change or a 100-to-0 percent load change. The output voltage shall recover to within 1 percent of nominal voltage within 1 cycle.
- E. Voltage Distortion: For linear loads, the output voltage total harmonic distortion (THD) shall not be greater than 1 percent. For 100 percent rated load of 3:1 crest factor nonlinear loads, the output voltage total harmonic distortion shall not be greater than 2.5 percent. The output rating is not to be derated in kVA or kW due to the 100 percent nonlinear load with 3:1 crest factor.
- F. Output Power Transformer: Where necessary, a dry-type power transformer shall be provided for the inverter AC output. It shall have copper wiring exclusively. The transformers hottest

spot winding temperature shall not exceed the temperature limit of the transformer insulation class of material when operating at full load at maximum ambient temperature.

- G. Phase Balance: Electronic controls shall be provided to regulate each phase so that an unbalanced loading will not cause the output voltage to go outside the specified voltage unbalance or phase displacement. With 100 percent load on one phase and 0 percent load on the other 2 phases or 100 percent load on 2 phases and 0 percent load on the other phase, the voltage balance is to be within 1 percent and the phase displacement is to be 120 degrees within plus or minus 1 degree.
- H. Fuse Failure Protection: Power semiconductors in the inverter unit shall be fused with fast-acting fuses, so that loss of any one power semiconductor will not cause cascading failures.
- I. Inverter Shutdown: For rapid removal of the inverter from the critical load, the inverter control electronics shall instantaneously turn off the inverter transistors. Simultaneously, the static transfer switch shall be turned on to maintain continuous power to the critical load.
- J. Inverter DC Protection: The inverter shall be protected by the following disconnect levels:
 - 1. DC Over-voltage Shutdown
 - 2. DC Undervoltage Warning (Low Battery Reserve), user adjustable from 1 to 99 minutes
 - 3. DC Undervoltage Shutdown (End of Discharge)
- K. Overdischarge Protection: To prevent battery damage from overdischarging, the UPS control logic shall automatically raise the shutdown voltage set point as discharge time increases beyond fifteen (15) minutes.
- L. Inverter Output Voltage Adjustment: The inverter shall use a software control to adjust the output voltage from plus or minus 5 percent of the nominal value.
- M. Output Frequency: The output frequency of the inverter shall be controlled by an oscillator. The oscillator shall be temperature compensated and hold the inverter output frequency to plus or minus 0.1 percent for steady state and transient conditions. Drift shall not exceed 0.1 percent during a 24-hour period. Total frequency deviation, including short time fluctuation and drift, shall not exceed 0.1 percent from the rated frequency.

2.9 STATIC BYPASS TRANSFER SWITCH

- A. General:
 - 1. A static transfer switch and bypass circuit shall be provided as an integral part of the UPS. The static switch shall be a naturally commutated high-speed static (SCR-type) device rated to conduct full load current continuously, The switch shall have an overload rating of 110 percent rated load continuously, 200 percent rated load for five seconds. The static transfer switch shall also have fault-clearing capabilities of 1100 amperes for 1 second, 3000 amperes for 10 cycles, and 6000 amperes peak for the first half cycle.
 - 2. The static transfer switch control logic shall contain an automatic transfer control circuit that senses the status of the inverter logic signals, and operating and alarm conditions. This control circuit shall provide an uninterrupted transfer of the load to an alternate bypass source, without exceeding the transient limits specified herein, when an overload or malfunction occurs within the UPS, or for bypassing the UPS for maintenance.
- B. Uninterrupted Transfer:

1. The transfer control logic shall automatically turn on the static transfer switch, transferring the critical AC load to the bypass source, after the transfer logic senses any of the following conditions:
 - a. Inverter overload capacity exceeded
 - b. Critical AC load overvoltage or undervoltage
 - c. Battery protection period expired
 - d. UPS fault condition
 2. The transfer control logic shall inhibit an automatic transfer of the critical load to the bypass source if any of the following conditions are present:
 - a. Inverter/bypass voltage difference exceeding preset limits
 - b. Bypass frequency out of limits
 - c. Bypass out-of-synchronization range with inverter output
- C. Uninterrupted Retransfer: Retransfer of the critical AC load from the bypass source to the inverter output shall be automatically initiated unless inhibited by manual control. The transfer control logic shall inhibit an automatic retransfer of the critical load to the inverter if one of the following conditions exists:
1. Bypass out of synchronization range with inverter output
 2. Inverter/bypass voltage difference exceeding preset limits
 3. Overload condition exists in excess of inverter full load rating
 4. UPS fault condition present

2.10 BATTERY

A. Description:

1. The battery power pack shall include sealed, lead-acid valve regulated battery cells housed in a separate cabinet that matches the UPS cabinet styling to form an integral system line-up.
2. Battery cells shall be mounted on slide-out trays for ease of maintenance.
3. A battery disconnect circuit breaker with undervoltage release (UVR) shall be included for isolation of the battery pack from the UPS module. The UPS shall automatically be disconnected from the battery by opening the breaker when the battery reaches the minimum discharge voltage level.
4. Casters and leveling feet shall also be provided with the battery power pack cabinet for ease of installation.
5. When the application calls for the battery cabinet to be bolted to the UPS cabinet, the interconnecting cables are to be provided, pre-cut and pre-lugged.

2.11 CONTROLS AND INDICATIONS

- A. Description: The UPS shall be provided with a microprocessor based unit status display and controls section designed for convenient and reliable user operation. A graphical display shall be used to show a single-line diagram of the UPS, and shall be provided as part of the monitoring and controls sections of the UPS. All of the operator controls and monitors shall be located on the front of the UPS cabinet. The monitoring functions such as metering, status and alarms shall be displayed on the graphical LCD display. Additional features of the monitoring system shall include:

1. Menu-driven display with pushbutton navigation

2. Real time clock (time and date)
 3. Alarm history with time and date stamp
 4. Battery backed-up memory
- B. Minimum displays, indicating devices, and controls include those in lists below. Provide sensors, transducers, terminals, relays, and wiring required to support listed items.
- C. Metering: The following parameters shall be displayed:
1. Input AC voltage for each phase, line-to-line
 2. Input AC current for each phase
 3. Input frequency
 4. Battery voltage
 5. Battery current and direction (charge/discharge)
 6. DC bus voltage
 7. Output AC voltage line-to-line and line-to-neutral for each phase
 8. Output AC current for each phase
 9. Output frequency
 10. Percent of rated load being supplied by the UPS
 11. Battery time left during battery operation
- D. Status Messages: The following UPS status messages shall be displayed:
1. Normal operation
 2. On Static Bypass Transfer Switch
 3. Load on UPS
 4. Load on bypass
 5. User Shutdown
 6. Battery Discharging
- E. Alarm Messages: The following alarm messages shall be displayed and an audible alarm shall be provided and activated by any of the following alarm conditions:
1. Input Line Fault
 2. Input Phase Rotation Error
 3. Input Over/Under Frequency
 4. Input Current Limit
 5. Rectifier Fail
 6. Battery Test Failed
 7. Battery Low Warning (Adjustable 1 to 99 Minutes)
 8. Battery Low Transfer
 9. DC Overvoltage Steady State
 10. Bypass Frequency Error
 11. Load On Bypass
 12. Excessive Auto Retransfers
 13. Static Bypass Transfer Switch SCR Shorted
 14. Bypass Sync Error
 15. Input Phase Loss
 16. DC Current Peak
 17. Output Undervoltage Transfer
 18. Output Overvoltage Transfer
 19. Inverter Overload
 20. Static Bypass Transfer Switch Overload
 21. Inverter Overload Transfer
 22. Transfer Failed Shutdown

23. Hardware Shutdown
24. Output Power Supply Fail
25. Inverter Control Fault Transfer
26. System Fan Fail
27. Ambient Overtemperature Limit
28. Overtemp Timeout Shutdown

F. Controls: UPS start-up, shutdown, and bypass operations shall be accomplished through the front-panel pushbutton controls. Menu-driven user prompts shall be provided to guide the operator through system operation without the use of additional manuals. Pushbuttons shall be provided to display the status of the UPS and to test and reset visual and audible alarms. A mimic screen shall be available on the LCD screen to depict a single-line diagram of the UPS, with switch positions and power flow. Pushbuttons shall include the following:

1. Inverter on-off.
2. UPS start.
3. Battery test.
4. Alarm silence/reset.
5. Output-voltage adjustment.

G. Programmable Relay Board: Eight sets of isolated Form C contacts shall be provided to indicate a change of status of any of the alarm conditions. Any of the UPS alarms can be programmed onto any channel of the programmable relay board.

H. On-Line Battery Test:

1. The UPS shall be provided with a menu-driven On-Line Battery Test Feature. The test shall ensure the capability of the battery to supply power to the inverter while the load is supplied power in the normal mode. If the battery fails the test, the system shall automatically do the following:
 - a. Maintain the load through the UPS
 - b. Display a warning message
 - c. Sound an audible alarm
2. The battery test feature shall have the following user selectable options:
 - a. Interval between tests (2 to 9 weeks)
 - b. Date and time of initial test
 - c. Enable/disable test

I. Emergency Power Off Switch: Capable of local operation and operation by means of activation by external dry contacts.

2.12 MAINTENANCE BYPASS/ISOLATION SWITCH

A. Description: A manually operated maintenance bypass switch shall be incorporated into the UPS cabinet to directly connect the critical load to the bypass AC input power source, bypassing the rectifier/charger, inverter, and static transfer switch.

1. Switch shall be electrically and mechanically interlocked to prevent interrupting power to the load when switching to bypass mode.

2. Switch shall electrically isolate other UPS components to permit safe servicing. A means to de-energize the static switch shall be provided when the UPS is in the maintenance bypass mode of operation

B. Switch Rating: Continuous duty at rated full UPS load current.

C. Maintenance Capability: With the critical load powered from the maintenance bypass circuit, it shall be possible to check out the operation of the rectifier/charger, inverter, battery, and static transfer switch.

2.13 MONITORING BY REMOTE STATUS AND ALARM PANEL

A. Description: Labeled LEDs on panel faceplate indicate the following basic status conditions. Audible signal indicates alarm conditions. Silencing switch in face of panel silences signal without altering visual indication. The panel shall also be provided with a lamp test/reset pushbutton. The remote status panel shall be provided in a NEMA Type 1 enclosure for wall mounting.

1. Load on UPS LED
2. Load on Bypass LED
3. Battery Discharge LED
4. Low Battery Reserve LED
5. UPS Alarm Condition LED
6. New Alarm Condition LED (for a second UPS alarm condition)

2.14 BATTERY CIRCUIT BREAKER

A. A battery circuit breaker shall be provided to isolate the battery from the UPS. This breaker shall have an undervoltage release (UVR) and auxiliary contacts, and shall be in the battery cabinet. The battery breaker provides a manual disconnecting means, short circuit protection, and overcurrent protection for the battery system. When opened, there shall be no battery voltage in the UPS enclosure. The UPS shall be automatically disconnected from the battery by opening the breaker when the battery reaches the minimum discharge voltage level.

2.15 SOURCE QUALITY CONTROL

A. Factory test complete UPS system before shipment. Use simulated battery testing. Include the following:

1. Test and demonstration of all functions, controls, indicators, sensors, and protective devices.
2. Full-load test.
3. Transient-load response test.
4. Overload test.
5. Power failure test.

B. Observation of Test: Give 14 days' advance notice of tests and provide opportunity for Owner's representative to observe tests at Owner's option.

C. Report test results. Include the following data:

1. Description of input source and output loads used. Describe actions required to simulate source load variation and various operating conditions and malfunctions.

2. List of indications, parameter values, and system responses considered satisfactory for each test action. Include tabulation of actual observations during test.
3. List of instruments and equipment used in factory tests.

2.16 EXTERNAL MAINTENANCE BYPASS CABINET

- A. A matching external maintenance bypass cabinet shall be provided to enable the UPS module to be completely isolated from the electrical system while the critical load is powered through the external maintenance bypass line. This optional cabinet shall provide make-before-break operation for transfers to and from the external maintenance bypass line with a single rotary switch. This matching cabinet shall bolt to the side of the UPS module with a barrier shield to separate the two cabinets. Only front access shall be required for installation and service.
- B. The following components shall be standard:
 1. Single rotary switch with auxiliary contacts
 2. Inter-cabinet wiring
 3. Casters
 4. Leveling feet
- C. The following components shall be optional:
 1. Input circuit breaker
 2. Shielded isolation transformer
 3. Load (output) circuit breaker

2.17 INTERNAL MODEM

- A. The UPS shall come with an internal modem capable of dialing out from the UPS to notify up to two remote computers, terminals, PC's, or pocket pagers when important events occur. The modem will also be capable of accepting incoming calls, with the appropriate security, and connecting to a remote terminal, computer or PC, to perform all those functions normally available on the front panel including viewing monitoring screens.

2.18 SNMP

- A. The UPS shall come equipped with an internal Simple Network Management Protocol (SNMP) adapter, which will connect the UPS directly to any I.P. based network using Ethernet communications. The UPS will become a managed device on the network. From a network management station the system administrator shall be capable of monitoring important system measurements, alarm status and alarm history data. In the event of a utility failure the SNMP shall continue with live communication without the requirement of additional or separate UPS equipment until such time as the UPS shuts down for Low battery. On resumption of Utility power the SNMP shall resume full SNMP communication automatically.

2.19 OUTPUT ISOLATION TRANSFORMER

- A. Description: Unit with low forward transfer impedance up to 3 kHz, minimum. Include the following features:
 1. Comply with applicable portions of UL 1561, including requirements for nonlinear load current-handling capability for a K-factor as specified on the drawings.
 2. Output Impedance at Fundamental Frequency: Between 3 and 4 percent.
 3. Regulation: 5 percent, maximum, at rated nonlinear load current.

4. Full-Load Efficiency at Rated Nonlinear Load Current: 96 percent, minimum.
5. Electrostatic Shielding of Windings: Independent for each winding.
6. Coil Leads: Physically arranged for minimum interlead capacitance.
7. Shield Grounding Terminal: Separately mounted; labeled "Shield Ground."
8. Capacitive Coupling between Primary and Secondary: 33 picofarads, maximum, over a frequency range of 20 Hz to 1 MHz.

2.20 OUTPUT DISTRIBUTION SECTION

- A. Panelboards: Comply with Division 26 Section "Panelboards," except provide assembly integral to UPS cabinet.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Maintain minimum clearances and workspace at equipment according to manufacturer's written instructions and all Federal, State, and local codes and guidelines.
- B. Connections: Interconnect system components. Make connections to supply and load circuits according to manufacturer's wiring diagrams, unless otherwise indicated.

3.2 IDENTIFICATION

- A. Identify components and wiring according to Division 26 Section "Identification for Electrical Systems."

3.3 BATTERY EQUALIZATION

- A. Equalize charging of battery cells according to manufacturer's written instructions. Record individual-cell voltages.

3.4 FIELD QUALITY CONTROL

- A. The following inspections and test procedures shall be performed by factory-trained field service personnel during the UPS startup.

1. Visual Inspection

- a. Inspect equipment for signs of damage
- b. Verify installation per drawings
- c. Inspect cabinets for foreign objects
- d. Verify neutral and ground conductors are properly sized and configured
- e. Inspect battery cases
- f. Inspect battery for proper polarity
- g. Verify all printed circuit boards are configured properly

2. Mechanical Inspection

- a. Check all control wiring connections for tightness
- b. Check all power wiring connections for tightness
- c. Check all terminal screws, nuts, and/or spade lugs for tightness

3. Electrical Inspection

- a. Check all fuses for continuity
 - b. Confirm input voltage and phase rotation is correct
 - c. Verify control transformer connections are correct for voltages being used
 - d. Assure connection and voltage of the battery string(s)
- B. Load Bank Testing: A 4-hour full load, load bank test shall be performed at the site. The resistive load bank shall be supplied by the contractor at no additional cost.
- C. Retest: Correct deficiencies and retest until specified requirements are met.
- D. Record of Tests and Inspections: Maintain and submit documentation of tests and inspections, including references to manufacturers' written instructions and other test and inspection criteria. Include results of tests, inspections, and retests.

3.5 MAINTENANCE CONTRACTS

- A. A complete offering of preventative and full service maintenance contracts for both the UPS system and battery system shall be available. An extended warranty and preventative maintenance package shall be available. Warranty and preventative maintenance service shall be performed by factory-trained service personnel.

3.6 CLEANING

- A. The contractor shall remove all paint spatters and other spots, dirt and debris from the equipment. Clean equipment and devices internally and externally using methods and materials recommended by the manufacturer.

3.7 CONTRACTOR STARTUP AND REPORTING

- A. Contractor shall prepare and submit a complete set of record drawings, test results, operation and maintenance data and certificates as outlined in this Section.

3.8 COMMISSIONING AND DEMONSTRATION

- A. Engage a factory-trained field service personnel to perform field quality control tests as outlined in this Section.
- B. After system checkout and adjustment, the contractor shall operate the system for the review of the owner and architect. Necessary adjustments or modifications shall be made as required by the owner or architect.
- C. UPS manufacturer representative to train Owner's maintenance personnel to adjust, operate, and maintain the UPS.

END OF SECTION 263353

SECTION 263600 - TRANSFER SWITCHES**PART 1 - GENERAL****1.01 SUMMARY**

- A. Work included: Materials, installation and testing of:
 - 1. Open Transition Transfer Switch
 - 2. Microprocessor Controller
 - 3. Accessories
 - 4. Automatic Sequence of Operation

1.02 RELATED SECTIONS

- A. Contents of Division 26, Electrical, and Division 01, General Requirements apply to this Section.

1.03 REFERENCES AND STANDARDS

- A. References and Standards as required by Section 26 00 00, Electrical Basic Requirements and Division 01, General Requirements.
- B. In addition, meet the following:
 - 1. UL 1008, Automatic Transfer Switches.
 - 2. NECA 1 - Standard for Good Workmanship in Electrical Construction; National Electrical Contractors Association
 - 3. NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum); National Electrical Manufacturers Association
 - 4. NEMA ICS 10 Part 1 - Industrial Control and Systems Part 1: Electromechanical AC Transfer Switch Equipment; National Electrical Manufacturers Association
 - 5. NETA ATS - Acceptance Testing Specifications for Electrical Power Equipment and Systems; International Electrical Testing Association (ANSI/NETA ATS)
 - 6. NFPA 70 - National Electrical Code; National Fire Protection Association
 - 7. NFPA 110 - Standard for Emergency and Standby Power Systems; National Fire Protection Association

1.04 SUBMITTALS

- A. Submittals as required by Section 26 00 00, Electrical Basic Requirements and Division 01, General Requirements
- B. In addition, provide:
 - 1. Product Data: Provide catalog sheets showing voltage, switch size, ratings and size of switching and overcurrent protective devices, operating logic, withstand current ratings, dimensions, and enclosure details.
 - 2. Operation Data: Instructions for operating equipment under emergency conditions when engine generator is running.
 - 3. Maintenance Data: Routine preventative maintenance and lubrication schedule. List special tools, maintenance materials, and replacement parts.

1.05 QUALITY ASSURANCE

- A. Quality assurance as required by Section 26 00 00, Electrical Basic Requirements and Division 01, General Requirements.
- B. In addition, meet the following:
 - 1. NFPA 70, National Electrical Code
 - 2. NFPA 110, Standard for Emergency and Standby Power Systems
- C. Manufacturer Qualifications: Company specializing in manufacturing the products specified in this section with minimum three years documented experience.
- D. Authorized service facilities located within 200 miles (320 km) of project site.

- E. Installer Qualifications: Company specializing in performing the work of this section with minimum three years documented experience with power transfer systems of similar size, type, and complexity; manufacturer's authorized installer.
- F. Product Listing Organization Qualifications: An organization recognized by OSHA as a Nationally Recognized Testing Laboratory (NRTL) and acceptable to authorities having jurisdiction.
- G. Field Conditions: Maintain field conditions within manufacturer's required service conditions during and after installation

1.06 WARRANTY

- A. Warranty of materials and workmanship as required by Section 26 00 00, Electrical Basic Requirements and Division 01, General Requirements.
- B. Provide a comprehensive warranty that includes parts, labor, and travel to the site.

1.07 COORDINATION

- A. Coordinate the work with other trades to avoid placement of ductwork, piping, equipment, or other potential obstructions within the dedicated equipment spaces and working clearances required by NFPA 70.
- B. Coordinate arrangement of equipment with the dimensions and clearance requirements of the actual equipment to be installed.
- C. Coordinate the work with placement of supports, anchors, etc. required for mounting.
- D. Notify Architect of any conflicts with or deviations from the contract documents. Obtain direction before proceeding with work.

1.08 DELIVERY, STORAGE AND HANDLING

- A. Receive, inspect, handle, and store transfer switches in accordance with manufacturer's instructions.
- B. Store in a clean, dry space. Maintain factory wrapping or provide an additional heavy canvas or heavy plastic cover to protect units from dirt, water, construction debris, and traffic.
- C. Handle carefully in accordance with manufacturer's instructions to avoid damage to transfer switch components, enclosure, and finish.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

- A. Approved manufacturers listed below are allowed on condition of meeting the specified conditions including available space allocated for the equipment (including code required working clearances) and functionality of system as described in drawings and specifications. Remove and replace electrical equipment installed and not meeting these conditions at no cost to Owner.
 - 1. ASCO Power Technologies
 - 2. Caterpillar
 - 3. Russelectric
 - 4. GE Zenith
 - 5. Or approved equivalent.
- B. Basis of Design: Transfer switches on drawings are designed based on ASCO product line.
- C. Source Limitations: Furnish transfer switches and accessories produced by a single manufacturer and obtained from a single supplier.

2.02 OPEN TRANSITION TRANSFER SWITCH

- A. Provide open transition transfer switches where indicated on the drawings.
- B. Provide type, 4 pole with, amperage, voltage, withstand, and close-on ratings as indicated on drawings.

- C. Provide transfer switch rated for available fault with upstream overcurrent protection.
- D. Minimum 3-cycle (0.05 seconds) Withstand Current Rating (WCR). If required, provide 18-cycle (0.3 seconds) or 30-cycle (0.5 seconds) to accommodate selective coordination and available fault current at the installed location as indicated on the drawings.
- E. Provide NEMA Type 3R - Outdoor enclosure.
- F. Transfer switch electrically operated and mechanically held. Electrical operator momentarily energized, single-solenoid mechanism. Mechanically interlocked to ensure only two possible positions, normal or emergency.
- G. Use only one type of main operator for ease of maintenance and commonality of parts.
- H. Lugs to be mechanical or tool applied compression type.
- I. Equip transfer switch(es) with bus stubs for compression lug connectors. Size and quantity of lugs as required.
- J. Positively locked and unaffected by momentary outages, so that contact pressure is maintained at constant value and contact temperature rise is minimized for maximum reliability and operating life.
- K. Contact structure to consist of main current carrying contact, silver alloy with minimum 50 percent silver content. Contacts protected by silver tungsten arching contacts on sizes above 400 amps.
- L. Inspection of contacts to be possible from front of switch without disassembly of operating linkages and without disconnection of power conductors.
- M. Designs utilizing components of molded-case circuit breakers, contactors, or parts thereof, which are not intended for continuous duty, repetitive switching or transfer between two active power sources are not acceptable.
- N. Each switch to have time delay in neutral position. Time delay adjustable from 0-5 seconds and initially set at 2 seconds. Provide means to permit neutral delay transfer switches to be driven to neutral position and held there.
- O. Construction Type: Only "contactor type" (open contact) transfer switches are acceptable. Do not use "breaker type" (enclosed contact) transfer switches.
- Q. Switching Methods:
 - 1. Open Transition:
 - a. Provide break-before-make transfer, and with interlocks to prevent simultaneous connection of the load to both sources.
 - b. Where in-phase transfer is indicated, utilize in-phase monitor to initiate transfer when phase angle difference between sources is near zero to limit in-rush currents.
- R. Enclosures:
 - 1. Environment Type per NEMA 250: Unless otherwise indicated, as specified for the following installation locations:
 - a. Outdoor Locations: Type 3R or Type 4.
 - 2. Provide lockable door(s) for outdoor locations.
 - 3. Finish: Manufacturer's standard unless otherwise indicated.
- S. Short Circuit Current Rating:
 - 1. Withstand and Closing Rating: Provide transfer switches, when protected by the supply side overcurrent protective devices to be installed, with listed withstand and closing rating not less than the available fault current at the installed location as indicated on the distribution manufacturer final short circuit calculation, protective devices and arc flash hazard study.
 - 2. Short Time Rating: Minimum 3-cycle (0.05 seconds) Withstand Current Rating (WCR). If required, provide 18-cycle (0.3 seconds) or 30-cycle (0.5 seconds) to accommodate selective coordination and available fault current at the installed location as indicated on the drawings.

- T. Control Functions:
 - 1. Manual source selection.
 - 2. Outputs:
 - a. Auxiliary contacts; one set for each switch position.
- U. Status Indications:
 - 1. Connected to alternate/emergency source.
 - 2. Connected to primary/normal source.
 - 3. Alternate/emergency source available.
 - 4. Primary/normal source available.
- V. Remote Annunciators:
 - 1. Remote Annunciator Mounting: Wall-mounted; Provide flush-mounted annunciator for finished areas and surface-mounted annunciator for non-finished areas unless otherwise indicated.
 - 2. Transfer Switch Status Indications:
 - a. Connected to alternate/emergency source.
 - b. Connected to primary/normal source.
 - c. Alternate/emergency source available.
 - d. Primary/normal source available.
- W. Source Quality Control:
 - 1. See Division 01, Quality Requirements for additional requirements.
 - 2. Perform production tests on transfer switches at factory to verify operation and performance characteristics prior to shipment. Include certified test report with submittals.
- X. Basis of Design: ASCO Power Technologies Series 7000 Open Transition Automatic Transfer Switch.

2.03 MICROPROCESSOR CONTROLLER

- A. Proved sensing and logic by single built-in microprocessor for maximum reliability, minimum maintenance, and ability to communicate serially through an optional serial communication module.
- B. Single controller to provide twelve selectable nominal voltages for maximum application flexibility and minimal spare part requirements.
 - 1. Voltage sensing to be true RMS type, accurate to ± 1 percent of nominal voltage.
 - 2. Frequency sensing accurate to ± 0.2 percent.
- C. Connect controller to transfer switch by an interconnecting wiring harness. Include keyed disconnect plug to enable controller to be disconnected from transfer switch for routine maintenance. Provide sensing and control logic on multi-layer printed circuit boards. Interfacing relays to be industrial grade plug-in type with dust covers. Enclose panel with protective cover, mounted separately from transfer switch unit for safety and ease of maintenance. Protective cover to include built-in pocket for storage of operator's manuals.
- D. Customer connections to be wired to common terminal block to simplify field-wiring connections.
- E. Controller Display and Keypad:
 - 1. A four line, 20 character LCD display and keypad to be an integral part of controller for viewing available data and setting desired operational parameters. Operational parameters to also be available for viewing and limited control through serial communications input port. Following parameters only to be adjustable via DIP switches on controller:
 - a. Nominal line voltage and frequency
 - b. Single or three phase sensing
 - c. Operating parameter protection
 - d. Transfer operating mode configuration (Open transition, Closed transition, or Delayed transition)

2. Instructions and controller settings to be easily accessible, readable and accomplished without use of codes, calculations, or instruction manuals.
3. Provide source status screens for both normal and emergency to provide digital readout of voltage on all 3 phases, frequency, and phase rotation.

2.04 ACCESSORIES

- A. Equip new transfer switches with the following time delays. Adjustable in 1 second increments, except extended parallel time which will be adjustable in 0.01 second increments.
 1. Time Delay - Adjustable 0 to 30 seconds on signal to start.
 2. Time Delay - Adjustable 0 to 60 minutes on transfer to emergency.
 3. Time Delay - Adjustable 0 to 60 minutes on re-transfer to normal after normal source failure.
 4. Time Delay - Adjustable 0 to 60 minutes on re-transfer to normal after a system test.
 5. Time Delay - Adjustable 0 to 60 minutes for unloaded cool down of engine generator.
 6. Time Delay - Adjustable 1 to 5 minute time delay on failure to synchronize normal and emergency sources prior to closed transition transfer.
 7. Time Delay - Adjustable 0.1 to 9.99 second time delay on an extended parallel condition of both power sources during closed transition operation.
- B. Continuously monitor voltage and frequency on both normal and emergency sources, with the following pickup, dropout, and trip setting capabilities (values shown as percentage of nominal unless otherwise specified):

Parameter	Sources	Dropout/Trip	Pickup/Reset
Undervoltage	N&E, 3	70 to 98%	85 to 100%
Overvoltage	N&E, 3	102 to 115%	2% below trip
Underfrequency	N&E	85 to 98%	90 to 100%
Overfrequency	N&E	102 to 110%	2% below trip
Voltage unbalance	N&E	5 to 20%	1% below dropout

- C. Provide three position momentary-type test switch for test/automatic/reset modes. Test position will simulate normal source failure. Reset position bypasses time delays on either transfer to emergency or retransfer to normal.
- D. Provide SPDT contact, rated 5 amps at 30 VDC, for a low-voltage engine start signal. Start signal prevents dry cranking of engine by requiring generator set to reach proper output, and run for duration of cool down setting, regardless of whether normal source restores before load is transferred.
- E. Provide auxiliary contacts, rated 10 amps, 250 VAC, consisting of two contacts, closed when ATS is connected to normal source and two contacts closed, when ATS is connected to emergency source, two contacts closed when normal source is available, two contacts closed when emergency source is available.
- F. Where indicated in documents, provide means to drive ATS to a center/off position upon receipt off a dry contact signal from the load shed controller.
- G. Provide signal to test transfer switches with elevator loads to prevent interruption of power during elevator operation.
- H. Provide LED indicating lights (16 mm industrial grade, type 12); one to indicate when ATS is connected to normal source (green) and one to indicate when ATS is connected to emergency source (red).
- I. Provide LED indicating lights (16 mm industrial grade, type 12), energized by controller outputs. Lights to provide true source availability of normal and emergency sources, as determined by voltage sensing trip and reset settings for each source.

- J. Data Logging - Controller to have ability to log data and to maintain last 99 events, even in event of total power loss. Following events to be time and date stamped and maintained in non-volatile memory:
1. Event Logging
 - a. Data and time and reason for transfer normal to emergency.
 - b. Data and time and reason for transfer emergency to normal.
 - c. Data and time and reason for engine start.
 - d. Data and time engine stopped.
 - e. Data and time emergency source available.
 - f. Data and time emergency source not available.
 2. Statistical Data
 - a. Total number of transfers.
 - b. Total number of transfers due to source failure.
 - c. Total number of days controller is energized.
 - d. Total number of hours both normal and emergency sources are available.
- K. Digital Power Meter: Provide a digital demand power meter at each transfer switch where indicated on plans. Install on the load side of the switch with the following functionality:
1. Provide 1 percent metering accuracy and data for the following:
 - a. Line-to-neutral voltages (VAN, VBN, VCN)
 - b. Line-to-neutral voltage average (VAVE)
 - c. Line-to-line voltages (VAB, VBC, VCA)
 - d. Line-to-line voltage average (VLAVE)
 - e. Current on each phase (IA, IB, IC)
 - f. Current in the neutral conductor (IN)
 - g. Average current (IAVE)
 - h. Active power, kW per phase and total (WA, WB, WC, WT)
 - i. Demand recording of peak kW; in intervals of: 15 minute, 30 minute, 1 hour, 1 day, 1, week, 1 month, and past 24 months
 - j. Reactive power, kVAR per phase and total (VARA, VARB, VARC, VART)
 - k. Apparent power, kVA per phase and total (VAA, VAB, VAC, VAT)
 - l. kWhours importing, exporting and net (kWhIMP, kWhEXP, kWhNET)
 - m. kVARhours leading, lagging and net (kVARhLEAD, kVARhLAG, kVARhNET)
 - n. kVAhours net (kVAhNET)
 - o. Power factor (PF)
 - p. Signal frequency (Hz)

2.05 AUTOMATIC SEQUENCE OF OPERATION

- A. Initiate Time Delay to Start Alternate Source Engine Generator: Upon initiation by normal source monitor.
- B. Time Delay To Start Alternate Source Engine Generator: 0 to 30 seconds, adjustable.
- C. Initiate Transfer Load to Alternate Source: Upon initiation by normal source monitor and permission by alternate source monitor.
- D. Time Delay Before Transfer to Alternate Power Source: 0 to 60 minutes, adjustable.
- E. Initiate Retransfer Load to Normal Source: Upon permission by normal source monitor.
- F. Time Delay Before Retransfer to Normal Power: 0 to 60 minutes, adjustable; bypass time delay in event of alternate source failure.
- G. Time Delay Before Engine Shut Down: 0 to 60 minutes, adjustable, of unloaded operation.
- H. The Standby ATS shall be provided with Shed function capability. When the Load on the generator is greater than 100% the ATS-SB to be shed and assure the ATS-LS will be always on.

PART 3 - EXECUTION

3.01 GENERAL INSTALLATION REQUIREMENTS

- A. Install in accordance with manufacturer's instructions and recommendations.
- B. Provide engraved plastic nameplates under provisions of Section 26 05 53, Identification for Electrical Systems.
- C. Install arc flash labels. See Section 26 05 73, Electrical Distribution System Studies.
- D. Transfer Switches:
 - 1. Perform work in a neat and workmanlike manner in accordance with NECA 1.
 - 2. Install transfer switches in accordance with manufacturer's instructions.
 - 3. Arrange equipment to provide minimum clearances and required maintenance access.
 - 4. Provide required support and attachment in accordance with Section 26 05 29, Hangers and Supports for Electrical Systems and Equipment.
 - 5. Install transfer switches plumb and level.
 - 6. Unless otherwise indicated, mount floor-mounted transfer switches on properly sized 3 inch (80 mm) high concrete pad constructed in accordance with Section 033000, Cast-in-Place Concrete.
 - 7. Provide grounding and bonding in accordance with Section 26 05 26, Grounding and Bonding for Electrical Systems.
- E. Transfer Switch Settings:
 - 1. Time delay on signal to start: 2 seconds
 - 2. Time delay on transfer to emergency: 0 seconds
 - 3. Time delay on retransfer to normal: 10 minutes
 - 4. Time delay for engine cool down: 15 minutes
 - 5. Time delay on failure to synchronize: 30 seconds
 - 6. Time delay on extended parallel: 0.1 seconds.
- F. Examination:
 - 1. Verify that field measurements are as shown on the drawings.
 - 2. Verify that the ratings and configurations of transfer switches are consistent with the indicated requirements.
 - 3. Verify that rough-in for field connections are in the proper locations.
 - 4. Verify that mounting surfaces are ready to receive transfer switches.
 - 5. Verify that conditions are satisfactory for installation prior to starting work.
- G. Acceptance and Inspection:
 - 1. Factory test complete ATS to ensure proper operation of individual components and correct overall sequence of operation and to ensure that operating transfer time, voltage, frequency and time delay settings are in compliance with specification requirements.
 - 2. Record the following for each switch:
 - a. Time delay on retransfer to normal.
 - b. Transfer switch time in neutral position during retransfer to normal for each switch with neutral delay.
 - c. Time delay to engine shut down for entire system.
 - 3. If transfer switch feeds UPS system, verify that voltage threshold for transfer switch is set to start generator before UPS would switch to battery power. This is to prevent running UPS on battery power during brownout that doesn't start generator.
 - 4. Prior to acceptance of installation, inspect and test equipment on site by equipment supplier and service tech employed and bonded by manufacturer, to show it is free of any defects and placed in service.
- H. Field Quality Control:
 - 1. Perform field inspection and testing in accordance with Division 01, General Requirements.
 - 2. Inspect and test in accordance with NETA STD ATS, except Section 4.

3. Perform inspections and tests listed in NETA STD ATS, Section 7.22.3.
 4. See Section 01 40 00, Quality Requirements, for additional requirements.
 5. Provide services of a manufacturer's authorized representative to observe installation and assist in inspection and testing. Include manufacturer's detailed testing procedures and field reports with submittals.
 6. Prepare and start system in accordance with manufacturer's instructions.
 7. Correct defective work, adjust for proper operation, and retest until entire system complies with contract documents.
 8. Submit detailed reports indicating inspection and testing results and corrective actions taken.
- I. Closeout:
1. Demonstrate operation of transfer switch in normal and emergency modes.
 2. See Division 01 for additional requirements for closeout submittals, demonstration and training.
 3. Demonstration: Demonstrate proper operation of transfer switches to Owner, and correct deficiencies or make adjustments as directed.
 4. Training: Train Owner's personnel on operation, adjustment, and maintenance of transfer switches.
 - a. Use operation and maintenance manual as training reference, supplemented with additional training materials as required.
 - b. Provide minimum of four hours of training.
 - c. Instructor: Manufacturer's authorized representative.
 - d. Location: At project site.
 - e. Date and Time: As selected by Owner.
- J. Cleaning and Maintenance:
1. See Division 01, General Requirements, for additional requirements relating to maintenance service.
 2. Clean exposed surfaces to remove dirt, paint, or other foreign material and restore to match original factory finish.
- K. Testing:
1. Reference Section 26 08 05, Electrical Acceptance Testing.

END OF SECTION 263600

SECTION 264113 - LIGHTNING PROTECTION FOR STRUCTURES**PART 1- GENERAL****1.1 WORK INCLUDES**

- A. A lightning protection Risk Assessment to be performed and determined if lightning protection system for structure is required in area of construction. If required, this specification applies.
- B. Base Bid:
 - 1. Electrical Contractor:
 - a. The Contractor shall provide all material and labor for the installation of a complete Lightning Protection System for all of the building(s) included in this project. This specification addresses the requirements of Lightning Protection Systems for buildings only.

1.2 RELATED WORK

- A. Specified elsewhere:
 - 1. Section 26 05 26 "Grounding and Bonding."

1.3 REFERENCES

- A. LPI: Lightning Protection Institute.
- B. NRTL: National recognized testing laboratory.

1.4 SUBMITTALS

- A. Product Data: For air terminals and mounting accessories.
- B. Shop Drawings: Detail lightning protection system layout and details, including air-terminal locations, conductor routing and connections, and bonding and grounding provisions. Include indications for use of raceway, data on how concealment requirements will be met, and calculations required by NFPA 780 for bonding of grounded and isolated metal bodies.
- C. Catalog Data with complete description of material components
- D. Qualification data for firms and persons specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include data on listing or certification by an NRTL or LPI.
- E. Certification, signed by Contractor, that roof adhesive for air terminals is approved by manufacturers of both the terminal assembly and the single-ply membrane roofing material.
- F. Field inspection reports indicating compliance with specified requirements.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an experienced installer who is an NRTL or who is certified by LPI as a Master Installer/Designer.

- B. Compliance Requirements:
 - 1. System Design: NFPA 780, latest edition.
 - 2. Component Design: UL 96 Standard, latest edition, L.P.I. 175.
 - 3. Certification: Lightning Protection Institute Certified System and Underwriters' Laboratories 96A Master Label, Lightning Protection Institute certification

1.6 COORDINATION

- A. Coordinate installation of lightning protection with installation of other building systems and components, including electrical wiring, supporting structures and building materials, metal bodies requiring bonding to lightning protection components, and building finishes.
- B. Coordinate installation of air terminals attached to roof systems with roofing manufacturer and Installer.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Harger Lightning Protection, Inc (IL).
 - 2. Automatic Lightning Protection.
 - 3. ERICO International Corporation.
 - 4. Heary Bros. Lightning Protection Co. Inc.
 - 5. Independent Protection Co.
 - 6. Robbins Lightning Inc.
 - 7. Thompson Lightning Protection, Inc.

2.2 LIGHTNING PROTECTION SYSTEM COMPONENTS

- A. Comply with UL 96.
- B. System to be designed and installed by a L.P.I. Certified Master Installer/Designer in good standing with the Lightning Protection Institute.
- C. System to consist of groundings, down conductors, air terminals, interconnecting conductors and bonding, designed to appear as a part of the building.
- D. Design to be complete per current NFPA 780 requirements.
 - 1. Class II Aluminum materials required.
 - 2. Class II Copper materials required.
 - 3. Aluminum Lightning Protection materials not to be embedded in concrete or masonry or installed on or below copper surfaces.
 - 4. Copper Lightning Protection materials not to be installed on aluminum surfaces.
 - 5. Grounding shall be suitable for the soil conditions per NFPA 780, this may include:
 - a. Ground loop only (full size cable) 2/0 buried 18 in. deep.
 - b. Ground loop combined with $\frac{3}{4}$ x 10 ft. rods buried 18 in. deep.

6. Strike termination devices (air terminals) required as follows, unless the area in question is located under a zone of protection.
 - a. Minimum 10 in. projection above the object projected.
 - b. Maximum 20 ft. spacing on roof ridges or edges.
 - c. Maximum 24 in. distance from ridge ends or roof edges & outside corners.
 - d. Penthouses, Protrusions and Mechanical roof top equipment, same guidelines.
 - e. Strike termination devices required on eaves of sloping roofs, when the eave is over 50 ft. in height.
 - f. Mid-roof areas are to be provided with Strike termination devices at either 50 ft. spacing or provided with Strike termination devices of sufficient quantity & height, to ensure the entire roof area is covered by a "zone of protection" as afforded by a 150 ft. radius sphere (per NFPA 780).
 - g. Strike termination safety devices to be provided in mid-roof areas and high traffic areas. Material to be the same as, or an alloy of, the point and base and to be one of the following:
 - 1) Safety tipped point
 7. Bonding is required in strict accordance with NFPA 780
 - a. Ground level potential equalization; below the 12 ft. elevation of the structure all grounded media to be interconnected.
 - b. Ground loop required for structures 60 ft. in height.
 - c. Lightning surge suppressors to be provided on electrical and communication service entrances and on communication antenna lead-ins
- E. Roof-Mounting Air Terminals: NFPA Class I copper, solid, unless otherwise indicated.
1. Single-Membrane, Roof-Mounting Air Terminals: Designed for single-membrane roof materials.
- F. Stack-Mounting Air Terminals: Solid copper.
- G. Ground Rods, Ground Loop Conductors, and Concrete-Encased Electrodes: Comply with Division 26 Section "Grounding and Bonding for Electrical Systems" and with standards referenced in this Section.
- H. Cable straps and supports: Cable straps shall be aluminum single-hole or two-hole with screws or fasteners. Cable holders for use with hot pitch roofing compound or commercial adhesive on built-up roof where holes for anchoring cable strap/holders are not allowed.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install lightning protection components and systems according to UL 96A and NFPA 780.
- B. L.P.I. Certified Master Installer required with a minimum of five (5) years experience.
- C. Complete per requirements of NFPA 780.
- D. Neat and inconspicuous manner.

- E. All mounting & penetration of roof surface shall be coordinated with roofing contractor to assure maximum roofing guarantee.
 - F. All through-roof penetration flashings to be furnished, sealed and guaranteed by the roofing contractor.
 - G. Install conductors with direct paths from air terminals to ground connections. Avoid sharp bends and narrow loops.
 - H. Conceal the following conductors:
 - 1. System conductors.
 - 2. Down conductors.
 - 3. Interior conductors.
 - 4. Conductors within normal view from exterior locations at grade within 200 feet of building.
 - 5. Notify Architect at least 48 hours in advance of inspection before concealing lightning protection components.
 - I. Cable Connections: Use approved exothermic-welded connections for all conductor splices and connections between conductors and other components, except those above single-ply membrane roofing.
 - J. Air Terminals on Single-Ply Membrane Roofing: Comply with adhesive manufacturer's written instructions.
 - K. Bond extremities of vertical metal bodies exceeding 60 feet in length to lightning protection components.
 - L. A counterpoise installation based on requirements in Division 26 Section "Grounding and Bonding " may be used as a ground loop required by NFPA 780, provided counterpoise conductor meets or exceeds minimum requirements in NFPA 780.
 - 1. Bond ground terminals to counterpoise conductor.
 - 2. Bond grounded metal bodies on building within 12 feet of ground to counterpoise conductor.
 - 3. Bond grounded metal bodies on building within 12 feet of roof to counterpoise conductor.
 - M. Bond lightning protection components with intermediate-level interconnection loop conductors to grounded metal bodies of building at 60-foot intervals.
 - N. Fasteners:
 - 1. At 3 ft. centers, maximum, on exposed conductor runs.
 - 2. As necessary to maintain position and hold permanently in place on concealed runs of conductor.
- 3.2 CORROSION PROTECTION
- A. Do not combine materials that can form an electrolytic couple that will accelerate corrosion in the presence of moisture unless moisture is permanently excluded from junction of such materials.

- B. Use conductors with protective coatings where conditions would cause deterioration or corrosion of conductors.

3.3 FIELD QUALITY CONTROL AND FINAL ACCEPTANCE

- A. Provide an inspection by an inspector certified by LPI to obtain an LPI certification.
- B. Procurement of L.P.I. Certification includes jobsite verification and completion of:
 - 1. Witness of Grounding System & Grade bonding (Stage I)
 - 2. Inspection of concealed equipment between roof & grade (Stage II)
 - 3. Final inspection of exposed equipment on roof (Stage III)
- C. Procurement of Underwriters' Laboratories Master Label indicating completion of;
 - 1. Show owner or his representative the type and manner of placing groundings and receiving his record of review.
 - 2. Completion of application form and submission to Underwriters' Laboratories for issuance of certification.
- D. Installation of Installer's Nameplate at location designated on UL application form.
- E. Any components or methods found to be not in accordance with this specification shall be repaired or replaced without cost to the owner.

END OF SECTION 264113

SECTION 264300 - SURGE PROTECTION DEVICES**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes Surge Protection Devices for low-voltage power, control, and communication equipment.
- B. Work under this section consists of furnishing all materials necessary for the execution and complete installation of Surge Protective Devices (SPDs).
- C. In addition to this section, the Contractor shall refer to other specification sections and drawings to ascertain the extent of work included.

1.2 DEFINITIONS

- A. ATS: Acceptance Testing Specifications.
- B. I_n : Nominal Discharge Current.
- C. NRTL: National Recognized Testing Lab.
- D. OCPD: Over Current Protective Device.
- E. RMS: Root Mean Square.
- F. SPD: Surge Protection Devices.
- G. VPR: Voltage Protection Rating.

1.3 SUBMITTALS

- A. Product Data: For each type of product indicated. Include rated capacities, operating weights, operating characteristics, furnished specialties, and accessories. Alternate / Substitute manufacturers must submit all information a minimum of 14 days prior to bid to be considered.
- B. Product Certificates: For SPD, signed by product manufacturer certifying compliance with the following standards:
 - 1. UL 1449, 3rd Edition.
 - 2. Submit proof that products are *UL listed and labeled by Underwriters Laboratories* to UL 1449 3rd Edition.
- C. Field quality-control test reports, including the following:
 - 1. Test procedures used.
 - 2. Test results that comply with requirements.
 - 3. Failed test results and corrective action taken to achieve requirements.
- D. Operation and Maintenance Data: For SPD to include in operation, and maintenance manuals.

- E. Warranties: Special warranties specified in this Section.
- F. See the electrical power riser diagram for Surge Protective Device (SPD) unit locations.

1.4 QUALITY ASSURANCE

- A. Manufacturer's Qualifications:
 - 1. Manufacturer shall be certified ISO 9001 or 9002.
 - 2. The manufacturer of this equipment shall have produced similar electrical equipment for a minimum period of five (15) years.
- B. Source Limitations: Obtain SPD and accessories through one source from a single manufacturer.
- C. Electrical Components, Devices, and Accessories: Listed and by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- D. Comply with IEEE C62.41, "IEEE Guide for Surge Voltages in Low Voltage AC Power Circuits," and test devices according to IEEE C62.45, "IEEE Guide on Surge Testing for Equipment Connected to Low-Voltage AC Power Circuits."
- E. Comply with UL 1449 3rd Edition, "Surge Protection Devices."
- F. UL1283 5th Edition: Electromagnetic Interference Filters.
- G. cUL – UL: Evaluation to Canadian Safety Requirements (UL 1449, 1283).
- H. Permanently affix surge ratings to the unit.
- I. UL 1449 3rd Edition Listed, bearing the official UL 3rd Edition gold hologram label.
- J. IEEE Compliance: Comply with Standard 1100 (Emerald Book) 2007 Edition.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Deliver SPD in original factory shipping cartons, with manufacturer's labels intact.
- B. Store SPD indoors in clean, dry space, protected from weather, with uniform temperature and humidity to prevent condensation. Protect SPD from exposure to dirt, fumes, water, corrosive substances, and physical damage.
- C. Handle SPD in accordance with manufacturer's instructions and to avoid damage.

1.6 PROJECT CONDITIONS

- A. Service Conditions: Rate surge protection devices for continuous operation under the following conditions, unless otherwise indicated:
 - 1. Maximum Continuous Operating Voltage: Not less than 115 percent of nominal system operating voltage.
 - 2. Operating Temperature: 30 to 120 deg F.
 - 3. Humidity: 0 to 85 percent, non-condensing.

1.7 ENCLOSURES

- A. NEMA Rated, compatible with panel or device being protected.

1.8 IDENTIFICATION

- A. General: Refer to Division 26 Section "Identification for Electrical Systems" for labeling requirements.

1.9 COORDINATION

- A. Coordinate location of field-mounted SPD to allow adequate clearances for maintenance.

1.10 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.
- B. All Surge Protective Devices (SPDs), associated hardware, and supporting components shall be warranted to be free from defects in materials and workmanship, under normal use and in accordance with the instructions provided, for a period of fifteen (15) years.
- C. Special Warranty for Cord-Connected, Plug-in Surge Suppressors: Manufacturer's standard form in which manufacturer agrees to repair or replace electronic equipment connected to circuits protected by surge suppressors.
- D. A detailed warranty statement shall be provided with each unit.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provided on **LiDL National Account List**:
 - 1. Schneider Electric

2.2 GENERAL

- A. Operating Frequency: 47 to 63 Hertz.
- B. The Surge Protective Device (SPD) shall be integral to module #1 and only for main service panel MDP1.
- C. The SPD shall be marked with a Short Circuit Current Rating (SCCR) and shall not be installed at a point on the system where the available fault current is in excess of that rating per the National Electric Code, Article 285, Section 6.
- D. SPD designs using a single fuse to protect two (2) or more surge paths shall not be acceptable.
- E. Accessories:
 - 1. Accessories listed shall be included for all types of SPDs.

2. Status Indicators. Integral status circuit to monitor the operational status of all modes of protection. No manual testing is required to confirm the integrity of the suppression and filter systems. If the unit does fail, the green LED will go out and the red LED will be lit.
3. Dry Contacts for remote monitoring.
4. Audible Alarm. Activated when any one or more of the surge current diversion components has failed. An alarm silence button shall be provided. The audible alarm and switch shall be located on the unit's hinged front cover.
5. Each individual module shall feature a green LED indicating the individual module has all surge protection devices active. If any module is taken off-line, the green LED will turn off and a red LED will illuminate, providing *individual module* as well as *total system* status indication.
6. The SPD shall provide EMI/RFI electrical noise attenuation up to 43dB in the range of 50kHz to 100MHz as defined by MIL-STD-220A test procedures.
7. Voltage Protection Ratings: The UL 1449 3rd Edition Voltage Protection Ratings "VPR" (6kV, 3000 Amps, 8/20μs waveform) shall not exceed the UL assigned values listed below.

Voltage Protection Ratings (VPR) 6kV, 3000A, 8/20μs Waveform	Voltage Rating	
	208/120V	480/277V
Line to Neutral	900V	1200V
Line to Ground	800V	1200V
Neutral to Ground	700V	1200V
Line to Line	1200V	2000V

2.3 SERVICE ENTRANCE, DISTRIBUTION AND PANELBOARD SUPPRESSORS

- A. Surge Protection Device Description: Modular solid state metal oxide varistor suppression modules. No series connected suppression components are allowed.
 1. Type 1 as defined by UL 1449, 3rd Edition.
 2. Short Circuit current rating complying with UL 1449.
 3. Voltage Protection Rating shall be in compliance with test and evaluation procedures described in UL 1449, 3rd Edition.
 4. Nominal Discharge Current Rating: 20 kA minimum as determined by the actual Nominal Discharge Current test.
 5. Externally mounted surge protection device within a NEMA 1 enclosure. Internally mounted surge protection devices shall be prohibited.
- B. Modular field repairable design with individually fused modules with the following features and additional accessories:
 1. Fabrication using bolted compression lugs for internal wiring.
 2. Integral disconnect switch or circuit breaker as required.
 3. Redundant suppression circuits.
 4. Arrangement with copper bus bars and for bolted connections to phase buses, neutral bus, and ground bus.
 5. Surge-event operations counter.

The SPD surge current ratings shall be based, in principle, on the electrical system ampacity listed in the table below. Use Risk Assessment Chart to more accurately define surge current rating specific to application.

Electrical System Ampacity @ SPD Install Point	Surge Protection (kA)	
	Per Mode	Per Phase
4100 – 6000A	200	400
2600 – 4000A	200	400
825 – 2500A	150	300
325 – 800A	100	200
175 – 300A	50	100
15 – 150A	50	100

- C. SPD modules shall be configured to isolate individual suppression component failures without causing total loss of surge protection in that mode.
- D. All primary transient path wiring shall be of copper minimum size as required by the manufacturer but not less than #2AWG, or via bus bar of equivalent capacity to provide equal impedance interconnection between phases. No plug-in modules, components or printed circuit boards shall be used in surge carrying path.
- E. Connection Means: Permanently wired, externally mounted adjacent to the switchboard: 60A-3P circuit breaker with 4 #6 and 1 #8 ground in a 3/4" emt conduit.
- F. Connection method Distribution Panels: 60A-3P circuit breaker with 4 #6 and 1 #8 ground in a 3/4" emt conduit.
- G. Connection Method Panel boards: 30A-3P circuit breaker with 4 #10 and 1 #10 ground in a 3/4" emt conduit.

PART 3 - EXECUTION

3.1 PREPARATION:

- A. Examine conditions for compliance with requirements for installation tolerances, power characteristics and other conditions affecting performance of the SPD device. Do not proceed until unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Install devices at service entrance on load side, with ground lead bonded to service entrance ground.
- B. Install devices for panelboard and auxiliary panels with conductors or buses between SPD and points of attachment as short and straight as possible. Do not exceed manufacturer's recommended lead length. Do not bond neutral and ground.
 - 1. Comply with manufacturer's written recommendation for conductor and circuit-breaker size for connecting SPD to distribution system. Match circuit-breaker size to conductor size. Coordinate with Drawings.

- C. SPD shall be installed with less than 3'-0" of lead length from the panels. Installation of leads shall avoid sharp and unnecessary bends. Where SPD cannot be mounted within 3'-0" of the panel, Low Impedance wire supplied by SPD manufacturer shall be utilized.
- D. Installation shall be in accordance with the NEC 280 & 285.
- E. System shall not require removal and replacement for warranty and/or other repairs. All internal component replacements shall be capable of being completed by a licensed electrician.
- F. Ground each SPD enclosure.
- G. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. Where manufacturer's torque values are not indicated, use those specified in UL 486A and UL 486B.

3.3 CLEANING

- A. Remove paint splatters and other spots, dirt and debris. Touch up scratches and mars of finish to match original finish.
- B. Clean devices internally using methods and materials recommended by the manufacturer.
- C. Do not use compressed air for cleaning.

3.4 CONTRACTOR START UP AND REPORTING

- A. Verify that electrical wiring installation complies with manufacturer's written installation requirements.
- B. Do not energize or connect service entrance equipment, panelboards, control terminals or data terminals to their sources until surge protection devices are installed and connected.
- C. Testing: Perform the following field tests and inspections and prepare test reports:
 - 1. After installing surge protection devices, but before electrical circuitry has been energized, test for compliance with manufacturer's requirements.
 - 2. Complete startup checks according to manufacturer's written instructions.
 - 3. Perform each visual and mechanical inspection and electrical test stated in NETA ATS, "Surge Arresters, Low-Voltage Surge Protection Devices" Section. Certify compliance with test parameters.
- D. Remove and replace malfunctioning units and retest as specified above.
- E. Technical assistance shall be provided by the manufacturer through the efforts of a factory representative or a local distributor.
- F. The unit shall be installed of the same voltage rating as the intended protected equipment.
- G. The unit shall be installed on the load side of the service equipment overcurrent device.
- H. The location of the field-mounted SPD shall allow adequate clearances for maintenance.
- I. Before energizing the SPD, the unit shall be verified to be correct as specified: manufacturer, product series, and model UL 3rd edition label shall be clearly visible.

- J. All voltage modes including L-L (Line-to-Line), L-G (Line-to-Ground), L-N (Line-to-Neutral), and N-G (Neutral-to-Ground) shall be measured and verified against the unit voltage ratings.
- K. Continuity measurements shall be made between the Neutral and Ground connections to verify the Neutral-to-Ground bond.

END OF SECTION 264300

SECTION 265100 - INTERIOR LIGHTING**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the following:
 - 1. Interior luminaires, lamps, ballasts, LED modules.
 - 2. Emergency lighting units.
 - 3. Illuminated exit signs.
 - 4. Luminaire supports.

1.2 DEFINITIONS

- A. ANSI: American National Standards Institute.
- B. Average Rated Lamp Life: The period of time after which 50 percent will have failed and 50 percent will have survived under normal conditions.
- C. BF: Ballast factor.
- D. CRI: Color-rendering index.
- E. Emergency Lighting Unit: A fixture with integral emergency battery power supply and the means for controlling and charging the battery. They are also known as an emergency light set. Emergency lighting units are available with and without integral heads.
- F. Fixture: A complete lighting unit, exit sign, or emergency lighting unit. Fixtures include lamps or LED modules, and parts required to distribute the light, position and protect lamps or LED modules, and connect lamps or LED modules to the power supply.
- G. IES: Illuminating Engineering Society of North America.
- H. LER: Luminaire efficacy rating.
- I. Luminaire: Complete lighting fixture, including optics, shielding, power supply or ballast housing if provided.
- J. Solid State Lighting/Luminaires: Light sources and their associated luminaires that produce light (radiant energy) without the use of electrical filaments, excited gas or plasma, such as LED, OLED, or PLED.
- K. UL: Underwriter's Laboratories, Inc.

1.3 SYSTEM DESCRIPTION

- A. The interior lighting system shall include all lighting fixtures, lamps or LED modules, switches, mounting, wiring, control equipment, and accessories required for complete system, whether or not they are indicated or specified, as indicated in the Drawings and as specified.
- B. The lighting fixture schedules in the Drawings indicate manufacturer, fixture design, appearance and performance desired.

- C. Verify locations of light fixtures indicated in Drawings and coordinate with other reference data and materials as required prior to installation to ensure locations will not interfere with other work. Verify space above luminaires and confirm non-interference with other equipment, such as ducts, pipes, conduit and cabling, and openings. Alert Architect and Commission Authorized Representative in writing to non-standard modifications required for compliance with the Contract Documents and for installation to coordinate with ceiling system before proceeding with the Work.
- D. Verify dimensions. Where discrepancies are found within the Contract Documents, or additional information is required, immediately contact Architect for clarifications and additional information.
- E. Coordinate installation of lighting system with other trades to prevent delays in the Work and to ensure the lighting fixtures and supports will not be damaged by subsequent construction operations.

1.4 SUBMITTALS

A. Action Submittals:

- 1. General: Provide separate submittal product data/shop drawings for each fixture type clearly indicating the fixture type designation used in the Drawings and all pertinent options and accessories. Do not group similar fixture types together on a single cut sheet. Submittals that do not indicate option data where multiple selections exist will be returned without being reviewed.
- 2. Product Data: For each type of lighting fixture, arranged in order of fixture designation. Provide separate submittal product data for each fixture type clearly indicating the fixture type designation used in the Drawings and all pertinent options and accessories. Do not group similar fixture types together on a single cut sheet. Submittals that do not indicate option selection where multiple selections exist will be returned without review. Include data on features, accessories, finishes, and the following:
 - a. Physical description of lighting fixture including materials, dimensions, and verification of indicated parameters.
 - b. Include information as to power factor, input watts, and ballast factor. Indicate mounting distance limitation and standard wire sizes for remote ballasts/transformers.
 - c. LED Drivers: Include information as to input watts. Indicate mounting distance limitation and standard wire sizes for remote drivers.
 - d. LED modules, per luminaire tag, including life, output, and energy-efficiency data.
 - e. Life, output (lumens, CCT, and CRI), and energy-efficiency data for luminaire.
 - f. Photometric performance data, based on laboratory testing, specific to the particular luminaire specified for this Project.

- B. Certification of Compatibility: Submit a written statement, signed by the Contractor, Installer, and Manufacturers, certifying that power supplies and drivers, automatic lighting control devices, including occupancy sensors and daylight harvesting controls, and related components being supplied for the Project are compatible equipment. Statement shall include that system components do not compromise the proper operation, or the design criterion, of the supplied ballasts, power supplies, drivers, and automatic control devices.

- C. Shop Drawings: For custom, non-standard, and continuous row/length luminaires. Include plans, elevations, sections, details, and attachments to other work.

- 1. Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.

2. Wiring Diagrams: For power, signal, and control wiring.

D. Informational Submittals:

1. Field quality-control reports.
2. Warranty: Sample of special warranty.

E. Closeout Submittals:

1. Operation and Maintenance Manuals: Provide a collection of manufacturer recommended maintenance practices for each lighting fixture type. Include the following:
 - a. Tools required.
 - b. Acceptable cleaners and recommended cleaning practices.
 - c. Replacement parts list.
 - d. Dates of installation.
 - e. Replacement parts order information.
 - f. Lamp and ballast re-ordering information and re-lamping schedule.
 - g. Contact information for manufacturer's service department or qualified service agencies.
 - h. Submittal data.
 - i. Operation data.
 - j. Intended operation narrative.
2. Record Drawings: Submit record drawings of the actual installation within thirty (30) days of date of Preliminary Acceptance / Substantial Completion.
3. Re-Commissioning Data: Submit manual containing all information required for re-commissioning of the installations.
 - a. Submit two (2) copies of manual within thirty (30) days of date of Preliminary Acceptance / Substantial Completion.
 - b. Submit re-commissioning manuals in heavy-duty, 3-ring binders. Submit manuals in accordance with Division 01 Section "Operations and Maintenance Data."

1.5 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- B. Regulatory Requirements:
 1. Comply with the local authority having jurisdiction.
 2. Comply with EPA, State of Maryland, and municipal regulations for proper recycling or disposal of existing lamps and ballasts removed from the Site.
- C. Commissioning: When required for the Project, Contractor shall assign representative(s) with expertise and authority to act on its behalf. The representative(s) shall perform commissioning activities including, but not limited to, the following:
 1. Review submittals relative to exterior lighting systems being commissioned.

1.6 DELIVERY, STORAGE AND HANDLING

- A. Protect luminaires, lamps, and accessories during transit, delivery, storage, and handling to prevent damage.

- B. Deliver luminaires to the Site factory-assembled and wired to the greatest extent possible and in accordance with approved submittals.
- C. Store luminaires, lamps, and accessories in accordance with manufacturer's instructions in a clean, dry location, protected from weather and away from dust generating construction activities. As required, cover materials with tarpaulin or polyethylene sheeting in a manner that allows air circulation and prevents condensation beneath the covering.
- D. Keep handling on site to a minimum. Exercise particular care to avoid damage to exposed finishes and materials.

1.7 COORDINATION

- A. Coordinate layout and installation of lighting fixtures and suspension system with other construction that penetrates ceilings or is supported by them, including HVAC equipment, fire-suppression system, and partition assemblies.
- B. Coordinate installation of lighting fixtures indicated to extend in continuous, wall to wall installation. Provide field, or established, dimensions to luminaire manufacturer in sufficient time so not to cause delays in the Work.

1.8 WARRANTY

- A. Special Warranty: Submit a written warranty, beginning from date of Preliminary Acceptance / Substantial Completion, and executed by the Contractor, manufacturer, and Installer agreeing to repair or replace products or components that fail in materials or workmanship within the specified warranty period. Failures shall include, but not be limited to, deterioration of metal finishes. Warranty shall include all materials and components, as well as labor and equipment required to remove existing and install new materials and components.
 - 1. Warranty Period for Electronic Ballasts: Seven (7) years from date of Preliminary Acceptance / Substantial Completion.
 - 2. Warranty Period for Electromagnetic Ballasts: Three (3) years from date of Preliminary Acceptance / Substantial Completion.
 - 3. Warranty Period for Metal Finishes: Seven (7) years from date of Preliminary Acceptance / Substantial Completion.
- B. Special Warranty for LED Luminaires: Submit a written warranty, beginning from date of Preliminary Acceptance / Substantial Completion, and executed by the Contractor, manufacturer, and Installer agreeing to repair or replace products or components that fail in materials or workmanship within the specified warranty period. Failures shall include, but not be limited to, a failure of the LED modules or LED drivers.
 - 1. Warranty Period for LED Luminaires: Seven (7) years from date of Preliminary Acceptance / Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers – Luminaires: Subject to compliance with requirements indicated and the design criteria specified in the Luminaire Schedule, provide one of the products specified in the Luminaire Schedule.

B. Manufacturers – Accessories: Subject to compliance with requirements, provided on **LIDL National Account List**:

1. LED Luminaires:
 - a. Nordeon Lighting Solutions
David Horowitz
National Accounts Sales Manager
T: (401) 255-7127

2.2 GENERAL REQUIREMENTS FOR LIGHTING FIXTURES AND COMPONENTS

- A. Recessed Fixtures: Comply with NEMA LE 4 for ceiling compatibility for recessed fixtures.
- B. Solid State Lighting Luminaires: Comply with UL 8750.
 1. Luminous flux, luminaire efficiency and chromaticity shall be tested, measured and reported in accordance with the most current versions of IES documents LM-79 and LM-80.
 2. Chromaticity ranges for "white light" products, with various correlated color temperatures, shall be provided in accordance with ANSI/NEMA-C78.377.
 3. Drivers and power supplies shall be provided in accordance with the requirements of ANSI/NEMA-C82.SSL1 and their maximum allowable harmonic emission limits shall be in accordance with ANSI/NEMA-C82.77.
 4. Shall be provided with a U.S. Department of Energy (DOE) "Lighting Facts" label indicating their specific performance characteristics, tested and reported in accordance with the requirements of the most current version of IES LM-79.
- C. Metal Parts: Free of burrs and sharp corners and edges.
- D. Sheet Metal Components: Steel, unless otherwise indicated. Form and support to prevent warping and sagging.
- E. Doors, Frames, and Other Internal Access: Smooth operating, free of light leakage under operating conditions, and designed to permit relamping without use of tools. Designed to prevent doors, frames, lenses, diffusers, and other components from falling accidentally during relamping and when secured in operating position.
- F. Luminaire Coatings and Finishes:
 1. Luminaire surfaces, components, trim, and housing shall be factory pre-treated, rustproof, primed and otherwise prepared to inhibit rust and corrosion. Exposed luminaire surfaces shall be factory pre-treated, primed and finish coated with a suitable rust and corrosion inhibiting product.
 2. Reflecting surfaces shall have minimum reflectance as follows, unless otherwise indicated:
 - a. White Surfaces: 85 percent.
 - b. Specular Surfaces: 83 percent.
 - c. Diffusing Specular Surfaces: 75 percent.
 - d. Laminated Silver Metallized Film: 90 percent.
 3. Luminaries shall receive manufacturer's standard finish, unless otherwise indicated. Color shall be as indicated or, if not indicated, as selected by Architect from manufacturer's standard range.
 4. Exposed finish shall be free of streaks, runs, holidays, stains, blisters, and similar defects.

G. Diffusers and Globes:

1. Acrylic Lighting Diffusers: 100 percent virgin acrylic plastic. High resistance to yellowing and other changes due to aging, exposure to heat, and UV radiation.
 - a. Lens Thickness: At least 0.125 inch minimum unless otherwise is indicated.
 - b. UV stabilized.

H. Louvers:

1. Provide louvers or baffles fabricated from aluminum reflector sheet free of marks including mars and indentations caused by fabrication and assembly techniques. No rivets, springs, or other hardware shall be visible after installation. Plastic louvers are not permitted.
2. Provide non-iridescent type louvers for fixtures using fluorescent lamps.
3. Provide louvers and baffles of first-quality polished, buffed, and anodized. Anodized finish shall be Alzak.

I. Reflectors and Trims:

1. Attach non-permanently affixed reflectors to housing by means of safety chains or spring clips, to prevent reflectors from falling. No part of the clip or chain shall be visible after installation, when viewed from any angle up to 45 degrees from horizontal.
2. Aluminum Reflectors:
 - a. Provide reflectors and reflecting cones fabricated from aluminum reflector sheet free of marks including spinning lines, mars, and indentations caused by fabrication and assembly techniques. No rivets, springs, or other hardware shall be visible after installation. Provide only reflectors free from blemishes, scratches, or indentations.
 - b. Provide reflectors of first-quality polished, buffed, and anodized. Anodized finish shall be Alzak.
 - c. Provide non-iridescent type louvers for fixtures using fluorescent lamps.
 - d. Provide polished self-flanged trim cones, color finish shall match that of the cone.
3. Painted Reflectors: Completely formed before application of primer and paint. Minimum of 87 percent reflectance white.

J. Product Description: Provide complete luminaire assemblies with features, options and accessories as scheduled and required for complete assembly, whether specified or not.

K. Provide fixtures constructed, wired, and installed in compliance with appropriate UL standards and applicable codes. Provide fixtures that are listed by UL for the applications and locations where they are shown. Provide all products with UL label.

L. Verify and provide luminaires that are appropriate for the mounting conditions and space structure.

M. All fixture components must operate within the temperature limits of their design.

N. Fixture Support Components: Comply with Division 26 Section "Hangers and Supports for Electrical Systems."

1. Single-Stem Hangers: 1/2-inch steel tubing with swivel ball fitting and ceiling canopy. Finish shall be the same as the fixture.
2. Twin-Stem Hangers: Two, 1/2-inch steel tubes with single canopy arranged to mount a single fixture. Finish shall be the same as the fixture.

3. Rod Hangers: 3/16-inch minimum diameter, cadmium-plated, threaded steel rod.
4. Hook Hanger: Integrated assembly matched to fixture and line voltage and equipped with threaded attachment, cord and locking-type plug.

2.3 LED DRIVERS

- A. LED drivers shall be factory provided by the respective luminaire manufacturers, and shall be suitable for their intended use, to operate the designated LED modules listed in the Luminaire Schedule, and as specified herein, to their full light output.
- A. Comply with the requirements of the Federal Communications Commission (FCC) rules and regulations, Title 47 CFR Part 18, Non-Consumer (Class A) for EMI/RFI (conducted and radiated).
- B. Provide identical drivers within each luminaire type.
- C. Provide UL listed and labeled drivers. Provide drivers with temperature ratings appropriate to the installation.
- D. Provide complete connection to LED type luminaires through both integrally installed and remote electronic drivers.
- E. Drivers shall be totally enclosed within a metallic enclosure, and shall be provided with integral leads color coded per ANSI C82.11, or with poke-in style wire retaining connectors.
- F. Fixtures intended to be dimmed shall have dimming driver compatible with the specified dimmer controls.
- G. Remote Drivers:
 1. Remote drivers are specifically not shown on the drawings. Contractor shall install remote drivers in a readily accessible, dry, indoor, concealed location, in accordance with the manufacturer's instructions.
 2. Provide ventilated metal enclosures for remote drivers furnished as loose equipment. All wiring to/from remote drivers and their associated LED luminaires shall be installed in conduit.
 3. Verify and comply with remote distance limitations specified by the luminaire/driver manufacturer.

2.4 EXIT SIGNS

- A. General: Comply with UL 924 and local codes and ordinances for sign colors, visibility, luminance, and lettering size.
- B. Internally Lighted Signs:
 1. Lamps for AC Operation: LED's, 70,000 hours minimum rated lamp life.

2.5 GENERAL REQUIREMENTS FOR LAMPS

- A. Source Limitations: Obtain each type of lamp from single source from single manufacturer.
- B. Reflector Lamp Beam Pattern: ANSI C78.379.

2.6 LENSES

- A. Source Limitations: Fixture lenses shall be provided by the light fixture manufacturer.
- B. Construct acrylic lenses, diffusers, covers, and globes of 100 percent virgin acrylic injection-molded plastic complying with ASTM D 788. Polystyrene lenses are not acceptable.
- C. All plastic shall be highly resistant to yellowing and other changes due to aging, exposure to heat and ultraviolet radiation.
- D. Destaticize all panels at the factory prior to shipping.
- E. Lens Patterns:
 - 1. Pattern 12 Lens: Pattern 12 lens shall be not less than 0.125-inch thick (overall panel thickness), with inverted conical prisms, in straight flat-sided prism shape, configured on 45-degree axis layout across the entire panel.
 - 2. Pattern 19 Lens: Pattern 19 lens shall be not less than 0.156-inch thick (overall panel thickness), with inverted conical prisms, in straight flat-sided prism shape, configured on parallel/perpendicular 90-degree axis layout across the entire panel. Provide pattern 19 lenses in kitchens, serveries, locker rooms, and toilet rooms.

2.7 LIGHTING FIXTURE SUPPORT COMPONENTS

- A. Comply with Division 26 Section "Hangers and Supports for Electrical Systems" for channel-supports.
- B. Single-Stem Hangers: 1/2-inch steel tubing with swivel ball fittings and ceiling canopy. Finish same as fixture.
- C. Wires: ASTM A 641, Class 3, soft temper, zinc-coated steel, 0.0808-inch (12 gage) minimum diameter.
- D. Rod Hangers: 3/16-inch minimum diameter, cadmium-plated, threaded steel rod.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Lighting Fixtures: Set level, plumb, and square with ceilings and walls. Install lamps in each luminaire.
- B. Lamps:
 - 1. Provide all lamps needed for all fixtures, including replacements for extinguished or defective lamps, until Final Acceptance. Replace lamps used in fixtures during construction prior to Final Acceptance.
 - 2. Install lamps according to manufacturer's instructions.
- C. Support for Luminaires in Lay-in/Accessible Ceilings: Use grid as a support element.
 - 1. Install a minimum of four ceiling support system rods or wires for each fixture. Locate not more than 6-inches from lighting fixture corners.
 - 2. Support Clips: Fasten to lighting fixtures and to ceiling grid members at or near each fixture corner with clips that are UL listed for the application.

3. Fixtures of Sizes Less Than Ceiling Grid: Install as indicated on reflected ceiling plans or center in acoustical panel, and support fixtures independently with at least two 3/4-inch metal channels spanning and secured to ceiling tees.
 4. Install at least two (2) independent support rods or wires from structure above to the light fixture housing. Supports shall be installed on opposite (diagonal) corners. Wires or rods shall be capable of supporting, without failure, a load equal to three times the weight of the light fixture.
- D. Suspended Lighting Fixture Support:
1. Pendants and Rods: Where longer than 48 inches, brace to limit swinging.
 2. Stem-Mounted, Single-Unit Fixtures: Suspend with twin-stem hangers.
 3. Continuous Rows: Use tubing or stem for wiring at one point and tubing or rod for suspension for each unit length of fixture chassis, including one at each end.
- E. Air-Handling Lighting Fixtures: Install with dampers closed and ready for adjustment.
- F. Adjust aimable lighting fixtures as required to provide required light intensities.
- G. Adjust aimable heads of emergency lighting units as required to provide required light intensities along egress paths.
- H. Connections:
1. Connect wiring according to Division 26 Section "Conductors and Cables for Electrical Systems."
 2. Ground lighting units. Tighten electrical connectors and terminals, including grounding connections, according to their manufacturer's published torque-tightening values. Where manufacturer's torque values are not indicated, use those specified in UL 486A and UL 486B.
- I. Lenses, Louvers and Reflectors:
1. Exercise particular care when installing fixtures and lamps in fixtures with specular reflector material to prevent smudging or damaging the reflector surface. Wear clean gloves as recommended by the fixture manufacturer.
 2. Do not install removable reflectors, louvers, diffusers, and decorative elements of lighting fixtures until completion of wet work, plastering, painting and general cleanup in the area of the fixtures, but not more than three days before date scheduled for inspections that establish date of Preliminary Acceptance / Substantial Completion.
 3. Parabolic luminaires shall be installed with protective covers, UL-listed for temporary lighting, over louvers. Upon completion of the Work, remove protective covers with clean gloves as recommended by fixture manufacturer.
- J. Accessibility: Install equipment such as junction and pull boxes, fixture housings, transformers, ballasts, switches and controls, and other apparatus that requires occasional maintenance to be accessible and appropriate for mounting and ceiling conditions.
- K. Install fixtures in mechanical areas after ductwork and piping installation. Locate and mount fixtures as indicated on Drawings unless mechanical equipment prohibits or makes it impractical to do so. In such cases, chain or wall mount fixtures so that serviceable equipment is illuminated.
- L. Locate recessed ceiling luminaires as indicated on Drawings. Mount fixtures at heights and locations indicated. Where heights are not indicated or conflicts exist, coordinate final locations with Architect.

- M. In mechanical fan housings, locate fixtures so as not to interfere with belts, filters, and similar items.

3.2 IDENTIFICATION

- A. Install labels with panel and circuit numbers on concealed junction and outlet boxes. Comply with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."

3.3 FIELD QUALITY CONTROL

- A. Normal Lighting: Verify normal operation of each fixture after fixtures have been installed and circuits have been energized with normal power source. Replace or repair malfunctioning fixtures and components.
- B. Test for Emergency Lighting: Interrupt power supply to demonstrate proper operation. Verify transfer from normal power to emergency power and retransfer to normal. Replace or repair malfunctioning fixtures and components, then retest. Repeat procedure until all units operate properly. Note and record the following:
 - 1. Duration of supply.
 - 2. Low battery voltage shutdown.
 - 3. Normal transfer to battery source and retransfer to normal.
 - 4. Low supply voltage transfer.

3.4 PROTECTION AND CLEANING

- A. Provide temporary protection as required to ensure light fixtures will be free of damage at date of Preliminary Acceptance / Substantial Completion.
- B. Clean fixtures internally and externally immediately before, but not more than three days before, date scheduled for inspections that establish dates of Preliminary Acceptance and Substantial Completion. Use methods and materials recommended by manufacturer.

3.5 ADJUSTMENT

- A. After system checkout and adjustment, operate the system for the review of the Commission and Architect. Make adjustments or modifications as required by the Commission or Architect.

END OF SECTION 265100

SECTION 265600 - EXTERIOR LIGHTING**PART 1 - GENERAL****1.1 SUMMARY**

- A. Section includes the following:
 - 1. Exterior luminaires and their associated LED modules, drivers and accessories.
 - 2. Poles, luminaire support structures and their associated accessories.

1.2 DEFINITIONS

- A. ANSI: American National Standards Institute.
- B. Average Rated LED Module Life: The period of time after which 50 percent will have failed and 50 percent will have survived under normal conditions.
- C. CCT: Correlated color temperature.
- D. CRI: Color-rendering index.
- E. IES: Illuminating Engineering Society of North America.
- F. Fixture: A complete lighting unit. Fixtures include LED modules and parts required to distribute the light, position and protect LED modules, and connect them to the power supply.
- G. Luminaire: A complete lighting assembly or product consisting of not less than a light source(s) and related devices to position the light source(s), housing or enclosure, light source control and optics and light source power supply control device(s).
- H. Pole: Luminaire support structure, including tower used for large area illumination.
- I. Solid State Lighting/Luminaires: Light sources and their associated luminaires that produce light (radiant energy) without the use of electrical filaments, excited gas or plasma, such as LED, OLED, or PLED.
- J. UL: Underwriter's Laboratories, Inc.

1.3 SYSTEM DESCRIPTION

- A. The exterior lighting system shall include all lighting fixtures, LED modules, switches, mounting, wiring, control equipment, and accessories, whether or not they are indicated or specified, required for a complete system, as indicated in the Drawings and as specified.
- B. The lighting fixture schedules in the Drawings indicate manufacturer, fixture design, appearance and performance desired.

- C. Verify locations of light fixtures indicated in Drawings and coordinate with other reference data and materials as required prior to installation to ensure locations will not interfere with underground utilities or openings. Alert Architect and Commission Authorized Representative in writing to non-standard modifications required for compliance with the Contract Documents prior to proceeding with the Work.
- D. Where discrepancies are found within the Contract Documents, or additional information is required, immediately contact Architect for clarifications and additional information.
- E. Coordinate installation of lighting system with other trades to prevent delays in the Work and to ensure the lighting fixtures and supports will not be damaged by subsequent construction operations.

1.4 STRUCTURAL ANALYSIS CRITERIA FOR POLE SELECTION

- A. Dead Load: Weight of luminaire and its horizontal and vertical supports, lowering devices, and supporting structure, applied as stated in AASHTO LTS-4.
- B. Live Load: Single load of 500 lbf, distributed as stated in AASHTO LTS-4.
- C. Ice Load: Load of 3 lbf/sq. ft. applied as stated in AASHTO LTS-4.
- D. Wind Load: Pressure of wind on pole and luminaire, calculated and applied as stated in AASHTO LTS-4.
 - 1. Wind speed for calculating wind load for poles exceeding 50 feet in height is 110 mph.
 - 2. Wind speed for calculating wind load for poles 50 feet or less in height is 110 mph.

1.5 SUBMITTALS

- A. Action Submittals:
 - 1. General: Provide separate submittal product data/shop drawings for each fixture type clearly indicating the fixture type designation used in the Drawings and all pertinent options and accessories. Do not group similar fixture types together on a single cut sheet. Submittals that do not indicate option data where multiple selections exist will be returned without being reviewed.
 - 2. Product Data: For each luminaire, pole, and support component, arranged in order of lighting unit designation. Provide separate submittal product data/shop drawings for each fixture type clearly indicating the fixture type designation used in the Drawings and all pertinent options and accessories. Do not group similar fixture types together on a single cut sheet. Submittals that do not indicate option selection where multiple selections exist will be returned without review. Include data on features, accessories, finishes, and the following:
 - a. Physical description of luminaire, including materials, dimensions, effective projected area, and verification of indicated parameters.
 - b. Details of attaching luminaires and accessories.
 - c. Details of installation and construction.
 - d. Luminaire materials such as finish and color information.
 - e. Photoelectric relays.

- f. LED Drivers: Include information as to input watts. Indicate mounting distance limitation and standard wire sizes for remote drivers.
 - g. LED modules, per luminaire tag, including life, output, and energy-efficiency data.
 - h. Materials, dimensions, accessories and finishes of poles.
 - i. Photometric data based on laboratory tests of each luminaire type, complete with indicated LED modules, drivers, and accessories.
 - j. Means of attaching luminaires to supports, and indication that attachment is suitable for components involved.
 - k. Pole and support structure anchor bolt information.
 - l. Manufactured pole foundations, if any.
- 3. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.
 - a. Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection. Include details that cannot be adequately represented to the satisfaction of the Architect in Product Data.
 - b. Anchor-bolt templates keyed to specific poles and certified by manufacturer.
 - c. Wiring Diagrams: For power, signal, and control wiring.
 - d. Photometric data, certified by the manufacturer in accordance with the recommended practices of the IES.
 - e. Quantity and type of LED modules.
- B. Informational Submittals:
 - 1. Pole and Support Component Certificates: Signed by manufacturers of poles, certifying that products are designed for indicated load requirements in AASHTO LTS-4-M and that load imposed by luminaire and attachments has been included in design. The certification shall be based on design calculations by a State of Maryland Registered Structural Engineer.
 - 2. Warranty: Sample of special warranty.
- C. Closeout Submittals:
 - 1. Operation and Maintenance Data: For luminaires and poles to include in emergency, operation, and maintenance manuals. Include manufacturers' recommended maintenance practices for each fixture type including, but not limited to, the following:
 - a. Tools required.
 - b. Acceptable cleaners and recommended cleaning practices.
 - c. Replacement parts list.
 - d. Manufacturer service department contact information/Qualified Service Agencies.
 - e. Submittal data.
 - f. Operation data.
 - g. Intended operation narrative.
 - 2. Record Drawings: Submit record drawings of the actual installation within thirty (30) days of date of Preliminary Acceptance / Substantial Completion.
 - 3. Re-Commissioning Data: Submit manual containing all information required for re-commissioning of the installations.
 - a. Submit two (2) copies of manual within thirty (30) days of date of Preliminary Acceptance / Substantial Completion.

- b. Submit re-commissioning manuals in heavy-duty, 3-ring binders. Submit manuals in accordance with Division 01 Section "Operations and Maintenance Data."

1.6 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled by a testing agency acceptable to authorities having jurisdiction, and marked for intended location and application.
- B. Solid State Lighting/Luminaires:
 - 1. Luminous flux, luminaire efficiency and chromaticity shall be tested, measured and reported in accordance with the most current versions of IES documents LM-79 and LM-80.
 - 2. Chromaticity ranges for "white light" products, with various correlated color temperatures, shall be provided in accordance with ANSI/NEMA-C78.377.
 - 3. Drivers and power supplies shall be provided in accordance with the requirements of ANSI/NEMA-C82.SSL1, and their maximum allowable harmonic emission limits shall be in accordance with ANSI/NEMA-C82.77.
 - 4. Shall be provided with a U.S. Department of Energy (DOE) "Lighting Facts" label indicating their specific performance characteristics, tested and reported in accordance with the requirements of the most current version of IES LM-79.
- C. Commissioning: When required for the Project, Contractor shall assign representative(s) with expertise and authority to act on its behalf. The representative(s) shall perform commissioning activities including, but not limited to, the following:
 - 1. Review submittals relative to exterior lighting systems being commissioned.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Package aluminum poles for shipping according to ASTM B 660.
- B. Store poles on decay-resistant-treated skids at least 12-inches above grade and vegetation. Support poles to prevent distortion and arrange to provide free air circulation.
- C. Handle poles so they will not be damaged. Do not use pointed tools that can indent pole surface more than 1/4-inch deep. Do not apply tools to section of pole to be installed below ground line.
- D. Retain factory-applied pole wrappings on metal poles until right before pole installation.
- E. Deliver exterior lighting fixtures individually wrapped in factory-fabricated fiber Commission type containers or equivalent.
- F. Handle exterior lighting fixtures carefully to prevent breakage, denting and scoring the fixture finish. Do not install damaged lighting fixtures; replace and return damaged units to equipment manufacturer.
- G. Store lighting fixtures in a clean, dry space. Store in original cartons and protect from dirt, physical damage, weather and construction traffic.

1.8 WARRANTY

- A. Special Warranty: Submit a written warranty, beginning from date of Preliminary Acceptance / Substantial Completion, and executed by the Contractor, manufacturer, and Installer agreeing to repair or replace products or components that fail in materials or workmanship; that corrode; or that fade, stain, perforate, erode, or chalk due to effects of weather or solar radiation within the specified warranty period. Damage due to lightning, hail, vandalism, abuse, or unauthorized repairs or alterations shall be excluded from special warranty coverage. Warranty shall include all materials and components.
1. Warranty Period for LED Luminaires: Five (5) years from date of Preliminary Acceptance / Substantial Completion.
 2. Warranty Period for Metal Corrosion: Five (5) years from date of Preliminary Acceptance / Substantial Completion.
 3. Warranty Period for Color Retention: Five (5) years from date of Preliminary Acceptance / Substantial Completion.
 4. Warranty Period for LED Modules and Fuses: Two (2) years from date of Preliminary Acceptance / Substantial Completion.
 - a. Replace LED modules and fuses that fail within 12 months from date of Preliminary Acceptance / Substantial Completion.
 - b. Furnish replacement LED modules and fuses that fail within the second 12 months from date of Preliminary Acceptance / Substantial Completion.
 5. Warranty Period for Ballasts and Drivers: Five (5) years from date of Preliminary Acceptance / Substantial Completion.
 6. Warranty Period for Poles: Three (3) years from date of Preliminary Acceptance / Substantial Completion. Warranty shall include pole finish.
 7. Warranty Period for LED Drivers: Five (5) years from date of Preliminary Acceptance / Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Products: Subject to compliance with requirements indicated and the design criteria specified in the Luminaire Schedule, provide one of the products specified in the Luminaire Schedule indicated in the civil drawings.
- B. Approved Manufacturers:
1. Trilux Lighting US Inc.

2.2 LUMINAIRES

- A. Luminaires shall comply with UL 1598 and be designed, manufactured, listed, and labeled for installation in wet locations and for outdoor service. Make components, including nuts, bolts, rivets, springs, and similar parts, of corrosion resistant materials. Fixtures that are aimed upward shall be listed and labeled for installation in wet locations in that position.

- B. Provide anodized aluminum for aluminum parts of exterior fixtures that are not specified as requiring a painted finish.
- C. Lateral Light Distribution Patterns: Comply with IES RP-8 for parameters of lateral light distribution patterns indicated for luminaires.
- D. Metal Parts: Free of burrs and sharp corners and edges.
- E. Sheet Metal Components: Corrosion-resistant aluminum, unless otherwise indicated. Form and support to prevent warping and sagging.
- F. Housings: Rigidly formed, weather- and light-tight enclosures that will not warp, sag, or deform in use. Provide filter/breather for enclosed luminaires.
- G. Doors, Frames, and Other Internal Access: Smooth operating, free of light leakage under operating conditions, and designed to permit relamping without use of tools. Designed to prevent doors, frames, lenses, diffusers, and other components from falling accidentally during relamping and when secured in operating position. Doors shall be removable for cleaning or replacing lenses. Designed to disconnect driver when door opens.
- H. Exposed Hardware Material: Stainless steel.
- I. Plastic Parts: High resistance to yellowing and other changes due to aging, exposure to heat, and UV radiation.
- J. Light Shields: Metal baffles, factory installed and field adjustable, arranged to block light distribution to indicated portion of normally illuminated area or field.
- K. Reflecting surfaces shall have minimum reflectance as follows, unless otherwise indicated:
 - 1. White Surfaces: 85 percent.
 - 2. Specular Surfaces: 83 percent.
 - 3. Diffusing Specular Surfaces: 75 percent.
- L. Lenses and Refractors Gaskets: Use heat- and aging-resistant resilient gaskets to seal and cushion lenses and refractors in luminaire doors.
- M. Luminaire Finish: Manufacturer's standard paint applied to factory-assembled and -tested luminaire before shipping. Where indicated, match finish process and color of pole or support materials.
 - 1. Where luminaire products are not indicated to be field painted and are not indicated to match the finish process and color of pole or support materials, provide the following:
 - a. Factory-Applied Finish for Aluminum Luminaires: Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.
 - 1) Finish designations prefixed by AA comply with the system established by the Aluminum Association for designating aluminum finishes.
 - 2) Class I, Clear Anodic Finish: AA-M32C22A41 (Mechanical Finish: medium satin; Chemical Finish: etched, medium matte; Anodic Coating: Architectural Class I, clear coating 0.018 mm or thicker) complying with AAMA 611.

- N. Factory-Applied Labels: Comply with UL 1598. Include recommended lamps and ballasts. Labels shall be located where they will be readily visible to service personnel, but not seen from normal viewing angles when lamps are in place.

1. Label shall include the following lamp and ballast characteristics:

- a. "USES ONLY" and include specific lamp type.
- b. CCT and CRI for all luminaires.

2.3 LED DRIVERS

- A. LED drivers shall be factory provided by the respective luminaire manufacturers, and shall be suitable for their intended use, to operate the designated LED modules listed in the Luminaire Schedule, and as specified herein, to their full light output.
- B. Comply with the requirements of the Federal Communications Commission (FCC) rules and regulations, Title 47 CFR part 18, Non-Consumer (Class A) for EMI/RFI (conducted and radiated).
- C. Provide complete connection to LED-type luminaires through both integrally installed and remote electronic drivers.
- D. Shall be totally enclosed within a metallic enclosure, and shall be provided with integral leads color coded per ANSI C82.11, or with poke-in style wire retaining connectors.
- E. Provide identical drivers within each luminaire type.
- F. Provide UL listed and labeled drivers. Provide drivers with temperature ratings appropriate to the installation.
- G. Fixtures intended to be dimmed shall have dimming driver compatible with the specified dimmer controls.
- H. Remote Drivers:
1. Remove drivers are specifically not indicated in Drawings. Install remove drivers in a readily accessible, dry, indoor, concealed location in accordance with the manufacturer's written instructions.
 2. Provide ventilated metal enclosures for remove drivers furnished as loose equipment. All wiring related to remove drivers and the related LED luminaires shall be installed in conduit.
 3. Verify and comply with the remote distance limitations specified by the luminaire/driver manufacturer.

2.4 POLES AND SUPPORT COMPONENTS

- A. Structural Characteristics: Comply with AASHTO LTS-4.
1. Wind-Load Strength of Poles: Adequate at indicated heights above grade without failure, permanent deflection, or whipping in steady winds of speed indicated in "Structural Analysis Criteria for Pole Selection" Article.
 2. Strength Analysis: For each pole, multiply the actual equivalent projected area of luminaires and brackets by a factor of 1.1 to obtain the equivalent projected area to be used in pole selection strength analysis.

- B. Luminaire Attachment Provisions: Comply with luminaire manufacturers' mounting requirements. Use stainless-steel fasteners and mounting bolts, unless otherwise indicated.
- C. Mountings, Fasteners, and Appurtenances: Corrosion-resistant items compatible with support components.
 - 1. Materials: Shall not cause galvanic action at contact points.
 - 2. Anchor Bolts, Leveling Nuts, Bolt Caps, and Washers: Hot-dip galvanized after fabrication, unless stainless-steel items are indicated.
 - 3. Anchor-Bolt Template: Plywood or steel.
- D. Concrete Pole Foundations: Cast in place, not less than 24-inches in diameter, and extending above and below grade as indicated in Drawings. Install with anchor bolts to match pole-base flange. Provide raceways, grounding electrodes, and other electric work relating to the pole foundations. Refer to Division 03 Section "Cast-in-Place Concrete" for concrete, reinforcement, and formwork.
- E. Power-Installed Screw Foundations: Factory fabricated by pole manufacturer, with structural steel complying with ASTM A 36 and hot-dip galvanized according to ASTM A 123; and with top-plate and mounting bolts to match pole base flange and strength required to support pole, luminaire, and accessories.
- F. Poles shall be provided with handhole, anchor bolt covers, and grounding lug.

2.5 POLE ACCESSORIES

- A. Duplex Receptacle: Where indicated, provide a 120V, 20A receptacle in a weatherproof assembly complying with Division 26 Section "Wiring Devices" for ground-fault circuit-interrupter type.
 - 1. Recessed, 12 inches above finished grade.
 - 2. Nonmetallic polycarbonate plastic or reinforced fiberglass cover, color to match pole, that when mounted results in NEMA 250, Type 3R enclosure.
 - 3. With cord opening.
 - 4. With lockable hasp and latch that complies with OSHA lockout and tag-out requirements.
- B. Base Covers: Manufacturers' standard metal units, arranged to cover pole's mounting bolts and nuts. Finish same as pole.

2.6 EXTERIOR LIGHTING DEVICES

- A. Lighting fixtures are scheduled in Drawings. Reference the Drawings for the specified and approved fixtures.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas and conditions, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 LUMINAIRE INSTALLATION

- A. LED Modules: Install LED modules in each luminaire in accordance with manufacturer's instructions.
- B. Fasten luminaire to indicated structural supports.
 - 1. Use fastening methods and materials approved in writing by manufacturer.
- C. Adjust luminaires that require field adjustment or aiming. Include adjustment of photoelectric device to prevent false operation of relay by artificial light sources.
- D. Where fixtures are wall-mounted and protrude from the wall surface, provide additional structural support within the wall framing to accommodate the extra moment force created by the fixture.

3.3 POLE INSTALLATION

- A. Align pole foundations and poles for optimum directional alignment of luminaires and their mounting provisions on the pole.
- B. Clearances: Maintain the following minimum horizontal distances of poles from surface and underground features, unless otherwise indicated on Drawings:
 - 1. Fire Hydrants and Storm Drainage Piping: 60 inches.
 - 2. Water, Gas, Electric, Communication, and Sewer Lines: 10 feet.
 - 3. Trees:
 - a. Less than 4-inch caliper: 15 feet.
 - b. Greater than 4-inch caliper: Locate outside dripline.
- C. Concrete Pole Foundations:
 - 1. Set anchor bolts according to anchor-bolt templates and requirements furnished by pole manufacturer.
 - 2. Exposed concrete surfaces of bases for all light poles and light fixtures shall be free of voids and tops of bases shall be level.
 - 3. Remove all rough edges from exposed surfaces. Leave exposed surfaces smooth.
 - 4. Remove all exposed forming materials.
 - 5. Comply with additional concrete materials, installation, and finishing requirements specified in Division 03 Section "Cast-in-Place Concrete."
- D. Foundation-Mounted Poles: Mount pole with leveling nuts, and tighten top nuts to torque level recommended by pole manufacturer.
 - 1. Use anchor bolts and nuts selected for the application and approved by manufacturer.
 - 2. Grout void between pole base and foundation. Use nonshrink or expanding concrete grout firmly packed to fill space.
 - 3. Install base covers, unless otherwise indicated.
 - 4. Use a short piece of 1/2-inch-diameter pipe to make a drain hole through grout. Arrange to drain condensation from interior of pole.

- E. Poles and Pole Foundations Set in Concrete Paved Areas: Install poles with minimum of 6-inch-wide, unpaved gap between the pole or pole foundation and the edge of adjacent concrete slab. Fill unpaved ring with pea gravel to a level 1 inch below top of concrete slab.
- F. Raise and set poles using web fabric slings (not chain or cable).

3.4 INSTALLATION OF INDIVIDUAL GROUND-MOUNTED LUMINAIRES

- A. Install on concrete base with top 4-inches above finished grade or surface at luminaire location. Cast conduit into base, and finish by troweling and rubbing smooth. Concrete materials, installation, and finishing are specified in Division 03 Section "Cast-in-Place Concrete."

3.5 CORROSION PREVENTION

- A. Aluminum: Do not use in contact with earth or concrete. When in direct contact with a dissimilar metal, protect aluminum by insulating fittings or treatment.
- B. Steel Conduits: Comply with Division 26 Section "Raceways and Boxes for Electrical Systems." In concrete foundations, wrap conduit with 0.010-inch-thick, pipe-wrapping plastic tape applied with a 50 percent overlap.

3.6 GROUNDING

- A. Ground metal poles and support structures according to Division 26 Section "Grounding and Bonding for Electrical Systems."
 - 1. Install a minimum of a 3/4-inch diameter by 8-foot copper grounding electrode for each pole, unless otherwise indicated, and installed as indicated in Drawings.
 - 2. Install grounding conductor pigtail in the base for connecting luminaire to grounding system.

3.7 RACEWAYS AND BOXES

- A. Plastic conduit shall be used where indicated in Drawings, in unpaved areas and lawn areas. The conduit shall be Schedule 40, UL Listed, polyvinyl chloride conduit, and not less than ~~1-1/2~~ 1 inch in diameter.
- B. Rigid PVC schedule 40 conduit shall be provided on all underground raceway.
- C. Provide pull line in empty conduit and duct.
- D. Comply with the additional requirements of Division 26 Section "Raceways and Boxes for Electrical Systems."

3.8 IDENTIFICATION

- A. Provide vinyl tagging with panel source and circuit number on wiring at handhole in each pole and at each exterior box.

- B. Identify each exterior box with 1-1/2 inch high black letters and numbers on yellow weatherproof, pressure-sensitive adhesive vinyl on the covers. Labels shall be Brady #1530.
- C. Identify each pole with 3-1/2 inch high black letters and numbers on yellow weatherproof, pressure sensitive adhesive vinyl placed 78-inches above grade facing the parking areas, or facing the road if the pole is at the road. Labels shall be Brady #1550.
- D. Comply with the additional requirements of Division 26 Section "Identification for Electrical Systems."

3.9 CLEANING

- A. Clean fixtures internally and externally after completion of installation. Use methods and materials recommended by manufacturer.

3.10 CONTRACTOR STARTUP AND REPORTING

- A. Inspect each installed fixture for damage. Replace damaged fixtures and components.
- B. Illumination Observations: Verify normal operation of lighting units after installing luminaires and energizing circuits with normal power source.
 - 1. Verify operation of photoelectric controls.
- C. Replace fixtures that show evidence of corrosion during project warranty period.
- D. Prepare a written report of tests, inspections, observations, and verifications indicating and interpreting results. If adjustments are made to lighting system, retest to demonstrate compliance with requirements.
- E. Contractor shall train the Commission's designated maintenance personnel on the proper operation and maintenance of all products prior to Final Completion.

3.11 COMMISSIONING AND DEMONSTRATION

- A. After system checkout and adjustment, the contractor shall operate the system for the review of Architect and Commission Authorized Representative, and shall make all adjustments and modifications as required by Architect and Commission Authorized Representative.

END OF SECTION 265600

SECTION 270500 - COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Provide and install a complete structured cabling, security and background music/overhead paging system support system for the LIDL Prototype Retail Grocery Store. The support systems shall include, but not be limited to, cabling pathways, grounding and bonding systems and appropriate mounting hardware and accessories. Build out of the telecommunications room and telecommunication enclosure shall be included in the scope of the project. The telecommunication equipment racks will be provided by LIDL through the "Emerson White Glove Service" and shall be installed within the scope of this project. Installation shall include setting or mounting the equipment racks in place, completion of grounding and bonding systems to properly ground and bond all equipment, members and items within the equipment racks and to connect the pre-mounted power strips within the equipment racks to the electrical receptacles. Power strip cables and all ground and bonding conductors shall be properly dressed and supported within the equipment racks and to the appropriate cable pathways. All support systems shall be installed to meet the industry standards of the cabling and connectivity being supported in addition to local and national codes.

1.3 REFERENCED STANDARDS AND CODES

- A. NFPA 70 National Electric Code (NEC).
- B. Authority Having Jurisdiction.
- C. UL® for wiring: UL® Standard 910 "Test method for fire and smoke characteristics of cable used in air handling spaces." Provide products that are UL® listed and labeled for such use. UL® testing bulletin. Underwriters Laboratories (UL®) cable certification and follow up program.
- D. American National Standards Institute/Telecommunications Industry Association. ANSI/TIA, including associated Addenda:
 - 1. ANSI/TIA-568-C.0 – Generic Telecommunications Cabling for Customer Premises.
 - 2. ANSI/TIA-568-C.0-1 – Generic Telecommunications Cabling for Customer Premises – Addendum 1, Updated References for Balanced Twisted-Pair.
 - 3. ANSI/TIA-568-C.0-2 – Generic Telecommunications Cabling for Customer Premises – Addendum 2, General Updates
 - 4. ANSI/TIA-568-C.1 – Commercial Building Telecommunications Cabling Standard.
 - 5. ANSI/TIA-568-C.1-1 – Commercial Building Telecommunications Cabling Standard – Addendum 1, Pathways and Spaces.
 - 6. ANSI/TIA-568-C.1-2 – Commercial Building Telecommunications Cabling Standard – Addendum 2, General Updates.
 - 7. ANSI/TIA-568-C.2 – Balanced Twisted Pair Telecommunications Cabling and Components Standard.
 - 8. ANSI/TIA-568-C.3 – Optical Fiber Cabling Components Standard.
 - 9. ANSI/TIA-568-C.3-1 – Optical Fiber Cabling Components Standard – Addendum 1, Addition of OM4 Cabled Optical Fiber and Array Connectivity.
 - 10. ANSI/TIA-569-C – Telecommunications Pathways and Spaces.
 - 11. TIA-590-A – Standard for Physical Location and Protection of Below-Ground Fiber Optic

- Cable Plant.
12. ANSI/TIA-606-AB – Administration Standard for Commercial Telecommunications Infrastructure.
 13. ANSI/TIA-606-A-1 – Administration of Equipment Rooms and Data Center Computer Rooms.
 14. ANSI/TIA-607-B – Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises.
 15. ANSI/TIA-607-B-1 – Generic Telecommunications Bonding and grounding (Earthing) for Customer Premises Addendum 1 – External Grounding
 16. ANSI/TIA-758-B – Customer Owned Outside Plant Telecommunications Infrastructure Standard.
 17. ANSI/TIA-1005-A – Telecommunications Infrastructure Standard for Industrial Premises.
 18. TIA TSB-155-A – Guidelines for the Assessment and Mitigation of Installed Category 6A Cabling to Support 10GBase-T
 19. TIA TSB-184 – Guidelines for Supporting Power Delivery Over Balanced Twisted-Pair Cabling.
 20. TIA-TSB-190 – Guidelines on Shared Pathways and Shared Sheaths.
- E. ANSI/NECA/BICSI 568-2006 – Installing Commercial Building Telecommunications Cabling.
- F. NECA/BICSI 607-2011 – Standard for Telecommunications Bonding and Grounding Planning and Installation Methods for Commercial Buildings.
- G. National Electrical Manufacturers Association (NEMA).
- H. Institute of Electrical and Electronic Engineers (IEEE).
- I. American Society for Testing Materials (ASTM)

1.4 RELATED SECTIONS

- | | | |
|----|--------|--|
| A. | 260526 | Grounding and Bonding for Electrical Systems |
| B. | 260529 | Hangers and Supports for Electrical Systems |
| C. | 260536 | Cable Trays for Electrical Systems |
| D. | 260539 | Underfloor Raceways for Electrical Systems |
| E. | 270526 | Grounding and Bonding for Communications Systems |
| F. | 270528 | Pathways for Communications Systems |
| G. | 270553 | Identification for Communications Systems |
| H. | 271100 | Communications Equipment Room Fittings |
| I. | 271500 | Communications Horizontal Cabling |

1.5 DEFINITIONS / ACRONYMS / ABBREVIATIONS

- | | | |
|----|------|---------------------------------------|
| A. | AHJ | Authority Having Jurisdiction |
| B. | ANSI | American National Standards Institute |

C.	AWG	American Wire Gauge
D.	CAT 6A	Category 6A performance as defined by ANSI/TIA-568-C.2
E.	EC	Electrical Contractor
F.	EIA	Electronic Industries Alliance
G.	ER	Equipment Room
H.	EOR	Engineer of Record
I.	GC	General Contractor
J.	IEC	International Electrotechnical Commission
K.	ISO	International Organization for Standardization
L.	MM	Multimode (Fiber Optic Cable)
M.	NECA	National Electrical Contractors Association
N.	NEMA	National Electrical Manufacturers Association
O.	OM4	Optical Multimode 4 "laser optimized" performance as defined by ISO/IEC 11801 Amd2 and ANSI/TIA-568-C.3-1
P.	OS1	Optical Singlemode 1 performance as defined by ISO/IEC 11801 and ANSI/TIA-568 C.3
Q.	OS2	Optical Singlemode 2 "low water peak" performance as defined by ISO/IEC 11801 Amd2 and ANSI/TIA-568-C.3
R.	SCS	Structured Cabling System
S.	SM	Singlemode (Fiber Optic Cable)
T.	STP	Shielded Twisted Pair Cable
U.	UTP	Unshielded Twisted Pair Cable
V.	TBB	Telecommunications Bonding Backbone
W.	TBBIBC	Telecommunications Bonding Backbone Interconnection Bonding Conductor
X.	TC	Telecommunications Contractor
Y.	TIA	Telecommunications Industry Association
Z.	TGB	Telecommunications Ground Bus Bar
AA.	TMGB	Telecommunications Main Ground Bus Bar
BB.	TO	Telecommunications Outlet

CC. TR Telecommunication Room

DD. UNO Unless Noted Otherwise

1.6 SUBMITTALS

A. Product Data

1. Submit the following in accordance with the general conditions and Division 01 and Division 20 specifications of contract. Submittals to include index with description, specification reference, manufacturer, part number, quantity and page number for manufacturer product specification sheet. Manufacturer's product specifications sheets shall indicate product part number. Where multiple products are indicated, products being submitted shall be clearly indicated on a single specification sheet. Products requiring submittals shall include but not be limited to the following:
 - a. All cable and wire including patch cables and cross-connect wire.
 - b. All connectors and required tools.
 - c. All termination system components for each cable type.
 - d. All equipment room and telecommunications room equipment racks, cabinets, horizontal and vertical cable management, ladder tray, installation hardware and other support equipment.
 - e. All grounding system components.
 - f. All fire seal systems, including manufacturer published installation requirements.
 - g. All cable raceways and support hardware.

B. Samples

1. Contractor shall provide samples to the Architect EOR for review and approval. Samples shall be provided for the following:
 - a. Each configuration of work area
 - b. Termination hardware labeling
 - c. Backbone labeling scheme
 - d. Horizontal cabling labeling scheme

C. Test Plans

1. Contractor shall provide detailed test plans and test documentation as specified for each sub-system.

D. Shop Drawings

1. Electrical Contractor shall produce, provide and coordinate shop drawings for all conduits, cable trays and other equipment. All shop drawings are to be coordinated with other trades, architect and EOR.
2. Telecommunications Contractor shall produce, provide and coordinate shop drawings for all ladder racks, equipment cabinets, equipment racks and other equipment. All shop drawings shall be coordinated with other trades, architect and the EOR.
3. All shop drawings are to be produced using AutoCAD version 2004 or later.
4. Cross-connection schedule is to be produced in a matrix / schedule format using the latest version of Microsoft Excel. Include soft copy and hard copies.

E. As-Built Documentation

1. Contractor shall provide complete as-built documentation. As-built documentation shall include AutoCAD drawings indicating all conduits, cable trays, ladder racks, equipment cabinets, equipment racks and other equipment. All TOs are to be indicated on as-built

- drawings including TO numbers. Cable paths for all backbone cables are to be indicated on as-built drawings.
2. Contractor shall provide complete as-built documentation. As-built documentation shall include AutoCAD drawings indicating all conduits, cable trays and other equipment.
 3. Contractor shall provide 3 hard copies and 1 soft copy on CD.
 4. As-built drawings are to be produced using AutoCAD version 2004 or later. The following criteria are to be used:
 - a. Floor/conduit/zone plans are to be produced at a scale of 1/8"=1'0".
 - b. TR and ER partial plans are to be produced at a scale of 1/4" = 1' 0" or 1/2" = 1' 0" as appropriate to the complexity and size of the room in being depicted.
 - c. Alignment targets (targets) are to be included in each file that references a building floor plan.
 - 1) The targets are to present a cross-hair registered to building column line intersections and labeled with the column intersection thus referenced.
 - 2) The targets shall be AutoCAD blocks with the origin at the intersection of the cross-hairs and attributes for the column information.
 - 3) The alignment targets are to reside on a separate layer created for this use.
 - 4) A minimum of two (2) such alignment targets are required for each contiguous area referenced, e.g. if two floors are shown in the same drawing, two alignment targets are required per floor.
 - d. Drawing files shall be prepared using English Architectural units.
 - e. No custom line-styles are to be used in the preparation of the drawing files.
 5. Provide a cross connect schedule for all backbone cables installed. Schedules shall indicate all pairs of the backbone cables and all installed cross connections. Schedules are to be created in latest version Microsoft Excel or other approved format. Include soft copy and hard copies.

F. Test Equipment

1. TC shall submit manufacturer data sheets for all test equipment to be utilized in the certification of the structured cabling system.
2. The TC shall submit current calibration certificates to EOR for all equipment to be utilized in the testing of the structured cabling system. All calibration certificates are to be issued by the test equipment manufacturer. Certificates are to be received by EOR no less than seven days prior to beginning any testing.

G. Test Data.

1. The contractor shall provide test data as specified for each sub-system.

H. Warranty

1. Contractor shall provide an extended 5-year warranty covering all work, including but not limited to: equipment, devices and labor.

1.7 CONTRACTOR QUALIFICATIONS

- A. The TC selected for this project shall be certified by the manufacturer(s) of the products specified in this scope of work, adhere to the design engineering, installation and testing procedures and utilize the authorized manufacturer(s) components in completing this project.
- B. The TC shall be experienced in all aspects of the work required to complete this project and shall be required to demonstrate direct experience on recently installed systems of similar size and type.
- C. The TC shall own and maintain equipment and tools required for the installation and testing of Category 6A copper and optical fiber structured cabling systems. The TC shall also employ

personnel who are adequately trained in the uses of required tools and equipment.

- D. TC shall submit letters of certification from the proposed system manufacture indicating the TC is an approved installer of proposed system and has had personnel trained in the proper installation and testing procedures of said system.

1.8 SCOPE OF WORK

A. Telecommunications Scope of Work.

1. The General Contractor (GC) and Telecommunications contractor (TC) shall carefully examine the contract documents and thoroughly become familiar with the building standards and local conditions relating to the work. Failure to do so will not relieve the contractor of the obligations of the contract.
2. Provide UL® listed fireseal systems for all telecommunications cabling pathways using caulk putty, pillows or similar devices per manufacturer's instructions to maintain existing and new fire ratings. Verify rating conditions and locations prior to final bids. All open sleeves, slots or other penetrations shall be fire sealed inside after all cabling is completely installed. Fireseal methods shall be submitted by contractor and shall be subject to the approval of the owner. Contractor shall provide caps and fireseal to maintain the original cores that are not to be used and/or reused. Contractor shall verify exact quantities in the field prior to submission of bids.
3. Provide all horizontal and backbone cabling grounding, and accessories as specified for a complete and operational system.
4. Properly ground telecommunications equipment, including but not limited to racks, cabinets, ladder tray, riser cables, etc. in accordance with manufacturer recommendations and ANSI/TIA-607-B, NFPA 70, all applicable building codes and specification section 27 05 26.
5. Label all telecommunications cables, cable pathways with permanent machine typed labels in accordance with ANSI/TIA-606-AB standard, telecommunications drawings and specifications.
6. Completely test the telecommunications system(s) in accordance with applicable standards as referenced in the telecommunication systems specifications.
7. Provide electronic formatted as-built drawings to owner upon completion of project.
8. Assist access providers with coordination of related work.
9. Provide all required voice type cross connects in the TR(s) from the backbone hardware to the workstation hardware and the main cross connect from the backbone hardware to the VoIP, entrance hardware, other backbone cables or other hardware as required for a complete and operational system.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 270500

SECTION 270526 - GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS**PART 1 – GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section includes the requirements for a uniform telecommunications bonding and grounding infrastructure system including grounding electrodes and conductors; equipment grounding conductors; bonding methods and materials; grounding of telecommunications systems and equipment, and requirements for grounding for protection of life, equipment, circuits and systems. Grounding requirements specified in this section may be supplemented in other sections of these specifications.

1.3 REFERENCED STANDARDS AND CODES

- A. NFPA 70 National Electric Code.
- B. Authority having jurisdiction.
- C. Local and State Building Codes.
- D. NFPA 262 "Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air- Handling Spaces" 2002 Edition Provide Products that are UL® listed and labeled for such use. Refer to UL® testing bulletin and Underwriters Laboratories (UL®) cable certification and follow up program.
- E. American National Standards Institute/Telecommunications Industry Association.
- F. ANSI/TIA, including associated Addenda:
 - 1. ANSI/TIA-568-C.0 – Generic Telecommunications Cabling for Customer Premises.
 - 2. ANSI/TIA-568-C.1 – Commercial Building Telecommunications Cabling Standard.
 - 3. ANSI/TIA-568-C.2 – Balanced Twisted Pair Telecommunications Cabling and Components Standard.
 - 4. ANSI/TIA-568-C.3 – Optical Fiber Cabling Components Standard.
 - 5. ANSI/TIA-569-C – Pathways and Spaces.
 - 6. TIA-590-A – Standard for Physical Location and Protection of Below-Ground Fiber Optic Cable Plant.
 - 7. ANSI/TIA-606-AB – Administration Standard for Commercial Telecommunications Infrastructure.
 - 8. ANSI/TIA-607-B – Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises.
 - 9. ANSI/TIA-607-B-1 – Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises – Addendum 1 – External Grounding
 - 10. ANSI/TIA-758-B – Customer Owned Outside Plant Telecommunications Infrastructure Standard.
 - 11. ANSI/TIA-942 – Telecommunications Standard for Data Centers.
 - 12. ANSI/TIA-1005-A – Telecommunications Infrastructure Standard for Industrial Premises.

- G. ANSI/NECA/BICSI 568-2006 Installing Commercial Building Telecommunications Cabling.
- H. NECA/BICSI 607-2011 – Standard for Telecommunications Bonding and Grounding Planning and Installation Methods for Commercial Buildings.
- I. National Electrical Manufacturers Association (NEMA).
- J. Institute of Electrical and Electronic Engineers (IEEE).
- K. American Society for Testing Materials (ASTM).
- L. NETA ATS (InterNational Electrical Testing Association) - Acceptance Testing Specifications for Electrical Power Distribution Equipment and Systems.
- M. ANSI/NFPA-780 Lightning Protection Code

1.4 RELATED SECTIONS

- A. 260526 Grounding and Bonding for Electrical Systems
- B. 260536 Cable Trays for Electrical Systems
- C. 260539 Underfloor Raceways for Electrical Systems
- D. 270500 Common Work Results for Communications Systems
- E. 270528 Pathways for Communications Systems
- F. 270553 Identification for Communications Systems
- G. 271100 Communications Equipment Room Fittings
- H. 271500 Communications Horizontal Cabling

1.5 DEFINITIONS/ACRONYMS/ABBREVIATIONS

Backbone:	A facility between telecommunications rooms, the facilities, and the equipment rooms within or between buildings.
Bonding:	The permanent joining of metallic parts to form an electrically conductive path that will assure electrical continuity and the capacity to conduct safely any current
Bonding Conductor for Telecommunications	A conductor that interconnects the telecommunications bonding infrastructure to the buildings service equipment (power) ground.
Commercial Building:	A building or portion thereof that is intended for office use.
Common Bonding Network (CBN):	The set of metallic components that are interconnected to form the principle means for effectively bonding equipment inside the building to the grounding electrode system.
Conduit:	A raceway of circular cross-section.
Conduit Run:	Multiple sections of conduit that are joined together with

Conduit System	Any combination of ducts, conduits, junction boxes, maintenance holes, handholes, and vaults joined to form an integrated whole.
Connecting Hardware:	A device providing mechanical cable terminations.
Earth:	See Ground.
Effectively Grounded:	Intentionally connected to earth through a ground connection or connections of sufficiently low impedance and having sufficient current-carrying capacity to prevent the buildup of voltages that may result in undue hazard to connected equipment or to persons.
Entrance Facility (EF):	An entrance to a building for both public and private network service cables including the entrance point at the building wall and continuing to and including the
Entrance Point:	The point of emergence of telecommunications conductors through an exterior wall, a concrete floor slab, or from a rigid metal conduit or an intermediate
Entrance Room or Space:	A space in which the joining of inter- or intra-building telecommunications backbone facilities takes place. An entrance room may also serve as an equipment room.
Equipment Room (ER):	An environmentally controlled centralized space for telecommunications equipment that usually houses a main or intermediate cross-connect.
Equipotential Bonding:	Bonding between metallic components to achieve a substantially equal potential.
Exothermic Weld:	A method of permanently bonding two metals together by a controlled heat reaction resulting in a molecular
Ground:	A conducting connection, whether intentional or incidental, between an electrical circuit or equipment and the earth, or to some conducting body that serves
Grounding Conductor:	A conductor used to connect the grounding electrode to the building's main grounding busbar.
Grounding Electrode:	A conductor, usually a rod, pipe or plate (or group of conductors) in direct contact with the earth for the purpose of providing a low impedance connection to the
Grounding Electrode Conductor:	The conductor used to connect the grounding electrode to the equipment grounding conductor, or to the grounded conductor of the circuit at the service equipment, or at the source of a separately derived system.
Grounding Electrode System:	One or more grounding electrodes that are connected together.
Grounding Equalizer (GE):	The conductor that interconnects elements of the telecommunications grounding infrastructure.
Infrastructure (telecommunications):	A collection of those telecommunications components, excluding equipment, that together provide the basic support for the distribution of information within a building
Listed:	Equipment or materials included in a list published by an organization acceptable to the authority having jurisdiction and concerned with product evaluation, that maintains periodic inspection of production of listed equipment or materials, and whose listing states either that the equipment or materials meet appropriate designated standards or has been tested and found

Mesh Bonding Network:	A bonding network to which all associated equipment (e.g.: cabinets, frames, racks, trays, pathways) are connected using a bonding grid, which is connected to multiple points on the common bonding network.
Pathway:	A facility for the placement of telecommunications cable.
Protector:	A device consisting of one or more protector units intended to limit abnormal voltages or currents on metallic communications circuits. Includes a mounting assembly
Protector Unit:	A device intended to protect against either over voltage or over current, or both. The unit may contain carbon arresters, gas tubes, solid state devices, heat coils, PTC devices, or a combination of these components for a
Sleeve:	An opening, usually circular, through the wall, ceiling, or floor to allow the passage of cables.
Telecommunications:	Any transmission, emission and reception of signs, signals, writings, images and sounds that is information of any nature by cable, radio, optical or other
Telecommunications Bonding Backbone (TBB):	A conductor that interconnects the telecommunications main grounding busbar (TMGB) to the telecommunications grounding busbar(s) (TGB).
Telecommunications Equipment Bonding	A conductor that connects the Telecommunications Main Grounding Busbar (TMGB) or Telecommunications Grounding Busbar (TGB) to equipment racks or cabinets.
Telecommunications Grounding Busbar (TGB):	The interface to the building telecommunications grounding system generally located in the telecommunications room (TR). A common point of connection for telecommunications system and equipment bonding to ground, and located in the TR or
Telecommunications Room (TR):	An enclosed space for housing telecommunications equipment, cable terminations, and cross-connect cabling, that is the recognized location of the horizontal
Telecommunications Main Grounding Busbar (TMGB):	A busbar placed in a convenient and accessible location and bonded by means of the bonding conductor for telecommunications to the building service equipment (power) ground.
AC	Alternating Current
ACEG	Alternating Current Equipment Ground
AHJ	Authority Having Jurisdiction
AWG	American Wire Gauge
BCT	Bonding Conductor for Telecommunications
BN	Bonding Network
BICSI	Building Industry Consulting Services International
CBC	Coupled Bonding Conductor
CBN	Common Bonding Network
DC	Direct Current
EF	Entrance Facility
EIA	Electronics Industries Alliance
EMT	Electrical Metallic Tubing
EOR	Engineer of Record
ESD	Electrostatic Discharge
GE	Ground Equalizer
IBN	Isolated Bonding Network

MTW	Machine Tool Wire
NRTL	National Recognized Testing Laboratory
NEC	National Electrical Code
NECA	National Electrical Contractors Association
NEMA	National Electrical Manufacturers Association
NESC	National Electrical Safety Code
NFPA	National Fire Protection Association
NRTL	Nationally Recognized Testing Laboratory
PBG	Principal Building Ground
SBTC	Solid Bare Tinned Copper
ScTP	Screened Twisted Pair
STP	Shielded Twisted Pair
TBB	Telecommunications Bonding Backbone
TEBC	Telecommunications Equipment Bonding Conductor
TEF	Telecommunications Entrance Facility
TER	Telecommunications Entrance Room
TGB	Telecommunications Grounding Busbar
TIA	Telecommunications Industry Association
TMGB	Telecommunications Main Grounding Busbar
UTP	Unshielded Twisted Pair

1.6 BONDING AND GROUNDING DESCRIPTION

- A. The basic principles, components and design methods of the telecommunications bonding and grounding infrastructure specified herein shall be followed.
- B. This clause specifies the telecommunications bonding and grounding infrastructure, describes and identifies with different elements that make up the infrastructure. The following clauses detail requirements associated with each element.
- C. The applicable electrical codes for bonding and grounding for telecommunications shall be met.
- D. The telecommunications bonding and grounding infrastructure originates at the electrical service equipment ground and extends throughout the building. The system shall include the following:
 1. Bonding Conductor for Telecommunications (BCT)
 2. Telecommunications main grounding busbar (TMGB)
 3. Telecommunications Bonding Backbone (TBB)
 4. Telecommunications Grounding Busbar (TGB)
 5. Telecommunications Equipment Bonding Conductor (TEBC)
- E. These telecommunications bonding and grounding components shall be installed in conjunction with ANSI/TIA-569-B, telecommunications pathways and spaces; ANSI/TIA-568-C series, cabling and ANSI/TIA-606-AB, administration including the following:
 1. Telecommunications Entrance Facility (TEF)
 2. Telecommunications Equipment Room (TER)
 3. Telecommunications Room (TR)
 4. Telecommunications cable and connecting hardware
 5. Telecommunications pathway components
 6. Telecommunications labeling and record keeping

- F. The electrical closets and associated panelboard(s) that may be located in the TEF, TER or TR are not part of the telecommunications infrastructure. They are referenced in this section because they are an integral to the telecommunications bonding and grounding system.

1.7 BONDING AND GROUNDING SYSTEM COMPONENTS:

- A. All bonding and grounding connectors shall be listed by a NRTL as required by NEC.
- B. TELECOMMUNICATIONS MAIN GROUNDING BUSBAR (TMGB)
 - 1. The telecommunications main grounding busbar (TMGB) is the dedicated extension of the building grounding electrode system for the telecommunications infrastructure. The TMGB is also the central attachment point for the telecommunications bonding backbone(s) and equipment.
- C. BONDING CONDUCTOR FOR TELECOMMUNICATIONS (BCT)
 - 1. The bonding conductor for telecommunications shall bond the TMGB to the electrical service equipment ground. The bonding conductor shall be Machine Tool Wire (MTW) and be sized, as a minimum, the same size as the TBB.
- D. TELECOMMUNICATIONS BONDING BACKBONE (TBB)
 - 1. The TMGB and interconnects all TGBs with the TMGB.
 - 2. The TBB extends throughout the building using the telecommunications backbone pathways and connects to the TGBs in all telecommunications rooms and equipment rooms.
- E. TELECOMMUNICATIONS GROUNDING BUSBAR (TGB)
 - 1. The TGB is the grounding connection point for telecommunications systems and equipment in the area served by that telecommunications room or equipment room.
 - 2. Each TGB shall be bonded to the GE where required by these specifications.
- F. TELECOMMUNICATIONS EQUIPMENT BONDING CONDUCTOR (TEBC)
 - 1. The Telecommunications Equipment Bonding Conductor(s) shall be used to bond all metallic components including conduits, raceways, racks, cabinets, etc. within each Equipment Room (ER) to the TGB or TMGB located in that space.
- G. TELECOMMUNICATIONS ENTRANCE FACILITY
 - 1. The telecommunications entrance facility (TEF) is the entrance point (room or space within the building) where:
 - a. The telecommunications facilities enter.
 - b. The joining of inter and intra-building backbone facilities takes place.
 - c. The grounding of these facilities is accomplished.
 - 2. The TEF also may include antenna cable entrances and carrier's electronic equipment serving the telecommunications functions of their systems.
 - 3. The TEF is the desirable location for the TMGB. The TMGB also may serve as the

TGB for the equipment located in the TEF.

1.8 SUBMITTALS

- A. All submittals shall be executed per Section 27 05 00 and Division 01
- B. Product Data: Submit grounding electrodes and connections; for fastening components and grounding fittings.
- C. Qualification data for firms and persons specified in Quality assurance Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.
- D. Field tests and observation reports certified by the testing organization and indicating and interpreting the test reports for compliance with performance requirements. Indicate overall resistance to ground (and resistance of each electrode). Submit certified test results stating ground resistance from service neutral at each service entrance.

1.9 QUALITY ASSURANCE

- A. Testing Agency Qualifications: A Nationally Recognized Testing Laboratory (NRTL) as defined in OSHA Regulation 1910, or a full member company of the National Electrical Testing Association (NETA).
 - 1. Testing Agency Field Supervision: Use persons currently certified by NETA or the National Institute for Certification in Engineering Technologies to supervise on-site testing specified in Part 3.
- B. Comply with NFPA 70.
- C. Comply with UL 467.
- D. Listing and Labeling: Provide products specified in this Section that are listed and labeled.
 - 1. The Terms Listed and Labeled As defined in the National Electrical Code, Article 100.
 - 2. Listing and Labeling Agency Qualifications: A Nationally Recognized Testing Laboratory (NRTL) as defined in OSHA Regulations 1910.7.

1.10 QUALIFICATIONS

- A. Manufacturer: Company specializing in manufacturing products specified in this section with minimum three years documented experience and with service facilities within 100 miles of project.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering

products that may be incorporated into the Work include, but are not limited to, the following.

- B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Erico Inc.; Electrical Products Group.
2. Harger
3. Panduit
4. O-Z/Gedney Co.
5. Thomas & Betts, Electrical
6. Engineer Approved Equal

2.2 BONDING AND GROUNDING SYSTEM COMPONENTS

A. TELECOMMUNICATIONS MAIN GROUNDING BUSBAR (TMGB)

1. The TMGB shall:
 - a. Be a predrilled copper busbar provided with holes conforming to NECA/BICSI 607-2011 and ANSI/TIA-607-B standards.
 - b. Have a minimum dimensions 0.25 inches thick X 4 inches wide X variable length (12 in. min.).
 - c. Be sized in accordance with the immediate application requirements plus a minimum of 50% spare capacity for future connections.
 - d. Be listed by a NRTL.
 - e. Be electro-tin plated copper or if not plated, the busbar shall be cleaned prior to attachment of conductors and an anti-oxidant should be applied to the contact area.
2. All lugs used to connect each BCT, TBB, GE or TEBC conductor to the TMGB shall be irreversible listed two-hole compression lugs.

B. TELECOMMUNICATIONS GROUNDING BUSBAR (TGB)

1. The TGB shall:
 - a. Be a predrilled copper busbar provided with holes conforming to NECA/BICSI 607-2011 and ANSI/TIA-607-B standards.
 - b. Have a minimum dimensions 0.25 inches thick X 2 inches wide X variable length (12 in. min.).
 - c. Be sized in accordance with the immediate application requirements plus a minimum of 50% spare capacity for future connections.
 - d. Be listed by a NRTL.
 - e. Be electro-tin plated copper or if not plated, the busbar shall be cleaned prior to attachment of conductors and an anti-oxidant should be applied to the contact area.
2. All lugs used to connect each BCT, TBB, GE or TEBC conductor to each TGB shall be irreversible listed two-hole compression lugs.

C. CONDUCTORS

1. All bonding and grounding conductors shall be copper and may be insulated. All insulated

conductors shall be listed for the application. The minimum bonding conductor size shall be No. 6 AWG MTW.

2. All Grounding Electrode Conductors shall be type: machine tool wire (MTW).
3. The BCT and each TBB and GE shall be insulated with green color insulation or marked with a distinctive green color.
4. Each TEBC shall be insulated with green color insulation.
5. All GE(s) shall be the same size as the largest TBB.
6. The minimum TBB size shall be a No. 6 AWG MTW. The TBB should be sized at 2 kcmil per linear foot of conductor length up to a maximum size of 750 kcmil (see chart below). If the TBB is insulated, the insulation shall meet the fire ratings of its pathway. The sizing of the TBB is not intended to account for the reduction or control of electromagnetic interference.

TBB Conductor size versus length:

TBB linear length ft (m)	TBB size (AWG)
Less than 13 (4)	6
14 – 20 (4 – 6)	4
21 – 26 (6 – 8)	3
27 – 33 (8 – 10)	2
34 – 41 (10 – 13)	1
42 – 52 (13 – 16)	1/0
53 – 66 (16 – 20)	2/0
67 – 84 (20 – 26)	3/0
85 – 105 (26 – 32)	4/0
106 – 125 (32 – 38)	250 kcmil
126 – 150 (38 – 46)	300 kcmil
151 – 175 (46 – 53)	350 kcmil
176 – 250 (53 – 76)	500 kcmil
251 – 300 (76 – 91)	600 kcmil
Greater than 300 (91)	750 kcmil

D. MECHANICAL CONNECTORS

1. Description: Bronze connectors, suitable for grounding and bonding applications, in configurations required for particular installation.
 - a. Pressure Connectors: High conductivity plated units.
 - b. Bolt Clamps: Heavy-duty type.
 - c. All connections shall use irreversible two-hole compression lugs listed for such use.

E. EXOTHERMIC CONNECTIONS

1. Product Description: Exothermic materials, accessories, and tools for preparing and making permanent field connections between grounding system components.

PART 3 - EXECUTION

3.1 BONDING AND GROUNDING DESCRIPTION

- A. If secondary protection is provided by this specification, the secondary protector grounding conductor shall be connected to the nearest telecommunications grounding busbar, using the shortest grounding conductor practicable.

3.2 BONDING AND GROUNDING SYSTEM COMPONENTS

A. TELECOMMUNICATIONS MAIN GROUNDING BUSBAR (TMGB)

1. The TMGB should be located to minimize the length of the bonding conductor for telecommunications. The TMGB shall be located as per contract drawings.
2. The TMGB should serve telecommunications equipment located within the same room or space.
3. If an electrical panelboard is located in the same room or space as the TMGB that panelboards alternating current equipment ground (ACEG) bus or the panelboard enclosure shall be bonded to the TMGB.
4. The TMGB shall be located as close to the panelboard as practicable and shall be installed per applicable codes.
5. The connections of the bonding conductor for telecommunications, the TBB and each GE to the TMGB shall utilize exothermic welding or irreversible listed two-hole compression lugs.
6. The connection of conductors for bonding telecommunications equipment to the TMGB shall utilize exothermic welding or irreversible listed two-hole compression connectors.
7. All metallic raceways for telecommunications cabling located within the same room or space as the TMGB shall be bonded to the TMGB. Metallic pathways containing grounding conductors that are bonded to the grounding conductor need not be bonded to the TMGB.
8. The TMGB shall be insulated from its support. A minimum of 2 inches of separation from the supporting wall is recommended to allow access to the rear of the busbar.
9. The TMGB shall be located so as to be accessible to telecommunications personnel. The mounting height should be adjusted to accommodate various cabling practices.

B. TELECOMMUNICATIONS GROUNDING BUSBAR (TGB)

1. The TBBs and other TGBs within the same space shall be bonded to the TGB with a conductor the same size as the TBB.
2. The bonding conductor between the TGB and a TBB shall be continuous and routed in the shortest possible straight-line path.
3. If an electrical panelboard is located in the same room or space as the TGB that panelboards alternating current equipment ground (ACEG) bus or the panelboard enclosure shall be bonded to the TGB.
4. The TGB shall be located as close to the panelboard as practicable and shall be installed per applicable codes.
5. Where panelboards that serve telecommunications equipment are not located within the same space or telecommunications rooms as the TGB, consideration should be given to bonding the panelboards ACEG bus or the enclosure to the TGB.
6. All metallic raceways for telecommunications cabling located with the same telecommunications room or space shall be bonded to the TGB.
7. Connections of the GE and TBBs to the TGB shall utilize exothermic welded connections, or irreversible listed two-hole compression lugs.
8. The TGB shall be insulated from its support. A minimum of 2 inches of separation from the supporting wall is recommended to allow access to the rear of the busbar.
9. A practical location for the TGB is to the side of the panelboard if one exists. The vertical location of the TGB should be taken into consideration whether the bonding conductors are routed in an access floor or overhead cable tray.

C. CONDUCTORS

1. Bonding and grounding conductors should not be placed in ferrous metallic conduit. If it is necessary or required by code to place bonding and grounding conductors in ferrous metallic conduit that exceeds 3 feet in length, the conductors shall be bonded to each end of the conduit using a grounding

bushing or a No. 6 AWG MTW conductor, minimum. Metal conduits that are bonded to the bonding conductor do not need an additional conductor connection at the TMGB or TGB.

2. Color-coding, Labeling and Marking
 - a. Each telecommunications bonding and grounding conductor shall be labeled. Labels shall be located on conductors as close as practicable to their point of termination in a readable position. Labels shall be nonmetallic and include the following:
 - 1) IF THIS CONNECTOR OR CABLE IS LOOSE OR MUST BE REMOVED, PLEASE CONTACT THE TELECOMMUNICATIONS MANAGER.
 - b. The bonding conductor for Telecommunications, each TBB conductor and each GE shall be green or marked with a distinctive green color.
 - c. Each TEBC shall be an insulated green MTW conductor.
3. TELECOMMUNICATIONS BONDING BACKBONE (TBB) AND GROUNDING EQUALIZER (GE)
 - a. The TBB should be installed with consideration given to the type of building construction, building size, general telecommunications requirements and the configuration of the telecommunications pathways and spaces.
 - b. If two or more TBBs are used within a multistory building, the TBBs shall be bonded together with a GE at the top floor and at a minimum of every third floor.
 - c. The TBB conductors should be installed without splices. If splices are required, the number should be kept to a minimum and they shall be accessible and located in telecommunications spaces. All joined segments of a TBB shall be connected using exothermal welding or equivalent. All joints shall be adequately supported and protected from damage. All proposed TBB splice locations shall be submitted to E.O.R. for approval prior to installation.
 - d. Permit multiple TBBs and GE as required by the size of the building
 - e. Address routing to minimize the lengths of the TBBs.
 - f. The interior water piping system of the building shall not be used as a TBB or GE.
 - g. The metallic cable shield of any case shall not be used as a TBB or GE.
4. TELECOMMUNICATIONS EQUIPMENT BONDING CONDUCTOR (TEBC)
 - a. Within each Equipment Room (ER) all metallic components including conduits, raceways, racks, cabinets, etc. shall be bonded to the TGB or TMGB using a minimum sized conductor of No. 6 AWG.
 - b. Each TEBC shall be green or marked with a distinctive green color.
 - c. All lugs used to connect each end of each TEBC conductor to the TMGB and metallic components shall be double bolt type and correctly matched and Listed for such use.

3.3 STRUCTURAL STEEL BONDING

- A. In a metal frame type building, where the steel framework is readily accessible within the room, each TGB and TMGB shall be bonded to the vertical steel frame using a minimum No. 6 AWG MTW conductor sized according to length based on the TBB sizing chart.
- B. All connectors used for bonding to the metal frame of a building shall be listed for the intended purpose.
- C. When horizontal steel members are permanently electrically bonded to the vertical column members, the TGB may be bonded to the horizontal members in lieu of the vertical column members.

3.4 TELECOMMUNICATIONS ENTRANCE FACILITY

- A. If primary protectors are located in the TEF, their grounding conductors shall be connected to the TMGB. This conductor is intended to conduct lightning and AC fault currents from the telecommunications primary protectors. A minimum of one (1) foot separation shall be maintained between this insulated conductor and any DC power cables, switchboard cable, or high frequency cables, even when placed in rigid metal conduit or EMT.
- B. When the outside plant cables in the TEF incorporate a cable shield isolation gap, the cable shield on the building side of the gap shall be bonded to the TMGB.
- C. If the backbone cables incorporate a shield or metal member, this shield or metallic member shall be bonded to the TMGB or TGB.
- D. The TMGB is intended to be the location for connecting grounding busbars of telecommunication equipment located in the TEF.

3.5 EQUIPMENT ROOM AND TELECOMMUNICATIONS ROOM

- A. Each equipment room and telecommunications room shall contain a TGB. The TGB shall be located inside the room and be located as to provide the greatest flexibility and accessibility for the telecommunications system grounding.
- B. Multiple TGBs may be installed within the same TR to aid in minimizing bonding conductor lengths and termination space. Multiple TGBs within the same room shall be bonded together with a conductor of the same size as the TBB.
- C. The TGB shall be placed per these specifications. The TGB should also be placed to provide for the shortest and straightest routing of the grounding conductors.
- D. All metallic raceways for telecommunications cabling located within the same telecommunications room or space shall be bonded to the TGB.
- E. Short metallic pathways (e.g. floor or wall sleeves not exceeding three (3) feet in length) are not required to be bonded.

3.6 FIELD MEASUREMENTS

- A. Verify field measurements prior to fabrication.

3.7 MECHANICAL CONNECTORS

- A. Grounding Bushings shall be malleable iron, threaded, with insulated liner and solderless lug. Manufacturer: O-Z/Gedney IBC-50L Series, Panduit type LCC or LCCX, T&B 3870 Series or Engineer Approved Equal.

3.8 EXOTHERMIC CONNECTIONS

- A. Exothermic-Welded Connections: Provided in kit form and selected per manufacturer's written instructions for specific types, sizes and combination of conductors and connected items.

3.9 EXAMINATION

- A. Section 270500 Common Work Results for Communications Systems.
- B. Verify that abandoned wiring and equipment serve only abandoned facilities.

3.10 INSTALLATION

- A. General: Ground electrical systems and equipment according to NEC requirements, except where Drawings or Specifications exceed NEC requirements.
- B. Provide bonding to meet Regulatory Requirements.
- C. Grounding Conductors: Route along the shortest and straightest paths possible, except as otherwise indicated. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.
- D. Underground Grounding Conductors: Use bare copper wire. Bury at least 24 inches (600mm) below grade.
- E. Bond together each metallic raceway, pipe, duct and other metal object entering space under access floors. Bond to underfloor ground grid. Use No. 6 AWG MTW bare copper conductor minimum.
- F. Ground metallic cable trays per NEC. The cable tray system may be used as the equipment ground provided all NEC requirements are satisfied. Bond metallic raceways which contain cables or conductors and which are routed through the cable tray system to the cable tray.

3.11 CONNECTIONS

- A. General: Make connections so possibility of galvanic action or electrolysis is minimized. Select connectors, connection hardware, conductors, and connection methods so metals in direct contact will be galvanically compatible.
 - 1. Use electroplated or hot-tin-coated materials to assure high conductivity and to make contact points closer in order of galvanic series.
 - 2. Make connections with clean, bare metal at points of contact.
 - 3. Make aluminum-to-steel connections with stainless-steel separators and mechanical clamps.
 - 4. Make aluminum-to-galvanized steel connections with tin-plated copper jumpers and

- mechanical clamps.
5. Coat and seal connections having dissimilar metals with inert material to prevent future penetration of moisture to contact surfaces.
- B. Exothermic-Welded Connections: Use for connections to structural steel and for underground connections, except those at test wells. Comply with manufacturer's written instructions. Welds that are puffed up or that show convex surfaces indicating improper cleaning are not acceptable.
- C. Equipment Grounding-Wire Terminations: For No. 6 AWG and larger, use irreversible compression type grounding 2-hole lugs listed for such use.
- D. Noncontact Metal Raceway Terminations: Where metallic raceways terminate at metal housings without mechanical and electrical connection to housing, terminate each conduit with a grounding bushing. Connect grounding bushings with a bare grounding conductor to grounding bus or terminal in housing. Bond electrically noncontinuous conduits at both entrances and exits with grounding bushings and bare grounding conductors, except as otherwise indicated.
- E. Connections at Test Wells: Use irreversible compression type connectors on conductors and make bolted and clamped-type connections between conductors and grounding rods.
- F. Tighten screws and bolts for grounding and bonding connectors and terminals according to manufacturers published torque-tightening values. Where these requirements are not available, use those specified in UL 486A and UL 486B.
- G. Compression-Type Connections: Use hydraulic compression tools to provide correct circumferential pressure for compression connectors. Use tools and dies recommended by manufacturer of connectors. Provide embossing die code or other standard method to make a visible indication that a connector has been adequately compressed on grounding conductor.
- H. Moisture Protection: Where insulated grounding conductors are connected to grounding rods or grounding buses, insulate entire area of connection and seal against moisture penetration of insulation and cable.

3.12 UNDERGROUND DISTRIBUTION SYSTEM GROUNDING

- A. Manholes and Handholes: Install a driven grounding rod close to wall and set rod depth so 4 inches (100 mm) will extend above finished floor. Where necessary, install grounding rod before manhole is placed and provide a minimum No. 3/0 AWG bare, tinned-copper conductor from grounding rod into manhole through a waterproof sleeve in manhole wall. Protect grounding rods passing through concrete floor with a double wrapping of pressure-sensitive tape or heat-shrunk insulating sleeve from 2 inches (50 mm) above to 6 inches (150 mm) below concrete. Seal floor opening with waterproof, nonshrink grout.
- B. Connections to Manhole Components: Connect exposed metal parts, such as inserts, cable racks, pulling irons, ladders and cable shields within each manhole or handhole, to grounding rod or grounding conductor. Make connections with minimum No. 4 AWG stranded, hard-drawn copper wire. Train conductors plumb or level around corners and fasten to manhole walls. Connect to cable armor and cable shields as recommended by manufacturer of splicing and termination kits.

3.13 FIELD QUALITY CONTROL

- A. Independent Testing Agency: Engage an independent electrical testing organization to

perform tests described below.

- B. Tests: Subject the completed grounding system to a megger test at each location where a maximum ground- resistance level is specified, at service disconnect enclosure grounding terminal, and at ground test wells. Measure ground resistance not less than 2 full days after the last trace of precipitation, and without the soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance. Perform tests by the 2-point method according to IEEE 81-1983 – IEEE Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System Part 1: Normal Measurements.
- C. Two-point ground continuity testing:
 - 1. This procedure will help determine if there is an acceptable maximum level of resistance between any point in the telecommunications bonding and grounding system and the building's electrical grounding electrode system. The test is performed using an earth ground resistance tester that is configured for a continuity test, otherwise known as a two-point test or a "dead earth" test.
 - 2. Prior to two-point ground testing, a visual inspection shall be performed to verify that the bonding and grounding system is installed according to the guidelines in ANSI/TIA-607-B.
 - 3. A voltage test should be performed prior to conducting the two-point continuity test due to the possibilities of ground faults traveling through the telecommunications bonding and grounding system. Consult with other contractors to ensure other electrical work does not interfere with this test.
 - 4. This test shall be performed in the following areas:
 - a. TMGB/TGB to the electrical ground at the TEF.
 - b. TMGB/TGB to the building steel (if present).
 - c. TMGB to TGB.
 - d. Building steel (if present) to the electrical ground.
 - 5. The recommended maximum value for resistance between any point in the telecommunications bonding and grounding system and the building's electrical grounding electrode system is 100 milliohms. In the case of long TBB and GE conductor runs, the resistance of the conductor must be factored into the total resistance.
- D. Excessive Ground Resistance: Where resistance to ground exceeds specified values, notify Owner promptly and include recommendations to reduce ground resistance and to accomplish recommended work.
- E. Report: Prepare test reports, certified by the testing organization, of ground resistance at each test location. Include observations of weather and other phenomena that may affect test results. Describe measures taken to improve test results.

END OF SECTION 270526

SECTION 270528 – PATHWAYS FOR COMMUNICATION SYSTEMS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. The work covered under this section consists of the furnishing of all necessary labor, supervision, materials, equipment, and services to completely execute the horizontal cable pathway system of non-continuous cable supports, conduit or cable tray as described in the drawings.
- B. The drawings indicate the general route of the raceway system. Data presented on the drawings is as accurate as preliminary surveys and planning can determine. Accuracy is not guaranteed and field verification of all dimensions, routing, etc. is required.
- C. Specifications and drawings are for assistance and guidance, but exact routing, locations, distances and levels will be governed by actual field conditions. Contractor is directed to make field surveys as part of his work prior to submitting system layout drawings.
- D. Section Includes:
 - 1. Metal conduits and fittings.
 - 2. Nonmetallic conduits and fittings.
 - 3. Cable pathways, supports and fittings.
 - 4. Boxes, enclosures, and cabinets.
- E. Related Requirements:
 - 1. Division 26 Section "Grounding and Bonding for Electrical Systems"
 - 2. Division 26 Section "Raceways and Boxes for Electrical Systems"
 - 3. Division 26 Section "Underfloor Raceways for Electrical Systems"
 - 4. Division 26 Section "Underground Ducts and Raceways for Electrical Systems"
 - 5. Division 26 Section "Sleeves and Sleeve Seals for Electrical Raceways and Cabling"
 - 6. Division 27 Section "Grounding and Bonding for Communications Systems"
 - 7. Division 27 Section "Communications Equipment Room Fittings".
 - 8. Division 27 Section "Communication Horizontal Cabling".

1.3 REFERENCES

- A. ANSI/NFPA 70 – National Electrical Code (NEC) 2008
- B. ASTM B633 – Specification for Electrodeposited Coatings of Zinc on Iron and Steel
- C. ASTM A653 – Specification for Steel Sheet, Zinc-Coated (Galvanized) by the Hot Dip Process
- D. ASTM A123 – Specification for Zinc (Hot Galvanized) Coatings on Iron and Steel

- E. ASTM A510 – Specification for General Requirements for Wire Rods and Coarse Round Wire, Carbon Steel
- F. ASTM A 641 – Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire
- G. ASTM D 769 – Standard Specification for Black Oxide Coatings
- H. NEMA VE 1-2002 – Metal Cable Tray Systems
- I. NEMA VE 2-2006 – Cable Tray Installation Guidelines
- J. UL Compliance - Provide products that are UL-classified and labeled
- K. ANSI/TIA-569-B Commercial Building Standard for Telecommunications Pathways and Spaces
- L. ANSI J-STD-607-A Commercial Building Grounding (Earthing) and Bonding Requirements for Telecommunications
- M. IEC 61537 (2001) – Cable Tray Systems and Cable Ladder Systems for Cable Management
- N. BICSI Telecommunications Distribution Methods Manual

1.4 DEFINITIONS

- A. ARC: Aluminum Rigid Conduit.
- B. EMT: Electrical Metal Tubing
- C. Engineer of Record: The engineer and engineering company that developed these contract documents, including the plans, diagrams and specifications.
- D. GRC: Galvanized Rigid Conduit.
- E. IMC: Intermediate Metal Conduit.
- F. PVC: Polyvinyl Chloride Conduit
- G. RMC: Rigid Metal Conduit.

1.5 DESCRIPTION

- A. This system consists of empty raceways as shown on the drawings and described herein.
- B. Horizontal pathways consist of structure that conceal, protect, support and provide access to horizontal cables between the telecommunication outlet/connector at the work area and the horizontal cross-connect in the serving area.
- C. Where ceiling areas are used as pathways for telecommunication cables and connecting hardware follow all applicable rules covering installations in both air plenums and non-plenum hollow-ceiling systems as found in the applicable electrical and building codes and standards.

1.6 RESPONSIBILITIES

- A. Responsibility for providing all equipment and associated support, splice, terminating hardware, etc. as necessary to provide a complete and functional cable support system shall be as outlined below:
 - 1. Wire Basket Tray/ Cable Basket Tray: Electrical Contractor
 - 2. Ladder Rack: Low Voltage
- B. The Electrical Contractor shall be responsible for providing all back-boxes, pull-boxes, junction boxes, conduit, and sleeves as indicated in the plans and specifications to support the low voltage systems
- C. The contractor installing cable pathways shall be responsible for coordinating installation with general contractor and other trades on the job to insure pathways are not impeding other systems.

1.7 ACTION SUBMITTALS

- A. Product Data
 - 1. Provide manufacturer's catalog information showing dimensions, colors, and configurations.
 - 2. Submittals shall include all items called for in PART 2 – PRODUCTS of this document and the manufacturers cut sheets for the following:
 - a. Continuous and non-continuous cable pathway and fittings
 - b. All boxes and enclosures.
- B. Shop Drawings:
 - 1. For custom enclosures and boxes. Include plans, elevations, sections, and attachment details.
 - 2. Cable pathway (continuous and non-continuous) layout, showing cable pathway route to scale, with relationship between the pathway and adjacent structural, electrical, and mechanical elements. Include the following:
 - a. Vertical and horizontal offsets and transitions.
 - b. Clearances for access above and to side of cable pathways.
 - c. Vertical elevation of cable trays above the floor or bottom of ceiling structure.
 - d. Load calculations to show dead and live loads as not exceeding manufacturer's rating for tray and its support elements.
- C. Samples: For custom color wire basket and tray ladder rack and for each color and texture specified, 12 inches (300 mm) long.

1.8 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Pathway routing plans, drawn to scale, on which the following items are shown and coordinated with each other, using input from the applicable contractors involved:
 - 1. Structural members in paths of pathway groups with common supports.

2. HVAC, plumbing, fire protection, and electrical items and architectural features in paths of conduit groups with common supports.
- B. Qualification Data: For Installer, qualified layout technician and installation.
- C. Seismic Qualification Certificates: Submit documentation on seismic properties for pathway tray, ladder rack, enclosures, cabinets, equipment racks and their mounting provisions including those for internal components from manufacturer to the structural engineer. Coordinate exact submittal requirements with the structural specifications.
- D. Source quality-control reports.

1.9 QUALITY ASSURANCE

- A. Installer Qualifications: Low Voltage Contractor must have personnel certified by BICSI on staff.
 1. Layout Responsibility: Preparation of Shop Drawings shall be under the direct supervision of Commercial Installer, Level 2.
 2. Installation Supervision: Installation shall be under the direct supervision of Registered Technician, who shall be present at all times when Work of this Section is performed at Project site.
 3. Field Inspector: Currently registered by BICSI as Commercial Installer, Level 2 to perform the on-site inspection.
- B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- C. Telecommunications Pathways and Spaces: Comply with ANSI/TIA-569-B and BICSI's Telecommunications Distribution Methods Manual.
- D. Grounding: Comply with NEC, the Authority Having Jurisdiction and ANSI-J-STD-607-A.

1.10 COORDINATION WITH OTHER TRADES

- A. Coordinate the support of cable pathways support systems with the work of other trades.
- B. Coordinate routing of cable pathways with the work of other trades to maintain adequate working clearances above, below and to the sides of cable pathways.
- C. Coordinate with other contractors during the BIM or installation coordination meetings to provide input on shared hangers planned for use to support cable pathway systems.

PART 2 - PRODUCTS

2.1 WIRE BASKET SUPPORT SYSTEM

- A. Acceptable Manufacturers
 1. Cablofil
 2. ~~Chatsworth Products, Inc.~~

3. ~~Cooper~~ Eaton B-line
4. G S Metals
5. MonoSystems
6. ~~WBT, LLC~~

B. General

1. Provide wire basket of types and sizes indicated; with connector assemblies, clamp assemblies, connector plates and splice bars. Construct units with rounded edges and smooth surfaces and in compliance with applicable standards.
2. Wire basket shall be made of high strength steel wires and formed into a standard 2 inch by 4 inch wire mesh pattern with intersecting wires welded together. All wire ends along wire basket sides (flanges) shall be rounded during manufacturing for safety of cables and installers.
3. Bond every section of cable tray to the adjoining tray as per manufacturer's specifications. Bond cable tray to ground at end of every run.
4. Provide cable drop out brackets at every location where cable exits the tray.
5. Material and finish specifications for each wire basket type are:
 - a. Electroplated Zinc: Straight sections shall be made from steel meeting the minimum mechanical properties of ASTM A510 and shall be electro-plated zinc in accordance with ASTB B633 SC2.
 - b. Pre-Galvanized Zinc: Straight sections, wall brackets and other pre-galvanized accessories shall be coated with zinc in accordance with ASTM A653. Materials must be cleaned after fabrication to remove any soot, manufacturing residue/ oils, or metallic particles.
 - c. Electro-Galvanized Zinc: Straight sections, accessories and miscellaneous hardware shall be coated in accordance with ASTM B633 SC3. All threaded components shall be coated in accordance with ASTM B SC1.
 - d. Paint: Straight sections shall be powder coated white over electrodeposited zinc.

C. Size

1. Wire basket shall conform to the following nominal criteria:
 - a. Wire Basket Support System shall be constructed of a minimum of 0.195" (5mm) steel wire.
 - b. Mesh construction: 2 x 4 inches.
 - c. Straight section lengths of standard 8' and 10'.
 - d. Wire basket shall have standard depths of 2, 4 and 6 inches.
 - e. Wire basket shall have standard widths of 4, 6, 8, 12, 18 and 24 inches.
 - f. Fittings shall be field formed or factory fittings.
 - g. All section splices shall be made per manufacturer's standards.

D. Support

1. Wire basket supports shall be trapeze hangers or wall brackets.
2. Trapeze hangers shall be supported by 1/4 inch or 3/8 inch diameter rods.
3. Special accessories shall be furnished as required to protect, support and install a complete wire basket support system.

E. Installation

1. Install wire basket as indicated in accordance with recognized industry practices (NEMA VE-2 2000), to ensure that the cable tray equipment complies with requirements of NEC, and applicable portions of NFPA 70B and NECA's "Standards of Installation" pertaining to general electrical practices.
2. Coordinate wire basket with other electrical work as necessary to properly interface installation of wire basket raceway with other work.
3. Provide dedicated space encompassing wire basket to permit access for installing and maintaining cables. Refer to Section 3.2 below for specific clearances required.
4. Test wire basket support system to ensure electrical continuity of bonding and grounding connections, and to demonstrate compliance with specified maximum grounding resistance. See NFPA 70B, Chapter 18, for testing and test method.
5. The cross sectional area of installed cables may not exceed 50% of the cross sectional area of the tray per National Electric Code per (Article 392.9 (B)). All wire basket tray shall be sized to provide a minimum of 50% growth once cable installation included in the scope of work of these documents is complete.

2.2 I-BEAM LADDER TRAY SUPPORT SYSTEM

A. Acceptable Manufacturers

1. Cablofil
2. Cooper Eaton B-line
3. G S Metals
4. MonoSystems

B. General

1. Ladder Cable Trays shall consist of two longitudinal members (side rails) with transverse members (rungs) welded to both side rails. Both side rails and rungs shall be I-beam configuration. Side rails shall have a splice retention groove to accept a splice plate. Rungs shall be spaced 12 inches on center. Rung spacing in radiused fittings shall be industry standard 9" and measured at the center of the tray's width. Each rung must be capable of supporting a 200 lb. concentrated load on top of the catalogued design load at the center of a 18" wide cable tray (with a safety factor of 1.5).
2. Solid Bottom Cable Trays shall consist of two longitudinal members (side rails) with a solid sheet over rungs spaced on 12" centers.
3. Cable tray loading depth shall be 4 inches per NEMA VE-1.

C. Size

1. Straight sections shall be supplied in standard 10 foot (3.05m) and 12 foot (3.65m) lengths.
2. Cable tray widths shall be available in 6, 9, 12, 18, 24, 30, 36 inches or as indicated on drawings.
3. Splice plates shall have a maximum of (2) two nuts and bolts per plate. The resistance of fixed splice connections between adjacent sections of tray shall not exceed 0.00033 ohms. Splice plates shall be furnished with straight sections and fittings.
4. All horizontal fittings (horizontal bend, horizontal tee, horizontal cross) to be installed utilizing matching fittings. All vertical fittings must have a minimum radius of 24 inches. All section splices shall be made per manufacturer's standards.

D. Support

1. The ladder racking shall be supported by the side rails or trapeze support by a minimum 3/8 inch threaded rod and at intervals of no more than 5'.
2. Special accessories shall be furnished as required to protect support and install a complete ladder rack support system.
3. The contractor shall provide manufacturer specific hardware for joining sections of ladder rack in straight lines and at right angles. Hardware shall be provided to mount and join the cable raceway in various configurations and attachment methods to walls, racks, and equipment.

2.3 LADDER TRAY SUPPORT SYSTEM

A. Acceptable Manufacturers

1. Chatsworth Products
2. Eaton B-line
3. G S Metals
4. MonoSystems

B. General

1. Provide metal; ladder racking, of types, classes, and size indicated; with splice plates, bolts, nuts and washers for connecting sections. Construct system maintaining rounded edges and smooth surfaces in compliance with the applicable standards. Ladder rack shall be installed according to the latest revision of NEMA VE-2.
2. Ladder tray shall consist of two longitudinal members (side rails) with transverse members (rungs) welded or mechanically fastened to the side rails.
3. Straight sections, fitting side rails, rungs and splice plates shall be extruded from aluminum (ASTM B 221 6063 alloy) or shall have stringers made of 16 gauge hot rolled steel tubing and cross members shall be made of 12 gauge cold rolled steel (ASTM A 569).
4. The ladder racking shall be available in black and gray painted finishes and clear anodized aluminum finish.
5. Bond the ladder rack to ground at end of every run.
6. Bond every section of cable rack to the adjoining section as per manufacturer's specification.
7. Provide cable drop out brackets at every location where cable exits the tray.
8. Provide plastic end caps on side rails ends.

C. Size

1. Ladder rack shall consist of two side rails with transverse rungs welded, or mechanically fastened to the side rails. Rungs shall be spaced at 9 inches on center and shall have a minimum width of 1" for cable laying.
2. Straight sections shall be supplied in standard 10 foot lengths.
3. Ladder Rack shall be available in 6, 9, 12, 18 and 24 inch widths (as indicated on the drawings).
4. All bends and elbows shall have a minimum radius of 12 inches
5. All section splices shall be made per manufacturer's standards.

D. Support

1. The ladder racking shall be supported by the side rails or trapeze support by a minimum 3/8 inch threaded rod and at intervals of no more than 5'.

2. Special accessories shall be furnished as required to protect support and install a complete ladder rack support system.
3. The ladder racking shall be capable of supporting 115 lb/ft when supported at 5' intervals.
4. The contractor shall provide manufacturer specific hardware for joining sections of ladder rack in straight lines and at right angles. Hardware shall be provided to mount and join the cable raceway in various configurations and attachment methods to walls, racks, and equipment.

E. Installation

1. Install ladder rack as indicated on drawings. Installation shall be in accordance with equipment manufacturer's instruction, utilizing recognized industry practices to ensure that ladder rack complies with requirements of NEC and applicable portions of NFPA 70B Reference NEMA-VE2 for general ladder rack installation guidelines.
2. Coordinate ladder rack installation with other electrical work as necessary to properly integrate installation with other work.
3. Provide dedicated space encompassing wire basket to permit access for installing and maintaining cables. Refer to Section 3.2 below for specific clearances required.
4. Ladder rack fitting supports shall be located such that they meet the strength requirements of straight sections. Install fitting supports per NEMA VE-2 guidelines, or in accordance with manufacturer's instruction.
5. Test ladder rack to ensure electrical continuity of bonding and grounding connections, and to demonstrate compliance with specified maximum grounding resistance. See NFPA 70B, Chapter 18, for testing and test methods.

2.4 BOXES AND ENCLOSURES

A. Acceptable Manufacturers:

1. Cooper Technologies Company; Cooper Crouse-Hinds.
2. Hoffman; a Pentair company.
3. Hubbell Incorporated; Killark Division.
4. RACO; a Hubbell company.
5. Wiremold / Legrand.

B. General Requirements for Boxes and Enclosures:

1. Comply with ANSI/TIA-569-B.
2. Boxes, enclosures and cabinets installed in wet locations shall be listed for use in wet locations.

C. Sheet-Metal Boxes: Comply with NEMA OS 1 and UL 514A.

D. Box extensions used to accommodate new building finishes shall be of same material as recessed box.

E. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.

F. Typical Device Box Dimensions: 4 inches square by 2-1/8 inches deep (101 mm square by 60 mm deep), unless otherwise noted.

G. Gangable boxes are prohibited.

H. Nonmetallic Outlet and Device Boxes: Comply with NEMA OS 2 and UL 514C.

I. Enclosures:

1. Comply with UL 50 and NEMA 250, Type 1, galvanized-steel box with removable interior panel and removable front, finished inside and out with manufacturer's standard enamel.
2. Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
3. Interior Panels: Steel; all sides finished with manufacturer's standard enamel.
4. Fully hinged door with flush latch and concealed hinge.
5. Keyed latch to match equipment racks/ cabinets.
6. Metal barriers to separate wiring of different systems and voltage.
7. Accessory feet where required for freestanding equipment.
8. Nonmetallic cabinets shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2.5 FLEXIBLE RACEWAY (INNERDUCT)

A. Acceptable Manufacturers

1. CANTEX Inc
2. Lamson & Sessions; Carlon Electrical Products.
3. Lamson & Sessions; Pyramid Industries.
4. RACO; a Hubbell company.
5. Thomas & Betts Corporation.

B. Flexible Raceway shall be plenum rated.

C. The rating of the flexible raceway shall match the rating of the surrounding environment which it is installed.

D. Flexible raceway shall meet the requirements:

1. NEC Articles 770 and 800 for Telecommunications Cables.
2. Flexible raceway shall be UL Listed under UL 910 Standard for Plenum Application for Optical Fiber Raceway.

E. Flexible raceway shall be provided with a pull (mule) tape routed throughout the raceway containing footage markings.

F. Flexible raceway fitting and accessories shall be manufactured by the same manufacturer as the flexible raceway.

2.6 CONDUITS/SLEEVES

A. General

1. Conduit/sleeve shall be EMT grade metallic conduit schedule 5 or heavier.
2. Any conduit/sleeve installed for communication cabling shall have a coupler on each end with a plastic bushing for cable protection.
3. Terminate metal conduit using connectors with plastic bushings.
4. Provide nylon or plastic pull strings in all conduit runs.

B. Fire Wall Penetrations

1. Apply firestopping to penetrations of fire-rated floor and wall assemblies for communications installations to restore original fire-resistance rating of assembly. Firestopping materials and installation requirements are specified in Division 07 Section "Penetration Firestopping."
2. Fire stop penetrations seal methods and materials shall be FM-approved and UL listed as applicable and as approved by the authorities having jurisdiction.
3. All sealing systems and methods shall be submitted to the engineer of record for review prior to installation and sealing.
4. Provide Fire Rated Pathway which is designed for easy installation in floors and walls. Tested and approved cable capacities range from 0 to 100% visual fill. EZ-PATH® Series 44+ Fire Rated Pathway when installed with available single wall plates is designed for new cable installations. In these installations, the device does not require mechanical attachment to either the wall or the wall framing and must be installed after the wallboard has been installed. Split floor plates and multi-gang wall and floor brackets permit installation around previously installed cables if so desired.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Pathways shall primarily be constructed from wire cable basket tray and conduit hung from dedicated communications supports installed with powder-actuated devices, wire basket trays supported by threaded rod or wall brackets, ladder rack supported by threaded rod or wall brackets or conduit supported as per the NEC.
- B. In finished areas, provide concealed conduits and flush mount boxes.
- C. Conduit installation shall be coordinated with their respective termination equipment layouts at each backboard location as required to provide adequate dedicated space for the mounting of equipment for each system identified on the plans.
- D. Cable pathway systems shall be supported by support systems specifically designed and manufactured for the support of cable pathway systems. The cable pathway systems shall not be supported by other installed building systems.
- E. Install all pathway systems as per manufacturers recommended practices and as per local governmental regulations and NEC, and BICSI regulations and practices.
- F. All cable pathway routes are to be parallel and/or perpendicular with the outside walls of the building. Alternate paths must be approved by the engineer of record prior to installation of the cabling.
- G. All vertical cable pathway transitions creating an elevation difference in the cable pathway system shall be made via sloping segments of the cable pathway system or via long sweep radius fittings.
- H. Waterfall cable supports shall be provided at each point that cabling exits wire cable basket tray or ladder tray. Coordinate the locations of cable exits with each cabling contractor and the owner's representative.

I. Grounding and Bonding

1. Ground conduits at terminal boards with grounding bushings.
2. Wire basket tray shall be bonded directly to the TMGB or TGB in the telecommunications equipment room that the wire basket tray originates.
3. Provide and install a grounding conductor routed through the entire run of all wire basket tray. The grounding conductor shall be bonded to each individual piece of cable basket tray.
 - a. Any splices in the grounding conductor shall be protected, readably visible and accessible upon completion of the cable installation.
 - b. Refer to specification section "Grounding and Bonding for Communication Systems" for grounding conductor requirements and sizing.
4. Ladder rack shall be bonded directly to the TMGB or TGB within the telecommunications equipment room the ladder rack is located.
5. Provide a bonding jumper connecting each rail of each piece of ladder rack to one another to form a continuous grounding path.

J. Cable pathways

1. A dedicated pathway shall be provided for each low voltage communications cabling sub-system, included but not limited to LIDL network cabling, Security (surveillance and electronic access control network cabling, electronic access control power cabling and HVAC control cabling.
2. Cable pathways supporting multiple low voltage communications cabling sub systems shall be divided into dedicated sections for each cabling sub system.
3. Coordinate locations and sizes with each system's Contractor, the General Contractor and the Owner's representative.

K. Conduit/ Sleeves

1. Any section of conduit longer than 150 feet or containing more than 180 total degrees of bends shall have pull boxes. The contractor shall identify the pull box location(s) on submitted shop drawings for the Architect and Engineer to review.
2. Each conduit bend shall be a long sweep radius wherever possible. In no instance shall the inside radius or bend be less than six (6) times the internal diameter of the conduit for conduits that are 2" in diameter or less, for conduits larger than 2" the bend radius shall be no less than 10 (ten) times the inside diameter.
3. All communications sleeves shall be of a sized to accommodate the appropriate NEC fill ratios and comply with UL system requirements in rated penetration instances.
4. All sleeves shall be of a size as to not have more than 40% of the sleeve filled with low voltage cable at the completion of the installation. Provide a minimum of 50% capacity for future growth (minimum of one open sleeve) at each sleeve location.
5. All conduits/sleeves shall be secured and strapped to building surfaces per National Electric Code Article 358.30 (A) and (B).

L. Pull Boxes

1. All pull boxes installed in low voltage communications conduit runs shall be sized per NEC or Table 4.7 of the BICSI Telecommunications Distribution Methods Manual, which ever requirement results in a larger pull box.
2. Provide dedicated space encompassing pull box to permit access for installing and maintaining cables. Refer to Section 3.2 below for specific clearances required.

M. Firestopping

1. Comply with requirements in Division 07 Section "Penetration Firestopping" and ANSI/TIA-569-B.
2. Responsibility for sealing of opening around the exterior of the low voltage system sleeves shall be by the contractor as described below:
 - a. Sleeves through fire rated and smoke walls created by the Low Voltage Contractor for cable pass through shall be the responsibility of the Low Voltage Contractor.
 - b. Sleeves through fire rated and smoke walls created by the Electrical Contractor for cable pass through shall be the responsibility of the Electrical Contractor.
 - c. Sleeves of openings between floors created by the Low Voltage Contractor for cable pass through shall be the responsibility of the Low Voltage Contractor.
 - d. Sleeves of openings between floors created by the Electrical Contractor for cable pass through shall be the responsibility of the Electrical Contractor.
3. Sealing of the space internal to all sleeves or openings specifically designated for telecommunications cabling shall be the responsibility of the Low Voltage Contractor.
4. Sealing material and application of this material shall be accomplished in such a manner which is acceptable to the local fire and building authorities having jurisdiction over this work.

N. Separation from EMI Sources:

1. Comply with BICSI TDMM and TIA/EIA-569-B recommendations for separating unshielded copper voice and data communication cable from potential EMI sources, including electrical power lines and equipment.
2. Separation between open communications cables or cables in nonmetallic raceways and unshielded power conductors and electrical equipment shall be as follows:
 - a. Electrical Equipment Rating Less Than 2 kVA: A minimum of **5 inches (127 mm)**.
 - b. Electrical Equipment Rating between 2 and 5 kVA: A minimum of **12 inches (300 mm)**.
 - c. Electrical Equipment Rating More Than 5 kVA: A minimum of **24 inches (610 mm)**.
3. Separation between communications cables in grounded metallic raceways and unshielded power lines or electrical equipment shall be as follows:
 - a. Electrical Equipment Rating Less Than 2 kVA: A minimum of **2-1/2 inches (64 mm)**.
 - b. Electrical Equipment Rating between 2 and 5 kVA: A minimum of **6 inches (150 mm)**.
 - c. Electrical Equipment Rating More Than 5 kVA: A minimum of **12 inches (300 mm)**.
4. Separation between communications cables in grounded metallic raceways and power lines and electrical equipment located in grounded metallic conduits or enclosures shall be as follows:
 - a. Electrical Equipment Rating Less Than 2 kVA: No requirement.
 - b. Electrical Equipment Rating between 2 and 5 kVA: A minimum of **3 inches (76 mm)**.
 - c. Electrical Equipment Rating More Than 5 kVA: A minimum of **6 inches (150 mm)**.

5. Separation between Communications Cables and Electrical Motors and Transformers, 5 kVA or HP and Larger: A minimum of 48 inches (1200 mm).
6. Separation between Communications Cables and Fluorescent Fixtures: A minimum of 5 inches (127 mm).

3.2 CLEARANCES AROUND CABLE PATHWAYS

- A. Clearances shall be provided around all cable pathways to provide continuous access to the cable pathway during and following the installation process. Coordinate required clearances with all other trades prior to beginning work.
- B. Provide a minimum clearance of 12" above the entire width of all cable basket tray and ladder tray.
- C. Provide a minimum clearance of 12" to one side of all cable basket tray and ladder tray.
- D. Provide a minimum clearance of 24" in front of each pull box location.

END OF SECTION 270528

SECTION 271100 - COMMUNICATIONS EQUIPMENT ROOM FITTINGS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section provides the requirements for the installation of Structured Cabling System Equipment Room Fittings and the associated equipment, hardware and accessories. The Equipment Room Support Hardware includes, but are not limited to equipment cabinets, equipment racks, cable management hardware and other hardware required in Equipment Rooms. Included in this section are the product requirements for these systems as well as the installation requirements.

1.3 REFERENCED STANDARDS AND CODES

- A. NFPA 70 National Electric Code.
- B. Authority having jurisdiction.
- C. Local and State Building Codes.
- D. UL® for wiring: UL® Standard 910 "Test method for fire and smoke characteristics of cable used in air handling spaces." Provide products that are UL® listed and labeled for such use. UL® testing bulletin. Underwriters Laboratories (UL®) cable certification and follow up program.
- E. American National Standards Institute/Telecommunications Industry Alliance. ANSI/TIA, including associated Addenda:
 - 1. ANSI/TIA-568-C.0 – Generic Telecommunications Cabling for Customer Premises.
 - 2. ANSI/TIA-568-C.1 – Commercial Building Telecommunications Cabling Standard.
 - 3. ANSI/TIA-568-C.2 – Balanced Twisted Pair Telecommunications Cabling and Components Standard.
 - 4. ANSI/TIA-568-C.3 – Optical Fiber Cabling Components Standard.
 - 5. ANSI/TIA-569-C – Pathways and Spaces.
 - 6. ANSI/TIA-606-AB – Administration Standard for Commercial Telecommunications Infrastructure.
 - 7. ANSI/TIA-606-A-1 – Administration of Equipment Rooms and Data Center Computer Rooms.
 - 8. ANSI/TIA-607-B – Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises.
 - 9. ANSI/TIA-758-B – Customer Owned Outside Plant Telecommunications Infrastructure Standard.

10. ANSI/TIA-1005-A – Telecommunications Infrastructure Standard for Industrial Premises.

F. ANSI/NECA/BICSI 568-2006 Installing Commercial Building Telecommunications Cabling.

G. NECA/BICSI 607-2011 – Standard for Telecommunications Bonding and Grounding Planning and Installation Methods for Commercial Buildings.

H. National Electrical Manufacturers Association (NEMA).

I. Institute of Electrical and Electronic Engineers (IEEE).

J. American Society for Testing Materials (ASTM).

1.4 RELATED SECTIONS

A. 260526 Grounding and Bonding for Electrical Systems

B. 260529 Hangers and Supports for Electrical Systems

C. 260536 Cable Trays for Electrical Systems

D. 270500 Common Work Results for Telecommunications Systems

E. 270526 Grounding and Bonding for Communications Systems

F. 270528 Pathways for Communications Systems

1.5 SUBMITTALS

A. Contractor shall submit product data sheets and samples for all products specified under this section, including but not limited to equipment racks, cabinets, wire managers and any other associated apparatus required for a complete and functional system. All submittals are to be made in accordance to section 270500 and Division 01.

PART 2 - PRODUCTS

2.1 Backboards:

A. Backboards shall be AC Grade Plywood 3/4 inch by 48 inches by 96 inches boards. Comply with requirements for plywood backing panels specified in Division 06 Section "Rough Carpentry."

B. The plywood backboard shall be constructed fire-retardant treated (FRT) plywood. When painting fire-retardant treated plywood, the paint shall not cover the re-retardant treaded stamp.

C. Mount the plywood backboard with the C-grade face of the plywood is facing the wall.

2.2 Cable Raceway Support Systems

A. Cable raceway systems shall be installed in all Equipment Rooms as indicated on drawings. Cable raceway systems shall include all cable raceways, and other devices including, but not limited to the following: splice plates, waterfalls, side posts, grounding connectors, clamps and other hangers and

support apparatus required for a complete system.

B. Ladder Rack system shall have the following features:

1. Tubular cross slots added to top of stringers
2. Rolled edge to protect cables.
3. UL Listed for the intended application.
4. Stringers are to be made of 3/8" x 1-1/2" x .065" wall rectangular steel tubing.
5. Cross rungs are to be made of 3/8" x 1 1/2" x .065" wall rectangular steel tubing.
6. Cross rungs are to be welded at 12 inch intervals.
7. Cable raceway is to be 18 24 inches Wide.
8. Cable raceway is to be black in color.

C. Acceptable manufactures:

1. B-Line
2. Cablofil
3. CPI.
4. Panduit
5. Engineer approved equal.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine the areas intended for the equipment rack/cabinet/raceway installation.
- B. Examine walls, floors and ceilings for suitable conditions where the specified equipment is to be installed.
- C. Proceed with the installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Assemble equipment rack/cabinets/raceways per the manufacturer's recommended procedures. Fasten and secure with mounting hardware as specified by the manufacturer.
- B. Assemble wire management assemblies per manufacturer's recommended procedures. Fasten and secure with mounting hardware as specified by the manufacturer.
- C. Assemble shelf assemblies per manufacturer's recommended procedures. Fasten and secure

with mounting hardware as specified by the manufacturer.

- D. Installation of equipment elements should not exceed the manufacturer's recommend weight load for the specific piece of support equipment.
- E. Assemble raceway components per manufacture's recommended procedures. Fasten and secure with mounting hardware as specified by the manufacturer for the specific application
- F. Do not exceed the manufacturer's recommend weight load for the raceway system.
- G. Verify and confirm that all mounting/ securing hardware is properly tightened and secured.
- H. Waterfall cable dropouts shall be provided to properly support cable exits from the cable pathways. A waterfall cable dropout shall be provided at each location that cabling exits a cable pathway. Coordinate the exact location and quantities of waterfall cable dropouts with the cabling contractor and owner's representatives.

END OF SECTION 271100

SECTION 271500 - COMMUNICATIONS HORIZONTAL CABLING**PART 1 – GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section provides the requirements for the installation of 4-pair unshielded twisted pair cabling dedicated to refrigeration control cabling. Included in this section are the product requirements and installation requirements.

1.3 REFERENCED STANDARDS AND CODES

- A. NFPA 70 National Electric Code.
- B. Authority having jurisdiction.
- C. Local and State Building Codes.
- D. UL® for wiring: UL® Standard 910 "Test method for fire and smoke characteristics of cable used in air handling spaces." Provide products that are UL® listed and labeled for such use. UL® testing bulletin. Underwriters Laboratories (UL®) cable certification and follow up program.
- E. American National Standards Institute/Telecommunications Industry.

ANSI/TIA, including associated Addenda:

- 1. ANSI/TIA-568-C.0 – Generic Telecommunications Cabling for Customer Premises.
- 2. ANSI/TIA-568-C.0-1 – Generic Telecommunications Cabling for Customer Premises – Addendum 1, Updated References for Balanced Twisted-Pair.
- 3. ANSI/TIA-568-C.0-2 – Generic Telecommunications Cabling for Customer Premises – Addendum 2, General Updates
- 4. ANSI/TIA-568-C.1 – Commercial Building Telecommunications Cabling Standard.
- 5. ANSI/TIA-568-C.2 – Balanced Twisted Pair Telecommunications Cabling and Components Standard.
- 6. ANSI/TIA-568-C.3 – Optical Fiber Cabling Components Standard.
- 7. ANSI/TIA-569-C – Telecommunications Pathways and Spaces.
- 8. ANSI/TIA-606-AB – Administration Standard for Commercial Telecommunications Infrastructure.
- 9. ANSI/TIA-606-A-1 – Administration of Equipment Rooms and Data Center Computer Rooms.
- 10. ANSI/TIA-607-B – Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises.
- 11. ANSI/TIA-942-A – Telecommunications Standard for Data Centers.
- 12. ANSI/TIA-1005-A – Telecommunications Infrastructure Standard for Industrial Premises.
- 13. TIA-1152 – Requirements for Field Test Instruments and Measurements for

- 14. Balanced Twisted-Pair Cabling.
TIA TSB-162 – Telecommunications Cabling Guidelines for Wireless Access Points.
- 15. TIA TSB-184 – Guidelines for Supporting Power Delivery Over Balanced Twisted-Pair Cabling.
- 16. TIA TSB-190 – Guidelines on Shared Pathways and Shared Sheaths.
- F. ANSI/NECA/BICSI 568-2006 Standard for Installing Commercial Building Telecommunications Cabling.
- G. NECA/BICSI 607-2011 – Standard for Telecommunications Bonding and Grounding Planning and Installation Methods for Commercial Buildings.
- H. National Electrical Manufacturers Association (NEMA).
- I. Institute of Electrical and Electronic Engineers (IEEE).
- J. American Society for Testing Materials (ASTM).

1.4 RELATED SECTIONS

- A. 260526 Grounding and Bonding for Electrical Systems
- B. 260536 Cable Trays for Electrical Systems
- C. 270500 Common Work Results for Communications Systems
- D. 270526 Grounding and Bonding for Communications Systems
- E. 270528 Pathways for Communications Systems
- F. 271100 Communications Equipment Room Fittings

1.5 SUBMITTALS

- A. Contractor shall submit product data sheets and samples for all products specified under this section including, but not limited to cables, information outlets, patch panels, face plates, termination hardware and any other apparatus or hardware for a complete and operational system. All submittals shall be made in accordance with the requirements as indicated in section 270500.

PART 2 - PRODUCTS

2.1 HORIZONTAL STRUCTURED CABLING SYSTEM

- A. Horizontal cabling shall be Category 6 4-pair unshielded twisted pair (UTP) cabling that meet the applicable industry standards
- B. The horizontal structured cabling system specified in this specification shall be manufactured either by a single manufacturer or two manufacturers holding a high level partnership capable of providing the extended warranty outlined above. Including:

1. 4-Pair UTP Cabling

C. Acceptable Horizontal Structured Cabling System Manufactures:

1. Commscope
2. General Cable
3. Panduit
4. Superior Essex

2.2 HORIZONTAL CABLING

A. Description: 100-ohm, four-pair UTP cable.

1. Comply with ANSI/TIA-568-C.2 Category 6.
2. All horizontal cabling shall be listed, labeled and rated for plenum use.
3. Listed and labeled by an NRTL acceptable to authorities having jurisdiction as complying with UL 444 and NFPA 70 for the following types:
 - a. Communications, General Purpose: Type CM or CMG.
 - b. Communications, Plenum Rated: Type CMP, complying with NFPA 262.
 - c. Communications, Riser Rated: Type CMR, complying with UL 1666.
 - d. Communications, Limited Purpose: Type CMX.
 - e. Multipurpose: Type MP or MPG.
 - f. Multipurpose, Plenum Rated: Type MPP, complying with NFPA 262.
 - g. Multipurpose, Riser Rated: Type MPR, complying with UL 1666.

2.3 GUARANTEED CHANNEL PERFORMANCE

A. Category 6 10 Gigabit Guaranteed Channel Performance

1. The Category 6, 4 pair UTP channel shall consist of all cable and components with four connections that comprise the full 100 meter (328 feet) length circuit from the Hub/Server/LAN Electronics port located in the ER and/or TR to the device port located at the User Work Station.
2. All copper cable and apparatus shall conform to the Category 6/ Class E Channel Performance Specification.
3. The Category 6, 4 pair UTP channel shall be capable of providing stable and continual performance up to 250 MHz over the entire swept frequency range.
4. The Category 6 cable and components shall be electrically compatible with future networks and backward compatible with existing Category 3, 5, 5e. Components of the Category 6 channel shall be engineered and manufactured to compensate for any Category 3, 5 or 5e component crosstalk and shall provide at least Category 3, 5 or 5e performance in all of the customer's existing installed base of voice/data/video. The Category 6 cable and components shall be physically compatible with existing installed base of equipment.
5. The Category 6 cable and components shall not require special cords, specialty tools or special installation requirements.
6. The Category 6, 4 pair UTP channel shall be capable of providing stable and continual performance from 0 MHz to 250 MHz over the entire swept frequency range.
7. The Delay Skew on the 100 meter channel shall not exceed 50 ns.
8. Each installed permanent installed link shall meet the performance requirements of ANSI/TIA-568-C.2 standard for a Category 6 four connector channel over the entire swept frequency range from 0 MHz to 250 MHz:

- a. Insertion Loss
- b. NEXT (Near End Cross Talk)
- c. PSNEXT (Power Sum Near End Cross Talk)
- d. ELFEXT (Equal Level Far End Cross Talk)
- e. ACR (Attenuation to Crosstalk Ratio)
- f. PSACR (Power Sum Attenuation to Crosstalk Ratio)
- g. PSELFEXT (Power Sum Equal Level Far End Cross Talk)
- h. Return Loss

PART 3 – EXECUTION

3.1 CABLE ROUTING AND INSTALLATION

- A. All communications cabling used throughout project shall comply with the requirements of the National Electrical Code (NFPA 70) articles 725, 760, 770 and 800 in addition to all local codes. All copper cabling shall bear CM, CMR, CMP and/or other appropriate markings for the environment in which they are installed.
- B. Contractor shall coordinate all work with other trades.
- C. Within suspended ceiling spaces where cable trays, conduit or duct are not available and where permitted, the Contractor shall bundle station wiring (50 cables or less or 12 cables or less for Category 6 cabling) with cable ties (plenum rated where required) snug, but not so tight as to deform the cable geometry. Cable shall be bundled at 2-3 foot intervals. Cable bundles shall be supported via cable support assemblies attached to the building structure at a maximum of five (5) foot intervals. The use of bridle rings is prohibited. The Contractor shall follow manufacturer's guidelines for bending radius and pulling tension of all telecommunications cables. Contractor shall not install any cables in plenum cavities that are not approved for such an installation by the AHJ.
- D. All cable shall be installed utilizing an independent cabling support system. Cables shall not be attached to ceiling grid supports and shall not be laid directly on the ceiling grid.
- E. Cable shall not be attached to or supported by fire sprinkler systems components or any environmental sensor located in the ceiling space.
- F. All cable runs between the termination hardware and the telecommunications outlet shall be continuous without any splices.
- G. Contractor shall schedule work under this contract in a manner so as to complete all above ceiling work prior to the installation of ceiling tile. If ceiling tiles are to be removed, the Contractor shall coordinate the activity with other trades.
- H. All telecommunications cabling installed within ceiling spaces shall be routed through these spaces at right angles to and a minimum of 12 inches from electrical circuits and devices.
- I. All telecommunications cabling installed within ceiling spaces or below raised access floors shall be routed parallel or perpendicular to building structure.
- J. The minimum bend radius, under no-load conditions, for 4-pair UTP cables shall be one (1) inch or four times the diameter of the cable across its major axis, whichever is greater.
- K. Tie and dress cables neatly using cable ties in all exposed areas where required. Cable

ties shall be snug, but not so tight as to deform the cable geometry. All cable ties installed in non-plenum environments shall be of the hook and loop type (e.g. Velcro).

- L. Avoid cable stress, as caused by the following:
 - 1. Cable twist during pulling or installation.
 - 2. Tension in suspended cable runs.
 - 3. Tightly cinched cable ties.
 - 4. Tight bend radii.
- M. Open cable tray fill ratio shall not exceed 50% or as allowed by code or the AHJ.
- N. Cable fill-in conduits and enclosed raceways shall not exceed 38% or as allowed by code or the AHJ.
- O. Additional installation requirements for Category 6 cabling:
 - 1. All structured cabling terminated at patch panels must be split evenly and routed to both the left and right sides of the equipment rack or cabinet.
 - 2. Within equipment racks and where exposed, the Contractor shall bundle not more than twelve (12) cables with cable ties, snug, but not so tight as to deform the cable geometry. All cable ties installed in non-plenum environments shall be of hook and loop type (e.g. Velcro).
 - 3. Unless noted otherwise, all cable placed in overhead or underfloor raceway or plenum trough shall NOT be combed and bundled, rather it shall be arranged loosely along its length.

3.2 HORIZONTAL CABLING

- A. Contractor shall provide horizontal cables to connect each telecommunications outlet to the backbone subsystem on the same floor unless noted otherwise.
- B. Unless noted otherwise on the telecommunications drawings or elsewhere within this document, the type of horizontal cables used for each telecommunications outlet shall be 4-pair unshielded twisted pair (UTP).

END OF SECTION 271500

SECTION 283100 - FIRE DETECTION AND ALARM**PART 1 - GENERAL****1.1 SUMMARY**

- A. Supervisory sprinkler and smoke duct detector systems shall be of the addressable type and shall be separately identified at the panel and the annunciator panel.
- B. Complete and operational fire alarm system will consist of the following:
 - 1. Fire-alarm control panel.
 - 2. Manual fire-alarm boxes.
 - 3. Smoke detectors.
 - 4. Duct detectors.
 - 5. Heat detectors.
 - 6. Notification appliances.
 - 7. Backlit remote annunciator.
 - 8. Addressable interface device.
 - 9. Sprinkler valve supervisory system, control panel and annunciator.
 - 10. Trouble bell stations.
 - 11. Data Gathering Panels
 - 12. Power Supplies/Battery System
- C. Duct smoke detectors shall be furnished, installed and wired by Division 26. Division 23 Contractor to interface with the fan systems to which they are connected, and shall be monitored through the Digital Control System where applicable. Refer to Division 23 Section "Building Automation System (BAS) - Basic Materials, Interface Devices and Sensors." Duct smoke detectors shall be connected to the fire alarm control panel by Division 26, Smoke Detectors shall initiate a supervisory alarm.
 - 1. Electrical contractor shall provide duct smoke detector on the return header duct with associated remote keyed test station / fan shut-down for HVAC equipment rated 2,000 CFM and greater.
 - 2. Electrical contractor shall provide duct smoke detector on the supply and return header duct with associated remote keyed test station / fan shut-down for HVAC equipment rated 15,000 CFM and greater.

1.2 DEFINITIONS

- A. Definitions in NFPA 72 apply to fire-alarm terms used in this Section.
- B. FCP: Fire alarm control panel.
- C. Addressable Analog System: Utilizing a signaling method characterized by the simultaneous or sequential transmission, or both, and the reception of multiple signals in a communication channel, including means for positively identifying each signal.
- D. FAAP: Fire alarm annunciator panel.
- E. FCP: Fire alarm control panel.
- F. NAC: Notification appliances circuit.

- G. IAC: Initiating appliances circuit.
- H. SSA: Sprinkler supervisory system annunciator (where applicable).
- I. SSCP: Sprinkler supervisory system control panel (where applicable).
- J. SLC: Signaling line circuit.
- K. LED: Light-emitting diode.
- L. NICET: National Institute for Certification in Engineering Technologies.
- M. Pathway Class Designations.
 - 1. Pathways shall be designated as Class A, Class B, Class C, Class D, Class E, or Class X, depending on their performance.
- N. Signaling Line Circuit – Addressable Loop
 - 1. Class A. A pathway shall be designated as Class A when it performs as follows:
 - a. It includes a redundant path.
 - b. Operational capability continues past a single open.
 - c. Conditions that affect the intended operation of the path are annunciated.
 - d. Isolator module located every 15 devices.
- O. Notification Appliance Circuits, and Initiating Device Circuits
 - 1. Class B. A pathway shall be designated as Class B when it performs as follows:
 - a. It does not include a redundant path.
 - b. Operational capability stops at a single open.
 - c. Conditions that affect the intended operation of the path are annunciated.
- P. WLAN, LAN, Internet Wireless or POTS
 - 1. Class C. A pathway shall be designated as Class C when it performs as follows:
 - a. It includes one or more pathways where operational capability is verified via end-to-end communication, but the integrity of individual paths is not monitored.
 - b. A loss of end-to-end communication is annunciated.
- Q. Failsafe Circuits (Door Holder where applicable)
 - 1. Class D. A pathway shall be designated as Class D when it has fail-safe operation, where no fault is annunciated, but the intended operation is performed in the event of a pathway failure.
- R. Unsupervised Circuits
 - 1. Class E. A pathway shall be designated as Class E when it is not monitored for integrity.
- S. Panel to Panel to Annunciator (SLC old Class A, Style 7)

1. Class X. A pathway shall be designated as Class X when it performs as follows:

- a. It includes a redundant path.
- b. Operational capability continues past a single open or short-circuit.
- c. Conditions that affect the intended operation of the path are annunciated.

1.3 SUBMITTALS

A. General Submittal Requirements:

1. All submittals shall be approved by the local authorities having jurisdiction prior to the beginning of any work.
2. Shop Drawings shall be prepared by persons with the following qualifications:
 - a. Trained and certified by manufacturer in fire-alarm system design.
 - b. NICET-certified fire-alarm technician, Level III minimum.

B. Submission to the local authorities of jurisdiction:

1. Submit detailed drawings in accordance with the BFP's procedures and requirements for the fire alarm plan review.
2. Provide identical submittal to the Architect for concurrent review.
3. Upon receipt of the comments from BFP, submit a copy of all the annotated drawings or correction sheets to the Architect.
4. Incorporate all comments into the detailed drawings and resubmit to the BFP until approval is obtained.

C. Product Data: For each type of product indicated. Provide manufacturers product data sheets, showing the types and models and manufacturer's installation/instruction of all equipment, devices and wire/cable proposed. Evidence of FM & ULI Listings and local approvals shall be submitted with the data sheets. Catalog numbers alone are not acceptable. Each document shall be highlighted indicating the model proposed, specific data, and cross out items that are not applicable for the work.

D. Shop Drawings: For fire-alarm system. Include plans, elevations, sections, details, and attachments to other work.

1. Comply with the recommendations in the "Documentation" Section of the "Fundamentals of Fire Alarm Systems" Chapter in NFPA 72.
2. Include voltage drop calculations for notification appliance circuits. Provide circuit calculations that include all system requirements including any requirements for notification appliance circuits, or any other auxiliary function powered by the system. All notification appliance circuit(s) shall be sized with 10% voltage drop at 24 volts.
3. Include battery-size calculations. Battery calculations shall list the type of devices and modules, quantities, unit and extended amperage draw for quiescent and alarm conditions, total amperage draw and battery amp/hour rating. For design criteria, the battery amp/hour rating listed by the manufacturer shall be de-rated by 20%. Include all system requirements including any requirements for visual signaling appliances, or any other auxiliary function powered by the system.
4. Include system operation description: Detailed description for this Project, including method of operation and supervision of each type of circuit and sequence of operations for manually and automatically initiated system inputs and outputs. Manufacturer's standard descriptions for generic systems are not acceptable.

5. Include system riser diagram with device addresses, conduit sizes, cable and wire type and sizes.
 6. Include wiring diagrams: power, signal and control wiring. Include diagrams for equipment and for system with all terminals and connections identified. Show wiring color code.
 7. Include performance parameters and installation details for each detector, verifying that each detector is listed for complete range of air velocity, temperature, and humidity possible when air-handling system is operating.
 8. Include device address list: Coordinate with final system programming.
 9. Point List: Provide complete zone list each initiating zone (device address), annunciator zone, custom message descriptor, notification appliance circuits, auxiliary function circuit, and the specific device associated with each function or zone. Descriptors shall be based on Room Signage Numbers.
 10. Include floor plans to indicate final outlet locations showing address of each addressable device. Show size and route of cable and conduits.
 11. Submit electronic copy of drawings, on CD or DVD, to Architect.
- E. Qualification Data: For Installer.
- F. Field Test Reports: Submit test reports documenting the activities and procedures specified. Submit reports within two weeks of completion of start-up procedures.
- G. Operation and Maintenance Data: For fire-alarm systems and components to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 01 Section "Operation and Maintenance Data," deliver copies to the Board Authorized Representative, Architect and to the authorities having jurisdiction. Provide six (6) copies of materials, both hardcopy and electronic, in format required by Board Authorized Representative. Include the following:
1. Comply with the "Records" Section of the "Inspection, Testing and Maintenance" Chapter in NFPA 72 and local requirements.
 2. Provide "Record of Completion Documents" according to NFPA 72 article "Permanent Records" in the "Records" Section of the "Inspection, Testing and Maintenance" Chapter.
 3. Provide "Maintenance, Inspection and Testing Records" according to NFPA 72 article of the same name and include the following:
 4. Frequency of testing of installed components.
 5. Frequency of inspection of installed components.
 6. Requirements and recommendations related to results of maintenance.
 7. Manufacturer's user training manuals.
 8. Manufacturer's required maintenance related to system warranty requirements.
 9. Provide spare parts data. Provide the name, addresses, and telephone numbers of service organizations that carry stock of repair parts for the system to be furnished.
 10. Abbreviated operating instructions for mounting at fire-alarm control unit.
- H. Training Reports: Submit reports on training documenting topic covered, dates and attendance.
- I. Software and Firmware Operational Documentation:
1. Software operating and upgrade manuals.
 2. Program Software Backup: Complete with data files, in format required by Board Authorized Representative.
 3. Device address list.
 4. Printout of software program.

1.4 QUALITY ASSURANCE

A. Installer Qualifications:

1. Installation shall be by personnel certified by NICET as fire-alarm Level III technician.
2. As a precondition to the contract, the Installer shall be regularly engaged in the design, servicing, installation and testing of fire detection and alarm notification systems, shall have at least five years of experience in the installation of fire detection and alarm notification systems.
3. Shop Drawings shall be prepared by persons with the following qualifications:
4. Trained and certified by manufacturer in fire-alarm system design.
5. NICET-certified fire-alarm technician, Level III minimum.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in the National Electrical Code, and marked for intended location and application.

C. Comply with International Building Code.

D. Comply with NFPA 72, current Edition.

E. 90A Standard for the Installation of Air Conditioning and Ventilating Systems

F. 101 Life Safety Code

G. National Recognized Testing Laboratories (NRTL); Underwriters Laboratories Inc. (UL), Intertek (ETL), Factory Mutual (FM):

1. No. 268 Smoke Detectors for Fire Protective Signaling Systems
2. No. 864, 9th Edition Control Units for Fire Protective Signaling Systems
3. No. 268A Smoke Detectors for Duct Applications.
4. No. 521 Heat Detectors for Fire Protective Signaling Systems
5. No. 464 Audible Signaling Appliances.
6. No. 38 Manually Actuated Signaling Boxes.
7. No. 346 Waterflow Indicators for Fire Protective Signaling Systems.
8. No. 1971 Visual Notification Appliances for the hearing impaired.
9. No. 1711 Amplifiers for Fire Protective Signaling Systems

H. FCC Part 15

I. The intelligent fire alarm shall have proper listing and/or approval to ANSI/UL-864 9th Edition.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver fire alarm system components and devices in original factory shipping cartons, with original labels intact.

B. Handle fire alarm system components and devices in accordance with manufacturer's written instructions, to avoid damage.

C. Store fire alarm system components and devices indoors in clean, dry space with uniform temperature to prevent condensation. Protect fire alarm system components and devices from exposure to dirt, fumes, water, corrosive substances, and physical damage.

- D. Install temporary plastic covers on smoke detectors. Remove covers immediately before final system testing.

1.6 EXTRA MATERIALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Lamps for Annunciator Unit: Quantity equal to 10 percent of amount installed, but no fewer than 1 unit.
 - 2. Smoke Detectors, Heat Detectors: Quantity equal to 10 percent of amount of each type installed, but no fewer than 1 unit of each type.
 - 3. Detector Bases: Quantity equal to 5 percent of amount of each type installed, but no fewer than 1 unit of each type.
 - 4. Keys and Tools: Two extra sets for access to locked and tamper-proof components.
 - 5. Audible and Visual Notification Appliances: Quantity equal to 5 percent of each type installed, but no fewer than 1 unit of each type.

1.7 MAINTENANCE SERVICE

- A. Maintenance Service Contract: Provide maintenance of fire alarm systems and equipment for a period of 12 months commencing with Preliminary Acceptance/Substantial Completion, using factory-authorized service representatives.
 - 1. Basic services: Respond to service calls within 24 hours of notification of system trouble. Adjust and replace defective parts and components with original manufacturer's replacement parts, components, and supplies.
 - 2. Additional Services: Perform services within the above 12-month period not classified as routine maintenance or as warranty work when authorized in writing. Compensation for additional services must be agreed upon in writing prior to performing services. Provide in the 11 month of the 12 months commencing with Substantial Completion a Test, Inspect and Report of the fire alarm system. Renewal of Maintenance Service Contract: No later than 60 days prior to the expiration of the 1 Year System Warranty contract, deliver to the Board a proposal to provide contract maintenance and repair services for an additional two-year term which includes labor rates for Year 2 and 3 of Warranty Period. The Board will be under no obligation to accept maintenance service contract renewal proposal.
 - 3. Software Service Agreement:
 - 4. Comply with UL 864.
 - 5. Technical Support: Beginning with Preliminary Acceptance/Substantial Completion, provide software support for one year.
 - 6. Upgrade Service: Update software to latest version at Project completion. Install and program software upgrades that become available within one year from date of Substantial Completion.
 - 7. Provide 30 days notice to Owner to allow scheduling and access to system.

1.8 WARRANTY

- A. Refer to Division 01 60 00 Product Requirements specification for warranty registration requirements.
- B. Warranty all materials, installation and workmanship for three (3) years from date of acceptance, unless otherwise specified. Provide a copy of the manufacturers' warranty with

closeout documentation the operation and installation manuals as required in Division 01 Sections "Closeout Procedures" and "Operation and Maintenance Data".

- C. The System Supplier shall maintain a service organization with adequate spare parts stock within 50 miles of the installation. Any defects that render the system inoperative shall be repaired within 24 hours of the Owner notifying the contractor.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by the following:

1. Notifier by Honeywell
2. Gamewell
3. Siemens

2.2 SYSTEMS OPERATIONAL DESCRIPTION

- A. Class I system fire-alarm signal initiation shall be by one or more of the following devices:

1. Manual stations.
2. Heat detectors.
3. Smoke detectors.
4. Automatic sprinkler system water flow.

- B. Fire-alarm signal shall initiate in the general alarm condition the following actions or sequence of operations:

1. Continuously operate alarm notification appliances.
2. Identify alarm at fire-alarm control unit and local remote annunciator.
3. Transmit an alarm signal to a UL Listed Central Station monitoring service via direct telephone connection.
4. Unlock electric door locks in designated egress paths.
5. Release fire and smoke doors held open by magnetic door holders.
6. Transmit signal to the Building Automation System to switch heating, ventilating, and air-conditioning equipment controls to fire-alarm mode.
7. Activate emergency lighting control.
8. Record events in the system memory.

- C. General alarm initiation on FACP, FAAP, SSCP and SSA shall be by the following conditions:

1. Fire pump run condition.
2. Flow Switch activation.

- D. Supervisory signal initiation on SSCP shall be by one or more of the following devices and actions:

1. Valve supervisory switch.
2. Fire-pump power failure, including a dead-phase or phase-reversal condition.
3. Provide a dedicated 20 amp, 120 volt, single-phase branch circuit with 2 #12 & 1 #12 ground in a ¾" conduit to fire pump controller for local supervisory power. This circuit shall be fed from an emergency panel where possible. The conduit shall be factory

painted RED and clearly identified to read "Fire Pump Local Supervisory 120VAC circuit. This is to power a local pump run/power failure alarm that is built into the controller, separate from flow bells & the building alarm system.

- E. Supervisory signal initiation on SSCP shall initiate the following actions:
 - 1. Initiate supervisory audible and visible signal indication on SSCP and SSA.
- F. System trouble signal initiation shall be by one or more of the following devices and actions:
 - 1. Open circuits, shorts, and grounds in designated circuits.
 - 2. Opening, tampering with, or removing alarm-initiating and supervisory signal-initiating devices.
 - 3. Loss of primary power at fire-alarm control unit.
 - 4. Ground or a single break in fire-alarm control unit internal circuits.
 - 5. Abnormal ac voltage at fire-alarm control unit.
 - 6. Break in standby battery circuitry.
 - 7. Failure of battery charging.
 - 8. Abnormal position of any switch at fire-alarm control unit or annunciator.
 - 9. System Trouble Signal Actions: Initiate notification appliance and annunciate at FCP, or SSCP and respective local remote annunciators.
- G. Priority of Signals: Automatic response functions shall be accomplished by the first zone initiated. Alarm functions resulting from initiation by the first zone shall not be altered by subsequent alarms. An alarm signal shall be the highest priority. Supervisory or trouble signals shall have second- and third-level priority. Signals of a higher level priority shall take precedence over signals of lower priority even though the lower priority condition occurred first. Annunciate all alarm signals regardless of priority or order received.
- H. Noninterfering: Provide zoned, powered, wired, and supervised system so a signal on one zone does not prevent the receipt of signals from any other zone. All zones shall be manually resettable from the FCP after the initiating device or devices have been restored to normal. Systems that require the use of batteries or battery backup for the programming function are not acceptable.
- I. Silencing at control panel: Switches shall provide capability for acknowledgment of alarm; supervisory, trouble, and other specified signals at the FCP or SSCP; and capability to silence the local audible signal and light an LED (light emitting diode). Subsequent zone alarms shall cause the audible signal to sound again until silenced in turn by switch operation. Restoration to normal of alarm, supervisory, and trouble conditions shall extinguish the associated LED and cause the audible signal to sound again until the restoration is acknowledged by switch operation.
- J. The system shall have spare installed capacity enabling it to support a twenty percent (20%) increase in initiating, control and indicating device circuits. Spare circuit capacity shall be evenly distributed throughout the system.
- K. Annunciators shall have an additional fifty percent (50%) spare spaces available for future building additions.
- L. Fire Alarm Control Panel shall monitor the following functions:
 - 1. Kitchen Fire-extinguishing system operation (where applicable).
 - 2. Monitor Duct Detectors in supervisory mode only.

2.3 FIRE-ALARM CONTROL UNITS (FCP AND SSCP)

A. General Requirements for Fire-Alarm Control Unit:

1. Field-programmable, microprocessor-based, modular, power-limited design with electronic modules, complying with UL 864 and listed and labeled by an NRTL.
2. System software and programs shall be held in flash electrically erasable programmable read-only memory (EEPROM), retaining the information through failure of primary and secondary power supplies.
3. Include a real-time clock for time annotation of events on the event recorder and printer.
4. Addressable initiation devices that communicate device identity and status.
5. Smoke sensors shall additionally communicate sensitivity setting.
6. Temperature sensors shall additionally test for and communicate the sensitivity range of the device.

B. Cabinet: Lockable steel enclosure. Arrange panel so all operations required for testing or for normal care and maintenance of the system are performed from the front of the enclosure. If more than a single unit is required to form a complete control panel, provide exactly matching modular unit enclosures. Provide cabinets large enough to accommodate all components and to allow ample gutter space for interconnection of panels as well as field wiring. Identify each enclosure and each component by an engraved red laminated phenolic resin nameplate. Lettering on the enclosure nameplate shall not be less than 1 inch high. Identify individual components and modules within the cabinets by engraved laminated phenolic resin nameplates.

C. The intelligent fire alarm shall contain multiprocessors for communications with and control of equipment used to make up the system: intelligent detectors, intelligent modules, printer, annunciators, and other system controlled devices.

D. System Capacity and General Operation

1. Each intelligent fire alarm and voice evacuation system shall be capable of providing up to 504 intelligent analog addressable devices, and shall be networkable up to a total of 200 intelligent fire alarm and emergency voice evacuation systems.
2. The intelligent fire alarm shall include a full featured user/operator interface control and annunciation panel that shall be composed of a 240 x 64 graphical LCD display, individual system status LEDs, and an alphanumeric keypad for field programming and control of the entire system.
3. All programming or editing of the existing program in the system shall be achieved without special equipment and without interrupting the alarm monitoring functions of the intelligent fire alarm and emergency voice evacuation system.
4. The intelligent fire alarm shall provide the following features; detector maintenance alert to warn of excessive dirt/dust within an individual detector, detector sensitivity read/test information, with real-time status reports capable of being viewed on the display or printed, intelligent smoke detector alarm verification and pre-signal capabilities, meeting NFPA 72 requirements, and rapid (priority) manual pull station reporting (< 3 seconds). Field programmability of various voice communications messages to meet various application requirements. An integral walk test feature shall also be provided capable of selecting with or without signals.
5. The Base Card of the system shall contain three Form-C relay contacts rated 2.0 amps @ 30VDC for: Alarm, Trouble (with or with/out AC delay), and Supervisory status conditions. These three contacts shall be field programmable to meet any installation requirement.

6. The Base Card shall contain two Notification Appliance Circuits (NACs) capable of being wired Class B (NFPA Style Y or Z). NAC circuits shall be 24VDC filtered and regulated and listed to operate with any Listed NAC device available or previously installed.
 7. The Base System shall as a minimum include a dual 40 Watt amplifier capable of being field configured for 80 Watts of power or 40 Watts of power with 100% backup. Additional amplifiers shall be capable of being added to the intelligent fire alarm to meet the total wattage requirement of the installation.
- E. Main System Microprocessors
1. The intelligent fire alarm shall include a minimum of four microprocessors; one for the main system, one for each Signaling Line Circuits (SLCs), one for the intelligent amplifier, and one for each switching regulated power supply.
 2. The microprocessors shall communicate with, monitor, and control all internal and external interfaces within the intelligent fire alarm on system. Each microprocessor shall include flash memory for program storage and a "watch-dog" timer circuit to detect and report microprocessor failure.
 3. The microprocessors shall contain and execute all programming for specific action to be taken based on system status changes. Such programming shall be held in non-volatile programmable memory and shall not be lost if both the system primary and secondary power supplies fail.
 4. The intelligent fire alarm shall provide a real-time clock for device programmable time functions, time stamping system displays, printers, and history files.
 5. The microprocessors shall contain flash memory capabilities for easy upload/download of future product upgrades and enhancements.
 6. All clock, date and history file information shall be maintained during AC and DC power loss.
- F. Alphanumeric Display and System Controls: Arranged for interface between human operator at fire-alarm control unit and addressable system components including annunciation and supervision. Display alarm, supervisory, and component status messages and the programming and control menu.
1. Annunciator and Display: Liquid-crystal type, 2 line(s) of 80 characters, minimum.
 2. Keypad: Arranged to permit entry and execution of programming, display, and control commands.
- G. Circuits:
1. Initiating Device, and Signaling Line Circuits: NFPA 72 2010.
 2. Initiating Device Circuits: Class B.
 3. Addressable Loop Signaling Line Circuits: Class A.
 4. Provide isolator module every 15 devices.
 5. Signaling Line Circuit (Network Data, Panels and Annunciators): Class X.
 6. Notification Appliance: Circuits: NFPA 72, Class B.
 7. Door holder Circuit: Class D.
- H. Notification Appliance Circuit: Operation pattern shall be as directed by the authorities having jurisdiction.
- I. Door Controls: Door hold-open devices shall be controlled by adjacent smoke detectors communicating with the fire-alarm system.

1. Fire Alarm Contractor shall supply door hold open devices. Provide fire alarm relay contacts or modules.
- J. Transmission to Remote Alarm Receiving Station: Automatically transmit alarm, supervisory, and trouble signals via a UL listed Central Monitoring Station using direct wire connection.
- K. Primary Power: 24-V dc obtained from 120-V ac service that is to be connected to the building source of the emergency power and a power-supply module. Initiating devices, notification appliances, signaling lines, trouble signals and supervisory signals shall be powered by 24-V dc source. The fire alarm control panel shall be fed from a 20 amp, 120VAC, single-phase branch circuit breaker. This circuit breaker must be permanently identified in "RED" and shall be provided with a permanent identification label that reads "FIRE ALARM CIRCUIT".
 1. Alarm current draw of entire fire-alarm system shall not exceed 80 percent of the power-supply module rating.
- L. Secondary Power: 24-V dc supply system with batteries, automatic battery charger, and automatic transfer switch.
- M. Automatic transfer switch: Transfer the load to the battery without loss of signals or status indications in the event of failure of primary power.
- N. Batteries: System control equipment shall receive secondary operating power from batteries integral to the equipment. Power Supply for Supervision Equipment: Supply for audible and visual equipment for supervision of the ac power shall be from a dedicated dc power supply, and power for the dc component shall be from the ac supply. Provide sufficient capacity to operate the complete system in normal or supervisory (non-alarm) mode for a period of 24 hours. Following this period of operation on battery power, the batteries shall have sufficient capacity to operate all components of the system, including all alarm indicating devices in alarm or supervisory mode for a period of 15 minutes.
- O. Battery Charger: Solid state, fully automatic, variable charging rate type. Provide for 150 percent of the connected system load while maintaining the batteries at full charge. In the event batteries are fully discharged the charger shall recharge them fully within four hours. Charger output shall be supervised as part of system power supply supervision.
- P. Surge Protection:
 1. Install surge protectors recommended by control panel manufacturer. Install on all system wiring external to the building housing the control panel.
- Q. Instructions: Computer printout or typewritten instruction card mounted behind a plastic or glass cover in a stainless-steel or aluminum frame. Include interpretation and describe appropriate response for displays and signals. Briefly describe the functional operation of the system under normal, alarm, and trouble conditions.
- R. Systems: Alarm and supervisory systems in FCP/SSCP shall be completely separate and independent. The alarm initiating zone boards in the FCP/SSCP shall consist of plug-in cards. Construction requiring removal of field wiring for module removal is not acceptable.
- S. Control Modules: Types and capacities to bmsperform all functions of the fire alarm system. Provide local, visible, and audible signals to notify of any alarm, supervisory, and trouble condition. Provide each type of audible alarm with a distinctly different sound.

- T. Indicating Lights: Provide individual alarm and trouble LED for each zone. Provide a System LED test switch for each FCP/SSCP section that will illuminate all LED devices on that section of the control panel. Alarm and supervisory signals shall light a red LED of the associated zone. Trouble signals shall light an amber LED for the associated zone.

U. Smoke/Fire Zones

1. Each floor shall be group zoned by building according to the fire partition construction with further breakdown as to the device type.
2. Additional zones shall be dedicated to:
 - a. Food Service Fire Extinguishing System.
 - b. Each electrical equipment room.
 - c. Each mechanical/pump room.
 - d. Each telephone/data room.

2.4 MANUAL FIRE-ALARM BOXES

- A. General Requirements for Manual Fire-Alarm Boxes: Comply with UL 38. Boxes shall be metallic finished in red with molded, raised-letter operating instructions in contrasting color; shall show visible indication of operation; and shall be mounted on recessed outlet box. If indicated as surface mounted, provide manufacturer's surface back box.

1. Double-action mechanism requiring two actions to initiate an alarm, pull-lever type; with integral addressable module arranged to communicate manual-station status (normal, alarm, or trouble) to fire-alarm control unit.
2. Approved by the authorities having jurisdiction.
3. Stations requiring the breaking of the glass panel are not acceptable.
4. Station Reset: Resetting the manual fire alarm stations after operation shall require the use of a key. Manual stations which use a screwdriver or allen wrench for reset are not acceptable. Keyed the same as FCP/SSCP.
5. Indoor Protective Shield: Factory-fabricated clear plastic enclosure hinged at the top to permit lifting for access to initiate an alarm. Lifting the cover actuates an integral battery-powered audible horn intended to discourage false-alarm operation.

2.5 SYSTEM SMOKE DETECTORS

- A. Requirements for Analog Addressable System Smoke Detectors:

1. Comply with UL 268; operating at 24-V dc, nominal.
2. Intelligent Addressable Analog Type: Designed to communicate detector status (normal, alarm, or trouble) to fire-alarm control unit.
3. Multipurpose type, containing the following:
 - a. Integrated Addressable Module: Arranged to communicate detector status (normal, alarm, dirty, trouble or maintenance) to the FCP.
 - b. Base Mounting: Detector and associated electronic components shall be mounted in a twist-lock module that connects to a fixed base. Provide terminals in the fixed base for connection to building wiring.
 - c. Self-Restoring: Detectors do not require resetting or readjustment after actuation to restore them to normal operation.
 - d. Integral Visual-Indicating Light: Blinking LED type indicating detector is communicating properly and turn RED in alarm.

4. Detector address shall be accessible from fire-alarm control unit and shall be able to identify the detector's location within the system and its sensitivity setting.
5. An operator at fire-alarm control unit, having the designated access level, shall be able to manually access the following for each detector:
 - a. Primary status.
 - b. Device type.
 - c. Present average value.
 - d. Present sensitivity selected.
 - e. Sensor range (normal, dirty, etc.).

B. Duct Detectors

1. Division 26 to provide addressable control relay for FCP monitoring.
2. Division 26 to provide remote alarm indicator/test switch, which indicates activation of detector.
3. Duct Smoke Detector Test Station
4. Flush wall mounted except surface wall mounted or group wall mounted or grouped FATC flush door mounted in mechanical equipment rooms.
5. Alarm/power LED.
6. Keyed test switch.
7. Stainless steel faceplate.
8. White lettering on red nameplate identifying associated duct smoke detector.

2.6 HEAT DETECTORS

- A. General Requirements for Analog Addressable Heat Detectors: Comply with UL 521.
- B. Heat Detector, Combination Type: Actuated by either a fixed temperature of 135 deg F or a rate of rise per minute. Fixed temperature sensing shall be independent from rate-of rise sensing.
 1. Mounting: Twist-lock base interchangeable with smoke-detector bases.
 2. Integrated Addressable Heat Detector: Arranged to communicate detector status (normal, alarm, or trouble) to fire-alarm control unit.
 3. Provide devices in ordinary ambient temperature rooms.
- C. Heat Detector, Fixed-Temperature Type: Actuated by temperature that exceeds a fixed temperature of 190 deg F.
 1. Mounting: Twist-lock base interchangeable with smoke-detector bases.
 2. Integrated Addressable Heat Detector: Arranged to communicate detector status (normal, alarm, or trouble) to fire-alarm control unit.
 3. Provide devices in high ambient temperature rooms.

2.7 NOTIFICATION APPLIANCES

- A. General Requirements for Notification Appliances: Connected to notification appliance signal circuits, zoned as indicated, equipped for mounting as indicated and with screw terminals for system connections.
 1. Combination Devices: Factory-integrated audible and visible devices in a single-mounting assembly, equipped for mounting as indicated and with screw terminals for system connections.

- B. Trouble Bells: Electric-vibrating, 24-V dc, under-dome type; with provision for housing the operating mechanism behind the bell. Bells shall produce a sound-pressure level of 94 dBA, measured 10 feet from the bell. 4-inch size, unless otherwise indicated. Bells are weatherproof where indicated.
- C. Audible Alarm-Indicating Devices (Horns): Comply with UL 464; factory finish: red. Devices shall be constructed for safe use in boiler rooms, kitchens and exterior locations without impairing the quality of tone reproduction in ambient conditions ranging from 30 degrees Fahrenheit to 150 degrees Fahrenheit. Horns shall be sound level setting of 92dB at 10 feet using the coded signal prescribed in UL 464 Test Protocol. Horn shall mount to surface or flush box. When flush mounted, the Horn shall extend less than one inch from the wall.
 - 1. Horns shall be field configurable for continuous, temporal, or march time pattern.
 - 2. In temporal or march time mode horn operation shall be synchronized.
- D. Visible Notification Appliances: Xenon strobe lights comply with UL 1971, with clear or nominal white polycarbonate lens. The word "FIRE" is engraved in minimum 1-inch- high letters on the lens.
 - 1. Rated Light Output: 15/30/75/110 cd, selectable in the field.
 - 2. Mounting: Wall mounted unless otherwise indicated.
 - 3. For units with guards to prevent physical damage, light output ratings shall be determined with guards in place.
 - 4. Flashing shall be synchronized with other units in common viewing areas per ADA and NFPA 72. Provide properly sized outlet box to accommodate synchronizing device on strobe circuit.
 - 5. Strobe Leads: Factory connected to screw terminals.
 - 6. Mounting Faceplate: Factory finished, red.

2.8 NAC POWER SUPPLY:

- A. The NAC power supply shall be independent unit that will provide power to visual strobe notification appliances. It shall be possible to configure NAC power supply activated from intelligent synchronized modules. The power supply NAC's must be configurable to operate independently at any one of the following rates: continuous synchronized, or 3-3-3 temporal. Fault conditions on the power supply shall not impede alarm activation of host NAC circuits or other power supplies. The NAC power supply must be able to provide concurrent power for notification devices, security devices, access control equipment and auxiliary devices such as door holders. . All the NAC Power Supplies shall be synchronized. The power supply shall support up to 24 amp hour batteries.
 - 1. Power supply shall be minimum of 8 amps and UL 864 Listed.
 - 2. Four independent 2 amp NAC circuits. Each being configurable as auxiliary power.
 - 3. All circuits shall be synchronized.

2.9 WATERFLOW SWITCHES AND AIR PRESSURE SWITCHES

- A. Provided and installed by Division 22, wired by Division 26.

2.10 SPRINKLER VALVE TAMPER SWITCHES

- A. Provided and installed by Division 22, wired by Division 26.

2.11 REMOTE ANNUNCIATORS: FAAP, SSA.

- A. Provide separate annunciators for FAAP and SSA, as required and approved by the authorities having jurisdiction.
- B. Description: Annunciator functions shall match those of fire-alarm control units (FCP and SSCP) for alarm, supervisory, and trouble indications.
- C. Annunciator shall be as approved by the authorities having jurisdiction.: indicate the alarm and supervisory zones by means of a white translucent rectangle illuminated from behind. Zone shall be identified by opaque black lettering on the rectangular indicator. Lettering shall be 1/2 inch minimum.
 - 1. Mounting: Flush cabinet, NEMA 250, Type 1 unless otherwise specifically indicated on the drawings.
- D. Display Type and Functional Performance: Alphanumeric display and LED indicating lights shall match those of fire-alarm control unit. Provide status indication with separate annunciation for each zone. Provide controls to acknowledge, silence, reset, and test functions for alarm, supervisory, and trouble signals.

2.12 ADDRESSABLE INTERFACE DEVICE

- A. Description: Microelectronic monitor module, NRTL listed for use in providing a system address for alarm-initiating devices for wired applications with normally open contacts.

2.13 SPRINKLER SYSTEM REMOTE INDICATORS

- A. Remote status and alarm indicator and test stations, with LED indicating lights. Light is connected to flash when the associated device is in an alarm or trouble mode. Lamp is flush mounted in a single-gang wall plate. A red, laminated, phenolic-resin identification plate at the indicating light identifies, in engraved white letters, device initiating the signal and room where the valve is located. The identification plate also designates protected spaces downstream from the water-flow switch.

2.14 DEVICE WIRE GUARDS

- A. Description: Welded wire mesh of size and shape for the notification device, smoke detector, or other device requiring protection.
 - 1. Factory fabricated and furnished by manufacturer of device.
 - 2. Finish: Paint of color to match the protected device.

2.15 WIRE AND CABLE

- A. Wire and cable for fire alarm systems shall be UL listed and labeled as complying with NEC, Article 760.
- B. Signaling Line Circuits: Twisted, shielded pair, or solid copper conductor, not less than No. 14 AWG, or size as recommended by system manufacturer.
- C. Non-Power-Limited Circuits: Solid-copper conductors with 600-V rated, 75 deg C, color-coded insulation.

1. Low-Voltage Circuits: No. 14 AWG, minimum.
2. Line-Voltage Circuits: No. 12 AWG, minimum.

2.16 SPARE CAPACITY

- A. Provide twenty percent (20%) spare capacity in all panels, annunciators, on every SLC and NAC for future expansion.

PART 3 - EXECUTION

3.1 EQUIPMENT INSTALLATION

- A. Comply with International Building Code and NFPA 72 for installation of fire-alarm equipment.
- B. Smoke- or Heat-Detector Spacing:
 1. Comply with NFPA 72, "Smoke-Sensing Fire Detectors" Section in the "Initiating Devices" Chapter, for smoke-detector spacing.
 2. Comply with NFPA 72, "Heat-Sensing Fire Detectors" Section in the "Initiating Devices" Chapter, for heat-detector spacing.
 3. Smooth ceiling spacing shall not exceed 30 feet.
 4. Spacing of detectors for irregular areas, for irregular ceiling construction, and for high ceiling areas shall be determined according to Appendix B in NFPA 72.
 5. HVAC: Locate detectors not closer than 3 feet from air-supply diffuser or return-air opening.
 6. Lighting Fixtures: Locate detectors not closer than 12 inches from any part of a lighting fixture.
- C. Remote Status and Alarm Indicators: Install near each sprinkler water-flow switch and valve-tamper switch that is not readily visible from normal viewing position.
- D. Audible Alarm-Indicating Devices: Install bells and horns on flush-mounted back boxes with the device-operating mechanism concealed behind a grille, at height indicated in the Drawings.
- E. Visible Alarm-Indicating Devices: Install at height indicated in the Drawings.
- F. Device Location-Indicating Lights: Locate in public space near the device they monitor.
- G. FAAP/SSA: Install with top of panel not more than 72 inches above the finished floor.

3.2 CONNECTIONS

- A. For fire-protection systems related to doors in fire-rated walls and partitions and to doors in smoke partitions, comply with requirements in Division 08 Section "Door Hardware." Connect hardware and devices to fire-alarm system.
 1. Verify that hardware and devices are NRTL listed for use with fire-alarm system in this Section before making connections.
- B. Make addressable connections with a supervised interface device to the following devices and systems. Install the interface device less than 3 feet from the device controlled. Make an addressable confirmation connection when such feedback is available at the device or system being controlled.

1. Alarm-initiating connection to activate emergency lighting control.
 2. Supervisory connections at valve supervisory switches.
 3. Supervisory connections at fire-pump power failure including a dead-phase or phase-reversal condition.
 4. Supervisory connection with Kitchen Fire-extinguishing system operation (where applicable).
 - a. Supervisory connection with Duct Detectors.
 - b. BAS system interconnections.
- C. System Trouble Bell: Furnish and install, where shown the drawings, a semi-flush mounted, 6-inch diameter, red trouble bell. Immediately adjacent to each trouble bell location, install a sign with a red background and 1/2-inch black lettering engraved as follows: "FCP (or SSCP or DSCP) TROUBLE BELL".
- D. Immediately adjacent to each trouble bell location, provide a trouble bell silence switch mounted on a single gang plate with the following features: A switch to silence the bell, a red light to indicate when the bell is silenced, and a momentary switch to test the trouble bell.

3.3 WIRING INSTALLATION

- A. Install wiring according to the following:
1. National Electrical Code
 2. NECA 1.
- B. Wiring Method: Install wiring in metal raceway according to Division 26 Section "Raceways and Boxes for Electrical Systems."
1. Fire alarm circuits and equipment control wiring associated with the fire alarm system shall be installed in a dedicated raceway system. This system shall not be used for any other wire or cable.
- C. Wiring within Enclosures: Separate power-limited and non-power-limited conductors as recommended by manufacturer. Install conductors parallel with or at right angles to sides and back of the enclosure. Bundle, lace, and train conductors to terminal points with no excess. Connect conductors that are terminated, spliced, or interrupted in any enclosure associated with the fire alarm system to terminal blocks. Mark each terminal according to the system's wiring diagrams. Make all connections with approved crimp-on terminal spade lugs, pressure-type terminal blocks, or plug connectors.
- D. Cable Taps: Use numbered terminal strips in junction, pull, and outlet boxes, cabinets, or equipment enclosures where circuit connections are made.
- E. Color-Coding: Color-code fire alarm conductors differently from the normal building power wiring. Use one color-code for alarm circuit wiring and a different color-code for supervisory circuits. Color-code audible alarm-indicating circuits differently from alarm-initiating circuits. Use different colors for visible alarm-indicating devices. Paint fire alarm system junction boxes and covers red. Identify the accessible raceway with color-coded, self-adhesive red vinyl tape applied in bands.
- F. Install separate and completely independent conduits for fire alarm and sprinkler supervisory systems.

- G. A maximum of five floors per one addressable loop shall be allowed.

3.4 IDENTIFICATION

- A. Identify system components, wiring, cabling, and terminals. Comply with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."
- B. Install framed instructions in a location visible from fire-alarm control unit.

3.5 GROUNDING

- A. Ground fire-alarm control unit and associated circuits; comply with IEEE 1100. Install a ground wire from main service ground to fire-alarm control unit.

3.6 CLEANING

- A. On completion of installation, vacuum dirt and debris from interiors; do not use compressed air to assist in cleaning.
- B. Inspect exposed surfaces and repair damaged finishes to match original factory finishes.
- C. Prior to the acceptance testing, remove plastic covers from smoke detectors.

3.7 CONTRACTOR START UP AND REPORTING

- A. Field tests shall be witnessed by Owner Representative and authorities having jurisdiction.
- B. Perform tests and inspections.
 - 1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.
- C. Tests and Inspections:
 - 1. Before requesting final approval of the installation, submit a written statement using the form for Record of Completion shown in NFPA 72 and as required by the authorities having jurisdiction.
 - 2. Perform each electrical test and visual and mechanical inspection listed in NFPA 72 and as required by the authorities having jurisdiction.
 - 3. Certify compliance with test parameters. All tests shall be conducted under the direct supervision of a NICET technician certified under the Fire Alarm Systems program at Level III.
 - 4. Visual Inspection: Conduct visual inspection prior to testing.
 - 5. Inspection shall be based on completed Record Drawings and system documentation that is required by NFPA 72 in its "Completion Documents, Preparation" Table in the "Documentation" Section of the "Fundamentals of Fire Alarm Systems" Chapter and as required by the authorities having jurisdiction.
 - a. Comply with "Visual Inspection Frequencies" Table in the "Inspection" Section of the "Inspection, Testing and Maintenance" Chapter in NFPA 72; retain the "Initial/Reacceptance" column and list only the installed components.
 - 6. Pretesting:

- a. Upon completing installation of the system, align, adjust, and balance the system and perform complete pretesting.
 - b. Determine, through pretesting, the conformance of the system to the requirements of the Drawings and Specifications.
 - c. Prepare forms for systematic recording of pretest results.
 - d. Report of Pretesting: After pretesting is complete, provide a letter certifying the installation is complete and fully operable. The letter shall include the names and titles of the witnesses to the preliminary tests.
7. Final Test Notice: Provide 10 days minimum notice in writing when the system is ready for final acceptance testing.
8. Minimum System Tests: Test the system in accordance with the procedures outlined in NFPA 72 Chapter 10 and as required by the authorities having jurisdiction.
9. Minimum required tests are but not limited to the ones listed below:
 - a. Correct deficiencies observed in pretesting.
 - b. Replace malfunctioning or damaged items with new and retest until satisfactory performance and conditions are achieved.
 - c. Verify the absence of unwanted voltages between circuit conductors and ground.
 - d. Test all conductors for short circuits utilizing an insulation testing device.
 - e. With each circuit pair, short circuit at the far end of the circuit and measure the circuit resistance with an ohmmeter. Record the circuit resistance of each circuit on the record drawings.
 - f. Verify the control units are in the normal condition as detailed in the manufacturer's operating and maintenance manual.
 - g. Test initiating and indicating circuits for proper signal transmission under open circuit conditions. One connection each should be opened at not less than 10 percent of the initiating and indicating devices. Proper signal transmission in accordance with class of wiring used shall be observed.
 - h. Test each initiating and indicating device for alarm operating and proper response at the control unit. Test smoke detectors with actual products of combustion.
 - i. Test the system for all specified functions in accordance with the manufacturer's operating and maintenance manual. Systematically initiate specified functional performance items at each station including making all possible alarm and monitoring initiations and using all communications options. For each item, observe related performance at all devices required to be affected by the item under all system sequences. Observe displays, signal tones, and annunciator indications.
 - j. Test both primary power and secondary power. Verify, by test, the secondary power system is capable of operating the system for the period and in the manner specified.
10. Factory-authorized service representative shall prepare the "Fire Alarm System Record of Completion" as required in the "Documentation" Section of the "Fundamentals of Fire Alarm Systems" Chapter in NFPA 72 and the "Inspection and Testing Form" as required in the "Records" Section of the "Inspection, Testing and Maintenance" Chapter in NFPA 72.
11. Provide all documentation as required by the authorities having jurisdiction.
12. Reacceptance Testing:
 - a. Perform reacceptance testing to verify the proper operation of added or replaced devices and appliances.
 - b. Tag all equipment and stations and other components at which tests have been satisfactorily completed. Place tags upon completion of tests.

- D. Fire-alarm system will be considered defective if it does not pass tests and inspections.
- E. Prepare test and inspection reports.

3.8 COMMISSIONING AND DEMONSTRATION

- A. Provide the services of a factory-authorized service representative to demonstrate and train The Board's maintenance personnel as specified below.
 - 1. Train The Board's maintenance personnel in the procedures and schedules involved in operating, troubleshooting, servicing, and preventive maintaining of the system. Include authorities having jurisdiction requirements. Provide a minimum of 8 hours' training.
 - 2. Schedule training with the Board at least seven days in advance.
 - 3. Occupancy Adjustments: When requested within one year of date of Substantial Completion, provide on-site assistance in adjusting sound levels and adjusting controls to suit actual occupied conditions. Provide up to three visits to the site for this purpose.
- B. Train Owner's maintenance personnel on procedures and schedules for starting up and shutting down, troubleshooting, servicing, and maintaining the equipment. The training shall occur after the startup report has been provided to the owner and the trainer will provide two (2) Installation and Operations manuals for the use of the owner's personnel during training. Provide a minimum of two (2) sessions of four (4) hours of training. Provide no more than one training session per day.
 - 1. Training Agenda:
 - a. Format shall be an outline broken up into (2) 4 hour sessions.
 - b. The material to be covered shall be sub divided into the description of the material to be covered for each 15 minutes intervals.
 - c. The descriptions shall include not only the material to be covered but also its location in the Installation and Operations manuals, including Section and page number.
 - 2. Documentation of training: After each 4 hour session the engineer shall initial and date the items on the training agenda that were covered that day.
- A. Review data in maintenance manuals. Refer to Division 01 Section "Operation and Maintenance Data." All required and recommended maintenance will be reviewed as well as operational trouble shooting. If the IOM does not include a written trouble shooting guide one will be provided.
- B. Demonstrate proper operation of equipment to commissioning agent or designated owners personnel.

END OF SECTION 283100

SECTION 311000 - SITE CLEARING**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Protection of existing trees.
 - 2. Clearing and grubbing.
 - 3. Removal of trees and other vegetation.
 - 4. Topsoil stripping.

1.3 DEFINITIONS

- A. Remove: Remove and legally dispose of items indicated. Removal includes digging out and off-site disposing of stumps and roots.
- B. Tree Protection Zone: The area surrounding individual trees or groups of trees to be protected during construction, and defined by the drip line of individual trees or the perimeter drip line of groups of trees, unless otherwise indicated.
- C. Topsoil: Friable, clay loam surface soil, found in varying depths.

1.4 MATERIALS OWNERSHIP

- A. Except for stripped topsoil or other materials indicated to remain Owner's property, cleared materials shall become Contractor's property and shall be removed from Project site.

1.5 SUBMITTALS

- A. Photographs or videotape, sufficiently detailed, of existing conditions of trees, plantings and other improvements adjoining the construction that might be misconstrued as damage caused by the Work.

1.6 PROJECT CONDITIONS

- A. Traffic: Conduct site clearing operations to ensure minimum interference with roads, streets, walks, and other adjacent occupied or used facilities. Do not close or obstruct streets, walks or other occupied or used facilities without permission from authorities having jurisdiction.
- B. Protection of Existing Improvements: Provide protections necessary to prevent damage to existing improvements indicated to remain in place.
 - 1. Protect existing improvements on adjoining properties and on Owner's property.
 - 2. Restore existing improvements damaged by clearing operations to their original condition.
- C. The conditions existing at the time of inspection for bidding purposes will be maintained by the Owner to the extent practical. However, minor variations may occur due to natural occurrences prior to the start of clearing work.
- D. Do not commence site-clearing operations until erosion and sedimentation control measures are in place.

PART 2 - PRODUCTS

2.1 TREE PROTECTION FENCING

- A. Tree protection fencing shall conform to standard and specification for State and local jurisdiction.

PART 3 – EXECUTION

3.1 PROTECTION OF EXISTING TREES AND VEGETATION

- A. Install tree protection fencing as indicated. Erect and maintain a temporary fence around the drip line of individual trees or around the perimeter drip line of groups of trees to remain.
 - 1. Do not store construction materials, debris, topsoil or other excavated material within the tree protection zone.
 - 2. Do not permit vehicles or other equipment within the tree protection zone.
 - 3. Maintain tree protection zones free of weeds and trash.
- B. Protect existing trees and other vegetation indicated to remain in place, against unnecessary cutting, breaking or skinning of roots, skinning or bruising of bark, smothering of trees by stockpiling construction materials or excavated materials within drip line, excess foot or vehicular traffic, or parking of vehicles within drip line.
- C. Provide protection for roots over 1-1/2 inch diameter that are cut during construction operations. Coat cut faces with emulsified asphalt, or other acceptable coating, formulated for use on damaged plant tissues. Temporarily cover exposed roots with wet burlap to prevent roots from drying out; cover with earth as soon as possible.
- D. Repair or replace trees and vegetation indicated to remain which are damaged by construction operations, in a manner acceptable to Architect and/or Engineer.

3.2 SITE CLEARING

- A. General: Remove trees, shrubs, grass and other vegetation as required to permit installation of the Work. Cut minor roots and branches of trees indicated to remain in a clean and careful manner, where such roots and branches obstruct installation of the Work.
- B. Clearing and Grubbing: Clear site of trees, shrubs and other vegetation within the clearing limits indicated.
 - 1. Completely remove stumps, roots, and other debris.
 - 2. Use only hand methods for grubbing inside drip line of trees indicated to remain.
 - 3. Fill depressions caused by clearing and grubbing operations with satisfactory soil material, unless further excavation or earthwork is indicated. Place fill material in horizontal layers not exceeding 6 inches loose depth, and thoroughly compact to a density equal to adjacent original ground.
 - 4. Utilize equipment with tracks to minimize the disturbance of the topsoil especially after the rain or after the snow starts melting. By doing so, the underlying subsoil will not intermingle with wet topsoil.
- C. Selective Clearing: Clear areas designated as "Selective Clearing" of all ground covers, underbrush and trees less than 6-inches in diameter at breast height. Coordinate extent of material removed with Architect and/or Engineer.
 - 1. Remove trees that appear to be dying or weakening for any reason and at any point during construction up to and including Substantial Completion at the Architect's and/or Engineer's direction.

3.3 TOPSOIL STRIPPING

- A. Remove heavy growths of grass from areas before stripping.
- B. Strip topsoil to whatever depths are encountered, but to a minimum of at least 4 inches.
- C. Strip topsoil in a manner to prevent intermingling with underlying subsoil or other material.
 - 1. Remove subsoil and un-suitable materials from topsoil, including trash, debris, weeds, roots, and other waste materials.
- D. Where existing trees are indicated to remain, leave existing topsoil in place within drip lines to prevent damage to root system.
- E. Temporarily stockpile topsoil in storage piles in areas indicated or directed. Construct storage piles to provide free drainage of surface water. Cover storage piles, if required, to prevent wind erosion.
 - 1. Do not stockpile topsoil within tree protection zones.
 - 2. Stockpile surplus topsoil to allow for resspreading.
- F. Dispose of unsuitable or excess topsoil in a legal manner off-site.

3.4 DISPOSAL OF WASTE MATERIALS

- A. Burning on Owner's Property: Burning may be allowed on this site subject to approval from the local Fire Marshall and other authorities having jurisdiction. Comply with all conditions of the burn permit, if it is obtained.
- B. Removal from Owner's Property: Remove waste materials generated by clearing operations from Owner's property and dispose of in a legal manner off-site.
 - 1. Remove waste materials and debris from the site in a manner to prevent spillage. Pavements and the area adjacent to the site shall remain free from mud, dirt and debris at all times.
 - 2. Clean up debris resulting from site clearing operations continuously with the progress of the work.

END OF SECTION 311000

SECTION 312000 - EARTHWORK**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.
- B. Refer to Section 01 2110 and the Bid Form for information concerning required allowances and unit prices.
- C. Refer to Section 31 1000 for topsoil stripping and Section 32 9200 for topsoil placement.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Excavation, filling, backfilling, and grading indicated and necessary for proper completion of the work.
 - 2. Preparing of subgrade for building slabs, walks, and pavements.
 - 3. Drainage/porous fill course for support of building slabs.
 - 4. Excavating and backfilling of trenches.
 - 5. Excavating and backfilling for underground mechanical and electrical utilities and buried mechanical and electrical appurtenances.
 - 6. Providing and monitoring settlement plates.

1.3 SUBMITTALS

- A. State approved Job Mix for stone.
- B. Imported fill (if required): Submit location of borrow pit and a sample of the soil for approval to the Owner's Geotechnical Engineer a minimum of five (5) working days prior to use
- C. Geotextile Fabric
- D. Copy of Blasting Permit, approved by authorities having jurisdiction, for record purposes.

1.4 DEFINITIONS

- A. Excavation: Removal of all material (except for rock) encountered to design subgrade elevations indicated for cut areas and to subsoil elevations in fill areas. Excavation also includes subsequent re-spreading, moisture conditioning, compaction, and grading of satisfactory materials removed.
- B. Unauthorized Excavation: Removal of materials beyond the limits indicated in the definition of "Excavation" without specific direction of Architect, Engineer and Owner.
- C. Additional Excavation: Removal, disposal and replacement of materials beyond the limits indicated in the definition of "Excavation" at the direction of the Architect, Engineer and Owner. Refer to Part 3 of this Section for requirements of Additional Excavation.
- D. Subgrade: The undisturbed earth (in cut) or the compacted soil layer (in fill) immediately below granular subbase, drainage fill, or topsoil materials.
- E. Subsoil: The undisturbed earth immediately below the existing topsoil layer.

- F. Building Pad: The area extending 10 feet beyond the exterior limits of the building/column footings and down to undisturbed soils at a one horizontal to one vertical slope.
- G. Structures: The area extending a minimum of ten (10) feet beyond the edge of foundations, slabs, curbs, underground tanks, piping or other man-made stationary features occurring above or below ground surface.
- H. Pavements: The area extending 10 feet beyond the exterior limits of paved areas and down to undisturbed soils at a one horizontal to one vertical slope. The area extending 3 feet beyond the exterior limits of walks and down to undisturbed soils at a one horizontal to one vertical slope
- I. Subbase Material: Artificially graded mixture of crushed gravel or crushed stone meeting local jurisdiction specifications. Material type is indicated on the drawings.
- J. Drainage/Porous Fill: Washed, evenly graded mixture of crushed stone, or crushed or uncrushed gravel meeting the requirements of the State DOT and Local jurisdiction.
- K. Rock: Hard bed rock, boulders or similar material requiring the use of rock drills and/or explosives for removal. The criteria for classification of general excavation as rock is any material which cannot be dislodged by a Caterpillar D-8 Tractor, or equivalent, equipped with a single tooth hydraulically operated power ripper. The criteria for trench rock shall be that a Caterpillar 345 Backhoe, or equivalent, with a proper width bucket cannot remove the material.

1.5 ADDITIONAL WORK

- A. Paragraph 4.3.4 of General Conditions refers to certain conditions that may require additional excavation work. This paragraph is further defined herein and, where there are conflicts, is superseded by this section.
- B. Claims for concealed, unknown, or unanticipated subsurface conditions are limited to those circumstances where:
 - 1. Additional excavation work is required below the contract limits indicated to provide acceptable bearing for building pad, structures or pavements.
 - 2. Additional excavation work is required to raise, lower, or revise the footings, foundations or other parts of the building to provide acceptable bearing.
 - 3. Additional excavation work below the utility trench design elevations, for utilities outside the limits of the building, as required to provide acceptable bearing for the utility.
 - 4. Rock is encountered between existing grade and design subgrade.
- C. The risks of concealed, unknown, or unanticipated subsurface conditions (except for rock) from existing ground surface to the design subgrade elevations in cut areas and to subsoil elevations in fill areas shall be included in the Contract Amount and shall not be considered as grounds for additional costs to the Contract. The risks of concealed, unknown, or unanticipated subsurface conditions below the elevations stated above shall be considered as Additional Excavation.
- D. During construction, if concealed, unknown, or unanticipated subsurface conditions are encountered which require that footings, foundations or other parts of the building be raised, lowered or revised to provide acceptable bearing for the building or if, outside the building limits, additional depth of utility trench excavation below the design subgrade or subsoil elevations is required, immediately notify the Architect and/or Engineer upon discovery of such condition prior to disturbing the material encountered.
- E. Payment for additional Work
 - 1. Additional excavation shall be counted toward the unit price allowances established in the Bid Form.

2. Lowering of footings shall be paid for at a negotiated amount. The additional excavation involved shall be counted toward the unit price allowance.
3. Rock removal, if required, shall be counted toward the unit price allowances established in the Bid Form. All rock removal required to complete work other than trenching shall be paid for at the unit price for mass rock removal. Rock payment lines are limited to the following:
 - a) Two feet outside of concrete work for which forms are required, except footings.
 - b) One foot outside perimeter of footings, two feet below bottom of footings.
 - c) In pipe trenches, 6 inches below invert elevation of pipe and 2 feet wider than outside diameter of pipe, but not less than 3 feet minimum trench width.
 - d) Outside dimensions of concrete work where no forms are required.
 - e) Under slabs on grade, 6 inches below bottom of concrete slab.
4. No payment will be made for unauthorized excavation.
5. The expense of surveying quantities of rock removal and additional excavation shall be included in the unit price allowances.

1.6 EARTHWORK BALANCE ADJUSTMENTS

- A. Adjustments of grades may be allowed with prior written approval of the Architect and/or Engineer in order to accommodate shortfall or surplus of material that may occur. Should adjustments be allowed, maintenance of designed drainage patterns and required adjustments to drainage structures shall be a Contract responsibility. **No additional payment will be made for these adjustments.**
- B. It is anticipated that some material will be required to be imported to achieve the finish grades indicated on the Drawings. Importation of the required material shall be a Contract responsibility. **No additional payment will be made for the importation of this material.**
- C. It is anticipated that some material will be required to be exported to achieve the finish grades indicated on the Drawings. Excavation and disposal of the required material off-site in a legal manner shall be a Contract responsibility. **No additional payment will be made for the export and disposal of this material.**

1.7 QUALITY ASSURANCE

- A. Codes and Standards: Perform excavation work in compliance with applicable requirements of authorities having jurisdiction.
- B. Environmental Compliance:
 1. Comply with the requirements of the State and Local Jurisdiction for erosion control during earthwork operations.
 2. Comply with the permit conditions for all work performed within wetlands.
- C. Testing and Inspection Service: Owner will employ and pay for an independent Geotechnical testing and inspection laboratory to perform soil testing and inspection service during earthwork operations. Cooperate with Owner's Geotechnical Engineer as required for testing and inspection of work. These services do not relieve the responsibility for compliance with Contract Document requirements.

1.8 PROJECT CONDITIONS

- A. Site Information: Data concerning subsurface materials or conditions, which are based on test borings, have been obtained by the Owner for his use in designing the project. This data is contained in a report titled "GEOTECHNICAL ENGINEERING STUDY, XXXXX County, Virginia" by XXXXXX. dated January 13, 1998. This report is included in this project manual for information only.
 - 1. The accuracy or completeness of the data is not warranted or guaranteed by the Owner or the Architect/Engineer, and in no event shall be considered part of the Contract Documents. The Owner and Architect/Engineer expressly disclaim any responsibility for the data as being representative of the conditions and materials that may be encountered.
- B. Bidders and interested parties (prior to receipt of bids) are encouraged to conduct their own soil and subsurface investigations, examinations, tests, and exploratory borings to determine the nature of the soil conditions underlying the project site. Contact the Owner's office to make an appointment to enter the site for the purpose of conducting your own investigation prior to bid.
- C. Existing Utilities: Do not interrupt existing utilities serving facilities occupied by the Owner or others except when permitted under the following conditions and then only after arranging to provide acceptable temporary utility services.
 - 1. Notify Architect/Engineer and Lidl not less than 48 hours in advance of proposed utility interruptions.
 - 2. Do not proceed with utility interruptions without receiving Architect's/Engineer's and Lidl written permission.
 - 3. Existing utilities across or along the line of work are indicated only in an approximate location. Locate all underground lines and structures. Call "Miss Utility" or equivalent locating service prior to construction. If utilities are marked that are not shown on the plans, locate utility vertically and horizontally and provide information to architect. Repair and correct any damage to underground lines and structures.

1.9 SAFETY

- A. Protection of Persons and Property: Barricade open excavations occurring as part of this work and post with warning lights.
 - 1. Operate warning lights as recommended by authorities having jurisdiction and governing regulations and standards.
 - 2. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.
- B. Work within the road right-of-way shall meet all requirements of the latest edition of the Virginia Department of Transportation Work Area Protection Manual or local equivalent.

PART 2 - PRODUCTS

2.1 SOIL MATERIALS

- A. All soil material shall be based on each individual site. All borrow sites must provide proctor and sample load to be approved before placing as fill

- B. Satisfactory soil materials are defined as those complying with ASTM D2487 soil classification groups CL, GC, SC, GW, GP, GM, SM, SW, and SP.
- C. Unsatisfactory soil materials are defined as those complying with ASTM D2487 soil classification groups CH, OL, OH, MH, ML and PT.
- D. Backfill and Fill Materials: Satisfactory soil materials free of clay, rock or gravel larger than 4 inches in any dimension (2 inches for material used in trench backfill), debris, waste, frozen materials, organics, vegetation and other deleterious matter.
- E. Imported material for structural fill shall comply with ASTM D2487 soil classification groups CL, ML, SC, SM, SP, SW, GC, GM, GP, or GW.

2.2 ACCESSORIES

- A. Non-woven Geotextile Fabric (for drainage): Mirafi 140N, or equivalent.
- B. Woven Geotextile Fabric (for reinforcement): PROPEX 2002, or equivalent.

PART 3 – EXECUTION

3.1 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.
- B. Preparation of subgrade for earthwork operations including removal of vegetation, topsoil, debris, obstructions, and deleterious materials from ground surface is specified in Section 02230 "Site Clearing."
- C. Protect and maintain erosion and sedimentation controls during earthwork operations.

3.2 DEWATERING

- A. Prevent surface water and subsurface or groundwater from flowing into excavations and from flooding project site and surrounding area.
 - 1. Do not allow water to accumulate in excavations. Remove water to prevent softening of foundation bottoms, undercutting footings, and soil changes detrimental to stability of subgrade and foundations. Provide and maintain pumps, well points, sumps, suction and discharge lines, and other dewatering system components necessary to convey water away from excavations.
 - 2. Establish and maintain temporary drainage ditches and other diversions outside excavation limits to convey rain water and water removed from excavations to collecting or runoff areas. Do not use utility trench excavations as temporary drainage ditches.
 - 3. Protection of the stockpile should be the contractor's responsibility. Contractors are required to cover the structural fill to prevent from saturation and seed it per the local jurisdiction requirements.
- B. Should any springs or running water be encountered in the excavation, notify the Architect/Engineer and provide discharge by trenches (or other acceptable means) and drain to an appropriate point of disposal. Provide temporary drainage facilities to minimize the flow of rainwater onto adjacent property. Repair any damage to property or to subgrade as a result of construction and/or

dewatering (or lack thereof) operations at no additional cost to the Contract. If permanent provision must be made for disposal of water other than as indicated, the Contract price shall be adjusted.

3.3 EXPLOSIVES

- A. Blasting may be done only if authorized by the Owner and local authorities having jurisdiction. When explosives are used, experienced powdermen or persons who are licensed or otherwise authorized to use explosives shall execute the work. Explosives shall be stored, handled, and used in accordance with local regulations and with the "Manual of Accident Prevention in Construction" of the Associated General Contractor of America, Inc. Correct any damage to foundations or other work caused by use of explosives. Meeting the requirements of the blasting permit, if issued, is a Contract responsibility.

3.4 EXCAVATION

- A. Excavation consists of removal, placement and disposal of material encountered when establishing required subgrade or finish grade elevations.
 - 1. Excavation includes removal and disposal of pavements and other obstructions visible on ground surface; underground structures, utilities and other items indicated to be demolished and removed; together with earth and other materials encountered that are not classified as rock or unauthorized excavation.
- B. Rock Excavation: If Rock is encountered the Owner's Geotechnical Engineer will verify that the material qualifies for classification as rock excavation.
 - 1. If rock is encountered in grading, remove to depths as follows:
 - a) Under surfaced areas, to 6" under the respective subgrade for such areas.
 - b) Under grass and planted areas - 12" minimum.
 - c) Under footings – Two feet below bottom of footing, One foot outside of perimeter of footing.
 - d) Under trenches – 6" below bottom of trench.
 - 2. Contractor shall employ a surveyor licensed in the Project State to calculate the quantity of material removed as Rock Excavation. The quantity of rock calculated shall not exceed the volume determined by the payment limits. The Owner's Project Representative shall review the quantity calculated within 48 hours of receiving the survey notes.

3.5 EXCAVATION FOR BUILDING PAD AND STRUCTURES

- A. Conform to elevations and dimensions indicated within a tolerance of plus or minus 0.10 foot, and extending a sufficient distance from footings and foundations to permit placing and removal of concrete formwork, installation of services, other construction and for review.
- B. Excavations for footings and foundations: Do not disturb bottoms of excavation. Excavate by hand to elevations required just before concrete reinforcement is placed. Trim bottoms to required lines and grades to leave solid base to receive other work.
 - 1. Where rock is encountered, carry excavation to required elevations and backfill with crushed stone prior to installation of footing.
- C. Excavation for Underground Tanks, Basins, and Mechanical or Electrical Structures: Conform to elevations and dimensions indicated within a tolerance of plus or minus 0.10 foot plus a sufficient distance to permit placing and removal of concrete formwork, installation of services, other construction and for review. Do not disturb bottom of excavations intended for bearing surface.

3.6 EXCAVATION FOR WALKS AND PAVEMENTS

- A. Cut surface under pavements to comply with cross-sections, elevations and grades as indicated.

3.7 EXCAVATION FOR UTILITY TRENCHES

- A. Excavate trenches to uniform width, sufficiently wide to provide ample working room and a minimum of 6 to 9 inches of clearance on both sides of pipe or conduit.
- B. Excavate trenches to depth indicated or required to establish indicated slope and invert elevations and to support bottom of pipe or conduit on undisturbed soil. Beyond building perimeter, excavate trenches to allow installation of top of pipe below frost line.
 - 1. Where rock is encountered, carry excavation to required elevations and backfill with open graded material equivalent to #57 crushed stone prior to installation of pipe.
 - 2. For pipes or conduit less than 6 inches in nominal size, and for flat-bottomed, multiple-duct conduit units, do not excavate beyond indicated depths. Hand-excavate bottom cut to accurate elevations and support pipe or conduit on undisturbed soil.
 - 3. For pipes and equipment 6 inches or larger in nominal size, shape bottom of trench to fit bottom of pipe for 90 degrees (bottom 1/4 of the circumference). Fill depressions with tamped sand backfill. At each pipe joint, dig bell holes to relieve pipe bell of loads ensure continuous bearing of pipe barrel on bearing surface.

3.8 EXCAVATION STABILITY

- A. General: Comply with local codes, ordinances, and requirements of agencies having jurisdiction.
- B. Slope sides of excavations to comply with local codes, ordinances, and requirements of agencies having jurisdiction. Shore and brace where sloping is not possible because of space restrictions or stability of material excavated. Maintain sides and slopes of excavations in safe condition until completion of backfilling.
- C. Shoring and Bracing: Provide materials for shoring and bracing, such as sheet piling, uprights, stringers, and cross braces, in good serviceable condition. Maintain shoring and bracing in excavations regardless of time period excavations will be open. Extend shoring and bracing as excavation progresses.

3.9 SUBGRADE INSPECTION

- A. Notify Architect/Engineer when mass, trench and footing excavations have reached required subgrade. The Architect/Engineer will arrange for an inspection of conditions by the Owner's Geotechnical Engineer. *Alternative procedures for arranging this review may be implemented at the Owner's written option.*
- B. If the Owner's Geotechnical Engineer determines that the subgrade bearing conditions are unacceptable, the Architect will authorize additional excavation until suitable bearing conditions are encountered.
- C. Proof-roll subgrade [**below the building slabs and pavements**] <Insert locations> with heavy pneumatic-tired equipment to identify soft pockets and areas of excess yielding. Do not proof-roll wet or saturated subgrades.
 - 1. Completely proof-roll subgrade in one direction[, **repeating proof-rolling in direction perpendicular to first direction**]. Limit vehicle speed to **3 mph (5 km/h)**.

2. Proof-roll with a loaded 10-wheel, tandem-axle dump truck weighing not less than **15 tons (13.6 tonnes)**.
3. Excavate soft spots, unsatisfactory soils, and areas of excessive pumping or rutting, as determined by Architect, and replace with compacted backfill or fill as directed.
- D. Under supervision of the Owner's Geotechnical Engineer, proofroll subgrade in cut areas below the building pad and pavement(s) with a loaded dump truck or other approved pneumatic tired vehicle. Should any unstable sub-soil be encountered below pavement or structures, break up the top eight inches of ground surface, pulverize, moisture-condition to optimum moisture content, and compact to percentage of maximum density as stated in Percentage of Maximum Density Requirements. Perform this work at no additional cost and/or time to the Contract.
- E. Reconstruct subgrades damaged by freezing temperatures, frost, rain, accumulated water, or construction activities, as directed by Architect, without additional compensation.

3.10 ADDITIONAL EXCAVATION

- A. Additional Excavation (Mass): Remove excavated materials and dispose of on-site as directed by the Architect/Engineer. Replace this excavated material with satisfactory material placed and compacted according to the requirements of the "Placement and Compaction" section.
- B. Additional Excavation in Trenches: Remove excavated materials and dispose of on-site as directed by the Architect/Engineer. Replace this excavated material with stone.
- C. Additional Excavation in Footings: Remove excavated materials and dispose of on-site as directed by the Architect/Engineer. Replace this excavated material with lean concrete/flowable fill or with stone extending 12 inches laterally beyond the footing in all directions.
- D. The quantity of material removed as Additional Excavation (Mass, Trench or Footing) shall be calculated (on an in-situ basis) by a surveyor licensed in the Project State and employed by the Contractor. The Owner's Project Representative shall review the quantity calculated within 48 hours of receiving the survey notes.
- E. Protect the subgrade during construction. During wet conditions, the subgrade soils may become saturated and soften, possibly resulting in damage to the subgrade if disturbed by equipment. Correct subgrade damaged in this manner. **No additional payment will be made to correct subgrade damaged in this manner.**

3.11 UNAUTHORIZED EXCAVATION

- A. Correct Unauthorized Excavation as follows:
 1. Under footings, foundation bases, or retaining walls, fill unauthorized excavation by extending indicated bottom elevation of footing or base to excavation bottom without altering required top elevation. Lean concrete fill may be used to bring elevations to proper position when acceptable to Architect/Engineer.
 2. Elsewhere, backfill and compact unauthorized excavations as indicated for authorized excavations of same classification unless otherwise directed by Architect/Engineer.

3.12 STORAGE OF EXCAVATED MATERIALS

- A. Temporarily stockpile excavated materials acceptable for use as backfill and fill. Place, grade, and shape stockpiles for proper drainage. Cover to prevent windblown dust.
 1. Stockpile excavated materials away from edge of excavations. Do not store within the drip line of trees to remain.

3.13 BACKFILL AND FILL

- A. Backfill excavations as promptly as work permits, but not until completion of the following:
 - 1. Acceptance by local authority having jurisdiction of construction below finished grade, including perimeter insulation.
 - 2. Review, approval, and recording of the locations of underground utilities.
 - 3. Removal of concrete formwork.
 - 4. Removal of shoring and bracing (including backfilling of voids with satisfactory materials).
 - 5. Removal of trash and debris from excavation.
 - 6. Permanent or temporary horizontal bracing is in place on horizontally supported walls.
- B. Place backfill on subgrades free of mud, frost, snow or ice.
- C. Ground Surface Preparation: Remove vegetation, debris, obstructions, and deleterious materials from ground surface prior to placement of fills.
- D. Bench sloped surfaces steeper than 1 vertical to 4 horizontal so fill material will bond with existing material. Plow, scarify, bench or break up sloped surfaces flatter than 1 vertical to 4 horizontal so fill material will bond with existing material.
- E. Place soil material in layers to required subgrade elevations, for each area classification listed below, using materials indicated in Part 2 of this Section.
 - 1. Under grassed areas, use satisfactory excavated or borrow material.
 - 2. Under walks, curbs, and pavements, use satisfactory excavated or borrow material.
 - 3. Under building slabs, use satisfactory excavated or borrow materials and drainage/porous fill material as indicated.

3.14 UTILITY TRENCH BACKFILL

- A. Place backfill on subgrades free of mud, frost, snow, or ice.
- B. Place and compact bedding course on trench bottoms and where indicated. Shape bedding course to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits.
- C. Backfill trenches with concrete where trench excavations pass within 18 inches of column or wall footings and that are carried below bottom of such footings or that pass under wall footings. Place concrete to level of bottom of adjacent footing.
- D. Provide 4-inch- (100-mm-) thick, concrete-base slab support for piping or conduit less than 30 inches (750 mm) below surface of roadways. After installing and testing, completely encase piping or conduit in a minimum of 4 inches (100 mm) of concrete before backfilling or placing roadway subbase.
- E. Place and compact initial backfill of [subbase material] [satisfactory soil], free of particles larger than 1 inch (25 mm) in any dimension, to a height of 12 inches (300 mm) over the utility pipe or conduit.
 - 1. Carefully compact initial backfill under pipe haunches and compact evenly up on both sides and along the full length of utility piping or conduit to avoid damage or displacement of piping or conduit. Coordinate backfilling with utilities testing.

- F. Controlled Low-Strength Material: Place initial backfill of controlled low-strength material to a height of **12 inches (300 mm)** over the utility pipe or conduit.
- G. Backfill voids with satisfactory soil while installing and removing shoring and bracing.
- H. Place and compact final backfill of satisfactory soil to final subgrade elevation.
- I. Controlled Low-Strength Material: Place final backfill of controlled low-strength material to final subgrade elevation.
- J. Install warning tape directly above utilities, **12 inches (300 mm)** below finished grade, except **6 inches (150 mm)** below subgrade under pavements and slabs.
- K. Do not backfill trenches until any required testing and inspections have been completed and Owner's Geotechnical Engineer authorizes backfilling. Backfill carefully to avoid damage or displacement of pipe systems.
- L. Under piping and conduit and equipment, use crushed stone where required over rock bearing surface and for correction of unauthorized excavation. Shape excavation bottom to fit bottom 90 degrees of cylinder.
- M. Place backfill and fill materials evenly adjacent to structures, piping, or conduit to required elevations. Prevent wedging action of backfill against structures or displacement of piping or conduit by carrying material uniformly around structure, piping, or conduit to approximately same elevation in each lift.

3.15 SOIL MOISTURE CONTROL

- A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill soil layer before compaction to within 2 percentage points of optimum moisture content.
 - 1. Do not place backfill or fill soil material on surfaces that are muddy, frozen, or contain frost or ice.
 - 2. Remove and replace, or scarify and air dry otherwise satisfactory soil material that exceeds optimum moisture content by 2 percentage points and is too wet to compact to specified dry unit weight.
- B. Moisture Control: Where subgrade or layer of soil material must be moisture conditioned before compaction, uniformly apply water to surface of subgrade or layer of soil material. Apply water in minimum quantity as necessary to prevent free water from appearing on surface during or subsequent to compaction operations. Maintain the moisture content of the structural fill materials to within 2 percentage points of the optimum moisture content until permanently covered.
- C. Remove and replace, or scarify and air dry, soil material that is too wet to permit compaction to required density.
 - 1. Stockpile or spread soil material that has been removed because it is too wet to permit compaction. Assist drying by discing, harrowing, or pulverizing until moisture content is reduced to a satisfactory value.
 - 2. Work wet materials as directed by the Owner's Geotechnical Engineer. Base bids on working material daily for a maximum of five days of acceptable weather.
 - 3. No additional payment will be made for these operations.

3.16 COMPACTION OF SOIL BACKFILL AND FILLS

- A. Place backfill and fill materials in layers not more than 8 inches in loose depth for material compacted by heavy compaction equipment, and not more than 4 inches in loose depth for material compacted by hand-operated tampers.
- B. Before compaction, moisten or aerate each layer as necessary to provide optimum moisture content. Compact each layer to required percentage of maximum dry density or relative dry density for each area classification. Do not place backfill or fill material on surfaces that are muddy, frozen, or contain frost or ice.
- C. Control soil and fill compaction, providing minimum percentage of density indicated for each area classification indicated below. Correct improperly compacted areas or lifts as directed by Architect if soil density tests indicate inadequate compaction.
- D. Percentage of Maximum Density Requirements: Compact soil to not less than the following percentages of maximum density at a moisture content within 2 percentage points of optimum in accordance with ASTM D698:
 - 1. Under structures, building pad and pavements, compact each layer of backfill or fill material at 95 percent maximum density. This includes ground under future expansion areas.
 - 2. Under grass or unpaved areas, compact each layer of backfill or fill material at 90 percent maximum density.
- E. Seal all fill areas at the end of each working day, utilizing a smooth drum roller.

3.17 GRADING

- A. General: Rough grading of areas within the Project, including cut and fill sections and adjacent transition areas, shall be reasonably smooth, compacted and free from irregular surface changes. The degree of finish shall be that ordinarily obtainable from either blade-grader or motor patrol except as otherwise indicated. The finished subgrade surface from the grassed areas generally shall be not more than 0.2 feet above or below the final grade or approved cross section, with due allowance for topsoil.
- B. The tolerance for areas within 10 feet of building perimeter, walks and all areas to be paved shall not exceed 0.10 feet above or below the established subgrade. Finish all ditches, swales and gutters to drain readily. Unless otherwise indicated, evenly slope the subgrade to provide drainage away from building walls in all directions at a grade not less than ¼ inch per foot. Provide rounding at top and bottom of cut and fill slopes and at other breaks in grade.
- C. Protection of Graded Areas: Protect newly graded areas and areas of cut, fill and design/subgrade elevations from the actions of the elements and from deterioration as a result of construction operations and weather conditions (frost, rains, snow, sleet, hail, etc.). Repair any settlement or washing that occurs prior to or after acceptance of the work. Fill to required subgrade levels any areas where settlement occurs. Protect trees to remain, and, at all areas of the Site where construction operations are in progress, provide protection for the safety of occupants of the existing facilities.
- D. General: Uniformly grade areas to a smooth surface, free of irregular surface changes. Comply with compaction requirements and grade to cross sections, lines, and elevations indicated.
 - 1. Provide a smooth transition between adjacent existing grades and new grades.
 - 2. Cut out soft spots, fill low spots, and trim high spots to comply with required surface tolerances.
- E. Site Grading: Slope grades to direct water away from buildings and to prevent ponding. Maximum slope of parking and pedestrian paved areas is 3.00%. Finish subgrades to required elevations within the following tolerances:
 - 1. Lawn or Unpaved Areas: Plus or minus [1 inch (25 mm)] <Insert tolerance>.

2. Walks: Plus or minus [**1 inch (25 mm)**] <Insert tolerance>.
3. Pavements: Plus or minus [**1/2 inch (13 mm)**] <Insert tolerance>.
- F. Grading inside Building Lines: Finish subgrade to a tolerance of **1/2 inch (13 mm)** when tested with a **10-foot (3-m)** straightedge.

3.18 PAVEMENT SUBBASE COURSE:

- A. General: Place subbase material, in layers of indicated thickness, over subgrade surface to support a pavement base course.
- B. Grade Control: During construction, maintain lines and grades including crown and cross-slope of subbase course.
- C. Shoulders: Place shoulders along edges of subbase course to prevent lateral movement. Construct shoulders of acceptable soil materials, placed in such quantity to compact to thickness of each subbase course layer. Compact and roll at least at 12" width of shoulder simultaneously with compacting and rolling each layer of subbase course.
- D. Placing: Place subbase course material on prepared subgrade in layers of uniform thickness, conforming to indicated cross-section and thickness. Maintain optimum moisture content for compacting subbase material during placement operations.
- E. When a compacted subbase course is 6" thick or less, place material in a single layer. When more than 6" thick, place material in equal layers, except no single layer more than 6" or less than 3" in thickness when compacted.
- F. Place subbase[**and base**] course on subgrades free of mud, frost, snow, or ice.
- G. On prepared subgrade, place subbase[**and base**] course under pavements and walks as follows:
 1. Place base course material over subbase course under hot-mix asphalt pavement.
 2. Shape subbase[**and base**] course to required crown elevations and cross-slope grades.
 3. Place subbase[**and base**] course **6 inches (150 mm)** or less in compacted thickness in a single layer.
 4. Place subbase[**and base**] course that exceeds **6 inches (150 mm)** in compacted thickness in layers of equal thickness, with no compacted layer more than **6 inches (150 mm)** thick or less than **3 inches (75 mm)** thick.
 5. Compact subbase[**and base**] course at optimum moisture content to required grades, lines, cross sections, and thickness to not less than **[95] <Insert percentage>** percent of maximum dry unit weight according to **[ASTM D 698] [ASTM D 1557]**.
- H. Pavement Shoulders: Place shoulders along edges of subbase and base course to prevent lateral movement. Construct shoulders, at least **12 inches (300 mm)** wide, of satisfactory soil materials and compact simultaneously with each subbase and base layer to not less than **[95] <Insert percentage>** percent of maximum dry unit weight according to **[ASTM D 698] [ASTM D 1557]**.

3.19 BUILDING SLAB DRAINAGE COURSE

- A. General: Place drainage/porous fill material, over subgrade surface to support concrete building slabs and sidewalks areas indicated.
- B. Place drainage course on subgrades free of mud, frost, snow, or ice.

- C. Placing: Place drainage/porous fill material on prepared subgrade in layers of uniform thickness, conforming to indicated cross-section and thickness. Maintain optimum moisture content for compacting material during placement operations.
- D. When a compacted drainage course is indicated to be 6 inches thick or less, place material in a single layer. When indicated to be more than 6 inches thick, place material in equal layers, except no single layer more than 6 inches or less than 3 inches in thickness when compacted.

3.20 FIELD QUALITY CONTROL

- A. Quality Control Testing During Construction: Allow testing service to inspect and approve each subgrade and fill layer before further backfill or construction work is performed.
 - 1. If in the opinion of the Architect, based on testing service reports and inspection, subgrade or fills have been placed that are below required density, perform additional compaction and testing until required density is obtained.
- B. The Owner will engage, and pay for, the services of a Geotechnical Engineer whose function shall be to afford complete engineering control by testing of the conditions of all footing subgrades, the placement of all structural fills under structures, building pad and pavement areas, and all compaction where required, and to observe the proof rolling of the building pad and pavement areas.
- C. The Owner's Geotechnical Engineer will be present as deemed necessary during all phases of the Work requiring filling, compaction operations or testing. The Geotechnical Engineer will provide the Architect with written certification that fill and compaction was completed with accepted materials in accordance with the Documents, and give a professional opinion regarding shrinkage or settlement of fill and safe load bearing capacity of fill.
- D. Site Preparation and Proofrolling: The Owner's Geotechnical Engineer will determine if any additional excavation or in-place densification is necessary to prepare a subgrade for fill placement for slab or pavement support.
- E. Fill Placement and Compaction: The Owner's Geotechnical Engineer will witness all fill operations and take sufficient in-place density tests to verify that the indicated degree of fill compaction is achieved. The Owner's Geotechnical Engineer will observe and approve borrow materials used and shall determine if their existing moisture contents are suitable/acceptable.
- F. Footing Excavation Review: The Owner's Geotechnical Engineer will review the footing excavations for the building foundations. He will verify that the design bearing pressures are available and that no loose or soft areas exist beneath the bearing surfaces of the footing excavations.
- G. The Owner's Geotechnical Engineer will submit two (2) copies each of his reports, recommendations and/or opinions to the Architect/Engineer and the Owner. Pertinent information will be provided to the Contractor as required.

3.21 EROSION CONTROL:

- A. Provide erosion control methods in accordance with requirements of authorities having jurisdiction , and as indicated in the Contract Documents.

3.22 PROTECTION

- A. Repair and reestablish grades in settled, eroded, and rutted areas to indicated tolerances.
- B. Reconditioning Compacted Areas: Where subsequent construction operations or adverse weather disturbs completed compacted areas, scarify surface, reshape, and compact to required density prior to further construction.
- C. Settling: Where settling is measurable or observable at excavated areas during general project warranty period, remove surface (pavement, lawn, or other finish), add backfill material, compact, and replace surface treatment. Restore appearance, quality, and condition of surface or finish to match adjacent work, and eliminate evidence of restoration to greatest extent possible.
- D. Protect excavation bottoms against freezing when atmospheric temperature is less than 35 degrees F.

3.23 DISPOSAL OF WASTE MATERIALS

- A. Removal from Owner's Property: Remove excess and/or waste materials, including trash and debris, and dispose of it off Owner's property in a legal manner.
- B. Dispose of excess material and materials not acceptable for use as backfill or fill legally offsite.
- C. Do not remove topsoil from site until it has been demonstrated to the Owner's satisfaction that it is excess.

3.24 SETTLEMENT PLATES *(DELETE SECTION IF NOT REQUIRED BY GEOTECH REPORT)*

- A. Provide and monitor three settlement plates to evaluate the settlement occurring during and after fill placement. Locate the settlement plates as indicated on Drawing C3.2 or as recommended by the Owners Geotechnical Engineer.
- B. Record the elevation of the top of the settlement plate daily until settlement has slowed to a point satisfactory to the Owners Geotechnical Engineer. Settlement plate readings shall be made to an accuracy of 0.01' and shall be referenced to a benchmark well beyond the influence of the fill being placed and protected from construction equipment disturbance.
- C. Take precautions to prevent damaging or disturbing the settlement plates during construction operations.
- D. Providing, maintenance and monitoring of the settlement plates is part of the Work.
- E. Base bids on a settlement period of 45 calendar days commencing at the time final subgrade elevations in the area are attained.

END OF SECTION 312000

SECTION 312000 - EARTHWORK**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.
- B. Refer to Section 01 2110 and the Bid Form for information concerning required allowances and unit prices.
- C. Refer to Section 31 1000 for topsoil stripping and Section 32 9200 for topsoil placement.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Excavation, filling, backfilling, and grading indicated and necessary for proper completion of the work.
 - 2. Preparing of subgrade for building slabs, walks, and pavements.
 - 3. Drainage/porous fill course for support of building slabs.
 - 4. Excavating and backfilling of trenches.
 - 5. Excavating and backfilling for underground mechanical and electrical utilities and buried mechanical and electrical appurtenances.
 - 6. Providing and monitoring settlement plates.

1.3 SUBMITTALS

- A. DOT approved Job Mix for stone.
- B. Imported fill (if required): Submit location of borrow pit and a sample of the soil for approval to the Owner's Geotechnical Engineer a minimum of fourteen (14) working days prior to use
- C. Geotextile Fabric
- D. Copy of Blasting Permit, approved by authorities having jurisdiction, for record purposes.

1.4 DEFINITIONS

- A. Excavation: Removal of all material encountered to design subgrade elevations indicated for cut areas and to subsoil elevations in fill areas. Excavation also includes subsequent resspreading, moisture conditioning, compaction, and grading of satisfactory materials removed.
- B. Unauthorized Excavation: Removal of materials beyond the limits indicated in the definition of "Excavation" without specific direction of Architect/Engineer/Owner.
- C. Additional Excavation: Removal, disposal and replacement of materials beyond the limits indicated in the definition of "Excavation" at the direction of the Architect. Refer to Part 3 of this Section for requirements of Additional Excavation.
- D. Subgrade: The undisturbed earth (in cut) or the compacted soil layer (in fill) immediately below granular subbase, drainage fill, or topsoil materials.
- E. Subsoil: The undisturbed earth immediately below the existing topsoil layer.
- F. Building Pad: The area extending 10 feet beyond the exterior limits of the building/column footings and down to undisturbed soils at a one horizontal to one vertical slope.

- G. Structures: The area extending a minimum of ten (10) feet beyond the edge of foundations, slabs, curbs, underground tanks, piping or other man-made stationary features occurring above or below ground surface.
- H. Pavements: The area extending 10 feet beyond the exterior limits of paved areas and down to undisturbed soils at a one horizontal to one vertical slope. The area extending 3 feet beyond the exterior limits of walks and down to undisturbed soils at a one horizontal to one vertical slope
- I. Subbase Material: Artificially graded mixture of crushed gravel or crushed stone meeting DOT specifications. Material type is indicated on the drawings.
- J. Drainage/Porous Fill: Washed, evenly graded mixture of crushed stone, or crushed or uncrushed gravel meeting the requirements of DOT required stone.
- K. Rock: Hard bed rock, boulders or similar material requiring the use of rock drills and/or explosives for removal. The criteria for classification of general excavation as rock is any material which cannot be dislodged by a Caterpillar D-8 Tractor, or equivalent, equipped with a single tooth hydraulically operated power ripper. The criteria for trench rock shall be that a Caterpillar 345 Backhoe, or equivalent, with a proper width bucket cannot remove the material.

1.5 ADDITIONAL WORK

- A. This Project is Unclassified.
- B. The risks of concealed, unknown, or unanticipated subsurface conditions from existing ground surface to the design subgrade elevations in cut areas and to subsoil elevations in fill areas shall be included in the Contract Amount and shall not be considered as grounds for additional costs to the Contract.
- C. The risks of concealed, unknown, or unanticipated subsurface conditions below the elevations stated above identified as additional excavation shall also be included in the Base Bid Contract amount and shall not be considered as grounds for additional cost to the Contract. The work includes the establishment of acceptable bearing conditions in both cut and fill situations.
- D. Rock Excavation: Any required rock excavation shall be included in the Base Bid Contract amount and shall not be considered as grounds for additional costs to the Contract.
- E. Time extensions will not be granted for Additional Excavation or Rock Excavation.

1.6 EARTHWORK BALANCE ADJUSTMENTS

- A. Adjustments of grades may be allowed with prior written approval of the Architect in order to accommodate shortfall or surplus of material that may occur. Should adjustments be allowed, maintenance of designed drainage patterns and required adjustments to drainage structures shall be a Contract responsibility. **No additional payment will be made for these adjustments.**
- B. Any surplus material remaining shall be removed from the site and disposed of in a legal manner.

1.7 QUALITY ASSURANCE

- A. Codes and Standards: Perform excavation work in compliance with applicable requirements of authorities having jurisdiction.
- B. Environmental Compliance:
 - 1. Comply with the requirements of the latest edition of the State and Local jurisdiction regulations for erosion control during earthwork operations.

2. Comply with the permit conditions for all work performed within wetlands.
- C. Testing and Inspection Service: Owner will employ and pay for an independent Geotechnical testing and inspection laboratory to perform soil testing and inspection service during earthwork operations. Cooperate with Owner's Geotechnical Engineer as required for testing and inspection of work. These services do not relieve the responsibility for compliance with Contract Document requirements.

1.8 PROJECT CONDITIONS

- A. Site Information: Data concerning subsurface materials or conditions, which are based on test borings, have been obtained by the Owner for his use in designing the project. This data is contained in a report titled "GEOTECHNICAL ENGINEERING STUDY, XXXXX County, Virginia" by XXXXXX, dated January 13, 1998. This report is included in this project manual for information only.
 1. The accuracy or completeness of the data is not warranted or guaranteed by the Owner or the Architect/Engineer, and in no event shall be considered part of the Contract Documents. The Owner and Architect/Engineer expressly disclaim any responsibility for the data as being representative of the conditions and materials that may be encountered.
- B. Bidders and interested parties (prior to receipt of bids) are encouraged to conduct their own soil and subsurface investigations, examinations, tests, and exploratory borings to determine the nature of the soil conditions underlying the project site. Contact the Owner's office to make an appointment to enter the site for the purpose of conducting your own investigation prior to bid.
- C. Existing Utilities: Do not interrupt existing utilities serving facilities occupied by the Owner or others except when permitted under the following conditions and then only after arranging to provide acceptable temporary utility services.
 1. Notify Architect not less than 48 hours in advance of proposed utility interruptions.
 2. Do not proceed with utility interruptions without receiving Architect's written permission.
 3. Existing utilities across or along the line of work are indicated only in an approximate location. Locate all underground lines and structures. Call "Miss Utility" or equivalent underground locating service prior to construction. If utilities are marked that are not shown on the plans, locate utility vertically and horizontally and provide information to architect. Repair and correct any damage to underground lines and structures.

1.9 SAFETY

- A. Protection of Persons and Property: Barricade open excavations occurring as part of this work and post with warning lights.
 1. Operate warning lights as recommended by authorities having jurisdiction and governing regulations and standards.
 2. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.
- B. Work within the road right-of-way shall meet all requirements of the latest edition of the State's Department of Transportation Work Area Protection Manual or local equivalent.

PART 2 - PRODUCTS

2.1 SOIL MATERIALS

- A. Satisfactory soil materials are defined as those complying with ASTM D2487 soil classification groups CL, GC, SC, GW, GP, GM, SM, SW, and SP.
- B. Unsatisfactory soil materials are defined as those complying with ASTM D2487 soil classification groups CH, OL, OH, MH, ML and PT.
- C. Backfill and Fill Materials: Satisfactory soil materials free of clay, rock or gravel larger than 4 inches in any dimension (2 inches for material used in trench backfill), debris, waste, frozen materials, organics, vegetation and other deleterious matter.
- D. Imported material for structural fill shall comply with ASTM D2487 soil classification groups CL, ML, SC, SM, SP, SW, GC, GM, GP, or GW.

2.2 ACCESSORIES

- A. Non-woven Geotextile Fabric (for drainage): Mirafi 140N, or equivalent.
- B. Woven Geotextile Fabric (for reinforcement): PROPEX 2002, or equivalent.

PART 3 – EXECUTION

3.1 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.
- B. Preparation of subgrade for earthwork operations including removal of vegetation, topsoil, debris, obstructions, and deleterious materials from ground surface is specified in Section 02230 "Site Clearing."
- C. Protect and maintain erosion and sedimentation controls during earthwork operations.

3.2 DEWATERING

- A. Prevent surface water and subsurface or groundwater from flowing into excavations and from flooding project site and surrounding area.
 - 1. Do not allow water to accumulate in excavations. Remove water to prevent softening of foundation bottoms, undercutting footings, and soil changes detrimental to stability of subgrade and foundations. Provide and maintain pumps, well points, sumps, suction and discharge lines, and other dewatering system components necessary to convey water away from excavations.
 - 2. Establish and maintain temporary drainage ditches and other diversions outside excavation limits to convey rain water and water removed from excavations to collecting or runoff areas. Do not use utility trench excavations as temporary drainage ditches.
- B. Should any springs or running water be encountered in the excavation, notify the Architect and provide discharge by trenches (or other acceptable means) and drain to an appropriate point of disposal. Provide temporary drainage facilities to minimize the flow of rainwater onto adjacent property. Repair any damage to property or to subgrade as a result of construction and/or

dewatering (or lack thereof) operations at no additional cost to the Contract. If permanent provision must be made for disposal of water other than as indicated, the Contract price shall be adjusted.

- C. Soils encountered may require disking, scarifying, moisture conditioning, harrowing, pulverizing or other special or careful handling when utilized as fill. No additional payment will be made for these operations.

3.3 EXPLOSIVES

- A. Blasting may be done only if authorized by the Owner and local authorities having jurisdiction. When explosives are used, experienced powdermen or persons who are licensed or otherwise authorized to use explosives shall execute the work. Explosives shall be stored, handled, and used in accordance with local regulations and with the "Manual of Accident Prevention in Construction" of the Associated General Contractor of America, Inc. Correct any damage to foundations or other work caused by use of explosives. Meeting the requirements of the blasting permit, if issued, is a Contract responsibility.
 - 1. Pre-blast inspections are required in accordance with all prevailing jurisdictional guidelines. These inspections and associated costs are the sole responsibility of the Contractor. Any damage resulting from blasting operations to onsite or offsite structures and appurtenances and costs associated therewith are the sole responsibility of the Contractor. Blasting operations in the vicinity of existing structures and new and/or in-progress construction shall also be performed in accordance with the constraints of the most recent edition of the Statewide Fire Prevention Code, NFPA 495 and/or local ordinances as applicable.

3.4 EXCAVATION

- A. Excavation consists of removal, placement and disposal of material encountered when establishing required subgrade or finish grade elevations.
 - 1. Excavation includes removal and disposal of pavements and other obstructions visible on ground surface; underground structures, utilities and other items indicated to be demolished and removed; together with earth and other materials encountered that are not classified as rock or unauthorized excavation.
- B. Rock Excavation: If Rock is encountered the Owner's Geotechnical Engineer will verify that the material qualifies for classification as rock excavation.
 - 1. If rock is encountered in grading, remove to depths as follows:
 - a) Under surfaced areas, to 6" under the respective subgrade for such areas.
 - b) Under grass and planted areas - 12" minimum.
 - c) Under footings – Two feet below bottom of footing, One foot outside of perimeter of footing.
 - d) Under trenches – 6" below bottom of trench.

3.5 EXCAVATION FOR BUILDING PAD AND STRUCTURES

- A. Conform to elevations and dimensions indicated within a tolerance of plus or minus 0.10 foot, and extending a sufficient distance from footings and foundations to permit placing and removal of concrete formwork, installation of services, other construction and for review.
- B. Excavations for footings and foundations: Do not disturb bottoms of excavation. Excavate by hand to elevations required just before concrete reinforcement is placed. Trim bottoms to required lines and grades to leave solid base to receive other work.

1. Where rock is encountered, carry excavation to required elevations and backfill with crushed stone prior to installation of footing.
- C. Excavation for Underground Tanks, Basins, and Mechanical or Electrical Structures: Conform to elevations and dimensions indicated within a tolerance of plus or minus 0.10 foot plus a sufficient distance to permit placing and removal of concrete formwork, installation of services, other construction and for review. Do not disturb bottom of excavations intended for bearing surface.

3.6 EXCAVATION FOR WALKS AND PAVEMENTS

- A. Cut surface under pavements to comply with cross-sections, elevations and grades as indicated.

3.7 EXCAVATION FOR UTILITY TRENCHES

- A. Excavate trenches to uniform width, sufficiently wide to provide ample working room and a minimum of 6 to 9 inches of clearance on both sides of pipe or conduit.
- B. Excavate trenches to depth indicated or required to establish indicated slope and invert elevations and to support bottom of pipe or conduit on undisturbed soil. Beyond building perimeter, excavate trenches to allow installation of top of pipe below frost line.
 1. Where rock is encountered, carry excavation to required elevations and backfill with required DOT crushed stone prior to installation of pipe.
 2. For pipes or conduit less than 6 inches in nominal size, and for flat-bottomed, multiple-duct conduit units, do not excavate beyond indicated depths. Hand-excavate bottom cut to accurate elevations and support pipe or conduit on undisturbed soil.
 3. For pipes and equipment 6 inches or larger in nominal size, shape bottom of trench to fit bottom of pipe for 90 degrees (bottom 1/4 of the circumference). Fill depressions with tamped sand backfill. At each pipe joint, dig bell holes to relieve pipe bell of loads ensure continuous bearing of pipe barrel on bearing surface.

3.8 EXCAVATION STABILITY

- A. General: Comply with local codes, ordinances, and requirements of agencies having jurisdiction.
- B. Slope sides of excavations to comply with local codes, ordinances, and requirements of agencies having jurisdiction. Shore and brace where sloping is not possible because of space restrictions or stability of material excavated. Maintain sides and slopes of excavations in safe condition until completion of backfilling.
- C. Shoring and Bracing: Provide materials for shoring and bracing, such as sheet piling, uprights, stringers, and cross braces, in good serviceable condition. Maintain shoring and bracing in excavations regardless of time period excavations will be open. Extend shoring and bracing as excavation progresses.

3.9 SUBGRADE INSPECTION

- A. Notify Architect/Engineer when mass, trench and footing excavations have reached required subgrade. The Architect/Engineer will arrange for an inspection of conditions by the Owner's Geotechnical Engineer. *Alternative procedures for arranging this review may be implemented at the Owner's written option.*

- B. If the Owner's Geotechnical Engineer determines that the subgrade bearing conditions are unacceptable, the Architect will authorize additional excavation until suitable bearing conditions are encountered.
- C. Proof-roll subgrade [**below the building slabs and pavements**] <Insert locations> with heavy pneumatic-tired equipment to identify soft pockets and areas of excess yielding. Do not proof-roll wet or saturated subgrades.
 - 1. Completely proof-roll subgrade in one direction[, **repeating proof-rolling in direction perpendicular to first direction**]. Limit vehicle speed to **3 mph (5 km/h)**.
 - 2. Proof-roll with a loaded 10-wheel, tandem-axle dump truck weighing not less than **15 tons (13.6 tonnes)**.
 - 3. Excavate soft spots, unsatisfactory soils, and areas of excessive pumping or rutting, as determined by Architect, and replace with compacted backfill or fill as directed.
- D. Under supervision of the Owner's Geotechnical Engineer, proofroll subgrade in cut areas below the building pad and pavement(s) with a loaded dump truck or other approved pneumatic tired vehicle. Should any unstable sub-soil be encountered below pavement or structures, break up the top eight inches of ground surface, pulverize, moisture-condition to optimum moisture content, and compact to percentage of maximum density as stated in Percentage of Maximum Density Requirements. Perform this work at no additional cost and/or time to the Contract.
- E. Reconstruct subgrades damaged by freezing temperatures, frost, rain, accumulated water, or construction activities, as directed by Architect, without additional compensation.

3.10 ADDITIONAL EXCAVATION

- A. Additional Excavation (Mass): Remove excavated materials and dispose of on-site as directed by the Architect/Engineer. Replace this excavated material with satisfactory material placed and compacted according to the requirements of the "Placement and Compaction" section.
- B. Additional Excavation in Trenches: Remove excavated materials and dispose of on-site as directed by the Architect/Engineer. Replace this excavated material with stone.
- C. Additional Excavation in Footings: Remove excavated materials and dispose of on-site as directed by the Architect/Engineer. Replace this excavated material with lean concrete/flowable fill or with stone extending 12 inches laterally beyond the footing in all directions.
- D. The quantity of material removed as Additional Excavation (Mass, Trench or Footing) shall be calculated (on an in-situ basis) by a surveyor licensed in the project state and employed by the Contractor. The Owner's Project Representative shall review the quantity calculated within 48 hours of receiving the survey notes.
- E. Protect the subgrade during construction. During wet conditions, the subgrade soils may become saturated and soften, possibly resulting in damage to the subgrade if disturbed by equipment. Correct subgrade damaged in this manner. **No additional payment will be made to correct subgrade damaged in this manner.**

3.11 UNAUTHORIZED EXCAVATION

- A. Correct Unauthorized Excavation as follows:
 - 1. Under footings, foundation bases, or retaining walls, fill unauthorized excavation by extending indicated bottom elevation of footing or base to excavation bottom without altering required top elevation. Lean concrete fill may be used to bring elevations to proper position when acceptable to Architect/Engineer/Owners Geotechnical Engineer.

2. Elsewhere, backfill and compact unauthorized excavations as indicated for authorized excavations of same classification unless otherwise directed by Architect/Engineer/Owners Geotechnical Engineer.

3.12 STORAGE OF EXCAVATED MATERIALS

- A. Temporarily stockpile excavated materials acceptable for use as backfill and fill. Place, grade, and shape stockpiles for proper drainage. Cover to prevent windblown dust.
 1. Stockpile excavated materials away from edge of excavations. Do not store within the drip line of trees to remain.

3.13 BACKFILL AND FILL

- A. Backfill excavations as promptly as work permits, but not until completion of the following:
 1. Acceptance by local authority having jurisdiction of construction below finished grade, including perimeter insulation.
 2. Review, approval, and recording of the locations of underground utilities.
 3. Removal of concrete formwork.
 4. Removal of shoring and bracing (including backfilling of voids with satisfactory materials).
 5. Removal of trash and debris from excavation.
 6. Permanent or temporary horizontal bracing is in place on horizontally supported walls.
- B. Place backfill on subgrades free of mud, frost, snow or ice.
- C. Ground Surface Preparation: Remove vegetation, debris, obstructions, and deleterious materials from ground surface prior to placement of fills.
- D. Bench sloped surfaces steeper than 1 vertical to 4 horizontal so fill material will bond with existing material. Plow, scarify, bench or break up sloped surfaces flatter than 1 vertical to 4 horizontal so fill material will bond with existing material.
- E. Place soil material in layers to required subgrade elevations, for each area classification listed below, using materials indicated in Part 2 of this Section.
 1. Under grassed areas, use satisfactory excavated or borrow material.
 2. Under walks, curbs, and pavements, use satisfactory excavated or borrow material.
 3. Under building slabs, use satisfactory excavated or borrow materials and drainage/porous fill material as indicated.

3.14 UTILITY TRENCH BACKFILL

- A. Place backfill on subgrades free of mud, frost, snow, or ice.
- B. Place and compact bedding course on trench bottoms and where indicated. Shape bedding course to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits.
- C. Backfill trenches with concrete where trench excavations pass within 18 inches of column or wall footings and that are carried below bottom of such footings or that pass under wall footings. Place concrete to level of bottom of adjacent footing.

- D. Provide 4-inch- (100-mm-) thick, concrete-base slab support for piping or conduit less than 30 inches (750 mm) below surface of roadways. After installing and testing, completely encase piping or conduit in a minimum of 4 inches (100 mm) of concrete before backfilling or placing roadway subbase.
- E. Place and compact initial backfill of [subbase material] [satisfactory soil], free of particles larger than 1 inch (25 mm) in any dimension, to a height of 12 inches (300 mm) over the utility pipe or conduit.
 - 1. Carefully compact initial backfill under pipe haunches and compact evenly up on both sides and along the full length of utility piping or conduit to avoid damage or displacement of piping or conduit. Coordinate backfilling with utilities testing.
- F. Controlled Low-Strength Material: Place initial backfill of controlled low-strength material to a height of 12 inches (300 mm) over the utility pipe or conduit.
- G. Backfill voids with satisfactory soil while installing and removing shoring and bracing.
- H. Place and compact final backfill of satisfactory soil to final subgrade elevation.
- I. Controlled Low-Strength Material: Place final backfill of controlled low-strength material to final subgrade elevation.
- J. Install warning tape directly above utilities, 12 inches (300 mm) below finished grade, except 6 inches (150 mm) below subgrade under pavements and slabs.
- K. Do not backfill trenches until any required testing and inspections have been completed and Architect authorizes backfilling. Backfill carefully to avoid damage or displacement of pipe systems.
- L. Under piping and conduit and equipment, use crushed stone where required over rock bearing surface and for correction of unauthorized excavation. Shape excavation bottom to fit bottom 90 degrees of cylinder.
- M. Place backfill and fill materials evenly adjacent to structures, piping, or conduit to required elevations. Prevent wedging action of backfill against structures or displacement of piping or conduit by carrying material uniformly around structure, piping, or conduit to approximately same elevation in each lift.

3.15 SOIL MOISTURE CONTROL

- A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill soil layer before compaction to within 2 percentage points of optimum moisture content.
 - 1. Do not place backfill or fill soil material on surfaces that are muddy, frozen, or contain frost or ice.
 - 2. Remove and replace, or scarify and air dry otherwise satisfactory soil material that exceeds optimum moisture content by 2 percentage points and is too wet to compact to specified dry unit weight.
- B. Moisture Control: Where subgrade or layer of soil material must be moisture conditioned before compaction, uniformly apply water to surface of subgrade or layer of soil material. Apply water in minimum quantity as necessary to prevent free water from appearing on surface during or subsequent to compaction operations. Maintain the moisture content of the structural fill materials to within 2 percentage points of the optimum moisture content until permanently covered.
- C. Remove and replace, or scarify and air dry, soil material that is too wet to permit compaction to required density.

1. Stockpile or spread soil material that has been removed because it is too wet to permit compaction. Assist drying by discing, harrowing, or pulverizing until moisture content is reduced to a satisfactory value.
2. Work wet materials as directed by the Owner's Geotechnical Engineer. Base bids on working material daily for a maximum of five days of acceptable weather.
3. No additional payment will be made for these operations.
4. Time extensions will not be granted for working wet material on site.

3.16 COMPACTION OF SOIL BACKFILL AND FILLS

- A. Place backfill and fill materials in layers not more than 8 inches in loose depth for material compacted by heavy compaction equipment, and not more than 4 inches in loose depth for material compacted by hand-operated tampers.
- B. Before compaction, moisten or aerate each layer as necessary to provide optimum moisture content. Compact each layer to required percentage of maximum dry density or relative dry density for each area classification. Do not place backfill or fill material on surfaces that are muddy, frozen, or contain frost or ice.
- C. Control soil and fill compaction, providing minimum percentage of density indicated for each area classification indicated below. Correct improperly compacted areas or lifts as directed by Architect if soil density tests indicate inadequate compaction.
- D. Percentage of Maximum Density Requirements: Compact soil to not less than the following percentages of maximum density at a moisture content within 2 percentage points of optimum in accordance with ASTM D698:
 1. Under structures, building pad and pavements, compact each layer of backfill or fill material at 95 percent maximum density. This includes ground under future expansion areas.
 2. Under grass or unpaved areas, compact each layer of backfill or fill material at 90 percent maximum density.
- E. Seal all fill areas at the end of each working day, utilizing a smooth drum roller.

3.17 GRADING

- A. General: Rough grading of areas within the Project, including cut and fill sections and adjacent transition areas, shall be reasonably smooth, compacted and free from irregular surface changes. The degree of finish shall be that ordinarily obtainable from either blade-grader or motor patrol except as otherwise indicated. The finished subgrade surface from the grassed areas generally shall be not more than 0.2 feet above or below the final grade or approved cross section, with due allowance for topsoil.
- B. The tolerance for areas within 10 feet of building perimeter, walks and all areas to be paved shall not exceed 0.10 feet above or below the established subgrade. Finish all ditches, swales and gutters to drain readily. Unless otherwise indicated, evenly slope the subgrade to provide drainage away from building walls in all directions at a grade not less than ¼ inch per foot. Provide rounding at top and bottom of cut and fill slopes and at other breaks in grade.
- C. Protection of Graded Areas: Protect newly graded areas and areas of cut, fill and design/subgrade elevations from the actions of the elements and from deterioration as a result of construction operations and weather conditions (frost, rains, snow, sleet, hail, etc.). Repair any settlement or washing that occurs prior to or after acceptance of the work. Fill to required subgrade levels any areas where settlement occurs. Protect trees to remain, and, at all areas of the Site where construction operations are in progress, provide protection for the safety of occupants of the existing facilities.

- D. General: Uniformly grade areas to a smooth surface, free of irregular surface changes. Comply with compaction requirements and grade to cross sections, lines, and elevations indicated.
 - 1. Provide a smooth transition between adjacent existing grades and new grades.
 - 2. Cut out soft spots, fill low spots, and trim high spots to comply with required surface tolerances.
- E. Site Grading: Slope grades to direct water away from buildings and to prevent ponding. Maximum slope of parking and pedestrian paved areas is 2.00%. Finish subgrades to required elevations within the following tolerances:
 - 1. Lawn or Unpaved Areas: Plus or minus [**1 inch (25 mm)**] <Insert tolerance>.
 - 2. Walks: Plus or minus [**1 inch (25 mm)**] <Insert tolerance>.
 - 3. Pavements: Plus or minus [**1/2 inch (13 mm)**] <Insert tolerance>.
- F. Grading inside Building Lines: Finish subgrade to a tolerance of **1/2 inch (13 mm)** when tested with a **10-foot (3-m)** straightedge.

3.18 PAVEMENT SUBBASE COURSE:

- A. General: Place subbase material, in layers of indicated thickness, over subgrade surface to support a pavement base course.
- B. Grade Control: During construction, maintain lines and grades including crown and cross-slope of subbase course.
- C. Shoulders: Place shoulders along edges of subbase course to prevent lateral movement. Construct shoulders of acceptable soil materials, placed in such quantity to compact to thickness of each subbase course layer. Compact and roll at least at 12" width of shoulder simultaneously with compacting and rolling each layer of subbase course.
- D. Placing: Place subbase course material on prepared subgrade in layers of uniform thickness, conforming to indicated cross-section and thickness. Maintain optimum moisture content for compacting subbase material during placement operations.
- E. When a compacted subbase course is 6" thick or less, place material in a single layer. When more than 6" thick, place material in equal layers, except no single layer more than 6" or less than 3" in thickness when compacted.
- F. Place subbase[**and base**] course on subgrades free of mud, frost, snow, or ice.
- G. On prepared subgrade, place subbase[**and base**] course under pavements and walks as follows:
 - 1. Place base course material over subbase course under hot-mix asphalt pavement.
 - 2. Shape subbase[**and base**] course to required crown elevations and cross-slope grades.
 - 3. Place subbase[**and base**] course **6 inches (150 mm)** or less in compacted thickness in a single layer.
 - 4. Place subbase[**and base**] course that exceeds **6 inches (150 mm)** in compacted thickness in layers of equal thickness, with no compacted layer more than **6 inches (150 mm)** thick or less than **3 inches (75 mm)** thick.
 - 5. Compact subbase[**and base**] course at optimum moisture content to required grades, lines, cross sections, and thickness to not less than [**95**] <Insert percentage> percent of maximum dry unit weight according to [**ASTM D 698**] [**ASTM D 1557**].
- H. Pavement Shoulders: Place shoulders along edges of subbase and base course to prevent lateral movement. Construct shoulders, at least **12 inches (300 mm)** wide, of satisfactory soil

materials and compact simultaneously with each subbase and base layer to not less than [95] **<Insert percentage>** percent of maximum dry unit weight according to [ASTM D 698] [ASTM D 1557].

3.19 BUILDING SLAB DRAINAGE COURSE

- A. General: Place drainage/porous fill material, over subgrade surface to support concrete building slabs and sidewalks areas indicated.
- B. Place drainage course on subgrades free of mud, frost, snow, or ice.
- C. Placing: Place drainage/porous fill material on prepared subgrade in layers of uniform thickness, conforming to indicated cross-section and thickness. Maintain optimum moisture content for compacting material during placement operations.
- D. When a compacted drainage course is indicated to be 6 inches thick or less, place material in a single layer. When indicated to be more than 6 inches thick, place material in equal layers, except no single layer more than 6 inches or less than 3 inches in thickness when compacted.

3.20 FIELD QUALITY CONTROL

- A. Quality Control Testing During Construction: Allow testing service to inspect and approve each subgrade and fill layer before further backfill or construction work is performed.
 - 1. If in the opinion of the Architect, based on testing service reports and inspection, subgrade or fills have been placed that are below required density, perform additional compaction and testing until required density is obtained.
- B. The Owner will engage, and pay for, the services of a Geotechnical Engineer whose function shall be to afford complete engineering control by testing of the conditions of all footing subgrades, the placement of all structural fills under structures, building pad and pavement areas, and all compaction where required, and to observe the proof rolling of the building pad and pavement areas.
- C. The Owner's Geotechnical Engineer will be present as deemed necessary during all phases of the Work requiring filling, compaction operations or testing. The Geotechnical Engineer will provide the Architect with written certification that fill and compaction was completed with accepted materials in accordance with the Documents, and give a professional opinion regarding shrinkage or settlement of fill and safe load bearing capacity of fill.
- D. Site Preparation and Proofrolling: The Owner's Geotechnical Engineer will determine if any additional excavation or in-place densification is necessary to prepare a subgrade for fill placement for slab or pavement support.
- E. Fill Placement and Compaction: The Owner's Geotechnical Engineer will witness all fill operations and take sufficient in-place density tests to verify that the indicated degree of fill compaction is achieved. The Owner's Geotechnical Engineer will observe and approve borrow materials used and shall determine if their existing moisture contents are suitable/acceptable.
- F. Footing Excavation Review: The Owner's Geotechnical Engineer will review the footing excavations for the building foundations. He will verify that the design bearing pressures are available and that no loose or soft areas exist beneath the bearing surfaces of the footing excavations.
- G. The Owner's Geotechnical Engineer will submit two (2) copies each of his reports, recommendations and/or opinions to the Architect/Engineer and the Owner. Pertinent information will be provided to the Contractor as required.

3.21 EROSION CONTROL:

- A. Provide erosion control methods in accordance with requirements of authorities having jurisdiction, and as indicated in the Contract Documents.

3.22 PROTECTION

- A. Repair and reestablish grades in settled, eroded, and rutted areas to indicated tolerances.
- B. Reconditioning Compacted Areas: Where subsequent construction operations or adverse weather disturbs completed compacted areas, scarify surface, reshape, and compact to required density prior to further construction.
- C. Settling: Where settling is measurable or observable at excavated areas during general project warranty period, remove surface (pavement, lawn, or other finish), add backfill material, compact, and replace surface treatment. Restore appearance, quality, and condition of surface or finish to match adjacent work, and eliminate evidence of restoration to greatest extent possible.
- D. Protect excavation bottoms against freezing when atmospheric temperature is less than 35 degrees F.
- E.

3.23 DISPOSAL OF WASTE MATERIALS

- A. Removal from Owner's Property: Remove excess and/or waste materials, including trash and debris, and dispose of it off Owner's property in a legal manner.
- B. Dispose of excess material and materials not acceptable for use as backfill or fill legally offsite.
- C. Do not remove topsoil from site until it has been demonstrated to the Owner's satisfaction that it is excess.

3.24 SETTLEMENT PLATES *(DELETE SECTION IF NOT REQUIRED BY GEOTECH REPORT)*

- A. Provide and monitor three settlement plates to evaluate the settlement occurring during and after fill placement. Locate the settlement plates as indicated on Drawing C3.2 or as recommended by the Owners Geotechnical Engineer.
- B. Record the elevation of the top of the settlement plate daily until settlement has slowed to a point satisfactory to the Owners Geotechnical Engineer. Settlement plate readings shall be made to an accuracy of 0.01' and shall be referenced to a benchmark well beyond the influence of the fill being placed and protected from construction equipment disturbance.
- C. Take precautions to prevent damaging or disturbing the settlement plates during construction operations.
- D. Providing, maintenance and monitoring of the settlement plates is part of the Work.
- E. Base bids on a settlement period of 45 calendar days commencing at the time final subgrade elevations in the area are attained.

END OF SECTION 312000

SECTION 312500 - EROSION CONTROL**PART 1 - GENERAL****1.1 RELATED DOCUMENTS:**

- A. The provisions of the Contract Documents apply to the work of this Section.
- B. The Project State and Local Jurisdiction Requirements.

1.2 SUMMARY

- A. This Section includes the installation, maintenance and removal of erosion control measures required for prevention of sediment leaving the project site.

1.3 EROSION AND SEDIMENT CONTROL PERMIT

- A. Prior to commencement of work, obtain a copy of the approved Erosion and Sediment Control Plan from the **Local Gov't Approval Agency**.
- B. Apply for the Land Disturbance Permit from the **Local Gov't Approval Agency**.
- C. Post Erosion and Sediment Control Bond in the amount of **\$XXXXXX** with the **Local Gov't Approval Agency**.
- D. Schedule a pre-construction conference on-site with the Architect and **Local Gov't Environmental Inspector**. Hold this meeting prior to the start of any construction activities.

1.4 VSMP REGISTRATION

- A. Department of Environmental Quality (DEQ)/local reviewing authority per the requirements of the Virginia Stormwater Management Program (VSMP). **The site must be registered to the Contractor's Responsible Land Disturber prior to the start of construction activities.** Contractor is responsible for all VSMP related fees.
- B. During construction, the following requirements shall be met:
 - 1. A copy of the Stormwater Pollution Prevention Plan (SWPPP) shall be kept at the job site at all times.
 - 2. Amend the SWPPP as necessary to account for significant changes in design, construction or maintenance that would increase the pollution potential of the site. File a copy of the amended plan with DEQ and with the Architect.
 - 3. The Responsible Land Disturber shall perform weekly inspections of the erosion and sediment control measures. Inspection reports shall be filed as an appendix to the SWPPP.
 - 4. Contractor is responsible for required VSMP renewals.
- C. Following final acceptance of the site by the Owner and sign off by the County Environmental Inspector, file a Notice of Termination with the Virginia Department of Environmental Quality/local reviewing authority.
- D. Prior to commencement of work, register /transfer the existing registration for the site with the State.

1.5 SUBMITTALS

- A. Responsible Land Disturber registration information.
- B. A copy of the VSMP registration application and a copy of the Stormwater Pollution Prevention Plan. A copy of the VSMP transfer application.
- C. Copies of the weekly Erosion Control Measure inspection reports. *Inspection reports should be maintained in the onsite SWPPP binder for review at the monthly progress meetings.*
- D. A copy of the Notice of Termination with the Virginia Department of Environmental Quality/ local

reviewing authority .

E. Silt Fence

F. Safety Fence

1.6 PAYMENT PROCEDURES FOR EROSION CONTROL MEASURES

- A. Establish a line item in the Schedule of Values for Erosion Control Maintenance. This line item shall represent a minimum of thirty percent (30%) of the total value of the erosion control for the project.
- B. Erosion control maintenance will be paid on a monthly basis, following the satisfactory installation and maintenance of the erosion control measures.

PART 2 - PRODUCTS

2.1 EROSION CONTROL PRODUCTS:

A. Safety Fence

- 1. Six foot high chain link fence, complying with Federal, State and Local jurisdictional requirements
- 2. Post appropriate warning signs along the Safety Fence.

B. Construction Entrance

- 1. Heavy-duty stone aggregate and filter fabric construction entrance, complying with the requirements of Federal, State and Local jurisdictional requirements Reinforced concrete wash-rack, draining to a sediment trap.
- 2. The water source for washing operations shall be the responsibility of the Contractor.

C. Silt Fence

- 1. Synthetic filter fabric, complying with the requirements of Federal, State and Local jurisdictional requirements
- 2. Wooden stakes shall be 2" oak, a minimum length of five feet.

D. Wire Reinforced Silt Fence

- 1. Synthetic filter fabric, complying with the requirements of Federal, State and Local jurisdictional requirements
- 2. Wooden stakes shall be 2" oak, a minimum length of five feet.
- 3. Wire fence reinforcement shall be a minimum of 14-gauge and have a maximum mesh spacing of six inches.

E. Storm Drain Inlet Protection

- 1. Block and Gravel Drop Inlet Sediment Filter, complying with the requirements of Federal, State and Local jurisdictional requirements.
- 2. Gravel Curb Inlet Sediment Filter {Block and Gravel Curb Inlet Sediment Filter}, complying with the requirements of Federal, State and Local jurisdictional requirements

F. Culvert Inlet Protection

- 1. Silt Fence {Stone Combination} Culvert Inlet Protection {Culvert Inlet Sediment Trap}, complying with the requirements of Federal, State and Local jurisdictional requirements.

G. Temporary Diversion Dike

- 1. Berm of compacted soil material, stabilized with vegetation, complying with the requirements of Federal, State and Local jurisdictional requirements

H. Temporary Fill Diversion

1. A channel with a berm on the lower side, placed at the top of a fill slope, complying with the requirements of Federal, State and Local jurisdictional requirements

I. Diversion

1. A channel with a berm on the lower side, stabilized with vegetation, complying with the requirements of Federal, State and Local jurisdictional requirements

J. Temporary Sediment Trap

1. A temporary ponding area, formed with an earthen embankment and a stone outlet structure, complying with the requirements of Federal, State and Local jurisdictional requirements

K. Temporary Sediment Basin

1. A temporary pond with a controlled outfall release, complying with the requirements Federal, State and Local jurisdictional requirements and to the details indicated on the Drawings.
2. The pond shall be constructed for use as a permanent stormwater management facility. Conversion of the pond from a temporary to a permanent facility is required. Refer to the Basin Conversion Narrative on the drawings.

L. Temporary Slope Drain

1. A flexible pipe extending from the top to the bottom of a fill slope, complying with the requirements of Federal, State and Local jurisdictional requirements
2. Pipe shall be smooth lined polyethylene, complying with the requirements of ASTM F667 or AASHTO M294.

M. Outlet Protection

1. A level area of riprap, placed over filter fabric, complying with the requirements of Standard and Federal, State and Local jurisdictional requirements

N. Riprap

1. Graded stone, placed over filter fabric, complying with the requirements of Federal, State and Local jurisdictional requirements
2. The size of the stone required is indicated on the drawings.

O. Rock Check Dams

1. Temporary stone dams constructed across a swale or ditch, complying with the requirements of Federal, State and Local jurisdictional requirements
2. Check dams shall be placed on filter fabric.

P. Dewatering Structure

1. A temporary filtering device used for dewatering operations, complying with the requirements of Federal, State and Local jurisdictional requirements

Q. Temporary Seeding

1. Temporary vegetative cover for disturbed areas, complying with the requirements of Federal, State and Local jurisdictional requirements

R. Permanent Seeding

1. Refer to Section "Lawns and Grasses" for permanent seeding requirements.

S. Soil Stabilization Blanket

1. Biodegradable stabilization blanket, complying with the requirements of Federal, State and

Local jurisdictional requirements

2. In lieu of plastic netting use East Coast Erosion Blankets (biodegradable single straw) or approved equal.

PART 3 - EXECUTION

3.1 INSTALLATION OF EROSION CONTROL MEASURES

- A. Install all erosion and sediment control measures per the requirements of Federal, State and Local jurisdictional requirements
- B. Protect all points of construction ingress and egress to the site to prevent tracking of mud onto public streets. Provide temporary construction entrances at all points of access to the site.
- C. Clear only those areas necessary for installation of the perimeter erosion control measures. The balance of the site shall not be cleared or otherwise disturbed until the perimeter erosion control measures are installed, functional and approved by the **Local Gov't Environmental Inspector**.
- D. Follow the construction sequence and install erosion control measures as indicated on the Drawings and as directed by the **Local Gov't Environmental Inspector**.
- E. Install additional measures as necessary to prevent sediment from leaving the project site.

3.2 MAINTENANCE OF EROSION CONTROL MEASURES

- A. Maintain all erosion and sediment control measures per the requirements of Federal, State and Local jurisdictional requirements.
- B. At a minimum, the following maintenance is required:
 1. Safety Fence
 - a) Review fence regularly for damage. Repair any damage immediately.
 - b) Secure the fence at the end of each working day. Repair or replace all locking devices as necessary.
 2. Construction Entrance
 - a) Wash and rework stone and/or place additional stone as required to prevent tracking of mud onto the roadways.
 - b) Clean out the sediment-trapping device for the washrack.
 - c) Remove all materials spilled, dropped, washed or otherwise tracked onto roadways or into storm sewers immediately. Do not use water trucks to wash the roadways.
 3. Silt Fence
 - a) Inspect immediately following each rainfall and at least daily during prolonged rainfall.
 - b) Make any required repairs immediately. Give special attention to damage resulting from end-runs and undercutting.
 - c) Replace fabric that is decomposing or is otherwise ineffective.
 - d) Clean out accumulated sediment following every storm event. Do not allow sediment to accumulate higher than one-half the height of the barrier.
 4. Wire Reinforced Silt Fence
 - a) Inspect immediately following each rainfall and at least daily during prolonged rainfall.
 - b) Make any required repairs immediately. Give special attention to damage resulting from end-runs and undercutting.
 - c) Replace fabric that is decomposing or is otherwise ineffective.

- d) Clean out accumulated sediment following every storm event. Do not allow sediment to accumulate higher than one-half the height of the barrier.
- 5. Storm Drain Inlet Protection
 - a) Inspect immediately following each rainfall and at least daily during prolonged rainfall.
 - b) Remove and clean or replace stone filters that have been clogged with sediment. Make any required repairs immediately
 - c) Remove accumulated sediment as required. Do not allow sediment to accumulate higher than one-half the height of the measure.
- 6. Culvert Inlet Protection
 - a) Inspect immediately following each rainfall and at least daily during prolonged rainfall.
 - b) Remove and clean or replace stone filters that have been clogged with sediment. Make any required repairs immediately
 - c) Remove accumulated sediment as required. Do not allow sediment to accumulate higher than one-half the height of the measure.
- 7. Temporary Diversion Dike
 - a) Inspect immediately following each rainfall and at least daily during prolonged rainfall. Inspect at least once every two weeks, whether or not it has rained. Make any necessary repairs immediately.
 - b) Repair damages caused by construction activities by the end of each working day.
- 8. Temporary Fill Diversion
 - a) Review measure at the end of each working day to ensure its effective operation.
- 9. Diversion
 - a) Inspect diversion following every rainfall and at least once every two weeks.
 - b) Remove accumulated sediment and make repairs as necessary.
 - c) Re-seed as necessary to maintain vegetative cover.
- 10. Temporary Sediment Trap
 - a) Remove sediment and restore the trap to its original dimensions once the sediment accumulates to the cleanout level. Refer to the drawings for the appropriate cleanout level elevations.
 - b) Any pumping shall be discharged through an approved dewatering structure.
 - c) Remove and clean or replace stone choked with sediment.
 - d) Regularly check the structure to ensure that it is structurally sound. Immediately repair any damage discovered.
- 11. Temporary Sediment Basin
 - a) Remove sediment and restore the basin to its original dimensions once the sediment accumulates to the cleanout level. Refer to the drawings for the appropriate cleanout level elevations.
 - b) Any pumping shall be discharged through an approved dewatering structure.
 - c) Regularly inspect the principal spillway and outfall for proper function. Regularly inspect the emergency spillway to ensure that its lining is well established and erosion resistant. Immediately repair any damage discovered.

- d) Regularly check the embankment to ensure that it is structurally sound. Immediately repair any damage discovered.

12. Temporary Slope Drain

- a) Inspect the temporary slope drains weekly and following every storm event. Immediately make any necessary repairs to ensure a free flow through the pipe.

13. Outlet Protection

- a) Inspect outlet protection following every storm event. Re-lay riprap as necessary to prevent concentrated flow from running across the outlet protection.

14. Riprap

- a) Inspect riprap following every storm event. Re-lay riprap as necessary to prevent concentrated flow from running under or around the riprap.
- b) Clean out accumulated sediment from the riprap.

15. Rock Check Dams

- a) Inspect immediately following each rainfall and at least daily during prolonged rainfall.
- b) Remove and clean or replace stone that has been clogged with sediment.
- c) Inspect for evidence of by-pass flows. Make any required repairs immediately.
- d) Remove accumulated sediment as required. Do not allow sediment to accumulate higher than one-half of the height of the dam.

16. Dewatering Structure

- a) Repair or replace the filtering media to prevent sediment accumulation from affecting the filtering capacity of the structure.

17. Temporary Seeding

- a) Re-seed and mulch areas where cover is inadequate to protect against erosion until adequate cover is obtained.

- C. Remove accumulated sediment as required and at appropriate intervals to maintain the effective function of all erosion control measures.
- D. Inspect, repair and remove accumulated sediment from erosion control measures following significant (greater than ½") rainfall events.
- E. If erosion control measures become clogged, causing the impoundment of water, restore the measures immediately. Ponded water poses a potential drowning hazard and shall be relieved immediately by either pumping (through an approved dewatering structure) or by removal of the blockage.

3.3 REMOVAL OF EROSION CONTROL MEASURES

- A. Remove all temporary erosion control measures following the stabilization of the site. Do not remove erosion control measures until authorized by the **Local Gov't Environmental Inspector**.
- B. Topsoil, permanently seed and stabilize areas occupied by erosion control measures.

END OF SECTION 312500

SECTION 313116 - TERMITE CONTROL**PART 1 - GENERAL****1.1 RELATED DOCUMENTS:**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes soil treatment for termite control.

1.3 SUBMITTALS

- A. Product data and application instructions.
- B. Certification that products used comply with U.S. Environmental Protection Agency (EPA) regulations for termiticides.

1.4 QUALITY ASSURANCE

- A. In addition to requirements of these specifications, comply with manufacturer's instructions and recommendations for preparing substrate and application.
- B. Engage a professional pest control operator who is licensed according to regulations of governing authorities to apply soil treatment solution.
- C. Use only termiticides that bear a federal registration number of the EPA and are approved by local authorities having jurisdiction.

1.5 JOB CONDITIONS

- A. Restrictions: Do not apply soil treatment solution until excavating, filling, and grading operations are completed, except as otherwise required in construction operations.
- B. To ensure penetration, do not apply soil treatment to frozen or excessively wet soils or during inclement weather. Comply with handling and application instructions of the soil toxicant manufacturer.

1.6 WARRANTY

- A. Warranty: Furnish written warranty, executed by Applicator and Contractor, certifying that applied soil termiticide treatment will prevent infestation of subterranean termites. If subterranean termite activity is discovered during warranty period, re-treat soil and repair or replace damage caused by termite infestation.
- B. Warranty Period: 5 years from date of final Completion. Also, include a renewable warranty for the Owner's future consideration.
- C. The warranty shall not deprive the Owner of other rights the Owner may have under other provisions of the Contract Documents and shall be in addition to and run concurrent with other warranties made by the Contractor under requirements of the Contract Documents.

PART 2 - PRODUCTS**2.1 SOIL TREATMENT SOLUTION:**

- A. Use an emusible concentrate insecticide for dilution with water, specially formulated to prevent infestation by termites. Fuel oil will not be permitted as a diluent. Provide a working solution of one of the following chemical elements and concentrations:
 - 1. Cypermethrin (Demon TC) 0.5% in water emulsion.
- B. Other solutions may be used as recommended by Applicator and if acceptable to local governing authorities. Use only soil treatment solutions that are not injurious to planting.

PART 3 - EXECUTION

3.1 APPLICATION

- A. Surface Preparation: Remove foreign matter that could decrease treatment effectiveness on areas to be treated. Loosen, rake, and level soil to be treated, except previously compacted areas under slabs and foundations. Toxicants may be applied before placing compacted fill under slabs if recommended by toxicant manufacturer.
- B. Application Rates: Apply soil treatment solution as follows:
1. Under slab-on-grade structures, treat soil before concrete slabs are placed, using the following application rates:
 - a) Apply 4 gallons of chemical solution per 10 linear feet (5.1 L of chemical solution per meter) to soil in critical areas under slab, including entire inside perimeter of foundation walls, along both sides of interior partition walls, around plumbing pipes and electric conduit penetrating slab, and around interior column footers.
 - b) Apply 1 gallon of chemical solution per 10 sq. ft. (4.1 L of chemical solution per sq. m) as an overall treatment under slab and attached slab areas where fill is soil or unwashed gravel. Apply 1-1/2 gallon of chemical solution per 10 sq. ft. (6.1 L of chemical solution per sq. m) to areas where fill is washed gravel or other coarse absorbent material.
 - c) Apply 4 gallons of chemical solution per 10 linear feet (5.1 L of chemical solution per meter) of trench for each 12 inches (300 mm) of depth from grade to footing, along outside edge of building. Dig a trench 6 to 8 inches (150 to 200 mm) wide along outside of foundation to a depth of not less than 12 inches (300 mm). Punch holes to top of footing at not more than 12 inches (300 mm) o.c. and apply chemical solution. Mix chemical solution with the soil as it is being replaced in the trench.
 2. Under crawlspace and basement structures, treat soil along exterior and interior walls of foundations with shallow footings as specified above for exterior of slab-on-grade structures.
 3. Treat soil under or around crawlspace structures as follows:
 - a) Apply 4 gallons of chemical solution per 10 linear feet (5.1 L of chemical solution per meter) of trench along inside of foundation walls, along both sides of interior partitions, and around piers and plumbing. Do not apply an overall treatment in crawlspaces.
 - b) Apply 4 gallons of chemical solution per 10 linear feet (5.1 L of chemical solution per meter) of trench, for each 12 inches (300 mm) of depth from grade to footing, along outside of foundation walls, including part beneath entrance platform porches, etc.
 - c) Apply 4 gallons of chemical solution per 10 linear feet (5.1 L of chemical solution per meter) along the inside and outside of foundation walls of porches.
 - d) Apply 1 gallon of chemical solution per 10 sq. ft. (4.1 L of chemical solution per sq. m) of soil surface as an overall treatment only where attached concrete platform and porches are on fill or ground.
 4. At hollow masonry foundations or grade beams, treat voids at rate of 2 gallons per 10 linear feet 2.6 L per meter, poured directly into the hollow spaces.
 5. At expansion joints, control joints, and areas where slabs will be penetrated, apply at rate of 4 gallons per 10 linear feet (5.1 L per linear m) of penetration.
- C. Post signs in areas of application to warn workers that soil termiticide treatment has been applied. Remove signs after areas are covered by other construction.
- D. Reapply soil treatment solution to areas disturbed by subsequent excavation, landscape grading, or other construction activities following application.
- E. Allow not less than 12 hours drying time after application before beginning concrete placement or other construction activities.

END OF SECTION 313116

SECTION 321216 - ASPHALT PAVEMENT**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes the following: *(SECTION TO BE UPDATED TO REFLECT REGIONAL PAVEMENT DESIGN REQUIREMENTS)*
1. Hot-mix asphalt paving over prepared subbase.
 2. Hot -mix asphalt patching.
 3. Hot-mix asphalt overlays.
 4. Asphalt surface treatments
 - a) Coal tar sealant

1.3 SUBMITTALS

- A. Job-Mix Designs: Certification, by authorities having jurisdiction, of approval of each job mix proposed for the Work.
- B. Material Certification: Certification signed by Contractor certifying that each material complies with requirements.
- C. Traffic maintenance and Work Area Protection Plan: Submit a plan indicating sequencing and measures to be used for the maintenance and protection of traffic during operations within or immediately adjacent to existing roadways open to vehicular traffic. The Architect/Engineer and Approving Local Jurisdiction must approve this plan prior to commencement of work within the Right-of-Way.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an experienced installer who has completed hot-mix asphalt paving similar in material, design, and extent to that indicated for this Project and with a record of successful in-service performance.
- B. Asphalt paving materials and installation shall conform to the requirements of the Local Approving Jurisdiction.

1.5 PROJECT CONDITIONS

- A. Environmental Limitations: Do not apply asphalt materials if substrate is wet or excessively damp or if the following conditions are not met:
1. Prime and Tack Coats: Minimum ambient temperature of 50 deg F (10 deg C), and when temperature has not been below 35 deg F (1 deg C) for 12 hours immediately prior to application.
 2. Asphalt Base Course: Minimum surface temperature of 40 deg F (4 deg C) and rising at time of placement.
 3. Asphalt Surface Course: Minimum surface temperature of 40 deg F (4 deg C) and rising at time of placement.

1.6 TESTING AND INSPECTION

- A. Within the road Right-of-Way the approving Jurisdiction shall inspect the work.
- B. The Owner's testing agency will observe the asphalt placement in the parking lots and on-site areas not in Right-of-Way.

PART 2 - PRODUCTS

2.1 ASPHALT-AGGREGATE MIXTURE

- A. General: Provide plant-mixed, hot-laid asphalt-aggregate mixture complying with the requirements of the local paving authority to suit project conditions.

2.2 ASPHALT MATERIALS

- A. Tack Coat: ASTM D 977, emulsified asphalt or ASTM D 2397, cationic emulsified asphalt, slow setting, factory diluted in water, of suitable grade and consistency for application.
- B. Prime Coat: Asphalt emulsion prime conforming to local jurisdiction requirements.

2.3 AUXILIARY MATERIALS

- A. Paving Geotextile: Nonwoven polypropylene, specifically designed for paving applications, resistant to chemical attack, rot, and mildew.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that subgrade is dry and in suitable condition to support paving and imposed loads.
- B. Proof-roll subbase using heavy, pneumatic-tired rollers to locate areas that are unstable or that require further compaction.
- C. Notify Architect/Engineer in writing of any unsatisfactory conditions. Do not begin paving installation until these conditions have been satisfactorily corrected.

3.2 MAINTENANCE AND PROTECTION OF TRAFFIC

- A. Utilize flagmen, barricades, warning signs and warning lights as required by the local jurisdiction Work Area Protection Manual.

3.3 PATCHING AND REPAIRS

- A. Patching: Saw cut perimeter of patch and excavate existing pavement section to sound base. Recompact new subgrade. Excavate rectangular or trapezoidal patches, extending 12 inches (300 mm) into adjacent sound pavement, unless otherwise indicated. Cut excavation faces vertically.
 - 1. Tack coat faces of excavation and allow to cure before paving.
 - 2. Fill excavation with dense-graded, hot-mix asphalt base mix and, while still hot, compact flush with adjacent surface.
- B. Leveling Course: Install and compact leveling course consisting of dense-graded, hot-mix asphalt surface course to level sags and fill depressions deeper than 1 inch (25 mm) in existing pavements.
 - 1. Install leveling wedges in compacted lifts not exceeding 3 inches (75 mm) thick.
- C. Crack and Joint Filling: Remove existing filler material from cracks or joints to a depth of 1/4 inch (6 mm). Refill with asphalt joint-filling material to restore watertight condition. Remove excess filler that has accumulated near cracks or joints.
- D. Tack Coat: Apply uniformly to existing surfaces of previously constructed asphalt or Portland cement concrete paving and to surfaces abutting or projecting into new, hot-mix asphalt pavement. Apply at a uniform rate of 0.05 to 0.15 gal./sq. yd. (0.2 to 0.7 L/sq. m) of surface.
 - 1. Allow tack coat to cure undisturbed before paving.
 - 2. Avoid smearing or staining adjoining surfaces, appurtenances, and surroundings. Remove spillage and clean affected surfaces.

3.4 SURFACE PREPARATION

- A. General: Immediately before placing asphalt materials, remove loose and deleterious material from substrate surfaces. Ensure that prepared subgrade is ready to receive paving.
- B. Sweep loose granular particles from surface of unbound-aggregate base course. Do not dislodge or disturb aggregate embedded in compacted surface of base course.
- C. Prime Coat: For asphalt sections less than 4" thick, apply uniformly over surface of compacted-aggregate base at a rate of 0.15 to 0.50 gal./sq. yd. (0.7 to 2.3 L/sq. m). Apply enough material to penetrate and seal, but not flood, surface. Allow prime coat to cure for 24 hours minimum.
 - 1. If prime coat is not entirely absorbed within 24 hours after application, spread sand over surface to blot excess asphalt. Use just enough sand to prevent pickup under traffic. Remove loose sand by sweeping before pavement is placed and after volatiles have evaporated.
 - 2. Protect primed substrate from damage until ready to receive paving.

3.5 GEOTEXTILE INSTALLATION

- A. Apply bond coat, consisting of asphalt cement, uniformly to existing surfaces at a rate of 0.20 to 0.30 gal./sq. yd. (0.8 to 1.2 L/sq. m).
- B. Place paving geotextile promptly according to manufacturer's written instructions. Broom or roll geotextile smooth and free of wrinkles and folds. Overlap longitudinal joints 4 inches (100 mm) and transverse joints 6 inches (150 mm).
 - 1. Protect paving geotextile from traffic and other damage and place overlay paving the same day.

3.6 HOT-MIX ASPHALT PLACING

- A. Machine place hot-mix asphalt mix on prepared surface, spread uniformly, and strike off. Place asphalt mix by hand to areas inaccessible to equipment in a manner that prevents segregation of mix. Place each course to required grade, cross section, and thickness, when compacted.
 - 1. Place hot-mix asphalt base course in number of lifts and thickness indicated.
 - 2. Spread mix at minimum temperature of 225 deg F (107 deg C).
- B. Place paving in consecutive strips not less than 10 feet (3 m) wide, except where infill edge strips of a lesser width are required.
 - 1. After first strip has been placed and rolled, place succeeding strips and extend rolling to overlap previous strips. Complete asphalt base course for a section before placing intermediate or surface courses.
- C. Promptly correct surface irregularities in paving course behind paver. Use suitable hand tools to remove excess material forming high spots. Fill depressions with hot-mix asphalt to prevent segregation of mix; use suitable hand tools to smooth surface.

3.7 JOINTS

- A. Construct joints between old and new pavement, or between successive days work, to ensure continuous bond between adjoining paving sections. Construct joints free of depressions with same texture and smoothness as other sections of hot-mix asphalt course.
 - 1. Clean contact surfaces and apply tack coat.
 - 2. Offset longitudinal joints in successive courses a minimum of 6 inches (150 mm).
 - 3. Offset transverse joints in successive courses a minimum of 24 inches (600 mm).
 - 4. Construct transverse joints as required by local jurisdiction.

5. Compact joints as soon as hot-mix asphalt will bear roller weight without excessive displacement.

3.8 COMPACTION

- A. General: Begin compaction as soon as placed hot-mix paving will bear roller weight without excessive displacement. Compact hot-mix paving with hot, hand tampers or vibratory-plate compactors in areas inaccessible to rollers.
 1. Complete compaction before mix temperature cools to 185 deg F (85 deg C).
- B. Breakdown Rolling: Accomplish breakdown or initial rolling immediately after rolling joints and outside edge. Examine surface immediately after breakdown rolling for indicated crown, grade, and smoothness. Repair surfaces by loosening displaced material, filling with hot-mix asphalt, and rerolling to required elevations.
- C. Intermediate Rolling: Begin intermediate rolling immediately after breakdown rolling, while hot-mix asphalt is still hot enough to achieve indicated density. Continue rolling until hot-mix asphalt course has been uniformly compacted to the following density:
 1. Average Density: 95 percent of reference laboratory density according to ASTM D 1559.
- D. Finish Rolling: Finish roll paved surfaces to remove roller marks while hot-mix asphalt is still warm. Surface course average density shall be 95 percent of reference laboratory density.
- E. Edge Shaping: While surface is being compacted and finished, trim edges of pavement to proper alignment. Bevel edges while still hot, with back of rake or smooth iron. Compact thoroughly using tamper or other satisfactory method. Edges adjacent to curbs and curb and gutter sections shall be flush with the edge of concrete.
- F. Repairs: Remove paved areas that are defective or contaminated with foreign materials. Remove paving course over area affected and replace with fresh, hot-mix asphalt. Compact by rolling to specified density and surface smoothness.
- G. Protection: After final rolling, do not permit vehicular traffic on pavement until it has cooled and hardened.
- H. Erect barricades to protect paving from traffic until mixture has cooled enough not to become marked.

3.9 INSTALLATION TOLERANCES

- A. Thickness: Compact each course to produce the thickness indicated within the following tolerances:
 1. Base Course: Plus or minus 1/2 inch (13 mm).
 2. Surface Course: Plus 1/4 inch (6 mm), no minus.
- B. Surface Smoothness: Compact each course to produce a surface smoothness within the following tolerances as determined by using a 10-foot (3-m) straightedge applied transversely or longitudinally to paved areas:
 1. Base Course: 1/4 inch (6 mm).
 2. Surface Course: 3/16 inch (3 mm).
 3. Crowned Surfaces: Test with crowned template centered and at right angle to crown. Maximum allowable variance from template is 1/4 inch (6 mm).
- C. Check surface areas at intervals as directed by Architect.

3.10 ASPHALT PAVEMENT OVERLAY

- A. Milling at edges.....

B. Subgrade repair.....

C. Tack Coat.....

3.11 FIELD QUALITY CONTROL

A. Within the Right-of-Way, coordinate with Local Jurisdiction for required inspections.

B. Testing Agency: Owner will engage a qualified independent testing agency to perform field inspections and tests and to prepare test reports.

1. Testing agency will conduct and interpret tests and state in each report whether tested Work complies with or deviates from requirements.

C. Additional testing, at Contractor's expense, will be performed to determine compliance of corrected Work with requirements.

D. Remove and replace or install additional hot-mix asphalt where test results or measurements indicate that it does not comply with requirements.

END OF SECTION 321216

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SECTION 321313 - SITE CONCRETE**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 DESCRIPTION OF WORK:

- A. Extent of Portland cement concrete paving is shown on drawings, including:

1. Curbs and gutters
2. Concrete Medians
3. Walkways
4. Service area pavement.
5. Paved Ditches

1.3 SUBMITTALS

- A. Job-Mix Designs: Certification, by authorities having jurisdiction, of approval of each job mix proposed for the Work.
- B. Material Certification: Certification signed by Contractor certifying that each material complies with requirements.

1.4 JOB CONDITIONS

- A. Traffic Control: Maintain access for vehicular and pedestrian traffic as required for other construction activities.

PART 2 - PRODUCTS**2.1 MATERIALS**

- A. Forms: Machined curbs shall be utilized to the maximum extent possible. Where machine formed curbs are not feasible, steel, wood, or other suitable material of size and strength to resist movement during concrete placement and to retain horizontal and vertical alignment until removal. Use straight forms, free of distortion and defects.
 1. Use flexible spring steel forms or laminated boards to form radius bends as required.
 2. Coat forms with a nonstaining form release agent that will not discolor or deface surface of concrete.
- B. Welded Wire Mesh: Welded plain cold-drawn steel wire fabric, ASTM A 185.
- C. Reinforcing Steel: ASTM A 615, Grade 60, deformed
- D. Concrete Materials: Comply with requirements of applicable Division 3 sections for concrete materials, admixtures, bonding materials, curing materials, and others as required.
- E. Expansion Joint Materials: Comply with requirements of applicable Division 7 sections for preformed expansion joint fillers and sealers.
- F. Antispalling Compound: Combination of boiled linseed oil and mineral spirits, complying with AASHTO M-233.
- G. Liquid-Membrane Forming and Sealing Curing Compound: Comply with local jurisdiction.

2.2 CONCRETE MIX, DESIGN, AND TESTING

- A. Comply with requirements of applicable Division 3 sections for concrete mix design, sampling and testing, and quality control or Local Jurisdiction/DOT Specifications whichever is more stringent.

- B. Design mix to produce normal-weight concrete consisting of Portland cement, aggregate, water-reducing or high-range water-reducing admixture (superplasticizer), air-entraining admixture, and water to produce the following properties:

- 1. Comply with the requirements of Local Jurisdiction/DOT, unless otherwise indicated.

PART 3 - EXECUTION

3.1 SURFACE PREPARATION

- A. Remove loose material from compacted subbase surface immediately before placing concrete.
- B. Proof-roll prepared subbase surface to check for unstable areas and need for additional compaction. Do not begin paving work until such conditions have been corrected and are ready to receive paving,

3.2 FORM CONSTRUCTION

- A. Set forms to required grades and lines, braced and secured. Install forms to allow continuous progress of work and so that forms can remain in place at least 24 hours after concrete placement.
- B. Check completed formwork for grade and alignment to following tolerances:
 - 1. Top of forms not more than 1/8 inch in 10 feet.
 - 2. Vertical face on longitudinal axis, not more than 1/4 inches in 10 feet.
- C. Clean forms after each use and coat with form release agent as required to ensure separation from concrete without damage.

3.3 REINFORCEMENT

- A. Locate, place and support reinforcement as specified in Division 3 sections, unless otherwise indicated.

3.4 CONCRETE PLACEMENT

- A. General: Comply with requirements of applicable Division 3 sections for mixing and placing concrete or Local Jurisdiction/DOT Specifications whichever is more stringent.
- B. Do not place concrete until subbase and forms have been checked for line and grade. Moisten subbase if required to provide a uniform dampened condition at time concrete is placed. Do not place concrete around manholes or other structures until they are at required finish elevation and alignment.
- C. Place concrete by methods that prevent segregation of mix. Consolidate concrete along face of forms and adjacent to transverse joints with internal vibrator. Keep vibrator away from joint assemblies, reinforcement, or side forms. Use only square-faced shovels for hand spreading and consolidation. Consolidate with care to prevent dislocation of reinforcing, dowels, and joint devices.
- D. Deposit and spread concrete in a continuous operation between transverse joints as far as possible. If interrupted for more than 1/2 hour, place a construction joint.
- E. Fabricated Bar Mats: Keep mats clean and free from excessive rust, and handle units to keep them flat and free of distortions. Straighten bends, kinks, and other irregularities or replace units as required before placement. Set mats for a minimum 2-inch overlap to adjacent mats.
- F. Place concrete in 2 operations; strike off initial pour for entire width of placement and to the required depth below finish surface. Lay fabricated bar mats immediately in final position. Place top layer of concrete, strike off, and screed.
- G. Remove and replace portions of bottom layer of concrete that have been placed more than 15 minutes without being covered by top layer or use bonding agent if acceptable to Architect/Engineer.

- H. Curbs and Gutters: Automatic machine may be used for curb and gutter placement. If machine placement is to be used, submit revised mix design and laboratory test results that meet or exceed minimums indicated. Machine placement must produce curbs and gutters to required cross-section, lines, grades, finish, and jointing as indicated for formed concrete. If results are not acceptable, remove and replace with formed concrete meeting requirements.

3.5 JOINTS

- A. General: Construct expansion, weakened-plane (contraction), and construction joints true to line with face perpendicular to surface of concrete. Construct transverse joints at right angles to the centerline, unless otherwise indicated.
- B. Weakened-Plane (Contraction) Joints: Provide weakened-plane (contraction) joints, sectioning concrete into areas as shown on drawings. Construct weakened-plane joints for a depth equal to at least 1/4 concrete thickness, as follows:
 - 1. Tooled Joints: Form weakened-plane joints in fresh concrete by grooving top portion with a recommended cutting tool and finishing edges with a jointer.
 - 2. Sawed Joints: Form weakened-plane joints with powered saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut joints into hardened concrete as soon as surface will not be torn, abraded, or otherwise damaged by cutting action.
 - 3. Inserts: Use embedded strips of metal or sealed wood to form weakened-plane joints. Set strips into plastic concrete and carefully remove strips after concrete has hardened.
- C. Construction Joints: Place construction joints at end of placements and at locations where placement operations are stopped for more than 1/2 hour, except where such placements terminate at expansion joints.
 - 1. Construct joints as indicated or, if not indicated, use standard metal keyway-section forms.
- D. Expansion Joints: Provide premolded joint filler for expansion joints abutting concrete curbs, catch basins, manholes, inlets, structures, walks, and other fixed objects, unless otherwise indicated.
- E. Locate expansion joints at 50 feet o.c. for each pavement lane unless otherwise indicated.
- F. Extend joint fillers full width and depth of joint, not less than 1/2 inch or more than 1 inch below finished surface where joint sealer is indicated. If no joint sealer, place top of joint filler flush with finished concrete surface.
- G. Provide joint fillers in one-piece lengths for full width being placed wherever possible. Where more than one length is required, lace or clip joint filler sections together.
- H. Protect top edge of joint filler during concrete placement with a metal cap or other temporary material. Remove protection after concrete has been placed on both sides of joint.
- I. Fillers and Sealants: Comply with requirements of applicable Division 7 sections for preparation of joints, materials, installation, and performance.
- J. Refer to Drawings for scoring patterns for:
 - 1. Selected sidewalk areas
 - 2. Service Areas
 - 3. Patios
 - 4. Courtyard

3.6 CONCRETE FINISHING

- A. After striking-off and consolidating concrete, smooth surface by screeding and floating. Use hand methods only where mechanical floating is not possible. Adjust floating to compact surface and produce uniform texture.

- B. After floating, test surface for trueness with a 10-ft. straightedge. Distribute concrete as required to remove surface irregularities, and refloat repaired areas to provide a continuous smooth finish.
- C. Work edges of slabs, gutters, back top edge of curb, and formed joints with an edging tool, and round to 1/2-inch radius, unless otherwise indicated. Eliminate tool marks on concrete surface.
- D. After completion of floating and when excess moisture or surface sheen has disappeared, complete troweling and finish surface as follows:
 - 1. Broom finish by drawing a fine-hair broom across concrete surface perpendicular to line of traffic. Repeat operation if required to provide a fine line texture acceptable to Architect.
 - 2. Exposed-Aggregate Finish: At handicap ramps and where indicated on drawings, by applying an approved retardant curing compound to the surface. Allow minimum 12 hours of setting time before washing surface to expose a maximum of (1/3) one-third of stone surface. Aggregate shall be brown Riverstone having a uniform size and color for each subsequent concrete pour. Aggregate size shall range between 1/2" and 3/4".
- E. Do not remove forms for 24 hours after concrete has been placed. After form removal, clean ends of joints and point-up any minor honeycombed areas. Remove and replace areas or sections with major defects, as directed by Architect.

3.7 CURING

- A. Protect and cure finished concrete paving in compliance with applicable requirements of Division 3 sections. Use membrane-forming curing and sealing compound or approved moist-curing methods.

3.8 REPAIRS AND PROTECTIONS

- A. Repair or replace cracked, broken or defective concrete curbs and curb and gutter, as directed by Architect/Engineer/Owner.
- B. Replace cracked, broken or defective concrete sidewalks.
- C. Repair or replace cracked, broken or defective concrete pavement, as directed by Architect/Engineer/Owner.
- D. Drill test cores where directed by Architect when necessary to determine magnitude of cracks or defective areas. Fill drilled core holes in satisfactory pavement areas with Portland cement concrete bonded to pavement with epoxy adhesive.
- E. Protect concrete from damage until acceptance of work. Exclude traffic from pavement for at least 14 days after placement. When construction traffic is permitted, maintain pavement as clean as possible by removing surface stains and spillage of materials as they occur.
- F. Sweep concrete pavement and wash free of stains, discolorations, dirt, and other foreign material just before final inspection.

END OF SECTION 321313

SECTION 321700 - PAVEMENT MARKINGS, SIGNS AND SPECIALTIES**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes, but is not limited to, the following:
 - 1. Establishing the location of pavement markings and applying pavement markings for parking space lines, traffic control, fire lane and accessible spaces.
 - 2. Installation of signs for traffic control and accessible spaces.
 - 3. Installation of wheel stops at parking spaces.

1.3 QUALITY ASSURANCE

- A. All work and materials shall conform to the requirements of State Department of Transportation (DOT).
- B. All materials for signs shall conform to the requirements of the State's DOT and to the requirements of the latest edition of the Manual of Uniform Traffic Control Devices for traffic signs.
- C. Installer Qualifications: Engage an experienced installer, who has successfully completed striping and signage projects similar in size and complexity to this project. The installer's primary business (defined as a minimum of 60% of total billings) shall be striping and signage.

1.4 SUBMITTALS

- A. Product Data and written confirmation that the following materials are included on The State's DOTs list of approved construction materials:
 - 1. Pavement marking paint
 - 2. Wheel stops
 - 3. Signs
 - 4. Posts
- B. **Installer Qualifications**

PART 2 - PRODUCTS**2.1 PAVEMENT MARKING PAINT**

- A. Paint shall be Type A, water emulsion base, traffic paint conforming to the requirements the State's DOT specification and Federal Specification TT-P-1952. Color shall be white unless otherwise indicated.
- B. Curb painting color along fire lanes shall be yellow, unless otherwise indicated.
- C. Lane markings within rights-of-way shall meet Local jurisdictions requirements

2.2 PAINT APPLICATOR

- A. Provide hand-operated push-type applicator machine of a type commonly used for application of paint to pavement surfaces. Paint applicator machine shall be acceptable for marking small street and parking areas. Applicator machine shall be equipped with the necessary paint tanks and spraying nozzles, and shall be capable of applying paint uniformly at coverage specified.

2.3 WHEEL STOPS

- A. Wheel stops shall be made of 3,000 psi precast concrete and be 6 inches high, 8 inches wide and approximately 6 feet long. Provide chamfered corners and edges and two holes for anchoring.
- B. Wheel Stops shall only be used as shown on the plans where required by the Authority Having Jurisdiction (AHJ).

2.4 SIGNS AND POSTS

- A. Signs shall conform to the requirements of the State's DOT. Signs shall be fabricated with encapsulated lens sheeting.
- B. Signposts for traffic control signage in Right of Way shall conform to the requirements of the State's DOT.
- C. Signposts or traffic control signage outside of the Right of Way shall be factory finished uni-strut Charcoal Grey.
- D. Utilize metal posts for fire-lane signage and for signage at accessible parking spaces.

2.5 CONCRETE

- A. Concrete shall be Class A3, General concrete, conforming to the requirements of the State's DOT .

PART 3 - EXECUTION

3.1 SURFACE PREPARATION FOR PAVEMENT MARKING

- A. Apply pavement markings only when the ambient temperatures is above 50°F and less than 95°F, unless otherwise approved.
- B. Allow pavement to cure for a period of not less than 7 days before applying pavement marking.
- C. Clean surfaces thoroughly before application of paint. Remove, dust, dirt and other granular surface deposits by sweeping, blowing with compressed air, rinsing with water, or a combination of these methods as required.
- D. Remove existing pavement markings, residual curing compounds and other coating adhering to the pavement with scrapers, wire brushes, water blasting, sandblasting or mechanical abrasion as required. Areas of existing pavement affected by oil or grease shall be scrubbed with an approved chemical and rinsed thoroughly. Seal oil soaked areas with shellac or primer after cleaning.
- E. Pavement surfaces shall be dry and clean prior to painting. Pavement markings shall not be applied within 24 hours following rain or other inclement weather or when rain is imminent.
- F. Apply seal coat across the existing pavement to provide a uniform surface appearance.

3.2 APPLICATION OF PAVEMENT MARKING

- A. Apply paint in accordance with the requirements of the State DOT.
- B. Lay out lines and markings to the width and length as indicated. All parking space lines shall be 4 inches wide.
- C. Apply paint with an approved paint applicator.
- D. Apply paint at manufacturer recommended rates to provide a minimum 15 mil wet thickness (depending on manufacturer, may require 2 applications).

3.3 FIRE LANE MARKINGS AND SIGNAGE

- A. Mark fire lanes and install fire lane signage in accordance with the requirements of the local Fire Marshall and as indicated on the drawings.

3.4 INSTALLATION OF WHEEL STOPS

- A. Secure wheel stops with two 1/2-inch diameter steel reinforcing rods. Rods shall be a minimum of 18 inches in length and be embedded into the pavement, base and subgrade a minimum of 12 inches and be flush with the top of the bumper block.
- B. Wheel Stops shall only be used as shown on the plans where required by the AHJ.

3.5 INSTALLATION OF SIGNS

- A. Install signs on signposts in accordance with the requirements of the State DOT.
- B. Install signposts in concrete foundation to a depth of 3 feet minimum by 12 inches in diameter.

END OF SECTION 321700

SECTION 323113 - CHAIN LINK FENCES AND GATES**PART 1 - GENERAL****1.1 SUMMARY****1.2 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.
- B. Section Includes:
 - 1. Chain-link fence and sliding gates at Staging/Backstock (interior).
 - 2. Chain-link fence and swinging gates at Trash Enclosure and Generator (exterior).

1.3 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at site.

1.4 CONFORMANCE AND ACTION SUBMITTALS

- A. Conformance Submittals are allowed for products described in this section; however, Conformance Submittals will not be reviewed by the Architect or Engineer: See Section 013300 "Submittal Procedures" for requirements of Conformance Submittals and Action Submittals.
- B. Submit the following Submittals via Owner's Project Management software platform.
 - 1. Product Data: For each type of product.
 - 2. Shop Drawings: For each type of fence and gate assembly.
 - a. Include plans, elevations, sections, details, and attachments to other work.
 - 3. Samples: For each exposed product and for each color and texture specified.

1.5 INFORMATIONAL SUBMITTALS

- A. Delegated-Design Submittal: For structural performance of chain-link fence and gate frameworks, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
- B. Product certificates.
- C. Product test reports.
- D. Sample warranty.

1.6 CLOSEOUT SUBMITTALS

- A. Operation and maintenance data.

1.7 WARRANTY

- A. Warranty Period: Two years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. Allied Consulting Inc.; Clarks Summit, PA; phone 570-430-1337
 2. Ameristar Fence Products; Tulsa, OK; phone 888-333-3422
 3. Chain Link Fence Manufacturers Institute (CLFMI); Old Columbia Rd., MD; phone 301-596-2583
 4. Iron World Manufacturing; Statesville, NC; phone 301-776-7448
 5. Lock Joint Tube; South Bend, TN; phone 800-257-6859
 6. Master Halco; Dallas, TX; phone 888-643-3623
 7. Sharon Fence Company; Sharon, PA; phone 724-981-6050

2.2 PERFORMANCE REQUIREMENTS

- A. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design chain-link fence and gate frameworks.
- B. Structural Performance: Chain-link fence and gate frameworks shall withstand the stresses for fence height(s) and under exposure conditions indicated according to ASCE/SEI 7.
1. Concentrated load of 50 lbf applied horizontally on an area of 1 sq. ft. at any location on a panel.
 2. Total load of 200 lbf applied uniformly over each panel.
 3. Concentrated load and total load need not be assumed to act concurrently.
- C. Seismic Performance: Wire mesh units shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.
1. Component Importance Factor: 1.0

2.3 CHAIN-LINK FENCE FABRIC

- A. General: Provide fabric in one-piece heights measured between top and bottom of outer edge of selvage knuckle or twist according to "CLFMI Product Manual" and requirements indicated below:
1. Fabric Height: As indicated on Drawings.
 2. Steel Wire for Fabric: Wire diameter of 0.120 inch (3.05 mm), 8 gauge.
 - a. Mesh Size: 2" by 2"
 - b. Zinc-Coated Steel Fabric: ASTM A392 hot dipped galvanized before weaving (GBW) or after weaving (GAW).
 - c. Coat selvage ends of metallic-coated fabric before the weaving process with manufacturer's standard clear protective coating.
 3. Selvage: Knuckled at both selvages.

2.4 FENCE FRAMEWORK

- A. Round steel pipe and rail: Schedule 40 standard weight pipe, in accordance with ASTM F1083, 1.8 oz/ ft² (550 g/m²) hot dip galvanized zinc exterior and 1.8 oz/ft² (550 g/m²) hot dip galvanized zinc interior coating.
 - 1. Regular Grade: Minimum steel yield strength 30,000 psi (205 MPa)
 - 2. Fence Height: As indicated on Drawings.
 - 3. Line Post: 3 inches (76 mm) in diameter
 - 4. End, Corner, and Pull Posts: 3 inches (76 mm) in diameter
- B. Horizontal Framework Members: top and bottom rails according to ASTM F 1043.
- C. Brace Rails: ASTM F 1043.
- D. Metallic Coating for Steel Framework:
 - 1. Type A zinc coating.
 - 2. Type B zinc with organic overcoat.
 - 3. External, Type B zinc with organic overcoat and internal, Type D zinc-pigmented coating.
 - 4. Type C, Zn-5-Al-MM alloy coating.
 - 5. Coatings: Any coating above.

2.5 HORIZONTAL SLIDE GATES AT STAGING/BACKSTOCK

- A. Slide Gates: Schedule 40 galvanized steel pipe welded fabrication.
 - 1. Gate Frame Members: 1.900 inch OD (48.3 mm).
 - 2. Frame members spaced no greater than 8 feet (2440 mm) apart vertically and horizontally.
 - 3. Protect welded joints with zinc-rich paint.
 - 4. Match gate fabric to that of the fence system.
- B. Pipe and Tubing:
 - 1. Zinc-Coated Steel: ASTM F 1043 and ASTM F 1083; manufacturer's standard protective coating and finish.
 - 2. Gate Posts: Round tubular steel.
 - 3. Gate Frames and Bracing: Round tubular steel.
 - 4. Fabric: Match fencing material.
- C. Frame Corner Construction: Assembled with corner fittings.
- D. Hardware:
 - 1. Overhead Sliding Track Kit: Pemko H300S-OH X LAR, galvanized steel, or approved equal.
 - 2. Latch: Permitting operation from one side of gate only.
 - 3. Include guides, hangers, roller assemblies, stops and end caps fabricated from galvanized steel.
 - 4. Provide Resettable Combination pad lock with 5 digits.

2.6 SWING GATES AT TRASH ENCLOSURE AND GENERATOR

- A. Swing Gates: Schedule 40 galvanized steel pipe welded fabrication in compliance with ASTM F900.

1. Gate Frame Members: 1.900 inch OD (48.3 mm).
2. Frame members spaced no greater than 8 feet (2440 mm) apart vertically and horizontally.
3. Protect welded joints with zinc-rich paint.
4. Match gate fabric to that of the fence system.

B. Pipe and Tubing:

1. Zinc-Coated Steel: ASTM F 1043 and ASTM F 1083; manufacturer's standard protective coating and finish.
2. Gate Posts: Round tubular steel per ASTM F 1083
3. Gate Frames and Bracing: Round tubular steel.

C. Frame Corner Construction: Assembled with corner fittings.

D. Hardware: Galvanized steel.

1. Hinges: 180-degree outward swing.
2. Latch: Permitting operation from one side of gate.
3. Positive locking gate latch.
4. Lockable drop bar and gate holdbacks with double gates.
5. See gate details on drawings for specified lockset.

2.7 FITTINGS

- A. Tension and Brace Bands: Galvanized pressed steel complying with ASTM F626, minimum steel thickness of 12 gauge (0.105 in.) (2.67 mm), minimum width of 3/4 in. (19 mm) and minimum zinc coating of 1.20 oz/ft² (366 g/m²). Secure bands with 5/16 in. (7.94 mm) galvanized steel carriage bolts.
- B. Terminal Post Caps, Line Post Loop Tops, Rail and Brace Ends, Boulevard Clamps, Rail Sleeves: In compliance to ASTM F626, pressed steel galvanized after fabrication having a minimum zinc coating of 1.20 oz/ft² (366 g/m²).
- C. Truss Rod Assembly: In compliance with ASTM F626, 3/8 in. (9.53 mm) or 5/16" (7.94 mm) diameter steel truss rod with a pressed steel tightener, minimum zinc coating of 1.2 oz/ft² (366 g/m²), assembly capable of withstanding a tension of 2,000 lbs. (970 kg).
- D. Tension Bars: In compliance with ASTM F626. Galvanized steel one-piece length 2 in. (50 mm) less than the fabric height. Minimum zinc coating 1.2 oz. /ft² (366 g/m²).
 1. Bars for 2 in. (50 mm) and 1 3/4 in. (44 mm) mesh shall have a minimum cross section of 3/16 in. (4.8 mm) by 3/4 in. (19 mm)
 2. Bars for 1 in. (25 mm) mesh shall have a cross section of 1/4 in. (6.4 mm) by 3/8 in. (9.5 mm)
 3. Small mesh 3/8 in. (10 mm), 1/2 in. (13 mm) and 5/8 in. (16 mm) shall be attached (sandwiched) to the terminal post using a galvanized steel strap having a minimum cross section of 2 in. (51 mm) by 3/16 in. (4.8 mm) with holes spaced 15 in. (381 mm) on center to accommodate 5/16 in. (7.9 mm) carriage bolts which are to be bolted thru the strap the mesh and thru the terminal post.
- E. Finish:
 1. Metallic Coating for Pressed Steel: Not less than 1.2 oz./sq. ft. (366 g/sq. m) of zinc.

- a. Galvanized over metallic coating.

2.8 PRIVACY SLATS

- A. Material: Polyethylene tubular slats, not less than 0.023 inch (0.58 mm) thick, manufactured for chain-link fences from virgin polyethylene containing UV inhibitor, sized to fit mesh specified for direction indicated.

- 1. Color: As selected by Architect from manufacturer's full range.

2.9 SLAB ANCHORING

- A. Concrete for post footings shall have a 28-day compressive strength of 2,500 psi. (17.2 MPa).
- B. Anchor 6" x 6" x 1/4" galvanized shop welded steel flanges with 5/8" holes at each posts to floor with 3/8-inch-diameter post installed expansion anchors. Shim flange as required to achieve level and plumb installation.

- 1. Anchors may be set with power-actuated fasteners instead of post installed expansion anchors if indicated on Shop Drawings.

2.10 GROUT AND ANCHORING CEMENT

- A. Nonshrink, Nonmetallic Grout: Factory-packaged, nonstaining, noncorrosive, nongaseous grout complying with ASTM C1107/C1107M. Provide grout, recommended in writing by manufacturer, for exterior applications.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Examine floors for suitable conditions where wire mesh items will be installed.
- C. Examine walls to which wire mesh items will be attached for properly located blocking, grounds, and other solid backing for attachment of support fasteners.
- D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Stake locations of fence lines, gates, and terminal posts.

3.3 CHAIN-LINK FENCE INSTALLATION

- A. Install chain-link fencing according to ASTM F 567 and more stringent requirements specified.
- B. Post Installation at Trash Enclosure: Set chain-link post in concrete footing as indicated on drawings.
 - 1. Posts Set into Sleeves in Concrete: Use steel pipe sleeves preset and anchored into concrete for installing posts. After posts are inserted into sleeves, fill annular space

between post and sleeve with nonshrink, nonmetallic grout, mixed and placed according to anchoring material manufacturer's written instructions. Finish anchorage joint to slope away from post to drain water.

- C. Verify that posts are set plumb, aligned, and at correct height and spacing, and hold in position during installation.
- D. Terminal Posts: Install terminal end, corner, and gate posts according to ASTM F 567 and terminal pull posts at changes in horizontal or vertical alignment of as indicated on Drawings.
- E. Line Posts: Space line posts uniformly as designed on drawings.
- F. Top Rail: Install according to ASTM F567, maintaining plumb position and alignment of fence posts. Run rail continuously through line post caps, bending to radius for curved runs and terminating into rail end attached to posts or post caps fabricated to receive rail at terminal posts. Provide expansion couplings as recommended in writing by fencing manufacturer.
- G. Bottom rail or intermediate rail shall be field cut and secured to the line posts using boulevard clamps or brace band with rail end. Fences 12 feet (3.66 m) high or higher require mid rail. Fabric in "Chain-Link Fabric".
- H. Chain-Link Fabric: Apply fabric to inside of enclosing framework. Leave 1-inch (25-mm) bottom clearance between finish grade or surface and bottom selvage unless otherwise indicated. Pull fabric taut and tie to posts and rails. Anchor to framework so fabric remains under tension after pulling force is released.
- I. Tie Wires: Use wire of proper length to firmly secure fabric to line posts and rails. Attach wire at one end to chain-link fabric, wrap wire around post a minimum of 180 degrees, and attach other end to chain-link fabric according to ASTM F626. Bend ends of wire to minimize hazard to individuals and clothing.
 - 1. Maximum Spacing: Tie fabric to line posts at 12 inches (300 mm) o.c. and to braces at 24 inches (610 mm) o.c.
- J. Privacy Slats: Install slats in direction indicated, securely locked in place.
 - 1. Vertically, for privacy factor of 70 to 75.

3.4 GATE INSTALLATION

- A. Install gates according to manufacturer's written instructions, level, plumb, and secure for full opening without interference. Attach fabric as for fencing. Attach hardware using tamper-resistant or concealed means. Install ground-set items in concrete for anchorage. Adjust hardware for smooth operation.

3.5 ADJUSTING AND CLEANING

- A. Adjust gates to operate smoothly and easily, without binding or warping. Adjust hardware to function smoothly. Confirm that latches and locks engage accurately and securely without forcing or binding.
- B. Remove and replace defective work, including doors and framing that are warped, bowed, or otherwise unacceptable.

- C. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas, and repair galvanizing to comply with ASTM A 780.

END OF SECTION 323113

SECTION 323113 – CHAIN-LINK FENCES AND GATES (PVC CLAD)**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 WORK INCLUDED

- A. Polyvinyl Chloride (PVC) clad chain link fence and gates

1.3 SUBMITTALS

- A. Product Data: Submit manufacturer's technical data and installation instructions for fencing, fabric, gates and accessories.
- B. Shop Drawings: Submit shop drawings indicating location of fence (with dimensions), height, post locations, details of post installation, gate swing, hardware and accessories. Identify PVC touch up paint.
- C. Samples: None required

1.4 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an experienced installer who has completed chain-link fences and gates similar in material, design, and extent to those indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.
- B. Source Limitations for Chain-Link Fences and Gates: Obtain each color, grade, finish, type, and variety of component for chain-link fences and gates from one source with resources to provide chain-link fences and gates of consistent quality in appearance and physical properties.

1.5 PROJECT CONDITIONS

- A. Field Measurements: Verify layout information for chain-link fences and gates indicated in relation to property survey and existing structures. Verify dimensions by field measurements.

PART 2 - PRODUCTS**2.1 GENERAL:**

- A. Subject to compliance with requirements, manufacturers offering products which may be incorporated in the work include, but are not limited to, the following:
 - 1. PVC Coated Steel Fencing and Fabric:
 - a) Colorguard Fence Products, Inc.
 - b) American Chain Link Fence Company
 - c) Semmerling Fence & Supply, Inc.
 - d) Anchor Fence, Inc.

2.2 FABRIC:

- A. Steel Fabric: Comply with Chain Link Fence Manufacturers Institute (CLMFI) Product Manual. Provide one-piece fabric widths. Wire size includes zinc coating.
- B. Size: 2-inch diamond mesh, 9-gauge (0.148-inch diameter) wire.
- C. PVC Coating: ASTM F668, Class 2B (fused and adhered) PVC coating, black color. Bonded or extruded & glued fabric may not be used.
- D. Selvage shall be knuckled at the top and bottom.

2.3 FRAMING:

- A. Strength requirements for posts and rails shall conform to ASTM F 669.
- B. Pipe shall be straight, true to section, material and sizes specified.
- C. Steel Framework, General: Posts, rails, braces and gate frames.
 - 1. Type II Pipe: Manufactured from steel conforming to ASTM A 569 or A 446, grade D, cold formed, electric welded with minimum yield strength of 50,000 p.s.i. and triple coated with minimum 0.9 oz. Zinc per square foot after welding, a chromatic conversion coating and a clear polymer overcoat. Corrosion protection on inside surfaces shall protect the metal from corrosion when subjected to the salt spray test of ASTM B 117 for 300 hours with the end point of 5% Red Rust.
 - 2. PVC-Coating finish: In accordance with ASTM F668, Class 2B (fused and adhered) apply supplemental color coating of 10 to 15 mils (0.254 – 0.38mm) of thermally fused PVC in color to match fabric.
- D. End, Corner and Pull Posts:
 - 1. For fabric height up to 6' - 2.375" OD Type II steel pipe (3.12 lb/ft). [2.875", 73mm, 5.79 lb/ft.]
 - 2. For fabric height over 6' - 2.875" OD Type II steel pipe.(4.64 lb/ft). [4.00", 101.6mm, 9.11 lb/ft]
- E. Line Posts:
 - 1. For fabric height up to 6' - 1.90" OD Type II steel pipe (2.28 lb/ft).
 - 2. For fabric height over 6' - 2.375" OD Type II steel pipe (3.65 lb/ft).
- F. Gate Posts:
 - 1. Provide posts for supporting single gate leaf, or one leaf of a double gate installation, for nominal gate widths as follows:
 - a) 6' or Under: 2.875" OD Type II steel pipe (4.64lb/ft).
 - b) Over 6': 4.000" OD Type II steel pipe (8.65 lb/ft).
- G. Top & Bottom Rail:
 - 1. Manufacturer's longest lengths, with expansion-type couplings, approximately 6" long, for each joint. Provide means for attaching rail securely to each gate corner, pull, & end post.
 - a) 1-1/4" NPS (1.66" OD) Type II steel pipe.
- H. Intermediate and/or Center Rail:
 - 1. Same material as top rail. Manufacturer's standard galvanized steel cap required for each end.

2.4 FITTINGS AND ACCESSORIES:

- A. Material: Comply with ASTM F 626. Mill finished galvanized steel, to suit manufacturer's standards.
 - 1. Zinc Coating: Unless specified otherwise, galvanize steel fence fittings and accessories in accordance with ASTM A 153, with zinc weights indicated.
 - 2. Supplemental Color Coating: In accordance with ASTM F668, Class 2B (fused and adhered), apply supplemental color coating of 10 to 15 mils (0.254 – 0.38mm) of thermally fused PVC in color to match fabric. Apply to exterior surfaces and, except inside cap shapes, to exposed interior surfaces. Color to match chain link fabric.
- B. Tension Wire: 7 gauge (0.177" diameter) metallic coated steel marcelled tension wire conforming to ASTM A 824 with finish to match fabric.

1. PVC-Coated finish: In accordance with ASTM F668, Class 2B (fused and adhered), apply supplemental color coating of 10 to 15 mils (0.254 – 0.38mm) of thermally fused PVC in color to match fabric.

C. Wire Ties:

1. 9 gauge [0.148" (3.76mm)] galvanized steel wire for attachment of fabric to line posts.
2. Double wrap 13 gauge [0.092" (2.324mm)] for rails and braces.
3. Hog ring ties of 12-1/2 gauge [0.0985" (2.502mm)] for attachment of fabric to tension wire

D. Post Brace Assembly:

1. Manufacturer's standard adjustable brace at end of gate posts and at both sides of corner and pull posts, with horizontal brace located at mid height of fabric. Provide same material as top rail for brace, and truss to line posts with 0.375" diameter rod and adjustable tightener. Manufacturer's standard galvanized steel cap required for each end.

E. Post and Line Caps: Weathertight closure cap required for each post. If top rail is required, use line post caps with loop.

F. Tension or Stretcher Bars: Hot-dip galvanized steel with minimum length 2" less than full height of fabric, minimum cross section of 3/16" by 3/4" and minimum 1.2 oz. zinc coating per sq. ft. of surface area. One bar is required for each gate and end post and two for each corner and pull post, except where fabric is integrally woven into post.

G. Tension and Brace Bands: Minimum 3/4" wide hot-dip galvanized steel with minimum 1.2 oz. zinc coating per sq. ft. of surface area.

1. Tension bands: Minimum 14 gauge (0.074") thick.
2. Tension and Brace bands: Minimum 12 gauge (0.105") thick.

H. Nuts and bolts shall be galvanized but not vinyl coated. Provide touch up paint and color coat nuts and bolts to match fabric.

2.5 POST SETTING MATERIALS

A. Comply with the requirements for the State Std. Class A3 concrete.

2.6 GATES:

A. Fabrication:

1. Fabricate perimeter frames of gates from metal and finish to match fence framework. Utilize Fusion or stainless steel welded connections to form a rigid one-piece unit. Assemble gate frames by welding, providing security against removal or breakage connections. Provide horizontal and vertical members to ensure proper gate operation and attachment of fabric, hardware and accessories. Space frame members maximum of 8' apart unless otherwise indicated.
2. Provide same fabric as for fence. Install fabric with stretcher bars at vertical edges and at top and bottom edges. Attach stretcher hooks to gate frame at not more than 15" o.c.
3. Install diagonal cross-bracing consisting of 3/8" diameter adjustable length truss rods on gates to ensure frame rigidity without sag or twist.

B. Swing Gates: Comply with ASTM F 900.

1. Fabricate perimeter frames of minimum 1.90" OD Type II steel pipe.

C. Gate Hardware: Provide hardware and accessories for each gate, galvanized per ASTM A 153, and in accordance with the following:

1. Hinges: Size and material to suit gate size, non-lift-off type, offset to permit 180 degrees gate opening. Provide 1-1/2 pair of hinges for each leaf over 6' nominal height.
2. Latch: Forked type to permit operation from either side of gate, with padlock eye as integral part of latch.
3. Keeper: Provide keeper that automatically engages gate leaf and holds it in open position until manually released.
4. Double Gates: Provide gate stops for double gates, consisting of mushroom type flush plate with anchors, set in concrete, and designed to engage center drop rod or plunger bar. Ensure plunger bar cannot be removed without tools. Include locking device and padlock eyes as integral part of latch, permitting both gate leaves to be locked with single padlock.
5. Hardware materials: Provide hot dipped galvanized steel or malleable iron shapes to suit gate size. Field coat hardware parts (e.g. hinges, latch, keeper and drop bar) with PVC touch up paint, provided by manufacturer, to match adjacent finishes.

PART 3 - EXECUTION

3.1 INSTALLATION:

- A. General: Install fence in compliance with ASTM F 567 and manufacturers recommendations. Do not begin installation and erection before final grading is completed, unless otherwise permitted. Apply fabric to outside of framework, unless otherwise indicated.
- B. Locate terminal post at each fence termination and change in horizontal or vertical direction of 30° or more, or as indicated on plans.
- C. Excavation:
 1. Drill or hand excavate (using post hole digger) holes for posts to diameters and spacing indicated, in firm, undisturbed or compacted soil.
 2. Holes in asphalt or concrete surfaces will be cut by core-drilling with a bit of diameter at least equal to the required hole diameter. Holes in concrete may be formed prior to placing concrete.
 3. Excavate holes for each post to minimum diameter recommended by fence manufacturer, but not less than 4 times largest cross-section of post.
 4. Excavate hole to depths approximately 6" lower than post bottom, with bottom of posts set not less than 36" below finish grade surface.
- D. Setting Posts:
 1. Space 10' o.c. maximum, unless otherwise indicated.
 2. Center and align posts in hole, 6" above bottom of excavation.
 3. Protect portions of concrete posts above ground from concrete splatter. Place concrete around post and vibrate or tamp for consolidation. Check each post for vertical and top alignment, and hold in position during placement and finishing operations.
 4. Extend concrete above grade and slope all around (dome) to allow for drainage away from post. Uniformly and neatly texture the concrete surface with a broom finish. Remove any spilled or splashed concrete from the post and surrounding area immediately.
- E. Top Rails:
 1. Run rail continuously through line post caps, bending to radius for curved runs and at other posts terminating into rail end attached to posts or post caps fabricated to receive rail. Provide expansion couplings as recommended by fencing manufacturer.
- F. Center Rails:

1. Install in one place between posts and flush with post on fabric side, using rail ends and special offset fittings where necessary. Provide center rails for fence 12' or taller, or as indicated on drawings.

G. Bottom Rails:

1. Install in one piece between posts and flush with post on fabric side, using rail ends and special offset fittings when necessary.

H. Brace Assemblies:

1. Install braces so posts are plumb when diagonal rod is under proper tension.

I. Top and Bottom Tension Wire:

1. Install tension wires through post cap loops before stretching fabric and tie to each post cap with not less than same gauge and type of wire. Pull wire taut, without sags. Fasten fabric to tension wire, using 11 - ga. galvanized steel hog rings spaced maximum 24" o.c. Install where top and/or bottom rails are not specified on plans.

J. Fabric:

1. Leave approximately 2" between finish grade and bottom selvage. Pull fabric taut and tie to posts, rails and tension wires. Attach fabric with wire ties to line posts at 12"-15" (381mm) o.c. and to rails, braces, and tension wire at 24" (600 mm) o.c. Install fabric on security side of fence, unless otherwise indicated, and anchor to framework so that fabric remains in tension after pulling force is released.
2. For athletic field fencing, install fabric on the field side of the fence unless otherwise indicated.

K. Stretcher Bars:

1. Thread through fabric 4" o.c., and secure to end, corner, pull and gate posts with tension bands spaced maximum 15" o.c.

L. Accessories

1. Tie Wires: Use U-shaped wire, conforming to diameter of pipe to which attached, clasp pipe and fabric firmly with ends twisted at least 2 full turns. Bend ends of wire to minimize hazard to persons or clothing.

M. Fasteners:

1. Install nuts for tension bands and hardware bolts on site of fence opposite fabric side. Peen ends of bolts or score threads to prevent removal of nuts.

N. Gates:

1. Install gates plumb, level, and secure for full opening without interference. Install ground-set items in concrete for anchorage. Adjust hardware for smooth operation and lubricate where necessary.

3.2 FINISHING

- A. Remove and replace sections of fence and fittings with damaged PVC coating. Minor aesthetic damage may be touched up with a suitable spray on material provided by the manufacturer.
- B. Clean up debris and unused material and remove from the site.

END OF SECTION 323113

SECTION 328000 - IRRIGATION SYSTEM (NON-PROPRIETARY)**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 QUALITY ASSURANCE

- A. Installer's qualifications: the contractor must provide the following, to include name of contact, name of project, address of project, & phone number of contact. If all of this criteria can't be met, then contractor will be disqualified:

1. **Virginia** Class A contractors license.
2. General liability insurance to \$1,000,000.
3. Five verifiable references in the project state to attest to the contractor's ability to install.

- B. Conform to applicable codes for piping and component requirements.

- C. Materials, equipment, and methods of installation shall comply with the following codes and standards:

1. **Authority Having Jurisdiction (AHJ) Standards and Requirements**
2. National Fire Protection Association, (NFPA): National Electrical Code.
3. American Society of Testing and Materials, (ASTM).
4. National Sanitation Foundation, (NSF).

- D. All system components and equipment must be consistent with respect to brand name and/or manufacturer.

- E. Obtain Architect's acceptance of installed and tested irrigation system prior to installing backfill materials. All mainline PVC must meet this criterion.

1.3 SUBMITTALS

- A. Submit irrigation design plans showing the landscape plans provided by the Civil Engineer.

- B. Submit manufacturer's product data and installation instructions for each of the system components to include electric valves, heads, controller, wire, pipe, fittings, valve boxes, manual valves, wire splices, and all other equipment installed on project. Submittals must be organized and semi-permanently bound in booklet form with a table of contents to each section. This book must be periodically updated as material changes on the project.

- C. Submit the following material samples:

1. Piping and fittings
2. Clamps
3. Wire connectors and sealer

- D. Submit the following equipment samples:

1. Sprinkler heads, 1 of each type, complete with housing
2. Valves and valve access boxes
3. Controller

- E. Approved equipment samples will be returned to contractor and may be used in the work.

- F. Upon irrigation system acceptance, submit written operating and maintenance instructions. Provide format and contents as directed by the Architect/Engineer.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Deliver irrigation system components in manufacturer's original undamaged and unopened containers with labels intact and legible.
- B. Deliver plastic piping in bundles, packaged to provide adequate protection of pipe ends, both threaded or plain.
- C. Store and handle materials to prevent damage and deterioration.
- D. Provide secure, locked storage for valves, sprinkler heads, and similar components that can not be immediately replaced, to prevent installation delays.

1.5 PROJECT CONDITIONS

- A. Known underground and surface utility lines are indicated on the utilities drawings.
- B. Protect existing trees, plants, lawns, and other features designated to remain as natural area.
- C. Promptly repair damage to adjacent facilities caused by irrigation system work operations. Cost of repairs at Contractor's expense.
- D. Promptly notify the Architect of unexpected sub-surface conditions.
- E. Irrigation system layout is diagrammatic. Exact locations of piping, sprinkler heads, valves, and other components shall be established by Contractor and the Architect in the field at time of installation.
 - 1. Stake mainline and all heads and valves and receive approval from architect prior to excavation. Provide 100% coverage on entire irrigated site.
- F. Space sprinkler components as indicated.
- G. Minor adjustments in system layout will be permitted to clear existing fixed obstructions. Final system layout shall be acceptable to the Architect.
- H. Cutting and patching:
 - 1. Cut through concrete and masonry with core drills. Jack hammers not permitted.
 - 2. Materials and finishes for patching shall match existing cut surface materials and finish. Exercise special care to provide patching at openings and exterior walls.
 - 3. Methods and materials used for cutting and patching shall be acceptable to the Architect.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. General:
 - 1. Provide only new materials, without flaws or defects and of the highest quality of their specified class and kind to be provided by an authorized Product Distributor. Product distributors must submit their factory authorization to sell and service all components that they provide.
 - 2. Comply with pipe sizes indicated. No substitution of smaller pipes will be permitted. Larger sizes may be used subject to acceptance of the Architect. Remove damaged and defective pipe.
 - 3. Provide pipe continuously and permanently marked with manufacturer's name or trademark, size schedule and type of pipe, working pressure at 73 degrees F. and National Sanitation Foundation (NSF) approval.
 - 4. All system components and equipment must be consistent with respect to brand name and/or manufacturer.

- B. Plastic pipe, fittings, and connections:
 - 1. Polyvinyl chloride pipe: ASTM D2241, rigid, un-plasticized PVC, extruded from virgin parent material. Provide pipe homogeneous throughout and free from visible cracks, holes, foreign materials, blisters, wrinkles, and dents.
 - 2. 2.5" diameter and under: SDR21, class 200, bell end PVC.
 - 3. 3" diameter and over: SDR 21, class 200, ring-tite gasketed bell end PVC.
- C. PVC pipe fittings: ASTM D2241 schedule 40 PVC molded fittings suitable for solvent weld connections. All fittings at all lateral or change in directions shall be ductile iron. Saddle and cross fittings not permitted.
- D. Insert fittings: ASTM D2466 insert type fittings.
- E. Sprinkler heads, valves, and associated equipment: Provide manufacturer's data on each type of equipment to obtain Architect's approval.
- F. Quick Couplet Outlets: solid bronze, 1.5" size to include swivel and key model #476-00 as manufactured by one of the following:
 - 1. Hunter
 - 2. Rainbird
 - 3. The Toro Company
- G. Sprinklers: 1" threaded inlet, stainless steel riser with the ability to mount ½" below grade. Sprinklers shall discharge 14.0 GPM @60 PSI base pressure. Sprinklers shall be model 64X-02-42 as manufactured by one of the following:
 - 1. Hunter
 - 2. Rainbird
 - 3. The Toro Company
- H. Automatic remote control valves (RCV) shall be sized as indicated on drawings. RCV's shall be installed per manufacturer's recommendations and shall be rated for pressure and flow as required. RCV's shall be low voltage. Valves shall be constructed of plastic. Valves shall be the 252-26-04, 1" and 252-26-08 2" model as manufactured by one of the following:
 - 1. Hunter
 - 2. Rainbird
 - 3. The Toro Company
- I. Wire
 - 1. Control wire shall be 14/1 gauge UL/UF, direct burial red in color. Common wire shall be 14/1 gauge UL/UF, direct burial white in color.
 - 2. All wire splices below grade shall be made water proof with correctly sized wire splice connectors, made for direct burial and rated for minimum of 30 volts.
 - 3. Wire shall be color coded. White is to be used for common, red is to be used for control, blue for extra, green for stub-out and yellow for sensor. Any deviation will be replaced.
- J. Field controller shall be installed in metal lockable cabinet. Controller shall have 4 independent programs, non-volatile memory, battery back-up, 1-7 day water delay, seasonal adjust. Controller shall conform to design and be Hunter, Rainbird or Toro as consistent with brand of remaining system.
- K. Sleeves shall be schedule 40 PVC as indicated on the plan. Any additional sleeves shall be two

sizes larger than the pipe routed through them.

2.2 ACCESSORIES

- A. Drainage fill: ½" washed pea gravel.
- B. Fill: Clean soil free of stones larger than 1" diameter foreign matter, organic material, and debris.
 - 1. Provide imported fill material as required to complete the work. Obtain rights and pay all costs for imported materials.
 - 2. Suitable excavated materials removed to accommodate the irrigation system work may be used as fill material subject to the Architect's review and acceptance.
- C. Clamps: Stainless steel, worm gear hose clamps with stainless steel screws or ear type clamps.
- D. Low voltage wire connectors: Socket seal type wire connectors and waterproof gel sealer with wire connectors 3M DBY's and DBR's or approved equal only.
- E. Valve access boxes shall be manufactured by Ametek, Carson Industries, or NDS. All valves shall be installed in 10" circular valve boxes.
- F. Backflow preventer shall be of the reduced pressure principal type and be installed as per Town of Culpeper and State codes. Backflow preventer shall be 2" model 825Y as manufactured by Febco or equivalent by Zurn Plumbing Products Group or Wilkins.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Examine final grades and installation conditions. Verify that field measurements are as shown on drawings. Do not start irrigation system work until unsatisfactory conditions are corrected.
- B. Verify the location of utilities, plant materials, shrub bed lines, and tree lines. Do not route irrigation lines within 5 feet of any tree on project. Contractor is responsible for coordinating the location of all trees with landscape contractor. Do not install irrigation heads until all trees have been installed or staked.
- C. Verify that required utilities are available, in proper location, and ready for use.
- D. Beginning of installation means installer accepts existing conditions.
- E. Provide at least one experienced on-site foreman or supervisor subject to approval of the architect, who must be on site at all times when a crew is working.

3.2 PREPARATION

- A. Notify utility companies prior to excavation in accordance with local codes and ordinances. Contact Miss Utility prior to beginning of installation. Follow all current Miss Utility rules for contact and digging.
- B. Contractor shall be responsible for damage to existing utilities and structures due to negligence and/or misuse by his crews or equipment.
- C. Layout and stake the location of each pipe run and all sprinkler heads and sprinkler valves. Obtain Architect's acceptance of layout prior to excavating.
- D. Strip sod for pipe trenches with a mechanical sod stripper uniformly 6" to 12" wide with clean cut edges.
- E. Place sleeves as necessary for installation of piping and control wire. All piping under walks and walls shall be within a Schedule 40 PVC sleeve 2 sizes larger than the pressure pipe.

3.3 INSTALLATION

- A. Excavating and back filling:
 - 1. All excavation shall be considered unclassified excavation and include all material encountered.
 - 2. Excavate trenches of sufficient depth and width to permit proper handling and installation of pipe and fittings.
 - 3. Excavate to depths required to provide 3" depth of debris free earth fill or sand bedding for piping when rock or other unsuitable bearing material is encountered.
 - 4. Fill to match adjacent grade elevations with approved earth fill material. Place and compact fill in layers not greater than 8" depth.
 - 5. Provide approved earth fill or sand to a point 4" above the top of pipe.
 - 6. Fill to within 6" of final grade with approved excavated or borrow fill materials free of lumps or rocks larger than 1" in any dimension.
 - 7. Provide clean topsoil fill free of rocks and debris for top 6" of fill.
- B. Install irrigation lateral lines with a minimum cover of 18" based on finished grades. Install irrigation mainline with a minimum of 24" based on finished grade.
- C. Excavate trenches and install piping and fill during the same working day. Do not leave open trenches or partially filled trenches open overnight.
- D. Replace stripped sod in sufficient time to allow for satisfactory sod recovery and growth. Water stripped and reinstalled sod until irrigation system is place in operation.
- E. Install valves boxes per manufacturer's specifications.
- F. Plastic pipe:
 - 1. Install plastic pipe in accordance with manufacturer's installation instructions. Provide for thermal expansion and contraction. Do not install plastic pipe if temperature is below 32 degrees.
 - 2. Saw cut plastic pipe. Use an electric miter saw, to ensure a square cut. Remove burrs and shavings at cut ends prior to installation. Use a hand beveler on all gasket pipe cuts.
 - 3. Make plastic to plastic joints with solvent weld joints or slip seal joints. Use only solvent recommended by the pipe manufacturer. Install plastic pipe fittings in accordance with pipe manufacturer's instructions. Make arrangements with pipe manufacturer for all necessary field assistance.
 - 4. Make plastic to metal joints with plastic male adapters.
 - 5. Make solvent weld joints in accordance with manufacturer's recommendations.
 - 6. Allow joints to set at least 24 hours before pressure is applied to the system.
 - 7. Maintain pipe interiors free of dirt and debris. Close open ends of pipe by acceptable methods (duct tape ends) when pipe installation is not in progress.
- G. Sprinklers, fittings, valves, and accessories:
 - 1. Install fittings, valves, sprinkler heads, risers, and accessories in accordance with manufacturer's instructions, except as otherwise indicated.
 - 2. Set sprinkler heads perpendicular to finished grades, except as otherwise indicated.
 - 3. Locate sprinkler heads to assure proper coverage of indicated areas. Do not exceed sprinkler head spacing distances indicated.
 - 4. Install quick-coupling valves with an adjustable double swing joint riser assembled by the use

of at least three (3) standard 90 degree elbows. Fabricate double swing joint risers of schedule 80 PVC nipples and schedule 40 PVC elbows. Pre-fabricated swing joints are acceptable.

5. Install controller in accordance with manufacturer's written instructions. All controller footings will be 12" deep concrete, minimum.
6. Install valve access boxes on a suitable base of bricks to provide a level foundation at proper grade and to provide drainage of the access box. Provide 1½" cubic feet of pea gravel below box for drainage.
7. Install automatic valves plumb with valve access box with all valve handles, bolts, connections and electrical splices accessible through the valves box opening (not less than 6" below valve box cover).
8. Seal all threaded connections of control valves with Teflon tape. Plastic joint type compound is not acceptable.

H. Control wiring:

1. Install electric control cable in the piping trenches wherever possible. Place wire in trench adjacent to pipe. Install wire with slack to allow for thermal expansion and contraction. Provide expansion joints in wire at minimum 200-foot intervals. Where necessary to run wire in a separate trench, provide a minimum cover of 15". Do not plow or pull wire with a vibratory plow.
 2. Provide sufficient slack at site connections at remote control valves in control boxes and at all wire splices to allow raising the valve bonnet or splice to the surface without disconnecting the wires when repair is required.
 3. Connect each remote control valve to one station of a controller except as otherwise indicated.
 4. Connect remote control valves for each controller to a common ground wire system, independent of all other controllers.
 5. Make wire connections to remote control electric valves and splices of wire in the field, using water proof wire connectors and sealing cement in accordance with manufacturer's recommendations, dry splices will not be accepted. Make wire splices in accessible splice or valve boxes 10" in diameter or greater.
 6. Provide tight joints to prevent leakage of water and corrosion build-up on the joint.
 7. When control wiring is in common trench with main line, wiring shall be below main line with 4" of fill dirt between pipe and wire.
 8. Install above-ground wire in conduit and/or in accordance with electrical codes.
 9. All materials and methods of installation shall conform to local electrical codes.
- I. Quick-Coupling valves: when installed in lawn area, provide a 10" valve box for location. Stabilize the valve with a stake and 1 ½" cubic feet of pea gravel (1/2" to ¾").
- J. Install field satellite per manufacturer's recommendation. Include all grounding equipment necessary.

3.4 FLUSHING, TESTING, AND ADJUSTMENT

- A. After sprinkler piping and risers are installed and before sprinkler heads are installed, open control valves and flush out the system with full head of water.
- B. Perform system testing upon completion of each section. Make necessary repairs and retest repaired sections as required.

- C. Adjust sprinklers after installation for proper and adequate distribution of the water over the coverage pattern. Adjust for the proper arc of coverage.
- D. Adjust all electric remote control valve flow control stems for system balance. Set pressure regulating valves prior to acceptance.
- E. Test and demonstrate the controller by operating all programs, day, hour, and station selection features as required to automatically start and shut down irrigation cycles to accommodate plant requirements and weather conditions.

3.5 SERVICE

- A. When requested, return to the site during the subsequent fall season and winterize the system. Drain all water from the system and blow out the system with compressed air.
- B. When requested, return to the site during the subsequent spring season and demonstrate to the Owner's Representative the proper procedures for the system start-up, operation, and maintenance.
- C. After three (3) months of continuous operation, after acceptance of job, return to the site and re-adjust all irrigation heads, by re-packing heads that have become dislodged due to resettling of ground, torque of spray, or vibration. Re-adjust radius, arc, and trajectory of spray of all heads.

3.6 DISPOSAL OF WASTE MATERIAL

- A. Stockpile, haul from site, and legally dispose of waste materials, including unsuitable excavated materials, rock, trash, and debris.
- B. Maintain disposal route clear, clean, and free of debris.

3.7 ACCEPTANCE

- A. Test and demonstrate to the Architect the satisfactory operation of the system free of leaks.
- B. Instruct the Owner's designated personnel in the operation of the system including adjustment of sprinklers, controller(s), valves, and pump controls.
- C. Upon acceptance, the Owner will assume operation of the system.
- D. Provide owner with three (3) quick coupler keys with swivels for use on project.
- E. Provide owner with all manuals for products used on project.
- F. Provide owner with one (1) key for every seven (7) manual valves used on project. More than one type of valve may require more than one type of key.
- G. Provide irrigation system record drawings:
 - 1. Legibly mark drawings to record actual construction. Mark all heads, valves, and wire splice boxes.
 - 2. Indicate a measured distance triangulated to locate each installed valve-automatic, manual, quick-coupling type.
 - 3. Identify field changes of dimension and detail and changes made by Change Order, if any.
 - 4. Submit three (3) copies of As-Built drawings to the owner on base drawings provided by the architect/Engineer.
 - 5. As-Builts shall also indicate control valve wiring routing paths, wire splice locations, and controller location.

3.8 GUARANTEE

- A. Guarantee all workmanship covered by this specifications to be free of defects for a period of one (1) year from the date of final acceptance of the project. Replace any part or parts found to be

defective within the period of guarantee at no cost to the owner, except repairs or replacement necessitated by damage by others.

- B. Guarantee back filling of all excavation. If, at any time during the first year of the guarantee period, trenches or heads should settle, repair any settling at no cost to the owner.

3.9 CLEANING

- A. Perform cleaning during installation of the work and upon completion of the work. Remove from site all excess materials, soil, debris, and equipment. Repair damage resulting from irrigation system installation.

END OF SECTION 328000

SECTION 329200 - LAWNS AND GRASSES *(SECTION TO BE MODIFIED AS REQUIRED FOR REGIONAL CONDITIONS)***PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes the following:
1. Fine grading and preparing lawn areas (including courtyards)
 2. Topsoil Placement
 3. Soil amendments
 4. Fertilizers
 5. Seeding
 6. Sodding
 7. Lawn Restoration

1.3 DEFINITIONS

- A. Finish Grade: Elevation of finished surface of planting soil.
- B. Lawns: All areas disturbed by construction and not otherwise covered by paving, buildings or other structures.

1.4 SUBMITTALS

- A. Certification by product manufacturer that the following products supplied comply with requirements:
1. Grass Seed
 - a) Certification of grass seed from seed vendor for each grass-seed mixture stating the botanical and common name and percentage by weight of each species and variety, and percentage of purity, germination, and weed seed. Include the year of production and date of packaging.
 - b) Blue Tag Certification tag for each bag of seed.
 2. Sod
 - a) Gold Tag Certification.
- B. Installers qualifications
1. Provide a list, with references, of the past three projects of a similar magnitude.
- C. Topsoil Amendment Plan.
1. Provide copy of topsoil testing report.
 2. List of amendments proposed for topsoil, including application rates.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an experienced installer, who has successfully completed lawn establishment projects similar in size and complexity to this project. The installer's primary business (defined as a minimum of 60% of total billings) shall be establishment of lawns.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Seed: Deliver seed in original sealed, labeled, and undamaged containers.
- B. Sod: Harvest, deliver, store and handle sod according to the requirements of the American Sod Producers Association (ASPA) "Specifications for Turfgrass Sod Materials and Transplanting/Installing".

1.7 COORDINATION AND SCHEDULING

- A. Planting Season: Sow lawn seed during normal planting seasons for type of lawn work required.
 - 1. Spring Planting Season: March 15 through May 15
 - 2. Fall Planting Season: September 15 through November 15
- B. Weather Limitations: Proceed with planting only when existing and forecast weather conditions are suitable for work.
- C. Lawn Seeding Schedule
 - 1. Refer to the drawings for early seeding requirements for specified lawn areas.
 - 2. If job completion schedule does not allow seeding within a normal planting season, provide interim temporary seeding necessary to stabilize site. Complete permanent seeding during the next planting season.

1.8 LIMITS OF SEEDING

- A. Spread topsoil and seed only lawn areas on plan indicated for seed.

1.9 LIMITS OF LAWN RENOVATION

- A. All existing lawn areas disturbed by construction activities.

1.10 LIMITS OF SODDING

- A. Provide sod in the following areas, and as indicated on the plans:
 - 1. Lawn Areas.

1.11 PAYMENT PROCEDURES FOR LAWNS AND GRASSES

- A. Establish a line item in the Schedule of Values for Lawn Maintenance. This line item shall represent a minimum of thirty percent (30%) of the total value of the seeding for the project.
- B. Lawn maintenance will be paid on a monthly basis, following the satisfactory maintenance of the lawns.

PART 2 – PRODUCTS

2.1 TOPSOIL

- A. Topsoil: ASTM D 5268, pH range of 5.5 to 7, a minimum of 4 percent organic material content; free of stones 1" or larger in any dimension and other extraneous materials harmful to plant growth.
 - 1. Topsoil Source: Reuse surface soil stockpiled on-site. Verify suitability of stockpiled surface soil to produce topsoil. Clean surface soil of roots, plants, sod, stones, clay lumps, and other extraneous materials harmful to plant growth.
 - a) Supplement with imported or manufactured topsoil from off-site sources when quantities are insufficient. Obtain topsoil displaced from naturally well-drained construction or mining sites where topsoil occurs at least 4 inches (100 mm) deep; do not obtain from agricultural land, bogs or marshes.
- B. Have topsoil tested by a certified soil testing laboratory to determine the type and quantity of soil amendments necessary. Add amendments to topsoil as necessary to meet these requirements.

2.2 INORGANIC SOIL AMENDMENTS

- A. If the topsoil analysis indicates the need for inorganic soil amendments, the following standards apply:
- B. Lime: ASTM C 602, agricultural limestone containing a minimum 80 percent calcium carbonate equivalent and as follows:
 - 1. Class: Class O, with a minimum 95 percent passing through No. 8 (2.36-mm) sieve and a minimum 55 percent passing through No. 60 (0.25-mm) sieve.
 - 2. Provide lime in form of dolomitic limestone.
- C. Sulfur: Granular, biodegradable, containing a minimum of 90 percent sulfur, with a minimum 99 percent passing through No. 6 (3.35-mm) sieve and a maximum 10 percent passing through No. 40 (0.425-mm) sieve.
- D. Iron Sulfate: Granulated ferrous sulfate containing a minimum of 20 percent iron and 10 percent sulfur.
- E. Aluminum Sulfate: Commercial grade, unadulterated.
- F. Perlite: Horticultural perlite, soil amendment grade.
- G. Agricultural Gypsum: Finely ground, containing a minimum of 90 percent calcium sulfate.
- H. Sand: Clean, washed, natural or manufactured, free of toxic materials.
- I. Diatomaceous Earth: Calcined, diatomaceous earth, 90 percent silica, with approximately 140 percent water absorption capacity by weight.
- J. Zeolites: Mineral clinoptilolite with at least 60 percent water absorption by weight.

2.3 ORGANIC SOIL AMENDMENTS

- A. If the topsoil analysis indicates the need for organic soil amendments, the following standards apply:
- B. Compost: Well-composted, stable, and weed-free organic matter, pH range of 5.5 to 8; moisture content 35 to 55 percent by weight; 100 percent passing through 3/4-inch (19-mm) sieve; soluble salt content of 5 to 10 decisiemens/m; not exceeding 0.5 percent inert contaminants and free of substances toxic to plantings; and as follows:
 - 1. Organic Matter Content: 50 percent of dry weight.
 - 2. Feedstock: Agricultural, food, or industrial residuals; biosolids; yard trimmings; or source-separated or compostable mixed solid waste.
 - 3. Peat: Finely divided or granular texture, with a pH range of 6 to 7.5, containing partially decomposed moss peat, native peat, or reed-sedge peat and having a water-absorbing capacity of 1100 to 2000 percent.
 - 4. Wood Derivatives: Decomposed, nitrogen-treated sawdust, ground bark, or wood waste; of uniform texture, free of chips, stones, sticks, soil, or toxic materials.
 - 5. Manure: Well-rotted, unleached, stable or cattle manure containing not more than 25 percent by volume of straw, sawdust, or other bedding materials; free of toxic substances, stones, sticks, soil, weed seed, and material harmful to plant growth.

2.4 HERBICIDES

- A. Pre-Emergent Herbicide: EPA registered and approved, of type recommended by manufacturer for application
- B. Selective Herbicides: EPA registered and approved, of type recommended by manufacturer for application.

2.5 FERTILIZER

- A. Bonemeal: Commercial, raw or steamed, finely ground; a minimum of 4 percent nitrogen and 20 percent phosphoric acid.
- B. Superphosphate: Commercial, phosphate mixture, soluble; a minimum of 20 percent available phosphoric acid.
- C. Commercial Fertilizer: Commercial-grade complete fertilizer of neutral character, consisting of fast- and slow-release nitrogen, 50 percent derived from natural organic sources of urea formaldehyde, phosphorous, and potassium in the following composition:
 - 1. Composition: Nitrogen, phosphorous, and potassium in amounts recommended in topsoil analysis reports from a qualified soil-testing agency.
 - 2. Minimum Composition: No less than 1 lb/1000 sq. ft. (0.45 kg/92.9 sq. m) of actual nitrogen, 4 percent phosphorous, and 2 percent potassium, by weight.

2.6 SEED

- A. Grass Seed: All grass seed must be fresh, clean, and dry.
- B. Seed Species

Proportion by Weight	Grass Species	Min. % Germination	Min. % Pure Seed	Max. % Weed Seed
10%	Kentucky bluegrass (<u>Poa pratensis</u>).	80	85	0.50
90%	Tall Fescue (<u>Festuca arundinacea</u>).	85	98	0.50

- A. Varieties shall be selected from the 2003-2004 list of recommended turfgrass varieties, published by Virginia Tech.
- B. All seed shall be Blue Tag certified by the Oregon State Seed Laboratory. Tags must be attached to each bag delivered on site.

2.7 TURFGRASS SOD

- A. Turfgrass Sod: Certified sod, complying with TPI's "Specifications for Turfgrass Sod Materials" in its "Guideline Specifications to Turfgrass Sodding." Comply with ASPA specifications for machine cut thickness, size, strength, moisture content, and mowed height and free of weeds and undesirable native grasses. Provide viable sod of uniform density, color, and texture, strongly rooted, and capable of vigorous growth and development when planted. Provide the following turfgrass species:

1. 90% Tall Fescue (Festuca arundinacea). 10% Kentucky bluegrass (Poa pratensis) mix.

- B. All sod shall be Gold Tag certified by the Virginia Crop Improvement Association.

2.8 MULCHES

- A. Straw Mulch: Provide air-dry, clean, mildew- and seed-free, salt hay or threshed straw of wheat, rye, oats, or barley.
- B. Peat Mulch: Finely divided or granular texture, with a pH range of 6 to 7.5, containing partially decomposed moss peat, native peat, or reed-sedge peat and having a water-absorbing capacity of 1100 to 2000 percent.
- C. Compost Mulch: Well-composted, stable, and weed-free organic matter, pH range of 5.5 to 8; moisture content 35 to 55 percent by weight; 100 percent passing through 1-inch (25-mm) sieve;

soluble salt content of 5 to 10 decisiemens/m; not exceeding 0.5 percent inert contaminants and free of substances toxic to plantings; and as follows:

1. Organic Matter Content: 50 percent of dry weight.

- D. Fiber Mulch: Biodegradable, dyed-wood, cellulose-fiber mulch; nontoxic; free of plant-growth or germination inhibitors; with maximum moisture content of 15 percent and a pH range of 4.5 to 6.5.

2.9 EROSION-CONTROL MATERIALS

- A. Erosion-Control Fiber Mesh: Biodegradable twisted jute or spun-coir mesh, a minimum of 0.92 lb/sq. yd. (0.5 kg/sq. m), with 50 to 65 percent open area. Include manufacturer's recommended steel wire staples, 6 inches (150 mm) long.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas to receive lawns and grass for compliance with requirements and for conditions affecting performance of the Work. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities, trees, shrubs, and plantings from damage caused by planting operations.
- B. Provide erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.
- C. Protect adjacent and adjoining areas from hydroseed overspraying.

3.3 TOPSOIL PLACEMENT FOR LAWNS

- A. Limit subgrade preparation to areas that will be planted in the immediate future.
- B. Loosen subgrade to a minimum depth of 4 inches. Remove stones, sticks and roots larger than 2 inches in any dimension from subgrade. Completely remove trash and other extraneous debris from subgrade.
- C. Have topsoil tested by a certified soil testing laboratory to determine the type and quantity of soil amendments necessary.
- D. Sift topsoil to remove stones and other objects larger than 1" in any dimension. Maximum object size for topsoil shall be achieved by sifting not by hand removal or raking following placement of topsoil.
- E. Mix soil amendments and fertilizers with topsoil at rates required by soil testing. Delay mixing fertilizer if planting does not follow placing of planting soil within 4 days. Either mix soil before spreading or apply soil amendments on surface of spread topsoil and mix thoroughly into top 4 inches (100 mm) of topsoil before planting.
- F. Mix lime with dry soil prior to mixing fertilizer.
- G. Spread topsoil to a minimum depth of six inches (6").

3.4 SEEDING LAWNS

- A. Sow seed with a spreader or a seeding machine. Do not broadcast or drop seed when wind velocity exceeds 5 mph (8 km/h). Evenly distribute seed by sowing equal quantities in 2 directions at right angles to each other.
- B. Do not use wet seed or seed that is moldy or otherwise damaged in transit or storage.
- C. Sow seed at the following rates:

1. Seeding Rate: 200 lbs./acre.

D. Rake seed lightly into top 1/4 inch of topsoil, roll lightly, and water with fine spray.

E. Hydroseed all slopes 3:1 or steeper.

F. Protect seeded areas 3:1 slope/grade or steeper against erosion by providing erosion-control blankets installed and stapled according to manufacturer's recommendations.

G. Protect seeded areas less than 3:1 slope/grade against erosion by spreading straw mulch after completion of seeding operations. Spread uniformly at a minimum rate of 2 tons per acre (45 kg per 100 sq. m) to form a continuous blanket 1-1/2 inches (38 mm) loose depth over seeded areas. Spread by hand, blower, or other suitable equipment.

1. Anchor straw mulch by crimping into topsoil by suitable mechanical equipment.

3.5 SODDING NEW LAWNS

A. Lay sod within 24 hours of stripping. Do not lay sod if dormant or if ground is frozen.

B. Lay sod to form a solid mass with tightly fitted joints. Butt ends and sides of sod; do not stretch or overlap. Stagger sod strips or pads to offset joints in adjacent courses. Avoid damage to subgrade or sod during installation. Tamp and roll lightly to ensure contact with subgrade, eliminate air pockets, and form a smooth surface. Work sifted soil or fine sand into minor cracks between pieces of sod; remove excess to avoid smothering sod and adjacent grass.

C. Lay sod across angle of slopes exceeding 3:1.

D. Anchor sod on slopes exceeding 6:1 with wood pegs spaced as recommended by sod manufacturer but not less than two anchors per sod strip to prevent slippage.

E. Saturate sod with fine water spray within 2 hours of planting. During first week, water daily or more frequently as necessary to maintain moist soil to a minimum depth of 1-1/2 inches below sod.

3.6 LAWN RENOVATION

A. Renovate existing lawn. Limits of existing lawn to be renovated are indicated on the drawings.

B. Renovate existing lawn damaged by Contractor's operations, such as storage of materials or equipment and movement of vehicles.

1. Reestablish lawn where settlement or washouts occur or where minor regrading is required.

C. Remove sod and vegetation from diseased or unsatisfactory lawn areas; do not bury in soil.

D. Remove topsoil containing foreign materials resulting from Contractor's operations, including oil drippings, fuel spills, stone, gravel, and other construction materials, and replace with new topsoil.

E. Mow, de-thatch, core aerate, and rake existing lawn.

F. Remove weeds before seeding. Where weeds are extensive, apply selective herbicides as required. Do not use pre-emergence herbicides.

G. Remove waste and foreign materials, including weeds, soil cores, grass, vegetation, and turf, and legally dispose of them off Owner's property.

H. Till stripped, bare, and compacted areas thoroughly to a soil depth of 6 inches.

I. Apply soil amendments and initial fertilizers required for establishing new lawns and mix thoroughly into top 4 inches of existing soil. Provide new planting soil to fill low spots and meet finish grades.

J. Apply seed and protect with straw mulch as required for new lawns.

K. Water newly planted areas and keep moist until new lawn is established.

3.7 MAINTENANCE OF NEW LAWNS

- A. Begin maintenance of lawns immediately after each area is planted and continue until acceptable lawn is established. Maintain seeded lawns until Substantial Completion. Maintain all grassed areas as necessary to ensure a satisfactory lawn is achieved at Substantial Completion.
- B. Maintain and establish lawns by watering, fertilizing, weeding, mowing, trimming, replanting, and other operations. Roll, regrade, and replant bare or eroded areas and remulch to produce a uniformly smooth lawn.
 - 1. Replant bare areas with same materials as for lawns.
 - 2. Replace disturbed mulch.
- C. Watering: Provide and maintain temporary hoses, and lawn-watering equipment to convey water from a water source to keep lawns uniformly moist to a depth of 4 inches.
 - 1. Provide a source of water for irrigation. Utilize temporary irrigation meters, a well or water trucks as necessary for the water source.
 - 2. Water seeded areas as necessary to promote vigorous growth of grass but at the minimum rate of 1 inch per week.
 - 3. Water sodded areas per the requirements of the grower. Maintain moist soil to a depth of at least four inches.
- D. At a minimum, the following fertilizer applications are required:
 - 1. By **November 30, 2001**, apply 15-5-10 commercial fertilizer at the rate of 200 lbs. per acre over all seeded and sodded areas.
 - 2. By **March 30, 2002**, apply 15-5-10 commercial fertilizer at the rate of 350 lbs. per acre over all seeded and sodded areas.
 - 3. By **November 30, 2002**, apply 15-5-10 commercial fertilizer at the rate of 200 lbs. per acre and apply lime at 2000 lbs. per acre over all seeded and sodded areas.
 - 4. Provide written acknowledgement that this requirement has been met prior to requesting Substantial Completion.
- E. Mow lawns as soon as there is enough top growth to cut with mower set at indicated height. Repeat mowing as required to maintain indicated height without cutting more than 40 percent of the grass height (minimum of 3 mowings). Remove no more than 40 percent of grass-leaf growth in initial or subsequent mowings. Do not delay mowing until grass blades bend over and become matted. Do not mow when grass is wet. Schedule initial and subsequent mowings to maintain following grass height:
 - 1. Mow grass to a finished height of **2 to 3** inches high.
- F. Apply pre-emergent herbicide to lawns areas. Apply 60 – 90 days after planting.**

3.8 SATISFACTORY LAWN

- A. Seeded lawns shall be considered satisfactory/acceptable provided requirements, including maintenance, have been met and a healthy, uniform, close stand of grass is established, free of weeds, bare spots exceeding 5 by 5 inches (125 by 125 mm), and surface irregularities.
- B. Sodded lawns shall be considered satisfactory/acceptable provided requirements, including maintenance, have been met and a healthy, well-rooted, even-colored, viable lawn is established, free of weeds, open joints, bare areas and surface irregularities.
- C. Replant lawns that do not meet requirements and continue maintenance until lawns are satisfactory/acceptable.

- D. Substantial Completion of the building and the remainder of the project may be achieved (pending prior Architect and Owner approval) before achieving a satisfactory/acceptable lawn. Continue to replant and maintain unsatisfactory/unacceptable lawn areas until acceptance is obtained. Warranties for lawns shall begin at the time of acceptance of the lawn.

3.9 CLEANUP AND PROTECTION

- A. Promptly remove soil and debris created by lawn work from sidewalks and paved areas. Clean wheels of vehicles before leaving site to avoid tracking soil onto surface of roads, walks, or other paved areas.
- B. Erect barricades and warning signs as required to protect newly planted areas from traffic. Maintain barricades throughout maintenance period until lawn is established.

END OF SECTION 329200

SECTION 329223 - SOD**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes the following:
1. Fine grading and preparing lawn areas (including courtyards)
 2. Topsoil Placement
 3. Fertilizers
 4. Sodding

1.3 DEFINITIONS

- A. Finish Grade: Elevation of finished surface of planting soil.
- B. Lawns: Areas indicated on drawings as such.

1.4 SUBMITTALS

- A. Certification by product manufacturer that the following products supplied comply with requirements:
1. Sod
- B. Installers qualifications
1. Provide a list, with references, of the past three projects of a similar magnitude.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an experienced installer, who has successfully completed lawn establishment projects similar in size and complexity to this project.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Sod: Harvest, deliver, store and handle sod according to the requirements of the American Sod Producers Association (ASPA) "Specifications for Turfgrass Sod Materials and Transplanting/Installing".

1.7 COORDINATION AND SCHEDULING

- A. Weather Limitations: Proceed with planting only when ground is not muddy or frozen.

1.8 LIMITS OF SODDING

- A. Provide sod in areas as indicated on the plans.

PART 2 – PRODUCTS**2.1 TOPSOIL**

- A. Topsoil: ASTM D 5268, pH range of 5.5 to 7, a minimum of 4 percent organic material content; free of stones 2" or larger in any dimension and other extraneous materials harmful to plant growth.
1. Topsoil Source: Reuse surface soil stockpiled on-site. Verify suitability of stockpiled surface soil to produce topsoil. Clean surface soil of roots, plants, sod, stones, clay lumps, and other extraneous materials harmful to plant growth.

- a) Supplement with imported or manufactured topsoil from off-site sources when quantities are insufficient. Obtain topsoil displaced from naturally well-drained construction or mining sites where topsoil occurs at least 4 inches (100 mm) deep; do not obtain from agricultural land, bogs or marshes.
- B. Have topsoil tested by a certified soil testing laboratory to determine the type and quantity of soil amendments necessary. Add amendments to topsoil as necessary to meet these requirements.

2.2 TURFGRASS SOD

- A. Turfgrass Sod: Certified sod, complying with TPI's "Specifications for Turfgrass Sod Materials" in its "Guideline Specifications to Turfgrass Sodding." Comply with ASPA specifications for machine cut thickness, size, strength, moisture content, and mowed height and free of weeds and undesirable native grasses. Provide viable sod of uniform density, color, and texture, strongly rooted, and capable of vigorous growth and development when planted. Provide the following turfgrass species:
 - 1. Centipede

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas to receive lawns and grass for compliance with requirements and for conditions affecting performance of the Work. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities, trees, shrubs, and plantings from damage caused by planting operations.
- B. Provide erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.

3.3 TOPSOIL PLACEMENT FOR LAWNS

- A. Limit subgrade preparation to areas that will be planted in the immediate future.
- B. Loosen subgrade. Remove stones, sticks and roots larger than 2 inches in any dimension from subgrade. Completely remove trash and other extraneous debris from subgrade.
- C. Rake topsoil to remove stones and other objects larger than 2" in any dimension.
- D. Lightly rake lime and fertilizers into top layer of topsoil during final grading. Delay mixing fertilizer if planting does not follow placing of planting soil within 4 days.

3.4 SODDING NEW LAWNS

- A. Lay sod within 24 hours of stripping. Do not lay sod if dormant or if ground is frozen.
- B. Lay sod to form a solid mass with tightly fitted joints. Butt ends and sides of sod; do not stretch or overlap. Stagger sod strips or pads to offset joints in adjacent courses. Avoid damage to subgrade or sod during installation. Tamp and roll lightly to ensure contact with subgrade, eliminate air pockets, and form a smooth surface. Work sifted soil or fine sand into minor cracks between pieces of sod; remove excess to avoid smothering sod and adjacent grass.
- C. Lay sod across angle of slopes exceeding 3:1.
- D. Anchor sod on slopes if necessary with sod staples with no less than two staples per sod strip to prevent slippage.

- E. Saturate sod with fine water spray within 2 hours of planting. During first week, water as necessary to maintain adequately moist soil.

3.5 MAINTENANCE OF NEW LAWNS

- A. Begin maintenance of lawns immediately after each area is planted and continue until sod is well rooted. Maintain and establish lawns by watering, fertilizing, weeding, mowing, trimming, replanting, and other operations. Roll, regrade, and replant bare or eroded areas and remulch to produce a uniformly smooth lawn.
 - 1. Replant bare areas with same materials as for lawns.
 - 2. Replace disturbed mulch.
- B. Watering: Utilize a water truck outside of business hours to keep lawns uniformly moist.
 - 1. Water sodded areas per the requirements of the grower.
- C. Mow lawns as soon as there is enough top growth to cut with mower set at indicated height. Mow as required to maintain indicated height without cutting more than 40 percent of the grass height. Remove no more than 40 percent of grass-leaf growth in initial or subsequent mowings. Do not delay mowing until grass blades bend over and become matted. Do not mow when grass is wet. Schedule initial and subsequent mowings to maintain following grass height:
 - 1. Mow grass to a finished height of 2 to 3 inches high.

3.6 SATISFACTORY LAWN

- A. Sodded lawns shall be considered satisfactory/acceptable provided requirements a healthy, well-rooted, viable lawn is established, free of open joints, bare areas and surface irregularities.
- B. Replant lawns that do not meet requirements and continue maintenance until lawns are satisfactory/acceptable.
- C. Substantial Completion of the building and the remainder of the project may be achieved (pending prior Architect and Owner approval) before achieving a satisfactory/acceptable lawn.

3.7 CLEANUP AND PROTECTION

- A. Promptly remove soil and debris created by lawn work from sidewalks and paved areas. Clean wheels of vehicles before leaving site to avoid tracking soil onto surface of roads, walks, or other paved areas.
- B. Erect barricades and warning signs as required to protect newly planted areas from traffic. Maintain barricades throughout maintenance period until lawn is established.

END OF SECTION 329223

SECTION 329300 - EXTERIOR PLANTS**PART 1 – GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes the following:

1. Trees
2. Shrubs
3. Groundcovers
4. Other Plant Materials
5. Stakes & Guys

1.3 SUBMITTALS

- A. Installers Qualifications: Provide a list, with references, of the past three projects of similar scope.
- B. Product Data: For each type of product indicated.
- C. Plant Material Certifications:
 1. Certificates of inspection as required by governmental authorities.
 2. Label data substantiating that plant materials comply with specified requirements.
- D. Planting Schedule:
 1. Typewritten planting schedule.
 2. Once accepted, revise dates only as approved in writing and submitted to Architect.
- E. Maintenance Schedules: Typewritten instructions recommending procedures for maintenance of landscape work for one full year. Submit prior to completion of project.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an experienced installer, who has successfully completed planting projects similar in size and complexity to this project. The installer's primary business (defined as a minimum of 60% of total billings) shall be exterior plant installation.
- B. Installer's Field Supervision: Installer to maintain an experienced full-time supervisor on the project site when exterior planting is in progress.
- C. Exterior Plant Materials:
 1. Provide plant materials of quantity, size, genus, species, and variety indicated on the Drawings.
 2. All plant materials and work shall comply with recommendations and requirements of ANSI Z60.1 "American Standard for Nursery Stock."
 3. Do not make substitutions. If specified landscape material is not obtainable, submit proof of non-availability to Architect, together with proposal for use of equivalent material.
 4. The Architect may inspect plant materials either at place of growth or on site before planting, for compliance with requirements for genus, species, variety, size, and quality. Architect retains right to further inspect trees for size and condition of balls and root systems, insects, injuries and latent defects, and to reject unsatisfactory or defective

material at any time during progress of work. Remove rejected trees immediately from project site.

- D. Preinstallation Conference: Conduct conference at Project site to comply with requirements in Division 1 Section "Project Management and Coordination."

1.5 DELIVERY, STORAGE AND HANDLING

A. Packaged Materials:

1. Deliver packaged materials in containers showing weight, analysis, and name of manufacturer or grower.
2. Protect materials from deterioration during delivery, and while stored at site.

B. Exterior Plant Materials

1. Protect bark, branches, and root systems from sun scald, drying, sweating, whipping, and other handling and tying damage. Do not bend or bind-tie trees or shrubs in such a manner as to destroy their natural shape. Provide protective covering of exterior plants during delivery. Do not drop exterior plants during delivery.
2. Deliver exterior plant materials after preparations for planting have been completed and plant immediately. If planting is delayed more than 6 hours after delivery, set plant materials in shade, protect from weather and mechanical damage, and keep roots moist and free from frost.
3. Do not remove container-grown stock from containers until planting time.
4. Balled and burlapped material shall be freshly dug.
5. Handle planting stock by root ball.

1.6 PROJECT CONDITIONS

- A. Examine the subgrade, verify the elevations, and observe the conditions under which work is to be performed. Do not proceed with the work until unsatisfactory conditions have been corrected in a manner acceptable to the installer.
- B. Determine location of underground utilities and perform work in a manner which will avoid possible damage. Hand excavate as required.
- C. When conditions detrimental to plant growth are encountered, such as rubble fill, adverse drainage conditions, or obstructions, notify Architect before planting.
- D. Provide all necessary safeguards for the protection of all planted areas until provisional inspection/acceptance is accomplished.
- E. Planting Restrictions: Plant during one of the following periods.
 1. Spring Planting: Unfrozen soil conditions March 1-June 1st.
 2. Fall Planting: September 1-November 1st or until frozen soil conditions prevent work.
 3. Summer Planting: June 1 – September 1 with approved irrigation system.
- F. Coordination with Lawns: Install plant materials after finish grades are established and before planting lawns, unless otherwise acceptable to the Architect.
 1. When planting exterior plants after lawns, protect lawn areas and promptly repair damage caused by planting operations.

1.7 WARRANTY

- A. Warranty exterior plant materials for a period of one year after date of Final Completion against defects including death and unsatisfactory growth, except for defects resulting from neglect by

Owner, abuse or damage by others, or unusual phenomena or incidents which are beyond Contractor's control.

1. The Contractor shall provide written notice to the Architect of any practice which will affect the warranty if not remedied promptly. The Architect will render an opinion of the conflict if necessary.
2. Make replacements of all dead plants or plants in impaired condition (more than 25% dead or dying) condition in early spring/fall following installation. Replacements of dead or rejected plants should again be made prior to the expiration of the warranty period.

1.8 MAINTENANCE

- A. The Owner is responsible for maintaining all exterior plant material throughout the warranty period according to the submitted Maintenance Schedule.
- B. Remove all stakes and guy wires at the end of the 12 month guarantee period.

PART 2 – PRODUCTS

2.1 EXTERIOR PLANT MATERIALS

- A. General: Provide nursery-grown plant materials complying with ANSI Z60.1, with healthy root systems developed by transplanting or root pruning. Provide well-shaped, fully branched, healthy, vigorous stock free of disease, insects, eggs, larvae, and defects such as knots, sun scald, injuries, abrasions, and disfigurement.
- B. Label at least one tree and one shrub of each variety and caliper with a securely attached, waterproof tag bearing legible designation of botanical and common name.

2.2 PLANTS

- A. Annuals: Provide healthy, disease-free plants of species and variety indicated. Provide only plants that are acclimated to outdoor conditions before delivery and that are in bud but not yet in bloom
- B. Perennials: Provide healthy, field-grown plants from a commercial nursery of species and variety shown or listed.
- C. Vines: Provide plants with heavy, well-branched tops, with not less than three runners and a vigorous well-developed root system.

2.3 FERTILIZER

- A. Commercial Fertilizer: Commercial-grade complete fertilizer of neutral character, consisting of fast- and slow-release nitrogen, 50 percent derived from natural organic sources of urea formaldehyde, phosphorous, and potassium. Revise fertilizer mix to remedy deficiencies found in soil.
 1. Composition: 1 lb/1000 sq. ft. (0.45 kg/92.9 sq. m. of actual nitrogen, 4 percent phosphorous, and 2 percent potassium, by weight.
 2. Composition: Nitrogen, phosphorous, and potassium in amounts recommended in soil reports from a qualified soil-testing agency.
- B. Slow-Release Fertilizer: Granular or pelleted fertilizer consisting of 50 percent water-insoluble nitrogen, phosphorus, and potassium. Revise fertilizer mix to remedy deficiencies found in soil.
 1. Composition: 20 percent nitrogen, 10 percent phosphorous, and 10 percent potassium, by weight.
 2. Composition: Nitrogen, phosphorous, and potassium in amounts recommended in soil reports from a qualified soil-testing agency.

2.4 MULCHES

1. Organic Mulch: Six (6) month old well rotted double shredded native hardwood bark mulch not larger than 4" in length and 1/2" in width, free of woodchips and sawdust.

2.5 WATER

1. Free of substances harmful to plant growth.

2.6 TOPSOIL

- A. Topsoil: ASTM D 5268, pH range of 5.5 to 7, a minimum of 4 percent organic material content. Topsoil shall be fertile, friable, natural topsoil of loamy character, without admixture of subsoil material, obtained from a well-drained arable site, reasonably free from clay, lumps, coarse sands, stones, plants, roots, sticks and other foreign materials.
- B. Topsoil Source:
 1. Reuse surface soil stockpiled on-site. Verify suitability of stockpiled surface soil to produce topsoil. Clean surface soil of roots, plants, sod, stones, clay lumps, and other extraneous materials harmful to plant growth.
 - a) Supplement with imported or manufactured topsoil from off-site sources when quantities are insufficient. Obtain topsoil displaced from naturally well-drained sites where topsoil occurs at least 4 inches (100 mm) deep; do not obtain from agricultural land, bogs or marshes.
 2. Import topsoil or manufactured topsoil from off-site sources. Obtain topsoil displaced from naturally well-drained sites where topsoil occurs at least 4 inches (100 mm) deep; do not obtain from agricultural land, bogs or marshes.
 3. Amend existing in-place surface soil to produce topsoil. Verify suitability of surface soil to produce topsoil. Clean surface soil of roots, plants, sod, stones, clay lumps, and other extraneous materials harmful to plant growth.
 - a) Surface soil may be supplemented with imported or manufactured topsoil from off-site sources. Obtain topsoil displaced from naturally well-drained sites where topsoil occurs at least 4 inches (100 mm) deep; do not obtain from agricultural land, bogs or marshes.

2.7 INORGANIC SOIL AMENDMENTS

- A. Lime: ASTM C 602, agricultural limestone containing a minimum 80 percent calcium carbonate equivalent and as follows:
 1. Class: Class T, with a minimum 99 percent passing through No. 8 (2.36-mm) sieve and a minimum 75 percent passing through No. 60 (0.25-mm) sieve.
 2. Class: Class O, with a minimum 95 percent passing through No. 8 (2.36-mm) sieve and a minimum 55 percent passing through No. 60 (0.25-mm) sieve.
 3. Provide lime in form of dolomitic limestone.
- B. Sulfur: Granular, biodegradable, containing a minimum of 90 percent sulfur, with a minimum 99 percent passing through No. 6 (3.35-mm) sieve and a maximum 10 percent passing through No. 40 (0.425-mm) sieve.
- C. Iron Sulfate: Granulated ferrous sulfate containing a minimum of 20 percent iron and 10 percent sulfur.
- D. Aluminum Sulfate: Commercial grade, unadulterated.
- E. Perlite: Horticultural perlite, soil amendment grade.
- F. Agricultural Gypsum: Finely ground, containing a minimum of 90 percent calcium sulfate.
- G. Sand: Clean, washed, natural or manufactured, free of toxic materials.

- H. Diatomaceous Earth: Calcined, diatomaceous earth, 90 percent silica, with approximately 140 percent water absorption capacity by weight.
- I. Zeolites: Mineral clinoptilolite with at least 60 percent water absorption by weight.

2.8 ORGANIC SOIL AMENDMENTS

- A. Compost: Well-composted, stable, and weed-free organic matter, pH range of 5.5 to 8; moisture content 35 to 55 percent by weight; 100 percent passing through 3/4-inch (19-mm) sieve; soluble salt content of 5 to 10 decisiemens/m; not exceeding 0.5 percent inert contaminants and free of substances toxic to plantings; and as follows:
 - 1. Organic Matter Content: 50 to 60 percent of dry weight.
 - 2. Feedstock: Agricultural, food, or industrial residuals; bio-solids; yard trimmings; or source-separated or compostable mixed solid waste.
- B. Sphagnum peat moss: Sphagnum peat moss shall be partially decomposed, finely divided or granular texture, with a pH range of 3.4 to 4.8.
- C. Peat: Finely divided or granular texture, with a pH range of 6 to 7.5, containing partially decomposed moss peat, native peat, or reed-sedge peat and having a water-absorbing capacity of 1100 to 2000 percent.
- D. Wood Derivatives: Decomposed, nitrogen-treated sawdust, ground bark, or wood waste; of uniform texture, free of chips, stones, sticks, soil, or toxic materials.
 - 1. In lieu of decomposed wood derivatives, mix partially decomposed wood derivatives with at least 0.15 lb (2.4 kg) of ammonium nitrate or 0.25 lb (4 kg) of ammonium sulfate per cubic foot (cubic meter) of loose sawdust or ground bark.
- E. Manure: Well-rotted, unleached, poultry, stable or cattle manure containing not more than 25 percent by volume of straw, sawdust, or other bedding materials; free of toxic substances, stones, sticks, soil, weed seed, and material harmful to plant growth.

2.9 MISCELLANEOUS PRODUCTS

- A. Antidesiccant: Water-insoluble emulsion, permeable moisture retarder, film forming, for trees and shrubs. Deliver in original, sealed, and fully labeled containers and mix according to manufacturer's written instructions.

PART 3 – EXECUTION

3.1 EXAMINATION

- A. Examine areas to receive exterior plants for compliance with requirements and conditions affecting installation and performance. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Tree save areas as indicated shall be tagged and approved by the Architect prior to any clearing and/or thinning.
- B. Protect structures, utilities, sidewalks, pavements, and other facilities, and lawns and existing exterior plants from damage caused by planting operations.
- C. Provide erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.
- D. Lay out individual tree and shrub locations and areas for multiple exterior plantings. Stake locations, outline areas, adjust locations when requested, and obtain Landscape Architect's acceptance of layout before planting. Make minor adjustments as required.

- E. Lay out exterior plants at locations indicated. Stake locations of individual trees and shrubs and outline areas for multiple plantings.
- F. Apply antidesiccant to trees and shrubs using power spray to provide an adequate film over trunks, branches, stems, twigs, and foliage to protect during digging, handling, and transportation.
 - 1. If deciduous trees or shrubs are moved in full leaf, spray with antidesiccant at nursery before moving and again two weeks after planting.

3.3 PLANTING BED ESTABLISHMENT

- A. Loosen subgrade of planting beds to a minimum depth of 4 inches (100 mm). Remove stones larger than 1 inch (25 mm) in any dimension and sticks, roots, rubbish, and other extraneous matter and legally dispose of them off of Owner's property.
 - 1. Apply fertilizer directly to subgrade before loosening.
 - 2. Spread topsoil, apply soil amendments and fertilizer on surface, and thoroughly blend planting soil mix.
 - a) Delay mixing fertilizer with planting soil if planting will not proceed within a few days.
 - b) Mix lime with dry soil before mixing fertilizer.
- B. Finish Grading: Grade planting beds to a smooth, uniform surface plane with loose, uniformly fine texture. Roll and rake, remove ridges, and fill depressions to meet finish grades.
- C. Restore planting beds if eroded or otherwise disturbed after finish grading and before planting.

3.4 TREE AND SHRUB PLANTING

- A. Set all plant materials plumb and in center of pit or trench as per detail.
 - 1. Remove burlap and wire baskets from tops of root balls and partially from sides, but do not remove from under root balls. Remove pallets, if any, before setting. Do not use planting stock if root ball is cracked or broken before or during planting operation.
 - 2. Carefully remove root ball from container without damaging root ball or plant.
 - 3. Place planting soil mix around root ball in layers, tamping to settle mix and eliminate voids and air pockets. When pit is approximately one-half backfilled, water thoroughly before placing remainder of backfill. Repeat watering until no more water is absorbed. Water again after placing and tamping final layer of planting soil mix.
 - 4. Spread roots without tangling or turning toward surface, and carefully work backfill around roots by hand. Puddle with water until backfill layers are completely saturated. Plumb before backfilling and maintain plumb while working backfill around roots and placing layers above roots. Tamp final layer of backfill. Remove injured roots by cutting cleanly, do not break.
 - 5. Dish top of backfill to allow for mulching.
- B. Organic Mulching: Apply 3-inch (75-mm.) average thickness of organic mulch extending 12 inches (300 mm) beyond edge of planting pit or trench. Do not place mulch within 3 inches (75 mm) of trunks or stems.

3.5 TREE AND SHRUB PRUNING

- A. Prune, thin, and shape trees and shrubs as indicated.

3.6 GROUND COVER AND PLANT PLANTING

- A. Set out and space ground cover and plants as indicated in details.
- B. Water thoroughly after planting, taking care not to cover plant crowns with wet soil.

3.7 CLEANUP AND PROTECTION

- A. During exterior planting, keep adjacent pavings and construction clean and work area in an orderly condition.
- B. Protect exterior plants from damage due to landscape operations, operations by other contractors and trades, and others. Maintain protection during installation and maintenance periods. Treat, repair, or replace damaged exterior planting.

3.8 DISPOSAL

- A. Disposal: Remove surplus soil and waste material, including excess subsoil, unsuitable soil, trash, and debris, and legally dispose of them off Owner's property.

END OF SECTION 329300

SECTION 331000 - EXTERIOR WATER SYSTEM**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This section includes water service piping, fire protection service mains and appurtenances from the source of water to a point 5 feet outside the building.

1.3 SUBMITTALS

- A. Product data for piping, valves, vaults, fire hydrants, and identification devices.

1.4 QUALITY ASSURANCE

- A. Comply with local utility department and fire department standards pertaining to materials, meter boxes, hose threads and installation.
- B. Comply with the requirements of the Authority Having Jurisdiction (AHJ)

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Preparation for Transport: Prepare valves, including fire hydrants, for shipping as follows:
 - 1. Ensure valves are dry and internally protected against rust and corrosion.
 - 2. Protect valves against damage to threaded ends, flange faces, and weld ends.
 - 3. Set valves in best position for handling. Set gate valves and fire hydrants closed to prevent rattling.
- B. Storage: Use the following precautions for valves, including fire hydrants, during storage:
 - 1. Do not remove end protectors unless necessary for inspection; then reinstall for storage.
 - 2. Protect valves from weather. Store valves indoors. Maintain valve temperature higher than the ambient dew point temperature. If outdoor storage is necessary, support valves off the ground or pavement in watertight enclosures.
- C. Handling: Use a sling to handle valves, including fire hydrants, whose size requires handling by crane or lift. Rig valves to avoid damage to exposed valve parts. Do not use handwheels or stems as lifting or rigging points.
- D. Deliver piping with factory-applied end caps. Maintain end caps through shipping, storage, and handling to prevent pipe-end damage and to prevent entrance of dirt, debris, and moisture.
- E. Protect stored piping from moisture and dirt. Elevate above grade. Do not exceed structural capacity of floor when storing inside.
- F. Protect flanges, fittings, and specialties from moisture and dirt.
- G. Store plastic piping protected from direct sunlight. Support to prevent sagging and bending.

1.6 PROJECT CONDITIONS

- A. Site Information: Perform site surveys, research public utility records, and verify existing utility locations. Verify that exterior water system may be installed in compliance with the original design and referenced standards. Notify Architect immediately of any discrepancies.
- B. Coordinate connection to the existing water service with

1.7 SEQUENCING AND SCHEDULING

- A. Coordinate with interior water piping and interior fire protection piping.

- B. Coordinate with other utility work.
- C. Do not interrupt utilities serving facilities occupied by Owner or others unless permitted under the following conditions:
 - 1. Notify Architect not less than two days in advance of the proposed interruption.
 - 2. Do not proceed with the interruption without Architects written permission.
 - 3. Provide temporary utility service to the facility.

PART 2 - PRODUCTS

2.1 GENERAL

- A. All piping, valves, fittings, fire hydrants, meters, meter vaults, appurtenances and other products shall conform to the requirements of the AHJ.

PART 3 - EXECUTION

3.1 GENERAL

- A. Installation of the exterior water system shall comply with the requirements of the AHJ.

3.2 PREPARATION OF BURIED PIPE FOUNDATION

- A. Grade trench bottom to provide a smooth, firm, stable, and rock-free foundation throughout the length of the pipe.
- B. Remove unstable, soft, and unsuitable materials at the surface upon which pipes are to be laid and backfill with clean sand or pea gravel to indicated level.
- C. Shape bottom of trench to fit bottom of pipe. Fill unevenness with tamped sand backfill. Dig bell holes at each pipe joint to relieve the bells of all loads and to ensure continuous bearing of the pipe barrel on the foundation.

3.3 PIPE AND PIPE FITTINGS INSTALLATION

- A. Depth of Cover: Provide minimum cover over piping of 18 inches below average local frost depth or 42 inches below finished grade, whichever is greater.
- B. Water Service Termination: Terminate water service piping 5'-0" from building foundation in location and invert as indicated. Coordinate location with interior water piping and interior fire service piping. Provide temporary pipe plug for piping extension into building.

3.4 IDENTIFICATION INSTALLATION

- A. Install underground warning tape and tracing wire for underground water service piping, as required by the AHJ.

3.5 FIELD QUALITY CONTROL

- A. Testing and Disinfection: Disinfect, flush and test in accordance with the requirements of the AHJ.
- B. Schedule tests, and their inspections by authorities having jurisdiction, with at least 24 hours' advance notice

END OF SECTION 331000

SECTION 333000 - SANITARY SEWERAGE**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes sanitary sewerage system piping and appurtenances from a point 5 feet outside the building to the point of disposal.

1.3 SUBMITTALS

- A. Product data for sewer piping specialties.
- B. Shop drawings for precast concrete sanitary manholes, including frames and covers.
- C. Shop drawings for cast-in-place concrete or field-erected masonry sanitary manholes, including frames and covers.
- D. Inspection and test reports specified in the "Field Quality Control" Article

1.4 QUALITY ASSURANCE

- A. Environmental Compliance: Comply with applicable portions of local environmental agency regulations pertaining to sanitary sewerage systems, and to the requirements of the State Sediment Control Handbook for erosion control during installation.
- B. Utility Compliance: Comply with the requirements of Authority Having Jurisdiction.

1.5 DELIVERY, STORAGE, AND HANDLING

- 1. Do not store plastic structures in direct sunlight.
- 2. Do not store plastic pipe or fittings in direct sunlight.
- 3. Protect pipe, pipe fittings, and seals from dirt and damage.
- 4. Handle precast concrete manholes and other structures according to manufacturer's rigging instructions.

1.6 PROJECT CONDITIONS

- A. Site Information: Perform site surveys, research public utility records, and verify existing utility locations. Verify that sanitary sewerage system piping may be installed in compliance with original design and referenced standards.
- B. Locate existing structures and piping to be closed and abandoned.
- C. Existing Utilities: Do not interrupt existing utilities serving facilities occupied by the Owner or others except when permitted under the following conditions and then only after arranging to provide acceptable temporary utility services.
 - 1. Notify Architect not less than 48 hours in advance of proposed utility interruptions.
 - 2. Do not proceed with utility interruptions without receiving Architect's written permission.

1.7 SEQUENCING AND SCHEDULING

- A. Coordinate with interior building sanitary drainage piping.
- B. Coordinate with other utility work.

PART 2 - PRODUCTS**2.1 PIPE AND FITTINGS**

- A. General: Provide pipe and pipe fitting materials compatible with each other. All materials shall comply with the requirements of the Authority Having Jurisdiction.
- B. PVC (Polyvinyl Chloride) Sewer Pipe and Fittings: ASTM D 3034, SDR 35, elastomeric gasket joints.
 - 1. Gaskets: ASTM F 477, elastomeric seal.
- C. Ductile-Iron Pipe: AWWA C151, Class 50 minimum, for push-on joints.
 - 1. Standard-Pattern, Ductile-Iron and Cast-Iron Fittings: AWWA C110, for push-on joints.
 - 2. Compact-Pattern, Ductile-Iron Fittings: AWWA C153, for push-on joints.
 - 3. Pipe and Fitting Interior Coating: AWWA C104, asphaltic-material seal coat, minimum 1-mil (0.025-mm) thickness.
 - 4. Gaskets: AWWA C111, rubber.

2.2 MANHOLES

- A. Precast Concrete Manholes: Shall comply with the requirements of the Authority Having Jurisdiction.
- B. Manhole Steps Shall comply with the requirements of the Authority Having Jurisdiction.
- C. Manhole Frames and Covers: Shall comply with the requirements of the Authority Having Jurisdiction.

2.3 CLEANOUTS

- A. General: Provide cast-iron ferrule and countersunk brass cleanout plug, with round cast-iron access frame and heavy-duty, secured, scoriated cast-iron cover. Shall comply with the requirements of the Authority Having Jurisdiction
- B. Sewer pipe fitting and riser to cleanout: 4" dia. ductile iron pipe. Shall comply with the requirements of the Authority Having Jurisdiction

PART 3 - EXECUTION

3.1 PREPARATION OF FOUNDATION FOR BURIED SANITARY SEWERAGE SYSTEMS

- A. Grade trench bottom to provide a smooth, firm, stable, and rock-free foundation, throughout the length of the pipe.
- B. Remove unstable, soft, and unsuitable materials at the surface upon which pipes are to be laid, and backfill with clean sand or pea gravel to indicated level.
- C. Install pipe bedding as required by the Authority Having Jurisdiction.

3.2 PIPE INSTALLATION

- A. Install the sanitary sewerage system in accordance with the requirements of the Authority Having Jurisdiction.
- B. Install piping beginning at low point of systems, true to grades and alignment indicated with unbroken continuity of invert. Place bell ends of piping facing upstream. Install gaskets, seals, sleeves, and couplings in accordance with manufacturer's recommendations for use of lubricants, cements, and other installation requirements. Maintain swab or drag in line and pull past each joint as it is completed.
- C. Use manholes for changes in direction, except where a fitting is indicated. Use fittings for branch connections, except where direct tap into existing sewer is indicated.
- D. Use proper size increasers, reducers, and couplings, where different size or material of pipes and fittings are connected. Reduction of the size of piping in the direction of flow is prohibited.
- E. Install piping pitched down in direction of flow, at minimum slope of 2 percent, except where indicated otherwise.

- F. Extend sanitary sewerage system piping to connect to building sanitary drains, of sizes and in locations indicated.

3.3 PIPE JOINT CONSTRUCTION AND INSTALLATION

- A. Join and install PVC pipe as follows:
 - 1. Pipe and gasketed fittings, joining with elastomeric seals, in accordance with ASTM D 3212.
 - 2. Installation in accordance with ASTM D 2321.
- B. Join different types of pipe with standard manufactured couplings and fittings intended for that purpose.

3.4 MANHOLES

- A. Install precast concrete manholes in accordance with the Authority Having Jurisdiction requirements.
- B. Provide rubber joint gasket complying with ASTM C 443 at joints of sections.
- C. Apply bituminous mastic coating at joints of sections.

3.5 CLEANOUTS

- A. Provide cleanouts as indicated, and as required by authorities having jurisdiction. Set cleanout frame and cover in concrete block 12 by 12 by 6 inches deep, except where location is in concrete paving. Set top of cleanout 1 inch above surrounding earth grades or flush with grade when installed in paving.

3.6 FIELD QUALITY CONTROL

- A. Cleaning: Clear interior of piping and structures of dirt and other superfluous material as work progresses. Maintain swab or drag in piping and pull past each joint as it is completed.
 - 1. In large, accessible piping, brushes and brooms may be used for cleaning.
 - 2. Place plugs in ends of uncompleted pipe at end of day or whenever work stops.
 - 3. Flush piping between manholes, if required by local authority, to remove collected debris.
- B. Interior Inspection: Inspect piping to determine whether line displacement or other damage has occurred.
 - 1. Make inspections after pipe between manholes and manhole locations has been installed and approximately 2 feet of backfill is in place, and again at completion of project.
 - 2. If inspection indicates poor alignment, debris, displaced pipe, infiltration or other defects correct such defects, and reinspect.
- C. Test new piping systems and parts of existing systems that have been altered, extended, or repaired for leaks and defects.
 - 1. Do not enclose, cover, or put into service before inspection and approval.
 - 2. Test completed piping systems according to the requirements of the Authority Having Jurisdiction.
 - 3. Schedule tests, and their inspections by authorities having jurisdiction, with at least 24 hours' advance notice.
 - 4. Submit separate reports for each test.

END OF SECTION 333000

SECTION 334100 - STORM DRAINAGE**PART 1 - GENERAL****1.1 RELATED DOCUMENTS:**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY:

- A. This Section includes the roof drainage collection system, the storm sewerage system piping and appurtenances from a point 5 feet outside the building to the point of disposal.

1.3 SUBMITTALS

- A. Product data for:

1. Concrete pipe
2. Polyethylene pipe
3. Ductile iron pipe
4. Frames and covers.
5. Grates

6. Couplings for connection into concrete pipe.

- B. Certification, signed by material producer and contractor, that standard precast and cast in place concrete storm drainage manholes and Drop Inlets comply with Authority Having Jurisdiction requirements.

- C. Authority Having Jurisdiction approved job mix for bedding stone.

- D. Shop drawings for:

1. Non-standard precast or cast-in-place concrete storm drainage manholes and Drop Inlets.
2. Trench drain system.
3. Cleanouts
4. Underdrains
5. Stormwater Management Basin Outlet structures, including: Riser pipe, outfall pipe, riser anchoring, anti-seep collars, trash rack and anti-vortex device.

- E. Record drawings of installed storm drainage system.

1.4 QUALITY ASSURANCE

- A. Environmental Compliance: Comply with applicable portions of local environmental agency regulations pertaining to storm sewerage systems.

- B. Utility Compliance: Comply with state and local regulations and standards pertaining to storm sewerage systems.

- C. All materials shall be new and free of defects (i.e. pipe shall not have chipped spigots or bells).

1.5 PROJECT CONDITIONS

- A. Site Information: Perform site surveys, research public utility records, and verify existing utility locations. Verify that storm sewerage system piping may be installed in compliance with original design and referenced standards.

- B. Locate existing structures and piping to be closed and abandoned.

- C. Existing Utilities: Do not interrupt existing storm sewer serving facilities occupied by the Owner of others except when permitted under the following conditions and then only after arranging to provide acceptable temporary storm sewer services.
 - 1. Notify Architect/Engineer not less than 48 hours in advance of proposed storm sewer interruptions.
 - 2. Do not proceed with storm sewer interruptions without receiving Architect's written permission.
- D. Existing utilities across or along the line of work are indicated only in an approximate location. Locate all underground lines and structures. Call "Miss Utility" prior to construction. If utilities are marked that are not shown on the plans, locate utility vertically and horizontally and provide information to architect.

1.6 SEQUENCING AND SCHEDULING

- A. Coordinate with interior building storm drainage piping.
- B. Coordinate with other utility work.

PART 2 - PRODUCTS

2.1 GENERAL

- A. All materials used for construction of the storm sewerage system shall comply with the requirements of the Authority Having Jurisdiction

2.2 PIPE AND FITTINGS

- A. Provide pipe and pipe fitting materials compatible with each other. Pipe materials are indicated on the drawings.
- B. Reinforced Concrete Pipe (RCP): Shall conform to the requirements of ASTM C76/AASHTO M170, Class III, unless otherwise indicated. All RCP shall be gasketed.
- C. O-Ring Gasket Reinforced Concrete Pipe: Shall conform to the requirements of ASTM C76/AASHTO M170, Class III, unless otherwise indicated. Joints shall conform to the requirements of ASTM C443/AASHTO M198.
- D. Corrugated Polyethylene Pipe (P.E.): Shall have a smooth lined interior and meet the requirements of ASTM F405 or AASHTO M252 for 10" diameter and smaller, and ASTM F667 or AASHTO M294 for 12" diameter and larger.
- E. PVC Storm Sewer Pipe: Shall conform to the requirements of ASTM D3034, SDR-35 with bell and spigot ends for gasketed joints with ASTM F 477 elastometric seals
 - a) Connections to the building downspouts shall be made with Schedule 40 PVC.
- F. Ductile Iron Storm Sewer Pipe: Shall conform to the requirements of AWWA C151, Class 52. Flanged joints shall conform to the requirements of AWWA C115.

2.3 MANHOLES

- A. Precast Concrete Manholes: Comply with the requirements of the Authority Having Jurisdiction.
- B. Cast-in-Place Manholes: Comply with the requirements of the Authority Having Jurisdiction.
- C. Manhole Steps, Safety Slabs and Inlet Shaping: Comply with the requirements of the Authority Having Jurisdiction.
- D. Manhole Frames and Covers: Comply with the requirements of the latest edition of the Authority Having Jurisdiction

2.4 CLEANOUTS

- A. Cast-iron ferrule and countersunk brass cleanout plug, with round cast-iron access frame and heavy-duty, secured, scoriated cast-iron cover.

2.5 DROP INLETS

- A. Precast Concrete Drop Inlets: Comply with the requirements of the Authority Having Jurisdiction.
- B. Cast-in-Place Drop Inlets: Comply with the requirements of the Authority Having Jurisdiction.
- C. Drop Inlet Steps, Safety Slabs and Inlet Shaping: Comply with the requirements of the Authority Having Jurisdiction.
- D. Drop Inlet Frames and Grates: Comply with the requirements of the latest edition of the Authority Having Jurisdiction.
- E. Plastic Drain Basins: **ADS 2812AG** or approved equal.

2.6 TRENCH DRAIN SYSTEM

- A. **Trench drainage system shall be manufactured by Zurn Industries, or approved equal.**
- B. **Trench drain grates shall be galvanized steel, heelproof grates and shall be reinforced to support heavy duty (H20) loads , unless otherwise indicated on the plans.**
- C. Provide all fittings and miscellaneous connections necessary for a complete the trench drainage system per the manufacturer requirements.

2.7 CONCRETE AND REINFORCEMENT

- A. **Concrete:** Conform to the requirements of the Authority Having Jurisdiction. **ASTM .**
- B. Reinforcement: Steel conforming to the following:
 - 1. Fabric: ASTM A 185 welded wire fabric, plain.
 - 2. Reinforcement Bars: ASTM A 615, Grade 60, deformed.

2.8 UNDERDRAINS

- A. Underdrains and combination underdrains: Conform to the requirements of the Authority Having Jurisdiction, unless otherwise indicated.
 - 1. PVC underdrains shall conform to the requirements of ASTM F758, Type PS 28 or ASTM F949.
 - 2. PE corrugated underdrain pipe shall conform to AASHTO M252.
- B. Provide a filter fabric "sock" wrapping for all underdrain pipe.

2.9 END WALLS AND END SECTIONS

- A. End walls: Conform to the requirements of the Authority Having Jurisdiction.
- B. End sections: Conform to the requirements of the Authority Having Jurisdiction for the size of pipe indicated.

PART 3 - EXECUTION

3.1 GENERAL

- A. Install the storm sewerage system in accordance with Authority Having Jurisdiction requirements.

3.2 PREPARATION OF FOUNDATION FOR BURIED STORM SEWERAGE SYSTEMS

- A. Grade trench bottom to provide a smooth, firm, stable, and rock-free foundation, throughout the length of the pipe.
- B. Remove unstable, soft, and unsuitable materials at the surface upon which pipes are to be laid, and backfill with clean sand or pea gravel to indicated level.

- C. Install pipe bedding conforming to the requirements of the Authority Having Jurisdiction.

3.3 PIPE INSTALLATION

- A. Install piping beginning at low point of systems, true to grades and alignment indicated with unbroken continuity of invert. Place bell ends of piping facing upstream. Install gaskets, seals, sleeves, and couplings in accordance with manufacturer's recommendations for use of lubricants, cements, and other installation requirements. Maintain swab or drag in line and pull past each joint as it is completed.
- B. Use proper size increasers, reducers, and couplings, where different size or material of pipes and fittings are connected. Reduction of the size of piping in the direction of flow is prohibited.
- C. Extend storm sewerage system piping to connect to building storm drains, of sizes and in locations indicated.
- D. Join and install concrete pipe and fittings per Authority Having Jurisdiction or State DOT specifications.
- E. Join and install PE pipe and fittings per manufacturer's recommendations.
- F. Join different types of pipe with standard manufactured couplings and fittings intended for that purpose.

3.4 MANHOLES

- A. General: Install manholes complete with accessories as indicated. Form continuous concrete or split pipe section channel and benches between inlets and outlet. Set tops of frames and covers flush with finish grade, unless otherwise indicated.
- B. Place precast concrete manhole sections as indicated, and install in accordance with ASTM C 891.
- C. Construct cast-in-place manholes as indicated.
- D. Apply bituminous mastic coating at joints of sections.

3.5 CLEANOUTS

- A. Install cleanouts and extension from sewer pipe to cleanout at grade as indicated. Set cleanout frame and cover in concrete block 12 by 12 by 6 inches deep or as indicated on the plans, except where location is in concrete paving. Set top of cleanout flush with finish grade.

3.6 DROP INLETS

- A. Construct drop inlets to sizes and shapes indicated.
- B. Set frames and grates to elevations indicated.

3.7 INLET SHAPING

- A. Construct inlet shaping conforming to Authority Having Jurisdiction Standards at all inlets and manholes.

3.8 TRENCH DRAIN INSTALLATION

- A. Installation of the trench drain shall comply with the manufacturer's recommendations.
- B. Verify connection to the storm sewer system. Utilize manufacturers standard outlet connections to make connection to the storm sewer system.
- C. Install trench drain system starting from the downstream end, working towards the upstream end.
- D. Verify proper placement and alignment prior to placement of concrete.
- E. Place concrete around suspended trench channel. Do not chute concrete directly against channel walls, as this may cause displacement. Work concrete under channels and vibrate with a finger-type vibrator.

F. Finish surface to be flush with the adjoining surfaces and to allow for positive drainage into the grates.

G. Install grate tops.

3.9 FIELD QUALITY CONTROL

A. Cleaning: Clear interior of piping and structures of dirt and other superfluous material as work progresses. Maintain swab or drag in piping and pull past each joint as it is completed.

1. In large, accessible piping, brushes and brooms may be used for cleaning.
2. Place plugs in ends of uncompleted pipe at end of day or whenever work stops.
3. Flush piping between manholes and drop inlets to remove collected debris. Flush pipes through an approved erosion and sediment control measure.

B. Interior Inspection: Inspect piping to determine whether line displacement or other damage has occurred.

1. Make inspections after pipe between manholes and manhole locations has been installed and approximately 2 feet of backfill is in place, and again at completion of project.
2. If inspection indicates poor alignment, debris, displaced pipe, infiltration, or other defects correct such defects and reinspect.

END OF SECTION 334100

SECTION 334613 - FOUNDATION DRAINAGE SYSTEMS**PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes underground foundation drainage systems, as follows:
 - 1. Footing drainage system.

1.3 SUBMITTALS

- A. Product data for each type of foundation drainage material required.
- B. Shop drawings for interfacing with total site drainage system.
- C. Certification signed by Contractor and foundation drainage system Installer that installed materials conform to requirements and system was successfully checked and tested prior to covering with filter fabric and drainage fill.

PART 2- PRODUCTS**2.1 DRAINAGE PIPE AND FITTINGS**

- A. Provide drainage pipes complete with bends, reducers, adapters, couplings, collars, and joint materials.
 - 1. Perforated Polyvinyl Chloride Pipe: ASTM D 2729.
 - 2. Perforated Polyethylene Pipe: ASTM F405.

2.2 SOIL & MISCELLANEOUS MATERIALS

- A. General: Bedding, backfill, and porous fill materials are indicated in Section 31 2000, Earthwork.
- B. Filter Fabric: Manufacturer's standard non-woven geotextile fabric of polypropylene or polyester fibers, or a combination thereof.

PART 3 - EXECUTION**3.1 INSPECTION**

- A. Examine adjacent surfaces to receive foundation drainage system to verify suitability. Do not begin installation until subsurface conditions are satisfactory to accept drainage system.

3.2 INSTALLATION

- A. Inspect pipe trench for line, level (or slope as required) and pipe support conditions.
- B. Laying Drain Pipe: Lay drain pipe solidly bedded. Provide full bearing for each pipe section throughout its length to true grades and alignment, and continuous slope in direction of flow.
 - 1. Lay perforated pipe with perforations down and joints tightly closed in accordance with pipe manufacturer's recommendations. Provide collars and couplings as required.
 - 2. Provide recesses in excavation bottom to receive bells for drain pipe having bell and spigot ends. Lay pipe with bells facing up slope and with spigot end entered fully into adjacent bell. Seal joint in accordance with requirements of local building official.
- C. Testing Drain Lines: Test and check lines before backfilling to assure free flow. Remove obstructions, replace damaged components, and retest system until satisfactory.
- D. Drainage Fill: Place drainage fill (material indicated in Section 02300) over drain lines after satisfactory testing. Completely cover drain lines to a width of at least 6 inches on each side and

above top of pipe to within 12 inches of finish grade. Place fill material in layers not exceeding 3 inches in loose depth and compact each layer placed.

1. Overlay drainage fill material with one layer of filter fabric, overlapping edges at least 4 inches.
- E. Fill to Grade: Install fill material over compacted drainage fill at footing drains, placing material in layers not exceeding 6 inches in loose depth and thoroughly compacting each layer, as indicated in Section 02300.

END OF SECTION 334613